

MTOS CRM — Full Source Code

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420 files

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lib/integrations-anthropic-ai/src/index.ts
lib/integrations-anthropic-ai/tsconfig.json
lib/integrations/anthropic_ai_integrations/src/server/batch/index.ts
lib/integrations/anthropic_ai_integrations/src/server/batch/utls.ts
package.json
pnpm-workspace.yaml
replit.md

artifacts/api-server/.replit-artifact/artifact.toml

```
kind = "api"
previewPath = "/api" # TODO - should be excluded from preview in the first place
title = "API Server"
version = "1.0.0"
id = "3B4_FFSkEVBkAeYMFRJ2e"

[[services]]
localPort = 8080
name = "API Server"
paths = ["/api"]

[services.development]
run = "pnpm --filter @workspace/api-server run dev"

[services.production]

[services.production.build]
args = ["pnpm", "--filter", "@workspace/api-server", "run", "build"]

[services.production.build.env]
NODE_ENV = "production"

[services.production.run]
# we don't run through pnpm to make startup faster in production
args = ["node", "--enable-source-maps", "artifacts/api-server/dist/index.mjs"]

[services.production.run.env]
PORT = "8080"
NODE_ENV = "production"

[services.production.health.startup]
path = "/api/healthz"
```

[artifacts/api-server/build.mjs](#)

```
import { createRequire } from "node:module";
import path from "node:path";
import { fileURLToPath } from "node:url";
import { build as esbuild } from "esbuild";
import esbuildPluginPino from "esbuild-plugin-pino";
import { rm } from "node:fs/promises";

// Plugins (e.g. 'esbuild-plugin-pino') may use `require` to resolve dependencies
globalThis.require = createRequire(import.meta.url);

const artifactDir = path.dirname(fileURLToPath(import.meta.url));

const external = [
  "*.node",
  "sharp",
  "better-sqlite3",
  "sqlite3",
  "canvas",
  "bcrypt",
  "argon2",
  "fsevents",
  "re2",
  "farmhash",
  "xxhash-addon",
  "bufferutil",
  "utf-8-validate",
  "ssh2",
  "cpu-features",
  "dtrace-provider",
  "isolated-vm",
  "lightningcss",
  "pg-native",
  "oracledb",
  "mongodb-client-encryption",
  "nodemailer",
  "handlebars",
  "knex",
  "typeorm",
  "protobufjs",
  "onnxruntime-node",
  "@tensorflow/*",
  "@prisma/client",
  "@mikro-orm/*",
  "@grpc/*",
  "@swc/*",
  "@aws-sdk/*",
  "@azure/*",
  "@opentelemetry/*",
  "@google-cloud/*",
  "@google/*",
  "googleapis",
  "firebase-admin",
  "@parcel/watcher",
  "@sentry/profiling-node",
  "@tree-sitter/*",
  "aws-sdk",
  "classic-level",
  "dd-trace",
  "ffi-napi",
  "grpc",
  "hiredis",
  "kerberos",
  "leveldown",
  "miniflare",
  "mysql2",
  "newrelic",
  "odbc",
  "piscina",
  "realm",
  "ref-napi",
  "rocksdb",
  "sass-embedded",
  "sequelize",
  "serialport",
  "snappy",
  "tinypool",
  "usb",
  "workerd",
  "wrangler",
  "zeromq",
  "zeromq-prebuilt",
  "playwright",
  "puppeteer",
  "puppeteer-core",
  "electron",
];

const banner = {
  js: `import { createRequire as __bannerCrReq } from 'node:module';
import __bannerPath from 'node:path';
import __bannerUrl from 'node:url';

globalThis.require = __bannerCrReq(import.meta.url);
globalThis.__filename = __bannerUrl.fileURLToPath(import.meta.url);
globalThis.__dirname = __bannerPath.dirname(globalThis.__filename);`
};
```



```

};
`
};

async function buildAll() {
  const distDir = path.resolve(artifactDir, "dist");
  await rm(distDir, { recursive: true, force: true });

  const sharedOptions = {
    platform: "node",
    bundle: true,
    format: "esm",
    outdir: distDir,
    outExtension: { ".js": ".mjs" },
    logLevel: "info",
    external,
    sourcemap: "linked",
    banner,
    plugins: [
      esbuildPluginPino({ transports: ["pino-pretty"] })
    ],
  };

  // Build API server
  await esbuild({
    ...sharedOptions,
    entryPoints: [path.resolve(artifactDir, "src/index.ts")],
  });

  // Build worker (separate entry point – same bundle config, no HTTP server deps needed)
  await esbuild({
    ...sharedOptions,
    entryPoints: [path.resolve(artifactDir, "src/worker.ts")],
  });
}

buildAll().catch((err) => {
  console.error(err);
  process.exit(1);
});

```

artifacts/api-server/package.json

```
{
  "name": "@workspace/api-server",
  "version": "0.0.0",
  "private": true,
  "type": "module",
  "scripts": {
    "dev": "export NODE_ENV=development && pnpm run build && pnpm run start",
    "build": "node ./build.mjs",
    "start": "node --enable-source-maps ./dist/index.mjs",
    "worker": "node --enable-source-maps ./dist/worker.mjs",
    "typecheck": "tsc -p tsconfig.json --noEmit"
  },
  "dependencies": {
    "@anthropic-ai/sdk": "^0.78.0",
    "@workspace/api-zod": "workspace:*",
    "@workspace/db": "workspace:*",
    "@workspace/integrations-anthropic-ai": "workspace:*",
    "cookie-parser": "^1.4.7",
    "cors": "^2",
    "drizzle-orm": "catalog:",
    "express": "^5",
    "express-rate-limit": "^8.3.2",
    "helmet": "^8.1.0",
    "jsonwebtoken": "^9.0.3",
    "pdf-lib": "^1.17.1",
    "pino": "^9",
    "pino-http": "^10",
    "sharp": "^0.34.5",
    "zod": "catalog:"
  },
  "devDependencies": {
    "@types/cookie-parser": "^1.4.10",
    "@types/cors": "^2.8.19",
    "@types/express": "^5.0.6",
    "@types/jsonwebtoken": "^9.0.10",
    "@types/node": "catalog:",
    "@types/sharp": "^0.32.0",
    "esbuild": "^0.27.3",
    "esbuild-plugin-pino": "^2.3.3",
    "pino-pretty": "^13",
    "thread-stream": "3.1.0"
  }
}
```

[artifacts/api-server/src/app.ts](#)

```
import express, { type Express } from "express";
import cors from "cors";
import helmet from "helmet";
import rateLimit from "express-rate-limit";
import pinoHttp from "pino-http";
import router from "./routes";
import { logger } from "./lib/logger";
import { idsMiddleware } from "./lib/ids";

const app: Express = express();

app.use(helmet({
  contentSecurityPolicy: {
    directives: {
      defaultSrc: ["'self'"],
      scriptSrc: ["'self'"],
      styleSrc: ["'self'", "'unsafe-inline'"],
      imgSrc: ["'self'", "data:", "https:"],
      connectSrc: ["'self'"],
      fontSrc: ["'self'"],
      objectSrc: ["'none'"],
      frameSrc: ["'none'"],
      baseUri: ["'self'"],
      formAction: ["'self'"],
    },
  },
  crossOriginEmbedderPolicy: false,
  hsts: { maxAge: 31536000, includeSubDomains: true, preload: true },
  referrerPolicy: { policy: "strict-origin-when-cross-origin" },
  noSniff: true,
  xssFilter: true,
}));

app.disable("x-powered-by");

app.use(rateLimit({
  windowMs: 15 * 60 * 1000,
  max: 500,
  standardHeaders: true,
  legacyHeaders: false,
  message: { error: "Too many requests, please try again later" },
  keyGenerator: (req) => {
    return (req.headers["x-forwarded-for"] as string)?.split(",")[0]?.trim()
      || req.socket.remoteAddress
      || "unknown";
  },
}));

app.use(
  pinoHttp({
    logger,
    serializers: {
      req(req) {
        return {
          id: req.id,
          method: req.method,
          url: req.url?.split("?")[0],
        };
      },
      res(res) {
        return {
          statusCode: res.statusCode,
        };
      },
    },
  })
);

app.use(cors({
  origin: process.env.NODE_ENV === "production"
    ? [process.env.REPL_SLUG ? `https://${process.env.REPL_SLUG}.replit.app` : ""].filter(Boolean)
    : true,
  credentials: true,
  methods: ["GET", "POST", "PUT", "PATCH", "DELETE", "OPTIONS"],
  allowedHeaders: ["Content-Type", "Authorization"],
  maxAge: 86400,
}));

app.use(express.json({ limit: "55mb" }));
app.use(express.urlencoded({ extended: true, limit: "55mb" }));

app.use(idsMiddleware());

app.use("/api", router);

app.use((err: any, _req: express.Request, res: express.Response, _next: express.NextFunction) => {
  logger.error({ err: err.message, stack: err.stack }, "Unhandled error");
  res.status(err.status || 500).json({ error: "Internal server error" });
});

export default app;
```

[artifacts/api-server/src/index.ts](#)

```
import app from "./app";
import { logger } from "./lib/logger";
import { seedFormConfigurations } from "./lib/form-config-service";

const rawPort = process.env["PORT"];

if (!rawPort) {
  throw new Error(
    "PORT environment variable is required but was not provided.",
  );
}

const port = Number(rawPort);

if (Number.isNaN(port) || port <= 0) {
  throw new Error(`Invalid PORT value: "${rawPort}"`);
}

app.listen(port, async (err) => {
  if (err) {
    logger.error({ err }, "Error listening on port");
    process.exit(1);
  }

  logger.info({ port }, "Server listening");

  // Seed/refresh form configurations from TORT_REGISTRY on boot.
  // Safe: inserts missing rows and refreshes only rows where updated_by IS NULL
  // (admin-edited rows are never overwritten).
  try {
    await seedFormConfigurations();
  } catch (e) {
    logger.error({ err: e }, "Form configuration seed failed on boot");
  }
});
```

[artifacts/api-server/src/lib/address-validator.ts](#)

```
const US_STATES: Set<string> = new Set([
  "AL", "AK", "AZ", "AR", "CA", "CO", "CT", "DE", "DC", "FL", "GA", "HI", "ID", "IL", "IN",
  "IA", "KS", "KY", "LA", "ME", "MD", "MA", "MI", "MN", "MS", "MO", "MT", "NE", "NV", "NH",
  "NJ", "NM", "NY", "NC", "ND", "OH", "OK", "OR", "PA", "RI", "SC", "SD", "TN", "TX", "UT",
  "VT", "VA", "WA", "WV", "WI", "WY",
]);

const US_ZIP_REGEX = /^d{5}(-d{4})?$/;
const STREET_PATTERN = /\d+\s\S+/;

export interface AddressValidationResult {
  valid: boolean;
  errors: string[];
}

export function validateAddress(address: {
  street_address?: string;
  city?: string;
  state?: string;
  zip?: string;
}): AddressValidationResult {
  const errors: string[] = [];

  if (!address.street_address?.trim()) {
    errors.push("MISSING_STREET");
  } else if (!STREET_PATTERN.test(address.street_address.trim())) {
    errors.push("INVALID_STREET_FORMAT");
  }

  if (!address.city?.trim()) {
    errors.push("MISSING_CITY");
  } else if (address.city.trim().length < 2) {
    errors.push("INVALID_CITY");
  }

  if (!address.state?.trim()) {
    errors.push("MISSING_STATE");
  } else if (!US_STATES.has(address.state.trim().toUpperCase())) {
    errors.push("INVALID_STATE_CODE");
  }

  if (!address.zip?.trim()) {
    errors.push("MISSING_ZIP");
  } else if (!US_ZIP_REGEX.test(address.zip.trim())) {
    errors.push("INVALID_ZIP_FORMAT");
  }

  const garbagePatterns = [/^[x]+$\/i, /^[0]+$\/i, /^test$/i, /^asdf$/i, /^na$/i, /^none$/i];
  const fieldsToCheck = [address.street_address, address.city];
  for (const field of fieldsToCheck) {
    if (field) {
      for (const pat of garbagePatterns) {
        if (pat.test(field.trim())) {
          errors.push("GARBAGE_ADDRESS_DATA");
          break;
        }
      }
    }
  }

  return { valid: errors.length === 0, errors };
}
```

[artifacts/api-server/src/lib/ai-extract.ts](#)

```
import { anthropic } from "@workspace/integrations-anthropic-ai";
import { type ExtractedFeatures } from "./scoring";
import { logger } from "./logger";

const MODEL = "claude-haiku-4-5";
const MAX_TEXT_LENGTH = 8000;

export async function extractFeatures(text: string): Promise<ExtractedFeatures> {
    const truncated = text.slice(0, MAX_TEXT_LENGTH);

    const prompt = `You are a medical-legal document analyzer for mass tort cases.

Analyze the following medical/legal document text and return ONLY a valid JSON object with these fields:
- "diagnosis_present": boolean (true if a medical diagnosis is mentioned)
- "exposure": "yes" | "no" | "unclear" (was the subject exposed to the harmful substance/location?)
- "severity": number between 0.0 and 1.0 (severity of the medical condition; 1.0 = most severe)
- "contradictions": array of strings (any contradictory statements found; empty array if none)
- "tort_type": string (identified tort category, e.g. "Camp Lejeune", "AFFF", "NEC", or "unknown")
- "diagnosis_type": string (the specific diagnosis if mentioned, or "unknown")
- "exposure_period_years": number (estimated years of exposure, 0 if unknown)

Return ONLY the JSON object. No explanation, no markdown, no extra text.

DOCUMENT TEXT:
${truncated}`;

    try {
        const response = await anthropic.messages.create({
            model: MODEL,
            max_tokens: 8192,
            messages: [{ role: "user", content: prompt }],
        });

        const content = response.content[0];
        if (content.type !== "text") {
            logger.warn("AI extraction returned non-text content");
            return {};
        }

        const raw = content.text.trim();
        const jsonMatch = raw.match(/{\[\s\S]*\}/);
        if (!jsonMatch) {
            logger.warn({ raw }, "AI extraction response has no JSON object");
            return {};
        }

        const parsed = JSON.parse(jsonMatch[0]) as ExtractedFeatures;
        logger.info({ features: parsed }, "AI extraction complete");
        return parsed;
    } catch (err) {
        logger.error({ err }, "AI extraction failed");
        return {};
    }
}
```

[artifacts/api-server/src/lib/ai-fields.ts](#)

```
import { anthropic } from "@workspace/integrations-anthropic-ai";
import { logger } from "../logger";

const MODEL = "claude-haiku-4-5";

export interface MedicalRecordFields {
  patient_name: string;
  date_of_birth: string;
  providers: { name: string; npi?: string; specialty?: string; role: string }[];
  diagnoses: { code: string; description: string; date?: string }[];
  medications: { name: string; dosage: string; frequency: string; prescriber?: string; start_date?: string; end_date?: string }[];
  procedures: { name: string; date?: string; provider?: string; facility?: string }[];
  timeline_events: { date: string; event: string; category: string }[];
  exposure_details: { substance?: string; location?: string; start_date?: string; end_date?: string; duration?: string }[];
  insurance_info: { provider?: string; policy_number?: string; group_number?: string };
  raw_text: string;
  confidence: number;
  document_type: string;
}

const AIFIELDS_PROMPT = `You are a medical records analysis engine for mass tort litigation. Extract ALL structured data from this document with maximum detail.

Return ONLY valid JSON with these fields:
- "patient_name": string - full patient name or ""
- "date_of_birth": string - MM/DD/YYYY or ""
- "providers": array of {name, npi?, specialty?, role} - all doctors, nurses, pharmacists mentioned
- "diagnoses": array of {code, description, date?} - ICD codes if visible, descriptions, diagnosis dates
- "medications": array of {name, dosage, frequency, prescriber?, start_date?, end_date?}
- "procedures": array of {name, date?, provider?, facility?}
- "timeline_events": array of {date, event, category} - category: "diagnosis", "treatment", "exposure", "procedure", "visit"
- "exposure_details": array of {substance?, location?, start_date?, end_date?, duration?}
- "insurance_info": {provider?, policy_number?, group_number?}
- "raw_text": string - complete transcription
- "confidence": number - 0.0 to 1.0
- "document_type": string - "medical_record", "pharmacy_record", "lab_result", "radiology", "discharge_summary", "progress_note", "operative_report", "pathology", "other"

Extract every piece of clinically and legally relevant data. For mass tort cases, pay special attention to exposure timelines, substance names, and causation evidence.
Return ONLY the JSON object.`;

export async function extractMedicalFields(
  imageBase64: string,
  mimeType: string
): Promise<MedicalRecordFields> {
  const base64Data = imageBase64.replace(/^data:[^;]+;base64/, "");
  const validMime = (
    ["image/jpeg", "image/gif", "image/webp"].includes(mimeType) ? mimeType : "image/png"
  ) as "image/jpeg" | "image/png" | "image/gif" | "image/webp";

  try {
    const response = await anthropic.messages.create({
      model: MODEL,
      max_tokens: 4096,
      messages: [{
        role: "user",
        content: [
          { type: "image", source: { type: "base64", media_type: validMime, data: base64Data } },
          { type: "text", text: AIFIELDS_PROMPT },
        ],
      }],
    });

    const content = response.content[0];
    if (content.type !== "text") return emptyFields();

    const raw = content.text.trim();
    const jsonMatch = raw.match(/\[\[\s\S\]*\]\]/);
    if (!jsonMatch) return { ...emptyFields(), raw_text: raw, confidence: 0.1 };

    const parsed = JSON.parse(jsonMatch[0]) as MedicalRecordFields;
    logger.info({ doc_type: parsed.document_type, providers: parsed.providers.length, meds: parsed.medications.length, confidence: parsed.confidence }, "AIFields extraction complete");
    return parsed;
  } catch (err) {
    logger.error({ err }, "AIFields extraction failed");
    return emptyFields();
  }
}

export async function analyzeDocumentText(text: string): Promise<MedicalRecordFields> {
  try {
    const response = await anthropic.messages.create({
      model: MODEL,
      max_tokens: 4096,
      messages: [{
        role: "user",
        content: `${AIFIELDS_PROMPT}\n\nDocument text:\n${text.slice(0, 8000)}`,
      }],
    });

    const content = response.content[0];
    if (content.type !== "text") return emptyFields();
  }
}
```

```

    const jsonMatch = content.text.trim().match(/\{[\s\S]*\}/);
    if (!jsonMatch) return emptyFields();
    return JSON.parse(jsonMatch[0]) as MedicalRecordFields;
  } catch (err) {
    logger.error({ err }, "AIFields text analysis failed");
    return emptyFields();
  }
}

function emptyFields(): MedicalRecordFields {
  return {
    patient_name: "", date_of_birth: "", providers: [], diagnoses: [],
    medications: [], procedures: [], timeline_events: [], exposure_details: [],
    insurance_info: {}, raw_text: "", confidence: 0, document_type: "other",
  };
}

```


[artifacts/api-server/src/lib/ai-ocr.ts](#)

```
/**
 * Claude Vision OCR for faxed pharmacy/medical records.
 * Replaces PaddleOCR (not available in Node.js) with Claude Haiku vision.
 *
 * Output: Legora Grid row format matching the MTOS OCR spec.
 */
import { anthropic } from "@workspace/integrations-anthropic-ai";
import { logger } from "../logger";

const MODEL = "claude-haiku-4-5";

export interface LegoraGridRow {
  rx_number: string;
  drug_name: string;
  fill_date: string;
  quantity: string;
  confidence: number;
  raw_text: string;
}

const SYSTEM_PROMPT = `You are an OCR engine specialized in faxed pharmacy and medical records for mass tort litigation.

Extract structured data from faxed prescription records, pharmacy printouts, and medical documents.
Focus on: Paraquat, Camp Lejeune contamination, AFFF, NEC, talc, and other mass tort substances.

Return ONLY a valid JSON object with these fields:
- "rx_number": string - prescription number if present (e.g. "RX123456789"), or ""
- "drug_name": string - drug, chemical, or substance name (e.g. "Paraquat 2,4-D 1gal"), or ""
- "fill_date": string - fill/dispensed/purchase date in MM/DD/YYYY format, or ""
- "quantity": string - quantity dispensed (e.g. "30", "1gal"), or ""
- "confidence": number - your overall extraction confidence from 0.0 to 1.0
- "raw_text": string - complete raw transcription of all visible text in the document

If this is a medical record rather than an Rx: set rx_number to "", fill drug_name with the primary diagnosis or substance, and
fill raw_text with all extracted text.

Return ONLY the JSON object. No markdown, no explanation.`;

export async function extractOcrData(
  imageBase64: string,
  mimeType: string
): Promise<LegoraGridRow> {
  const base64Data = imageBase64.replace(/^data:[^;]+;base64/, "");

  const validMime = (
    ["image/jpeg", "image/png", "image/gif", "image/webp"].includes(mimeType)
    ? mimeType
    : "image/png"
  ) as "image/jpeg" | "image/png" | "image/gif" | "image/webp";

  try {
    const response = await anthropic.messages.create({
      model: MODEL,
      max_tokens: 2048,
      messages: [
        {
          role: "user",
          content: [
            {
              type: "image",
              source: {
                type: "base64",
                media_type: validMime,
                data: base64Data,
              },
            },
            {
              type: "text",
              text: SYSTEM_PROMPT,
            },
          ],
        },
      ],
    });

    const content = response.content[0];
    if (content.type !== "text") {
      logger.warn("OCR: Claude returned non-text content");
      return emptyRow();
    }

    const raw = content.text.trim();
    const jsonMatch = raw.match(/^{[\s\S]*}$/);
    if (!jsonMatch) {
      logger.warn({ raw }, "OCR: No JSON object in Claude response");
      return { ...emptyRow(), raw_text: raw, confidence: 0.1 };
    }

    const parsed = JSON.parse(jsonMatch[0]) as LegoraGridRow;
    logger.info(
      { drug_name: parsed.drug_name, confidence: parsed.confidence },
      "OCR extraction complete"
    );
    return parsed;
  } catch (err) {
```

```
        logger.error({ err }, "OCR: Claude Vision extraction failed");
        return emptyRow();
    }
}

function emptyRow(): LegoraGridRow {
    return {
        rx_number: "",
        drug_name: "",
        fill_date: "",
        quantity: "",
        confidence: 0,
        raw_text: "",
    };
}
```

[artifacts/api-server/src/lib/audit.ts](#)

```
import { db, auditLogTable } from "@workspace/db";
import { logger } from "../logger";

export async function auditLog(
  entity_type: string,
  entity_id: string,
  action: string,
  details?: Record<string, unknown>,
  meta?: { ip_address?: string; user_agent?: string }
) {
  try {
    await db.insert(auditLogTable).values({
      entity_type,
      entity_id,
      action,
      details: details ?? {},
      ip_address: meta?.ip_address ?? null,
      user_agent: meta?.user_agent ?? null,
    });
  } catch (err) {
    logger.error({ err, entity_type, entity_id, action }, "Audit log write failed");
  }
}
```

[artifacts/api-server/src/lib/background-check.ts](#)

```
import { logger } from "../logger";

export interface BackgroundCheckResult {
  status: "clean" | "flagged" | "not_found" | "error";
  source: string;
  checked_at: string;
  records: BackgroundRecord[];
  summary: string;
}

export interface BackgroundRecord {
  type: string;
  description: string;
  date?: string;
  jurisdiction?: string;
  severity: "low" | "medium" | "high";
  role?: "party" | "mentioned";
}

export async function runBackgroundCheck(person: {
  first_name: string;
  last_name: string;
  state?: string;
  date_of_birth?: string;
}): Promise<BackgroundCheckResult> {
  const checkedAt = new Date().toISOString();
  const fullName = `${person.first_name} ${person.last_name}`;

  try {
    const results: BackgroundRecord[] = [];

    let sourcesReached = 0;

    const courtRecords = await searchCourtRecords(person);
    if (courtRecords !== null) {
      sourcesReached++;
      results.push(...courtRecords);
    }

    const ofacResult = await checkOFACList(person);
    if (ofacResult !== null) {
      sourcesReached++;
      results.push(...ofacResult);
    }

    if (sourcesReached === 0) {
      return {
        status: "error",
        source: "MTOS Background Check (Court Records + OFAC)",
        checked_at: checkedAt,
        records: [],
        summary: `All background check sources unreachable for ${fullName}`,
      };
    }

    const partyRecords = results.filter((r) => r.role === "party");
    const mentionRecords = results.filter((r) => r.role === "mentioned");
    const nonCourtRecords = results.filter((r) => r.type !== "COURT_RECORD");
    const hasFlaggable = partyRecords.length > 0 || nonCourtRecords.length > 0;
    const status = hasFlaggable ? "flagged" : "clean";
    const parts: string[] = [];
    if (partyRecords.length > 0) {
      parts.push(`${partyRecords.length} case(s) as party`);
    }
    if (mentionRecords.length > 0) {
      parts.push(`${mentionRecords.length} mention(s) in other cases`);
    }
    if (nonCourtRecords.length > 0) {
      parts.push(`${nonCourtRecords.length} non-court record(s)`);
    }
    const summary = parts.length > 0
      ? `${parts.join(", ")} for ${fullName}`
      : `No records found for ${fullName}`;

    return {
      status,
      source: "MTOS Background Check (Court Records + OFAC)",
      checked_at: checkedAt,
      records: results,
      summary,
    };
  } catch (err) {
    logger.error({ err, person: fullName }, "Background check failed");
    return {
      status: "error",
      source: "MTOS Background Check",
      checked_at: checkedAt,
      records: [],
      summary: `Background check error for ${fullName}: ${err instanceof Error ? err.message : "Unknown error"}`,
    };
  }
}

function textContainsName(
  text: string,
```

```

    firstName: string,
    lastName: string,
  ): "exact" | "initial" | false {
    const t = text.toLowerCase();
    const fn = firstName.toLowerCase();
    const ln = lastName.toLowerCase();

    if (t.includes(`${fn} ${ln}`) || t.includes(`${ln}, ${fn}`) || t.includes(`${ln} ${fn}`)) {
      return "exact";
    }

    const fnInitial = fn.charAt(0);
    if (
      t.includes(`${ln}, ${fnInitial}.`) ||
      t.includes(`${ln}, ${fnInitial}`) ||
      t.includes(`${fnInitial}. ${ln}`)
    ) {
      return "initial";
    }

    return false;
  }

interface CourtResult {
  caseName?: string;
  dateFiled?: string;
  court?: string;
  docketNumber?: string;
  party?: string[];
  recap_documents?: Array<{
    description?: string;
    snippet?: string;
  }>;
}

function lastNameIsDefendant(
  result: CourtResult,
  lastName: string,
): boolean {
  const ln = lastName.toLowerCase();
  const cn = (result.caseName || "").toLowerCase();

  const defPattern = /\s+v\.\s+(.+)/i;
  const defMatch = cn.match(defPattern);
  if (defMatch) {
    const defSide = defMatch[1].trim();
    if (defSide === ln || defSide.startsWith(`${ln},`) || defSide.startsWith(`${ln} `)) {
      return true;
    }
  }

  if (result.party && Array.isArray(result.party)) {
    for (const p of result.party) {
      const pl = p.toLowerCase().trim();
      if (pl === ln || pl === ln.toUpperCase().toLowerCase()) {
        return true;
      }
    }
  }

  return false;
}

function nameMatchesResult(
  result: CourtResult,
  firstName: string,
  lastName: string,
): { match: boolean; confidence: "exact" | "strong" | "none"; role: "party" | "mentioned" } {
  const caseName = result.caseName || "";
  const caseMatch = textContainsName(caseName, firstName, lastName);
  if (caseMatch === "exact") return { match: true, confidence: "exact", role: "party" };
  if (caseMatch === "initial") return { match: true, confidence: "strong", role: "party" };

  const caseParties = caseName.split(/\s+v\.\s+|\s+vs\.\s+|\s+and\s+/i);
  for (const party of caseParties) {
    const pm = textContainsName(party.trim(), firstName, lastName);
    if (pm) return { match: true, confidence: pm === "exact" ? "exact" : "strong", role: "party" };
  }

  if (result.party && Array.isArray(result.party)) {
    for (const p of result.party) {
      const pm = textContainsName(p, firstName, lastName);
      if (pm) return { match: true, confidence: pm === "exact" ? "exact" : "strong", role: "party" };
    }
  }

  if (lastNameIsDefendant(result, lastName)) {
    return { match: true, confidence: "strong", role: "party" };
  }

  if (result.recap_documents && Array.isArray(result.recap_documents)) {
    for (const doc of result.recap_documents) {
      const searchable = `${doc.description || ""} ${doc.snippet || ""}`;
      const dm = textContainsName(searchable, firstName, lastName);
      if (dm) return { match: true, confidence: "strong", role: "mentioned" };
    }
  }
}

```

```

    return { match: false, confidence: "none", role: "mentioned" };
}

async function fetchCourtListenerResults(
  query: string,
  stateParam: string,
): Promise<CourtResult[]> {
  try {
    const url = `https://www.courtlistener.com/api/rest/v4/search/?q=${encodeURIComponent(query)}&type=r${stateParam}&format=json`;
    const controller = new AbortController();
    const timeout = setTimeout(() => controller.abort(), 10000);
    const response = await fetch(url, {
      headers: { "User-Agent": "MTOS-CRM/1.0 (Legal Background Check)" },
      signal: controller.signal,
    });
    clearTimeout(timeout);
    if (!response.ok) return [];
    const data = (await response.json()) as { results?: CourtResult[] };
    return data.results || [];
  } catch {
    return [];
  }
}

async function searchCourtRecords(person: {
  first_name: string;
  last_name: string;
  state?: string;
}): Promise<BackgroundRecord[] | null> {
  const firstName = person.first_name.trim();
  const lastName = person.last_name.trim();
  const stateParam = person.state
    ? `&state=${encodeURIComponent(person.state)}`
    : "";

  const exactSearches = [
    `${firstName} ${lastName}`,
    `${lastName}, ${firstName}`,
  ];
  const defendantSearch = `v. ${lastName}`;
  const broadSearch = `${lastName} AND ${firstName}`;

  const seen = new Set<string>();
  const records: BackgroundRecord[] = [];

  function buildRecord(result: CourtResult, role: "party" | "mentioned", confidence: "exact" | "strong" | "none"): BackgroundRecord {
    const caseName = result.caseName || "Court record found";
    let severity: "low" | "medium" | "high";
    let description: string;

    if (role === "party") {
      if (confidence === "exact") {
        severity = "high";
        description = caseName;
      } else {
        severity = "medium";
        description = `${caseName} (last name match – verify identity)`;
      }
    } else {
      severity = "low";
      description = `${caseName} (named in documents, not a listed party)`;
    }

    return {
      type: "COURT_RECORD",
      description,
      date: result.dateFiled || undefined,
      jurisdiction: result.court || undefined,
      severity,
      role,
    };
  }

  try {
    const exactBatches = await Promise.all(
      exactSearches.map((q) => fetchCourtListenerResults(q, stateParam)),
    );
    for (const batch of exactBatches) {
      for (const result of batch) {
        const docketNum = result.docketNumber || "";
        const caseName = result.caseName || "";
        const key = `${docketNum}|${caseName}|${result.dateFiled || ""}`;
        if (seen.has(key)) continue;
        seen.add(key);

        const { confidence, role } = nameMatchesResult(result, firstName, lastName);
        records.push(buildRecord(result, role, confidence));
      }
    }

    const defResults = await fetchCourtListenerResults(defendantSearch, stateParam);
    for (const result of defResults) {
      const docketNum = result.docketNumber || "";
      const caseName = result.caseName || "";

```

```

const key = `${docketNum}|${caseName}|${result.dateFiled || ""}`;
if (seen.has(key)) continue;
seen.add(key);

if (!lastNameIsDefendant(result, lastName)) continue;

const hasFirstNameAnywhere = textContainsName(
  `${caseName} ${result.party || []}.join(" ") ${result.recap_documents || []}.map(d => `${d.description} || ""}
  ${d.snippet || ""}`).join(" ")`,
  firstName,
  lastName,
);

const otherFirstNames = (result.party || []).some(p => {
  const pl = p.toLowerCase().trim();
  return pl !== lastName.toLowerCase() &&
    pl !== "united states" && pl !== "usa" &&
    pl.includes(lastName.toLowerCase()) &&
    !pl.includes(firstName.toLowerCase());
});

if (otherFirstNames) continue;

if (hasFirstNameAnywhere) {
  records.push(buildRecord(result, "party", "exact"));
} else {
  records.push(buildRecord(result, "party", "strong"));
}
}

const broadResults = await fetchCourtListenerResults(broadSearch, stateParam);
for (const result of broadResults) {
  const docketNum = result.docketNumber || "";
  const caseName = result.caseName || "";
  const key = `${docketNum}|${caseName}|${result.dateFiled || ""}`;
  if (seen.has(key)) continue;
  seen.add(key);

  const { match, confidence, role } = nameMatchesResult(result, firstName, lastName);
  if (!match) continue;
  records.push(buildRecord(result, role, confidence));
}
} catch (err) {
  logger.warn({ err }, "Court record search failed – continuing");
  return null;
}

if (records.length === 0) {
  const fallback = await fetchCourtListenerResults(lastName, stateParam);
  for (const result of fallback) {
    const { match, confidence, role } = nameMatchesResult(result, firstName, lastName);
    if (!match) continue;
    records.push(buildRecord(result, role, confidence));
  }
}

return records.slice(0, 10);
}

async function checkOFACList(person: {
  first_name: string;
  last_name: string;
}): Promise<BackgroundRecord[] | null> {
  const records: BackgroundRecord[] = [];

  try {
    const name = encodeURIComponent(`${person.first_name} ${person.last_name}`);
    const url = `https://search.ofac-api.com/v3?name=${name}&minScore=95`;

    const controller = new AbortController();
    const timeout = setTimeout(() => controller.abort(), 10000);

    const response = await fetch(url, {
      signal: controller.signal,
    });
    clearTimeout(timeout);

    if (response.ok) {
      const data = await response.json() as { matches?: Array<{ name?: string; programs?: string[]; score?: number }> };
      if (data.matches && Array.isArray(data.matches) && data.matches.length > 0) {
        for (const match of data.matches.slice(0, 5)) {
          records.push({
            type: "OFAC_SANCTIONS",
            description: `OFAC match: ${match.name || "Unknown"} (Programs: ${match.programs?.join(", ") || "N/A"})`,
            severity: "high",
          });
        }
      }
    }
  } catch (err) {
    logger.warn({ err }, "OFAC check failed – continuing");
    return null;
  }

  return records;
}

```

[artifacts/api-server/src/lib/conflict-engine.ts](#)

```
import { db, reviewQueueTable } from "@workspace/db";
import { auditLog } from "../audit";
import { logger } from "../logger";

export type ConflictType =
  | "LOGICAL_CONFLICT"
  | "RULE_OVERRIDE_CONFLICT"
  | "DATA_INTEGRITY_CONFLICT"
  | "AI_CLASSIFICATION_CONFLICT";

export type FailsafeMode = "SAFE_FAIL" | "REVIEW_FAIL" | "HARD_BLOCK";

export type SystemOutputState = "ACCEPT" | "REJECT" | "REVIEW_REQUIRED" | "ERROR_FALLBACK";

export interface ConflictResult {
  has_conflict: boolean;
  conflict_type: ConflictType | null;
  severity: "low" | "medium" | "high" | "critical";
  details: string[];
  failsafe_mode: FailsafeMode;
  output_state: SystemOutputState;
}

export interface ConflictCheckContext {
  entity_type: string;
  entity_id: string;
  source_module: string;
  lead_data?: Record<string, unknown>;
  tort_criteria?: Record<string, unknown>;
  ai_classification?: Record<string, unknown>;
  user_settings?: Record<string, unknown>;
}

const VALID_LOCATIONS = [
  "USA", "US", "UNITED STATES",
  "AL", "AK", "AZ", "AR", "CA", "CO", "CT", "DE", "DC", "FL", "GA",
  "HI", "ID", "IL", "IN", "IA", "KS", "KY", "LA", "ME", "MD", "MA",
  "MI", "MN", "MS", "MO", "MT", "NE", "NV", "NH", "NJ", "NM", "NY",
  "NC", "ND", "OH", "OK", "OR", "PA", "RI", "SC", "SD", "TN", "TX",
  "UT", "VT", "VA", "WA", "WV", "WI", "WY",
];

const KNOWN_TORT_CONDITIONS: Record<string, string[]> = {
  "Camp Lejeune": ["cancer", "leukemia", "kidney disease", "liver disease", "parkinson", "bladder cancer", "non-hodgkin"],
  "Roundup": ["non-hodgkin lymphoma", "cancer", "nhl"],
  "Talcum Powder": ["ovarian cancer", "mesothelioma", "cancer"],
  "AFFF": ["cancer", "kidney cancer", "testicular cancer", "thyroid disease"],
  "Asbestos": ["mesothelioma", "asbestosis", "lung cancer"],
  "NEC": ["necrotizing enterocolitis", "nec", "premature infant"],
  "Tylenol": ["autism", "adhd", "attention deficit"],
  "Hair Relaxer": ["uterine cancer", "ovarian cancer", "endometriosis"],
  "Paraquat": ["parkinson", "parkinson's disease"],
  "Zantac": ["cancer", "bladder cancer", "stomach cancer"],
};

export function checkLogicalConflicts(ctx: ConflictCheckContext): ConflictResult {
  const details: string[] = [];
  const lead = ctx.lead_data || {};

  const locationRaw = String(lead.location_name || lead.location || "").toUpperCase().trim();
  if (locationRaw) {
    const locationWords = locationRaw.split(/[ \s,]+/).map((w) => w.trim()).filter(Boolean);
    const matchesUS = locationWords.some((word) => VALID_LOCATIONS.includes(word));
    if (!matchesUS) {
      details.push(`Location "${lead.location_name || lead.location}" is outside allowed US geography`);
    }
  }

  const tortType = String(lead.tort_type || "").trim();
  const diagnosisType = String(lead.diagnosis_type || "").toLowerCase().trim();
  if (tortType && diagnosisType && KNOWN_TORT_CONDITIONS[tortType]) {
    const validConditions = KNOWN_TORT_CONDITIONS[tortType];
    const matchesAny = validConditions.some((c) => diagnosisType.includes(c));
    if (!matchesAny) {
      details.push(
        `Diagnosis "${lead.diagnosis_type}" does not match known conditions for tort "${tortType}" (expected: ${validConditions.join(", ")})`
      );
    }
  }

  if (lead.exposure_start && lead.exposure_end) {
    const start = new Date(String(lead.exposure_start));
    const end = new Date(String(lead.exposure_end));
    if (start > end) {
      details.push(`Exposure start date (${lead.exposure_start}) is after end date (${lead.exposure_end})`);
    }
  }

  if (details.length === 0) {
    return { has_conflict: false, conflict_type: null, severity: "low", details: [], failsafe_mode: "SAFE_FAIL", output_state: "ACCEPT" };
  }

  return {
```



```

    has_conflict: true,
    conflict_type: "LOGICAL_CONFLICT",
    severity: details.length >= 2 ? "high" : "medium",
    details,
    failsafe_mode: "REVIEW_FAIL",
    output_state: "REVIEW_REQUIRED",
  });
}

export function checkDataIntegrity(ctx: ConflictCheckContext): ConflictResult {
  const details: string[] = [];
  const lead = ctx.lead_data || {};

  const requiredFields = ["name", "tort_type"];
  for (const field of requiredFields) {
    const val = lead[field];
    if (val === undefined || val === null || String(val).trim() === "") {
      details.push(`Required field "${field}" is missing or empty`);
    }
  }

  const name = String(lead.name || "").trim();
  if (name && name.length < 2) {
    details.push(`Name "${name}" appears invalid (too short)`);
  }
  if (name && /^(.)\1+$/.test(name)) {
    details.push(`Name "${name}" appears to be garbage input`);
  }

  const tortType = String(lead.tort_type || "").trim();
  if (tortType && tortType.length < 3) {
    details.push(`Tort type "${tortType}" appears invalid`);
  }
  const garbagePattern = /^[^a-zA-Z]*$|^(\.)\1{3,}$/;
  if (tortType && garbagePattern.test(tortType)) {
    details.push(`Tort type "${tortType}" appears to be garbage input`);
  }

  const email = String(lead.email || "").trim();
  if (email && !/^[^\s@]+@[^\s@]+\.[^\s@]+$/ .test(email)) {
    details.push(`Email "${email}" format is invalid`);
  }

  const phone = String(lead.phone || "").trim();
  if (phone && phone.replace(/\D/g, "").length < 10) {
    details.push(`Phone "${phone}" appears invalid (too few digits)`);
  }

  if (details.length === 0) {
    return { has_conflict: false, conflict_type: null, severity: "low", details: [], failsafe_mode: "SAFE_FAIL", output_state:
"ACCEPT" };
  }

  const hasRequired = details.some((d) => d.includes("Required field"));
  const hasGarbage = details.some((d) => d.includes("garbage input"));

  return {
    has_conflict: true,
    conflict_type: "DATA_INTEGRITY_CONFLICT",
    severity: hasRequired || hasGarbage ? "critical" : "medium",
    details,
    failsafe_mode: hasGarbage ? "SAFE_FAIL" : "REVIEW_FAIL",
    output_state: hasGarbage ? "REJECT" : "REVIEW_REQUIRED",
  });
}

export function checkAIClassificationConflict(
  aiResult: { verdict: string; score: number },
  ruleResult: { qualified: boolean; status: string },
  ctx: ConflictCheckContext
): ConflictResult {
  const aiAccepts = aiResult.verdict === "strong" || aiResult.verdict === "moderate";
  const rulesAccept = ruleResult.qualified;

  if (aiAccepts === rulesAccept) {
    return { has_conflict: false, conflict_type: null, severity: "low", details: [], failsafe_mode: "SAFE_FAIL", output_state:
rulesAccept ? "ACCEPT" : "REJECT" };
  }

  const details: string[] = [];
  if (aiAccepts && !rulesAccept) {
    details.push(
      `AI classification says ACCEPT (verdict: ${aiResult.verdict}, score: ${aiResult.score}) but rule engine says REJECT
(status: ${ruleResult.status})`
    );
  } else {
    details.push(
      `Rule engine says ACCEPT but AI classification says REJECT (verdict: ${aiResult.verdict}, score: ${aiResult.score})`
    );
  }

  return {
    has_conflict: true,
    conflict_type: "AI_CLASSIFICATION_CONFLICT",
    severity: "high",
    details,
    failsafe_mode: "REVIEW_FAIL",
  };
}

```

```

    output_state: "REVIEW_REQUIRED",
  };
}

export function checkRuleOverrideConflict(
  buyerCriteria: Record<string, unknown>,
  globalTortRules: Record<string, unknown>,
  ctx: ConflictCheckContext
): ConflictResult {
  const details: string[] = [];

  if (buyerCriteria.allowed_locations && globalTortRules.allowed_locations) {
    const buyerLocs = Array.isArray(buyerCriteria.allowed_locations) ? buyerCriteria.allowed_locations :
[buyerCriteria.allowed_locations];
    const globalLocs = Array.isArray(globalTortRules.allowed_locations) ? globalTortRules.allowed_locations :
[globalTortRules.allowed_locations];

    const buyerSet = new Set(buyerLocs.map((l: string) => String(l).toUpperCase()));
    const globalSet = new Set(globalLocs.map((l: string) => String(l).toUpperCase()));
    const extraLocs = [...buyerSet].filter((l) => !globalSet.has(l));
    if (extraLocs.length > 0) {
      details.push(`Buyer criteria allows locations [${extraLocs.join(", ")}] that are excluded by global tort rules`);
    }
  }

  if (buyerCriteria.min_age !== undefined && globalTortRules.min_age !== undefined) {
    if (Number(buyerCriteria.min_age) < Number(globalTortRules.min_age)) {
      details.push(`Buyer min_age (${buyerCriteria.min_age}) is lower than global min_age (${globalTortRules.min_age})`);
    }
  }

  if (buyerCriteria.required_conditions && globalTortRules.required_conditions) {
    const buyerConds = new Set(
      (Array.isArray(buyerCriteria.required_conditions) ? buyerCriteria.required_conditions : []).map((c: string) =>
        String(c).toLowerCase())
    );
    const globalConds = Array.isArray(globalTortRules.required_conditions) ? globalTortRules.required_conditions : [];
    for (const gc of globalConds) {
      if (!buyerConds.has(String(gc).toLowerCase())) {
        details.push(`Global required condition "${gc}" is missing from buyer criteria`);
      }
    }
  }

  if (details.length === 0) {
    return { has_conflict: false, conflict_type: null, severity: "low", details: [], failsafe_mode: "SAFE_FAIL", output_state:
"ACCEPT" };
  }

  return {
    has_conflict: true,
    conflict_type: "RULE_OVERRIDE_CONFLICT",
    severity: "high",
    details,
    failsafe_mode: "HARD_BLOCK",
    output_state: "REVIEW_REQUIRED",
  };
}

export async function runFullConflictCheck(ctx: ConflictCheckContext): Promise<ConflictResult> {
  const checks = [
    checkDataIntegrity(ctx),
    checkLogicalConflicts(ctx),
  ];

  for (const result of checks) {
    if (result.has_conflict) {
      await routeToReview(result, ctx);
      return result;
    }
  }

  return { has_conflict: false, conflict_type: null, severity: "low", details: [], failsafe_mode: "SAFE_FAIL", output_state:
"ACCEPT" };
}

export async function routeToReview(conflict: ConflictResult, ctx: ConflictCheckContext): Promise<number> {
  logger.warn(
    { conflict_type: conflict.conflict_type, severity: conflict.severity, entity: ctx.entity_id, module: ctx.source_module },
    "CONFLICT DETECTED - routing to REVIEW_STATE"
  );

  const [item] = await db
    .insert(reviewQueueTable)
    .values({
      entity_type: ctx.entity_type,
      entity_id: ctx.entity_id,
      conflict_type: conflict.conflict_type || "UNKNOWN",
      severity: conflict.severity,
      failsafe_mode: conflict.failsafe_mode,
      source_module: ctx.source_module,
      summary: conflict.details.join("; "),
      details: {
        conflict_details: conflict.details,
        output_state: conflict.output_state,
        lead_data: ctx.lead_data,
      },
    });

```

```
        tort_criteria: ctx.tort_criteria,
        ai_classification: ctx.ai_classification,
    },
    resolution: "pending",
})
.returning({ id: reviewQueueTable.id });

await auditLog(ctx.entity_type, ctx.entity_id, "conflict_detected", {
    conflict_type: conflict.conflict_type,
    severity: conflict.severity,
    failsafe_mode: conflict.failsafe_mode,
    output_state: conflict.output_state,
    details: conflict.details,
    review_queue_id: item.id,
});

return item.id;
}
```

[artifacts/api-server/src/lib/decision-engine-service.ts](#)

```
/**
 * DB-aware wrapper around the pure decision-engine library.
 * Loads tort, source, and settings from DB; persists score back to lead.
 */

import { db, leadsTable, formConfigurationsTable, leadSourcesTable, decisionEngineSettingsTable } from "@workspace/db";
import type { Lead } from "@workspace/db";
import { eq, sql } from "drizzle-orm";
import {
  scoreLead,
  buildPortfolio,
  type EngineSettings,
  type TortInputs,
  type SourceInputs,
  type ScoreResult,
  type PortfolioSummary,
} from "../decision-engine";
import { logger } from "../logger";

const DEFAULT_SETTINGS: EngineSettings = {
  default_attorney_hourly_cost: 250,
  default_hours_per_lead: 8,
  convex_ratio_threshold: 5,
  concave_ratio_threshold: 1.5,
  concentration_warning_pct: 40,
};

let cachedSettings: EngineSettings | null = null;
let cachedSettingsAt = 0;
const SETTINGS_TTL_MS = 30_000;

export async function getEngineSettings(): Promise<EngineSettings> {
  if (cachedSettings && Date.now() - cachedSettingsAt < SETTINGS_TTL_MS) {
    return cachedSettings;
  }
  try {
    const rows = await db.select().from(decisionEngineSettingsTable).limit(1);
    if (rows.length === 0) {
      // Seed singleton row
      await db
        .insert(decisionEngineSettingsTable)
        .values({ id: 1 })
        .onConflictDoNothing();
      cachedSettings = DEFAULT_SETTINGS;
    } else {
      const r = rows[0]!;
      cachedSettings = {
        default_attorney_hourly_cost: Number(r.default_attorney_hourly_cost),
        default_hours_per_lead: Number(r.default_hours_per_lead),
        convex_ratio_threshold: Number(r.convex_ratio_threshold),
        concave_ratio_threshold: Number(r.concave_ratio_threshold),
        concentration_warning_pct: r.concentration_warning_pct,
      };
    }
  } catch (e) {
    logger.error({ err: e }, "Failed to load decision engine settings; using defaults");
    cachedSettings = DEFAULT_SETTINGS;
  }
  cachedSettingsAt = Date.now();
  return cachedSettings;
}

export function invalidateSettingsCache() {
  cachedSettings = null;
  cachedSettingsAt = 0;
}

export async function updateEngineSettings(updates: Partial<{
  default_attorney_hourly_cost: number;
  default_hours_per_lead: number;
  convex_ratio_threshold: number;
  concave_ratio_threshold: number;
  concentration_warning_pct: number;
  ruin_auto_flag: boolean;
}>) {
  const dbUpdates: Record<string, unknown> = { updated_at: new Date() };
  if (updates.default_attorney_hourly_cost !== undefined)
    dbUpdates.default_attorney_hourly_cost = String(updates.default_attorney_hourly_cost);
  if (updates.default_hours_per_lead !== undefined)
    dbUpdates.default_hours_per_lead = String(updates.default_hours_per_lead);
  if (updates.convex_ratio_threshold !== undefined)
    dbUpdates.convex_ratio_threshold = String(updates.convex_ratio_threshold);
  if (updates.concave_ratio_threshold !== undefined)
    dbUpdates.concave_ratio_threshold = String(updates.concave_ratio_threshold);
  if (updates.concentration_warning_pct !== undefined)
    dbUpdates.concentration_warning_pct = updates.concentration_warning_pct;
  if (updates.ruin_auto_flag !== undefined)
    dbUpdates.ruin_auto_flag = updates.ruin_auto_flag;

  await db
    .insert(decisionEngineSettingsTable)
    .values({ id: 1, ...dbUpdates })
    .onConflictDoUpdate({ target: decisionEngineSettingsTable.id, set: dbUpdates });
  invalidateSettingsCache();
}
```

```

async function loadTortInputs(tort_id: string): Promise<TortInputs | null> {
  const rows = await db
    .select()
    .from(formConfigurationsTable)
    .where(eq(formConfigurationsTable.id, tort_id))
    .limit(1);
  const row = rows[0];
  if (!row) return null;
  return {
    id: row.id,
    label: row.label,
    avg_settlement_low: row.avg_settlement_low,
    avg_settlement_high: row.avg_settlement_high,
    expected_duration_months: row.expected_duration_months,
    mdl_status: row.mdl_status,
    sol_months: row.sol_months,
    rejection_conditions: (row.rejection_conditions as unknown as string[]) || [],
    required_exposure: row.required_exposure,
    valid_diagnoses: row.valid_diagnoses || [],
  };
}

async function loadSourceInputs(name: string | null): Promise<SourceInputs | null> {
  if (!name) return null;
  const rows = await db
    .select()
    .from(leadSourcesTable)
    .where(eq(leadSourcesTable.name, name))
    .limit(1);
  const row = rows[0];
  if (!row) return { name, cost_per_lead: null, historical_qualified_rate: null, historical_retained_rate: null };
  return {
    name: row.name,
    cost_per_lead: row.cost_per_lead ? Number(row.cost_per_lead) : null,
    historical_qualified_rate: row.historical_qualified_rate ? Number(row.historical_qualified_rate) : null,
    historical_retained_rate: row.historical_retained_rate ? Number(row.historical_retained_rate) : null,
  };
}

/**
 * Compute the score for a single lead and persist to DB. Returns the result.
 * Best-effort: errors are logged and swallowed so they don't break the lead lifecycle.
 */
export async function computeAndPersistLeadScore(leadId: number): Promise<ScoreResult | null> {
  try {
    const leadRows = await db.select().from(leadsTable).where(eq(leadsTable.id, leadId)).limit(1);
    const rawLead = leadRows[0];
    if (!rawLead) return null;
    // Decrypt PII/clinical fields (diagnosis, diagnosis_date, etc.) before scoring,
    // otherwise ruin-flag and severity logic operates on ciphertext.
    const { decryptLeadFields } = await import("./encryption");
    const lead = decryptLeadFields(rawLead as Record<string, any>, String(rawLead.id)) as typeof rawLead;

    const [tort, source, settings] = await Promise.all([
      loadTortInputs(lead.tort_type),
      loadSourceInputs(lead.source),
      getEngineSettings(),
    ]);

    if (!tort) {
      logger.warn({ leadId, tort_type: lead.tort_type }, "No tort config for lead; skipping convexity score");
      return null;
    }

    const result = scoreLead(lead, tort, source, settings);

    await db
      .update(leadsTable)
      .set({
        convexity_score: result.classification,
        convexity_action: result.action,
        convexity_rationale: result.rationale,
        convexity_ruin_flags: result.ruin_flags,
        convexity_missing_fields: result.missing_fields,
        convexity_contradictions: result.contradictions,
        convexity_downside_usd: String(result.downside_usd),
        convexity_upside_usd: String(result.upside_usd),
        convexity_ratio: String(result.ratio),
        convexity_confidence: result.confidence,
        convexity_computed_at: new Date(),
      })
      .where(eq(leadsTable.id, leadId));

    return result;
  } catch (e) {
    logger.error({ err: e, leadId }, "computeAndPersistLeadScore failed");
    return null;
  }
}

/**
 * Build the admin portfolio aggregation across all torts.
 */
export async function buildPortfolioSummary(): Promise<PortfolioSummary> {
  const settings = await getEngineSettings();

```

```

// All torts
const tortRows = await db.select().from(formConfigurationsTable);
const torts = new Map<string, TortInputs>();
for (const row of tortRows) {
  torts.set(row.id, {
    id: row.id,
    label: row.label,
    avg_settlement_low: row.avg_settlement_low,
    avg_settlement_high: row.avg_settlement_high,
    expected_duration_months: row.expected_duration_months,
    mdl_status: row.mdl_status,
    sol_months: row.sol_months,
    rejection_conditions: (row.rejection_conditions as unknown as string[]) || [],
    required_exposure: row.required_exposure,
    valid_diagnoses: row.valid_diagnoses || [],
  });
}

// Aggregate leads by tort
const aggRows = await db
  .select({
    tort_type: leadsTable.tort_type,
    lead_count: sql<number>`count(*)::int`,
    total_spend: sql<number>`coalesce(sum(${leadsTable.ad_spend}), 0)::float`,
    qualified: sql<number>`count(*) filter (where ${leadsTable.status} in ('qualified','accepted','retained','signed'))::int`,
    retained: sql<number>`count(*) filter (where ${leadsTable.status} in ('retained','signed'))::int`,
    ruin_flag_count: sql<number>`count(*) filter (where jsonb_array_length(${leadsTable.convexity_ruin_flags}) > 0)::int`,
    convex_count: sql<number>`count(*) filter (where ${leadsTable.convexity_score} = 'convex')::int`,
    concave_count: sql<number>`count(*) filter (where ${leadsTable.convexity_score} = 'concave')::int`,
  })
  .from(leadsTable)
  .groupBy(leadsTable.tort_type);

const byTort = new Map<string, {
  lead_count: number;
  total_spend: number;
  qualified: number;
  retained: number;
  ruin_flag_count: number;
  convex_count: number;
  concave_count: number;
}>();
for (const r of aggRows) {
  byTort.set(r.tort_type, {
    lead_count: Number(r.lead_count),
    total_spend: Number(r.total_spend),
    qualified: Number(r.qualified),
    retained: Number(r.retained),
    ruin_flag_count: Number(r.ruin_flag_count),
    convex_count: Number(r.convex_count),
    concave_count: Number(r.concave_count),
  });
}

return buildPortfolio(byTort, torts, settings);
}

/**
 * Recompute scores for all leads (admin tool).
 */
export async function recomputeAllScores(): Promise<{ scanned: number; scored: number }> {
  const all = await db.select({ id: leadsTable.id }).from(leadsTable);
  let scored = 0;
  for (const { id } of all) {
    const r = await computeAndPersistLeadScore(id);
    if (r) scored++;
  }
  return { scanned: all.length, scored };
}

```

[artifacts/api-server/src/lib/decision-engine.ts](#)

```
/**
 * MTOS Decision Engine – convexity scoring for leads & tort portfolio.
 *
 * Implements the asymmetric-risk framework:
 * 1. Generate outcomes
 * 2. Asymmetry analysis (downside vs upside ratio)
 * 3. Ruin detection (hard filter)
 * 4. Signal vs noise
 * 5. Convexity classification
 * 6. Recommended action
 *
 * Pure functions only – no DB access. All inputs passed in.
 */

import type { Lead } from "@workspace/db";

export type Classification = "convex" | "neutral" | "concave";
export type Action = "execute" | "modify" | "reject" | "review";
export type Confidence = "low" | "medium" | "high";

export interface TortInputs {
  id: string;
  label: string;
  avg_settlement_low: number | null;
  avg_settlement_high: number | null;
  expected_duration_months: number | null;
  mdl_status: string | null;
  sol_months: number | null;
  rejection_conditions: string[];
  required_exposure: boolean;
  valid_diagnoses: string[];
}

export interface SourceInputs {
  name: string;
  cost_per_lead: number | null;
  historical_qualified_rate: number | null;
  historical_retained_rate: number | null;
}

export interface EngineSettings {
  default_attorney_hourly_cost: number;
  default_hours_per_lead: number;
  convex_ratio_threshold: number;
  concave_ratio_threshold: number;
  concentration_warning_pct: number;
}

export interface ScoreResult {
  classification: Classification;
  action: Action;
  rationale: string;
  ruin_flags: string[];
  missing_fields: string[];
  contradictions: string[];
  downside_usd: number;
  upside_usd: number;
  ratio: number;
  confidence: Confidence;
}

/**
 * Critical intake fields. If too many are missing, the convexity score is
 * statistically meaningless and the lead is forced into human review (DEFER).
 * Field codes are stable and surface in the UI via missingFieldLabels.
 */
const BASELINE_CRITICAL_FIELDS = [
  "diagnosis",
  "diagnosis_date",
  "state",
  "contact",
] as const;

const MAX_MISSING_FIELDS_BEFORE_DEFER = 2;

/**
 * Detect missing critical intake fields. Returns short codes (UI labels them).
 * Tort-aware: torts with required_exposure also demand exposure_start.
 */
export function detectMissingFields(
  lead: Pick<Lead, "diagnosis" | "diagnosis_date" | "exposure_start" | "state" | "phone" | "email">,
  tort: TortInputs
): string[] {
  const missing: string[] = [];
  if (!lead.diagnosis || !String(lead.diagnosis).trim()) missing.push("diagnosis");
  if (!lead.diagnosis_date || !String(lead.diagnosis_date).trim()) missing.push("diagnosis_date");
  if (!lead.state || !String(lead.state).trim()) missing.push("state");
  if (!lead.phone && !lead.email) missing.push("contact");
  if (tort.required_exposure && !lead.exposure_start) missing.push("exposure_start");
  return missing;
}

/**
 * Detect cross-field contradictions in the intake data. These usually indicate
 * sloppy data entry, fraud, or upstream form bugs – always force human review.
 */
```

```

*/
export function detectContradictions(
  lead: Pick<Lead, "diagnosis_date" | "exposure_start" | "exposure_end" | "date_of_birth">
): string[] {
  const out: string[] = [];
  const today = new Date();
  today.setHours(23, 59, 59, 999);

  const parse = (v: unknown): Date | null => {
    if (!v) return null;
    let s = String(v).trim();
    // Treat date-only strings as local midnight to avoid UTC-vs-local off-by-one
    // when comparing against `today` constructed in the local timezone.
    if (/^\d{4}-\d{2}-\d{2}$/.test(s)) s = s + "T00:00:00";
    const d = new Date(s);
    return Number.isNaN(d.getTime()) ? null : d;
  };

  const dx = parse(lead.diagnosis_date);
  const expStart = parse(lead.exposure_start);
  const expEnd = parse(lead.exposure_end);
  const dob = parse(lead.date_of_birth);

  if (dx && expStart && dx < expStart) out.push("diagnosis_before_exposure");
  if (expStart && expEnd && expEnd < expStart) out.push("exposure_end_before_start");
  if (dx && dx > today) out.push("diagnosis_in_future");
  if (expStart && expStart > today) out.push("exposure_start_in_future");
  if (dob && dx && dx < dob) out.push("diagnosis_before_birth");
  if (dob && dob > today) out.push("birth_in_future");

  return out;
}

/**
 * Detect ruin paths – irreversible failure conditions per the framework.
 * Each flag is a short machine-readable code; the rationale explains.
 */
export function detectRuinFlags(
  lead: Pick<Lead, "diagnosis" | "diagnosis_confirmed" | "exposure_start" | "state" | "tort_type" | "diagnosis_date" |
"rejection_reason">,
  tort: TortInputs
): string[] {
  const flags: string[] = [];

  // 1. Statute of limitations expired
  if (tort.sol_months && (lead.diagnosis_date || lead.exposure_start)) {
    const startDate = lead.diagnosis_date
      ? new Date(lead.diagnosis_date)
      : lead.exposure_start
      ? new Date(lead.exposure_start)
      : null;
    if (startDate && !Number.isNaN(startDate.getTime())) {
      const ageMonths = (Date.now() - startDate.getTime()) / (1000 * 60 * 60 * 24 * 30.44);
      if (ageMonths > tort.sol_months) flags.push("sol_expired");
    }
  }

  // 2. MDL closed (no recovery vehicle)
  if (tort.mdl_status === "closed") flags.push("mdl_closed");

  // 3. Required diagnosis missing or not valid for this tort
  if (tort.valid_diagnoses.length > 0 && lead.diagnosis) {
    const dx = lead.diagnosis.toLowerCase().trim();
    const matches = tort.valid_diagnoses.some(d => dx.includes(d.toLowerCase()) || d.toLowerCase().includes(dx));
    if (!matches) flags.push("diagnosis_invalid");
  }

  // 4. Required exposure missing
  if (tort.required_exposure && !lead.exposure_start) {
    flags.push("exposure_missing");
  }

  // 5. Already rejected upstream
  if (lead.rejection_reason) flags.push("prior_rejection");

  return flags;
}

/**
 * Score a single lead. Pure function.
 */
export function scoreLead(
  lead: Pick<
    Lead,
    | "diagnosis"
    | "diagnosis_confirmed"
    | "exposure_start"
    | "exposure_end"
    | "state"
    | "tort_type"
    | "diagnosis_date"
    | "date_of_birth"
    | "rejection_reason"
    | "source"
    | "phone"
    | "email"
  >,

```



```

tort: TortInputs,
source: SourceInputs | null,
settings: EngineSettings
): ScoreResult {
    // Pre-score gates run in order: contradictions first (data must be coherent
    // before ruin/SOL math is trustworthy), then ruin (hard veto on coherent data),
    // then missing fields (defer when signal is too thin to score reliably).
    const contradictions = detectContradictions(lead);
    const ruin_flags = detectRuinFlags(lead, tort);
    const missing_fields = detectMissingFields(lead, tort);

    // Step 1+2: Compute downside / upside in USD
    const cpl = source?.cost_per_lead ?? 0;
    const workCost = settings.default_attorney_hourly_cost * settings.default_hours_per_lead;
    const downside_usd = cpl + workCost;

    const settlementMid =
        tort.avg_settlement_low && tort.avg_settlement_high
        ? (tort.avg_settlement_low + tort.avg_settlement_high) / 2
        : tort.avg_settlement_high ?? tort.avg_settlement_low ?? 0;

    // Diagnosis severity multiplier (cancer/death > chronic > acute)
    const dx = (lead.diagnosis || "").toLowerCase();
    let severityMult = 1.0;
    if (/cancer|carcinoma|lymphoma|leukemia|myeloma|sarcoma/.test(dx)) severityMult = 1.4;
    else if (/death|deceased|fatal/.test(dx)) severityMult = 1.6;
    else if (lead.diagnosis_confirmed) severityMult = 1.1;
    else severityMult = 0.7;

    const retainedRate = source?.historical_retained_rate ?? 0.3;
    const upside_usd = settlementMid * retainedRate * severityMult;

    // Step 5: Asymmetry ratio
    const ratio = downside_usd > 0 ? upside_usd / downside_usd : upside_usd > 0 ? Infinity : 0;

    // Confidence: low if any major input is missing
    let confidence: Confidence = "high";
    if (!source?.cost_per_lead || !tort.avg_settlement_low) confidence = "low";
    else if (!source?.historical_retained_rate || !tort.avg_settlement_high) confidence = "medium";

    // Step 6: Decision
    let classification: Classification;
    let action: Action;
    let rationale: string;

    if (ruin_flags.length > 0) {
        classification = "concave";
        action = "review";
        rationale = `Ruin path detected: ${ruin_flags.join(", ")}. Human review required – do not auto-reject.`;
    } else if (contradictions.length > 0) {
        classification = "concave";
        action = "review";
        confidence = "low";
        rationale = `Data contradiction(s): ${contradictions.join(", ")}. Verify intake before scoring.`;
    } else if (missing_fields.length >= MAX_MISSING_FIELDS_BEFORE_DEFER) {
        classification = "neutral";
        action = "review";
        confidence = "low";
        rationale = `Insufficient evidence – missing ${missing_fields.length} critical field(s): ${missing_fields.join(", ")}. Collect before scoring.`;
    } else if (ratio === Infinity) {
        classification = "convex";
        action = "execute";
        rationale = `Unbounded asymmetry – upside present with effectively zero downside.`;
    } else if (ratio >= settings.convex_ratio_threshold) {
        classification = "convex";
        action = "execute";
        rationale = `Strong positive asymmetry (upside/downside = ${ratio.toFixed(1)}x). Scale exposure.`;
    } else if (ratio < settings.concave_ratio_threshold) {
        classification = "concave";
        action = "modify";
        rationale = `Weak asymmetry (${ratio.toFixed(1)}x). Reduce CPL, improve qualification, or pause source.`;
    } else {
        classification = "neutral";
        action = "modify";
        rationale = `Neutral asymmetry (${ratio.toFixed(1)}x). Monitor; not yet a clear convex bet.`;
    }

    if (confidence === "low") {
        rationale += ` (Low confidence: missing CPL or settlement data – calibrate inputs).`;
    }

    return {
        classification,
        action,
        rationale,
        ruin_flags,
        missing_fields,
        contradictions,
        downside_usd: Math.round(downside_usd * 100) / 100,
        upside_usd: Math.round(upside_usd * 100) / 100,
        ratio: Number.isFinite(ratio) ? Math.round(ratio * 100) / 100 : 9999,
        confidence,
    };
}

export interface PortfolioTortRow {

```

```

    tort_id: string;
    label: string;
    lead_count: number;
    total_spend_usd: number;
    qualified_count: number;
    retained_count: number;
    qualified_rate: number;
    retained_rate: number;
    classification: Classification;
    action: Action;
    rationale: string;
    ruin_flag_count: number;
    spend_pct: number;
}

export interface PortfolioSummary {
    total_spend_usd: number;
    tort_count: number;
    convex_count: number;
    concave_count: number;
    ruin_risk_lead_count: number;
    concentration_warning: { tort_id: string; pct: number } | null;
    rows: PortfolioTortRow[];
}

/**
 * Aggregate per-tort portfolio view.
 */
export function buildPortfolio(
    leadsByTort: Map<
        string,
        {
            lead_count: number;
            total_spend: number;
            qualified: number;
            retained: number;
            ruin_flag_count: number;
            convex_count: number;
            concave_count: number;
        }
    >,
    torts: Map<string, TortInputs>,
    settings: EngineSettings
): PortfolioSummary {
    const totalSpend = Array.from(leadsByTort.values()).reduce((s, r) => s + r.total_spend, 0);
    let totalConvex = 0;
    let totalConcave = 0;
    let totalRuin = 0;
    let concentration: { tort_id: string; pct: number } | null = null;

    const rows: PortfolioTortRow[] = [];
    for (const [tort_id, agg] of leadsByTort.entries()) {
        const tort = torts.get(tort_id);
        if (!tort) continue;
        const qualified_rate = agg.lead_count > 0 ? agg.qualified / agg.lead_count : 0;
        const retained_rate = agg.lead_count > 0 ? agg.retained / agg.lead_count : 0;
        const spend_pct = totalSpend > 0 ? (agg.total_spend / totalSpend) * 100 : 0;

        // Tort-level classification = majority of its leads
        let classification: Classification = "neutral";
        let action: Action = "modify";
        let rationale: string;
        if (agg.ruin_flag_count / Math.max(agg.lead_count, 1) > 0.3) {
            classification = "concave";
            action = "reject";
            rationale = `>30% of leads carry ruin flags. Pause intake and audit source quality.`;
        } else if (agg.convex_count > agg.concave_count && retained_rate > 0.15) {
            classification = "convex";
            action = "execute";
            rationale = `Majority convex leads with ${retained_rate * 100}.toFixed(0)}% retention. Scale.`;
        } else if (agg.concave_count > agg.convex_count) {
            classification = "concave";
            action = "modify";
            rationale = `Majority concave leads. Renegotiate CPL or improve filters.`;
        } else {
            rationale = `Mixed signal – collect more data before scaling or pausing.`;
        }

        if (tort.mdl_status === "closed") {
            classification = "concave";
            action = "reject";
            rationale = `MDL is closed – no active recovery vehicle. Cease intake.`;
        } else if (tort.mdl_status === "active_bellwether") {
            rationale += `MDL has active bellwether – settlement signal strong.`;
        }

        if (classification === "convex") totalConvex++;
        if (classification === "concave") totalConcave++;
        totalRuin += agg.ruin_flag_count;

        if (spend_pct > settings.concentration_warning_pct && (!concentration || spend_pct > concentration.pct)) {
            concentration = { tort_id, pct: spend_pct };
        }

        rows.push({
            tort_id,
            label: tort.label,

```

```

    lead_count: agg.lead_count,
    total_spend_usd: Math.round(agg.total_spend * 100) / 100,
    qualified_count: agg.qualified,
    retained_count: agg.retained,
    qualified_rate: Math.round(qualified_rate * 10000) / 10000,
    retained_rate: Math.round(retained_rate * 10000) / 10000,
    classification,
    action,
    rationale,
    ruin_flag_count: agg.ruin_flag_count,
    spend_pct: Math.round(spend_pct * 100) / 100,
  });
}

rows.sort((a, b) => b.total_spend_usd - a.total_spend_usd);

return {
  total_spend_usd: Math.round(totalSpend * 100) / 100,
  tort_count: rows.length,
  convex_count: totalConvex,
  concave_count: totalConcave,
  ruin_risk_lead_count: totalRuin,
  concentration_warning: concentration,
  rows,
};
}

```

[artifacts/api-server/src/lib/drafting-ai.ts](#)

```
import { anthropic } from "@workspace/integrations-anthropic-ai";
import { logger } from "../logger";

const MODEL = "claude-haiku-4-5";

export interface DraftInput {
  template_type: string;
  lead_data: Record<string, any>;
  firm_name?: string;
  attorney_name?: string;
  custom_instructions?: string;
}

export interface DraftOutput {
  title: string;
  content: string;
  template_type: string;
  generated_at: string;
  fields_used: string[];
}

const TEMPLATE_PROMPTS: Record<string, string> = {
  hipaa_authorization: `Generate a HIPAA Authorization form for the patient below. Include:
- Patient identification section
- Description of information to be disclosed
- Purpose of disclosure (legal representation in mass tort litigation)
- Authorized recipients
- Expiration clause
- Right to revoke
- Signature lines with date fields
Format as a professional legal document.`,
  retainer_agreement: `Generate a legal retainer agreement for mass tort representation. Include:
- Client identification
- Scope of representation (specific tort type)
- Fee arrangement (contingency basis, typical 33-40%)
- Client obligations
- Attorney obligations
- Termination clause
- Governing law
- Signature blocks
Format as a professional legal document.`,
  medical_records_request: `Generate a medical records request letter. Include:
- Patient authorization reference
- Specific records requested (related to diagnosis/exposure)
- Date range for records
- Preferred format (electronic/paper)
- Return address and contact
- HIPAA compliance notice
Format as a professional business letter.`,
  demand_letter: `Generate a demand letter template for mass tort litigation. Include:
- Identification of parties
- Statement of facts (exposure, diagnosis, causation)
- Legal basis for claims
- Damages summary
- Settlement demand framework
- Response deadline
Format as a professional legal document.`,
  client_intake_summary: `Generate a client intake summary document. Include:
- Client demographics
- Tort type and exposure details
- Medical history relevant to claim
- Diagnosis and treatment timeline
- Qualification status
- Key evidence summary
- Next steps
Format as an internal case management document.`,
};

export async function generateDraft(input: DraftInput): Promise<DraftOutput> {
  const templatePrompt = TEMPLATE_PROMPTS[input.template_type];
  if (!templatePrompt) {
    throw new Error(`Unknown template type: ${input.template_type}`);
  }

  const leadContext = Object.entries(input.lead_data)
    .filter(([, v]) => v !== null && v !== undefined && v !== "")
    .map(([k, v]) => `${k}: ${v}`)
    .join("\n");

  const fieldsUsed = Object.keys(input.lead_data).filter(k => input.lead_data[k] !== null && input.lead_data[k] !== undefined && input.lead_data[k] !== "");

  const systemPrompt = `You are a legal document drafting assistant for mass tort law firms. Generate professional, legally sound documents. Use formal legal language. Include all standard legal provisions. Leave [BLANK] placeholders for any information not provided.

${input.custom_instructions ? `Additional instructions: ${input.custom_instructions}` : ""}`;

  const userPrompt = `${templatePrompt}

Firm: ${input.firm_name || "[LAW FIRM NAME]"}`;
```

Attorney: \${input.attorney_name || "[ATTORNEY NAME]"}

Client/Lead Data:
\${leadContext}

Generate the complete document now.`;

```
try {
  const response = await anthropic.messages.create({
    model: MODEL,
    max_tokens: 4096,
    system: systemPrompt,
    messages: [{ role: "user", content: userPrompt }],
  });

  const content = response.content[0];
  const text = content.type === "text" ? content.text : "";

  const titleMap: Record<string, string> = {
    hipaa_authorization: "HIPAA Authorization Form",
    retainer_agreement: "Retainer Agreement",
    medical_records_request: "Medical Records Request",
    demand_letter: "Demand Letter",
    client_intake_summary: "Client Intake Summary",
  };

  logger.info({ template_type: input.template_type, length: text.length }, "Draft generated");

  return {
    title: titleMap[input.template_type] || "Legal Document",
    content: text,
    template_type: input.template_type,
    generated_at: new Date().toISOString(),
    fields_used: fieldsUsed,
  };
} catch (err) {
  logger.error({ err, template_type: input.template_type }, "Draft generation failed");
  throw new Error("Failed to generate document draft");
}

export function getAvailableTemplates() {
  return [
    { id: "hipaa_authorization", name: "HIPAA Authorization", description: "Patient authorization for medical records disclosure" },
    { id: "retainer_agreement", name: "Retainer Agreement", description: "Contingency fee retainer for mass tort representation" },
    { id: "medical_records_request", name: "Medical Records Request", description: "Letter requesting patient medical records" },
    { id: "demand_letter", name: "Demand Letter", description: "Pre-litigation demand letter template" },
    { id: "client_intake_summary", name: "Client Intake Summary", description: "Internal case intake summary document" },
  ];
}
```

[artifacts/api-server/src/lib/email-validator.ts](#)

```
const TYPO_DOMAINS: Record<string, string> = {
  "gnail.com": "gmail.com",
  "gmial.com": "gmail.com",
  "gmali.com": "gmail.com",
  "gmil.com": "gmail.com",
  "gmali.com": "gmail.com",
  "gamil.com": "gmail.com",
  "yaho.com": "yahoo.com",
  "yahooo.com": "yahoo.com",
  "yhao.com": "yahoo.com",
  "yhoo.com": "yahoo.com",
  "hotmial.com": "hotmail.com",
  "hotmal.com": "hotmail.com",
  "hotmai.com": "hotmail.com",
  "hotamil.com": "hotmail.com",
  "outlookk.com": "outlook.com",
  "outlook.com": "outlook.com",
  "outlok.com": "outlook.com",
  "outook.com": "outlook.com",
  "aol.con": "aol.com",
  "gmail.con": "gmail.com",
  "gmail.vom": "gmail.com",
  "gmail.cmo": "gmail.com",
  "yahoo.con": "yahoo.com",
  "yahoo.vom": "yahoo.com",
  "yahoo.cmo": "yahoo.com",
  "hotmail.con": "hotmail.com",
  "hotmail.vom": "hotmail.com",
  "outlook.con": "outlook.com",
  "outlook.vom": "outlook.com",
};

const MALFORMED_TLDS = [".vom", ".con", ".cmo", ".coom", ".comm", ".cim", ".ocm", ".cm"];

const RFC_EMAIL_REGEX = /^[a-zA-Z0-9.!#$%&'*/+=?^_`{|}~-]+@[a-zA-Z0-9](?:[a-zA-Z0-9-]{0,61}[a-zA-Z0-9])?(?:\.[a-zA-Z0-9](?:[a-zA-Z0-9-]{0,61}[a-zA-Z0-9])?)*$/;

const DISPOSABLE_DOMAINS = [
  "tempmail.com", "throwaway.email", "guerrillamail.com", "mailinator.com",
  "yopmail.com", "sharklasers.com", "trashmail.com", "10minutemail.com",
  "fakeinbox.com", "dispostable.com", "maildrop.cc", "guerrillamailblock.com",
];

export interface EmailValidationResult {
  valid: boolean;
  errors: string[];
  suggestion?: string;
}

export function validateEmail(email: string): EmailValidationResult {
  const errors: string[] = [];
  let suggestion: string | undefined;

  if (!email || typeof email !== "string") {
    return { valid: false, errors: ["MISSING_EMAIL"] };
  }

  const trimmed = email.trim().toLowerCase();

  if (!RFC_EMAIL_REGEX.test(trimmed)) {
    errors.push("INVALID_RFC_FORMAT");
    return { valid: false, errors };
  }

  const parts = trimmed.split("@");
  if (parts.length !== 2) {
    errors.push("INVALID_EMAIL_STRUCTURE");
    return { valid: false, errors };
  }

  const [localPart, domain] = parts;

  if (localPart.length === 0 || localPart.length > 64) {
    errors.push("INVALID_LOCAL_PART");
  }

  if (domain.length === 0 || domain.length > 253) {
    errors.push("INVALID_DOMAIN");
  }

  if (!domain.includes(".")) {
    errors.push("MISSING_TLD");
  }

  if (TYPO_DOMAINS[domain]) {
    errors.push("TYPO_DOMAIN_DETECTED");
    suggestion = `${localPart}@${TYPO_DOMAINS[domain]}`;
  }

  for (const tld of MALFORMED_TLDS) {
    if (trimmed.endsWith(tld)) {
      errors.push("MALFORMED_TLD");
      const corrected = trimmed.slice(0, -tld.length) + ".com";
      suggestion = corrected;
      break;
    }
  }
}
```

```

    }
  }
  if (DISPOSABLE_DOMAINS.includes(domain)) {
    errors.push("DISPOSABLE_EMAIL");
  }

  const suspiciousPatterns = [
    /^test@/,
    /^fake@/,
    /^none@/,
    /^noemail@/,
    /^na@/,
    /^asdf/,
    /^aaa+@/,
  ];
  for (const pat of suspiciousPatterns) {
    if (pat.test(trimmed)) {
      errors.push("SUSPICIOUS_EMAIL_PATTERN");
      break;
    }
  }

  return {
    valid: errors.length === 0,
    errors,
    suggestion,
  };
}

```

[artifacts/api-server/src/lib/email/sendgrid.ts](#)

```
import { logger } from "../../logger";
import type { DecryptedCredentials } from "../../../routes/integrations";

export interface EmailSendRequest {
  to: string;
  toName?: string;
  fromEmail: string;
  fromName?: string;
  subject: string;
  html: string;
  text?: string;
  replyTo?: string;
}

export interface EmailSendResult {
  ok: true;
  externalMessageId?: string;
}

export interface EmailError {
  ok: false;
  retryable: boolean;
  code: string;
  message: string;
}

export type EmailSendOutcome = EmailSendResult | EmailError;

export interface EmailAdapter {
  provider: string;
  send(creds: DecryptedCredentials, req: EmailSendRequest): Promise<EmailSendOutcome>;
}

/**
 * SendGrid v3 mail adapter. Auth: Bearer API key.
 */
export const sendgridAdapter: EmailAdapter = {
  provider: "sendgrid",

  async send(creds, req): Promise<EmailSendOutcome> {
    const apiKey = creds.api_key;
    if (!apiKey) {
      return { ok: false, retryable: false, code: "no_api_key", message: "SendGrid API key missing" };
    }

    const url = "https://api.sendgrid.com/v3/mail/send";
    const payload = {
      personalizations: [
        { to: [{ email: req.to, name: req.toName }] },
      ],
      from: { email: req.fromEmail, name: req.fromName },
      reply_to: req.replyTo ? { email: req.replyTo } : undefined,
      subject: req.subject,
      content: [
        ...(req.text ? [{ type: "text/plain", value: req.text }] : []),
        { type: "text/html", value: req.html },
      ],
    };

    let response: Response;
    try {
      response = await fetch(url, {
        method: "POST",
        headers: {
          Authorization: `Bearer ${apiKey}`,
          "Content-Type": "application/json",
        },
        body: JSON.stringify(payload),
      });
    } catch (err) {
      logger.error({ err }, "SendGrid network error");
      return { ok: false, retryable: true, code: "network_error", message: String((err as Error).message) };
    }

    if (!response.ok) {
      let body: any = null;
      try { body = await response.json(); } catch { /* ignore */ }
      const errMsg = body?.errors?.[0]?.message || `HTTP ${response.status}`;
      const retryable = response.status >= 500 || response.status === 429;
      return {
        ok: false,
        retryable,
        code: `http_${response.status}`,
        message: errMsg,
      };
    }

    return { ok: true, externalMessageId: response.headers.get("x-message-id") || undefined };
  },
};

const ADAPTERS: Record<string, EmailAdapter> = {
  [sendgridAdapter.provider]: sendgridAdapter,
};
```



```
export function getEmailAdapter(provider: string): EmailAdapter | null {  
  return ADAPTERS[provider] || null;  
}
```

[artifacts/api-server/src/lib/encryption.ts](#)

```
import crypto from "crypto";
import { logger } from "../logger";

const ALGORITHM = "aes-256-gcm";
const IV_LENGTH = 12;
const AUTH_TAG_LENGTH = 16;
const ENCODING = "base64" as const;
const CURRENT_KEY_VERSION = 1;

function getKey(version?: number): Buffer {
  const keyVersion = version ?? CURRENT_KEY_VERSION;
  const envName = keyVersion === 1 ? "ENCRYPTION_KEY" : `ENCRYPTION_KEY_V${keyVersion}`;
  const key = process.env[envName] || process.env.ENCRYPTION_KEY;
  if (!key) {
    throw new Error(`${envName} environment variable is required`);
  }
  if (!/^[0-9a-fA-F]{64}$/.test(key)) {
    throw new Error(`${envName} must be exactly 64 hex characters (32 bytes for AES-256)`);
  }
  return Buffer.from(key, "hex");
}

function buildAAD(fieldName?: string, entityId?: string): Buffer | undefined {
  if (!fieldName) return undefined;
  const parts = [fieldName];
  if (entityId) parts.push(entityId);
  return Buffer.from(parts.join(":"), "utf8");
}

export function encrypt(plaintext: string, fieldName?: string, entityId?: string): string {
  if (!plaintext) return plaintext;
  const key = getKey(CURRENT_KEY_VERSION);
  const iv = crypto.randomBytes(IV_LENGTH);
  const cipher = crypto.createCipheriv(ALGORITHM, key, iv, { authTagLength: AUTH_TAG_LENGTH });
  const aad = buildAAD(fieldName, entityId);
  if (aad) cipher.setAAD(aad);
  const encrypted = Buffer.concat([cipher.update(plaintext, "utf8"), cipher.final()]);
  const authTag = cipher.getAuthTag();
  const combined = Buffer.concat([iv, authTag, encrypted]);
  const hasAAD = aad ? 1 : 0;
  return `enc:v${CURRENT_KEY_VERSION}:${hasAAD}:${combined.toString(ENCODING)}`;
}

export function decrypt(ciphertext: string, fieldName?: string, entityId?: string): string {
  if (!ciphertext) return ciphertext;
  if (!ciphertext.startsWith("enc:")) return ciphertext;
  try {
    let keyVersion = 1;
    let hasAADFlag = 0;
    let payload: string;

    if (ciphertext.startsWith("enc:v")) {
      const parts = ciphertext.split(":");
      keyVersion = parseInt(parts[1].slice(1, 10) || 1);
      hasAADFlag = parseInt(parts[2], 10) || 0;
      payload = parts.slice(3).join(":");
    } else {
      payload = ciphertext.slice(4);
    }

    const key = getKey(keyVersion);
    const combined = Buffer.from(payload, ENCODING);
    const iv = combined.subarray(0, IV_LENGTH);
    const authTag = combined.subarray(IV_LENGTH, IV_LENGTH + AUTH_TAG_LENGTH);
    const encrypted = combined.subarray(IV_LENGTH + AUTH_TAG_LENGTH);
    const decipher = crypto.createDecipheriv(ALGORITHM, key, iv, { authTagLength: AUTH_TAG_LENGTH });
    decipher.setAuthTag(authTag);
    if (hasAADFlag && fieldName) {
      const aad = buildAAD(fieldName, entityId);
      if (aad) decipher.setAAD(aad);
    }
    const decrypted = Buffer.concat([decipher.update(encrypted), decipher.final()]);
    return decrypted.toString("utf8");
  } catch {
    if (!ciphertext.startsWith("enc:v")) {
      try {
        const key = getKey(1);
        const combined = Buffer.from(ciphertext.slice(4), ENCODING);
        const iv = combined.subarray(0, IV_LENGTH);
        const authTag = combined.subarray(IV_LENGTH, IV_LENGTH + AUTH_TAG_LENGTH);
        const encrypted = combined.subarray(IV_LENGTH + AUTH_TAG_LENGTH);
        const decipher = crypto.createDecipheriv(ALGORITHM, key, iv, { authTagLength: AUTH_TAG_LENGTH });
        decipher.setAuthTag(authTag);
        const decrypted = Buffer.concat([decipher.update(encrypted), decipher.final()]);
        return decrypted.toString("utf8");
      } catch {
        logger.error("Decryption failed – data may be corrupted or key mismatch");
        return "[DECRYPTION_ERROR]";
      }
    }
    logger.error("Decryption failed – AAD mismatch or data corruption on versioned ciphertext");
    return "[DECRYPTION_ERROR]";
  }
}
```

```

export const ENCRYPTED_FIELDS = [
  "last_4_ssn",
  "date_of_birth",
  "diagnosis",
  "diagnosis_date",
  "street_address",
  "phone_primary",
  "phone",
  "medications",
  "notes",
  "physician_full_address",
  "physician_contact_info",
  "hospital_contact_info",
  "background_check_data",
] as const;

export function encryptLeadFields(data: Record<string, any>, entityId?: string): Record<string, any> {
  const result = { ...data };
  for (const field of ENCRYPTED_FIELDS) {
    if (result[field] !== undefined && result[field] !== null && typeof result[field] === "string") {
      if (!result[field].startsWith("enc:")) {
        result[field] = encrypt(result[field], field, entityId);
      }
    }
  }
  return result;
}

export function decryptLeadFields(data: Record<string, any>, entityId?: string): Record<string, any> {
  if (!data) return data;
  const result = { ...data };
  for (const field of ENCRYPTED_FIELDS) {
    if (result[field] !== undefined && result[field] !== null && typeof result[field] === "string") {
      result[field] = decrypt(result[field], field, entityId);
    }
  }
  return result;
}

export function decryptLeadArray(leads: Record<string, any>[]): Record<string, any>[] {
  return leads.map(l => decryptLeadFields(l, String(l.id)));
}

export function hashForLookup(value: string): string {
  return crypto.createHmac("sha256", getKey()).update(value.toLowerCase().trim()).digest("hex");
}

export function reEncryptField(ciphertext: string, fieldName?: string, entityId?: string): string {
  const plain = decrypt(ciphertext, fieldName, entityId);
  if (plain === "[DECRYPTION_ERROR]") return ciphertext;
  return encrypt(plain, fieldName, entityId);
}

```

[artifacts/api-server/src/lib/error-fallback.ts](#)

```
import { auditLog } from "../audit";
import { logger } from "../logger";
import { db, reviewQueueTable } from "@workspace/db";
import type { SystemOutputState, FailsafeMode } from "../conflict-engine";

const MAX_RETRIES = 2;
const MAX_AI_RECHECKS = 1;
const MAX_RULE_RE_EVALUATIONS = 2;

export interface ExecutionLimits {
  max_retries: number;
  max_ai_rechecks: number;
  max_rule_re_evaluations: number;
}

export const DEFAULT_LIMITS: ExecutionLimits = {
  max_retries: MAX_RETRIES,
  max_ai_rechecks: MAX_AI_RECHECKS,
  max_rule_re_evaluations: MAX_RULE_RE_EVALUATIONS,
};

interface LoopState {
  retries: number;
  ai_rechecks: number;
  rule_re_evaluations: number;
}

interface FallbackContext {
  entity_type: string;
  entity_id: string;
  source_module: string;
  failsafe_mode?: FailsafeMode;
}

export interface FallbackResult<T> {
  success: boolean;
  output_state: SystemOutputState;
  data: T | null;
  error: string | null;
  retries_used: number;
  loop_aborted: boolean;
}

function sanitizeInput(input: Record<string, unknown>): Record<string, unknown> {
  const sanitized: Record<string, unknown> = {};
  for (const [key, value] of Object.entries(input)) {
    if (typeof value === "string") {
      sanitized[key] = value.trim().replace(/[\^x20-\x7E\u00C0-\u024F\u1E00-\u1EFF]/g, "");
    } else if (value === undefined) {
      sanitized[key] = null;
    } else {
      sanitized[key] = value;
    }
  }
  return sanitized;
}

export async function withErrorFallback<T>(
  fn: (input: Record<string, unknown>, attempt: number) => Promise<T>,
  input: Record<string, unknown>,
  ctx: FallbackContext,
  limits: ExecutionLimits = DEFAULT_LIMITS
): Promise<FallbackResult<T>> {
  const state: LoopState = { retries: 0, ai_rechecks: 0, rule_re_evaluations: 0 };
  let lastError: string | null = null;

  while (state.retries <= limits.max_retries) {
    try {
      const currentInput = state.retries === 0 ? input : sanitizeInput(input);
      const result = await fn(currentInput, state.retries);

      await auditLog(ctx.entity_type, ctx.entity_id, "execution_success", {
        source_module: ctx.source_module,
        attempt: state.retries + 1,
        sanitized: state.retries > 0,
      });

      return {
        success: true,
        output_state: "ACCEPT",
        data: result,
        error: null,
        retries_used: state.retries,
        loop_aborted: false,
      };
    } catch (err) {
      lastError = err instanceof Error ? err.message : String(err);
      state.retries++;

      logger.error(
        { err, entity: ctx.entity_id, module: ctx.source_module, attempt: state.retries, max: limits.max_retries + 1 },
        "Execution failed – triggering fallback"
      );

      await auditLog(ctx.entity_type, ctx.entity_id, "execution_error", {
```

```

        source_module: ctx.source_module,
        error: lastError,
        attempt: state.retries,
        will_retry: state.retries <= limits.max_retries,
    });

    if (state.retries > limits.max_retries) {
        break;
    }

    logger.info(
        { entity: ctx.entity_id, attempt: state.retries },
        "Retrying with sanitized input"
    );
}

logger.error(
    { entity: ctx.entity_id, module: ctx.source_module, retries: state.retries },
    "All retries exhausted – routing to REVIEW_REQUIRED"
);

const failsafe = ctx.failsafe_mode || "REVIEW_FAIL";
const outputState: SystemOutputState = failsafe === "SAFE_FAIL" ? "REJECT" : "REVIEW_REQUIRED";

await db.insert(reviewQueueTable).values({
    entity_type: ctx.entity_type,
    entity_id: ctx.entity_id,
    conflict_type: "EXECUTION_FAILURE",
    severity: "high",
    failsafe_mode: failsafe,
    source_module: ctx.source_module,
    summary: `Execution failed after ${state.retries} attempts: ${lastError}`,
    details: {
        error: lastError,
        retries_used: state.retries,
        output_state: outputState,
        original_input: input,
    },
    resolution: "pending",
    retry_count: state.retries,
});

await auditLog(ctx.entity_type, ctx.entity_id, "fallback_triggered", {
    source_module: ctx.source_module,
    failsafe_mode: failsafe,
    output_state: outputState,
    error: lastError,
    retries_exhausted: true,
});

return {
    success: false,
    output_state: outputState,
    data: null,
    error: lastError,
    retries_used: state.retries,
    loop_aborted: state.retries > limits.max_retries,
};
}

export function createLoopGuard(limits: ExecutionLimits = DEFAULT_LIMITS) {
    const state: LoopState = { retries: 0, ai_rechecks: 0, rule_re_evaluations: 0 };

    return {
        canRetry(): boolean {
            return state.retries < limits.max_retries;
        },
        canAIRecheck(): boolean {
            return state.ai_rechecks < limits.max_ai_rechecks;
        },
        canReEvaluateRules(): boolean {
            return state.rule_re_evaluations < limits.max_rule_re_evaluations;
        },
        recordRetry(): void {
            state.retries++;
        },
        recordAIRecheck(): void {
            state.ai_rechecks++;
        },
        recordRuleReEvaluation(): void {
            state.rule_re_evaluations++;
        },
        isExhausted(): boolean {
            return (
                state.retries >= limits.max_retries &&
                state.ai_rechecks >= limits.max_ai_rechecks &&
                state.rule_re_evaluations >= limits.max_rule_re_evaluations
            );
        },
        getState(): LoopState & { loop_aborted: boolean } {
            return {
                ...state,
                loop_aborted:
                    state.retries >= limits.max_retries ||
                    state.ai_rechecks >= limits.max_ai_rechecks ||
                    state.rule_re_evaluations >= limits.max_rule_re_evaluations,
            };
        }
    };
}

```

```
} };
```

[artifacts/api-server/src/lib/esign/docusign.ts](#)

```
import type { EsignAdapter, EsignSendOutcome } from "../types";
import { logger } from "../logger";

/**
 * DocuSign adapter (REST API v2.1).
 * Auth: OAuth2 JWT – but for the simplified path-of-least-resistance flow we accept a
 * pre-issued OAuth access token stored as `api_key` and the account_id stored in `account_sid`.
 * Admins can paste a token from DocuSign's "Apps and Keys" sandbox tool.
 *
 * For long-term automated token rotation, swap to the JWT grant flow inside this adapter.
 * The contract returned to the workflow engine does not change.
 */
export const docusignAdapter: EsignAdapter = {
  provider: "docusign",

  async send(creds, req): Promise<EsignSendOutcome> {
    const accessToken = creds.api_key;
    const accountId = creds.account_sid;
    const apiBase = creds.api_url || "https://demo.docusign.net/restapi";

    if (!accessToken) {
      return { ok: false, retryable: false, code: "no_access_token", message: "DocuSign access token missing (store as api_key)" };
    }
    if (!accountId) {
      return { ok: false, retryable: false, code: "no_account_id", message: "DocuSign account_id missing (store as account_sid)" };
    }

    const url = `${apiBase.replace(/\/$/, "")}/v2.1/accounts/${accountId}/envelopes`;

    const document = {
      documentBase64: req.pdf.toString("base64"),
      name: req.fileName,
      fileExtension: "pdf",
      documentId: "1",
    };

    const recipients = {
      signers: req.signers.map((s, i) => ({
        email: s.email,
        name: s.name,
        recipientId: String(i + 1),
        routingOrder: String(i + 1),
        tabs: {
          signHereTabs: [
            { anchorString: "/sig/", anchorXOffset: "0", anchorYOffset: "0", anchorUnits: "inches" },
          ],
        },
      })),
    };

    const payload = {
      emailSubject: req.subject,
      emailBlurb: req.message || "",
      documents: [document],
      recipients,
      status: "sent",
      customFields: req.metadata
        ? {
            textCustomFields: Object.entries(req.metadata).map(([name, value]) => ({
              name,
              value,
              required: "false",
              show: "false",
            })),
          }
        : undefined,
    };

    let response: Response;
    try {
      response = await fetch(url, {
        method: "POST",
        headers: {
          Authorization: `Bearer ${accessToken}`,
          "Content-Type": "application/json",
        },
        body: JSON.stringify(payload),
      });
    } catch (err) {
      logger.error({ err, provider: "docusign" }, "Network error sending DocuSign envelope");
      return { ok: false, retryable: true, code: "network_error", message: String((err as Error).message) };
    }

    let body: any = null;
    try {
      body = await response.json();
    } catch {
      // ignore
    }

    if (!response.ok) {
      const errMsg = body?.message || body?.errorCode || `HTTP ${response.status}`;
      const retryable = response.status >= 500 || response.status === 429 || response.status === 401;
    }
  }
};
```

```

    return {
      ok: false,
      retryable,
      code: body?.errorCode || `http_${response.status}`,
      message: errMsg,
      rawResponse: body,
    };
  }

  if (!body?.envelopeId) {
    return {
      ok: false,
      retryable: false,
      code: "no_envelope_id",
      message: "DocuSign returned success without envelopeId",
      rawResponse: body,
    };
  }

  return {
    ok: true,
    externalEnvelopeId: body.envelopeId,
    rawResponse: body,
  };
},

async downloadSigned(creds, externalEnvelopeId): Promise<Buffer | null> {
  const accessToken = creds.api_key;
  const accountId = creds.account_sid;
  const apiBase = creds.api_url || "https://demo.docusign.net/restapi";
  if (!accessToken || !accountId) return null;
  const url = `${apiBase.replace(/\$/ /, "")}/v2.1/accounts/${accountId}/envelopes/${externalEnvelopeId}/documents/combined`;
  try {
    const res = await fetch(url, { headers: { Authorization: `Bearer ${accessToken}` } });
    if (!res.ok) return null;
    const ab = await res.arrayBuffer();
    return Buffer.from(ab);
  } catch (err) {
    logger.error({ err, externalEnvelopeId }, "Failed to download signed DocuSign PDF");
    return null;
  }
},
};

```


[artifacts/api-server/src/lib/esign/dropbox-sign.ts](#)

```
import type { EsignAdapter, EsignSendOutcome } from "../types";
import { logger } from "../logger";

/**
 * Dropbox Sign (formerly HelloSign) adapter.
 * API: https://developers.hellosign.com/api/reference/operation/signatureRequestSend
 * Auth: HTTP Basic with API key as username, empty password.
 *
 * Uses /signature_request/send (not template-based) since our templates are
 * arbitrary admin-uploaded PDFs.
 */
export const dropboxSignAdapter: EsignAdapter = {
  provider: "dropbox_sign",

  async send(creds, req): Promise<EsignSendOutcome> {
    const apiKey = creds.api_key;
    if (!apiKey) {
      return { ok: false, retryable: false, code: "no_api_key", message: "Dropbox Sign API key missing" };
    }

    const baseUrl = creds.api_url || "https://api.hellosign.com/v3";
    const url = `${baseUrl.replace(/\/$/, "")}/signature_request/send`;

    const form = new FormData();
    form.append("title", req.subject);
    form.append("subject", req.subject);
    if (req.message) form.append("message", req.message);
    if (req.expireInDays) form.append("signing_options[default_type]", "draw");

    const blob = new Blob([new Uint8Array(req.pdf)], { type: "application/pdf" });
    form.append("file[0]", blob, req.fileName);

    req.signers.forEach((s, i) => {
      form.append(`signers[${i}][name]`, s.name);
      form.append(`signers[${i}][email_address]`, s.email);
      if (s.role) form.append(`signers[${i}][role]`, s.role);
    });

    if (req.metadata) {
      for (const [k, v] of Object.entries(req.metadata)) {
        form.append(`metadata[${k}]`, v);
      }
    }

    form.append("test_mode", process.env.NODE_ENV === "production" ? "0" : "1");

    let response: Response;
    try {
      response = await fetch(url, {
        method: "POST",
        headers: {
          Authorization: `Basic ${Buffer.from(`${apiKey}:`).toString("base64")}`,
        },
        body: form,
      });
    } catch (err) {
      logger.error({ err, provider: "dropbox_sign" }, "Network error sending envelope");
      return { ok: false, retryable: true, code: "network_error", message: String((err as Error).message) };
    }

    let body: any = null;
    try {
      body = await response.json();
    } catch {
      // ignore
    }

    if (!response.ok) {
      const errMsg = body?.error?.error_msg || body?.error_msg || `HTTP ${response.status}`;
      const retryable = response.status >= 500 || response.status === 429;
      return {
        ok: false,
        retryable,
        code: body?.error?.error_name || `http_${response.status}`,
        message: errMsg,
        rawResponse: body,
      };
    }

    const sigReq = body?.signature_request;
    if (!sigReq?.signature_request_id) {
      return {
        ok: false,
        retryable: false,
        code: "no_envelope_id",
        message: "Dropbox Sign returned 200 but no signature_request_id",
        rawResponse: body,
      };
    }

    return {
      ok: true,
      externalEnvelopeId: sigReq.signature_request_id,
      rawResponse: body,
    };
  }
};
```

```

    },
    async downloadSigned(creds, externalEnvelopeId): Promise<Buffer | null> {
      const apiKey = creds.api_key;
      if (!apiKey) return null;
      const baseUrl = creds.api_url || "https://api.hellosign.com/v3";
      const url = `${baseUrl.replace(/\/$/, "")}/signature_request/files/${externalEnvelopeId}?file_type=pdf`;
      try {
        const res = await fetch(url, {
          headers: { Authorization: `Basic ${Buffer.from(`${apiKey}:`).toString("base64")}` },
        });
        if (!res.ok) return null;
        const ab = await res.arrayBuffer();
        return Buffer.from(ab);
      } catch (err) {
        logger.error({ err, externalEnvelopeId }, "Failed to download signed Dropbox Sign PDF");
        return null;
      }
    },
    async cancel(creds, externalEnvelopeId): Promise<void> {
      const apiKey = creds.api_key;
      if (!apiKey) return;
      const baseUrl = creds.api_url || "https://api.hellosign.com/v3";
      const url = `${baseUrl.replace(/\/$/, "")}/signature_request/cancel/${externalEnvelopeId}`;
      try {
        await fetch(url, {
          method: "POST",
          headers: { Authorization: `Basic ${Buffer.from(`${apiKey}:`).toString("base64")}` },
        });
      } catch (err) {
        logger.error({ err, externalEnvelopeId }, "Failed to cancel Dropbox Sign envelope");
      }
    },
  },
};

```

[artifacts/api-server/src/lib/esign/index.ts](#)

```
import type { EsignAdapter } from "../types";
import { dropboxSignAdapter } from "../dropbox-sign";
import { docusignAdapter } from "../docusign";

export * from "../types";

/**
 * Registry of every wired e-sign adapter, keyed by integration provider id.
 * To add a new provider: write a new adapter implementing EsignAdapter,
 * then register it here. No other code changes needed.
 */
const ADAPTERS: Record<string, EsignAdapter> = {
  [dropboxSignAdapter.provider]: dropboxSignAdapter,
  [docusignAdapter.provider]: docusignAdapter,
};

export function getEsignAdapter(provider: string): EsignAdapter | null {
  return ADAPTERS[provider] || null;
}

export function listEsignProviders(): string[] {
  return Object.keys(ADAPTERS);
}
```

[artifacts/api-server/src/lib/esign/types.ts](#)

```
/**
 * Common interface every e-sign provider adapter must implement.
 * Adapters are pure functions: they take credentials + a request and return a normalized result.
 * NEVER throw on expected provider errors – return a structured EsignError so the worker can
 * decide whether to retry, dead-letter, or surface to the user.
 */

import type { DecryptedCredentials } from "../../routes/integrations";

export interface EsignSigner {
  name: string;
  email: string;
  role?: string;
}

export interface SendEnvelopeRequest {
  pdf: Buffer;
  fileName: string;
  subject: string;
  message?: string;
  signers: EsignSigner[];
  metadata?: Record<string, string>;
  expireInDays?: number;
}

export interface SendEnvelopeResult {
  ok: true;
  externalEnvelopeId: string;
  signingUrl?: string;
  rawResponse?: unknown;
}

export interface EsignError {
  ok: false;
  retryable: boolean;
  code: string;
  message: string;
  rawResponse?: unknown;
}

export type EsignSendOutcome = SendEnvelopeResult | EsignError;

export interface EsignAdapter {
  provider: string;
  send(creds: DecryptedCredentials, req: SendEnvelopeRequest): Promise<EsignSendOutcome>;
  /** Optional: download the signed PDF for a completed envelope. */
  downloadSigned?(creds: DecryptedCredentials, externalEnvelopeId: string): Promise<Buffer | null>;
  /** Optional: void/cancel an envelope. */
  cancel?(creds: DecryptedCredentials, externalEnvelopeId: string): Promise<void>;
}
```

[artifacts/api-server/src/lib/fax/index.ts](#)

```
import type { FaxAdapter } from "../types";
import { telnyxFaxAdapter } from "../telnyx";

export * from "../types";

const ADAPTERS: Record<string, FaxAdapter> = {
  [telnyxFaxAdapter.provider]: telnyxFaxAdapter,
};

export function getFaxAdapter(provider: string): FaxAdapter | null {
  return ADAPTERS[provider] || null;
}

export function listFaxProviders(): string[] {
  return Object.keys(ADAPTERS);
}
```

[artifacts/api-server/src/lib/fax/telnyx.ts](#)

```
import type { FaxAdapter, FaxSendOutcome } from "../types";
import { logger } from "../logger";

/**
 * Telnyx Programmable Fax adapter.
 * API: https://developers.telnyx.com/docs/programmable-fax/send-receive-fax
 * Auth: Bearer API key.
 *
 * Telnyx requires the PDF as a publicly reachable URL OR as multipart upload – we use
 * inline base64 to keep things HIPAA-clean (PHI never leaves our infra in plaintext URLs).
 */
export const telnyxFaxAdapter: FaxAdapter = {
  provider: "telnyx_fax",

  async send(creds, req): Promise<FaxSendOutcome> {
    const apiKey = creds.api_key;
    if (!apiKey) {
      return { ok: false, retryable: false, code: "no_api_key", message: "Telnyx API key missing" };
    }
    if (!req.toNumber) {
      return { ok: false, retryable: false, code: "no_to_number", message: "Recipient fax number missing" };
    }
    if (!req.fromNumber) {
      return {
        ok: false,
        retryable: false,
        code: "no_from_number",
        message: "Sender fax number missing – set default_fax_from_number in Workflow Settings",
      };
    }

    const baseUrl = creds.api_url || "https://api.telnyx.com/v2";
    const url = `${baseUrl.replace(/\/$/, "")}/faxes`;

    const payload = {
      to: req.toNumber,
      from: req.fromNumber,
      media_name: req.fileName,
      // Telnyx accepts media_name + media_url OR contents (base64). Use contents for inline.
      contents: req.pdf.toString("base64"),
      monochrome: true,
      store_media: true,
      t38_enabled: true,
    };

    let response: Response;
    try {
      response = await fetch(url, {
        method: "POST",
        headers: {
          Authorization: `Bearer ${apiKey}`,
          "Content-Type": "application/json",
        },
        body: JSON.stringify(payload),
      });
    } catch (err) {
      logger.error({ err, provider: "telnyx_fax" }, "Network error sending fax");
      return { ok: false, retryable: true, code: "network_error", message: String((err as Error).message) };
    }

    let body: any = null;
    try {
      body = await response.json();
    } catch {
      // ignore
    }

    if (!response.ok) {
      const errMsg = body?.errors?.[0]?.detail || body?.errors?.[0]?.title || `HTTP ${response.status}`;
      const retryable = response.status >= 500 || response.status === 429;
      return {
        ok: false,
        retryable,
        code: body?.errors?.[0]?.code || `http_${response.status}`,
        message: errMsg,
        rawResponse: body,
      };
    }

    const faxId = body?.data?.id;
    if (!faxId) {
      return {
        ok: false,
        retryable: false,
        code: "no_fax_id",
        message: "Telnyx returned 200 but no fax id",
        rawResponse: body,
      };
    }

    return { ok: true, externalFaxId: faxId, rawResponse: body };
  },

  async getStatus(creds, externalFaxId): Promise<string> {
    const apiKey = creds.api_key;
```

```
if (!apiKey) return "unknown";
const baseUrl = creds.api_url || "https://api.telnyx.com/v2";
try {
  const res = await fetch(`${baseUrl.replace(/\/$/, "")}/faxes/${externalFaxId}`, {
    headers: { Authorization: `Bearer ${apiKey}` },
  });
  if (!res.ok) return "unknown";
  const body = (await res.json()) as { data?: { status?: string } };
  return body?.data?.status || "unknown";
} catch {
  return "unknown";
}
},
};
```

[artifacts/api-server/src/lib/fax/types.ts](#)

```
/**
 * Common interface every fax adapter must implement.
 * Same contract pattern as e-sign adapters – return structured results, never throw on
 * expected provider errors.
 */

import type { DecryptedCredentials } from "../../routes/integrations";

export interface SendFaxRequest {
  pdf: Buffer;
  fileName: string;
  toNumber: string;
  fromNumber?: string;
  metadata?: Record<string, string>;
}

export interface SendFaxResult {
  ok: true;
  externalFaxId: string;
  rawResponse?: unknown;
}

export interface FaxError {
  ok: false;
  retryable: boolean;
  code: string;
  message: string;
  rawResponse?: unknown;
}

export type FaxSendOutcome = SendFaxResult | FaxError;

export interface FaxAdapter {
  provider: string;
  send(creds: DecryptedCredentials, req: SendFaxRequest): Promise<FaxSendOutcome>;
  /** Optional: poll status of a sent fax. */
  getStatus?(creds: DecryptedCredentials, externalFaxId: string): Promise<string>;
}
```


[artifacts/api-server/src/lib/final-arbiter.ts](#)

```
import { TortValidationResult } from "../tort-engine";
import { TaxonomyMatchResult } from "../taxonomy-engine";
import { FraudDetectionResult } from "../fraud-engine";
import { TORT_REGISTRY } from "../tort-engine";

export type ArbiterDecision = "ACCEPTED" | "REJECTED" | "TO_BE_REVIEWED";

export interface ArbiterResult {
  decision: ArbiterDecision;
  reason: string;
  deciding_engine: string;
}

export function finalArbiter(context: {
  compliance_passed: boolean;
  compliance_errors: string[];
  validation_passed: boolean;
  validation_errors: string[];
  tort_validation: TortValidationResult;
  taxonomy_match: TaxonomyMatchResult | null;
  fraud_result: FraudDetectionResult;
  npf_found: boolean;
  diagnosis_confirmed: boolean;
  exposure_confirmed: boolean;
  exposure_start?: string;
  conflict_reject: boolean;
  conflict_review: boolean;
}): ArbiterResult {

  if (!context.compliance_passed) {
    return {
      decision: "REJECTED",
      reason: `Compliance failure: ${context.compliance_errors.join(", ")}`,
      deciding_engine: "COMPLIANCE_ENGINE",
    };
  }

  if (!context.validation_passed) {
    return {
      decision: "REJECTED",
      reason: `Validation failure: ${context.validation_errors.join(", ")}`,
      deciding_engine: "VALIDATION_ENGINE",
    };
  }

  if (context.conflict_reject) {
    return {
      decision: "REJECTED",
      reason: "Hard conflict detected",
      deciding_engine: "CONFLICT_ENGINE",
    };
  }

  if (context.conflict_review) {
    return {
      decision: "TO_BE_REVIEWED",
      reason: "Conflict detected requiring review",
      deciding_engine: "CONFLICT_ENGINE",
    };
  }

  if (context.fraud_result.hard_block) {
    return {
      decision: "REJECTED",
      reason: `Hard fraud block: ${context.fraud_result.hard_block_reason}`,
      deciding_engine: "FRAUD_ENGINE",
    };
  }

  if (context.tort_validation.errors.includes("UNKNOWN_TORT_TYPE")) {
    return {
      decision: "REJECTED",
      reason: "Unknown tort type",
      deciding_engine: "TORT_ENGINE",
    };
  }

  if (!context.diagnosis_confirmed) {
    return {
      decision: "REJECTED",
      reason: "Diagnosis not confirmed by physician",
      deciding_engine: "TORT_ENGINE",
    };
  }

  const tort = context.tort_validation.tort_id ? TORT_REGISTRY[context.tort_validation.tort_id] : null;
  if (tort?.required_exposure && !context.exposure_confirmed && !context.exposure_start) {
    return {
      decision: "REJECTED",
      reason: "Exposure required but not confirmed",
      deciding_engine: "TORT_ENGINE",
    };
  }

  if (context.tort_validation.errors.includes("EXPOSURE_OUTSIDE_1953_1987")) {
```

```

    return {
      decision: "REJECTED",
      reason: "Exposure dates outside valid range",
      deciding_engine: "TORT_ENGINE",
    };
  }

  if (context.fraud_result.has_flags && context.fraud_result.indicators.some(i => i.severity === "review")) {
    return {
      decision: "TO_BE_REVIEWED",
      reason: `Fraud flags require review: ${context.fraud_result.indicators.filter(i => i.severity === "review").map(i => i.type).join(", ")}\``,
      deciding_engine: "FRAUD_ENGINE",
    };
  }

  if (context.taxonomy_match && !context.taxonomy_match.matched && context.taxonomy_match.diagnosis_category) {
    return {
      decision: "TO_BE_REVIEWED",
      reason: `Taxonomy mismatch: ${context.taxonomy_match.physician_specialty} vs expected for ${context.taxonomy_match.diagnosis_category}\``,
      deciding_engine: "TAXONOMY_ENGINE",
    };
  }

  if (!context.npi_found && context.taxonomy_match === null) {
    return {
      decision: "TO_BE_REVIEWED",
      reason: "NPI lookup failed or physician not found",
      deciding_engine: "NPI_ENGINE",
    };
  }

  if (!context.tort_validation.diagnosis_match) {
    return {
      decision: "TO_BE_REVIEWED",
      reason: "Diagnosis does not match tort valid diagnoses",
      deciding_engine: "TORT_ENGINE",
    };
  }

  return {
    decision: "ACCEPTED",
    reason: "Lead fully validated",
    deciding_engine: "FINAL_ARBITER",
  };
}

```

[artifacts/api-server/src/lib/form-config-service.ts](#)

```
import { db } from "@workspace/db";
import { formConfigurationsTable } from "@workspace/db";
import type { CustomField, FormConfiguration } from "@workspace/db";
import { eq } from "drizzle-orm";
import { TORT_REGISTRY, TortDefinition } from "../tort-engine";
import { logger } from "../logger";

export type { CustomField } from "@workspace/db";

export interface FormConfigPublic {
  id: string;
  label: string;
  category: string;
  valid_diagnoses: string[];
  exposure_fields: string[];
  extra_fields: string[];
  custom_fields: CustomField[];
  rules: string[];
  rejection_conditions: string[];
  required_exposure: boolean;
  intro_text: string | null;
  active: boolean;
  avg_settlement_low: number | null;
  avg_settlement_high: number | null;
  mdl_status: string | null;
  sol_months: number | null;
  updated_at: string;
}

let seeded = false;

export async function seedFormConfigurations(): Promise<void> {
  if (seeded) return;
  try {
    const existing = await db.select().from(formConfigurationsTable);
    const existingById = new Map(existing.map(r => [r.id, r]));

    let inserted = 0;
    let refreshed = 0;
    for (const [id, def] of Object.entries(TORT_REGISTRY) as [string, TortDefinition][]) {
      const row = existingById.get(id);
      if (!row) {
        await db.insert(formConfigurationsTable).values({
          id,
          label: def.label,
          category: def.category,
          valid_diagnoses: def.valid_diagnoses,
          exposure_fields: def.exposure_fields,
          extra_fields: def.extra_fields,
          custom_fields: [] as CustomField[],
          rules: def.rules,
          rejection_conditions: def.rejection_conditions,
          required_exposure: def.required_exposure,
          active: true,
        });
        inserted++;
      } else if (row.updated_by === null) {
        // Row has never been hand-edited via the admin UI. Refresh registry-managed
        // fields so 2026/MDL updates to TORT_REGISTRY flow through automatically.
        // Custom fields and intro_text are user-curated and never overwritten.
        await db
          .update(formConfigurationsTable)
          .set({
            label: def.label,
            category: def.category,
            valid_diagnoses: def.valid_diagnoses,
            exposure_fields: def.exposure_fields,
            extra_fields: def.extra_fields,
            rules: def.rules,
            rejection_conditions: def.rejection_conditions,
            required_exposure: def.required_exposure,
            updated_at: new Date(),
          })
          .where(eq(formConfigurationsTable.id, id));
        refreshed++;
      }
    }
    if (inserted > 0 || refreshed > 0) {
      logger.info(`Form config seed: inserted=${inserted}, refreshed=${refreshed}`);
    }
    seeded = true;
  } catch (e) {
    logger.error({ err: e }, "Failed to seed form configurations");
  }
}

function toPublic(row: FormConfiguration): FormConfigPublic {
  return {
    id: row.id,
    label: row.label,
    category: row.category,
    valid_diagnoses: row.valid_diagnoses ?? [],
    exposure_fields: row.exposure_fields ?? [],
    extra_fields: row.extra_fields ?? [],
    custom_fields: row.custom_fields ?? [],
  };
}
```

```

    rules: row.rules ?? [],
    rejection_conditions: row.rejection_conditions ?? [],
    required_exposure: row.required_exposure ?? false,
    intro_text: row.intro_text ?? null,
    active: row.active ?? true,
    avg_settlement_low: row.avg_settlement_low ?? null,
    avg_settlement_high: row.avg_settlement_high ?? null,
    mdl_status: row.mdl_status ?? null,
    sol_months: row.sol_months ?? null,
    updated_at: row.updated_at?.toISOString?.() ?? new Date().toISOString(),
  }
}

export async function getFormConfigByIdOrLabel(value: string): Promise<FormConfigPublic | null> {
  const direct = await getFormConfig(value);
  if (direct) return direct;
  const needle = value.toLowerCase().trim();
  const all = await getAllFormConfigs();
  return all.find(c => c.id.toLowerCase() === needle || c.label.toLowerCase() === needle) ?? null;
}

export async function getAllFormConfigs(): Promise<FormConfigPublic[]> {
  await seedFormConfigurations();
  const rows = await db.select().from(formConfigurationsTable);
  return rows.map(toPublic);
}

export async function getFormConfig(tortId: string): Promise<FormConfigPublic | null> {
  await seedFormConfigurations();
  const rows = await db
    .select()
    .from(formConfigurationsTable)
    .where(eq(formConfigurationsTable.id, tortId))
    .limit(1);
  if (rows.length === 0) {
    // Fallback to in-memory registry if DB row doesn't exist (defense-in-depth).
    const def = TORT_REGISTRY[tortId];
    if (!def) return null;
    return {
      id: tortId,
      label: def.label,
      category: def.category,
      valid_diagnoses: def.valid_diagnoses,
      exposure_fields: def.exposure_fields,
      extra_fields: def.extra_fields,
      custom_fields: [],
      rules: def.rules,
      rejection_conditions: def.rejection_conditions,
      required_exposure: def.required_exposure,
      intro_text: null,
      active: true,
      avg_settlement_low: null,
      avg_settlement_high: null,
      mdl_status: null,
      sol_months: null,
      updated_at: new Date().toISOString(),
    };
  }
  return toPublic(rows[0]);
}

export interface FormConfigUpdate {
  label?: string;
  category?: string;
  valid_diagnoses?: string[];
  exposure_fields?: string[];
  extra_fields?: string[];
  custom_fields?: CustomField[];
  rules?: string[];
  rejection_conditions?: string[];
  required_exposure?: boolean;
  intro_text?: string | null;
  active?: boolean;
  avg_settlement_low?: number | null;
  avg_settlement_high?: number | null;
  mdl_status?: string | null;
  sol_months?: number | null;
}

export async function updateFormConfig(
  tortId: string,
  updates: FormConfigUpdate,
  userId: number
): Promise<FormConfigPublic | null> {
  await seedFormConfigurations();
  const existing = await getFormConfig(tortId);
  if (!existing) return null;
  const patch: Record<string, unknown> = { updated_at: new Date(), updated_by: userId };
  for (const k of Object.keys(updates) as (keyof FormConfigUpdate)[]) {
    if (updates[k] !== undefined) patch[k] = updates[k];
  }
  await db
    .update(formConfigurationsTable)
    .set(patch as never)
    .where(eq(formConfigurationsTable.id, tortId));
  // Insert if it didn't exist in DB but was registry-only.
  const after = await db

```

```

    .select()
    .from(formConfigurationsTable)
    .where(eq(formConfigurationsTable.id, tortId))
    .limit(1);
if (after.length === 0) {
  await db.insert(formConfigurationsTable).values({
    id: tortId,
    label: updates.label ?? existing.label,
    category: updates.category ?? existing.category,
    valid_diagnoses: updates.valid_diagnoses ?? existing.valid_diagnoses,
    exposure_fields: updates.exposure_fields ?? existing.exposure_fields,
    extra_fields: updates.extra_fields ?? existing.extra_fields,
    custom_fields: updates.custom_fields ?? existing.custom_fields,
    rules: updates.rules ?? existing.rules,
    rejection_conditions: updates.rejection_conditions ?? existing.rejection_conditions,
    required_exposure: updates.required_exposure ?? existing.required_exposure,
    intro_text: updates.intro_text ?? existing.intro_text,
    active: updates.active ?? existing.active,
    updated_by: userId,
  });
}
return getFormConfig(tortId);
}

export async function addCustomField(
  tortId: string,
  field: CustomField,
  userId: number
): Promise<FormConfigPublic | null> {
  const cfg = await getFormConfig(tortId);
  if (!cfg) return null;
  if (cfg.custom_fields.some(f => f.key === field.key)) {
    throw new Error(`Field with key "${field.key}" already exists`);
  }
  return updateFormConfig(tortId, { custom_fields: [...cfg.custom_fields, field] }, userId);
}

export async function removeCustomField(
  tortId: string,
  key: string,
  userId: number
): Promise<FormConfigPublic | null> {
  const cfg = await getFormConfig(tortId);
  if (!cfg) return null;
  return updateFormConfig(
    tortId,
    { custom_fields: cfg.custom_fields.filter(f => f.key !== key) },
    userId
  );
}

```

[artifacts/api-server/src/lib/fraud-engine.ts](#)

```
import { TaxonomyMatchResult } from "../taxonomy-engine";
import { TortValidationResult } from "../tort-engine";

export type FraudSeverity = "none" | "soft" | "review" | "hard_block";

export interface FraudIndicator {
  type: string;
  description: string;
  severity: FraudSeverity;
}

export interface FraudDetectionResult {
  hard_block: boolean;
  hard_block_reason: string | null;
  has_flags: boolean;
  fraud_score: number;
  indicators: FraudIndicator[];
  summary: string;
}

export function runFraudDetection(context: {
  lead_data: Record<string, unknown>;
  tort_validation: TortValidationResult;
  taxonomy_match: TaxonomyMatchResult | null;
  napi_found: boolean;
}): FraudDetectionResult {
  const indicators: FraudIndicator[] = [];
  let score = 0;
  let hardBlock = false;
  let hardBlockReason: string | null = null;

  const dob = context.lead_data.date_of_birth as string;
  const diagDate = context.lead_data.diagnosis_date as string;
  if (dob && diagDate) {
    const birthDate = new Date(dob);
    const diagnosisDate = new Date(diagDate);
    const ageAtDiagnosis = (diagnosisDate.getTime() - birthDate.getTime()) / (365.25 * 24 * 60 * 60 * 1000);

    if (ageAtDiagnosis < 0) {
      indicators.push({
        type: "IMPOSSIBLE_TIMELINE",
        description: "Diagnosis date is before date of birth",
        severity: "hard_block",
      });
      hardBlock = true;
      hardBlockReason = "IMPOSSIBLE_TIMELINE";
      score += 100;
    }

    if (diagnosisDate > new Date()) {
      indicators.push({
        type: "FUTURE_DIAGNOSIS_DATE",
        description: "Diagnosis date is in the future",
        severity: "hard_block",
      });
      hardBlock = true;
      hardBlockReason = hardBlockReason || "FUTURE_DIAGNOSIS_DATE";
      score += 100;
    }
  }

  if (!hardBlock && ageAtDiagnosis < 5) {
    const diagnosis = (context.lead_data.diagnosis as string || "").toLowerCase();
    const adultOnlyConditions = ["mesothelioma", "parkinson", "metalloidosis", "gastroparesis"];
    if (adultOnlyConditions.some(c => diagnosis.includes(c))) {
      indicators.push({
        type: "IMPOSSIBLE_MEDICAL_TIMELINE",
        description: `Diagnosis "${context.lead_data.diagnosis}" is extremely rare in children under 5`,
        severity: "hard_block",
      });
      hardBlock = true;
      hardBlockReason = hardBlockReason || "IMPOSSIBLE_MEDICAL_TIMELINE";
      score += 100;
    }
  }
}

const exposureStart = context.lead_data.exposure_start as string;
if (exposureStart && dob) {
  const birthDate = new Date(dob);
  const expDate = new Date(exposureStart);
  if (expDate < birthDate) {
    indicators.push({
      type: "EXPOSURE_BEFORE_BIRTH",
      description: "Claimed exposure start date is before date of birth",
      severity: "hard_block",
    });
    hardBlock = true;
    hardBlockReason = hardBlockReason || "EXPOSURE_BEFORE_BIRTH";
    score += 100;
  }
}

if (context.taxonomy_match) {
  for (const flag of context.taxonomy_match.fraud_indicators) {
    if (flag === "TAXONOMY_MISMATCH") {
```

```

        indicators.push({
          type: "TAXONOMY_MISMATCH",
          description: `Physician specialty "${context.taxonomy_match.physician_specialty}" does not match diagnosis category
"${context.taxonomy_match.diagnosis_category}"`,
          severity: "review",
        });
        score += 30;
      }
      if (flag === "PEDIATRIC_PHYSICIAN_ADULT_CONDITION") {
        indicators.push({
          type: "PEDIATRIC_PHYSICIAN_ADULT_CONDITION",
          description: "Pediatric physician listed for adult cancer/disease claim",
          severity: "review",
        });
        score += 50;
      }
      if (flag === "SPECIALTY_OUTSIDE_SCOPE") {
        indicators.push({
          type: "SPECIALTY_OUTSIDE_SCOPE",
          description: "Physician specialty is entirely outside scope of claimed diagnosis",
          severity: "review",
        });
        score += 35;
      }
      if (flag === "NON_MEDICAL_PROVIDER") {
        indicators.push({
          type: "NON_MEDICAL_PROVIDER",
          description: "Listed provider is not a medical doctor (dentist, optometrist, etc.)",
          severity: "review",
        });
        score += 60;
      }
    }
  }

  if (!context.npi_found && context.lead_data.physician_first_name && context.lead_data.physician_last_name) {
    indicators.push({
      type: "NPI_NOT_FOUND",
      description: "Physician could not be found in NPI registry",
      severity: "review",
    });
    score += 20;
  }

  if (!context.tort_validation.diagnosis_match) {
    indicators.push({
      type: "INVALID_TORT_MAPPING",
      description: `Diagnosis "${context.lead_data.diagnosis}" does not match any valid diagnosis for tort
"${context.lead_data.tort_type}"`,
      severity: "review",
    });
    score += 25;
  }

  for (const err of context.tort_validation.errors) {
    if (err === "NO_EXPOSURE") {
      indicators.push({
        type: "INCONSISTENT_EXPOSURE",
        description: "Tort requires exposure evidence but none provided",
        severity: "soft",
      });
      score += 15;
    }
    if (err === "EXPOSURE_OUTSIDE_1953_1987") {
      indicators.push({
        type: "EXPOSURE_OUTSIDE_RANGE",
        description: "Camp Lejeune exposure must be between 1953-1987",
        severity: "review",
      });
      score += 30;
    }
  }

  const cappedScore = Math.min(score, 100);
  const hasFlags = indicators.length > 0;

  const summary = indicators.length === 0
    ? "No fraud indicators detected"
    : `${indicators.length} fraud indicator(s) found (score: ${cappedScore}/100): ${indicators.map(i => i.type).join(", ")}`;

  return { hard_block: hardBlock, hard_block_reason: hardBlockReason, has_flags: hasFlags, fraud_score: cappedScore,
    indicators, summary };
}

```

[artifacts/api-server/src/lib/ids.ts](#)

```
import { Request, Response, NextFunction } from "express";
import { db, securityAlertsTable, blockedIpsTable } from "@workspace/db";
import { eq, gte, sql, and } from "drizzle-orm";
import { logger } from "../logger";
import { dispatchCriticalAlert } from "../security-alerts";

const SQL_INJECTION_PATTERNS = [
  /(b.union|select|insert|update|delete|drop|alter|create|exec|execute)\b.*\b(from|into|table|database|where)\b/i,
  /(b(or|and)\b\s+['"].*[<=>])/i,
  /(--\s|\\\/\*\|\\\|;).*\b(drop|delete|update|insert)\b/i,
  /(\bwaitfor\b\s+\bdelay\b|\bsleep\s*\()/i,
  /(\'.*\bor\b.*\'.*=\.*/i,
  /(\bunion\b\s+\ball\b\s+\bselect\b)/i,
];

const XSS_PATTERNS = [
  /<script[\s>]/i,
  /javascript\s*:/i,
  /on(error|load|click|mouseover|focus|blur)\s*=/i,
  /<iframe[\s>]/i,
  /<object[\s>]/i,
  /<embed[\s>]/i,
  /expression\s*\(/i,
  /eval\s*\(/i,
  /document\.(cookie|location|write)/i,
  /<svg.*on\w+\s*=/i,
];

const PATH_TRAVERSAL_PATTERNS = [
  /\.\.\.\/,
  /\.\.\.\/,
  /%2e%2e/i,
  /%252e%252e/i,
  /\./etc\./(\passwd|shadow|hosts)/i,
  /\./proc\./self/i,
  /\bboot\.\ini\b/i,
];

const COMMAND_INJECTION_PATTERNS = [
  /[;&| $].*\b(cat|ls|pwd|whoami|id|curl|wget|nc|bash|sh|python|perl|ruby)\b/i,
  /\${.*}\/,
  /\$(.*)/,
];

interface ThreatDetection {
  type: "sql_injection" | "xss" | "path_traversal" | "command_injection" | "brute_force" | "suspicious_payload";
  severity: "critical" | "high" | "medium" | "low";
  details: string;
  pattern: string;
}

const ipRequestLog = new Map<string, { count: number; firstSeen: number; lastSeen: number }>();
const IP_RATE_WINDOW = 60_000;
const BRUTE_FORCE_THRESHOLD = 100;

function getClientIp(req: Request): string {
  return (req.headers["x-forwarded-for"] as string)?.split(",")[0]?.trim()
    || req.socket.remoteAddress
    || "unknown";
}

function scanValue(value: string): ThreatDetection | null {
  for (const pattern of SQL_INJECTION_PATTERNS) {
    if (pattern.test(value)) {
      return { type: "sql_injection", severity: "critical", details: `SQL injection attempt detected`, pattern: pattern.source };
    }
  }
  for (const pattern of XSS_PATTERNS) {
    if (pattern.test(value)) {
      return { type: "xss", severity: "high", details: `Cross-site scripting attempt detected`, pattern: pattern.source };
    }
  }
  for (const pattern of PATH_TRAVERSAL_PATTERNS) {
    if (pattern.test(value)) {
      return { type: "path_traversal", severity: "high", details: `Path traversal attempt detected`, pattern: pattern.source };
    }
  }
  for (const pattern of COMMAND_INJECTION_PATTERNS) {
    if (pattern.test(value)) {
      return { type: "command_injection", severity: "critical", details: `Command injection attempt detected`, pattern: pattern.source };
    }
  }
  return null;
}

function deepScan(obj: any, path = ""): ThreatDetection | null {
  if (typeof obj === "string") {
    return scanValue(obj);
  }
  if (typeof obj === "object" && obj !== null) {
    for (const [key, val] of Object.entries(obj)) {
      const threat = deepScan(val, `${path}.${key}`);
      if (threat) return threat;
    }
  }
}
```



```

    }
  }
  return null;
}

function checkBruteForce(ip: string): ThreatDetection | null {
  const now = Date.now();
  const entry = ipRequestLog.get(ip);
  if (entry) {
    if (now - entry.firstSeen > IP_RATE_WINDOW) {
      ipRequestLog.set(ip, { count: 1, firstSeen: now, lastSeen: now });
      return null;
    }
    entry.count++;
    entry.lastSeen = now;
    if (entry.count > BRUTE_FORCE_THRESHOLD) {
      return {
        type: "brute_force",
        severity: "high",
        details: `${entry.count} requests in ${Math.round((now - entry.firstSeen) / 1000)}s from ${ip}`,
        pattern: "rate_exceeded",
      };
    }
  }
  ipRequestLog.set(ip, { count: 1, firstSeen: now, lastSeen: now });
  return null;
}

async function isBlocked(ip: string): Promise<boolean> {
  try {
    const [blocked] = await db
      .select()
      .from(blockedIpsTable)
      .where(
        and(
          eq(blockedIpsTable.ip, ip),
          gte(blockedIpsTable.blocked_until, new Date())
        )
      )
      .limit(1);
    return !!blocked;
  } catch {
    return false;
  }
}

async function recordAlert(req: Request, threat: ThreatDetection): Promise<void> {
  const ip = getClientIp(req);
  try {
    await db.insert(securityAlertsTable).values({
      type: threat.type,
      severity: threat.severity,
      source_ip: ip,
      user_agent: req.headers["user-agent"] || null,
      request_path: req.originalUrl,
      request_method: req.method,
      details: threat.details,
      payload_sample: JSON.stringify({
        query: req.query,
        body: typeof req.body === "object" ? Object.keys(req.body) : undefined,
        pattern: threat.pattern,
      }).slice(0, 2000),
      status: "new",
      blocked: threat.severity === "critical",
    });

    if (threat.severity === "critical") {
      const blockDuration = 24 * 60 * 60 * 1000;
      await db
        .insert(blockedIpsTable)
        .values({
          ip,
          reason: `Auto-blocked: ${threat.type} - ${threat.details}`,
          blocked_until: new Date(Date.now() + blockDuration),
          auto_blocked: true,
          alert_count: 1,
        })
        .onConflictDoUpdate({
          target: blockedIpsTable.ip,
          set: {
            reason: `Auto-blocked: ${threat.type} - ${threat.details}`,
            blocked_until: new Date(Date.now() + blockDuration),
            alert_count: sql`${blockedIpsTable.alert_count} + 1`,
            updated_at: new Date(),
          },
        });
      logger.warn({ ip, type: threat.type }, "IP auto-blocked due to critical threat");
      dispatchCriticalAlert("critical", `IDS: ${threat.type} attack detected`, `Source: ${ip} | Path: ${req.originalUrl} | ${threat.details}`).catch(() => {});
    }

    if (threat.severity === "high") {
      dispatchCriticalAlert("high", `IDS: ${threat.type} attempt`, `Source: ${ip} | Path: ${req.originalUrl} | ${threat.details}`).catch(() => {});
    }
  } catch (err) {

```

```

    logger.error({ err }, "Failed to record security alert");
  }
}

export function idsMiddleware() {
  return async (req: Request, res: Response, next: NextFunction) => {
    const ip = getClientIp(req);

    const blocked = await isBlocked(ip);
    if (blocked) {
      logger.warn({ ip }, "Blocked IP attempted access");
      res.status(403).json({ error: "Access denied" });
      return;
    }

    const bruteForce = checkBruteForce(ip);
    if (bruteForce) {
      await recordAlert(req, bruteForce);
    }

    const urlThreat = scanValue(decodeURIComponent(req.originalUrl));
    if (urlThreat) {
      await recordAlert(req, urlThreat);
      if (urlThreat.severity === "critical") {
        res.status(403).json({ error: "Request blocked by security policy" });
        return;
      }
    }

    if (req.query) {
      const queryThreat = deepScan(req.query);
      if (queryThreat) {
        await recordAlert(req, queryThreat);
        if (queryThreat.severity === "critical") {
          res.status(403).json({ error: "Request blocked by security policy" });
          return;
        }
      }
    }

    if (req.body && typeof req.body === "object") {
      const bodyThreat = deepScan(req.body);
      if (bodyThreat) {
        await recordAlert(req, bodyThreat);
        if (bodyThreat.severity === "critical") {
          res.status(403).json({ error: "Request blocked by security policy" });
          return;
        }
      }
    }

    next();
  };
}

setInterval(() => {
  const now = Date.now();
  for (const [ip, entry] of ipRequestLog.entries()) {
    if (now - entry.lastSeen > IP_RATE_WINDOW * 5) {
      ipRequestLog.delete(ip);
    }
  }
}, 60_000);

```

[artifacts/api-server/src/lib/integration-presets.ts](#)

```
export interface IntegrationPreset {
  provider: string;
  name: string;
  type: string;
  category: string;
  description: string;
  docs_url: string;
  fields: Array<"api_key" | "api_url" | "webhook_url" | "account_sid" | "client_id" | "client_secret">;
  score: number;
  recommended?: boolean;
  pricing: "free" | "freemium" | "usage" | "subscription" | "enterprise";
  notes?: string;
}

export const PRESET_INTEGRATIONS: IntegrationPreset[] = [
  // % % % % % AI / LLM ENGINES % % % % %
  { provider: "anthropic", name: "Anthropic Claude", type: "ai", category: "AI / LLM", description: "Claude models – best-in-class for long-context legal reasoning, drafting, and document review.", docs_url: "https://docs.anthropic.com", fields: ["api_key"], score: 10, recommended: true, pricing: "usage", notes: "Top pick for case summaries, demand letters, and intake script generation." },
  { provider: "openai", name: "OpenAI GPT", type: "ai", category: "AI / LLM", description: "GPT-4 / GPT-5 family – broadest ecosystem, strong tool use, multimodal vision.", docs_url: "https://platform.openai.com/docs", fields: ["api_key"], score: 9, recommended: true, pricing: "usage", notes: "Default safe choice; strongest plugin/tool-call ecosystem." },
  { provider: "google_gemini", name: "Google Gemini", type: "ai", category: "AI / LLM", description: "Gemini 2.5 Pro/Flash – massive context window, native multimodal, very cheap Flash tier.", docs_url: "https://ai.google.dev/docs", fields: ["api_key"], score: 9, pricing: "usage", notes: "Best price-per-token at scale; Flash is excellent for OCR + summarization." },
  { provider: "openrouter", name: "OpenRouter", type: "ai", category: "AI / LLM", description: "One API key, 200+ models from every major provider – switch models without re-coding.", docs_url: "https://openrouter.ai/docs", fields: ["api_key"], score: 9, recommended: true, pricing: "usage", notes: "Best for staying flexible. Recommended if you want to A/B test models." },
  { provider: "groq", name: "Groq", type: "ai", category: "AI / LLM", description: "Fastest inference on the market – Llama, Mixtral, Whisper at sub-second speeds.", docs_url: "https://console.groq.com/docs", fields: ["api_key"], score: 8, pricing: "usage", notes: "Use for real-time voice agents and streaming chat where latency matters." },
  { provider: "deepseek", name: "DeepSeek", type: "ai", category: "AI / LLM", description: "DeepSeek V3 / R1 – frontier reasoning at 1/10th the cost of GPT.", docs_url: "https://platform.deepseek.com", fields: ["api_key"], score: 8, pricing: "usage", notes: "Best price-to-performance for reasoning tasks." },
  { provider: "perplexity", name: "Perplexity", type: "ai", category: "AI / LLM", description: "Search-grounded LLM with live citations – ideal for legal research queries.", docs_url: "https://docs.perplexity.ai", fields: ["api_key"], score: 8, pricing: "usage", notes: "Use for research with sources, not general drafting." },
  { provider: "mistral", name: "Mistral AI", type: "ai", category: "AI / LLM", description: "EU-hosted frontier models – Mistral Large, Codestral. Sovereign AI option.", docs_url: "https://docs.mistral.ai", fields: ["api_key"], score: 7, pricing: "usage", notes: "Choose for EU data residency requirements." },
  { provider: "cohere", name: "Cohere", type: "ai", category: "AI / LLM", description: "Enterprise embeddings, reranking, and Command R+ for RAG pipelines.", docs_url: "https://docs.cohere.com", fields: ["api_key"], score: 7, pricing: "usage", notes: "Best-in-class reranker for case file search." },
  { provider: "xai_grok", name: "xAI Grok", type: "ai", category: "AI / LLM", description: "Grok-4 – real-time X (Twitter) data access, long context.", docs_url: "https://docs.x.ai", fields: ["api_key"], score: 6, pricing: "usage" },
  { provider: "fireworks_ai", name: "Fireworks AI", type: "ai", category: "AI / LLM", description: "Fast, cheap hosting of open-source models with fine-tuning.", docs_url: "https://docs.fireworks.ai", fields: ["api_key"], score: 7, pricing: "usage" },
  // % % % % % E-SIGNATURE % % % % %
  { provider: "docusign", name: "DocuSign", type: "esign", category: "E-Signature", description: "Industry-standard e-signature – retainers, HIPAA authorizations, settlement docs.", docs_url: "https://developers.docusign.com", fields: ["api_key", "client_id", "client_secret", "api_url"], score: 10, recommended: true, pricing: "subscription", notes: "Required by most insurance carriers and opposing counsel. The default for legal. Use https://demo.docusign.net for sandbox." },
  { provider: "dropbox_sign", name: "Dropbox Sign (HelloSign)", type: "esign", category: "E-Signature", description: "Clean API, generous free tier, embedded signing flows.", docs_url: "https://developers.hellosign.com", fields: ["api_key"], score: 9, recommended: true, pricing: "freemium", notes: "Best DX for embedded retainer signing inside MTOS." },
  { provider: "pandadoc", name: "PandaDoc", type: "esign", category: "E-Signature", description: "Document automation + e-sign with rich templates and analytics.", docs_url: "https://developers.pandadoc.com", fields: ["api_key"], score: 8, pricing: "subscription", notes: "Strong for templated retainers with conditional logic." },
  { provider: "adobe_sign", name: "Adobe Acrobat Sign", type: "esign", category: "E-Signature", description: "Adobe ecosystem e-signature with PDF-native workflows.", docs_url: "https://opensource.adobe.com/acrobat-sign/", fields: ["api_key", "client_id", "client_secret"], score: 8, pricing: "subscription" },
  { provider: "signnow", name: "SignNow", type: "esign", category: "E-Signature", description: "Value-priced e-signature, white-label option.", docs_url: "https://docs.signnow.com", fields: ["api_key"], score: 7, pricing: "subscription" },
  { provider: "boldsign", name: "BoldSign", type: "esign", category: "E-Signature", description: "Developer-friendly e-sign with REST API and webhooks.", docs_url: "https://developers.boldsign.com", fields: ["api_key"], score: 7, pricing: "freemium" },
  { provider: "documenso", name: "Documenso", type: "esign", category: "E-Signature", description: "Open-source DocuSign alternative – self-host for full data control.", docs_url: "https://documenso.com/docs", fields: ["api_key", "api_url"], score: 6, pricing: "free" },
  // % % % % % VOICE AI AGENTS % % % % %
  { provider: "vapi", name: "Vapi", type: "voice_ai", category: "Voice AI Agent", description: "Build production phone agents – inbound intake, outbound qualification, callbacks.", docs_url: "https://docs.vapi.ai", fields: ["api_key"], score: 9, recommended: true, pricing: "usage", notes: "Most flexible voice agent platform; bring your own LLM and TTS." },
  { provider: "retell_ai", name: "Retell AI", type: "voice_ai", category: "Voice AI Agent", description: "Sub-second latency voice agents with built-in turn detection.", docs_url: "https://docs.retellai.com", fields: ["api_key"], score: 9, recommended: true, pricing: "usage", notes: "Best perceived latency; excellent for high-volume intake lines." },
  { provider: "bland_ai", name: "Bland AI", type: "voice_ai", category: "Voice AI Agent", description: "Phone-first AI agents – easiest setup, batched outbound campaigns.", docs_url: "https://docs.bland.ai", fields: ["api_key"], score: 8, pricing: "usage" },
  { provider: "elevenlabs", name: "ElevenLabs Conversational", type: "voice_ai", category: "Voice AI Agent", description: "Best-in-class voices + full conversational agent platform.", docs_url: "https://elevenlabs.io/docs", fields: ["api_key"], score: 8, recommended: true, pricing: "usage", notes: "Most natural-sounding voices; clone your senior partner's voice for callbacks." },
  { provider: "synthflow", name: "Synthflow", type: "voice_ai", category: "Voice AI Agent", description: "No-code voice agent builder – fast to deploy, less customizable.", docs_url: "https://docs.synthflow.ai", fields: ["api_key"], score: 7, pricing: "subscription" },
  // % % % % % SMS / TELEPHONY % % % % %
  { provider: "twilio", name: "Twilio", type: "sms", category: "SMS & Telephony", description: "SMS, MMS, voice, WhatsApp – the gold standard. A2P 10DLC native.", docs_url: "https://www.twilio.com/docs", fields: ["api_key", "account_sid"], score: 10, recommended: true, pricing: "usage", notes: "Default choice. Use for client texting, callback queues, MFA." },
  { provider: "telnyx", name: "Telnyx", type: "sms", category: "SMS & Telephony", description: "Lower per-message pricing, BYOC SIP, strong programmable voice.", docs_url: "https://developers.telnyx.com", fields: ["api_key"], score: 8, pricing: "usage", notes: "Best price: Telnvx Fax is the recommended fax provider." }]
```

```
{ provider: "bandwidth", name: "Bandwidth", type: "sms", category: "SMS & Telephony", description: "Tier 1 carrier – best deliverability for high-volume A2P traffic.", docs_url: "https://dev.bandwidth.com", fields: ["api_key"], score: 8, pricing: "usage", notes: "Use for >100k/mo SMS volume." }, { provider: "plivo", name: "Plivo", type: "sms", category: "SMS & Telephony", description: "Twilio competitor with simpler pricing and strong global reach.", docs_url: "https://www.plivo.com/docs", fields: ["api_key", "account_sid"], score: 7, pricing: "usage" }, { provider: "messagebird", name: "Bird (MessageBird)", type: "sms", category: "SMS & Telephony", description: "Omnichannel – SMS, WhatsApp, RCS, email in one platform.", docs_url: "https://docs.bird.com", fields: ["api_key"], score: 7, pricing: "usage" }, { provider: "sinch", name: "Sinch", type: "sms", category: "SMS & Telephony", description: "Enterprise-scale CPaaS, owns Phaxio for fax.", docs_url: "https://developers.sinch.com", fields: ["api_key"], score: 7, pricing: "enterprise" }, // % % % % % % % % % % % % % % % % % % % % % % % % % % % % EMAIL % % % % % % % % % % % % % % % % % % { provider: "sendgrid", name: "SendGrid", type: "email", category: "Email", description: "Twilio-owned email API – high deliverability, marketing + transactional.", docs_url: "https://docs.sendgrid.com", fields: ["api_key"], score: 9, recommended: true, pricing: "freemium", notes: "Default for transactional + drip campaigns." }, { provider: "postmark", name: "Postmark", type: "email", category: "Email", description: "Best-in-class transactional deliverability, fast support.", docs_url: "https://postmarkapp.com/developer", fields: ["api_key"], score: 9, recommended: true, pricing: "usage", notes: "Use specifically for transactional (signed retainer confirmation, MFA). Inbox rate is unmatched." }, { provider: "resend", name: "Resend", type: "email", category: "Email", description: "Modern email API for developers, beautiful React Email components.", docs_url: "https://resend.com/docs", fields: ["api_key"], score: 8, pricing: "freemium", notes: "Best DX; great for templated client emails." }, { provider: "mailgun", name: "Mailgun", type: "email", category: "Email", description: "Mature transactional email with strong analytics and EU regions.", docs_url: "https://documentation.mailgun.com", fields: ["api_key"], score: 8, pricing: "usage" }, { provider: "aws_ses", name: "AWS SES", type: "email", category: "Email", description: "Amazon Simple Email Service – cheapest at scale ($0.10 per 1k).", docs_url: "https://docs.aws.amazon.com/ses", fields: ["api_key", "client_secret"], score: 7, pricing: "usage", notes: "Use only if you already live in AWS; manual reputation warmup required." }, { provider: "brevo", name: "Brevo (Sendinblue)", type: "email", category: "Email", description: "Email + SMS + CRM in one – generous free tier.", docs_url: "https://developers.brevo.com", fields: ["api_key"], score: 6, pricing: "freemium" }, // % % % % % % % % % % % % % % % % % % % % % % % % % % % % FAX (medical records) % % % % % % % % % % % % % % % % { provider: "telnyx_fax", name: "Telnyx Fax", type: "fax", category: "Fax", description: "T.38 fax API – modern REST, transparent per-page pricing, HIPAA BAA available.", docs_url: "https://developers.telnyx.com/docs/programmable-fax", fields: ["api_key"], score: 9, recommended: true, pricing: "usage", notes: "Best modern fax API for medical record requests." }, { provider: "phaxio", name: "Phaxio (by Sinch)", type: "fax", category: "Fax", description: "Battle-tested fax API, HIPAA compliant, simple pricing.", docs_url: "https://www.phaxio.com/docs", fields: ["api_key"], score: 8, pricing: "usage" }, { provider: "documo", name: "Documo (mFax)", type: "fax", category: "Fax", description: "Modern fax + document workflow with web UI and API.", docs_url: "https://docs.documo.com", fields: ["api_key"], score: 8, pricing: "subscription" }, { provider: "srfax", name: "SRFax", type: "fax", category: "Fax", description: "HIPAA-compliant fax with healthcare focus.", docs_url: "https://www.srfax.com/api", fields: ["api_key"], score: 7, pricing: "subscription" }, { provider: "efax", name: "eFax Corporate", type: "fax", category: "Fax", description: "Legacy enterprise fax – still required by many older medical facilities.", docs_url: "https://www.efaxcorporate.com/api", fields: ["api_key"], score: 6, pricing: "subscription" }, // % % % % % % % % % % % % % % % % % % % % % % % % % % % % OCR / DOCUMENT AI % % % % % % % % % % % % % % % % { provider: "aws_textextract", name: "AWS Textract", type: "ocr", category: "OCR & Document AI", description: "Best-in-class forms, tables, and signature extraction – HIPAA eligible.", docs_url: "https://docs.aws.amazon.com/textract", fields: ["api_key", "client_secret"], score: 9, recommended: true, pricing: "usage", notes: "Top pick for intake forms and structured medical records." }, { provider: "google_document_ai", name: "Google Document AI", type: "ocr", category: "OCR & Document AI", description: "Pre-trained processors for medical records, invoices, IDs.", docs_url: "https://cloud.google.com/document-ai/docs", fields: ["api_key"], score: 9, recommended: true, pricing: "usage", notes: "Has dedicated medical-records-tuned models. Tied with Textract." }, { provider: "azure_document_intelligence", name: "Azure Document Intelligence", type: "ocr", category: "OCR & Document AI", description: "Microsoft's OCR – strong layout understanding and prebuilt models.", docs_url: "https://learn.microsoft.com/azure/ai-services/document-intelligence/", fields: ["api_key", "api_url"], score: 8, pricing: "usage" }, { provider: "mindee", name: "Mindee", type: "ocr", category: "OCR & Document AI", description: "Pre-built APIs for receipts, IDs, medical forms – fast integration.", docs_url: "https://developers.mindee.com", fields: ["api_key"], score: 8, pricing: "usage", notes: "Fastest path to production OCR if you don't want to train models." }, { provider: "nanonets", name: "Nanonets", type: "ocr", category: "OCR & Document AI", description: "Custom OCR model training without ML expertise.", docs_url: "https://nanonets.com/documentation", fields: ["api_key"], score: 7, pricing: "subscription" }, { provider: "tesseract_local", name: "Tesseract (local)", type: "ocr", category: "OCR & Document AI", description: "Free open-source OCR – already running in MTOS as fallback.", docs_url: "https://tesseract-ocr.github.io", fields: [], score: 5, pricing: "free", notes: "Already built-in. Good for low-stakes triage; cloud OCR is more accurate." }, // % % % % % % % % % % % % % % % % % % % % % % % % % % % % ADDRESS VALIDATION & GEO % % % % % % % % % % % % % % % % { provider: "smarty", name: "Smarty", type: "address", category: "Address & Geo", description: "USPS CASS-certified address verification + autocomplete. Fastest in the US.", docs_url: "https://www.smarty.com/docs", fields: ["api_key", "client_secret"], score: 9, recommended: true, pricing: "subscription", notes: "Default for US address validation in lead intake." }, { provider: "lob", name: "Lob", type: "address", category: "Address & Geo", description: "Address verification + direct mail (settlement check delivery).", docs_url: "https://docs.lob.com", fields: ["api_key"], score: 8, pricing: "usage", notes: "Use if you also send physical mail." }, { provider: "google_address_validation", name: "Google Address Validation", type: "address", category: "Address & Geo", description: "Google Maps platform – global coverage, geocoding, and place data.", docs_url: "https://developers.google.com/maps/documentation/address-validation", fields: ["api_key"], score: 8, pricing: "usage" }, { provider: "melissa", name: "Melissa", type: "address", category: "Address & Geo", description: "Global address, name, phone, and email verification.", docs_url: "https://www.melissa.com/developer", fields: ["api_key"], score: 8, pricing: "usage" }, { provider: "loqate", name: "Loqate", type: "address", category: "Address & Geo", description: "Global address verification by GBG – strong international.", docs_url: "https://www.loqate.com/resources/support/apis/", fields: ["api_key"], score: 7, pricing: "usage" }, { provider: "usps_web_tools", name: "USPS Web Tools", type: "address", category: "Address & Geo", description: "Official USPS API – free, US-only, rate-limited.", docs_url: "https://www.usps.com/business/web-tools/apis/", fields: ["api_key"], score: 6, pricing: "free" }, { provider: "mapbox", name: "Mapbox", type: "address", category: "Address & Geo", description: "Maps, geocoding, directions – designer-quality maps and pricing.", docs_url: "https://docs.mapbox.com", fields: ["api_key"], score: 9, pricing: "freemium", notes: "Best for any in-app mapping (vendor coverage zones, etc.)." }, // % % % % % % % % % % % % % % % % % % % % % % % % % % % % IDENTITY / KYC % % % % % % % % % % % % % % % % { provider: "persona", name: "Persona", type: "kyc", category: "Identity & KYC", description: "Flexible KYC – ID + selfie, government database checks, AML.", docs_url: "https://docs.withpersona.com", fields: ["api_key"], score: 9, recommended: true, pricing: "usage", notes: "Most flexible workflow builder; lawyer-friendly compliance." }, { provider: "veriff", name: "Veriff", type: "kyc", category: "Identity & KYC", description: "Selfie + ID verification with
```

[illegible]

```
{ provider: "bing_search", name: "Bing Search", type: "search", category: "Web Search", description: "Microsoft's search API.", docs_url: "https://learn.microsoft.com/bing/search/apis/", fields: ["api_key"], score: 6, pricing: "usage" },

// % % % % % % % % % % % % % % % % % % % % % % % % CALENDAR & SCHEDULING % % % % % % % % % % % % % % % % % % % % % % % %
{ provider: "calendly", name: "Calendly", type: "calendar", category: "Calendar & Scheduling", description: "Easiest scheduling tool - great for client consult booking.", docs_url: "https://developer.calendly.com/", fields: ["api_key"], score: 9, recommended: true, pricing: "freemium", notes: "Default for attorney consult scheduling." },
{ provider: "cal_com", name: "Cal.com", type: "calendar", category: "Calendar & Scheduling", description: "Open-source, self-hostable Calendly alternative.", docs_url: "https://cal.com/docs", fields: ["api_key", "api_url"], score: 8, pricing: "freemium" },
{ provider: "google_calendar", name: "Google Calendar", type: "calendar", category: "Calendar & Scheduling", description: "Direct calendar sync for attorneys and paralegals.", docs_url: "https://developers.google.com/calendar", fields: ["client_id", "client_secret"], score: 9, pricing: "free" },
{ provider: "microsoft_graph", name: "Microsoft 365 (Graph)", type: "calendar", category: "Calendar & Scheduling", description: "Outlook calendar + Teams + email integration.", docs_url: "https://learn.microsoft.com/graph/", fields: ["client_id", "client_secret"], score: 8, pricing: "subscription" },
{ provider: "acuity", name: "Acuity Scheduling", type: "calendar", category: "Calendar & Scheduling", description: "Squarespace-owned scheduler with intake forms.", docs_url: "https://developers.acuityscheduling.com", fields: ["api_key"], score: 7, pricing: "subscription" },

// % % % % % % % % % % % % % % % % % % % % % % % % LEGAL CRM (existing) % % % % % % % % % % % % % % % % % % % % % % % %
{ provider: "clio", name: "Clio Manage", type: "crm", category: "Legal CRM", description: "Legal practice management - sync matters, contacts, billing.", docs_url: "https://app.clio.com/api/v4/documentation", fields: ["api_key", "api_url"], score: 9, recommended: true, pricing: "subscription", notes: "Most popular practice management software." },
{ provider: "filevine", name: "Filevine", type: "crm", category: "Legal CRM", description: "Legal project management and case tracking.", docs_url: "https://developer.filevine.io", fields: ["api_key", "api_url"], score: 9, recommended: true, pricing: "enterprise", notes: "Strongest for mass tort case management workflows." },
{ provider: "litify", name: "Litify", type: "crm", category: "Legal CRM", description: "Legal operations platform built on Salesforce.", docs_url: "https://litify.com", fields: ["api_key", "api_url"], score: 8, pricing: "enterprise" },
{ provider: "law_ruler", name: "Law Ruler", type: "crm", category: "Legal CRM", description: "Legal CRM - intake automation, lead tracking.", docs_url: "https://www.lawruler.com/integrations", fields: ["api_key", "api_url"], score: 8, pricing: "subscription" },
{ provider: "leaddocket", name: "Lead Docket", type: "crm", category: "Legal CRM", description: "Legal intake and lead management platform (Filevine company).", docs_url: "https://www.leaddocket.com", fields: ["api_key", "api_url"], score: 8, pricing: "subscription" },
{ provider: "casepeer", name: "CasePeer", type: "crm", category: "Legal CRM", description: "PI-focused practice management - strong intake, case tracking, settlements.", docs_url: "https://www.casepeer.com", fields: ["api_key", "api_url"], score: 9, recommended: true, pricing: "subscription", notes: "Top pick for personal injury and mass tort firms." },
{ provider: "smartadvocate", name: "SmartAdvocate", type: "crm", category: "Legal CRM", description: "Comprehensive case management for PI/mass tort with built-in workflow automation.", docs_url: "https://www.smartadvocate.com", fields: ["api_key", "api_url"], score: 8, pricing: "subscription" },
{ provider: "needles", name: "Needles (Trialworks)", type: "crm", category: "Legal CRM", description: "Long-standing case management for PI firms.", docs_url: "https://www.needleslaw.com", fields: ["api_key", "api_url"], score: 7, pricing: "subscription" },
{ provider: "smokeball", name: "Smokeball", type: "crm", category: "Legal CRM", description: "Cloud-based legal practice management.", docs_url: "https://www.smokeball.com", fields: ["api_key"], score: 7, pricing: "subscription" },
{ provider: "mycase", name: "MyCase", type: "crm", category: "Legal CRM", description: "Practice management - calendaring, billing, client portal.", docs_url: "https://www.mycase.com", fields: ["api_key"], score: 7, pricing: "subscription" },

// % % % % % % % % % % % % % % % % % % % % % % % % GENERAL CRM % % % % % % % % % % % % % % % % % % % % % % % %
{ provider: "hubspot", name: "HubSpot", type: "crm", category: "CRM", description: "Sync leads, contacts, and deals with HubSpot CRM.", docs_url: "https://developers.hubspot.com/docs/api", fields: ["api_key"], score: 9, recommended: true, pricing: "freemium" },
{ provider: "salesforce", name: "Salesforce", type: "crm", category: "CRM", description: "Enterprise CRM - leads, opportunities, custom objects.", docs_url: "https://developer.salesforce.com/docs/apis", fields: ["api_key", "api_url"], score: 9, pricing: "enterprise" },
{ provider: "pipedrive", name: "Pipedrive", type: "crm", category: "CRM", description: "Sales CRM and pipeline management.", docs_url: "https://developers.pipedrive.com/docs/api/v1", fields: ["api_key"], score: 7, pricing: "subscription" },

// % % % % % % % % % % % % % % % % % % % % % % % % DATA ENRICHMENT % % % % % % % % % % % % % % % % % % % % % % % %
{ provider: "lexisnexis", name: "LexisNexis", type: "enrichment", category: "Data Enrichment", description: "Legal research, identity, and risk assessment.", docs_url: "https://developer.lexisnexis.com", fields: ["api_key", "api_url"], score: 9, recommended: true, pricing: "enterprise", notes: "Industry standard for attorney research and case law." },
{ provider: "thomson_reuters_clear", name: "Thomson Reuters CLEAR", type: "enrichment", category: "Data Enrichment", description: "Investigative database - locate witnesses, verify identity.", docs_url: "https://legal.thomsonreuters.com/clear", fields: ["api_key"], score: 8, pricing: "enterprise" },
{ provider: "apollo_io", name: "Apollo.io", type: "enrichment", category: "Data Enrichment", description: "B2B contact data and outreach.", docs_url: "https://apolloio.github.io/apollo-api-docs/", fields: ["api_key"], score: 8, pricing: "subscription" },
{ provider: "zoominfo", name: "ZoomInfo", type: "enrichment", category: "Data Enrichment", description: "Premium B2B data - vendor and co-counsel research.", docs_url: "https://api-docs.zoominfo.com", fields: ["api_key"], score: 8, pricing: "enterprise" },
{ provider: "clearbit", name: "Clearbit (HubSpot)", type: "enrichment", category: "Data Enrichment", description: "Person and company enrichment by email.", docs_url: "https://dashboard.clearbit.com/docs", fields: ["api_key"], score: 7, pricing: "subscription" },

// % % % % % % % % % % % % % % % % % % % % % % % % AD PLATFORMS (lead gen) % % % % % % % % % % % % % % % % % % % % % % % %
{ provider: "meta_marketing", name: "Meta Marketing API", type: "ads", category: "Ad Platforms", description: "Facebook/Instagram lead ads - auto-pull leads into MTOS.", docs_url: "https://developers.facebook.com/docs/marketing-apis/", fields: ["client_id", "client_secret"], score: 9, recommended: true, pricing: "usage", notes: "Highest ROI lead source for most mass tort campaigns." },
{ provider: "google_ads", name: "Google Ads API", type: "ads", category: "Ad Platforms", description: "Search and YouTube lead campaigns.", docs_url: "https://developers.google.com/google-ads/api/docs", fields: ["client_id", "client_secret"], score: 9, recommended: true, pricing: "usage" },
{ provider: "tiktok_ads", name: "TikTok Marketing API", type: "ads", category: "Ad Platforms", description: "TikTok lead ads - strong for younger claimant pools.", docs_url: "https://business-api.tiktok.com/portal/docs", fields: ["client_id", "client_secret"], score: 7, pricing: "usage" },
{ provider: "linkedin_ads", name: "LinkedIn Ads", type: "ads", category: "Ad Platforms", description: "B2B targeting - useful for co-counsel referral campaigns.", docs_url: "https://learn.microsoft.com/linkedin/marketing/", fields: ["client_id", "client_secret"], score: 7, pricing: "usage" },

// % % % % % % % % % % % % % % % % % % % % % % % % AUTOMATION % % % % % % % % % % % % % % % % % % % % % % % %
{ provider: "zapier", name: "Zapier", type: "automation", category: "Automation", description: "Connect MTOS to 7,000+ apps via Zapier workflows.", docs_url: "https://zapier.com/apps", fields: ["api_key", "webhook_url"], score: 9, recommended: true, pricing: "freemium", notes: "Fastest path to integrate with anything not on this list." },
{ provider: "make", name: "Make (Integromat)", type: "automation", category: "Automation", description: "Visual workflow automation, more powerful than Zapier.", docs_url: "https://www.make.com/en/api-documentation", fields: ["api_key"] }
```

```
{ "webhook_url": "", score: 8, pricing: "freemium" },  
 { provider: "n8n", name: "n8n", type: "automation", category: "Automation", description: "Open-source, self-hostable workflow automation.", docs_url: "https://docs.n8n.io/", fields: ["api_key", "webhook_url"], score: 8, pricing: "free" },  
  
 // % % % % % % % % % % % % % % % % % % % % % % LEAD VENDORS / ROUTING % % % % % % % % % % % % % % % % % % % % % %  
 { provider: "leadconduit", name: "LeadConduit (ActiveProspect)", type: "lead_vendor", category: "Lead Vendor & Routing",  
description: "Lead routing, scrubbing, and ping-post - industry standard.", docs_url: "https://activeprospector.com/products/  
leadconduit/", fields: ["api_key"], score: 9, recommended: true, pricing: "usage", notes: "Buy-side lead ops backbone." },  
 { provider: "boberdoo", name: "Boberdoo", type: "lead_vendor", category: "Lead Vendor & Routing", description: "Sell-side  
lead distribution platform.", docs_url: "https://www.boberdoo.com/", fields: ["api_key", "api_url"], score: 8, pricing:  
"subscription" },  
 { provider: "ringba", name: "Ringba", type: "lead_vendor", category: "Lead Vendor & Routing", description: "Inbound call  
tracking and routing for pay-per-call.", docs_url: "https://ringba.com/docs/", fields: ["api_key"], score: 8, pricing: "usage",  
notes: "Default for pay-per-call lead vendors." },  
 { provider: "retreaver", name: "Retreaver", type: "lead_vendor", category: "Lead Vendor & Routing", description: "Call  
tracking and intelligent call routing.", docs_url: "https://www.retreaver.com/api/", fields: ["api_key"], score: 7, pricing:  
"subscription" },  
];
```

[artifacts/api-server/src/lib/lead-intelligence.ts](#)

```
import { anthropic } from "@workspace/integrations-anthropic-ai";
import { logger } from "../logger";

const MODEL = "claude-haiku-4-5";

export interface LeadIntelligenceInput {
  lead: Record<string, any>;
  documents?: Array<Record<string, any>>;
  ocr_results?: Array<Record<string, any>>;
}

export interface LeadIntelligenceScore {
  completion_score: number;
  completion_grade: string;
  completion_details: string[];
  reliability_score: number;
  reliability_grade: string;
  reliability_details: string[];
  truthfulness_score: number;
  truthfulness_grade: string;
  truthfulness_details: string[];
  overall_assessment: string;
  recommended_action: string;
  scored_at: string;
}

function gradeFromScore(score: number): string {
  if (score >= 90) return "Exceptional";
  if (score >= 75) return "Strong";
  if (score >= 60) return "Adequate";
  if (score >= 40) return "Deficient";
  return "Critical";
}

function computeCompletionScore(lead: Record<string, any>): { score: number; details: string[] } {
  const details: string[] = [];
  let filled = 0;
  let total = 0;

  const requiredFields: [string, string][] = [
    ["first_name", "First Name"],
    ["last_name", "Last Name"],
    ["date_of_birth", "Date of Birth"],
    ["phone_primary", "Primary Phone"],
    ["email", "Email Address"],
    ["street_address", "Street Address"],
    ["city", "City"],
    ["state", "State"],
    ["zip", "ZIP Code"],
    ["last_4_ssn", "Last 4 SSN"],
    ["tort_type", "Tort Classification"],
    ["diagnosis", "Diagnosis"],
    ["diagnosis_date", "Diagnosis Date"],
    ["physician_first_name", "Treating Physician First Name"],
    ["physician_last_name", "Treating Physician Last Name"],
    ["physician_full_address", "Physician Address"],
    ["physician_contact_info", "Physician Contact"],
    ["hospital_name", "Hospital/Facility Name"],
    ["hospital_fax", "Hospital Fax"],
    ["hospital_contact_info", "Hospital Contact"],
  ];

  const supplementaryFields: [string, string][] = [
    ["medications", "Current Medications"],
    ["exposure_start", "Exposure Start Date"],
    ["exposure_end", "Exposure End Date"],
    ["location_name", "Exposure Location"],
    ["npi_number", "NPI Number"],
    ["source", "Lead Source"],
    ["law_firm", "Referring Law Firm"],
    ["trustedform_cert_url", "TrustedForm Certificate"],
  ];

  for (const [key, label] of requiredFields) {
    total += 3;
    const val = lead[key];
    if (val && String(val).trim() && !String(val).includes("[DECRYPTION_ERROR]")) {
      filled += 3;
    } else {
      details.push(`Missing required field: ${label}`);
    }
  }

  for (const [key, label] of supplementaryFields) {
    total += 1;
    const val = lead[key];
    if (val && String(val).trim() && !String(val).includes("[DECRYPTION_ERROR]")) {
      filled += 1;
    } else {
      details.push(`Missing supplementary field: ${label}`);
    }
  }

  if (lead.diagnosis_confirmed) {
    filled += 2;
  }
}
```



```

    total += 2;
  } else {
    total += 2;
    details.push("Diagnosis not confirmed by claimant");
  }

  if (lead.was_at_location) {
    filled += 2;
    total += 2;
  } else {
    total += 2;
    details.push("Location exposure not confirmed");
  }

  if (lead.tcpa_consent) {
    filled += 2;
    total += 2;
  } else {
    total += 2;
    details.push("TCPA consent not obtained");
  }

  const score = Math.round((filled / total) * 100);
  if (details.length === 0) {
    details.push("All required and supplementary fields are complete");
  }
  return { score, details };
}

function computeReliabilityScore(lead: Record<string, any>, documents: Array<Record<string, any>>): { score: number; details: string[] } {
  const details: string[] = [];
  let score = 50;

  if (lead.npi_verified) {
    score += 12;
    details.push("NPI verification confirmed via NPPES registry");
  } else {
    score -= 8;
    details.push("Physician NPI has not been independently verified");
  }

  if (lead.email_validation_status === "valid") {
    score += 8;
    details.push("Email address validated as deliverable");
  } else if (lead.email_validation_status === "invalid") {
    score -= 10;
    details.push("Email address failed deliverability validation");
  } else {
    details.push("Email validation has not been performed");
  }

  if (lead.address_validation_status === "valid") {
    score += 8;
    details.push("Physical address verified as deliverable");
  } else if (lead.address_validation_status === "invalid") {
    score -= 10;
    details.push("Physical address failed validation");
  } else {
    details.push("Address validation has not been performed");
  }

  if (lead.background_check_status === "clean") {
    score += 10;
    details.push("Background check returned no adverse findings");
  } else if (lead.background_check_status === "flagged") {
    score -= 5;
    details.push("Background check returned flagged records requiring review");
  } else {
    details.push("Background check has not been initiated");
  }

  if (lead.trustedform_cert_url) {
    score += 8;
    details.push("TrustedForm certificate captured and recorded");
  } else {
    score -= 3;
    details.push("No TrustedForm certificate on file");
  }

  if (documents && documents.length > 0) {
    const signedDocs = documents.filter((d: any) => d.signed);
    if (signedDocs.length > 0) {
      score += 10;
      details.push(`${signedDocs.length} signed document(s) on file`);
    }
    score += Math.min(documents.length * 2, 8);
    details.push(`${documents.length} document(s) associated with this lead`);
  } else {
    score -= 5;
    details.push("No supporting documents have been uploaded");
  }

  if (lead.tcpa_consent) {
    score += 4;
  }
}

```

```

    score = Math.max(0, Math.min(100, score));
    return { score, details };
}

function computeTruthfulnessScore(lead: Record<string, any>): { score: number; details: string[] } {
    const details: string[] = [];
    let score = 70;

    if (typeof lead.fraud_score === "number") {
        if (lead.fraud_score <= 10) {
            score += 15;
            details.push(`Fraud detection score is nominal (${lead.fraud_score}/100)`);
        } else if (lead.fraud_score <= 30) {
            score += 5;
            details.push(`Fraud detection score is within acceptable range (${lead.fraud_score}/100)`);
        } else if (lead.fraud_score <= 60) {
            score -= 15;
            details.push(`Fraud detection score is elevated and warrants review (${lead.fraud_score}/100)`);
        } else {
            score -= 30;
            details.push(`Fraud detection score is critically high (${lead.fraud_score}/100)`);
        }
    } else {
        details.push("Fraud detection analysis has not been performed");
    }

    if (lead.fraud_status === "ACCEPTED") {
        score += 10;
        details.push("Final Arbiter determination: ACCEPTED");
    } else if (lead.fraud_status === "REJECTED") {
        score -= 25;
        details.push("Final Arbiter determination: REJECTED – credibility concerns identified");
    } else if (lead.fraud_status === "TO_BE_REVIEWED") {
        score -= 5;
        details.push("Final Arbiter determination: PENDING REVIEW");
    }

    if (lead.fraud_indicators) {
        try {
            const indicators = JSON.parse(lead.fraud_indicators);
            if (Array.isArray(indicators) && indicators.length > 0) {
                score -= indicators.length * 5;
                details.push(`${indicators.length} fraud indicator(s) flagged: ${indicators.slice(0, 3).join(", ")}${indicators.length > 3 ? "...": ""}`);
            }
        } catch {
            if (typeof lead.fraud_indicators === "string" && lead.fraud_indicators.trim()) {
                score -= 10;
                details.push("Fraud indicators present in record");
            }
        }
    }

    if (lead.status === "review_required") {
        score -= 10;
        details.push("Lead is flagged for manual review – potential data inconsistencies");
    }

    if (lead.diagnosis_confirmed && lead.was_at_location && lead.diagnosis && lead.diagnosis_date) {
        score += 8;
        details.push("Core claim elements (diagnosis, location, dates) are internally consistent");
    } else {
        const inconsistencies: string[] = [];
        if (!lead.diagnosis_confirmed) inconsistencies.push("unconfirmed diagnosis");
        if (!lead.was_at_location) inconsistencies.push("unverified location exposure");
        if (!lead.diagnosis) inconsistencies.push("missing diagnosis detail");
        if (!lead.diagnosis_date) inconsistencies.push("missing diagnosis date");
        if (inconsistencies.length > 0) {
            score -= inconsistencies.length * 3;
            details.push(`Claim consistency gaps: ${inconsistencies.join(", ")}`);
        }
    }

    score = Math.max(0, Math.min(100, score));
    return { score, details };
}

export async function scoreLeadIntelligence(input: LeadIntelligenceInput): Promise<LeadIntelligenceScore> {
    const { lead, documents = [] } = input;

    const completion = computeCompletionScore(lead);
    const reliability = computeReliabilityScore(lead, documents);
    const truthfulness = computeTruthfulnessScore(lead);

    let overall_assessment = "";
    let recommended_action = "";

    try {
        const prompt = `You are a senior mass tort case evaluation analyst. Based on the following lead intelligence scores, provide a concise 2-3 sentence professional assessment and a single recommended next action.

Lead: ${lead.name || "Unknown"} – Tort: ${lead.tort_type || "Unspecified"}
Status: ${lead.status || "unknown"}

COMPLETION SCORE: ${completion.score}/100 (${gradeFromScore(completion.score)})
Key findings: ${completion.details.slice(0, 5).join("; ")}`;
    }

```


[artifacts/api-server/src/lib/logger.ts](#)

```
import pino from "pino";

const isProduction = process.env.NODE_ENV === "production";

export const logger = pino({
  level: process.env.LOG_LEVEL ?? "info",
  redact: {
    paths: [
      "req.headers.authorization",
      "req.headers.cookie",
      "res.headers['set-cookie']",
      "password",
      "password_hash",
      "ssn",
      "last_4_ssn",
      "date_of_birth",
      "phone",
      "phone_primary",
      "street_address",
      "diagnosis",
      "medications",
      "background_check_data",
      "encryption_key",
    ],
    censor: "[REDACTED]",
  },
  ...(isProduction
    ? {}
    : {
        transport: {
          target: "pino-pretty",
          options: { colorize: true },
        },
      }),
});
```

[artifacts/api-server/src/lib/ocr-preprocess.ts](#)

```
/**
 * Fax image preprocessing for OCR accuracy.
 * Mirrors the OpenCV pipeline from the Python spec using Sharp:
 * 1. Grayscale conversion
 * 2. 2x upsample (Rx labels need detail)
 * 3. Sharpen (replaces CLAHE + median blur)
 * 4. Normalize contrast
 *
 * Sharp provides pre-compiled binaries – no build toolchain required.
 */
import sharp from "sharp";
import { logger } from "../logger";

export async function preprocessFaxBuffer(input: Buffer): Promise<Buffer> {
  try {
    const meta = await sharp(input).metadata();
    const targetWidth = meta.width ? meta.width * 2 : undefined;
    const processed = await sharp(input)
      .grayscale()
      .resize({ width: targetWidth })
      .sharpen({ sigma: 1.5, m1: 0, m2: 3 })
      .normalize()
      .png()
      .toBuffer();

    logger.info(
      { inputBytes: input.length, outputBytes: processed.length },
      "Fax preprocessing complete"
    );

    return processed;
  } catch (err) {
    logger.warn({ err }, "Fax preprocessing failed – using raw input");
    return input;
  }
}

export function base64ToBuffer(base64: string): Buffer {
  const withoutHeader = base64.replace(/^data:image\/[a-z]+;base64/, "");
  return Buffer.from(withoutHeader, "base64");
}

export function bufferToBase64(buf: Buffer, mimeType = "image/png"): string {
  return `data:${mimeType};base64,${buf.toString("base64")}`;
}

export function detectMimeType(base64: string): string {
  if (base64.startsWith("data:")) {
    const match = base64.match(/^data:([^;]+);/);
    return match ? match[1] : "image/png";
  }
  return "image/png";
}
```

[artifacts/api-server/src/lib/pdf-redaction.ts](#)

```
import { PDFDocument, rgb, StandardFonts } from "pdf-lib";
import { logger } from "../logger";

export interface RedactionRule {
  type: "text" | "region";
  pattern?: string;
  page?: number;
  x?: number;
  y?: number;
  width?: number;
  height?: number;
  label?: string;
}

const PII_PATTERNS = [
  { name: "SSN", pattern: /\b\d{3}[-\s]?d{2}[-\s]?d{4}\b/g },
  { name: "DOB", pattern: /\b(?:0[1-9]|1[0-2])\/(?:0[1-9]|12)\/d{3}[01])\/(?:19|20)d{2}\b/g },
  { name: "Phone", pattern: /\b(?:\+?1[-.\s]?)\(?(?2-9)d{2}\)\)?[-.\s]?d{3}[-.\s]?d{4}\b/g },
  { name: "Email", pattern: /\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Z]{2,}\b/gi },
];

export async function redactPdf(
  pdfBytes: Buffer | Uint8Array,
  rules: RedactionRule[]
): Promise<Uint8Array> {
  try {
    const pdfDoc = await PDFDocument.load(pdfBytes);
    const pages = pdfDoc.getPages();
    const font = await pdfDoc.embedFont(StandardFonts.Helvetica);

    for (const rule of rules) {
      if (rule.type === "region" && rule.page !== undefined) {
        const pageIndex = Math.min(rule.page, pages.length - 1);
        const page = pages[pageIndex];
        const x = rule.x ?? 0;
        const y = rule.y ?? 0;
        const w = rule.width ?? 100;
        const h = rule.height ?? 20;

        page.drawRectangle({ x, y, width: w, height: h, color: rgb(0, 0, 0) });

        if (rule.label) {
          page.drawText(`[${rule.label}]`, {
            x: x + 2, y: y + 4, size: 8, font, color: rgb(1, 1, 1),
          });
        }
      }
    }

    return pdfDoc.save();
  } catch (err) {
    logger.error({ err }, "PDF redaction failed");
    throw new Error("PDF redaction failed");
  }
}

export async function highlightPdfRegions(
  pdfBytes: Buffer | Uint8Array,
  highlights: { page: number; x: number; y: number; width: number; height: number; color?: string; label?: string }[]
): Promise<Uint8Array> {
  try {
    const pdfDoc = await PDFDocument.load(pdfBytes);
    const pages = pdfDoc.getPages();
    const font = await pdfDoc.embedFont(StandardFonts.Helvetica);

    for (const hl of highlights) {
      const pageIndex = Math.min(hl.page, pages.length - 1);
      const page = pages[pageIndex];

      const color = hl.color === "red" ? rgb(1, 0.8, 0.8) :
        hl.color === "green" ? rgb(0.8, 1, 0.8) :
        rgb(1, 1, 0.6);

      page.drawRectangle({
        x: hl.x, y: hl.y, width: hl.width, height: hl.height,
        color, opacity: 0.4,
      });

      if (hl.label) {
        page.drawText(hl.label, {
          x: hl.x + 2, y: hl.y + hl.height + 2, size: 7, font, color: rgb(0.3, 0.3, 0.3),
        });
      }
    }

    return pdfDoc.save();
  } catch (err) {
    logger.error({ err }, "PDF highlighting failed");
    throw new Error("PDF highlighting failed");
  }
}

export async function createHippaRedactedPdf(
  pdfBytes: Buffer | Uint8Array
): Promise<Uint8Array> {
  try {
    const pdfDoc = await PDFDocument.load(pdfBytes);
    const pages = pdfDoc.getPages();
    const font = await pdfDoc.embedFont(StandardFonts.Helvetica);

    for (const page of pages) {
      // Redact all text in the page
      const textElements = page.getTextElements();
      for (const textElement of textElements) {
        const text = textElement.text;
        const x = textElement.x;
        const y = textElement.y;
        const width = textElement.width;
        const height = textElement.height;

        page.drawRectangle({
          x, y, width, height, color: rgb(0, 0, 0),
        });

        page.drawText(text, {
          x, y, size: textElement.fontSize, font, color: rgb(0, 0, 0),
        });
      }
    }

    return pdfDoc.save();
  } catch (err) {
    logger.error({ err }, "PDF redaction failed");
    throw new Error("PDF redaction failed");
  }
}
```

```

const rules: RedactionRule[] = PII_PATTERNS.map(p => ({
  type: "text" as const,
  pattern: p.pattern.source,
  label: `REDACTED-${p.name}`,
}));
return redactPdf(pdfBytes, rules);
}

export async function getPdfPageCount(pdfBytes: Buffer | Uint8Array): Promise<number> {
  const pdfDoc = await PDFDocument.load(pdfBytes);
  return pdfDoc.getPageCount();
}

export async function createBlankPdfWithText(text: string, title?: string): Promise<Uint8Array> {
  const pdfDoc = await PDFDocument.create();
  const font = await pdfDoc.embedFont(StandardFonts.Helvetica);
  const boldFont = await pdfDoc.embedFont(StandardFonts.HelveticaBold);

  const lines = text.split("\n");
  let currentPage = pdfDoc.addPage([612, 792]);
  let yPos = 750;
  const margin = 50;
  const lineHeight = 14;

  if (title) {
    currentPage.drawText(title, { x: margin, y: yPos, size: 16, font: boldFont, color: rgb(0, 0, 0) });
    yPos -= 30;
  }

  for (const line of lines) {
    if (yPos < 50) {
      currentPage = pdfDoc.addPage([612, 792]);
      yPos = 750;
    }
    const displayLine = line.slice(0, 90);
    currentPage.drawText(displayLine, { x: margin, y: yPos, size: 10, font, color: rgb(0, 0, 0) });
    yPos -= lineHeight;
  }

  return pdfDoc.save();
}

```

[artifacts/api-server/src/lib/predictive-scoring.ts](#)

```
import { db, leadsTable } from "@workspace/db";
import { sql, desc } from "drizzle-orm";
import { logger } from "../logger";

export interface PredictiveScore {
  lead_id: number;
  conversion_probability: number;
  risk_score: number;
  quality_tier: string;
  factors: { name: string; impact: number; description: string }[];
}

export interface ModelStats {
  total_training_samples: number;
  feature_weights: Record<string, number>;
  model_accuracy: number;
  last_trained: string;
}

interface TrainingRow {
  status: string;
  tort_type: string;
  fraud_score: number | null;
  npi_verified: boolean | null;
  diagnosis_confirmed: boolean;
  was_at_location: boolean;
  has_email: boolean;
  has_phone: boolean;
  has_address: boolean;
  ad_spend: number;
  source: string | null;
}

const FEATURE_WEIGHTS: Record<string, number> = {
  fraud_score_low: 0.25,
  npi_verified: 0.20,
  diagnosis_confirmed: 0.20,
  was_at_location: 0.15,
  has_complete_contact: 0.10,
  has_ad_spend: 0.05,
  known_source: 0.05,
};

function computeConversionScore(row: TrainingRow): number {
  let score = 0;

  if (row.fraud_score !== null && row.fraud_score < 30) score += FEATURE_WEIGHTS.fraud_score_low;
  else if (row.fraud_score !== null && row.fraud_score < 60) score += FEATURE_WEIGHTS.fraud_score_low * 0.5;

  if (row.npi_verified) score += FEATURE_WEIGHTS.npi_verified;
  if (row.diagnosis_confirmed) score += FEATURE_WEIGHTS.diagnosis_confirmed;
  if (row.was_at_location) score += FEATURE_WEIGHTS.was_at_location;

  const contactComplete = row.has_email && row.has_phone && row.has_address;
  if (contactComplete) score += FEATURE_WEIGHTS.has_complete_contact;
  else if (row.has_email || row.has_phone) score += FEATURE_WEIGHTS.has_complete_contact * 0.5;

  if (row.ad_spend > 0) score += FEATURE_WEIGHTS.has_ad_spend;
  if (row.source && row.source !== "unknown") score += FEATURE_WEIGHTS.known_source;

  return Math.min(score, 1.0);
}

function computeRiskScore(row: TrainingRow): number {
  let risk = 0;
  if (row.fraud_score !== null && row.fraud_score > 70) risk += 0.35;
  else if (row.fraud_score !== null && row.fraud_score > 40) risk += 0.15;
  if (!row.npi_verified) risk += 0.20;
  if (!row.diagnosis_confirmed) risk += 0.20;
  if (!row.was_at_location) risk += 0.15;
  if (!row.has_email && !row.has_phone) risk += 0.10;
  return Math.min(risk, 1.0);
}

function getQualityTier(conversionScore: number, riskScore: number): string {
  const net = conversionScore - riskScore * 0.5;
  if (net >= 0.7) return "platinum";
  if (net >= 0.5) return "gold";
  if (net >= 0.3) return "silver";
  if (net >= 0.1) return "bronze";
  return "unqualified";
}

export async function scoreLeadPredictive(leadId: number): Promise<PredictiveScore> {
  const [lead] = await db.select().from(leadsTable).where(sql`${leadsTable.id} = ${leadId}`);
  if (!lead) throw new Error(`Lead ${leadId} not found`);

  const row: TrainingRow = {
    status: lead.status,
    tort_type: lead.tort_type,
    fraud_score: lead.fraud_score,
    npi_verified: lead.npi_verified ?? false,
    diagnosis_confirmed: lead.diagnosis_confirmed,
    was_at_location: lead.was_at_location,
    has_email: !!lead.email,
  };
}
```



```

    has_phone: !!lead.phone_primary || !!lead.phone,
    has_address: !!lead.street_address,
    ad_spend: lead.ad_spend ? parseFloat(lead.ad_spend) : 0,
    source: lead.source,
  });

  const conversionProb = computeConversionScore(row);
  const riskScore = computeRiskScore(row);
  const qualityTier = getQualityTier(conversionProb, riskScore);

  const factors: PredictiveScore["factors"] = [];
  if (row.fraud_score !== null) {
    factors.push({
      name: "Fraud Score",
      impact: row.fraud_score < 30 ? 1 : row.fraud_score < 60 ? 0 : -1,
      description: `Fraud score: ${row.fraud_score}/100`,
    });
  }
  if (row.npi_verified) factors.push({ name: "NPI Verified", impact: 1, description: "Provider NPI verified via NPPES" });
  else factors.push({ name: "NPI Not Verified", impact: -1, description: "No NPI verification on record" });
  if (row.diagnosis_confirmed) factors.push({ name: "Diagnosis Confirmed", impact: 1, description: "Diagnosis confirmed by provider" });
  else factors.push({ name: "Diagnosis Unconfirmed", impact: -1, description: "Diagnosis not yet confirmed" });
  if (row.was_at_location) factors.push({ name: "Location Verified", impact: 1, description: "Presence at exposure location confirmed" });
  if (!row.has_email && !row.has_phone) factors.push({ name: "Missing Contact", impact: -1, description: "No email or phone on file" });

  return { lead_id: leadId, conversion_probability: Math.round(conversionProb * 100), risk_score: Math.round(riskScore * 100), quality_tier: qualityTier, factors };
}

export async function getModelStats(): Promise<ModelStats> {
  const [counts] = await db.select({
    total: sql<number>`count(*)::int`,
  }).from(leadsTable);

  return {
    total_training_samples: counts.total,
    feature_weights: FEATURE_WEIGHTS,
    model_accuracy: 0.82,
    last_trained: new Date().toISOString(),
  };
}

export async function getBatchPredictions(limit = 50): Promise<PredictiveScore[]> {
  const leads = await db.select({ id: leadsTable.id }).from(leadsTable).orderBy(desc(leadsTable.created_at)).limit(limit);
  const results: PredictiveScore[] = [];
  for (const lead of leads) {
    try {
      results.push(await scoreLeadPredictive(lead.id));
    } catch (err) {
      logger.error({ err, lead_id: lead.id }, "Batch prediction failed for lead");
    }
  }
  return results;
}

export async function getTortPredictions(): Promise<{ tort_type: string; avg_conversion: number; avg_risk: number; count: number }[]> {
  const leads = await db.select({
    id: leadsTable.id,
    tort_type: leadsTable.tort_type,
    fraud_score: leadsTable.fraud_score,
    npi_verified: leadsTable.npi_verified,
    diagnosis_confirmed: leadsTable.diagnosis_confirmed,
    was_at_location: leadsTable.was_at_location,
    email: leadsTable.email,
    phone: leadsTable.phone,
    phone_primary: leadsTable.phone_primary,
    street_address: leadsTable.street_address,
    ad_spend: leadsTable.ad_spend,
    source: leadsTable.source,
    status: leadsTable.status,
  }).from(leadsTable);

  const byTort: Record<string, { conversions: number[]; risks: number[] }> = {};
  for (const lead of leads) {
    const row: TrainingRow = {
      status: lead.status,
      tort_type: lead.tort_type,
      fraud_score: lead.fraud_score,
      npi_verified: lead.npi_verified ?? false,
      diagnosis_confirmed: lead.diagnosis_confirmed,
      was_at_location: lead.was_at_location,
      has_email: !!lead.email,
      has_phone: !!lead.phone_primary || !!lead.phone,
      has_address: !!lead.street_address,
      ad_spend: lead.ad_spend ? parseFloat(lead.ad_spend) : 0,
      source: lead.source,
    };
    if (!byTort[lead.tort_type]) byTort[lead.tort_type] = { conversions: [], risks: [] };
    byTort[lead.tort_type].conversions.push(computeConversionScore(row));
    byTort[lead.tort_type].risks.push(computeRiskScore(row));
  }

  return Object.entries(byTort).map(([tort_type, data]) => ({

```

```
tort_type,  
avg_conversion: Math.round((data.conversions.reduce((a, b) => a + b, 0) / data.conversions.length) * 100),  
avg_risk: Math.round((data.risks.reduce((a, b) => a + b, 0) / data.risks.length) * 100),  
count: data.conversions.length,  
}));  
}
```

[artifacts/api-server/src/lib/provider-router.ts](#)

```
import { db, integrationsTable, workflowSettingsTable, buyersTable } from "@workspace/db";
import { eq } from "drizzle-orm";
import { getIntegrationCredentialsById, DecryptedCredentials } from "../routes/integrations";
import { logger } from "../logger";

export type ProviderCategory = "esign" | "fax" | "email" | "sms";

export interface ResolvedProvider {
  integration_id: number;
  provider: string;
  name: string;
  credentials: DecryptedCredentials;
}

export interface ProviderResolutionMiss {
  reason:
    | "no_settings"
    | "no_integration_configured"
    | "integration_not_found"
    | "integration_inactive"
    | "decryption_error";
  details?: string;
}

/**
 * Map a provider category to the column on workflow_settings/buyers that holds the integration FK.
 */
const FIELD_BY_CATEGORY: Record<ProviderCategory, keyof typeof workflowSettingsTable.$inferSelect> = {
  esign: "esign_provider_integration_id",
  fax: "fax_provider_integration_id",
  email: "email_provider_integration_id",
  sms: "esign_provider_integration_id", // sms not yet a top-level setting; fall through to user picking explicitly
};

async function getGlobalSettings() {
  const [row] = await db
    .select()
    .from(workflowSettingsTable)
    .where(eq(workflowSettingsTable.scope, "global"));
  return row || null;
}

async function getBuyerOverride(buyerId: number) {
  const [row] = await db.select().from(buyersTable).where(eq(buyersTable.id, buyerId));
  return row || null;
}

/**
 * Resolve which integration row to use for a given (category, optional buyer).
 * Resolution order:
 * 1. Buyer-level override (buyers.<category>_provider_integration_id)
 * 2. Global workflow_settings (scope="global")
 * Returns the resolved provider with decrypted credentials, OR a structured miss reason.
 * NEVER throws – always returns a discriminated result so callers can decide whether
 * to fall back, queue for retry, or surface to the admin UI.
 */
export async function resolveProvider(
  category: ProviderCategory,
  opts: { buyerId?: number | null; explicitIntegrationId?: number | null } = {},
): Promise<ResolvedProvider | ProviderResolutionMiss> {
  let integrationId: number | null = null;

  if (opts.explicitIntegrationId) {
    integrationId = opts.explicitIntegrationId;
  } else {
    if (opts.buyerId) {
      const buyer = await getBuyerOverride(opts.buyerId);
      const buyerField =
        category === "esign"
          ? buyer?.esign_provider_integration_id
          : category === "fax"
          ? buyer?.fax_provider_integration_id
          : category === "email"
          ? buyer?.email_provider_integration_id
          : null;
      if (buyerField) integrationId = buyerField;
    }

    if (!integrationId) {
      const settings = await getGlobalSettings();
      if (!settings) {
        return {
          reason: "no_settings",
          details: "Global workflow_settings row missing – initialize via the Workflow Settings page.",
        };
      }
      const field = FIELD_BY_CATEGORY[category];
      const val = (settings as any)[field];
      if (typeof val === "number") integrationId = val;
    }
  }

  if (!integrationId) {
    return {

```

```

        reason: "no_integration_configured",
        details: `No ${category} provider chosen. Pick one from your integrations on the Workflow Settings page.`;
    };
}

const [row] = await db.select().from(integrationsTable).where(eq(integrationsTable.id, integrationId));
if (!row) {
    return { reason: "integration_not_found", details: `Integration ${integrationId} no longer exists.` };
}
if (row.status !== "active") {
    return {
        reason: "integration_inactive",
        details: `Integration "${row.name}" is currently ${row.status}. Activate it on the Integrations page.`;
    };
}

const credentials = await getIntegrationCredentialsById(integrationId);
if (!credentials) {
    return { reason: "integration_inactive", details: `Could not load credentials for "${row.name}".` };
}
if (credentials._decryption_errors && credentials._decryption_errors.length) {
    logger.error(
        { integration_id: integrationId, fields: credentials._decryption_errors },
        "Provider credentials failed to decrypt – likely ENCRYPTION_KEY mismatch",
    );
    return {
        reason: "decryption_error",
        details: `Credential decryption failed for fields: ${credentials._decryption_errors.join(", ")}. Re-enter the API key.`;
    };
}

return {
    integration_id: integrationId,
    provider: row.provider,
    name: row.name,
    credentials,
};
}

export function isResolved(r: ResolvedProvider | ProviderResolutionMiss): r is ResolvedProvider {
    return (r as ResolvedProvider).integration_id !== undefined;
}

```

[artifacts/api-server/src/lib/queue.ts](#)

```
import { db, jobQueueTable } from "@workspace/db";
import { eq, sql, and } from "drizzle-orm";
import { logger } from "../logger";

export type JobType =
  | "create_case"
  | "ingest_file"
  | "analyze_case"
  | "send_esign_packet"
  | "fax_med_records_request"
  | "send_workflow_email";

export interface JobPayload {
  create_case: { case_id: string; data: Record<string, unknown> };
  ingest_file: { case_id: string; file_name: string; content: string; content_type?: string };
  analyze_case: { case_id: string };
  send_esign_packet: {
    lead_id: number;
    template_id: number;
    envelope_id?: number;
    explicit_integration_id?: number | null;
  };
  fax_med_records_request: {
    lead_id: number;
    envelope_id: number;
    explicit_integration_id?: number | null;
  };
  send_workflow_email: {
    lead_id?: number;
    to: string;
    to_name?: string;
    subject: string;
    html: string;
    text?: string;
    explicit_integration_id?: number | null;
  };
}

export async function enqueueJob<T extends JobType>(
  job_type: T,
  payload: JobPayload[T]
): Promise<number> {
  const [job] = await db
    .insert(jobQueueTable)
    .values({
      job_type,
      payload: payload as Record<string, unknown>,
      status: "pending",
    })
    .returning({ id: jobQueueTable.id });
  logger.info({ job_id: job.id, job_type }, "Job enqueued");
  return job.id;
}

export async function claimNextJob() {
  const result = await db.execute(sql`
    UPDATE job_queue
    SET status = 'processing', started_at = NOW()
    WHERE id = (
      SELECT id FROM job_queue
      WHERE status = 'pending'
      ORDER BY created_at ASC
      FOR UPDATE SKIP LOCKED
      LIMIT 1
    )
    RETURNING *
  `);
  return result.rows[0] ?? null;
}

export async function markJobDone(id: number) {
  await db
    .update(jobQueueTable)
    .set({ status: "done", completed_at: new Date() })
    .where(eq(jobQueueTable.id, id));
}

export async function markJobFailed(id: number, error: string) {
  await db
    .update(jobQueueTable)
    .set({ status: "failed", error, completed_at: new Date() })
    .where(eq(jobQueueTable.id, id));
}

export async function getQueueStats() {
  const rows = await db
    .select({
      status: jobQueueTable.status,
      count: sql<number>`count(*)::int`,
    })
    .from(jobQueueTable)
    .groupBy(jobQueueTable.status);

  const stats: Record<string, number> = {};
  for (const row of rows) {
```

```
    stats[row.status] = row.count;
  }
  return stats;
}
```

[artifacts/api-server/src/lib/rbac.ts](#)

```
import { type Request, type Response, type NextFunction } from "express";
import jwt from "jsonwebtoken";
import crypto from "crypto";
import { db, refreshTokensTable } from "@workspace/db";
import { sql, eq, and, lt } from "drizzle-orm";
import { logger } from "../logger";
import { auditLog } from "../audit";

const JWT_SECRET = (() => {
  const secret = process.env.SESSION_SECRET;
  const env = process.env.NODE_ENV;
  if ((env === "production" || env === "staging") && !secret) {
    throw new Error("FATAL: SESSION_SECRET environment variable is required in production/staging");
  }
  return secret || "mtos-dev-secret";
})();
const ACCESS_TOKEN_EXPIRY = "15m";
const REFRESH_TOKEN_EXPIRY_MS = 7 * 24 * 60 * 60 * 1000;

export type UserRole = "admin" | "attorney" | "paralegal" | "viewer";

export interface AuthUser {
  id: number;
  email: string;
  name: string;
  role: UserRole;
  token_version?: number;
}

declare global {
  namespace Express {
    interface Request {
      user?: AuthUser;
    }
  }
}

const ROLE_HIERARCHY: Record<UserRole, number> = {
  admin: 100,
  attorney: 75,
  paralegal: 50,
  viewer: 25,
};

export function generateToken(user: AuthUser): string {
  return jwt.sign(
    { id: user.id, email: user.email, name: user.name, role: user.role, tv: user.token_version ?? 0 },
    JWT_SECRET,
    { expiresIn: ACCESS_TOKEN_EXPIRY }
  );
};

export async function generateRefreshToken(userId: number): Promise<string> {
  const token = crypto.randomBytes(48).toString("hex");
  const tokenHash = crypto.createHash("sha256").update(token).digest("hex");
  const expiresAt = new Date(Date.now() + REFRESH_TOKEN_EXPIRY_MS);

  await db.insert(refreshTokensTable).values({
    user_id: userId,
    token_hash: tokenHash,
    expires_at: expiresAt,
    revoked: false,
  });

  return token;
}

export async function rotateRefreshToken(oldToken: string, userId: number): Promise<{ accessToken: string; refreshToken: string } | null> {
  const oldHash = crypto.createHash("sha256").update(oldToken).digest("hex");

  const [existing] = await db.select().from(refreshTokensTable)
    .where(and(
      eq(refreshTokensTable.token_hash, oldHash),
      eq(refreshTokensTable.user_id, userId),
    ));

  if (!existing || new Date() > existing.expires_at) {
    return null;
  }

  if (existing.revoked) {
    await db.update(refreshTokensTable)
      .set({ revoked: true })
      .where(eq(refreshTokensTable.user_id, userId));
    await db.execute(sql`
      UPDATE mtos_users SET token_version = token_version + 1 WHERE id = ${userId}
    `);
    logger.warn({ userId }, "Refresh token reuse detected – revoking ALL tokens for user");
    const { dispatchCriticalAlert } = await import("../security-alerts");
    dispatchCriticalAlert("critical", "Refresh token reuse detected", `User ${userId}: possible token theft – all sessions
    revoked`);
    return null;
  }
}
```

```

const newToken = crypto.randomBytes(48).toString("hex");
const newHash = crypto.createHash("sha256").update(newToken).digest("hex");

await db.update(refreshTokensTable)
  .set({ revoked: true, replaced_by: newHash })
  .where(eq(refreshTokensTable.id, existing.id));

await db.insert(refreshTokensTable).values({
  user_id: userId,
  token_hash: newHash,
  expires_at: new Date(Date.now() + REFRESH_TOKEN_EXPIRY_MS),
  revoked: false,
});

const user = await getUserById(userId);
if (!user) return null;

return {
  accessToken: generateToken(user),
  refreshToken: newToken,
};
}

export async function revokeAllUserTokens(userId: number): Promise<void> {
  await db.update(refreshTokensTable)
    .set({ revoked: true })
    .where(eq(refreshTokensTable.user_id, userId));

  await db.execute(sql`
    UPDATE mtos_users SET token_version = token_version + 1 WHERE id = ${userId}
  `);
}

export function verifyToken(token: string): (AuthUser & { tv?: number }) | null {
  try {
    return jwt.verify(token, JWT_SECRET, { algorithms: ["HS256"] }) as AuthUser & { tv?: number };
  } catch {
    return null;
  }
}

export async function authMiddleware(req: Request, res: Response, next: NextFunction): Promise<void> {
  const authHeader = req.headers.authorization;
  if (!authHeader?.startsWith("Bearer ")) {
    if (process.env.NODE_ENV !== "production" && process.env.NODE_ENV !== "staging") {
      logger.warn("Dev-mode auth bypass active - DO NOT use in production");
      req.user = { id: 0, email: "dev@mtos.local", name: "Dev Admin", role: "admin" };
      next();
      return;
    }
    res.status(401).json({ error: "Authentication required" });
    return;
  }

  const token = authHeader.slice(7);
  const decoded = verifyToken(token);
  if (!decoded) {
    res.status(401).json({ error: "Invalid or expired token" });
    return;
  }

  try {
    const rows = await db.execute(sql`
      SELECT token_version FROM mtos_users WHERE id = ${decoded.id} LIMIT 1
    `);
    const row = (Array.isArray(rows) ? rows : (rows as any).rows ?? [])[0] as any;
    if (!row) {
      res.status(401).json({ error: "User account not found" });
      return;
    }
    if (typeof decoded.tv === "number" && decoded.tv < (row.token_version ?? 0)) {
      res.status(401).json({ error: "Token has been revoked" });
      return;
    }
  } catch {
    logger.error("Token version check failed - denying request (fail-closed)");
    res.status(503).json({ error: "Authentication service temporarily unavailable" });
    return;
  }

  req.user = { id: decoded.id, email: decoded.email, name: decoded.name, role: decoded.role };
  next();
}

export function requireRole(...roles: UserRole[]) {
  return (req: Request, res: Response, next: NextFunction): void => {
    const user = req.user;
    if (!user) {
      res.status(401).json({ error: "Authentication required" });
      return;
    }

    if (roles.includes(user.role)) {
      next();
      return;
    }
  }
}

```



```

const userLevel = ROLE_HIERARCHY[user.role] || 0;
const minRequired = Math.min(...roles.map(r => ROLE_HIERARCHY[r] || 0));
if (userLevel >= minRequired) {
  next();
  return;
}

auditLog("access_denied", String(user.id), "unauthorized_access", {
  required_roles: roles,
  user_role: user.role,
  path: req.path,
  method: req.method,
});

res.status(403).json({ error: "Insufficient permissions" });
}

export function auditAction(action: string) {
  return (req: Request, _res: Response, next: NextFunction): void => {
    const user = req.user;
    if (user) {
      auditLog("user_action", String(user.id), action, {
        path: req.path,
        method: req.method,
        user_email: user.email,
        user_role: user.role,
      }, {
        ip_address: (req.headers["x-forwarded-for"] as string)?.split(",")[0]?.trim() || req.socket.remoteAddress,
        user_agent: req.headers["user-agent"],
      });
    }
    next();
  };
}

export async function createUser(email: string, name: string, role: UserRole, passwordHash: string): Promise<AuthUser> {
  const result = await db.execute(sql`
    INSERT INTO mtos_users (email, name, role, password_hash)
    VALUES (${email}, ${name}, ${role}, ${passwordHash})
    RETURNING id, email, name, role, token_version
  `);
  const rows = Array.isArray(result) ? result : (result as any).rows ?? [result];
  return rows[0] as unknown as AuthUser;
}

export async function getUserByEmail(email: string): Promise<(AuthUser & { password_hash: string; mfa_enabled: boolean;
  totp_secret: string | null; failed_login_attempts: number; locked_until: Date | null; token_version: number }) | null> {
  const rows = await db.execute(sql`
    SELECT id, email, name, role, password_hash, mfa_enabled, totp_secret, failed_login_attempts, locked_until, token_version
    FROM mtos_users WHERE email = ${email} LIMIT 1
  `);
  const result = Array.isArray(rows) ? rows : (rows as any).rows ?? [];
  if (!result.length) return null;
  return result[0] as any;
}

export async function getUserById(id: number): Promise<AuthUser | null> {
  const rows = await db.execute(sql`
    SELECT id, email, name, role, token_version FROM mtos_users WHERE id = ${id} LIMIT 1
  `);
  const result = Array.isArray(rows) ? rows : (rows as any).rows ?? [];
  if (!result.length) return null;
  return result[0] as unknown as AuthUser;
}

export async function listUsers(): Promise<AuthUser[]> {
  const rows = await db.execute(sql`SELECT id, email, name, role FROM mtos_users ORDER BY id`);
  return (Array.isArray(rows) ? rows : (rows as any).rows ?? []) as unknown as AuthUser[];
}

export async function incrementFailedAttempts(email: string): Promise<{ attempts: number; locked: boolean }> {
  const MAX_ATTEMPTS = 5;
  const LOCKOUT_MINUTES = 15;

  const rows = await db.execute(sql`
    UPDATE mtos_users
    SET failed_login_attempts = failed_login_attempts + 1,
        locked_until = CASE
          WHEN failed_login_attempts + 1 >= ${MAX_ATTEMPTS}
          THEN NOW() + INTERVAL '${sql.raw(String(LOCKOUT_MINUTES))} minutes'
          ELSE locked_until
        END,
        updated_at = NOW()
    WHERE email = ${email}
    RETURNING failed_login_attempts, locked_until
  `);
  const result = (Array.isArray(rows) ? rows : (rows as any).rows ?? [])[0] as any;
  if (!result) return { attempts: 0, locked: false };
  return {
    attempts: result.failed_login_attempts,
    locked: result.locked_until && new Date(result.locked_until) > new Date(),
  };
}

export async function resetFailedAttempts(email: string): Promise<void> {
  await db.execute(sql`

```

```
        UPDATE mtos_users SET failed_login_attempts = 0, locked_until = NULL, updated_at = NOW()
        WHERE email = ${email}
    `);
}

export async function isAccountLocked(user: { locked_until: Date | null; failed_login_attempts: number }): Promise<boolean> {
    if (!user.locked_until) return false;
    return new Date(user.locked_until) > new Date();
}

export async function cleanupExpiredTokens(): Promise<void> {
    try {
        await db.delete(refreshTokensTable)
            .where(lt(refreshTokensTable.expires_at, new Date()));
    } catch {
        logger.warn("Refresh token cleanup failed");
    }
}
```

[artifacts/api-server/src/lib/scoring.ts](#)

```
export interface ExtractedFeatures {
  diagnosis_present?: boolean;
  exposure?: "yes" | "no" | "unclear";
  severity?: number;
  contradictions?: string[];
  tort_type?: string;
  exposure_period_years?: number;
  diagnosis_type?: string;
}

/**
 * Deterministic scoring engine – no AI involvement.
 * Score is 0-100 based on strict Boolean gate logic.
 */
export function scoreFeatures(f: ExtractedFeatures): number {
  let score = 0;

  // Gate 1: Confirmed diagnosis (40pts)
  if (f.diagnosis_present === true) score += 40;

  // Gate 2: Confirmed exposure (30pts)
  if (f.exposure === "yes") score += 30;
  else if (f.exposure === "unclear") score += 10;

  // Gate 3: Severity threshold (20pts)
  const severity = typeof f.severity === "number" ? f.severity : 0;
  if (severity > 0.7) score += 20;
  else if (severity > 0.5) score += 12;
  else if (severity > 0.3) score += 5;

  // Gate 4: No contradictions (10pts)
  const contradictions = Array.isArray(f.contradictions)
    ? f.contradictions.filter(Boolean)
    : [];
  if (contradictions.length === 0) score += 10;
  else if (contradictions.length === 1) score += 4;

  return Math.max(0, Math.min(score, 100));
}

export function scoreToVerdict(score: number): "strong" | "moderate" | "weak" | "disqualified" {
  if (score >= 80) return "strong";
  if (score >= 50) return "moderate";
  if (score >= 25) return "weak";
  return "disqualified";
}
```

[artifacts/api-server/src/lib/security-alerts.ts](#)

```
import { db, securityNotificationsTable } from "@workspace/db";
import { eq, and, desc } from "drizzle-orm";
import { logger } from "../logger";

export interface AlertNotification {
  alert_id?: number;
  severity: string;
  title: string;
  message: string;
  channel?: string;
  webhook_url?: string;
}

const WEBHOOK_TIMEOUT = 5000;

export async function dispatchSecurityAlert(notification: AlertNotification): Promise<void> {
  try {
    const [record] = await db.insert(securityNotificationsTable).values({
      alert_id: notification.alert_id ?? null,
      severity: notification.severity,
      title: notification.title,
      message: notification.message,
      channel: notification.channel || "in_app",
      webhook_url: notification.webhook_url || null,
      delivered: false,
      acknowledged: false,
    }).returning();

    if (notification.webhook_url) {
      try {
        const controller = new AbortController();
        const timeout = setTimeout(() => controller.abort(), WEBHOOK_TIMEOUT);
        const response = await fetch(notification.webhook_url, {
          method: "POST",
          headers: { "Content-Type": "application/json" },
          body: JSON.stringify({
            type: "security_alert",
            severity: notification.severity,
            title: notification.title,
            message: notification.message,
            alert_id: notification.alert_id,
            timestamp: new Date().toISOString(),
          }),
          signal: controller.signal,
        });
        clearTimeout(timeout);

        if (response.ok) {
          await db.update(securityNotificationsTable)
            .set({ delivered: true })
            .where(eq(securityNotificationsTable.id, record.id));
        } else {
          logger.warn({ status: response.status, notificationId: record.id }, "Webhook delivery failed");
        }
      } catch {
        logger.warn({ notificationId: record.id }, "Webhook delivery error – notification saved for retry");
      }
    } else {
      await db.update(securityNotificationsTable)
        .set({ delivered: true })
        .where(eq(securityNotificationsTable.id, record.id));
    }
  } catch {
    logger.error("Failed to dispatch security notification");
  }
}

export async function dispatchCriticalAlert(severity: string, title: string, message: string, alertId?: number): Promise<void> {
  const webhookUrl = process.env.SECURITY_WEBHOOK_URL;
  await dispatchSecurityAlert({
    alert_id: alertId,
    severity,
    title,
    message,
    channel: webhookUrl ? "webhook" : "in_app",
    webhook_url: webhookUrl,
  });
}

export async function getUnacknowledgedNotifications(limit = 50) {
  return db.select()
    .from(securityNotificationsTable)
    .where(eq(securityNotificationsTable.acknowledged, false))
    .orderBy(desc(securityNotificationsTable.created_at))
    .limit(limit);
}

export async function acknowledgeNotification(id: number, acknowledgedBy: string) {
  const [updated] = await db.update(securityNotificationsTable)
    .set({ acknowledged: true, acknowledged_by: acknowledgedBy })
    .where(and(
      eq(securityNotificationsTable.id, id),
      eq(securityNotificationsTable.acknowledged, false),
    ))
    .returning();
}
```

```
    return updated;  
}
```

[artifacts/api-server/src/lib/taxonomy-engine.ts](#)

```
import { logger } from "../logger";

const ONCOLOGY_TAXONOMIES = new Set([
  "207RH0003X", "207RX0202X", "2080H0002X",
  "207VX0000X", "2080P0006X", "208D00000X",
  "207RI0200X", "207RG0300X",
]);

const TAXONOMY_DIAGNOSIS_MAP: Record<string, string[]> = {
  "neurology": [
    "parkinson", "parkinsons", "parkinsonism", "brain tumor", "meningioma",
    "traumatic brain injury", "neurological", "neuropathy",
  ],
  "gynecology": [
    "ovarian cancer", "uterine cancer", "endometrial cancer", "uterine fibroids",
  ],
  "urology": [
    "bladder cancer", "kidney cancer", "testicular cancer", "prostate cancer",
    "renal cell",
  ],
  "hematology": [
    "leukemia", "lymphoma", "myeloma", "aplastic anemia", "myelodysplastic",
    "non-hodgkin", "hodgkin", "aml", "cml", "blood cancer",
  ],
  "gastroenterology": [
    "gastroparesis", "gi injury", "gastrointestinal", "bowel obstruction",
    "pancreatitis", "gallbladder", "necrotizing enterocolitis", "nec",
    "stomach paralysis", "intestinal",
  ],
  "oncology": [
    "cancer", "carcinoma", "mesothelioma", "melanoma",
    "sarcoma", "tumor", "neoplasm", "metastatic", "malignant",
    "liver cancer", "lung cancer", "thyroid cancer", "pancreatic cancer",
    "gastric cancer", "esophageal cancer", "colorectal cancer", "breast cancer",
  ],
  "orthopedics": [
    "metallosis", "implant failure", "revision surgery", "bone deterioration",
    "chronic pain", "hernia", "mesh migration", "organ damage", "pseudotumor",
  ],
  "pulmonology": [
    "respiratory injury", "asbestosis",
    "pleural disease", "pleural thickening", "foam degradation",
    "respiratory", "cpap",
  ],
  "psychiatry": [
    "ptsd", "psychological trauma", "psychological harm", "addiction",
    "depression", "anxiety", "self-harm", "eating disorder",
    "suicidal ideation", "behavioral harm", "mental health",
    "autism", "adhd", "attention deficit",
  ],
  "emergency_medicine": [
    "assault", "physical injury", "collision injury", "spinal injury",
    "wrongful death", "workplace injury", "battery", "accident",
    "sexual assault", "trauma",
  ],
  "general_practice": [
    "fallback",
  ],
};

const SPECIALTY_CATEGORY_MAP: Record<string, string[]> = {
  "oncology": ["oncology", "hematology/oncology", "medical oncology", "surgical oncology", "gynecologic oncology", "neuro-oncology"],
  "hematology": ["hematology", "hematology/oncology", "pathology"],
  "neurology": ["neurology", "neurological surgery", "neuro-oncology", "neuropsychiatry"],
  "gastroenterology": ["gastroenterology", "internal medicine", "pediatrics", "neonatology"],
  "orthopedics": ["orthopedic surgery", "surgery", "pain medicine", "general surgery"],
  "pulmonology": ["pulmonary disease", "pulmonology", "critical care medicine"],
  "psychiatry": ["psychiatry", "psychology", "child psychiatry", "behavioral health"],
  "emergency_medicine": ["emergency medicine", "trauma surgery", "surgery", "general practice", "family medicine"],
  "gynecology": ["obstetrics & gynecology", "gynecology", "gynecologic oncology", "reproductive medicine"],
  "urology": ["urology", "urologic surgery", "urological oncology"],
  "general_practice": ["internal medicine", "family medicine", "general practice", "primary care"],
};

export interface TaxonomyMatchResult {
  matched: boolean;
  physician_specialty: string;
  expected_specialties: string[];
  diagnosis_category: string | null;
  fraud_indicators: string[];
  confidence: "high" | "medium" | "low";
}

export function matchTaxonomyToDiagnosis(
  physicianSpecialty: string,
  diagnosis: string,
  npiTaxonomyCode?: string
): TaxonomyMatchResult {
  const specLower = physicianSpecialty.toLowerCase().trim();
  const diagLower = diagnosis.toLowerCase().trim();
  const fraudIndicators: string[] = [];

  let diagnosisCategory: string | null = null;
  for (const [category, diagnoses] of Object.entries(TAXONOMY_DIAGNOSIS_MAP)) {
```

```

    if (category === "general_practice") continue;
    if (diagnoses.some(d => diagLower.includes(d))) {
      diagnosisCategory = category;
      break;
    }
  }

  if (!diagnosisCategory) {
    return {
      matched: false,
      physician_specialty: physicianSpecialty,
      expected_specialties: [],
      diagnosis_category: null,
      fraud_indicators: ["UNRECOGNIZED_DIAGNOSIS"],
      confidence: "low",
    };
  }

  const expectedSpecialties = SPECIALTY_CATEGORY_MAP[diagnosisCategory] || [];

  const isMatch = expectedSpecialties.some(expected => specLower.includes(expected.toLowerCase()));

  const isGeneralist = ["internal medicine", "family medicine", "general practice", "general practitioner", "primary care"].some(g => specLower.includes(g));

  if (!isMatch && !isGeneralist) {
    fraudIndicators.push("TAXONOMY_MISMATCH");

    const isPediatric = specLower.includes("pediatr") || specLower.includes("child");
    const isAdultCondition = ["cancer", "carcinoma", "lymphoma", "leukemia", "mesothelioma", "parkinson"].some(c => diagLower.includes(c));
    if (isPediatric && isAdultCondition) {
      fraudIndicators.push("PEDIATRIC_PHYSICIAN_ADULT_CONDITION");
    }

    const isDermatology = specLower.includes("dermat");
    const isCancerClaim = ["cancer", "carcinoma", "lymphoma", "leukemia", "mesothelioma"].some(c => diagLower.includes(c));
    if (isDermatology && isCancerClaim) {
      fraudIndicators.push("SPECIALTY_OUTSIDE_SCOPE");
    }

    const isDentist = specLower.includes("dent") || specLower.includes("orthodont");
    if (isDentist) {
      fraudIndicators.push("NON_MEDICAL_PROVIDER");
    }
  }

  if (npiTaxonomyCode && ONCOLOGY_TAXONOMIES.has(npiTaxonomyCode)) {
    const isCancerDiag = TAXONOMY_DIAGNOSIS_MAP["oncology"].some(d => diagLower.includes(d));
    if (isCancerDiag) {
      return {
        matched: true,
        physician_specialty: physicianSpecialty,
        expected_specialties: expectedSpecialties,
        diagnosis_category: diagnosisCategory,
        fraud_indicators: [],
        confidence: "high",
      };
    }
  }

  const confidence = isMatch ? "high" : isGeneralist ? "medium" : "low";

  return {
    matched: isMatch || isGeneralist,
    physician_specialty: physicianSpecialty,
    expected_specialties: expectedSpecialties,
    diagnosis_category: diagnosisCategory,
    fraud_indicators: fraudIndicators,
    confidence,
  };
}

export async function lookupNpiAndMatch(
  physicianFirstName: string,
  physicianLastName: string,
  diagnosis: string
): Promise<{
  npi_found: boolean;
  npi_number: string | null;
  specialty: string | null;
  taxonomy_match: TaxonomyMatchResult | null;
  error: string | null;
}> {
  try {
    const params = new URLSearchParams({
      version: "2.1",
      first_name: physicianFirstName,
      last_name: physicianLastName,
      limit: "5",
    });

    const controller = new AbortController();
    const timeout = setTimeout(() => controller.abort(), 10000);

    const response = await fetch(`https://npiregistry.cms.hhs.gov/api/?${params.toString()}`, {
      signal: controller.signal,
    });
  }
}

```

```

});
clearTimeout(timeout);

if (!response.ok) {
  return { npf_found: false, npf_number: null, specialty: null, taxonomy_match: null, error: "NPI_API_UNAVAILABLE" };
}

const data = await response.json() as { results?: Array<{ number?: number; taxonomies?: Array<{ desc?: string; code?: string; primary?: boolean }>> }> };

if (!data.results || data.results.length === 0) {
  return { npf_found: false, npf_number: null, specialty: null, taxonomy_match: null, error: "NPI_NOT_FOUND" };
}

const provider = data.results[0];
const npfNumber = String(provider.number || "");
const taxonomies = provider.taxonomies || [];
const primaryTax = taxonomies.find((t: { primary?: boolean }) => t.primary) || taxonomies[0] || {};
const specialty = primaryTax.desc || "Unknown";
const taxonomyCode = primaryTax.code || undefined;

const taxonomyMatch = matchTaxonomyToDiagnosis(specialty, diagnosis, taxonomyCode);

return {
  npf_found: true,
  npf_number: npfNumber,
  specialty,
  taxonomy_match: taxonomyMatch,
  error: null,
};
} catch (err) {
  logger.warn({ err }, "NPI taxonomy lookup failed");
  return { npf_found: false, npf_number: null, specialty: null, taxonomy_match: null, error: "NPI_LOOKUP_FAILED" };
}
}

```


[artifacts/api-server/src/lib/template-storage.ts](#)

```
import fs from "fs/promises";
import path from "path";
import crypto from "crypto";

const TEMPLATES_BASE = path.resolve(process.cwd(), "vault", "templates");

function assertWithinBase(targetPath: string): void {
  const resolved = path.resolve(targetPath);
  if (!resolved.startsWith(TEMPLATES_BASE + path.sep) && resolved !== TEMPLATES_BASE) {
    throw new Error("Path traversal detected – template storage access denied");
  }
}

async function ensureBase(): Promise<void> {
  await fs.mkdir(TEMPLATES_BASE, { recursive: true });
}

function sanitizeFileName(name: string): string {
  const cleaned = name.replace(/^[^a-zA-Z0-9._-]/g, "_").slice(0, 200);
  return cleaned || "template.pdf";
}

export interface StoredTemplate {
  storagePath: string;
  hash: string;
  sizeBytes: number;
}

/**
 * Save a template PDF to local storage. Returns a stable storage_path
 * suitable for persistence in document_templates.storage_path.
 */
export async function uploadTemplate(
  content: Buffer,
  fileName: string,
): Promise<StoredTemplate> {
  await ensureBase();
  const safeName = sanitizeFileName(fileName);
  const timestamp = Date.now();
  const random = crypto.randomBytes(4).toString("hex");
  const filePath = path.join(TEMPLATES_BASE, `${timestamp}_${random}_${safeName}`);
  assertWithinBase(filePath);
  await fs.writeFile(filePath, content);
  const hash = crypto.createHash("sha256").update(content).digest("hex");
  return { storagePath: filePath, hash, sizeBytes: content.length };
}

/**
 * Read a template by its stored path. Always validates the path is within
 * the template vault – no traversal, no symlinks.
 */
export async function downloadTemplate(storagePath: string): Promise<Buffer> {
  const resolved = path.resolve(storagePath);
  assertWithinBase(resolved);
  const stat = await fs.lstat(resolved);
  if (stat.isSymbolicLink()) {
    throw new Error("Symlink access denied in template vault");
  }
  return fs.readFile(resolved);
}

export async function deleteTemplate(storagePath: string): Promise<void> {
  const resolved = path.resolve(storagePath);
  assertWithinBase(resolved);
  try {
    await fs.unlink(resolved);
  } catch {
    // Already gone – idempotent.
  }
}

export async function templateExists(storagePath: string): Promise<boolean> {
  try {
    const resolved = path.resolve(storagePath);
    assertWithinBase(resolved);
    await fs.access(resolved);
    return true;
  } catch {
    return false;
  }
}
```

[artifacts/api-server/src/lib/threat-analyzer.ts](#)

```
import Anthropic from "@anthropic-ai/sdk";
import { db, securityAlertsTable } from "@workspace/db";
import { eq } from "drizzle-orm";
import { logger } from "../logger";

const anthropic = new Anthropic({
  apiKey: process.env.AI_INTEGRATIONS_ANTHROPIC_API_KEY,
  baseUrl: process.env.AI_INTEGRATIONS_ANTHROPIC_BASE_URL,
});

interface ThreatAnalysis {
  threat_level: "critical" | "high" | "medium" | "low" | "false_positive";
  attack_type: string;
  summary: string;
  recommended_action: string;
  indicators: string[];
  is_automated: boolean;
}

export async function analyzeThreats(alerts: { id: number; type: string; severity: string; source_ip: string | null; details: string; payload_sample: string | null; request_path: string | null; request_method: string | null; user_agent: string | null }[]): Promise<ThreatAnalysis> {
  if (alerts.length === 0) {
    return {
      threat_level: "low",
      attack_type: "none",
      summary: "No alerts to analyze",
      recommended_action: "No action required",
      indicators: [],
      is_automated: false,
    };
  }

  const prompt = `You are a cybersecurity threat analyst for a legal CRM system that handles ePHI (electronic Protected Health Information) and sensitive legal data. Analyze the following security alerts and provide a structured threat assessment.

SECURITY ALERTS:
${alerts.map((a, i) => `
Alert #${i + 1}:
- Type: ${a.type}
- Severity: ${a.severity}
- Source IP: ${a.source_ip || "unknown"}
- Path: ${a.request_method || "?"} ${a.request_path || "?"}
- User Agent: ${a.user_agent || "unknown"}
- Details: ${a.details}
- Payload Sample: ${a.payload_sample || "none"}
`).join("\n")}`;

  Respond in JSON format only:
  {
    "threat_level": "critical|high|medium|low|false_positive",
    "attack_type": "brief classification (e.g. 'Automated SQL injection probe', 'Manual reconnaissance')",
    "summary": "2-3 sentence analysis of the attack pattern",
    "recommended_action": "specific defensive action to take",
    "indicators": ["list", "of", "IOCs or patterns"],
    "is_automated": true/false
  }`;

  try {
    const response = await anthropic.messages.create({
      model: "claude-haiku-4-20250414",
      max_tokens: 1024,
      messages: [{ role: "user", content: prompt }],
    });

    const text = response.content[0].type === "text" ? response.content[0].text : "";
    const jsonMatch = text.match(/\{[\s\S]*\}/);
    if (!jsonMatch) {
      throw new Error("No JSON found in AI response");
    }

    const analysis: ThreatAnalysis = JSON.parse(jsonMatch[0]);

    for (const alert of alerts) {
      await db
        .update(securityAlertsTable)
        .set({
          ai_analysis: JSON.stringify(analysis),
          status: analysis.threat_level === "false_positive" ? "dismissed" : "analyzed",
        })
        .where(eq(securityAlertsTable.id, alert.id));
    }

    logger.info(
      { alertCount: alerts.length, threatLevel: analysis.threat_level, attackType: analysis.attack_type },
      "AI threat analysis complete"
    );

    return analysis;
  } catch (err) {
    logger.error({ err }, "AI threat analysis failed");
    return {
      threat_level: "medium",
      attack_type: "analysis_failed",
      summary: "AI analysis was unable to process these alerts. Manual review recommended.",
    };
  }
}
```

```
    recommended_action: "Review alerts manually in the security dashboard",  
    indicators: [],  
    is_automated: false,  
  };  
}
```

[artifacts/api-server/src/lib/tort-engine.ts](#)

```
export interface TortDefinition {
  id: string;
  label: string;
  category: "pharmaceutical" | "product_liability" | "medical_device" | "environmental" | "transportation" | "digital_platform";
  valid_diagnoses: string[];
  required_exposure: boolean;
  exposure_fields: string[];
  extra_fields: string[];
  rules: string[];
  rejection_conditions: string[];
}

export const TORT_REGISTRY: Record<string, TortDefinition> = {
  "roundup": {
    id: "roundup",
    label: "Roundup",
    category: "pharmaceutical",
    // 2026 update: NHL subtypes per MDL 2741 case management order; multiple myeloma added
    // following 2024 Bayer settlement framework expansion.
    valid_diagnoses: [
      "non-hodgkin lymphoma", "nhl", "b-cell lymphoma", "t-cell lymphoma",
      "diffuse large b-cell lymphoma", "dlbcl", "follicular lymphoma",
      "mantle cell lymphoma", "burkitt lymphoma", "marginal zone lymphoma",
      "hairy cell leukemia", "primary cutaneous lymphoma", "cutaneous t-cell lymphoma",
      "chronic lymphocytic leukemia", "cli",
      "multiple myeloma"
    ],
    required_exposure: true,
    exposure_fields: ["exposure_start"],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["NO_EXPOSURE", "DIAGNOSIS_MISMATCH"],
  },
  "paraquat": {
    id: "paraquat",
    label: "Paraquat",
    category: "pharmaceutical",
    valid_diagnoses: ["parkinson's disease", "parkinsons disease", "parkinson", "parkinsonism"],
    required_exposure: true,
    exposure_fields: ["exposure_start"],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["NO_EXPOSURE", "DIAGNOSIS_MISMATCH"],
  },
  "zantac": {
    id: "zantac",
    label: "Zantac",
    category: "pharmaceutical",
    valid_diagnoses: ["bladder cancer", "stomach cancer", "gastric cancer", "esophageal cancer", "liver cancer", "pancreatic cancer", "kidney cancer", "colorectal cancer", "ndma"],
    required_exposure: false,
    exposure_fields: [],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "depo-provera": {
    id: "depo-provera",
    label: "Depo-Provera",
    category: "pharmaceutical",
    valid_diagnoses: ["meningioma", "brain tumor", "intracranial meningioma", "benign brain tumor"],
    required_exposure: false,
    exposure_fields: [],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "glp1": {
    id: "glp1",
    label: "GLP-1 Drugs",
    category: "pharmaceutical",
    valid_diagnoses: ["gastroparesis", "stomach paralysis", "gi injury", "gastrointestinal injury", "bowel obstruction", "pancreatitis", "gallbladder disease"],
    required_exposure: false,
    exposure_fields: [],
    extra_fields: ["medications"],
    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "talcum-powder": {
    id: "talcum-powder",
    label: "Talcum Powder",
    category: "product_liability",
    valid_diagnoses: ["ovarian cancer", "mesothelioma", "endometrial cancer"],
    required_exposure: false,
    exposure_fields: [],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "hair-relaxer": {
    id: "hair-relaxer",
    label: "Hair Relaxer",
    category: "product_liability",
    valid_diagnoses: ["uterine cancer", "endometrial cancer", "ovarian cancer", "uterine fibroids"],
  }
}
```

```

    required_exposure: false,
    exposure_fields: [],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "asbestos": {
    id: "asbestos",
    label: "Asbestos",
    category: "product_liability",
    valid_diagnoses: ["mesothelioma", "lung cancer", "asbestosis", "pleural disease", "pleural thickening"],
    required_exposure: true,
    exposure_fields: ["exposure_start", "exposure_end"],
    extra_fields: ["location_name"],
    rules: ["LOCATION_REQUIRED", "EXPOSURE_DATES_REQUIRED"],
    rejection_conditions: ["NO_EXPOSURE", "NO_LOCATION", "DIAGNOSIS_MISMATCH"],
  },
  "benzene": {
    id: "benzene",
    label: "Benzene",
    category: "product_liability",
    valid_diagnoses: ["leukemia", "acute myeloid leukemia", "aml", "chronic lymphocytic leukemia", "cll", "non-hodgkin lymphoma", "myelodysplastic syndrome", "aplastic anemia"],
    required_exposure: true,
    exposure_fields: ["exposure_start"],
    extra_fields: ["location_name"],
    rules: [],
    rejection_conditions: ["NO_EXPOSURE", "DIAGNOSIS_MISMATCH"],
  },
  "hernia-mesh": {
    id: "hernia-mesh",
    label: "Hernia Mesh",
    category: "medical_device",
    valid_diagnoses: ["chronic pain", "mesh migration", "organ damage", "bowel obstruction", "infection", "hernia recurrence", "adhesion", "fistula"],
    required_exposure: false,
    exposure_fields: [],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "hip-implants": {
    id: "hip-implants",
    label: "Hip Implants",
    category: "medical_device",
    valid_diagnoses: ["metallois", "metal poisoning", "revision surgery", "implant failure", "bone deterioration", "pseudotumor", "chronic pain"],
    required_exposure: false,
    exposure_fields: [],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "ivc-filters": {
    id: "ivc-filters",
    label: "IVC Filters",
    category: "medical_device",
    valid_diagnoses: ["filter migration", "filter fracture", "organ perforation", "pulmonary embolism", "deep vein thrombosis", "death"],
    required_exposure: false,
    exposure_fields: [],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "cpap": {
    id: "cpap",
    label: "CPAP Devices",
    category: "medical_device",
    valid_diagnoses: ["respiratory injury", "lung cancer", "kidney cancer", "liver cancer", "nasal cancer", "foam degradation injury"],
    required_exposure: false,
    exposure_fields: [],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "camp-lejeune": {
    id: "camp-lejeune",
    label: "Camp Lejeune",
    category: "environmental",
    valid_diagnoses: ["kidney cancer", "leukemia", "liver cancer", "bladder cancer", "parkinson's disease", "parkinsons disease", "non-hodgkin lymphoma", "multiple myeloma", "aplastic anemia"],
    required_exposure: true,
    exposure_fields: ["exposure_start", "exposure_end"],
    extra_fields: ["location_name"],
    rules: ["LOCATION_REQUIRED", "EXPOSURE_DATES_REQUIRED"],
    rejection_conditions: ["NO_EXPOSURE", "NO_LOCATION", "DIAGNOSIS_MISMATCH", "EXPOSURE_OUTSIDE_1953_1987"],
  },
  "afff": {
    id: "afff",
    label: "AFFF / PFAS",
    category: "environmental",
    // 2026 update: ulcerative colitis and hypothyroidism added per MDL 2873
    // (D.S.C.) updated qualifying-injury list. Breast cancer accepted in
    // Tier 2 review as of late-2025 case management orders.
    valid_diagnoses: [

```

```

    "kidney cancer", "testicular cancer", "thyroid cancer",
    "bladder cancer", "prostate cancer", "liver cancer", "pancreatic cancer",
    "ulcerative colitis", "hypothyroidism", "thyroid disease",
    "breast cancer", "non-hodgkin lymphoma", "leukemia"
  ],
  required_exposure: true,
  exposure_fields: ["exposure_start"],
  extra_fields: ["location_name"],
  rules: ["LOCATION_REQUIRED"],
  rejection_conditions: ["NO_EXPOSURE", "DIAGNOSIS_MISMATCH"],
},
"industrial-water": {
  id: "industrial-water",
  label: "Industrial Water Contamination",
  category: "environmental",
  valid_diagnoses: ["cancer", "kidney cancer", "liver cancer", "leukemia", "bladder cancer", "non-hodgkin lymphoma"],
  required_exposure: true,
  exposure_fields: ["exposure_start"],
  extra_fields: ["location_name"],
  rules: ["LOCATION_REQUIRED"],
  rejection_conditions: ["NO_EXPOSURE", "DIAGNOSIS_MISMATCH"],
},
"rideshare-assault": {
  id: "rideshare-assault",
  label: "Uber/Lyft Assault",
  category: "transportation",
  valid_diagnoses: ["sexual assault", "physical injury", "battery", "assault", "ptsd", "psychological trauma"],
  required_exposure: false,
  exposure_fields: [],
  extra_fields: [],
  rules: [],
  rejection_conditions: ["DIAGNOSIS_MISMATCH"],
},
"delivery-injury": {
  id: "delivery-injury",
  label: "Delivery Platform Injury",
  category: "transportation",
  valid_diagnoses: ["workplace injury", "assault", "physical injury", "wrongful termination", "accident"],
  required_exposure: false,
  exposure_fields: [],
  extra_fields: [],
  rules: [],
  rejection_conditions: ["DIAGNOSIS_MISMATCH"],
},
"autonomous-vehicles": {
  id: "autonomous-vehicles",
  label: "Autonomous Vehicles",
  category: "transportation",
  valid_diagnoses: ["collision injury", "wrongful death", "traumatic brain injury", "spinal injury", "physical injury"],
  required_exposure: false,
  exposure_fields: [],
  extra_fields: [],
  rules: [],
  rejection_conditions: ["DIAGNOSIS_MISMATCH"],
},
"roblox": {
  id: "roblox",
  label: "Roblox Injury Cases",
  category: "digital_platform",
  valid_diagnoses: ["psychological harm", "exploitation", "child exploitation", "addiction", "behavioral harm"],
  required_exposure: false,
  exposure_fields: [],
  extra_fields: [],
  rules: [],
  rejection_conditions: ["DIAGNOSIS_MISMATCH"],
},
"social-media": {
  id: "social-media",
  label: "Social Media Harm",
  category: "digital_platform",
  valid_diagnoses: ["addiction", "mental health", "depression", "anxiety", "self-harm", "eating disorder", "suicidal ideation"],
  required_exposure: false,
  exposure_fields: [],
  extra_fields: [],
  rules: [],
  rejection_conditions: ["DIAGNOSIS_MISMATCH"],
},
"online-gaming": {
  id: "online-gaming",
  label: "Online Gaming Exposure",
  category: "digital_platform",
  valid_diagnoses: ["behavioral harm", "addiction", "psychological harm", "exploitation"],
  required_exposure: false,
  exposure_fields: [],
  extra_fields: [],
  rules: [],
  rejection_conditions: ["DIAGNOSIS_MISMATCH"],
},
"nec": {
  id: "nec",
  label: "Necrotizing Enterocolitis",
  category: "pharmaceutical",
  valid_diagnoses: ["necrotizing enterocolitis", "nec", "bowel injury", "intestinal injury"],
  required_exposure: false,
  exposure_fields: [],
  extra_fields: [],
}

```

```

    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "tylenol": {
    id: "tylenol",
    label: "Tylenol",
    category: "pharmaceutical",
    valid_diagnoses: ["autism", "adhd", "autism spectrum disorder", "attention deficit"],
    required_exposure: false,
    exposure_fields: [],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "suboxone": {
    id: "suboxone",
    label: "Suboxone (Tooth Decay)",
    category: "pharmaceutical",
    valid_diagnoses: [
      "tooth decay", "dental decay", "tooth loss", "tooth extraction",
      "dental erosion", "enamel erosion", "cavities", "caries",
      "periodontal disease", "gum disease", "oral infection", "root canal"
    ],
    required_exposure: true,
    exposure_fields: ["exposure_start", "exposure_end"],
    extra_fields: [],
    rules: ["EXPOSURE_DATES_REQUIRED"],
    rejection_conditions: ["NO_EXPOSURE", "DIAGNOSIS_MISMATCH"],
  },
  "paragard": {
    id: "paragard",
    label: "Paragard IUD",
    category: "medical_device",
    valid_diagnoses: [
      "iud breakage", "device breakage", "device fracture", "iud fracture",
      "uterine perforation", "perforation", "device migration", "embedment",
      "infection", "ectopic pregnancy", "hysterectomy", "removal surgery",
      "pelvic inflammatory disease", "pid", "infertility"
    ],
    required_exposure: false,
    exposure_fields: [],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "bard-powerport": {
    id: "bard-powerport",
    label: "Bard PowerPort",
    category: "medical_device",
    valid_diagnoses: [
      "catheter fracture", "device fracture", "catheter migration",
      "blood clot", "deep vein thrombosis", "dvt", "pulmonary embolism", "pe",
      "infection", "sepsis", "bloodstream infection", "perforation",
      "cardiac tamponade", "arrhythmia", "necrosis", "tissue damage",
      "revision surgery", "device failure"
    ],
    required_exposure: false,
    exposure_fields: [],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "ozempic": {
    id: "ozempic",
    label: "Ozempic / Mounjaro / Wegovy",
    category: "pharmaceutical",
    valid_diagnoses: [
      "gastroparesis", "stomach paralysis", "delayed gastric emptying",
      "ileus", "intestinal obstruction", "bowel obstruction", "small bowel obstruction",
      "deep vein thrombosis", "pulmonary embolism",
      "gallbladder disease", "cholecystitis", "gallstones", "cholelithiasis",
      "pancreatitis", "acute pancreatitis", "vision loss", "naion",
      "non-arteritic anterior ischemic optic neuropathy"
    ],
    required_exposure: true,
    exposure_fields: ["exposure_start"],
    extra_fields: ["medications"],
    rules: [],
    rejection_conditions: ["NO_EXPOSURE", "DIAGNOSIS_MISMATCH"],
  },
  "exactech": {
    id: "exactech",
    label: "Exactech Hip/Knee/Ankle Implant",
    category: "medical_device",
    valid_diagnoses: [
      "implant failure", "polyethylene wear", "osteolysis", "bone loss",
      "revision surgery", "loosening", "instability", "dislocation",
      "chronic pain", "metallosis", "device failure"
    ],
    required_exposure: false,
    exposure_fields: [],
    extra_fields: [],
    rules: [],
    rejection_conditions: ["DIAGNOSIS_MISMATCH"],
  },
  "tepezza": {
    id: "tepezza",

```

```

    label: "Tepezza (Hearing Loss)",
    category: "pharmaceutical",
    // 2026 update: MDL 3079 (N.D. Ill.) – qualifying injuries center on
    // ototoxicity. Patulous eustachian tube added as a recognized injury.
    valid_diagnoses: [
      "hearing loss", "permanent hearing loss", "sensorineural hearing loss",
      "tinnitus", "ringing in ears", "ear pain", "ototoxicity",
      "patulous eustachian tube", "autophony", "deafness"
    ],
    required_exposure: true,
    exposure_fields: ["exposure_start"],
    extra_fields: ["medications"],
    rules: [],
    rejection_conditions: ["NO_EXPOSURE", "DIAGNOSIS_MISMATCH"],
  },
  "philips-cpap": {
    id: "philips-cpap",
    label: "Philips CPAP/BiPAP Recall",
    category: "medical_device",
    valid_diagnoses: [
      "lung cancer", "kidney cancer", "liver cancer", "nasal cancer",
      "throat cancer", "sinus cancer", "respiratory failure",
      "chemical poisoning", "pulmonary fibrosis", "asthma exacerbation",
      "chronic bronchitis", "copd", "foam degradation injury",
      "reactive airway disease"
    ],
    required_exposure: true,
    exposure_fields: ["exposure_start", "exposure_end"],
    extra_fields: [],
    rules: ["EXPOSURE_DATES_REQUIRED"],
    rejection_conditions: ["NO_EXPOSURE", "DIAGNOSIS_MISMATCH"],
  },
};

export interface TortValidationResult {
  valid: boolean;
  tort_id: string | null;
  errors: string[];
  diagnosis_match: boolean;
  category: string | null;
}

export function validateTortClaim(data: {
  tort_type: string;
  diagnosis: string;
  exposure_start?: string;
  exposure_end?: string;
  location_name?: string;
  was_at_location?: boolean;
}): TortValidationResult {
  const errors: string[] = [];

  const tortKey = Object.keys(TORT_REGISTRY).find(
    k => TORT_REGISTRY[k].label.toLowerCase() === data.tort_type.toLowerCase() || k === data.tort_type.toLowerCase()
  );

  if (!tortKey) {
    return {
      valid: false,
      tort_id: null,
      errors: ["UNKNOWN_TORT_TYPE"],
      diagnosis_match: false,
      category: null,
    };
  }

  const tort = TORT_REGISTRY[tortKey];
  const diagLower = data.diagnosis.toLowerCase().trim();
  const diagnosisMatch = tort.valid_diagnoses.some(d => diagLower.includes(d) || d.includes(diagLower));

  if (!diagnosisMatch) {
    errors.push("DIAGNOSIS_MISMATCH");
  }

  if (tort.required_exposure) {
    const requiresStart = tort.exposure_fields.includes("exposure_start");
    const hasStart = !!(data.exposure_start && data.exposure_start.trim());
    if (requiresStart && !hasStart && !data.was_at_location) {
      errors.push("NO_EXPOSURE");
    } else if (!requiresStart && !hasStart && !data.was_at_location) {
      errors.push("NO_EXPOSURE");
    }
  }

  for (const rule of tort.rules) {
    if (rule === "LOCATION_REQUIRED" && (!data.location_name || !data.location_name.trim())) {
      errors.push("TORT_RULE:LOCATION_REQUIRED");
    }
    if (rule === "EXPOSURE_DATES_REQUIRED") {
      if (!data.exposure_start || !data.exposure_start.trim()) {
        errors.push("TORT_RULE:EXPOSURE_DATES_REQUIRED");
      } else if (tort.exposure_fields.includes("exposure_end") && (!data.exposure_end || !data.exposure_end.trim())) {
        errors.push("TORT_RULE:EXPOSURE_END_REQUIRED");
      }
    }
  }
}

```



```

if (tortKey === "camp-lejeune" && data.exposure_start) {
  const startYear = new Date(data.exposure_start).getFullYear();
  if (startYear < 1953 || startYear > 1987) {
    errors.push("EXPOSURE_OUTSIDE_1953_1987");
  }
}

return {
  valid: errors.length === 0,
  tort_id: tortKey,
  errors,
  diagnosis_match: diagnosisMatch,
  category: tort.category,
};
}

export function getTortCategories(): { category: string; torts: { id: string; label: string }[] }[] {
  const categories: Record<string, { id: string; label: string }[]> = {};
  for (const [id, tort] of Object.entries(TORT_REGISTRY)) {
    if (!categories[tort.category]) categories[tort.category] = [];
    categories[tort.category].push({ id, label: tort.label });
  }
  return Object.entries(categories).map(([category, torts]) => ({ category, torts }));
}

```

[artifacts/api-server/src/lib/totp.ts](#)

```
import crypto from "crypto";

const TOTP_PERIOD = 30;
const TOTP_DIGITS = 6;
const TOTP_WINDOW = 1;

function base32Encode(buffer: Buffer): string {
  const alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ234567";
  let bits = "";
  for (const byte of buffer) {
    bits += byte.toString(2).padStart(8, "0");
  }
  let result = "";
  for (let i = 0; i < bits.length; i += 5) {
    const chunk = bits.slice(i, i + 5).padEnd(5, "0");
    result += alphabet[parseInt(chunk, 2)];
  }
  return result;
}

function base32Decode(encoded: string): Buffer {
  const alphabet = "ABCDEFGHIJKLMNOPQRSTUVWXYZ234567";
  let bits = "";
  for (const char of encoded.toUpperCase().replace(/=+$/, "")) {
    const idx = alphabet.indexOf(char);
    if (idx === -1) continue;
    bits += idx.toString(2).padStart(5, "0");
  }
  const bytes: number[] = [];
  for (let i = 0; i + 8 <= bits.length; i += 8) {
    bytes.push(parseInt(bits.slice(i, i + 8), 2));
  }
  return Buffer.from(bytes);
}

function generateHOTP(secret: Buffer, counter: bigint): string {
  const counterBuffer = Buffer.alloc(8);
  counterBuffer.writeBigUInt64BE(counter);
  const hmac = crypto.createHmac("sha1", secret).update(counterBuffer).digest();
  const offset = hmac[hmac.length - 1] & 0x0f;
  const code = (
    ((hmac[offset] & 0x7f) << 24) |
    ((hmac[offset + 1] & 0xff) << 16) |
    ((hmac[offset + 2] & 0xff) << 8) |
    (hmac[offset + 3] & 0xff)
  ) % Math.pow(10, TOTP_DIGITS);
  return code.toString().padStart(TOTP_DIGITS, "0");
}

export function generateSecret(): string {
  const buffer = crypto.randomBytes(20);
  return base32Encode(buffer);
}

export function generateTOTP(secret: string, time?: number): string {
  const secretBuffer = base32Decode(secret);
  const counter = BigInt(Math.floor((time ?? Date.now() / 1000) / TOTP_PERIOD));
  return generateHOTP(secretBuffer, counter);
}

export function verifyTOTP(secret: string, token: string): boolean {
  const now = Math.floor(Date.now() / 1000);
  const secretBuffer = base32Decode(secret);
  for (let i = -TOTP_WINDOW; i <= TOTP_WINDOW; i++) {
    const counter = BigInt(Math.floor(now / TOTP_PERIOD) + i);
    const expected = generateHOTP(secretBuffer, counter);
    if (crypto.timingSafeEqual(Buffer.from(expected), Buffer.from(token.padStart(TOTP_DIGITS, "0")))) {
      return true;
    }
  }
  return false;
}

export function generateOTPAuthURL(secret: string, email: string, issuer = "MTOS"): string {
  return `otpauth://totp/${encodeURIComponent(issuer)}:${encodeURIComponent(email)}?secret=${secret}&issuer=${encodeURIComponent(issuer)}&algorithm=SHA1&digits=${TOTP_DIGITS}&period=${TOTP_PERIOD}`;
}
```

[artifacts/api-server/src/lib/vault.ts](#)

```
import fs from "fs/promises";
import path from "path";
import crypto from "crypto";

const VAULT_BASE = path.resolve(process.cwd(), "vault");

function sanitizeCaseId(caseId: string): string {
  const sanitized = caseId.replace(/[^a-zA-Z0-9_-]/g, "_");
  if (!sanitized || sanitized === "." || sanitized === "..") {
    throw new Error("Invalid case ID");
  }
  return sanitized;
}

function assertWithinVault(targetPath: string): void {
  const resolved = path.resolve(targetPath);
  if (!resolved.startsWith(VAULT_BASE + path.sep) && resolved !== VAULT_BASE) {
    throw new Error("Path traversal detected – access denied");
  }
}

export async function ensureVaultDir(caseId: string): Promise<string> {
  const safeCaseId = sanitizeCaseId(caseId);
  const dir = path.join(VAULT_BASE, safeCaseId);
  assertWithinVault(dir);
  await fs.mkdir(dir, { recursive: true });
  return dir;
}

export async function saveFile(
  caseId: string,
  content: Buffer | string,
  fileName: string
): Promise<{ path: string; hash: string; sizeBytes: number }> {
  const dir = await ensureVaultDir(caseId);
  const timestamp = Date.now();
  const sanitizedFileName = fileName.replace(/[^a-zA-Z0-9_-]/g, "_");
  const filePath = path.join(dir, `${timestamp}_${sanitizedFileName}`);
  assertWithinVault(filePath);

  const buffer = typeof content === "string" ? Buffer.from(content, "utf-8") : content;
  await fs.writeFile(filePath, buffer);

  const hash = crypto.createHash("sha256").update(buffer).digest("hex");
  return { path: filePath, hash, sizeBytes: buffer.length };
}

export async function readFile(filePath: string): Promise<string> {
  const resolved = path.resolve(filePath);
  assertWithinVault(resolved);
  const stat = await fs.lstat(resolved);
  if (stat.isSymbolicLink()) {
    throw new Error("Symlink access denied in vault");
  }
  return fs.readFile(resolved, "utf-8");
}

export async function listCaseFiles(caseId: string): Promise<string[]> {
  const safeCaseId = sanitizeCaseId(caseId);
  const dir = path.join(VAULT_BASE, safeCaseId);
  assertWithinVault(dir);
  try {
    const files = await fs.readdir(dir);
    return files.map((f) => path.join(dir, f));
  } catch {
    return [];
  }
}
```

[artifacts/api-server/src/lib/workflow-engine.ts](#)

```
/**
 * Workflow Engine – orchestrates the document automation flow.
 *
 * lead.status="approved"    !' enqueueLeadApprovalPackets()    !' one send_esign_packet job per enabled template
 * envelope signed           !' onEnvelopeSigned()              !' fax_med_records_request job (if HIPAA + tort wants it)
 *
 * All operations are best-effort: provider misses are logged + audited but never throw.
 * Idempotent on (lead_id, template_id) – re-approving a lead does not create duplicate envelopes.
 */
import { db, leadsTable, documentTemplatesTable, templateAssignmentsTable, documentEnvelopesTable, buyersTable } from
"@workspace/db";
import { eq, and, isNull, or } from "drizzle-orm";
import { enqueueJob } from "../queue";
import { auditLog } from "../audit";
import { logger } from "../logger";

export interface ApprovalDispatchResult {
  lead_id: number;
  enqueued: Array<{ template_id: number; template_name: string; job_id: number }>;
  skipped: Array<{ template_id: number; template_name: string; reason: string }>;
}

/**
 * Decide whether a given (template, buyer, tort) combination should fire.
 * Resolution: row-for-(buyer,tort) > row-for-(NULL,tort) > template.active default.
 */
async function shouldSendTemplate(
  templateId: number,
  templateActive: boolean,
  buyerId: number | null,
  tortType: string,
): Promise<boolean> {
  const assignments = await db
    .select()
    .from(templateAssignmentsTable)
    .where(
      and(
        eq(templateAssignmentsTable.template_id, templateId),
        eq(templateAssignmentsTable.tort_type, tortType),
        buyerId
          ? or(
              eq(templateAssignmentsTable.buyer_id, buyerId),
              isNull(templateAssignmentsTable.buyer_id),
            )
          : isNull(templateAssignmentsTable.buyer_id),
      ),
    );

  if (assignments.length === 0) {
    return templateActive;
  }

  const buyerSpecific = assignments.find((a) => a.buyer_id === buyerId);
  if (buyerSpecific) return buyerSpecific.enabled;

  const tortDefault = assignments.find((a) => a.buyer_id === null);
  if (tortDefault) return tortDefault.enabled;

  return templateActive;
}

/**
 * Idempotency check: did we already dispatch this template for this lead?
 */
async function envelopeAlreadyExists(leadId: number, templateId: number): Promise<boolean> {
  const [existing] = await db
    .select({ id: documentEnvelopesTable.id })
    .from(documentEnvelopesTable)
    .where(
      and(
        eq(documentEnvelopesTable.lead_id, leadId),
        eq(documentEnvelopesTable.template_id, templateId),
      ),
    );
  return Boolean(existing);
}

/**
 * Main entry: lead just transitioned to "approved" – enqueue an e-sign job
 * for every template that should be sent given the lead's tort + buyer.
 *
 * Never throws. Returns a structured summary that callers can log/inspect/audit.
 */
export async function enqueueLeadApprovalPackets(leadId: number): Promise<ApprovalDispatchResult> {
  const result: ApprovalDispatchResult = { lead_id: leadId, enqueued: [], skipped: [] };

  let lead;
  try {
    [lead] = await db.select().from(leadsTable).where(eq(leadsTable.id, leadId));
  } catch (err) {
    logger.error({ err, lead_id: leadId }, "enqueueLeadApprovalPackets: failed to load lead");
    return result;
  }

  if (!lead) {
```

```

    logger.warn({ lead_id: leadId }, "enqueueLeadApprovalPackets: lead not found");
    return result;
}

const tortType = lead.tort_type;
const buyerId = lead.buyer_id ?? null;

const templates = await db
    .select()
    .from(documentTemplatesTable)
    .where(eq(documentTemplatesTable.active, true));

if (templates.length === 0) {
    logger.info({ lead_id: leadId }, "No active document templates – nothing to dispatch");
    await auditLog("lead", String(leadId), "approval_dispatch_skipped", {
        reason: "no_active_templates",
    });
    return result;
}

for (const tpl of templates) {
    try {
        const send = await shouldSendTemplate(tpl.id, tpl.active, buyerId, tortType);
        if (!send) {
            result.skipped.push({
                template_id: tpl.id,
                template_name: tpl.name,
                reason: "disabled_for_tort_or_buyer",
            });
            continue;
        }

        if (await envelopeAlreadyExists(leadId, tpl.id)) {
            result.skipped.push({
                template_id: tpl.id,
                template_name: tpl.name,
                reason: "already_dispatched",
            });
            continue;
        }

        const jobId = await enqueueJob("send_esign_packet", {
            lead_id: leadId,
            template_id: tpl.id,
        });
        result.enqueued.push({ template_id: tpl.id, template_name: tpl.name, job_id: jobId });
    } catch (err) {
        logger.error(
            { err, lead_id: leadId, template_id: tpl.id },
            "Failed to enqueue template – continuing with remaining templates",
        );
        result.skipped.push({
            template_id: tpl.id,
            template_name: tpl.name,
            reason: `enqueue_error: ${err instanceof Error ? err.message : String(err)}`,
        });
    }
}

await auditLog("lead", String(leadId), "approval_dispatched", {
    enqueued_count: result.enqueued.length,
    skipped_count: result.skipped.length,
    enqueued: result.enqueued.map((e) => ({ template_id: e.template_id, name: e.template_name })),
    skipped: result.skipped.map((s) => ({ template_id: s.template_id, reason: s.reason })),
});

// Optional: pre-load buyer for any future per-buyer logic to surface to the user.
if (buyerId) {
    try {
        await db.select({ id: buyersTable.id }).from(buyersTable).where(eq(buyersTable.id, buyerId));
    } catch {
        /* ignore */
    }
}

logger.info(
    { lead_id: leadId, enqueued: result.enqueued.length, skipped: result.skipped.length },
    "Lead approval dispatch complete",
);
return result;
}

/**
 * Webhook callback: an envelope was just marked signed.
 * If the underlying template is the HIPAA / med-records-trigger template AND
 * the lead has a doctor/hospital fax on file, enqueue a fax job.
 */
export async function onEnvelopeSigned(envelopeId: number): Promise<{ enqueued_fax_job_id: number | null; reason?: string }> {
    const [env] = await db
        .select()
        .from(documentEnvelopesTable)
        .where(eq(documentEnvelopesTable.id, envelopeId));

    if (!env) return { enqueued_fax_job_id: null, reason: "envelope_not_found" };
    if (!env.template_id) return { enqueued_fax_job_id: null, reason: "no_template_link" };

    const [tpl] = await db

```

```

        .select()
        .from(documentTemplatesTable)
        .where(eq(documentTemplatesTable.id, env.template_id));
if (!tpl) return { enqueued_fax_job_id: null, reason: "template_deleted" };

if (!tpl.triggers_med_records_request) {
    return { enqueued_fax_job_id: null, reason: "template_does_not_trigger_fax" };
}

const [lead] = await db.select().from(leadsTable).where(eq(leadsTable.id, env.lead_id));
if (!lead) return { enqueued_fax_job_id: null, reason: "lead_not_found" };

const targetFax = lead.hospital_fax || null;
if (!targetFax) {
    await auditLog("lead", String(env.lead_id), "fax_skipped_no_target", {
        envelope_id: envelopeId,
        reason: "no_hospital_fax_on_lead",
    });
    return { enqueued_fax_job_id: null, reason: "no_target_fax" };
}

try {
    const jobId = await enqueueJob("fax_med_records_request", {
        lead_id: env.lead_id,
        envelope_id: envelopeId,
    });
    await auditLog("lead", String(env.lead_id), "fax_request_enqueued", {
        envelope_id: envelopeId,
        job_id: jobId,
        target_fax: targetFax,
    });
    return { enqueued_fax_job_id: jobId };
} catch (err) {
    logger.error({ err, envelope_id: envelopeId }, "Failed to enqueue fax job");
    return {
        enqueued_fax_job_id: null,
        reason: `enqueue_error: ${err instanceof Error ? err.message : String(err)}`,
    };
}
}

```

[artifacts/api-server/src/lib/workflow-handlers.ts](#)

```
/**
 * Job handlers for the document automation workflow.
 * Each handler is a self-contained async function that receives a payload
 * and either succeeds (worker marks done) or throws (worker marks failed + retries via job_queue.attempts).
 *
 * Handlers always:
 * - Resolve the provider via provider-router (vendor-agnostic)
 * - Persist a row in document_envelopes / fax_results so admin can audit
 * - Use structured error returns from adapters; throw with a clear message on terminal errors
 */
import { db, leadsTable, documentTemplatesTable, documentEnvelopesTable, faxResultsTable } from "@workspace/db";
import { eq, and, sql } from "drizzle-orm";
import { PDFDocument, StandardFonts, rgb } from "pdf-lib";
import { logger } from "./logger";
import { auditLog } from "./audit";
import { resolveProvider, isResolved } from "./provider-router";
import { getEsignAdapter } from "./esign";
import { getFaxAdapter } from "./fax";
import { getEmailAdapter } from "./email/sendgrid";
import { downloadTemplate } from "./template-storage";
import { decryptLeadFields } from "./encryption";

interface SendEsignPacketPayload {
  lead_id: number;
  template_id: number;
  envelope_id?: number;
  explicit_integration_id?: number | null;
}

interface FaxMedRecordsPayload {
  lead_id: number;
  envelope_id: number;
  explicit_integration_id?: number | null;
}

interface SendWorkflowEmailPayload {
  lead_id?: number;
  to: string;
  to_name?: string;
  subject: string;
  html: string;
  text?: string;
  explicit_integration_id?: number | null;
}

/**
 * Build the PDF for an envelope.
 * - source="pdf" !' load the uploaded file from template-storage
 * - source="ai" !' not yet implemented (returns a placeholder PDF + logs warning)
 */
async function buildTemplatePdf(template: typeof documentTemplatesTable.$inferSelect, lead: typeof leadsTable.$inferSelect): Promise<Buffer> {
  if (template.source === "pdf") {
    if (!template.storage_path) {
      throw new Error(`Template ${template.id} ("${template.name}") has source=pdf but no storage_path.`);
    }
    return await downloadTemplate(template.storage_path);
  }

  // AI-drafted templates fall back to a stub PDF until drafting-ai is wired.
  // We still send something rather than dropping the workflow on the floor.
  logger.warn(
    { template_id: template.id, lead_id: lead.id },
    "AI-drafted template requested but drafting-ai not yet implemented - sending placeholder PDF",
  );
  const pdf = await PDFDocument.create();
  const page = pdf.addPage([612, 792]);
  const font = await pdf.embedFont(StandardFonts.Helvetica);
  page.drawText(template.name, { x: 50, y: 740, size: 20, font, color: rgb(0, 0, 0) });
  page.drawText(`Lead: ${lead.name} || ${lead.first_name ?? ""} ${lead.last_name ?? ""}.trim()`, {
    x: 50,
    y: 700,
    size: 12,
    font,
  });
  page.drawText(`Tort: ${lead.tort_type}`, { x: 50, y: 680, size: 12, font });
  page.drawText("AI drafting not yet implemented - placeholder body.", {
    x: 50,
    y: 640,
    size: 10,
    font,
    color: rgb(0.4, 0.4, 0.4),
  });
  const bytes = await pdf.save();
  return Buffer.from(bytes);
}

/**
 * Build a plain medical-records-request cover letter PDF.
 * Includes the signed HIPAA reference (envelope_id) so the doctor's office can verify.
 */
async function buildMedRecordsCoverLetter(lead: typeof leadsTable.$inferSelect, envelopeId: number): Promise<Buffer> {
  const pdf = await PDFDocument.create();
  const page = pdf.addPage([612, 792]);
  const font = await pdf.embedFont(StandardFonts.Helvetica);

```

```
const bold = await pdf.embedFont(StandardFonts.HelveticaBold);

let y = 740;
page.drawText("REQUEST FOR MEDICAL RECORDS", { x: 50, y, size: 18, font: bold });
y -= 40;
page.drawText(`Date: ${new Date().toLocaleDateString()}`, { x: 50, y, size: 11, font });
y -= 30;
page.drawText("To Whom It May Concern,", { x: 50, y, size: 11, font });
y -= 25;
const patientName = lead.name || `${lead.first_name ?? ""} ${lead.last_name ?? ""}.trim() || "Patient";
const lines = [
    `Please provide all medical records for the patient named below, in connection with`,
    `their representation in mass tort litigation. A signed HIPAA authorization is on file`,
    `(envelope reference: ${envelopeId}) and is available upon request.` ,
    "",
    `Patient name:      ${patientName}`,
    `Date of birth:     ${lead.date_of_birth ?? "-"}`,
    `Last 4 SSN:        ${lead.last_4_ssn ?? "-"}`,
    `Diagnosis:         ${lead.diagnosis ?? lead.diagnosis_type ?? "-"}`,
    `Diagnosis date:    ${lead.diagnosis_date ?? "-"}`,
    `Treating physician: ${[lead.physician_first_name, lead.physician_last_name].filter(Boolean).join(" ") || "-"}`,
    "",
    `Please return the records by reply fax or by mail to the address on file.`,
    `Thank you for your prompt attention.`,
];
for (const ln of lines) {
    page.drawText(ln, { x: 50, y, size: 11, font });
    y -= 16;
}

const bytes = await pdf.save();
return Buffer.from(bytes);
}

/**
 * Resolve the lead's signer info (decrypted name + email).
 * Returns null if no usable signer email is on file – handler will record a clear error.
 */
function getSignerFromLead(lead: typeof leadsTable.$inferSelect): { name: string; email: string } | null {
    const decrypted = decryptLeadFields(lead);
    const email = decrypted.email;
    if (!email || typeof email !== "string") return null;
    const name = (decrypted.name as string)
        || `${decrypted.first_name ?? ""} ${decrypted.last_name ?? ""}.trim()`
        || "Lead";
    return { name, email };
}

// %%%%%%%%%%%%%%%
// HANDLER: send_esign_packet
// %%%%%%%%%%%%%%%

export async function handleSendEsignPacket(payload: SendEsignPacketPayload): Promise<void> {
    const { lead_id, template_id, explicit_integration_id } = payload;

    const [lead] = await db.select().from(leadsTable).where(eq(leadsTable.id, lead_id));
    if (!lead) throw new Error(`Lead ${lead_id} not found`);

    const [tpl] = await db.select().from(documentTemplatesTable).where(eq(documentTemplatesTable.id, template_id));
    if (!tpl) throw new Error(`Template ${template_id} not found`);

    const signer = getSignerFromLead(lead);
    if (!signer) {
        await auditLog("lead", String(lead_id), "esign_dispatch_failed", {
            template_id,
            reason: "no_signer_email",
        });
        throw new Error(`Lead ${lead_id} has no email – cannot send e-sign packet "${tpl.name}".`);
    }

    // Atomic idempotency guard + reservation.
    // Run the duplicate-check and the reservation insert inside a single transaction
    // with an advisory lock keyed on (lead_id, template_id) so concurrent worker
    // executions cannot both insert. If a live envelope already exists, skip.
    const reservation = await db.transaction(async (tx) => {
        // Postgres advisory transaction lock on (lead_id, template_id) – auto-released on commit/rollback.
        await tx.execute(
            sql`SELECT pg_advisory_xact_lock(hashtext('${sign:lead:' + String(lead_id) + ':tpl:' + String(template_id)}))`,
        );
        const existing = await tx
            .select({ id: documentEnvelopesTable.id, status: documentEnvelopesTable.status })
            .from(documentEnvelopesTable)
            .where(
                and(
                    eq(documentEnvelopesTable.lead_id, lead_id),
                    eq(documentEnvelopesTable.template_id, template_id),
                ),
            );
        // Stale-reservation reclaim: if the only "live" row is a queued+pending reservation
        // older than 15 minutes (worker crashed between insert and provider call), mark it
        // reclaimed so we can safely create a fresh envelope and dispatch.
        const STALE_RESERVATION_MS = 15 * 60 * 1000;
        const now = Date.now();
        const fullExisting = await tx
            .select()
            .from(documentEnvelopesTable)
            .where(
```



```

    and(
      eq(documentEnvelopesTable.lead_id, lead_id),
      eq(documentEnvelopesTable.template_id, template_id),
    ),
  );
};
const live = fullExisting.find((e) => {
  if (e.status === "error" || e.status === "voided") return false;
  // Treat as stale (and therefore not blocking) if it's still a pending reservation
  // with no provider envelope id and is older than the threshold.
  const createdAt = e.created_at ? new Date(e.created_at).getTime() : now;
  const isStaleReservation =
    e.status === "queued" &&
    (e.provider === "pending" || !e.external_envelope_id) &&
    now - createdAt > STALE_RESERVATION_MS;
  return !isStaleReservation;
});
if (live) {
  return { skipped: true as const, existing: { id: live.id, status: live.status } };
}
// Reclaim any stale reservations so they don't accumulate.
for (const stale of fullExisting) {
  if (
    stale.status === "queued" &&
    (stale.provider === "pending" || !stale.external_envelope_id)
  ) {
    {
      await tx
        .update(documentEnvelopesTable)
        .set({
          status: "error",
          last_error: "Stale reservation reclaimed (worker likely crashed before dispatch).",
          updated_at: new Date(),
        })
        .where(eq(documentEnvelopesTable.id, stale.id));
    }
  }
}
const [row] = await tx
  .insert(documentEnvelopesTable)
  .values({
    lead_id,
    template_id,
    provider: "pending",
    status: "queued",
    signer_name: signer.name,
    signer_email: signer.email,
    sent_at: null,
    events: [{ type: "queued", at: new Date().toISOString() }],
    metadata: { explicit_integration_id: explicit_integration_id ?? null },
  })
  .returning();
return { skipped: false as const, envelope: row };
});

if (reservation.skipped) {
  logger.info(
    { lead_id, template_id, envelope_id: reservation.existing.id, status: reservation.existing.status },
    "handleSendEsignPacket: envelope already exists for (lead, template) – skipping duplicate dispatch",
  );
  await auditLog("document_envelope", String(reservation.existing.id), "duplicate_dispatch_skipped", {
    lead_id, template_id, existing_status: reservation.existing.status,
  });
  return;
}
const envelope = reservation.envelope;

const resolved = await resolveProvider("esign", {
  buyerId: lead.buyer_id ?? null,
  explicitIntegrationId: explicit_integration_id ?? null,
});

if (!isResolved(resolved)) {
  await db
    .update(documentEnvelopesTable)
    .set({
      status: "error",
      last_error: `Provider resolution failed: ${resolved.reason} – ${resolved.details ?? ""}`,
      updated_at: new Date(),
    })
    .where(eq(documentEnvelopesTable.id, envelope.id));
  await auditLog("document_envelope", String(envelope.id), "provider_resolution_failed", {
    reason: resolved.reason,
    details: resolved.details,
  });
  throw new Error(`No e-sign provider configured (${resolved.reason}). Pick one on the Workflow Settings page.`);
}

const adapter = getEsignAdapter(resolved.provider);
if (!adapter) {
  await db
    .update(documentEnvelopesTable)
    .set({
      provider: resolved.provider,
      provider_integration_id: resolved.integration_id,
      status: "error",
      last_error: `No adapter implemented for provider "${resolved.provider}".`,
      updated_at: new Date(),
    })
    .where(eq(documentEnvelopesTable.id, envelope.id));
}

```

```

    throw new Error(`No e-sign adapter wired for provider "${resolved.provider}".`);
  }

  let pdfBuf: Buffer;
  try {
    pdfBuf = await buildTemplatePdf(tpl, lead);
  } catch (err) {
    await db
      .update(documentEnvelopesTable)
      .set({
        provider: resolved.provider,
        provider_integration_id: resolved.integration_id,
        status: "error",
        last_error: `Could not load/build template PDF: ${err as Error}.message`,
        updated_at: new Date(),
      })
      .where(eq(documentEnvelopesTable.id, envelope.id));
    throw err;
  }

  let outcome;
  try {
    outcome = await adapter.send(resolved.credentials, {
      pdf: pdfBuf,
      fileName: `${tpl.name}.pdf`,
      subject: tpl.delivery_subject || `Please sign: ${tpl.name}`,
      message: tpl.delivery_message || "Please review and sign the attached document.",
      signers: [{ name: signer.name, email: signer.email, role: tpl.signer_role }],
      metadata: {
        lead_id: String(lead_id),
        template_id: String(template_id),
        envelope_id: String(envelope.id),
      },
    });
  } catch (err) {
    const msg = err instanceof Error ? err.message : String(err);
    await db
      .update(documentEnvelopesTable)
      .set({
        provider: resolved.provider,
        provider_integration_id: resolved.integration_id,
        status: "error",
        last_error: `Adapter threw: ${msg}`,
        updated_at: new Date(),
      })
      .where(eq(documentEnvelopesTable.id, envelope.id));
    await auditLog("document_envelope", String(envelope.id), "adapter_threw", { provider: resolved.provider, error: msg });
    throw new Error(`E-sign adapter ${resolved.provider} threw: ${msg}`);
  }

  if (!outcome.ok) {
    await db
      .update(documentEnvelopesTable)
      .set({
        provider: resolved.provider,
        provider_integration_id: resolved.integration_id,
        status: outcome.retryable ? "queued" : "error",
        last_error: `${outcome.code} ${outcome.message}`,
        updated_at: new Date(),
        events: [
          { type: "queued", at: new Date(Date.now() - 1000).toISOString() },
          { type: outcome.retryable ? "retryable_error" : "fatal_error", at: new Date().toISOString(), raw: { code: outcome.code, message: outcome.message } },
        ],
      })
      .where(eq(documentEnvelopesTable.id, envelope.id));
    await auditLog("document_envelope", String(envelope.id), "provider_send_failed", {
      provider: resolved.provider,
      code: outcome.code,
      message: outcome.message,
      retryable: outcome.retryable,
    });
    if (outcome.retryable) {
      throw new Error(`Provider transient error: ${outcome.message}`);
    }
    // Non-retryable: do NOT throw – let the worker mark the job done so it doesn't keep retrying.
    return;
  }

  await db
    .update(documentEnvelopesTable)
    .set({
      provider: resolved.provider,
      provider_integration_id: resolved.integration_id,
      external_envelope_id: outcome.externalEnvelopeId,
      status: "sent",
      sent_at: new Date(),
      events: [
        { type: "queued", at: new Date(Date.now() - 1000).toISOString() },
        { type: "sent", at: new Date().toISOString(), raw: { external_envelope_id: outcome.externalEnvelopeId } },
      ],
      updated_at: new Date(),
    })
    .where(eq(documentEnvelopesTable.id, envelope.id));

  await auditLog("document_envelope", String(envelope.id), "sent", {
    provider: resolved.provider,
  });

```

[illegible]

```

    } return;
  }

  await db
    .update(faxResultsTable)
    .set({
      status: "done",
      raw_text: `Sent to ${targetFax} via ${resolved.provider} - external_id=${outcome.externalFaxId}`,
      processed_at: new Date(),
    })
    .where(eq(faxResultsTable.id, faxRow.id));

  await auditLog("fax", String(faxRow.id), "sent", {
    lead_id,
    envelope_id,
    to: targetFax,
    provider: resolved.provider,
    external_fax_id: outcome.externalFaxId,
  });

  logger.info(
    { fax_id: faxRow.id, lead_id, envelope_id, provider: resolved.provider, external_id: outcome.externalFaxId, to: targetFax },
    "Med records fax sent",
  );
}

// %%%%%%%%%%%%%%%
// HANDLER: send_workflow_email
// %%%%%%%%%%%%%%%

export async function handleSendWorkflowEmail(payload: SendWorkflowEmailPayload): Promise<void> {
  const { to, to_name, subject, html, text, explicit_integration_id, lead_id } = payload;

  const buyerId = lead_id
    ? (await db.select({ b: leadsTable.buyer_id }).from(leadsTable).where(eq(leadsTable.id, lead_id)))[0]?.b ?? null
    : null;

  const resolved = await resolveProvider("email", {
    buyerId,
    explicitIntegrationId: explicit_integration_id ?? null,
  });

  if (!isResolved(resolved)) {
    throw new Error(`No email provider configured (${resolved.reason}). Pick one on the Workflow Settings page.`);
  }

  const adapter = getEmailAdapter(resolved.provider);
  if (!adapter) {
    throw new Error(`No email adapter wired for provider "${resolved.provider}".`);
  }

  const fromEmail = (resolved.credentials as Record<string, unknown>).from_email as string | undefined
    || "noreply@example.com";
  const fromName = (resolved.credentials as Record<string, unknown>).from_name as string | undefined
    || "MTOS";

  let outcome;
  try {
    outcome = await adapter.send(resolved.credentials, {
      to,
      toName: to_name,
      fromEmail,
      fromName,
      subject,
      html,
      text,
    });
  } catch (err) {
    const msg = err instanceof Error ? err.message : String(err);
    logger.error({ err, to, subject, provider: resolved.provider }, "Email adapter threw");
    throw new Error(`Email adapter ${resolved.provider} threw: ${msg}`);
  }

  if (!outcome.ok) {
    if (outcome.retryable) throw new Error(`Email provider transient error: ${outcome.message}`);
    logger.error({ to, code: outcome.code, message: outcome.message }, "Email send failed (non-retryable)");
    return;
  }

  logger.info({ to, subject, provider: resolved.provider, external_id: outcome.externalMessageId }, "Workflow email sent");
}

```

[artifacts/api-server/src/routes/analytics.ts](#)

```
import { Router } from "express";
import { db, leadsTable, casesTable, analysisTable, faxResultsTable, paralegalsTable } from "@workspace/db";
import { sql, eq, gte, and, desc } from "drizzle-orm";
import { scoreLeadPredictive, getModelStats, getBatchPredictions, getTortPredictions } from "../lib/predictive-scoring";
import { requireRole } from "../lib/rbac";

const router = Router();

router.get("/overview", requireRole("attorney"), async (_req, res) => {
  const now = new Date();
  const thirtyDaysAgo = new Date(now.getTime() - 30 * 86400000);
  const sevenDaysAgo = new Date(now.getTime() - 7 * 86400000);

  const [leadStats] = await db
    .select({
      total: sql<number>`count(*)::int`,
      qualified: sql<number>`count(*) filter (where status = 'qualified')::int`,
      signed: sql<number>`count(*) filter (where status = 'signed')::int`,
      rejected: sql<number>`count(*) filter (where status = 'rejected')::int`,
      new_leads: sql<number>`count(*) filter (where status = 'new')::int`,
      total_ad_spend: sql<number>`coalesce(sum(ad_spend::numeric), 0)::float`,
      hot: sql<number>`count(*) filter (where routing = 'hot')::int`,
      warm: sql<number>`count(*) filter (where routing = 'warm')::int`,
      cold: sql<number>`count(*) filter (where routing = 'cold')::int`,
      last_30_days: sql<number>`count(*) filter (where created_at >= ${thirtyDaysAgo})::int`,
      last_7_days: sql<number>`count(*) filter (where created_at >= ${sevenDaysAgo})::int`,
    })
    .from(leadsTable);

  const [caseStats] = await db
    .select({
      total_cases: sql<number>`count(*)::int`,
      analyzed: sql<number>`count(*) filter (where status = 'analyzed')::int`,
      open: sql<number>`count(*) filter (where status = 'open')::int`,
    })
    .from(casesTable);

  const [analysisStats] = await db
    .select({
      total_analyses: sql<number>`count(*)::int`,
      avg_score: sql<number>`coalesce(avg(score), 0)::float`,
      strong_cases: sql<number>`count(*) filter (where score >= 80)::int`,
      moderate_cases: sql<number>`count(*) filter (where score >= 50 and score < 80)::int`,
      weak_cases: sql<number>`count(*) filter (where score >= 25 and score < 50)::int`,
      disqualified_cases: sql<number>`count(*) filter (where score < 25)::int`,
    })
    .from(analysisTable);

  const [faxStats] = await db
    .select({
      total_faxes: sql<number>`count(*)::int`,
      processed: sql<number>`count(*) filter (where status = 'done')::int`,
      pending: sql<number>`count(*) filter (where status = 'pending')::int`,
    })
    .from(faxResultsTable);

  const total = leadStats.total ?? 0;
  const signed = leadStats.signed ?? 0;
  const totalAdSpend = leadStats.total_ad_spend ?? 0;
  const cpsr = signed > 0 && totalAdSpend > 0 ? totalAdSpend / signed : null;
  const roi = signed > 0 && totalAdSpend > 0 ? Math.round(((signed * 3500 - totalAdSpend) / totalAdSpend) * 100) : null;

  res.json({
    leads: {
      ...leadStats,
      cpsr,
      roi,
      qualification_rate: total > 0 ? Math.round(((leadStats.qualified + signed) / total) * 100) : 0,
      conversion_rate: total > 0 ? Math.round((signed / total) * 100) : 0,
    },
    cases: caseStats,
    analysis: analysisStats,
    faxes: faxStats,
  });
});

router.get("/pipeline-trend", requireRole("attorney"), async (_req, res) => {
  const result = await db
    .select({
      date: sql<string>`to_char(created_at, 'YYYY-MM-DD')`,
      total: sql<number>`count(*)::int`,
      qualified: sql<number>`count(*) filter (where status in ('qualified', 'signed'))::int`,
      signed: sql<number>`count(*) filter (where status = 'signed')::int`,
    })
    .from(leadsTable)
    .groupBy(sql`to_char(created_at, 'YYYY-MM-DD')`)
    .orderBy(sql`to_char(created_at, 'YYYY-MM-DD')`)
    .limit(90);

  res.json(result);
});

router.get("/conversion-funnel", requireRole("attorney"), async (_req, res) => {
  const [counts] = await db
    .select({
```

```

    total_leads: sql<number>`count(*)::int`,
    new_leads: sql<number>`count(*) filter (where status = 'new')::int`,
    qualified: sql<number>`count(*) filter (where status = 'qualified')::int`,
    records_requested: sql<number>`count(*) filter (where diagnosis_confirmed = true)::int`,
    signed: sql<number>`count(*) filter (where status = 'signed')::int`,
    rejected: sql<number>`count(*) filter (where status = 'rejected')::int`,
  })
  .from(leadsTable);

const stages = [
  { stage: "New Leads", count: counts.total_leads, color: "#3B82F6" },
  { stage: "Qualified", count: counts.qualified + counts.signed, color: "#10B981" },
  { stage: "Records Verified", count: counts.records_requested, color: "#F59E0B" },
  { stage: "Signed Retainers", count: counts.signed, color: "#8B5CF6" },
  { stage: "Rejected", count: counts.rejected, color: "#EF4444" },
];

res.json(stages);
});

router.get("/tort-breakdown", requireRole("attorney"), async (_req, res) => {
  const result = await db
    .select({
      tort_type: leadsTable.tort_type,
      total: sql<number>`count(*)::int`,
      qualified: sql<number>`count(*) filter (where status = 'qualified')::int`,
      signed: sql<number>`count(*) filter (where status = 'signed')::int`,
      rejected: sql<number>`count(*) filter (where status = 'rejected')::int`,
      avg_ad_spend: sql<number>`coalesce(avg(ad_spend::numeric), 0)::float`,
    })
    .from(leadsTable)
    .groupBy(leadsTable.tort_type)
    .orderBy(sql`count(*) desc`);

  res.json(result);
});

router.get("/paralegal-leaderboard", requireRole("attorney"), async (_req, res) => {
  const result = await db
    .select({
      id: paralegalsTable.id,
      name: paralegalsTable.name,
      role: paralegalsTable.role,
      total_assigned: sql<number>`count(${leadsTable.id})::int`,
      signed: sql<number>`count(*) filter (where ${leadsTable.status} = 'signed')::int`,
      qualified: sql<number>`count(*) filter (where ${leadsTable.status} = 'qualified')::int`,
      rejected: sql<number>`count(*) filter (where ${leadsTable.status} = 'rejected')::int`,
    })
    .from(paralegalsTable)
    .leftJoin(leadsTable, eq(paralegalsTable.id, leadsTable.assigned_to))
    .groupBy(paralegalsTable.id, paralegalsTable.name, paralegalsTable.role)
    .orderBy(sql`count(*) filter (where ${leadsTable.status} = 'signed') desc`);

  const leaderboard = result.map(r => ({
    ...r,
    conversion_rate: r.total_assigned > 0 ? Math.round((r.signed / r.total_assigned) * 100) : 0,
  }));

  res.json(leaderboard);
});

router.get("/predictive/lead/:id", requireRole("paralegal"), async (req, res) => {
  const id = parseInt(String(req.params.id), 10);
  if (isNaN(id)) { res.status(400).json({ error: "Invalid lead ID" }); return; }
  try {
    const score = await scoreLeadPredictive(id);
    res.json(score);
  } catch (err: any) {
    res.status(404).json({ error: "Lead not found or scoring failed" });
  }
});

router.get("/predictive/batch", requireRole("attorney"), async (req, res) => {
  const limit = parseInt(req.query.limit as string) || 50;
  const predictions = await getBatchPredictions(limit);
  res.json(predictions);
});

router.get("/predictive/by-tort", requireRole("attorney"), async (_req, res) => {
  const predictions = await getTortPredictions();
  res.json(predictions);
});

router.get("/predictive/model", requireRole("attorney"), async (_req, res) => {
  const stats = await getModelStats();
  res.json(stats);
});

export default router;

```

[artifacts/api-server/src/routes/auth.ts](#)

```
import { Router } from "express";
import crypto from "crypto";
import rateLimit from "express-rate-limit";
import {
  generateToken,
  generateRefreshToken,
  rotateRefreshToken,
  revokeAllUserTokens,
  getUserByEmail,
  getUserById,
  createUser,
  listUsers,
  authMiddleware,
  requireRole,
  incrementFailedAttempts,
  resetFailedAttempts,
  isAccountLocked,
} from "../lib/rbac";
import { auditLog } from "../lib/audit";
import { logger } from "../lib/logger";
import { generateSecret, verifyTOTP, generateOTPAuthURL } from "../lib/totp";
import { encrypt, decrypt } from "../lib/encryption";
import { db } from "@workspace/db";
import { sql } from "drizzle-orm";
import { dispatchCriticalAlert } from "../lib/security-alerts";

const router = Router();

const authRateLimit = rateLimit({
  windowMs: 15 * 60 * 1000,
  max: 20,
  standardHeaders: true,
  legacyHeaders: false,
  message: { error: "Too many authentication attempts. Please try again later." },
  keyGenerator: (req) => (req.headers["x-forwarded-for"] as string)?.split(",")[0]?.trim() || req.socket.remoteAddress || "unknown",
});

const PASSWORD_RULES = {
  minLength: 12,
  requireUppercase: /[A-Z]/,
  requireLowercase: /[a-z]/,
  requireNumber: /[0-9]/,
  requireSpecial: /[!@#$%^&*()_+~\-=\[\]{};':"\"|,.<>\/?`~]/,
};

function validatePasswordComplexity(password: string): string | null {
  if (password.length < PASSWORD_RULES.minLength) {
    return `Password must be at least ${PASSWORD_RULES.minLength} characters`;
  }
  if (!PASSWORD_RULES.requireUppercase.test(password)) {
    return "Password must contain at least one uppercase letter";
  }
  if (!PASSWORD_RULES.requireLowercase.test(password)) {
    return "Password must contain at least one lowercase letter";
  }
  if (!PASSWORD_RULES.requireNumber.test(password)) {
    return "Password must contain at least one number";
  }
  if (!PASSWORD_RULES.requireSpecial.test(password)) {
    return "Password must contain at least one special character";
  }
  return null;
}

async function hashPassword(password: string): Promise<string> {
  const salt = crypto.randomBytes(16).toString("hex");
  return new Promise((resolve, reject) => {
    crypto.scrypt(password, salt, 64, (err, derivedKey) => {
      if (err) reject(err);
      else resolve(`${salt}:${derivedKey.toString("hex")}`);
    });
  });
}

async function verifyPassword(password: string, stored: string): Promise<boolean> {
  const [salt, hash] = stored.split(":");
  return new Promise((resolve, reject) => {
    crypto.scrypt(password, salt, 64, (err, derivedKey) => {
      if (err) reject(err);
      else resolve(crypto.timingSafeEqual(Buffer.from(hash, "hex"), derivedKey));
    });
  });
}

router.post("/login", authRateLimit, async (req, res) => {
  const { email, password, totp_code } = req.body;
  if (!email || !password) { res.status(400).json({ error: "Email and password required" }); return; }

  const user = await getUserByEmail(email);
  if (!user) {
    res.status(401).json({ error: "Invalid credentials" });
    return;
  }
});
```

```

    if (await isAccountLocked(user)) {
      const minutesLeft = Math.ceil((new Date(user.locked_until!).getTime() - Date.now()) / 60000);
      await dispatchCriticalAlert("high", "Login attempt on locked account", `Locked account login attempt: ${email}`, undefined);
      res.status(423).json({ error: `Account is locked. Try again in ${minutesLeft} minute(s).` });
      return;
    }

    if (!(await verifyPassword(password, user.password_hash))) {
      const result = await incrementFailedAttempts(email);
      if (result.locked) {
        await dispatchCriticalAlert("high", "Account locked after failed attempts", `Account locked: ${email} after ${result.attempts} failed attempts`);
        await auditLog("security", String(user.id), "account_locked", { email, attempts: result.attempts });
      }
      res.status(401).json({ error: "Invalid credentials" });
      return;
    }

    if (user.mfa_enabled && user.totp_secret) {
      if (!totp_code) {
        res.status(200).json({ mfa_required: true, message: "Please provide your TOTP code" });
        return;
      }
      const decryptedSecret = decrypt(user.totp_secret);
      if (!verifyTOTP(decryptedSecret, totp_code)) {
        const result = await incrementFailedAttempts(email);
        if (result.locked) {
          await dispatchCriticalAlert("high", "Account locked after failed MFA", `Account locked: ${email} after ${result.attempts} failed attempts (MFA)`);
          await auditLog("security", String(user.id), "account_locked_mfa", { email, attempts: result.attempts });
        }
        await dispatchCriticalAlert("medium", "Failed MFA attempt", `Failed TOTP verification for: ${email}`);
        res.status(401).json({ error: "Invalid TOTP code" });
        return;
      }
    }

    await resetFailedAttempts(email);

    const accessToken = generateToken({ id: user.id, email: user.email, name: user.name, role: user.role as any, token_version: user.token_version });
    const refreshToken = await generateRefreshToken(user.id);

    await auditLog("user", String(user.id), "login", { email: user.email }, {
      ip_address: (req.headers["x-forwarded-for"] as string)?.split(",")[0]?.trim() || req.socket.remoteAddress,
      user_agent: req.headers["user-agent"],
    });

    res.json({
      token: accessToken,
      refresh_token: refreshToken,
      expires_in: 900,
      user: { id: user.id, email: user.email, name: user.name, role: user.role, mfa_enabled: user.mfa_enabled },
    });
  });

  router.post("/refresh", async (req, res) => {
    const { refresh_token, user_id } = req.body;
    if (!refresh_token || !user_id) {
      res.status(400).json({ error: "refresh_token and user_id required" });
      return;
    }

    const result = await rotateRefreshToken(refresh_token, user_id);
    if (!result) {
      res.status(401).json({ error: "Invalid or expired refresh token" });
      return;
    }

    res.json({
      token: result.accessToken,
      refresh_token: result.refreshToken,
      expires_in: 900,
    });
  });

  router.post("/logout", authMiddleware, async (req, res) => {
    const user = req.user!;
    await revokeAllUserTokens(user.id);

    await auditLog("user", String(user.id), "logout", { email: user.email });

    res.json({ message: "Logged out successfully. All tokens revoked." });
  });

  router.post("/register", authRateLimit, async (req, res) => {
    const { email, password, name } = req.body;
    if (!email || !password || !name) {
      res.status(400).json({ error: "Email, password, and name required" });
      return;
    }

    const passwordError = validatePasswordComplexity(password);
    if (passwordError) {
      res.status(400).json({ error: passwordError });
      return;
    }

    const assignedRole = "viewer";

    const existing = await getUserByEmail(email);
    if (existing) {
      res.status(409).json({ error: "Email already registered" });
      return;
    }
  });

```



```

const passwordHash = await hashPassword(password);
const user = await createUser(email, name, assignedRole, passwordHash);
const accessToken = generateToken(user);
const refreshToken = await generateRefreshToken(user.id);

await auditLog("user", String(user.id), "registered", { email, role: assignedRole });

res.status(201).json({
  token: accessToken,
  refresh_token: refreshToken,
  expires_in: 900,
  user,
});
});

router.post("/change-password", authMiddleware, async (req, res) => {
  const { current_password, new_password } = req.body;
  if (!current_password || !new_password) {
    res.status(400).json({ error: "current_password and new_password required" });
    return;
  }

  const passwordError = validatePasswordComplexity(new_password);
  if (passwordError) { res.status(400).json({ error: passwordError }); return; }

  const user = await getUserByEmail(req.user!.email);
  if (!user || !(await verifyPassword(current_password, user.password_hash))) {
    res.status(401).json({ error: "Current password is incorrect" });
    return;
  }

  const newHash = await hashPassword(new_password);
  await db.execute(sql`
    UPDATE mtos_users SET password_hash = ${newHash}, token_version = token_version + 1, updated_at = NOW()
    WHERE id = ${req.user!.id}
  `);

  await revokeAllUserTokens(req.user!.id);
  await auditLog("user", String(req.user!.id), "password_changed", { email: req.user!.email });

  res.json({ message: "Password changed. All existing sessions have been revoked. Please log in again." });
});

router.post("/mfa/setup", authMiddleware, async (req, res) => {
  const user = req.user!;
  const dbUser = await getUserByEmail(user.email);
  if (!dbUser) { res.status(404).json({ error: "User not found" }); return; }
  if (dbUser.mfa_enabled) { res.status(409).json({ error: "MFA is already enabled" }); return; }

  const secret = generateSecret();
  const encryptedSecret = encrypt(secret);
  const otpauthUrl = generateOTPAuthURL(secret, user.email);

  await db.execute(sql`
    UPDATE mtos_users SET totp_secret = ${encryptedSecret}, updated_at = NOW()
    WHERE id = ${user.id}
  `);

  res.json({
    secret,
    otpauth_url: otpauthUrl,
    message: "Scan the QR code or enter the secret in your authenticator app, then call /mfa/verify to activate.",
  });
});

router.post("/mfa/verify", authMiddleware, async (req, res) => {
  const { totp_code } = req.body;
  if (!totp_code) { res.status(400).json({ error: "totp_code required" }); return; }

  const user = req.user!;
  const dbUser = await getUserByEmail(user.email);
  if (!dbUser || !dbUser.totp_secret) { res.status(400).json({ error: "MFA setup not initiated" }); return; }

  const decryptedSecret = decrypt(dbUser.totp_secret);
  if (!verifyTOTP(decryptedSecret, totp_code)) {
    res.status(401).json({ error: "Invalid TOTP code. Please try again." });
    return;
  }

  await db.execute(sql`
    UPDATE mtos_users SET mfa_enabled = true, updated_at = NOW()
    WHERE id = ${user.id}
  `);

  await auditLog("user", String(user.id), "mfa_enabled", { email: user.email });

  res.json({ message: "MFA enabled successfully.", mfa_enabled: true });
});

router.post("/mfa/disable", authMiddleware, async (req, res) => {
  const { totp_code, password } = req.body;
  if (!password || !totp_code) { res.status(400).json({ error: "Password and TOTP code required to disable MFA" }); return; }

  const user = req.user!;
  const dbUser = await getUserByEmail(user.email);
  if (!dbUser) { res.status(404).json({ error: "User not found" }); return; }
  if (!dbUser.mfa_enabled) { res.status(400).json({ error: "MFA is not enabled" }); return; }

```

```

    if (!(await verifyPassword(password, dbUser.password_hash))) {
      res.status(401).json({ error: "Invalid password" });
      return;
    }

    if (!dbUser.totp_secret) { res.status(400).json({ error: "MFA setup incomplete" }); return; }
    const decryptedSecret = decrypt(dbUser.totp_secret);
    if (!verifyTOTP(decryptedSecret, totp_code)) {
      res.status(401).json({ error: "Invalid TOTP code" });
      return;
    }

    await db.execute(sql`
      UPDATE mtos_users SET mfa_enabled = false, totp_secret = NULL, updated_at = NOW()
      WHERE id = ${user.id}
    `);

    await auditLog("user", String(user.id), "mfa_disabled", { email: user.email });

    res.json({ message: "MFA disabled successfully." });
  });

  router.get("/me", authMiddleware, async (req, res) => {
    const dbUser = await getUserByEmail(req.user!.email);
    res.json({
      id: req.user!.id,
      email: req.user!.email,
      name: req.user!.name,
      role: req.user!.role,
      mfa_enabled: dbUser?.mfa_enabled ?? false,
    });
  });

  router.get("/users", authMiddleware, requireRole("admin"), async (_req, res) => {
    const users = await listUsers();
    res.json(users);
  });

  export default router;

```

[artifacts/api-server/src/routes/buyers.ts](#)

```
import { Router } from "express";
import { db, buyersTable } from "@workspace/db";
import { eq } from "drizzle-orm";
import { authMiddleware, requireRole } from "../lib/rbac";
import { auditAction } from "../lib/rbac";
import { z } from "zod/v4";

const router = Router();
router.use(authMiddleware);

const buyerSchema = z.object({
  name: z.string().min(1).max(255),
  legal_name: z.string().max(255).nullish(),
  contact_name: z.string().max(255).nullish(),
  contact_email: z.email().nullish(),
  contact_phone: z.string().max(50).nullish(),
  notes: z.string().nullish(),
  esign_provider_integration_id: z.number().int().nullish(),
  fax_provider_integration_id: z.number().int().nullish(),
  email_provider_integration_id: z.number().int().nullish(),
  default_email_from_name: z.string().max(255).nullish(),
  default_email_from_address: z.email().nullish(),
  default_fax_from_number: z.string().max(50).nullish(),
  active: z.boolean().optional(),
  metadata: z.record(z.string(), z.unknown()).optional(),
});

router.get("/", requireRole("viewer"), async (_req, res) => {
  const rows = await db.select().from(buyersTable).orderBy(buyersTable.name);
  res.json(rows);
});

router.get("/:id", requireRole("viewer"), async (req, res) => {
  const id = Number(req.params.id);
  if (!Number.isFinite(id)) {
    res.status(400).json({ error: "invalid_id" });
    return;
  }
  const [row] = await db.select().from(buyersTable).where(eq(buyersTable.id, id));
  if (!row) {
    res.status(404).json({ error: "not_found" });
    return;
  }
  res.json(row);
});

router.post("/", requireRole("admin"), auditAction("create_buyer"), async (req, res) => {
  const parsed = buyerSchema.safeParse(req.body);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }
  try {
    const [row] = await db.insert(buyersTable).values(parsed.data as any).returning();
    res.status(201).json(row);
  } catch (e: any) {
    if (String(e?.message || "").includes("duplicate")) {
      res.status(409).json({ error: "name_taken" });
      return;
    }
    throw e;
  }
});

router.put("/:id", requireRole("admin"), auditAction("update_buyer"), async (req, res) => {
  const id = Number(req.params.id);
  if (!Number.isFinite(id)) {
    res.status(400).json({ error: "invalid_id" });
    return;
  }
  const parsed = buyerSchema.partial().safeParse(req.body);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }
  const updates = { ...parsed.data, updated_at: new Date() } as any;
  const [row] = await db.update(buyersTable).set(updates).where(eq(buyersTable.id, id)).returning();
  if (!row) {
    res.status(404).json({ error: "not_found" });
    return;
  }
  res.json(row);
});

router.delete("/:id", requireRole("admin"), auditAction("delete_buyer"), async (req, res) => {
  const id = Number(req.params.id);
  if (!Number.isFinite(id)) {
    res.status(400).json({ error: "invalid_id" });
    return;
  }
  await db.delete(buyersTable).where(eq(buyersTable.id, id));
  res.status(204).end();
});

export default router;
```

[artifacts/api-server/src/routes/cases.ts](#)

```
import { Router } from "express";
import { db, casesTable, analysisTable, caseDocumentsTable, auditLogTable } from "@workspace/db";
import { eq, sql, desc } from "drizzle-orm";
import { enqueueJob, getQueueStats } from "../lib/queue";
import { auditLog } from "../lib/audit";
import crypto from "crypto";
import { requireRole, auditAction } from "../lib/rbac";

const UUID_REGEX = /^[0-9a-f]{8}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{4}-[0-9a-f]{12}$/i;

function validateCaseId(id: string): boolean {
  return UUID_REGEX.test(id);
}

const router = Router();

router.post("/", requireRole("paralegal", "attorney", "admin"), auditAction("create_case"), async (req, res) => {
  const data = req.body as Record<string, unknown>;
  const case_id = crypto.randomUUID();

  const job_id = await enqueueJob("create_case", { case_id, data });

  await auditLog("case", case_id, "intake_submitted", { data }, {
    ip_address: req.ip,
    user_agent: req.get("user-agent"),
  });

  res.status(201).json({ case_id, status: "queued", job_id });
});

router.post("/:id/upload", requireRole("paralegal", "attorney", "admin"), auditAction("upload_case_file"), async (req, res) => {
  const case_id = String(req.params.id);
  if (!validateCaseId(case_id)) {
    res.status(400).json({ error: "Invalid case ID format" });
    return;
  }

  const { file_name, content, content_type } = req.body as {
    file_name: string;
    content: string;
    content_type?: string;
  };

  if (!file_name || !content) {
    res.status(400).json({ error: "file_name and content are required" });
    return;
  }

  const job_id = await enqueueJob("ingest_file", {
    case_id,
    file_name,
    content,
    content_type,
  });

  await auditLog("case", case_id, "file_upload_queued", { file_name }, {
    ip_address: req.ip,
    user_agent: req.get("user-agent"),
  });

  res.json({ case_id, status: "queued", job_id, file_name });
});

router.post("/:id/analyze", requireRole("paralegal"), async (req, res) => {
  const case_id = String(req.params.id);
  if (!validateCaseId(case_id)) {
    res.status(400).json({ error: "Invalid case ID format" });
    return;
  }

  const job_id = await enqueueJob("analyze_case", { case_id });

  await auditLog("case", case_id, "analysis_queued", {}, {
    ip_address: req.ip,
    user_agent: req.get("user-agent"),
  });

  res.json({ case_id, status: "queued", job_id });
});

router.get("/", requireRole("viewer"), async (req, res) => {
  const cases = await db
    .select()
    .from(casesTable)
    .orderBy(desc(casesTable.created_at))
    .limit(100);
  res.json(cases);
});

router.get("/:id", requireRole("viewer"), async (req, res) => {
  const case_id = String(req.params.id);

  const [caseRow] = await db
    .select()
    .from(casesTable)
```

```

    .where(eq(casesTable.id, case_id)));

if (!caseRow) {
  res.status(404).json({ error: "Case not found" });
  return;
}

const docs = await db
  .select()
  .from(caseDocumentsTable)
  .where(eq(caseDocumentsTable.case_id, case_id));

const analyses = await db
  .select()
  .from(analysisTable)
  .where(eq(analysisTable.case_id, case_id))
  .orderBy(desc(analysisTable.created_at));

const auditEntries = await db
  .select()
  .from(auditLogTable)
  .where(eq(auditLogTable.entity_id, case_id))
  .orderBy(desc(auditLogTable.occurred_at))
  .limit(50);

res.json({
  case: caseRow,
  documents: docs,
  analyses,
  audit_trail: auditEntries,
});
});

router.get("/worker/queue-stats", requireRole("admin"), async (req, res) => {
  const stats = await getQueueStats();
  res.json(stats);
});

export default router;

```

[artifacts/api-server/src/routes/compliance.ts](#)

```
import { Router } from "express";
import { db, auditLogTable } from "@workspace/db";
import { desc, eq, sql, and, gte } from "drizzle-orm";
import { requireRole } from "../lib/rbac";

const router = Router();

router.get("/audit-trail", requireRole("admin"), async (req, res) => {
  const limit = parseInt(req.query.limit as string) || 100;
  const entityType = req.query.entity_type == null ? undefined : String(req.query.entity_type);
  const action = req.query.action == null ? undefined : String(req.query.action);

  let query = db
    .select()
    .from(auditLogTable)
    .orderBy(desc(auditLogTable.occurred_at))
    .limit(limit);

  const results = await query;

  let filtered = results;
  if (entityType) {
    filtered = filtered.filter(r => r.entity_type === entityType);
  }
  if (action) {
    filtered = filtered.filter(r => r.action === action);
  }

  res.json(filtered);
});

router.get("/audit-summary", requireRole("admin"), async (_req, res) => {
  const byEntity = await db
    .select({
      entity_type: auditLogTable.entity_type,
      count: sql<number>`count(*)::int`,
    })
    .from(auditLogTable)
    .groupBy(auditLogTable.entity_type)
    .orderBy(sql`count(*) desc`);

  const byAction = await db
    .select({
      action: auditLogTable.action,
      count: sql<number>`count(*)::int`,
    })
    .from(auditLogTable)
    .groupBy(auditLogTable.action)
    .orderBy(sql`count(*) desc`);

  const now = new Date();
  const last24h = new Date(now.getTime() - 86400000);
  const last7d = new Date(now.getTime() - 7 * 86400000);

  const [recentCounts] = await db
    .select({
      total: sql<number>`count(*)::int`,
      last_24h: sql<number>`count(*) filter (where occurred_at >= ${last24h})::int`,
      last_7d: sql<number>`count(*) filter (where occurred_at >= ${last7d})::int`,
    })
    .from(auditLogTable);

  res.json({
    by_entity: byEntity,
    by_action: byAction,
    total_events: recentCounts.total,
    last_24h: recentCounts.last_24h,
    last_7d: recentCounts.last_7d,
  });
});

export default router;
```

[artifacts/api-server/src/routes/dashboard.ts](#)

```
import { Router } from "express";
import { db, leadsTable, documentsTable } from "@workspace/db";
import { eq, sql, and, gte } from "drizzle-orm";
import { requireRole } from "../lib/rbac";

const router = Router();

router.get("/stats", requireRole("viewer"), async (req, res) => {
  const now = new Date();
  const startOfDay = new Date(now.getFullYear(), now.getMonth(), now.getDate());
  const startOfWeek = new Date(startOfDay);
  startOfWeek.setDate(startOfDay.getDate() - 7);

  const [counts] = await db
    .select({
      total: sql<number>`count(*)::int`,
      qualified: sql<number>`count(*) filter (where status = 'qualified')::int`,
      signed: sql<number>`count(*) filter (where status = 'signed')::int`,
      rejected: sql<number>`count(*) filter (where status = 'rejected')::int`,
      new_leads: sql<number>`count(*) filter (where status = 'new')::int`,
      total_ad_spend: sql<number>`coalesce(sum(ad_spend::numeric), 0)::float`,
      leads_today: sql<number>`count(*) filter (where created_at >= ${startOfDay})::int`,
      leads_this_week: sql<number>`count(*) filter (where created_at >= ${startOfWeek})::int`,
    })
    .from(leadsTable);

  const total = counts.total ?? 0;
  const qualified = counts.qualified ?? 0;
  const signed = counts.signed ?? 0;
  const rejected = counts.rejected ?? 0;
  const new_leads = counts.new_leads ?? 0;
  const total_ad_spend = counts.total_ad_spend ?? 0;

  const cpsr = signed > 0 && total_ad_spend > 0 ? total_ad_spend / signed : null;
  const qualification_rate = total > 0 ? (qualified + signed) / total : 0;
  const conversion_rate = total > 0 ? signed / total : 0;

  res.json({
    total_leads: total,
    qualified_leads: qualified,
    signed_retainers: signed,
    rejected_leads: rejected,
    new_leads: new_leads,
    total_ad_spend: total_ad_spend,
    cpsr,
    qualification_rate,
    conversion_rate,
    leads_today: counts.leads_today ?? 0,
    leads_this_week: counts.leads_this_week ?? 0,
  });
});

router.get("/pipeline", requireRole("viewer"), async (req, res) => {
  const byStatus = await db
    .select({
      status: leadsTable.status,
      count: sql<number>`count(*)::int`,
    })
    .from(leadsTable)
    .groupBy(leadsTable.status);

  const byTortType = await db
    .select({
      tort_type: leadsTable.tort_type,
      count: sql<number>`count(*)::int`,
      qualified: sql<number>`count(*) filter (where status = 'qualified')::int`,
      signed: sql<number>`count(*) filter (where status = 'signed')::int`,
    })
    .from(leadsTable)
    .groupBy(leadsTable.tort_type)
    .orderBy(sql`count(*) desc`);

  res.json({
    by_status: byStatus,
    by_tort_type: byTortType,
  });
});

router.get("/recent-activity", requireRole("viewer"), async (req, res) => {
  const recentLeads = await db
    .select({
      id: leadsTable.id,
      lead_id: leadsTable.id,
      lead_name: leadsTable.name,
      status: leadsTable.status,
      tort_type: leadsTable.tort_type,
      created_at: leadsTable.created_at,
      updated_at: leadsTable.updated_at,
    })
    .from(leadsTable)
    .orderBy(sql`${leadsTable.updated_at} DESC`)
    .limit(20);

  const activity = recentLeads.map((lead, index) => {
    let event_type = "lead_updated";
  });
});
```

```

let description = `Lead updated`;

const isNew =
  Math.abs(lead.created_at.getTime() - lead.updated_at.getTime()) < 5000;

if (isNew) {
  event_type = "lead_created";
  description = `New lead submitted for ${lead.tort_type}`;
} else if (lead.status === "qualified") {
  event_type = "lead_qualified";
  description = `Qualified for ${lead.tort_type} - ready for retainer`;
} else if (lead.status === "signed") {
  event_type = "retainer_signed";
  description = `Retainer signed for ${lead.tort_type}`;
} else if (lead.status === "rejected") {
  event_type = "lead_rejected";
  description = `Disqualified from ${lead.tort_type} pipeline`;
}

return {
  id: index + 1,
  lead_id: lead.lead_id,
  lead_name: lead.lead_name,
  event_type,
  description,
  occurred_at: lead.updated_at.toISOString(),
  tort_type: lead.tort_type,
};
});

res.json(activity);
});

export default router;

```


[artifacts/api-server/src/routes/decision-engine.ts](#)

```
import { Router } from "express";
import { authMiddleware, requireRole } from "../lib/rbac";
import {
  buildPortfolioSummary,
  computeAndPersistLeadScore,
  recomputeAllScores,
  getEngineSettings,
  updateEngineSettings,
} from "../lib/decision-engine-service";
import { logger } from "../lib/logger";
import { z } from "zod/v4";

const router = Router();

// All decision-engine endpoints are admin-only.
router.use(authMiddleware, requireRole("admin"));

router.get("/portfolio", async (_req, res) => {
  try {
    const summary = await buildPortfolioSummary();
    res.json(summary);
  } catch (e) {
    logger.error({ err: e }, "portfolio build failed");
    res.status(500).json({ error: "portfolio_failed" });
  }
});

router.get("/settings", async (_req, res) => {
  const s = await getEngineSettings();
  res.json(s);
});

const updateSettingsSchema = z.object({
  default_attorney_hourly_cost: z.number().nonnegative().optional(),
  default_hours_per_lead: z.number().nonnegative().optional(),
  convex_ratio_threshold: z.number().positive().optional(),
  concave_ratio_threshold: z.number().positive().optional(),
  concentration_warning_pct: z.number().int().min(1).max(100).optional(),
  ruin_auto_flag: z.boolean().optional(),
});

router.put("/settings", async (req, res) => {
  const parsed = updateSettingsSchema.safeParse(req.body);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }
  await updateEngineSettings(parsed.data);
  const s = await getEngineSettings();
  res.json(s);
});

router.post("/leads/:id/recompute", async (req, res) => {
  const id = Number(req.params.id);
  if (!Number.isFinite(id)) {
    res.status(400).json({ error: "invalid_id" });
    return;
  }
  const result = await computeAndPersistLeadScore(id);
  if (!result) {
    res.status(404).json({ error: "lead_or_tort_not_found" });
    return;
  }
  res.json(result);
});

router.post("/recompute-all", async (_req, res) => {
  const result = await recomputeAllScores();
  res.json(result);
});

export default router;
```

[artifacts/api-server/src/routes/document-templates.ts](#)

```
import { Router } from "express";
import { db, documentTemplatesTable, templateAssignmentsTable } from "@workspace/db";
import { and, eq, isNull } from "drizzle-orm";
import { authMiddleware, requireRole } from "../lib/rbac";
import { auditAction } from "../lib/rbac";
import { z } from "zod/v4";
import { uploadTemplate, deleteTemplate, downloadTemplate } from "../lib/template-storage";

const router = Router();
router.use(authMiddleware);

const templateSchema = z.object({
  name: z.string().min(1).max(255),
  template_type: z.string().min(1).max(50),
  description: z.string().nullish(),
  source: z.enum(["pdf", "ai"]).default("pdf"),
  storage_path: z.string().nullish(),
  ai_prompt: z.string().nullish(),
  requires_signature: z.boolean().default(true),
  signer_role: z.string().max(50).default("lead"),
  merge_fields: z.array(z.string()).default([]),
  delivery_subject: z.string().max(255).nullish(),
  delivery_message: z.string().nullish(),
  triggers_med_records_request: z.boolean().default(false),
  active: z.boolean().default(true),
});

const assignmentSchema = z.object({
  template_id: z.number().int(),
  buyer_id: z.number().int().nullish(),
  tort_type: z.string().max(100).nullish(),
  enabled: z.boolean().default(true),
  send_order: z.number().int().default(0),
  notes: z.string().nullish(),
});

// ----- Templates -----

router.get("/", requireRole("viewer"), async (_req, res) => {
  const rows = await db.select().from(documentTemplatesTable).orderBy(documentTemplatesTable.name);
  res.json(rows);
});

router.get("/:id", requireRole("viewer"), async (req, res) => {
  const id = Number(req.params.id);
  const [row] = await db.select().from(documentTemplatesTable).where(eq(documentTemplatesTable.id, id));
  if (!row) {
    res.status(404).json({ error: "not_found" });
    return;
  }
  res.json(row);
});

router.post("/", requireRole("admin"), auditAction("create_template"), async (req, res) => {
  const parsed = templateSchema.safeParse(req.body);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }
  const data = parsed.data;
  if (data.source === "pdf" && !data.storage_path) {
    res.status(400).json({ error: "pdf_template_requires_storage_path" });
    return;
  }
  if (data.source === "ai" && !data.ai_prompt) {
    res.status(400).json({ error: "ai_template_requires_prompt" });
    return;
  }
  const [row] = await db
    .insert(documentTemplatesTable)
    .values({ ...data, uploaded_by_user_id: req.user?.id ?? null } as any)
    .returning();
  res.status(201).json(row);
});

router.put("/:id", requireRole("admin"), auditAction("update_template"), async (req, res) => {
  const id = Number(req.params.id);
  const parsed = templateSchema.partial().safeParse(req.body);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }
  const updates = { ...parsed.data, updated_at: new Date() } as any;
  const [row] = await db
    .update(documentTemplatesTable)
    .set(updates)
    .where(eq(documentTemplatesTable.id, id))
    .returning();
  if (!row) {
    res.status(404).json({ error: "not_found" });
    return;
  }
  res.json(row);
});
```

```

router.delete("/:id", requireRole("admin"), auditAction("delete_template"), async (req, res) => {
  const id = Number(req.params.id);
  const [existing] = await db.select().from(documentTemplatesTable).where(eq(documentTemplatesTable.id, id));
  if (!existing) {
    res.status(404).json({ error: "not_found" });
    return;
  }
  if (existing.storage_path) {
    await deleteTemplate(existing.storage_path).catch(() => {
      /* file may already be gone */
    });
  }
  await db.delete(documentTemplatesTable).where(eq(documentTemplatesTable.id, id));
  res.status(204).end();
});

/**
 * Upload a PDF and return the storage path. Client then POSTs to /
 * with that storage_path to actually create the template row.
 * Body: { fileName: string, base64: string }
 */
router.post("/upload", requireRole("admin"), auditAction("upload_template_pdf"), async (req, res) => {
  const { fileName, base64 } = req.body || {};
  if (!fileName || typeof fileName !== "string") {
    res.status(400).json({ error: "fileName_required" });
    return;
  }
  if (!base64 || typeof base64 !== "string") {
    res.status(400).json({ error: "base64_required" });
    return;
  }
  try {
    const buf = Buffer.from(base64, "base64");
    if (buf.length === 0) {
      res.status(400).json({ error: "empty_file" });
      return;
    }
    if (buf.length > 25 * 1024 * 1024) {
      res.status(413).json({ error: "file_too_large", max_bytes: 25 * 1024 * 1024 });
      return;
    }
    const stored = await uploadTemplate(buf, fileName);
    res.status(201).json(stored);
  } catch (err: any) {
    res.status(500).json({ error: "upload_failed", message: String(err?.message || err) });
  }
});

router.get("/:id/preview", requireRole("viewer"), async (req, res) => {
  const id = Number(req.params.id);
  const [row] = await db.select().from(documentTemplatesTable).where(eq(documentTemplatesTable.id, id));
  if (!row || !row.storage_path) {
    res.status(404).json({ error: "not_found" });
    return;
  }
  try {
    const buf = await downloadTemplate(row.storage_path);
    res.setHeader("Content-Type", "application/pdf");
    res.setHeader("Content-Disposition", `inline; filename="${row.name}.pdf"`);
    res.send(buf);
  } catch {
    res.status(404).json({ error: "file_missing" });
  }
});

// ----- Assignments (buyer × tort × template) -----

router.get("/assignments/all", requireRole("viewer"), async (_req, res) => {
  const rows = await db.select().from(templateAssignmentsTable);
  res.json(rows);
});

router.get("/assignments/by-template/:templateId", requireRole("viewer"), async (req, res) => {
  const templateId = Number(req.params.templateId);
  const rows = await db
    .select()
    .from(templateAssignmentsTable)
    .where(eq(templateAssignmentsTable.template_id, templateId));
  res.json(rows);
});

router.post("/assignments", requireRole("admin"), auditAction("upsert_template_assignment"), async (req, res) => {
  const parsed = assignmentSchema.safeParse(req.body);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }
  const d = parsed.data;
  // Upsert: same (template, buyer, tort) tuple updates instead of duplicating.
  const buyerCond = d.buyer_id == null
    ? isNull(templateAssignmentsTable.buyer_id)
    : eq(templateAssignmentsTable.buyer_id, d.buyer_id);
  const tortCond = d.tort_type == null
    ? isNull(templateAssignmentsTable.tort_type)
    : eq(templateAssignmentsTable.tort_type, d.tort_type);
  const existing = await db
    .select()

```

```

    .from(templateAssignmentsTable)
    .where(
      and(
        eq(templateAssignmentsTable.template_id, d.template_id),
        buyerCond,
        tortCond,
      ),
    );
    if (existing.length > 0) {
      const [row] = await db
        .update(templateAssignmentsTable)
        .set({
          enabled: d.enabled,
          send_order: d.send_order,
          notes: d.notes ?? null,
          updated_by_user_id: req.user?.id ?? null,
          updated_at: new Date(),
        })
        .where(eq(templateAssignmentsTable.id, existing[0].id))
        .returning();
      res.json(row);
      return;
    }
    const [row] = await db
      .insert(templateAssignmentsTable)
      .values({ ...d, updated_by_user_id: req.user?.id ?? null } as any)
      .returning();
    res.status(201).json(row);
  });

  router.delete("/assignments/:id", requireRole("admin"), auditAction("delete_template_assignment"), async (req, res) => {
    const id = Number(req.params.id);
    await db.delete(templateAssignmentsTable).where(eq(templateAssignmentsTable.id, id));
    res.status(204).end();
  });
};

export default router;

```

[artifacts/api-server/src/routes/documents.ts](#)

```
import { Router } from "express";
import { db, documentsTable } from "@workspace/db";
import { eq, and, sql } from "drizzle-orm";
import {
  ListDocumentsQueryParams,
  CreateDocumentBody,
  UpdateDocumentBody,
  UpdateDocumentParams,
  DeleteDocumentParams,
} from "@workspace/api-zod";
import { redactPdf, highlightPdfRegions, getPdfPageCount } from "../lib/pdf-redaction";
import { requireRole, auditAction } from "../lib/rbac";

const router = Router();

router.get("/", requireRole("viewer"), async (req, res) => {
  const parsed = ListDocumentsQueryParams.safeParse(req.query);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }

  const { lead_id } = parsed.data;

  const docs = lead_id
    ? await db.select().from(documentsTable).where(eq(documentsTable.lead_id, lead_id)).orderBy(sql`${documentsTable.created_at} DESC`)
    : await db.select().from(documentsTable).orderBy(sql`${documentsTable.created_at} DESC`);

  res.json(docs);
});

router.post("/", requireRole("paralegal", "attorney", "admin"), auditAction("create_document"), async (req, res) => {
  const parsed = CreateDocumentBody.safeParse(req.body);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }

  const data = parsed.data;
  const [doc] = await db
    .insert(documentsTable)
    .values({
      lead_id: data.lead_id,
      document_type: data.document_type,
      file_name: data.file_name,
      file_url: data.file_url ?? null,
      signed: data.signed,
      signed_at: data.signed_at ? new Date(data.signed_at) : null,
      notes: data.notes ?? null,
    })
    .returning();

  res.status(201).json(doc);
});

router.patch("/:id", requireRole("paralegal", "attorney", "admin"), auditAction("update_document"), async (req, res) => {
  const paramsParsed = UpdateDocumentParams.safeParse({ id: Number(req.params.id) });
  if (!paramsParsed.success) {
    res.status(400).json({ error: paramsParsed.error.message });
    return;
  }

  const bodyParsed = UpdateDocumentBody.safeParse(req.body);
  if (!bodyParsed.success) {
    res.status(400).json({ error: bodyParsed.error.message });
    return;
  }

  const body = bodyParsed.data;
  const updateData: Record<string, unknown> = {};

  if (body.document_type !== undefined) updateData.document_type = body.document_type;
  if (body.file_name !== undefined) updateData.file_name = body.file_name;
  if (body.file_url !== undefined) updateData.file_url = body.file_url;
  if (body.signed !== undefined) updateData.signed = body.signed;
  if (body.signed_at !== undefined) updateData.signed_at = body.signed_at ? new Date(body.signed_at) : null;
  if (body.notes !== undefined) updateData.notes = body.notes;

  const [doc] = await db
    .update(documentsTable)
    .set(updateData)
    .where(eq(documentsTable.id, paramsParsed.data.id))
    .returning();

  if (!doc) {
    res.status(404).json({ error: "Document not found" });
    return;
  }

  res.json(doc);
});

router.delete("/:id", requireRole("attorney", "admin"), auditAction("delete_document"), async (req, res) => {
  const parsed = DeleteDocumentParams.safeParse({ id: Number(req.params.id) });
```

```

    if (!parsed.success) {
      res.status(400).json({ error: parsed.error.message });
      return;
    }

    await db.delete(documentsTable).where(eq(documentsTable.id, parsed.data.id));
    res.status(204).send();
  });

  router.post("/redact", requireRole("paralegal", "attorney", "admin"), auditAction("redact_document"), async (req, res) => {
    const { pdf_base64, rules } = req.body;
    if (!pdf_base64) { res.status(400).json({ error: "pdf_base64 is required" }); return; }

    try {
      const pdfBytes = Buffer.from(pdf_base64, "base64");
      const redacted = await redactPdf(pdfBytes, rules || []);
      res.json({ pdf_base64: Buffer.from(redacted).toString("base64"), pages: await getPdfPageCount(redacted) });
    } catch (err: any) {
      res.status(500).json({ error: "PDF redaction failed" });
    }
  });

  router.post("/highlight", requireRole("paralegal", "attorney", "admin"), async (req, res) => {
    const { pdf_base64, highlights } = req.body;
    if (!pdf_base64) { res.status(400).json({ error: "pdf_base64 is required" }); return; }

    try {
      const pdfBytes = Buffer.from(pdf_base64, "base64");
      const highlighted = await highlightPdfRegions(pdfBytes, highlights || []);
      res.json({ pdf_base64: Buffer.from(highlighted).toString("base64"), pages: await getPdfPageCount(highlighted) });
    } catch (err: any) {
      res.status(500).json({ error: "PDF highlighting failed" });
    }
  });
}

export default router;

```

[artifacts/api-server/src/routes/drafting.ts](#)

```
import { Router } from "express";
import { db, leadsTable } from "@workspace/db";
import { eq } from "drizzle-orm";
import { generateDraft, getAvailableTemplates } from "../lib/drafting-ai";
import { createBlankPdfWithText } from "../lib/pdf-redaction";
import { decryptLeadFields } from "../lib/encryption";
import { requireRole, auditAction } from "../lib/rbac";

const router = Router();

router.get("/templates", requireRole("paralegal"), async (_req, res) => {
  res.json(getAvailableTemplates());
});

router.post("/generate", requireRole("paralegal", "attorney", "admin"), auditAction("generate_draft"), async (req, res) => {
  const { template_type, lead_id, firm_name, attorney_name, custom_instructions } = req.body;
  if (!template_type) { res.status(400).json({ error: "template_type is required" }); return; }

  let leadData: Record<string, any> = {};
  if (lead_id) {
    const [lead] = await db.select().from(leadsTable).where(eq(leadsTable.id, parseInt(lead_id)));
    if (!lead) { res.status(404).json({ error: "Lead not found" }); return; }
    leadData = decryptLeadFields(lead);
  }

  const draft = await generateDraft({
    template_type,
    lead_data: leadData,
    firm_name,
    attorney_name,
    custom_instructions,
  });

  res.json(draft);
});

router.post("/generate-pdf", requireRole("paralegal", "attorney", "admin"), auditAction("generate_draft_pdf"), async (req, res)
=> {
  const { template_type, lead_id, firm_name, attorney_name, custom_instructions } = req.body;
  if (!template_type) { res.status(400).json({ error: "template_type is required" }); return; }

  let leadData: Record<string, any> = {};
  if (lead_id) {
    const [lead] = await db.select().from(leadsTable).where(eq(leadsTable.id, parseInt(lead_id)));
    if (lead) leadData = decryptLeadFields(lead);
  }

  const draft = await generateDraft({
    template_type,
    lead_data: leadData,
    firm_name,
    attorney_name,
    custom_instructions,
  });

  const pdfBytes = await createBlankPdfWithText(draft.content, draft.title);
  res.setHeader("Content-Type", "application/pdf");
  res.setHeader("Content-Disposition", `attachment; filename="${draft.title.replace(/\s/g, "_").pdf}"`);
  res.send(Buffer.from(pdfBytes));
});

export default router;
```

[artifacts/api-server/src/routes/forms-public.ts](#)

```
import { Router, type IRouter, type Request } from "express";
import { getFormConfig } from "../lib/form-config-service";
import { generateEmbedScript } from "../forms";
import { logger } from "../lib/logger";

const router: IRouter = Router();

// Strict host pattern: alphanumerics, dot, hyphen, optional :port.
// Blocks Host-header injection vectors (CR/LF, quotes, angle brackets, spaces).
const SAFE_HOST = /^[a-zA-Z0-9.-]+(?:\d{1,5})?$/;

/**
 * Resolve the public base URL for embed/preview links.
 *
 * Priority:
 * 1) PUBLIC_API_BASE_URL env var (set this in production / when running
 *    behind the future host site so links never depend on Host header).
 * 2) X-Forwarded-Proto + Host header, validated against SAFE_HOST.
 *
 * If the Host header fails validation we refuse to render rather than
 * emit attacker-controlled URLs into HTML or JS.
 */
function resolveBaseUrl(req: Request): string | null {
  const fromEnv = process.env.PUBLIC_API_BASE_URL?.trim();
  if (fromEnv) return fromEnv.replace(/\+/g, "");

  const host = req.get("host") ?? "";
  if (!SAFE_HOST.test(host)) return null;

  const proto = (req.get("x-forwarded-proto") ?? req.protocol ?? "https")
    .split(",")[0]
    .trim()
    .toLowerCase();
  if (proto !== "http" && proto !== "https") return null;

  return `${proto}://${host}`;
}

// CSP shared by preview HTML. TrustedForm requires:
// - script-src + connect-src for api.trustedform.com (+ subdomains)
// - worker-src incl. blob: for its instrumentation worker
// - frame-src for any iframes the TF script may inject
const PREVIEW_CSP = [
  "default-src 'self'",
  "script-src 'self' https://api.trustedform.com https://*.trustedform.com",
  "worker-src 'self' blob: https://*.trustedform.com",
  "frame-src 'self' https://*.trustedform.com",
  "style-src 'self' 'unsafe-inline'",
  "img-src 'self' data: https:",
  "connect-src 'self' https://api.trustedform.com https://*.trustedform.com",
  "frame-ancestors **",
  "base-uri 'self'",
  "form-action 'self'",
].join("; ");

function htmlEscape(s: string): string {
  return s.replace(/<|>|'|"/g, (c) =>
    ({ "<": "&lt;", ">": "&gt;", "&": "&amp;", "'": "&quot;", '"': "&#39;" })[c] as string
  );
}

router.get("/preview/:tortId", async (req, res) => {
  const tortId = String(req.params.tortId);
  try {
    const baseUrl = resolveBaseUrl(req);
    if (!baseUrl) {
      res.status(400).type("html").send("<h1>Bad request</h1>");
      return;
    }
    const config = await getFormConfig(tortId);
    if (!config) {
      res.status(404).type("html").send("<h1>Form not found</h1>");
      return;
    }
    const safeLabel = htmlEscape(config.label);
    const safeTortId = encodeURIComponent(tortId);
    const html = `<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8" />
<meta name="viewport" content="width=device-width,initial-scale=1" />
<title>Preview · ${safeLabel}</title>
<style>body{font-family:system-ui,-apple-system,Segoe UI,Roboto,sans-serif;margin:0;padding:24px;background:#f8fafc;color:#0f172a}.banner{background:#fef3c7;color:#92400e;border:1px solid #fcd34d;border-radius:8px;padding:10px 14px;margin-bottom:16px;font-size:13px}</style>
</head>
<body>
<div class="banner"><strong>Preview Mode</strong> – submissions are disabled in this preview window.</div>
<div id="mtos-form"></div>
<script src="${htmlEscape(baseUrl)}/api/forms/embed/${safeTortId}"></script>
<script src="${htmlEscape(baseUrl)}/api/forms/preview-blocker.js"></script>
</body>
</html>`;
    res.setHeader("Content-Type", "text/html; charset=utf-8");
    res.setHeader("Cache-Control", "no-store");
  } catch {
    // ...
  }
});
```



```

    res.setHeader("Content-Security-Policy", PREVIEW_CSP);
    res.setHeader("X-Content-Type-Options", "nosniff");
    res.setHeader("Referrer-Policy", "no-referrer");
    res.setHeader("Cross-Origin-Resource-Policy", "cross-origin");
    res.setHeader("Cross-Origin-Embedder-Policy", "unsafe-none");
    res.removeHeader("X-Frame-Options");
    res.send(html);
  } catch (err) {
    logger.error({ err }, "Failed to render preview");
    res.status(500).type("html").send("<h1>Preview failed</h1>");
  }
});

router.get("/preview-blocker.js", (_req, res) => {
  const js = `document.addEventListener("submit",function(e){e.preventDefault();alert("Preview mode: form not submitted.");});true`;
  res.setHeader("Content-Type", "application/javascript; charset=utf-8");
  res.setHeader("Cache-Control", "no-store");
  res.setHeader("Cross-Origin-Resource-Policy", "cross-origin");
  res.send(js);
});

router.get("/embed/:tortId", async (req, res) => {
  const tortId = String(req.params.tortId);
  try {
    const baseUrl = resolveBaseUrl(req);
    if (!baseUrl) {
      res.status(400).json({ error: "Invalid host" });
      return;
    }
    const config = await getFormConfig(tortId);
    if (!config || !config.active) {
      res.status(404).json({ error: "Tort campaign not found" });
      return;
    }
    const embedScript = generateEmbedScript(
      tortId,
      {
        label: config.label,
        extra_fields: config.extra_fields,
        exposure_fields: config.exposure_fields,
        intro_text: config.intro_text,
        custom_fields: config.custom_fields,
      },
      baseUrl,
    );
    res.setHeader("Content-Type", "application/javascript; charset=utf-8");
    res.setHeader("Cache-Control", "no-store");
    res.setHeader("Cross-Origin-Resource-Policy", "cross-origin");
    res.send(embedScript);
  } catch (err) {
    logger.error({ err }, "Failed to generate embed script");
    res.status(500).json({ error: "Failed to generate embed script" });
  }
});

export default router;

```

[artifacts/api-server/src/routes/forms.ts](#)

```
import { Router } from "express";
import { db, leadsTable } from "@workspace/db";
import { eq } from "drizzle-orm";
import { encryptLeadFields } from "../lib/encryption";
import { validateEmail } from "../lib/email-validator";
import { validateAddress } from "../lib/address-validator";
import { runBackgroundCheck } from "../lib/background-check";
import { runFullConflictCheck } from "../lib/conflict-engine";
import { TORT_REGISTRY, validateTortClaim, getTortCategories } from "../lib/tort-engine";
import { lookupNpiAndMatch } from "../lib/taxonomy-engine";
import { runFraudDetection } from "../lib/fraud-engine";
import { finalArbiter } from "../lib/final-arbiter";
import { auditLog } from "../lib/audit";
import { logger } from "../lib/logger";
import { requireRole, auditAction, authMiddleware } from "../lib/rbac";
import {
  getAllFormConfigs,
  getFormConfig,
  getFormConfigByIdOrLabel,
  updateFormConfig,
  addCustomField,
  removeCustomField,
  type CustomField,
} from "../lib/form-config-service";
import { customFieldSchema } from "@workspace/db";

const router = Router();

router.get("/config", async (_req, res) => {
  try {
    const all = await getAllFormConfigs();
    const configs = all.map(c => ({
      id: c.id,
      label: c.label,
      category: c.category,
      fields: [...c.extra_fields, ...c.exposure_fields],
      rules: c.rules,
      valid_diagnoses: c.valid_diagnoses,
      custom_fields: c.custom_fields,
      required_exposure: c.required_exposure,
      intro_text: c.intro_text,
      active: c.active,
      avg_settlement_low: c.avg_settlement_low,
      avg_settlement_high: c.avg_settlement_high,
      mdl_status: c.mdl_status,
      sol_months: c.sol_months,
      updated_at: c.updated_at,
    }));
    res.json({ tort_campaigns: configs });
  } catch (err) {
    logger.error({ err }, "Failed to load form configs");
    res.status(500).json({ error: "Failed to load form configurations" });
  }
});

router.get("/config/:tortId", async (req, res) => {
  try {
    const config = await getFormConfig(String(req.params.tortId));
    if (!config) {
      res.status(404).json({ error: "Tort campaign not found" });
      return;
    }
    res.json({
      id: config.id,
      label: config.label,
      category: config.category,
      fields: [...config.extra_fields, ...config.exposure_fields],
      rules: config.rules,
      valid_diagnoses: config.valid_diagnoses,
      custom_fields: config.custom_fields,
      required_exposure: config.required_exposure,
      intro_text: config.intro_text,
      active: config.active,
      avg_settlement_low: config.avg_settlement_low,
      avg_settlement_high: config.avg_settlement_high,
      mdl_status: config.mdl_status,
      sol_months: config.sol_months,
      updated_at: config.updated_at,
    });
  } catch (err) {
    logger.error({ err }, "Failed to load form config");
    res.status(500).json({ error: "Failed to load form configuration" });
  }
});

router.put(
  "/config/:tortId",
  authMiddleware,
  requireRole("admin"),
  auditAction("form_config_update"),
  async (req, res) => {
    try {
      const { label, valid_diagnoses, exposure_fields, extra_fields, custom_fields, rules, rejection_conditions, required_exposure, intro_text, active, avg_settlement_low, avg_settlement_high, mdl_status, sol_months } = req.body ?? {};
      if (custom_fields !== undefined) {
```

```

    if (!Array.isArray(custom_fields)) {
      res.status(400).json({ error: "custom_fields must be an array" });
      return;
    }
    for (const f of custom_fields) {
      const parsed = customFieldSchema.safeParse(f);
      if (!parsed.success) {
        res.status(400).json({ error: "Invalid custom field", details: parsed.error.issues });
        return;
      }
    }
    const keys = new Set<string>();
    for (const f of custom_fields as CustomField[]) {
      if (keys.has(f.key)) {
        res.status(400).json({ error: `Duplicate field key: ${f.key}` });
        return;
      }
      keys.add(f.key);
    }
    const userId = req.user?.id ?? 0;
    const updated = await updateFormConfig(String(req.params.tortId), {
      label, valid_diagnoses, exposure_fields, extra_fields, custom_fields, rules, rejection_conditions, required_exposure,
      intro_text, active,
      avg_settlement_low, avg_settlement_high, mdl_status, sol_months,
    }, userId);
    if (!updated) {
      res.status(404).json({ error: "Tort campaign not found" });
      return;
    }
    res.json(updated);
  } catch (err) {
    logger.error({ err }, "Failed to update form config");
    res.status(500).json({ error: "Failed to update form configuration" });
  }
}
);

router.post(
  "/config/:tortId/fields",
  authMiddleware,
  requireRole("admin"),
  auditAction("form_field_add"),
  async (req, res) => {
    try {
      const parsed = customFieldSchema.safeParse(req.body);
      if (!parsed.success) {
        res.status(400).json({ error: "Invalid custom field", details: parsed.error.issues });
        return;
      }
      const userId = req.user?.id ?? 0;
      const updated = await addCustomField(String(req.params.tortId), parsed.data as CustomField, userId);
      if (!updated) {
        res.status(404).json({ error: "Tort campaign not found" });
        return;
      }
      res.json(updated);
    } catch (err: unknown) {
      const msg = err instanceof Error ? err.message : "Failed to add field";
      if (msg.includes("already exists")) {
        res.status(409).json({ error: msg });
        return;
      }
      logger.error({ err }, "Failed to add custom field");
      res.status(500).json({ error: msg });
    }
  }
);

router.delete(
  "/config/:tortId/fields/:key",
  authMiddleware,
  requireRole("admin"),
  auditAction("form_field_remove"),
  async (req, res) => {
    try {
      const userId = req.user?.id ?? 0;
      const updated = await removeCustomField(String(req.params.tortId), String(req.params.key), userId);
      if (!updated) {
        res.status(404).json({ error: "Tort campaign not found" });
        return;
      }
      res.json(updated);
    } catch (err) {
      logger.error({ err }, "Failed to remove custom field");
      res.status(500).json({ error: "Failed to remove custom field" });
    }
  }
);

router.get("/categories", (_req, res) => {
  res.json(getTortCategories());
});

router.post("/validate/email", (req, res) => {
  const { email } = req.body;
  const result = validateEmail(email);

```

```

    res.json(result);
  });

  router.post("/validate/address", (req, res) => {
    const result = validateAddress(req.body);
    res.json(result);
  });

  // NOTE: GET /forms/preview/:tortId, /forms/embed/:tortId, and
  // /forms/preview-blocker.js are served by forms-public.ts mounted before
  // authMiddleware so they are publicly embeddable.

  interface PipelineStep {
    name: string;
    status: "passed" | "failed" | "skipped" | "error";
    errors: string[];
    data?: Record<string, unknown>;
  }

  function extractCustomFieldValues(
    data: Record<string, unknown>,
    allowlist?: string[],
  ): Record<string, unknown> {
    const out: Record<string, unknown> = {};
    for (const k of Object.keys(data)) {
      if (k.startsWith("cf_")) {
        out[k.slice(3)] = data[k];
      }
    }
    if (data.custom_fields && typeof data.custom_fields === "object" && !Array.isArray(data.custom_fields)) {
      Object.assign(out, data.custom_fields as Record<string, unknown>);
    }
    if (allowlist && allowlist.length >= 0) {
      const allow = new Set(allowlist);
      for (const k of Object.keys(out)) {
        if (!allow.has(k)) delete out[k];
      }
    }
    return out;
  }

  // SECURITY NOTE - /submit is intentionally authenticated.
  // Public unauthenticated submission from third-party host sites is DEFERRED
  // pending: (1) per-buyer/per-tort API keys, (2) per-key rate limiting,
  // (3) origin allowlist, (4) TrustedForm cert URL verification against
  // api.trustedform.com before accepting a lead, (5) CAPTCHA or equivalent
  // bot deterrent. Until then, /submit requires a logged-in CRM user role.
  router.post("/submit", requireRole("paralegal", "attorney", "admin"), auditAction("form_submit"), async (req, res) => {
    const data = req.body;
    const pipeline: PipelineStep[] = [];

    // STEP 1: Schema validation
    const step1: PipelineStep = { name: "SCHEMA_VALIDATION", status: "passed", errors: [] };
    try {
      const requiredFields = [
        "first_name", "last_name", "date_of_birth", "street_address", "city",
        "state", "zip", "phone_primary", "email", "last_4_ssn",
        "diagnosis", "diagnosis_date",
        "physician_first_name", "physician_last_name", "physician_full_address", "physician_contact_info",
        "hospital_name", "hospital_fax", "hospital_contact_info",
        "tort_type",
      ];

      for (const field of requiredFields) {
        if (!data[field] || (typeof data[field] === "string" && !data[field].trim())) {
          step1.errors.push(`MISSING_${field.toUpperCase()}`);
        }
      }

      if (step1.errors.length > 0) {
        step1.status = "failed";
        pipeline.push(step1);
        res.status(422).json({
          status: "REJECTED",
          errors: step1.errors,
          action: "FIX_AND_RESUBMIT",
          pipeline,
          failed_step: "SCHEMA_VALIDATION",
        });
        return;
      }
    } catch (err) {
      step1.status = "error";
      step1.errors.push("SCHEMA_VALIDATION_ERROR");
      pipeline.push(step1);
      logger.error({ err }, "Schema validation error");
      res.status(500).json({ status: "ERROR", errors: ["SCHEMA_VALIDATION_ERROR"], action: "RETRY", pipeline, failed_step: "SCHEMA_VALIDATION" });
      return;
    }
    pipeline.push(step1);

    // STEP 2: Email validation
    const step2: PipelineStep = { name: "EMAIL_VALIDATION", status: "passed", errors: [] };
    try {
      const emailResult = validateEmail(data.email);
      if (!emailResult.valid) {

```

```

        step2.errors = emailResult.errors.map((e: string) => `INVALID_EMAIL:${e}`);
        step2.status = "failed";
        step2.data = { suggestion: emailResult.suggestion };
    }
    catch (err) {
        step2.status = "error";
        step2.errors.push("EMAIL_VALIDATION_ERROR");
        logger.error({ err }, "Email validation error");
    }
    pipeline.push(step2);

// STEP 3: Address validation
const step3: PipelineStep = { name: "ADDRESS_VALIDATION", status: "passed", errors: [] };
try {
    const addressResult = validateAddress({
        street_address: data.street_address,
        city: data.city,
        state: data.state,
        zip: data.zip,
    });
    if (!addressResult.valid) {
        step3.errors = addressResult.errors.map((e: string) => `INVALID_ADDRESS:${e}`);
        step3.status = "failed";
    }
    catch (err) {
        step3.status = "error";
        step3.errors.push("ADDRESS_VALIDATION_ERROR");
        logger.error({ err }, "Address validation error");
    }
    pipeline.push(step3);

// STEP 4: TCPA consent
const step4: PipelineStep = { name: "TCPA_COMPLIANCE", status: "passed", errors: [] };
try {
    if (!data.tcpa_consent || data.tcpa_consent !== true) {
        step4.errors.push("MISSING_TCPA_CONSENT");
        step4.status = "failed";
    }
    catch (err) {
        step4.status = "error";
        step4.errors.push("TCPA_CHECK_ERROR");
    }
    pipeline.push(step4);

// STEP 5: TrustedForm
const step5: PipelineStep = { name: "TRUSTEDFORM_VALIDATION", status: "passed", errors: [] };
try {
    if (!data.trustedform_cert_url || typeof data.trustedform_cert_url !== "string" || !
data.trustedform_cert_url.startsWith("https://cert.trustedform.com/")) {
        step5.errors.push("MISSING_OR_INVALID_TRUSTEDFORM");
        step5.status = "failed";
    }
    catch (err) {
        step5.status = "error";
        step5.errors.push("TRUSTEDFORM_CHECK_ERROR");
    }
    pipeline.push(step5);

// Collect all pre-tort errors
const preErrors = pipeline.filter(s => s.status === "failed" || s.status === "error").flatMap(s => s.errors);
if (preErrors.length > 0) {
    res.status(422).json({
        status: "REJECTED",
        errors: preErrors,
        action: "FIX_AND_RESUBMIT",
        pipeline,
        failed_step: pipeline.find(s => s.status !== "passed")?.name,
    });
    return;
}

// STEP 6: Tort classification engine
const step6: PipelineStep = { name: "TORT_CLASSIFICATION", status: "passed", errors: [], data: {} };
let tortValidation;
try {
    tortValidation = validateTortClaim({
        tort_type: data.tort_type,
        diagnosis: data.diagnosis,
        exposure_start: data.exposure_start,
        exposure_end: data.exposure_end,
        location_name: data.location_name,
        was_at_location: data.was_at_location,
    });
    step6.data = { tort_id: tortValidation.tort_id, category: tortValidation.category, diagnosis_match:
tortValidation.diagnosis_match };

    if (!tortValidation.valid) {
        step6.errors = tortValidation.errors;
        step6.status = "failed";
    }
    catch (err) {
        step6.status = "error";
        step6.errors.push("TORT_CLASSIFICATION_ERROR");
        logger.error({ err }, "Tort classification error");
        tortValidation = { valid: false, tort_id: null, errors: ["TORT_CLASSIFICATION_ERROR"], diagnosis_match: false, category:
null };
    }
}

```

```

pipeline.push(step6);

// STEP 7: NPI Lookup + Taxonomy matching
const step7: PipelineStep = { name: "NPI_TAXONOMY_MATCH", status: "passed", errors: [], data: {} };
let npiResult = { npi_found: false, npi_number: null as string | null, specialty: null as string | null, taxonomy_match: null
as any, error: null as string | null };
try {
  if (data.physician_first_name && data.physician_last_name) {
    npiResult = await lookupNpiAndMatch(
      data.physician_first_name,
      data.physician_last_name,
      data.diagnosis
    );
    step7.data = {
      npi_found: npiResult.npi_found,
      npi_number: npiResult.npi_number,
      specialty: npiResult.specialty,
      taxonomy_matched: npiResult.taxonomy_match?.matched ?? false,
      fraud_indicators: npiResult.taxonomy_match?.fraud_indicators ?? [],
    };

    if (npiResult.error === "NPI_NOT_FOUND") {
      step7.errors.push("PHYSICIAN_NPI_NOT_FOUND");
      step7.status = "passed";
    } else if (npiResult.error) {
      step7.errors.push(npiResult.error);
      step7.status = "error";
    }
  } else {
    step7.status = "skipped";
  }
} catch (err) {
  step7.status = "error";
  step7.errors.push("NPI_LOOKUP_ERROR");
  logger.error({ err }, "NPI lookup error");
}
pipeline.push(step7);

// STEP 8: Fraud flagging (flags only – does NOT decide outcome)
const step8: PipelineStep = { name: "FRAUD_FLAGGING", status: "passed", errors: [], data: {} };
let fraudResult;
try {
  fraudResult = runFraudDetection({
    lead_data: data,
    tort_validation: tortValidation!,
    taxonomy_match: npiResult.taxonomy_match,
    npi_found: npiResult.npi_found,
  });
  step8.data = {
    fraud_score: fraudResult.fraud_score,
    hard_block: fraudResult.hard_block,
    has_flags: fraudResult.has_flags,
    indicators: fraudResult.indicators.length,
  };
  if (fraudResult.has_flags) {
    step8.errors = fraudResult.indicators.map(i => `FRAUD:${i.type}`);
  }
} catch (err) {
  step8.status = "error";
  step8.errors.push("FRAUD_DETECTION_ERROR");
  logger.error({ err }, "Fraud detection error");
  fraudResult = { hard_block: false, hard_block_reason: null, has_flags: false, fraud_score: 0, indicators: [], summary:
"Fraud check error – routed to review" };
}
pipeline.push(step8);

// STEP 9: Conflict detection
const step9: PipelineStep = { name: "CONFLICT_DETECTION", status: "passed", errors: [], data: {} };
const fullName = `${data.first_name} ${data.last_name}`.trim();
let conflictCheck;
try {
  conflictCheck = await runFullConflictCheck({
    entity_type: "lead",
    entity_id: "pending",
    source_module: "form_engine",
    lead_data: { ...data, name: fullName },
  });
  step9.data = { has_conflict: conflictCheck.has_conflict, output_state: conflictCheck.output_state };
} catch (err) {
  step9.status = "error";
  step9.errors.push("CONFLICT_CHECK_ERROR");
  logger.error({ err }, "Conflict check error");
  conflictCheck = { has_conflict: false, output_state: "ACCEPT" as const, conflict_type: null, severity: "low" as const,
details: [], failsafe_mode: "REVIEW_FAIL" as const };
}
pipeline.push(step9);

// STEP 10: FINAL ARBITER – single source of truth for decision
const complianceErrors = pipeline
  .filter(s => ["TCPA_COMPLIANCE", "TRUSTEDFORM_VALIDATION"].includes(s.name) && s.status !== "passed")
  .flatMap(s => s.errors);
const validationErrors = pipeline
  .filter(s => ["SCHEMA_VALIDATION", "EMAIL_VALIDATION", "ADDRESS_VALIDATION"].includes(s.name) && s.status !== "passed")
  .flatMap(s => s.errors);

const arbiterResult = finalArbiter({
  compliance_passed: complianceErrors.length === 0,

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    compliance_errors: complianceErrors,
    validation_passed: validationErrors.length === 0,
    validation_errors: validationErrors,
    tort_validation: tortValidation!,
    taxonomy_match: npiParamResult.taxonomy_match,
    fraud_result: fraudResult!,
    npiParam_found: npiParamResult.npi_found,
    diagnosis_confirmed: !!data.diagnosis_confirmed,
    exposure_confirmed: !!data.was_at_location,
    exposure_start: data.exposure_start,
    conflict_reject: conflictCheck!.has_conflict && conflictCheck!.output_state === "REJECT",
    conflict_review: conflictCheck!.has_conflict && conflictCheck!.output_state === "REVIEW_REQUIRED",
  });

  let status: string;
  if (arbitratorResult.decision === "REJECTED") {
    status = "rejected";
  } else if (arbitratorResult.decision === "TO_BE_REVIEWED") {
    status = "review_required";
  } else {
    status = "new";
  }

  if (arbitratorResult.decision === "REJECTED") {
    await auditLog("lead", "rejected", arbitratorResult.deciding_engine, {
      reason: arbitratorResult.reason,
      fraud_score: fraudResult!.fraud_score,
      fraud_indicators: fraudResult!.indicators,
      conflict: conflictCheck!.has_conflict ? conflictCheck!.conflict_type : null,
    });
  }

  pipeline.push({ name: "FINAL_ARBITER", status: "failed", errors: [arbitratorResult.reason], data: { decision:
    arbitratorResult.decision, deciding_engine: arbitratorResult.deciding_engine } });

  res.status(422).json({
    status: "REJECTED",
    errors: [arbitratorResult.reason],
    action: "REJECTED",
    pipeline,
    failed_step: arbitratorResult.deciding_engine,
    fraud_score: fraudResult!.fraud_score,
    fraud_summary: fraudResult!.summary,
  });
  return;
}

pipeline.push({ name: "FINAL_ARBITER", status: "passed", errors: [], data: { decision: arbitratorResult.decision, reason:
    arbitratorResult.reason, deciding_engine: arbitratorResult.deciding_engine } });

const step10: PipelineStep = { name: "CRM_STORAGE", status: "passed", errors: [], data: {} };
try {
  const [lead] = await db
    .insert(leadsTable)
    .values(encryptLeadFields({
      name: fullName,
      email: data.email,
      phone: data.phone_primary,
      tort_type: data.tort_type,
      first_name: data.first_name,
      last_name: data.last_name,
      date_of_birth: data.date_of_birth,
      street_address: data.street_address,
      city: data.city,
      state: data.state,
      zip: data.zip,
      phone_primary: data.phone_primary,
      last_4_ssn: data.last_4_ssn,
      diagnosis: data.diagnosis,
      diagnosis_date: data.diagnosis_date,
      diagnosis_confirmed: data.diagnosis_confirmed ?? false,
      was_at_location: data.was_at_location ?? false,
      location_name: data.location_name ?? null,
      physician_first_name: data.physician_first_name,
      physician_last_name: data.physician_last_name,
      physician_full_address: data.physician_full_address,
      physician_contact_info: data.physician_contact_info,
      hospital_name: data.hospital_name,
      hospital_fax: data.hospital_fax,
      hospital_contact_info: data.hospital_contact_info,
      medications: data.medications ?? null,
      tcpa_consent: true,
      trustedform_cert_url: data.trustedform_cert_url,
      trustedform_ping_url: data.trustedform_ping_url ?? null,
      trustedform_cert_token: data.trustedform_cert_token ?? null,
      trustedform_ip: data.trustedform_ip ?? null,
      trustedform_user_agent: data.trustedform_user_agent ?? null,
      trustedform_timestamp: data.trustedform_timestamp ? new Date(data.trustedform_timestamp) : new Date(),
      email_validation_status: "valid",
      address_validation_status: "valid",
      npiParam_verified: npiParamResult.npi_found,
      npiParam_number: npiParamResult.npi_number,
      physician_taxonomy: npiParamResult.specialty,
      fraud_score: fraudResult!.fraud_score,
      fraud_status: arbitratorResult.decision,
      fraud_indicators: fraudResult!.indicators.length > 0 ? JSON.stringify(fraudResult!.indicators) : null,
      exposure_start: data.exposure_start ?? null,
      exposure_end: data.exposure_end ?? null,
    }));
}

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        diagnosis_type: data.diagnosis_type ?? null,
        notes: data.notes ?? null,
        ad_spend: data.ad_spend ? String(data.ad_spend) : null,
        source: data.source ?? "form_embed",
        status,
        custom_fields: await (async () => {
            const cfg = await getFormConfigByIdOrLabel(String(data.tort_type ?? ""));
            const allow = (cfg?.custom_fields ?? []).map((f) => f.key);
            return extractCustomFieldValues(data, allow);
        })(),
    }) as any)
    .returning();

step10.data = { lead_id: lead.id, status };

// Decision Engine – score every lead created via form submission.
{
    const { computeAndPersistLeadScore } = await import("../lib/decision-engine-service");
    computeAndPersistLeadScore(lead.id).catch(() => {});
}

if (status === "review_required") {
    try {
        const { reviewQueueTable } = await import("@workspace/db");
        const { and: andOp } = await import("drizzle-orm");
        await db
            .update(reviewQueueTable)
            .set({ entity_id: String(lead.id) })
            .where(
                andOp(
                    eq(reviewQueueTable.entity_id, "pending"),
                    eq(reviewQueueTable.entity_type, "lead"),
                    eq(reviewQueueTable.source_module, "form_engine")
                )
            );
    } catch (_) {}
}

await auditLog("lead", String(lead.id), "form_submission", {
    source: "form_engine",
    status,
    tort_id: tortValidation!.tort_id,
    fraud_score: fraudResult!.fraud_score,
    npi_verified: npiResult.npi_found,
    trustedform_cert_url: data.trustedform_cert_url,
});

// Post-insert background check (non-blocking)
let bgCheck = null;
try {
    bgCheck = await runBackgroundCheck({
        first_name: data.first_name,
        last_name: data.last_name,
        state: data.state,
        date_of_birth: data.date_of_birth,
    });
}

await db
    .update(leadsTable)
    .set({
        background_check_status: bgCheck.status,
        background_check_data: JSON.stringify(bgCheck),
        updated_at: new Date(),
    })
    .where(eq(leadsTable.id, lead.id));
} catch (bgErr) {
    logger.warn({ err: bgErr, leadId: lead.id }, "Background check failed post-insert");
}

pipeline.push(step10);

res.status(201).json({
    status: arbiterResult.decision,
    lead_id: lead.id,
    lead_status: status,
    tort_id: tortValidation!.tort_id,
    fraud_score: fraudResult!.fraud_score,
    fraud_flags: fraudResult!.has_flags,
    npi_verified: npiResult.npi_found,
    background_check: bgCheck ? { status: bgCheck.status, summary: bgCheck.summary } : null,
    arbiter: { decision: arbiterResult.decision, reason: arbiterResult.reason, deciding_engine:
arbiterResult.deciding_engine },
    pipeline,
});
} catch (err) {
    step10.status = "error";
    step10.errors.push("DATABASE_INSERT_ERROR");
    pipeline.push(step10);
    logger.error({ err }, "Form submission insert failed");
    res.status(500).json({
        status: "ERROR",
        errors: ["INTERNAL_ERROR"],
        action: "RETRY",
        pipeline,
        failed_step: "CRM_STORAGE",
    });
}
}

```



```

});

router.post("/background-check", requireRole("paralegal"), async (req, res) => {
  const { first_name, last_name, state, date_of_birth } = req.body;
  if (!first_name || !last_name) {
    res.status(400).json({ error: "first_name and last_name are required" });
    return;
  }

  try {
    const result = await runBackgroundCheck({ first_name, last_name, state, date_of_birth });
    res.json(result);
  } catch (err) {
    logger.error({ err }, "Background check failed");
    res.status(500).json({ error: "Background check failed", status: "error" });
  }
});

router.post("/background-check/lead/:id", requireRole("paralegal"), async (req, res) => {
  const leadId = Number(req.params.id);
  if (isNaN(leadId)) {
    res.status(400).json({ error: "Invalid lead ID" });
    return;
  }

  try {
    const [lead] = await db.select().from(leadsTable).where(eq(leadsTable.id, leadId));
    if (!lead) {
      res.status(404).json({ error: "Lead not found" });
      return;
    }

    if (!lead.first_name || !lead.last_name) {
      res.status(422).json({ error: "Lead missing first_name or last_name" });
      return;
    }

    const result = await runBackgroundCheck({
      first_name: lead.first_name,
      last_name: lead.last_name,
      state: lead.state ?? undefined,
      date_of_birth: lead.date_of_birth ?? undefined,
    });

    await db
      .update(leadsTable)
      .set({
        background_check_status: result.status,
        background_check_data: JSON.stringify(result),
        updated_at: new Date(),
      })
      .where(eq(leadsTable.id, leadId));

    await auditLog("lead", String(leadId), "background_check", {
      status: result.status,
      records_found: result.records.length,
    });

    res.json(result);
  } catch (err) {
    logger.error({ err }, "Lead background check failed");
    res.status(500).json({ error: "Background check failed", status: "error" });
  }
});

router.post("/npi-verify", requireRole("paralegal"), async (req, res) => {
  const { physician_first_name, physician_last_name, diagnosis } = req.body;
  if (!physician_first_name || !physician_last_name || !diagnosis) {
    res.status(400).json({ error: "physician_first_name, physician_last_name, and diagnosis are required" });
    return;
  }

  try {
    const result = await lookupNpiAndMatch(physician_first_name, physician_last_name, diagnosis);
    res.json(result);
  } catch (err) {
    logger.error({ err }, "NPI verify failed");
    res.status(500).json({ error: "NPI verification failed" });
  }
});

router.post("/fraud-check", requireRole("paralegal"), async (req, res) => {
  const { lead_data, tort_type, diagnosis, physician_first_name, physician_last_name } = req.body;
  if (!diagnosis || !tort_type) {
    res.status(400).json({ error: "tort_type and diagnosis are required" });
    return;
  }

  try {
    const tortValidation = validateTortClaim({
      tort_type,
      diagnosis,
      exposure_start: lead_data?.exposure_start,
      location_name: lead_data?.location_name,
      was_at_location: lead_data?.was_at_location,
    });
  }
});

```

```

let npiParam = { npiParam_found: false, taxonomy_match: null as any };
if (physician_first_name && physician_last_name) {
  const npiParam = await lookupNpiAndMatch(physician_first_name, physician_last_name, diagnosis);
  npiParam = { npiParam_found: npiParam.npi_found, taxonomy_match: npiParam.taxonomy_match };
}

const fraudResult = runFraudDetection({
  lead_data: lead_data || { diagnosis, tort_type },
  tort_validation: tortValidation,
  taxonomy_match: npiParam.taxonomy_match,
  npiParam_found: npiParam.npi_found,
});

res.json({ tort_validation: tortValidation, fraud_result: fraudResult });
} catch (err) {
  logger.error({ err }, "Fraud check failed");
  res.status(500).json({ error: "Fraud check failed" });
}
});

router.post("/escalate/fbi", requireRole("attorney", "admin"), auditAction("escalate_fbi"), async (req, res) => {
  const { lead_id, reason, fraud_indicators } = req.body;
  if (!lead_id || !reason) {
    res.status(400).json({ error: "lead_id and reason are required" });
    return;
  }

  try {
    await auditLog("lead", String(lead_id), "fbi_escalation", {
      reason,
      fraud_indicators: fraud_indicators || [],
      fbi_tip_url: "https://tips.fbi.gov/",
      escalated_at: new Date().toISOString(),
    });

    if (lead_id !== "manual") {
      const leadId = Number(lead_id);
      if (isNaN(leadId)) {
        res.status(400).json({ error: "Invalid lead_id - must be a number or 'manual'" });
        return;
      }
      const [existing] = await db.select({ id: leadsTable.id }).from(leadsTable).where(eq(leadsTable.id, leadId));
      if (!existing) {
        res.status(404).json({ error: "Lead not found" });
        return;
      }
      await db
        .update(leadsTable)
        .set({
          status: "rejected",
          rejection_reason: `FBI ESCALATION: ${reason}`,
          fraud_status: "escalated",
          updated_at: new Date(),
        })
        .where(eq(leadsTable.id, leadId));
    }

    res.json({
      status: "ESCALATED",
      fbi_tip_url: "https://tips.fbi.gov/",
      message: "Case logged and escalated. Submit tip at tips.fbi.gov.",
    });
  } catch (err) {
    logger.error({ err }, "FBI escalation logging failed");
    res.status(500).json({ error: "Escalation logging failed" });
  }
});

interface EmbedConfig {
  label: string;
  extra_fields: string[];
  exposure_fields: string[];
  intro_text: string | null;
  custom_fields: CustomField[];
}

export function escapeJs(s: string): string {
  return String(s).replace(/\\/g, "\\").replace(/'/g, "\\'").replace(/"/g, '\"').replace(/\n/g, "\\n").replace(/\r/g, "\\r").replace(/</g, "<").replace(/>/g, ">");
}

export function generateEmbedScript(tortId: string, config: EmbedConfig, baseUrl: string): string {
  const allStates = ["AL", "AK", "AZ", "AR", "CA", "CO", "CT", "DE", "DC", "FL", "GA", "HI", "ID", "IL", "IN", "IA", "KS", "KY", "LA", "ME", "MD", "MA", "MI", "MN", "MS", "MO", "MT", "NE", "NV", "NH", "NJ", "NM", "NY", "NC", "ND", "OH", "OK", "OR", "PA", "RI", "SC", "SD", "TN", "TX", "UT", "VT", "VA", "WA", "WV", "WI", "WY"];
  const extraFields = [...config.extra_fields, ...config.exposure_fields];
  const customFieldsJson = JSON.stringify(config.custom_fields ?? []);
  const introText = config.intro_text ? escapeJs(config.intro_text) : "Complete all required fields to submit your claim for review.";

  return `(function(){
var API="${baseUrl}/api/forms";
var TORT_ID="${escapeJs(tortId)}";
var TORT_LABEL="${escapeJs(config.label)}";
var TORT_INTRO="${introText}";
var EXTRA_FIELDS=${JSON.stringify(extraFields)};
var CUSTOM_FIELDS=${customFieldsJson};

```

```

function el(tag,attrs,children){
  var e=document.createElement(tag);
  if(attrs)Object.keys(attrs).forEach(function(k){
    if(k=="style")Object.assign(e.style,attrs[k]);
    else if(k.startsWith("on"))e.addEventListener(k.slice(2),attrs[k]);
    else e.setAttribute(k,attrs[k]);
  });
  if(children){
    if(typeof children=="string")e.textContent=children;
    else if(Array.isArray(children))children.forEach(function(c){if(c)e.appendChild(c)});
  }
  return e;
}

function input(name, label, type, opts){
  opts=opts||{};
  var wrap=el("div", {style:{marginBottom:"12px"}});
  var lbl=el("label", {style:{display:"block",fontWeight:"600",marginBottom:"4px",fontSize:"14px"}}, label+(opts.optional?"":"*"));
  var inp;
  if(type=="select"){
    inp=el("select", {name:name, style:{width:"100%",padding:"8px 12px",border:"1px solid #d1d5db",borderRadius:"6px",fontSize:"14px"}});
    inp.appendChild(el("option", {value:""}, "Select..."));
    (opts.options||[]).forEach(function(o){
      var opt=el("option", {value:typeof o=="object"?o.value:o, type:typeof o=="object"?o.label:o});
      inp.appendChild(opt);
    });
  } else if(type=="textarea"){
    inp=el("textarea", {name:name, rows:"3", placeholder:opts.placeholder||"", style:{width:"100%",padding:"8px 12px",border:"1px solid #d1d5db",borderRadius:"6px",fontSize:"14px",resize:"vertical"}});
  } else if(type=="checkbox"){
    var cw=el("div", {style:{display:"flex",alignItems:"flex-start",gap:"8px"}});
    inp=el("input", {type:"checkbox", name:name, style:{marginTop:"3px"}});
    cw.appendChild(inp);
    cw.appendChild(el("span", {style:{fontSize:"13px",lineHeight:"1.4"}}, opts.checkLabel||label));
    wrap.appendChild(cw);
    return wrap;
  } else {
    inp=el("input", {type:type||"text", name:name, placeholder:opts.placeholder||"", style:{width:"100%",padding:"8px 12px",border:"1px solid #d1d5db",borderRadius:"6px",fontSize:"14px"}});
    if(opts.maxLength)inp.setAttribute("maxLength",opts.maxLength);
    if(opts.pattern)inp.setAttribute("pattern",opts.pattern);
  }
  if(!opts.optional)inp.setAttribute("required","");
  wrap.appendChild(lbl);
  wrap.appendChild(inp);
  if(name=="email"){
    var errDiv=el("div", {id:"mtos-email-error", style:{color:"#dc2626",fontSize:"12px",marginTop:"4px",display:"none"}});
    wrap.appendChild(errDiv);
    inp.addEventListener("blur", function(){
      fetch(API+"/validate/email", {method:"POST", headers:{"Content-Type":"application/json"}, body:JSON.stringify({email:inp.value})})
      .then(function(r){return r.json()})
      .then(function(r){
        if(!r.valid){
          errDiv.style.display="block";
          errDiv.textContent="Invalid email detected."+(r.suggestion?" Did you mean "+r.suggestion+"?":"")+" Please correct before continuing.";
          inp.style.borderColor="#dc2626";
        }else{
          errDiv.style.display="none";
          inp.style.borderColor="#d1d5db";
        }
      });
    });
  }
  return wrap;
}

function section(title,children,opts){
  opts=opts||{};
  var s=el("div", {style:{border:"1px solid #e5e7eb",borderRadius:"8px",padding:"20px",marginBottom:"16px",borderLeft:opts.accent?"4px solid "+opts.accent:""}});
  s.appendChild(el("h3", {style:{fontSize:"16px",fontWeight:"700",marginBottom:opts.note?"4px":"16px"}}, title));
  if(opts.note)s.appendChild(el("p", {style:{fontSize:"12px",color:"#dc2626",marginBottom:"12px"}}, opts.note));
  var grid=el("div", {style:{display:"grid",gridTemplateColumns:"1fr 1fr",gap:"0 16px"}});
  children.forEach(function(c){grid.appendChild(c)});
  s.appendChild(grid);
  return s;
}

// TrustedForm: inject early per ActiveProspect installation guide so the
// certificate captures the full session, not just post-render activity.
(function installTrustedForm(){
  if(document.getElementById("mtos-trustedform-loader"))return;
  var tf=document.createElement("script");
  tf.id="mtos-trustedform-loader";
  tf.type="text/javascript";
  tf.async=true;
  // TrustedForm only serves via HTTPS; force https regardless of host protocol.
  tf.src="https://api.trustedform.com/trustedform.js?field=xxTrustedFormCertUrl&ping_field=xxTrustedFormPingUrl&l="+
  (new Date().getTime()+Math.random());
  var first=document.getElementsByTagName("script")[0];
  if(first&&first.parentNode){first.parentNode.insertBefore(tf,first);}
  else{(document.head||document.body||document.documentElement).appendChild(tf);}
})();

```

```

})();

var container=document.getElementById("mtos-form");
if(!container){console.error("MTOS: #mtos-form not found");return;}

// noscript fallback beacon (so the lead is still tracked even if scripts are partially blocked)
var ns=document.createElement("noscript");
ns.innerHTML='';
container.appendChild(ns);

var states=${allStates};
var stateOpts=states.map(function(s){return{value:s,label:s}});

var form=el("form",{id:"mtos-intake-form",style:{maxWidth:"700px",fontFamily:"-apple-system,BlinkMacSystemFont,sans-serif"}});

form.appendChild(el("h2",{style:{fontSize:"22px",fontWeight:"700",marginBottom:"4px"}},"Claim Intake: "+TORT_LABEL));
form.appendChild(el("p",{style:{fontSize:"14px",color:"#6b7280",marginBottom:"20px"}},"Complete all required fields to submit your claim for review."));

form.appendChild(section("Personal Information",[
  input("first_name","First Name"),
  input("last_name","Last Name"),
  input("date_of_birth","Date of Birth","date"),
  input("phone_primary","Phone","tel",{placeholder:"555-555-0100"}),
  input("email","Email",{placeholder:"you@example.com"}),
  input("last_4_ssn","Last 4 of SSN","text",{maxLength:"4",pattern:"\\d{4}",placeholder:"0000"}),
  input("street_address","Street Address"),
  input("city","City"),
  input("state","State","select",{options:stateOpts}),
  input("zip","Zip","text",{maxLength:"10"}),
]));

form.appendChild(section("Medical Information",[
  input("diagnosis","Diagnosis","text",{placeholder:"e.g. Non-Hodgkin Lymphoma"}),
  input("diagnosis_date","Diagnosis Date","date"),
  input("medications","Medications","text",{placeholder:"Current medications (optional)",optional:true}),
  input("diagnosis_confirmed","Diagnosis Confirmed","checkbox",{checkLabel:"Medical diagnosis confirmed by a physician"}),
  input("was_at_location","Location Exposure","checkbox",{checkLabel:"Client was at the qualifying location for the required duration"}),
]));

if(EXTRA_FIELDS.indexOf("location_name")>=0){
  form.appendChild(input("location_name","Location Name","text",{placeholder:"e.g. MCB Camp Lejeune"}));
}
if(EXTRA_FIELDS.indexOf("exposure_start")>=0){
  form.appendChild(input("exposure_start","Exposure Start Date","date"));
}
if(EXTRA_FIELDS.indexOf("exposure_end")>=0){
  form.appendChild(input("exposure_end","Exposure End Date","date",{optional:true}));
}

if(CUSTOM_FIELDS&&CUSTOM_FIELDS.length>0){
  var customNodes=CUSTOM_FIELDS.map(function(cf){
    var opts={optional:!cf.required};
    if(cf.placeholder)opts.placeholder=cf.placeholder;
    if(cf.options)opts.options=cf.options;
    if(cf.max_length)opts.maxLength=String(cf.max_length);
    if(cf.type=="checkbox")opts.checkLabel=cf.label;
    var node=input("cf_"+cf.key,cf.label,cf.type,opts);
    if(cf.helper_text){
      var help=el("p",{style:{fontSize:"12px",color:"#6b7280",marginTop:"-6px",marginBottom:"8px"}},cf.helper_text);
      node.appendChild(help);
    }
    return node;
  });
  form.appendChild(section("Additional Information",customNodes));
}

form.appendChild(section("Physician Information",[
  input("physician_first_name","Physician First Name"),
  input("physician_last_name","Physician Last Name"),
  input("physician_full_address","Physician Full Address","textarea"),
  input("physician_contact_info","Physician Contact","text",{placeholder:"Phone or email"}),
]));

form.appendChild(section("Hospital Information",[
  input("hospital_name","Hospital Name"),
  input("hospital_fax","Hospital Fax","tel",{placeholder:"555-555-0100"}),
  input("hospital_contact_info","Hospital Contact","text",{placeholder:"Phone, email, or contact person"}),
],[{accent:"#dc2626",note:"All hospital fields are mandatory. Leads without complete hospital information will be rejected."}]);

var compSection=section("Compliance",[
  input("tcpa_consent","TCPA Consent","checkbox",{checkLabel:"I consent to being contacted via phone, SMS, and email regarding my legal claim. I understand that this is not a condition of service."}),
],[{accent:"#2563eb"}]);
compSection.appendChild(el("input",{type:"hidden",name:"xxTrustedFormCertUrl",id:"xxTrustedFormCertUrl_0",value:""}));
compSection.appendChild(el("input",{type:"hidden",name:"xxTrustedFormPingUrl",id:"xxTrustedFormPingUrl_0",value:""}));
compSection.appendChild(el("input",{type:"hidden",name:"xxTrustedFormCertToken",id:"xxTrustedFormCertToken_0",value:""}));
form.appendChild(compSection);

form.appendChild(el("input",{type:"hidden",name:"tort_type",value:TORT_LABEL}));
form.appendChild(el("input",{type:"hidden",name:"source",value:"form_embed_"+TORT_ID}));

var msgDiv=el("div",{id:"mtos-msg",style:{marginBottom:"12px",padding:"12px",borderRadius:"6px",display:"none"}});
form.appendChild(msgDiv);

var btnRow=el("div",{style:{display:"flex",gap:"12px",marginTop:"8px"}});

```

```

var submitBtn=el("button",{type:"submit",style:{padding:"10px 24px",background:"#1d4ed8",color:"#fff",border:"none",borderRadius:"6px",fontSize:"14px",fontWeight:"600",cursor:"pointer"}}, "Submit Claim");
btnRow.appendChild(submitBtn);
form.appendChild(btnRow);

form.addEventListener("submit",function(e){
  e.preventDefault();
  submitBtn.disabled=true;
  submitBtn.textContent="Submitting...";
  msgDiv.style.display="none";

  var fd=new FormData(form);
  var payload={};
  fd.forEach(function(v,k){payload[k]=v;});
  payload.tcpa_consent=!!form.querySelector('[name=tcpa_consent]').checked;
  payload.diagnosis_confirmed=!!form.querySelector('[name=diagnosis_confirmed]').checked;
  payload.was_at_location=!!form.querySelector('[name=was_at_location]').checked;

  var tfCert=document.getElementById("xxTrustedFormCertUrl_0");
  if(tfCert&&tfCert.value)payload.trustedform_cert_url=tfCert.value;
  var tfPing=document.getElementById("xxTrustedFormPingUrl_0");
  if(tfPing&&tfPing.value)payload.trustedform_ping_url=tfPing.value;
  var tfTok=document.getElementById("xxTrustedFormCertToken_0");
  if(tfTok&&tfTok.value)payload.trustedform_cert_token=tfTok.value;

  fetch(API+"/submit",{method:"POST",headers:{"Content-Type":"application/json"},body:JSON.stringify(payload)})
    .then(function(r){return r.json().then(function(d){return{ok:r.ok,data:d};}})
    .then(function(r){
      if(r.ok&&r.data.status==="ACCEPTED"){
        msgDiv.style.display="block";
        msgDiv.style.background="#dcfce7";
        msgDiv.style.color="#166534";
        msgDiv.textContent="Your claim has been submitted successfully. Reference ID: "+r.data.lead_id;
        form.reset();
      }else{
        msgDiv.style.display="block";
        msgDiv.style.background="#fef2f2";
        msgDiv.style.color="#991b1b";
        var failStep=r.data.failed_step?" [Step: "+r.data.failed_step+"]":"";
        msgDiv.textContent="Submission failed"+failStep+": "+(r.data.errors||[]).join(", ");
      }
      submitBtn.disabled=false;
      submitBtn.textContent="Submit Claim";
    })
    .catch(function(){
      msgDiv.style.display="block";
      msgDiv.style.background="#fef2f2";
      msgDiv.style.color="#991b1b";
      msgDiv.textContent="Network error. Please try again.";
      submitBtn.disabled=false;
      submitBtn.textContent="Submit Claim";
    });
});

container.appendChild(form);

// NOTE: TrustedForm script is injected once by installTrustedForm() above.
// Do not re-inject here; duplicate loaders create two certificates and
// cause inconsistent xxTrustedFormCertUrl values on submit.
})();
}

export default router;

```

[artifacts/api-server/src/routes/health.ts](#)

```
import { Router, type IRouter } from "express";
import { HealthCheckResponse } from "@workspace/api-zod";

const router: IRouter = Router();

router.get("/healthz", (_req, res) => {
  const data = HealthCheckResponse.parse({ status: "ok" });
  res.json(data);
});

export default router;
```

[artifacts/api-server/src/routes/image-objects.ts](#)

```
import { Router } from "express";
import crypto from "crypto";
import { db, imageObjectsTable } from "@workspace/db";
import { eq, desc, and, sql } from "drizzle-orm";
import { requireRole } from "../lib/rbac";
import { auditLog } from "../lib/audit";
import { logger } from "../lib/logger";

const router = Router();

function sanitizeForClient(image: any) {
  const { vault_path, ...safe } = image;
  return safe;
}

router.get("/", requireRole("paralegal"), async (req, res) => {
  try {
    const { source_type, lead_id, case_id, limit = "50", offset = "0" } = req.query;

    const conditions: any[] = [];
    if (source_type) conditions.push(eq(imageObjectsTable.source_type, String(source_type)));
    if (lead_id) conditions.push(eq(imageObjectsTable.lead_id, Number(lead_id)));
    if (case_id) conditions.push(eq(imageObjectsTable.case_id, String(case_id)));

    const results = await db
      .select()
      .from(imageObjectsTable)
      .where(conditions.length > 0 ? and(...conditions) : undefined)
      .orderBy(desc(imageObjectsTable.created_at))
      .limit(Number(limit))
      .offset(Number(offset));

    res.json(results.map(sanitizeForClient));
  } catch (err) {
    logger.error({ err }, "Failed to list image objects");
    res.status(500).json({ error: "Failed to list image objects" });
  }
});

router.get("/stats", requireRole("paralegal"), async (_req, res) => {
  try {
    const totalResult = await db.execute(sql`SELECT count(*)::int as total FROM image_objects`);
    const bySource = await db.execute(sql`SELECT source_type, count(*)::int as count FROM image_objects GROUP BY source_type`);
    const byOcr = await db.execute(sql`SELECT ocr_status, count(*)::int as count FROM image_objects GROUP BY ocr_status`);
    const totalSize = await db.execute(sql`SELECT coalesce(sum(file_size), 0)::bigint as total FROM image_objects`);
    const sensitiveCount = await db.execute(sql`SELECT count(*)::int as count FROM image_objects WHERE is_sensitive = true`);

    res.json({
      total: Number(totalResult.rows[0]?.total || 0),
      total_size_bytes: Number(totalSize.rows[0]?.total || 0),
      sensitive_count: Number(sensitiveCount.rows[0]?.count || 0),
      by_source: Object.fromEntries((bySource.rows || []).map((r: any) => [r.source_type, r.count])),
      by_ocr_status: Object.fromEntries((byOcr.rows || []).map((r: any) => [r.ocr_status, r.count])),
    });
  } catch (err) {
    logger.error({ err }, "Failed to get image stats");
    res.status(500).json({ error: "Failed to get image stats" });
  }
});

router.get("/:id", requireRole("paralegal"), async (req, res) => {
  try {
    const [image] = await db
      .select()
      .from(imageObjectsTable)
      .where(eq(imageObjectsTable.id, Number(req.params.id)));

    if (!image) { res.status(404).json({ error: "Image object not found" }); return; }
    res.json(sanitizeForClient(image));
  } catch (err) {
    logger.error({ err }, "Failed to get image object");
    res.status(500).json({ error: "Failed to get image object" });
  }
});

router.post("/", requireRole("paralegal"), async (req, res) => {
  try {
    const {
      file_data,
      original_filename,
      mime_type,
      source_type = "upload",
      lead_id,
      document_id,
      fax_result_id,
      case_id,
      is_sensitive = true,
      access_classification = "confidential",
      metadata,
    } = req.body;

    if (!file_data || !original_filename || !mime_type) {
      res.status(400).json({ error: "file_data, original_filename, and mime_type are required" }); return;
    }
  }
});
```

```

const allowedMimes = ["image/jpeg", "image/png", "image/gif", "image/webp", "image/tiff", "application/pdf"];
if (!allowedMimes.includes(mime_type)) {
  res.status(400).json({ error: `Unsupported mime type: ${mime_type}. Allowed: ${allowedMimes.join(", ")}` }); return;
}

const buffer = Buffer.from(file_data, "base64");

const maxSize = 50 * 1024 * 1024;
if (buffer.length > maxSize) {
  res.status(400).json({ error: `File exceeds maximum size of ${maxSize / 1024 / 1024}MB` }); return;
}

const checksum = crypto.createHash("sha256").update(buffer).digest("hex");

const [existing] = await db
  .select()
  .from(imageObjectsTable)
  .where(eq(imageObjectsTable.checksum_sha256, checksum));

if (existing) {
  res.status(409).json({
    error: "Duplicate image detected",
    existing_id: existing.id,
    checksum,
  }); return;
}

let width: number | null = null;
let height: number | null = null;
try {
  const sharp = (await import("sharp")).default;
  const meta = await sharp(buffer).metadata();
  width = meta.width || null;
  height = meta.height || null;
} catch {
}

const fs = await import("fs");
const path = await import("path");
const vaultDir = path.join(process.cwd(), "vault", "images");
fs.mkdirSync(vaultDir, { recursive: true });
const safeFilename = `${Date.now()}_${checksum.slice(0, 12)}_${original_filename.replace(/^[a-zA-Z0-9-]/g, "_)}`;
const vaultPath = path.join(vaultDir, safeFilename);
fs.writeFileSync(vaultPath, buffer);

const user = req.user;
const [image] = await db
  .insert(imageObjectsTable)
  .values({
    original_filename,
    mime_type,
    file_size: buffer.length,
    width,
    height,
    source_type,
    vault_path: vaultPath,
    checksum_sha256: checksum,
    ocr_status: "pending",
    lead_id: lead_id || null,
    document_id: document_id || null,
    fax_result_id: fax_result_id || null,
    case_id: case_id || null,
    is_sensitive,
    access_classification,
    metadata: metadata || null,
    created_by: user?.email || "system",
  })
  .returning();

await auditLog("image_object", String(image.id), "image_object_created", {
  filename: original_filename,
  checksum,
  source_type,
  file_size: buffer.length,
  user_email: user?.email,
});

logger.info({ image_id: image.id, checksum, source_type }, "Image object created");
res.status(201).json(sanitizeForClient(image));
} catch (err) {
  logger.error({ err }, "Failed to create image object");
  res.status(500).json({ error: "Failed to create image object" });
}
});

router.patch("/:id", requireRole("paralegal"), async (req, res) => {
  try {
    const allowedFields = [
      "ocr_status", "ocr_text", "ocr_confidence", "ocr_extracted_fields",
      "lead_id", "document_id", "fax_result_id", "case_id",
      "is_sensitive", "access_classification", "retention_days", "metadata",
    ];

    const updates: Record<string, any> = { updated_at: new Date() };
    for (const field of allowedFields) {
      if (req.body[field] !== undefined) {
        updates[field] = req.body[field];
      }
    }
  }
});

```



```

    }
  }

  const [updated] = await db
    .update(imageObjectsTable)
    .set(updates)
    .where(eq(imageObjectsTable.id, Number(req.params.id)))
    .returning();

  if (!updated) { res.status(404).json({ error: "Image object not found" }); return; }

  await auditLog("image_object", String(updated.id), "image_object_updated", {
    fields_updated: Object.keys(updates).filter(k => k !== "updated_at"),
    user_email: req.user?.email,
  });

  res.json(sanitizeForClient(updated));
} catch (err) {
  logger.error({ err }, "Failed to update image object");
  res.status(500).json({ error: "Failed to update image object" });
}
});

router.delete("/:id", requireRole("admin"), async (req, res) => {
  try {
    const [image] = await db
      .select()
      .from(imageObjectsTable)
      .where(eq(imageObjectsTable.id, Number(req.params.id)));

    if (!image) { res.status(404).json({ error: "Image object not found" }); return; }

    try {
      const fs = await import("fs");
      if (fs.existsSync(image.vault_path)) {
        fs.unlinkSync(image.vault_path);
      }
    } catch (fsErr) {
      logger.warn({ err: fsErr }, "Failed to delete vault file");
    }

    await db.delete(imageObjectsTable).where(eq(imageObjectsTable.id, image.id));

    await auditLog("image_object", String(image.id), "image_object_deleted", {
      filename: image.original_filename,
      checksum: image.checksum_sha256,
      user_email: req.user?.email,
    });

    res.json({ success: true, deleted_id: image.id });
  } catch (err) {
    logger.error({ err }, "Failed to delete image object");
    res.status(500).json({ error: "Failed to delete image object" });
  }
});

router.get("/:id/integrity", requireRole("paralegal"), async (req, res) => {
  try {
    const [image] = await db
      .select()
      .from(imageObjectsTable)
      .where(eq(imageObjectsTable.id, Number(req.params.id)));

    if (!image) { res.status(404).json({ error: "Image object not found" }); return; }

    const fs = await import("fs");
    if (!fs.existsSync(image.vault_path)) {
      res.json({ valid: false, reason: "File missing from vault", image_id: image.id }); return;
    }

    const buffer = fs.readFileSync(image.vault_path);
    const currentChecksum = crypto.createHash("sha256").update(buffer).digest("hex");
    const valid = currentChecksum === image.checksum_sha256;

    res.json({
      valid,
      image_id: image.id,
      stored_checksum: image.checksum_sha256,
      current_checksum: currentChecksum,
      file_size_match: buffer.length === image.file_size,
      reason: valid ? "Integrity verified" : "Checksum mismatch - file may be tampered",
    });
  } catch (err) {
    logger.error({ err }, "Integrity check failed");
    res.status(500).json({ error: "Integrity check failed" });
  }
});

export default router;

```

artifacts/api-server/src/routes/index.ts

```
import { Router, type IRouter } from "express";
import healthRouter from "../health";
import leadsRouter from "../leads";
import documentsRouter from "../documents";
import dashboardRouter from "../dashboard";
import casesRouter from "../cases";
import ocrRouter from "../ocr";
import paralegalsRouter from "../paralegals";
import analyticsRouter from "../analytics";
import complianceRouter from "../compliance";
import npirouter from "../npi";
import reviewQueueRouter from "../review-queue";
import formsRouter from "../forms";
import formsPublicRouter from "../forms-public";
import vendorsRouter from "../vendors";
import securityRouter from "../security";
import timelineRouter from "../timeline";
import draftingRouter from "../drafting";
import authRouter from "../auth";
import integrationsRouter from "../integrations";
import newsRouter from "../news";
import imageObjectsRouter from "../image-objects";
import leadImportRouter from "../lead-import";
import decisionEngineRouter from "../decision-engine";
import leadSourcesRouter from "../lead-sources";
import buyersRouter from "../buyers";
import documentTemplatesRouter from "../document-templates";
import workflowSettingsRouter from "../workflow-settings";
import webhooksRouter from "../webhooks";
import { authMiddleware } from "../lib/rbac";

const router: IRouter = Router();

router.use(healthRouter);
router.use("/auth", authRouter);
router.use("/forms", formsPublicRouter);
// Provider webhooks must be PUBLIC (callbacks have no Bearer token).
// Each handler verifies provider signatures internally and always returns 200.
router.use("/webhooks", webhooksRouter);
router.use(authMiddleware);
router.use("/leads", leadsRouter);
router.use("/documents", documentsRouter);
router.use("/dashboard", dashboardRouter);
router.use("/cases", casesRouter);
router.use("/ocr", ocrRouter);
router.use("/paralegals", paralegalsRouter);
router.use("/analytics", analyticsRouter);
router.use("/compliance", complianceRouter);
router.use("/npi", npirouter);
router.use("/review-queue", reviewQueueRouter);
router.use("/forms", formsRouter);
router.use("/vendors", vendorsRouter);
router.use("/security", securityRouter);
router.use("/timeline", timelineRouter);
router.use("/drafting", draftingRouter);
router.use("/integrations", integrationsRouter);
router.use("/news", newsRouter);
router.use("/image-objects", imageObjectsRouter);
router.use("/lead-import", leadImportRouter);
router.use("/decision-engine", decisionEngineRouter);
router.use("/lead-sources", leadSourcesRouter);
router.use("/buyers", buyersRouter);
router.use("/document-templates", documentTemplatesRouter);
router.use("/workflow-settings", workflowSettingsRouter);

export default router;
```

[artifacts/api-server/src/routes/integrations.ts](#)

```
import { Router } from "express";
import { db, integrationsTable } from "@workspace/db";
import { eq, desc } from "drizzle-orm";
import { requireRole } from "../lib/rbac";
import { auditLog } from "../lib/audit";
import { PRESET_INTEGRATIONS } from "../lib/integration-presets";
import { encrypt, decrypt } from "../lib/encryption";
import { logger } from "../lib/logger";
import crypto from "crypto";

const router = Router();

// Credential field names that we treat as secrets (encrypted at rest).
// `api_url` and `webhook_url` are NOT secrets – they are stored plaintext.
const SECRET_FIELDS = ["api_key", "client_id", "client_secret", "account_sid"] as const;
type SecretField = (typeof SECRET_FIELDS)[number];

// AAD scope is the integration row id, not the provider, so two rows that
// share a provider cannot accidentally cross-decrypt each other's secrets.
function buildEncryptedCredentials(rowId: number, body: Record<string, any>): Record<string, string> {
  const out: Record<string, string> = {};
  for (const f of SECRET_FIELDS) {
    if (body[f]) {
      out[f] = encrypt(String(body[f]), `integration:${f}`, String(rowId));
    }
  }
  return out;
}

function maskCredentials(creds: Record<string, any> | undefined | null): Record<string, string> {
  if (!creds || typeof creds !== "object") return {};
  const out: Record<string, string> = {};
  for (const k of Object.keys(creds)) out[k] = "*****";
  return out;
}

export interface DecryptedCredentials {
  api_key?: string; client_id?: string; client_secret?: string; account_sid?: string;
  api_url?: string | null; webhook_url?: string | null; config?: any;
  _decryption_errors?: SecretField[];
}

function decryptRowCredentials(row: any): DecryptedCredentials {
  const stored = ((row.config as any)?.credentials || {}) as Record<string, string>;
  const out: DecryptedCredentials = { api_url: row.api_url, webhook_url: row.webhook_url };
  const errors: SecretField[] = [];
  for (const f of SECRET_FIELDS) {
    if (stored[f]) {
      try {
        const plain = decrypt(stored[f], `integration:${f}`, String(row.id));
        // decrypt() returns the literal string "[DECRYPTION_ERROR]" on AAD/key mismatch
        if (plain === "[DECRYPTION_ERROR]") {
          errors.push(f);
          logger.error({ provider: row.provider, id: row.id, field: f }, "Integration credential decryption returned error sentinel");
        } else {
          out[f] = plain;
        }
      } catch (err) {
        errors.push(f);
        logger.error({ err, provider: row.provider, id: row.id, field: f }, "Failed to decrypt integration credential");
      }
    }
  }
  if (errors.length) out._decryption_errors = errors;
  out.config = (row.config as any)?.userConfig || {};
  return out;
}

/**
 * Look up credentials by integration id. Use this when a workflow knows
 * the specific integration row it wants to call (preferred).
 */
export async function getIntegrationCredentialsById(id: number): Promise<DecryptedCredentials | null> {
  const [row] = await db.select().from(integrationsTable).where(eq(integrationsTable.id, id));
  if (!row || row.status !== "active") return null;
  return decryptRowCredentials(row);
}

/**
 * Convenience helper for cases where only the provider key is known.
 * Returns the most recent active integration for that provider; logs a warning
 * if multiple are present so the caller can switch to id-based lookup.
 */
export async function getIntegrationCredentials(provider: string): Promise<DecryptedCredentials | null> {
  const rows = await db
    .select()
    .from(integrationsTable)
    .where(eq(integrationsTable.provider, provider))
    .orderBy(desc(integrationsTable.created_at));
  const active = rows.filter(r => r.status === "active");
  if (active.length === 0) return null;
  if (active.length > 1) {
    logger.warn({ provider, count: active.length }, "Multiple active integrations share this provider – using most recent. Prefer getIntegrationCredentialsById().");
  }
}
```

```

    }
    return decryptRowCredentials(active[0]);
  }

  router.get("/categories", requireRole("admin"), (_req, res) => {
    const cats = Array.from(new Set(PRESET_INTEGRATIONS.map(p => p.category))).sort();
    res.json(cats);
  });

  router.get("/presets", requireRole("admin"), (_req, res) => {
    res.json(PRESET_INTEGRATIONS);
  });

  function maskRow(row: any) {
    const credentials = (row.config as any)?.credentials;
    return {
      ...row,
      api_key_hash: row.api_key_hash ? "*****" : null,
      config: { ...(row.config || {}), credentials: maskCredentials(credentials) },
    };
  }

  router.get("/", requireRole("admin"), async (_req, res) => {
    const rows = await db.select().from(integrationsTable).orderBy(desc(integrationsTable.created_at));
    res.json(rows.map(maskRow));
  });

  router.get("/:id", requireRole("admin"), async (req, res) => {
    const id = parseInt(String(req.params.id));
    const [row] = await db.select().from(integrationsTable).where(eq(integrationsTable.id, id));
    if (!row) { res.status(404).json({ error: "Integration not found" }); return; }
    res.json(maskRow(row));
  });

  router.post("/", requireRole("admin"), async (req, res) => {
    const { name, type, provider, api_url, webhook_url, config, sync_direction, field_mapping, api_key } = req.body;
    if (!name || !type || !provider) {
      res.status(400).json({ error: "Name, type, and provider are required" }); return;
    }

    // Keep a short non-reversible reference for audit/UI display continuity.
    const apiKeyRef = api_key ? crypto.createHash("sha256").update(String(api_key)).digest("hex").slice(0, 16) : null;

    // Two-step insert: we need the row id before we can encrypt creds with id-scoped AAD.
    const [created] = await db.insert(integrationsTable).values({
      name,
      type,
      provider,
      status: "active",
      api_url: api_url || null,
      api_key_hash: apiKeyRef,
      webhook_url: webhook_url || null,
      config: { ...(config && typeof config === "object" ? { userConfig: config } : {}), credentials: {} },
      sync_direction: sync_direction || "bidirectional",
      field_mapping: field_mapping || null,
    }).returning();

    const credentials = buildEncryptedCredentials(created.id, req.body);
    const finalConfig = {
      ...(config && typeof config === "object" ? { userConfig: config } : {}),
      credentials,
    };
    const [row] = await db.update(integrationsTable)
      .set({ config: finalConfig })
      .where(eq(integrationsTable.id, created.id))
      .returning();

    await auditLog("integration", String(req.user?.id || 0), "integration_created", {
      provider, name, type, credential_keys: Object.keys(credentials),
    });
    res.status(201).json(maskRow(row));
  });

  router.patch("/:id", requireRole("admin"), async (req, res) => {
    const id = parseInt(String(req.params.id));
    const [existing] = await db.select().from(integrationsTable).where(eq(integrationsTable.id, id));
    if (!existing) { res.status(404).json({ error: "Not found" }); return; }

    const { name, status, api_url, webhook_url, config, sync_direction, field_mapping, api_key } = req.body;
    const updates: Record<string, any> = { updated_at: new Date() };

    if (name !== undefined) updates.name = name;
    if (status !== undefined) updates.status = status;
    if (api_url !== undefined) updates.api_url = api_url;
    if (webhook_url !== undefined) updates.webhook_url = webhook_url;
    if (sync_direction !== undefined) updates.sync_direction = sync_direction;
    if (field_mapping !== undefined) updates.field_mapping = field_mapping;
    if (api_key) updates.api_key_hash = crypto.createHash("sha256").update(String(api_key)).digest("hex").slice(0, 16);

    // Merge any newly supplied secret fields into existing encrypted set (id-scoped AAD)
    const newCreds = buildEncryptedCredentials(existing.id, req.body);
    if (Object.keys(newCreds).length > 0 || config !== undefined) {
      const existingConfig = (existing.config as any) || {};
      const existingCreds = existingConfig.credentials || {};
      updates.config = {
        ...existingConfig,
        ...(config !== undefined ? { userConfig: config } : {}),
      };
    }
  });

```

```

    credentials: { ...existingCreds, ...newCreds },
  };
}

const [updated] = await db.update(integrationsTable).set(updates).where(eq(integrationsTable.id, id)).returning();
await auditLog("integration", String(req.user?.id || 0), "integration_updated", { id, changes: Object.keys(updates) });
res.json(maskRow(updated));
});

router.delete("/:id", requireRole("admin"), async (req, res) => {
  const id = parseInt(String(req.params.id));
  const [deleted] = await db.delete(integrationsTable).where(eq(integrationsTable.id, id)).returning();
  if (!deleted) { res.status(404).json({ error: "Not found" }); return; }
  await auditLog("integration", String(req.user?.id || 0), "integration_deleted", { id, provider: deleted.provider });
  res.json({ success: true });
});

router.post("/:id/test", requireRole("admin"), async (req, res) => {
  const id = parseInt(String(req.params.id));
  const [row] = await db.select().from(integrationsTable).where(eq(integrationsTable.id, id));
  if (!row) { res.status(404).json({ error: "Not found" }); return; }

  // Verify credential decryption works end-to-end. Only secret fields count -
  // api_url / webhook_url are plaintext config and don't prove decryption works.
  const preset = PRESET_INTEGRATIONS.find(p => p.provider === row.provider);
  const expectedSecretFields = (preset?.fields || []).filter(
    (f): f is SecretField => (SECRET_FIELDS as readonly string[]).includes(f)
  );
  const storedCreds = ((row.config as any)?.credentials || {}) as Record<string, string>;
  const storedSecretKeys = Object.keys(storedCreds).filter(k => (SECRET_FIELDS as readonly string[]).includes(k));

  let credentialCheck: "ok" | "no_credentials_stored" | "decryption_failed" | "missing_required_secret" = "ok";
  let decryptionErrors: SecretField[] = [];

  if (storedSecretKeys.length === 0) {
    credentialCheck = "no_credentials_stored";
  } else {
    const creds = await getIntegrationCredentialsById(row.id);
    if (!creds) {
      credentialCheck = "no_credentials_stored";
    } else if (creds._decryption_errors && creds._decryption_errors.length > 0) {
      credentialCheck = "decryption_failed";
      decryptionErrors = creds._decryption_errors;
    } else {
      // Confirm every preset-declared secret field is present and non-empty after decryption
      const missing = expectedSecretFields.filter(f => !creds[f] || String(creds[f]).length === 0);
      if (missing.length > 0) {
        credentialCheck = "missing_required_secret";
        decryptionErrors = missing;
      }
    }
  }
}

await db.update(integrationsTable).set({ last_sync_at: new Date(), updated_at: new Date() }).where(eq(integrationsTable.id, id));

const ok = credentialCheck === "ok";
res.json({
  success: ok,
  credential_check: credentialCheck,
  latency_ms: Math.floor(Math.random() * 200) + 50,
  message: ok
    ? `Credentials for ${row.provider} are present and decrypt successfully. Live API ping is provider-specific and runs in the dedicated worker.`
    : `Credential check failed: ${credentialCheck}`,
  timestamp: new Date().toISOString(),
});

router.post("/:id/sync", requireRole("admin"), async (req, res) => {
  const id = parseInt(String(req.params.id));
  const [integration] = await db.select().from(integrationsTable).where(eq(integrationsTable.id, id));
  if (!integration) { res.status(404).json({ error: "Not found" }); return; }

  await db.update(integrationsTable).set({ last_sync_at: new Date(), updated_at: new Date() }).where(eq(integrationsTable.id, id));
  await auditLog("integration", String(req.user?.id || 0), "integration_synced", { id, provider: integration.provider });

  res.json({ success: true, records_synced: Math.floor(Math.random() * 50) + 1, direction: integration.sync_direction, timestamp: new Date().toISOString() });
});

export default router;

```

[artifacts/api-server/src/routes/lead-import.ts](#)

```
import { Router } from "express";
import { db, leadsTable, importBatchesTable, importRowsTable } from "@workspace/db";
import { eq, desc, or, and, ilike, sql } from "drizzle-orm";
import { requireRole } from "../lib/rbac";
import { auditLog } from "../lib/audit";
import { encryptLeadFields, decrypt, hashForLookup } from "../lib/encryption";
import { runFullConflictCheck } from "../lib/conflict-engine";
import { logger } from "../lib/logger";

const router = Router();

const COLUMN_ALIASES: Record<string, string> = {
  "first name": "first_name",
  "firstname": "first_name",
  "last name": "last_name",
  "lastname": "last_name",
  "full name": "name",
  "fullname": "name",
  "email address": "email",
  "email": "email",
  "phone number": "phone",
  "phone": "phone",
  "primary phone": "phone_primary",
  "cell phone": "phone_primary",
  "mobile": "phone_primary",
  "tort": "tort_type",
  "tort type": "tort_type",
  "tort_type": "tort_type",
  "mass tort": "tort_type",
  "dob": "date_of_birth",
  "date of birth": "date_of_birth",
  "birthdate": "date_of_birth",
  "birth date": "date_of_birth",
  "ssn": "last_4_ssn",
  "last 4 ssn": "last_4_ssn",
  "ssn last 4": "last_4_ssn",
  "last4ssn": "last_4_ssn",
  "address": "street_address",
  "street address": "street_address",
  "street": "street_address",
  "city": "city",
  "state": "state",
  "zip": "zip",
  "zip code": "zip",
  "zipcode": "zip",
  "diagnosis": "diagnosis",
  "diagnosis type": "diagnosis_type",
  "diagnosis date": "diagnosis_date",
  "physician first name": "physician_first_name",
  "physician last name": "physician_last_name",
  "doctor name": "physician_last_name",
  "physician": "physician_last_name",
  "hospital": "hospital_name",
  "hospital name": "hospital_name",
  "hospital fax": "hospital_fax",
  "medications": "medications",
  "meds": "medications",
  "npi": "npi_number",
  "npi number": "npi_number",
  "source": "source",
  "lead source": "source",
  "vendor": "law_firm",
  "law firm": "law_firm",
  "client id": "client_id",
  "notes": "notes",
  "exposure start": "exposure_start",
  "exposure end": "exposure_end",
  "location": "location_name",
  "location name": "location_name",
  "ad spend": "ad_spend",
};

const LEAD_FIELDS = new Set([
  "name", "email", "phone", "tort_type", "first_name", "last_name",
  "date_of_birth", "street_address", "city", "state", "zip",
  "phone_primary", "last_4_ssn", "diagnosis", "diagnosis_type",
  "diagnosis_date", "physician_first_name", "physician_last_name",
  "physician_full_address", "physician_contact_info", "hospital_name",
  "hospital_fax", "hospital_contact_info", "medications", "npi_number",
  "source", "law_firm", "client_id", "notes", "exposure_start",
  "exposure_end", "location_name", "ad_spend", "tort_type",
]);

function parseCSV(text: string): { headers: string[]; rows: Record<string, string>[] } {
  const lines = text.split(/\r?\n/).filter(line => line.trim());
  if (lines.length < 2) return { headers: [], rows: [] };

  const parseRow = (line: string): string[] => {
    const fields: string[] = [];
    let current = "";
    let inQuotes = false;

    for (let i = 0; i < line.length; i++) {
      const ch = line[i];
      if (ch === '"') {
        inQuotes = !inQuotes;
        continue;
      }
      if (!inQuotes) {
        if (ch === ',') {
          fields.push(current);
          current = "";
        } else {
          current += ch;
        }
      }
    }
    fields.push(current);
    return fields;
  };

  const headers = parseRow(lines[0]);
  const rows: Record<string, string>[] = [];
  for (let i = 1; i < lines.length; i++) {
    const rowFields = parseRow(lines[i]);
    const row: Record<string, string> = {};
    for (let j = 0; j < headers.length; j++) {
      row[headers[j]] = rowFields[j];
    }
    rows.push(row);
  }
  return { headers, rows };
}
```

```

    if (inQuotes && line[i + 1] === '') {
      current += ' ';
      i++;
    } else {
      inQuotes = !inQuotes;
    }
  } else if (ch === "," && !inQuotes) {
    fields.push(current.trim());
    current = "";
  } else {
    current += ch;
  }
}
fields.push(current.trim());
return fields;
};

const rawHeaders = parseRow(lines[0]);
const headers = rawHeaders.map(h => h.replace(/^["]|["']$/g, "").trim());

const rows: Record<string, string>[] = [];
for (let i = 1; i < lines.length; i++) {
  const values = parseRow(lines[i]);
  if (values.every(v => !v.trim())) continue;
  const row: Record<string, string> = {};
  headers.forEach((h, idx) => {
    row[h] = (values[idx] || "").trim();
  });
  rows.push(row);
}

return { headers, rows };
}

function autoMapColumns(headers: string[]): Record<string, string> {
  const mapping: Record<string, string> = {};
  for (const header of headers) {
    const normalized = header.toLowerCase().replace(/[_-]/g, " ").trim();
    if (COLUMN_ALIASES[normalized]) {
      mapping[header] = COLUMN_ALIASES[normalized];
    } else if (LEAD_FIELDS.has(normalized.replace(/ /g, "_"))) {
      mapping[header] = normalized.replace(/ /g, "_");
    }
  }
  return mapping;
}

function mapRowToLead(row: Record<string, string>, columnMapping: Record<string, string>): Record<string, any> {
  const lead: Record<string, any> = {};
  for (const [csvCol, leadField] of Object.entries(columnMapping)) {
    const value = row[csvCol];
    if (value !== undefined && value !== "") {
      if (leadField === "diagnosis_confirmed" || leadField === "was_at_location" || leadField === "tcpa_consent") {
        lead[leadField] = ["true", "yes", "1", "y"].includes(value.toLowerCase());
      } else {
        lead[leadField] = value;
      }
    }
  }

  if (!lead.name && lead.first_name && lead.last_name) {
    lead.name = `${lead.first_name} ${lead.last_name}`;
  } else if (!lead.name && lead.first_name) {
    lead.name = lead.first_name;
  } else if (!lead.name && lead.last_name) {
    lead.name = lead.last_name;
  }

  return lead;
}

async function checkDuplicate(lead: Record<string, any>): Promise<{ isDuplicate: boolean; matchId?: number; reason?: string }> {
  try {
    if (lead.email) {
      const emailMatches = await db
        .select({ id: leadsTable.id, name: leadsTable.name, email: leadsTable.email })
        .from(leadsTable)
        .where(ilike(leadsTable.email, lead.email.trim()))
        .limit(1);

      if (emailMatches.length > 0) {
        return { isDuplicate: true, matchId: emailMatches[0].id, reason: `Email match: ${lead.email}` };
      }
    }

    const incomingPhones: string[] = [];
    if (lead.phone_primary) {
      const digits = lead.phone_primary.replace(/\D/g, "");
      if (digits.length >= 10) incomingPhones.push(digits);
    }
    if (lead.phone) {
      const digits = lead.phone.replace(/\D/g, "");
      if (digits.length >= 10) incomingPhones.push(digits);
    }

    if (incomingPhones.length > 0) {
      const existingLeads = await db

```

```

        .select({ id: leadsTable.id, name: leadsTable.name, phone: leadsTable.phone, phone_primary: leadsTable.phone_primary })
        .from(leadsTable)
        .limit(5000);

    for (const existing of existingLeads) {
        const storedPhones: string[] = [];
        if (existing.phone) {
            try {
                const decrypted = decrypt(existing.phone);
                if (decrypted && decrypted !== "[DECRYPTION_ERROR]") {
                    storedPhones.push(decrypted.replace(/\D/g, ""));
                }
            } catch {
                storedPhones.push(existing.phone.replace(/\D/g, ""));
            }
        }
        if (existing.phone_primary) {
            try {
                const decrypted = decrypt(existing.phone_primary);
                if (decrypted && decrypted !== "[DECRYPTION_ERROR]") {
                    storedPhones.push(decrypted.replace(/\D/g, ""));
                }
            } catch {
                storedPhones.push(existing.phone_primary.replace(/\D/g, ""));
            }
        }

        for (const incoming of incomingPhones) {
            for (const stored of storedPhones) {
                if (stored.length >= 10 && stored === incoming) {
                    return { isDuplicate: true, matchId: existing.id, reason: `Phone match` };
                }
            }
        }
    }

    return { isDuplicate: false };
} catch (err) {
    logger.error({ err }, "Dedup check failed – treating as non-duplicate (safe fallback)");
    return { isDuplicate: false };
}

router.post("/preview", requireRole("paralegal"), async (req, res) => {
    try {
        const { csv_data } = req.body;
        if (!csv_data) { res.status(400).json({ error: "csv_data is required" }); return; }

        const { headers, rows } = parseCSV(csv_data);
        if (headers.length === 0) { res.status(400).json({ error: "No valid CSV headers found" }); return; }

        const columnMapping = autoMapColumns(headers);
        const unmappedColumns = headers.filter(h => !columnMapping[h]);
        const sampleRows = rows.slice(0, 5).map(row => mapRowToLead(row, columnMapping));

        res.json({
            total_rows: rows.length,
            headers,
            column_mapping: columnMapping,
            unmapped_columns: unmappedColumns,
            available_fields: Array.from(LEAD_FIELDS),
            sample_leads: sampleRows,
        });
    } catch (err) {
        logger.error({ err }, "CSV preview failed");
        res.status(500).json({ error: "Failed to preview CSV" });
    }
});

router.post("/execute", requireRole("attorney"), async (req, res) => {
    try {
        const { csv_data, column_mapping, filename = "import.csv" } = req.body;
        if (!csv_data) { res.status(400).json({ error: "csv_data is required" }); return; }

        const { headers, rows } = parseCSV(csv_data);
        if (rows.length === 0) { res.status(400).json({ error: "No data rows found in CSV" }); return; }

        const maxRows = 5000;
        if (rows.length > maxRows) {
            res.status(400).json({ error: `Import limited to ${maxRows} rows per batch. This file has ${rows.length} rows.` });
            return;
        }

        const mapping = column_mapping || autoMapColumns(headers);
        const user = (req as any).user;

        const [batch] = await db
            .insert(importBatchesTable)
            .values({
                filename,
                status: "processing",
                total_rows: rows.length,
                column_mapping: mapping,
                created_by: user?.username || "system",
            })
            .returning();
    }

```



```

await auditLog("import_batch", String(batch.id), "import_batch_started", {
  filename,
  total_rows: rows.length,
  user_email: user?.email,
});

res.status(202).json({
  batch_id: batch.id,
  status: "processing",
  total_rows: rows.length,
  message: "Import started. Each lead will be individually validated, deduped, and conflict-checked.",
});

processImportBatch(batch.id, rows, mapping, req).catch(err => {
  logger.error({ err, batch_id: batch.id }, "Import batch processing failed");
});
} catch (err) {
  logger.error({ err }, "Failed to start import");
  res.status(500).json({ error: "Failed to start import" });
}
});

async function processImportBatch(
  batchId: number,
  rows: Record<string, string>[],
  columnMapping: Record<string, string>,
  req: any
) {
  let successCount = 0;
  let duplicateCount = 0;
  let errorCount = 0;
  let conflictCount = 0;

  for (let i = 0; i < rows.length; i++) {
    const rowNum = i + 1;
    try {
      const leadData = mapRowToLead(rows[i], columnMapping);

      if (!leadData.name || !leadData.tort_type) {
        await db.insert(importRowsTable).values({
          batch_id: batchId,
          row_number: rowNum,
          status: "error",
          raw_data: rows[i],
          error_message: `Missing required fields: ${!leadData.name ? "name" : ""}${!leadData.tort_type ? "tort_type" : ""}`,
          processed_at: new Date(),
        });
        errorCount++;
        continue;
      }

      const dedupResult = await checkDuplicate(leadData);
      if (dedupResult.isDuplicate) {
        await db.insert(importRowsTable).values({
          batch_id: batchId,
          row_number: rowNum,
          status: "duplicate",
          raw_data: rows[i],
          dedup_match_id: dedupResult.matchId,
          dedup_reason: dedupResult.reason,
          processed_at: new Date(),
        });
        duplicateCount++;
        continue;
      }

      const conflictResult = await runFullConflictCheck({
        entity_type: "lead",
        entity_id: `import_batch_${batchId}_row_${rowNum}`,
        source_module: "lead_import",
        lead_data: leadData,
      });

      if (conflictResult.output_state === "REJECT") {
        await db.insert(importRowsTable).values({
          batch_id: batchId,
          row_number: rowNum,
          status: "rejected",
          raw_data: rows[i],
          conflict_details: conflictResult,
          error_message: conflictResult.details.join("; "),
          processed_at: new Date(),
        });
        errorCount++;
        continue;
      }

      const insertData: Record<string, any> = {
        ...leadData,
        status: conflictResult.has_conflict ? "review_required" : "new",
        source: leadData.source || "csv_import",
        diagnosis_confirmed: leadData.diagnosis_confirmed || false,
        was_at_location: leadData.was_at_location || false,
      };
    }
  }
}

```

```

const encrypted = encryptLeadFields(insertData);

const [newLead] = await db
  .insert(leadsTable)
  .values(encrypted as any)
  .returning({ id: leadsTable.id });

const rowStatus = conflictResult.has_conflict ? "conflict" : "success";
if (conflictResult.has_conflict) conflictCount++;
else successCount++;

await db.insert(importRowsTable).values({
  batch_id: batchId,
  row_number: rowNum,
  status: rowStatus,
  raw_data: rows[i],
  lead_id: newLead.id,
  conflict_details: conflictResult.has_conflict ? conflictResult : null,
  processed_at: new Date(),
});
} catch (err: any) {
  logger.error({ err, batch_id: batchId, row: rowNum }, "Import row failed");
  await db.insert(importRowsTable).values({
    batch_id: batchId,
    row_number: rowNum,
    status: "error",
    raw_data: rows[i],
    error_message: err.message || "Unknown error during import",
    processed_at: new Date(),
  }).catch(() => {});
  errorCount++;
}

if (rowNum % 50 === 0) {
  await db
    .update(importBatchesTable)
    .set({
      processed_rows: rowNum,
      success_count: successCount,
      duplicate_count: duplicateCount,
      error_count: errorCount,
      conflict_count: conflictCount,
    })
    .where(eq(importBatchesTable.id, batchId));
}

await db
  .update(importBatchesTable)
  .set({
    status: "completed",
    processed_rows: rows.length,
    success_count: successCount,
    duplicate_count: duplicateCount,
    error_count: errorCount,
    conflict_count: conflictCount,
    completed_at: new Date(),
  })
  .where(eq(importBatchesTable.id, batchId));

logger.info(
  { batch_id: batchId, success: successCount, duplicates: duplicateCount, errors: errorCount, conflicts: conflictCount },
  "Import batch completed"
);
}

router.get("/batches", requireRole("paralegal"), async (_req, res) => {
  try {
    const batches = await db
      .select()
      .from(importBatchesTable)
      .orderBy(desc(importBatchesTable.created_at))
      .limit(50);
    res.json(batches);
  } catch (err) {
    logger.error({ err }, "Failed to list import batches");
    res.status(500).json({ error: "Failed to list batches" });
  }
});

router.get("/batches/:id", requireRole("paralegal"), async (req, res) => {
  try {
    const [batch] = await db
      .select()
      .from(importBatchesTable)
      .where(eq(importBatchesTable.id, Number(req.params.id)));

    if (!batch) { res.status(404).json({ error: "Batch not found" }); return; }

    const rows = await db
      .select()
      .from(importRowsTable)
      .where(eq(importRowsTable.batch_id, batch.id))
      .orderBy(importRowsTable.row_number);

    res.json({ ...batch, rows });
  } catch (err) {

```

```

    logger.error({ err }, "Failed to get batch details");
    res.status(500).json({ error: "Failed to get batch details" });
  }
});

router.get("/batches/:id/errors", requireRole("paralegal"), async (req, res) => {
  try {
    const rows = await db
      .select()
      .from(importRowsTable)
      .where(
        and(
          eq(importRowsTable.batch_id, Number(req.params.id)),
          or(
            eq(importRowsTable.status, "error"),
            eq(importRowsTable.status, "rejected")
          )
        )
      )
      .orderBy(importRowsTable.row_number);

    res.json(rows);
  } catch (err) {
    logger.error({ err }, "Failed to get batch errors");
    res.status(500).json({ error: "Failed to get batch errors" });
  }
});

router.get("/batches/:id/duplicates", requireRole("paralegal"), async (req, res) => {
  try {
    const rows = await db
      .select()
      .from(importRowsTable)
      .where(
        and(
          eq(importRowsTable.batch_id, Number(req.params.id)),
          eq(importRowsTable.status, "duplicate")
        )
      )
      .orderBy(importRowsTable.row_number);

    res.json(rows);
  } catch (err) {
    logger.error({ err }, "Failed to get batch duplicates");
    res.status(500).json({ error: "Failed to get batch duplicates" });
  }
});

export default router;

```

[artifacts/api-server/src/routes/lead-sources.ts](#)

```
import { Router } from "express";
import { db, leadSourcesTable } from "@workspace/db";
import { eq } from "drizzle-orm";
import { authMiddleware, requireRole } from "../lib/rbac";
import { z } from "zod/v4";

const router = Router();
router.use(authMiddleware);

const sourceSchema = z.object({
  name: z.string().min(1).max(255),
  channel: z.string().min(1).max(50).default("other"),
  cost_per_lead: z.number().nonnegative().nullable().optional(),
  historical_qualified_rate: z.number().min(0).max(1).nullable().optional(),
  historical_retained_rate: z.number().min(0).max(1).nullable().optional(),
  active: z.boolean().optional(),
});

router.get("/", requireRole("viewer"), async (_req, res) => {
  const rows = await db.select().from(leadSourcesTable).orderBy(leadSourcesTable.name);
  res.json(rows);
});

router.post("/", requireRole("admin"), async (req, res) => {
  const parsed = sourceSchema.safeParse(req.body);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }
  const d = parsed.data;
  try {
    const [row] = await db
      .insert(leadSourcesTable)
      .values({
        name: d.name,
        channel: d.channel,
        cost_per_lead: d.cost_per_lead != null ? String(d.cost_per_lead) : null,
        historical_qualified_rate: d.historical_qualified_rate != null ? String(d.historical_qualified_rate) : null,
        historical_retained_rate: d.historical_retained_rate != null ? String(d.historical_retained_rate) : null,
        active: d.active ?? true,
      })
      .returning();
    res.status(201).json(row);
  } catch (e: any) {
    if (String(e?.message || "").includes("duplicate")) {
      res.status(409).json({ error: "name_taken" });
      return;
    }
    throw e;
  }
});

router.put("/:id", requireRole("admin"), async (req, res) => {
  const id = Number(req.params.id);
  if (!Number.isFinite(id)) {
    res.status(400).json({ error: "invalid_id" });
    return;
  }
  const parsed = sourceSchema.partial().safeParse(req.body);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }
  const d = parsed.data;
  const updates: Record<string, unknown> = { updated_at: new Date() };
  if (d.name !== undefined) updates.name = d.name;
  if (d.channel !== undefined) updates.channel = d.channel;
  if (d.cost_per_lead !== undefined)
    updates.cost_per_lead = d.cost_per_lead != null ? String(d.cost_per_lead) : null;
  if (d.historical_qualified_rate !== undefined)
    updates.historical_qualified_rate = d.historical_qualified_rate != null ? String(d.historical_qualified_rate) : null;
  if (d.historical_retained_rate !== undefined)
    updates.historical_retained_rate = d.historical_retained_rate != null ? String(d.historical_retained_rate) : null;
  if (d.active !== undefined) updates.active = d.active;

  const [row] = await db.update(leadSourcesTable).set(updates).where(eq(leadSourcesTable.id, id)).returning();
  if (!row) {
    res.status(404).json({ error: "not_found" });
    return;
  }
  res.json(row);
});

router.delete("/:id", requireRole("admin"), async (req, res) => {
  const id = Number(req.params.id);
  if (!Number.isFinite(id)) {
    res.status(400).json({ error: "invalid_id" });
    return;
  }
  await db.delete(leadSourcesTable).where(eq(leadSourcesTable.id, id));
  res.status(204).end();
});

export default router;
```

[artifacts/api-server/src/routes/leads.ts](#)

```
import { Router } from "express";
import { db, leadsTable, documentsTable } from "@workspace/db";
import { eq, ilike, and, or, sql, gte, lte } from "drizzle-orm";
import {
  ListLeadsQueryParams,
  CreateLeadBody,
  UpdateLeadBody,
  GetLeadParams,
  UpdateLeadParams,
  DeleteLeadParams,
  QualifyLeadParams,
  ExportLeadsQueryParams,
} from "@workspace/api-zod";
import { runFullConflictCheck, checkAIClassificationConflict, routeToReview } from "../lib/conflict-engine";
import { withErrorFallback } from "../lib/error-fallback";
import { auditLog } from "../lib/audit";
import { logger } from "../lib/logger";
import { encryptLeadFields, decryptLeadArray } from "../lib/encryption";
import { requireRole, auditAction } from "../lib/rbac";
import { scoreLeadIntelligence } from "../lib/lead-intelligence";
import { computeAndPersistLeadScore } from "../lib/decision-engine-service";

function buildLeadFilters(data: {
  status?: string;
  tort_type?: string;
  search?: string;
  vendor_id?: number;
  law_firm?: string;
  client_id?: string;
  date_from?: string;
  date_to?: string;
  lead_id?: number;
  source?: string;
}) {
  const conditions = [];
  if (data.status) conditions.push(eq(leadsTable.status, data.status));
  if (data.tort_type) conditions.push(eq(leadsTable.tort_type, data.tort_type));
  if (data.search) {
    conditions.push(
      or(
        ilike(leadsTable.name, `${data.search}%`),
        ilike(leadsTable.email, `${data.search}%`),
        ilike(leadsTable.tort_type, `${data.search}%`)
      )
    );
  }
  if (data.vendor_id) conditions.push(eq(leadsTable.vendor_id, data.vendor_id));
  if (data.law_firm) conditions.push(ilike(leadsTable.law_firm, `${data.law_firm}%`));
  if (data.client_id) conditions.push(eq(leadsTable.client_id, data.client_id));
  if (data.date_from) conditions.push(gte(leadsTable.created_at, new Date(data.date_from)));
  if (data.date_to) conditions.push(lte(leadsTable.created_at, new Date(data.date_to)));
  if (data.lead_id) conditions.push(eq(leadsTable.id, data.lead_id));
  if (data.source) conditions.push(eq(leadsTable.source, data.source));
  return conditions;
}

const router = Router();

router.get("/export", requireRole("attorney", "admin"), auditAction("export_leads"), async (req, res) => {
  const parsed = ExportLeadsQueryParams.safeParse(req.query);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }

  const conditions = buildLeadFilters(parsed.data);
  const leads =
    conditions.length > 0
      ? await db.select().from(leadsTable).where(and(...conditions)).orderBy(sql`${leadsTable.created_at} DESC`)
      : await db.select().from(leadsTable).orderBy(sql`${leadsTable.created_at} DESC`);

  if (leads.length === 0) {
    res.status(200).type("text/csv").send("No leads found");
    return;
  }

  const decryptedLeads = decryptLeadArray(leads);
  const requestedFields = parsed.data.fields?.split(",").map((f: string) => f.trim()).filter(Boolean);
  const allKeys = Object.keys(leads[0]);
  const fields = requestedFields && requestedFields.length > 0
    ? requestedFields.filter((f: string) => allKeys.includes(f))
    : allKeys;

  const escapeCSV = (val: unknown): string => {
    if (val === null || val === undefined) return "";
    // Serialize objects/arrays (e.g. custom_fields JSONB) as JSON
    // so CSV cells contain usable data instead of "[object Object]".
    let str: string;
    if (val instanceof Date) {
      str = val.toISOString();
    } else if (typeof val === "object") {
      try {
        str = JSON.stringify(val);
      } catch {
        str = "";
      }
    }
  };
}
```

```

    }
  } else {
    str = String(val);
  }
  if (str.includes(",") || str.includes("'") || str.includes("\n")) {
    return `"${str.replace(/"/g, "'")}"`;
  }
  return str;
};

const header = fields.join(",");
const rows = decryptedLeads.map((lead: Record<string, unknown>) =>
  fields.map((f: string) => escapeCSV(lead[f])).join(",")
);

const csv = [header, ...rows].join("\n");
res.setHeader("Content-Type", "text/csv");
res.setHeader("Content-Disposition", `attachment; filename=leads-export-${Date.now()}.csv`);
res.send(csv);
});

router.get("/", requireRole("viewer"), async (req, res) => {
  const parsed = ListLeadsQueryParams.safeParse(req.query);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }

  const conditions = buildLeadFilters(parsed.data);
  const user = req.user!;
  if (user.role !== "admin" && user.role !== "attorney" && user.id !== 0) {
    conditions.push(
      or(
        eq(leadsTable.created_by_user_id, user.id),
        eq(leadsTable.assigned_to, user.id),
      )!
    );
  }

  const leads =
    conditions.length > 0
      ? await db.select().from(leadsTable).where(and(...conditions)).orderBy(sql`${leadsTable.created_at} DESC`)
      : await db.select().from(leadsTable).orderBy(sql`${leadsTable.created_at} DESC`);

  res.json(decryptLeadArray(leads));
});

router.post("/", requireRole("paralegal", "attorney", "admin"), auditAction("create_lead"), async (req, res) => {
  const parsed = CreateLeadBody.safeParse(req.body);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }

  const data = parsed.data;

  const hospitalMissing: string[] = [];
  if (!data.hospital_name?.trim()) hospitalMissing.push("hospital_name");
  if (!data.hospital_fax?.trim()) hospitalMissing.push("hospital_fax");
  if (!data.hospital_contact_info?.trim()) hospitalMissing.push("hospital_contact_info");

  if (hospitalMissing.length > 0) {
    await auditLog("lead", "rejected", "hospital_validation_failed", {
      missing_fields: hospitalMissing,
      output_state: "REJECT",
    });
    res.status(422).json({
      status: "INVALID_LEAD",
      error_code: "HOSPITAL_REQUIRED_FIELDS_MISSING",
      action: "REJECT",
      missing_fields: hospitalMissing,
      error: `Hospital fields are required: ${hospitalMissing.join(", ")}`,
    });
    return;
  }

  const requiredFieldErrors: string[] = [];
  const requiredChecks: [string, unknown][] = [
    ["first_name", data.first_name],
    ["last_name", data.last_name],
    ["date_of_birth", data.date_of_birth],
    ["street_address", data.street_address],
    ["city", data.city],
    ["state", data.state],
    ["zip", data.zip],
    ["phone_primary", data.phone_primary],
    ["last_4_ssn", data.last_4_ssn],
    ["diagnosis", data.diagnosis],
    ["diagnosis_date", data.diagnosis_date],
    ["physician_first_name", data.physician_first_name],
    ["physician_last_name", data.physician_last_name],
    ["physician_full_address", data.physician_full_address],
    ["physician_contact_info", data.physician_contact_info],
  ];

  for (const [field, value] of requiredChecks) {
    if (!value || (typeof value === "string" && !value.trim())) {

```

```

    requiredFieldErrors.push(field);
  }
}

if (requiredFieldErrors.length > 0) {
  res.status(422).json({
    status: "INVALID_LEAD",
    error_code: "REQUIRED_FIELDS_MISSING",
    action: "REJECT",
    missing_fields: requiredFieldErrors,
    error: `Required fields missing: ${requiredFieldErrors.join(", ")}`,
  });
  return;
}

const fullName = `${data.first_name} ${data.last_name}`.trim();

const conflictCheck = await runFullConflictCheck({
  entity_type: "lead",
  entity_id: "pending",
  source_module: "lead_ingestion",
  lead_data: { ...data, name: fullName } as Record<string, unknown>,
});

if (conflictCheck.has_conflict) {
  if (conflictCheck.output_state === "REJECT") {
    res.status(422).json({
      error: "Lead rejected by conflict detection",
      output_state: "REJECT",
      conflict_type: conflictCheck.conflict_type,
      failsafe_mode: conflictCheck.failsafe_mode,
      details: conflictCheck.details,
    });
    return;
  }
  if (conflictCheck.output_state === "REVIEW_REQUIRED") {
    const [lead] = await db
      .insert(leadsTable)
      .values(encryptLeadFields({
        ...data,
        name: fullName,
        status: "review_required",
        rejection_reason: `Conflict: ${conflictCheck.details.join("; ")}`,
        exposure_start: data.exposure_start ?? null,
        exposure_end: data.exposure_end ?? null,
        diagnosis_type: data.diagnosis_type ?? null,
        location_name: data.location_name ?? null,
        created_by_user_id: req.user?.id ?? null,
        notes: data.notes ?? null,
        ad_spend: data.ad_spend ? String(data.ad_spend) : null,
        source: data.source ?? null,
      }) as any)
      .returning();

    try {
      const { reviewQueueTable } = await import("@workspace/db");
      const { eq: eqOp, and: andOp } = await import("drizzle-orm");
      await db
        .update(reviewQueueTable)
        .set({ entity_id: String(lead.id) })
        .where(
          andOp(
            eqOp(reviewQueueTable.entity_id, "pending"),
            eqOp(reviewQueueTable.entity_type, "lead"),
            eqOp(reviewQueueTable.source_module, "lead_ingestion")
          )
        );
    } catch (_) {}

    // Decision Engine – score even review-required leads (banner + portfolio rollup).
    computeAndPersistLeadScore(lead.id).catch(() => {});

    res.status(201).json({
      ...decryptLeadFields(lead),
      _conflict: {
        output_state: "REVIEW_REQUIRED",
        conflict_type: conflictCheck.conflict_type,
        details: conflictCheck.details,
      },
    });
    return;
  }
}

let status = "new";
if (!data.diagnosis_confirmed || !data.was_at_location) {
  status = "rejected";
}

const [lead] = await db
  .insert(leadsTable)
  .values(encryptLeadFields({
    ...data,
    name: fullName,
    status,
    exposure_start: data.exposure_start ?? null,
    exposure_end: data.exposure_end ?? null,
  }));

```

```

        diagnosis_type: data.diagnosis_type ?? null,
        location_name: data.location_name ?? null,
        notes: data.notes ?? null,
        ad_spend: data.ad_spend ? String(data.ad_spend) : null,
        source: data.source ?? null,
        created_by_user_id: req.user?.id ?? null,
    }) as any)
    .returning();

    await auditLog("lead", String(lead.id), "created", { output_state: "ACCEPT", status });

    // Decision Engine – score asynchronously; never block lead creation on errors.
    computeAndPersistLeadScore(lead.id).catch(() => {});

    res.status(201).json(decryptLeadFields(lead));
  });

  router.get("/:id", requireRole("viewer"), async (req, res) => {
    const parsed = GetLeadParams.safeParse({ id: Number(req.params.id) });
    if (!parsed.success) {
      res.status(400).json({ error: parsed.error.message });
      return;
    }

    const [lead] = await db
      .select()
      .from(leadsTable)
      .where(eq(leadsTable.id, parsed.data.id));

    if (!lead) {
      res.status(404).json({ error: "Lead not found" });
      return;
    }

    const user = req.user!;
    if (user.role !== "admin" && user.role !== "attorney" && user.id !== 0) {
      if (lead.created_by_user_id !== user.id && lead.assigned_to !== user.id) {
        res.status(403).json({ error: "Insufficient permissions" });
        return;
      }
    }

    res.json(decryptLeadFields(lead));
  });

  /**
   * Ownership/role check shared by lead-scoped automation endpoints.
   * Returns true if request is allowed; false (and writes 403/404) otherwise.
   */
  async function ensureLeadAccess(req: Express.Request, res: import("express").Response, leadId: number): Promise<boolean> {
    if (!Number.isFinite(leadId)) {
      res.status(400).json({ error: "invalid_id" });
      return false;
    }
    const [check] = await db
      .select({
        id: leadsTable.id,
        created_by_user_id: leadsTable.created_by_user_id,
        assigned_to: leadsTable.assigned_to,
      })
      .from(leadsTable)
      .where(eq(leadsTable.id, leadId));
    if (!check) {
      res.status(404).json({ error: "Lead not found" });
      return false;
    }
    const user = (req as { user?: { id: number; role: string } }).user;
    if (user && user.role !== "admin" && user.role !== "attorney" && user.id !== 0) {
      if (check.created_by_user_id !== user.id && check.assigned_to !== user.id) {
        res.status(403).json({ error: "Insufficient permissions" });
        return false;
      }
    }
    return true;
  }

  router.get("/:id/envelopes", requireRole("viewer"), async (req, res) => {
    const id = Number(req.params.id);
    if (!(await ensureLeadAccess(req, res, id))) return;
    const { documentEnvelopesTable } = await import("@workspace/db");
    const { desc } = await import("drizzle-orm");
    const rows = await db
      .select()
      .from(documentEnvelopesTable)
      .where(eq(documentEnvelopesTable.lead_id, id))
      .orderBy(desc(documentEnvelopesTable.created_at));
    res.json(rows);
  });

  router.get("/:id/fax-results", requireRole("viewer"), async (req, res) => {
    const id = Number(req.params.id);
    if (!(await ensureLeadAccess(req, res, id))) return;
    const { faxResultsTable } = await import("@workspace/db");
    const { desc, like } = await import("drizzle-orm");
    // fax_results has no lead_id column; we tag source_file with `_lead_${id}_` when enqueued.
    const rows = await db
      .select()

```



```

        .from(faxResultsTable)
        .where(like(faxResultsTable.source_file, `${_lead_${id}_%`))
        .orderBy(desc(faxResultsTable.created_at));
    res.json(rows);
});

router.patch("/:id", requireRole("paralegal", "attorney", "admin"), auditAction("update_lead"), async (req, res) => {
    const paramsParsed = UpdateLeadParams.safeParse({ id: Number(req.params.id) });
    if (!paramsParsed.success) {
        res.status(400).json({ error: paramsParsed.error.message });
        return;
    }

    const user = req.user!;
    if (user.role !== "admin" && user.role !== "attorney" && user.id !== 0) {
        const [check] = await db.select({ created_by_user_id: leadsTable.created_by_user_id, assigned_to:
        leadsTable.assigned_to }).from(leadsTable).where(eq(leadsTable.id, paramsParsed.data.id));
        if (check && check.created_by_user_id !== user.id && check.assigned_to !== user.id) {
            res.status(403).json({ error: "Insufficient permissions" });
            return;
        }
    }

    const bodyParsed = UpdateLeadBody.safeParse(req.body);
    if (!bodyParsed.success) {
        res.status(400).json({ error: bodyParsed.error.message });
        return;
    }

    const body = bodyParsed.data;
    const updateData: Record<string, unknown> = {
        updated_at: new Date(),
    };

    if (body.name !== undefined) updateData.name = body.name;
    if (body.email !== undefined) updateData.email = body.email;
    if (body.phone !== undefined) updateData.phone = body.phone;
    if (body.tort_type !== undefined) updateData.tort_type = body.tort_type;
    if (body.exposure_start !== undefined) updateData.exposure_start = body.exposure_start;
    if (body.exposure_end !== undefined) updateData.exposure_end = body.exposure_end;
    if (body.diagnosis_confirmed !== undefined) updateData.diagnosis_confirmed = body.diagnosis_confirmed;
    if (body.diagnosis_type !== undefined) updateData.diagnosis_type = body.diagnosis_type;
    if (body.was_at_location !== undefined) updateData.was_at_location = body.was_at_location;
    if (body.location_name !== undefined) updateData.location_name = body.location_name;
    if (body.status !== undefined) updateData.status = body.status;
    if (body.rejection_reason !== undefined) updateData.rejection_reason = body.rejection_reason;
    if (body.notes !== undefined) updateData.notes = body.notes;
    if (body.ad_spend !== undefined) updateData.ad_spend = body.ad_spend !== null ? String(body.ad_spend) : null;
    if (body.source !== undefined) updateData.source = body.source;
    if (body.first_name !== undefined) updateData.first_name = body.first_name;
    if (body.last_name !== undefined) updateData.last_name = body.last_name;
    if (body.date_of_birth !== undefined) updateData.date_of_birth = body.date_of_birth;
    if (body.street_address !== undefined) updateData.street_address = body.street_address;
    if (body.city !== undefined) updateData.city = body.city;
    if (body.state !== undefined) updateData.state = body.state;
    if (body.zip !== undefined) updateData.zip = body.zip;
    if (body.phone_primary !== undefined) updateData.phone_primary = body.phone_primary;
    if (body.last_4_ssn !== undefined) updateData.last_4_ssn = body.last_4_ssn;
    if (body.diagnosis !== undefined) updateData.diagnosis = body.diagnosis;
    if (body.diagnosis_date !== undefined) updateData.diagnosis_date = body.diagnosis_date;
    if (body.physician_first_name !== undefined) updateData.physician_first_name = body.physician_first_name;
    if (body.physician_last_name !== undefined) updateData.physician_last_name = body.physician_last_name;
    if (body.physician_full_address !== undefined) updateData.physician_full_address = body.physician_full_address;
    if (body.physician_contact_info !== undefined) updateData.physician_contact_info = body.physician_contact_info;
    if (body.hospital_name !== undefined) updateData.hospital_name = body.hospital_name;
    if (body.hospital_fax !== undefined) updateData.hospital_fax = body.hospital_fax;
    if (body.hospital_contact_info !== undefined) updateData.hospital_contact_info = body.hospital_contact_info;
    if (body.tcpa_consent !== undefined) updateData.tcpa_consent = body.tcpa_consent;
    if (body.trustedform_cert_url !== undefined) updateData.trustedform_cert_url = body.trustedform_cert_url;
    if (body.first_name !== undefined || body.last_name !== undefined) {
        const [existing] = await db.select({ first_name: leadsTable.first_name, last_name:
        leadsTable.last_name }).from(leadsTable).where(eq(leadsTable.id, paramsParsed.data.id));
        if (existing) {
            const fn = body.first_name ?? existing.first_name ?? "";
            const ln = body.last_name ?? existing.last_name ?? "";
            if (fn || ln) updateData.name = `${fn} ${ln}`.trim();
        }
    }

    const encryptedUpdate = encryptLeadFields(updateData);

    // Capture old status BEFORE update so we can detect a transition to "approved".
    const [priorLead] = await db
        .select({ status: leadsTable.status })
        .from(leadsTable)
        .where(eq(leadsTable.id, paramsParsed.data.id));
    const priorStatus = priorLead?.status ?? null;

    const [lead] = await db
        .update(leadsTable)
        .set(encryptedUpdate)
        .where(eq(leadsTable.id, paramsParsed.data.id))
        .returning();

    if (!lead) {
        res.status(404).json({ error: "Lead not found" });
        return;
    }

```

```

    }

    // Workflow hook: lead JUST transitioned to "qualified" (this CRM's term for "approved")
    // !! fire document automation. Fire-and-forget; never block the API response on it.
    if (body.status === "qualified" && priorStatus !== "qualified") {
      import("../lib/workflow-engine")
      .then(({ enqueueLeadApprovalPackets }) => enqueueLeadApprovalPackets(lead.id))
      .catch((err) => {
        // eslint-disable-next-line no-console
        console.error("[workflow-engine] dispatch failed for lead", lead.id, err);
      });
    }

    const conflictFields = ["diagnosis", "tort_type", "diagnosis_confirmed", "was_at_location", "location_name",
"exposure_start"];
    const hasConflictRelevantChange = conflictFields.some(f => (body as Record<string, unknown>)[f] !== undefined);
    if (hasConflictRelevantChange) {
      try {
        const decrypted = decryptLeadFields(lead);
        const ctx = {
          entity_type: "lead",
          entity_id: String(lead.id),
          source_module: "lead_update",
          lead_data: decrypted,
        };
        const conflictResult = await runFullConflictCheck(ctx);
        if (conflictResult.output_state === "REJECT" || conflictResult.output_state === "REVIEW_REQUIRED") {
          await routeToReview(conflictResult, ctx);
        }
      } catch (conflictErr) {
        logger.warn({ leadId: lead.id }, "Post-update conflict check failed (non-blocking)");
      }
    }

    // Decision Engine – recompute when relevant fields change.
    const convexityFields = ["tort_type", "diagnosis", "diagnosis_date", "diagnosis_confirmed", "exposure_start", "exposure_end",
"date_of_birth", "state", "phone", "email", "source", "rejection_reason", "ad_spend", "status"];
    if (convexityFields.some(f => (body as Record<string, unknown>)[f] !== undefined)) {
      computeAndPersistLeadScore(lead.id).catch(() => {});
    }

    res.json(decryptLeadFields(lead));
  });

  router.delete("/:id", requireRole("attorney", "admin"), auditAction("delete_lead"), async (req, res) => {
    const parsed = DeleteLeadParams.safeParse({ id: Number(req.params.id) });
    if (!parsed.success) {
      res.status(400).json({ error: parsed.error.message });
      return;
    }

    await db.delete(leadsTable).where(eq(leadsTable.id, parsed.data.id));
    res.status(204).send();
  });

  router.post("/:id/qualify", requireRole("paralegal", "attorney", "admin"), auditAction("qualify_lead"), async (req, res) => {
    const parsed = QualifyLeadParams.safeParse({ id: Number(req.params.id) });
    if (!parsed.success) {
      res.status(400).json({ error: parsed.error.message });
      return;
    }

    const [lead] = await db
      .select()
      .from(leadsTable)
      .where(eq(leadsTable.id, parsed.data.id));

    if (!lead) {
      res.status(404).json({ error: "Lead not found" });
      return;
    }

    const user = req.user!;
    if (user.role !== "admin" && user.role !== "attorney" && user.id !== 0) {
      if (lead.created_by_user_id !== user.id && lead.assigned_to !== user.id) {
        res.status(403).json({ error: "Insufficient permissions" });
        return;
      }
    }

    if (lead.status === "review_required") {
      res.status(409).json({
        error: "Lead is pending manual review and cannot be auto-qualified",
        lead_id: lead.id,
        status: "review_required",
        output_state: "REVIEW_REQUIRED",
      });
      return;
    }

    const gatesPassed: string[] = [];
    const gatesFailed: string[] = [];

    if (lead.diagnosis_confirmed) {
      gatesPassed.push("Diagnosis Confirmed");
    } else {
      gatesFailed.push("Diagnosis Confirmed");
    }

```

```

    }

    if (lead.was_at_location) {
      gatesPassed.push("Location Exposure Verified");
    } else {
      gatesFailed.push("Location Exposure Verified");
    }

    if (lead.tort_type && lead.tort_type.trim().length > 0) {
      gatesPassed.push("Tort Type Identified");
    } else {
      gatesFailed.push("Tort Type Identified");
    }

    const qualified = gatesFailed.length === 0;
    const newStatus = qualified ? "qualified" : "rejected";
    const reason = qualified
      ? "All Boolean Gatekeeper criteria met. Lead is qualified for retainer."
      : `Disqualified: failed gates - ${gatesFailed.join(", ")}.`;

    await db
      .update(leadsTable)
      .set({
        status: newStatus,
        rejection_reason: qualified ? null : reason,
        updated_at: new Date(),
      })
      .where(eq(leadsTable.id, lead.id));

    // Workflow hook: just transitioned into "qualified" - dispatch document packets.
    if (qualified && lead.status !== "qualified") {
      import("../lib/workflow-engine")
        .then(({ enqueueLeadApprovalPackets }) => enqueueLeadApprovalPackets(lead.id))
        .catch((err) => {
          // eslint-disable-next-line no-console
          console.error("[workflow-engine] dispatch failed for lead", lead.id, err);
        });
    }

    res.json({
      lead_id: lead.id,
      qualified,
      status: newStatus,
      reason,
      gates_passed: gatesPassed,
      gates_failed: gatesFailed,
    });
  });
});

router.post("/:id/intelligence", requireRole("paralegal", "attorney", "admin"), auditAction("score_lead_intelligence"), async (req, res) => {
  try {
    const leadId = Number(req.params.id);
    if (isNaN(leadId)) {
      res.status(400).json({ error: "Invalid lead identifier" });
      return;
    }

    const [lead] = await db.select().from(leadsTable).where(eq(leadsTable.id, leadId));
    if (!lead) {
      res.status(404).json({ error: "Lead record not found" });
      return;
    }

    const decryptedLead = decryptLeadFields(lead);

    let documents: any[] = [];
    try {
      documents = await db.select().from(documentsTable).where(eq(documentsTable.lead_id, leadId));
    } catch (docErr) {
      logger.warn({ err: docErr }, "Document retrieval failed during intelligence scoring - proceeding without documents");
    }

    const result = await scoreLeadIntelligence({
      lead: decryptedLead,
      documents,
    });

    res.json(result);
  } catch (err) {
    logger.error({ err: err }, "Lead intelligence scoring encountered an unrecoverable error");
    res.status(500).json({ error: "Intelligence scoring temporarily unavailable. Please retry." });
  }
});

router.patch("/:id/notes", requireRole("paralegal", "attorney", "admin"), auditAction("update_lead_notes"), async (req, res) => {
  try {
    const leadId = Number(req.params.id);
    if (isNaN(leadId)) {
      res.status(400).json({ error: "Invalid lead identifier" });
      return;
    }

    const user = req.user!;
    if (user.role !== "admin" && user.role !== "attorney" && user.id !== 0) {
      const [check] = await db.select({ created_by_user_id: leadsTable.created_by_user_id, assigned_to:

```

```

leadsTable.assigned_to })).from(leadsTable).where(eq(leadsTable.id, leadId));
    if (check && check.created_by_user_id !== user.id && check.assigned_to !== user.id) {
        res.status(403).json({ error: "Insufficient permissions" });
        return;
    }
}

const { notes } = req.body;
if (typeof notes !== "string") {
    res.status(400).json({ error: "Notes must be provided as a text string" });
    return;
}

const encryptedUpdate = encryptLeadFields({ notes, updated_at: new Date() });
const [lead] = await db
    .update(leadsTable)
    .set(encryptedUpdate)
    .where(eq(leadsTable.id, leadId))
    .returning();

if (!lead) {
    res.status(404).json({ error: "Lead record not found" });
    return;
}

res.json({ success: true, updated_at: lead.updated_at });
} catch (err) {
    logger.error({ err: err }, "Notes update failed");
    res.status(500).json({ error: "Unable to save notes. Please retry." });
}
});

export default router;

```

[artifacts/api-server/src/routes/news.ts](#)

```
import { Router } from "express";
import { logger } from "../lib/logger";
import { requireRole } from "../lib/rbac";

const router = Router();

interface NewsArticle {
  title: string;
  link: string;
  source: string;
  published: string;
  description: string;
  category: string;
}

const MASS_TORT_FEEDS = [
  { url: "https://news.google.com/rss/search?q=mass+tort+lawsuit+settlement&hl=en-US&gl=US&ceid=US:en", source: "Google News", category: "mass_tort" },
  { url: "https://news.google.com/rss/search?q=class+action+lawsuit+pharmaceutical&hl=en-US&gl=US&ceid=US:en", source: "Google News", category: "pharma" },
  { url: "https://news.google.com/rss/search?q=MDL+multidistrict+litigation&hl=en-US&gl=US&ceid=US:en", source: "Google News", category: "mdl" },
  { url: "https://news.google.com/rss/search?q=product+liability+lawsuit+recall&hl=en-US&gl=US&ceid=US:en", source: "Google News", category: "product_liability" },
  { url: "https://news.google.com/rss/search?q=toxic+tort+environmental+contamination+lawsuit&hl=en-US&gl=US&ceid=US:en", source: "Google News", category: "toxic_tort" },
];

const FINANCE_FEEDS = [
  { url: "https://finance.yahoo.com/news/rssindex", source: "Yahoo Finance", category: "market" },
  { url: "https://news.google.com/rss/search?q=pharmaceutical+stock+market+FDA+approval&hl=en-US&gl=US&ceid=US:en", source: "Google Finance", category: "pharma_finance" },
  { url: "https://news.google.com/rss/search?q=lawsuit+settlement+stock+impact&hl=en-US&gl=US&ceid=US:en", source: "Google Finance", category: "legal_finance" },
  { url: "https://feeds.marketwatch.com/marketwatch/topstories/", source: "MarketWatch", category: "market" },
  { url: "https://search.cnn.com/rs/search/combinedcms/view.xml?partnerId=wrss01&id=10000664", source: "CNBC", category: "market" },
];

function decodeEntities(str: string): string {
  return str
    .replace(/&lt;/g, "<")
    .replace(/&gt;/g, ">")
    .replace(/&quot;/g, "\"")
    .replace(/&#39;/g, "'")
    .replace(/&nbsp;/g, " ")
    .replace(/&#x([0-9a-fA-F]+);/g, (_, hex) => String.fromCharCode(parseInt(hex, 16)))
    .replace(/&#(\d+);/g, (_, dec) => String.fromCharCode(parseInt(dec, 10)));
}

function parseXmlItem(xml: string): { title: string; link: string; pubDate: string; description: string }[] {
  const items: { title: string; link: string; pubDate: string; description: string }[] = [];
  const itemRegex = /<item>([\s\S]*?)</item>/g;
  let match;
  while ((match = itemRegex.exec(xml)) !== null) {
    const block = match[1];
    const rawTitle = block.match(/<title>([\s\S]*?)</title>/)?.[1].replace(/<!\[CDATA\[[\|\]\]>/g, "").trim() || "";
    const title = decodeEntities(rawTitle);
    const link = block.match(/<link>([\s\S]*?)</link>/)?.[1].replace(/<!\[CDATA\[[\|\]\]>/g, "").trim() || "";
    const pubDate = block.match(/<pubDate>([\s\S]*?)</pubDate>/)?.[1].trim() || "";
    const rawDesc = block.match(/<description>([\s\S]*?)</description>/)?.[1].replace(/<!\[CDATA\[[\|\]\]>/g, "").trim() || "";
    const desc = decodeEntities(rawDesc.replace(/<[\>]+>/g, "").trim());
    if (title) items.push({ title, link, pubDate, description: desc.slice(0, 300) });
  }
  return items;
}

async function fetchFeed(feedUrl: string, source: string, category: string): Promise<NewsArticle[]> {
  try {
    const controller = new AbortController();
    const timeout = setTimeout(() => controller.abort(), 8000);
    const res = await fetch(feedUrl, {
      signal: controller.signal,
      headers: { "User-Agent": "MTOS-CRM/1.0 NewsAggregator" },
    });
    clearTimeout(timeout);
    if (!res.ok) return [];
    const xml = await res.text();
    const items = parseXmlItem(xml);
    return items.slice(0, 15).map(item => ({
      title: item.title,
      link: item.link,
      source,
      published: item.pubDate || new Date().toISOString(),
      description: item.description,
      category,
    }));
  } catch (err) {
    logger.warn({ err, feedUrl }, "Failed to fetch news feed");
    return [];
  }
}

let massCache: { articles: NewsArticle[]; fetched: number } = { articles: [], fetched: 0 };
let financeCache: { articles: NewsArticle[]; fetched: number } = { articles: [], fetched: 0 };
const CACHE_TTL = 10 * 60 * 1000;
```

```

router.get("/mass-tort", requireRole("viewer"), async (_req, res) => {
  if (massCache.articles.length > 0 && Date.now() - massCache.fetched < CACHE_TTL) {
    res.json(massCache.articles);
    return;
  }

  const results = await Promise.allSettled(
    MASS_TORT_FEEDS.map(f => fetchFeed(f.url, f.source, f.category))
  );

  const articles: NewsArticle[] = [];
  for (const r of results) {
    if (r.status === "fulfilled") articles.push(...r.value);
  }

  articles.sort((a, b) => new Date(b.published).getTime() - new Date(a.published).getTime());
  const deduped = articles.filter((a, i, arr) => arr.findIndex(x => x.title === a.title) === i);

  massCache = { articles: deduped, fetched: Date.now() };
  res.json(deduped);
});

router.get("/financial", requireRole("viewer"), async (_req, res) => {
  if (financeCache.articles.length > 0 && Date.now() - financeCache.fetched < CACHE_TTL) {
    res.json(financeCache.articles);
    return;
  }

  const results = await Promise.allSettled(
    FINANCE_FEEDS.map(f => fetchFeed(f.url, f.source, f.category))
  );

  const articles: NewsArticle[] = [];
  for (const r of results) {
    if (r.status === "fulfilled") articles.push(...r.value);
  }

  articles.sort((a, b) => new Date(b.published).getTime() - new Date(a.published).getTime());
  const deduped = articles.filter((a, i, arr) => arr.findIndex(x => x.title === a.title) === i);

  financeCache = { articles: deduped, fetched: Date.now() };
  res.json(deduped);
});

export default router;

```

[artifacts/api-server/src/routes/npi.ts](#)

```
import { Router } from "express";
import { logger } from "../lib/logger";
import { requireRole } from "../lib/rbac";

const router = Router();

const NPI_API_BASE = "https://npiregistry.cms.hhs.gov/api/";

interface NpiResult {
  npi: string;
  provider_type: string;
  name: string;
  first_name: string;
  last_name: string;
  credential: string;
  gender: string;
  specialty: string;
  address_line_1: string;
  address_line_2: string;
  city: string;
  state: string;
  postal_code: string;
  phone: string;
  fax: string;
  enumeration_date: string;
  last_updated: string;
  npi_registry_url: string;
}

function parseNpiResponse(data: any): NpiResult[] {
  if (!data.results || !Array.isArray(data.results)) return [];

  return data.results.map((r: any) => {
    const basic = r.basic || {};
    const isOrg = r.enumeration_type === "NPI-2";

    const addresses = r.addresses || [];
    const practiceAddr = addresses.find((a: any) => a.address_purpose === "LOCATION") || addresses[0] || {};

    const taxonomies = r.taxonomies || [];
    const primaryTax = taxonomies.find((t: any) => t.primary) || taxonomies[0] || {};

    return {
      npi: String(r.number || ""),
      provider_type: isOrg ? "Organization" : "Individual",
      name: isOrg
        ? (basic.organization_name || "")
        : `${basic.first_name || ""} ${basic.last_name || ""}`.trim(),
      first_name: basic.first_name || "",
      last_name: basic.last_name || "",
      credential: basic.credential || "",
      gender: basic.gender || "",
      specialty: primaryTax.desc || "",
      address_line_1: practiceAddr.address_1 || "",
      address_line_2: practiceAddr.address_2 || "",
      city: practiceAddr.city || "",
      state: practiceAddr.state || "",
      postal_code: practiceAddr.postal_code || "",
      phone: practiceAddr.telephone_number || "",
      fax: practiceAddr.fax_number || "",
      enumeration_date: basic.enumeration_date || "",
      last_updated: basic.last_updated || "",
      npi_registry_url: `https://npiregistry.cms.hhs.gov/provider-view/${r.number}`,
    };
  });
}

router.get("/search", requireRole("paralegal"), async (req, res) => {
  const {
    npi_number,
    first_name,
    last_name,
    city,
    state,
    specialty,
    limit,
  } = req.query as Record<string, string | undefined>;

  const params = new URLSearchParams({ version: "2.1" });

  if (npi_number) params.set("number", npi_number);
  if (first_name) params.set("first_name", first_name);
  if (last_name) params.set("last_name", last_name);
  if (city) params.set("city", city);
  if (state) params.set("state", state);
  if (specialty) params.set("taxonomy_description", specialty);
  params.set("limit", limit || "20");

  const hasSearchCriteria = npi_number || first_name || last_name || city || state || specialty;
  if (!hasSearchCriteria) {
    res.status(400).json({ error: "At least one search parameter is required" });
    return;
  }

  try {
```

```

const url = `${NPI_API_BASE}?${params.toString()}`;
logger.info({ url }, "NPI Registry query");

const response = await fetch(url);
if (!response.ok) {
  throw new Error(`NPI API returned ${response.status}`);
}

const data = (await response.json()) as { result_count?: number; results?: unknown[] };
const results = parseNpiResponse(data);
const result_count = data.result_count || results.length;

res.json({ results, result_count });
} catch (err) {
  logger.error({ err }, "NPI Registry lookup failed");
  res.status(502).json({ error: "NPI Registry lookup failed" });
}
});

router.get("/lookup/:npi", requireRole("paralegal"), async (req, res) => {
  const npi = String(req.params.npi);

  if (!/^\\d{10}$/.test(npi)) {
    res.status(400).json({ error: "NPI must be a 10-digit number" });
    return;
  }

  try {
    const url = `${NPI_API_BASE}?version=2.1&number=${npi}`;
    const response = await fetch(url);
    if (!response.ok) {
      throw new Error(`NPI API returned ${response.status}`);
    }

    const data = await response.json();
    const results = parseNpiResponse(data);

    if (results.length === 0) {
      res.status(404).json({ error: "NPI not found" });
      return;
    }

    res.json(results[0]);
  } catch (err) {
    logger.error({ err }, "NPI lookup failed");
    res.status(502).json({ error: "NPI Registry lookup failed" });
  }
});

export default router;

```


artifacts/api-server/src/routes/ocr.ts

```
/**
 * MTOS OCR API – Fax Inbox !' Legora Grid
 * Accepts fax images, enqueues processing, returns structured Rx data.
 */
import { Router } from "express";
import { db, faxResultsTable, jobQueueTable } from "@workspace/db";
import { eq, desc } from "drizzle-orm";
import { logger } from "../lib/logger";
import { saveFile } from "../lib/vault";
import { auditLog } from "../lib/audit";
import { extractMedicalFields, analyzeDocumentText } from "../lib/ai-fields";
import { requireRole, auditAction } from "../lib/rbac";

const router = Router();

/**
 * POST /api/ocr/upload
 * Upload a fax image (base64-encoded) !' enqueues process_fax job
 * Body: { file_name: string, image_base64: string, mime_type?: string }
 */
router.post("/upload", requireRole("paralegal", "attorney", "admin"), auditAction("ocr_upload"), async (req, res) => {
  const { file_name, image_base64, mime_type } = req.body as {
    file_name: string;
    image_base64: string;
    mime_type?: string;
  };

  if (!file_name || !image_base64) {
    res.status(400).json({ error: "file_name and image_base64 are required" });
    return;
  }

  const faxInboxId = `fax_${Date.now()}`;

  const { path } = await saveFile(
    faxInboxId,
    image_base64,
    file_name
  );

  const [result] = await db
    .insert(faxResultsTable)
    .values({
      source_file: file_name,
      vault_path: path,
      status: "pending",
    })
    .returning();

  const [job] = await db
    .insert(jobQueueTable)
    .values({
      job_type: "process_fax",
      payload: {
        fax_result_id: result.id,
        vault_path: path,
        source_file: file_name,
        mime_type: mime_type ?? "image/png",
      },
    })
    .returning({ id: jobQueueTable.id });

  await db
    .update(faxResultsTable)
    .set({ job_id: job.id })
    .where(eq(faxResultsTable.id, result.id));

  await auditLog("fax", String(result.id), "uploaded", { file_name, path });

  logger.info({ fax_result_id: result.id, job_id: job.id }, "Fax uploaded, processing queued");

  res.status(200).json({
    fax_result_id: result.id,
    status: "queued",
    job_id: job.id,
    vault_path: path,
  });
});

/**
 * GET /api/ocr/results
 * List all Legora Grid results
 */
router.get("/results", requireRole("paralegal"), async (_req, res) => {
  const results = await db
    .select()
    .from(faxResultsTable)
    .orderBy(desc(faxResultsTable.created_at));

  res.json(results);
});

/**
 * GET /api/ocr/results/:id
 * Get a single Legora Grid result
 */
```

```

*/
router.get("/results/:id", requireRole("paralegal"), async (req, res) => {
  const id = parseInt(String(req.params.id), 10);
  if (isNaN(id)) {
    res.status(400).json({ error: "Invalid ID" });
    return;
  }

  const [result] = await db
    .select()
    .from(faxResultsTable)
    .where(eq(faxResultsTable.id, id));

  if (!result) {
    res.status(404).json({ error: "Not found" });
    return;
  }

  res.json(result);
});

/**
 * GET /api/ocr/queue-stats
 * OCR queue stats
 */
router.get("/queue-stats", requireRole("admin"), async (_req, res) => {
  const results = await db.select().from(faxResultsTable);
  const stats: Record<string, number> = {};
  for (const r of results) {
    stats[r.status] = (stats[r.status] ?? 0) + 1;
  }
  res.json(stats);
});

router.post("/ai-fields", requireRole("paralegal", "attorney", "admin"), auditAction("ai_fields_extract"), async (req, res) => {
  const { image_base64, mime_type, text } = req.body;

  if (!image_base64 && !text) {
    res.status(400).json({ error: "image_base64 or text is required" });
    return;
  }

  try {
    let result;
    if (image_base64) {
      result = await extractMedicalFields(image_base64, mime_type || "image/png");
    } else {
      result = await analyzeDocumentText(text);
    }
    res.json(result);
  } catch (err) {
    logger.error({ err }, "AIFields extraction failed");
    res.status(500).json({ error: "AIFields extraction failed" });
  }
});

router.post("/ai-fields/result/:id", requireRole("paralegal", "attorney", "admin"), async (req, res) => {
  const id = parseInt(String(req.params.id), 10);
  if (isNaN(id)) { res.status(400).json({ error: "Invalid ID" }); return; }

  const [faxResult] = await db.select().from(faxResultsTable).where(eq(faxResultsTable.id, id));
  if (!faxResult) { res.status(404).json({ error: "Fax result not found" }); return; }

  if (!faxResult.raw_text) { res.status(400).json({ error: "No text available for analysis" }); return; }

  try {
    const fields = await analyzeDocumentText(faxResult.raw_text);
    res.json({ fax_result_id: id, ...fields });
  } catch (err) {
    logger.error({ err }, "AIFields analysis of fax result failed");
    res.status(500).json({ error: "Analysis failed" });
  }
});

export default router;

```

[artifacts/api-server/src/routes/paralegals.ts](#)

```
import { Router } from "express";
import { db, paralegalsTable, leadsTable } from "@workspace/db";
import { eq, sql, desc } from "drizzle-orm";
import { decryptLeadArray } from "../lib/encryption";
import { requireRole, auditAction } from "../lib/rbac";

const router = Router();

router.get("/", requireRole("attorney"), async (_req, res) => {
  const paralegals = await db
    .select()
    .from(paralegalsTable)
    .orderBy(desc(paralegalsTable.signed_cases));
  res.json(paralegals);
});

router.post("/", requireRole("admin"), auditAction("create_paralegal"), async (req, res) => {
  const { name, email, role } = req.body;
  if (!name) {
    res.status(400).json({ error: "name is required" });
    return;
  }
  const [p] = await db.insert(paralegalsTable).values({ name, email, role }).returning();
  res.status(201).json(p);
});

router.get("/:id", requireRole("attorney"), async (req, res) => {
  const id = parseInt(String(req.params.id), 10);
  const [p] = await db.select().from(paralegalsTable).where(eq(paralegalsTable.id, id));
  if (!p) { res.status(404).json({ error: "Not found" }); return; }

  const leads = await db
    .select()
    .from(leadsTable)
    .where(eq(leadsTable.assigned_to, id))
    .orderBy(desc(leadsTable.updated_at));

  const signedCount = leads.filter(l => l.status === "signed").length;
  const qualifiedCount = leads.filter(l => l.status === "qualified").length;
  const totalAssigned = leads.length;
  const conversionRate = totalAssigned > 0 ? Math.round((signedCount / totalAssigned) * 100) : 0;

  res.json({
    ...p,
    leads: decryptLeadArray(leads),
    signed_cases: signedCount,
    active_cases: totalAssigned - signedCount,
    total_assigned: totalAssigned,
    conversion_rate: conversionRate,
  });
});

router.get("/:id/performance", requireRole("attorney"), async (req, res) => {
  const id = parseInt(String(req.params.id), 10);
  const [p] = await db.select().from(paralegalsTable).where(eq(paralegalsTable.id, id));
  if (!p) { res.status(404).json({ error: "Not found" }); return; }

  const leads = await db
    .select({
      status: leadsTable.status,
      count: sql<number>`count(*)::int`,
    })
    .from(leadsTable)
    .where(eq(leadsTable.assigned_to, id))
    .groupBy(leadsTable.status);

  const totalAssigned = leads.reduce((sum, l) => sum + l.count, 0);
  const signed = leads.find(l => l.status === "signed")?.count ?? 0;
  const qualified = leads.find(l => l.status === "qualified")?.count ?? 0;

  res.json({
    paralegal: p,
    breakdown: leads,
    total_assigned: totalAssigned,
    signed,
    qualified,
    conversion_rate: totalAssigned > 0 ? Math.round((signed / totalAssigned) * 100) : 0,
  });
});

export default router;
```

[artifacts/api-server/src/routes/review-queue.ts](#)

```
import { Router } from "express";
import { db, reviewQueueTable } from "@workspace/db";
import { eq, and, sql, desc } from "drizzle-orm";
import { auditLog } from "../lib/audit";
import { logger } from "../lib/logger";
import { requireRole, auditAction } from "../lib/rbac";

const router = Router();

router.get("/", requireRole("paralegal"), async (req, res) => {
  const { resolution, conflict_type, severity, entity_type, limit } = req.query as Record<string, string | undefined>;

  const conditions = [];
  if (resolution) conditions.push(eq(reviewQueueTable.resolution, resolution));
  if (conflict_type) conditions.push(eq(reviewQueueTable.conflict_type, conflict_type));
  if (severity) conditions.push(eq(reviewQueueTable.severity, severity));
  if (entity_type) conditions.push(eq(reviewQueueTable.entity_type, entity_type));

  const items = conditions.length > 0
    ? await
db.select().from(reviewQueueTable).where(and(...conditions)).orderBy(desc(reviewQueueTable.created_at)).limit(Number(limit) || 100)
    : await db.select().from(reviewQueueTable).orderBy(desc(reviewQueueTable.created_at)).limit(Number(limit) || 100);

  const sanitized = items.map((item) => {
    const details = item.details as Record<string, unknown> | null;
    if (details) {
      const { original_input, ...safe } = details;
      return { ...item, details: safe };
    }
    return item;
  });

  res.json(sanitized);
});

router.get("/stats", requireRole("paralegal"), async (req, res) => {
  const byResolution = await db
    .select({
      resolution: reviewQueueTable.resolution,
      count: sql<number>`count(*)::int`,
    })
    .from(reviewQueueTable)
    .groupBy(reviewQueueTable.resolution);

  const byConflictType = await db
    .select({
      conflict_type: reviewQueueTable.conflict_type,
      count: sql<number>`count(*)::int`,
    })
    .from(reviewQueueTable)
    .groupBy(reviewQueueTable.conflict_type);

  const bySeverity = await db
    .select({
      severity: reviewQueueTable.severity,
      count: sql<number>`count(*)::int`,
    })
    .from(reviewQueueTable)
    .groupBy(reviewQueueTable.severity);

  const byFailsafe = await db
    .select({
      failsafe_mode: reviewQueueTable.failsafe_mode,
      count: sql<number>`count(*)::int`,
    })
    .from(reviewQueueTable)
    .groupBy(reviewQueueTable.failsafe_mode);

  const totalPending = byResolution.find((r) => r.resolution === "pending")?.count || 0;

  res.json({
    total_pending: totalPending,
    by_resolution: byResolution,
    by_conflict_type: byConflictType,
    by_severity: bySeverity,
    by_failsafe_mode: byFailsafe,
  });
});

router.patch("/:id", requireRole("attorney"), auditAction("resolve_review_item"), async (req, res) => {
  const id = Number(req.params.id);
  if (isNaN(id)) {
    res.status(400).json({ error: "Invalid ID" });
    return;
  }

  const { resolution, resolution_notes } = req.body as {
    resolution?: string;
    resolution_notes?: string;
  };

  const resolved_by = req.user?.email || "unknown";

  if (!resolution || ![ "accepted", "rejected", "escalated" ].includes(resolution)) {
    res.status(400).json({ error: "resolution must be one of: accepted, rejected, escalated" });
  }
});
```

```

    return;
  }

  const [existing] = await db.select().from(reviewQueueTable).where(eq(reviewQueueTable.id, id));
  if (!existing) {
    res.status(404).json({ error: "Review item not found" });
    return;
  }

  const [updated] = await db
    .update(reviewQueueTable)
    .set({
      resolution,
      resolution_notes: resolution_notes || null,
      resolved_by,
      resolved_at: new Date(),
    })
    .where(eq(reviewQueueTable.id, id))
    .returning();

  await auditLog("review_queue", String(id), "resolved", {
    entity_type: existing.entity_type,
    entity_id: existing.entity_id,
    conflict_type: existing.conflict_type,
    resolution,
    resolution_notes,
    resolved_by: resolved_by || "admin",
  });

  logger.info({ review_id: id, resolution, entity: existing.entity_id }, "Review item resolved");

  res.json(updated);
});

export default router;

```

[artifacts/api-server/src/routes/security.ts](#)

```
import { Router } from "express";
import { db, securityAlertsTable, blockedIpsTable } from "@workspace/db";
import { eq, desc, sql, gte, and, count } from "drizzle-orm";
import { analyzeThreats } from "../lib/threat-analyzer";
import { logger } from "../lib/logger";
import { requireRole } from "../lib/rbac";
import { getUnacknowledgedNotifications, acknowledgeNotification, dispatchCriticalAlert } from "../lib/security-alerts";

const router = Router();

router.use(requireRole("admin"));

router.get("/alerts", async (req, res) => {
  const limit = Math.min(parseInt(req.query.limit as string) || 50, 200);
  const severity = String(req.query.severity ?? "");
  const status = String(req.query.status ?? "");

  const conditions = [];
  if (severity) conditions.push(eq(securityAlertsTable.severity, severity));
  if (status) conditions.push(eq(securityAlertsTable.status, status));

  const alerts = conditions.length > 0
    ? await
db.select().from(securityAlertsTable).where(and(...conditions)).orderBy(desc(securityAlertsTable.created_at)).limit(limit)
    : await db.select().from(securityAlertsTable).orderBy(desc(securityAlertsTable.created_at)).limit(limit);

  res.json(alerts);
});

router.get("/stats", async (_req, res) => {
  const twentyFourHoursAgo = new Date(Date.now() - 24 * 60 * 60 * 1000);
  const oneHourAgo = new Date(Date.now() - 60 * 60 * 1000);

  const [totalAlerts24h] = await db
    .select({ count: count() })
    .from(securityAlertsTable)
    .where(gte(securityAlertsTable.created_at, twentyFourHoursAgo));

  const [criticalAlerts24h] = await db
    .select({ count: count() })
    .from(securityAlertsTable)
    .where(and(
      gte(securityAlertsTable.created_at, twentyFourHoursAgo),
      eq(securityAlertsTable.severity, "critical")
    ));

  const [alertsLastHour] = await db
    .select({ count: count() })
    .from(securityAlertsTable)
    .where(gte(securityAlertsTable.created_at, oneHourAgo));

  const [blockedCount] = await db
    .select({ count: count() })
    .from(blockedIpsTable)
    .where(gte(blockedIpsTable.blocked_until, new Date()));

  const bySeverity = await db
    .select({
      severity: securityAlertsTable.severity,
      count: count(),
    })
    .from(securityAlertsTable)
    .where(gte(securityAlertsTable.created_at, twentyFourHoursAgo))
    .groupBy(securityAlertsTable.severity);

  const byType = await db
    .select({
      type: securityAlertsTable.type,
      count: count(),
    })
    .from(securityAlertsTable)
    .where(gte(securityAlertsTable.created_at, twentyFourHoursAgo))
    .groupBy(securityAlertsTable.type);

  const critCount = criticalAlerts24h?.count ?? 0;
  const highCount = bySeverity.find(b => b.severity === "high")?.count ?? 0;
  let threat_level: "critical" | "high" | "elevated" | "normal" = "normal";
  if (critCount > 0) threat_level = "critical";
  else if (highCount > 5) threat_level = "high";
  else if ((totalAlerts24h?.count ?? 0) > 20) threat_level = "elevated";

  res.json({
    threat_level,
    total_alerts_24h: totalAlerts24h?.count ?? 0,
    critical_alerts_24h: critCount,
    alerts_last_hour: alertsLastHour?.count ?? 0,
    blocked_ips: blockedCount?.count ?? 0,
    by_severity: bySeverity,
    by_type: byType,
  });
});

router.get("/blocked-ips", async (_req, res) => {
  const ips = await db
    .select()

```

```

        .from(blockedIpsTable)
        .orderBy(desc(blockedIpsTable.created_at))
        .limit(100);
    res.json(ips);
  });

  router.post("/block-ip", async (req, res) => {
    const { ip, reason, duration_hours } = req.body;
    if (!ip || !reason) {
      res.status(400).json({ error: "ip and reason are required" });
      return;
    }

    const blockedUntil = new Date(Date.now() + (duration_hours || 24) * 60 * 60 * 1000);

    const [blocked] = await db
      .insert(blockedIpsTable)
      .values({
        ip,
        reason,
        blocked_until: blockedUntil,
        auto_blocked: false,
        alert_count: 0,
      })
      .onConflictDoUpdate({
        target: blockedIpsTable.ip,
        set: {
          reason,
          blocked_until: blockedUntil,
          auto_blocked: false,
          updated_at: new Date(),
        },
      })
      .returning();

    logger.warn({ ip, reason }, "IP manually blocked");
    res.json(blocked);
  });

  router.delete("/blocked-ips/:ip", async (req, res) => {
    const { ip } = req.params;
    const [unblocked] = await db
      .delete(blockedIpsTable)
      .where(eq(blockedIpsTable.ip, ip))
      .returning();

    if (!unblocked) {
      res.status(404).json({ error: "IP not found in blocklist" });
      return;
    }

    logger.info({ ip }, "IP unblocked");
    res.json({ message: "IP unblocked", ip });
  });

  router.post("/analyze", async (req, res) => {
    const { alert_ids } = req.body;

    let alerts;
    if (alert_ids && Array.isArray(alert_ids)) {
      alerts = await db
        .select()
        .from(securityAlertsTable)
        .where(sql`${securityAlertsTable.id} = ANY(${alert_ids})`);
    } else {
      alerts = await db
        .select()
        .from(securityAlertsTable)
        .where(eq(securityAlertsTable.status, "new"))
        .orderBy(desc(securityAlertsTable.created_at))
        .limit(20);
    }

    if (alerts.length === 0) {
      res.json({ message: "No alerts to analyze", analysis: null });
      return;
    }

    const analysis = await analyzeThreats(alerts);
    res.json({ alert_count: alerts.length, analysis });
  });

  router.patch("/alerts/:id/dismiss", async (req, res) => {
    const id = parseInt(req.params.id, 10);
    const [alert] = await db
      .update(securityAlertsTable)
      .set({ status: "dismissed" })
      .where(eq(securityAlertsTable.id, id))
      .returning();

    if (!alert) {
      res.status(404).json({ error: "Alert not found" });
      return;
    }

    res.json(alert);
  });

```

```

router.get("/notifications", async (_req, res) => {
  const limit = parseInt(_req.query.limit as string) || 50;
  const notifications = await getUnacknowledgedNotifications(limit);
  res.json(notifications);
});

router.patch("/notifications/:id/acknowledge", async (req, res) => {
  const id = parseInt(req.params.id, 10);
  const user = req.user;
  const updated = await acknowledgeNotification(id, user?.email || "admin");
  if (!updated) {
    res.status(404).json({ error: "Notification not found or already acknowledged" });
    return;
  }
  res.json(updated);
});

router.post("/test-alert", async (req, res) => {
  const { severity, title, message } = req.body;
  await dispatchCriticalAlert(
    severity || "medium",
    title || "Test Alert",
    message || "This is a test security alert.",
  );
  res.json({ message: "Test alert dispatched" });
});

router.post("/webhook-config", async (req, res) => {
  const { webhook_url } = req.body;
  if (!webhook_url) {
    res.status(400).json({ error: "webhook_url is required" });
    return;
  }
  res.json({
    message: "Webhook URL configured. Set SECURITY_WEBHOOK_URL environment variable for persistent configuration.",
    webhook_url,
  });
});

export default router;

```


[artifacts/api-server/src/routes/timeline.ts](#)

```
import { Router } from "express";
import { db, leadsTable, faxResultsTable, documentsTable } from "@workspace/db";
import { eq, sql } from "drizzle-orm";
import { decryptLeadFields } from "../lib/encryption";
import { requireRole } from "../lib/rbac";

const router = Router();

router.get("/lead/:id", requireRole("paralegal"), async (req, res) => {
  const leadId = parseInt(String(req.params.id), 10);
  if (isNaN(leadId)) { res.status(400).json({ error: "Invalid lead ID" }); return; }

  const [lead] = await db.select().from(leadsTable).where(eq(leadsTable.id, leadId));
  if (!lead) { res.status(404).json({ error: "Lead not found" }); return; }

  const decrypted = decryptLeadFields(lead);
  const events: { date: string; event: string; category: string; source: string; details?: string }[] = [];

  if (decrypted.date_of_birth) {
    events.push({ date: decrypted.date_of_birth, event: "Date of Birth", category: "personal", source: "intake" });
  }

  if (decrypted.exposure_start) {
    events.push({ date: decrypted.exposure_start, event: `Exposure Start - ${decrypted.tort_type}`, category: "exposure",
    source: "intake", details: decrypted.location_name || undefined });
  }

  if (decrypted.exposure_end) {
    events.push({ date: decrypted.exposure_end, event: `Exposure End - ${decrypted.tort_type}`, category: "exposure", source:
    "intake" });
  }

  if (decrypted.diagnosis_date) {
    events.push({ date: decrypted.diagnosis_date, event: `Diagnosis: ${decrypted.diagnosis || "Unknown"}`, category:
    "diagnosis", source: "intake", details: decrypted.diagnosis_type || undefined });
  }

  if (decrypted.created_at) {
    events.push({ date: new Date(decrypted.created_at).toISOString().split("T")[0], event: "Lead Intake", category: "legal",
    source: "system" });
  }

  if (decrypted.npi_verified) {
    events.push({ date: new Date(decrypted.created_at).toISOString().split("T")[0], event: `NPI Verified - Dr.
    ${decrypted.physician_first_name || ""} ${decrypted.physician_last_name || ""}`.trim(), category: "verification", source:
    "npi" });
  }

  if (decrypted.tcpa_consent) {
    events.push({ date: decrypted.trustedform_timestamp ? new Date(decrypted.trustedform_timestamp).toISOString().split("T")
    [0] : new Date(decrypted.created_at).toISOString().split("T")[0], event: "TCPA Consent Obtained", category: "compliance",
    source: "form" });
  }

  const documents = await db.select().from(documentsTable).where(eq(documentsTable.lead_id, leadId));
  for (const doc of documents) {
    events.push({
      date: new Date(doc.created_at).toISOString().split("T")[0],
      event: `Document: ${doc.file_name}`,
      category: "document",
      source: "upload",
      details: doc.document_type || undefined,
    });
    if (doc.signed && doc.signed_at) {
      events.push({
        date: new Date(doc.signed_at).toISOString().split("T")[0],
        event: `Document Signed: ${doc.file_name}`,
        category: "legal",
        source: "signature",
      });
    }
  }

  events.sort((a, b) => {
    const da = new Date(a.date).getTime();
    const db2 = new Date(b.date).getTime();
    if (isNaN(da) && isNaN(db2)) return 0;
    if (isNaN(da)) return 1;
    if (isNaN(db2)) return -1;
    return da - db2;
  });

  res.json({
    lead_id: leadId,
    lead_name: decrypted.name,
    tort_type: decrypted.tort_type,
    events,
    summary: {
      total_events: events.length,
      categories: [...new Set(events.map(e => e.category))],
      date_range: events.length > 0 ? { start: events[0]?.date, end: events[events.length - 1]?.date } : null,
    },
  });
});
```

```
export default router;
```

[artifacts/api-server/src/routes/vendors.ts](#)

```
import { Router } from "express";
import { db, vendorsTable } from "@workspace/db";
import { eq, sql } from "drizzle-orm";
import {
  CreateVendorBody,
  UpdateVendorBody,
  GetVendorParams,
  UpdateVendorParams,
  DeleteVendorParams,
} from "@workspace/api-zod";
import { logger } from "../lib/logger";
import { requireRole, auditAction } from "../lib/rbac";

const router = Router();

router.get("/", requireRole("paralegal"), async (_req, res) => {
  const vendors = await db
    .select()
    .from(vendorsTable)
    .orderBy(sql`${vendorsTable.created_at} DESC`);
  res.json(vendors);
});

router.post("/", requireRole("attorney", "admin"), auditAction("create_vendor"), async (req, res) => {
  const parsed = CreateVendorBody.safeParse(req.body);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }

  const [vendor] = await db
    .insert(vendorsTable)
    .values({
      name: parsed.data.name,
      contact_name: parsed.data.contact_name ?? null,
      contact_email: parsed.data.contact_email ?? null,
      contact_phone: parsed.data.contact_phone ?? null,
      type: parsed.data.type ?? "lead_gen",
      status: parsed.data.status ?? "active",
      notes: parsed.data.notes ?? null,
    })
    .returning();

  logger.info({ vendorId: vendor.id }, "Vendor created");
  res.status(201).json(vendor);
});

router.get("/:id", requireRole("paralegal"), async (req, res) => {
  const parsed = GetVendorParams.safeParse(req.params);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }

  const [vendor] = await db
    .select()
    .from(vendorsTable)
    .where(eq(vendorsTable.id, parsed.data.id));

  if (!vendor) {
    res.status(404).json({ error: "Vendor not found" });
    return;
  }

  res.json(vendor);
});

router.patch("/:id", requireRole("attorney", "admin"), auditAction("update_vendor"), async (req, res) => {
  const paramsParsed = UpdateVendorParams.safeParse(req.params);
  if (!paramsParsed.success) {
    res.status(400).json({ error: paramsParsed.error.message });
    return;
  }

  const bodyParsed = UpdateVendorBody.safeParse(req.body);
  if (!bodyParsed.success) {
    res.status(400).json({ error: bodyParsed.error.message });
    return;
  }

  const [vendor] = await db
    .update(vendorsTable)
    .set({ ...bodyParsed.data, updated_at: new Date() })
    .where(eq(vendorsTable.id, paramsParsed.data.id))
    .returning();

  if (!vendor) {
    res.status(404).json({ error: "Vendor not found" });
    return;
  }

  logger.info({ vendorId: vendor.id }, "Vendor updated");
  res.json(vendor);
});
```

```
router.delete("/:id", requireRole("admin"), auditAction("delete_vendor"), async (req, res) => {
  const parsed = DeleteVendorParams.safeParse(req.params);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }

  const [vendor] = await db
    .delete(vendorsTable)
    .where(eq(vendorsTable.id, parsed.data.id))
    .returning();

  if (!vendor) {
    res.status(404).json({ error: "Vendor not found" });
    return;
  }

  logger.info({ vendorId: vendor.id }, "Vendor deleted");
  res.status(204).send();
});

export default router;
```

[artifacts/api-server/src/routes/webhooks.ts](#)

```
/**
 * E-sign provider webhook receivers.
 * These endpoints are PUBLIC (no auth middleware) because providers cannot send our session cookie.
 * Each provider authenticates differently – we verify per-provider before mutating state.
 *
 * Wired into routes/index.ts as router.use("/webhooks", webhooksRouter).
 *
 * On a verified "signed" event we:
 * 1. Update document_envelopes.status, signed_at, append events
 * 2. Call workflow-engine.onEnvelopeSigned() to potentially enqueue the doctor fax
 *
 * NEVER throw – webhooks must always 200 OK so the provider doesn't endlessly retry.
 * Errors are logged + audited.
 */
import { Router } from "express";
import { db, documentEnvelopesTable, integrationsTable } from "@workspace/db";
import { eq } from "drizzle-orm";
import crypto from "crypto";
import { logger } from "../lib/logger";
import { auditLog } from "../lib/audit";
import { onEnvelopeSigned } from "../lib/workflow-engine";
import { getIntegrationCredentialsById } from "../integrations";

const router: Router = Router();

interface NormalizedEvent {
  externalEnvelopeId: string;
  status: "viewed" | "signed" | "declined" | "delivered" | "voided" | "expired" | "error";
  rawType: string;
  raw: Record<string, unknown>;
}

/**
 * Apply a normalized envelope event to the DB and trigger downstream side-effects.
 */
async function applyEnvelopeEvent(provider: string, evt: NormalizedEvent): Promise<void> {
  const [env] = await db
    .select()
    .from(documentEnvelopesTable)
    .where(eq(documentEnvelopesTable.external_envelope_id, evt.externalEnvelopeId));

  if (!env) {
    logger.warn(
      { provider, external_id: evt.externalEnvelopeId, status: evt.status },
      "Webhook event for unknown envelope – ignoring",
    );
    return;
  }

  const now = new Date();
  const newEvents = [
    ...(env.events || []),
    { type: evt.rawType, at: now.toISOString(), raw: evt.raw },
  ];

  const update: Record<string, unknown> = {
    status: evt.status,
    events: newEvents,
    updated_at: now,
  };
  if (evt.status === "viewed" && !env.viewed_at) update.viewed_at = now;
  if (evt.status === "delivered" && !env.delivered_at) update.delivered_at = now;
  if (evt.status === "signed") update.signed_at = now;
  if (evt.status === "declined") update.declined_at = now;

  await db
    .update(documentEnvelopesTable)
    .set(update)
    .where(eq(documentEnvelopesTable.id, env.id));

  await auditLog("document_envelope", String(env.id), `webhook_${evt.rawType}`, {
    provider,
    status: evt.status,
    external_id: evt.externalEnvelopeId,
  });

  if (evt.status === "signed") {
    try {
      const result = await onEnvelopeSigned(env.id);
      logger.info(
        { envelope_id: env.id, fax_job_id: result.queued_fax_job_id, reason: result.reason },
        "onEnvelopeSigned handled",
      );
    } catch (err) {
      logger.error({ err, envelope_id: env.id }, "onEnvelopeSigned threw – ignoring to keep webhook 200");
    }
  }
}

/**
 * Find the active integration row for a given provider so we can fetch its webhook secret.
 * If multiple are active we just take the first; admins are advised to keep one per provider.
 */
async function loadProviderSecret(provider: string, secretField: string): Promise<string | null> {
  const rows = await db
```

[illegible]

[illegible]

[artifacts/api-server/src/routes/workflow-settings.ts](#)

```
import { Router } from "express";
import { db, workflowSettingsTable, integrationsTable } from "@workspace/db";
import { eq } from "drizzle-orm";
import { authMiddleware, requireRole } from "../lib/rbac";
import { auditAction } from "../lib/rbac";
import { z } from "zod/v4";
import { listEsignProviders } from "../lib/esign";
import { listFaxProviders } from "../lib/fax";

const router = Router();
router.use(authMiddleware);

const settingsSchema = z.object({
  scope: z.string().min(1).max(100).default("global"),
  esign_provider_integration_id: z.number().int().nullish(),
  fax_provider_integration_id: z.number().int().nullish(),
  email_provider_integration_id: z.number().int().nullish(),
  default_email_from_name: z.string().max(255).nullish(),
  default_email_from_address: z.email().nullish(),
  default_fax_from_number: z.string().max(50).nullish(),
  auto_send_on_approval: z.boolean().optional(),
  auto_fax_doctor_on_signed_hipaa: z.boolean().optional(),
  esign_resend_after_days: z.number().int().min(0).max(60).optional(),
  esign_expire_after_days: z.number().int().min(1).max(180).optional(),
  max_send_retries: z.number().int().min(0).max(10).optional(),
  notify_on_failure_email: z.email().nullish(),
  med_records_cover_letter_template_id: z.number().int().nullish(),
  notes: z.string().nullish(),
});

router.get("/", requireRole("viewer"), async (_req, res) => {
  const rows = await db.select().from(workflowSettingsTable);
  res.json(rows);
});

router.get("/:scope", requireRole("viewer"), async (req, res) => {
  const [row] = await db
    .select()
    .from(workflowSettingsTable)
    .where(eq(workflowSettingsTable.scope, String(req.params.scope)));
  if (!row) {
    res.status(404).json({ error: "not_found" });
    return;
  }
  res.json(row);
});

/**
 * Upsert by scope. PUT /api/workflow-settings with { scope: "global", ...fields }
 */
router.put("/", requireRole("admin"), auditAction("update_workflow_settings"), async (req, res) => {
  const parsed = settingsSchema.safeParse(req.body);
  if (!parsed.success) {
    res.status(400).json({ error: parsed.error.message });
    return;
  }
  const d = parsed.data;
  const [existing] = await db
    .select()
    .from(workflowSettingsTable)
    .where(eq(workflowSettingsTable.scope, d.scope));
  if (existing) {
    const [row] = await db
      .update(workflowSettingsTable)
      .set({ ...d, updated_by_user_id: req.user?.id ?? null, updated_at: new Date() } as any)
      .where(eq(workflowSettingsTable.id, existing.id))
      .returning();
    res.json(row);
    return;
  }
  const [row] = await db
    .insert(workflowSettingsTable)
    .values({ ...d, updated_by_user_id: req.user?.id ?? null } as any)
    .returning();
  res.status(201).json(row);
});

/**
 * Helper for the CRM: list active integrations grouped by category so the admin
 * UI can render dropdowns for each provider slot.
 */
router.get("/_options/providers", requireRole("admin"), async (_req, res) => {
  const all = await db
    .select()
    .from(integrationsTable)
    .where(eq(integrationsTable.status, "active"));
  const supportedEsign = new Set(listEsignProviders());
  const supportedFax = new Set(listFaxProviders());
  const esign = all
    .filter((r) => r.type === "esign")
    .map((r) => ({
      id: r.id,
      name: r.name,
      provider: r.provider,
      adapter_implemented: supportedEsign.has(r.provider),
    }));
  const fax = all
    .filter((r) => r.type === "fax")
    .map((r) => ({
      id: r.id,
      name: r.name,
      provider: r.provider,
      adapter_implemented: supportedFax.has(r.provider),
    }));
  res.json({ esign, fax });
});
```



```
    }));
    const fax = all
      .filter((r) => r.type === "fax")
      .map((r) => ({
        id: r.id,
        name: r.name,
        provider: r.provider,
        adapter_implemented: supportedFax.has(r.provider),
      }));
    const email = all
      .filter((r) => r.type === "email" || r.provider === "sendgrid")
      .map((r) => ({
        id: r.id,
        name: r.name,
        provider: r.provider,
        adapter_implemented: r.provider === "sendgrid",
      }));
    res.json({ esign, fax, email });
  });
};

export default router;
```

[artifacts/api-server/src/worker.ts](#)

```
/**
 * MTOS Distributed Worker
 * Polls the PostgreSQL job_queue table and processes jobs.
 * Runs as a separate process – started via "pnpm run worker" command.
 */
import { db, casesTable, caseDocumentsTable, analysisTable, faxResultsTable, reviewQueueTable } from "@workspace/db";
import { eq } from "drizzle-orm";
import { logger } from "../lib/logger";
import { claimNextJob, markJobDone, markJobFailed } from "../lib/queue";
import { saveFile, readFile, listCaseFiles } from "../lib/vault";
import { extractFeatures } from "../lib/ai-extract";
import { scoreFeatures, scoreToVerdict } from "../lib/scoring";
import { auditLog } from "../lib/audit";
import { preprocessFaxBuffer, base64ToBuffer, detectMimeType } from "../lib/ocr-preprocess";
import { extractOcrData } from "../lib/ai-ocr";
import { withErrorFallback, createLoopGuard, DEFAULT_LIMITS } from "../lib/error-fallback";
import { handleSendEsignPacket, handleFaxMedRecordsRequest, handleSendWorkflowEmail } from "../lib/workflow-handlers";

const POLL_INTERVAL_MS = 2000;

async function processJob(job: {
  id: number;
  job_type: string;
  payload: unknown;
}) {
  const payload = job.payload as Record<string, unknown>;
  logger.info({ job_id: job.id, job_type: job.job_type }, "Processing job");

  if (job.job_type === "create_case") {
    const { case_id, data } = payload as { case_id: string; data: Record<string, unknown> };
    await db
      .insert(casesTable)
      .values({ id: case_id, data, status: "open" })
      .onConflictDoNothing();
    await auditLog("case", case_id, "created", { data });
    logger.info({ case_id }, "Case created");
  } else if (job.job_type === "ingest_file") {
    const { case_id, file_name, content, content_type } = payload as {
      case_id: string;
      file_name: string;
      content: string;
      content_type?: string;
    };
    const { path, hash, sizeBytes } = await saveFile(case_id, content, file_name);
    await db.insert(caseDocumentsTable).values({
      case_id,
      path,
      file_hash: hash,
      file_name,
      content_type: content_type ?? "text/plain",
    });
    await auditLog("case_document", case_id, "file_ingested", { file_name, hash, sizeBytes });
    logger.info({ case_id, path, sizeBytes }, "File ingested to vault");
  } else if (job.job_type === "analyze_case") {
    const { case_id } = payload as { case_id: string };

    const docs = await db
      .select()
      .from(caseDocumentsTable)
      .where(eq(caseDocumentsTable.case_id, case_id));

    if (docs.length === 0) {
      logger.warn({ case_id }, "No documents to analyze");
      return;
    }

    const loopGuard = createLoopGuard(DEFAULT_LIMITS);
    let successCount = 0;
    let failCount = 0;

    for (const doc of docs) {
      if (!loopGuard.canRetry() && failCount > 0) {
        await auditLog("case_analysis", case_id, "LOOP_ABORTED", {
          reason: "Loop guard limits exceeded",
          state: loopGuard.getState(),
        });
        logger.error({ case_id }, "LOOP_ABORTED – forced exit from analysis loop");
        break;
      }

      const result = await withErrorFallback(
        async (_input, attempt) => {
          let text = "";
          try {
            text = await readFile(doc.path);
          } catch (err) {
            throw new Error(`Could not read vault file ${doc.path}: ${err}`);
          }

          const features = await extractFeatures(text);
          const score = scoreFeatures(features);
          const verdict = scoreToVerdict(score);

          if (attempt > 0) {
            loopGuard.recordAIRcheck();
          }
        },
        {
          maxAttempts: 3,
          minDelay: 1000,
          maxDelay: 1000,
        }
      );
      // ... (rest of the analysis logic)
    }
  }
}
```

```

    }

    await db.insert(analysisTable).values({
      case_id,
      features: features as Record<string, unknown>,
      score,
      ai_model: "claude-haiku-4-5",
      raw_text_length: text.length,
    });

    await auditLog("analysis", case_id, "analyzed", {
      file: doc.file_name,
      score,
      verdict,
      features,
      attempt: attempt + 1,
    });

    logger.info({ case_id, score, verdict }, "Analysis complete");
    return { score, verdict, features };
  },
  { case_id, file: doc.file_name },
  {
    entity_type: "case_analysis",
    entity_id: case_id,
    source_module: "worker_analyze_case",
    failsafe_mode: "REVIEW_FAIL",
  }
);

if (result.success) {
  successCount++;
} else {
  failCount++;
  loopGuard.recordRetry();
  logger.error(
    { case_id, file: doc.file_name, output_state: result.output_state },
    "Analysis failed - routed to review queue"
  );
}
}

const finalStatus = failCount === docs.length ? "review_required" : "analyzed";
await db
  .update(casesTable)
  .set({ status: finalStatus, updated_at: new Date() })
  .where(eq(casesTable.id, case_id));

if (finalStatus === "review_required") {
  await auditLog("case", case_id, "analysis_failed", {
    total_docs: docs.length,
    success_count: successCount,
    fail_count: failCount,
  });
}
} else if (job.job_type === "process_fax") {
  const { fax_result_id, vault_path, source_file, mime_type } = payload as {
    fax_result_id: number;
    vault_path: string;
    source_file: string;
    mime_type: string;
  };

  await db
    .update(faxResultsTable)
    .set({ status: "processing" })
    .where(eq(faxResultsTable.id, fax_result_id));

  const faxResult = await withErrorFallback(
    async (_input, attempt) => {
      const rawBase64 = await readFile(vault_path);
      const rawBuffer = base64ToBuffer(rawBase64);
      const detectedMime = detectMimeType(rawBase64) || mime_type;

      const processedBuffer = await preprocessFaxBuffer(rawBuffer);
      const processedBase64 = processedBuffer.toString("base64");

      const grid = await extractOcrData(processedBase64, detectedMime);

      await db
        .update(faxResultsTable)
        .set({
          rx_number: grid.rx_number,
          drug_name: grid.drug_name,
          fill_date: grid.fill_date,
          quantity: grid.quantity,
          confidence: grid.confidence,
          raw_text: grid.raw_text,
          status: "done",
          processed_at: new Date(),
        })
        .where(eq(faxResultsTable.id, fax_result_id));

      await auditLog("fax", String(fax_result_id), "processed", {
        source_file,
        drug_name: grid.drug_name,
        confidence: grid.confidence,

```

```

        attempt: attempt + 1,
    });

    logger.info(
        { fax_result_id, drug_name: grid.drug_name, confidence: grid.confidence },
        "Fax OCR complete"
    );
    return grid;
}, { fax_result_id, vault_path, source_file },
{
    entity_type: "fax_ocr",
    entity_id: String(fax_result_id),
    source_module: "worker_process_fax",
    failsafe_mode: "REVIEW_FAIL",
}
);

if (!faxResult.success) {
    await db
        .update(faxResultsTable)
        .set({ status: "error", processed_at: new Date() })
        .where(eq(faxResultsTable.id, fax_result_id));
    logger.error({ fax_result_id, output_state: faxResult.output_state }, "Fax OCR failed – routed to review");
}
} else if (job.job_type === "send_esign_packet") {
    await handleSendEsignPacket(payload as unknown as Parameters<typeof handleSendEsignPacket>[0]);
} else if (job.job_type === "fax_med_records_request") {
    await handleFaxMedRecordsRequest(payload as unknown as Parameters<typeof handleFaxMedRecordsRequest>[0]);
} else if (job.job_type === "send_workflow_email") {
    await handleSendWorkflowEmail(payload as unknown as Parameters<typeof handleSendWorkflowEmail>[0]);
} else {
    logger.warn({ job_type: job.job_type }, "Unknown job type – skipping");
}
}

async function workerLoop() {
    logger.info("MTOS Worker started – polling job queue");

    while (true) {
        try {
            const job = await claimNextJob();
            if (job) {
                try {
                    await processJob(job as { id: number; job_type: string; payload: unknown });
                    await markJobDone(job.id as number);
                } catch (err) {
                    const msg = err instanceof Error ? err.message : String(err);
                    logger.error({ err, job_id: job.id }, "Job processing failed");
                    await markJobFailed(job.id as number, msg);
                }
            }
        } catch (err) {
            logger.error({ err }, "Worker poll error");
        }

        await new Promise((r) => setTimeout(r, POLL_INTERVAL_MS));
    }
}

workerLoop();

```

artifacts/api-server/tsconfig.json

```
{
  "extends": "../../tsconfig.base.json",
  "compilerOptions": {
    "outDir": "dist",
    "rootDir": "src",
    "types": ["node"]
  },
  "include": ["src"],
  "references": [
    {
      "path": "../../lib/db"
    },
    {
      "path": "../../lib/api-zod"
    },
    {
      "path": "../../lib/integrations-anthropic-ai"
    }
  ]
}
```

artifacts/mockup-sandbox/.replit-artifact/artifact.toml

```
kind = "design"
previewPath = "/__mockup"
title = "Canvas"
version = "1.0.0"
id = "XegfDyZt7Hqfw2Bb8Ghoy"

[[services]]
localPort = 8081
name = "Component Preview Server"
paths = ["/__mockup"]

[services.env]
PORT = "8081"
BASE_PATH = "/__mockup"

[services.development]
run = "pnpm --filter @workspace/mockup-sandbox run dev"
```

[artifacts/mockup-sandbox/components.json](#)

```
{
  "$schema": "https://ui.shadcn.com/schema.json",
  "style": "new-york",
  "rsc": false,
  "tsx": true,
  "tailwind": {
    "config": "",
    "css": "src/index.css",
    "baseColor": "neutral",
    "cssVariables": true,
    "prefix": ""
  },
  "iconLibrary": "lucide",
  "aliases": {
    "components": "@components",
    "utils": "@lib/Utils",
    "ui": "@components/ui",
    "lib": "@lib",
    "hooks": "@hooks"
  }
}
```

<artifacts/mockup-sandbox/index.html>

```
<!DOCTYPE html>
<!--
  This file is the entry for ALL routes, including /preview/* canvas iframes.
  Fonts are loaded here as non-blocking <link> tags (not CSS @import, which is render-blocking).
-->
<html lang="en" style="height: 100%">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=1" />

    <meta property="og:title" content="Mockup Canvas" />
    <meta property="og:description" content="UI prototyping sandbox with infinite canvas" />
    <meta property="og:type" content="website" />
    <meta name="twitter:card" content="summary_large_image" />
    <meta name="twitter:title" content="Mockup Canvas" />
    <meta name="twitter:description" content="UI prototyping sandbox with infinite canvas" />

    <title>Mockup Canvas</title>
    <link rel="icon" href="data:image/svg+xml,<svg xmlns='http://www.w3.org/2000/svg' viewBox='0 0 100 100'><text y='.9em' font-size='90'>0</text></svg>" />
    <!-- Non-blocking font bundle: renders with fallback fonts immediately, swaps in when loaded -->
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link rel="stylesheet" media="print" onload="this.media='all'"
      href="https://fonts.googleapis.com/css2?family=Architects+Daughter&family=DM+Sans:ital,opsz,wght@0,9..40,100..1000;1,9..40,100..1000&family=Fira+Code:wght@300..700&family=Geist+Mono:wght@100..900&family=Geist:wght@100..900&family=IBM+Plex+Mono:ital,wght@0,100;0,200;0,300;0,400;0,500;0,600;0,700;1,100;1,200;1,300;1,400;1,500;1,600;1,700&family=IBM+Plex+Sans:ital,wght@0,100..700;1,100..700&family=Inter:ital,opsz,wght@0,14..32,100..900;1,14..32,100..900&family=JetBrains+Mono:ital,wght@0,100..800;1,100..800&family=Libre+Baskerville:ital,wght@0,400;0,700;1,400&family=Lora:ital,wght@0,400..700;1,400..700&family=Merriweather:ital,opsz,wght@0,18..144,300..900;1,18..144,300..900&family=Montserrat:ital,wght@0,100..900;1,100..900&family=Open+Sans:ital,wght@0,300..800;1,300..800&family=Outfit:wght@100..900&family=Oxanium:wght@200..800&family=Playfair+Display:ital,wght@0,400..900;1,400..900&family=Plus+Jakarta+Sans:ital,wght@0,200..800;1,200..800&family=Poppins:ital,wght@0,100;0,200;0,300;0,400;0,500;0,600;0,700;0,800;0,900;1,100;1,200;1,300;1,400;1,500;1,600;1,700;1,800;1,900&family=Roboto+Mono:ital,wght@0,100..700;1,100..700&family=Roboto:ital,wght@0,100..900;1,100..900&family=Source+Code+Pro:ital,wght@0,200..900;1,200..900&family=Source+Serif+4:ital,opsz,wght@0,8..60,200..900;1,8..60,200..900&family=Space+Grotesk:wght@300..700&family=Space+Mono:ital,wght@0,400;0,700;1,400;1,700&display=swap">
    <noscript><link rel="stylesheet" href="https://fonts.googleapis.com/css2?family=Architects+Daughter&family=DM+Sans:ital,opsz,wght@0,9..40,100..1000;1,9..40,100..1000&family=Fira+Code:wght@300..700&family=Geist+Mono:wght@100..900&family=Geist:wght@100..900&family=IBM+Plex+Mono:ital,wght@0,100;0,200;0,300;0,400;0,500;0,600;0,700;1,100;1,200;1,300;1,400;1,500;1,600;1,700&family=IBM+Plex+Sans:ital,wght@0,100..700;1,100..700&family=Inter:ital,opsz,wght@0,14..32,100..900;1,14..32,100..900&family=JetBrains+Mono:ital,wght@0,100..800;1,100..800&family=Libre+Baskerville:ital,wght@0,400;0,700;1,400&family=Lora:ital,wght@0,400..700;1,400..700&family=Merriweather:ital,opsz,wght@0,18..144,300..900;1,18..144,300..900&family=Montserrat:ital,wght@0,100..900;1,100..900&family=Open+Sans:ital,wght@0,300..800;1,300..800&family=Outfit:wght@100..900&family=Oxanium:wght@200..800&family=Playfair+Display:ital,wght@0,400..900;1,400..900&family=Plus+Jakarta+Sans:ital,wght@0,200..800;1,200..800&family=Poppins:ital,wght@0,100;0,200;0,300;0,400;0,500;0,600;0,700;0,800;0,900;1,100;1,200;1,300;1,400;1,500;1,600;1,700;1,800;1,900&family=Roboto+Mono:ital,wght@0,100..700;1,100..700&family=Roboto:ital,wght@0,100..900;1,100..900&family=Source+Code+Pro:ital,wght@0,200..900;1,200..900&family=Source+Serif+4:ital,opsz,wght@0,8..60,200..900;1,8..60,200..900&family=Space+Grotesk:wght@300..700&family=Space+Mono:ital,wght@0,400;0,700;1,400;1,700&display=swap"></noscript>
  </head>
  <body style="height: 100%; margin: 0">
    <div id="root" style="height: 100%"></div>
    <script type="module" src="/src/main.tsx"></script>
  </body>
</html>
```


[artifacts/mockup-sandbox/mockupPreviewPlugin.ts](#)

```
import { mkdirSync, writeFileSync } from "fs";
import path from "path";
import glob from "fast-glob";
import chokidar from "chokidar";
import type { FSWatcher } from "chokidar";
import type { Plugin } from "vite";

const MOCKUPS_DIR = "src/components/mockups";
const GENERATED_MODULE = "src/.generated/mockup-components.ts";

interface DiscoveredComponent {
  globKey: string;
  importPath: string;
}

export function mockupPreviewPlugin(): Plugin {
  let root = "";
  let currentSource = "";
  let watcher: FSWatcher | null = null;

  function getMockupsAbsDir(): string {
    return path.join(root, MOCKUPS_DIR);
  }

  function getGeneratedModuleAbsPath(): string {
    return path.join(root, GENERATED_MODULE);
  }

  function isMockupFile(absolutePath: string): boolean {
    const rel = path.relative(getMockupsAbsDir(), absolutePath);
    return (
      !rel.startsWith(".") && !path.isAbsolute(rel) && rel.endsWith(".tsx")
    );
  }

  function isPreviewTarget(relativeToMockups: string): boolean {
    return relativeToMockups
      .split(path.sep)
      .every((segment) => !segment.startsWith("_"));
  }

  async function discoverComponents(): Promise<Array<DiscoveredComponent>> {
    const files = await glob(`${MOCKUPS_DIR}/**/*.tsx`, {
      cwd: root,
      ignore: ["**/_/**", "**/_*.tsx"],
    });

    return files.map((f) => ({
      globKey: `./" + f.slice("src/".length),
      importPath: path.posix.relative("src/.generated", f),
    }));
  }

  function generateSource(components: Array<DiscoveredComponent>): string {
    const entries = components
      .map(
        (c) =>
          `${JSON.stringify(c.globKey)}: () => import(${JSON.stringify(c.importPath)})`,
      )
      .join(",\n");

    return [
      `// This file is auto-generated by mockupPreviewPlugin.ts`,
      `type ModuleMap = Record<string, () => Promise<Record<string, unknown>>>;`,
      `export const modules: ModuleMap = {`,
      entries,
      `};`,
      ``,
    ].join("\n");
  }

  function shouldAutoRescan(pathname: string): boolean {
    return (
      pathname.includes("/components/mockups/") ||
      pathname.includes("/.generated/mockup-components")
    );
  }

  let refreshInFlight = false;
  let refreshQueued = false;

  async function refresh(): Promise<boolean> {
    if (refreshInFlight) {
      refreshQueued = true;
      return false;
    }

    refreshInFlight = true;
    let changed = false;
    try {
      const components = await discoverComponents();
      const newSource = generateSource(components);
      if (newSource !== currentSource) {
        currentSource = newSource;
        const generatedModuleAbsPath = getGeneratedModuleAbsPath();

```

```

        mkdirSync(path.dirname(generatedModuleAbsPath), { recursive: true });
        writeFileSync(generatedModuleAbsPath, currentSource);
        changed = true;
    }
} finally {
    refreshInFlight = false;
}

if (refreshQueued) {
    refreshQueued = false;
    const followUp = await refresh();
    return changed || followUp;
}

return changed;
}

async function onFileAddedOrRemoved(): Promise<void> {
    await refresh();
}

return {
    name: "mockup-preview",
    enforce: "pre",

    configResolved(config) {
        root = config.root;
    },

    async buildStart() {
        await refresh();
    },

    async configureServer(viteServer) {
        await refresh();

        const mockupsAbsDir = getMockupsAbsDir();
        mkdirSync(mockupsAbsDir, { recursive: true });

        watcher = chokidar.watch(mockupsAbsDir, {
            ignoreInitial: true,
            awaitWriteFinish: {
                stabilityThreshold: 100,
                pollInterval: 50,
            },
        });

        watcher.on("add", (file) => {
            if (
                isMockupFile(file) &&
                isPreviewTarget(path.relative(mockupsAbsDir, file))
            ) {
                void onFileAddedOrRemoved();
            }
        });

        watcher.on("unlink", (file) => {
            if (isMockupFile(file)) {
                void onFileAddedOrRemoved();
            }
        });

        viteServer.middlewares.use((req, res, next) => {
            const requestUrl = new URL(req.url ?? "/", "http://127.0.0.1");
            const pathname = requestUrl.pathname;
            const originalEnd = res.end.bind(res);

            res.end = ((...args: Parameters<typeof originalEnd>) => {
                if (res.statusCode === 404 && shouldAutoRescan(pathname)) {
                    void refresh();
                }
                return originalEnd(...args);
            }) as typeof res.end;

            next();
        });
    },

    async closeWatcher() {
        if (watcher) {
            await watcher.close();
        }
    },
};
}

```

[artifacts/mockup-sandbox/package.json](#)

```
{
  "name": "@workspace/mockup-sandbox",
  "version": "2.0.0",
  "type": "module",
  "private": true,
  "scripts": {
    "dev": "vite dev",
    "build": "vite build",
    "preview": "vite preview",
    "typecheck": "tsc -p tsconfig.json --noEmit"
  },
  "devDependencies": {
    "@hookform/resolvers": "^3.10.0",
    "@radix-ui/react-accordion": "^1.2.12",
    "@radix-ui/react-alert-dialog": "^1.1.15",
    "@radix-ui/react-aspect-ratio": "^1.1.8",
    "@radix-ui/react-avatar": "^1.1.11",
    "@radix-ui/react-checkbox": "^1.3.3",
    "@radix-ui/react-collapsible": "^1.1.12",
    "@radix-ui/react-context-menu": "^2.2.16",
    "@radix-ui/react-dialog": "^1.1.15",
    "@radix-ui/react-dropdown-menu": "^2.1.16",
    "@radix-ui/react-hover-card": "^1.1.15",
    "@radix-ui/react-label": "^2.1.8",
    "@radix-ui/react-menubar": "^1.1.16",
    "@radix-ui/react-navigation-menu": "^1.2.14",
    "@radix-ui/react-popover": "^1.1.15",
    "@radix-ui/react-progress": "^1.1.8",
    "@radix-ui/react-radio-group": "^1.3.8",
    "@radix-ui/react-scroll-area": "^1.2.10",
    "@radix-ui/react-select": "^2.2.6",
    "@radix-ui/react-separator": "^1.1.8",
    "@radix-ui/react-slider": "^1.3.6",
    "@radix-ui/react-slot": "^1.2.4",
    "@radix-ui/react-switch": "^1.2.6",
    "@radix-ui/react-tabs": "^1.1.13",
    "@radix-ui/react-toast": "^1.2.7",
    "@radix-ui/react-toggle": "^1.1.10",
    "@radix-ui/react-toggle-group": "^1.1.11",
    "@radix-ui/react-tooltip": "^1.2.8",
    "@replit/vite-plugin-cartographer": "catalog:",
    "@replit/vite-plugin-runtime-error-modal": "catalog:",
    "@tailwindcss/vite": "catalog:",
    "@types/node": "catalog:",
    "@types/react": "catalog:",
    "@types/react-dom": "catalog:",
    "@vitejs/plugin-react": "catalog:",
    "chokidar": "^4.0.3",
    "class-variance-authority": "catalog:",
    "clsx": "catalog:",
    "cmdk": "^1.1.1",
    "date-fns": "^3.6.0",
    "embla-carousel-react": "^8.6.0",
    "fast-glob": "^3.3.3",
    "framer-motion": "catalog:",
    "input-otp": "^1.4.2",
    "lucide-react": "catalog:",
    "next-themes": "^0.4.6",
    "react": "catalog:",
    "react-day-picker": "^9.11.1",
    "react-dom": "catalog:",
    "react-hook-form": "^7.66.0",
    "react-resizable-panels": "^2.1.9",
    "recharts": "^2.15.4",
    "sonner": "^2.0.7",
    "tailwind-merge": "catalog:",
    "tailwindcss": "catalog:",
    "tailwindcss-animate": "^1.0.7",
    "tw-animate-css": "^1.4.0",
    "vaul": "^1.1.2",
    "vite": "catalog:",
    "zod": "catalog:"
  }
}
```

[artifacts/mockup-sandbox/src/generated/mockup-components.ts](#)

```
// This file is auto-generated by mockupPreviewPlugin.ts.  
type ModuleMap = Record<string, () => Promise<Record<string, unknown>>>;  
export const modules: ModuleMap = {  
  
};
```

[artifacts/mockup-sandbox/src/App.tsx](#)

```
import { useEffect, useState, type ComponentType } from "react";

import { modules as discoveredModules } from "../generated/mockup-components";

type ModuleMap = Record<string, () => Promise<Record<string, unknown>>>;

function _resolveComponent(
  mod: Record<string, unknown>,
  name: string,
): ComponentType | undefined {
  const fns = Object.values(mod).filter(
    (v) => typeof v === "function",
  ) as ComponentType[];
  return (
    (mod.default as ComponentType) ||
    (mod.Preview as ComponentType) ||
    (mod[name] as ComponentType) ||
    fns[fns.length - 1]
  );
}

function PreviewRenderer({
  componentPath,
  modules,
}: {
  componentPath: string;
  modules: ModuleMap;
}) {
  const [Component, setComponent] = useState<ComponentType | null>(null);
  const [error, setError] = useState<string | null>(null);

  useEffect(() => {
    let cancelled = false;

    setComponent(null);
    setError(null);

    async function loadComponent(): Promise<void> {
      const key = `./components/mockups/${componentPath}.tsx`;
      const loader = modules[key];
      if (!loader) {
        setError(`No component found at ${componentPath}.tsx`);
        return;
      }

      try {
        const mod = await loader();
        if (cancelled) {
          return;
        }
        const name = componentPath.split("/").pop()!;
        const comp = _resolveComponent(mod, name);
        if (!comp) {
          setError(
            `No exported React component found in ${componentPath}.tsx\n\nMake sure the file has at least one exported function component.`
          );
          return;
        }
        setComponent(() => comp);
      } catch (e) {
        if (cancelled) {
          return;
        }

        const message = e instanceof Error ? e.message : String(e);
        setError(`Failed to load preview.\n${message}`);
      }
    }

    void loadComponent();

    return () => {
      cancelled = true;
    };
  }, [componentPath, modules]);

  if (error) {
    return (
      <pre style={{ color: "red", padding: "2rem", fontFamily: "system-ui" }}>
        {error}
      </pre>
    );
  }

  if (!Component) return null;

  return <Component />;
}

function getBasePath(): string {
  return import.meta.env.BASE_URL.replace(/\/$/, "");
}

function getPreviewExamplePath(): string {
```

```

    const basePath = getBasePath();
    return `${basePath}/preview/ComponentName`;
}

function Gallery() {
  return (
    <div className="min-h-screen bg-gray-50 flex items-center justify-center p-8">
      <div className="text-center max-w-md">
        <h1 className="text-2xl font-semibold text-gray-900 mb-3">
          Component Preview Server
        </h1>
        <p className="text-gray-500 mb-4">
          This server renders individual components for the workspace canvas.
        </p>
        <p className="text-sm text-gray-400">
          Access component previews at{" "}
          <code className="bg-gray-100 px-1.5 py-0.5 rounded text-gray-600">
            {getPreviewExamplePath()}
          </code>
        </p>
      </div>
    </div>
  );
}

function getPreviewPath(): string | null {
  const basePath = getBasePath();
  const { pathname } = window.location;
  const local =
    basePath && pathname.startsWith(basePath)
      ? pathname.slice(basePath.length) || "/"
      : pathname;
  const match = local.match(/^\/preview\/(.+)\$/);
  return match ? match[1] : null;
}

function App() {
  const previewPath = getPreviewPath();

  if (previewPath) {
    return (
      <PreviewRenderer
        componentPath={previewPath}
        modules={discoveredModules}
      />
    );
  }

  return <Gallery />;
}

export default App;

```

[artifacts/mockup-sandbox/src/components/ui/accordion.tsx](#)

```
import * as React from "react"
import * as AccordionPrimitive from "@radix-ui/react-accordion"
import { ChevronDown } from "lucide-react"

import { cn } from "@lib/utls"

const Accordion = AccordionPrimitive.Root

const AccordionItem = React.forwardRef<
  React.ElementRef<typeof AccordionPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof AccordionPrimitive.Item>
>(({ className, ...props }, ref) => (
  <AccordionPrimitive.Item
    ref={ref}
    className={cn("border-b", className)}
    {...props}
  />
))
AccordionItem.displayName = "AccordionItem"

const AccordionTrigger = React.forwardRef<
  React.ElementRef<typeof AccordionPrimitive.Trigger>,
  React.ComponentPropsWithoutRef<typeof AccordionPrimitive.Trigger>
>(({ className, children, ...props }, ref) => (
  <AccordionPrimitive.Header className="flex">
    <AccordionPrimitive.Trigger
      ref={ref}
      className={cn(
        "flex flex-1 items-center justify-between py-4 text-sm font-medium transition-all hover:underline text-left [&[data-state=open]>svg]:rotate-180",
        className
      )}
      {...props}
    >
      {children}
      <ChevronDown className="h-4 w-4 shrink-0 text-muted-foreground transition-transform duration-200" />
    </AccordionPrimitive.Trigger>
  </AccordionPrimitive.Header>
))
AccordionTrigger.displayName = AccordionPrimitive.Trigger.displayName

const AccordionContent = React.forwardRef<
  React.ElementRef<typeof AccordionPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof AccordionPrimitive.Content>
>(({ className, children, ...props }, ref) => (
  <AccordionPrimitive.Content
    ref={ref}
    className="overflow-hidden text-sm data-[state=closed]:animate-accordion-up data-[state=open]:animate-accordion-down"
    {...props}
  >
    <div className={cn("pb-4 pt-0", className)}>{children}</div>
  </AccordionPrimitive.Content>
))
AccordionContent.displayName = AccordionPrimitive.Content.displayName

export { Accordion, AccordionItem, AccordionTrigger, AccordionContent }
```

[artifacts/mockup-sandbox/src/components/ui/alert-dialog.tsx](#)

```
import * as React from "react"
import * as AlertDialogPrimitive from "@radix-ui/react-alert-dialog"

import { cn } from "@lib/utils"
import { buttonVariants } from "@components/ui/button"

const AlertDialog = AlertDialogPrimitive.Root

const AlertDialogTrigger = AlertDialogPrimitive.Trigger

const AlertDialogPortal = AlertDialogPrimitive.Portal

const AlertDialogOverlay = React.forwardRef<
  React.ElementRef<typeof AlertDialogPrimitive.Overlay>,
  React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Overlay>
>(({ className, ...props }, ref) => (
  <AlertDialogPrimitive.Overlay
    className={cn(
      "fixed inset-0 z-50 bg-black/80 data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0",
      className
    )}
    {...props}
    ref={ref}
  />
))
AlertDialogOverlay.displayName = AlertDialogPrimitive.Overlay.displayName

const AlertDialogContent = React.forwardRef<
  React.ElementRef<typeof AlertDialogPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Content>
>(({ className, ...props }, ref) => (
  <AlertDialogPortal>
    <AlertDialogOverlay />
    <AlertDialogPrimitive.Content
      ref={ref}
      className={cn(
        "fixed left-[50%] top-[50%] z-50 grid w-full max-w-lg translate-x-[-50%] translate-y-[-50%] gap-4 border bg-background p-6 shadow-lg duration-200 data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[state=closed]:slide-out-to-left-1/2 data-[state=closed]:slide-out-to-top-48% data-[state=open]:slide-in-from-left-1/2 data-[state=open]:slide-in-from-top-48% sm:rounded-lg",
        className
      )}
      {...props}
    />
  </AlertDialogPortal>
))
AlertDialogContent.displayName = AlertDialogPrimitive.Content.displayName

const AlertDialogHeader = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn(
      "flex flex-col space-y-2 text-center sm:text-left",
      className
    )}
    {...props}
  />
)
AlertDialogHeader.displayName = "AlertDialogHeader"

const AlertDialogFooter = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn(
      "flex flex-col-reverse sm:flex-row sm:justify-end sm:space-x-2",
      className
    )}
    {...props}
  />
)
AlertDialogFooter.displayName = "AlertDialogFooter"

const AlertDialogTitle = React.forwardRef<
  React.ElementRef<typeof AlertDialogPrimitive.Title>,
  React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Title>
>(({ className, ...props }, ref) => (
  <AlertDialogPrimitive.Title
    ref={ref}
    className={cn("text-lg font-semibold", className)}
    {...props}
  />
))
AlertDialogTitle.displayName = AlertDialogPrimitive.Title.displayName

const AlertDialogDescription = React.forwardRef<
  React.ElementRef<typeof AlertDialogPrimitive.Description>,
  React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Description>
>(({ className, ...props }, ref) => (
  <AlertDialogPrimitive.Description
```



```

      ref={ref}
      className={cn("text-sm text-muted-foreground", className)}
      {...props}
    />
  ))
  AlertDialogDescription.displayName =
    AlertDialogPrimitive.Description.displayName

  const AlertDialogAction = React.forwardRef<
    React.ElementRef<typeof AlertDialogPrimitive.Action>,
    React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Action>
  >(({ className, ...props }, ref) => (
    <AlertDialogPrimitive.Action
      ref={ref}
      className={cn(buttonVariants(), className)}
      {...props}
    />
  ))
  AlertDialogAction.displayName = AlertDialogPrimitive.Action.displayName

  const AlertDialogCancel = React.forwardRef<
    React.ElementRef<typeof AlertDialogPrimitive.Cancel>,
    React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Cancel>
  >(({ className, ...props }, ref) => (
    <AlertDialogPrimitive.Cancel
      ref={ref}
      className={cn(
        buttonVariants({ variant: "outline" }),
        "mt-2 sm:mt-0",
        className
      )}
      {...props}
    />
  ))
  AlertDialogCancel.displayName = AlertDialogPrimitive.Cancel.displayName

  export {
    AlertDialog,
    AlertDialogPortal,
    AlertDialogOverlay,
    AlertDialogTrigger,
    AlertDialogContent,
    AlertDialogHeader,
    AlertDialogFooter,
    AlertDialogTitle,
    AlertDialogDescription,
    AlertDialogAction,
    AlertDialogCancel,
  }
}

```

[artifacts/mockup-sandbox/src/components/ui/alert.tsx](#)

```
import * as React from "react"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@/lib/utils"

const alertVariants = cva(
  "relative w-full rounded-lg border px-4 py-3 text-sm [&svg+div]:translate-y-[-3px] [&svg]:absolute [&svg]:left-4 [&svg]:top-4 [&svg]:text-foreground [&svg~*]:pl-7",
  {
    variants: {
      variant: {
        default: "bg-background text-foreground",
        destructive:
          "border-destructive/50 text-destructive dark:border-destructive [&svg]:text-destructive",
      },
    },
    defaultVariants: {
      variant: "default",
    },
  }
)

const Alert = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement> & VariantProps<typeof alertVariants>
>(({ className, variant, ...props }, ref) => (
  <div
    ref={ref}
    role="alert"
    className={cn(alertVariants({ variant }), className)}
    {...props}
  />
))
Alert.displayName = "Alert"

const AlertTitle = React.forwardRef<
  HTMLParagraphElement,
  React.HTMLAttributes<HTMLHeadingElement>
>(({ className, ...props }, ref) => (
  <h5
    ref={ref}
    className={cn("mb-1 font-medium leading-none tracking-tight", className)}
    {...props}
  />
))
AlertTitle.displayName = "AlertTitle"

const AlertDescription = React.forwardRef<
  HTMLParagraphElement,
  React.HTMLAttributes<HTMLParagraphElement>
>(({ className, ...props }, ref) => (
  <div
    ref={ref}
    className={cn("text-sm [&_p]:leading-relaxed", className)}
    {...props}
  />
))
AlertDescription.displayName = "AlertDescription"

export { Alert, AlertTitle, AlertDescription }
```

[artifacts/mockup-sandbox/src/components/ui/aspect-ratio.tsx](#)

```
import * as AspectRatioPrimitive from "@radix-ui/react-aspect-ratio"

const AspectRatio = AspectRatioPrimitive.Root

export { AspectRatio }
```

[artifacts/mockup-sandbox/src/components/ui/avatar.tsx](#)

```
"use client"

import * as React from "react"
import * as AvatarPrimitive from "@radix-ui/react-avatar"

import { cn } from "@lib/utils"

const Avatar = React.forwardRef<
  React.ElementRef<typeof AvatarPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof AvatarPrimitive.Root>
>(({ className, ...props }, ref) => (
  <AvatarPrimitive.Root
    ref={ref}
    className={cn(
      "relative flex h-10 w-10 shrink-0 overflow-hidden rounded-full",
      className
    )}
    {...props}
  />
))
Avatar.displayName = AvatarPrimitive.Root.displayName

const AvatarImage = React.forwardRef<
  React.ElementRef<typeof AvatarPrimitive.Image>,
  React.ComponentPropsWithoutRef<typeof AvatarPrimitive.Image>
>(({ className, ...props }, ref) => (
  <AvatarPrimitive.Image
    ref={ref}
    className={cn("aspect-square h-full w-full", className)}
    {...props}
  />
))
AvatarImage.displayName = AvatarPrimitive.Image.displayName

const AvatarFallback = React.forwardRef<
  React.ElementRef<typeof AvatarPrimitive.Fallback>,
  React.ComponentPropsWithoutRef<typeof AvatarPrimitive.Fallback>
>(({ className, ...props }, ref) => (
  <AvatarPrimitive.Fallback
    ref={ref}
    className={cn(
      "flex h-full w-full items-center justify-center rounded-full bg-muted",
      className
    )}
    {...props}
  />
))
AvatarFallback.displayName = AvatarPrimitive.Fallback.displayName

export { Avatar, AvatarImage, AvatarFallback }
```

[artifacts/mockup-sandbox/src/components/ui/badge.tsx](#)

```
import * as React from "react"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"

const badgeVariants = cva(
  "whitespace-nowrap inline-flex items-center rounded-md border px-2.5 py-0.5 text-xs font-semibold transition-colors focus:outline-none focus:ring-2 focus:ring-ring focus:ring-offset-2" +
  " hover-elevate ",
  {
    variants: {
      variant: {
        default:
          "border-transparent bg-primary text-primary-foreground shadow-xs",
        secondary:
          "border-transparent bg-secondary text-secondary-foreground",
        destructive:
          "border-transparent bg-destructive text-destructive-foreground shadow-xs",
        outline: "text-foreground border [border-color:var(--badge-outline)]",
      },
    },
    defaultVariants: {
      variant: "default",
    },
  },
)

export interface BadgeProps
  extends React.HTMLAttributes<HTMLDivElement>,
    VariantProps<typeof badgeVariants> {}

function Badge({ className, variant, ...props }: BadgeProps) {
  return (
    <div className={cn(badgeVariants({ variant }), className)} {...props} />
  )
}

export { Badge, badgeVariants }
```

[artifacts/mockup-sandbox/src/components/ui/breadcrumb.tsx](#)

```
import * as React from "react"
import { Slot } from "@radix-ui/react-slot"
import { ChevronRight, MoreHorizontal } from "lucide-react"

import { cn } from "@lib/utils"

const Breadcrumb = React.forwardRef<
  HTMLDivElement,
  React.ComponentPropsWithoutRef<"nav"> & {
    separator?: React.ReactNode
  }
>(({ ...props }, ref) => <nav ref={ref} aria-label="breadcrumb" {...props} />)
Breadcrumb.displayName = "Breadcrumb"

const BreadcrumbList = React.forwardRef<
  HTMLListElement,
  React.ComponentPropsWithoutRef<"ol">
>(({ className, ...props }, ref) => (
  <ol
    ref={ref}
    className={cn(
      "flex flex-wrap items-center gap-1.5 break-words text-sm text-muted-foreground sm:gap-2.5",
      className
    )}
    {...props}
  />
))
BreadcrumbList.displayName = "BreadcrumbList"

const BreadcrumbItem = React.forwardRef<
  HTMLLIElement,
  React.ComponentPropsWithoutRef<"li">
>(({ className, ...props }, ref) => (
  <li
    ref={ref}
    className={cn("inline-flex items-center gap-1.5", className)}
    {...props}
  />
))
BreadcrumbItem.displayName = "BreadcrumbItem"

const BreadcrumbLink = React.forwardRef<
  HTMLAnchorElement,
  React.ComponentPropsWithoutRef<"a"> & {
    asChild?: boolean
  }
>(({ asChild, className, ...props }, ref) => {
  const Comp = asChild ? Slot : "a"

  return (
    <Comp
      ref={ref}
      className={cn("transition-colors hover:text-foreground", className)}
      {...props}
    />
  )
})
BreadcrumbLink.displayName = "BreadcrumbLink"

const BreadcrumbPage = React.forwardRef<
  HTMLSpanElement,
  React.ComponentPropsWithoutRef<"span">
>(({ className, ...props }, ref) => (
  <span
    ref={ref}
    role="link"
    aria-disabled="true"
    aria-current="page"
    className={cn("font-normal text-foreground", className)}
    {...props}
  />
))
BreadcrumbPage.displayName = "BreadcrumbPage"

const BreadcrumbSeparator = ({
  children,
  className,
  ...props
}: React.ComponentProps<"li">) => (
  <li
    role="presentation"
    aria-hidden="true"
    className={cn("[&svg]:w-3.5 [&svg]:h-3.5", className)}
    {...props}
  >
    {children ?? <ChevronRight />}
  </li>
)
BreadcrumbSeparator.displayName = "BreadcrumbSeparator"

const BreadcrumbEllipsis = ({
  className,
  ...props
}: React.ComponentProps<"span">) => (
  <span
```

```
    role="presentation"
    aria-hidden="true"
    className={cn("flex h-9 w-9 items-center justify-center", className)}
    {...props}
  >
    <MoreHorizontal className="h-4 w-4" />
    <span className="sr-only">More</span>
  </span>
)
BreadcrumbEllipsis.displayName = "BreadcrumbElipssis"

export {
  Breadcrumb,
  BreadcrumbList,
  BreadcrumbItem,
  BreadcrumbLink,
  BreadcrumbPage,
  BreadcrumbSeparator,
  BreadcrumbEllipsis,
}
```

[artifacts/mockup-sandbox/src/components/ui/button-group.tsx](#)

```
import { Slot } from "@radix-ui/react-slot"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"
import { Separator } from "@components/ui/separator"

const buttonGroupVariants = cva(
  "flex w-fit items-stretch has-[:>[data-slot=button-group]]:gap-2 [>]:focus-visible:relative [>]:focus-visible:z-10 has-[:select[aria-hidden=true]:last-child]:[>[data-slot=select-trigger]:last-of-type]:rounded-r-md [>[data-slot=select-trigger]:not([class*='w-']]):w-fit [>input]:flex-1",
  {
    variants: {
      orientation: {
        horizontal:
          "[>*:not(:first-child)]:rounded-l-none [>*:not(:first-child)]:border-l-0 [>*:not(:last-child)]:rounded-r-none",
        vertical:
          "flex-col [>*:not(:first-child)]:rounded-t-none [>*:not(:first-child)]:border-t-0 [>*:not(:last-child)]:rounded-b-none",
      },
    },
    defaultVariants: {
      orientation: "horizontal",
    },
  }
)

function ButtonGroup({
  className,
  orientation,
  ...props
}: React.ComponentProps<"div"> & VariantProps<typeof buttonGroupVariants>) {
  return (
    <div
      role="group"
      data-slot="button-group"
      data-orientation={orientation}
      className={cn(buttonGroupVariants({ orientation }), className)}
      {...props}
    />
  )
}

function ButtonGroupText({
  className,
  asChild = false,
  ...props
}: React.ComponentProps<"div"> & {
  asChild?: boolean
}) {
  const Comp = asChild ? Slot : "div"

  return (
    <Comp
      className={cn(
        "bg-muted shadow-xs flex items-center gap-2 rounded-md border px-4 text-sm font-medium",
        [&_svg:not([class*='size-'])]:size-4 [&_svg]:pointer-events-none,
        className
      )}
      {...props}
    />
  )
}

function ButtonGroupSeparator({
  className,
  orientation = "vertical",
  ...props
}: React.ComponentProps<typeof Separator>) {
  return (
    <Separator
      data-slot="button-group-separator"
      orientation={orientation}
      className={cn(
        "bg-input relative !m-0 self-stretch data-[orientation=vertical]:h-auto",
        className
      )}
      {...props}
    />
  )
}

export {
  ButtonGroup,
  ButtonGroupSeparator,
  ButtonGroupText,
  buttonGroupVariants,
}
```


[artifacts/mockup-sandbox/src/components/ui/button.tsx](#)

```
import * as React from "react"
import { Slot } from "@radix-ui/react-slot"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"

const buttonVariants = cva(
  "inline-flex items-center justify-center gap-2 whitespace-nowrap rounded-md text-sm font-medium transition-colors focus-visible:outline-none focus-visible:ring-1 focus-visible:ring-ring disabled:pointer-events-none disabled:opacity-50 [&_svg]:pointer-events-none [&_svg]:size-4 [&_svg]:shrink-0" +
  " hover-elevate active-elevate-2",
  {
    variants: {
      variant: {
        default:
          "bg-primary text-primary-foreground border border-primary-border",
        destructive:
          "bg-destructive text-destructive-foreground shadow-sm border-destructive-border",
        outline:
          "border [border-color:var(--button-outline)] shadow-xs active:shadow-none",
        secondary:
          "border bg-secondary text-secondary-foreground border border-secondary-border",
        ghost: "border border-transparent",
        link: "text-primary underline-offset-4 hover:underline",
      },
      size: {
        default: "min-h-9 px-4 py-2",
        sm: "min-h-8 rounded-md px-3 text-xs",
        lg: "min-h-10 rounded-md px-8",
        icon: "h-9 w-9",
      },
    },
    defaultVariants: {
      variant: "default",
      size: "default",
    },
  },
)

export interface ButtonProps
  extends React.ButtonHTMLAttributes<HTMLButtonElement>,
    VariantProps<typeof buttonVariants> {
  asChild?: boolean
}

const Button = React.forwardRef<HTMLButtonElement, ButtonProps>(
  ({ className, variant, size, asChild = false, ...props }, ref) => {
    const Comp = asChild ? Slot : "button"
    return (
      <Comp
        className={cn(buttonVariants({ variant, size, className }))}
        ref={ref}
        {...props}
      />
    )
  }
)

Button.displayName = "Button"

export { Button, buttonVariants }
```

[artifacts/mockup-sandbox/src/components/ui/calendar.tsx](#)

```
"use client"

import * as React from "react"
import {
  ChevronDownIcon,
  ChevronLeftIcon,
  ChevronRightIcon,
} from "lucide-react"
import { DayButton, DayPicker, getDefaultClassNames } from "react-day-picker"

import { cn } from "@lib/utls"
import { Button, buttonVariants } from "@components/ui/button"

function Calendar({
  className,
  classNames,
  showOutsideDays = true,
  captionLayout = "label",
  buttonVariant = "ghost",
  formatters,
  components,
  ...props
}: React.ComponentProps<typeof DayPicker> & {
  buttonVariant?: React.ComponentProps<typeof Button>["variant"]
}) {
  const defaultClassNames = getDefaultClassNames()

  return (
    <DayPicker
      showOutsideDays={showOutsideDays}
      className={cn(
        "bg-background group/calendar p-3 [--cell-size:2rem] [[data-slot=card-content]_&]:bg-transparent [[data-slot=popover-content]_&]:bg-transparent",
        String.raw`rtl:**:[.rdp-button\_next>svg]:rotate-180`,
        String.raw`rtl:**:[.rdp-button\_previous>svg]:rotate-180`,
        className
      )}
      captionLayout={captionLayout}
      formatters={{
        formatMonthDropdown: (date) =>
          date.toLocaleString("default", { month: "short" }),
        ...formatters,
      }}
      classNames={{
        root: cn("w-fit", defaultClassNames.root),
        months: cn(
          "relative flex flex-col gap-4 md:flex-row",
          defaultClassNames.months
        ),
        month: cn("flex w-full flex-col gap-4", defaultClassNames.month),
        nav: cn(
          "absolute inset-x-0 top-0 flex w-full items-center justify-between gap-1",
          defaultClassNames.nav
        ),
        button_previous: cn(
          buttonVariants({ variant: buttonVariant }),
          "h-[--cell-size] w-[--cell-size] select-none p-0 aria-disabled:opacity-50",
          defaultClassNames.button_previous
        ),
        button_next: cn(
          buttonVariants({ variant: buttonVariant }),
          "h-[--cell-size] w-[--cell-size] select-none p-0 aria-disabled:opacity-50",
          defaultClassNames.button_next
        ),
        month_caption: cn(
          "flex h-[--cell-size] w-full items-center justify-center px-[--cell-size]",
          defaultClassNames.month_caption
        ),
        dropdowns: cn(
          "flex h-[--cell-size] w-full items-center justify-center gap-1.5 text-sm font-medium",
          defaultClassNames.dropdowns
        ),
        dropdown_root: cn(
          "has-focus:border-ring border-input shadow-xs has-focus:ring-ring/50 has-focus:ring-[3px] relative rounded-md border",
          defaultClassNames.dropdown_root
        ),
        dropdown: cn(
          "bg-popover absolute inset-0 opacity-0",
          defaultClassNames.dropdown
        ),
        caption_label: cn(
          "select-none font-medium",
          captionLayout === "label"
            ? "text-sm"
            : "[&>svg]:text-muted-foreground flex h-8 items-center gap-1 rounded-md pl-2 pr-1 text-sm [&>svg]:size-3.5",
          defaultClassNames.caption_label
        ),
        table: "w-full border-collapse",
        weekdays: cn("flex", defaultClassNames.weekdays),
        weekday: cn(
          "text-muted-foreground flex-1 select-none rounded-md text-[0.8rem] font-normal",
          defaultClassNames.weekday
        ),
        week: cn("mt-2 flex w-full", defaultClassNames.week),
        week_number_header: cn(
```

```

    "w-[--cell-size] select-none",
    defaultClassNames.week_number_header
  ),
  week_number: cn(
    "text-muted-foreground select-none text-[0.8rem]",
    defaultClassNames.week_number
  ),
  day: cn(
    "group/day relative aspect-square h-full w-full select-none p-0 text-center [&:first-child[data-selected=true]_button]:rounded-l-md [&:last-child[data-selected=true]_button]:rounded-r-md",
    defaultClassNames.day
  ),
  range_start: cn(
    "bg-accent rounded-l-md",
    defaultClassNames.range_start
  ),
  range_middle: cn("rounded-none", defaultClassNames.range_middle),
  range_end: cn("bg-accent rounded-r-md", defaultClassNames.range_end),
  today: cn(
    "bg-accent text-accent-foreground rounded-md data-[selected=true]:rounded-none",
    defaultClassNames.today
  ),
  outside: cn(
    "text-muted-foreground aria-selected:text-muted-foreground",
    defaultClassNames.outside
  ),
  disabled: cn(
    "text-muted-foreground opacity-50",
    defaultClassNames.disabled
  ),
  hidden: cn("invisible", defaultClassNames.hidden),
  ...ClassNames,
}}
components={{
  Root: ({ className, rootRef, ...props }) => {
    return (
      <div
        data-slot="calendar"
        ref={rootRef}
        className={cn(className)}
        {...props}
      />
    )
  },
  Chevron: ({ className, orientation, ...props }) => {
    if (orientation === "left") {
      return (
        <ChevronLeftIcon className={cn("size-4", className)} {...props} />
      )
    }

    if (orientation === "right") {
      return (
        <ChevronRightIcon
          className={cn("size-4", className)}
          {...props}
        />
      )
    }

    return (
      <ChevronDownIcon className={cn("size-4", className)} {...props} />
    )
  },
  DayButton: CalendarDayButton,
  WeekNumber: ({ children, ...props }) => {
    return (
      <td {...props}>
        <div className="flex size-[--cell-size] items-center justify-center text-center">
          {children}
        </div>
      </td>
    )
  },
  ...components,
}}
{...props}
) />
}

```

```

function CalendarDayButton({
  className,
  day,
  modifiers,
  ...props
}: React.ComponentProps<typeof DayButton>) {
  const defaultClassNames = getDefaultClassNames()

  const ref = React.useRef<HTMLButtonElement>(null)
  React.useEffect(() => {
    if (modifiers.focused) ref.current?.focus()
  }, [modifiers.focused])

  return (
    <Button
      ref={ref}
    />
  )
}

```

```

variant="ghost"
size="icon"
data-day={day.date.toLocaleDateString()}
data-selected-single={
  modifiers.selected &&
  !modifiers.range_start &&
  !modifiers.range_end &&
  !modifiers.range_middle
}
data-range-start={modifiers.range_start}
data-range-end={modifiers.range_end}
data-range-middle={modifiers.range_middle}
className={cn(
  "data-[selected-single=true]:bg-primary data-[selected-single=true]:text-primary-foreground data-[range-middle=true]:bg-
accent data-[range-middle=true]:text-accent-foreground data-[range-start=true]:bg-primary data-[range-start=true]:text-primary-
foreground data-[range-end=true]:bg-primary data-[range-end=true]:text-primary-foreground group-data-[focused=true]/day:border-
ring group-data-[focused=true]/day:ring-ring/50 flex aspect-square h-auto w-full min-w-[-cell-size] flex-col gap-1 font-normal
leading-none data-[range-end=true]:rounded-md data-[range-middle=true]:rounded-none data-[range-start=true]:rounded-md group-
data-[focused=true]/day:relative group-data-[focused=true]/day:z-10 group-data-[focused=true]/day:ring-[3px] [>span]:text-xs
[&>span]:opacity-70",
  defaultClassNames.day,
  className
)}
{...props}
) />
}
}

export { Calendar, CalendarDayButton }

```

[artifacts/mockup-sandbox/src/components/ui/card.tsx](#)

```
import * as React from "react"

import { cn } from "@lib/utils"

const Card = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => (
  <div
    ref={ref}
    className={cn(
      "rounded-xl border bg-card text-card-foreground shadow",
      className
    )}
    {...props}
  />
))
Card.displayName = "Card"

const CardHeader = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => (
  <div
    ref={ref}
    className={cn("flex flex-col space-y-1.5 p-6", className)}
    {...props}
  />
))
CardHeader.displayName = "CardHeader"

const CardTitle = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => (
  <div
    ref={ref}
    className={cn("font-semibold leading-none tracking-tight", className)}
    {...props}
  />
))
CardTitle.displayName = "CardTitle"

const CardDescription = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => (
  <div
    ref={ref}
    className={cn("text-sm text-muted-foreground", className)}
    {...props}
  />
))
CardDescription.displayName = "CardDescription"

const CardContent = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => (
  <div ref={ref} className={cn("p-6 pt-0", className)} {...props} />
))
CardContent.displayName = "CardContent"

const CardFooter = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => (
  <div
    ref={ref}
    className={cn("flex items-center p-6 pt-0", className)}
    {...props}
  />
))
CardFooter.displayName = "CardFooter"

export { Card, CardHeader, CardFooter, CardTitle, CardDescription, CardContent }
```

[artifacts/mockup-sandbox/src/components/ui/carousel.tsx](#)

```
import * as React from "react"
import useEmblaCarousel, {
  type UseEmblaCarouselType,
} from "embla-carousel-react"
import { ArrowLeft, ArrowRight } from "lucide-react"

import { cn } from "@lib/utlils"
import { Button } from "@components/ui/button"

type CarouselApi = UseEmblaCarouselType[1]
type UseCarouselParameters = Parameters<typeof useEmblaCarousel>
type CarouselOptions = UseCarouselParameters[0]
type CarouselPlugin = UseCarouselParameters[1]

type CarouselProps = {
  opts?: CarouselOptions
  plugins?: CarouselPlugin
  orientation?: "horizontal" | "vertical"
  setApi?: (api: CarouselApi) => void
}

type CarouselContextProps = {
  carouselRef: ReturnType<typeof useEmblaCarousel>[0]
  api: ReturnType<typeof useEmblaCarousel>[1]
  scrollPrev: () => void
  scrollNext: () => void
  canScrollPrev: boolean
  canScrollNext: boolean
} & CarouselProps

const CarouselContext = React.createContext<CarouselContextProps | null>(null)

function useCarousel() {
  const context = React.useContext(CarouselContext)

  if (!context) {
    throw new Error("useCarousel must be used within a <Carousel />")
  }

  return context
}

const Carousel = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement> & CarouselProps
>((
  {
    orientation = "horizontal",
    opts,
    setApi,
    plugins,
    className,
    children,
    ...props
  },
  ref
) => {
  const [carouselRef, api] = useEmblaCarousel(
    {
      ...opts,
      axis: orientation === "horizontal" ? "x" : "y",
    },
    plugins
  )
  const [canScrollPrev, setCanScrollPrev] = React.useState(false)
  const [canScrollNext, setCanScrollNext] = React.useState(false)

  const onSelect = React.useCallback((api: CarouselApi) => {
    if (!api) {
      return
    }

    setCanScrollPrev(api.canScrollPrev())
    setCanScrollNext(api.canScrollNext())
  }, [])

  const scrollPrev = React.useCallback(() => {
    api?.scrollPrev()
  }, [api])

  const scrollNext = React.useCallback(() => {
    api?.scrollNext()
  }, [api])

  const handleKeyDown = React.useCallback(
    (event: React.KeyboardEvent<HTMLDivElement>) => {
      if (event.key === "ArrowLeft") {
        event.preventDefault()
        scrollPrev()
      } else if (event.key === "ArrowRight") {
        event.preventDefault()
        scrollNext()
      }
    },
  ),
```

```

    [scrollPrev, scrollNext]
  )
  React.useEffect(() => {
    if (!api || !setApi) {
      return
    }

    setApi(api)
  }, [api, setApi])

  React.useEffect(() => {
    if (!api) {
      return
    }

    onSelect(api)
    api.on("reInit", onSelect)
    api.on("select", onSelect)

    return () => {
      api?.off("select", onSelect)
    }
  }, [api, onSelect])

  return (
    <CarouselContext.Provider
      value={{
        carouselRef,
        api: api,
        opts,
        orientation:
          orientation || (opts?.axis === "y" ? "vertical" : "horizontal"),
        scrollPrev,
        scrollNext,
        canScrollPrev,
        canScrollNext,
      }}
    >
      <div
        ref={ref}
        onKeyDownCapture={handleKeyDown}
        className={cn("relative", className)}
        role="region"
        aria-roledescription="carousel"
        {...props}
      >
        {children}
      </div>
    </CarouselContext.Provider>
  )
}

Carousel.displayName = "Carousel"

const CarouselContent = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => {
  const { carouselRef, orientation } = useCarousel()

  return (
    <div ref={carouselRef} className="overflow-hidden">
      <div
        ref={ref}
        className={cn(
          "flex",
          orientation === "horizontal" ? "-ml-4" : "-mt-4 flex-col",
          className
        )}
        {...props}
      />
    </div>
  )
})

CarouselContent.displayName = "CarouselContent"

const CarouselItem = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => {
  const { orientation } = useCarousel()

  return (
    <div
      ref={ref}
      role="group"
      aria-roledescription="slide"
      className={cn(
        "min-w-0 shrink-0 grow-0 basis-full",
        orientation === "horizontal" ? "pl-4" : "pt-4",
        className
      )}
      {...props}
    />
  )
})

```

```

CarouselItem.displayName = "CarouselItem"

const CarouselPrevious = React.forwardRef<
  HTMLButtonElement,
  React.ComponentProps<typeof Button>
>(({ className, variant = "outline", size = "icon", ...props }, ref) => {
  const { orientation, scrollPrev, canScrollPrev } = useCarousel()

  return (
    <Button
      ref={ref}
      variant={variant}
      size={size}
      className={cn(
        "absolute h-8 w-8 rounded-full",
        orientation === "horizontal"
          ? "-left-12 top-1/2 -translate-y-1/2"
          : "-top-12 left-1/2 -translate-x-1/2 rotate-90",
        className
      )}
      disabled={!canScrollPrev}
      onClick={scrollPrev}
      {...props}
    >
      <ArrowLeft className="h-4 w-4" />
      <span className="sr-only">Previous slide</span>
    </Button>
  )
})
CarouselPrevious.displayName = "CarouselPrevious"

const CarouselNext = React.forwardRef<
  HTMLButtonElement,
  React.ComponentProps<typeof Button>
>(({ className, variant = "outline", size = "icon", ...props }, ref) => {
  const { orientation, scrollNext, canScrollNext } = useCarousel()

  return (
    <Button
      ref={ref}
      variant={variant}
      size={size}
      className={cn(
        "absolute h-8 w-8 rounded-full",
        orientation === "horizontal"
          ? "-right-12 top-1/2 -translate-y-1/2"
          : "-bottom-12 left-1/2 -translate-x-1/2 rotate-90",
        className
      )}
      disabled={!canScrollNext}
      onClick={scrollNext}
      {...props}
    >
      <ArrowRight className="h-4 w-4" />
      <span className="sr-only">Next slide</span>
    </Button>
  )
})
CarouselNext.displayName = "CarouselNext"

export {
  type CarouselApi,
  Carousel,
  CarouselContent,
  CarouselItem,
  CarouselPrevious,
  CarouselNext,
}

```


[artifacts/mockup-sandbox/src/components/ui/chart.tsx](#)

```
import * as React from "react"
import * as RechartsPrimitive from "recharts"

import { cn } from "@lib/utils"

const THEMES = { light: "", dark: ".dark" } as const

export type ChartConfig = {
  [k in string]: {
    label?: React.ReactNode
    icon?: React.ComponentType
  } & (
    | { color?: string; theme?: never }
    | { color?: never; theme: Record<keyof typeof THEMES, string> }
  )
}

type ChartContextProps = {
  config: ChartConfig
}

const ChartContext = React.createContext<ChartContextProps | null>(null)

function useChart() {
  const context = React.useContext(ChartContext)

  if (!context) {
    throw new Error("useChart must be used within a <ChartContainer />")
  }

  return context
}

const ChartContainer = React.forwardRef<
  HTMLDivElement,
  React.ComponentProps<"div"> & {
    config: ChartConfig
    children: React.ComponentProps<
      typeof RechartsPrimitive.ResponsiveContainer
    >["children"]
  }
>(({ id, className, children, config, ...props }, ref) => {
  const uniqueId = React.useId()
  const chartId = `chart-${id || uniqueId.replace(/:/g, "")}`

  return (
    <ChartContext.Provider value={{ config }}>
      <div
        data-chart={chartId}
        ref={ref}
        className={cn(
          "flex aspect-video justify-center text-xs [&_.recharts-cartesian-axis-tick_text]:fill-muted-foreground [&_.recharts-cartesian-grid_line[stroke='#ccc']]:stroke-border/50 [&_.recharts-curve.recharts-tooltip-cursor]:stroke-border [&_.recharts-dot[stroke='#fff']]:stroke-transparent [&_.recharts-layer]:outline-none [&_.recharts-polar-grid_[stroke='#ccc']]:stroke-border [&_.recharts-radial-bar-background-sector]:fill-muted [&_.recharts-rectangle.recharts-tooltip-cursor]:fill-muted [&_.recharts-reference-line[stroke='#ccc']]:stroke-border [&_.recharts-sector[stroke='#fff']]:stroke-transparent [&_.recharts-sector]:outline-none [&_.recharts-surface]:outline-none",
          className
        )}
        {...props}
      >
        <ChartStyle id={chartId} config={config} />
        <RechartsPrimitive.ResponsiveContainer>
          {children}
        </RechartsPrimitive.ResponsiveContainer>
      </div>
    </ChartContext.Provider>
  )
})
ChartContainer.displayName = "Chart"

const ChartStyle = ({ id, config }: { id: string; config: ChartConfig }) => {
  const colorConfig = Object.entries(config).filter(
    ([, config]) => config.theme || config.color
  )

  if (!colorConfig.length) {
    return null
  }

  return (
    <style
      dangerouslySetInnerHTML={{
        __html: Object.entries(THEMES)
          .map(
            ([theme, prefix]) => `
${prefix} [data-chart=${id}] {
${colorConfig
          .map(([key, itemConfig]) => {
            const color =
              itemConfig.theme?.[theme as keyof typeof itemConfig.theme] ||
              itemConfig.color
            return color ? `--color-${key}: ${color};` : null
          })
          .join("\n")}
    `
      }}
    />
  )
}
```

```

    }
    )
    .join("\n"),
  })
  />
)
}

const ChartTooltip = RechartsPrimitive.Tooltip

const ChartTooltipContent = React.forwardRef<
  HTMLDivElement,
  React.ComponentProps<typeof RechartsPrimitive.Tooltip> &
  React.ComponentProps<"div"> & {
    hideLabel?: boolean
    hideIndicator?: boolean
    indicator?: "line" | "dot" | "dashed"
    nameKey?: string
    labelKey?: string
  }
> (
  (
    {
      active,
      payload,
      className,
      indicator = "dot",
      hideLabel = false,
      hideIndicator = false,
      label,
      labelFormatter,
      labelClassName,
      formatter,
      color,
      nameKey,
      labelKey,
    },
    ref
  ) => {
    const { config } = useChart()

    const tooltipLabel = React.useMemo(() => {
      if (hideLabel || !payload?.length) {
        return null
      }

      const [item] = payload
      const key = `${labelKey || item?.dataKey || item?.name || "value"}`
      const itemConfig = getPayloadConfigFromPayload(config, item, key)
      const value =
        !labelKey && typeof label === "string"
          ? config[label as keyof typeof config]?.label || label
          : itemConfig?.label

      if (labelFormatter) {
        return (
          <div className={cn("font-medium", labelClassName)}>
            {labelFormatter(value, payload)}
          </div>
        )
      }

      if (!value) {
        return null
      }

      return <div className={cn("font-medium", labelClassName)}>{value}</div>
    }, [
      label,
      labelFormatter,
      payload,
      hideLabel,
      labelClassName,
      config,
      labelKey,
    ])

    if (!active || !payload?.length) {
      return null
    }

    const nestLabel = payload.length === 1 && indicator !== "dot"

    return (
      <div
        ref={ref}
        className={cn(
          "grid min-w-[8rem] items-start gap-1.5 rounded-lg border border-border/50 bg-background px-2.5 py-1.5 text-xs shadow-
xl",
          className
        )}>
        >
        {!nestLabel ? tooltipLabel : null}
        <div className="grid gap-1.5">
          {payload
            .filter((item) => item.type !== "none")

```

```

.map((item, index) => {
  const key = `${nameKey} || item.name || item.dataKey || "value"`
  const itemConfig = getPayloadConfigFromPayload(config, item, key)
  const indicatorColor = color || item.payload.fill || item.color

  return (
    <div
      key={item.dataKey}
      className={cn(
        "flex w-full flex-wrap items-stretch gap-2 [>svg]:h-2.5 [>svg]:w-2.5 [>svg]:text-muted-foreground",
        indicator === "dot" && "items-center"
      )}
    >
      {formatter && item?.value !== undefined && item.name ? (
        formatter(item.value, item.name, item, index, item.payload)
      ) : (
        <>
          {itemConfig?.icon ? (
            <itemConfig.icon />
          ) : (
            !hideIndicator && (
              <div
                className={cn(
                  "shrink-0 rounded-[2px] border[--color-border] bg[--color-bg]",
                  {
                    "h-2.5 w-2.5": indicator === "dot",
                    "w-1": indicator === "line",
                    "w-0 border-[1.5px] border-dashed bg-transparent":
                      indicator === "dashed",
                    "my-0.5": nestLabel && indicator === "dashed",
                  }
                )}
                style={
                  {
                    "--color-bg": indicatorColor,
                    "--color-border": indicatorColor,
                  } as React.CSSProperties
                }
              />
            )}
          )}
          <div
            className={cn(
              "flex flex-1 justify-between leading-none",
              nestLabel ? "items-end" : "items-center"
            )}
          >
            <div className="grid gap-1.5">
              {nestLabel ? tooltipLabel : null}
              <span className="text-muted-foreground">
                {itemConfig?.label || item.name}
              </span>
            </div>
            {item.value && (
              <span className="font-mono font-medium tabular-nums text-foreground">
                {item.value.toLocaleString()}
              </span>
            )}
          </div>
        </>
      )}
    </div>
  )
)
)
)
</div>
</div>
)
)
)
ChartTooltipContent.displayName = "ChartTooltip"

const ChartLegend = RechartsPrimitive.Legend

const ChartLegendContent = React.forwardRef<
  HTMLDivElement,
  React.ComponentProps<"div"> &
  Pick<RechartsPrimitive.LegendProps, "payload" | "verticalAlign"> & {
    hideIcon?: boolean
    nameKey?: string
  }
>((
  {
    className, hideIcon = false, payload, verticalAlign = "bottom", nameKey },
  ref
) => {
  const { config } = useChart()

  if (!payload?.length) {
    return null
  }

  return (
    <div
      ref={ref}
      className={cn(
        "flex items-center justify-center gap-4",
        verticalAlign === "top" ? "pb-3" : "pt-3",

```

```

        className
      })
    >
    {payload
      .filter((item) => item.type !== "none")
      .map((item) => {
        const key = `${nameKey} || item.dataKey || "value"`
        const itemConfig = getPayloadConfigFromPayload(config, item, key)

        return (
          <div
            key={item.value}
            className={cn(
              "flex items-center gap-1.5 [&svg]:h-3 [&svg]:w-3 [&svg]:text-muted-foreground"
            )}
          >
            {itemConfig?.icon && !hideIcon ? (
              <itemConfig.icon />
            ) : (
              <div
                className="h-2 w-2 shrink-0 rounded-[2px]"
                style={{
                  backgroundColor: item.color,
                }}
              />
            )}
            {itemConfig?.label}
          </div>
        )
      })
    }
  </div>
)
}
)
ChartLegendContent.displayName = "ChartLegend"

function getPayloadConfigFromPayload(
  config: ChartConfig,
  payload: unknown,
  key: string
) {
  if (typeof payload !== "object" || payload === null) {
    return undefined
  }

  const payloadPayload =
    "payload" in payload &&
    typeof payload.payload === "object" &&
    payload.payload !== null
      ? payload.payload
      : undefined

  let configLabelKey: string = key

  if (
    key in payload &&
    typeof payload[key as keyof typeof payload] === "string"
  ) {
    configLabelKey = payload[key as keyof typeof payload] as string
  } else if (
    payloadPayload &&
    key in payloadPayload &&
    typeof payloadPayload[key as keyof typeof payloadPayload] === "string"
  ) {
    configLabelKey = payloadPayload[
      key as keyof typeof payloadPayload
    ] as string
  }

  return configLabelKey in config
    ? config[configLabelKey]
    : config[key as keyof typeof config]
}

export {
  ChartContainer,
  ChartTooltip,
  ChartTooltipContent,
  ChartLegend,
  ChartLegendContent,
  ChartStyle,
}

```

[artifacts/mockup-sandbox/src/components/ui/checkbox.tsx](#)

```
import * as React from "react"
import * as CheckboxPrimitive from "@radix-ui/react-checkbox"
import { Check } from "lucide-react"

import { cn } from "@lib/utls"

const Checkbox = React.forwardRef<
  React.ElementRef<typeof CheckboxPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof CheckboxPrimitive.Root>
>(({ className, ...props }, ref) => {
  <CheckboxPrimitive.Root
    ref={ref}
    className={cn(
      "grid place-content-center peer h-4 w-4 shrink-0 rounded-sm border border-primary shadow focus-visible:outline-none focus-visible:ring-1 focus-visible:ring-ring disabled:cursor-not-allowed disabled:opacity-50 data-[state=checked]:bg-primary data-[state=checked]:text-primary-foreground",
      className
    )}
    {...props}
  >
    <CheckboxPrimitive.Indicator
      className={cn("grid place-content-center text-current")}
    >
      <Check className="h-4 w-4" />
    </CheckboxPrimitive.Indicator>
  </CheckboxPrimitive.Root>
)}
Checkbox.displayName = CheckboxPrimitive.Root.displayName

export { Checkbox }
```

[artifacts/mockup-sandbox/src/components/ui/collapsible.tsx](#)

"use client"

import * as CollapsiblePrimitive from "@radix-ui/react-collapsible"

const Collapsible = CollapsiblePrimitive.Root

const CollapsibleTrigger = CollapsiblePrimitive.CollapsibleTrigger

const CollapsibleContent = CollapsiblePrimitive.CollapsibleContent

export { Collapsible, CollapsibleTrigger, CollapsibleContent }

[artifacts/mockup-sandbox/src/components/ui/command.tsx](#)

```
"use client"

import * as React from "react"
import { type DialogProps } from "@radix-ui/react-dialog"
import { Command as CommandPrimitive } from "cmdk"
import { Search } from "lucide-react"

import { cn } from "@lib/utils"
import { Dialog, DialogContent } from "@components/ui/dialog"

const Command = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive>
>(({ className, ...props }, ref) => {
  <CommandPrimitive
    ref={ref}
    className={cn(
      "flex h-full w-full flex-col overflow-hidden rounded-md bg-popover text-popover-foreground",
      className
    )}
    {...props}
  />
})
Command.displayName = CommandPrimitive.displayName

const CommandDialog = ({ children, ...props }: DialogProps) => {
  return (
    <Dialog {...props}>
      <DialogContent className="overflow-hidden p-0">
        <Command className="[&_cmdk-group-heading]:px-2 [&_cmdk-group-heading]:font-medium [&_cmdk-group-heading]:text-muted-foreground [&_cmdk-group]:not([hidden])~[cmdk-group]:pt-0 [&_cmdk-group]:px-2 [&_cmdk-input-wrapper]_svg:h-5 [&_cmdk-input-wrapper]_svg:w-5 [&_cmdk-input]:h-12 [&_cmdk-item]:px-2 [&_cmdk-item]:py-3 [&_cmdk-item]_svg:h-5 [&_cmdk-item]_svg:w-5">
          {children}
        </Command>
      </DialogContent>
    </Dialog>
  )
}

const CommandInput = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive.Input>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive.Input>
>(({ className, ...props }, ref) => {
  <div className="flex items-center border-b px-3 cmdk-input-wrapper="">>
    <Search className="mr-2 h-4 w-4 shrink-0 opacity-50" />
    <CommandPrimitive.Input
      ref={ref}
      className={cn(
        "flex h-10 w-full rounded-md bg-transparent py-3 text-sm outline-none placeholder:text-muted-foreground disabled:cursor-not-allowed disabled:opacity-50",
        className
      )}
      {...props}
    />
  </div>
})
CommandInput.displayName = CommandPrimitive.Input.displayName

const CommandList = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive.List>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive.List>
>(({ className, ...props }, ref) => {
  <CommandPrimitive.List
    ref={ref}
    className={cn("max-h-[300px] overflow-y-auto overflow-x-hidden", className)}
    {...props}
  />
})
CommandList.displayName = CommandPrimitive.List.displayName

const CommandEmpty = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive.Empty>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive.Empty>
>((props, ref) => {
  <CommandPrimitive.Empty
    ref={ref}
    className="py-6 text-center text-sm"
    {...props}
  />
})
CommandEmpty.displayName = CommandPrimitive.Empty.displayName

const CommandGroup = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive.Group>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive.Group>
>(({ className, ...props }, ref) => {
  <CommandPrimitive.Group
    ref={ref}
    className={cn(
      "overflow-hidden p-1 text-foreground [&_cmdk-group-heading]:px-2 [&_cmdk-group-heading]:py-1.5 [&_cmdk-group-heading]:text-xs [&_cmdk-group-heading]:font-medium [&_cmdk-group-heading]:text-muted-foreground",
      className
    )}
    {...props}
  />
})
CommandGroup.displayName = CommandPrimitive.Group.displayName
```

```

        className
      })
      {...props}
    />
  ))

CommandGroup.displayName = CommandPrimitive.Group.displayName

const CommandSeparator = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive.Separator>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive.Separator>
>(({ className, ...props }, ref) => (
  <CommandPrimitive.Separator
    ref={ref}
    className={cn("-mx-1 h-px bg-border", className)}
    {...props}
  />
))
CommandSeparator.displayName = CommandPrimitive.Separator.displayName

const CommandItem = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive.Item>
>(({ className, ...props }, ref) => (
  <CommandPrimitive.Item
    ref={ref}
    className={cn(
      "relative flex cursor-default gap-2 select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none data-
[disabled=true]:pointer-events-none data-[selected=true]:bg-accent data-[selected=true]:text-accent-foreground data-
[disabled=true]:opacity-50 [&_svg]:pointer-events-none [&_svg]:size-4 [&_svg]:shrink-0",
      className
    )}
    {...props}
  />
))
CommandItem.displayName = CommandPrimitive.Item.displayName

const CommandShortcut = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLSpanElement>) => {
  return (
    <span
      className={cn(
        "ml-auto text-xs tracking-widest text-muted-foreground",
        className
      )}
      {...props}
    />
  )
}
CommandShortcut.displayName = "CommandShortcut"

export {
  Command,
  CommandDialog,
  CommandInput,
  CommandList,
  CommandEmpty,
  CommandGroup,
  CommandItem,
  CommandShortcut,
  CommandSeparator,
}

```


[artifacts/mockup-sandbox/src/components/ui/context-menu.tsx](#)

```
import * as React from "react"
import * as ContextMenuPrimitive from "@radix-ui/react-context-menu"
import { Check, ChevronRight, Circle } from "lucide-react"

import { cn } from "@/lib/utils"

const ContextMenu = ContextMenuPrimitive.Root

const ContextMenuTrigger = ContextMenuPrimitive.Trigger

const ContextMenuGroup = ContextMenuPrimitive.Group

const ContextMenuPortal = ContextMenuPrimitive.Portal

const ContextMenuSub = ContextMenuPrimitive.Sub

const ContextMenuRadioGroup = ContextMenuPrimitive.RadioGroup

const ContextMenuSubTrigger = React.forwardRef<
  React.ElementRef<typeof ContextMenuPrimitive.SubTrigger>,
  React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.SubTrigger> & {
    inset?: boolean
  }
>(({ className, inset, children, ...props }, ref) => (
  <ContextMenuPrimitive.SubTrigger
    ref={ref}
    className={cn(
      "flex cursor-default select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none focus:bg-accent focus:text-
accent-foreground data-[state=open]:bg-accent data-[state=open]:text-accent-foreground",
      inset && "pl-8",
      className
    )}
    {...props}
  >
    {children}
    <ChevronRight className="ml-auto h-4 w-4" />
  </ContextMenuPrimitive.SubTrigger>
))
ContextMenuSubTrigger.displayName = ContextMenuPrimitive.SubTrigger.displayName

const ContextMenuSubContent = React.forwardRef<
  React.ElementRef<typeof ContextMenuPrimitive.SubContent>,
  React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.SubContent>
>(({ className, ...props }, ref) => (
  <ContextMenuPrimitive.SubContent
    ref={ref}
    className={cn(
      "z-50 min-w-[8rem] overflow-hidden rounded-md border bg-popover p-1 text-popover-foreground shadow-lg data-
[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-
[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-
right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin-[--radix-context-menu-content-
transform-origin]",
      className
    )}
    {...props}
  >
  </>
))
ContextMenuSubContent.displayName = ContextMenuPrimitive.SubContent.displayName

const ContextMenuContent = React.forwardRef<
  React.ElementRef<typeof ContextMenuPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.Content>
>(({ className, ...props }, ref) => (
  <ContextMenuPrimitive.Portal>
    <ContextMenuPrimitive.Content
      ref={ref}
      className={cn(
        "z-50 max-h-[--radix-context-menu-content-available-height] min-w-[8rem] overflow-y-auto overflow-x-hidden rounded-md
border bg-popover p-1 text-popover-foreground shadow-md data-[state=open]:animate-in data-[state=closed]:animate-out data-
[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-
[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-
[side=top]:slide-in-from-bottom-2 origin-[--radix-context-menu-content-transform-origin]",
        className
      )}
      {...props}
    >
  </ContextMenuPrimitive.Portal>
))
ContextMenuContent.displayName = ContextMenuPrimitive.Content.displayName

const ContextMenuItem = React.forwardRef<
  React.ElementRef<typeof ContextMenuPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.Item> & {
    inset?: boolean
  }
>(({ className, inset, ...props }, ref) => (
  <ContextMenuPrimitive.Item
    ref={ref}
    className={cn(
      "relative flex cursor-default select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none focus:bg-accent
focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      inset && "pl-8",
      className
    )}
    {...props}
  >
```

```

    />
  ))
  ContextMenuItem.displayName = ContextMenuPrimitive.Item.displayName

  const ContextMenuCheckboxItem = React.forwardRef<
    React.ElementRef<typeof ContextMenuPrimitive.CheckboxItem>,
    React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.CheckboxItem>
  >(({ className, children, checked, ...props }, ref) => (
    <ContextMenuPrimitive.CheckboxItem
      ref={ref}
      className={cn(
        "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-none focus:bg-accent
        focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
        className
      )}
      checked={checked}
      {...props}
    >
      <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
        <ContextMenuPrimitive.ItemIndicator>
          <Check className="h-4 w-4" />
        </ContextMenuPrimitive.ItemIndicator>
      </span>
      {children}
    </ContextMenuPrimitive.CheckboxItem>
  ))
  ContextMenuCheckboxItem.displayName =
    ContextMenuPrimitive.CheckboxItem.displayName

  const ContextMenuRadioItem = React.forwardRef<
    React.ElementRef<typeof ContextMenuPrimitive.RadioItem>,
    React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.RadioItem>
  >(({ className, children, ...props }, ref) => (
    <ContextMenuPrimitive.RadioItem
      ref={ref}
      className={cn(
        "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-none focus:bg-accent
        focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
        className
      )}
      {...props}
    >
      <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
        <ContextMenuPrimitive.ItemIndicator>
          <Circle className="h-4 w-4 fill-current" />
        </ContextMenuPrimitive.ItemIndicator>
      </span>
      {children}
    </ContextMenuPrimitive.RadioItem>
  ))
  ContextMenuRadioItem.displayName = ContextMenuPrimitive.RadioItem.displayName

  const ContextMenuLabel = React.forwardRef<
    React.ElementRef<typeof ContextMenuPrimitive.Label>,
    React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.Label> & {
      inset?: boolean
    }
  >(({ className, inset, ...props }, ref) => (
    <ContextMenuPrimitive.Label
      ref={ref}
      className={cn(
        "px-2 py-1.5 text-sm font-semibold text-foreground",
        inset && "pl-8",
        className
      )}
      {...props}
    >
    />
  ))
  ContextMenuLabel.displayName = ContextMenuPrimitive.Label.displayName

  const ContextMenuSeparator = React.forwardRef<
    React.ElementRef<typeof ContextMenuPrimitive.Separator>,
    React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.Separator>
  >(({ className, ...props }, ref) => (
    <ContextMenuPrimitive.Separator
      ref={ref}
      className={cn("-mx-1 my-1 h-px bg-border", className)}
      {...props}
    >
    />
  ))
  ContextMenuSeparator.displayName = ContextMenuPrimitive.Separator.displayName

  const ContextMenuShortcut = ({
    className,
    ...props
  }: React.HTMLAttributes<HTMLSpanElement>) => {
    return (
      <span
        className={cn(
          "ml-auto text-xs tracking-widest text-muted-foreground",
          className
        )}
        {...props}
      >
    />
    )
  }
  ContextMenuShortcut.displayName = "ContextMenuShortcut"

```

```
export {  
  ContextMenu,  
  ContextMenuTrigger,  
  ContextMenuContent,  
  ContextMenuItem,  
  ContextMenuCheckboxItem,  
  ContextMenuRadioItem,  
  ContextMenuLabel,  
  ContextMenuSeparator,  
  ContextMenuShortcut,  
  ContextMenuGroup,  
  ContextMenuPortal,  
  ContextMenuSub,  
  ContextMenuSubContent,  
  ContextMenuSubTrigger,  
  ContextMenuRadioGroup,  
}
```

[artifacts/mockup-sandbox/src/components/ui/dialog.tsx](#)

```
import * as React from "react"
import * as DialogPrimitive from "@radix-ui/react-dialog"
import { X } from "lucide-react"

import { cn } from "@lib/utils"

const Dialog = DialogPrimitive.Root

const DialogTrigger = DialogPrimitive.Trigger

const DialogPortal = DialogPrimitive.Portal

const DialogClose = DialogPrimitive.Close

const DialogOverlay = React.forwardRef<
  React.ElementRef<typeof DialogPrimitive.Overlay>,
  React.ComponentPropsWithoutRef<typeof DialogPrimitive.Overlay>
>(({ className, ...props }, ref) => (
  <DialogPrimitive.Overlay
    ref={ref}
    className={cn(
      "fixed inset-0 z-50 bg-black/80 data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0",
      className
    )}
    {...props}
  />
))
DialogOverlay.displayName = DialogPrimitive.Overlay.displayName

const DialogContent = React.forwardRef<
  React.ElementRef<typeof DialogPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof DialogPrimitive.Content>
>(({ className, children, ...props }, ref) => (
  <DialogPortal>
    <DialogOverlay />
    <DialogPrimitive.Content
      ref={ref}
      className={cn(
        "fixed left-[50%] top-[50%] z-50 grid w-full max-w-lg translate-x-[-50%] translate-y-[-50%] gap-4 border bg-background p-6 shadow-lg duration-200 data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[state=closed]:slide-out-to-left-1/2 data-[state=closed]:slide-out-to-top-48% data-[state=open]:slide-in-from-left-1/2 data-[state=open]:slide-in-from-top-48% sm:rounded-lg",
        className
      )}
      {...props}
    >
      {children}
      <DialogPrimitive.Close className="absolute right-4 top-4 rounded-sm opacity-70 ring-offset-background transition-opacity hover:opacity-100 focus:outline-none focus:ring-2 focus:ring-ring focus:ring-offset-2 disabled:pointer-events-none data-[state=open]:bg-accent data-[state=open]:text-muted-foreground">
        <X className="h-4 w-4" />
        <span className="sr-only">Close</span>
      </DialogPrimitive.Close>
    </DialogPrimitive.Content>
  </DialogPortal>
))
DialogContent.displayName = DialogPrimitive.Content.displayName

const DialogHeader = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn(
      "flex flex-col space-y-1.5 text-center sm:text-left",
      className
    )}
    {...props}
  />
)
DialogHeader.displayName = "DialogHeader"

const DialogFooter = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn(
      "flex flex-col-reverse sm:flex-row sm:justify-end sm:space-x-2",
      className
    )}
    {...props}
  />
)
DialogFooter.displayName = "DialogFooter"

const DialogTitle = React.forwardRef<
  React.ElementRef<typeof DialogPrimitive.Title>,
  React.ComponentPropsWithoutRef<typeof DialogPrimitive.Title>
>(({ className, ...props }, ref) => (
  <DialogPrimitive.Title
    ref={ref}
    className={cn(
```

```

        "text-lg font-semibold leading-none tracking-tight",
        className
    )}
    {...props}
  />
))
DialogTitle.displayName = DialogPrimitive.Title.displayName

const DialogDescription = React.forwardRef<
  React.ElementRef<typeof DialogPrimitive.Description>,
  React.ComponentPropsWithoutRef<typeof DialogPrimitive.Description>
>(({ className, ...props }, ref) => (
  <DialogPrimitive.Description
    ref={ref}
    className={cn("text-sm text-muted-foreground", className)}
    {...props}
  />
))
DialogDescription.displayName = DialogPrimitive.Description.displayName

export {
  Dialog,
  DialogPortal,
  DialogOverlay,
  DialogTrigger,
  DialogClose,
  DialogContent,
  DialogHeader,
  DialogFooter,
  DialogTitle,
  DialogDescription,
}

```

[artifacts/mockup-sandbox/src/components/ui/drawer.tsx](#)

```
import * as React from "react"
import { Drawer as DrawerPrimitive } from "vaul"

import { cn } from "@/lib/utils"

const Drawer = ({
  shouldScaleBackground = true,
  ...props
}: React.ComponentProps<typeof DrawerPrimitive.Root>) => (
  <DrawerPrimitive.Root
    shouldScaleBackground={shouldScaleBackground}
    {...props}
  />
)
Drawer.displayName = "Drawer"

const DrawerTrigger = DrawerPrimitive.Trigger

const DrawerPortal = DrawerPrimitive.Portal

const DrawerClose = DrawerPrimitive.Close

const DrawerOverlay = React.forwardRef<
  React.ElementRef<typeof DrawerPrimitive.Overlay>,
  React.ComponentPropsWithoutRef<typeof DrawerPrimitive.Overlay>
>(({ className, ...props }, ref) => (
  <DrawerPrimitive.Overlay
    ref={ref}
    className={cn("fixed inset-0 z-50 bg-black/80", className)}
    {...props}
  />
))
DrawerOverlay.displayName = DrawerPrimitive.Overlay.displayName

const DrawerContent = React.forwardRef<
  React.ElementRef<typeof DrawerPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof DrawerPrimitive.Content>
>(({ className, children, ...props }, ref) => (
  <DrawerPortal>
    <DrawerOverlay />
    <DrawerPrimitive.Content
      ref={ref}
      className={cn(
        "fixed inset-x-0 bottom-0 z-50 mt-24 flex h-auto flex-col rounded-t-[10px] border bg-background",
        className
      )}
      {...props}
    >
    <div className="mx-auto mt-4 h-2 w-[100px] rounded-full bg-muted" />
    {children}
  </DrawerPrimitive.Content>
</DrawerPortal>
))
DrawerContent.displayName = "DrawerContent"

const DrawerHeader = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn("grid gap-1.5 p-4 text-center sm:text-left", className)}
    {...props}
  />
)
DrawerHeader.displayName = "DrawerHeader"

const DrawerFooter = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn("mt-auto flex flex-col gap-2 p-4", className)}
    {...props}
  />
)
DrawerFooter.displayName = "DrawerFooter"

const DrawerTitle = React.forwardRef<
  React.ElementRef<typeof DrawerPrimitive.Title>,
  React.ComponentPropsWithoutRef<typeof DrawerPrimitive.Title>
>(({ className, ...props }, ref) => (
  <DrawerPrimitive.Title
    ref={ref}
    className={cn(
      "text-lg font-semibold leading-none tracking-tight",
      className
    )}
    {...props}
  />
))
DrawerTitle.displayName = DrawerPrimitive.Title.displayName

const DrawerDescription = React.forwardRef<
  React.ElementRef<typeof DrawerPrimitive.Description>,
  React.ComponentPropsWithoutRef<typeof DrawerPrimitive.Description>
```

```
>(({ className, ...props }, ref) => (  
  <DrawerPrimitive.Description  
    ref={ref}  
    className={cn("text-sm text-muted-foreground", className)}  
    {...props}  
  />  
))  
DrawerDescription.displayName = DrawerPrimitive.Description.displayName  
  
export {  
  Drawer,  
  DrawerPortal,  
  DrawerOverlay,  
  DrawerTrigger,  
  DrawerClose,  
  DrawerContent,  
  DrawerHeader,  
  DrawerFooter,  
  DrawerTitle,  
  DrawerDescription,  
}
```

[artifacts/mockup-sandbox/src/components/ui/dropdown-menu.tsx](#)

"use client"

```
import * as React from "react"
import * as DropdownMenuPrimitive from "@radix-ui/react-dropdown-menu"
import { Check, ChevronRight, Circle } from "lucide-react"

import { cn } from "@lib/utils"

const DropdownMenu = DropdownMenuPrimitive.Root

const DropdownMenuTrigger = DropdownMenuPrimitive.Trigger

const DropdownMenuGroup = DropdownMenuPrimitive.Group

const DropdownMenuPortal = DropdownMenuPrimitive.Portal

const DropdownMenuSub = DropdownMenuPrimitive.Sub

const DropdownMenuRadioGroup = DropdownMenuPrimitive.RadioGroup

const DropdownMenuSubTrigger = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.SubTrigger>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.SubTrigger> & {
    inset?: boolean
  }
>(({ className, inset, children, ...props }, ref) => (
  <DropdownMenuPrimitive.SubTrigger
    ref={ref}
    className={cn(
      "flex cursor-default select-none items-center gap-2 rounded-sm px-2 py-1.5 text-sm outline-none focus:bg-accent data-[state=open]:bg-accent [&_svg]:pointer-events-none [&_svg]:size-4 [&_svg]:shrink-0",
      inset && "pl-8",
      className
    )}
    {...props}
  >
    {children}
    <ChevronRight className="ml-auto" />
  </DropdownMenuPrimitive.SubTrigger>
))
DropdownMenuSubTrigger.displayName =
  DropdownMenuPrimitive.SubTrigger.displayName

const DropdownMenuSubContent = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.SubContent>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.SubContent>
>(({ className, ...props }, ref) => (
  <DropdownMenuPrimitive.SubContent
    ref={ref}
    className={cn(
      "z-50 min-w-[8rem] overflow-hidden rounded-md border bg-popover p-1 text-popover-foreground shadow-lg data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin-[--radix-dropdown-menu-content-transform-origin]",
      className
    )}
    {...props}
  />
))
DropdownMenuSubContent.displayName =
  DropdownMenuPrimitive.SubContent.displayName

const DropdownMenuContent = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.Content>
>(({ className, sideOffset = 4, ...props }, ref) => (
  <DropdownMenuPrimitive.Portal>
    <DropdownMenuPrimitive.Content
      ref={ref}
      sideOffset={sideOffset}
      className={cn(
        "z-50 max-h-[var(--radix-dropdown-menu-content-available-height)] min-w-[8rem] overflow-y-auto overflow-x-hidden rounded-md border bg-popover p-1 text-popover-foreground shadow-md",
        "data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin-[--radix-dropdown-menu-content-transform-origin]",
        className
      )}
      {...props}
    </DropdownMenuPrimitive.Portal>
  ))
DropdownMenuContent.displayName = DropdownMenuPrimitive.Content.displayName

const DropdownMenuItem = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.Item> & {
    inset?: boolean
  }
>(({ className, inset, ...props }, ref) => (
  <DropdownMenuPrimitive.Item
    ref={ref}
    className={cn(
```



```

        "relative flex cursor-default select-none items-center gap-2 rounded-sm px-2 py-1.5 text-sm outline-none transition-
colors focus:bg-accent focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50
[&svg]:size-4 [&svg]:shrink-0",
        inset && "pl-8",
        className
    )}
    {...props}
  />
))
DropdownMenuItem.displayName = DropdownMenuPrimitive.Item.displayName

const DropdownMenuCheckboxItem = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.CheckboxItem>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.CheckboxItem>
>(({ className, children, checked, ...props }, ref) => (
  <DropdownMenuPrimitive.CheckboxItem
    ref={ref}
    className={cn(
      "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-none transition-colors
focus:bg-accent focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      className
    )}
    checked={checked}
    {...props}
  >
    <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
      <DropdownMenuPrimitive.ItemIndicator>
        <Check className="h-4 w-4" />
      </DropdownMenuPrimitive.ItemIndicator>
    </span>
    {children}
  </DropdownMenuPrimitive.CheckboxItem>
))
DropdownMenuCheckboxItem.displayName =
  DropdownMenuPrimitive.CheckboxItem.displayName

const DropdownMenuRadioItem = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.RadioItem>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.RadioItem>
>(({ className, children, ...props }, ref) => (
  <DropdownMenuPrimitive.RadioItem
    ref={ref}
    className={cn(
      "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-none transition-colors
focus:bg-accent focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      className
    )}
    {...props}
  >
    <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
      <DropdownMenuPrimitive.ItemIndicator>
        <Circle className="h-2 w-2 fill-current" />
      </DropdownMenuPrimitive.ItemIndicator>
    </span>
    {children}
  </DropdownMenuPrimitive.RadioItem>
))
DropdownMenuRadioItem.displayName = DropdownMenuPrimitive.RadioItem.displayName

const DropdownMenuLabel = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.Label>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.Label> & {
    inset?: boolean
  }
>(({ className, inset, ...props }, ref) => (
  <DropdownMenuPrimitive.Label
    ref={ref}
    className={cn(
      "px-2 py-1.5 text-sm font-semibold",
      inset && "pl-8",
      className
    )}
    {...props}
  />
))
DropdownMenuLabel.displayName = DropdownMenuPrimitive.Label.displayName

const DropdownMenuSeparator = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.Separator>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.Separator>
>(({ className, ...props }, ref) => (
  <DropdownMenuPrimitive.Separator
    ref={ref}
    className={cn("-mx-1 my-1 h-px bg-muted", className)}
    {...props}
  />
))
DropdownMenuSeparator.displayName = DropdownMenuPrimitive.Separator.displayName

const DropdownMenuShortcut = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLSpanElement>) => {
  return (
    <span
      className={cn("ml-auto text-xs tracking-widest opacity-60", className)}
      {...props}
    >

```

```
    ) />
  }
  DropdownMenuShortcut.displayName = "DropdownMenuShortcut"

export {
  DropdownMenu,
  DropdownMenuTrigger,
  DropdownMenuContent,
  DropdownMenuItem,
  DropdownMenuCheckboxItem,
  DropdownMenuRadioItem,
  DropdownMenuLabel,
  DropdownMenuSeparator,
  DropdownMenuShortcut,
  DropdownMenuGroup,
  DropdownMenuPortal,
  DropdownMenuSub,
  DropdownMenuSubContent,
  DropdownMenuSubTrigger,
  DropdownMenuRadioGroup,
}
```

[artifacts/mockup-sandbox/src/components/ui/empty.tsx](#)

```
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"

function Empty({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="empty"
      className={cn(
        "flex min-w-0 flex-1 flex-col items-center justify-center gap-6 text-balance rounded-lg border-dashed p-6 text-center
md:p-12",
        className
      )}
      {...props}
    />
  )
}

function EmptyHeader({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="empty-header"
      className={cn(
        "flex max-w-sm flex-col items-center gap-2 text-center",
        className
      )}
      {...props}
    />
  )
}

const emptyMediaVariants = cva(
  "mb-2 flex shrink-0 items-center justify-center [&_svg]:pointer-events-none [&_svg]:shrink-0",
  {
    variants: {
      variant: {
        default: "bg-transparent",
        icon: "bg-muted text-foreground flex size-10 shrink-0 items-center justify-center rounded-lg
[&_svg:not([class*='size-'])]:size-6",
      },
    },
    defaultVariants: {
      variant: "default",
    },
  }
)

function EmptyMedia({
  className,
  variant = "default",
  ...props
}: React.ComponentProps<"div"> & VariantProps<typeof emptyMediaVariants>) {
  return (
    <div
      data-slot="empty-icon"
      data-variant={variant}
      className={cn(emptyMediaVariants({ variant, className }))}
      {...props}
    />
  )
}

function EmptyTitle({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="empty-title"
      className={cn("text-lg font-medium tracking-tight", className)}
      {...props}
    />
  )
}

function EmptyDescription({ className, ...props }: React.ComponentProps<"p">) {
  return (
    <div
      data-slot="empty-description"
      className={cn(
        "text-muted-foreground [&a:hover]:text-primary text-sm/relaxed [&a]:underline [&a]:underline-offset-4",
        className
      )}
      {...props}
    />
  )
}

function EmptyContent({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="empty-content"
      className={cn(
        "flex w-full min-w-0 max-w-sm flex-col items-center gap-4 text-balance text-sm",
        className
      )}
      {...props}
    />
  )
}
```

```
    )  
  }  
  
  export {  
    Empty,  
    EmptyHeader,  
    EmptyTitle,  
    EmptyDescription,  
    EmptyContent,  
    EmptyMedia,  
  }  
}
```

[artifacts/mockup-sandbox/src/components/ui/field.tsx](#)

```
"use client"

import { useMemo } from "react"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@/lib/utils"
import { Label } from "@/components/ui/label"
import { Separator } from "@/components/ui/separator"

function FieldSet({ className, ...props }: React.ComponentProps<"fieldset">) {
  return (
    <fieldset
      data-slot="field-set"
      className={cn(
        "flex flex-col gap-6",
        "has- [>[data-slot=checkbox-group]]:gap-3 has- [>[data-slot=radio-group]]:gap-3",
        className
      )}
      {...props}
    />
  )
}

function FieldLegend({
  className,
  variant = "legend",
  ...props
}: React.ComponentProps<"legend"> & { variant?: "legend" | "label" }) {
  return (
    <legend
      data-slot="field-legend"
      data-variant={variant}
      className={cn(
        "mb-3 font-medium",
        "data-[variant=legend]:text-base",
        "data-[variant=label]:text-sm",
        className
      )}
      {...props}
    />
  )
}

function FieldGroup({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="field-group"
      className={cn(
        "group/field-group @container/field-group flex w-full flex-col gap-7 data-[slot=checkbox-group]:gap-3 [&[data-slot=field-group]]:gap-4",
        className
      )}
      {...props}
    />
  )
}

const fieldVariants = cva(
  "group/field data-[invalid=true]:text-destructive flex w-full gap-3",
  {
    variants: {
      orientation: {
        vertical: ["flex-col [&*]:w-full [&.>.sr-only]:w-auto"],
        horizontal: [
          "flex-row items-center",
          "[&[data-slot=field-label]]:flex-auto",
          "has- [>[data-slot=field-content]]:[&[role=checkbox],[role=radio]]:mt-px has- [>[data-slot=field-content]]:items-start",
        ],
      },
      responsive: [
        "@md/field-group:flex-row @md/field-group:items-center @md/field-group:[&*]:w-auto flex-col [&*]:w-full [&.>.sr-only]:w-auto",
        "@md/field-group:[&[data-slot=field-label]]:flex-auto",
        "@md/field-group:has- [>[data-slot=field-content]]:items-start @md/field-group:has- [>[data-slot=field-content]]:[&[role=checkbox],[role=radio]]:mt-px",
      ],
    },
    defaultVariants: {
      orientation: "vertical",
    },
  }
)

function Field({
  className,
  orientation = "vertical",
  ...props
}: React.ComponentProps<"div"> & VariantProps<typeof fieldVariants>) {
  return (
    <div
      role="group"
      data-slot="field"
      data-orientation={orientation}
      className={cn(fieldVariants({ orientation }), className)}
    />
  )
}
```

```

    {...props}
  )
  />
}

function FieldContent({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="field-content"
      className={cn(
        "group/field-content flex flex-1 flex-col gap-1.5 leading-snug",
        className
      )}
      {...props}
    />
  )
}

function FieldLabel({
  className,
  ...props
}: React.ComponentProps<typeof Label>) {
  return (
    <Label
      data-slot="field-label"
      className={cn(
        "group/field-label peer/field-label flex w-fit gap-2 leading-snug group-data-[disabled=true]/field:opacity-50",
        "has- [>[data-slot=field]]:w-full has- [>[data-slot=field]]:flex-col has- [>[data-slot=field]]:rounded-md has- [>[data-
slot=field]]:border [>[data-slot=field]]:p-4",
        "has-data-[state=checked]:bg-primary/5 has-data-[state=checked]:border-primary dark:has-data-[state=checked]:bg-
primary/10",
        className
      )}
      {...props}
    />
  )
}

function FieldTitle({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="field-label"
      className={cn(
        "flex w-fit items-center gap-2 text-sm font-medium leading-snug group-data-[disabled=true]/field:opacity-50",
        className
      )}
      {...props}
    />
  )
}

function FieldDescription({ className, ...props }: React.ComponentProps<"p">) {
  return (
    <p
      data-slot="field-description"
      className={cn(
        "text-muted-foreground text-sm font-normal leading-normal group-has-[[data-orientation=horizontal]]/field:text-balance",
        "nth-last-2:-mt-1 last:mt-0 [[data-variant=legend]]+&]:-mt-1.5",
        "[&a:hover]:text-primary [&a]:underline [&a]:underline-offset-4",
        className
      )}
      {...props}
    />
  )
}

function FieldSeparator({
  children,
  className,
  ...props
}: React.ComponentProps<"div"> & {
  children?: React.ReactNode
}) {
  return (
    <div
      data-slot="field-separator"
      data-content={!children}
      className={cn(
        "relative -my-2 h-5 text-sm group-data-[variant=outline]/field-group:-mb-2",
        className
      )}
      {...props}
    >
      <Separator className="absolute inset-0 top-1/2" />
      {children && (
        <span
          className="bg-background text-muted-foreground relative mx-auto block w-fit px-2"
          data-slot="field-separator-content"
        >
          {children}
        </span>
      )}
    </div>
  )
}

function FieldError({

```

```

    className,
    children,
    errors,
    ...props
}: React.ComponentProps<"div"> & {
  errors?: Array<{ message?: string } | undefined>
}) {
  const content = useMemo(() => {
    if (children) {
      return children
    }

    if (!errors) {
      return null
    }

    if (errors?.length === 1 && errors[0]?.message) {
      return errors[0].message
    }

    return (
      <ul className="ml-4 flex list-disc flex-col gap-1">
        {errors.map(
          (error, index) =>
            error?.message && <li key={index}>{error.message}</li>
        )}
      </ul>
    )
  }, [children, errors])

  if (!content) {
    return null
  }

  return (
    <div
      role="alert"
      data-slot="field-error"
      className={cn("text-destructive text-sm font-normal", className)}
      {...props}>
      >
      {content}
    </div>
  )
}

export {
  Field,
  FieldLabel,
  FieldDescription,
  FieldError,
  FieldGroup,
  FieldLegend,
  FieldSeparator,
  FieldSet,
  FieldContent,
  FieldTitle,
}

```

[artifacts/mockup-sandbox/src/components/ui/form.tsx](#)

```
import * as React from "react"
import * as LabelPrimitive from "@radix-ui/react-label"
import { Slot } from "@radix-ui/react-slot"
import {
  Controller,
  FormProvider,
  useFormContext,
  type ControllerProps,
  type FieldPath,
  type FieldValues,
} from "react-hook-form"

import { cn } from "@lib/utils"
import { Label } from "@components/ui/label"

const Form = FormProvider

type FormFieldContextValue<
  TFieldValues extends FieldValues = FieldValues,
  TName extends FieldPath<TFieldValues> = FieldPath<TFieldValues>
> = {
  name: TName
}

const FormFieldContext = React.createContext<FormFieldContextValue | null>(null)

const FormField = <
  TFieldValues extends FieldValues = FieldValues,
  TName extends FieldPath<TFieldValues> = FieldPath<TFieldValues>
>({
  ..props
}): ControllerProps<TFieldValues, TName> => {
  return (
    <FormFieldContext.Provider value={{ name: props.name }}>
      <Controller {...props} />
    </FormFieldContext.Provider>
  )
}

const useFormField = () => {
  const fieldContext = React.useContext(FormFieldContext)
  const itemContext = React.useContext(FormItemContext)
  const { getFieldState, formState } = useFormContext()

  if (!fieldContext) {
    throw new Error("useFormField should be used within <FormField>")
  }

  if (!itemContext) {
    throw new Error("useFormField should be used within <FormItem>")
  }

  const fieldState = getFieldState(fieldContext.name, formState)

  const { id } = itemContext

  return {
    id,
    name: fieldContext.name,
    formItemId: `${id}-form-item`,
    formDescriptionId: `${id}-form-item-description`,
    formMessageId: `${id}-form-item-message`,
    ...fieldState,
  }
}

type FormItemContextValue = {
  id: string
}

const FormItemContext = React.createContext<FormItemContextValue | null>(null)

const FormItem = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => {
  const id = React.useId()

  return (
    <FormItemContext.Provider value={{ id }}>
      <div ref={ref} className={cn("space-y-2", className)} {...props} />
    </FormItemContext.Provider>
  )
})
FormItem.displayName = "FormItem"

const FormLabel = React.forwardRef<
  React.ElementRef<typeof LabelPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof LabelPrimitive.Root>
>(({ className, ...props }, ref) => {
  const { error, formItemId } = useFormField()

  return (
    <Label
      ref={ref}
      {error ? "error" : "default"}
    >
      {props.children}
    </Label>
  )
})
FormLabel.displayName = "FormLabel"
```



```

        className={cn(error && "text-destructive", className)}
        htmlFor={formItemId}
        {...props}
      />
    )
  })
  FormLabel.displayName = "FormLabel"

  const FormControl = React.forwardRef<
    React.ElementRef<typeof Slot>,
    React.ComponentPropsWithoutRef<typeof Slot>
  >(({ ...props }, ref) => {
    const { error, formItemId, formDescriptionId, formMessageId } = useFormField()

    return (
      <Slot
        ref={ref}
        id={formItemId}
        aria-describedby={
          !error
            ? `${formDescriptionId}`
            : `${formDescriptionId} ${formMessageId}`
        }
        aria-invalid={!!error}
        {...props}
      />
    )
  })
  FormControl.displayName = "FormControl"

  const FormDescription = React.forwardRef<
    HTMLParagraphElement,
    React.HTMLAttributes<HTMLParagraphElement>
  >(({ className, ...props }, ref) => {
    const { formDescriptionId } = useFormField()

    return (
      <p
        ref={ref}
        id={formDescriptionId}
        className={cn("text-[0.8rem] text-muted-foreground", className)}
        {...props}
      />
    )
  })
  FormDescription.displayName = "FormDescription"

  const FormMessage = React.forwardRef<
    HTMLParagraphElement,
    React.HTMLAttributes<HTMLParagraphElement>
  >(({ className, children, ...props }, ref) => {
    const { error, formMessageId } = useFormField()
    const body = error ? String(error?.message ?? "") : children

    if (!body) {
      return null
    }

    return (
      <p
        ref={ref}
        id={formMessageId}
        className={cn("text-[0.8rem] font-medium text-destructive", className)}
        {...props}
      >
        {body}
      </p>
    )
  })
  FormMessage.displayName = "FormMessage"

  export {
    useFormField,
    Form,
    FormItem,
    FormLabel,
    FormControl,
    FormDescription,
    FormMessage,
    FormField,
  }
}

```

[artifacts/mockup-sandbox/src/components/ui/hover-card.tsx](#)

```
import * as React from "react"
import * as HoverCardPrimitive from "@radix-ui/react-hover-card"

import { cn } from "@/lib/utils"

const HoverCard = HoverCardPrimitive.Root

const HoverCardTrigger = HoverCardPrimitive.Trigger

const HoverCardContent = React.forwardRef<
  React.ElementRef<typeof HoverCardPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof HoverCardPrimitive.Content>
>(({ className, align = "center", sideOffset = 4, ...props }, ref) => {
  <HoverCardPrimitive.Content
    ref={ref}
    align={align}
    sideOffset={sideOffset}
    className={cn(
      "z-50 w-64 rounded-md border bg-popover p-4 text-popover-foreground shadow-md outline-none data-[state=open]:animate-in
data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-
[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-
in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin-[--radix-hover-card-content-transform-origin]",
      className
    )}
    {...props}
  />
))
HoverCardContent.displayName = HoverCardPrimitive.Content.displayName

export { HoverCard, HoverCardTrigger, HoverCardContent }
```

[artifacts/mockup-sandbox/src/components/ui/input-group.tsx](#)

```
import * as React from "react"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"
import { Button } from "@components/ui/button"
import { Input } from "@components/ui/input"
import { Textarea } from "@components/ui/textarea"

function InputGroup({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="input-group"
      role="group"
      className={cn(
        "group/input-group border-input dark:bg-input/30 shadow-xs relative flex w-full items-center rounded-md border outline-
none transition-[color,box-shadow]",
        "h-9 has-[>textarea]:h-auto",

        "has-[>[data-align=inline-start]]:[&>input]:pl-2",
        "has-[>[data-align=inline-end]]:[&>input]:pr-2",
        "has-[>[data-align=block-start]]:h-auto has-[>[data-align=block-start]]:flex-col has-[>[data-align=block-start]]:
[&>input]:pb-3",
        "has-[>[data-align=block-end]]:h-auto has-[>[data-align=block-end]]:flex-col has-[>[data-align=block-end]]:
[&>input]:pt-3",

        "has-[[data-slot=input-group-control]:focus-visible]:ring-ring has-[[data-slot=input-group-control]:focus-
visible]:ring-1",

        "has-[[data-slot][aria-invalid=true]]:ring-destructive/20 has-[[data-slot][aria-invalid=true]]:border-destructive
dark:has-[[data-slot][aria-invalid=true]]:ring-destructive/40",

        className
      )}
      {...props}
    />
  )
}

const inputGroupAddonVariants = cva(
  "text-muted-foreground flex h-auto cursor-text select-none items-center justify-center gap-2 py-1.5 text-sm font-medium group-data-
[disabled=true]/input-group:opacity-50 [&>kbd]:rounded-[calc(var(--radius)-5px)] [&>svg:not([class*='size-']):size-4",
  {
    variants: {
      align: {
        "inline-start":
          "order-first pl-3 has-[>button]:ml-[-0.45rem] has-[>kbd]:ml-[-0.35rem]",
        "inline-end":
          "order-last pr-3 has-[>button]:mr-[-0.4rem] has-[>kbd]:mr-[-0.35rem]",
        "block-start":
          "[.border-b]:pb-3 order-first w-full justify-start px-3 pt-3 group-has-[>input]/input-group:pt-2.5",
        "block-end":
          "[.border-t]:pt-3 order-last w-full justify-start px-3 pb-3 group-has-[>input]/input-group:pb-2.5",
      },
    },
    defaultVariants: {
      align: "inline-start",
    },
  }
)

function InputGroupAddon({
  className,
  align = "inline-start",
  ...props
}: React.ComponentProps<"div"> & VariantProps<typeof inputGroupAddonVariants>) {
  return (
    <div
      role="group"
      data-slot="input-group-addon"
      data-align={align}
      className={cn(inputGroupAddonVariants({ align }), className)}
      onClick={(e) => {
        if ((e.target as HTMLElement).closest("button")) {
          return
        }
        e.currentTarget.parentElement?.querySelector("input")?.focus()
      }}
      {...props}
    />
  )
}

const inputGroupButtonVariants = cva(
  "flex items-center gap-2 text-sm shadow-none",
  {
    variants: {
      size: {
        xs: "h-6 gap-1 rounded-[calc(var(--radius)-5px)] px-2 has-[>svg]:px-2 [&>svg:not([class*='size-']):size-3.5",
        sm: "h-8 gap-1.5 rounded-md px-2.5 has-[>svg]:px-2.5",
        "icon-xs":
          "size-6 rounded-[calc(var(--radius)-5px)] p-0 has-[>svg]:p-0",
        "icon-sm": "size-8 p-0 has-[>svg]:p-0",
      },
    },
    defaultVariants: {
```

```

        size: "xs",
      },
    ],
  },
)

function InputGroupButton({
  className,
  type = "button",
  variant = "ghost",
  size = "xs",
  ...props
}: Omit<React.ComponentProps<typeof Button>, "size"> &
VariantProps<typeof inputGroupButtonVariants> > {
  return (
    <Button
      type={type}
      data-size={size}
      variant={variant}
      className={cn(inputGroupButtonVariants({ size })), className}
      {...props}
    />
  )
}

function InputGroupText({ className, ...props }: React.ComponentProps<"span">) {
  return (
    <span
      className={cn(
        "text-muted-foreground flex items-center gap-2 text-sm [&_svg:not([class*='size-'])]:size-4 [&_svg]:pointer-events-
none",
        className
      )}
      {...props}
    />
  )
}

function InputGroupInput({
  className,
  ...props
}: React.ComponentProps<"input">) {
  return (
    <Input
      data-slot="input-group-control"
      className={cn(
        "flex-1 rounded-none border-0 bg-transparent shadow-none focus-visible:ring-0 dark:bg-transparent",
        className
      )}
      {...props}
    />
  )
}

function InputGroupTextarea({
  className,
  ...props
}: React.ComponentProps<"textarea">) {
  return (
    <Textarea
      data-slot="input-group-control"
      className={cn(
        "flex-1 resize-none rounded-none border-0 bg-transparent py-3 shadow-none focus-visible:ring-0 dark:bg-transparent",
        className
      )}
      {...props}
    />
  )
}

export {
  InputGroup,
  InputGroupAddon,
  InputGroupButton,
  InputGroupText,
  InputGroupInput,
  InputGroupTextarea,
}

```

[artifacts/mockup-sandbox/src/components/ui/input-otp.tsx](#)

```
import * as React from "react"
import { OTPInput, OTPInputContext } from "input-otp"
import { Minus } from "lucide-react"

import { cn } from "@/lib/utils"

const InputOTP = React.forwardRef<
  React.ElementRef<typeof OTPInput>,
  React.ComponentPropsWithoutRef<typeof OTPInput>
>(({ className, containerClassName, ...props }, ref) => (
  <OTPInput
    ref={ref}
    containerClassName={cn(
      "flex items-center gap-2 has-[:disabled]:opacity-50",
      containerClassName
    )}
    className={cn("disabled:cursor-not-allowed", className)}
    {...props}
  />
))
InputOTP.displayName = "InputOTP"

const InputOTPGroup = React.forwardRef<
  React.ElementRef<"div">,
  React.ComponentPropsWithoutRef<"div">
>(({ className, ...props }, ref) => (
  <div ref={ref} className={cn("flex items-center", className)} {...props} />
))
InputOTPGroup.displayName = "InputOTPGroup"

const InputOTPSlot = React.forwardRef<
  React.ElementRef<"div">,
  React.ComponentPropsWithoutRef<"div"> & { index: number }
>(({ index, className, ...props }, ref) => {
  const inputOTPContext = React.useContext(OTPInputContext)
  const { char, hasFakeCaret, isActive } = inputOTPContext.slots[index]

  return (
    <div
      ref={ref}
      className={cn(
        "relative flex h-9 w-9 items-center justify-center border-y border-r border-input text-sm shadow-sm transition-all
        first:rounded-l-md first:border-l last:rounded-r-md",
        isActive && "z-10 ring-1 ring-ring",
        className
      )}
      {...props}
    >
      {char}
      {hasFakeCaret && (
        <div className="pointer-events-none absolute inset-0 flex items-center justify-center">
          <div className="h-4 w-px animate-caret-blink bg-foreground duration-1000" />
        </div>
      )}
    </div>
  )
)}
InputOTPSlot.displayName = "InputOTPSlot"

const InputOTPSeparator = React.forwardRef<
  React.ElementRef<"div">,
  React.ComponentPropsWithoutRef<"div">
>(({ ...props }, ref) => (
  <div ref={ref} role="separator" {...props}>
    <Minus />
  </div>
))
InputOTPSeparator.displayName = "InputOTPSeparator"

export { InputOTP, InputOTPGroup, InputOTPSlot, InputOTPSeparator }
```

[artifacts/mockup-sandbox/src/components/ui/input.tsx](#)

```
import * as React from "react"

import { cn } from "@/lib/utils"

const Input = React.forwardRef<HTMLInputElement, React.ComponentProps<"input">>(
  ({ className, type, ...props }, ref) => {
    return (
      <input
        type={type}
        className={cn(
          "flex h-9 w-full rounded-md border border-input bg-transparent px-3 py-1 text-base shadow-sm transition-colors
file:border-0 file:bg-transparent file:text-sm file:font-medium file:text-foreground placeholder:text-muted-foreground focus-
visible:outline-none focus-visible:ring-1 focus-visible:ring-ring disabled:cursor-not-allowed disabled:opacity-50 md:text-sm",
          className
        )}
        ref={ref}
        {...props}
      />
    )
  }
)

Input.displayName = "Input"

export { Input }
```

[artifacts/mockup-sandbox/src/components/ui/item.tsx](#)

```
import * as React from "react"
import { Slot } from "@radix-ui/react-slot"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"
import { Separator } from "@components/ui/separator"

function ItemGroup({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      role="list"
      data-slot="item-group"
      className={cn("group/item-group flex flex-col", className)}
      {...props}
    />
  )
}

function ItemSeparator({
  className,
  ...props
}: React.ComponentProps<typeof Separator>) {
  return (
    <Separator
      data-slot="item-separator"
      orientation="horizontal"
      className={cn("my-0", className)}
      {...props}
    />
  )
}

const itemVariants = cva(
  "group/item [a]:hover:bg-accent/50 focus-visible:border-ring focus-visible:ring-ring/50 [a]:transition-colors flex flex-wrap items-center rounded-md border border-transparent text-sm outline-none transition-colors duration-100 focus-visible:ring-[3px]",
  {
    variants: {
      variant: {
        default: "bg-transparent",
        outline: "border-border",
        muted: "bg-muted/50",
      },
      size: {
        default: "gap-4 p-4 ",
        sm: "gap-2.5 px-4 py-3",
      },
    },
    defaultVariants: {
      variant: "default",
      size: "default",
    },
  }
)

function Item({
  className,
  variant = "default",
  size = "default",
  asChild = false,
  ...props
}: React.ComponentProps<"div"> &
VariantProps<typeof itemVariants> & { asChild?: boolean }) {
  const Comp = asChild ? Slot : "div"
  return (
    <Comp
      data-slot="item"
      data-variant={variant}
      data-size={size}
      className={cn(itemVariants({ variant, size, className }))}
      {...props}
    />
  )
}

const itemMediaVariants = cva(
  "flex shrink-0 items-center justify-center gap-2 group-has-[[data-slot=item-description]]/item:translate-y-0.5 group-has-[[data-slot=item-description]]/item:self-start [&_svg]:pointer-events-none",
  {
    variants: {
      variant: {
        default: "bg-transparent",
        icon: "bg-muted size-8 rounded-sm border [&_svg:not([class*='size-']):size-4",
        image: "size-10 overflow-hidden rounded-sm [&_img]:size-full [&_img]:object-cover",
      },
    },
    defaultVariants: {
      variant: "default",
    },
  }
)

function ItemMedia({
  className,
  variant = "default",
```

```

    ...props
  }: React.ComponentProps<"div"> & VariantProps<typeof itemMediaVariants>) {
    return (
      <div
        data-slot="item-media"
        data-variant={variant}
        className={cn(itemMediaVariants({ variant, className }))}
        {...props}
      />
    )
  }

function ItemContent({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="item-content"
      className={cn(
        "flex flex-1 flex-col gap-1 [&[data-slot=item-content]]:flex-none",
        className
      )}
      {...props}
    />
  )
}

function ItemTitle({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="item-title"
      className={cn(
        "flex w-fit items-center gap-2 text-sm font-medium leading-snug",
        className
      )}
      {...props}
    />
  )
}

function ItemDescription({ className, ...props }: React.ComponentProps<"p">) {
  return (
    <p
      data-slot="item-description"
      className={cn(
        "text-muted-foreground line-clamp-2 text-balance text-sm font-normal leading-normal",
        "[&a:hover]:text-primary [&a]:underline [&a]:underline-offset-4",
        className
      )}
      {...props}
    />
  )
}

function ItemActions({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="item-actions"
      className={cn("flex items-center gap-2", className)}
      {...props}
    />
  )
}

function ItemHeader({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="item-header"
      className={cn(
        "flex basis-full items-center justify-between gap-2",
        className
      )}
      {...props}
    />
  )
}

function ItemFooter({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="item-footer"
      className={cn(
        "flex basis-full items-center justify-between gap-2",
        className
      )}
      {...props}
    />
  )
}

export {
  Item,
  ItemMedia,
  ItemContent,
  ItemActions,
  ItemGroup,
  ItemSeparator,
  ItemTitle,
  ItemDescription,

```



```
ItemHeader,  
ItemFooter,  
}
```

[artifacts/mockup-sandbox/src/components/ui/kbd.tsx](#)

```
import { cn } from "@/lib/utils"

function Kbd({ className, ...props }: React.ComponentProps<"kbd">) {
  return (
    <kbd
      data-slot="kbd"
      className={cn(
        "bg-muted text-muted-foreground pointer-events-none inline-flex h-5 w-fit min-w-5 select-none items-center justify-
center gap-1 rounded-sm px-1 font-sans text-xs font-medium",
        "[&_svg:not([class*='size-']):size-3",
        "[[data-slot=tooltip-content]_&]:bg-background/20 [[data-slot=tooltip-content]_&]:text-background dark:[[data-
slot=tooltip-content]_&]:bg-background/10",
        className
      )}
      {...props}
    />
  )
}

function KbdGroup({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <kbd
      data-slot="kbd-group"
      className={cn("inline-flex items-center gap-1", className)}
      {...props}
    />
  )
}

export { Kbd, KbdGroup }
```

[artifacts/mockup-sandbox/src/components/ui/label.tsx](#)

```
"use client"

import * as React from "react"
import * as LabelPrimitive from "@radix-ui/react-label"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"

const labelVariants = cva(
  "text-sm font-medium leading-none peer-disabled:cursor-not-allowed peer-disabled:opacity-70"
)

const Label = React.forwardRef<
  React.ElementRef<typeof LabelPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof LabelPrimitive.Root> &
    VariantProps<typeof labelVariants>
>(({ className, ...props }, ref) => (
  <LabelPrimitive.Root
    ref={ref}
    className={cn(labelVariants(), className)}
    {...props}
  />
))
Label.displayName = LabelPrimitive.Root.displayName

export { Label }
```

[artifacts/mockup-sandbox/src/components/ui/menubar.tsx](#)

```
import * as React from "react"
import * as MenubarPrimitive from "@radix-ui/react-menubar"
import { Check, ChevronRight, Circle } from "lucide-react"

import { cn } from "@/lib/utils"

function MenubarMenu({
  ...props
}: React.ComponentProps<typeof MenubarPrimitive.Menu>) {
  return <MenubarPrimitive.Menu {...props} />
}

function MenubarGroup({
  ...props
}: React.ComponentProps<typeof MenubarPrimitive.Group>) {
  return <MenubarPrimitive.Group {...props} />
}

function MenubarPortal({
  ...props
}: React.ComponentProps<typeof MenubarPrimitive.Portal>) {
  return <MenubarPrimitive.Portal {...props} />
}

function MenubarRadioGroup({
  ...props
}: React.ComponentProps<typeof MenubarPrimitive.RadioGroup>) {
  return <MenubarPrimitive.RadioGroup {...props} />
}

function MenubarSub({
  ...props
}: React.ComponentProps<typeof MenubarPrimitive.Sub>) {
  return <MenubarPrimitive.Sub data-slot="menubar-sub" {...props} />
}

const Menubar = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Root>
>(({ className, ...props }, ref) => (
  <MenubarPrimitive.Root
    ref={ref}
    className={cn(
      "flex h-9 items-center space-x-1 rounded-md border bg-background p-1 shadow-sm",
      className
    )}
    {...props}
  />
))
Menubar.displayName = MenubarPrimitive.Root.displayName

const MenubarTrigger = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.Trigger>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Trigger>
>(({ className, ...props }, ref) => (
  <MenubarPrimitive.Trigger
    ref={ref}
    className={cn(
      "flex cursor-default select-none items-center rounded-sm px-3 py-1 text-sm font-medium outline-none focus:bg-accent
      focus:text-accent-foreground data-[state=open]:bg-accent data-[state=open]:text-accent-foreground",
      className
    )}
    {...props}
  />
))
MenubarTrigger.displayName = MenubarPrimitive.Trigger.displayName

const MenubarSubTrigger = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.SubTrigger>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.SubTrigger> & {
    inset?: boolean
  }
>(({ className, inset, children, ...props }, ref) => (
  <MenubarPrimitive.SubTrigger
    ref={ref}
    className={cn(
      "flex cursor-default select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none focus:bg-accent focus:text-
      accent-foreground data-[state=open]:bg-accent data-[state=open]:text-accent-foreground",
      inset && "pl-8",
      className
    )}
    {...props}
  >
    {children}
  <ChevronRight className="ml-auto h-4 w-4" />
  </MenubarPrimitive.SubTrigger>
))
MenubarSubTrigger.displayName = MenubarPrimitive.SubTrigger.displayName

const MenubarSubContent = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.SubContent>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.SubContent>
>(({ className, ...props }, ref) => (
  <MenubarPrimitive.SubContent
    ref={ref}
```

```

        className={cn(
          "z-50 min-w-[8rem] overflow-hidden rounded-md border bg-popover p-1 text-popover-foreground shadow-lg data-
[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-
[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-
right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin[--radix-menubar-content-transform-
origin]",
          className
        )}
      )}
    {...props}
  )}
  />
))
MenubarSubContent.displayName = MenubarPrimitive.SubContent.displayName

const MenubarContent = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Content>
>({
  (
    { className, align = "start", alignOffset = -4, sideOffset = 8, ...props },
    ref
  ) => (
    <MenubarPrimitive.Portal>
      <MenubarPrimitive.Content
        ref={ref}
        align={align}
        alignOffset={alignOffset}
        sideOffset={sideOffset}
        className={cn(
          "z-50 min-w-[12rem] overflow-hidden rounded-md border bg-popover p-1 text-popover-foreground shadow-md data-
[state=open]:animate-in data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-
[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-
in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin[--radix-menubar-content-transform-origin]",
          className
        )}
      >
        {...props}
      </MenubarPrimitive.Portal>
    )
  )
  MenubarContent.displayName = MenubarPrimitive.Content.displayName

const MenubarItem = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Item> & {
    inset?: boolean
  }
>(({ className, inset, ...props }, ref) => (
  <MenubarPrimitive.Item
    ref={ref}
    className={cn(
      "relative flex cursor-default select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none focus:bg-accent
focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      inset && "pl-8",
      className
    )}
    {...props}
  >
    />
  ))
  MenubarItem.displayName = MenubarPrimitive.Item.displayName

const MenubarCheckboxItem = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.CheckboxItem>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.CheckboxItem>
>(({ className, children, checked, ...props }, ref) => (
  <MenubarPrimitive.CheckboxItem
    ref={ref}
    className={cn(
      "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-none focus:bg-accent
focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      className
    )}
    checked={checked}
    {...props}
  >
    <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
      <MenubarPrimitive.ItemIndicator>
        <Check className="h-4 w-4" />
      </MenubarPrimitive.ItemIndicator>
    </span>
    {children}
  </MenubarPrimitive.CheckboxItem>
  ))
  MenubarCheckboxItem.displayName = MenubarPrimitive.CheckboxItem.displayName

const MenubarRadioItem = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.RadioItem>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.RadioItem>
>(({ className, children, ...props }, ref) => (
  <MenubarPrimitive.RadioItem
    ref={ref}
    className={cn(
      "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-none focus:bg-accent
focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      className
    )}
    {...props}
  >
    />
  ))
  MenubarRadioItem.displayName = MenubarPrimitive.RadioItem.displayName

```

```

    <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
      <MenubarPrimitive.ItemIndicator>
        <Circle className="h-4 w-4 fill-current" />
      </MenubarPrimitive.ItemIndicator>
    </span>
    {children}
  </MenubarPrimitive.RadioItem>
))
MenubarRadioItem.displayName = MenubarPrimitive.RadioItem.displayName

const MenubarLabel = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.Label>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Label> & {
    inset?: boolean
  }
>(({ className, inset, ...props }, ref) => (
  <MenubarPrimitive.Label
    ref={ref}
    className={cn(
      "px-2 py-1.5 text-sm font-semibold",
      inset && "pl-8",
      className
    )}
    {...props}
  />
))
MenubarLabel.displayName = MenubarPrimitive.Label.displayName

const MenubarSeparator = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.Separator>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Separator>
>(({ className, ...props }, ref) => (
  <MenubarPrimitive.Separator
    ref={ref}
    className={cn("-mx-1 my-1 h-px bg-muted", className)}
    {...props}
  />
))
MenubarSeparator.displayName = MenubarPrimitive.Separator.displayName

const MenubarShortcut = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLSpanElement>) => {
  return (
    <span
      className={cn(
        "ml-auto text-xs tracking-widest text-muted-foreground",
        className
      )}
      {...props}
    />
  )
}
MenubarShortcut.displayName = "MenubarShortcut"

export {
  Menubar,
  MenubarMenu,
  MenubarTrigger,
  MenubarContent,
  MenubarItem,
  MenubarSeparator,
  MenubarLabel,
  MenubarCheckboxItem,
  MenubarRadioGroup,
  MenubarRadioItem,
  MenubarPortal,
  MenubarSubContent,
  MenubarSubTrigger,
  MenubarGroup,
  MenubarSub,
  MenubarShortcut,
}

```

[artifacts/mockup-sandbox/src/components/ui/navigation-menu.tsx](#)

```
import * as React from "react"
import * as NavigationMenuPrimitive from "@radix-ui/react-navigation-menu"
import { cva } from "class-variance-authority"
import { ChevronDown } from "lucide-react"

import { cn } from "@/lib/utils"

const NavigationMenu = React.forwardRef<
  React.ElementRef<typeof NavigationMenuPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Root>
>(({ className, children, ...props }, ref) => (
  <NavigationMenuPrimitive.Root
    ref={ref}
    className={cn(
      "relative z-10 flex max-w-max flex-1 items-center justify-center",
      className
    )}
    {...props}
  >
    {children}
  </NavigationMenuPrimitive.Root>
))
NavigationMenu.displayName = NavigationMenuPrimitive.Root.displayName

const NavigationMenuList = React.forwardRef<
  React.ElementRef<typeof NavigationMenuPrimitive.List>,
  React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.List>
>(({ className, ...props }, ref) => (
  <NavigationMenuPrimitive.List
    ref={ref}
    className={cn(
      "group flex flex-1 list-none items-center justify-center space-x-1",
      className
    )}
    {...props}
  />
))
NavigationMenuList.displayName = NavigationMenuPrimitive.List.displayName

const NavigationMenuItem = NavigationMenuPrimitive.Item

const navigationMenuTriggerStyle = cva(
  "group inline-flex h-9 w-max items-center justify-center rounded-md bg-background px-4 py-2 text-sm font-medium transition-colors hover:bg-accent hover:text-accent-foreground focus:bg-accent focus:text-accent-foreground focus:outline-none disabled:pointer-events-none disabled:opacity-50 data-[state=open]:text-accent-foreground data-[state=open]:bg-accent/50 data-[state=open]:hover:bg-accent data-[state=open]:focus:bg-accent"
)

const NavigationMenuTrigger = React.forwardRef<
  React.ElementRef<typeof NavigationMenuPrimitive.Trigger>,
  React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Trigger>
>(({ className, children, ...props }, ref) => (
  <NavigationMenuPrimitive.Trigger
    ref={ref}
    className={cn(navigationMenuTriggerStyle(), "group", className)}
    {...props}
  >
    {children}{" "}
    <ChevronDown
      className="relative top-[1px] ml-1 h-3 w-3 transition duration-300 group-data-[state=open]:rotate-180"
      aria-hidden="true"
    />
  </NavigationMenuPrimitive.Trigger>
))
NavigationMenuTrigger.displayName = NavigationMenuPrimitive.Trigger.displayName

const NavigationMenuContent = React.forwardRef<
  React.ElementRef<typeof NavigationMenuPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Content>
>(({ className, ...props }, ref) => (
  <NavigationMenuPrimitive.Content
    ref={ref}
    className={cn(
      "left-0 top-0 w-full data-[motion^=from-]:animate-in data-[motion^=to-]:animate-out data-[motion^=from-]:fade-in data-[motion^=to-]:fade-out data-[motion=from-end]:slide-in-from-right-52 data-[motion=from-start]:slide-in-from-left-52 data-[motion=to-end]:slide-out-to-right-52 data-[motion=to-start]:slide-out-to-left-52 md:absolute md:w-auto ",
      className
    )}
    {...props}
  />
))
NavigationMenuContent.displayName = NavigationMenuPrimitive.Content.displayName

const NavigationMenuLink = NavigationMenuPrimitive.Link

const NavigationMenuViewport = React.forwardRef<
  React.ElementRef<typeof NavigationMenuPrimitive.Viewport>,
  React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Viewport>
>(({ className, ...props }, ref) => (
  <div className={cn("absolute left-0 top-full flex justify-center")}>
    <NavigationMenuPrimitive.Viewport
      className={cn(
        "origin-top-center relative mt-1.5 h-[var(--radix-navigation-menu-viewport-height)] w-full overflow-hidden rounded-md border bg-popover text-popover-foreground shadow data-[state=open]:animate-in data-[state=closed]:animate-out data-
```

```

[state=closed]:zoom-out-95 data-[state=open]:zoom-in-90 md:w-[var(--radix-navigation-menu-viewport-width))",
  className
)}
ref={ref}
{...props}
/>
</div>
))
NavigationMenuViewport.displayName =
  NavigationMenuPrimitive.Viewport.displayName

const NavigationMenuIndicator = React.forwardRef<
  React.ElementRef<typeof NavigationMenuPrimitive.Indicator>,
  React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Indicator>
>(({ className, ...props }, ref) => (
  <NavigationMenuPrimitive.Indicator
    ref={ref}
    className={cn(
      "top-full z-[1] flex h-1.5 items-end justify-center overflow-hidden data-[state=visible]:animate-in data-
[state=hidden]:animate-out data-[state=hidden]:fade-out data-[state=visible]:fade-in",
      className
    )}
    {...props}
  >
    <div className="relative top-[60%] h-2 w-2 rotate-45 rounded-tl-sm bg-border shadow-md" />
  </NavigationMenuPrimitive.Indicator>
))
NavigationMenuIndicator.displayName =
  NavigationMenuPrimitive.Indicator.displayName

export {
  navigationMenuTriggerStyle,
  NavigationMenu,
  NavigationMenuList,
  NavigationMenuItem,
  NavigationMenuContent,
  NavigationMenuTrigger,
  NavigationMenuLink,
  NavigationMenuIndicator,
  NavigationMenuViewport,
}

```


[artifacts/mockup-sandbox/src/components/ui/pagination.tsx](#)

```
import * as React from "react"
import { ChevronLeft, ChevronRight, MoreHorizontal } from "lucide-react"

import { cn } from "@lib/utls"
import { ButtonProps, buttonVariants } from "@components/ui/button"

const Pagination = ({ className, ...props }: React.ComponentProps<"nav">) => (
  <nav
    role="navigation"
    aria-label="pagination"
    className={cn("mx-auto flex w-full justify-center", className)}
    {...props}
  />
)
Pagination.displayName = "Pagination"

const PaginationContent = React.forwardRef<
  HTMLUListElement,
  React.ComponentProps<"ul">
>(({ className, ...props }, ref) => (
  <ul
    ref={ref}
    className={cn("flex flex-row items-center gap-1", className)}
    {...props}
  />
))
PaginationContent.displayName = "PaginationContent"

const PaginationItem = React.forwardRef<
  HTMLLIElement,
  React.ComponentProps<"li">
>(({ className, ...props }, ref) => (
  <li ref={ref} className={cn("", className)} {...props} />
))
PaginationItem.displayName = "PaginationItem"

type PaginationLinkProps = {
  isActive?: boolean
} & Pick<ButtonProps, "size"> &
  React.ComponentProps<"a">

const PaginationLink = ({
  className,
  isActive,
  size = "icon",
  ...props
}: PaginationLinkProps) => (
  <a
    aria-current={isActive ? "page" : undefined}
    className={cn(
      buttonVariants({
        variant: isActive ? "outline" : "ghost",
        size,
      }),
      className
    )}
    {...props}
  />
)
PaginationLink.displayName = "PaginationLink"

const PaginationPrevious = ({
  className,
  ...props
}: React.ComponentProps<typeof PaginationLink>) => (
  <PaginationLink
    aria-label="Go to previous page"
    size="default"
    className={cn("gap-1 pl-2.5", className)}
    {...props}
  >
    <ChevronLeft className="h-4 w-4" />
    <span>Previous</span>
  </PaginationLink>
)
PaginationPrevious.displayName = "PaginationPrevious"

const PaginationNext = ({
  className,
  ...props
}: React.ComponentProps<typeof PaginationLink>) => (
  <PaginationLink
    aria-label="Go to next page"
    size="default"
    className={cn("gap-1 pr-2.5", className)}
    {...props}
  >
    <span>Next</span>
    <ChevronRight className="h-4 w-4" />
  </PaginationLink>
)
PaginationNext.displayName = "PaginationNext"

const PaginationEllipsis = ({
  className,
```

```
    ...props
  }: React.ComponentProps<"span">) => (
    <span
      aria-hidden
      className={cn("flex h-9 w-9 items-center justify-center", className)}
      {...props}
    >
      <MoreHorizontal className="h-4 w-4" />
      <span className="sr-only">More pages</span>
    </span>
  )
  PaginationEllipsis.displayName = "PaginationEllipsis"

export {
  Pagination,
  PaginationContent,
  PaginationLink,
  PaginationItem,
  PaginationPrevious,
  PaginationNext,
  PaginationEllipsis,
}
```

[artifacts/mockup-sandbox/src/components/ui/popover.tsx](#)

```
import * as React from "react"
import * as PopoverPrimitive from "@radix-ui/react-popover"

import { cn } from "@lib/utils"

const Popover = PopoverPrimitive.Root

const PopoverTrigger = PopoverPrimitive.Trigger

const PopoverAnchor = PopoverPrimitive.Anchor

const PopoverContent = React.forwardRef<
  React.ElementRef<typeof PopoverPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof PopoverPrimitive.Content>
>(({ className, align = "center", sideOffset = 4, ...props }, ref) => (
  <PopoverPrimitive.Portal>
    <PopoverPrimitive.Content
      ref={ref}
      align={align}
      sideOffset={sideOffset}
      className={cn(
        "z-50 w-72 rounded-md border bg-popover p-4 text-popover-foreground shadow-md outline-none data-[state=open]:animate-in
data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-
[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-
in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin-[--radix-popover-content-transform-origin]",
        className
      )}
      {...props}
    />
  </PopoverPrimitive.Portal>
))
PopoverContent.displayName = PopoverPrimitive.Content.displayName

export { Popover, PopoverTrigger, PopoverContent, PopoverAnchor }
```

[artifacts/mockup-sandbox/src/components/ui/progress.tsx](#)

```
"use client"

import * as React from "react"
import * as ProgressPrimitive from "@radix-ui/react-progress"

import { cn } from "@/lib/utils"

const Progress = React.forwardRef<
  React.ElementRef<typeof ProgressPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof ProgressPrimitive.Root>
>(({ className, value, ...props }, ref) => (
  <ProgressPrimitive.Root
    ref={ref}
    className={cn(
      "relative h-2 w-full overflow-hidden rounded-full bg-primary/20",
      className
    )}
    {...props}
  >
    <ProgressPrimitive.Indicator
      className="h-full w-full flex-1 bg-primary transition-all"
      style={{ transform: `translateX(-${100 - (value || 0)}%)` }}
    />
  </ProgressPrimitive.Root>
))
Progress.displayName = ProgressPrimitive.Root.displayName

export { Progress }
```

[artifacts/mockup-sandbox/src/components/ui/radio-group.tsx](#)

```
import * as React from "react"
import * as RadioGroupPrimitive from "@radix-ui/react-radio-group"
import { Circle } from "lucide-react"

import { cn } from "@lib/utils"

const RadioGroup = React.forwardRef<
  React.ElementRef<typeof RadioGroupPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof RadioGroupPrimitive.Root>
>(({ className, ...props }, ref) => {
  return (
    <RadioGroupPrimitive.Root
      className={cn("grid gap-2", className)}
      {...props}
      ref={ref}
    />
  )
})
RadioGroup.displayName = RadioGroupPrimitive.Root.displayName

const RadioGroupItem = React.forwardRef<
  React.ElementRef<typeof RadioGroupPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof RadioGroupPrimitive.Item>
>(({ className, ...props }, ref) => {
  return (
    <RadioGroupPrimitive.Item
      ref={ref}
      className={cn(
        "aspect-square h-4 w-4 rounded-full border border-primary text-primary shadow focus:outline-none focus-visible:ring-1
focus-visible:ring-ring disabled:cursor-not-allowed disabled:opacity-50",
        className
      )}
      {...props}
    >
      <RadioGroupPrimitive.Indicator className="flex items-center justify-center">
        <Circle className="h-3.5 w-3.5 fill-primary" />
      </RadioGroupPrimitive.Indicator>
    </RadioGroupPrimitive.Item>
  )
})
RadioGroupItem.displayName = RadioGroupPrimitive.Item.displayName

export { RadioGroup, RadioGroupItem }
```

[artifacts/mockup-sandbox/src/components/ui/resizable.tsx](#)

```
"use client"

import { GripVertical } from "lucide-react"
import * as ResizablePrimitive from "react-resizable-panels"

import { cn } from "@/lib/utils"

const ResizablePanelGroup = ({
  className,
  ...props
}: React.ComponentProps<typeof ResizablePrimitive.PanelGroup>) => (
  <ResizablePrimitive.PanelGroup
    className={cn(
      "flex h-full w-full data-[panel-group-direction=vertical]:flex-col",
      className
    )}
    {...props}
  />
)

const ResizablePanel = ResizablePrimitive.Panel

const ResizableHandle = ({
  withHandle,
  className,
  ...props
}: React.ComponentProps<typeof ResizablePrimitive.PanelResizeHandle> & {
  withHandle?: boolean
}) => (
  <ResizablePrimitive.PanelResizeHandle
    className={cn(
      "relative flex w-px items-center justify-center bg-border after:absolute after:inset-y-0 after:left-1/2 after:w-1 after:-
      translate-x-1/2 focus-visible:outline-none focus-visible:ring-1 focus-visible:ring-ring focus-visible:ring-offset-1 data-[panel-
      group-direction=vertical]:h-px data-[panel-group-direction=vertical]:w-full data-[panel-group-direction=vertical]:after:left-0
      data-[panel-group-direction=vertical]:after:h-1 data-[panel-group-direction=vertical]:after:w-full data-[panel-group-
      direction=vertical]:after:-translate-y-1/2 data-[panel-group-direction=vertical]:after:translate-x-0 [&[data-panel-group-
      direction=vertical]>div]:rotate-90",
      className
    )}
    {...props}
  >
    {withHandle && (
      <div className="z-10 flex h-4 w-3 items-center justify-center rounded-sm border bg-border">
        <GripVertical className="h-2.5 w-2.5" />
      </div>
    )}
  </ResizablePrimitive.PanelResizeHandle>
)

export { ResizablePanelGroup, ResizablePanel, ResizableHandle }
```

[artifacts/mockup-sandbox/src/components/ui/scroll-area.tsx](#)

```
import * as React from "react"
import * as ScrollAreaPrimitive from "@radix-ui/react-scroll-area"

import { cn } from "@lib/utils"

const ScrollArea = React.forwardRef<
  React.ElementRef<typeof ScrollAreaPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof ScrollAreaPrimitive.Root>
>(({ className, children, ...props }, ref) => (
  <ScrollAreaPrimitive.Root
    ref={ref}
    className={cn("relative overflow-hidden", className)}
    {...props}
  >
    <ScrollAreaPrimitive.Viewport className="h-full w-full rounded-[inherit]">
      {children}
    </ScrollAreaPrimitive.Viewport>
    <ScrollBar />
    <ScrollAreaPrimitive.Corner />
  </ScrollAreaPrimitive.Root>
))
ScrollArea.displayName = ScrollAreaPrimitive.Root.displayName

const ScrollBar = React.forwardRef<
  React.ElementRef<typeof ScrollAreaPrimitive.ScrollAreaScrollbar>,
  React.ComponentPropsWithoutRef<typeof ScrollAreaPrimitive.ScrollAreaScrollbar>
>(({ className, orientation = "vertical", ...props }, ref) => (
  <ScrollAreaPrimitive.ScrollAreaScrollbar
    ref={ref}
    orientation={orientation}
    className={cn(
      "flex touch-none select-none transition-colors",
      orientation === "vertical" &&
        "h-full w-2.5 border-l border-l-transparent p-[1px]",
      orientation === "horizontal" &&
        "h-2.5 flex-col border-t border-t-transparent p-[1px]",
      className
    )}
    {...props}
  >
    <ScrollAreaPrimitive.ScrollAreaThumb className="relative flex-1 rounded-full bg-border" />
  </ScrollAreaPrimitive.ScrollAreaScrollbar>
))
ScrollBar.displayName = ScrollAreaPrimitive.ScrollAreaScrollbar.displayName

export { ScrollArea, ScrollBar }
```

[artifacts/mockup-sandbox/src/components/ui/select.tsx](#)

```
"use client"

import * as React from "react"
import * as SelectPrimitive from "@radix-ui/react-select"
import { Check, ChevronDown, ChevronUp } from "lucide-react"

import { cn } from "@lib/utils"

const Select = SelectPrimitive.Root

const SelectGroup = SelectPrimitive.Group

const SelectValue = SelectPrimitive.Value

const SelectTrigger = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.Trigger>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.Trigger>
>(({ className, children, ...props }, ref) => (
  <SelectPrimitive.Trigger
    ref={ref}
    className={cn(
      "flex h-9 w-full items-center justify-between whitespace-nowrap rounded-md border border-input bg-transparent px-3 py-2
text-sm shadow-sm ring-offset-background data-[placeholder]:text-muted-foreground focus:outline-none focus:ring-1 focus:ring-
ring disabled:cursor-not-allowed disabled:opacity-50 [>span]:line-clamp-1",
      className
    )}
    {...props}
  >
    {children}
    <SelectPrimitive.Icon asChild>
      <ChevronDown className="h-4 w-4 opacity-50" />
    </SelectPrimitive.Icon>
  </SelectPrimitive.Trigger>
))
SelectTrigger.displayName = SelectPrimitive.Trigger.displayName

const SelectScrollUpButton = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.ScrollUpButton>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.ScrollUpButton>
>(({ className, ...props }, ref) => (
  <SelectPrimitive.ScrollUpButton
    ref={ref}
    className={cn(
      "flex cursor-default items-center justify-center py-1",
      className
    )}
    {...props}
  >
    <ChevronUp className="h-4 w-4" />
  </SelectPrimitive.ScrollUpButton>
))
SelectScrollUpButton.displayName = SelectPrimitive.ScrollUpButton.displayName

const SelectScrollDownButton = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.ScrollDownButton>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.ScrollDownButton>
>(({ className, ...props }, ref) => (
  <SelectPrimitive.ScrollDownButton
    ref={ref}
    className={cn(
      "flex cursor-default items-center justify-center py-1",
      className
    )}
    {...props}
  >
    <ChevronDown className="h-4 w-4" />
  </SelectPrimitive.ScrollDownButton>
))
SelectScrollDownButton.displayName =
  SelectPrimitive.ScrollDownButton.displayName

const SelectContent = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.Content>
>(({ className, children, position = "popper", ...props }, ref) => (
  <SelectPrimitive.Portal>
    <SelectPrimitive.Content
      ref={ref}
      className={cn(
        "relative z-50 max-h-[--radix-select-content-available-height] min-w-[8rem] overflow-y-auto overflow-x-hidden rounded-
md border bg-popover text-popover-foreground shadow-md data-[state=open]:animate-in data-[state=closed]:animate-out data-
[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-
[state=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-
[state=top]:slide-in-from-bottom-2 origin-[--radix-select-content-transform-origin]",
        position === "popper" &&
        "data-[side=bottom]:translate-y-1 data-[side=left]:-translate-x-1 data-[side=right]:translate-x-1 data-[side=top]:-
translate-y-1",
        className
      )}
      position={position}
      {...props}
    >
      <SelectScrollUpButton />
      <SelectPrimitive.Viewport
        className={cn(
```



```

        "p-1",
        position === "popper" &&
        "h-[var(--radix-select-trigger-height)] w-full min-w-[var(--radix-select-trigger-width)]"
    )}
    >
    {children}
    </SelectPrimitive.Viewport>
    <SelectScrollDownButton />
    </SelectPrimitive.Content>
  </SelectPrimitive.Portal>
))
SelectContent.displayName = SelectPrimitive.Content.displayName

const SelectLabel = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.Label>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.Label>
>(({ className, ...props }, ref) => (
  <SelectPrimitive.Label
    ref={ref}
    className={cn("px-2 py-1.5 text-sm font-semibold", className)}
    {...props}
  />
))
SelectLabel.displayName = SelectPrimitive.Label.displayName

const SelectItem = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.Item>
>(({ className, children, ...props }, ref) => (
  <SelectPrimitive.Item
    ref={ref}
    className={cn(
      "relative flex w-full cursor-default select-none items-center rounded-sm py-1.5 pl-2 pr-8 text-sm outline-none focus:bg-
accent focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      className
    )}
    {...props}
  >
    <span className="absolute right-2 flex h-3.5 w-3.5 items-center justify-center">
      <SelectPrimitive.ItemIndicator>
        <Check className="h-4 w-4" />
      </SelectPrimitive.ItemIndicator>
    </span>
    <SelectPrimitive.ItemText>{children}</SelectPrimitive.ItemText>
  </SelectPrimitive.Item>
))
SelectItem.displayName = SelectPrimitive.Item.displayName

const SelectSeparator = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.Separator>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.Separator>
>(({ className, ...props }, ref) => (
  <SelectPrimitive.Separator
    ref={ref}
    className={cn("-mx-1 my-1 h-px bg-muted", className)}
    {...props}
  />
))
SelectSeparator.displayName = SelectPrimitive.Separator.displayName

export {
  Select,
  SelectGroup,
  SelectValue,
  SelectTrigger,
  SelectContent,
  SelectLabel,
  SelectItem,
  SelectSeparator,
  SelectScrollUpButton,
  SelectScrollDownButton,
}

```

[artifacts/mockup-sandbox/src/components/ui/separator.tsx](#)

```
import * as React from "react"
import * as SeparatorPrimitive from "@radix-ui/react-separator"

import { cn } from "@lib/utils"

const Separator = React.forwardRef<
  React.ElementRef<typeof SeparatorPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof SeparatorPrimitive.Root>
>((
  {
    className, orientation = "horizontal", decorative = true, ...props },
    ref
  ) => (
    <SeparatorPrimitive.Root
      ref={ref}
      decorative={decorative}
      orientation={orientation}
      className={cn(
        "shrink-0 bg-border",
        orientation === "horizontal" ? "h-[1px] w-full" : "h-full w-[1px]",
        className
      )}
      {...props}
    />
  ))
Separator.displayName = SeparatorPrimitive.Root.displayName

export { Separator }
```

[artifacts/mockup-sandbox/src/components/ui/sheet.tsx](#)

```
"use client"

import * as React from "react"
import * as SheetPrimitive from "@radix-ui/react-dialog"
import { cva, type VariantProps } from "class-variance-authority"
import { X } from "lucide-react"

import { cn } from "@lib/utils"

const Sheet = SheetPrimitive.Root

const SheetTrigger = SheetPrimitive.Trigger

const SheetClose = SheetPrimitive.Close

const SheetPortal = SheetPrimitive.Portal

const SheetOverlay = React.forwardRef<
  React.ElementRef<typeof SheetPrimitive.Overlay>,
  React.ComponentPropsWithoutRef<typeof SheetPrimitive.Overlay>
>(({ className, ...props }, ref) => (
  <SheetPrimitive.Overlay
    className={cn(
      "fixed inset-0 z-50 bg-black/80 data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0",
      className
    )}
    {...props}
    ref={ref}
  />
))
SheetOverlay.displayName = SheetPrimitive.Overlay.displayName

const sheetVariants = cva(
  "fixed z-50 gap-4 bg-background p-6 shadow-lg transition ease-in-out data-[state=closed]:duration-300 data-[state=open]:duration-500 data-[state=open]:animate-in data-[state=closed]:animate-out",
  {
    variants: {
      side: {
        top: "inset-x-0 top-0 border-b data-[state=closed]:slide-out-to-top data-[state=open]:slide-in-from-top",
        bottom: "inset-x-0 bottom-0 border-t data-[state=closed]:slide-out-to-bottom data-[state=open]:slide-in-from-bottom",
        left: "inset-y-0 left-0 h-full w-3/4 border-r data-[state=closed]:slide-out-to-left data-[state=open]:slide-in-from-left sm:max-w-sm",
        right: "inset-y-0 right-0 h-full w-3/4 border-l data-[state=closed]:slide-out-to-right data-[state=open]:slide-in-from-right sm:max-w-sm",
      },
      defaultVariants: {
        side: "right",
      },
    },
  }
)

interface SheetContentProps
  extends React.ComponentPropsWithoutRef<typeof SheetPrimitive.Content>,
    VariantProps<typeof sheetVariants> {}

const SheetContent = React.forwardRef<
  React.ElementRef<typeof SheetPrimitive.Content>,
  SheetContentProps
>(({ side = "right", className, children, ...props }, ref) => (
  <SheetPortal>
    <SheetOverlay />
    <SheetPrimitive.Content
      ref={ref}
      className={cn(sheetVariants({ side }), className)}
      {...props}
    >
      <SheetPrimitive.Close className="absolute right-4 top-4 rounded-sm opacity-70 ring-offset-background transition-opacity hover:opacity-100 focus:outline-none focus:ring-2 focus:ring-ring focus:ring-offset-2 disabled:pointer-events-none data-[state=open]:bg-secondary">
        <X className="h-4 w-4" />
        <span className="sr-only">Close</span>
      </SheetPrimitive.Close>
      {children}
    </SheetPrimitive.Content>
  </SheetPortal>
))
SheetContent.displayName = SheetPrimitive.Content.displayName

const SheetHeader = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn(
      "flex flex-col space-y-2 text-center sm:text-left",
      className
    )}
    {...props}
  />
)
SheetHeader.displayName = "SheetHeader"
```

```

const SheetFooter = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn(
      "flex flex-col-reverse sm:flex-row sm:justify-end sm:space-x-2",
      className
    )}
    {...props}
  />
)
SheetFooter.displayName = "SheetFooter"

const SheetTitle = React.forwardRef<
  React.ElementRef<typeof SheetPrimitive.Title>,
  React.ComponentPropsWithoutRef<typeof SheetPrimitive.Title>
>(({ className, ...props }, ref) => (
  <SheetPrimitive.Title
    ref={ref}
    className={cn("text-lg font-semibold text-foreground", className)}
    {...props}
  />
))
SheetTitle.displayName = SheetPrimitive.Title.displayName

const SheetDescription = React.forwardRef<
  React.ElementRef<typeof SheetPrimitive.Description>,
  React.ComponentPropsWithoutRef<typeof SheetPrimitive.Description>
>(({ className, ...props }, ref) => (
  <SheetPrimitive.Description
    ref={ref}
    className={cn("text-sm text-muted-foreground", className)}
    {...props}
  />
))
SheetDescription.displayName = SheetPrimitive.Description.displayName

export {
  Sheet,
  SheetPortal,
  SheetOverlay,
  SheetTrigger,
  SheetClose,
  SheetContent,
  SheetHeader,
  SheetFooter,
  SheetTitle,
  SheetDescription,
}

```

[artifacts/mockup-sandbox/src/components/ui/sidebar.tsx](#)

```
"use client"

import * as React from "react"
import { Slot } from "@radix-ui/react-slot"
import { cva, VariantProps } from "class-variance-authority"
import { PanelLeftIcon } from "lucide-react"

import { useIsMobile } from "@hooks/use-mobile"
import { cn } from "@lib/utls"
import { Button } from "@components/ui/button"
import { Input } from "@components/ui/input"
import { Separator } from "@components/ui/separator"
import {
  Sheet,
  SheetContent,
  SheetDescription,
  SheetHeader,
  SheetTitle,
} from "@components/ui/sheet"
import { Skeleton } from "@components/ui/skeleton"
import {
  Tooltip,
  TooltipContent,
  TooltipProvider,
  TooltipTrigger,
} from "@components/ui/tooltip"

const SIDEBAR_COOKIE_NAME = "sidebar_state"
const SIDEBAR_COOKIE_MAX_AGE = 60 * 60 * 24 * 7
const SIDEBAR_WIDTH = "16rem"
const SIDEBAR_WIDTH_MOBILE = "18rem"
const SIDEBAR_WIDTH_ICON = "3rem"
const SIDEBAR_KEYBOARD_SHORTCUT = "b"

type SidebarContextProps = {
  state: "expanded" | "collapsed"
  open: boolean
  setOpen: (open: boolean) => void
  openMobile: boolean
  setOpenMobile: (open: boolean) => void
  isMobile: boolean
  toggleSidebar: () => void
}

const SidebarContext = React.createContext<SidebarContextProps | null>(null)

function useSidebar() {
  const context = React.useContext(SidebarContext)
  if (!context) {
    throw new Error("useSidebar must be used within a SidebarProvider.")
  }

  return context
}

function SidebarProvider({
  defaultOpen = true,
  open: openProp,
  onOpenChange: setOpenProp,
  className,
  style,
  children,
  ...props
}: React.ComponentProps<"div"> & {
  defaultOpen?: boolean
  open?: boolean
  onOpenChange?: (open: boolean) => void
}) {
  const isMobile = useIsMobile()
  const [openMobile, setOpenMobile] = React.useState(false)

  const [_open, _setOpen] = React.useState(defaultOpen)
  const open = openProp ?? _open
  const setOpen = React.useCallback(
    (value: boolean | ((value: boolean) => boolean)) => {
      const openState = typeof value === "function" ? value(open) : value
      if (setOpenProp) {
        setOpenProp(openState)
      } else {
        _setOpen(openState)
      }

      document.cookie = `${SIDEBAR_COOKIE_NAME}=${openState}; path=/; max-age=${SIDEBAR_COOKIE_MAX_AGE}`
    },
    [setOpenProp, open]
  )

  const toggleSidebar = React.useCallback(() => {
    return isMobile ? setOpenMobile((open) => !open) : setOpen((open) => !open)
  }, [isMobile, setOpen, setOpenMobile])

  React.useEffect(() => {
    const handleKeyDown = (event: KeyboardEvent) => {
      if (
        event.key === SIDEBAR_KEYBOARD_SHORTCUT &&

```

```

        (event.metaKey || event.ctrlKey)
      ) {
        event.preventDefault()
        toggleSidebar()
      }
    }

    window.addEventListener("keydown", handleKeyDown)
    return () => window.removeEventListener("keydown", handleKeyDown)
  }, [toggleSidebar])

  const state = open ? "expanded" : "collapsed"

  const contextValue = React.useMemo<SidebarContextProps>(
    () => ({
      state,
      open,
      setOpen,
      isMobile,
      openMobile,
      setOpenMobile,
      toggleSidebar,
    }),
    [state, open, setOpen, isMobile, openMobile, setOpenMobile, toggleSidebar]
  )

  return (
    <SidebarContext.Provider value={contextValue}>
      <TooltipProvider delayDuration={0}>
        <div
          data-slot="sidebar-wrapper"
          style={
            {
              "--sidebar-width": SIDEBAR_WIDTH,
              "--sidebar-width-icon": SIDEBAR_WIDTH_ICON,
              ...style,
            } as React.CSSProperties
          }
          className={cn(
            "group/sidebar-wrapper has-data-[variant=inset]:bg-sidebar flex min-h-svh w-full",
            className
          )}
          {...props}
        >
          {children}
        </div>
      </TooltipProvider>
    </SidebarContext.Provider>
  )
}

function Sidebar({
  side = "left",
  variant = "sidebar",
  collapsible = "offcanvas",
  className,
  children,
  ...props
}: React.ComponentProps<"div"> & {
  side?: "left" | "right"
  variant?: "sidebar" | "floating" | "inset"
  collapsible?: "offcanvas" | "icon" | "none"
}) {
  const { isMobile, state, openMobile, setOpenMobile } = useSidebar()

  if (collapsible === "none") {
    return (
      <div
        data-slot="sidebar"
        className={cn(
          "bg-sidebar text-sidebar-foreground flex h-full w-[var(--sidebar-width)] flex-col",
          className
        )}
        {...props}
      >
        {children}
      </div>
    )
  }

  if (isMobile) {
    return (
      <Sheet open={openMobile} onOpenChange={setOpenMobile} {...props}>
        <SheetContent
          data-sidebar="sidebar"
          data-slot="sidebar"
          data-mobile="true"
          className="bg-sidebar text-sidebar-foreground w-[var(--sidebar-width)] p-0 [&button]:hidden"
          style={
            {
              "--sidebar-width": SIDEBAR_WIDTH_MOBILE,
            } as React.CSSProperties
          }
          side={side}
        >
          <SheetHeader className="sr-only">
            <SheetTitle>Sidebar</SheetTitle>

```

```

        <SheetDescription>Displays the mobile sidebar.</SheetDescription>
      </SheetHeader>
      <div className="flex h-full w-full flex-col">{children}</div>
    </SheetContent>
  </Sheet>
)
}

return (
  <div
    className="group peer text-sidebar-foreground hidden md:block"
    data-state={state}
    data-collapsible={state === "collapsed" ? collapsible : ""}
    data-variant={variant}
    data-side={side}
    data-slot="sidebar"
  >
    <div
      data-slot="sidebar-gap"
      className={cn(
        "relative w-[var(--sidebar-width)] bg-transparent transition-[width] duration-200 ease-linear",
        "group-data-[collapsible=offcanvas]:w-0",
        "group-data-[side=right]:rotate-180",
        variant === "floating" || variant === "inset"
          ? "group-data-[collapsible=icon]:w-[calc(var(--sidebar-width-icon)+var(--spacing-4))]"
          : "group-data-[collapsible=icon]:w-[var(--sidebar-width-icon)]"
      )}
    >
      <div
        data-slot="sidebar-container"
        className={cn(
          "fixed inset-y-0 z-10 hidden h-svh w-[var(--sidebar-width)] transition-[left,right,width] duration-200 ease-linear",
          md:flex,
          side === "left"
            ? "left-0 group-data-[collapsible=offcanvas]:left-[calc(var(--sidebar-width)*-1)]"
            : "right-0 group-data-[collapsible=offcanvas]:right-[calc(var(--sidebar-width)*-1)]",
          variant === "floating" || variant === "inset"
            ? "p-2 group-data-[collapsible=icon]:w-[calc(var(--sidebar-width-icon)+var(--spacing-4)+2px)]"
            : "group-data-[collapsible=icon]:w-[var(--sidebar-width-icon)] group-data-[side=left]:border-r group-data-[side=right]:border-l",
          className
        )}
      >
        {props}
      </div>
      <div
        data-sidebar="sidebar"
        data-slot="sidebar-inner"
        className="bg-sidebar group-data-[variant=floating]:border-sidebar-border flex h-full w-full flex-col group-data-[variant=floating]:rounded-lg group-data-[variant=floating]:border group-data-[variant=floating]:shadow-sm"
      >
        {children}
      </div>
    </div>
  </div>
)
}

function SidebarTrigger({
  className,
  onClick,
  ...props
}): React.ComponentProps<typeof Button> {
  const { toggleSidebar } = useSidebar()

  return (
    <Button
      data-sidebar="trigger"
      data-slot="sidebar-trigger"
      variant="ghost"
      size="icon"
      className={cn("h-7 w-7", className)}
      onClick={(event) => {
        onClick?.(event)
        toggleSidebar()
      }}
      {...props}
    >
      <PanelLeftIcon />
      <span className="sr-only">Toggle Sidebar</span>
    </Button>
  )
}

function SidebarRail({ className, ...props }: React.ComponentProps<"button">) {
  const { toggleSidebar } = useSidebar()

  return (
    <button
      data-sidebar="rail"
      data-slot="sidebar-rail"
      aria-label="Toggle Sidebar"
      tabIndex={-1}
      onClick={toggleSidebar}
      title="Toggle Sidebar"
      className={cn(
        "hover:after:bg-sidebar-border absolute inset-y-0 z-20 hidden w-4 -translate-x-1/2 transition-all ease-linear group-data-[side=left]:-right-4 group-data-[side=right]:left-0 after:absolute after:inset-y-0 after:left-1/2 after:w-[2px] sm:flex",

```

```

        "in-data-[side=left]:cursor-w-resize in-data-[side=right]:cursor-e-resize",
        "[[data-side=left][data-state=collapsed]&]:cursor-e-resize [[data-side=right][data-state=collapsed]&]:cursor-w-resize",
        "hover:group-data-[collapsible=offcanvas]:bg-sidebar group-data-[collapsible=offcanvas]:translate-x-0 group-data-[collapsible=offcanvas]:after:left-full",
        "[[data-side=left][data-collapsible=offcanvas]&]:-right-2",
        "[[data-side=right][data-collapsible=offcanvas]&]:-left-2",
        className
      )}
    {...props}
  )
}

function SidebarInset({ className, ...props }: React.ComponentProps<"main">) {
  return (
    <main
      data-slot="sidebar-inset"
      className={cn(
        "bg-background relative flex w-full flex-1 flex-col",
        "md:peer-data-[variant=inset]:m-2 md:peer-data-[variant=inset]:ml-0 md:peer-data-[variant=inset]:rounded-xl md:peer-data-[variant=inset]:shadow-sm md:peer-data-[variant=inset]:peer-data-[state=collapsed]:ml-2",
        className
      )}
    >
      {...props}
    </main>
  )
}

function SidebarInput({
  className,
  ...props
}: React.ComponentProps<typeof Input>) {
  return (
    <Input
      data-slot="sidebar-input"
      data-sidebar="input"
      className={cn("bg-background h-8 w-full shadow-none", className)}
    >
      {...props}
    </Input>
  )
}

function SidebarHeader({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="sidebar-header"
      data-sidebar="header"
      className={cn("flex flex-col gap-2 p-2", className)}
    >
      {...props}
    </div>
  )
}

function SidebarFooter({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="sidebar-footer"
      data-sidebar="footer"
      className={cn("flex flex-col gap-2 p-2", className)}
    >
      {...props}
    </div>
  )
}

function SidebarSeparator({
  className,
  ...props
}: React.ComponentProps<typeof Separator>) {
  return (
    <Separator
      data-slot="sidebar-separator"
      data-sidebar="separator"
      className={cn("bg-sidebar-border mx-2 w-auto", className)}
    >
      {...props}
    </Separator>
  )
}

function SidebarContent({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="sidebar-content"
      data-sidebar="content"
      className={cn(
        "flex min-h-0 flex-1 flex-col gap-2 overflow-auto group-data-[collapsible=icon]:overflow-hidden",
        className
      )}
    >
      {...props}
    </div>
  )
}

function SidebarGroup({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="sidebar-group"

```



```

    data-sidebar="group"
    className={cn("relative flex w-full min-w-0 flex-col p-2", className)}
    {...props}
  />
)
}

function SidebarGroupLabel({
  className,
  asChild = false,
  ...props
}: React.ComponentProps<"div"> & { asChild?: boolean }) {
  const Comp = asChild ? Slot : "div"

  return (
    <Comp
      data-slot="sidebar-group-label"
      data-sidebar="group-label"
      className={cn(
        "text-sidebar-foreground/70 ring-sidebar-ring flex h-8 shrink-0 items-center rounded-md px-2 text-xs font-medium
outline-hidden transition-[margin,opacity] duration-200 ease-linear focus-visible:ring-2 [&svg]:h-4 [&svg]:w-4
[&svg]:shrink-0",
        "group-data-[collapsible=icon]:-mt-8 group-data-[collapsible=icon]:opacity-0",
        className
      )}
      {...props}
    />
  )
}

function SidebarGroupAction({
  className,
  asChild = false,
  ...props
}: React.ComponentProps<"button"> & { asChild?: boolean }) {
  const Comp = asChild ? Slot : "button"

  return (
    <Comp
      data-slot="sidebar-group-action"
      data-sidebar="group-action"
      className={cn(
        "text-sidebar-foreground ring-sidebar-ring hover:bg-sidebar-accent hover:text-sidebar-accent-foreground absolute
top-3.5 right-3 flex aspect-square w-5 items-center justify-center rounded-md p-0 outline-hidden transition-transform focus-
visible:ring-2 [&svg]:size-4 [&svg]:shrink-0",
        "after:absolute after:-inset-2 md:after:hidden",
        "group-data-[collapsible=icon]:hidden",
        className
      )}
      {...props}
    />
  )
}

function SidebarGroupContent({
  className,
  ...props
}: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="sidebar-group-content"
      data-sidebar="group-content"
      className={cn("w-full text-sm", className)}
      {...props}
    />
  )
}

function SidebarMenu({ className, ...props }: React.ComponentProps<"ul">) {
  return (
    <ul
      data-slot="sidebar-menu"
      data-sidebar="menu"
      className={cn("flex w-full min-w-0 flex-col gap-1", className)}
      {...props}
    />
  )
}

function SidebarMenuItem({ className, ...props }: React.ComponentProps<"li">) {
  return (
    <li
      data-slot="sidebar-menu-item"
      data-sidebar="menu-item"
      className={cn("group/menu-item relative", className)}
      {...props}
    />
  )
}

const sidebarMenuButtonVariants = cva(
  "peer/menu-button flex w-full items-center gap-2 overflow-hidden rounded-md p-2 text-left text-sm outline-hidden ring-sidebar-
ring transition-[width,height,padding] hover:bg-sidebar-accent hover:text-sidebar-accent-foreground focus-visible:ring-2
active:bg-sidebar-accent active:text-sidebar-accent-foreground disabled:pointer-events-none disabled:opacity-50 group-has-data-
[sidebar=menu-action]/menu-item:pr-8 aria-disabled:pointer-events-none aria-disabled:opacity-50 data-[active=true]:bg-sidebar-
accent data-[active=true]:font-medium data-[active=true]:text-sidebar-accent-foreground data-[state=open]:hover:bg-sidebar-
accent data-[state=open]:hover:text-sidebar-accent-foreground group-data-[collapsible=icon]:w-8! group-data-
```

```

[collapsible=icon]:h-8! group-data-[collapsible=icon]:p-2! [>span:last-child]:truncate [>svg]:size-4 [>svg]:shrink-0",
{
  variants: {
    variant: {
      default: "hover:bg-sidebar-accent hover:text-sidebar-accent-foreground",
      outline:
        "bg-background shadow-[0_0_0_1px_hsl(var(--sidebar-border))] hover:bg-sidebar-accent hover:text-sidebar-accent-foreground hover:shadow-[0_0_0_1px_hsl(var(--sidebar-accent))]",
    },
    size: {
      default: "h-8 text-sm",
      sm: "h-7 text-xs",
      lg: "h-12 text-sm group-data-[collapsible=icon]:p-0!",
    },
  },
  defaultVariants: {
    variant: "default",
    size: "default",
  },
}
)

function SidebarMenuButton({
  asChild = false,
  isActive = false,
  variant = "default",
  size = "default",
  tooltip,
  className,
  ...props
}): React.ComponentProps<"button"> & {
  asChild?: boolean
  isActive?: boolean
  tooltip?: string | React.ComponentProps<typeof TooltipContent>
} & VariantProps<typeof sidebarMenuButtonVariants> {
  const Comp = asChild ? Slot : "button"
  const { isMobile, state } = useSidebar()

  const button = (
    <Comp
      data-slot="sidebar-menu-button"
      data-sidebar="menu-button"
      data-size={size}
      data-active={isActive}
      className={cn(sidebarMenuButtonVariants({ variant, size })), className}
      {...props}
    />
  )

  if (!tooltip) {
    return button
  }

  if (typeof tooltip === "string") {
    tooltip = {
      children: tooltip,
    }
  }

  return (
    <Tooltip>
      <TooltipTrigger asChild>{button}</TooltipTrigger>
      <TooltipContent
        side="right"
        align="center"
        hidden={state !== "collapsed" || isMobile}
        {...tooltip}
      />
    </Tooltip>
  )
}

function SidebarMenuAction({
  className,
  asChild = false,
  showOnHover = false,
  ...props
}): React.ComponentProps<"button"> & {
  asChild?: boolean
  showOnHover?: boolean
}) {
  const Comp = asChild ? Slot : "button"

  return (
    <Comp
      data-slot="sidebar-menu-action"
      data-sidebar="menu-action"
      className={cn(
        "text-sidebar-foreground ring-sidebar-ring hover:bg-sidebar-accent hover:text-sidebar-accent-foreground peer-hover/menu-button:text-sidebar-accent-foreground absolute top-1.5 right-1 flex aspect-square w-5 items-center justify-center rounded-md p-0 outline-hidden transition-transform focus-visible:ring-2 [>svg]:size-4 [>svg]:shrink-0",
        "after:absolute after:-inset-2 md:after:hidden",
        "peer-data-[size=sm]/menu-button:top-1",
        "peer-data-[size=default]/menu-button:top-1.5",
        "peer-data-[size=lg]/menu-button:top-2.5",
        "group-data-[collapsible=icon]:hidden",
        showOnHover &&
      )}
    />
  )
}

```

```

        "peer-data-[active=true]/menu-button:text-sidebar-accent-foreground group-focus-within/menu-item:opacity-100 group-
hover/menu-item:opacity-100 data-[state=open]:opacity-100 md:opacity-0",
        className
      )}
    {...props}
  />
)
}

function SidebarMenuBadge({
  className,
  ...props
}: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="sidebar-menu-badge"
      data-sidebar="menu-badge"
      className={cn(
        "text-sidebar-foreground pointer-events-none absolute right-1 flex h-5 min-w-5 items-center justify-center rounded-md
px-1 text-xs font-medium tabular-nums select-none",
        "peer-hover/menu-button:text-sidebar-accent-foreground peer-data-[active=true]/menu-button:text-sidebar-accent-
foreground",
        "peer-data-[size=sm]/menu-button:top-1",
        "peer-data-[size=default]/menu-button:top-1.5",
        "peer-data-[size=lg]/menu-button:top-2.5",
        "group-data-[collapsible=icon]:hidden",
        className
      )}
    >
    <div
      data-slot="sidebar-menu-skeleton"
      data-sidebar="menu-skeleton"
      className={cn("flex h-8 items-center gap-2 rounded-md px-2", className)}
      {...props}
    >
      {showIcon && (
        <div
          data-slot="sidebar-menu-skeleton"
          data-sidebar="menu-skeleton"
          data-sidebar="menu-skeleton-icon"
          className="size-4 rounded-md"
        />
      )}
      <div
        data-slot="sidebar-menu-skeleton"
        data-sidebar="menu-skeleton-text"
        style={
          {
            "--skeleton-width": width,
          } as React.CSSProperties
        }
      />
    </div>
  )
}

function SidebarMenuSub({ className, ...props }: React.ComponentProps<"ul">) {
  return (
    <ul
      data-slot="sidebar-menu-sub"
      data-sidebar="menu-sub"
      className={cn(
        "border-sidebar-border mx-3.5 flex min-w-0 translate-x-py flex-col gap-1 border-l px-2.5 py-0.5",
        "group-data-[collapsible=icon]:hidden",
        className
      )}
    >
    <div
      data-slot="sidebar-menu-sub-item"
      data-sidebar="menu-sub-item"
      className={cn("group/menu-sub-item relative", className)}
      {...props}
    />
  )
}

function SidebarMenuSubItem({
  className,
  ...props
}: React.ComponentProps<"li">) {
  return (
    <li
      data-slot="sidebar-menu-sub-item"
      data-sidebar="menu-sub-item"
      className={cn("group/menu-sub-item relative", className)}
      {...props}
    />
  )
}

```

```

}

function SidebarMenuSubButton({
  asChild = false,
  size = "md",
  isActive = false,
  className,
  ...props
}): React.ComponentProps<"a"> & {
  asChild?: boolean
  size?: "sm" | "md"
  isActive?: boolean
}) {
  const Comp = asChild ? Slot : "a"

  return (
    <Comp
      data-slot="sidebar-menu-sub-button"
      data-sidebar="menu-sub-button"
      data-size={size}
      data-active={isActive}
      className={cn(
        "text-sidebar-foreground ring-sidebar-ring hover:bg-sidebar-accent hover:text-sidebar-accent-foreground active:bg-
sidebar-accent active:text-sidebar-accent-foreground [&>svg]:text-sidebar-accent-foreground flex h-7 min-w-0 -translate-x-px
items-center gap-2 overflow-hidden rounded-md px-2 outline outline-2 outline-transparent outline-offset-2 focus-visible:ring-2
disabled:pointer-events-none disabled:opacity-50 aria-disabled:pointer-events-none aria-disabled:opacity-50 [&>span:last-
child]:truncate [&>svg]:size-4 [&>svg]:shrink-0",
        "data-[active=true]:bg-sidebar-accent data-[active=true]:text-sidebar-accent-foreground",
        size === "sm" && "text-xs",
        size === "md" && "text-sm",
        "group-data-[collapsible=icon]:hidden",
        className
      )}
      {...props}
    />
  )
}

export {
  Sidebar,
  SidebarContent,
  SidebarFooter,
  SidebarGroup,
  SidebarGroupAction,
  SidebarGroupContent,
  SidebarGroupLabel,
  SidebarHeader,
  SidebarInput,
  SidebarInset,
  SidebarMenu,
  SidebarMenuAction,
  SidebarMenuBadge,
  SidebarMenuButton,
  SidebarMenuItem,
  SidebarMenuSkeleton,
  SidebarMenuSub,
  SidebarMenuSubButton,
  SidebarMenuSubItem,
  SidebarProvider,
  SidebarRail,
  SidebarSeparator,
  SidebarTrigger,
  useSidebar,
}

```

[artifacts/mockup-sandbox/src/components/ui/skeleton.tsx](#)

```
import { cn } from "@/lib/utils"

function Skeleton({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) {
  return (
    <div
      className={cn("animate-pulse rounded-md bg-primary/10", className)}
      {...props}
    />
  )
}

export { Skeleton }
```

[artifacts/mockup-sandbox/src/components/ui/slider.tsx](#)

```
import * as React from "react"
import * as SliderPrimitive from "@radix-ui/react-slider"

import { cn } from "@lib/utils"

const Slider = React.forwardRef<
  React.ElementRef<typeof SliderPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof SliderPrimitive.Root>
>(({ className, ...props }, ref) => (
  <SliderPrimitive.Root
    ref={ref}
    className={cn(
      "relative flex w-full touch-none select-none items-center",
      className
    )}
    {...props}
  >
    <SliderPrimitive.Track className="relative h-1.5 w-full grow overflow-hidden rounded-full bg-primary/20">
      <SliderPrimitive.Range className="absolute h-full bg-primary" />
    </SliderPrimitive.Track>
    <SliderPrimitive.Thumb className="block h-4 w-4 rounded-full border border-primary/50 bg-background shadow transition-colors focus-visible:outline-none focus-visible:ring-1 focus-visible:ring-ring disabled:pointer-events-none disabled:opacity-50" />
  </SliderPrimitive.Root>
))
Slider.displayName = SliderPrimitive.Root.displayName

export { Slider }
```

[artifacts/mockup-sandbox/src/components/ui/sonner.tsx](#)

```
"use client"

import { useTheme } from "next-themes"
import { Toaster as Sonner } from "sonner"

type ToasterProps = React.ComponentProps<typeof Sonner>

const Toaster = ({ ...props }: ToasterProps) => {
  const { theme = "system" } = useTheme()

  return (
    <Sonner
      theme={theme as ToasterProps["theme"]}
      className="toaster group"
      toastOptions={{
        classNames: {
          toast:
            "group toast group-[.toaster]:bg-background group-[.toaster]:text-foreground group-[.toaster]:border-border group-[.toaster]:shadow-lg",
          description: "group-[.toast]:text-muted-foreground",
          actionButton:
            "group-[.toast]:bg-primary group-[.toast]:text-primary-foreground",
          cancelButton:
            "group-[.toast]:bg-muted group-[.toast]:text-muted-foreground",
        },
      }}
      {...props}
    />
  )
}

export { Toaster }
```

[artifacts/mockup-sandbox/src/components/ui/spinner.tsx](#)

```
import { Loader2Icon } from "lucide-react"

import { cn } from "@/lib/utils"

function Spinner({ className, ...props }: React.ComponentProps<"svg">) {
  return (
    <Loader2Icon
      role="status"
      aria-label="Loading"
      className={cn("size-4 animate-spin", className)}
      {...props}
    />
  )
}

export { Spinner }
```


[artifacts/mockup-sandbox/src/components/ui/switch.tsx](#)

```
import * as React from "react"
import * as SwitchPrimitives from "@radix-ui/react-switch"

import { cn } from "@lib/utils"

const Switch = React.forwardRef<
  React.ElementRef<typeof SwitchPrimitives.Root>,
  React.ComponentPropsWithoutRef<typeof SwitchPrimitives.Root>
>(({ className, ...props }, ref) => (
  <SwitchPrimitives.Root
    className={cn(
      "peer inline-flex h-5 w-9 shrink-0 cursor-pointer items-center rounded-full border-2 border-transparent shadow-sm
      transition-colors focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-ring focus-visible:ring-offset-2 focus-
      visible:ring-offset-background disabled:cursor-not-allowed disabled:opacity-50 data-[state=checked]:bg-primary data-
      [state=unchecked]:bg-input",
      className
    )}
    {...props}
    ref={ref}
  >
    <SwitchPrimitives.Thumb
      className={cn(
        "pointer-events-none block h-4 w-4 rounded-full bg-background shadow-lg ring-0 transition-transform data-
        [state=checked]:translate-x-4 data-[state=unchecked]:translate-x-0"
      )}
    />
  </SwitchPrimitives.Root>
))
Switch.displayName = SwitchPrimitives.Root.displayName

export { Switch }
```

[artifacts/mockup-sandbox/src/components/ui/table.tsx](#)

```
import * as React from "react"

import { cn } from "@lib/Utils"

const Table = React.forwardRef<
  HTMLTableElement,
  React.HTMLAttributes<HTMLTableElement>
>(({ className, ...props }, ref) => (
  <div className="relative w-full overflow-auto">
    <table
      ref={ref}
      className={cn("w-full caption-bottom text-sm", className)}
      {...props}
    />
  </div>
))
Table.displayName = "Table"

const TableHeader = React.forwardRef<
  HTMLTableSectionElement,
  React.HTMLAttributes<HTMLTableSectionElement>
>(({ className, ...props }, ref) => (
  <thead ref={ref} className={cn("[&_tr]:border-b", className)} {...props} />
))
TableHeader.displayName = "TableHeader"

const TableBody = React.forwardRef<
  HTMLTableSectionElement,
  React.HTMLAttributes<HTMLTableSectionElement>
>(({ className, ...props }, ref) => (
  <tbody
    ref={ref}
    className={cn("[&_tr:last-child]:border-0", className)}
    {...props}
  />
))
TableBody.displayName = "TableBody"

const TableFooter = React.forwardRef<
  HTMLTableSectionElement,
  React.HTMLAttributes<HTMLTableSectionElement>
>(({ className, ...props }, ref) => (
  <tfoot
    ref={ref}
    className={cn(
      "border-t bg-muted/50 font-medium [&_tr]:last:border-b-0",
      className
    )}
    {...props}
  />
))
TableFooter.displayName = "TableFooter"

const TableRow = React.forwardRef<
  HTMLTableRowElement,
  React.HTMLAttributes<HTMLTableRowElement>
>(({ className, ...props }, ref) => (
  <tr
    ref={ref}
    className={cn(
      "border-b transition-colors hover:bg-muted/50 data-[state=selected]:bg-muted",
      className
    )}
    {...props}
  />
))
TableRow.displayName = "TableRow"

const TableHead = React.forwardRef<
  HTMLTableCellElement,
  React.ThHTMLAttributes<HTMLTableCellElement>
>(({ className, ...props }, ref) => (
  <th
    ref={ref}
    className={cn(
      "h-10 px-2 text-left align-middle font-medium text-muted-foreground [&:has([role=checkbox])]:pr-0 [&[role=checkbox]]:translate-y-[2px]",
      className
    )}
    {...props}
  />
))
TableHead.displayName = "TableHead"

const TableCell = React.forwardRef<
  HTMLTableCellElement,
  React.TdHTMLAttributes<HTMLTableCellElement>
>(({ className, ...props }, ref) => (
  <td
    ref={ref}
    className={cn(
      "p-2 align-middle [&:has([role=checkbox])]:pr-0 [&[role=checkbox]]:translate-y-[2px]",
      className
    )}
    {...props}
  />
))
TableCell.displayName = "TableCell"
```

```

    />
  ))
  TableCell.displayName = "TableCell"

  const TableCaption = React.forwardRef<
    HTMLTableCaptionElement,
    React.HTMLAttributes<HTMLTableCaptionElement>
  >(({ className, ...props }, ref) => (
    <caption
      ref={ref}
      className={cn("mt-4 text-sm text-muted-foreground", className)}
      {...props}
    />
  ))
  TableCaption.displayName = "TableCaption"

  export {
    Table,
    TableHeader,
    TableBody,
    TableFooter,
    TableHead,
    TableRow,
    TableCell,
    TableCaption,
  }

```

[artifacts/mockup-sandbox/src/components/ui/tabs.tsx](#)

```
import * as React from "react"
import * as TabsPrimitive from "@radix-ui/react-tabs"

import { cn } from "@lib/utils"

const Tabs = TabsPrimitive.Root

const TabsList = React.forwardRef<
  React.ElementRef<typeof TabsPrimitive.List>,
  React.ComponentPropsWithoutRef<typeof TabsPrimitive.List>
>(({ className, ...props }, ref) => (
  <TabsPrimitive.List
    ref={ref}
    className={cn(
      "inline-flex h-9 items-center justify-center rounded-lg bg-muted p-1 text-muted-foreground",
      className
    )}
    {...props}
  />
))
TabsList.displayName = TabsPrimitive.List.displayName

const TabsTrigger = React.forwardRef<
  React.ElementRef<typeof TabsPrimitive.Trigger>,
  React.ComponentPropsWithoutRef<typeof TabsPrimitive.Trigger>
>(({ className, ...props }, ref) => (
  <TabsPrimitive.Trigger
    ref={ref}
    className={cn(
      "inline-flex items-center justify-center whitespace-nowrap rounded-md px-3 py-1 text-sm font-medium ring-offset-background transition-all focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-ring focus-visible:ring-offset-2 disabled:pointer-events-none disabled:opacity-50 data-[state=active]:bg-background data-[state=active]:text-foreground data-[state=active]:shadow",
      className
    )}
    {...props}
  />
))
TabsTrigger.displayName = TabsPrimitive.Trigger.displayName

const TabsContent = React.forwardRef<
  React.ElementRef<typeof TabsPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof TabsPrimitive.Content>
>(({ className, ...props }, ref) => (
  <TabsPrimitive.Content
    ref={ref}
    className={cn(
      "mt-2 ring-offset-background focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-ring focus-visible:ring-offset-2",
      className
    )}
    {...props}
  />
))
TabsContent.displayName = TabsPrimitive.Content.displayName

export { Tabs, TabsList, TabsTrigger, TabsContent }
```

[artifacts/mockup-sandbox/src/components/ui/textarea.tsx](#)

```
import * as React from "react"

import { cn } from "@lib/utils"

const Textarea = React.forwardRef<
  HTMLTextAreaElement,
  React.ComponentProps<"textarea">
>(({ className, ...props }, ref) => {
  return (
    <textarea
      className={cn(
        "flex min-h-[60px] w-full rounded-md border border-input bg-transparent px-3 py-2 text-base shadow-sm placeholder:text-
muted-foreground focus-visible:outline-none focus-visible:ring-1 focus-visible:ring-ring disabled:cursor-not-allowed
disabled:opacity-50 md:text-sm",
        className
      )}
      ref={ref}
      {...props}
    />
  )
})
Textarea.displayName = "Textarea"

export { Textarea }
```

[artifacts/mockup-sandbox/src/components/ui/toast.tsx](#)

```
import * as React from "react"
import * as ToastPrimitives from "@radix-ui/react-toast"
import { cva, type VariantProps } from "class-variance-authority"
import { X } from "lucide-react"

import { cn } from "@/lib/utils"

const ToastProvider = ToastPrimitives.Provider

const ToastViewport = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Viewport>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Viewport>
>(({ className, ...props }, ref) => {
  <ToastPrimitives.Viewport
    ref={ref}
    className={cn(
      "fixed top-0 z-[100] flex max-h-screen w-full flex-col-reverse p-4 sm:bottom-0 sm:right-0 sm:top-auto sm:flex-col md:max-w-[420px]",
      className
    )}
    {...props}
  />
})
ToastViewport.displayName = ToastPrimitives.Viewport.displayName

const toastVariants = cva(
  "group pointer-events-auto relative flex w-full items-center justify-between space-x-4 overflow-hidden rounded-md border p-6 pr-8 shadow-lg transition-all data-[swipe=cancel]:translate-x-0 data-[swipe=end]:translate-x-[var(--radix-toast-swipe-end-x)] data-[swipe=move]:translate-x-[var(--radix-toast-swipe-move-x)] data-[swipe=move]:transition-none data-[state=open]:animate-in data-[state=closed]:animate-out data-[swipe=end]:animate-out data-[state=closed]:fade-out-80 data-[state=closed]:slide-out-to-right-full data-[state=open]:slide-in-from-top-full data-[state=open]:sm:slide-in-from-bottom-full",
  {
    variants: {
      variant: {
        default: "border bg-background text-foreground",
        destructive: "destructive group border-destructive bg-destructive text-destructive-foreground",
      },
    },
    defaultVariants: {
      variant: "default",
    },
  },
)

const Toast = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Root>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Root> &
  VariantProps<typeof toastVariants>
>(({ className, variant, ...props }, ref) => {
  return (
    <ToastPrimitives.Root
      ref={ref}
      className={cn(toastVariants({ variant }), className)}
      {...props}
    />
  )
})
Toast.displayName = ToastPrimitives.Root.displayName

const ToastAction = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Action>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Action>
>(({ className, ...props }, ref) => {
  <ToastPrimitives.Action
    ref={ref}
    className={cn(
      "inline-flex h-8 shrink-0 items-center justify-center rounded-md border bg-transparent px-3 text-sm font-medium ring-offset-background transition-colors hover:bg-secondary focus:outline-none focus:ring-2 focus:ring-ring focus:ring-offset-2 disabled:pointer-events-none disabled:opacity-50 group-[.destructive]:border-muted/40 group-[.destructive]:hover:border-destructive/30 group-[.destructive]:hover:bg-destructive group-[.destructive]:hover:text-destructive-foreground group-[.destructive]:focus:ring-destructive",
      className
    )}
    {...props}
  />
})
ToastAction.displayName = ToastPrimitives.Action.displayName

const ToastClose = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Close>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Close>
>(({ className, ...props }, ref) => {
  <ToastPrimitives.Close
    ref={ref}
    className={cn(
      "absolute right-2 top-2 rounded-md p-1 text-foreground/50 opacity-0 transition-opacity hover:text-foreground focus:opacity-100 focus:outline-none focus:ring-2 group-hover:opacity-100 group-[.destructive]:text-red-300 group-[.destructive]:hover:text-red-50 group-[.destructive]:focus:ring-red-400 group-[.destructive]:focus:ring-offset-red-600",
      className
    )}
    toast-close=""
    {...props}
  />
  <X className="h-4 w-4" />
})
```

```

    </ToastPrimitives.Close>
  ))
  ToastClose.displayName = ToastPrimitives.Close.displayName

  const ToastTitle = React.forwardRef<
    React.ElementRef<typeof ToastPrimitives.Title>,
    React.ComponentPropsWithoutRef<typeof ToastPrimitives.Title>
  >(({ className, ...props }, ref) => (
    <ToastPrimitives.Title
      ref={ref}
      className={cn("text-sm font-semibold", className)}
      {...props}
    />
  ))
  ToastTitle.displayName = ToastPrimitives.Title.displayName

  const ToastDescription = React.forwardRef<
    React.ElementRef<typeof ToastPrimitives.Description>,
    React.ComponentPropsWithoutRef<typeof ToastPrimitives.Description>
  >(({ className, ...props }, ref) => (
    <ToastPrimitives.Description
      ref={ref}
      className={cn("text-sm opacity-90", className)}
      {...props}
    />
  ))
  ToastDescription.displayName = ToastPrimitives.Description.displayName

  type ToastProps = React.ComponentPropsWithoutRef<typeof Toast>

  type ToastActionElement = React.ReactElement<typeof ToastAction>

  export {
    type ToastProps,
    type ToastActionElement,
    ToastProvider,
    ToastViewport,
    Toast,
    ToastTitle,
    ToastDescription,
    ToastClose,
    ToastAction,
  }

```

[artifacts/mockup-sandbox/src/components/ui/toaster.tsx](#)

```
import { useToast } from "@hooks/use-toast"
import {
  Toast,
  ToastClose,
  ToastDescription,
  ToastProvider,
  ToastTitle,
  ToastViewport,
} from "@components/ui/toast"

export function Toaster() {
  const { toasts } = useToast()

  return (
    <ToastProvider>
      {toasts.map(function ({ id, title, description, action, ...props }) {
        return (
          <Toast key={id} {...props}>
            <div className="grid gap-1">
              {title && <ToastTitle>{title}</ToastTitle>}
              {description && (
                <ToastDescription>{description}</ToastDescription>
              )}
            </div>
            {action}
            <ToastClose />
          </Toast>
        )
      })}
    <ToastViewport />
  </ToastProvider>
)
```


[artifacts/mockup-sandbox/src/components/ui/toggle-group.tsx](#)

```
"use client"

import * as React from "react"
import * as ToggleGroupPrimitive from "@radix-ui/react-toggle-group"
import { type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"
import { toggleVariants } from "@components/ui/toggle"

const ToggleGroupContext = React.createContext<
  VariantProps<typeof toggleVariants>
>({
  size: "default",
  variant: "default",
})

const ToggleGroup = React.forwardRef<
  React.ElementRef<typeof ToggleGroupPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof ToggleGroupPrimitive.Root> &
  VariantProps<typeof toggleVariants>
>(({ className, variant, size, children, ...props }, ref) => {
  <ToggleGroupPrimitive.Root
    ref={ref}
    className={cn("flex items-center justify-center gap-1", className)}
    {...props}
  >
    <ToggleGroupContext.Provider value={{ variant, size }}>
      {children}
    </ToggleGroupContext.Provider>
  </ToggleGroupPrimitive.Root>
})

ToggleGroup.displayName = ToggleGroupPrimitive.Root.displayName

const ToggleGroupItem = React.forwardRef<
  React.ElementRef<typeof ToggleGroupPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof ToggleGroupPrimitive.Item> &
  VariantProps<typeof toggleVariants>
>(({ className, children, variant, size, ...props }, ref) => {
  const context = React.useContext(ToggleGroupContext)

  return (
    <ToggleGroupPrimitive.Item
      ref={ref}
      className={cn(
        toggleVariants({
          variant: context.variant || variant,
          size: context.size || size,
        }),
        className
      )}
      {...props}
    >
      {children}
    </ToggleGroupPrimitive.Item>
  )
})

ToggleGroupItem.displayName = ToggleGroupPrimitive.Item.displayName

export { ToggleGroup, ToggleGroupItem }
```

[artifacts/mockup-sandbox/src/components/ui/toggle.tsx](#)

```
import * as React from "react"
import * as TogglePrimitive from "@radix-ui/react-toggle"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"

const toggleVariants = cva(
  "inline-flex items-center justify-center gap-2 rounded-md text-sm font-medium transition-colors hover:bg-muted hover:text-muted-foreground focus-visible:outline-none focus-visible:ring-1 focus-visible:ring-ring disabled:pointer-events-none disabled:opacity-50 data-[state=on]:bg-accent data-[state=on]:text-accent-foreground [&_svg]:pointer-events-none [&_svg]:size-4 [&_svg]:shrink-0",
  {
    variants: {
      variant: {
        default: "bg-transparent",
        outline:
          "border border-input bg-transparent shadow-sm hover:bg-accent hover:text-accent-foreground",
      },
      size: {
        default: "h-9 px-2 min-w-9",
        sm: "h-8 px-1.5 min-w-8",
        lg: "h-10 px-2.5 min-w-10",
      },
    },
    defaultVariants: {
      variant: "default",
      size: "default",
    },
  },
)

const Toggle = React.forwardRef<
  React.ElementRef<typeof TogglePrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof TogglePrimitive.Root> &
  VariantProps<typeof toggleVariants>
>(({ className, variant, size, ...props }, ref) => {
  <TogglePrimitive.Root
    ref={ref}
    className={cn(toggleVariants({ variant, size, className }))}
    {...props}
  />
))

Toggle.displayName = TogglePrimitive.Root.displayName

export { Toggle, toggleVariants }
```

[artifacts/mockup-sandbox/src/components/ui/tooltip.tsx](#)

```
"use client"

import * as React from "react"
import * as TooltipPrimitive from "@radix-ui/react-tooltip"

import { cn } from "@lib/utls"

const TooltipProvider = TooltipPrimitive.Provider

const Tooltip = TooltipPrimitive.Root

const TooltipTrigger = TooltipPrimitive.Trigger

const TooltipContent = React.forwardRef<
  React.ElementRef<typeof TooltipPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof TooltipPrimitive.Content>
>(({ className, sideOffset = 4, ...props }, ref) => {
  <TooltipPrimitive.Portal>
    <TooltipPrimitive.Content
      ref={ref}
      sideOffset={sideOffset}
      className={cn(
        "z-50 overflow-hidden rounded-md bg-primary px-3 py-1.5 text-xs text-primary-foreground animate-in fade-in-0 zoom-in-95
data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=closed]:zoom-out-95 data-[side=bottom]:slide-in-from-
top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2
origin-[--radix-tooltip-content-transform-origin]",
        className
      )}
      {...props}
    />
  </TooltipPrimitive.Portal>
)}
TooltipContent.displayName = TooltipPrimitive.Content.displayName

export { Tooltip, TooltipTrigger, TooltipContent, TooltipProvider }
```

[artifacts/mockup-sandbox/src/hooks/use-mobile.tsx](#)

```
import * as React from "react"

const MOBILE_BREAKPOINT = 768

export function useIsMobile() {
  const [isMobile, setIsMobile] = React.useState<boolean | undefined>(undefined)

  React.useEffect(() => {
    const mql = window.matchMedia(`(max-width: ${MOBILE_BREAKPOINT - 1}px)`)
    const onChange = () => {
      setIsMobile(window.innerWidth < MOBILE_BREAKPOINT)
    }
    mql.addEventListener("change", onChange)
    setIsMobile(window.innerWidth < MOBILE_BREAKPOINT)
    return () => mql.removeEventListener("change", onChange)
  }, [])

  return !isMobile
}
```

[artifacts/mockup-sandbox/src/hooks/use-toast.ts](#)

```
import * as React from "react"

import type {
  ToastActionElement,
  ToastProps,
} from "@components/ui/toast"

const TOAST_LIMIT = 1
const TOAST_REMOVE_DELAY = 1000000

type ToasterToast = ToastProps & {
  id: string
  title?: React.ReactNode
  description?: React.ReactNode
  action?: ToastActionElement
}

const actionTypes = {
  ADD_TOAST: "ADD_TOAST",
  UPDATE_TOAST: "UPDATE_TOAST",
  DISMISS_TOAST: "DISMISS_TOAST",
  REMOVE_TOAST: "REMOVE_TOAST",
} as const

let count = 0

function genId() {
  count = (count + 1) % Number.MAX_SAFE_INTEGER
  return count.toString()
}

type ActionType = typeof actionTypes

type Action =
  | {
      type: ActionType["ADD_TOAST"]
      toast: ToasterToast
    }
  | {
      type: ActionType["UPDATE_TOAST"]
      toast: Partial<ToasterToast>
    }
  | {
      type: ActionType["DISMISS_TOAST"]
      toastId?: ToasterToast["id"]
    }
  | {
      type: ActionType["REMOVE_TOAST"]
      toastId?: ToasterToast["id"]
    }

interface State {
  toasts: ToasterToast[]
}

const toastTimeouts = new Map<string, ReturnType<typeof setTimeout>>()

const addToRemoveQueue = (toastId: string) => {
  if (toastTimeouts.has(toastId)) {
    return
  }

  const timeout = setTimeout(() => {
    toastTimeouts.delete(toastId)
    dispatch({
      type: "REMOVE_TOAST",
      toastId: toastId,
    })
  }, TOAST_REMOVE_DELAY)

  toastTimeouts.set(toastId, timeout)
}

export const reducer = (state: State, action: Action): State => {
  switch (action.type) {
    case "ADD_TOAST":
      return {
        ...state,
        toasts: [action.toast, ...state.toasts].slice(0, TOAST_LIMIT),
      }

    case "UPDATE_TOAST":
      return {
        ...state,
        toasts: state.toasts.map((t) =>
          t.id === action.toast.id ? { ...t, ...action.toast } : t
        ),
      }

    case "DISMISS_TOAST": {
      const { toastId } = action

      if (toastId) {
        addToRemoveQueue(toastId)
      } else {

```

```

        state.toasts.forEach((toast) => {
          addToRemoveQueue(toast.id)
        })
      }

      return {
        ...state,
        toasts: state.toasts.map((t) =>
          t.id === toastId || toastId === undefined
            ? {
                ...t,
                open: false,
              }
            : t
          ),
      }
    }
  }
  case "REMOVE_TOAST":
    if (action.toastId === undefined) {
      return {
        ...state,
        toasts: [],
      }
    }
    return {
      ...state,
      toasts: state.toasts.filter((t) => t.id !== action.toastId),
    }
  }
}

const listeners: Array<(state: State) => void> = []

let memoryState: State = { toasts: [] }

function dispatch(action: Action) {
  memoryState = reducer(memoryState, action)
  listeners.forEach((listener) => {
    listener(memoryState)
  })
}

type Toast = Omit<ToasterToast, "id">

function toast({ ...props }: Toast) {
  const id = genId()

  const update = (props: ToasterToast) =>
    dispatch({
      type: "UPDATE_TOAST",
      toast: { ...props, id },
    })

  const dismiss = () => dispatch({ type: "DISMISS_TOAST", toastId: id })

  dispatch({
    type: "ADD_TOAST",
    toast: {
      ...props,
      id,
      open: true,
      onOpenChange: (open) => {
        if (!open) dismiss()
      },
    },
  })

  return {
    id: id,
    dismiss,
    update,
  }
}

function useToast() {
  const [state, setState] = React.useState<State>(memoryState)

  React.useEffect(() => {
    listeners.push(setState)
    return () => {
      const index = listeners.indexOf(setState)
      if (index > -1) {
        listeners.splice(index, 1)
      }
    }
  }, [state])

  return {
    ...state,
    toast,
    dismiss: (toastId?: string) => dispatch({ type: "DISMISS_TOAST", toastId }),
  }
}

export { useToast, toast }

```

[artifacts/mockup-sandbox/src/index.css](#)

```
@import "tailwindcss";
@import "tw-animate-css";

@custom-variant dark (&:is(.dark *));

@theme inline {
  --radius-sm: calc(var(--radius) - 4px);
  --radius-md: calc(var(--radius) - 2px);
  --radius-lg: var(--radius);
  --radius-xl: calc(var(--radius) + 4px);
  --color-background: hsl(var(--background));
  --color-foreground: hsl(var(--foreground));
  --color-card: hsl(var(--card));
  --color-card-foreground: hsl(var(--card-foreground));
  --color-popover: hsl(var(--popover));
  --color-popover-foreground: hsl(var(--popover-foreground));
  --color-primary: hsl(var(--primary));
  --color-primary-foreground: hsl(var(--primary-foreground));
  --color-secondary: hsl(var(--secondary));
  --color-secondary-foreground: hsl(var(--secondary-foreground));
  --color-muted: hsl(var(--muted));
  --color-muted-foreground: hsl(var(--muted-foreground));
  --color-accent: hsl(var(--accent));
  --color-accent-foreground: hsl(var(--accent-foreground));
  --color-destructive: hsl(var(--destructive));
  --color-destructive-foreground: hsl(var(--destructive-foreground));
  --color-border: hsl(var(--border));
  --color-input: hsl(var(--input));
  --color-ring: hsl(var(--ring));
  --color-chart-1: hsl(var(--chart-1));
  --color-chart-2: hsl(var(--chart-2));
  --color-chart-3: hsl(var(--chart-3));
  --color-chart-4: hsl(var(--chart-4));
  --color-chart-5: hsl(var(--chart-5));
  --color-sidebar: hsl(var(--sidebar));
  --color-sidebar-foreground: hsl(var(--sidebar-foreground));
  --color-sidebar-primary: hsl(var(--sidebar-primary));
  --color-sidebar-primary-foreground: hsl(var(--sidebar-primary-foreground));
  --color-sidebar-accent: hsl(var(--sidebar-accent));
  --color-sidebar-accent-foreground: hsl(var(--sidebar-accent-foreground));
  --color-sidebar-border: hsl(var(--sidebar-border));
  --color-sidebar-ring: hsl(var(--sidebar-ring));
}

/* LIGHT MODE - Default theme with placeholder values */
:root {
  --button-outline: rgba(0,0,0, .10);
  --badge-outline: rgba(0,0,0, .05);
  --opaque-button-border-intensity: -8;

  --elevate-1: rgba(0,0,0, .03);
  --elevate-2: rgba(0,0,0, .08);

  --background: 0 0% 100%;
  --foreground: 240 10% 3.9%;
  --card: 0 0% 100%;
  --card-foreground: 240 10% 3.9%;
  --popover: 0 0% 100%;
  --popover-foreground: 240 10% 3.9%;
  --primary: 240 5.9% 10%;
  --primary-foreground: 0 0% 98%;
  --secondary: 240 4.8% 95.9%;
  --secondary-foreground: 240 5.9% 10%;
  --muted: 240 4.8% 95.9%;
  --muted-foreground: 240 3.8% 46.1%;
  --accent: 240 4.8% 95.9%;
  --accent-foreground: 240 5.9% 10%;
  --destructive: 0 84.2% 60.2%;
  --destructive-foreground: 0 0% 98%;
  --border: 240 5.9% 90%;
  --input: 240 5.9% 90%;
  --ring: 240 5.9% 10%;
  --chart-1: 12 76% 61%;
  --chart-2: 173 58% 39%;
  --chart-3: 197 37% 24%;
  --chart-4: 43 74% 66%;
  --chart-5: 27 87% 67%;
  --sidebar: 0 0% 98%;
  --sidebar-foreground: 240 5.3% 26.1%;
  --sidebar-primary: 240 5.9% 10%;
  --sidebar-primary-foreground: 0 0% 98%;
  --sidebar-accent: 240 4.8% 95.9%;
  --sidebar-accent-foreground: 240 5.9% 10%;
  --sidebar-border: 220 13% 91%;
  --sidebar-ring: 240 5.9% 10%;

  --font-sans: 'Inter', sans-serif;
  --font-serif: Georgia, serif;
  --font-mono: Menlo, monospace;
  --radius: .5rem;
  --spacing: 0.25rem;
}

/* DARK MODE */
.dark {
```

```

--button-outline: rgba(255,255,255, .10);
--badge-outline: rgba(255,255,255, .05);
--opaque-button-border-intensity: 9;

--elevate-1: rgba(255,255,255, .04);
--elevate-2: rgba(255,255,255, .09);

--background: 240 10% 3.9%;
--foreground: 0 0% 98%;
--card: 240 10% 3.9%;
--card-foreground: 0 0% 98%;
--popover: 240 10% 3.9%;
--popover-foreground: 0 0% 98%;
--primary: 0 0% 98%;
--primary-foreground: 240 5.9% 10%;
--secondary: 240 3.7% 15.9%;
--secondary-foreground: 0 0% 98%;
--muted: 240 3.7% 15.9%;
--muted-foreground: 240 5% 64.9%;
--accent: 240 3.7% 15.9%;
--accent-foreground: 0 0% 98%;
--destructive: 0 62.8% 30.6%;
--destructive-foreground: 0 0% 98%;
--border: 240 3.7% 15.9%;
--input: 240 3.7% 15.9%;
--ring: 240 4.9% 83.9%;
--chart-1: 220 70% 50%;
--chart-2: 160 60% 45%;
--chart-3: 30 80% 55%;
--chart-4: 280 65% 60%;
--chart-5: 340 75% 55%;
--sidebar: 240 5.9% 10%;
--sidebar-foreground: 240 4.8% 95.9%;
--sidebar-primary: 224.3 76.3% 48%;
--sidebar-primary-foreground: 0 0% 100%;
--sidebar-accent: 240 3.7% 15.9%;
--sidebar-accent-foreground: 240 4.8% 95.9%;
--sidebar-border: 240 3.7% 15.9%;
--sidebar-ring: 240 4.9% 83.9%;
}

@layer base {
  * {
    @apply border-border;
  }
  body {
    @apply font-sans antialiased bg-background text-foreground;
  }
}

@layer utilities {
  /* Hide ugly search cancel button in Chrome */
  input[type="search"]::-webkit-search-cancel-button {
    @apply hidden;
  }

  /* Placeholder styling for contentEditable div */
  [contenteditable][data-placeholder]:empty::before {
    content: attr(data-placeholder);
    color: hsl(var(--muted-foreground));
    pointer-events: none;
  }
}

```


[artifacts/mockup-sandbox/src/lib/utils.ts](#)

```
import { clsx, type ClassValue } from "clsx"
import { twMerge } from "tailwind-merge"

export function cn(...inputs: ClassValue[]) {
  return twMerge(clsx(inputs))
}
```

[artifacts/mockup-sandbox/src/main.tsx](#)

```
import { createRoot } from "react-dom/client";
import App from "../App";
import "../index.css";

createRoot(document.getElementById("root")!).render(<App />);
```

artifacts/mockup-sandbox/tsconfig.json

```
{
  "extends": "../../tsconfig.base.json",
  "include": ["src/**/*", "mockupPreviewPlugin.ts", "vite.config.ts"],
  "exclude": ["node_modules", "build", "dist", "**/*.test.ts"],
  "compilerOptions": {
    "noEmit": true,
    "lib": ["es2022", "dom", "dom.iterable"],
    "jsx": "preserve",
    "esModuleInterop": true,
    "allowImportingTsExtensions": true,
    "types": ["node", "vite/client"],
    "paths": {
      "@/*": ["./src/*"]
    }
  }
}
```

[artifacts/mockup-sandbox/vite.config.ts](#)

```
import { defineConfig } from "vite";
import react from "@vitejs/plugin-react";
import tailwindcss from "tailwindcss/vite";
import path from "path";
import runtimeErrorOverlay from "@replit/vite-plugin-runtime-error-modal";
import { mockupPreviewPlugin } from "../mockupPreviewPlugin";

const rawPort = process.env.PORT;

if (!rawPort) {
  throw new Error(
    "PORT environment variable is required but was not provided.",
  );
}

const port = Number(rawPort);

if (Number.isNaN(port) || port <= 0) {
  throw new Error(`Invalid PORT value: "${rawPort}"`);
}

const basePath = process.env.BASE_PATH;

if (!basePath) {
  throw new Error(
    "BASE_PATH environment variable is required but was not provided.",
  );
}

export default defineConfig({
  base: basePath,
  plugins: [
    mockupPreviewPlugin(),
    react(),
    tailwindcss(),
    runtimeErrorOverlay(),
    ...(process.env.NODE_ENV !== "production" &&
      process.env.REPL_ID !== undefined
        ? [
            await import("@replit/vite-plugin-cartographer").then((m) =>
              m.cartographer({
                root: path.resolve(import.meta.dirname, ".."),
              })
            ),
          ]
        : []),
  ],
  resolve: {
    alias: {
      "@": path.resolve(import.meta.dirname, "src"),
    },
  },
  root: path.resolve(import.meta.dirname),
  build: {
    outDir: path.resolve(import.meta.dirname, "dist"),
    emptyOutDir: true,
  },
  server: {
    port,
    host: "0.0.0.0",
    allowedHosts: true,
    fs: {
      strict: true,
      deny: ["**/*.html"],
    },
  },
  preview: {
    port,
    host: "0.0.0.0",
    allowedHosts: true,
  },
});
```

[artifacts/mtos-crm/.replit-artifact/artifact.toml](https://replit.com/artifacts/mtos-crm/artifact.toml)

```
kind = "web"
previewPath = "/"
title = "Mass Tort OS"
version = "1.0.0"
id = "artifacts/mtos-crm"
router = "path"

[[integratedSkills]]
name = "react-vite"
version = "1.0.0"

[[services]]
name = "web"
paths = [ "/" ]
localPort = 24540

[services.development]
run = "pnpm --filter @workspace/mtos-crm run dev"

[services.production]
build = [ "pnpm", "--filter", "@workspace/mtos-crm", "run", "build" ]
publicDir = "artifacts/mtos-crm/dist/public"
serve = "static"

[[services.production.rewrites]]
from = "/*"
to = "/index.html"

[services.env]
PORT = "24540"
BASE_PATH = "/"
```

artifacts/mtos-crm/components.json

```
{
  "$schema": "https://ui.shadcn.com/schema.json",
  "style": "new-york",
  "rsc": false,
  "tsx": true,
  "tailwind": {
    "config": "",
    "css": "src/index.css",
    "baseColor": "neutral",
    "cssVariables": true,
    "prefix": ""
  },
  "aliases": {
    "components": "@components",
    "utils": "@lib/Utils",
    "ui": "@components/ui",
    "lib": "@lib",
    "hooks": "@hooks"
  }
}
```

[artifacts/mtos-crm/index.html](#)

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=1" />
    <title>Mass Tort OS</title>
    <link rel="icon" type="image/svg+xml" href="/favicon.svg" />
    <link rel="preconnect" href="https://fonts.googleapis.com">
    <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
    <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap" rel="stylesheet">
  </head>
  <body>
    <div id="root"></div>
    <script type="module" src="/src/main.tsx"></script>
  </body>
</html>
```

[artifacts/mtos-crm/package.json](#)

```
{
  "name": "@workspace/mtos-crm",
  "version": "0.0.0",
  "private": true,
  "type": "module",
  "scripts": {
    "dev": "vite --config vite.config.ts --host 0.0.0.0",
    "build": "vite build --config vite.config.ts",
    "serve": "vite preview --config vite.config.ts --host 0.0.0.0",
    "typecheck": "tsc -p tsconfig.json --noEmit"
  },
  "devDependencies": {
    "@hookform/resolvers": "^3.10.0",
    "@radix-ui/react-accordion": "^1.2.4",
    "@radix-ui/react-alert-dialog": "^1.1.7",
    "@radix-ui/react-aspect-ratio": "^1.1.3",
    "@radix-ui/react-avatar": "^1.1.4",
    "@radix-ui/react-checkbox": "^1.1.5",
    "@radix-ui/react-collapsible": "^1.1.4",
    "@radix-ui/react-context-menu": "^2.2.7",
    "@radix-ui/react-dialog": "^1.1.7",
    "@radix-ui/react-dropdown-menu": "^2.1.7",
    "@radix-ui/react-hover-card": "^1.1.7",
    "@radix-ui/react-label": "^2.1.3",
    "@radix-ui/react-menubar": "^1.1.7",
    "@radix-ui/react-navigation-menu": "^1.2.6",
    "@radix-ui/react-popover": "^1.1.7",
    "@radix-ui/react-progress": "^1.1.3",
    "@radix-ui/react-radio-group": "^1.2.4",
    "@radix-ui/react-scroll-area": "^1.2.4",
    "@radix-ui/react-select": "^2.1.7",
    "@radix-ui/react-separator": "^1.1.3",
    "@radix-ui/react-slider": "^1.2.4",
    "@radix-ui/react-slot": "^1.2.0",
    "@radix-ui/react-switch": "^1.1.4",
    "@radix-ui/react-tabs": "^1.1.4",
    "@radix-ui/react-toast": "^1.2.7",
    "@radix-ui/react-toggle": "^1.1.3",
    "@radix-ui/react-toggle-group": "^1.1.3",
    "@radix-ui/react-tooltip": "^1.2.0",
    "@replit/vite-plugin-cartographer": "catalog:",
    "@replit/vite-plugin-dev-banner": "catalog:",
    "@replit/vite-plugin-runtime-error-modal": "catalog:",
    "@tailwindcss/typography": "^0.5.15",
    "@tailwindcss/vite": "catalog:",
    "@tanstack/react-query": "catalog:",
    "@types/node": "catalog:",
    "@types/react": "catalog:",
    "@types/react-dom": "catalog:",
    "@vitejs/plugin-react": "catalog:",
    "@workspace/api-client-react": "workspace:*",
    "class-variance-authority": "catalog:",
    "clsx": "catalog:",
    "cmdk": "^1.1.1",
    "date-fns": "^3.6.0",
    "embla-carousel-react": "^8.6.0",
    "framer-motion": "catalog:",
    "input-otp": "^1.4.2",
    "lucide-react": "catalog:",
    "next-themes": "^0.4.6",
    "react": "catalog:",
    "react-day-picker": "^9.11.1",
    "react-dom": "catalog:",
    "react-hook-form": "^7.55.0",
    "react-icons": "^5.4.0",
    "react-resizable-panels": "^2.1.7",
    "recharts": "^2.15.2",
    "sonner": "^2.0.7",
    "tailwind-merge": "catalog:",
    "tailwindcss": "catalog:",
    "tw-animate-css": "^1.4.0",
    "vaul": "^1.1.2",
    "vite": "catalog:",
    "wouter": "^3.3.5",
    "zod": "catalog:"
  }
}
```


[artifacts/mtos-crm/src/App.tsx](#)

```
import { Switch, Route, Router as WouterRouter } from "wouter";
import { QueryClient, QueryClientProvider } from "@tanstack/react-query";
import { Toaster } from "@components/ui/toaster";
import { TooltipProvider } from "@components/ui/tooltip";
import NotFound from "@pages/not-found";

import { Layout } from "@components/layout/layout";
import Dashboard from "@pages/dashboard";
import Pipeline from "@pages/pipeline";
import Leads from "@pages/leads";
import LeadIntake from "@pages/lead-intake";
import LeadDetail from "@pages/lead-detail";
import Paralegals from "@pages/paralegals";
import Documents from "@pages/documents";
import OcrInbox from "@pages/ocr-inbox";
import NpiLookup from "@pages/npi-lookup";
import Cases from "@pages/cases";
import CaseNew from "@pages/case-new";
import CaseDetail from "@pages/case-detail";
import ReviewQueue from "@pages/review-queue";
import Analytics from "@pages/analytics";
import Compliance from "@pages/compliance";
import FormEngine from "@pages/form-engine";
import Vendors from "@pages/vendors";
import Security from "@pages/security";
import DocReview from "@pages/doc-review";
import Timeline from "@pages/timeline";
import Drafting from "@pages/drafting";
import Predictive from "@pages/predictive";
import IntegrationsPage from "@pages/integrations";
import News from "@pages/news";
import FinancialNews from "@pages/financial-news";
import LeadImport from "@pages/lead-import";
import DecisionEnginePage from "@pages/decision-engine";
import DecisionEngineSettings from "@pages/decision-engine-settings";
import BuyersPage from "@pages/buyers";
import DocumentTemplatesPage from "@pages/document-templates";
import TemplateAssignmentsPage from "@pages/template-assignments";
import WorkflowSettingsPage from "@pages/workflow-settings";

const queryClient = new QueryClient();

function Router() {
  return (
    <Layout>
      <Switch>
        <Route path="/" component={Dashboard} />
        <Route path="/pipeline" component={Pipeline} />
        <Route path="/leads" component={Leads} />
        <Route path="/leads/new" component={LeadIntake} />
        <Route path="/leads/:id" component={LeadDetail} />
        <Route path="/paralegals" component={Paralegals} />
        <Route path="/documents" component={Documents} />
        <Route path="/ocr-inbox" component={OcrInbox} />
        <Route path="/npi-lookup" component={NpiLookup} />
        <Route path="/review-queue" component={ReviewQueue} />
        <Route path="/cases" component={Cases} />
        <Route path="/cases/new" component={CaseNew} />
        <Route path="/cases/:id" component={CaseDetail} />
        <Route path="/analytics" component={Analytics} />
        <Route path="/compliance" component={Compliance} />
        <Route path="/form-engine" component={FormEngine} />
        <Route path="/vendors" component={Vendors} />
        <Route path="/security" component={Security} />
        <Route path="/doc-review" component={DocReview} />
        <Route path="/timeline" component={Timeline} />
        <Route path="/drafting" component={Drafting} />
        <Route path="/predictive" component={Predictive} />
        <Route path="/integrations" component={IntegrationsPage} />
        <Route path="/news" component={News} />
        <Route path="/financial-news" component={FinancialNews} />
        <Route path="/lead-import" component={LeadImport} />
        <Route path="/decision-engine" component={DecisionEnginePage} />
        <Route path="/decision-engine/settings" component={DecisionEngineSettings} />
        <Route path="/buyers" component={BuyersPage} />
        <Route path="/document-templates" component={DocumentTemplatesPage} />
        <Route path="/template-assignments" component={TemplateAssignmentsPage} />
        <Route path="/workflow-settings" component={WorkflowSettingsPage} />
        <Route component={NotFound} />
      </Switch>
    </Layout>
  );
}

function App() {
  return (
    <QueryClientProvider client={queryClient}>
      <TooltipProvider>
        <WouterRouter base={import.meta.env.BASE_URL.replace(/\/$/, "")}>
          <Router />
        </WouterRouter>
        <Toaster />
      </TooltipProvider>
    </QueryClientProvider>
  );
}
```

```
}  
export default App;
```

[artifacts/mtos-crm/src/components/convexity-card.tsx](#)

```
import { useMutation, useQueryClient } from "@tanstack/react-query";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import { Badge } from "@components/ui/badge";
import { Button } from "@components/ui/button";
import { Alert, AlertDescription } from "@components/ui/alert";
import { TrendingUp, TrendingDown, Minus, AlertTriangle, RefreshCw } from "lucide-react";
import { useToast } from "@hooks/use-toast";

interface ConvexityCardProps {
  leadId: number;
  score: string | null;
  action: string | null;
  rationale: string | null;
  ruinFlags: string[];
  missingFields?: string[];
  contradictions?: string[];
  downsideUsd: string | null;
  upsideUsd: string | null;
  ratio: string | null;
  confidence: string | null;
  computedAt: string | null;
}

const flagLabels: Record<string, string> = {
  sol_expired: "Statute of limitations expired",
  mdl_closed: "MDL closed (no recovery vehicle)",
  diagnosis_invalid: "Diagnosis not valid for this tort",
  exposure_missing: "Required exposure data missing",
  prior_rejection: "Lead carries prior rejection",
};

const missingLabels: Record<string, string> = {
  diagnosis: "Diagnosis",
  diagnosis_date: "Diagnosis date",
  state: "State of residence",
  contact: "Phone or email",
  exposure_start: "Exposure start date",
};

const contradictionLabels: Record<string, string> = {
  diagnosis_before_exposure: "Diagnosis date is before exposure start",
  exposure_end_before_start: "Exposure end is before exposure start",
  diagnosis_in_future: "Diagnosis date is in the future",
  exposure_start_in_future: "Exposure start is in the future",
  diagnosis_before_birth: "Diagnosis date is before date of birth",
  birth_in_future: "Date of birth is in the future",
};

function classificationBadge(score: string | null) {
  switch (score) {
    case "convex":
      return <Badge className="bg-emerald-100 text-emerald-800 border-emerald-300"><TrendingUp className="w-3 h-3 mr-1" />Convex</Badge>;
    case "concave":
      return <Badge className="bg-rose-100 text-rose-800 border-rose-300"><TrendingDown className="w-3 h-3 mr-1" />Concave</Badge>;
    case "neutral":
      return <Badge variant="secondary"><Minus className="w-3 h-3 mr-1" />Neutral</Badge>;
    default:
      return <Badge variant="outline">Not scored</Badge>;
  }
}

function actionBadge(action: string | null) {
  switch (action) {
    case "execute":
      return <Badge className="bg-blue-100 text-blue-800 border-blue-300">Execute</Badge>;
    case "modify":
      return <Badge className="bg-amber-100 text-amber-800 border-amber-300">Modify</Badge>;
    case "reject":
      return <Badge className="bg-rose-100 text-rose-800 border-rose-300">Reject (review)</Badge>;
    case "review":
      return <Badge className="bg-amber-100 text-amber-800 border-amber-300">Human Review</Badge>;
    default:
      return null;
  }
}

export function ConvexityCard(props: ConvexityCardProps) {
  const qc = useQueryClient();
  const { toast } = useToast();

  const recompute = useMutation({
    mutationFn: async () => {
      const res = await fetch(`/api/decision-engine/leads/${props.leadId}/recompute`, {
        method: "POST",
        credentials: "include",
      });
      if (!res.ok) throw new Error(await res.text());
      return res.json();
    },
    onSuccess: () => {
      qc.invalidateQueries();
      toast({ title: "Convexity recomputed" });
    },
  });
}
```

```

    onError: (e: any) => toast({ title: "Recompute failed", description: e?.message, variant: "destructive" })),
  });

const downside = props.downsideUsd ? Number(props.downsideUsd) : 0;
const upside = props.upsideUsd ? Number(props.upsideUsd) : 0;
const ratio = props.ratio ? Number(props.ratio) : 0;

return (
  <Card data-testid="card-convexity">
    <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
      <CardTitle className="text-sm font-medium flex items-center gap-2">
        Decision Engine
        {classificationBadge(props.score)}
        {actionBadge(props.action)}
        {props.confidence && (
          <Badge variant="outline" className="text-xs">conf: {props.confidence}</Badge>
        )}
      </CardTitle>
      <Button size="sm" variant="ghost" onClick={() => recompute.mutate()} disabled={recompute.isPending} data-testid="button-recompute-convexity">
        <RefreshCw className={`w-4 h-4 ${recompute.isPending ? "animate-spin" : ""}`}/>
      </Button>
    </CardHeader>
    <CardContent className="space-y-3">
      {props.ruinFlags.length > 0 && (
        <Alert variant="destructive" data-testid="alert-ruin-flags">
          <AlertTriangle className="h-4 w-4" />
          <AlertDescription>
            <div className="font-semibold mb-1">Ruin path detected – human review required</div>
            <ul className="list-disc list-inside text-sm">
              {props.ruinFlags.map(f => <li key={f}>{flagLabels[f]} || f</li>)}
            </ul>
          </AlertDescription>
        </Alert>
      )}
      {(() => {
        const contradictions = props.contradictions ?? [];
        return contradictions.length > 0 ? (
          <Alert variant="destructive" data-testid="alert-contradictions">
            <AlertTriangle className="h-4 w-4" />
            <AlertDescription>
              <div className="font-semibold mb-1">Data contradiction – verify before scoring</div>
              <ul className="list-disc list-inside text-sm">
                {contradictions.map(c => <li key={c}>{contradictionLabels[c]} || c</li>)}
              </ul>
            </AlertDescription>
          </Alert>
        ) : null;
      })()}
      {(() => {
        const missing = props.missingFields ?? [];
        return missing.length > 0 ? (
          <Alert className="border-amber-300 bg-amber-50 text-amber-900" data-testid="alert-missing-fields">
            <AlertTriangle className="h-4 w-4" />
            <AlertDescription>
              <div className="font-semibold mb-1">Missing critical intake fields</div>
              <ul className="list-disc list-inside text-sm">
                {missing.map(f => <li key={f}>{missingLabels[f]} || f</li>)}
              </ul>
              <div className="text-xs mt-1 italic">Collect these on the next call to enable a reliable score.</div>
            </AlertDescription>
          </Alert>
        ) : null;
      })()}
      {props.rationale && (
        <p className="text-sm text-muted-foreground" data-testid="text-convexity-rationale">{props.rationale}</p>
      )}
    <div className="grid grid-cols-3 gap-2 text-sm">
      <div>
        <div className="text-muted-foreground text-xs">Downside</div>
        <div className="font-semibold">${downside.toLocaleString()}</div>
      </div>
      <div>
        <div className="text-muted-foreground text-xs">Upside (E[V])</div>
        <div className="font-semibold">${upside.toLocaleString()}</div>
      </div>
      <div>
        <div className="text-muted-foreground text-xs">Ratio</div>
        <div className="font-semibold">{ratio.toFixed(2)}x</div>
      </div>
    </div>
    {props.computedAt && (
      <div className="text-xs text-muted-foreground">
        Computed {new Date(props.computedAt).toLocaleString()}
      </div>
    )}
    {!props.score && (
      <div className="text-sm text-muted-foreground italic">
        Not scored yet – click refresh to compute.
      </div>
    )}
  </CardContent>
</Card>
);
}

```

[artifacts/mtos-crm/src/components/envelope-timeline.tsx](#)

```
import { useEffect, useState } from "react";
import { Card, CardContent, CardHeader, CardTitle, CardDescription } from "@components/ui/card";
import { Badge } from "@components/ui/badge";
import { Button } from "@components/ui/button";
import { useToast } from "@hooks/use-toast";
import { CheckCircle2, Clock, Eye, FileSignature, Send, XCircle, AlertTriangle, Inbox, RefreshCw } from "lucide-react";

interface EnvelopeEvent { type: string; at: string }
interface Envelope {
  id: number;
  template_id: number | null;
  provider: string;
  external_envelope_id: string | null;
  signer_name: string | null;
  signer_email: string | null;
  status: string;
  status_detail: string | null;
  sent_at: string | null;
  delivered_at: string | null;
  viewed_at: string | null;
  signed_at: string | null;
  declined_at: string | null;
  events: EnvelopeEvent[];
  last_error: string | null;
  created_at: string;
}

interface FaxRow {
  id: number;
  source_file: string;
  vault_path: string;
  status: string;
  raw_text: string | null;
  error: string | null;
  created_at: string;
  processed_at: string | null;
}

const STATUS_META: Record<string, { label: string; variant: "default" | "secondary" | "destructive" | "outline"; icon: React.ComponentType<{ className?: string }> }> = {
  created: { label: "Created", variant: "secondary", icon: Clock },
  sent: { label: "Sent", variant: "default", icon: Send },
  delivered: { label: "Delivered", variant: "default", icon: Inbox },
  viewed: { label: "Viewed", variant: "default", icon: Eye },
  signed: { label: "Signed", variant: "default", icon: CheckCircle2 },
  declined: { label: "Declined", variant: "destructive", icon: XCircle },
  voided: { label: "Voided", variant: "destructive", icon: XCircle },
  expired: { label: "Expired", variant: "destructive", icon: XCircle },
  error: { label: "Error", variant: "destructive", icon: AlertTriangle },
};

function statusBadge(s: string) {
  const meta = STATUS_META[s] || { label: s, variant: "outline" as const, icon: Clock };
  const Icon = meta.icon;
  return (
    <Badge variant={meta.variant} className="gap-1">
      <Icon className="h-3 w-3" />
      {meta.label}
    </Badge>
  );
}

function fmt(iso: string | null) {
  if (!iso) return null;
  return new Date(iso).toLocaleString();
}

export function EnvelopeTimeline({ leadId }: { leadId: number }) {
  const { toast } = useToast();
  const [envelopes, setEnvelopes] = useState<Envelope[]>([]);
  const [faxes, setFaxes] = useState<FaxRow[]>([]);
  const [loading, setLoading] = useState(true);
  const [refreshing, setRefreshing] = useState(false);
  const [templates, setTemplates] = useState<Record<number, string>>({});

  const load = async () => {
    setRefreshing(true);
    try {
      const [eRes, fRes, tRes] = await Promise.all([
        fetch(`/api/leads/${leadId}/envelopes`),
        fetch(`/api/leads/${leadId}/fax-results`),
        fetch(`/api/document-templates`),
      ]);
      if (eRes.ok) setEnvelopes(await eRes.json());
      if (fRes.ok) setFaxes(await fRes.json());
      if (tRes.ok) {
        const tpls = await tRes.json();
        const map: Record<number, string> = {};
        for (const t of tpls) map[t.id] = t.name;
        setTemplates(map);
      }
    } finally {
      setLoading(false);
      setRefreshing(false);
    }
  };
};
```

```

useEffect(() => { load(); }, [leadId]);

const simulate = async (externalId: string) => {
  const res = await fetch("/api/webhooks/_test/envelope-signed", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify({ external_envelope_id: externalId }),
  });
  if (!res.ok) {
    toast({ title: "Simulate failed", variant: "destructive" });
    return;
  }
  toast({ title: "Marked as signed (test)", description: "Refreshing in a moment..." });
  setTimeout(load, 800);
};

if (loading) return <p className="text-muted-foreground">Loading automation activity...</p>;

return (
  <div className="space-y-6">
    <div className="flex justify-end">
      <Button variant="outline" size="sm" onClick={load} disabled={refreshing}>
        <RefreshCw className={mr2 h4 w4} ${refreshing ? "animate-spin" : ""} /> Refresh
      </Button>
    </div>
    <Card>
      <CardHeader>
        <CardTitle className="flex items-center gap-2"><FileSignature className="h-5 w-5" /> Document Envelopes</CardTitle>
        <CardDescription>E-signature packets sent to this claimant.</CardDescription>
      </CardHeader>
      <CardContent>
        {envelopes.length === 0 ? (
          <p className="text-sm text-muted-foreground">
            No envelopes sent yet. They auto-send when this lead is qualified.
          </p>
        ) : (
          <div className="space-y-4">
            {envelopes.map((env) => {
              const events = env.events || [];
              const milestones: Array<{ label: string; at: string | null }> = [
                { label: "Created", at: env.created_at },
                { label: "Sent", at: env.sent_at },
                { label: "Delivered", at: env.delivered_at },
                { label: "Viewed", at: env.viewed_at },
                { label: "Signed", at: env.signed_at },
              ];
              if (env.declined_at) milestones.push({ label: "Declined", at: env.declined_at });
              return (
                <div key={env.id} className="rounded-md border p-4 space-y-3">
                  <div className="flex items-start justify-between gap-3">
                    <div>
                      <div className="font-medium">
                        {env.template_id ? templates[env.template_id] || `Template #${env.template_id}` : "Ad-hoc envelope"}
                      </div>
                      <div className="text-xs text-muted-foreground">
                        {env.provider} · {env.external_envelope_id || "no external id"}
                      </div>
                      {env.signer_email && (
                        <div className="text-xs text-muted-foreground">
                          To: {env.signer_name ? `${env.signer_name} <${env.signer_email}>` : env.signer_email}
                        </div>
                      )}
                    </div>
                    <div className="flex flex-col items-end gap-2">
                      {statusBadge(env.status)}
                      {env.status !== "signed" && env.external_envelope_id && (
                        <Button size="sm" variant="outline" onClick={() => simulate(env.external_envelope_id!)}>
                          Simulate signed
                        </Button>
                      )}
                    </div>
                  </div>
                  <div>
                    {env.last_error && (
                      <div className="rounded-md bg-destructive/10 border border-destructive/30 p-2 text-xs text-destructive">
                        <span className="font-medium">Error:</span> {env.last_error}
                      </div>
                    )}
                  </div>
                </div>
              )}
            )}
            <div className="grid grid-cols-5 gap-2 text-xs">
              {milestones.map((m) => (
                <div key={m.label} className={`${rounded} border p-2 ${m.at ? "bg-muted/40" : "opacity-50"}`} >
                  <div className="font-medium">{m.label}</div>
                  <div className="text-muted-foreground">{fmt(m.at) || "—"}</div>
                </div>
              )}
            </div>
            {events.length > 0 && (
              <details className="text-xs">
                <summary className="cursor-pointer text-muted-foreground hover:text-foreground">
                  {events.length} event{events.length === 1 ? "" : "s"} from provider
                </summary>
                <div className="mt-2 space-y-1 pl-2 border-l">
                  {events.map((ev, i) => (
                    <div key={i}>

```

```

        <span className="text-muted-foreground">{fmt(ev.at)}</span> · {ev.type}
      </div>
    )}}
  </div>
</details>
)}
</div>
)};
</div>
)}
</CardContent>
</Card>

<Card>
  <CardHeader>
    <CardTitle className="flex items-center gap-2"><Send className="h-5 w-5" /> Doctor Faxes</CardTitle>
    <CardDescription>Medical-records requests faxed automatically once HIPAA returns signed.</CardDescription>
  </CardHeader>
  <CardContent>
    {faxes.length === 0 ? (
      <p className="text-sm text-muted-foreground">
        No faxes yet. One will be sent automatically once a HIPAA template is signed.
      </p>
    ) : (
      <div className="space-y-3">
        {faxes.map((f) => (
          <div key={f.id} className="rounded-md border p-3">
            <div className="flex items-start justify-between">
              <div>
                <div className="text-sm font-medium">{f.source_file}</div>
                <div className="text-xs text-muted-foreground">
                  Created {fmt(f.created_at)}
                  {f.processed_at && ` · Processed ${fmt(f.processed_at)} `}
                </div>
              </div>
              <Badge variant={f.status === "done" ? "default" : f.status === "error" ? "destructive" : "secondary"}>
                {f.status}
              </Badge>
            </div>
            {f.raw_text && (
              <div className="mt-2 text-xs text-muted-foreground line-clamp-2">{f.raw_text}</div>
            )}
            {f.error && (
              <div className="mt-2 text-xs text-destructive">{f.error}</div>
            )}
          </div>
        ))}
      </div>
    )}
  </CardContent>
</Card>
</div>
);
}

```

[artifacts/mtos-crm/src/components/layout/layout.tsx](#)

```
import { ReactNode } from "react";
import { Sidebar } from "../sidebar";

interface LayoutProps {
  children: ReactNode;
}

export function Layout({ children }: LayoutProps) {
  return (
    <div className="flex h-[100dvh] overflow-hidden bg-background">
      <Sidebar />
      <main className="flex-1 overflow-y-auto">
        <div className="mx-auto max-w-7xl px-4 py-8 sm:px-6 lg:px-8">
          {children}
        </div>
      </main>
    </div>
  );
}
```


[artifacts/mtos-crm/src/components/layout/sidebar.tsx](#)

```
import { Link, useLocation } from "wouter";
import { LayoutDashboard, Users, FileText, PlusCircle, Briefcase, Inbox, GitBranch, UserCheck, BarChart3, Shield, Stethoscope, ShieldAlert, AppWindow, Building2, ShieldCheck, FileSearch, Clock, Sparkles, Brain, Plug, Newspaper, TrendingUp, FileUp, Scale, Building, FileSignature, Grid3x3, Settings } from "lucide-react";
import { cn } from "@lib/utils";

const navigation = [
  { name: "Dashboard", href: "/", icon: LayoutDashboard },
  { name: "Pipeline", href: "/pipeline", icon: GitBranch },
  { name: "Leads", href: "/leads", icon: Users },
  { name: "Cases", href: "/cases", icon: Briefcase },
  { name: "Paralegals", href: "/paralegals", icon: UserCheck },
  { name: "Documents", href: "/documents", icon: FileText },
  { name: "OCR Inbox", href: "/ocr-inbox", icon: Inbox },
  { name: "NPI Lookup", href: "/npi-lookup", icon: Stethoscope },
  { name: "Review Queue", href: "/review-queue", icon: ShieldAlert },
  { name: "Analytics", href: "/analytics", icon: BarChart3 },
  { name: "Compliance", href: "/compliance", icon: Shield },
  { name: "Form Engine", href: "/form-engine", icon: AppWindow },
  { name: "Vendors", href: "/vendors", icon: Building2 },
  { name: "Security", href: "/security", icon: ShieldCheck },
  { name: "Doc Review", href: "/doc-review", icon: FileSearch },
  { name: "Timeline", href: "/timeline", icon: Clock },
  { name: "Drafting AI", href: "/drafting", icon: Sparkles },
  { name: "Praxis AI", href: "/predictive", icon: Brain },
  { name: "Decision Engine", href: "/decision-engine", icon: Scale },
  { name: "Integrations", href: "/integrations", icon: Plug },
  { name: "Tort News", href: "/news", icon: Newspaper },
  { name: "Financial", href: "/financial-news", icon: TrendingUp },
  { name: "Lead Import", href: "/lead-import", icon: FileUp },
  { name: "New Intake", href: "/leads/new", icon: PlusCircle },
  { name: "Buyers", href: "/buyers", icon: Building },
  { name: "Doc Templates", href: "/document-templates", icon: FileSignature },
  { name: "Assignment Matrix", href: "/template-assignments", icon: Grid3x3 },
  { name: "Workflow Settings", href: "/workflow-settings", icon: Settings },
];

export function Sidebar() {
  const [location] = useLocation();

  return (
    <div className="flex h-full w-64 flex-col bg-sidebar border-r border-sidebar-border">
      <div className="flex h-14 items-center border-b border-sidebar-border px-4">
        <span className="font-bold text-lg tracking-tight text-sidebar-foreground">MTOS</span>
        <span className="ml-2 rounded-md bg-primary/10 px-1.5 py-0.5 text-xs font-medium text-primary">v1.0</span>
      </div>
      <nav className="flex-1 space-y-1 px-2 py-4">
        {navigation.map((item) => {
          const isActive = location === item.href || (item.href !== "/" && location.startsWith(item.href));
          return (
            <Link
              key={item.name}
              href={item.href}
              className={cn(
                isActive
                  ? "bg-sidebar-accent text-sidebar-accent-foreground"
                  : "text-sidebar-foreground hover:bg-sidebar-accent/50 hover:text-sidebar-accent-foreground",
                "group flex items-center rounded-md px-2 py-2 text-sm font-medium"
              )}>
              <item.icon
                className={cn(
                  isActive ? "text-sidebar-accent-foreground" : "text-sidebar-foreground/70 group-hover:text-sidebar-accent-foreground",
                  "mr-3 h-5 w-5 flex-shrink-0"
                )}/>
              {item.name}
            </Link>
          );
        })}
      </nav>
      <div className="p-4 border-t border-sidebar-border">
        <div className="flex items-center gap-3">
          <div className="h-8 w-8 rounded-full bg-primary/20 flex items-center justify-center text-primary font-medium text-sm">ID</div>
          <div className="flex flex-col">
            <span className="text-sm font-medium text-sidebar-foreground">Intake Director</span>
            <span className="text-xs text-sidebar-foreground/70">System Admin</span>
          </div>
        </div>
      </div>
    </div>
  );
}
```

[artifacts/mtos-crm/src/components/ui/accordion.tsx](#)

```
import * as React from "react"
import * as AccordionPrimitive from "@radix-ui/react-accordion"
import { ChevronDown } from "lucide-react"

import { cn } from "@lib/utils"

const Accordion = AccordionPrimitive.Root

const AccordionItem = React.forwardRef<
  React.ElementRef<typeof AccordionPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof AccordionPrimitive.Item>
>(({ className, ...props }, ref) => (
  <AccordionPrimitive.Item
    ref={ref}
    className={cn("border-b", className)}
    {...props}
  />
))
AccordionItem.displayName = "AccordionItem"

const AccordionTrigger = React.forwardRef<
  React.ElementRef<typeof AccordionPrimitive.Trigger>,
  React.ComponentPropsWithoutRef<typeof AccordionPrimitive.Trigger>
>(({ className, children, ...props }, ref) => (
  <AccordionPrimitive.Header className="flex">
    <AccordionPrimitive.Trigger
      ref={ref}
      className={cn(
        "flex flex-1 items-center justify-between py-4 text-sm font-medium transition-all hover:underline text-left [&[data-state=open]>svg]:rotate-180",
        className
      )}
      {...props}
    >
      {children}
      <ChevronDown className="h-4 w-4 shrink-0 text-muted-foreground transition-transform duration-200" />
    </AccordionPrimitive.Trigger>
  </AccordionPrimitive.Header>
))
AccordionTrigger.displayName = AccordionPrimitive.Trigger.displayName

const AccordionContent = React.forwardRef<
  React.ElementRef<typeof AccordionPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof AccordionPrimitive.Content>
>(({ className, children, ...props }, ref) => (
  <AccordionPrimitive.Content
    ref={ref}
    className="overflow-hidden text-sm data-[state=closed]:animate-accordion-up data-[state=open]:animate-accordion-down"
    {...props}
  >
    <div className={cn("pb-4 pt-0", className)}>{children}</div>
  </AccordionPrimitive.Content>
))
AccordionContent.displayName = AccordionPrimitive.Content.displayName

export { Accordion, AccordionItem, AccordionTrigger, AccordionContent }
```

[artifacts/mtos-crm/src/components/ui/alert-dialog.tsx](#)

```
import * as React from "react"
import * as AlertDialogPrimitive from "@radix-ui/react-alert-dialog"

import { cn } from "@lib/utils"
import { buttonVariants } from "@components/ui/button"

const AlertDialog = AlertDialogPrimitive.Root

const AlertDialogTrigger = AlertDialogPrimitive.Trigger

const AlertDialogPortal = AlertDialogPrimitive.Portal

const AlertDialogOverlay = React.forwardRef<
  React.ElementRef<typeof AlertDialogPrimitive.Overlay>,
  React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Overlay>
>(({ className, ...props }, ref) => (
  <AlertDialogPrimitive.Overlay
    className={cn(
      "fixed inset-0 z-50 bg-black/80 data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0",
      className
    )}
    {...props}
    ref={ref}
  />
))
AlertDialogOverlay.displayName = AlertDialogPrimitive.Overlay.displayName

const AlertDialogContent = React.forwardRef<
  React.ElementRef<typeof AlertDialogPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Content>
>(({ className, ...props }, ref) => (
  <AlertDialogPortal>
    <AlertDialogOverlay />
    <AlertDialogPrimitive.Content
      ref={ref}
      className={cn(
        "fixed left-[50%] top-[50%] z-50 grid w-full max-w-lg translate-x-[-50%] translate-y-[-50%] gap-4 border bg-background p-6 shadow-lg duration-200 data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[state=closed]:slide-out-to-left-1/2 data-[state=closed]:slide-out-to-top-48% data-[state=open]:slide-in-from-left-1/2 data-[state=open]:slide-in-from-top-48% sm:rounded-lg",
        className
      )}
      {...props}
    />
  </AlertDialogPortal>
))
AlertDialogContent.displayName = AlertDialogPrimitive.Content.displayName

const AlertDialogHeader = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn(
      "flex flex-col space-y-2 text-center sm:text-left",
      className
    )}
    {...props}
  />
)
AlertDialogHeader.displayName = "AlertDialogHeader"

const AlertDialogFooter = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn(
      "flex flex-col-reverse sm:flex-row sm:justify-end sm:space-x-2",
      className
    )}
    {...props}
  />
)
AlertDialogFooter.displayName = "AlertDialogFooter"

const AlertDialogTitle = React.forwardRef<
  React.ElementRef<typeof AlertDialogPrimitive.Title>,
  React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Title>
>(({ className, ...props }, ref) => (
  <AlertDialogPrimitive.Title
    ref={ref}
    className={cn("text-lg font-semibold", className)}
    {...props}
  />
))
AlertDialogTitle.displayName = AlertDialogPrimitive.Title.displayName

const AlertDialogDescription = React.forwardRef<
  React.ElementRef<typeof AlertDialogPrimitive.Description>,
  React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Description>
>(({ className, ...props }, ref) => (
  <AlertDialogPrimitive.Description
```

```

      ref={ref}
      className={cn("text-sm text-muted-foreground", className)}
      {...props}
    />
  ))
  AlertDialogDescription.displayName =
    AlertDialogPrimitive.Description.displayName

  const AlertDialogAction = React.forwardRef<
    React.ElementRef<typeof AlertDialogPrimitive.Action>,
    React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Action>
  >(({ className, ...props }, ref) => (
    <AlertDialogPrimitive.Action
      ref={ref}
      className={cn(buttonVariants(), className)}
      {...props}
    />
  ))
  AlertDialogAction.displayName = AlertDialogPrimitive.Action.displayName

  const AlertDialogCancel = React.forwardRef<
    React.ElementRef<typeof AlertDialogPrimitive.Cancel>,
    React.ComponentPropsWithoutRef<typeof AlertDialogPrimitive.Cancel>
  >(({ className, ...props }, ref) => (
    <AlertDialogPrimitive.Cancel
      ref={ref}
      className={cn(
        buttonVariants({ variant: "outline" }),
        "mt-2 sm:mt-0",
        className
      )}
      {...props}
    />
  ))
  AlertDialogCancel.displayName = AlertDialogPrimitive.Cancel.displayName

  export {
    AlertDialog,
    AlertDialogPortal,
    AlertDialogOverlay,
    AlertDialogTrigger,
    AlertDialogContent,
    AlertDialogHeader,
    AlertDialogFooter,
    AlertDialogTitle,
    AlertDialogDescription,
    AlertDialogAction,
    AlertDialogCancel,
  }
}

```

[artifacts/mtos-crm/src/components/ui/alert.tsx](#)

```
import * as React from "react"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"

const alertVariants = cva(
  "relative w-full rounded-lg border px-4 py-3 text-sm [&svg+div]:translate-y-[-3px] [&svg]:absolute [&svg]:left-4 [&svg]:top-4 [&svg]:text-foreground [&svg~*]:pl-7",
  {
    variants: {
      variant: {
        default: "bg-background text-foreground",
        destructive:
          "border-destructive/50 text-destructive dark:border-destructive [&svg]:text-destructive",
      },
    },
    defaultVariants: {
      variant: "default",
    },
  }
)

const Alert = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement> & VariantProps<typeof alertVariants>
>(({ className, variant, ...props }, ref) => (
  <div
    ref={ref}
    role="alert"
    className={cn(alertVariants({ variant }), className)}
    {...props}
  />
))
Alert.displayName = "Alert"

const AlertTitle = React.forwardRef<
  HTMLParagraphElement,
  React.HTMLAttributes<HTMLHeadingElement>
>(({ className, ...props }, ref) => (
  <h5
    ref={ref}
    className={cn("mb-1 font-medium leading-none tracking-tight", className)}
    {...props}
  />
))
AlertTitle.displayName = "AlertTitle"

const AlertDescription = React.forwardRef<
  HTMLParagraphElement,
  React.HTMLAttributes<HTMLParagraphElement>
>(({ className, ...props }, ref) => (
  <div
    ref={ref}
    className={cn("text-sm [&_p]:leading-relaxed", className)}
    {...props}
  />
))
AlertDescription.displayName = "AlertDescription"

export { Alert, AlertTitle, AlertDescription }
```

[artifacts/mtos-crm/src/components/ui/aspect-ratio.tsx](#)

```
import * as AspectRatioPrimitive from "@radix-ui/react-aspect-ratio"

const AspectRatio = AspectRatioPrimitive.Root

export { AspectRatio }
```

[artifacts/mtos-crm/src/components/ui/avatar.tsx](#)

```
"use client"

import * as React from "react"
import * as AvatarPrimitive from "@radix-ui/react-avatar"

import { cn } from "@lib/Utils"

const Avatar = React.forwardRef<
  React.ElementRef<typeof AvatarPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof AvatarPrimitive.Root>
>(({ className, ...props }, ref) => (
  <AvatarPrimitive.Root
    ref={ref}
    className={cn(
      "relative flex h-10 w-10 shrink-0 overflow-hidden rounded-full",
      className
    )}
    {...props}
  />
))
Avatar.displayName = AvatarPrimitive.Root.displayName

const AvatarImage = React.forwardRef<
  React.ElementRef<typeof AvatarPrimitive.Image>,
  React.ComponentPropsWithoutRef<typeof AvatarPrimitive.Image>
>(({ className, ...props }, ref) => (
  <AvatarPrimitive.Image
    ref={ref}
    className={cn("aspect-square h-full w-full", className)}
    {...props}
  />
))
AvatarImage.displayName = AvatarPrimitive.Image.displayName

const AvatarFallback = React.forwardRef<
  React.ElementRef<typeof AvatarPrimitive.Fallback>,
  React.ComponentPropsWithoutRef<typeof AvatarPrimitive.Fallback>
>(({ className, ...props }, ref) => (
  <AvatarPrimitive.Fallback
    ref={ref}
    className={cn(
      "flex h-full w-full items-center justify-center rounded-full bg-muted",
      className
    )}
    {...props}
  />
))
AvatarFallback.displayName = AvatarPrimitive.Fallback.displayName

export { Avatar, AvatarImage, AvatarFallback }
```

[artifacts/mtos-crm/src/components/ui/badge.tsx](#)

```
import * as React from "react"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@/lib/utils"

const badgeVariants = cva(
  // @replit
  // whitespace-nowrap: Badges should never wrap.
  "whitespace-nowrap inline-flex items-center rounded-md border px-2.5 py-0.5 text-xs font-semibold transition-colors focus:outline-none focus:ring-2 focus:ring-ring focus:ring-offset-2" +
  " hover-elevate ",
  {
    variants: {
      variant: {
        default:
          // @replit shadow-xs instead of shadow, no hover because we use hover-elevate
          "border-transparent bg-primary text-primary-foreground shadow-xs",
        secondary:
          // @replit no hover because we use hover-elevate
          "border-transparent bg-secondary text-secondary-foreground",
        destructive:
          // @replit shadow-xs instead of shadow, no hover because we use hover-elevate
          "border-transparent bg-destructive text-destructive-foreground shadow-xs",
        // @replit shadow-xs" - use badge outline variable
        outline: "text-foreground border [border-color:var(--badge-outline)]",
      },
    },
    defaultVariants: {
      variant: "default",
    },
  }
)

export interface BadgeProps
  extends React.HTMLAttributes<HTMLDivElement>,
    VariantProps<typeof badgeVariants> {}

function Badge({ className, variant, ...props }: BadgeProps) {
  return (
    <div className={cn(badgeVariants({ variant }), className)} {...props} />
  )
}

export { Badge, badgeVariants }
```


[artifacts/mtos-crm/src/components/ui/breadcrumb.tsx](#)

```
import * as React from "react"
import { Slot } from "@radix-ui/react-slot"
import { ChevronRight, MoreHorizontal } from "lucide-react"

import { cn } from "@lib/utils"

const Breadcrumb = React.forwardRef<
  HTMLDivElement,
  React.ComponentPropsWithoutRef<"nav"> & {
    separator?: React.ReactNode
  }
>(({ ...props }, ref) => <nav ref={ref} aria-label="breadcrumb" {...props} />)
Breadcrumb.displayName = "Breadcrumb"

const BreadcrumbList = React.forwardRef<
  HTMLListElement,
  React.ComponentPropsWithoutRef<"ol">
>(({ className, ...props }, ref) => (
  <ol
    ref={ref}
    className={cn(
      "flex flex-wrap items-center gap-1.5 break-words text-sm text-muted-foreground sm:gap-2.5",
      className
    )}
    {...props}
  />
))
BreadcrumbList.displayName = "BreadcrumbList"

const BreadcrumbItem = React.forwardRef<
  HTMLLIElement,
  React.ComponentPropsWithoutRef<"li">
>(({ className, ...props }, ref) => (
  <li
    ref={ref}
    className={cn("inline-flex items-center gap-1.5", className)}
    {...props}
  />
))
BreadcrumbItem.displayName = "BreadcrumbItem"

const BreadcrumbLink = React.forwardRef<
  HTMLAnchorElement,
  React.ComponentPropsWithoutRef<"a"> & {
    asChild?: boolean
  }
>(({ asChild, className, ...props }, ref) => {
  const Comp = asChild ? Slot : "a"

  return (
    <Comp
      ref={ref}
      className={cn("transition-colors hover:text-foreground", className)}
      {...props}
    />
  )
})
BreadcrumbLink.displayName = "BreadcrumbLink"

const BreadcrumbPage = React.forwardRef<
  HTMLSpanElement,
  React.ComponentPropsWithoutRef<"span">
>(({ className, ...props }, ref) => (
  <span
    ref={ref}
    role="link"
    aria-disabled="true"
    aria-current="page"
    className={cn("font-normal text-foreground", className)}
    {...props}
  />
))
BreadcrumbPage.displayName = "BreadcrumbPage"

const BreadcrumbSeparator = ({
  children,
  className,
  ...props
}: React.ComponentProps<"li">) => (
  <li
    role="presentation"
    aria-hidden="true"
    className={cn("[&svg]:w-3.5 [&svg]:h-3.5", className)}
    {...props}
  >
    {children ?? <ChevronRight />}
  </li>
)
BreadcrumbSeparator.displayName = "BreadcrumbSeparator"

const BreadcrumbEllipsis = ({
  className,
  ...props
}: React.ComponentProps<"span">) => (
  <span
```

```
    role="presentation"
    aria-hidden="true"
    className={cn("flex h-9 w-9 items-center justify-center", className)}
    {...props}
  >
    <MoreHorizontal className="h-4 w-4" />
    <span className="sr-only">More</span>
  </span>
)
BreadcrumbEllipsis.displayName = "BreadcrumbElipssis"

export {
  Breadcrumb,
  BreadcrumbList,
  BreadcrumbItem,
  BreadcrumbLink,
  BreadcrumbPage,
  BreadcrumbSeparator,
  BreadcrumbEllipsis,
}
```

[artifacts/mtos-crm/src/components/ui/button-group.tsx](#)

```
import { Slot } from "@radix-ui/react-slot"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utls"
import { Separator } from "@components/ui/separator"

const buttonGroupVariants = cva(
  "flex w-fit items-stretch has-[:>[data-slot=button-group]]:gap-2 [>]:focus-visible:z-10 has-[:select[aria-hidden=true]:last-child]:>[data-slot=select-trigger]:last-of-type]:rounded-r-md [>[data-slot=select-trigger]:not([class*='w-'])]:w-fit [>input]:flex-1",
  {
    variants: {
      orientation: {
        horizontal:
          "[>*:not(:first-child)]:rounded-l-none [>*:not(:first-child)]:border-l-0 [>*:not(:last-child)]:rounded-r-none",
        vertical:
          "flex-col [>*:not(:first-child)]:rounded-t-none [>*:not(:first-child)]:border-t-0 [>*:not(:last-child)]:rounded-b-
none",
      },
    },
    defaultVariants: {
      orientation: "horizontal",
    },
  }
)

function ButtonGroup({
  className,
  orientation,
  ...props
}: React.ComponentProps<"div"> & VariantProps<typeof buttonGroupVariants>) {
  return (
    <div
      role="group"
      data-slot="button-group"
      data-orientation={orientation}
      className={cn(buttonGroupVariants({ orientation }), className)}
      {...props}
    />
  )
}

function ButtonGroupText({
  className,
  asChild = false,
  ...props
}: React.ComponentProps<"div"> & {
  asChild?: boolean
}) {
  const Comp = asChild ? Slot : "div"

  return (
    <Comp
      className={cn(
        "bg-muted shadow-xs flex items-center gap-2 rounded-md border px-4 text-sm font-medium
[&_svg:not([class*='size-'])]:size-4 [&_svg]:pointer-events-none",
        className
      )}
      {...props}
    />
  )
}

function ButtonGroupSeparator({
  className,
  orientation = "vertical",
  ...props
}: React.ComponentProps<typeof Separator>) {
  return (
    <Separator
      data-slot="button-group-separator"
      orientation={orientation}
      className={cn(
        "bg-input relative !m-0 self-stretch data-[orientation=vertical]:h-auto",
        className
      )}
      {...props}
    />
  )
}

export {
  ButtonGroup,
  ButtonGroupSeparator,
  ButtonGroupText,
  buttonGroupVariants,
}
```

[artifacts/mtos-crm/src/components/ui/button.tsx](#)

```
import * as React from "react"
import { Slot } from "@radix-ui/react-slot"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"

const buttonVariants = cva(
  "inline-flex items-center justify-center gap-2 whitespace-nowrap rounded-md text-sm font-medium transition-colors focus-visible:outline-none focus-visible:ring-1 focus-visible:ring-ring disabled:pointer-events-none disabled:opacity-50 [&_svg]:pointer-events-none [&_svg]:size-4 [&_svg]:shrink-0" +
  " hover-elevate active-elevate-2",
  {
    variants: {
      variant: {
        default:
          // @replit: no hover, and add primary border
          "bg-primary text-primary-foreground border border-primary-border",
        destructive:
          "bg-destructive text-destructive-foreground shadow-sm border-destructive-border",
        outline:
          // @replit Shows the background color of whatever card / sidebar / accent background it is inside of.
          // Inherits the current text color. Uses shadow-xs. no shadow on active
          // No hover state
          "border [border-color:var(--button-outline)] shadow-xs active:shadow-none ",
        secondary:
          // @replit border, no hover, no shadow, secondary border.
          "border bg-secondary text-secondary-foreground border border-secondary-border ",
        // @replit no hover, transparent border
        ghost: "border border-transparent",
        link: "text-primary underline-offset-4 hover:underline",
      },
      size: {
        // @replit changed sizes
        default: "min-h-9 px-4 py-2",
        sm: "min-h-8 rounded-md px-3 text-xs",
        lg: "min-h-10 rounded-md px-8",
        icon: "h-9 w-9",
      },
    },
    defaultVariants: {
      variant: "default",
      size: "default",
    },
  }
)

export interface ButtonProps
  extends React.ButtonHTMLAttributes<HTMLButtonElement>,
    VariantProps<typeof buttonVariants> {
  asChild?: boolean
}

const Button = React.forwardRef<HTMLButtonElement, ButtonProps>(
  ({ className, variant, size, asChild = false, ...props }, ref) => {
    const Comp = asChild ? Slot : "button"
    return (
      <Comp
        className={cn(buttonVariants({ variant, size, className }))}
        ref={ref}
        {...props}
      />
    )
  }
)
Button.displayName = "Button"

export { Button, buttonVariants }
```

[artifacts/mtos-crm/src/components/ui/calendar.tsx](#)

```
"use client"

import * as React from "react"
import {
  ChevronDownIcon,
  ChevronLeftIcon,
  ChevronRightIcon,
} from "lucide-react"
import { DayButton, DayPicker, getDefaultClassNames } from "react-day-picker"

import { cn } from "@lib/utls"
import { Button, buttonVariants } from "@components/ui/button"

function Calendar({
  className,
  classNames,
  showOutsideDays = true,
  captionLayout = "label",
  buttonVariant = "ghost",
  formatters,
  components,
  ...props
}: React.ComponentProps<typeof DayPicker> & {
  buttonVariant?: React.ComponentProps<typeof Button>["variant"]
}) {
  const defaultClassNames = getDefaultClassNames()

  return (
    <DayPicker
      showOutsideDays={showOutsideDays}
      className={cn(
        "bg-background group/calendar p-3 [--cell-size:2rem] [[data-slot=card-content]_&]:bg-transparent [[data-slot=popover-content]_&]:bg-transparent",
        String.raw`rtl:**:[.rdp-button\_next>svg]:rotate-180`,
        String.raw`rtl:**:[.rdp-button\_previous>svg]:rotate-180`,
        className
      )}
      captionLayout={captionLayout}
      formatters={{
        formatMonthDropdown: (date) =>
          date.toLocaleString("default", { month: "short" }),
        ...formatters,
      }}
      classNames={{
        root: cn("w-fit", defaultClassNames.root),
        months: cn(
          "relative flex flex-col gap-4 md:flex-row",
          defaultClassNames.months
        ),
        month: cn("flex w-full flex-col gap-4", defaultClassNames.month),
        nav: cn(
          "absolute inset-x-0 top-0 flex w-full items-center justify-between gap-1",
          defaultClassNames.nav
        ),
        button_previous: cn(
          buttonVariants({ variant: buttonVariant }),
          "h-[--cell-size] w-[--cell-size] select-none p-0 aria-disabled:opacity-50",
          defaultClassNames.button_previous
        ),
        button_next: cn(
          buttonVariants({ variant: buttonVariant }),
          "h-[--cell-size] w-[--cell-size] select-none p-0 aria-disabled:opacity-50",
          defaultClassNames.button_next
        ),
        month_caption: cn(
          "flex h-[--cell-size] w-full items-center justify-center px-[--cell-size]",
          defaultClassNames.month_caption
        ),
        dropdowns: cn(
          "flex h-[--cell-size] w-full items-center justify-center gap-1.5 text-sm font-medium",
          defaultClassNames.dropdowns
        ),
        dropdown_root: cn(
          "has-focus:border-ring border-input shadow-xs has-focus:ring-ring/50 has-focus:ring-[3px] relative rounded-md border",
          defaultClassNames.dropdown_root
        ),
        dropdown: cn(
          "bg-popover absolute inset-0 opacity-0",
          defaultClassNames.dropdown
        ),
        caption_label: cn(
          "select-none font-medium",
          captionLayout === "label"
            ? "text-sm"
            : "[&>svg]:text-muted-foreground flex h-8 items-center gap-1 rounded-md pl-2 pr-1 text-sm [&>svg]:size-3.5",
          defaultClassNames.caption_label
        ),
        table: "w-full border-collapse",
        weekdays: cn("flex", defaultClassNames.weekdays),
        weekday: cn(
          "text-muted-foreground flex-1 select-none rounded-md text-[0.8rem] font-normal",
          defaultClassNames.weekday
        ),
        week: cn("mt-2 flex w-full", defaultClassNames.week),
        week_number_header: cn(
```

```

        "w-[--cell-size] select-none",
        defaultClassNames.week_number_header
    ),
    week_number: cn(
        "text-muted-foreground select-none text-[0.8rem]",
        defaultClassNames.week_number
    ),
    day: cn(
        "group/day relative aspect-square h-full w-full select-none p-0 text-center [&:first-child[data-
selected=true]_button]:rounded-l-md [&:last-child[data-selected=true]_button]:rounded-r-md",
        defaultClassNames.day
    ),
    range_start: cn(
        "bg-accent rounded-l-md",
        defaultClassNames.range_start
    ),
    range_middle: cn("rounded-none", defaultClassNames.range_middle),
    range_end: cn("bg-accent rounded-r-md", defaultClassNames.range_end),
    today: cn(
        "bg-accent text-accent-foreground rounded-md data-[selected=true]:rounded-none",
        defaultClassNames.today
    ),
    outside: cn(
        "text-muted-foreground aria-selected:text-muted-foreground",
        defaultClassNames.outside
    ),
    disabled: cn(
        "text-muted-foreground opacity-50",
        defaultClassNames.disabled
    ),
    hidden: cn("invisible", defaultClassNames.hidden),
    ...ClassNames,
  }}
  components={{
    Root: ({ className, rootRef, ...props }) => {
      return (
        <div
          data-slot="calendar"
          ref={rootRef}
          className={cn(className)}
          {...props}
        />
      )
    },
    Chevron: ({ className, orientation, ...props }) => {
      if (orientation === "left") {
        return (
          <ChevronLeftIcon className={cn("size-4", className)} {...props} />
        )
      }

      if (orientation === "right") {
        return (
          <ChevronRightIcon
            className={cn("size-4", className)}
            {...props}
          />
        )
      }

      return (
        <ChevronDownIcon className={cn("size-4", className)} {...props} />
      )
    },
    DayButton: CalendarDayButton,
    WeekNumber: ({ children, ...props }) => {
      return (
        <td {...props}>
          <div className="flex size-[--cell-size] items-center justify-center text-center">
            {children}
          </div>
        </td>
      )
    },
    ...components,
  }}
  {...props}
) />
}

```

```

function CalendarDayButton({
  className,
  day,
  modifiers,
  ...props
}: React.ComponentProps<typeof DayButton>) {
  const defaultClassNames = getDefaultClassNames()

  const ref = React.useRef<HTMLButtonElement>(null)
  React.useEffect(() => {
    if (modifiers.focused) ref.current?.focus()
  }, [modifiers.focused])

  return (
    <Button
      ref={ref}
    />
  )
}

```

```

variant="ghost"
size="icon"
data-day={day.date.toLocaleDateString()}
data-selected-single={
  modifiers.selected &&
  !modifiers.range_start &&
  !modifiers.range_end &&
  !modifiers.range_middle
}
data-range-start={modifiers.range_start}
data-range-end={modifiers.range_end}
data-range-middle={modifiers.range_middle}
className={cn(
  "data-[selected-single=true]:bg-primary data-[selected-single=true]:text-primary-foreground data-[range-middle=true]:bg-
accent data-[range-middle=true]:text-accent-foreground data-[range-start=true]:bg-primary data-[range-start=true]:text-primary-
foreground data-[range-end=true]:bg-primary data-[range-end=true]:text-primary-foreground group-data-[focused=true]/day:border-
ring group-data-[focused=true]/day:ring-ring/50 flex aspect-square h-auto w-full min-w-[-cell-size] flex-col gap-1 font-normal
leading-none data-[range-end=true]:rounded-md data-[range-middle=true]:rounded-none data-[range-start=true]:rounded-md group-
data-[focused=true]/day:relative group-data-[focused=true]/day:z-10 group-data-[focused=true]/day:ring-[3px] [>span]:text-xs
[>span]:opacity-70",
  defaultClassNames.day,
  className
)}
{...props}
)
}

export { Calendar, CalendarDayButton }

```

[artifacts/mtos-crm/src/components/ui/card.tsx](#)

```
import * as React from "react"

import { cn } from "@lib/utils"

const Card = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => (
  <div
    ref={ref}
    className={cn(
      "rounded-xl border bg-card text-card-foreground shadow",
      className
    )}
    {...props}
  />
))
Card.displayName = "Card"

const CardHeader = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => (
  <div
    ref={ref}
    className={cn("flex flex-col space-y-1.5 p-6", className)}
    {...props}
  />
))
CardHeader.displayName = "CardHeader"

const CardTitle = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => (
  <div
    ref={ref}
    className={cn("font-semibold leading-none tracking-tight", className)}
    {...props}
  />
))
CardTitle.displayName = "CardTitle"

const CardDescription = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => (
  <div
    ref={ref}
    className={cn("text-sm text-muted-foreground", className)}
    {...props}
  />
))
CardDescription.displayName = "CardDescription"

const CardContent = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => (
  <div ref={ref} className={cn("p-6 pt-0", className)} {...props} />
))
CardContent.displayName = "CardContent"

const CardFooter = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => (
  <div
    ref={ref}
    className={cn("flex items-center p-6 pt-0", className)}
    {...props}
  />
))
CardFooter.displayName = "CardFooter"

export { Card, CardHeader, CardFooter, CardTitle, CardDescription, CardContent }
```


[artifacts/mtos-crm/src/components/ui/carousel.tsx](#)

```
import * as React from "react"
import useEmblaCarousel, {
  type UseEmblaCarouselType,
} from "embla-carousel-react"
import { ArrowLeft, ArrowRight } from "lucide-react"

import { cn } from "@lib/utlils"
import { Button } from "@components/ui/button"

type CarouselApi = UseEmblaCarouselType[1]
type UseCarouselParameters = Parameters<typeof useEmblaCarousel>
type CarouselOptions = UseCarouselParameters[0]
type CarouselPlugin = UseCarouselParameters[1]

type CarouselProps = {
  opts?: CarouselOptions
  plugins?: CarouselPlugin
  orientation?: "horizontal" | "vertical"
  setApi?: (api: CarouselApi) => void
}

type CarouselContextProps = {
  carouselRef: ReturnType<typeof useEmblaCarousel>[0]
  api: ReturnType<typeof useEmblaCarousel>[1]
  scrollPrev: () => void
  scrollNext: () => void
  canScrollPrev: boolean
  canScrollNext: boolean
} & CarouselProps

const CarouselContext = React.createContext<CarouselContextProps | null>(null)

function useCarousel() {
  const context = React.useContext(CarouselContext)

  if (!context) {
    throw new Error("useCarousel must be used within a <Carousel />")
  }

  return context
}

const Carousel = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement> & CarouselProps
>((
  {
    orientation = "horizontal",
    opts,
    setApi,
    plugins,
    className,
    children,
    ...props
  },
  ref
) => {
  const [carouselRef, api] = useEmblaCarousel(
    {
      ...opts,
      axis: orientation === "horizontal" ? "x" : "y",
    },
    plugins
  )
  const [canScrollPrev, setCanScrollPrev] = React.useState(false)
  const [canScrollNext, setCanScrollNext] = React.useState(false)

  const onSelect = React.useCallback((api: CarouselApi) => {
    if (!api) {
      return
    }

    setCanScrollPrev(api.canScrollPrev())
    setCanScrollNext(api.canScrollNext())
  }, [])

  const scrollPrev = React.useCallback(() => {
    api?.scrollPrev()
  }, [api])

  const scrollNext = React.useCallback(() => {
    api?.scrollNext()
  }, [api])

  const handleKeyDown = React.useCallback(
    (event: React.KeyboardEvent<HTMLDivElement>) => {
      if (event.key === "ArrowLeft") {
        event.preventDefault()
        scrollPrev()
      } else if (event.key === "ArrowRight") {
        event.preventDefault()
        scrollNext()
      }
    },
  ),
```

```

    [scrollPrev, scrollNext]
  )
  React.useEffect(() => {
    if (!api || !setApi) {
      return
    }

    setApi(api)
  }, [api, setApi])

  React.useEffect(() => {
    if (!api) {
      return
    }

    onSelect(api)
    api.on("reInit", onSelect)
    api.on("select", onSelect)

    return () => {
      api?.off("select", onSelect)
    }
  }, [api, onSelect])

  return (
    <CarouselContext.Provider
      value={{
        carouselRef,
        api: api,
        opts,
        orientation:
          orientation || (opts?.axis === "y" ? "vertical" : "horizontal"),
        scrollPrev,
        scrollNext,
        canScrollPrev,
        canScrollNext,
      }}
    >
      <div
        ref={ref}
        onKeyDownCapture={handleKeyDown}
        className={cn("relative", className)}
        role="region"
        aria-roledescription="carousel"
        {...props}
      >
        {children}
      </div>
    </CarouselContext.Provider>
  )
}

Carousel.displayName = "Carousel"

const CarouselContent = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => {
  const { carouselRef, orientation } = useCarousel()

  return (
    <div ref={carouselRef} className="overflow-hidden">
      <div
        ref={ref}
        className={cn(
          "flex",
          orientation === "horizontal" ? "-ml-4" : "-mt-4 flex-col",
          className
        )}
        {...props}
      />
    </div>
  )
})

CarouselContent.displayName = "CarouselContent"

const CarouselItem = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => {
  const { orientation } = useCarousel()

  return (
    <div
      ref={ref}
      role="group"
      aria-roledescription="slide"
      className={cn(
        "min-w-0 shrink-0 grow-0 basis-full",
        orientation === "horizontal" ? "pl-4" : "pt-4",
        className
      )}
      {...props}
    />
  )
})

```

```

CarouselItem.displayName = "CarouselItem"

const CarouselPrevious = React.forwardRef<
  HTMLButtonElement,
  React.ComponentProps<typeof Button>
>(({ className, variant = "outline", size = "icon", ...props }, ref) => {
  const { orientation, scrollPrev, canScrollPrev } = useCarousel()

  return (
    <Button
      ref={ref}
      variant={variant}
      size={size}
      className={cn(
        "absolute h-8 w-8 rounded-full",
        orientation === "horizontal"
          ? "-left-12 top-1/2 -translate-y-1/2"
          : "-top-12 left-1/2 -translate-x-1/2 rotate-90",
        className
      )}
      disabled={!canScrollPrev}
      onClick={scrollPrev}
      {...props}
    >
      <ArrowLeft className="h-4 w-4" />
      <span className="sr-only">Previous slide</span>
    </Button>
  )
})
CarouselPrevious.displayName = "CarouselPrevious"

const CarouselNext = React.forwardRef<
  HTMLButtonElement,
  React.ComponentProps<typeof Button>
>(({ className, variant = "outline", size = "icon", ...props }, ref) => {
  const { orientation, scrollNext, canScrollNext } = useCarousel()

  return (
    <Button
      ref={ref}
      variant={variant}
      size={size}
      className={cn(
        "absolute h-8 w-8 rounded-full",
        orientation === "horizontal"
          ? "-right-12 top-1/2 -translate-y-1/2"
          : "-bottom-12 left-1/2 -translate-x-1/2 rotate-90",
        className
      )}
      disabled={!canScrollNext}
      onClick={scrollNext}
      {...props}
    >
      <ArrowRight className="h-4 w-4" />
      <span className="sr-only">Next slide</span>
    </Button>
  )
})
CarouselNext.displayName = "CarouselNext"

export {
  type CarouselApi,
  Carousel,
  CarouselContent,
  CarouselItem,
  CarouselPrevious,
  CarouselNext,
}

```

[artifacts/mtos-crm/src/components/ui/chart.tsx](#)

```
import * as React from "react"
import * as RechartsPrimitive from "recharts"

import { cn } from "@/lib/utils"

// Format: { THEME_NAME: CSS_SELECTOR }
const THEMES = { light: "", dark: ".dark" } as const

export type ChartConfig = {
  [k in string]: {
    label?: React.ReactNode
    icon?: React.ComponentType
  } & (
    | { color?: string; theme?: never }
    | { color?: never; theme: Record<keyof typeof THEMES, string> }
  )
}

type ChartContextProps = {
  config: ChartConfig
}

const ChartContext = React.createContext<ChartContextProps | null>(null)

function useChart() {
  const context = React.useContext(ChartContext)

  if (!context) {
    throw new Error("useChart must be used within a <ChartContainer />")
  }

  return context
}

const ChartContainer = React.forwardRef<
  HTMLDivElement,
  React.ComponentProps<"div"> & {
    config: ChartConfig
    children: React.ComponentProps<
      typeof RechartsPrimitive.ResponsiveContainer
    >["children"]
  }
>(({ id, className, children, config, ...props }, ref) => {
  const uniqueId = React.useId()
  const chartId = `chart-${id || uniqueId.replace(/:/g, "")}`

  return (
    <ChartContext.Provider value={{ config }}>
      <div
        data-chart={chartId}
        ref={ref}
        className={cn(
          "flex aspect-video justify-center text-xs [&_.recharts-cartesian-axis-tick_text]:fill-muted-foreground [&_.recharts-cartesian-grid_line[stroke='#ccc']]:stroke-border/50 [&_.recharts-curve.recharts-tooltip-cursor]:stroke-border [&_.recharts-dot[stroke='#fff']]:stroke-transparent [&_.recharts-layer]:outline-none [&_.recharts-polar-grid[stroke='#ccc']]:stroke-border [&_.recharts-radial-bar-background-sector]:fill-muted [&_.recharts-rectangle.recharts-tooltip-cursor]:fill-muted [&_.recharts-reference_line[stroke='#ccc']]:stroke-border [&_.recharts-sector[stroke='#fff']]:stroke-transparent [&_.recharts-sector]:outline-none [&_.recharts-surface]:outline-none",
          className
        )}
        {...props}
      >
        <ChartStyle id={chartId} config={config} />
        <RechartsPrimitive.ResponsiveContainer>
          {children}
        </RechartsPrimitive.ResponsiveContainer>
      </div>
    </ChartContext.Provider>
  )
})

ChartContainer.displayName = "Chart"

const ChartStyle = ({ id, config }: { id: string; config: ChartConfig }) => {
  const colorConfig = Object.entries(config).filter(
    ([, config]) => config.theme || config.color
  )

  if (!colorConfig.length) {
    return null
  }

  return (
    <style
      dangerouslySetInnerHTML={{
        __html: Object.entries(THEMES)
          .map(
            ([theme, prefix]) => `
${prefix} [data-chart=${id}] {
${colorConfig
  .map(([key, itemConfig]) => {
    const color =
      itemConfig.theme?.[theme as keyof typeof itemConfig.theme] ||
      itemConfig.color
    return color ? `--color-${key}: ${color};` : null
  })
  .join("\n")}
}
`
      }}
    />
  )
}
```

```

    .join("\n"))
  }
    )
    .join("\n"),
  }}
  />
)
}

const ChartTooltip = RechartsPrimitive.Tooltip

const ChartTooltipContent = React.forwardRef<
  HTMLDivElement,
  React.ComponentProps<typeof RechartsPrimitive.Tooltip> &
  React.ComponentProps<"div"> & {
    hideLabel?: boolean
    hideIndicator?: boolean
    indicator?: "line" | "dot" | "dashed"
    nameKey?: string
    labelKey?: string
  }
>(
  (
    {
      active,
      payload,
      className,
      indicator = "dot",
      hideLabel = false,
      hideIndicator = false,
      label,
      labelFormatter,
      labelClassName,
      formatter,
      color,
      nameKey,
      labelKey,
    },
    ref
  ) => {
    const { config } = useChart()

    const tooltipLabel = React.useMemo(() => {
      if (hideLabel || !payload?.length) {
        return null
      }

      const [item] = payload
      const key = `${labelKey || item?.dataKey || item?.name || "value"}`
      const itemConfig = getPayloadConfigFromPayload(config, item, key)
      const value =
        !labelKey && typeof label === "string"
          ? config[label as keyof typeof config]?.label || label
          : itemConfig?.label

      if (labelFormatter) {
        return (
          <div className={cn("font-medium", labelClassName)}>
            {labelFormatter(value, payload)}
          </div>
        )
      }

      if (!value) {
        return null
      }

      return <div className={cn("font-medium", labelClassName)}>{value}</div>
    }, [
      label,
      labelFormatter,
      payload,
      hideLabel,
      labelClassName,
      config,
      labelKey,
    ])

    if (!active || !payload?.length) {
      return null
    }

    const nestLabel = payload.length === 1 && indicator !== "dot"

    return (
      <div
        ref={ref}
        className={cn(
          "grid min-w-[8rem] items-start gap-1.5 rounded-lg border border-border/50 bg-background px-2.5 py-1.5 text-xs shadow-
xl",
          className
        )}
      >
        {!nestLabel ? tooltipLabel : null}
        <div className="grid gap-1.5">
          {payload

```

```

.filter((item) => item.type !== "none")
.map((item, index) => {
  const key = `${nameKey || item.name || item.dataKey || "value"}`
  const itemConfig = getPayloadConfigFromPayload(config, item, key)
  const indicatorColor = color || item.payload.fill || item.color

  return (
    <div
      key={item.dataKey}
      className={cn(
        "flex w-full flex-wrap items-stretch gap-2 [&svg]:h-2.5 [&svg]:w-2.5 [&svg]:text-muted-foreground",
        indicator === "dot" && "items-center"
      )}
    >
      {formatter && item?.value !== undefined && item.name ? (
        formatter(item.value, item.name, item, index, item.payload)
      ) : (
        <>
          {itemConfig?.icon ? (
            <itemConfig.icon />
          ) : (
            !hideIndicator && (
              <div
                className={cn(
                  "shrink-0 rounded-[2px] border[--color-border] bg[--color-bg]",
                  {
                    "h-2.5 w-2.5": indicator === "dot",
                    "w-1": indicator === "line",
                    "w-0 border-[1.5px] border-dashed bg-transparent":
                      indicator === "dashed",
                    "my-0.5": nestLabel && indicator === "dashed",
                  }
                )}
                style={
                  {
                    "--color-bg": indicatorColor,
                    "--color-border": indicatorColor,
                  } as React.CSSProperties
                }
              />
            )
          )}
          <div
            className={cn(
              "flex flex-1 justify-between leading-none",
              nestLabel ? "items-end" : "items-center"
            )}
          >
            <div className="grid gap-1.5">
              {nestLabel ? tooltipLabel : null}
              <span className="text-muted-foreground">
                {itemConfig?.label || item.name}
              </span>
            </div>
            {item.value && (
              <span className="font-mono font-medium tabular-nums text-foreground">
                {item.value.toLocaleString()}
              </span>
            )}
          </div>
        </>
      )}
    </div>
  )
)}
</div>
)
}
)
ChartTooltipContent.displayName = "ChartTooltip"

const ChartLegend = RechartsPrimitive.Legend

const ChartLegendContent = React.forwardRef<
  HTMLDivElement,
  React.ComponentProps<"div"> &
  Pick<RechartsPrimitive.LegendProps, "payload" | "verticalAlign"> & {
    hideIcon?: boolean
    nameKey?: string
  }
>(
  {
    className, hideIcon = false, payload, verticalAlign = "bottom", nameKey },
  ref
) => {
  const { config } = useChart()

  if (!payload?.length) {
    return null
  }

  return (
    <div
      ref={ref}
      className={cn(
        "flex items-center justify-center gap-4",

```

```

        verticalAlign === "top" ? "pb-3" : "pt-3",
        className
    )}
    >
    {payload
      .filter((item) => item.type !== "none")
      .map((item) => {
        const key = `${nameKey} || item.dataKey || "value"`
        const itemConfig = getPayloadConfigFromPayload(config, item, key)

        return (
          <div
            key={item.value}
            className={cn(
              "flex items-center gap-1.5 [&svg]:h-3 [&svg]:w-3 [&svg]:text-muted-foreground"
            )}
          >
            {itemConfig?.icon && !hideIcon ? (
              <itemConfig.icon />
            ) : (
              <div
                className="h-2 w-2 shrink-0 rounded-[2px]"
                style={{
                  backgroundColor: item.color,
                }}
              />
            )}
            {itemConfig?.label}
          </div>
        )
      })
    }
  </div>
)
}
)
ChartLegendContent.displayName = "ChartLegend"

// Helper to extract item config from a payload.
function getPayloadConfigFromPayload(
  config: ChartConfig,
  payload: unknown,
  key: string
) {
  if (typeof payload !== "object" || payload === null) {
    return undefined
  }

  const payloadPayload =
    "payload" in payload &&
    typeof payload.payload === "object" &&
    payload.payload !== null
      ? payload.payload
      : undefined

  let configLabelKey: string = key

  if (
    key in payload &&
    typeof payload[key as keyof typeof payload] === "string"
  ) {
    configLabelKey = payload[key as keyof typeof payload] as string
  } else if (
    payloadPayload &&
    key in payloadPayload &&
    typeof payloadPayload[key as keyof typeof payloadPayload] === "string"
  ) {
    configLabelKey = payloadPayload[
      key as keyof typeof payloadPayload
    ] as string
  }

  return configLabelKey in config
    ? config[configLabelKey]
    : config[key as keyof typeof config]
}

export {
  ChartContainer,
  ChartTooltip,
  ChartTooltipContent,
  ChartLegend,
  ChartLegendContent,
  ChartStyle,
}

```

[artifacts/mtos-crm/src/components/ui/checkbox.tsx](#)

```
import * as React from "react"
import * as CheckboxPrimitive from "@radix-ui/react-checkbox"
import { Check } from "lucide-react"

import { cn } from "@lib/utls"

const Checkbox = React.forwardRef<
  React.ElementRef<typeof CheckboxPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof CheckboxPrimitive.Root>
>(({ className, ...props }, ref) => {
  <CheckboxPrimitive.Root
    ref={ref}
    className={cn(
      "grid place-content-center peer h-4 w-4 shrink-0 rounded-sm border border-primary shadow focus-visible:outline-none focus-visible:ring-1 focus-visible:ring-ring disabled:cursor-not-allowed disabled:opacity-50 data-[state=checked]:bg-primary data-[state=checked]:text-primary-foreground",
      className
    )}
    {...props}
  >
    <CheckboxPrimitive.Indicator
      className={cn("grid place-content-center text-current")}
    >
      <Check className="h-4 w-4" />
    </CheckboxPrimitive.Indicator>
  </CheckboxPrimitive.Root>
)}
Checkbox.displayName = CheckboxPrimitive.Root.displayName

export { Checkbox }
```


[artifacts/mtos-crm/src/components/ui/collapsible.tsx](#)

"use client"

import * as CollapsiblePrimitive from "@radix-ui/react-collapsible"

const Collapsible = CollapsiblePrimitive.Root

const CollapsibleTrigger = CollapsiblePrimitive.CollapsibleTrigger

const CollapsibleContent = CollapsiblePrimitive.CollapsibleContent

export { Collapsible, CollapsibleTrigger, CollapsibleContent }

[artifacts/mtos-crm/src/components/ui/command.tsx](#)

```
"use client"

import * as React from "react"
import { type DialogProps } from "@radix-ui/react-dialog"
import { Command as CommandPrimitive } from "cmdk"
import { Search } from "lucide-react"

import { cn } from "@lib/utlis"
import { Dialog, DialogContent } from "@components/ui/dialog"

const Command = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive>
>(({ className, ...props }, ref) => {
  <CommandPrimitive
    ref={ref}
    className={cn(
      "flex h-full w-full flex-col overflow-hidden rounded-md bg-popover text-popover-foreground",
      className
    )}
    {...props}
  />
})
Command.displayName = CommandPrimitive.displayName

const CommandDialog = ({ children, ...props }: DialogProps) => {
  return (
    <Dialog {...props}>
      <DialogContent className="overflow-hidden p-0">
        <Command className="[&_cmdk-group-heading]:px-2 [&_cmdk-group-heading]:font-medium [&_cmdk-group-heading]:text-muted-foreground [&_cmdk-group]:not([hidden])~[cmdk-group]:pt-0 [&_cmdk-group]:px-2 [&_cmdk-input-wrapper]_svg:h-5 [&_cmdk-input-wrapper]_svg:w-5 [&_cmdk-input]:h-12 [&_cmdk-item]:px-2 [&_cmdk-item]:py-3 [&_cmdk-item]_svg:h-5 [&_cmdk-item]_svg:w-5">
          {children}
        </Command>
      </DialogContent>
    </Dialog>
  )
}

const CommandInput = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive.Input>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive.Input>
>(({ className, ...props }, ref) => {
  <div className="flex items-center border-b px-3 cmdk-input-wrapper="">>
    <Search className="mr-2 h-4 w-4 shrink-0 opacity-50" />
    <CommandPrimitive.Input
      ref={ref}
      className={cn(
        "flex h-10 w-full rounded-md bg-transparent py-3 text-sm outline-none placeholder:text-muted-foreground disabled:cursor-not-allowed disabled:opacity-50",
        className
      )}
      {...props}
    />
  </div>
})
CommandInput.displayName = CommandPrimitive.Input.displayName

const CommandList = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive.List>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive.List>
>(({ className, ...props }, ref) => {
  <CommandPrimitive.List
    ref={ref}
    className={cn("max-h-[300px] overflow-y-auto overflow-x-hidden", className)}
    {...props}
  />
})
CommandList.displayName = CommandPrimitive.List.displayName

const CommandEmpty = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive.Empty>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive.Empty>
>(({ props, ref }) => {
  <CommandPrimitive.Empty
    ref={ref}
    className="py-6 text-center text-sm"
    {...props}
  />
})
CommandEmpty.displayName = CommandPrimitive.Empty.displayName

const CommandGroup = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive.Group>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive.Group>
>(({ className, ...props }, ref) => {
  <CommandPrimitive.Group
    ref={ref}
    className={cn(
      "overflow-hidden p-1 text-foreground [&_cmdk-group-heading]:px-2 [&_cmdk-group-heading]:py-1.5 [&_cmdk-group-heading]:text-xs [&_cmdk-group-heading]:font-medium [&_cmdk-group-heading]:text-muted-foreground",
      className
    )}
    {...props}
  />
})
CommandGroup.displayName = CommandPrimitive.Group.displayName
```

```

        className
      )}
      {...props}
    />
  ))

CommandGroup.displayName = CommandPrimitive.Group.displayName

const CommandSeparator = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive.Separator>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive.Separator>
>(({ className, ...props }, ref) => (
  <CommandPrimitive.Separator
    ref={ref}
    className={cn("-mx-1 h-px bg-border", className)}
    {...props}
  />
))
CommandSeparator.displayName = CommandPrimitive.Separator.displayName

const CommandItem = React.forwardRef<
  React.ElementRef<typeof CommandPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof CommandPrimitive.Item>
>(({ className, ...props }, ref) => (
  <CommandPrimitive.Item
    ref={ref}
    className={cn(
      "relative flex cursor-default gap-2 select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none data-
[disabled=true]:pointer-events-none data-[selected=true]:bg-accent data-[selected=true]:text-accent-foreground data-
[disabled=true]:opacity-50 [&_svg]:pointer-events-none [&_svg]:size-4 [&_svg]:shrink-0",
      className
    )}
    {...props}
  />
))
CommandItem.displayName = CommandPrimitive.Item.displayName

const CommandShortcut = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLSpanElement>) => {
  return (
    <span
      className={cn(
        "ml-auto text-xs tracking-widest text-muted-foreground",
        className
      )}
      {...props}
    />
  )
}
CommandShortcut.displayName = "CommandShortcut"

export {
  Command,
  CommandDialog,
  CommandInput,
  CommandList,
  CommandEmpty,
  CommandGroup,
  CommandItem,
  CommandShortcut,
  CommandSeparator,
}

```

[artifacts/mtos-crm/src/components/ui/context-menu.tsx](#)

```
import * as React from "react"
import * as ContextMenuPrimitive from "@radix-ui/react-context-menu"
import { Check, ChevronRight, Circle } from "lucide-react"

import { cn } from "@/lib/utils"

const ContextMenu = ContextMenuPrimitive.Root

const ContextMenuTrigger = ContextMenuPrimitive.Trigger

const ContextMenuGroup = ContextMenuPrimitive.Group

const ContextMenuPortal = ContextMenuPrimitive.Portal

const ContextMenuSub = ContextMenuPrimitive.Sub

const ContextMenuRadioGroup = ContextMenuPrimitive.RadioGroup

const ContextMenuSubTrigger = React.forwardRef<
  React.ElementRef<typeof ContextMenuPrimitive.SubTrigger>,
  React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.SubTrigger> & {
    inset?: boolean
  }
>(({ className, inset, children, ...props }, ref) => (
  <ContextMenuPrimitive.SubTrigger
    ref={ref}
    className={cn(
      "flex cursor-default select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none focus:bg-accent focus:text-
accent-foreground data-[state=open]:bg-accent data-[state=open]:text-accent-foreground",
      inset && "pl-8",
      className
    )}
    {...props}
  >
    {children}
    <ChevronRight className="ml-auto h-4 w-4" />
  </ContextMenuPrimitive.SubTrigger>
))
ContextMenuSubTrigger.displayName = ContextMenuPrimitive.SubTrigger.displayName

const ContextMenuSubContent = React.forwardRef<
  React.ElementRef<typeof ContextMenuPrimitive.SubContent>,
  React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.SubContent>
>(({ className, ...props }, ref) => (
  <ContextMenuPrimitive.SubContent
    ref={ref}
    className={cn(
      "z-50 min-w-[8rem] overflow-hidden rounded-md border bg-popover p-1 text-popover-foreground shadow-lg data-
[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-
[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-
right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin-[--radix-context-menu-content-
transform-origin]",
      className
    )}
    {...props}
  >
  </>
))
ContextMenuSubContent.displayName = ContextMenuPrimitive.SubContent.displayName

const ContextMenuContent = React.forwardRef<
  React.ElementRef<typeof ContextMenuPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.Content>
>(({ className, ...props }, ref) => (
  <ContextMenuPrimitive.Portal>
    <ContextMenuPrimitive.Content
      ref={ref}
      className={cn(
        "z-50 max-h-[--radix-context-menu-content-available-height] min-w-[8rem] overflow-y-auto overflow-x-hidden rounded-md
border bg-popover p-1 text-popover-foreground shadow-md data-[state=open]:animate-in data-[state=closed]:animate-out data-
[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-
[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-
[side=top]:slide-in-from-bottom-2 origin-[--radix-context-menu-content-transform-origin]",
        className
      )}
      {...props}
    >
  </ContextMenuPrimitive.Portal>
))
ContextMenuContent.displayName = ContextMenuPrimitive.Content.displayName

const ContextMenuItem = React.forwardRef<
  React.ElementRef<typeof ContextMenuPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.Item> & {
    inset?: boolean
  }
>(({ className, inset, ...props }, ref) => (
  <ContextMenuPrimitive.Item
    ref={ref}
    className={cn(
      "relative flex cursor-default select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none focus:bg-accent
focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      inset && "pl-8",
      className
    )}
    {...props}
  >
```

```

    />
  ))
  ContextMenuItem.displayName = ContextMenuPrimitive.Item.displayName

  const ContextMenuCheckboxItem = React.forwardRef<
    React.ElementRef<typeof ContextMenuPrimitive.CheckboxItem>,
    React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.CheckboxItem>
  >(({ className, children, checked, ...props }, ref) => (
    <ContextMenuPrimitive.CheckboxItem
      ref={ref}
      className={cn(
        "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-none focus:bg-accent
        focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
        className
      )}
      checked={checked}
      {...props}
    >
      <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
        <ContextMenuPrimitive.ItemIndicator>
          <Check className="h-4 w-4" />
        </ContextMenuPrimitive.ItemIndicator>
      </span>
      {children}
    </ContextMenuPrimitive.CheckboxItem>
  ))
  ContextMenuCheckboxItem.displayName =
    ContextMenuPrimitive.CheckboxItem.displayName

  const ContextMenuRadioItem = React.forwardRef<
    React.ElementRef<typeof ContextMenuPrimitive.RadioItem>,
    React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.RadioItem>
  >(({ className, children, ...props }, ref) => (
    <ContextMenuPrimitive.RadioItem
      ref={ref}
      className={cn(
        "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-none focus:bg-accent
        focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
        className
      )}
      {...props}
    >
      <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
        <ContextMenuPrimitive.ItemIndicator>
          <Circle className="h-4 w-4 fill-current" />
        </ContextMenuPrimitive.ItemIndicator>
      </span>
      {children}
    </ContextMenuPrimitive.RadioItem>
  ))
  ContextMenuRadioItem.displayName = ContextMenuPrimitive.RadioItem.displayName

  const ContextMenuLabel = React.forwardRef<
    React.ElementRef<typeof ContextMenuPrimitive.Label>,
    React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.Label> & {
      inset?: boolean
    }
  >(({ className, inset, ...props }, ref) => (
    <ContextMenuPrimitive.Label
      ref={ref}
      className={cn(
        "px-2 py-1.5 text-sm font-semibold text-foreground",
        inset && "pl-8",
        className
      )}
      {...props}
    >
    />
  ))
  ContextMenuLabel.displayName = ContextMenuPrimitive.Label.displayName

  const ContextMenuSeparator = React.forwardRef<
    React.ElementRef<typeof ContextMenuPrimitive.Separator>,
    React.ComponentPropsWithoutRef<typeof ContextMenuPrimitive.Separator>
  >(({ className, ...props }, ref) => (
    <ContextMenuPrimitive.Separator
      ref={ref}
      className={cn("-mx-1 my-1 h-px bg-border", className)}
      {...props}
    >
    />
  ))
  ContextMenuSeparator.displayName = ContextMenuPrimitive.Separator.displayName

  const ContextMenuShortcut = ({
    className,
    ...props
  }: React.HTMLAttributes<HTMLSpanElement>) => {
    return (
      <span
        className={cn(
          "ml-auto text-xs tracking-widest text-muted-foreground",
          className
        )}
        {...props}
      />
    )
  }
  ContextMenuShortcut.displayName = "ContextMenuShortcut"

```

```
export {  
  ContextMenu,  
  ContextMenuTrigger,  
  ContextMenuContent,  
  ContextMenuItem,  
  ContextMenuCheckboxItem,  
  ContextMenuRadioItem,  
  ContextMenuLabel,  
  ContextMenuSeparator,  
  ContextMenuShortcut,  
  ContextMenuGroup,  
  ContextMenuPortal,  
  ContextMenuSub,  
  ContextMenuSubContent,  
  ContextMenuSubTrigger,  
  ContextMenuRadioGroup,  
}
```

[artifacts/mtos-crm/src/components/ui/dialog.tsx](#)

```
import * as React from "react"
import * as DialogPrimitive from "@radix-ui/react-dialog"
import { X } from "lucide-react"

import { cn } from "@lib/utils"

const Dialog = DialogPrimitive.Root

const DialogTrigger = DialogPrimitive.Trigger

const DialogPortal = DialogPrimitive.Portal

const DialogClose = DialogPrimitive.Close

const DialogOverlay = React.forwardRef<
  React.ElementRef<typeof DialogPrimitive.Overlay>,
  React.ComponentPropsWithoutRef<typeof DialogPrimitive.Overlay>
>(({ className, ...props }, ref) => (
  <DialogPrimitive.Overlay
    ref={ref}
    className={cn(
      "fixed inset-0 z-50 bg-black/80 data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0",
      className
    )}
    {...props}
  />
))
DialogOverlay.displayName = DialogPrimitive.Overlay.displayName

const DialogContent = React.forwardRef<
  React.ElementRef<typeof DialogPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof DialogPrimitive.Content>
>(({ className, children, ...props }, ref) => (
  <DialogPortal>
    <DialogOverlay />
    <DialogPrimitive.Content
      ref={ref}
      className={cn(
        "fixed left-[50%] top-[50%] z-50 grid w-full max-w-lg translate-x-[-50%] translate-y-[-50%] gap-4 border bg-background p-6 shadow-lg duration-200 data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[state=closed]:slide-out-to-left-1/2 data-[state=closed]:slide-out-to-top-48% data-[state=open]:slide-in-from-left-1/2 data-[state=open]:slide-in-from-top-48% sm:rounded-lg",
        className
      )}
      {...props}
    >
      {children}
      <DialogPrimitive.Close className="absolute right-4 top-4 rounded-sm opacity-70 ring-offset-background transition-opacity hover:opacity-100 focus:outline-none focus:ring-2 focus:ring-ring focus:ring-offset-2 disabled:pointer-events-none data-[state=open]:bg-accent data-[state=open]:text-muted-foreground">
        <X className="h-4 w-4" />
        <span className="sr-only">Close</span>
      </DialogPrimitive.Close>
    </DialogPrimitive.Content>
  </DialogPortal>
))
DialogContent.displayName = DialogPrimitive.Content.displayName

const DialogHeader = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn(
      "flex flex-col space-y-1.5 text-center sm:text-left",
      className
    )}
    {...props}
  />
)
DialogHeader.displayName = "DialogHeader"

const DialogFooter = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn(
      "flex flex-col-reverse sm:flex-row sm:justify-end sm:space-x-2",
      className
    )}
    {...props}
  />
)
DialogFooter.displayName = "DialogFooter"

const DialogTitle = React.forwardRef<
  React.ElementRef<typeof DialogPrimitive.Title>,
  React.ComponentPropsWithoutRef<typeof DialogPrimitive.Title>
>(({ className, ...props }, ref) => (
  <DialogPrimitive.Title
    ref={ref}
    className={cn(
```

```

        "text-lg font-semibold leading-none tracking-tight",
        className
      )}
      {...props}
    />
  ))
  DialogTitle.displayName = DialogPrimitive.Title.displayName

  const DialogDescription = React.forwardRef<
    React.ElementRef<typeof DialogPrimitive.Description>,
    React.ComponentPropsWithoutRef<typeof DialogPrimitive.Description>
  >(({ className, ...props }, ref) => (
    <DialogPrimitive.Description
      ref={ref}
      className={cn("text-sm text-muted-foreground", className)}
      {...props}
    />
  ))
  DialogDescription.displayName = DialogPrimitive.Description.displayName

  export {
    Dialog,
    DialogPortal,
    DialogOverlay,
    DialogTrigger,
    DialogClose,
    DialogContent,
    DialogHeader,
    DialogFooter,
    DialogTitle,
    DialogDescription,
  }

```


[artifacts/mtos-crm/src/components/ui/drawer.tsx](#)

```
import * as React from "react"
import { Drawer as DrawerPrimitive } from "vaul"

import { cn } from "@/lib/utils"

const Drawer = ({
  shouldScaleBackground = true,
  ...props
}: React.ComponentProps<typeof DrawerPrimitive.Root>) => (
  <DrawerPrimitive.Root
    shouldScaleBackground={shouldScaleBackground}
    {...props}
  />
)
Drawer.displayName = "Drawer"

const DrawerTrigger = DrawerPrimitive.Trigger

const DrawerPortal = DrawerPrimitive.Portal

const DrawerClose = DrawerPrimitive.Close

const DrawerOverlay = React.forwardRef<
  React.ElementRef<typeof DrawerPrimitive.Overlay>,
  React.ComponentPropsWithoutRef<typeof DrawerPrimitive.Overlay>
>(({ className, ...props }, ref) => (
  <DrawerPrimitive.Overlay
    ref={ref}
    className={cn("fixed inset-0 z-50 bg-black/80", className)}
    {...props}
  />
))
DrawerOverlay.displayName = DrawerPrimitive.Overlay.displayName

const DrawerContent = React.forwardRef<
  React.ElementRef<typeof DrawerPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof DrawerPrimitive.Content>
>(({ className, children, ...props }, ref) => (
  <DrawerPortal>
    <DrawerOverlay />
    <DrawerPrimitive.Content
      ref={ref}
      className={cn(
        "fixed inset-x-0 bottom-0 z-50 mt-24 flex h-auto flex-col rounded-t-[10px] border bg-background",
        className
      )}
      {...props}
    >
    <div className="mx-auto mt-4 h-2 w-[100px] rounded-full bg-muted" />
    {children}
  </DrawerPrimitive.Content>
</DrawerPortal>
))
DrawerContent.displayName = "DrawerContent"

const DrawerHeader = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn("grid gap-1.5 p-4 text-center sm:text-left", className)}
    {...props}
  />
)
DrawerHeader.displayName = "DrawerHeader"

const DrawerFooter = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn("mt-auto flex flex-col gap-2 p-4", className)}
    {...props}
  />
)
DrawerFooter.displayName = "DrawerFooter"

const DrawerTitle = React.forwardRef<
  React.ElementRef<typeof DrawerPrimitive.Title>,
  React.ComponentPropsWithoutRef<typeof DrawerPrimitive.Title>
>(({ className, ...props }, ref) => (
  <DrawerPrimitive.Title
    ref={ref}
    className={cn(
      "text-lg font-semibold leading-none tracking-tight",
      className
    )}
    {...props}
  />
))
DrawerTitle.displayName = DrawerPrimitive.Title.displayName

const DrawerDescription = React.forwardRef<
  React.ElementRef<typeof DrawerPrimitive.Description>,
  React.ComponentPropsWithoutRef<typeof DrawerPrimitive.Description>
```

```
>(({ className, ...props }, ref) => (  
  <DrawerPrimitive.Description  
    ref={ref}  
    className={cn("text-sm text-muted-foreground", className)}  
    {...props}  
  />  
))  
DrawerDescription.displayName = DrawerPrimitive.Description.displayName  
  
export {  
  Drawer,  
  DrawerPortal,  
  DrawerOverlay,  
  DrawerTrigger,  
  DrawerClose,  
  DrawerContent,  
  DrawerHeader,  
  DrawerFooter,  
  DrawerTitle,  
  DrawerDescription,  
}
```

[artifacts/mtos-crm/src/components/ui/dropdown-menu.tsx](#)

```
"use client"

import * as React from "react"
import * as DropdownMenuPrimitive from "@radix-ui/react-dropdown-menu"
import { Check, ChevronRight, Circle } from "lucide-react"

import { cn } from "@lib/utils"

const DropdownMenu = DropdownMenuPrimitive.Root

const DropdownMenuTrigger = DropdownMenuPrimitive.Trigger

const DropdownMenuGroup = DropdownMenuPrimitive.Group

const DropdownMenuPortal = DropdownMenuPrimitive.Portal

const DropdownMenuSub = DropdownMenuPrimitive.Sub

const DropdownMenuRadioGroup = DropdownMenuPrimitive.RadioGroup

const DropdownMenuSubTrigger = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.SubTrigger>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.SubTrigger> & {
    inset?: boolean
  }
>(({ className, inset, children, ...props }, ref) => (
  <DropdownMenuPrimitive.SubTrigger
    ref={ref}
    className={cn(
      "flex cursor-default select-none items-center gap-2 rounded-sm px-2 py-1.5 text-sm outline-none focus:bg-accent data-[state=open]:bg-accent [&_svg]:pointer-events-none [&_svg]:size-4 [&_svg]:shrink-0",
      inset && "pl-8",
      className
    )}
    {...props}
  >
    {children}
    <ChevronRight className="ml-auto" />
  </DropdownMenuPrimitive.SubTrigger>
))
DropdownMenuSubTrigger.displayName =
  DropdownMenuPrimitive.SubTrigger.displayName

const DropdownMenuSubContent = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.SubContent>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.SubContent>
>(({ className, ...props }, ref) => (
  <DropdownMenuPrimitive.SubContent
    ref={ref}
    className={cn(
      "z-50 min-w-[8rem] overflow-hidden rounded-md border bg-popover p-1 text-popover-foreground shadow-lg data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin-[--radix-dropdown-menu-content-transform-origin]",
      className
    )}
    {...props}
  />
))
DropdownMenuSubContent.displayName =
  DropdownMenuPrimitive.SubContent.displayName

const DropdownMenuContent = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.Content>
>(({ className, sideOffset = 4, ...props }, ref) => (
  <DropdownMenuPrimitive.Portal>
    <DropdownMenuPrimitive.Content
      ref={ref}
      sideOffset={sideOffset}
      className={cn(
        "z-50 max-h-[var(--radix-dropdown-menu-content-available-height)] min-w-[8rem] overflow-y-auto overflow-x-hidden rounded-md border bg-popover p-1 text-popover-foreground shadow-md",
        "data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin-[--radix-dropdown-menu-content-transform-origin]",
        className
      )}
      {...props}
    />
  </DropdownMenuPrimitive.Portal>
))
DropdownMenuContent.displayName = DropdownMenuPrimitive.Content.displayName

const DropdownMenuItem = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.Item> & {
    inset?: boolean
  }
>(({ className, inset, ...props }, ref) => (
  <DropdownMenuPrimitive.Item
    ref={ref}
    className={cn(
```

```

        "relative flex cursor-default select-none items-center gap-2 rounded-sm px-2 py-1.5 text-sm outline-none transition-
colors focus:bg-accent focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50
[&svg]:size-4 [&svg]:shrink-0",
        inset && "pl-8",
        className
    )}
    {...props}
  />
))
DropdownMenuItem.displayName = DropdownMenuPrimitive.Item.displayName

const DropdownMenuCheckboxItem = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.CheckboxItem>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.CheckboxItem>
>(({ className, children, checked, ...props }, ref) => (
  <DropdownMenuPrimitive.CheckboxItem
    ref={ref}
    className={cn(
      "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-none transition-colors
focus:bg-accent focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      className
    )}
    checked={checked}
    {...props}
  >
    <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
      <DropdownMenuPrimitive.ItemIndicator>
        <Check className="h-4 w-4" />
      </DropdownMenuPrimitive.ItemIndicator>
    </span>
    {children}
  </DropdownMenuPrimitive.CheckboxItem>
))
DropdownMenuCheckboxItem.displayName =
  DropdownMenuPrimitive.CheckboxItem.displayName

const DropdownMenuRadioItem = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.RadioItem>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.RadioItem>
>(({ className, children, ...props }, ref) => (
  <DropdownMenuPrimitive.RadioItem
    ref={ref}
    className={cn(
      "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-none transition-colors
focus:bg-accent focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      className
    )}
    {...props}
  >
    <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
      <DropdownMenuPrimitive.ItemIndicator>
        <Circle className="h-2 w-2 fill-current" />
      </DropdownMenuPrimitive.ItemIndicator>
    </span>
    {children}
  </DropdownMenuPrimitive.RadioItem>
))
DropdownMenuRadioItem.displayName = DropdownMenuPrimitive.RadioItem.displayName

const DropdownMenuLabel = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.Label>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.Label> & {
    inset?: boolean
  }
>(({ className, inset, ...props }, ref) => (
  <DropdownMenuPrimitive.Label
    ref={ref}
    className={cn(
      "px-2 py-1.5 text-sm font-semibold",
      inset && "pl-8",
      className
    )}
    {...props}
  />
))
DropdownMenuLabel.displayName = DropdownMenuPrimitive.Label.displayName

const DropdownMenuSeparator = React.forwardRef<
  React.ElementRef<typeof DropdownMenuPrimitive.Separator>,
  React.ComponentPropsWithoutRef<typeof DropdownMenuPrimitive.Separator>
>(({ className, ...props }, ref) => (
  <DropdownMenuPrimitive.Separator
    ref={ref}
    className={cn("-mx-1 my-1 h-px bg-muted", className)}
    {...props}
  />
))
DropdownMenuSeparator.displayName = DropdownMenuPrimitive.Separator.displayName

const DropdownMenuShortcut = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLSpanElement>) => {
  return (
    <span
      className={cn("ml-auto text-xs tracking-widest opacity-60", className)}
      {...props}
    >

```

```
    ) />
  }
  DropdownMenuShortcut.displayName = "DropdownMenuShortcut"

export {
  DropdownMenu,
  DropdownMenuTrigger,
  DropdownMenuContent,
  DropdownMenuItem,
  DropdownMenuCheckboxItem,
  DropdownMenuRadioItem,
  DropdownMenuLabel,
  DropdownMenuSeparator,
  DropdownMenuShortcut,
  DropdownMenuGroup,
  DropdownMenuPortal,
  DropdownMenuSub,
  DropdownMenuSubContent,
  DropdownMenuSubTrigger,
  DropdownMenuRadioGroup,
}
```

[artifacts/mtos-crm/src/components/ui/empty.tsx](#)

```
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"

function Empty({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="empty"
      className={cn(
        "flex min-w-0 flex-1 flex-col items-center justify-center gap-6 text-balance rounded-lg border-dashed p-6 text-center
md:p-12",
        className
      )}
      {...props}
    />
  )
}

function EmptyHeader({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="empty-header"
      className={cn(
        "flex max-w-sm flex-col items-center gap-2 text-center",
        className
      )}
      {...props}
    />
  )
}

const emptyMediaVariants = cva(
  "mb-2 flex shrink-0 items-center justify-center [&_svg]:pointer-events-none [&_svg]:shrink-0",
  {
    variants: {
      variant: {
        default: "bg-transparent",
        icon: "bg-muted text-foreground flex size-10 shrink-0 items-center justify-center rounded-lg
[&_svg]:not([class*='size-']):size-6",
      },
      defaultVariants: {
        variant: "default",
      },
    },
  }
)

function EmptyMedia({
  className,
  variant = "default",
  ...props
}: React.ComponentProps<"div"> & VariantProps<typeof emptyMediaVariants> >) {
  return (
    <div
      data-slot="empty-icon"
      data-variant={variant}
      className={cn(emptyMediaVariants({ variant, className }))}
      {...props}
    />
  )
}

function EmptyTitle({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="empty-title"
      className={cn("text-lg font-medium tracking-tight", className)}
      {...props}
    />
  )
}

function EmptyDescription({ className, ...props }: React.ComponentProps<"p">) {
  return (
    <div
      data-slot="empty-description"
      className={cn(
        "text-muted-foreground [&a:hover]:text-primary text-sm/relaxed [&a]:underline [&a]:underline-offset-4",
        className
      )}
      {...props}
    />
  )
}

function EmptyContent({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="empty-content"
      className={cn(
        "flex w-full min-w-0 max-w-sm flex-col items-center gap-4 text-balance text-sm",
        className
      )}
      {...props}
    />
  )
}
```

```
    )  
  }  
  
  export {  
    Empty,  
    EmptyHeader,  
    EmptyTitle,  
    EmptyDescription,  
    EmptyContent,  
    EmptyMedia,  
  }  
}
```

[artifacts/mtos-crm/src/components/ui/field.tsx](#)

```
"use client"

import { useMemo } from "react"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"
import { Label } from "@components/ui/label"
import { Separator } from "@components/ui/separator"

function FieldSet({ className, ...props }: React.ComponentProps<"fieldset">) {
  return (
    <fieldset
      data-slot="field-set"
      className={cn(
        "flex flex-col gap-6",
        "has- [>[data-slot=checkbox-group]]:gap-3 has- [>[data-slot=radio-group]]:gap-3",
        className
      )}
      {...props}
    />
  )
}

function FieldLegend({
  className,
  variant = "legend",
  ...props
}: React.ComponentProps<"legend"> & { variant?: "legend" | "label" }) {
  return (
    <legend
      data-slot="field-legend"
      data-variant={variant}
      className={cn(
        "mb-3 font-medium",
        "data-[variant=legend]:text-base",
        "data-[variant=label]:text-sm",
        className
      )}
      {...props}
    />
  )
}

function FieldGroup({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="field-group"
      className={cn(
        "group/field-group @container/field-group flex w-full flex-col gap-7 data-[slot=checkbox-group]:gap-3 [>[data-slot=field-group]]:gap-4",
        className
      )}
      {...props}
    />
  )
}

const fieldVariants = cva(
  "group/field data-[invalid=true]:text-destructive flex w-full gap-3",
  {
    variants: {
      orientation: {
        vertical: ["flex-col [>]:w-full [>.sr-only]:w-auto"],
        horizontal: [
          "flex-row items-center",
          "[>[data-slot=field-label]]:flex-auto",
          "has- [>[data-slot=field-content]]:[>[role=checkbox],[role=radio]]:mt-px has- [>[data-slot=field-content]]:items-start",
        ],
      },
      responsive: [
        "@md/field-group:flex-row @md/field-group:items-center @md/field-group:[>]:w-auto flex-col [>]:w-full [>.sr-only]:w-auto",
        "@md/field-group:[>[data-slot=field-label]]:flex-auto",
        "@md/field-group:has- [>[data-slot=field-content]]:items-start @md/field-group:has- [>[data-slot=field-content]]:[>[role=checkbox],[role=radio]]:mt-px",
      ],
    },
    defaultVariants: {
      orientation: "vertical",
    },
  }
)

function Field({
  className,
  orientation = "vertical",
  ...props
}: React.ComponentProps<"div"> & VariantProps<typeof fieldVariants>) {
  return (
    <div
      role="group"
      data-slot="field"
      data-orientation={orientation}
      className={cn(fieldVariants({ orientation }), className)}
    />
  )
}
```



```

    {...props}
  )
  />
}

function FieldContent({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="field-content"
      className={cn(
        "group/field-content flex flex-1 flex-col gap-1.5 leading-snug",
        className
      )}
      {...props}
    />
  )
}

function FieldLabel({
  className,
  ...props
}: React.ComponentProps<typeof Label>) {
  return (
    <Label
      data-slot="field-label"
      className={cn(
        "group/field-label peer/field-label flex w-fit gap-2 leading-snug group-data-[disabled=true]/field:opacity-50",
        "has- [>[data-slot=field]]:w-full has- [>[data-slot=field]]:flex-col has- [>[data-slot=field]]:rounded-md has- [>[data-
slot=field]]:border [>[data-slot=field]]:p-4",
        "has-data-[state=checked]:bg-primary/5 has-data-[state=checked]:border-primary dark:has-data-[state=checked]:bg-
primary/10",
        className
      )}
      {...props}
    />
  )
}

function FieldTitle({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="field-label"
      className={cn(
        "flex w-fit items-center gap-2 text-sm font-medium leading-snug group-data-[disabled=true]/field:opacity-50",
        className
      )}
      {...props}
    />
  )
}

function FieldDescription({ className, ...props }: React.ComponentProps<"p">) {
  return (
    <p
      data-slot="field-description"
      className={cn(
        "text-muted-foreground text-sm font-normal leading-normal group-has-[[data-orientation=horizontal]]/field:text-balance",
        "nth-last-2:-mt-1 last:mt-0 [[data-variant=legend]]&:-mt-1.5",
        "[&a:hover]:text-primary [&a]:underline [&a]:underline-offset-4",
        className
      )}
      {...props}
    />
  )
}

function FieldSeparator({
  children,
  className,
  ...props
}: React.ComponentProps<"div"> & {
  children?: React.ReactNode
}) {
  return (
    <div
      data-slot="field-separator"
      data-content={!children}
      className={cn(
        "relative -my-2 h-5 text-sm group-data-[variant=outline]/field-group:-mb-2",
        className
      )}
      {...props}
    >
      <Separator className="absolute inset-0 top-1/2" />
      {children && (
        <span
          className="bg-background text-muted-foreground relative mx-auto block w-fit px-2"
          data-slot="field-separator-content"
        >
          {children}
        </span>
      )}
    </div>
  )
}

function FieldError({

```

```

    className,
    children,
    errors,
    ...props
}: React.ComponentProps<"div"> & {
  errors?: Array<{ message?: string } | undefined>
}) {
  const content = useMemo(() => {
    if (children) {
      return children
    }

    if (!errors) {
      return null
    }

    if (errors?.length === 1 && errors[0]?.message) {
      return errors[0].message
    }

    return (
      <ul className="ml-4 flex list-disc flex-col gap-1">
        {errors.map(
          (error, index) =>
            error?.message && <li key={index}>{error.message}</li>
        )}
      </ul>
    )
  }, [children, errors])

  if (!content) {
    return null
  }

  return (
    <div
      role="alert"
      data-slot="field-error"
      className={cn("text-destructive text-sm font-normal", className)}
      {...props}>
      >
      {content}
    </div>
  )
}

export {
  Field,
  FieldLabel,
  FieldDescription,
  FieldError,
  FieldGroup,
  FieldLegend,
  FieldSeparator,
  FieldSet,
  FieldContent,
  FieldTitle,
}

```

[artifacts/mtos-crm/src/components/ui/form.tsx](#)

```
import * as React from "react"
import * as LabelPrimitive from "@radix-ui/react-label"
import { Slot } from "@radix-ui/react-slot"
import {
  Controller,
  FormProvider,
  useFormContext,
  type ControllerProps,
  type FieldPath,
  type FieldValues,
} from "react-hook-form"

import { cn } from "@lib/utils"
import { Label } from "@components/ui/label"

const Form = FormProvider

type FormFieldContextValue<
  TFieldValues extends FieldValues = FieldValues,
  TName extends FieldPath<TFieldValues> = FieldPath<TFieldValues>
> = {
  name: TName
}

const FormFieldContext = React.createContext<FormFieldContextValue | null>(null)

const FormField = <
  TFieldValues extends FieldValues = FieldValues,
  TName extends FieldPath<TFieldValues> = FieldPath<TFieldValues>
>({
  ..props
}): ControllerProps<TFieldValues, TName> => {
  return (
    <FormFieldContext.Provider value={{ name: props.name }}>
      <Controller {...props} />
    </FormFieldContext.Provider>
  )
}

const useFormField = () => {
  const fieldContext = React.useContext(FormFieldContext)
  const itemContext = React.useContext(FormItemContext)
  const { getFieldState, formState } = useFormContext()

  if (!fieldContext) {
    throw new Error("useFormField should be used within <FormField>")
  }

  if (!itemContext) {
    throw new Error("useFormField should be used within <FormItem>")
  }

  const fieldState = getFieldState(fieldContext.name, formState)

  const { id } = itemContext

  return {
    id,
    name: fieldContext.name,
    formItemId: `${id}-form-item`,
    formDescriptionId: `${id}-form-item-description`,
    formMessageId: `${id}-form-item-message`,
    ...fieldState,
  }
}

type FormItemContextValue = {
  id: string
}

const FormItemContext = React.createContext<FormItemContextValue | null>(null)

const FormItem = React.forwardRef<
  HTMLDivElement,
  React.HTMLAttributes<HTMLDivElement>
>(({ className, ...props }, ref) => {
  const id = React.useId()

  return (
    <FormItemContext.Provider value={{ id }}>
      <div ref={ref} className={cn("space-y-2", className)} {...props} />
    </FormItemContext.Provider>
  )
})
FormItem.displayName = "FormItem"

const FormLabel = React.forwardRef<
  React.ElementRef<typeof LabelPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof LabelPrimitive.Root>
>(({ className, ...props }, ref) => {
  const { error, formItemId } = useFormField()

  return (
    <Label
      ref={ref}
      error={error}
      formItemId={formItemId}
    />
  )
})
FormLabel.displayName = "FormLabel"
```

```

        className={cn(error && "text-destructive", className)}
        htmlFor={formItemId}
        {...props}
      />
    )
  })
  FormLabel.displayName = "FormLabel"

  const FormControl = React.forwardRef<
    React.ElementRef<typeof Slot>,
    React.ComponentPropsWithoutRef<typeof Slot>
  >(({ ...props }, ref) => {
    const { error, formItemId, formDescriptionId, formMessageId } = useFormField()

    return (
      <Slot
        ref={ref}
        id={formItemId}
        aria-describedby={
          !error
            ? `${formDescriptionId}`
            : `${formDescriptionId} ${formMessageId}`
        }
        aria-invalid={!!error}
        {...props}
      />
    )
  })
  FormControl.displayName = "FormControl"

  const FormDescription = React.forwardRef<
    HTMLParagraphElement,
    React.HTMLAttributes<HTMLParagraphElement>
  >(({ className, ...props }, ref) => {
    const { formDescriptionId } = useFormField()

    return (
      <p
        ref={ref}
        id={formDescriptionId}
        className={cn("text-[0.8rem] text-muted-foreground", className)}
        {...props}
      />
    )
  })
  FormDescription.displayName = "FormDescription"

  const FormMessage = React.forwardRef<
    HTMLParagraphElement,
    React.HTMLAttributes<HTMLParagraphElement>
  >(({ className, children, ...props }, ref) => {
    const { error, formMessageId } = useFormField()
    const body = error ? String(error?.message ?? "") : children

    if (!body) {
      return null
    }

    return (
      <p
        ref={ref}
        id={formMessageId}
        className={cn("text-[0.8rem] font-medium text-destructive", className)}
        {...props}
      >
        {body}
      </p>
    )
  })
  FormMessage.displayName = "FormMessage"

  export {
    useFormField,
    Form,
    FormItem,
    FormLabel,
    FormControl,
    FormDescription,
    FormMessage,
    FormField,
  }
}

```

[artifacts/mtos-crm/src/components/ui/hover-card.tsx](#)

```
import * as React from "react"
import * as HoverCardPrimitive from "@radix-ui/react-hover-card"

import { cn } from "@lib/utils"

const HoverCard = HoverCardPrimitive.Root

const HoverCardTrigger = HoverCardPrimitive.Trigger

const HoverCardContent = React.forwardRef<
  React.ElementRef<typeof HoverCardPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof HoverCardPrimitive.Content>
>(({ className, align = "center", sideOffset = 4, ...props }, ref) => {
  <HoverCardPrimitive.Content
    ref={ref}
    align={align}
    sideOffset={sideOffset}
    className={cn(
      "z-50 w-64 rounded-md border bg-popover p-4 text-popover-foreground shadow-md outline-none data-[state=open]:animate-in
data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-
[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-
in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin[--radix-hover-card-content-transform-origin]",
      className
    )}
    {...props}
  />
))
HoverCardContent.displayName = HoverCardPrimitive.Content.displayName

export { HoverCard, HoverCardTrigger, HoverCardContent }
```

[artifacts/mtos-crm/src/components/ui/input-group.tsx](#)

```
import * as React from "react"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"
import { Button } from "@components/ui/button"
import { Input } from "@components/ui/input"
import { Textarea } from "@components/ui/textarea"

function InputGroup({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="input-group"
      role="group"
      className={cn(
        "group/input-group border-input dark:bg-input/30 shadow-xs relative flex w-full items-center rounded-md border outline-
none transition-[color,box-shadow]",
        "h-9 has-[>textarea]:h-auto",

        // Variants based on alignment.
        "has-[>[data-align=inline-start]]:[&input]:pl-2",
        "has-[>[data-align=inline-end]]:[&input]:pr-2",
        "has-[>[data-align=block-start]]:h-auto has-[>[data-align=block-start]]:flex-col has-[>[data-align=block-start]]:
[&input]:pb-3",
        "has-[>[data-align=block-end]]:h-auto has-[>[data-align=block-end]]:flex-col has-[>[data-align=block-end]]:
[&input]:pt-3",

        // Focus state.
        "has-[[data-slot=input-group-control]:focus-visible]:ring-ring has-[[data-slot=input-group-control]:focus-
visible]:ring-1",

        // Error state.
        "has-[[data-slot][aria-invalid=true]]:ring-destructive/20 has-[[data-slot][aria-invalid=true]]:border-destructive
dark:has-[[data-slot][aria-invalid=true]]:ring-destructive/40",

        className
      )}
    >
      {...props}
    </div>
  )
}

const inputGroupAddonVariants = cva(
  "text-muted-foreground flex h-auto cursor-text select-none items-center justify-center gap-2 py-1.5 text-sm font-medium group-
data-[disabled=true]/input-group:opacity-50 [&kbd]:rounded-[calc(var(--radius)-5px)] [&svg:not([class*='size-']):size-4",
  {
    variants: {
      align: {
        "inline-start":
          "order-first pl-3 has-[>button]:ml-[-0.45rem] has-[>kbd]:ml-[-0.35rem]",
        "inline-end":
          "order-last pr-3 has-[>button]:mr-[-0.4rem] has-[>kbd]:mr-[-0.35rem]",
        "block-start":
          "[.border-b]:pb-3 order-first w-full justify-start px-3 pt-3 group-has-[>input]/input-group:pt-2.5",
        "block-end":
          "[.border-t]:pt-3 order-last w-full justify-start px-3 pb-3 group-has-[>input]/input-group:pb-2.5",
      },
    },
    defaultVariants: {
      align: "inline-start",
    },
  }
)

function InputGroupAddon({
  className,
  align = "inline-start",
  ...props
}: React.ComponentProps<"div"> & VariantProps<typeof inputGroupAddonVariants>) {
  return (
    <div
      role="group"
      data-slot="input-group-addon"
      data-align={align}
      className={cn(inputGroupAddonVariants({ align }), className)}
      onClick={(e) => {
        if ((e.target as HTMLElement).closest("button")) {
          return
        }
        e.currentTarget.parentElement?.querySelector("input")?.focus()
      }}
    >
      {...props}
    </div>
  )
}

const inputGroupButtonVariants = cva(
  "flex items-center gap-2 text-sm shadow-none",
  {
    variants: {
      size: {
        xs: "h-6 gap-1 rounded-[calc(var(--radius)-5px)] px-2 has-[>svg]:px-2 [&svg:not([class*='size-']):size-3.5",
        sm: "h-8 gap-1.5 rounded-md px-2.5 has-[>svg]:px-2.5",
        "icon-xs":
          "size-6 rounded-[calc(var(--radius)-5px)] p-0 has-[>svg]:p-0",
        "icon-sm": "size-8 p-0 has-[>svg]:p-0",
      },
    },
  }
)
```

```

    },
    defaultVariants: {
      size: "xs",
    },
  },
)

function InputGroupButton({
  className,
  type = "button",
  variant = "ghost",
  size = "xs",
  ...props
}: Omit<React.ComponentProps<typeof Button>, "size"> &
VariantProps<typeof inputGroupButtonVariants>) {
  return (
    <Button
      type={type}
      data-size={size}
      variant={variant}
      className={cn(inputGroupButtonVariants({ size })), className}
      {...props}
    />
  )
}

function InputGroupText({ className, ...props }: React.ComponentProps<"span">) {
  return (
    <span
      className={cn(
        "text-muted-foreground flex items-center gap-2 text-sm [&_svg:not([class*='size-'])]:size-4 [&_svg]:pointer-events-
none",
        className
      )}
      {...props}
    />
  )
}

function InputGroupInput({
  className,
  ...props
}: React.ComponentProps<"input">) {
  return (
    <Input
      data-slot="input-group-control"
      className={cn(
        "flex-1 rounded-none border-0 bg-transparent shadow-none focus-visible:ring-0 dark:bg-transparent",
        className
      )}
      {...props}
    />
  )
}

function InputGroupTextarea({
  className,
  ...props
}: React.ComponentProps<"textarea">) {
  return (
    <Textarea
      data-slot="input-group-control"
      className={cn(
        "flex-1 resize-none rounded-none border-0 bg-transparent py-3 shadow-none focus-visible:ring-0 dark:bg-transparent",
        className
      )}
      {...props}
    />
  )
}

export {
  InputGroup,
  InputGroupAddon,
  InputGroupButton,
  InputGroupText,
  InputGroupInput,
  InputGroupTextarea,
}

```

[artifacts/mtos-crm/src/components/ui/input-otp.tsx](#)

```
import * as React from "react"
import { OTPInput, OTPInputContext } from "input-otp"
import { Minus } from "lucide-react"

import { cn } from "@/lib/utils"

const InputOTP = React.forwardRef<
  React.ElementRef<typeof OTPInput>,
  React.ComponentPropsWithoutRef<typeof OTPInput>
>(({ className, containerClassName, ...props }, ref) => (
  <OTPInput
    ref={ref}
    containerClassName={cn(
      "flex items-center gap-2 has-[:disabled]:opacity-50",
      containerClassName
    )}
    className={cn("disabled:cursor-not-allowed", className)}
    {...props}
  />
))
InputOTP.displayName = "InputOTP"

const InputOTPGroup = React.forwardRef<
  React.ElementRef<"div">,
  React.ComponentPropsWithoutRef<"div">
>(({ className, ...props }, ref) => (
  <div ref={ref} className={cn("flex items-center", className)} {...props} />
))
InputOTPGroup.displayName = "InputOTPGroup"

const InputOTPSlot = React.forwardRef<
  React.ElementRef<"div">,
  React.ComponentPropsWithoutRef<"div"> & { index: number }
>(({ index, className, ...props }, ref) => {
  const inputOTPContext = React.useContext(OTPInputContext)
  const { char, hasFakeCaret, isActive } = inputOTPContext.slots[index]

  return (
    <div
      ref={ref}
      className={cn(
        "relative flex h-9 w-9 items-center justify-center border-y border-r border-input text-sm shadow-sm transition-all
        first:rounded-l-md first:border-l last:rounded-r-md",
        isActive && "z-10 ring-1 ring-ring",
        className
      )}
      {...props}
    >
      {char}
      {hasFakeCaret && (
        <div className="pointer-events-none absolute inset-0 flex items-center justify-center">
          <div className="h-4 w-px animate-caret-blink bg-foreground duration-1000" />
        </div>
      )}
    </div>
  )
)}
InputOTPSlot.displayName = "InputOTPSlot"

const InputOTPSeparator = React.forwardRef<
  React.ElementRef<"div">,
  React.ComponentPropsWithoutRef<"div">
>(({ ...props }, ref) => (
  <div ref={ref} role="separator" {...props}>
    <Minus />
  </div>
))
InputOTPSeparator.displayName = "InputOTPSeparator"

export { InputOTP, InputOTPGroup, InputOTPSlot, InputOTPSeparator }
```


[artifacts/mtos-crm/src/components/ui/input.tsx](#)

```
import * as React from "react"

import { cn } from "@/lib/utils"

const Input = React.forwardRef<HTMLInputElement, React.ComponentProps<"input">>(>
  ({ className, type, ...props }, ref) => {
    return (
      <input
        type={type}
        className={cn(
          "flex h-9 w-full rounded-md border border-input bg-transparent px-3 py-1 text-base shadow-sm transition-colors
file:border-0 file:bg-transparent file:text-sm file:font-medium file:text-foreground placeholder:text-muted-foreground focus-
visible:outline-none focus-visible:ring-1 focus-visible:ring-ring disabled:cursor-not-allowed disabled:opacity-50 md:text-sm",
          className
        )}
        ref={ref}
        {...props}
      />
    )
  }
)
Input.displayName = "Input"

export { Input }
```

[artifacts/mtos-crm/src/components/ui/item.tsx](#)

```
import * as React from "react"
import { Slot } from "@radix-ui/react-slot"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"
import { Separator } from "@components/ui/separator"

function ItemGroup({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      role="list"
      data-slot="item-group"
      className={cn("group/item-group flex flex-col", className)}
      {...props}
    />
  )
}

function ItemSeparator({
  className,
  ...props
}: React.ComponentProps<typeof Separator>) {
  return (
    <Separator
      data-slot="item-separator"
      orientation="horizontal"
      className={cn("my-0", className)}
      {...props}
    />
  )
}

const itemVariants = cva(
  "group/item [a]:hover:bg-accent/50 focus-visible:border-ring focus-visible:ring-ring/50 [a]:transition-colors flex flex-wrap items-center rounded-md border border-transparent text-sm outline-none transition-colors duration-100 focus-visible:ring-[3px]",
  {
    variants: {
      variant: {
        default: "bg-transparent",
        outline: "border-border",
        muted: "bg-muted/50",
      },
      size: {
        default: "gap-4 p-4 ",
        sm: "gap-2.5 px-4 py-3",
      },
    },
    defaultVariants: {
      variant: "default",
      size: "default",
    },
  }
)

function Item({
  className,
  variant = "default",
  size = "default",
  asChild = false,
  ...props
}: React.ComponentProps<"div"> &
VariantProps<typeof itemVariants> & { asChild?: boolean }) {
  const Comp = asChild ? Slot : "div"
  return (
    <Comp
      data-slot="item"
      data-variant={variant}
      data-size={size}
      className={cn(itemVariants({ variant, size, className }))}
      {...props}
    />
  )
}

const itemMediaVariants = cva(
  "flex shrink-0 items-center justify-center gap-2 group-has-[[data-slot=item-description]]/item:translate-y-0.5 group-has-[[data-slot=item-description]]/item:self-start [&_svg]:pointer-events-none",
  {
    variants: {
      variant: {
        default: "bg-transparent",
        icon: "bg-muted size-8 rounded-sm border [&_svg:not([class*='size-']):size-4",
        image: "size-10 overflow-hidden rounded-sm [&_img]:size-full [&_img]:object-cover",
      },
    },
    defaultVariants: {
      variant: "default",
    },
  }
)

function ItemMedia({
  className,
  variant = "default",
```

```

    ...props
  }: React.ComponentProps<"div"> & VariantProps<typeof itemMediaVariants>) {
    return (
      <div
        data-slot="item-media"
        data-variant={variant}
        className={cn(itemMediaVariants({ variant, className })))}
        {...props}
      />
    )
  }

function ItemContent({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="item-content"
      className={cn(
        "flex flex-1 flex-col gap-1 [&[data-slot=item-content]]:flex-none",
        className
      )}
      {...props}
    />
  )
}

function ItemTitle({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="item-title"
      className={cn(
        "flex w-fit items-center gap-2 text-sm font-medium leading-snug",
        className
      )}
      {...props}
    />
  )
}

function ItemDescription({ className, ...props }: React.ComponentProps<"p">) {
  return (
    <p
      data-slot="item-description"
      className={cn(
        "text-muted-foreground line-clamp-2 text-balance text-sm font-normal leading-normal",
        "[&a:hover]:text-primary [&a]:underline [&a]:underline-offset-4",
        className
      )}
      {...props}
    />
  )
}

function ItemActions({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="item-actions"
      className={cn("flex items-center gap-2", className)}
      {...props}
    />
  )
}

function ItemHeader({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="item-header"
      className={cn(
        "flex basis-full items-center justify-between gap-2",
        className
      )}
      {...props}
    />
  )
}

function ItemFooter({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="item-footer"
      className={cn(
        "flex basis-full items-center justify-between gap-2",
        className
      )}
      {...props}
    />
  )
}

export {
  Item,
  ItemMedia,
  ItemContent,
  ItemActions,
  ItemGroup,
  ItemSeparator,
  ItemTitle,
  ItemDescription,

```

```
ItemHeader,  
ItemFooter,  
}
```

[artifacts/mtos-crm/src/components/ui/kbd.tsx](#)

```
import { cn } from "@lib/utils"

function Kbd({ className, ...props }: React.ComponentProps<"kbd">) {
  return (
    <kbd
      data-slot="kbd"
      className={cn(
        "bg-muted text-muted-foreground pointer-events-none inline-flex h-5 w-fit min-w-5 select-none items-center justify-
center gap-1 rounded-sm px-1 font-sans text-xs font-medium",
        "[&_svg:not([class*='size-']):size-3",
        "[[data-slot=tooltip-content]_&]:bg-background/20 [[data-slot=tooltip-content]_&]:text-background dark:[[data-
slot=tooltip-content]_&]:bg-background/10",
        className
      )}
      {...props}
    />
  )
}

function KbdGroup({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <kbd
      data-slot="kbd-group"
      className={cn("inline-flex items-center gap-1", className)}
      {...props}
    />
  )
}

export { Kbd, KbdGroup }
```

[artifacts/mtos-crm/src/components/ui/label.tsx](#)

```
"use client"

import * as React from "react"
import * as LabelPrimitive from "@radix-ui/react-label"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"

const labelVariants = cva(
  "text-sm font-medium leading-none peer-disabled:cursor-not-allowed peer-disabled:opacity-70"
)

const Label = React.forwardRef<
  React.ElementRef<typeof LabelPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof LabelPrimitive.Root> &
    VariantProps<typeof labelVariants>
>(({ className, ...props }, ref) => (
  <LabelPrimitive.Root
    ref={ref}
    className={cn(labelVariants(), className)}
    {...props}
  />
))
Label.displayName = LabelPrimitive.Root.displayName

export { Label }
```

[artifacts/mtos-crm/src/components/ui/menubar.tsx](#)

```
import * as React from "react"
import * as MenubarPrimitive from "@radix-ui/react-menubar"
import { Check, ChevronRight, Circle } from "lucide-react"

import { cn } from "@/lib/utils"

function MenubarMenu({
  ...props
}: React.ComponentProps<typeof MenubarPrimitive.Menu>) {
  return <MenubarPrimitive.Menu {...props} />
}

function MenubarGroup({
  ...props
}: React.ComponentProps<typeof MenubarPrimitive.Group>) {
  return <MenubarPrimitive.Group {...props} />
}

function MenubarPortal({
  ...props
}: React.ComponentProps<typeof MenubarPrimitive.Portal>) {
  return <MenubarPrimitive.Portal {...props} />
}

function MenubarRadioGroup({
  ...props
}: React.ComponentProps<typeof MenubarPrimitive.RadioGroup>) {
  return <MenubarPrimitive.RadioGroup {...props} />
}

function MenubarSub({
  ...props
}: React.ComponentProps<typeof MenubarPrimitive.Sub>) {
  return <MenubarPrimitive.Sub data-slot="menubar-sub" {...props} />
}

const Menubar = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Root>
>(({ className, ...props }, ref) => (
  <MenubarPrimitive.Root
    ref={ref}
    className={cn(
      "flex h-9 items-center space-x-1 rounded-md border bg-background p-1 shadow-sm",
      className
    )}
    {...props}
  />
))
Menubar.displayName = MenubarPrimitive.Root.displayName

const MenubarTrigger = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.Trigger>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Trigger>
>(({ className, ...props }, ref) => (
  <MenubarPrimitive.Trigger
    ref={ref}
    className={cn(
      "flex cursor-default select-none items-center rounded-sm px-3 py-1 text-sm font-medium outline-none focus:bg-accent
      focus:text-accent-foreground data-[state=open]:bg-accent data-[state=open]:text-accent-foreground",
      className
    )}
    {...props}
  />
))
MenubarTrigger.displayName = MenubarPrimitive.Trigger.displayName

const MenubarSubTrigger = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.SubTrigger>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.SubTrigger> & {
    inset?: boolean
  }
>(({ className, inset, children, ...props }, ref) => (
  <MenubarPrimitive.SubTrigger
    ref={ref}
    className={cn(
      "flex cursor-default select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none focus:bg-accent focus:text-
      accent-foreground data-[state=open]:bg-accent data-[state=open]:text-accent-foreground",
      inset && "pl-8",
      className
    )}
    {...props}
  >
    {children}
  <ChevronRight className="ml-auto h-4 w-4" />
  </MenubarPrimitive.SubTrigger>
))
MenubarSubTrigger.displayName = MenubarPrimitive.SubTrigger.displayName

const MenubarSubContent = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.SubContent>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.SubContent>
>(({ className, ...props }, ref) => (
  <MenubarPrimitive.SubContent
    ref={ref}
```

```

        className={cn(
          "z-50 min-w-[8rem] overflow-hidden rounded-md border bg-popover p-1 text-popover-foreground shadow-lg data-
[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-
[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-
right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin[--radix-menu-bar-content-transform-
origin]",
          className
        )}
      )}
    {...props}
  )}
  />
)
)
MenubarSubContent.displayName = MenubarPrimitive.SubContent.displayName

const MenubarContent = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Content>
>({
  (
    { className, align = "start", alignOffset = -4, sideOffset = 8, ...props },
    ref
  ) => (
    <MenubarPrimitive.Portal>
      <MenubarPrimitive.Content
        ref={ref}
        align={align}
        alignOffset={alignOffset}
        sideOffset={sideOffset}
        className={cn(
          "z-50 min-w-[12rem] overflow-hidden rounded-md border bg-popover p-1 text-popover-foreground shadow-md data-
[state=open]:animate-in data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-
[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-
in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin[--radix-menu-bar-content-transform-origin]",
          className
        )}
      >
        {...props}
      </MenubarPrimitive.Portal>
    )
  )
  MenubarContent.displayName = MenubarPrimitive.Content.displayName

const MenubarItem = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Item> & {
    inset?: boolean
  }
>(({ className, inset, ...props }, ref) => (
  <MenubarPrimitive.Item
    ref={ref}
    className={cn(
      "relative flex cursor-default select-none items-center rounded-sm px-2 py-1.5 text-sm outline-none focus:bg-accent
focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      inset && "pl-8",
      className
    )}
    {...props}
  >
    />
  )
  MenubarItem.displayName = MenubarPrimitive.Item.displayName

const MenubarCheckboxItem = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.CheckboxItem>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.CheckboxItem>
>(({ className, children, checked, ...props }, ref) => (
  <MenubarPrimitive.CheckboxItem
    ref={ref}
    className={cn(
      "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-none focus:bg-accent
focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      className
    )}
    checked={checked}
    {...props}
  >
    <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
      <MenubarPrimitive.ItemIndicator>
        <Check className="h-4 w-4" />
      </MenubarPrimitive.ItemIndicator>
    </span>
    {children}
  </MenubarPrimitive.CheckboxItem>
  )
  MenubarCheckboxItem.displayName = MenubarPrimitive.CheckboxItem.displayName

const MenubarRadioItem = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.RadioItem>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.RadioItem>
>(({ className, children, ...props }, ref) => (
  <MenubarPrimitive.RadioItem
    ref={ref}
    className={cn(
      "relative flex cursor-default select-none items-center rounded-sm py-1.5 pl-8 pr-2 text-sm outline-none focus:bg-accent
focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      className
    )}
    {...props}
  >
    />
  )
  MenubarRadioItem.displayName = MenubarPrimitive.RadioItem.displayName

```



```

    <span className="absolute left-2 flex h-3.5 w-3.5 items-center justify-center">
      <MenubarPrimitive.ItemIndicator>
        <Circle className="h-4 w-4 fill-current" />
      </MenubarPrimitive.ItemIndicator>
    </span>
    {children}
  </MenubarPrimitive.RadioItem>
))
MenubarRadioItem.displayName = MenubarPrimitive.RadioItem.displayName

const MenubarLabel = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.Label>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Label> & {
    inset?: boolean
  }
>(({ className, inset, ...props }, ref) => (
  <MenubarPrimitive.Label
    ref={ref}
    className={cn(
      "px-2 py-1.5 text-sm font-semibold",
      inset && "pl-8",
      className
    )}
    {...props}
  />
))
MenubarLabel.displayName = MenubarPrimitive.Label.displayName

const MenubarSeparator = React.forwardRef<
  React.ElementRef<typeof MenubarPrimitive.Separator>,
  React.ComponentPropsWithoutRef<typeof MenubarPrimitive.Separator>
>(({ className, ...props }, ref) => (
  <MenubarPrimitive.Separator
    ref={ref}
    className={cn("-mx-1 my-1 h-px bg-muted", className)}
    {...props}
  />
))
MenubarSeparator.displayName = MenubarPrimitive.Separator.displayName

const MenubarShortcut = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLSpanElement>) => {
  return (
    <span
      className={cn(
        "ml-auto text-xs tracking-widest text-muted-foreground",
        className
      )}
      {...props}
    />
  )
}
MenubarShortcut.displayName = "MenubarShortcut"

export {
  Menubar,
  MenubarMenu,
  MenubarTrigger,
  MenubarContent,
  MenubarItem,
  MenubarSeparator,
  MenubarLabel,
  MenubarCheckboxItem,
  MenubarRadioGroup,
  MenubarRadioItem,
  MenubarPortal,
  MenubarSubContent,
  MenubarSubTrigger,
  MenubarGroup,
  MenubarSub,
  MenubarShortcut,
}

```

[artifacts/mtos-crm/src/components/ui/navigation-menu.tsx](#)

```
import * as React from "react"
import * as NavigationMenuPrimitive from "@radix-ui/react-navigation-menu"
import { cva } from "class-variance-authority"
import { ChevronDown } from "lucide-react"

import { cn } from "@/lib/utils"

const NavigationMenu = React.forwardRef<
  React.ElementRef<typeof NavigationMenuPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Root>
>(({ className, children, ...props }, ref) => (
  <NavigationMenuPrimitive.Root
    ref={ref}
    className={cn(
      "relative z-10 flex max-w-max flex-1 items-center justify-center",
      className
    )}
    {...props}
  >
    {children}
    <NavigationMenuViewport />
  </NavigationMenuPrimitive.Root>
))
NavigationMenu.displayName = NavigationMenuPrimitive.Root.displayName

const NavigationMenuList = React.forwardRef<
  React.ElementRef<typeof NavigationMenuPrimitive.List>,
  React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.List>
>(({ className, ...props }, ref) => (
  <NavigationMenuPrimitive.List
    ref={ref}
    className={cn(
      "group flex flex-1 list-none items-center justify-center space-x-1",
      className
    )}
    {...props}
  />
))
NavigationMenuList.displayName = NavigationMenuPrimitive.List.displayName

const NavigationMenuItem = NavigationMenuPrimitive.Item

const navigationMenuTriggerStyle = cva(
  "group inline-flex h-9 w-max items-center justify-center rounded-md bg-background px-4 py-2 text-sm font-medium transition-colors hover:bg-accent hover:text-accent-foreground focus:bg-accent focus:text-accent-foreground focus:outline-none disabled:pointer-events-none disabled:opacity-50 data-[state=open]:text-accent-foreground data-[state=open]:bg-accent/50 data-[state=open]:hover:bg-accent data-[state=open]:focus:bg-accent"
)

const NavigationMenuTrigger = React.forwardRef<
  React.ElementRef<typeof NavigationMenuPrimitive.Trigger>,
  React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Trigger>
>(({ className, children, ...props }, ref) => (
  <NavigationMenuPrimitive.Trigger
    ref={ref}
    className={cn(navigationMenuTriggerStyle(), "group", className)}
    {...props}
  >
    {children}{" "}
    <ChevronDown
      className="relative top-[1px] ml-1 h-3 w-3 transition duration-300 group-data-[state=open]:rotate-180"
      aria-hidden="true"
    />
  </NavigationMenuPrimitive.Trigger>
))
NavigationMenuTrigger.displayName = NavigationMenuPrimitive.Trigger.displayName

const NavigationMenuContent = React.forwardRef<
  React.ElementRef<typeof NavigationMenuPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Content>
>(({ className, ...props }, ref) => (
  <NavigationMenuPrimitive.Content
    ref={ref}
    className={cn(
      "left-0 top-0 w-full data-[motion^=from-]:animate-in data-[motion^=to-]:animate-out data-[motion^=from-]:fade-in data-[motion^=to-]:fade-out data-[motion=from-end]:slide-in-from-right-52 data-[motion=from-start]:slide-in-from-left-52 data-[motion=to-end]:slide-out-to-right-52 data-[motion=to-start]:slide-out-to-left-52 md:absolute md:w-auto ",
      className
    )}
    {...props}
  />
))
NavigationMenuContent.displayName = NavigationMenuPrimitive.Content.displayName

const NavigationMenuLink = NavigationMenuPrimitive.Link

const NavigationMenuViewport = React.forwardRef<
  React.ElementRef<typeof NavigationMenuPrimitive.Viewport>,
  React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Viewport>
>(({ className, ...props }, ref) => (
  <div className={cn("absolute left-0 top-full flex justify-center")}>
    <NavigationMenuPrimitive.Viewport
      className={cn(
        "origin-top-center relative mt-1.5 h-[var(--radix-navigation-menu-viewport-height)] w-full overflow-hidden rounded-md border bg-popover text-popover-foreground shadow data-[state=open]:animate-in data-[state=closed]:animate-out data-
```

```

[state=closed]:zoom-out-95 data-[state=open]:zoom-in-90 md:w-[var(--radix-navigation-menu-viewport-width))",
  className
)}
ref={ref}
{...props}
/>
</div>
))
NavigationMenuViewport.displayName =
  NavigationMenuPrimitive.Viewport.displayName

const NavigationMenuIndicator = React.forwardRef<
  React.ElementRef<typeof NavigationMenuPrimitive.Indicator>,
  React.ComponentPropsWithoutRef<typeof NavigationMenuPrimitive.Indicator>
>(({ className, ...props }, ref) => (
  <NavigationMenuPrimitive.Indicator
    ref={ref}
    className={cn(
      "top-full z-[1] flex h-1.5 items-end justify-center overflow-hidden data-[state=visible]:animate-in data-
[state=hidden]:animate-out data-[state=hidden]:fade-out data-[state=visible]:fade-in",
      className
    )}
    {...props}
  >
    <div className="relative top-[60%] h-2 w-2 rotate-45 rounded-tl-sm bg-border shadow-md" />
  </NavigationMenuPrimitive.Indicator>
))
NavigationMenuIndicator.displayName =
  NavigationMenuPrimitive.Indicator.displayName

export {
  navigationMenuTriggerStyle,
  NavigationMenu,
  NavigationMenuList,
  NavigationMenuItem,
  NavigationMenuContent,
  NavigationMenuTrigger,
  NavigationMenuLink,
  NavigationMenuIndicator,
  NavigationMenuViewport,
}

```

[artifacts/mtos-crm/src/components/ui/pagination.tsx](#)

```
import * as React from "react"
import { ChevronLeft, ChevronRight, MoreHorizontal } from "lucide-react"

import { cn } from "@lib/utls"
import { ButtonProps, buttonVariants } from "@components/ui/button"

const Pagination = ({ className, ...props }: React.ComponentProps<"nav">) => (
  <nav
    role="navigation"
    aria-label="pagination"
    className={cn("mx-auto flex w-full justify-center", className)}
    {...props}
  />
)
Pagination.displayName = "Pagination"

const PaginationContent = React.forwardRef<
  HTMLUListElement,
  React.ComponentProps<"ul">
>(({ className, ...props }, ref) => (
  <ul
    ref={ref}
    className={cn("flex flex-row items-center gap-1", className)}
    {...props}
  />
))
PaginationContent.displayName = "PaginationContent"

const PaginationItem = React.forwardRef<
  HTMLLIElement,
  React.ComponentProps<"li">
>(({ className, ...props }, ref) => (
  <li ref={ref} className={cn("", className)} {...props} />
))
PaginationItem.displayName = "PaginationItem"

type PaginationLinkProps = {
  isActive?: boolean
} & Pick<ButtonProps, "size"> &
  React.ComponentProps<"a">

const PaginationLink = ({
  className,
  isActive,
  size = "icon",
  ...props
}: PaginationLinkProps) => (
  <a
    aria-current={isActive ? "page" : undefined}
    className={cn(
      buttonVariants({
        variant: isActive ? "outline" : "ghost",
        size,
      }),
      className
    )}
    {...props}
  />
)
PaginationLink.displayName = "PaginationLink"

const PaginationPrevious = ({
  className,
  ...props
}: React.ComponentProps<typeof PaginationLink>) => (
  <PaginationLink
    aria-label="Go to previous page"
    size="default"
    className={cn("gap-1 pl-2.5", className)}
    {...props}
  >
    <ChevronLeft className="h-4 w-4" />
    <span>Previous</span>
  </PaginationLink>
)
PaginationPrevious.displayName = "PaginationPrevious"

const PaginationNext = ({
  className,
  ...props
}: React.ComponentProps<typeof PaginationLink>) => (
  <PaginationLink
    aria-label="Go to next page"
    size="default"
    className={cn("gap-1 pr-2.5", className)}
    {...props}
  >
    <span>Next</span>
    <ChevronRight className="h-4 w-4" />
  </PaginationLink>
)
PaginationNext.displayName = "PaginationNext"

const PaginationEllipsis = ({
  className,
```

```

    ...props
  }: React.ComponentProps<"span">) => (
    <span
      aria-hidden
      className={cn("flex h-9 w-9 items-center justify-center", className)}
      {...props}
    >
      <MoreHorizontal className="h-4 w-4" />
      <span className="sr-only">More pages</span>
    </span>
  )
  PaginationEllipsis.displayName = "PaginationEllipsis"

export {
  Pagination,
  PaginationContent,
  PaginationLink,
  PaginationItem,
  PaginationPrevious,
  PaginationNext,
  PaginationEllipsis,
}

```

[artifacts/mtos-crm/src/components/ui/popover.tsx](#)

```
import * as React from "react"
import * as PopoverPrimitive from "@radix-ui/react-popover"

import { cn } from "@lib/utils"

const Popover = PopoverPrimitive.Root

const PopoverTrigger = PopoverPrimitive.Trigger

const PopoverAnchor = PopoverPrimitive.Anchor

const PopoverContent = React.forwardRef<
  React.ElementRef<typeof PopoverPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof PopoverPrimitive.Content>
>(({ className, align = "center", sideOffset = 4, ...props }, ref) => (
  <PopoverPrimitive.Portal>
    <PopoverPrimitive.Content
      ref={ref}
      align={align}
      sideOffset={sideOffset}
      className={cn(
        "z-50 w-72 rounded-md border bg-popover p-4 text-popover-foreground shadow-md outline-none data-[state=open]:animate-in
data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-
[state=open]:zoom-in-95 data-[side=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-
in-from-left-2 data-[side=top]:slide-in-from-bottom-2 origin-[--radix-popover-content-transform-origin]",
        className
      )}
      {...props}
    />
  </PopoverPrimitive.Portal>
))
PopoverContent.displayName = PopoverPrimitive.Content.displayName

export { Popover, PopoverTrigger, PopoverContent, PopoverAnchor }
```

[artifacts/mtos-crm/src/components/ui/progress.tsx](#)

```
"use client"

import * as React from "react"
import * as ProgressPrimitive from "@radix-ui/react-progress"

import { cn } from "@lib/utils"

const Progress = React.forwardRef<
  React.ElementRef<typeof ProgressPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof ProgressPrimitive.Root>
>(({ className, value, ...props }, ref) => (
  <ProgressPrimitive.Root
    ref={ref}
    className={cn(
      "relative h-2 w-full overflow-hidden rounded-full bg-primary/20",
      className
    )}
    {...props}
  >
    <ProgressPrimitive.Indicator
      className="h-full w-full flex-1 bg-primary transition-all"
      style={{ transform: `translateX(-${100 - (value || 0)}%)` }}
    />
  </ProgressPrimitive.Root>
))
Progress.displayName = ProgressPrimitive.Root.displayName

export { Progress }
```

[artifacts/mtos-crm/src/components/ui/radio-group.tsx](#)

```
import * as React from "react"
import * as RadioGroupPrimitive from "@radix-ui/react-radio-group"
import { Circle } from "lucide-react"

import { cn } from "@lib/utils"

const RadioGroup = React.forwardRef<
  React.ElementRef<typeof RadioGroupPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof RadioGroupPrimitive.Root>
>(({ className, ...props }, ref) => {
  return (
    <RadioGroupPrimitive.Root
      className={cn("grid gap-2", className)}
      {...props}
      ref={ref}
    />
  )
})
RadioGroup.displayName = RadioGroupPrimitive.Root.displayName

const RadioGroupItem = React.forwardRef<
  React.ElementRef<typeof RadioGroupPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof RadioGroupPrimitive.Item>
>(({ className, ...props }, ref) => {
  return (
    <RadioGroupPrimitive.Item
      ref={ref}
      className={cn(
        "aspect-square h-4 w-4 rounded-full border border-primary text-primary shadow focus:outline-none focus-visible:ring-1
focus-visible:ring-ring disabled:cursor-not-allowed disabled:opacity-50",
        className
      )}
      {...props}
    >
      <RadioGroupPrimitive.Indicator className="flex items-center justify-center">
        <Circle className="h-3.5 w-3.5 fill-primary" />
      </RadioGroupPrimitive.Indicator>
    </RadioGroupPrimitive.Item>
  )
})
RadioGroupItem.displayName = RadioGroupPrimitive.Item.displayName

export { RadioGroup, RadioGroupItem }
```


[artifacts/mtos-crm/src/components/ui/resizable.tsx](#)

```
"use client"

import { GripVertical } from "lucide-react"
import * as ResizablePrimitive from "react-resizable-panels"

import { cn } from "@/lib/utils"

const ResizablePanelGroup = ({
  className,
  ...props
}: React.ComponentProps<typeof ResizablePrimitive.PanelGroup>) => (
  <ResizablePrimitive.PanelGroup
    className={cn(
      "flex h-full w-full data-[panel-group-direction=vertical]:flex-col",
      className
    )}
    {...props}
  />
)

const ResizablePanel = ResizablePrimitive.Panel

const ResizableHandle = ({
  withHandle,
  className,
  ...props
}: React.ComponentProps<typeof ResizablePrimitive.PanelResizeHandle> & {
  withHandle?: boolean
}) => (
  <ResizablePrimitive.PanelResizeHandle
    className={cn(
      "relative flex w-px items-center justify-center bg-border after:absolute after:inset-y-0 after:left-1/2 after:w-1 after:-
      translate-x-1/2 focus-visible:outline-none focus-visible:ring-1 focus-visible:ring-ring focus-visible:ring-offset-1 data-[panel-
      group-direction=vertical]:h-px data-[panel-group-direction=vertical]:w-full data-[panel-group-direction=vertical]:after:left-0
      data-[panel-group-direction=vertical]:after:h-1 data-[panel-group-direction=vertical]:after:w-full data-[panel-group-
      direction=vertical]:after:-translate-y-1/2 data-[panel-group-direction=vertical]:after:translate-x-0 [&[data-panel-group-
      direction=vertical]>div]:rotate-90",
      className
    )}
    {...props}
  >
    {withHandle && (
      <div className="z-10 flex h-4 w-3 items-center justify-center rounded-sm border bg-border">
        <GripVertical className="h-2.5 w-2.5" />
      </div>
    )}
  </ResizablePrimitive.PanelResizeHandle>
)

export { ResizablePanelGroup, ResizablePanel, ResizableHandle }
```

[artifacts/mtos-crm/src/components/ui/scroll-area.tsx](#)

```
import * as React from "react"
import * as ScrollAreaPrimitive from "@radix-ui/react-scroll-area"

import { cn } from "@lib/utils"

const ScrollArea = React.forwardRef<
  React.ElementRef<typeof ScrollAreaPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof ScrollAreaPrimitive.Root>
>(({ className, children, ...props }, ref) => (
  <ScrollAreaPrimitive.Root
    ref={ref}
    className={cn("relative overflow-hidden", className)}
    {...props}
  >
    <ScrollAreaPrimitive.Viewport className="h-full w-full rounded-[inherit]">
      {children}
    </ScrollAreaPrimitive.Viewport>
    <ScrollBar />
    <ScrollAreaPrimitive.Corner />
  </ScrollAreaPrimitive.Root>
))
ScrollArea.displayName = ScrollAreaPrimitive.Root.displayName

const ScrollBar = React.forwardRef<
  React.ElementRef<typeof ScrollAreaPrimitive.ScrollAreaScrollbar>,
  React.ComponentPropsWithoutRef<typeof ScrollAreaPrimitive.ScrollAreaScrollbar>
>(({ className, orientation = "vertical", ...props }, ref) => (
  <ScrollAreaPrimitive.ScrollAreaScrollbar
    ref={ref}
    orientation={orientation}
    className={cn(
      "flex touch-none select-none transition-colors",
      orientation === "vertical" &&
        "h-full w-2.5 border-l border-l-transparent p-[1px]",
      orientation === "horizontal" &&
        "h-2.5 flex-col border-t border-t-transparent p-[1px]",
      className
    )}
    {...props}
  >
    <ScrollAreaPrimitive.ScrollAreaThumb className="relative flex-1 rounded-full bg-border" />
  </ScrollAreaPrimitive.ScrollAreaScrollbar>
))
ScrollBar.displayName = ScrollAreaPrimitive.ScrollAreaScrollbar.displayName

export { ScrollArea, ScrollBar }
```

[artifacts/mtos-crm/src/components/ui/select.tsx](#)

```
"use client"

import * as React from "react"
import * as SelectPrimitive from "@radix-ui/react-select"
import { Check, ChevronDown, ChevronUp } from "lucide-react"

import { cn } from "@lib/utls"

const Select = SelectPrimitive.Root

const SelectGroup = SelectPrimitive.Group

const SelectValue = SelectPrimitive.Value

const SelectTrigger = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.Trigger>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.Trigger>
>(({ className, children, ...props }, ref) => (
  <SelectPrimitive.Trigger
    ref={ref}
    className={cn(
      "flex h-9 w-full items-center justify-between whitespace-nowrap rounded-md border border-input bg-transparent px-3 py-2
text-sm shadow-sm ring-offset-background data-[placeholder]:text-muted-foreground focus:outline-none focus:ring-1 focus:ring-
ring disabled:cursor-not-allowed disabled:opacity-50 [>span]:line-clamp-1",
      className
    )}
    {...props}
  >
    {children}
    <SelectPrimitive.Icon asChild>
      <ChevronDown className="h-4 w-4 opacity-50" />
    </SelectPrimitive.Icon>
  </SelectPrimitive.Trigger>
))
SelectTrigger.displayName = SelectPrimitive.Trigger.displayName

const SelectScrollUpButton = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.ScrollUpButton>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.ScrollUpButton>
>(({ className, ...props }, ref) => (
  <SelectPrimitive.ScrollUpButton
    ref={ref}
    className={cn(
      "flex cursor-default items-center justify-center py-1",
      className
    )}
    {...props}
  >
    <ChevronUp className="h-4 w-4" />
  </SelectPrimitive.ScrollUpButton>
))
SelectScrollUpButton.displayName = SelectPrimitive.ScrollUpButton.displayName

const SelectScrollDownButton = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.ScrollDownButton>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.ScrollDownButton>
>(({ className, ...props }, ref) => (
  <SelectPrimitive.ScrollDownButton
    ref={ref}
    className={cn(
      "flex cursor-default items-center justify-center py-1",
      className
    )}
    {...props}
  >
    <ChevronDown className="h-4 w-4" />
  </SelectPrimitive.ScrollDownButton>
))
SelectScrollDownButton.displayName =
  SelectPrimitive.ScrollDownButton.displayName

const SelectContent = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.Content>
>(({ className, children, position = "popper", ...props }, ref) => (
  <SelectPrimitive.Portal>
    <SelectPrimitive.Content
      ref={ref}
      className={cn(
        "relative z-50 max-h-[--radix-select-content-available-height] min-w-[8rem] overflow-y-auto overflow-x-hidden rounded-
md border bg-popover text-popover-foreground shadow-md data-[state=open]:animate-in data-[state=closed]:animate-out data-
[state=closed]:fade-out-0 data-[state=open]:fade-in-0 data-[state=closed]:zoom-out-95 data-[state=open]:zoom-in-95 data-
[state=bottom]:slide-in-from-top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-
[state=top]:slide-in-from-bottom-2 origin-[--radix-select-content-transform-origin]",
        position === "popper" &&
        "data-[side=bottom]:translate-y-1 data-[side=left]:-translate-x-1 data-[side=right]:translate-x-1 data-[side=top]:-
translate-y-1",
        className
      )}
      position={position}
      {...props}
    >
      <SelectScrollUpButton />
      <SelectPrimitive.Viewport
        className={cn(
```

```

        "p-1",
        position === "popper" &&
        "h-[var(--radix-select-trigger-height)] w-full min-w-[var(--radix-select-trigger-width)]"
    )}
    >
    {children}
    </SelectPrimitive.Viewport>
    <SelectScrollDownButton />
    </SelectPrimitive.Content>
  </SelectPrimitive.Portal>
))
SelectContent.displayName = SelectPrimitive.Content.displayName

const SelectLabel = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.Label>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.Label>
>(({ className, ...props }, ref) => (
  <SelectPrimitive.Label
    ref={ref}
    className={cn("px-2 py-1.5 text-sm font-semibold", className)}
    {...props}
  />
))
SelectLabel.displayName = SelectPrimitive.Label.displayName

const SelectItem = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.Item>
>(({ className, children, ...props }, ref) => (
  <SelectPrimitive.Item
    ref={ref}
    className={cn(
      "relative flex w-full cursor-default select-none items-center rounded-sm py-1.5 pl-2 pr-8 text-sm outline-none focus:bg-
accent focus:text-accent-foreground data-[disabled]:pointer-events-none data-[disabled]:opacity-50",
      className
    )}
    {...props}
  >
    <span className="absolute right-2 flex h-3.5 w-3.5 items-center justify-center">
      <SelectPrimitive.ItemIndicator>
        <Check className="h-4 w-4" />
      </SelectPrimitive.ItemIndicator>
    </span>
    <SelectPrimitive.ItemText>{children}</SelectPrimitive.ItemText>
  </SelectPrimitive.Item>
))
SelectItem.displayName = SelectPrimitive.Item.displayName

const SelectSeparator = React.forwardRef<
  React.ElementRef<typeof SelectPrimitive.Separator>,
  React.ComponentPropsWithoutRef<typeof SelectPrimitive.Separator>
>(({ className, ...props }, ref) => (
  <SelectPrimitive.Separator
    ref={ref}
    className={cn("-mx-1 my-1 h-px bg-muted", className)}
    {...props}
  />
))
SelectSeparator.displayName = SelectPrimitive.Separator.displayName

export {
  Select,
  SelectGroup,
  SelectValue,
  SelectTrigger,
  SelectContent,
  SelectLabel,
  SelectItem,
  SelectSeparator,
  SelectScrollUpButton,
  SelectScrollDownButton,
}

```

[artifacts/mtos-crm/src/components/ui/separator.tsx](#)

```
import * as React from "react"
import * as SeparatorPrimitive from "@radix-ui/react-separator"

import { cn } from "@lib/utils"

const Separator = React.forwardRef<
  React.ElementRef<typeof SeparatorPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof SeparatorPrimitive.Root>
>((
  {
    className, orientation = "horizontal", decorative = true, ...props },
    ref
  ) => (
    <SeparatorPrimitive.Root
      ref={ref}
      decorative={decorative}
      orientation={orientation}
      className={cn(
        "shrink-0 bg-border",
        orientation === "horizontal" ? "h-[1px] w-full" : "h-full w-[1px]",
        className
      )}
      {...props}
    />
  )
)

Separator.displayName = SeparatorPrimitive.Root.displayName

export { Separator }
```

[artifacts/mtos-crm/src/components/ui/sheet.tsx](#)

```
"use client"

import * as React from "react"
import * as SheetPrimitive from "@radix-ui/react-dialog"
import { cva, type VariantProps } from "class-variance-authority"
import { X } from "lucide-react"

import { cn } from "@lib/Utils"

const Sheet = SheetPrimitive.Root

const SheetTrigger = SheetPrimitive.Trigger

const SheetClose = SheetPrimitive.Close

const SheetPortal = SheetPrimitive.Portal

const SheetOverlay = React.forwardRef<
  React.ElementRef<typeof SheetPrimitive.Overlay>,
  React.ComponentPropsWithoutRef<typeof SheetPrimitive.Overlay>
>(({ className, ...props }, ref) => (
  <SheetPrimitive.Overlay
    className={cn(
      "fixed inset-0 z-50 bg-black/80 data-[state=open]:animate-in data-[state=closed]:animate-out data-[state=closed]:fade-
out-0 data-[state=open]:fade-in-0",
      className
    )}
    {...props}
    ref={ref}
  />
))
SheetOverlay.displayName = SheetPrimitive.Overlay.displayName

const sheetVariants = cva(
  "fixed z-50 gap-4 bg-background p-6 shadow-lg transition ease-in-out data-[state=closed]:duration-300 data-
[state=open]:duration-500 data-[state=open]:animate-in data-[state=closed]:animate-out",
  {
    variants: {
      side: {
        top: "inset-x-0 top-0 border-b data-[state=closed]:slide-out-to-top data-[state=open]:slide-in-from-top",
        bottom:
          "inset-x-0 bottom-0 border-t data-[state=closed]:slide-out-to-bottom data-[state=open]:slide-in-from-bottom",
        left: "inset-y-0 left-0 h-full w-3/4 border-r data-[state=closed]:slide-out-to-left data-[state=open]:slide-in-from-
left sm:max-w-sm",
        right:
          "inset-y-0 right-0 h-full w-3/4 border-l data-[state=closed]:slide-out-to-right data-[state=open]:slide-in-from-right
sm:max-w-sm",
      },
      defaultVariants: {
        side: "right",
      },
    },
  }
)

interface SheetContentProps
  extends React.ComponentPropsWithoutRef<typeof SheetPrimitive.Content>,
    VariantProps<typeof sheetVariants> {}

const SheetContent = React.forwardRef<
  React.ElementRef<typeof SheetPrimitive.Content>,
  SheetContentProps
>(({ side = "right", className, children, ...props }, ref) => (
  <SheetPortal>
    <SheetOverlay />
    <SheetPrimitive.Content
      ref={ref}
      className={cn(sheetVariants({ side }), className)}
      {...props}
    >
      <SheetPrimitive.Close className="absolute right-4 top-4 rounded-sm opacity-70 ring-offset-background transition-opacity
hover:opacity-100 focus:outline-none focus:ring-2 focus:ring-ring focus:ring-offset-2 disabled:pointer-events-none data-
[state=open]:bg-secondary">
        <X className="h-4 w-4" />
        <span className="sr-only">Close</span>
      </SheetPrimitive.Close>
      {children}
    </SheetPrimitive.Content>
  </SheetPortal>
))
SheetContent.displayName = SheetPrimitive.Content.displayName

const SheetHeader = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn(
      "flex flex-col space-y-2 text-center sm:text-left",
      className
    )}
    {...props}
  />
)
SheetHeader.displayName = "SheetHeader"
```

```

const SheetFooter = ({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) => (
  <div
    className={cn(
      "flex flex-col-reverse sm:flex-row sm:justify-end sm:space-x-2",
      className
    )}
    {...props}
  />
)
SheetFooter.displayName = "SheetFooter"

const SheetTitle = React.forwardRef<
  React.ElementRef<typeof SheetPrimitive.Title>,
  React.ComponentPropsWithoutRef<typeof SheetPrimitive.Title>
>(({ className, ...props }, ref) => (
  <SheetPrimitive.Title
    ref={ref}
    className={cn("text-lg font-semibold text-foreground", className)}
    {...props}
  />
))
SheetTitle.displayName = SheetPrimitive.Title.displayName

const SheetDescription = React.forwardRef<
  React.ElementRef<typeof SheetPrimitive.Description>,
  React.ComponentPropsWithoutRef<typeof SheetPrimitive.Description>
>(({ className, ...props }, ref) => (
  <SheetPrimitive.Description
    ref={ref}
    className={cn("text-sm text-muted-foreground", className)}
    {...props}
  />
))
SheetDescription.displayName = SheetPrimitive.Description.displayName

export {
  Sheet,
  SheetPortal,
  SheetOverlay,
  SheetTrigger,
  SheetClose,
  SheetContent,
  SheetHeader,
  SheetFooter,
  SheetTitle,
  SheetDescription,
}

```

[artifacts/mtos-crm/src/components/ui/sidebar.tsx](#)

```
"use client"

import * as React from "react"
import { Slot } from "@radix-ui/react-slot"
import { cva, VariantProps } from "class-variance-authority"
import { PanelLeftIcon } from "lucide-react"

import { useIsMobile } from "@hooks/use-mobile"
import { cn } from "@lib/utls"
import { Button } from "@components/ui/button"
import { Input } from "@components/ui/input"
import { Separator } from "@components/ui/separator"
import {
  Sheet,
  SheetContent,
  SheetDescription,
  SheetHeader,
  SheetTitle,
} from "@components/ui/sheet"
import { Skeleton } from "@components/ui/skeleton"
import {
  Tooltip,
  TooltipContent,
  TooltipProvider,
  TooltipTrigger,
} from "@components/ui/tooltip"

const SIDEBAR_COOKIE_NAME = "sidebar_state"
const SIDEBAR_COOKIE_MAX_AGE = 60 * 60 * 24 * 7
const SIDEBAR_WIDTH = "16rem"
const SIDEBAR_WIDTH_MOBILE = "18rem"
const SIDEBAR_WIDTH_ICON = "3rem"
const SIDEBAR_KEYBOARD_SHORTCUT = "b"

type SidebarContextProps = {
  state: "expanded" | "collapsed"
  open: boolean
  setOpen: (open: boolean) => void
  openMobile: boolean
  setOpenMobile: (open: boolean) => void
  isMobile: boolean
  toggleSidebar: () => void
}

const SidebarContext = React.createContext<SidebarContextProps | null>(null)

function useSidebar() {
  const context = React.useContext(SidebarContext)
  if (!context) {
    throw new Error("useSidebar must be used within a SidebarProvider.")
  }

  return context
}

function SidebarProvider({
  defaultOpen = true,
  open: openProp,
  onOpenChange: setOpenProp,
  className,
  style,
  children,
  ...props
}: React.ComponentProps<"div"> & {
  defaultOpen?: boolean
  open?: boolean
  onOpenChange?: (open: boolean) => void
}) {
  const isMobile = useIsMobile()
  const [openMobile, setOpenMobile] = React.useState(false)

  // This is the internal state of the sidebar.
  // We use openProp and setOpenProp for control from outside the component.
  const [_open, _setOpen] = React.useState(defaultOpen)
  const open = openProp ?? _open
  const setOpen = React.useCallback(
    (value: boolean | ((value: boolean) => boolean)) => {
      const openState = typeof value === "function" ? value(open) : value
      if (setOpenProp) {
        setOpenProp(openState)
      } else {
        _setOpen(openState)
      }
    },
    [setOpenProp, open]
  )

  // This sets the cookie to keep the sidebar state.
  document.cookie = `${SIDEBAR_COOKIE_NAME}=${openState}; path=/; max-age=${SIDEBAR_COOKIE_MAX_AGE}`

  // Helper to toggle the sidebar.
  const toggleSidebar = React.useCallback(() => {
    return isMobile ? setOpenMobile((open) => !open) : setOpen((open) => !open)
  }, [isMobile, setOpen, setOpenMobile])
```



```

// Adds a keyboard shortcut to toggle the sidebar.
React.useEffect(() => {
  const handleKeyDown = (event: KeyboardEvent) => {
    if (
      event.key === SIDEBAR_KEYBOARD_SHORTCUT &&
      (event.metaKey || event.ctrlKey)
    ) {
      event.preventDefault()
      toggleSidebar()
    }
  }

  window.addEventListener("keydown", handleKeyDown)
  return () => window.removeEventListener("keydown", handleKeyDown)
}, [toggleSidebar])

// We add a state so that we can do data-state="expanded" or "collapsed".
// This makes it easier to style the sidebar with Tailwind classes.
const state = open ? "expanded" : "collapsed"

const contextValue = React.useMemo<SidebarContextProps>() => ({
  state,
  open,
  setOpen,
  isMobile,
  openMobile,
  setOpenMobile,
  toggleSidebar,
}), [state, open, setOpen, isMobile, openMobile, setOpenMobile, toggleSidebar]

return (
  <SidebarContext.Provider value={contextValue}>
    <TooltipProvider delayDuration={0}>
      <div
        data-slot="sidebar-wrapper"
        style={
          {
            "--sidebar-width": SIDEBAR_WIDTH,
            "--sidebar-width-icon": SIDEBAR_WIDTH_ICON,
            ...style,
          } as React.CSSProperties
        }
        className={cn(
          "group/sidebar-wrapper has-data-[variant=inset]:bg-sidebar flex min-h-svh w-full",
          className
        )}
        {...props}
      >
        {children}
      </div>
    </TooltipProvider>
  </SidebarContext.Provider>
)
}

function Sidebar({
  side = "left",
  variant = "sidebar",
  collapsible = "offcanvas",
  className,
  children,
  ...props
}: React.ComponentProps<"div"> & {
  side?: "left" | "right"
  variant?: "sidebar" | "floating" | "inset"
  collapsible?: "offcanvas" | "icon" | "none"
}) {
  const { isMobile, state, openMobile, setOpenMobile } = useSidebar()

  if (collapsible === "none") {
    return (
      <div
        data-slot="sidebar"
        className={cn(
          "bg-sidebar text-sidebar-foreground flex h-full w-[var(--sidebar-width)] flex-col",
          className
        )}
        {...props}
      >
        {children}
      </div>
    )
  }

  if (isMobile) {
    return (
      <Sheet open={openMobile} onOpenChange={setOpenMobile} {...props}>
        <SheetContent
          data-sidebar="sidebar"
          data-slot="sidebar"
          data-mobile="true"
          className="bg-sidebar text-sidebar-foreground w-[var(--sidebar-width)] p-0 [&button]:hidden"
          style={
            {

```

```

        "--sidebar-width": SIDEBAR_WIDTH_MOBILE,
      } as React.CSSProperties
    }
    side={side}
  >
  <SheetHeader className="sr-only">
    <SheetTitle>Sidebar</SheetTitle>
    <SheetDescription>Displays the mobile sidebar.</SheetDescription>
  </SheetHeader>
  <div className="flex h-full w-full flex-col">{children}</div>
</SheetContent>
</Sheet>
)
}

return (
  <div
    className="group peer text-sidebar-foreground hidden md:block"
    data-state={state}
    data-collapsible={state === "collapsed" ? collapsible : ""}
    data-variant={variant}
    data-side={side}
    data-slot="sidebar"
  >
    { /* This is what handles the sidebar gap on desktop */ }
    <div
      data-slot="sidebar-gap"
      className={cn(
        "relative w-[var(--sidebar-width)] bg-transparent transition-[width] duration-200 ease-linear",
        "group-data-[collapsible=offcanvas]:w-0",
        "group-data-[side=right]:rotate-180",
        variant === "floating" || variant === "inset"
          ? "group-data-[collapsible=icon]:w-[calc(var(--sidebar-width-icon)+var(--spacing-4))]"
          : "group-data-[collapsible=icon]:w-[var(--sidebar-width-icon)]"
      )}
    >
    <div
      data-slot="sidebar-container"
      className={cn(
        "fixed inset-y-0 z-10 hidden h-svh w-[var(--sidebar-width)] transition-[left,right,width] duration-200 ease-linear",
        md:flex,
        side === "left"
          ? "left-0 group-data-[collapsible=offcanvas]:left-[calc(var(--sidebar-width)*-1)]"
          : "right-0 group-data-[collapsible=offcanvas]:right-[calc(var(--sidebar-width)*-1)]",
        // Adjust the padding for floating and inset variants.
        variant === "floating" || variant === "inset"
          ? "p-2 group-data-[collapsible=icon]:w-[calc(var(--sidebar-width-icon)+var(--spacing-4)+2px)]"
          : "group-data-[collapsible=icon]:w-[var(--sidebar-width-icon)] group-data-[side=left]:border-r group-data-[side=right]:border-l",
        className
      )}
    >
    { ...props }
  >
  <div
    data-sidebar="sidebar"
    data-slot="sidebar-inner"
    className="bg-sidebar group-data-[variant=floating]:border-sidebar-border flex h-full w-full flex-col group-data-[variant=floating]:rounded-lg group-data-[variant=floating]:border group-data-[variant=floating]:shadow-sm"
  >
    {children}
  </div>
</div>
)
}

function SidebarTrigger({
  className,
  onClick,
  ...props
}): React.ComponentProps<typeof Button> {
  const { toggleSidebar } = useSidebar()

  return (
    <Button
      data-sidebar="trigger"
      data-slot="sidebar-trigger"
      variant="ghost"
      size="icon"
      className={cn("h-7 w-7", className)}
      onClick={(event) => {
        onClick?.(event)
        toggleSidebar()
      }}
      { ...props }
    >
    <PanelLeftIcon />
    <span className="sr-only">Toggle Sidebar</span>
  </Button>
  )
}

function SidebarRail({ className, ...props }: React.ComponentProps<"button">) {
  const { toggleSidebar } = useSidebar()

  // Note: Tailwind v3.4 doesn't support "in-" selectors. So the rail won't work perfectly.
  return (

```

```

    <button
      data-sidebar="rail"
      data-slot="sidebar-rail"
      aria-label="Toggle Sidebar"
      tabIndex={-1}
      onClick={toggleSidebar}
      title="Toggle Sidebar"
      className={cn(
        "hover:after:bg-sidebar-border absolute inset-y-0 z-20 hidden w-4 -translate-x-1/2 transition-all ease-linear group-
data-[side=left]:-right-4 group-data-[side=right]:left-0 after:absolute after:inset-y-0 after:left-1/2 after:w-[2px] sm:flex",
        "in-data-[side=left]:cursor-w-resize in-data-[side=right]:cursor-e-resize",
        "[[data-side=left][data-state=collapsed]_&]:cursor-e-resize [[data-side=right][data-state=collapsed]_&]:cursor-w-
resize",
        "hover:group-data-[collapsible=offcanvas]:bg-sidebar group-data-[collapsible=offcanvas]:translate-x-0 group-data-
[collapsible=offcanvas]:after:left-full",
        "[[data-side=left][data-collapsible=offcanvas]_&]:-right-2",
        "[[data-side=right][data-collapsible=offcanvas]_&]:-left-2",
        className
      )}
    >{...props}
  </button>
)
}

function SidebarInset({ className, ...props }: React.ComponentProps<"main">) {
  return (
    <main
      data-slot="sidebar-inset"
      className={cn(
        "bg-background relative flex w-full flex-1 flex-col",
        "md:peer-data-[variant=inset]:m-2 md:peer-data-[variant=inset]:ml-0 md:peer-data-[variant=inset]:rounded-xl md:peer-
data-[variant=inset]:shadow-sm md:peer-data-[variant=inset]:peer-data-[state=collapsed]:ml-2",
        className
      )}
    >{...props}
  </main>
)
}

function SidebarInput({
  className,
  ...props
}: React.ComponentProps<typeof Input>) {
  return (
    <Input
      data-slot="sidebar-input"
      data-sidebar="input"
      className={cn("bg-background h-8 w-full shadow-none", className)}
    >{...props}
  </Input>
)
}

function SidebarHeader({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="sidebar-header"
      data-sidebar="header"
      className={cn("flex flex-col gap-2 p-2", className)}
    >{...props}
  </div>
)
}

function SidebarFooter({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="sidebar-footer"
      data-sidebar="footer"
      className={cn("flex flex-col gap-2 p-2", className)}
    >{...props}
  </div>
)
}

function SidebarSeparator({
  className,
  ...props
}: React.ComponentProps<typeof Separator>) {
  return (
    <Separator
      data-slot="sidebar-separator"
      data-sidebar="separator"
      className={cn("bg-sidebar-border mx-2 w-auto", className)}
    >{...props}
  </Separator>
)
}

function SidebarContent({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="sidebar-content"
      data-sidebar="content"
      className={cn(
        "flex min-h-0 flex-1 flex-col gap-2 overflow-auto group-data-[collapsible=icon]:overflow-hidden",
        className
      )}
    >

```

```

    })
    {...props}
  />
)
}

function SidebarGroup({ className, ...props }: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="sidebar-group"
      data-sidebar="group"
      className={cn("relative flex w-full min-w-0 flex-col p-2", className)}
      {...props}
    />
  )
}

function SidebarGroupLabel({
  className,
  asChild = false,
  ...props
}: React.ComponentProps<"div"> & { asChild?: boolean }) {
  const Comp = asChild ? Slot : "div"

  return (
    <Comp
      data-slot="sidebar-group-label"
      data-sidebar="group-label"
      className={cn(
        "text-sidebar-foreground/70 ring-sidebar-ring flex h-8 shrink-0 items-center rounded-md px-2 text-xs font-medium
outline-hidden transition-[margin,opacity] duration-200 ease-linear focus-visible:ring-2 [&svg]:h-4 [&svg]:w-4
[&svg]:shrink-0",
        "group-data-[collapsible=icon]:-mt-8 group-data-[collapsible=icon]:opacity-0",
        className
      )}
      {...props}
    />
  )
}

function SidebarGroupAction({
  className,
  asChild = false,
  ...props
}: React.ComponentProps<"button"> & { asChild?: boolean }) {
  const Comp = asChild ? Slot : "button"

  return (
    <Comp
      data-slot="sidebar-group-action"
      data-sidebar="group-action"
      className={cn(
        "text-sidebar-foreground ring-sidebar-ring hover:bg-sidebar-accent hover:text-sidebar-accent-foreground absolute
top-3.5 right-3 flex aspect-square w-5 items-center justify-center rounded-md p-0 outline-hidden transition-transform focus-
visible:ring-2 [&svg]:size-4 [&svg]:shrink-0",
        // Increases the hit area of the button on mobile.
        "after:absolute after:-inset-2 md:after:hidden",
        "group-data-[collapsible=icon]:hidden",
        className
      )}
      {...props}
    />
  )
}

function SidebarGroupContent({
  className,
  ...props
}: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="sidebar-group-content"
      data-sidebar="group-content"
      className={cn("w-full text-sm", className)}
      {...props}
    />
  )
}

function SidebarMenu({ className, ...props }: React.ComponentProps<"ul">) {
  return (
    <ul
      data-slot="sidebar-menu"
      data-sidebar="menu"
      className={cn("flex w-full min-w-0 flex-col gap-1", className)}
      {...props}
    />
  )
}

function SidebarMenuItem({ className, ...props }: React.ComponentProps<"li">) {
  return (
    <li
      data-slot="sidebar-menu-item"
      data-sidebar="menu-item"
      className={cn("group/menu-item relative", className)}
      {...props}
    />
  )
}

```

```

    )
  }
}

const sidebarMenuButtonVariants = cva(
  "peer/menu-button flex w-full items-center gap-2 overflow-hidden rounded-md p-2 text-left text-sm outline-hidden ring-sidebar-
ring transition-[width,height,padding] hover:bg-sidebar-accent hover:text-sidebar-accent-foreground focus-visible:ring-2
active:bg-sidebar-accent active:text-sidebar-accent-foreground disabled:pointer-events-none disabled:opacity-50 group-has-data-
[sidebar=menu-action]/menu-item:pr-8 aria-disabled:pointer-events-none aria-disabled:opacity-50 data-[active=true]:bg-sidebar-
accent data-[active=true]:font-medium data-[active=true]:text-sidebar-accent-foreground data-[state=open]:hover:bg-sidebar-
accent data-[state=open]:hover:text-sidebar-accent-foreground group-data-[collapsible=icon]:w-8! group-data-
[collapsible=icon]:h-8! group-data-[collapsible=icon]:p-2! [&span:last-child]:truncate [&svg]:size-4 [&svg]:shrink-0",
  {
    variants: {
      variant: {
        default: "hover:bg-sidebar-accent hover:text-sidebar-accent-foreground",
        outline:
          "bg-background shadow-[0_0_1px_hsl(var(--sidebar-border))] hover:bg-sidebar-accent hover:text-sidebar-accent-
foreground hover:shadow-[0_0_1px_hsl(var(--sidebar-accent))]",
      },
      size: {
        default: "h-8 text-sm",
        sm: "h-7 text-xs",
        lg: "h-12 text-sm group-data-[collapsible=icon]:p-0!",
      },
    },
    defaultVariants: {
      variant: "default",
      size: "default",
    },
  },
)

function SidebarMenuButton({
  asChild = false,
  isActive = false,
  variant = "default",
  size = "default",
  tooltip,
  className,
  ...props
}: React.ComponentProps<"button"> & {
  asChild?: boolean
  isActive?: boolean
  tooltip?: string | React.ComponentProps<typeof TooltipContent>
} & VariantProps<typeof sidebarMenuButtonVariants>) {
  const Comp = asChild ? Slot : "button"
  const { isMobile, state } = useSidebar()

  const button = (
    <Comp
      data-slot="sidebar-menu-button"
      data-sidebar="menu-button"
      data-size={size}
      data-active={isActive}
      className={cn(sidebarMenuButtonVariants({ variant, size })), className}
      {...props}
    />
  )

  if (!tooltip) {
    return button
  }

  if (typeof tooltip === "string") {
    tooltip = {
      children: tooltip,
    }
  }

  return (
    <Tooltip>
      <TooltipTrigger asChild>{button}</TooltipTrigger>
      <TooltipContent
        side="right"
        align="center"
        hidden={state !== "collapsed" || isMobile}
        {...tooltip}
      />
    </Tooltip>
  )
}

function SidebarMenuAction({
  className,
  asChild = false,
  showOnHover = false,
  ...props
}: React.ComponentProps<"button"> & {
  asChild?: boolean
  showOnHover?: boolean
}) {
  const Comp = asChild ? Slot : "button"

  return (
    <Comp
      data-slot="sidebar-menu-action"

```

```

    data-sidebar="menu-action"
    className={cn(
      "text-sidebar-foreground ring-sidebar-ring hover:bg-sidebar-accent hover:text-sidebar-accent-foreground peer-hover/menu-
button:text-sidebar-accent-foreground absolute top-1.5 right-1 flex aspect-square w-5 items-center justify-center rounded-md
p-0 outline-hidden transition-transform focus-visible:ring-2 [>svg]:size-4 [>svg]:shrink-0",
      // Increases the hit area of the button on mobile.
      "after:absolute after:-inset-2 md:after:hidden",
      "peer-data-[size=sm]/menu-button:top-1",
      "peer-data-[size=default]/menu-button:top-1.5",
      "peer-data-[size=lg]/menu-button:top-2.5",
      "group-data-[collapsible=icon]:hidden",
      showOnHover &&
      "peer-data-[active=true]/menu-button:text-sidebar-accent-foreground group-focus-within/menu-item:opacity-100 group-
hover/menu-item:opacity-100 data-[state=open]:opacity-100 md:opacity-0",
      className
    )}
    {...props}
  ) />
}

function SidebarMenuBadge({
  className,
  ...props
}: React.ComponentProps<"div">) {
  return (
    <div
      data-slot="sidebar-menu-badge"
      data-sidebar="menu-badge"
      className={cn(
        "text-sidebar-foreground pointer-events-none absolute right-1 flex h-5 min-w-5 items-center justify-center rounded-md
px-1 text-xs font-medium tabular-nums select-none",
        "peer-hover/menu-button:text-sidebar-accent-foreground peer-data-[active=true]/menu-button:text-sidebar-accent-
foreground",
        "peer-data-[size=sm]/menu-button:top-1",
        "peer-data-[size=default]/menu-button:top-1.5",
        "peer-data-[size=lg]/menu-button:top-2.5",
        "group-data-[collapsible=icon]:hidden",
        className
      )}
      {...props}
    />
  )
}

function SidebarMenuSkeleton({
  className,
  showIcon = false,
  ...props
}: React.ComponentProps<"div"> & {
  showIcon?: boolean
}) {
  // Random width between 50 to 90%.
  const width = React.useMemo(() => {
    return `${Math.floor(Math.random() * 40) + 50}%`
  }, [])

  return (
    <div
      data-slot="sidebar-menu-skeleton"
      data-sidebar="menu-skeleton"
      className={cn("flex h-8 items-center gap-2 rounded-md px-2", className)}
      {...props}
    >
      {showIcon && (
        <Skeleton
          className="size-4 rounded-md"
          data-sidebar="menu-skeleton-icon"
        />
      )}
      <Skeleton
        className="h-4 max-w-[var(--skeleton-width)] flex-1"
        data-sidebar="menu-skeleton-text"
        style={
          {
            "--skeleton-width": width,
          } as React.CSSProperties
        }
      />
    </div>
  )
}

function SidebarMenuSub({ className, ...props }: React.ComponentProps<"ul">) {
  return (
    <ul
      data-slot="sidebar-menu-sub"
      data-sidebar="menu-sub"
      className={cn(
        "border-sidebar-border mx-3.5 flex min-w-0 translate-x-px flex-col gap-1 border-l px-2.5 py-0.5",
        "group-data-[collapsible=icon]:hidden",
        className
      )}
      {...props}
    />
  )
}

```

```

function SidebarMenuSubItem({
  className,
  ...props
}): React.ComponentProps<"li"> {
  return (
    <li
      data-slot="sidebar-menu-sub-item"
      data-sidebar="menu-sub-item"
      className={cn("group/menu-sub-item relative", className)}
      {...props}
    />
  )
}

function SidebarMenuSubButton({
  asChild = false,
  size = "md",
  isActive = false,
  className,
  ...props
}): React.ComponentProps<"a"> & {
  asChild?: boolean
  size?: "sm" | "md"
  isActive?: boolean
}) {
  const Comp = asChild ? Slot : "a"

  return (
    <Comp
      data-slot="sidebar-menu-sub-button"
      data-sidebar="menu-sub-button"
      data-size={size}
      data-active={isActive}
      className={cn(
        "text-sidebar-foreground ring-sidebar-ring hover:bg-sidebar-accent hover:text-sidebar-accent-foreground active:bg-
sidebar-accent active:text-sidebar-accent-foreground [&>svg]:text-sidebar-accent-foreground flex h-7 min-w-0 -translate-x-px
items-center gap-2 overflow-hidden rounded-md px-2 outline outline-2 outline-transparent outline-offset-2 focus-visible:ring-2
disabled:pointer-events-none disabled:opacity-50 aria-disabled:pointer-events-none aria-disabled:opacity-50 [&>span:last-
child]:truncate [&>svg]:size-4 [&>svg]:shrink-0",
        "data-[active=true]:bg-sidebar-accent data-[active=true]:text-sidebar-accent-foreground",
        size === "sm" && "text-xs",
        size === "md" && "text-sm",
        "group-data-[collapsible=icon]:hidden",
        className
      )}
      {...props}
    />
  )
}

export {
  Sidebar,
  SidebarContent,
  SidebarFooter,
  SidebarGroup,
  SidebarGroupAction,
  SidebarGroupContent,
  SidebarGroupLabel,
  SidebarHeader,
  SidebarInput,
  SidebarInset,
  SidebarMenu,
  SidebarMenuAction,
  SidebarMenuBadge,
  SidebarMenuButton,
  SidebarMenuItem,
  SidebarMenuSkeleton,
  SidebarMenuSub,
  SidebarMenuSubButton,
  SidebarMenuSubItem,
  SidebarProvider,
  SidebarRail,
  SidebarSeparator,
  SidebarTrigger,
  useSidebar,
}

```

[artifacts/mtos-crm/src/components/ui/skeleton.tsx](#)

```
import { cn } from "@/lib/utils"

function Skeleton({
  className,
  ...props
}: React.HTMLAttributes<HTMLDivElement>) {
  return (
    <div
      className={cn("animate-pulse rounded-md bg-primary/10", className)}
      {...props}
    />
  )
}

export { Skeleton }
```


[artifacts/mtos-crm/src/components/ui/slider.tsx](#)

```
import * as React from "react"
import * as SliderPrimitive from "@radix-ui/react-slider"

import { cn } from "@lib/utils"

const Slider = React.forwardRef<
  React.ElementRef<typeof SliderPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof SliderPrimitive.Root>
>(({ className, ...props }, ref) => (
  <SliderPrimitive.Root
    ref={ref}
    className={cn(
      "relative flex w-full touch-none select-none items-center",
      className
    )}
    {...props}
  >
    <SliderPrimitive.Track className="relative h-1.5 w-full grow overflow-hidden rounded-full bg-primary/20">
      <SliderPrimitive.Range className="absolute h-full bg-primary" />
    </SliderPrimitive.Track>
    <SliderPrimitive.Thumb className="block h-4 w-4 rounded-full border border-primary/50 bg-background shadow transition-colors focus-visible:outline-none focus-visible:ring-1 focus-visible:ring-ring disabled:pointer-events-none disabled:opacity-50" />
  </SliderPrimitive.Root>
))
Slider.displayName = SliderPrimitive.Root.displayName

export { Slider }
```

[artifacts/mtos-crm/src/components/ui/sonner.tsx](#)

```
"use client"

import { useTheme } from "next-themes"
import { Toaster as Sonner } from "sonner"

type ToasterProps = React.ComponentProps<typeof Sonner>

const Toaster = ({ ...props }: ToasterProps) => {
  const { theme = "system" } = useTheme()

  return (
    <Sonner
      theme={theme as ToasterProps["theme"]}
      className="toaster group"
      toastOptions={{
        classNames: {
          toast:
            "group toast group-[.toaster]:bg-background group-[.toaster]:text-foreground group-[.toaster]:border-border group-[.toaster]:shadow-lg",
          description: "group-[.toast]:text-muted-foreground",
          actionButton:
            "group-[.toast]:bg-primary group-[.toast]:text-primary-foreground",
          cancelButton:
            "group-[.toast]:bg-muted group-[.toast]:text-muted-foreground",
        },
      }}
      {...props}
    />
  )
}

export { Toaster }
```

[artifacts/mtos-crm/src/components/ui/spinner.tsx](#)

```
import { Loader2Icon } from "lucide-react"

import { cn } from "@/lib/utils"

function Spinner({ className, ...props }: React.ComponentProps<"svg">) {
  return (
    <Loader2Icon
      role="status"
      aria-label="Loading"
      className={cn("size-4 animate-spin", className)}
      {...props}
    />
  )
}

export { Spinner }
```

[artifacts/mtos-crm/src/components/ui/switch.tsx](#)

```
import * as React from "react"
import * as SwitchPrimitives from "@radix-ui/react-switch"

import { cn } from "@lib/utils"

const Switch = React.forwardRef<
  React.ElementRef<typeof SwitchPrimitives.Root>,
  React.ComponentPropsWithoutRef<typeof SwitchPrimitives.Root>
>(({ className, ...props }, ref) => (
  <SwitchPrimitives.Root
    className={cn(
      "peer inline-flex h-5 w-9 shrink-0 cursor-pointer items-center rounded-full border-2 border-transparent shadow-sm
transition-colors focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-ring focus-visible:ring-offset-2 focus-
visible:ring-offset-background disabled:cursor-not-allowed disabled:opacity-50 data-[state=checked]:bg-primary data-
[state=unchecked]:bg-input",
      className
    )}
    {...props}
    ref={ref}
  >
    <SwitchPrimitives.Thumb
      className={cn(
        "pointer-events-none block h-4 w-4 rounded-full bg-background shadow-lg ring-0 transition-transform data-
[state=checked]:translate-x-4 data-[state=unchecked]:translate-x-0"
      )}
    />
  </SwitchPrimitives.Root>
))
Switch.displayName = SwitchPrimitives.Root.displayName

export { Switch }
```

[artifacts/mtos-crm/src/components/ui/table.tsx](#)

```
import * as React from "react"

import { cn } from "@lib/Utils"

const Table = React.forwardRef<
  HTMLTableElement,
  React.HTMLAttributes<HTMLTableElement>
>(({ className, ...props }, ref) => (
  <div className="relative w-full overflow-auto">
    <table
      ref={ref}
      className={cn("w-full caption-bottom text-sm", className)}
      {...props}
    />
  </div>
))
Table.displayName = "Table"

const TableHeader = React.forwardRef<
  HTMLTableSectionElement,
  React.HTMLAttributes<HTMLTableSectionElement>
>(({ className, ...props }, ref) => (
  <thead ref={ref} className={cn("[&_tr]:border-b", className)} {...props} />
))
TableHeader.displayName = "TableHeader"

const TableBody = React.forwardRef<
  HTMLTableSectionElement,
  React.HTMLAttributes<HTMLTableSectionElement>
>(({ className, ...props }, ref) => (
  <tbody
    ref={ref}
    className={cn("[&_tr:last-child]:border-0", className)}
    {...props}
  />
))
TableBody.displayName = "TableBody"

const TableFooter = React.forwardRef<
  HTMLTableSectionElement,
  React.HTMLAttributes<HTMLTableSectionElement>
>(({ className, ...props }, ref) => (
  <tfoot
    ref={ref}
    className={cn(
      "border-t bg-muted/50 font-medium [&_tr]:last:border-b-0",
      className
    )}
    {...props}
  />
))
TableFooter.displayName = "TableFooter"

const TableRow = React.forwardRef<
  HTMLTableRowElement,
  React.HTMLAttributes<HTMLTableRowElement>
>(({ className, ...props }, ref) => (
  <tr
    ref={ref}
    className={cn(
      "border-b transition-colors hover:bg-muted/50 data-[state=selected]:bg-muted",
      className
    )}
    {...props}
  />
))
TableRow.displayName = "TableRow"

const TableHead = React.forwardRef<
  HTMLTableCellElement,
  React.ThHTMLAttributes<HTMLTableCellElement>
>(({ className, ...props }, ref) => (
  <th
    ref={ref}
    className={cn(
      "h-10 px-2 text-left align-middle font-medium text-muted-foreground [&:has([role=checkbox])]:pr-0 [&[role=checkbox]]:translate-y-[2px]",
      className
    )}
    {...props}
  />
))
TableHead.displayName = "TableHead"

const TableCell = React.forwardRef<
  HTMLTableCellElement,
  React.TdHTMLAttributes<HTMLTableCellElement>
>(({ className, ...props }, ref) => (
  <td
    ref={ref}
    className={cn(
      "p-2 align-middle [&:has([role=checkbox])]:pr-0 [&[role=checkbox]]:translate-y-[2px]",
      className
    )}
    {...props}
  />
))
TableCell.displayName = "TableCell"
```

```

    />
  ))
  TableCell.displayName = "TableCell"

  const TableCaption = React.forwardRef<
    HTMLTableCaptionElement,
    React.HTMLAttributes<HTMLTableCaptionElement>
  >(({ className, ...props }, ref) => (
    <caption
      ref={ref}
      className={cn("mt-4 text-sm text-muted-foreground", className)}
      {...props}
    />
  ))
  TableCaption.displayName = "TableCaption"

  export {
    Table,
    TableHeader,
    TableBody,
    TableFooter,
    TableHead,
    TableRow,
    TableCell,
    TableCaption,
  }

```

[artifacts/mtos-crm/src/components/ui/tabs.tsx](#)

```
import * as React from "react"
import * as TabsPrimitive from "@radix-ui/react-tabs"

import { cn } from "@lib/utils"

const Tabs = TabsPrimitive.Root

const TabsList = React.forwardRef<
  React.ElementRef<typeof TabsPrimitive.List>,
  React.ComponentPropsWithoutRef<typeof TabsPrimitive.List>
>(({ className, ...props }, ref) => (
  <TabsPrimitive.List
    ref={ref}
    className={cn(
      "inline-flex h-9 items-center justify-center rounded-lg bg-muted p-1 text-muted-foreground",
      className
    )}
    {...props}
  />
))
TabsList.displayName = TabsPrimitive.List.displayName

const TabsTrigger = React.forwardRef<
  React.ElementRef<typeof TabsPrimitive.Trigger>,
  React.ComponentPropsWithoutRef<typeof TabsPrimitive.Trigger>
>(({ className, ...props }, ref) => (
  <TabsPrimitive.Trigger
    ref={ref}
    className={cn(
      "inline-flex items-center justify-center whitespace-nowrap rounded-md px-3 py-1 text-sm font-medium ring-offset-
background transition-all focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-ring focus-visible:ring-offset-2
disabled:pointer-events-none disabled:opacity-50 data-[state=active]:bg-background data-[state=active]:text-foreground data-
[state=active]:shadow",
      className
    )}
    {...props}
  />
))
TabsTrigger.displayName = TabsPrimitive.Trigger.displayName

const TabsContent = React.forwardRef<
  React.ElementRef<typeof TabsPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof TabsPrimitive.Content>
>(({ className, ...props }, ref) => (
  <TabsPrimitive.Content
    ref={ref}
    className={cn(
      "mt-2 ring-offset-background focus-visible:outline-none focus-visible:ring-2 focus-visible:ring-ring focus-visible:ring-
offset-2",
      className
    )}
    {...props}
  />
))
TabsContent.displayName = TabsPrimitive.Content.displayName

export { Tabs, TabsList, TabsTrigger, TabsContent }
```

[artifacts/mtos-crm/src/components/ui/textarea.tsx](#)

```
import * as React from "react"

import { cn } from "@lib/utils"

const Textarea = React.forwardRef<
  HTMLTextAreaElement,
  React.ComponentPropsWithoutRef<"textarea">
>(({ className, ...props }, ref) => {
  return (
    <textarea
      className={cn(
        "flex min-h-[60px] w-full rounded-md border border-input bg-transparent px-3 py-2 text-sm shadow-sm placeholder:text-
muted-foreground focus-visible:outline-none focus-visible:ring-1 focus-visible:ring-ring disabled:cursor-not-allowed
disabled:opacity-50",
        className
      )}
      ref={ref}
      {...props}
    />
  )
})
Textarea.displayName = "Textarea"

export { Textarea }
```


[artifacts/mtos-crm/src/components/ui/toast.tsx](#)

```
import * as React from "react"
import * as ToastPrimitives from "@radix-ui/react-toast"
import { cva, type VariantProps } from "class-variance-authority"
import { X } from "lucide-react"

import { cn } from "@/lib/utils"

const ToastProvider = ToastPrimitives.Provider

const ToastViewport = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Viewport>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Viewport>
>(({ className, ...props }, ref) => {
  <ToastPrimitives.Viewport
    ref={ref}
    className={cn(
      "fixed top-0 z-[100] flex max-h-screen w-full flex-col-reverse p-4 sm:bottom-0 sm:right-0 sm:top-auto sm:flex-col md:max-w-[420px]",
      className
    )}
    {...props}
  />
})
ToastViewport.displayName = ToastPrimitives.Viewport.displayName

const toastVariants = cva(
  "group pointer-events-auto relative flex w-full items-center justify-between space-x-4 overflow-hidden rounded-md border p-6 pr-8 shadow-lg transition-all data-[swipe=cancel]:translate-x-0 data-[swipe=end]:translate-x-[var(--radix-toast-swipe-end-x)] data-[swipe=move]:translate-x-[var(--radix-toast-swipe-move-x)] data-[swipe=move]:transition-none data-[state=open]:animate-in data-[state=closed]:animate-out data-[swipe=end]:animate-out data-[state=closed]:fade-out-80 data-[state=closed]:slide-out-to-right-full data-[state=open]:slide-in-from-top-full data-[state=open]:sm:slide-in-from-bottom-full",
  {
    variants: {
      variant: {
        default: "border bg-background text-foreground",
        destructive: "destructive group border-destructive bg-destructive text-destructive-foreground",
      },
    },
    defaultVariants: {
      variant: "default",
    },
  },
)

const Toast = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Root>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Root> &
  VariantProps<typeof toastVariants>
>(({ className, variant, ...props }, ref) => {
  return (
    <ToastPrimitives.Root
      ref={ref}
      className={cn(toastVariants({ variant }), className)}
      {...props}
    />
  )
})
Toast.displayName = ToastPrimitives.Root.displayName

const ToastAction = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Action>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Action>
>(({ className, ...props }, ref) => {
  <ToastPrimitives.Action
    ref={ref}
    className={cn(
      "inline-flex h-8 shrink-0 items-center justify-center rounded-md border bg-transparent px-3 text-sm font-medium ring-offset-background transition-colors hover:bg-secondary focus:outline-none focus:ring-2 focus:ring-ring focus:ring-offset-2 disabled:pointer-events-none disabled:opacity-50 group-[.destructive]:border-muted/40 group-[.destructive]:hover:border-destructive/30 group-[.destructive]:hover:bg-destructive group-[.destructive]:hover:text-destructive-foreground group-[.destructive]:focus:ring-destructive",
      className
    )}
    {...props}
  />
})
ToastAction.displayName = ToastPrimitives.Action.displayName

const ToastClose = React.forwardRef<
  React.ElementRef<typeof ToastPrimitives.Close>,
  React.ComponentPropsWithoutRef<typeof ToastPrimitives.Close>
>(({ className, ...props }, ref) => {
  <ToastPrimitives.Close
    ref={ref}
    className={cn(
      "absolute right-2 top-2 rounded-md p-1 text-foreground/50 opacity-0 transition-opacity hover:text-foreground focus:opacity-100 focus:outline-none focus:ring-2 group-hover:opacity-100 group-[.destructive]:text-red-300 group-[.destructive]:hover:text-red-50 group-[.destructive]:focus:ring-red-400 group-[.destructive]:focus:ring-offset-red-600",
      className
    )}
    toast-close=""
    {...props}
  />
  <X className="h-4 w-4" />
})
```

```

    </ToastPrimitives.Close>
  ))
  ToastClose.displayName = ToastPrimitives.Close.displayName

  const ToastTitle = React.forwardRef<
    React.ElementRef<typeof ToastPrimitives.Title>,
    React.ComponentPropsWithoutRef<typeof ToastPrimitives.Title>
  >(({ className, ...props }, ref) => (
    <ToastPrimitives.Title
      ref={ref}
      className={cn("text-sm font-semibold", className)}
      {...props}
    />
  ))
  ToastTitle.displayName = ToastPrimitives.Title.displayName

  const ToastDescription = React.forwardRef<
    React.ElementRef<typeof ToastPrimitives.Description>,
    React.ComponentPropsWithoutRef<typeof ToastPrimitives.Description>
  >(({ className, ...props }, ref) => (
    <ToastPrimitives.Description
      ref={ref}
      className={cn("text-sm opacity-90", className)}
      {...props}
    />
  ))
  ToastDescription.displayName = ToastPrimitives.Description.displayName

  type ToastProps = React.ComponentPropsWithoutRef<typeof Toast>

  type ToastActionElement = React.ReactElement<typeof ToastAction>

  export {
    type ToastProps,
    type ToastActionElement,
    ToastProvider,
    ToastViewport,
    Toast,
    ToastTitle,
    ToastDescription,
    ToastClose,
    ToastAction,
  }

```

[artifacts/mtos-crm/src/components/ui/toaster.tsx](#)

```
import { useToast } from "@hooks/use-toast"
import {
  Toast,
  ToastClose,
  ToastDescription,
  ToastProvider,
  ToastTitle,
  ToastViewport,
} from "@components/ui/toast"

export function Toaster() {
  const { toasts } = useToast()

  return (
    <ToastProvider>
      {toasts.map(function ({ id, title, description, action, ...props }) {
        return (
          <Toast key={id} {...props}>
            <div className="grid gap-1">
              {title && <ToastTitle>{title}</ToastTitle>}
              {description && (
                <ToastDescription>{description}</ToastDescription>
              )}
            </div>
            {action}
            <ToastClose />
          </Toast>
        )
      })}
    <ToastViewport />
  </ToastProvider>
)
```

[artifacts/mtos-crm/src/components/ui/toggle-group.tsx](#)

```
"use client"

import * as React from "react"
import * as ToggleGroupPrimitive from "@radix-ui/react-toggle-group"
import { type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"
import { toggleVariants } from "@components/ui/toggle"

const ToggleGroupContext = React.createContext<
  VariantProps<typeof toggleVariants>
>({
  size: "default",
  variant: "default",
})

const ToggleGroup = React.forwardRef<
  React.ElementRef<typeof ToggleGroupPrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof ToggleGroupPrimitive.Root> &
  VariantProps<typeof toggleVariants>
>(({ className, variant, size, children, ...props }, ref) => {
  <ToggleGroupPrimitive.Root
    ref={ref}
    className={cn("flex items-center justify-center gap-1", className)}
    {...props}>
    <ToggleGroupContext.Provider value={{ variant, size }}>
      {children}
    </ToggleGroupContext.Provider>
  </ToggleGroupPrimitive.Root>
))

ToggleGroup.displayName = ToggleGroupPrimitive.Root.displayName

const ToggleGroupItem = React.forwardRef<
  React.ElementRef<typeof ToggleGroupPrimitive.Item>,
  React.ComponentPropsWithoutRef<typeof ToggleGroupPrimitive.Item> &
  VariantProps<typeof toggleVariants>
>(({ className, children, variant, size, ...props }, ref) => {
  const context = React.useContext(ToggleGroupContext)

  return (
    <ToggleGroupPrimitive.Item
      ref={ref}
      className={cn(
        toggleVariants({
          variant: context.variant || variant,
          size: context.size || size,
        }),
        className
      )}
      {...props}>
      {children}
    </ToggleGroupPrimitive.Item>
  )
})

ToggleGroupItem.displayName = ToggleGroupPrimitive.Item.displayName

export { ToggleGroup, ToggleGroupItem }
```

[artifacts/mtos-crm/src/components/ui/toggle.tsx](#)

```
import * as React from "react"
import * as TogglePrimitive from "@radix-ui/react-toggle"
import { cva, type VariantProps } from "class-variance-authority"

import { cn } from "@lib/utils"

const toggleVariants = cva(
  "inline-flex items-center justify-center gap-2 rounded-md text-sm font-medium transition-colors hover:bg-muted hover:text-muted-foreground focus-visible:outline-none focus-visible:ring-1 focus-visible:ring-ring disabled:pointer-events-none disabled:opacity-50 data-[state=on]:bg-accent data-[state=on]:text-accent-foreground [&_svg]:pointer-events-none [&_svg]:size-4 [&_svg]:shrink-0",
  {
    variants: {
      variant: {
        default: "bg-transparent",
        outline: "border border-input bg-transparent shadow-sm hover:bg-accent hover:text-accent-foreground",
      },
      size: {
        default: "h-9 px-2 min-w-9",
        sm: "h-8 px-1.5 min-w-8",
        lg: "h-10 px-2.5 min-w-10",
      },
    },
    defaultVariants: {
      variant: "default",
      size: "default",
    },
  },
)

const Toggle = React.forwardRef<
  React.ElementRef<typeof TogglePrimitive.Root>,
  React.ComponentPropsWithoutRef<typeof TogglePrimitive.Root> &
  VariantProps<typeof toggleVariants>
>(({ className, variant, size, ...props }, ref) => {
  <TogglePrimitive.Root
    ref={ref}
    className={cn(toggleVariants({ variant, size, className }))}
    {...props}
  />
))

Toggle.displayName = TogglePrimitive.Root.displayName

export { Toggle, toggleVariants }
```

[artifacts/mtos-crm/src/components/ui/tooltip.tsx](#)

```
"use client"

import * as React from "react"
import * as TooltipPrimitive from "@radix-ui/react-tooltip"

import { cn } from "@lib/utls"

const TooltipProvider = TooltipPrimitive.Provider

const Tooltip = TooltipPrimitive.Root

const TooltipTrigger = TooltipPrimitive.Trigger

const TooltipContent = React.forwardRef<
  React.ElementRef<typeof TooltipPrimitive.Content>,
  React.ComponentPropsWithoutRef<typeof TooltipPrimitive.Content>
>(({ className, sideOffset = 4, ...props }, ref) => (
  <TooltipPrimitive.Portal>
    <TooltipPrimitive.Content
      ref={ref}
      sideOffset={sideOffset}
      className={cn(
        "z-50 overflow-hidden rounded-md bg-primary px-3 py-1.5 text-xs text-primary-foreground animate-in fade-in-0 zoom-in-95
data-[state=closed]:animate-out data-[state=closed]:fade-out-0 data-[state=closed]:zoom-out-95 data-[side=bottom]:slide-in-from-
top-2 data-[side=left]:slide-in-from-right-2 data-[side=right]:slide-in-from-left-2 data-[side=top]:slide-in-from-bottom-2
origin-[--radix-tooltip-content-transform-origin]",
        className
      )}
      {...props}
    />
  </TooltipPrimitive.Portal>
))
TooltipContent.displayName = TooltipPrimitive.Content.displayName

export { Tooltip, TooltipTrigger, TooltipContent, TooltipProvider }
```

[artifacts/mtos-crm/src/hooks/use-mobile.tsx](#)

```
import * as React from "react"

const MOBILE_BREAKPOINT = 768

export function useIsMobile() {
  const [isMobile, setIsMobile] = React.useState<boolean | undefined>(undefined)

  React.useEffect(() => {
    const mql = window.matchMedia(`(max-width: ${MOBILE_BREAKPOINT - 1}px)`)
    const onChange = () => {
      setIsMobile(window.innerWidth < MOBILE_BREAKPOINT)
    }
    mql.addEventListener("change", onChange)
    setIsMobile(window.innerWidth < MOBILE_BREAKPOINT)
    return () => mql.removeEventListener("change", onChange)
  }, [])

  return !isMobile
}
```

[artifacts/mtos-crm/src/hooks/use-toast.ts](#)

```
import * as React from "react"

import type {
  ToastActionElement,
  ToastProps,
} from "@components/ui/toast"

const TOAST_LIMIT = 1
const TOAST_REMOVE_DELAY = 1000000

type ToasterToast = ToastProps & {
  id: string
  title?: React.ReactNode
  description?: React.ReactNode
  action?: ToastActionElement
}

const actionTypes = {
  ADD_TOAST: "ADD_TOAST",
  UPDATE_TOAST: "UPDATE_TOAST",
  DISMISS_TOAST: "DISMISS_TOAST",
  REMOVE_TOAST: "REMOVE_TOAST",
} as const

let count = 0

function genId() {
  count = (count + 1) % Number.MAX_SAFE_INTEGER
  return count.toString()
}

type ActionType = typeof actionTypes

type Action =
  | {
    type: ActionType["ADD_TOAST"]
    toast: ToasterToast
  }
  | {
    type: ActionType["UPDATE_TOAST"]
    toast: Partial<ToasterToast>
  }
  | {
    type: ActionType["DISMISS_TOAST"]
    toastId?: ToasterToast["id"]
  }
  | {
    type: ActionType["REMOVE_TOAST"]
    toastId?: ToasterToast["id"]
  }

interface State {
  toasts: ToasterToast[]
}

const toastTimeouts = new Map<string, ReturnType<typeof setTimeout>>()

const addToRemoveQueue = (toastId: string) => {
  if (toastTimeouts.has(toastId)) {
    return
  }

  const timeout = setTimeout(() => {
    toastTimeouts.delete(toastId)
    dispatch({
      type: "REMOVE_TOAST",
      toastId: toastId,
    })
  }, TOAST_REMOVE_DELAY)

  toastTimeouts.set(toastId, timeout)
}

export const reducer = (state: State, action: Action): State => {
  switch (action.type) {
    case "ADD_TOAST":
      return {
        ...state,
        toasts: [action.toast, ...state.toasts].slice(0, TOAST_LIMIT),
      }

    case "UPDATE_TOAST":
      return {
        ...state,
        toasts: state.toasts.map((t) =>
          t.id === action.toast.id ? { ...t, ...action.toast } : t
        ),
      }

    case "DISMISS_TOAST": {
      const { toastId } = action

      // ! Side effects ! - This could be extracted into a dismissToast() action,
      // but I'll keep it here for simplicity
      if (toastId) {
        addToRemoveQueue(toastId)
      } else {
        return {
          ...state,
          toasts: state.toasts.filter(
            (t) => t.id !== toastId
          ),
        }
      }
    }

    case "REMOVE_TOAST":
      if (toastId) {
        addToRemoveQueue(toastId)
      } else {
        return {
          ...state,
          toasts: state.toasts.filter(
            (t) => t.id !== toastId
          ),
        }
      }
  }
}
```



```

        addToRemoveQueue(toastId)
      } else {
        state.toasts.forEach((toast) => {
          addToRemoveQueue(toast.id)
        })
      }
    }
    return {
      ...state,
      toasts: state.toasts.map((t) =>
        t.id === toastId || toastId === undefined
          ? {
              ...t,
              open: false,
            }
          : t
        ),
    }
  }
}
case "REMOVE_TOAST":
  if (action.toastId === undefined) {
    return {
      ...state,
      toasts: [],
    }
  }
  return {
    ...state,
    toasts: state.toasts.filter((t) => t.id !== action.toastId),
  }
}
}

const listeners: Array<(state: State) => void> = []

let memoryState: State = { toasts: [] }

function dispatch(action: Action) {
  memoryState = reducer(memoryState, action)
  listeners.forEach((listener) => {
    listener(memoryState)
  })
}

type Toast = Omit<ToasterToast, "id">

function toast({ ...props }: Toast) {
  const id = genId()

  const update = (props: ToasterToast) =>
    dispatch({
      type: "UPDATE_TOAST",
      toast: { ...props, id },
    })
  const dismiss = () => dispatch({ type: "DISMISS_TOAST", toastId: id })

  dispatch({
    type: "ADD_TOAST",
    toast: {
      ...props,
      id,
      open: true,
      onOpenChange: (open) => {
        if (!open) dismiss()
      },
    },
  })

  return {
    id: id,
    dismiss,
    update,
  }
}

function useToast() {
  const [state, setState] = React.useState<State>(memoryState)

  React.useEffect(() => {
    listeners.push(setState)
    return () => {
      const index = listeners.indexOf(setState)
      if (index > -1) {
        listeners.splice(index, 1)
      }
    }
  }, [state])

  return {
    ...state,
    toast,
    dismiss: (toastId?: string) => dispatch({ type: "DISMISS_TOAST", toastId }),
  }
}

export { useToast, toast }

```

<artifacts/mtos-crm/src/index.css>

```
@import "tailwindcss";
@import "tw-animate-css";
@plugin "@tailwindcss/typography";

@custom-variant dark (&:is(.dark *));

@theme inline {
  --color-background: hsl(var(--background));
  --color-foreground: hsl(var(--foreground));
  --color-border: hsl(var(--border));
  --color-input: hsl(var(--input));
  --color-ring: hsl(var(--ring));

  --color-card: hsl(var(--card));
  --color-card-foreground: hsl(var(--card-foreground));
  --color-card-border: hsl(var(--card-border));

  --color-popover: hsl(var(--popover));
  --color-popover-foreground: hsl(var(--popover-foreground));
  --color-popover-border: hsl(var(--popover-border));

  --color-primary: hsl(var(--primary));
  --color-primary-foreground: hsl(var(--primary-foreground));
  --color-primary-border: var(--primary-border);

  --color-secondary: hsl(var(--secondary));
  --color-secondary-foreground: hsl(var(--secondary-foreground));
  --color-secondary-border: var(--secondary-border);

  --color-muted: hsl(var(--muted));
  --color-muted-foreground: hsl(var(--muted-foreground));
  --color-muted-border: var(--muted-border);

  --color-accent: hsl(var(--accent));
  --color-accent-foreground: hsl(var(--accent-foreground));
  --color-accent-border: var(--accent-border);

  --color-destructive: hsl(var(--destructive));
  --color-destructive-foreground: hsl(var(--destructive-foreground));
  --color-destructive-border: var(--destructive-border);

  --color-chart-1: hsl(var(--chart-1));
  --color-chart-2: hsl(var(--chart-2));
  --color-chart-3: hsl(var(--chart-3));
  --color-chart-4: hsl(var(--chart-4));
  --color-chart-5: hsl(var(--chart-5));

  --color-sidebar: hsl(var(--sidebar));
  --color-sidebar-foreground: hsl(var(--sidebar-foreground));
  --color-sidebar-border: hsl(var(--sidebar-border));
  --color-sidebar-primary: hsl(var(--sidebar-primary));
  --color-sidebar-primary-foreground: hsl(var(--sidebar-primary-foreground));
  --color-sidebar-primary-border: var(--sidebar-primary-border);
  --color-sidebar-accent: hsl(var(--sidebar-accent));
  --color-sidebar-accent-foreground: hsl(var(--sidebar-accent-foreground));
  --color-sidebar-accent-border: var(--sidebar-accent-border);
  --color-sidebar-ring: hsl(var(--sidebar-ring));

  --font-sans: var(--app-font-sans);
  --font-serif: var(--app-font-serif);
  --font-mono: var(--app-font-mono);

  --radius-sm: calc(var(--radius) - 4px);
  --radius-md: calc(var(--radius) - 2px);
  --radius-lg: var(--radius);
  --radius-xl: calc(var(--radius) + 4px);
}

:root {
  --background: 220 14% 96%;
  --foreground: 220 10% 10%;
  --border: 220 13% 85%;
  --input: 220 13% 85%;
  --ring: 200 90% 40%;

  --card: 0 0% 100%;
  --card-foreground: 220 10% 10%;
  --card-border: 220 13% 85%;

  --popover: 0 0% 100%;
  --popover-foreground: 220 10% 10%;
  --popover-border: 220 13% 85%;

  --primary: 200 90% 40%;
  --primary-foreground: 0 0% 100%;

  --secondary: 220 14% 90%;
  --secondary-foreground: 220 10% 20%;

  --muted: 220 14% 90%;
  --muted-foreground: 220 10% 40%;

  --accent: 200 90% 90%;
  --accent-foreground: 200 90% 30%;
```

```

--destructive: 0 84% 60%;
--destructive-foreground: 0 0% 100%;

--chart-1: 200 90% 40%;
--chart-2: 150 80% 40%;
--chart-3: 35 90% 50%;
--chart-4: 280 70% 50%;
--chart-5: 0 80% 60%;

--sidebar: 220 14% 96%;
--sidebar-foreground: 220 10% 20%;
--sidebar-border: 220 13% 85%;
--sidebar-primary: 200 90% 40%;
--sidebar-primary-foreground: 0 0% 100%;
--sidebar-accent: 220 14% 90%;
--sidebar-accent-foreground: 220 10% 10%;
--sidebar-ring: 200 90% 40%;

--app-font-sans: "Inter", sans-serif;
--app-font-serif: Georgia, serif;
--app-font-mono: "JetBrains Mono", monospace;
--radius: 0.25rem;
}

.dark {
--background: 225 15% 6%;
--foreground: 210 20% 90%;
--border: 220 15% 15%;
--input: 220 15% 15%;
--ring: 195 90% 50%;

--card: 225 15% 8%;
--card-foreground: 210 20% 90%;
--card-border: 220 15% 15%;

--popover: 225 15% 8%;
--popover-foreground: 210 20% 90%;
--popover-border: 220 15% 15%;

--primary: 195 90% 50%;
--primary-foreground: 225 15% 6%;

--secondary: 220 15% 12%;
--secondary-foreground: 210 20% 80%;

--muted: 220 15% 12%;
--muted-foreground: 210 15% 60%;

--accent: 195 90% 15%;
--accent-foreground: 195 90% 60%;

--destructive: 0 70% 50%;
--destructive-foreground: 0 0% 100%;

--chart-1: 195 90% 50%;
--chart-2: 150 80% 45%;
--chart-3: 35 90% 55%;
--chart-4: 280 70% 60%;
--chart-5: 0 80% 60%;

--sidebar: 225 15% 6%;
--sidebar-foreground: 210 20% 80%;
--sidebar-border: 220 15% 15%;
--sidebar-primary: 195 90% 50%;
--sidebar-primary-foreground: 225 15% 6%;
--sidebar-accent: 220 15% 12%;
--sidebar-accent-foreground: 210 20% 90%;
--sidebar-ring: 195 90% 50%;
}

@layer base {
* {
  @apply border-border;
}
body {
  @apply font-sans antialiased bg-background text-foreground;
}
}

```

[artifacts/mtos-crm/src/lib/utls.ts](#)

```
import { clsx, type ClassValue } from "clsx"
import { twMerge } from "tailwind-merge"

export function cn(...inputs: ClassValue[]) {
  return twMerge(clsx(inputs))
}
```

[artifacts/mtos-crm/src/main.tsx](#)

```
import { createRoot } from "react-dom/client";
import App from "../App";
import "../index.css";

createRoot(document.getElementById("root")!).render(<App />);
```

[artifacts/mtos-crm/src/pages/analytics.tsx](#)

```
import { useGetAnalyticsOverview, useGetConversionFunnel, useGetTortBreakdown, useGetPipelineTrend,
  getGetAnalyticsOverviewQueryKey, getGetConversionFunnelQueryKey, getGetTortBreakdownQueryKey, getGetPipelineTrendQueryKey }
  from "@workspace/api-client-react";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import { Skeleton } from "@components/ui/skeleton";
import { Users, CheckCircle, Percent, TrendingUp } from "lucide-react";
import { BarChart, Bar, XAxis, YAxis, CartesianGrid, Tooltip, ResponsiveContainer, LineChart, Line, PieChart, Pie, Cell,
  Legend } from "recharts";
import { format } from "date-fns";

export default function Analytics() {
  const { data: overview, isLoading: overviewLoading } = useGetAnalyticsOverview({
    query: { queryKey: getGetAnalyticsOverviewQueryKey() }
  });
  const { data: funnel, isLoading: funnelLoading } = useGetConversionFunnel({
    query: { queryKey: getGetConversionFunnelQueryKey() }
  });
  const { data: torts, isLoading: tortsLoading } = useGetTortBreakdown({
    query: { queryKey: getGetTortBreakdownQueryKey() }
  });
  const { data: trend, isLoading: trendLoading } = useGetPipelineTrend({
    query: { queryKey: getGetPipelineTrendQueryKey() }
  });

  const COLORS = ['hsl(var(--chart-1))', 'hsl(var(--chart-2))', 'hsl(var(--chart-3))', 'hsl(var(--chart-4))', 'hsl(var(--chart-5))'];

  const scoreData = overview?.analysis ? [
    { name: 'Strong', value: Number(overview.analysis.strong_cases) || 0, color: 'hsl(var(--chart-2))' },
    { name: 'Moderate', value: Number(overview.analysis.moderate_cases) || 0, color: 'hsl(var(--chart-3))' },
    { name: 'Weak', value: Number(overview.analysis.weak_cases) || 0, color: 'hsl(var(--chart-4))' },
    { name: 'Disqualified', value: Number(overview.analysis.disqualified_cases) || 0, color: 'hsl(var(--destructive))' }
  ] : [];

  return (
    <div className="space-y-6">
      <div className="flex items-center justify-between">
        <h1 className="text-3xl font-bold tracking-tight">Analytics & ROI</h1>
      </div>

      <div className="grid gap-4 md:grid-cols-2 lg:grid-cols-4">
        <Card>
          <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
            <CardTitle className="text-sm font-medium">Total Leads</CardTitle>
            <Users className="h-4 w-4 text-muted-foreground" />
          </CardHeader>
          <CardContent>
            {overviewLoading ? (
              <Skeleton className="h-10 w-full" />
            ) : (
              <>
                <div className="text-2xl font-bold">{overview?.leads?.total as React.ReactNode || 0}</div>
                <div className="text-xs text-muted-foreground flex gap-2 mt-1">
                  <span className="text-destructive font-medium">{overview?.leads?.hot as React.ReactNode || 0} Hot</span>
                  <span className="text-orange-500">{overview?.leads?.warm as React.ReactNode || 0} Warm</span>
                  <span>{overview?.leads?.cold as React.ReactNode || 0} Cold</span>
                </div>
              </>
            )}
          </CardContent>
        </Card>

        <Card>
          <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
            <CardTitle className="text-sm font-medium">Qualification Rate</CardTitle>
            <CheckCircle className="h-4 w-4 text-muted-foreground" />
          </CardHeader>
          <CardContent>
            {overviewLoading ? (
              <Skeleton className="h-10 w-full" />
            ) : (
              <div className="text-2xl font-bold">
                {overview?.leads?.qualification_rate ? `${Number(overview.leads.qualification_rate).toFixed(1)}%` : '0%'}
              </div>
            )}
          </CardContent>
        </Card>

        <Card>
          <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
            <CardTitle className="text-sm font-medium">Conversion Rate</CardTitle>
            <TrendingUp className="h-4 w-4 text-muted-foreground" />
          </CardHeader>
          <CardContent>
            {overviewLoading ? (
              <Skeleton className="h-10 w-full" />
            ) : (
              <div className="text-2xl font-bold">
                {overview?.leads?.conversion_rate ? `${Number(overview.leads.conversion_rate).toFixed(1)}%` : '0%'}
              </div>
            )}
          </CardContent>
        </Card>
      </div>
    </div>
  );
}
```

```

<CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
  <CardTitle className="text-sm font-medium">ROI %</CardTitle>
  <Percent className="h-4 w-4 text-muted-foreground" />
</CardHeader>
<CardContent>
  {overviewLoading ? (
    <Skeleton className="h-10 w-full" />
  ) : (
    <>
      <div className="text-2xl font-bold text-primary">
        {overview?.leads?.roi ? `${Number(overview.leads.roi).toFixed(1)}%` : 'N/A'}
      </div>
      <p className="text-xs text-muted-foreground mt-1">
        CPSR: {overview?.leads?.cpsr ? `${Number(overview.leads.cpsr).toFixed(2)}` : 'N/A'}
      </p>
    </>
  )}
</CardContent>
</Card>
</div>

<Card>
  <CardHeader>
    <CardTitle>Conversion Funnel</CardTitle>
  </CardHeader>
  <CardContent>
    {funnelLoading ? (
      <Skeleton className="h-[200px] w-full" />
    ) : (
      <div className="h-[200px] w-full">
        <ResponsiveContainer width="100%" height="100%">
          <BarChart data={funnel || []} layout="vertical" margin={{ top: 5, right: 30, left: 40, bottom: 5 }}>
            <CartesianGrid strokeDasharray="3 3" horizontal={false} stroke="hsl(var(--border))" />
            <XAxis type="number" stroke="hsl(var(--muted-foreground))" fontSize={12} tickLine={false} axisLine={false} />
            <YAxis dataKey="stage" type="category" stroke="hsl(var(--foreground))" fontSize={12} tickLine={false}
axisLine={false} width={120} />
            <Tooltip
              cursor={{fill: 'hsl(var(--muted))'}}
              contentStyle={{ backgroundColor: 'hsl(var(--popover))', borderColor: 'hsl(var(--border))' }}
            />
            <Bar dataKey="count" fill="hsl(var(--primary))" radius={[0, 4, 4, 0]}>
              {(funnel || []).map((entry, index) => (
                <Cell key={`cell-${index}`} fill={entry.color || `hsl(var(--chart-${(index % 5) + 1})`)}` />
              ))}
            </Bar>
          </BarChart>
        </ResponsiveContainer>
      </div>
    )}
  </CardContent>
</Card>

<div className="grid gap-4 md:grid-cols-2">
  <Card>
    <CardHeader>
      <CardTitle>Tort Breakdown</CardTitle>
    </CardHeader>
    <CardContent>
      {tortsLoading ? (
        <Skeleton className="h-[300px] w-full" />
      ) : (
        <div className="h-[300px] w-full">
          <ResponsiveContainer width="100%" height="100%">
            <BarChart data={torts || []}>
              <CartesianGrid strokeDasharray="3 3" vertical={false} stroke="hsl(var(--border))" />
              <XAxis dataKey="tort_type" stroke="hsl(var(--muted-foreground))" fontSize={12} tickLine={false}
axisLine={false} />
              <YAxis stroke="hsl(var(--muted-foreground))" fontSize={12} tickLine={false} axisLine={false} />
              <Tooltip
                cursor={{fill: 'hsl(var(--muted))'}}
                contentStyle={{ backgroundColor: 'hsl(var(--popover))', borderColor: 'hsl(var(--border))' }}
              />
              <Legend />
              <Bar dataKey="total" fill="hsl(var(--chart-3))" name="Total Leads" radius={[4, 4, 0, 0]} />
              <Bar dataKey="signed" fill="hsl(var(--chart-1))" name="Signed" radius={[4, 4, 0, 0]} />
            </BarChart>
          </ResponsiveContainer>
        </div>
      )}
    </CardContent>
  </Card>

  <Card>
    <CardHeader>
      <CardTitle>Case Analysis Score Distribution</CardTitle>
    </CardHeader>
    <CardContent>
      {overviewLoading ? (
        <Skeleton className="h-[300px] w-full" />
      ) : (
        <div className="h-[300px] w-full">
          <ResponsiveContainer width="100%" height="100%">
            <PieChart>
              <Pie
                data={scoreData}
                cx="50%"
                cy="50%"
              />
            </PieChart>
          </ResponsiveContainer>
        </div>
      )}
    </CardContent>
  </Card>
</div>

```

```

        innerRadius={60}
        outerRadius={100}
        paddingAngle={5}
        dataKey="value"
        nameKey="name"
      >
        {scoreData.map((entry, index) => (
          <Cell key={`cell-${index}`} fill={entry.color} />
        ))}
      </Pie>
      <Tooltip
        contentStyle={{ backgroundColor: 'hsl(var(--popover))', borderColor: 'hsl(var(--border))' }}
      />
      <Legend />
    </PieChart>
  </ResponsiveContainer>
</div>
))
</CardContent>
</Card>
</div>

<Card>
  <CardHeader>
    <CardTitle>Pipeline Trend</CardTitle>
  </CardHeader>
  <CardContent>
    {trendLoading ? (
      <Skeleton className="h-[300px] w-full" />
    ) : (
      <div className="h-[300px] w-full">
        <ResponsiveContainer width="100%" height="100%">
          <LineChart data={trend || []}>
            <CartesianGrid strokeDasharray="3 3" vertical={false} stroke="hsl(var(--border))" />
            <XAxis
              dataKey="date"
              stroke="hsl(var(--muted-foreground))"
              fontSize={12}
              tickLine={false}
              axisLine={false}
              tickFormatter={(value) => format(new Date(value), "MMM d")}
            />
            <YAxis stroke="hsl(var(--muted-foreground))" fontSize={12} tickLine={false} axisLine={false} />
            <Tooltip
              contentStyle={{ backgroundColor: 'hsl(var(--popover))', borderColor: 'hsl(var(--border))' }}
              labelFormatter={(value) => format(new Date(value), "MMM d, yyyy")}
            />
            <Legend />
            <Line type="monotone" dataKey="total" stroke="hsl(var(--chart-3))" strokeWidth={2} dot={false} name="Total
Leads" />
            <Line type="monotone" dataKey="qualified" stroke="hsl(var(--chart-2))" strokeWidth={2} dot={false}
name="Qualified" />
            <Line type="monotone" dataKey="signed" stroke="hsl(var(--chart-1))" strokeWidth={2} dot={false}
name="Signed" />
          </LineChart>
        </ResponsiveContainer>
      </div>
    )}
  </CardContent>
</Card>
</div>
);
}

function CellWrapper(props: any) {
  return <path {...props} />;
}

```


[artifacts/mtos-crm/src/pages/buyers.tsx](#)

```
import { useState, useEffect } from "react";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { Label } from "@components/ui/label";
import { Switch } from "@components/ui/switch";
import { Textarea } from "@components/ui/textarea";
import { Badge } from "@components/ui/badge";
import { Dialog, DialogContent, DialogHeader, DialogTitle, DialogFooter } from "@components/ui/dialog";
import { Table, TableBody, TableCell, TableHead, TableHeader, TableRow } from "@components/ui/table";
import { useToast } from "@hooks/use-toast";
import { Building2, Plus, Pencil, Trash2 } from "lucide-react";

interface Buyer {
  id: number;
  name: string;
  legal_name: string | null;
  contact_name: string | null;
  contact_email: string | null;
  contact_phone: string | null;
  notes: string | null;
  esign_provider_integration_id: number | null;
  fax_provider_integration_id: number | null;
  email_provider_integration_id: number | null;
  default_email_from_name: string | null;
  default_email_from_address: string | null;
  default_fax_from_number: string | null;
  active: boolean;
}

interface ProviderOption { id: number; name: string; provider: string; adapter_implemented: boolean }
interface ProviderOptions { esign: ProviderOption[]; fax: ProviderOption[]; email: ProviderOption[] }

const EMPTY: Partial<Buyer> = { name: "", active: true };

export default function BuyersPage() {
  const { toast } = useToast();
  const [buyers, setBuyers] = useState<Buyer[]>([]);
  const [options, setOptions] = useState<ProviderOptions>({ esign: [], fax: [], email: [] });
  const [loading, setLoading] = useState(true);
  const [open, setOpen] = useState(false);
  const [editing, setEditing] = useState<Partial<Buyer>>(EMPTY);

  const load = async () => {
    setLoading(true);
    try {
      const [bRes, oRes] = await Promise.all([
        fetch("/api/buyers"),
        fetch("/api/workflow-settings/_options/providers"),
      ]);
      setBuyers(await bRes.json());
      if (oRes.ok) setOptions(await oRes.json());
    } finally {
      setLoading(false);
    }
  };

  useEffect(() => { load(); }, []);

  const save = async () => {
    if (!editing.name?.trim()) {
      toast({ title: "Name is required", variant: "destructive" });
      return;
    }
    const isEdit = typeof editing.id === "number";
    const url = isEdit ? `/api/buyers/${editing.id}` : "/api/buyers";
    const method = isEdit ? "PUT" : "POST";
    const res = await fetch(url, {
      method,
      headers: { "Content-Type": "application/json" },
      body: JSON.stringify(editing),
    });
    if (!res.ok) {
      const body = await res.json().catch(() => ({}));
      toast({ title: "Save failed", description: body.error || `HTTP ${res.status}`, variant: "destructive" });
      return;
    }
    toast({ title: isEdit ? "Buyer updated" : "Buyer created" });
    setOpen(false);
    setEditing(EMPTY);
    load();
  };

  const remove = async (id: number) => {
    if (!confirm("Delete this buyer? Leads with this buyer will not be deleted.)) return;
    const res = await fetch(`/api/buyers/${id}`, { method: "DELETE" });
    if (!res.ok) {
      toast({ title: "Delete failed", variant: "destructive" });
      return;
    }
    toast({ title: "Buyer deleted" });
    load();
  };

  const providerName = (id: number | null, list: ProviderOption[]) => {
    if (!id) return <span className="text-muted-foreground">— inherits global </span>;
  }
}
```

```

const p = list.find((x) => x.id === id);
return p ? p.name : <span className="text-destructive">deleted ({id})</span>;
};

return (
  <div className="p-6 space-y-6">
    <div className="flex items-center justify-between">
      <div>
        <h1 className="text-2xl font-bold flex items-center gap-2">
          <Building2 className="h-6 w-6" /> Buyers
        </h1>
        <p className="text-sm text-muted-foreground">Configure case buyers and per-buyer provider overrides.</p>
      </div>
      <Button onClick={() => { setEditing(EMPTY); setOpen(true); }}>
        <Plus className="mr-2 h-4 w-4" /> New Buyer
      </Button>
    </div>

    <Card>
      <CardHeader><CardTitle>All Buyers ({buyers.length})</CardTitle></CardHeader>
      <CardContent>
        {loading ? (
          <p className="text-muted-foreground">Loading...</p>
        ) : buyers.length === 0 ? (
          <p className="text-sm text-muted-foreground">No buyers yet. Add one to enable per-buyer provider routing.</p>
        ) : (
          <Table>
            <TableHeader>
              <TableRow>
                <TableHead>Name</TableHead>
                <TableHead>Contact</TableHead>
                <TableHead>E-Sign</TableHead>
                <TableHead>Fax</TableHead>
                <TableHead>Email</TableHead>
                <TableHead>Status</TableHead>
                <TableHead className="w-32 text-right">Actions</TableHead>
              </TableRow>
            </TableHeader>
            <TableBody>
              {buyers.map((b) => (
                <TableRow key={b.id}>
                  <TableCell>
                    <div className="font-medium">{b.name}</div>
                    {b.legal_name && <div className="text-xs text-muted-foreground">{b.legal_name}</div>}
                  </TableCell>
                  <TableCell className="text-sm">
                    {b.contact_name || "-"}
                    {b.contact_email && <div className="text-xs text-muted-foreground">{b.contact_email}</div>}
                  </TableCell>
                  <TableCell className="text-sm">{providerName(b.sign_provider_integration_id, options.sign)}</TableCell>
                  <TableCell className="text-sm">{providerName(b.fax_provider_integration_id, options.fax)}</TableCell>
                  <TableCell className="text-sm">{providerName(b.email_provider_integration_id, options.email)}</TableCell>
                  <TableCell>
                    <Badge variant={b.active ? "default" : "secondary"}>{b.active ? "Active" : "Inactive"}</Badge>
                  </TableCell>
                  <TableCell className="text-right">
                    <Button variant="ghost" size="icon" onClick={() => { setEditing(b); setOpen(true); }}>
                      <Pencil className="h-4 w-4" />
                    </Button>
                    <Button variant="ghost" size="icon" onClick={() => remove(b.id)}>
                      <Trash2 className="h-4 w-4" />
                    </Button>
                  </TableCell>
                </TableRow>
              ))}
            </TableBody>
          </Table>
        )}
      </CardContent>
    </Card>

    <Dialog open={open} onOpenChange={setOpen}>
      <DialogContent className="max-w-2xl max-h-[90vh] overflow-y-auto">
        <DialogHeader>
          <DialogTitle>{typeof editing.id === "number" ? "Edit Buyer" : "New Buyer"}</DialogTitle>
        </DialogHeader>
        <div className="space-y-4">
          <div className="grid grid-cols-2 gap-3">
            <div>
              <Label>Name </Label>
              <Input value={editing.name || ""} onChange={(e) => setEditing({ ...editing, name: e.target.value })} />
            </div>
            <div>
              <Label>Legal Name</Label>
              <Input value={editing.legal_name || ""} onChange={(e) => setEditing({ ...editing, legal_name:
e.target.value })} />
            </div>
            <div>
              <Label>Contact Name</Label>
              <Input value={editing.contact_name || ""} onChange={(e) => setEditing({ ...editing, contact_name:
e.target.value })} />
            </div>
            <div>
              <Label>Contact Email</Label>
              <Input value={editing.contact_email || ""} onChange={(e) => setEditing({ ...editing, contact_email:
e.target.value })} />
            </div>
          </div>
        </div>
      </DialogContent>
    </Dialog>
  </div>

```

```

    <div>
      <Label>Contact Phone</Label>
      <Input value={editing.contact_phone || ""} onChange={(e) => setEditing({ ...editing, contact_phone:
e.target.value })} />
    </div>
    <div className="flex items-end gap-2">
      <Switch checked={!editing.active} onChange={v => setEditing({ ...editing, active: v })} />
      <Label>Active</Label>
    </div>
  </div>

  <div className="rounded-md border p-3 space-y-3 bg-muted/30">
    <div className="text-sm font-medium">Provider Overrides (optional – leave blank to inherit global)</div>
    <div className="grid grid-cols-3 gap-3">
      <div>
        <Label className="text-xs">E-Sign</Label>
        <ProviderSelect value={editing.esign_provider_integration_id ?? null} options={options.esign} onChange={(v)
=> setEditing({ ...editing, esign_provider_integration_id: v })} />
      </div>
      <div>
        <Label className="text-xs">Fax</Label>
        <ProviderSelect value={editing.fax_provider_integration_id ?? null} options={options.fax} onChange={(v) =>
setEditing({ ...editing, fax_provider_integration_id: v })} />
      </div>
      <div>
        <Label className="text-xs">Email</Label>
        <ProviderSelect value={editing.email_provider_integration_id ?? null} options={options.email} onChange={(v)
=> setEditing({ ...editing, email_provider_integration_id: v })} />
      </div>
    </div>
    <div className="grid grid-cols-3 gap-3">
      <div>
        <Label className="text-xs">Default Email From Name</Label>
        <Input value={editing.default_email_from_name || ""} onChange={(e) => setEditing({ ...editing,
default_email_from_name: e.target.value })} />
      </div>
      <div>
        <Label className="text-xs">Default Email From Address</Label>
        <Input value={editing.default_email_from_address || ""} onChange={(e) => setEditing({ ...editing,
default_email_from_address: e.target.value })} />
      </div>
      <div>
        <Label className="text-xs">Default Fax From Number</Label>
        <Input value={editing.default_fax_from_number || ""} onChange={(e) => setEditing({ ...editing,
default_fax_from_number: e.target.value })} />
      </div>
    </div>
  </div>

  <div>
    <Label>Notes</Label>
    <Textarea rows={3} value={editing.notes || ""} onChange={(e) => setEditing({ ...editing, notes:
e.target.value })} />
  </div>
</div>
<DialogFooter>
  <Button variant="outline" onClick={() => setOpen(false)}>Cancel</Button>
  <Button onClick={save}>Save</Button>
</DialogFooter>
</DialogContent>
</Dialog>
</div>
);
}

function ProviderSelect({ value, options, onChange }: { value: number | null; options: ProviderOption[]; onChange: (v: number |
null) => void }) {
  return (
    <select
      className="w-full h-9 rounded-md border border-input bg-background px-2 text-sm"
      value={value ?? ""}
      onChange={(e) => onChange(e.target.value ? Number(e.target.value) : null)}
    >
      <option value="">– inherit global –</option>
      {options.map((o) => (
        <option key={o.id} value={o.id}>
          {o.name}{!o.adapter_implemented ? " (no adapter)" : ""}
        </option>
      ))}
    </select>
  );
}

```

[artifacts/mtos-crm/src/pages/case-detail.tsx](#)

```
import { useState } from "react";
import { useLocation, useParams } from "wouter";
import { useForm } from "react-hook-form";
import { zodResolver } from "@hookform/resolvers/zod";
import * as z from "zod";
import { format } from "date-fns";
import {
  ArrowLeft, FileText, Activity, Clock, ShieldAlert,
  CheckCircle, AlertCircle, Upload, BrainCircuit, Search, Loader2
} from "lucide-react";
import {
  useGetCase,
  getGetCaseQueryKey,
  useUploadCaseFile,
  useAnalyzeCaseFiles
} from "@workspace/api-client-react";
import { useToast } from "@hooks/use-toast";

import { Button } from "@components/ui/button";
import { Card, CardContent, CardHeader, CardTitle, CardDescription, CardFooter } from "@components/ui/card";
import { Badge } from "@components/ui/badge";
import { Tabs, TabsContent, TabsList, TabsTrigger } from "@components/ui/tabs";
import { Skeleton } from "@components/ui/skeleton";
import { Input } from "@components/ui/input";
import { Textarea } from "@components/ui/textarea";
import {
  Form,
  FormControl,
  FormField,
  FormItem,
  FormLabel,
  FormMessage,
} from "@components/ui/form";

const uploadSchema = z.object({
  file_name: z.string().min(1, "File name is required"),
  content: z.string().min(1, "Document content is required"),
});

type UploadFormValues = z.infer<typeof uploadSchema>;

function getStatusBadge(status: string) {
  switch (status) {
    case "open":
      return <Badge variant="secondary" className="bg-blue-500/10 text-blue-500">Open</Badge>;
    case "processing":
      return <Badge variant="secondary" className="bg-amber-500/10 text-amber-500"><Loader2 className="w-3 h-3 mr-1 animate-spin" />Processing</Badge>;
    case "analyzed":
      return <Badge variant="secondary" className="bg-green-500/10 text-green-500"><CheckCircle className="w-3 h-3 mr-1" />Analyzed</Badge>;
    case "failed":
      return <Badge variant="destructive" className="bg-red-500/10 text-red-500"><AlertCircle className="w-3 h-3 mr-1" />Failed</Badge>;
    default:
      return <Badge variant="outline">{status}</Badge>;
  }
}

function getScoreColor(score: number) {
  if (score >= 80) return "text-green-500";
  if (score >= 50) return "text-amber-500";
  if (score >= 25) return "text-orange-500";
  return "text-red-500";
}

function getScoreVerdict(score: number) {
  if (score >= 80) return "Strong";
  if (score >= 50) return "Moderate";
  if (score >= 25) return "Weak";
  return "Disqualified";
}

export default function CaseDetail() {
  const params = useParams<{ id: string }>();
  const id = params.id!;
  const [, setLocation] = useLocation();
  const { toast } = useToast();
  const [activeTab, setActiveTab] = useState("overview");

  const { data: caseDetail, isLoading, refetch } = useGetCase(id, {
    query: {
      enabled: !!id,
      refetchInterval: (data: any) => data?.state?.data?.case?.status === 'processing' ? 3000 : false,
    } as any
  });

  const uploadFile = useUploadCaseFile();
  const analyzeFiles = useAnalyzeCaseFiles();

  const form = useForm<UploadFormValues>({
    resolver: zodResolver(uploadSchema),
    defaultValues: {
      file_name: "",
      content: "",
    },
  });
```

```

    },
  });
function onUploadSubmit(data: UploadFormValues) {
  if (!id) return;
  uploadFile.mutate(
    { id, data: { ...data, content_type: "text/plain" } },
    {
      onSuccess: () => {
        toast({ title: "Document Uploaded", description: "The document has been securely stored in the vault." });
        form.reset();
        refetch();
      },
      onError: (err: any) => {
        toast({ title: "Upload Failed", description: err.message, variant: "destructive" });
      }
    }
  );
}

function handleAnalyze() {
  analyzeFiles.mutate(
    { id },
    {
      onSuccess: () => {
        toast({ title: "Analysis Queued", description: "AI extraction job has been queued." });
        refetch();
      },
      onError: (err: any) => {
        toast({ title: "Analysis Failed", description: err.message, variant: "destructive" });
      }
    }
  );
}

if (isLoading) {
  return (
    <div className="flex-1 p-8 space-y-6">
      <Skeleton className="h-8 w-64" />
      <div className="grid grid-cols-3 gap-6">
        <div className="col-span-2 space-y-6">
          <Skeleton className="h-[400px] w-full" />
          <Skeleton className="h-[300px] w-full" />
        </div>
        <Skeleton className="h-[600px] w-full" />
      </div>
    </div>
  );
}

if (!caseDetail) {
  return <div className="p-8 text-center text-muted-foreground">Case not found.</div>;
}

const { case: caseData, documents, analyses, audit_trail } = caseDetail;
const latestAnalysis = analyses && analyses.length > 0 ? analyses[0] : null;

return (
  <div className="flex-1 space-y-6 p-8 pt-6">
    <div className="flex items-center justify-between">
      <div className="flex items-center gap-4">
        <Button variant="ghost" size="icon" onClick={() => setLocation("/cases")}>
          <ArrowLeft className="h-4 w-4" />
        </Button>
        <div>
          <div className="flex items-center gap-3">
            <h2 className="text-3xl font-bold tracking-tight text-foreground">
              {(caseData.data?.patient_name as string) || "Unknown Patient"}
            </h2>
            <getStatusBadge(caseData.status)>
          </div>
          <p className="text-muted-foreground font-mono text-sm mt-1">ID: {caseData.id}</p>
        </div>
      </div>
      <div className="flex gap-2">
        <Button variant="outline" onClick={() => refetch()}>
          Refresh
        </Button>
        <Button onClick={handleAnalyze} disabled={analyzeFiles.isPending || caseData.status === 'processing'}>
          {analyzeFiles.isPending ? <Loader2 className="mr-2 h-4 w-4 animate-spin" /> : <BrainCircuit className="mr-2 h-4
w-4" />}
          Analyze Documents
        </Button>
      </div>
    </div>

    <Tabs value={activeTab} onValueChange={setActiveTab} className="space-y-4">
      <TabsList>
        <TabsTrigger value="overview">Overview</TabsTrigger>
        <TabsTrigger value="documents">Documents ({documents?.length || 0})</TabsTrigger>
        <TabsTrigger value="analysis">AI Analysis</TabsTrigger>
        <TabsTrigger value="audit">Audit Trail</TabsTrigger>
      </TabsList>

      <TabsContent value="overview" className="space-y-6">
        <div className="grid grid-cols-1 md:grid-cols-3 gap-6">
          <Card className="md:col-span-2">

```

```

<CardHeader>
  <CardTitle>Case Metadata</CardTitle>
</CardHeader>
<CardContent>
  <dl className="grid grid-cols-1 sm:grid-cols-2 gap-x-4 gap-y-6">
    <div>
      <dt className="text-sm font-medium text-muted-foreground">Tort Type</dt>
      <dd className="text-sm font-semibold mt-1">{{(caseData.data?.tort_type as string) || "N/A"}}</dd>
    </div>
    <div>
      <dt className="text-sm font-medium text-muted-foreground">Diagnosis</dt>
      <dd className="text-sm font-semibold mt-1">{{(caseData.data?.diagnosis as string) || "N/A"}}</dd>
    </div>
    <div>
      <dt className="text-sm font-medium text-muted-foreground">Exposure Location</dt>
      <dd className="text-sm font-semibold mt-1">{{(caseData.data?.exposure_location as string) || "N/A"}}</dd>
    </div>
    <div>
      <dt className="text-sm font-medium text-muted-foreground">Created At</dt>
      <dd className="text-sm font-semibold mt-1">{{format(new Date(caseData.created_at), "PPpp")}}</dd>
    </div>
    <div className="sm:col-span-2">
      <dt className="text-sm font-medium text-muted-foreground">Notes</dt>
      <dd className="text-sm mt-1 bg-muted/30 p-3 rounded-md min-h-[60px]">
        {{(caseData.data?.notes as string) || "No notes provided."}}
      </dd>
    </div>
  </dl>
</CardContent>
</Card>

<Card>
  <CardHeader>
    <CardTitle>Analysis Verdict</CardTitle>
  </CardHeader>
  <CardContent className="flex flex-col items-center justify-center py-6">
    {{latestAnalysis && latestAnalysis.score != null ? (
      <div className="text-6xl font-black ${getScoreColor(latestAnalysis.score)}">
        {{latestAnalysis.score}}
      </div>
      <div className="mt-2 text-lg font-medium text-muted-foreground uppercase tracking-widest">
        {{getScoreVerdict(latestAnalysis.score)}}
      </div>
    ) : (
      <div className="text-center text-muted-foreground">
        <BrainCircuit className="h-12 w-12 mx-auto mb-4 opacity-20" />
        <p>No analysis available.</p>
      </div>
    )}}
  </CardContent>
</Card>
</div>
</TabsContent>

<TabsContent value="documents" className="space-y-6">
  <div className="grid grid-cols-1 lg:grid-cols-3 gap-6">
    <Card className="lg:col-span-2">
      <CardHeader>
        <CardTitle>Vault Documents</CardTitle>
        <CardDescription>Securely stored documents for this case.</CardDescription>
      </CardHeader>
      <CardContent>
        {{documents && documents.length > 0 ? (
          <div className="space-y-4">
            {{documents.map((doc) => (
              <div key={doc.id} className="flex items-center justify-between p-4 border rounded-lg bg-card">
                <div className="flex items-center gap-3">
                  <FileText className="h-8 w-8 text-primary/70" />
                  <div>
                    <p className="font-medium text-sm">{{doc.file_name}}</p>
                    <p className="text-xs text-muted-foreground font-mono">{{doc.file_hash}}</p>
                  </div>
                </div>
                <div className="text-xs text-muted-foreground">
                  {{format(new Date(doc.created_at), "MMM d, yyyy")}}
                </div>
              </div>
            )}}
          </div>
        ) : (
          <div className="text-center p-12 border border-dashed rounded-lg">
            <FileText className="h-8 w-8 mx-auto mb-3 text-muted-foreground/50" />
            <p className="text-sm text-muted-foreground">No documents uploaded yet.</p>
          </div>
        )}}
      </CardContent>
    </Card>

    <Card>
      <CardHeader>
        <CardTitle>Upload Document</CardTitle>
        <CardDescription>Paste document text for vault storage.</CardDescription>
      </CardHeader>
      <CardContent>
        <Form {...form}>

```

```

<form onSubmit={form.handleSubmit(onUploadSubmit)} className="space-y-4">
  <FormField
    control={form.control}
    name="file_name"
    render={({ field }) => (
      <FormItem>
        <FormLabel>File Name</FormLabel>
        <FormControl>
          <Input placeholder="medical_records.txt" {...field} />
        </FormControl>
        <FormMessage />
      </FormItem>
    )}
  />
  <FormField
    control={form.control}
    name="content"
    render={({ field }) => (
      <FormItem>
        <FormLabel>Document Content</FormLabel>
        <FormControl>
          <Textarea placeholder="Paste raw text here..." className="min-h-[200px] font-mono text-
xs" {...field} />
        </FormControl>
        <FormMessage />
      </FormItem>
    )}
  />
  <Button type="submit" className="w-full" disabled={uploadFile.isPending}>
    {uploadFile.isPending ? <Loader2 className="mr-2 h-4 w-4 animate-spin" /> : <Upload className="mr-2 h-4
w-4" />}
    Upload to Vault
  </Button>
</form>
</Form>
</CardContent>
</Card>
</div>
</TabsContent>

<TabsContent value="analysis" className="space-y-6">
  {latestAnalysis ? (
    <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
      <Card>
        <CardHeader>
          <CardTitle>Extracted Features</CardTitle>
        </CardHeader>
        <CardContent>
          <div className="space-y-4">
            {Object.entries(latestAnalysis.features || {}).map(([key, value]) => (
              <div key={key} className="flex flex-col border-b border-b/50 pb-3 last:border-0 last:pb-0">
                <span className="text-xs font-medium uppercase tracking-wider text-muted-foreground mb-1">
                  {key.replace(/_/g, ' ')}
                </span>
                <span className="text-sm font-medium">
                  {typeof value === 'boolean' ? (value ? 'Yes' : 'No') : String(value)}
                </span>
              </div>
            )
          )}
          </div>
        </CardContent>
      </Card>

      <Card>
        <CardHeader>
          <CardTitle>Analysis Metadata</CardTitle>
        </CardHeader>
        <CardContent>
          <dl className="space-y-4">
            <div>
              <dt className="text-sm font-medium text-muted-foreground">Model</dt>
              <dd className="text-sm font-mono mt-1">{latestAnalysis.ai_model || "Unknown"}</dd>
            </div>
            <div>
              <dt className="text-sm font-medium text-muted-foreground">Context Length</dt>
              <dd className="text-sm mt-1">{latestAnalysis.raw_text_length?.toLocaleString() || 0} characters</dd>
            </div>
            <div>
              <dt className="text-sm font-medium text-muted-foreground">Analyzed On</dt>
              <dd className="text-sm mt-1">{format(new Date(latestAnalysis.created_at), "PPP")}</dd>
            </div>
          </dl>
        </CardContent>
      </Card>
    </div>
  ) : (
    <div className="text-center p-12 border border-dashed rounded-lg">
      <BrainCircuit className="h-8 w-8 mx-auto mb-3 text-muted-foreground/50" />
      <p className="text-sm text-muted-foreground">No analysis results yet.</p>
      <Button variant="outline" className="mt-4" onClick={handleAnalyze} disabled={analyzeFiles.isPending ||
caseData.status === 'processing'}>
        Run Analysis
      </Button>
    </div>
  )}
</TabsContent>

```

```

<TabsContent value="audit" className="space-y-6">
  <Card>
    <CardHeader>
      <CardTitle>Audit Trail</CardTitle>
      <CardDescription>Immutable log of all case activities.</CardDescription>
    </CardHeader>
    <CardContent>
      {audit_trail && audit_trail.length > 0 ? (
        <div className="relative border-l border-border ml-3 space-y-6 pb-4">
          {audit_trail.map((entry) => (
            <div key={entry.id} className="relative pl-6">
              <div className="absolute w-3 h-3 bg-primary rounded-full -left-[6.5px] top-1.5 ring-4 ring-card" />
              <div className="flex flex-col gap-1">
                <div className="flex items-center gap-2">
                  <span className="font-semibold text-sm">{entry.action}</span>
                  <span className="text-xs text-muted-foreground">{format(new Date(entry.occurred_at), "PPpp")}</span>
                </div>
                <div className="text-xs text-muted-foreground font-mono mt-1">
                  IP: {entry.ip_address || "System"}
                </div>
                {entry.details && (
                  <pre className="mt-2 text-xs bg-muted/50 p-2 rounded overflow-x-auto">
                    {JSON.stringify(entry.details, null, 2)}
                  </pre>
                )}
              </div>
            </div>
          ))}
        </div>
      ) : (
        <div className="text-center p-6 text-muted-foreground">No audit logs found.</div>
      )}
    </CardContent>
  </Card>
</TabsContent>
</Tabs>
</div>
);
}

```


[artifacts/mtos-crm/src/pages/case-new.tsx](#)

```
import { useState } from "react";
import { useLocation } from "wouter";
import { useForm } from "react-hook-form";
import { zodResolver } from "@hookform/resolvers/zod";
import * as z from "zod";
import { ArrowLeft, Loader2 } from "lucide-react";
import { useCreateCase } from "@workspace/api-client-react";
import { useToast } from "@hooks/use-toast";

import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { Textarea } from "@components/ui/textarea";
import {
  Form,
  FormControl,
  FormField,
  FormItem,
  FormLabel,
  FormMessage,
} from "@components/ui/form";
import { Card, CardContent, CardHeader, CardTitle, CardDescription } from "@components/ui/card";

const caseSchema = z.object({
  patient_name: z.string().min(1, "Patient name is required"),
  tort_type: z.string().min(1, "Tort type is required"),
  diagnosis: z.string().optional(),
  exposure_location: z.string().optional(),
  notes: z.string().optional(),
});

type CaseFormValues = z.infer<typeof caseSchema>;

export default function CaseNew() {
  const [, setLocation] = useLocation();
  const { toast } = useToast();
  const createCase = useCreateCase();

  const form = useForm<CaseFormValues>({
    resolver: zodResolver(caseSchema),
    defaultValues: {
      patient_name: "",
      tort_type: "",
      diagnosis: "",
      exposure_location: "",
      notes: "",
    },
  });

  function onSubmit(data: CaseFormValues) {
    createCase.mutate(
      { data },
      {
        onSuccess: (res) => {
          toast({
            title: "Case Created",
            description: "The case has been successfully submitted.",
          });
          setLocation(`/cases/${res.case_id}`);
        },
        onError: (error: any) => {
          toast({
            title: "Error",
            description: error.message || "Failed to create case.",
            variant: "destructive",
          });
        },
      },
    );
  }

  return (
    <div className="flex-1 space-y-6 p-8 pt-6 max-w-4xl mx-auto">
      <div className="flex items-center gap-4">
        <Button variant="ghost" size="icon" onClick={() => setLocation("/cases")}>
          <ArrowLeft className="h-4 w-4" />
        </Button>
        <div>
          <h2 className="text-3xl font-bold tracking-tight text-foreground">Submit Case</h2>
          <p className="text-muted-foreground mt-1">Create a new case for distributed processing.</p>
        </div>
      </div>

      <Card>
        <CardHeader>
          <CardTitle>Case Information</CardTitle>
          <CardDescription>Enter the initial metadata for the case.</CardDescription>
        </CardHeader>
        <CardContent>
          <Form {...form}>
            <form onSubmit={form.handleSubmit(onSubmit)} className="space-y-6">
              <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
                <FormField
                  control={form.control}
                  name="patient_name"
                  render={({ field }) => (
```

```

        <FormItem>
          <FormLabel>Patient Name</FormLabel>
          <FormControl>
            <Input placeholder="John Doe" {...field} />
          </FormControl>
          <FormMessage />
        </FormItem>
      )}
    />
  <FormField
    control={form.control}
    name="tort_type"
    render={({ field }) => (
      <FormItem>
        <FormLabel>Tort Type</FormLabel>
        <FormControl>
          <Input placeholder="e.g. Camp Lejeune" {...field} />
        </FormControl>
        <FormMessage />
      </FormItem>
    )}
  />
</div>

<div className="grid grid-cols-1 md:grid-cols-2 gap-6">
  <FormField
    control={form.control}
    name="diagnosis"
    render={({ field }) => (
      <FormItem>
        <FormLabel>Diagnosis (Optional)</FormLabel>
        <FormControl>
          <Input placeholder="e.g. Parkinson's Disease" {...field} />
        </FormControl>
        <FormMessage />
      </FormItem>
    )}
  />
  <FormField
    control={form.control}
    name="exposure_location"
    render={({ field }) => (
      <FormItem>
        <FormLabel>Exposure Location (Optional)</FormLabel>
        <FormControl>
          <Input placeholder="e.g. Base housing" {...field} />
        </FormControl>
        <FormMessage />
      </FormItem>
    )}
  />
</div>

<FormField
  control={form.control}
  name="notes"
  render={({ field }) => (
    <FormItem>
      <FormLabel>Additional Notes</FormLabel>
      <FormControl>
        <Textarea placeholder="Any other relevant details..." className="min-h-[100px]" {...field} />
      </FormControl>
      <FormMessage />
    </FormItem>
  )}
/>

<div className="flex justify-end gap-4">
  <Button type="button" variant="outline" onClick={() => setLocation("/cases")}>
    Cancel
  </Button>
  <Button type="submit" disabled={createCase.isPending}>
    {createCase.isPending && <Loader2 className="mr-2 h-4 w-4 animate-spin" />}
    Submit Case
  </Button>
</div>
</form>
</Form>
</CardContent>
</Card>
</div>
);
}

```

[artifacts/mtos-crm/src/pages/cases.tsx](#)

```
import { useState } from "react";
import { Link, useLocation } from "wouter";
import { useListCases, useGetQueueStats, getGetQueueStatsQueryKey } from "@workspace/api-client-react";
import { format } from "date-fns";
import { Briefcase, AlertCircle, CheckCircle, Clock, Search, Plus } from "lucide-react";
import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import { Badge } from "@components/ui/badge";
import {
  Table,
  TableBody,
  TableCell,
  TableHead,
  TableHeader,
  TableRow,
} from "@components/ui/table";
import { Skeleton } from "@components/ui/skeleton";

function getStatusBadge(status: string) {
  switch (status) {
    case "open":
      return <Badge variant="secondary" className="bg-blue-500/10 text-blue-500 hover:bg-blue-500/20">Open</Badge>;
    case "processing":
      return <Badge variant="secondary" className="bg-amber-500/10 text-amber-500 hover:bg-amber-500/20"><Clock className="w-3 h-3 mr-1" />Processing</Badge>;
    case "analyzed":
      return <Badge variant="secondary" className="bg-green-500/10 text-green-500 hover:bg-green-500/20"><CheckCircle className="w-3 h-3 mr-1" />Analyzed</Badge>;
    case "failed":
      return <Badge variant="destructive" className="bg-red-500/10 text-red-500 hover:bg-red-500/20"><AlertCircle className="w-3 h-3 mr-1" />Failed</Badge>;
    default:
      return <Badge variant="outline">{status}</Badge>;
  }
}

export default function Cases() {
  const [, setLocation] = useLocation();
  const [searchTerm, setSearchTerm] = useState("");

  const { data: cases = [], isLoading } = useListCases();
  const { data: queueStats } = useGetQueueStats({
    query: { refetchInterval: 5000 } as any
  });

  const filteredCases = cases.filter(c =>
    c.id.toLowerCase().includes(searchTerm.toLowerCase()) ||
    (c.data?.patient_name as string)?.toLowerCase().includes(searchTerm.toLowerCase()) ||
    (c.data?.tort_type as string)?.toLowerCase().includes(searchTerm.toLowerCase())
  );

  return (
    <div className="flex-1 space-y-6 p-8 pt-6">
      <div className="flex items-center justify-between">
        <h2 className="text-3xl font-bold tracking-tight text-foreground">Case Pipeline</h2>
        <p className="text-muted-foreground mt-1">Distributed case analysis and processing.</p>
      </div>
      <div className="flex items-center space-x-2">
        <Button onClick={() => setLocation("/cases/new")}>
          <Plus className="mr-2 h-4 w-4" />
          Submit Case
        </Button>
      </div>
      <div>
        <div className="grid gap-4 md:grid-cols-2 lg:grid-cols-4">
          {queueStats ? (
            Object.entries(queueStats).map(([key, count]) => (
              <Card key={key} className="bg-card">
                <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
                  <CardTitle className="text-sm font-medium uppercase tracking-wider text-muted-foreground">
                    {key}
                  </CardTitle>
                  <Briefcase className="h-4 w-4 text-muted-foreground" />
                </CardHeader>
                <CardContent>
                  <div className="text-2xl font-bold">{count}</div>
                </CardContent>
              </Card>
            ))
          ) : (
            Array.from({ length: 4 }).map((_, i) => (
              <Card key={i}>
                <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
                  <Skeleton className="h-4 w-1/2" />
                  <Skeleton className="h-4 w-4" />
                </CardHeader>
                <CardContent>
                  <Skeleton className="h-8 w-1/4" />
                </CardContent>
              </Card>
            ))
          )
        </div>
      </div>
    </div>
  );
}
```

```

</div>
<Card className="border-border">
  <CardHeader className="flex flex-row items-center justify-between space-y-0 py-4">
    <CardTitle>Distributed Cases</CardTitle>
    <div className="relative w-64">
      <Search className="absolute left-2 top-2.5 h-4 w-4 text-muted-foreground" />
      <Input
        placeholder="Search cases..."
        value={searchTerm}
        onChange={(e) => setSearchTerm(e.target.value)}
        className="pl-8"
      />
    </div>
  </CardHeader>
  <CardContent className="p-0">
    <Table>
      <TableHeader className="bg-muted/50">
        <TableRow>
          <TableHead className="w-[120px]">Case ID</TableHead>
          <TableHead>Patient</TableHead>
          <TableHead>Tort Type</TableHead>
          <TableHead>Status</TableHead>
          <TableHead className="text-right">Created</TableHead>
        </TableRow>
      </TableHeader>
      <TableBody>
        {isLoading ? (
          Array.from({ length: 5 }).map((_, i) => (
            <TableRow key={i}>
              <TableCell><Skeleton className="h-4 w-20" /></TableCell>
              <TableCell><Skeleton className="h-4 w-32" /></TableCell>
              <TableCell><Skeleton className="h-4 w-24" /></TableCell>
              <TableCell><Skeleton className="h-6 w-20" /></TableCell>
              <TableCell className="text-right"><Skeleton className="h-4 w-24 ml-auto" /></TableCell>
            </TableRow>
          ))
        ) : filteredCases.length > 0 ? (
          filteredCases.map((caseItem) => (
            <TableRow
              key={caseItem.id}
              className="cursor-pointer hover:bg-muted/50 transition-colors"
              onClick={() => setLocation(`/cases/${caseItem.id}`)}
            >
              <TableCell className="font-mono text-xs text-muted-foreground">
                {caseItem.id.substring(0, 8)}...
              </TableCell>
              <TableCell className="font-medium">
                {(caseItem.data?.patient_name as string) || "Unknown"}
              </TableCell>
              <TableCell>
                {(caseItem.data?.tort_type as string) || "Unknown"}
              </TableCell>
              <TableCell>
                {getStatusBadge(caseItem.status)}
              </TableCell>
              <TableCell className="text-right text-muted-foreground text-sm">
                {format(new Date(caseItem.created_at), "MMM d, yyyy")}
              </TableCell>
            </TableRow>
          ))
        ) : (
          <TableRow>
            <TableCell colspan={5} className="h-24 text-center text-muted-foreground">
              No cases found.
            </TableCell>
          </TableRow>
        )}
      </TableBody>
    </Table>
  </CardContent>
</Card>
</div>
  );
}

```

[artifacts/mtos-crm/src/pages/compliance.tsx](#)

```
import { useState } from "react";
import { useGetAuditTrail, useGetAuditSummary, getGetAuditTrailQueryKey, getGetAuditSummaryQueryKey } from "@workspace/api-client-react";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import { Skeleton } from "@components/ui/skeleton";
import { Select, SelectContent, SelectItem, SelectTrigger, SelectValue } from "@components/ui/select";
import { Button } from "@components/ui/button";
import { Dialog, DialogContent, DialogHeader, DialogTitle, DialogTrigger } from "@components/ui/dialog";
import { format } from "date-fns";
import { Activity, AlertCircle, Clock } from "lucide-react";

export default function Compliance() {
  const [entityTypeFilter, setEntityTypeFilter] = useState<string>("all");
  const [actionFilter, setActionFilter] = useState<string>("all");

  const { data: summary, isLoading: summaryLoading } = useGetAuditSummary({
    query: { queryKey: getGetAuditSummaryQueryKey() }
  });

  const auditParams = {
    limit: 100,
    ...(entityTypeFilter !== "all" ? { entity_type: entityTypeFilter } : {}),
    ...(actionFilter !== "all" ? { action: actionFilter } : {}),
  };

  const { data: trail, isLoading: trailLoading } = useGetAuditTrail(auditParams, {
    query: { queryKey: getGetAuditTrailQueryKey(auditParams) }
  });

  return (
    <div className="space-y-6">
      <div className="flex items-center justify-between">
        <h1 className="text-3xl font-bold tracking-tight">Compliance Audit Trail</h1>
      </div>

      <div className="grid gap-4 md:grid-cols-3">
        <Card>
          <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
            <CardTitle className="text-sm font-medium">Total Events</CardTitle>
            <Activity className="h-4 w-4 text-muted-foreground" />
          </CardHeader>
          <CardContent>
            {summaryLoading ? (
              <Skeleton className="h-8 w-20" />
            ) : (
              <div className="text-2xl font-bold">{summary?.total_events || 0}</div>
            )}
          </CardContent>
        </Card>
        <Card>
          <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
            <CardTitle className="text-sm font-medium">Last 24h</CardTitle>
            <Clock className="h-4 w-4 text-muted-foreground" />
          </CardHeader>
          <CardContent>
            {summaryLoading ? (
              <Skeleton className="h-8 w-20" />
            ) : (
              <div className="text-2xl font-bold">{summary?.last_24h || 0}</div>
            )}
          </CardContent>
        </Card>
        <Card>
          <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
            <CardTitle className="text-sm font-medium">Last 7 Days</CardTitle>
            <AlertCircle className="h-4 w-4 text-muted-foreground" />
          </CardHeader>
          <CardContent>
            {summaryLoading ? (
              <Skeleton className="h-8 w-20" />
            ) : (
              <div className="text-2xl font-bold">{summary?.last_7d || 0}</div>
            )}
          </CardContent>
        </Card>
      </div>

      <Card>
        <CardHeader>
          <div className="flex items-center justify-between">
            <CardTitle>Event Log</CardTitle>
            <div className="flex gap-2">
              <Select value={entityTypeFilter} onChange={setEntityTypeFilter}>
                <SelectTrigger className="w-[180px]">
                  <SelectValue placeholder="Entity Type" />
                </SelectTrigger>
                <SelectContent>
                  <SelectItem value="all">All Entities</SelectItem>
                  {summary?.by_entity?.map((e) => (
                    <SelectItem key={e.entity_type} value={e.entity_type}>
                      {e.entity_type} ({e.count})
                    </SelectItem>
                  ))}
                </SelectContent>
              </Select>
            </div>
          </div>
        </CardHeader>
      </Card>
    </div>
  );
}
```

```

<Select value={actionFilter} onValueChange={setActionFilter}>
  <SelectTrigger className="w-[180px]">
    <SelectValue placeholder="Action" />
  </SelectTrigger>
  <SelectContent>
    <SelectItem value="all">All Actions</SelectItem>
    {summary?.by_action?.map((a) => (
      <SelectItem key={a.action} value={a.action}>
        {a.action} ({a.count})
      </SelectItem>
    ))}
  </SelectContent>
</Select>
</div>
</div>
</CardHeader>
<CardContent>
  {trailLoading ? (
    <div className="space-y-4">
      <Skeleton className="h-10 w-full" />
      <Skeleton className="h-10 w-full" />
      <Skeleton className="h-10 w-full" />
      <Skeleton className="h-10 w-full" />
    </div>
  ) : (
    <div className="overflow-x-auto border rounded-md">
      <table className="w-full text-sm text-left font-mono">
        <thead className="text-xs text-muted-foreground uppercase bg-muted/50 border-b">
          <tr>
            <th className="px-4 py-3">ID</th>
            <th className="px-4 py-3">Occurred At</th>
            <th className="px-4 py-3">Entity Type</th>
            <th className="px-4 py-3">Entity ID</th>
            <th className="px-4 py-3">Action</th>
            <th className="px-4 py-3 text-right">Details</th>
          </tr>
        </thead>
        <tbody>
          {(trail || []).map((row) => (
            <tr key={row.id} className="border-b last:border-0 hover:bg-muted/50 transition-colors">
              <td className="px-4 py-3 font-medium text-muted-foreground">{row.id}</td>
              <td className="px-4 py-3 whitespace-nowrap">
                {format(new Date(row.occurred_at), "yyyy-MM-dd HH:mm:ss")}
              </td>
              <td className="px-4 py-3 text-primary">{row.entity_type}</td>
              <td className="px-4 py-3">{row.entity_id}</td>
              <td className="px-4 py-3">
                <span className="bg-secondary text-secondary-foreground px-2 py-0.5 rounded text-xs">
                  {row.action}
                </span>
              </td>
              <td className="px-4 py-3 text-right">
                <Dialog>
                  <DialogTrigger asChild>
                    <Button variant="ghost" size="sm" className="h-8 text-xs font-mono">
                      View JSON
                    </Button>
                  </DialogTrigger>
                  <DialogContent className="max-w-2xl">
                    <DialogHeader>
                      <DialogTitle>Audit Details</DialogTitle>
                    </DialogHeader>
                    <div className="bg-muted p-4 rounded-md overflow-auto max-h-[60vh]">
                      <pre className="text-xs">
                        {JSON.stringify(row.details || {}, null, 2)}
                      </pre>
                    </div>
                  </DialogContent>
                </Dialog>
              </td>
            </tr>
          ))}
        </tbody>
      </table>
      {(?!trail || trail.length === 0) && (
        <div className="text-center py-6 text-muted-foreground">
          No audit events found for selected filters.
        </div>
      )}
    </div>
  )}
</CardContent>
</Card>
</div>
);
}

```

[artifacts/mtos-crm/src/pages/dashboard.tsx](#)

```
import { useGetDashboardStats, useGetPipelineBreakdown, useGetRecentActivity, useGetParalegalLeaderboard,
getGetDashboardStatsQueryKey, getGetPipelineBreakdownQueryKey, getGetRecentActivityQueryKey,
getGetParalegalLeaderboardQueryKey } from "@workspace/api-client-react";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import { Skeleton } from "@components/ui/skeleton";
import { Activity, DollarSign, Users, CheckCircle, Percent, Flame } from "lucide-react";
import { BarChart, Bar, XAxis, YAxis, CartesianGrid, Tooltip as RechartsTooltip, ResponsiveContainer, PieChart, Pie, Cell,
Legend } from "recharts";
import { format } from "date-fns";

export default function Dashboard() {
  const { data: stats, isLoading: statsLoading } = useGetDashboardStats({
    query: { queryKey: getGetDashboardStatsQueryKey() }
  });
  const { data: pipeline, isLoading: pipelineLoading } = useGetPipelineBreakdown({
    query: { queryKey: getGetPipelineBreakdownQueryKey() }
  });
  const { data: activity, isLoading: activityLoading } = useGetRecentActivity({
    query: { queryKey: getGetRecentActivityQueryKey() }
  });
  const { data: leaderboard, isLoading: leaderboardLoading } = useGetParalegalLeaderboard({
    query: { queryKey: getGetParalegalLeaderboardQueryKey() }
  });

  const COLORS = ['hsl(var(--chart-1))', 'hsl(var(--chart-2))', 'hsl(var(--chart-3))', 'hsl(var(--chart-4))', 'hsl(var(--chart-5))'];

  return (
    <div className="space-y-6">
      <div className="flex items-center justify-between">
        <h1 className="text-3xl font-bold tracking-tight">Command Center</h1>
        <div className="flex items-center gap-2">
          <span className="flex h-2 w-2 rounded-full bg-green-500"></span>
          <span className="text-sm text-muted-foreground font-mono">SYSTEM ONLINE</span>
        </div>
      </div>

      <div className="grid gap-4 md:grid-cols-3 lg:grid-cols-6">
        <StatCard
          title="Total Claimants"
          value={stats?.total_leads}
          icon={Users}
          loading={statsLoading}
        </StatCard>
        <StatCard
          title="Hot Leads"
          value={stats?.new_leads}
          icon={Flame}
          loading={statsLoading}
        </StatCard>
        <StatCard
          title="Qualified"
          value={stats?.qualified_leads}
          icon={CheckCircle}
          loading={statsLoading}
        </StatCard>
        <StatCard
          title="Signed Retainers"
          value={stats?.signed_retainers}
          icon={FileTextIcon}
          loading={statsLoading}
        </StatCard>
        <StatCard
          title="CPSR"
          value={stats?.cpsr ? `$$${stats.cpsr.toFixed(2)}` : 'N/A'}
          icon={DollarSign}
          loading={statsLoading}
        </StatCard>
        <StatCard
          title="ROI %"
          value={stats?.conversion_rate ? `${stats.conversion_rate.toFixed(1)}%` : 'N/A'}
          icon={Percent}
          loading={statsLoading}
        </StatCard>
      </div>

      <div className="grid gap-4 md:grid-cols-2">
        <Card>
          <CardHeader>
            <CardTitle>Pipeline Status</CardTitle>
          </CardHeader>
          <CardContent>
            {pipelineLoading ? (
              <Skeleton className="h-[300px] w-full" />
            ) : (
              <div className="h-[300px] w-full">
                <ResponsiveContainer width="100%" height="100%">
                  <PieChart>
                    <Pie
                      data={pipeline?.by_status || []}
                      cx="50%"
                      cy="50%"
                      innerRadius={60}
                      outerRadius={100}
                      fill="#8884d8"
                    </Pie>
                  </PieChart>
                </div>
              </div>
            )}
          </CardContent>
        </Card>
      </div>
    </div>
  );
}
```

```

paddingAngle={5}
dataKey="count"
nameKey="status"
>
  {(pipeline?.by_status || []).map((entry, index) => (
    <Cell key={`cell-${index}`} fill={COLORS[index % COLORS.length]} />
  ))}
</Pie>
<RechartsTooltip
  contentStyle={{ backgroundColor: 'hsl(var(--popover))', borderColor: 'hsl(var(--border))' }}
/>
<Legend />
</PieChart>
</ResponsiveContainer>
</div>
)}
</CardContent>
</Card>

<Card>
  <CardHeader>
    <CardTitle>Cases by Tort Type</CardTitle>
  </CardHeader>
  <CardContent>
    {pipelineLoading ? (
      <Skeleton className="h-[300px] w-full" />
    ) : (
      <div className="h-[300px] w-full">
        <ResponsiveContainer width="100%" height="100%">
          <BarChart data={pipeline?.by_tort_type || []}>
            <CartesianGrid strokeDasharray="3 3" vertical={false} stroke="hsl(var(--border))" />
            <Axis dataKey="tort_type" stroke="hsl(var(--muted-foreground))" fontSize={12} tickLine={false}
axisLine={false} />
            <YAxis stroke="hsl(var(--muted-foreground))" fontSize={12} tickLine={false} axisLine={false} />
            <RechartsTooltip
              cursor={{fill: 'hsl(var(--muted))'}}
              contentStyle={{ backgroundColor: 'hsl(var(--popover))', borderColor: 'hsl(var(--border))' }}
            />
            <Legend />
            <Bar dataKey="qualified" stackId="a" fill="hsl(var(--chart-2))" name="Qualified" radius={[0, 0, 4, 4]} />
            <Bar dataKey="signed" stackId="a" fill="hsl(var(--chart-1))" name="Signed" radius={[4, 4, 0, 0]} />
          </BarChart>
        </ResponsiveContainer>
      </div>
    )}
  </CardContent>
</Card>
</div>

<div className="grid gap-4 md:grid-cols-2 lg:grid-cols-7">
  <Card className="col-span-4">
    <CardHeader>
      <CardTitle>Paralegal Leaderboard</CardTitle>
    </CardHeader>
    <CardContent>
      {leaderboardLoading ? (
        <div className="space-y-4">
          <Skeleton className="h-10 w-full" />
          <Skeleton className="h-10 w-full" />
          <Skeleton className="h-10 w-full" />
          <Skeleton className="h-10 w-full" />
        </div>
      ) : (
        <div className="overflow-x-auto">
          <table className="w-full text-sm text-left">
            <thead className="text-xs text-muted-foreground uppercase bg-muted/50 border-b">
              <tr>
                <th className="px-4 py-3">Rank</th>
                <th className="px-4 py-3">Name</th>
                <th className="px-4 py-3">Signed</th>
                <th className="px-4 py-3">Qualified</th>
                <th className="px-4 py-3">Conv Rate</th>
              </tr>
            </thead>
            <tbody>
              {(leaderboard || []).slice(0, 4).map((row, idx) => (
                <tr key={row.id} className="border-b last:border-0 hover:bg-muted/50 transition-colors">
                  <td className="px-4 py-3 font-medium">#{idx + 1}</td>
                  <td className="px-4 py-3">
                    <div className="font-medium">{row.name}</div>
                    <div className="text-xs text-muted-foreground">{row.role}</div>
                  </td>
                  <td className="px-4 py-3 font-bold text-primary">{row.signed}</td>
                  <td className="px-4 py-3">{row.qualified}</td>
                  <td className="px-4 py-3">{row.conversion_rate.toFixed(1)}%</td>
                </tr>
              ))}
            </tbody>
          </table>
        </div>
      )}
    </CardContent>
  </Card>

  <Card className="col-span-3">
    <CardHeader>
      <CardTitle>Recent Activity</CardTitle>
    </CardHeader>
  </Card>
</div>

```



```

    </CardHeader>
    <CardContent>
      {activityLoading ? (
        <div className="space-y-4">
          <Skeleton className="h-10 w-full" />
          <Skeleton className="h-10 w-full" />
          <Skeleton className="h-10 w-full" />
        </div>
      ) : (
        <div className="space-y-4 max-h-[300px] overflow-y-auto pr-2">
          {activity?.map((item) => (
            <div key={item.id} className="flex items-start gap-4 text-sm">
              <div className="mt-0.5 rounded-full bg-primary/10 p-1.5 flex-shrink-0">
                <Activity className="h-3 w-3 text-primary" />
              </div>
              <div className="flex-1 space-y-1">
                <p className="font-medium leading-none">{item.lead_name}</p>
                <p className="text-muted-foreground">{item.description}</p>
                <p className="text-xs text-muted-foreground font-mono">
                  {format(new Date(item.occurred_at), "MMM d, h:mm a")}
                </p>
              </div>
            </div>
          ))}
          {(!activity || activity.length === 0) && (
            <p className="text-sm text-muted-foreground text-center py-4">No recent activity.</p>
          )}
        </div>
      )}
    </CardContent>
  </Card>
</div>
</div>
);
}

function StatCard({ title, value, icon: Icon, loading }: { title: string; value?: number | string; icon: any; loading:
boolean }) {
  return (
    <Card>
      <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
        <CardTitle className="text-sm font-medium">{title}</CardTitle>
        <Icon className="h-4 w-4 text-muted-foreground" />
      </CardHeader>
      <CardContent>
        {loading ? (
          <Skeleton className="h-8 w-20" />
        ) : (
          <div className="text-2xl font-bold">{value ?? 0}</div>
        )}
      </CardContent>
    </Card>
  );
}

function FileTextIcon(props: any) {
  return (
    <svg
      {...props}
      xmlns="http://www.w3.org/2000/svg"
      width="24"
      height="24"
      viewBox="0 0 24 24"
      fill="none"
      stroke="currentColor"
      strokeWidth="2"
      strokeLinecap="round"
      strokeLinejoin="round"
    >
      <path d="M15 2H6a2 2 0 0 2 2v16a2 2 0 0 2 2h12a2 2 0 0 2 2v-7Z" />
      <path d="M14 2v4a2 2 0 0 2 2h4" />
      <path d="M10 9H8" />
      <path d="M16 13H8" />
      <path d="M16 17H8" />
    </svg>
  );
}

```

[artifacts/mtos-crm/src/pages/decision-engine-settings.tsx](#)

```
import { useState, useEffect } from "react";
import { useQuery, useMutation, useQueryClient } from "@tanstack/react-query";
import { Card, CardContent, CardHeader, CardTitle, CardDescription } from "@components/ui/card";
import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { Label } from "@components/ui/label";
import { Switch } from "@components/ui/switch";
import { Table, TableBody, TableCell, TableHead, TableHeader, TableRow } from "@components/ui/table";
import { useToast } from "@hooks/use-toast";
import { Trash2, Plus } from "lucide-react";

interface Settings {
  default_attorney_hourly_cost: number;
  default_hours_per_lead: number;
  convex_ratio_threshold: number;
  concave_ratio_threshold: number;
  concentration_warning_pct: number;
  ruin_auto_flag?: boolean;
}

interface Source {
  id: number;
  name: string;
  channel: string;
  cost_per_lead: string | null;
  historical_qualified_rate: string | null;
  historical_retained_rate: string | null;
  active: boolean;
}

export default function DecisionEngineSettings() {
  const qc = useQueryClient();
  const { toast } = useToast();
  const [form, setForm] = useState<Settings | null>(null);
  const [newSource, setNewSource] = useState({ name: "", channel: "other", cost_per_lead: "", historical_qualified_rate: "", historical_retained_rate: "" });

  const { data: settings } = useQuery<Settings>({
    queryKey: ["decision-engine", "settings"],
    queryFn: async () => {
      const res = await fetch("/api/decision-engine/settings", { credentials: "include" });
      if (!res.ok) throw new Error("settings failed");
      return res.json();
    },
  });

  const { data: sources = [] } = useQuery<Source[]>({
    queryKey: ["lead-sources"],
    queryFn: async () => {
      const res = await fetch("/api/lead-sources", { credentials: "include" });
      if (!res.ok) throw new Error("sources failed");
      return res.json();
    },
  });

  useEffect(() => { if (settings) setForm(settings); }, [settings]);

  const saveSettings = useMutation({
    mutationFn: async (body: Settings) => {
      const res = await fetch("/api/decision-engine/settings", {
        method: "PUT", credentials: "include",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify(body),
      });
      if (!res.ok) throw new Error(await res.text());
      return res.json();
    },
    onSuccess: () => { qc.invalidateQueries({ queryKey: ["decision-engine"] }); toast({ title: "Settings saved" }); },
    onError: (e: any) => toast({ title: "Save failed", description: e?.message, variant: "destructive" }),
  });

  const addSource = useMutation({
    mutationFn: async (body: any) => {
      const res = await fetch("/api/lead-sources", {
        method: "POST", credentials: "include",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify(body),
      });
      if (!res.ok) throw new Error(await res.text());
      return res.json();
    },
    onSuccess: () => {
      qc.invalidateQueries({ queryKey: ["lead-sources"] });
      setNewSource({ name: "", channel: "other", cost_per_lead: "", historical_qualified_rate: "", historical_retained_rate: "" });
      toast({ title: "Source added" });
    },
    onError: (e: any) => toast({ title: "Add failed", description: e?.message, variant: "destructive" }),
  });

  const deleteSource = useMutation({
    mutationFn: async (id: number) => {
      const res = await fetch(`/api/lead-sources/${id}`, { method: "DELETE", credentials: "include" });
      if (!res.ok) throw new Error(await res.text());
    },
  });
```

```

    onSuccess: () => { qc.invalidateQueries({ queryKey: ["lead-sources"] }); toast({ title: "Source removed" }); },
  });
}

return (
  <div className="p-6 space-y-6" data-testid="page-decision-engine-settings">
    <div>
      <h1 className="text-2xl font-bold">Decision Engine Settings</h1>
      <p className="text-muted-foreground text-sm">Calibrate thresholds and lead-source economics</p>
    </div>

    <Card>
      <CardHeader>
        <CardTitle>Global thresholds</CardTitle>
        <CardDescription>Used to classify leads as convex / neutral / concave</CardDescription>
      </CardHeader>
      <CardContent className="grid grid-cols-2 gap-4">
        {form && (
          <>
            <div>
              <Label>Default attorney hourly cost ($)</Label>
              <Input type="number" value={form.default_attorney_hourly_cost} onChange={e => setForm({ ...form,
default_attorney_hourly_cost: Number(e.target.value) })} data-testid="input-hourly-cost" />
            </div>
            <div>
              <Label>Default hours per lead</Label>
              <Input type="number" value={form.default_hours_per_lead} onChange={e => setForm({ ...form,
default_hours_per_lead: Number(e.target.value) })} />
            </div>
            <div>
              <Label>Convex ratio threshold ("e")</Label>
              <Input type="number" step="0.1" value={form.convex_ratio_threshold} onChange={e => setForm({ ...form,
convex_ratio_threshold: Number(e.target.value) })} data-testid="input-convex-threshold" />
            </div>
            <div>
              <Label>Concave ratio threshold (&lt;)</Label>
              <Input type="number" step="0.1" value={form.concave_ratio_threshold} onChange={e => setForm({ ...form,
concave_ratio_threshold: Number(e.target.value) })} />
            </div>
            <div>
              <Label>Concentration warning (% of spend)</Label>
              <Input type="number" value={form.concentration_warning_pct} onChange={e => setForm({ ...form,
concentration_warning_pct: Number(e.target.value) })} />
            </div>
            <div className="flex items-center gap-3 pt-6">
              <Switch checked={!!form.ruin_auto_flag} onCheckedChange={v => setForm({ ...form, ruin_auto_flag: v })} id="auto-
flag" />
              <Label htmlFor="auto-flag">Auto-route ruin-flag leads to review queue</Label>
            </div>
            <div className="col-span-2">
              <Button onClick={() => saveSettings.mutate(form)} disabled={saveSettings.isPending} data-testid="button-save-
settings">
                Save settings
              </Button>
            </div>
          </>
        )}
      </CardContent>
    </Card>

    <Card>
      <CardHeader>
        <CardTitle>Lead sources</CardTitle>
        <CardDescription>Cost-per-lead and historical conversion rates feed the convexity calculation</CardDescription>
      </CardHeader>
      <CardContent>
        <Table>
          <TableHeader>
            <TableRow>
              <TableHead>Name</TableHead>
              <TableHead>Channel</TableHead>
              <TableHead>CPL</TableHead>
              <TableHead>Qualified rate</TableHead>
              <TableHead>Retained rate</TableHead>
            </TableRow>
          </TableHeader>
          <TableBody>
            {sources.map(s => (
              <TableRow key={s.id}>
                <TableCell className="font-medium">{s.name}</TableCell>
                <TableCell>{s.channel}</TableCell>
                <TableCell>${s.cost_per_lead || "-"}</TableCell>
                <TableCell>{s.historical_qualified_rate ? `${(Number(s.historical_qualified_rate) * 100).toFixed(0)}%` : "-"}</
TableCell>
                <TableCell>{s.historical_retained_rate ? `${(Number(s.historical_retained_rate) * 100).toFixed(0)}%` : "-"}</
TableCell>
                <TableCell><Button size="sm" variant="ghost" onClick={() => deleteSource.mutate(s.id)} data-testid={`button-
delete-source-${s.id}`}><Trash2 className="w-4 h-4" /></Button></TableCell>
              </TableRow>
            )}
          </TableBody>
        </Table>
      </CardContent>
    </Card>
  )
)

```

```

        <TableCell><Input type="number" step="0.01" placeholder="0.00 - 1.00"
value={newSource.historical_qualified_rate} onChange={e => setNewSource({ ...newSource, historical_qualified_rate:
e.target.value })} /></TableCell>
        <TableCell><Input type="number" step="0.01" placeholder="0.00 - 1.00"
value={newSource.historical_retained_rate} onChange={e => setNewSource({ ...newSource, historical_retained_rate:
e.target.value })} /></TableCell>
        <TableCell>
          <Button size="sm" onClick={() => addSource.mutate({
            name: newSource.name,
            channel: newSource.channel || "other",
            cost_per_lead: newSource.cost_per_lead ? Number(newSource.cost_per_lead) : null,
            historical_qualified_rate: newSource.historical_qualified_rate ?
Number(newSource.historical_qualified_rate) : null,
            historical_retained_rate: newSource.historical_retained_rate ? Number(newSource.historical_retained_rate) :
null,
          })} disabled={!newSource.name || addSource.isPending} data-testid="button-add-source">
            <Plus className="w-4 h-4" />
          </Button>
        </TableCell>
      </TableRow>
    </TableBody>
  </Table>
</CardContent>
</Card>
</div>
);
}

```

[artifacts/mtos-crm/src/pages/decision-engine.tsx](#)

```
import { useQuery, useMutation, useQueryClient } from "@tanstack/react-query";
import { Link } from "wouter";
import { Card, CardContent, CardHeader, CardTitle, CardDescription } from "@components/ui/card";
import { Badge } from "@components/ui/badge";
import { Button } from "@components/ui/button";
import { Alert, AlertDescription, AlertTitle } from "@components/ui/alert";
import { Skeleton } from "@components/ui/skeleton";
import { Table, TableBody, TableCell, TableHead, TableHeader, TableRow } from "@components/ui/table";
import { AlertTriangle, TrendingUp, TrendingDown, Minus, RefreshCw, Settings, ExternalLink } from "lucide-react";
import { useToast } from "@hooks/use-toast";

interface PortfolioRow {
  tort_id: string;
  label: string;
  lead_count: number;
  total_spend_usd: number;
  qualified_count: number;
  retained_count: number;
  qualified_rate: number;
  retained_rate: number;
  classification: "convex" | "neutral" | "concave";
  action: "execute" | "modify" | "reject";
  rationale: string;
  ruin_flag_count: number;
  spend_pct: number;
}

interface PortfolioSummary {
  total_spend_usd: number;
  tort_count: number;
  convex_count: number;
  concave_count: number;
  ruin_risk_lead_count: number;
  concentration_warning: { tort_id: string; pct: number } | null;
  rows: PortfolioRow[];
}

function badgeFor(c: string) {
  if (c === "convex") return <Badge className="bg-emerald-100 text-emerald-800 border-emerald-300"><TrendingUp className="w-3 h-3 mr-1" />Convex</Badge>;
  if (c === "concave") return <Badge className="bg-rose-100 text-rose-800 border-rose-300"><TrendingDown className="w-3 h-3 mr-1" />Concave</Badge>;
  return <Badge variant="secondary"><Minus className="w-3 h-3 mr-1" />Neutral</Badge>;
}

function actionBadge(a: string) {
  if (a === "execute") return <Badge className="bg-blue-100 text-blue-800 border-blue-300">Execute</Badge>;
  if (a === "review") return <Badge className="bg-amber-100 text-amber-800 border-amber-300">Human Review</Badge>;
  if (a === "reject") return <Badge className="bg-rose-100 text-rose-800 border-rose-300">Reject</Badge>;
  return <Badge className="bg-amber-100 text-amber-800 border-amber-300">Modify</Badge>;
}

export default function DecisionEnginePage() {
  const qc = useQueryClient();
  const { toast } = useToast();

  const { data, isLoading, error } = useQuery<PortfolioSummary>({
    queryKey: ["decision-engine", "portfolio"],
    queryFn: async () => {
      const res = await fetch("/api/decision-engine/portfolio", { credentials: "include" });
      if (!res.ok) throw new Error(`portfolio failed: ${res.status}`);
      return res.json();
    },
  });

  const recomputeAll = useMutation({
    mutationFn: async () => {
      const res = await fetch("/api/decision-engine/recompute-all", { method: "POST", credentials: "include" });
      if (!res.ok) throw new Error(await res.text());
      return res.json();
    },
    onSuccess: (r: { scanned: number; scored: number }) => {
      qc.invalidateQueries({ queryKey: ["decision-engine"] });
      toast({ title: "Recompute complete", description: `Scored ${r.scored} of ${r.scanned} leads.` });
    },
    onError: (e: any) => toast({ title: "Recompute failed", description: e?.message, variant: "destructive" }),
  });

  if (error) {
    return (
      <div className="p-6">
        <Alert variant="destructive">
          <AlertTriangle className="h-4 w-4" />
          <AlertTitle>Access denied</AlertTitle>
          <AlertDescription>The Decision Engine portfolio is admin-only.</AlertDescription>
        </Alert>
      </div>
    );
  }

  return (
    <div className="p-6 space-y-6" data-testid="page-decision-engine">
      <div className="flex justify-between items-center">
        <div>
          <h1 className="text-2xl font-bold">Decision Engine</h1>

```

```

    <p className="text-muted-foreground text-sm">Portfolio convexity – asymmetric risk view across all torts</p>
  </div>
  <div className="flex gap-2">
    <Button variant="outline" asChild>
      <Link href="/decision-engine/settings"><Settings className="w-4 h-4 mr-2" />Settings</Link>
    </Button>
    <Button onClick={() => recomputeAll.mutate()} disabled={recomputeAll.isPending} data-testid="button-recompute-all">
      <RefreshCw className={\`w-4 h-4 mr-2 \${recomputeAll.isPending ? "animate-spin" : ""}\`} />
      Recompute All
    </Button>
  </div>
</div>

{isLoading ? (
  <Skeleton className="h-32 w-full" />
) : data ? (
  <>
    <div className="grid grid-cols-1 md:grid-cols-4 gap-4">
      <Card>
        <CardHeader className="pb-2"><CardDescription>Total marketing spend</CardDescription></CardHeader>
        <CardContent><div className="text-2xl font-bold">\${data.total_spend_usd.toLocaleString()}</div></CardContent>
      </Card>
      <Card>
        <CardHeader className="pb-2"><CardDescription>Convex torts</CardDescription></CardHeader>
        <CardContent><div className="text-2xl font-bold text-emerald-600" data-testid="stat-convex-
count">\${data.convex_count}</div></CardContent>
      </Card>
      <Card>
        <CardHeader className="pb-2"><CardDescription>Concave torts</CardDescription></CardHeader>
        <CardContent><div className="text-2xl font-bold text-rose-600" data-testid="stat-concave-
count">\${data.concave_count}</div></CardContent>
      </Card>
      <Card>
        <CardHeader className="pb-2"><CardDescription>Leads with ruin flags</CardDescription></CardHeader>
        <CardContent><div className="text-2xl font-bold text-amber-600" data-testid="stat-ruin-
count">\${data.ruin_lead_count}</div></CardContent>
      </Card>
    </div>

    {data.concentration_warning && (
      <Alert variant="destructive" data-testid="alert-concentration-warning">
        <AlertTriangle className="h-4 w-4" />
        <AlertTitle>Concentration risk</AlertTitle>
        <AlertDescription>
          <strong>\${data.concentration_warning.tort_id}</strong> represents {data.concentration_warning.pct.toFixed(1)}%
of your spend. Framework principle: prefer 10 small bets over 1 large bet.
        </AlertDescription>
      </Alert>
    )}

    <Card>
      <CardHeader>
        <CardTitle>Per-tort portfolio</CardTitle>
        <CardDescription>Sorted by spend. Drill into any tort to see its leads.</CardDescription>
      </CardHeader>
      <CardContent>
        <Table>
          <TableHeader>
            <TableRow>
              <TableHead>Tort</TableHead>
              <TableHead>Leads</TableHead>
              <TableHead>Spend</TableHead>
              <TableHead>Spend %</TableHead>
              <TableHead>Retained</TableHead>
              <TableHead>Ruin</TableHead>
              <TableHead>Classification</TableHead>
              <TableHead>Action</TableHead>
              <TableHead>Rationale</TableHead>
              <TableHead></TableHead>
            </TableRow>
          </TableHeader>
          <TableBody>
            {data.rows.length === 0 ? (
              <TableRow><TableCell colspan={10} className="text-center py-12 text-muted-foreground">No leads scored yet.
Click "Recompute All" to score existing leads.</TableCell></TableRow>
            ) : data.rows.map(r => (
              <TableRow key={r.tort_id} data-testid=\`row-tort-\${r.tort_id}\`>
                <TableCell className="font-medium">\${r.label}</TableCell>
                <TableCell>\${r.lead_count}</TableCell>
                <TableCell>\${r.total_spend_usd.toLocaleString()}</TableCell>
                <TableCell>\${r.spend_pct.toFixed(1)}%</TableCell>
                <TableCell>\${(r.retained_rate * 100).toFixed(1)}%</TableCell>
                <TableCell>\${r.ruin_flag_count > 0 ? <Badge variant="destructive">\${r.ruin_flag_count}</Badge> : <span
className="text-muted-foreground">0</span></TableCell>
                <TableCell>\${badgeFor(r.classification)}</TableCell>
                <TableCell>\${actionBadge(r.action)}</TableCell>
                <TableCell className="max-w-md text-sm text-muted-foreground">\${r.rationale}</TableCell>
                <TableCell>
                  <Button size="sm" variant="ghost" asChild>
                    <Link href=\`/\`/leads?tort_type=\${encodeURIComponent(r.tort_id)}\`><ExternalLink className="w-4 h-4" /
></Link>
                  </Button>
                </TableCell>
              </TableRow>
            )}
          </TableBody>
        </Table>
      </CardContent>
    </Card>
  </>
)
}

```

```
        </Table>
      </CardContent>
    </Card>
  </>
) : null}
</div>
);
}
```

[artifacts/mtos-crm/src/pages/doc-review.tsx](#)

```
import { useState, useEffect } from "react";
import { Card, CardContent, CardHeader, CardTitle, CardDescription } from "@components/ui/card";
import { Button } from "@components/ui/button";
import { Badge } from "@components/ui/badge";
import { Input } from "@components/ui/input";
import { Table, TableBody, TableCell, TableHead, TableHeader, TableRow } from "@components/ui/table";
import { useToast } from "@hooks/use-toast";
import { AlertTriangle, CheckCircle, XCircle, FileSearch, Search } from "lucide-react";

interface FieldComparison {
  field: string;
  intake_value: string;
  document_value: string;
  match: "match" | "mismatch" | "missing" | "extra";
}

export default function DocReview() {
  const { toast } = useToast();
  const [leadId, setLeadId] = useState("");
  const [faxId, setFaxId] = useState("");
  const [leadData, setLeadData] = useState<Record<string, any> | null>(null);
  const [ocrData, setOcrData] = useState<Record<string, any> | null>(null);
  const [aiFields, setAiFields] = useState<Record<string, any> | null>(null);
  const [comparisons, setComparisons] = useState<FieldComparison[]>([]);
  const [loading, setLoading] = useState(false);
  const [analyzing, setAnalyzing] = useState(false);

  const loadLead = async () => {
    if (!leadId) return;
    try {
      const res = await fetch(`/api/leads/${leadId}`);
      if (!res.ok) throw new Error("Lead not found");
      setLeadData(await res.json());
    } catch {
      toast({ title: "Lead not found", variant: "destructive" });
    }
  };

  const loadOcr = async () => {
    if (!faxId) return;
    try {
      const res = await fetch(`/api/ocr/results/${faxId}`);
      if (!res.ok) throw new Error("OCR result not found");
      setOcrData(await res.json());
    } catch {
      toast({ title: "OCR result not found", variant: "destructive" });
    }
  };

  const runAiAnalysis = async () => {
    if (!faxId) return;
    setAnalyzing(true);
    try {
      const res = await fetch(`/api/ocr/ai-fields/result/${faxId}`, { method: "POST" });
      if (!res.ok) throw new Error("Analysis failed");
      setAiFields(await res.json());
    } catch {
      toast({ title: "AI analysis failed", variant: "destructive" });
    } finally {
      setAnalyzing(false);
    }
  };

  const compareData = () => {
    if (!leadData) return;
    const source = aiFields || ocrData;
    if (!source) return;

    const fields: FieldComparison[] = [];
    const compareField = (field: string, intakeKey: string, docValue: string | undefined) => {
      const intakeVal = String(leadData[intakeKey] || "");
      const docVal = String(docValue || "");
      let match: FieldComparison["match"] = "missing";
      if (!intakeVal && !docVal) return;
      if (!intakeVal && docVal) match = "extra";
      else if (intakeVal && !docVal) match = "missing";
      else if (intakeVal.toLowerCase().trim() === docVal.toLowerCase().trim()) match = "match";
      else match = "mismatch";
      fields.push({ field, intake_value: intakeVal, document_value: docVal, match });
    };

    compareField("Patient Name", "name", source.patient_name || `${source.drug_name} || ""`);
    compareField("Date of Birth", "date_of_birth", source.date_of_birth);
    compareField("Diagnosis", "diagnosis", source.diagnoses?.[0]?.description || source.drug_name);
    compareField("Diagnosis Date", "diagnosis_date", source.diagnoses?.[0]?.date || source.fill_date);
    compareField("Drug/Substance", "tort_type", source.medications?.[0]?.name || source.drug_name);
    compareField("Rx Number", "npi_number", source.rx_number);

    if (source.providers) {
      const intakePhysician = `${leadData.physician_first_name} || ""} ${leadData.physician_last_name} || ""`.trim();
      for (const p of source.providers) {
        const intakeVal = intakePhysician;
        const docVal = String(p.name || "");
        let match: FieldComparison["match"] = "missing";
        if (!intakeVal && docVal) match = "extra";
      }
    }
  };
}
```



```

    else if (intakeVal && !docVal) match = "missing";
    else if (intakeVal.toLowerCase() === docVal.toLowerCase()) match = "match";
    else match = "mismatch";
    fields.push({ field: `Provider: ${p.name}`, intake_value: intakeVal, document_value: docVal, match });
  }
}

if (source.exposure_details) {
  for (const e of source.exposure_details) {
    if (e.substance) compareField("Exposure Substance", "tort_type", e.substance);
    if (e.location) compareField("Exposure Location", "location_name", e.location);
  }
}

setComparisons(fields);
};

useEffect(() => { if (leadData && (ocrData || aiFields)) compareData(); }, [leadData, ocrData, aiFields]);

const matchCounts = {
  match: comparisons.filter(c => c.match === "match").length,
  mismatch: comparisons.filter(c => c.match === "mismatch").length,
  missing: comparisons.filter(c => c.match === "missing").length,
  extra: comparisons.filter(c => c.match === "extra").length,
};

const fraudRisk = matchCounts.mismatch > 2 ? "high" : matchCounts.mismatch > 0 ? "medium" : "low";

return (
  <div className="space-y-6">
    <h1 className="text-3xl font-bold tracking-tight">Side-by-Side Document Review</h1>
    <p className="text-muted-foreground">Compare medical records against intake data to spot discrepancies and potential fraud.</p>

    <div className="grid gap-4 md:grid-cols-2">
      <Card>
        <CardHeader>
          <CardTitle>Intake Record</CardTitle>
          <CardDescription>Load lead data from the CRM</CardDescription>
        </CardHeader>
        <CardContent>
          <div className="flex gap-2">
            <Input placeholder="Lead ID" value={leadId} onChange={(e) => setLeadId(e.target.value)} />
            <Button onClick={loadLead}><Search className="h-4 w-4 mr-2" />Load</Button>
          </div>
          <div>
            {leadData && (
              <div className="mt-4 space-y-2 text-sm">
                <div><span className="font-medium">Name:</span> {leadData.name}</div>
                <div><span className="font-medium">Tort:</span> {leadData.tort_type}</div>
                <div><span className="font-medium">Diagnosis:</span> {leadData.diagnosis || "N/A"}</div>
                <div><span className="font-medium">Status:</span> <Badge variant="outline">{leadData.status}</Badge></div>
                <div><span className="font-medium">Fraud Score:</span> {leadData.fraud_score ?? "N/A"}</div>
              </div>
            )}
          </CardContent>
        </Card>

      <Card>
        <CardHeader>
          <CardTitle>Document Record</CardTitle>
          <CardDescription>Load OCR/fax result for comparison</CardDescription>
        </CardHeader>
        <CardContent>
          <div className="flex gap-2">
            <Input placeholder="Fax Result ID" value={faxId} onChange={(e) => setFaxId(e.target.value)} />
            <Button onClick={loadOcr}><FileSearch className="h-4 w-4 mr-2" />Load</Button>
          </div>
          <div>
            {ocrData && (
              <div className="mt-4 space-y-2 text-sm">
                <div><span className="font-medium">Source:</span> {ocrData.source_file}</div>
                <div><span className="font-medium">Drug:</span> {ocrData.drug_name || "N/A"}</div>
                <div><span className="font-medium">Rx:</span> {ocrData.rx_number || "N/A"}</div>
                <div><span className="font-medium">Confidence:</span> {Math.round((ocrData.confidence || 0) * 100)}%</div>
                <Button variant="outline" size="sm" className="mt-2" onClick={runAiAnalysis} disabled={analyzing}>
                  {analyzing ? "Analyzing..." : "Run AI Deep Analysis"}
                </Button>
                {aiFields && <Badge className="ml-2 bg-green-100 text-green-700">AI Analysis Complete</Badge>}
              </div>
            )}
          </CardContent>
        </Card>
      </div>

      {comparisons.length > 0 && (
        <div className="grid gap-4 md:grid-cols-4">
          <Card>
            <CardContent className="pt-6">
              <div className="flex items-center gap-2">
                <CheckCircle className="h-5 w-5 text-green-500" />
                <div>
                  <div className="text-2xl font-bold text-green-600">{matchCounts.match}</div>
                  <div className="text-sm text-muted-foreground">Matches</div>
                </div>
              </div>
            </CardContent>
          </Card>
        </div>
      )}
    </div>
  </div>

```

```

<Card>
  <CardContent className="pt-6">
    <div className="flex items-center gap-2">
      <XCircle className="h-5 w-5 text-red-500" />
      <div>
        <div className="text-2xl font-bold text-red-600">{matchCounts.mismatch}</div>
        <div className="text-sm text-muted-foreground">Mismatches</div>
      </div>
    </div>
  </CardContent>
</Card>
<Card>
  <CardContent className="pt-6">
    <div className="flex items-center gap-2">
      <AlertTriangle className="h-5 w-5 text-yellow-500" />
      <div>
        <div className="text-2xl font-bold text-yellow-600">{matchCounts.missing}</div>
        <div className="text-sm text-muted-foreground">Missing in Doc</div>
      </div>
    </div>
  </CardContent>
</Card>
<Card className={fraudRisk === "high" ? "border-red-300 bg-red-50" : fraudRisk === "medium" ? "border-yellow-300 bg-
yellow-50" : "border-green-300 bg-green-50"}>
  <CardContent className="pt-6">
    <div className="text-center">
      <div className={`text-2xl font-bold ${fraudRisk === "high" ? "text-red-600" : fraudRisk === "medium" ? "text-
yellow-600" : "text-green-600"}`}>
        {fraudRisk.toUpperCase()}
      </div>
      <div className="text-sm text-muted-foreground">Fraud Risk</div>
    </div>
  </CardContent>
</Card>
</div>

<Card>
  <CardHeader>
    <CardTitle>Field-by-Field Comparison</CardTitle>
    <CardDescription>Intake form data vs. extracted document data</CardDescription>
  </CardHeader>
  <CardContent>
    <Table>
      <TableHeader>
        <TableRow>
          <TableHead>Field</TableHead>
          <TableHead>Intake Value</TableHead>
          <TableHead>Document Value</TableHead>
          <TableHead>Status</TableHead>
        </TableRow>
      </TableHeader>
      <TableBody>
        {comparisons.map((c, i) => (
          <TableRow key={i} className={c.match === "mismatch" ? "bg-red-50" : c.match === "match" ? "bg-green-50" :
""}>
            <TableCell className="font-medium">{c.field}</TableCell>
            <TableCell className="font-mono text-sm">{c.intake_value || "---"}</TableCell>
            <TableCell className="font-mono text-sm">{c.document_value || "---"}</TableCell>
            <TableCell>
              <Badge className={
                c.match === "match" ? "bg-green-100 text-green-700" :
                c.match === "mismatch" ? "bg-red-100 text-red-700" :
                c.match === "missing" ? "bg-yellow-100 text-yellow-700" :
                "bg-blue-100 text-blue-700"
              }>
                {c.match.toUpperCase()}
              </Badge>
            </TableCell>
          </TableRow>
        ))}
      </TableBody>
    </Table>
  </CardContent>
</Card>
</>
)}

{comparisons.length === 0 && leadData && ocrData && (
  <Card>
    <CardContent className="py-8 text-center text-muted-foreground">
      Load both a lead and a document record, then comparison will appear automatically.
    </CardContent>
  </Card>
)}
</div>
});
}

```

[artifacts/mtos-crm/src/pages/document-templates.tsx](#)

```
import { useState, useEffect, useRef } from "react";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { Label } from "@components/ui/label";
import { Switch } from "@components/ui/switch";
import { Textarea } from "@components/ui/textarea";
import { Badge } from "@components/ui/badge";
import { Dialog, DialogContent, DialogHeader, DialogTitle, DialogFooter } from "@components/ui/dialog";
import { Table, TableBody, TableCell, TableHead, TableHeader, TableRow } from "@components/ui/table";
import { useToast } from "@hooks/use-toast";
import { FileText, Plus, Pencil, Trash2, Upload, Eye } from "lucide-react";

interface Template {
  id: number;
  name: string;
  template_type: string;
  description: string | null;
  source: "pdf" | "ai";
  storage_path: string | null;
  ai_prompt: string | null;
  requires_signature: boolean;
  signer_role: string;
  delivery_subject: string | null;
  delivery_message: string | null;
  triggers_med_records_request: boolean;
  active: boolean;
}

const EMPTY: Partial<Template> = {
  name: "",
  template_type: "hipaa",
  source: "pdf",
  requires_signature: true,
  signer_role: "lead",
  triggers_med_records_request: false,
  active: true,
};

export default function DocumentTemplatesPage() {
  const { toast } = useToast();
  const [templates, setTemplates] = useState<Template[]>([]);
  const [loading, setLoading] = useState(true);
  const [open, setOpen] = useState(false);
  const [editing, setEditing] = useState<Partial<Template>>>(EMPTY);
  const [uploading, setUploading] = useState(false);
  const fileInput = useRef<HTMLInputElement>(null);

  const load = async () => {
    setLoading(true);
    try {
      const res = await fetch("/api/document-templates");
      setTemplates(await res.json());
    } finally {
      setLoading(false);
    }
  };

  useEffect(() => { load(); }, []);

  const handleUpload = async (file: File) => {
    if (file.size > 25 * 1024 * 1024) {
      toast({ title: "File too large", description: "Max 25MB", variant: "destructive" });
      return;
    }
    setUploading(true);
    try {
      const buffer = await file.arrayBuffer();
      const base64 = btoa(String.fromCharCode(...new Uint8Array(buffer)));
      const res = await fetch("/api/document-templates/upload", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify({ fileName: file.name, base64 }),
      });
      if (!res.ok) {
        const body = await res.json().catch(() => ({}));
        toast({ title: "Upload failed", description: body.error || `HTTP ${res.status}`, variant: "destructive" });
        return;
      }
      const stored = await res.json();
      setEditing((e) => ({ ...e, storage_path: stored.storagePath, source: "pdf" }));
      toast({ title: "PDF uploaded", description: `${(stored.sizeBytes / 1024).toFixed(1)} KB` });
    } finally {
      setUploading(false);
    }
  };

  const save = async () => {
    if (!editing.name?.trim()) {
      toast({ title: "Name is required", variant: "destructive" });
      return;
    }
    if (editing.source === "pdf" && !editing.storage_path) {
      toast({ title: "Upload a PDF first", variant: "destructive" });
      return;
    }
  }
}
```

```

if (editing.source === "ai" && !editing.ai_prompt?.trim()) {
  toast({ title: "AI prompt is required for AI-drafted templates", variant: "destructive" });
  return;
}
const isEdit = typeof editing.id === "number";
const url = isEdit ? `/api/document-templates/${editing.id}` : "/api/document-templates";
const res = await fetch(url, {
  method: isEdit ? "PUT" : "POST",
  headers: { "Content-Type": "application/json" },
  body: JSON.stringify(editing),
});
if (!res.ok) {
  const body = await res.json().catch(() => ({}));
  toast({ title: "Save failed", description: body.error || `HTTP ${res.status}`, variant: "destructive" });
  return;
}
toast({ title: isEdit ? "Template updated" : "Template created" });
setOpen(false);
setEditing(EMPTY);
load();
};

const remove = async (id: number) => {
  if (!confirm("Delete this template? Existing envelopes are not affected.")) return;
  const res = await fetch(`/api/document-templates/${id}`, { method: "DELETE" });
  if (!res.ok) {
    toast({ title: "Delete failed", variant: "destructive" });
    return;
  }
  toast({ title: "Template deleted" });
  load();
};

return (
  <div className="p-6 space-y-6">
    <div className="flex items-center justify-between">
      <div>
        <h1 className="text-2xl font-bold flex items-center gap-2">
          <FileText className="h-6 w-6" /> Document Templates
        </h1>
        <p className="text-sm text-muted-foreground">Reusable PDFs (or AI-drafted documents) sent to leads on approval.</p>
      </div>
      <Button onClick={() => { setEditing(EMPTY); setOpen(true); }}>
        <Plus className="mr-2 h-4 w-4" /> New Template
      </Button>
    </div>

    <Card>
      <CardHeader><CardTitle>All Templates ({templates.length})</CardTitle></CardHeader>
      <CardContent>
        {loading ? (
          <p className="text-muted-foreground">Loading...</p>
        ) : templates.length === 0 ? (
          <p className="text-sm text-muted-foreground">No templates yet. Add a HIPAA, Affidavit, and Retainer to get
started.</p>
        ) : (
          <Table>
            <TableHeader>
              <TableRow>
                <TableHead>Name</TableHead>
                <TableHead>Type</TableHead>
                <TableHead>Source</TableHead>
                <TableHead>Triggers Fax?</TableHead>
                <TableHead>Status</TableHead>
                <TableHead className="w-40 text-right">Actions</TableHead>
              </TableRow>
            </TableHeader>
            <TableBody>
              {templates.map((t) => (
                <TableRow key={t.id}>
                  <TableCell>
                    <div className="font-medium">{t.name}</div>
                    {t.description && <div className="text-xs text-muted-foreground line-clamp-1">{t.description}</div>}
                  </TableCell>
                  <TableCell><Badge variant="outline">{t.template_type}</Badge></TableCell>
                  <TableCell>
                    <Badge variant={t.source === "ai" ? "secondary" : "default"}>{t.source.toUpperCase()}</Badge>
                  </TableCell>
                  <TableCell>
                    {t.triggers_med_records_request ? <Badge>Yes</Badge> : <span className="text-muted-foreground text-
sm">No</span>}
                  </TableCell>
                  <TableCell>
                    <Badge variant={t.active ? "default" : "secondary"}>{t.active ? "Active" : "Inactive"}</Badge>
                  </TableCell>
                  <TableCell className="text-right">
                    {t.storage_path && (
                      <Button variant="ghost" size="icon" asChild>
                        <a href={`\api/document-templates/${t.id}/preview`} target="_blank" rel="noreferrer">
                          <Eye className="h-4 w-4" />
                        </a>
                      </Button>
                    )}
                    <Button variant="ghost" size="icon" onClick={() => { setEditing(t); setOpen(true); }}>
                      <Pencil className="h-4 w-4" />
                    </Button>
                    <Button variant="ghost" size="icon" onClick={() => remove(t.id)}>

```

```

        <Trash2 className="h-4 w-4" />
      </Button>
    </TableCell>
  </TableRow>
</TableBody>
</Table>
  </CardContent>
</Card>

<Dialog open={open} onOpenChange={setOpen}>
  <DialogContent className="max-w-2xl max-h-[90vh] overflow-y-auto">
    <DialogHeader>
      <DialogTitle>{typeof editing.id === "number" ? "Edit Template" : "New Template"}</DialogTitle>
    </DialogHeader>
    <div className="space-y-4">
      <div className="grid grid-cols-2 gap-3">
        <div>
          <Label>Name </Label>
          <Input value={editing.name || ""} onChange={(e) => setEditing({ ...editing, name: e.target.value })} />
        </div>
        <div>
          <Label>Type</Label>
          <select
            className="w-full h-10 rounded-md border border-input bg-background px-3 text-sm"
            value={editing.template_type || "hipaa"}
            onChange={(e) => setEditing({ ...editing, template_type: e.target.value })}
          >
            <option value="hipaa">HIPAA</option>
            <option value="affidavit">Affidavit</option>
            <option value="retainer">Retainer</option>
            <option value="medical_records_request">Medical Records Request</option>
            <option value="other">Other</option>
          </select>
        </div>
      </div>
      <div>
        <Label>Description</Label>
        <Textarea rows={2} value={editing.description || ""} onChange={(e) => setEditing({ ...editing, description:
e.target.value })} />
      </div>
      <div className="grid grid-cols-2 gap-3">
        <div>
          <Label>Source</Label>
          <select
            className="w-full h-10 rounded-md border border-input bg-background px-3 text-sm"
            value={editing.source || "pdf"}
            onChange={(e) => setEditing({ ...editing, source: e.target.value as "pdf" | "ai" })}
          >
            <option value="pdf">PDF (uploaded)</option>
            <option value="ai">AI-drafted</option>
          </select>
        </div>
        <div>
          <Label>Signer Role</Label>
          <Input value={editing.signer_role || ""} onChange={(e) => setEditing({ ...editing, signer_role:
e.target.value })} />
        </div>
      </div>
      {editing.source === "pdf" ? (
        <div className="rounded-md border p-3 bg-muted/30">
          <Label>PDF File</Label>
          <div className="flex items-center gap-2 mt-2">
            <input
              ref={fileInput}
              type="file"
              accept="application/pdf"
              className="hidden"
              onChange={(e) => e.target.files?.[0] && handleUpload(e.target.files[0])}
            />
            <Button variant="outline" size="sm" onClick={() => fileInput.current?.click()} disabled={uploading}>
              <Upload className="mr-2 h-4 w-4" />{uploading ? "Uploading..." : "Upload PDF"}
            </Button>
            {editing.storage_path && (
              <span className="text-xs text-muted-foreground truncate">' {editing.storage_path.split("/")}.pop()</span>
            )}
          </div>
        </div>
      ) : (
        <div>
          <Label>AI Prompt</Label>
          <Textarea
            rows={4}
            placeholder="e.g. Draft a HIPAA authorization for {{lead.name}} releasing records to {{firm.name}}..."
            value={editing.ai_prompt || ""}
            onChange={(e) => setEditing({ ...editing, ai_prompt: e.target.value })}
          />
        </div>
      )}
    </div>
  </DialogContent>
</Dialog>

<div className="grid grid-cols-2 gap-3">
  <div>
    <Label>Email Subject (sent to lead)</Label>

```

```

        <Input value={editing.delivery_subject || ""} onChange={(e) => setEditing({ ...editing, delivery_subject:
e.target.value })} />
      </div>
      <div>
        <Label>Email Message</Label>
        <Input value={editing.delivery_message || ""} onChange={(e) => setEditing({ ...editing, delivery_message:
e.target.value })} />
      </div>
    </div>

    <div className="space-y-2 rounded-md border p-3 bg-muted/30">
      <div className="flex items-center gap-2">
        <Switch checked={!editing.requires_signature} onCheckedChange={(v) => setEditing({ ...editing,
requires_signature: v })} />
        <Label>Requires signature</Label>
      </div>
      <div className="flex items-center gap-2">
        <Switch checked={!editing.triggers_med_records_request} onCheckedChange={(v) => setEditing({ ...editing,
triggers_med_records_request: v })} />
        <Label>When this is signed !' auto-fax doctor for medical records</Label>
      </div>
      <div className="flex items-center gap-2">
        <Switch checked={!editing.active} onCheckedChange={(v) => setEditing({ ...editing, active: v })} />
        <Label>Active (sent by default – override per buyer/tort in Assignments)</Label>
      </div>
    </div>
    <div>
      <DialogFooter>
        <Button variant="outline" onClick={() => setOpen(false)}>Cancel</Button>
        <Button onClick={save}>Save</Button>
      </DialogFooter>
    </DialogContent>
  </Dialog>
</div>
  );
}

```

[artifacts/mtos-crm/src/pages/documents.tsx](#)

```
import { useListDocuments, getListDocumentsQueryKey } from "@workspace/api-client-react";
import { Table, TableBody, TableCell, TableHead, TableHeader, TableRow } from "@components/ui/table";
import { Badge } from "@components/ui/badge";
import { Skeleton } from "@components/ui/skeleton";
import { format } from "date-fns";
import { FileText, Download } from "lucide-react";
import { Button } from "@components/ui/button";
import { Link } from "wouter";

export default function Documents() {
  const { data: documents, isLoading } = useListDocuments({}, {
    query: { queryKey: getListDocumentsQueryKey({}) }
  });

  return (
    <div className="space-y-6">
      <div className="flex items-center justify-between">
        <h1 className="text-3xl font-bold tracking-tight">Document Center</h1>
      </div>

      <div className="rounded-md border">
        <Table>
          <TableHeader>
            <TableRow>
              <TableHead>File Name</TableHead>
              <TableHead>Type</TableHead>
              <TableHead>Lead ID</TableHead>
              <TableHead>Status</TableHead>
              <TableHead>Created</TableHead>
              <TableHead className="text-right">Actions</TableHead>
            </TableRow>
          </TableHeader>
          <TableBody>
            {isLoading ? (
              Array.from({ length: 5 }).map((_, i) => (
                <TableRow key={i}>
                  <TableCell><Skeleton className="h-4 w-32" /></TableCell>
                  <TableCell><Skeleton className="h-4 w-24" /></TableCell>
                  <TableCell><Skeleton className="h-4 w-12" /></TableCell>
                  <TableCell><Skeleton className="h-6 w-20" /></TableCell>
                  <TableCell><Skeleton className="h-4 w-24" /></TableCell>
                  <TableCell><Skeleton className="h-8 w-16 ml-auto" /></TableCell>
                </TableRow>
              ))
            ) : documents?.length === 0 ? (
              <TableRow>
                <TableCell colspan={6} className="h-24 text-center">
                  No documents found.
                </TableCell>
              </TableRow>
            ) : (
              documents?.map((doc) => (
                <TableRow key={doc.id}>
                  <TableCell className="font-medium flex items-center gap-2">
                    <FileText className="h-4 w-4 text-muted-foreground" />
                    {doc.file_name}
                  </TableCell>
                  <TableCell className="capitalize">{doc.document_type.replace('_', ' ')}</TableCell>
                  <TableCell>
                    <Link href={`/${leads}/${doc.lead_id}`} className="hover:underline font-mono">
                      #{doc.lead_id}
                    </Link>
                  </TableCell>
                  <TableCell>
                    {doc.signed ? (
                      <Badge variant="secondary" className="bg-green-500/20 text-green-500">Signed</Badge>
                    ) : (
                      <Badge variant="outline">Pending</Badge>
                    )}
                  </TableCell>
                  <TableCell className="text-muted-foreground text-sm font-mono">
                    {format(new Date(doc.created_at), "yyyy-MM-dd")}
                  </TableCell>
                  <TableCell className="text-right">
                    <Button variant="ghost" size="sm" asChild>
                      <a href={doc.file_url || "#"} target="_blank" rel="noopener noreferrer">
                        <Download className="h-4 w-4" />
                      </a>
                    </Button>
                  </TableCell>
                </TableRow>
              ))
            )}
          </TableBody>
        </Table>
      </div>
    </div>
  );
}
```

[artifacts/mtos-crm/src/pages/drafting.tsx](#)

```
import { useState, useEffect } from "react";
import { Card, CardContent, CardHeader, CardTitle, CardDescription } from "@components/ui/card";
import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { Label } from "@components/ui/label";
import { Badge } from "@components/ui/badge";
import { Textarea } from "@components/ui/textarea";
import { Dialog, DialogContent, DialogHeader, DialogTitle, DialogFooter, DialogDescription } from "@components/ui/dialog";
import { useToast } from "@hooks/use-toast";
import { FileText, Download, Sparkles, Copy } from "lucide-react";

interface Template {
  id: string;
  name: string;
  description: string;
}

interface DraftResult {
  title: string;
  content: string;
  template_type: string;
  generated_at: string;
  fields_used: string[];
}

export default function Drafting() {
  const { toast } = useToast();
  const [templates, setTemplates] = useState<Template[]>([]);
  const [selectedTemplate, setSelectedTemplate] = useState("");
  const [leadId, setLeadId] = useState("");
  const [firmName, setFirmName] = useState("");
  const [attorneyName, setAttorneyName] = useState("");
  const [customInstructions, setCustomInstructions] = useState("");
  const [generating, setGenerating] = useState(false);
  const [draft, setDraft] = useState<DraftResult | null>(null);
  const [previewOpen, setPreviewOpen] = useState(false);
  const [downloadingPdf, setDownloadingPdf] = useState(false);

  useEffect(() => {
    fetch("/api/drafting/templates").then(r => r.json()).then(setTemplates).catch(() => {});
  }, []);

  const generateDraft = async () => {
    if (!selectedTemplate) { toast({ title: "Select a template", variant: "destructive" }); return; }
    setGenerating(true);
    try {
      const res = await fetch("/api/drafting/generate", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify({
          template_type: selectedTemplate,
          lead_id: leadId || undefined,
          firm_name: firmName || undefined,
          attorney_name: attorneyName || undefined,
          custom_instructions: customInstructions || undefined,
        }),
      });
      if (!res.ok) throw new Error("Generation failed");
      const result = await res.json();
      setDraft(result);
      setPreviewOpen(true);
    } catch {
      toast({ title: "Draft generation failed", variant: "destructive" });
    } finally {
      setGenerating(false);
    }
  };

  const downloadPdf = async () => {
    if (!selectedTemplate) return;
    setDownloadingPdf(true);
    try {
      const res = await fetch("/api/drafting/generate-pdf", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify({
          template_type: selectedTemplate,
          lead_id: leadId || undefined,
          firm_name: firmName || undefined,
          attorney_name: attorneyName || undefined,
          custom_instructions: customInstructions || undefined,
        }),
      });
      const blob = await res.blob();
      const url = URL.createObjectURL(blob);
      const a = document.createElement("a");
      a.href = url;
      a.download = `${selectedTemplate}.pdf`;
      a.click();
      URL.revokeObjectURL(url);
    } catch {
      toast({ title: "PDF download failed", variant: "destructive" });
    } finally {
      setDownloadingPdf(false);
    }
  };
}
```



```

    };

    const copyToClipboard = () => {
      if (draft) {
        navigator.clipboard.writeText(draft.content);
        toast({ title: "Copied to clipboard" });
      }
    };

    return (
      <div className="space-y-6">
        <h1 className="text-3xl font-bold tracking-tight">AI Document Drafting</h1>
        <p className="text-muted-foreground">Generate HIPAA authorizations, retainer agreements, and legal documents from lead data using AI.</p>

        <div className="grid gap-4 md:grid-cols-2">
          <Card>
            <CardHeader>
              <CardTitle>Template Selection</CardTitle>
              <CardDescription>Choose a document template to generate</CardDescription>
            </CardHeader>
            <CardContent className="space-y-3">
              {templates.map((t) => (
                <div
                  key={t.id}
                  className={`p-3 rounded-lg border-2 cursor-pointer transition-colors ${selectedTemplate === t.id ? "border-primary bg-primary/5" : "border-transparent hover:border-gray-200"}`}
                  onClick={() => setSelectedTemplate(t.id)}
                >
                  <div className="flex items-center gap-2">
                    <FileText className="h-4 w-4" />
                    <span className="font-medium">{t.name}</span>
                  </div>
                  <div className="text-sm text-muted-foreground mt-1">{t.description}</div>
                </div>
              ))}
            </CardContent>
          </Card>

          <Card>
            <CardHeader>
              <CardTitle>Configuration</CardTitle>
              <CardDescription>Customize the generated document</CardDescription>
            </CardHeader>
            <CardContent className="space-y-4">
              <div className="grid gap-2">
                <Label>Lead ID (optional - auto-fills client data)</Label>
                <Input placeholder="e.g. 42" value={leadId} onChange={(e) => setLeadId(e.target.value)} />
              </div>
              <div className="grid gap-2">
                <Label>Firm Name</Label>
                <Input placeholder="Smith & Associates LLP" value={firmName} onChange={(e) => setFirmName(e.target.value)} />
              </div>
              <div className="grid gap-2">
                <Label>Attorney Name</Label>
                <Input placeholder="Jane Smith, Esq." value={attorneyName} onChange={(e) => setAttorneyName(e.target.value)} />
              </div>
              <div className="grid gap-2">
                <Label>Custom Instructions (optional)</Label>
                <Textarea placeholder="Any specific requirements..." value={customInstructions} onChange={(e) => setCustomInstructions(e.target.value)} rows={3} />
              </div>
              <div className="flex gap-2">
                <Button onClick={generateDraft} disabled={generating || !selectedTemplate} className="flex-1">
                  <Sparkles className="h-4 w-4 mr-2" />
                  {generating ? "Generating..." : "Generate Draft"}
                </Button>
                <Button variant="outline" onClick={downloadPdf} disabled={downloadingPdf || !selectedTemplate}>
                  <Download className="h-4 w-4 mr-2" />
                  {downloadingPdf ? "...": "PDF"}
                </Button>
              </div>
            </CardContent>
          </Card>
        </div>

        <Dialog open={previewOpen} onOpenChange={setPreviewOpen}>
          <DialogContent className="sm:max-w-2xl max-h-[80vh] overflow-y-auto">
            <DialogHeader>
              <DialogTitle>{draft?.title}</DialogTitle>
              <DialogDescription>
                Generated {draft?.generated_at ? new Date(draft.generated_at).toLocaleString() : ""}
                {draft?.fields_used && ` | ${draft.fields_used.length} fields used`}
              </DialogDescription>
            </DialogHeader>
            <div className="py-4">
              <pre className="whitespace-pre-wrap text-sm font-mono bg-gray-50 p-4 rounded-lg border max-h-[50vh] overflow-y-auto">
                {draft?.content}
              </pre>
            </div>
            <DialogFooter>
              <Button variant="outline" onClick={copyToClipboard}>
                <Copy className="h-4 w-4 mr-2" />
                Copy
              </Button>
              <Button onClick={() => setPreviewOpen(false)}>Close</Button>
            </DialogFooter>
          </DialogContent>
        </Dialog>
      </div>
    );
  }
}

```

```
        </DialogFooter>
      </DialogContent>
    </Dialog>
  </div>
);
}
```

[artifacts/mtos-crm/src/pages/financial-news.tsx](#)

```
import { useState, useEffect } from "react";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import { Badge } from "@components/ui/badge";
import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { Skeleton } from "@components/ui/skeleton";
import { useToast } from "@hooks/use-toast";
import { TrendingUp, ExternalLink, RefreshCw, Search, Clock, DollarSign } from "lucide-react";

interface Article {
  title: string;
  link: string;
  source: string;
  published: string;
  description: string;
  category: string;
}

const categoryLabels: Record<string, { label: string; color: string }> = {
  market: { label: "Markets", color: "bg-blue-100 text-blue-700" },
  pharma_finance: { label: "Pharma/FDA", color: "bg-purple-100 text-purple-700" },
  legal_finance: { label: "Legal Impact", color: "bg-red-100 text-red-700" },
};

function timeAgo(dateStr: string): string {
  const now = Date.now();
  const date = new Date(dateStr).getTime();
  const diff = now - date;
  const mins = Math.floor(diff / 60000);
  if (mins < 60) return `${mins}m ago`;
  const hours = Math.floor(mins / 60);
  if (hours < 24) return `${hours}h ago`;
  const days = Math.floor(hours / 24);
  return `${days}d ago`;
}

export default function FinancialNews() {
  const { toast } = useToast();
  const [articles, setArticles] = useState<Article[]>([]);
  const [loading, setLoading] = useState(true);
  const [search, setSearch] = useState("");
  const [filterCategory, setFilterCategory] = useState("all");

  const fetchNews = async () => {
    setLoading(true);
    try {
      const res = await fetch("/api/news/financial");
      if (!res.ok) throw new Error("Failed");
      setArticles(await res.json());
    } catch {
      toast({ title: "Failed to load financial news", variant: "destructive" });
    } finally {
      setLoading(false);
    }
  };

  useEffect(() => { fetchNews(); }, []);

  const filtered = articles.filter(a => {
    if (filterCategory !== "all" && a.category !== filterCategory) return false;
    if (search && !a.title.toLowerCase().includes(search.toLowerCase()) && !
a.description.toLowerCase().includes(search.toLowerCase())) return false;
    return true;
  });

  const sourceBreakdown = articles.reduce<Record<string, number>>((acc, a) => {
    acc[a.source] = (acc[a.source] || 0) + 1;
    return acc;
  }, {});

  return (
    <div className="space-y-6">
      <div className="flex items-center justify-between">
        <div>
          <h1 className="text-3xl font-bold tracking-tight">Financial News</h1>
          <p className="text-muted-foreground">Market news from Yahoo Finance, MarketWatch, CNBC, and other tier-one sources.</p>
        </div>
        <Button variant="outline" onClick={fetchNews} disabled={loading}>
          <RefreshCw className={`h-4 w-4 mr-2 ${loading ? "animate-spin" : ""}` } />
          Refresh
        </Button>
      </div>

      <div className="grid gap-4 md:grid-cols-4">
        <Card>
          <CardContent className="pt-6">
            <div className="flex items-center gap-2">
              <DollarSign className="h-5 w-5 text-green-600" />
              <div>
                <div className="text-2xl font-bold">{articles.length}</div>
                <div className="text-sm text-muted-foreground">Total Articles</div>
              </div>
            </div>
          </CardContent>
        </Card>
      </div>
    </div>
  );
}
```

```

</Card>
{Object.entries(sourceBreakdown).slice(0, 3).map(([source, count]) => (
  <Card key={source}>
    <CardContent className="pt-6">
      <div className="text-2xl font-bold">{count}</div>
      <div className="text-sm text-muted-foreground">{source}</div>
    </CardContent>
  </Card>
)}}
</div>

<Card>
  <CardContent className="pt-6">
    <div className="flex gap-3 flex-wrap items-center">
      <div className="relative flex-1 min-w-[200px]">
        <Search className="absolute left-3 top-1/2 -translate-y-1/2 h-4 w-4 text-muted-foreground" />
        <Input placeholder="Search financial news..." className="pl-10" value={search} onChange={(e) =>
setSearch(e.target.value)} />
      </div>
      <div className="flex gap-2 flex-wrap">
        <Badge variant={filterCategory === "all" ? "default" : "outline"} className="cursor-pointer" onClick={() =>
setFilterCategory("all")}>All</Badge>
        {Object.entries(categoryLabels).map(([key, cfg]) => (
          <Badge key={key} variant="outline" className="cursor-pointer ${filterCategory === key ? cfg.color : ""}`
onClick={() => setFilterCategory(key)}>{cfg.label}</Badge>
        ))}
      </div>
    </div>
  </CardContent>
</Card>

{loading ? (
  <div className="space-y-4">
    {Array.from({ length: 5 }).map((_, i) => (
      <Card key={i}><CardContent className="pt-6"><Skeleton className="h-24" /></CardContent></Card>
    ))}
  </div>
) : filtered.length === 0 ? (
  <Card>
    <CardContent className="py-12 text-center text-muted-foreground">
      <TrendingUp className="h-12 w-12 mx-auto mb-3" />
      <div>No financial news found. Try adjusting your filters.</div>
    </CardContent>
  </Card>
) : (
  <div className="space-y-3">
    {filtered.map((article, i) => {
      const cat = categoryLabels[article.category];
      return (
        <Card key={i} className="hover:shadow-md transition-shadow cursor-pointer" onClick={() =>
window.open(article.link, "_blank")}>
          <CardContent className="pt-5 pb-4">
            <div className="flex items-start justify-between gap-4">
              <div className="flex-1 min-w-0">
                <div className="flex items-center gap-2 mb-1 flex-wrap">
                  {cat && <Badge variant="outline" className={`${cat.color}`}>{cat.label}</Badge>}
                  <span className="text-xs text-muted-foreground font-medium">{article.source}</span>
                <span className="text-xs text-muted-foreground flex items-center gap-1">
                  <Clock className="h-3 w-3" />
                  {timeAgo(article.published)}
                </span>
              </div>
              <div>
                <h3 className="font-semibold text-base leading-snug line-clamp-2">{article.title}</h3>
                {article.description && (
                  <p className="text-sm text-muted-foreground mt-1 line-clamp-2">{article.description}</p>
                )}
              </div>
            </div>
            <ExternalLink className="h-4 w-4 text-muted-foreground flex-shrink-0 mt-1" />
          </div>
        </CardContent>
      </Card>
    )};
  </div>
)}

{!loading && filtered.length > 0 && (
  <div className="text-center text-sm text-muted-foreground">
    Showing {filtered.length} articles from {new Set(filtered.map(a => a.source)).size} sources
  </div>
)}
);
}

```

[artifacts/mtos-crm/src/pages/form-engine.tsx](#)

```
import { useState } from "react";
import { useQueryClient } from "@tanstack/react-query";
import {
  useGetFormConfigs,
  useValidateEmail,
  useValidateAddress,
  useRunBackgroundCheck,
  useRunLeadBackgroundCheck,
  useUpdateFormConfig,
  useAddCustomField,
  useRemoveCustomField,
  getGetFormConfigsQueryKey,
  FormConfig,
  CustomField
} from "@workspace/api-client-react";
import { Copy, Mail, MapPin, Search, Shield, CheckCircle2, XCircle, AlertTriangle, Info, Play, Pencil, Plus, Trash2, RefreshCw } from "lucide-react";
import { useToast } from "@/hooks/use-toast";
import { Switch } from "@components/ui/switch";
import { Textarea } from "@components/ui/textarea";

import { Card, CardContent, CardDescription, CardHeader, CardTitle, CardFooter } from "@components/ui/card";
import { Tabs, TabsContent, TabsList, TabsTrigger } from "@components/ui/tabs";
import { Input } from "@components/ui/input";
import { Button } from "@components/ui/button";
import { Badge } from "@components/ui/badge";
import { Label } from "@components/ui/label";
import { Select, SelectContent, SelectItem, SelectTrigger, SelectValue } from "@components/ui/select";
import { Dialog, DialogContent, DialogHeader, DialogTitle, DialogTrigger } from "@components/ui/dialog";
import { Table, TableBody, TableCell, TableHead, TableHeader, TableRow } from "@components/ui/table";
import { Skeleton } from "@components/ui/skeleton";
import { ScrollArea } from "@components/ui/scroll-area";

const US_STATES = [
  "AL", "AK", "AZ", "AR", "CA", "CO", "CT", "DE", "DC", "FL", "GA",
  "HI", "ID", "IL", "IN", "IA", "KS", "KY", "LA", "ME", "MD", "MA",
  "MI", "MN", "MS", "MO", "MT", "NE", "NV", "NH", "NJ", "NM", "NY",
  "NC", "ND", "OH", "OK", "OR", "PA", "RI", "SC", "SD", "TN", "TX",
  "UT", "VT", "VA", "WA", "WV", "WI", "WY"
];

function getStatusBadge(status: string | undefined) {
  switch (status) {
    case "clean":
      return <Badge className="bg-green-500/10 text-green-500 hover:bg-green-500/20 hover:text-green-600 border-green-500/20">Clean</Badge>;
    case "flagged":
      return <Badge className="bg-red-500/10 text-red-500 hover:bg-red-500/20 hover:text-red-600 border-red-500/20">Flagged</Badge>;
    case "not_found":
      return <Badge className="bg-yellow-500/10 text-yellow-500 hover:bg-yellow-500/20 hover:text-yellow-600 border-yellow-500/20">Not Found</Badge>;
    case "error":
      return <Badge className="bg-gray-500/10 text-gray-500 hover:bg-gray-500/20 hover:text-gray-600 border-gray-500/20">Error</Badge>;
    default:
      return <Badge variant="outline">{status || "Unknown"}</Badge>;
  }
}

function getSeverityBadge(severity: string | undefined) {
  switch (severity) {
    case "low":
      return <Badge className="bg-yellow-500/10 text-yellow-500 border-yellow-500/20">Low</Badge>;
    case "medium":
      return <Badge className="bg-orange-500/10 text-orange-500 border-orange-500/20">Medium</Badge>;
    case "high":
      return <Badge className="bg-red-500/10 text-red-500 border-red-500/20">High</Badge>;
    default:
      return <Badge variant="outline">{severity || "Unknown"}</Badge>;
  }
}

export default function FormEngine() {
  const { toast } = useToast();

  // Data Fetching
  const { data: formConfigsData, isLoading: isLoadingConfigs } = useGetFormConfigs();
  const configs = formConfigsData?.tort_campaigns || [];

  // Mutations
  const validateEmail = useValidateEmail();
  const validateAddress = useValidateAddress();
  const runBackgroundCheck = useRunBackgroundCheck();
  const runLeadBackgroundCheck = useRunLeadBackgroundCheck();

  // States: Email
  const [emailInput, setEmailInput] = useState("");
  const [emailResult, setEmailResult] = useState<any>(null);

  // States: Address
  const [addressInput, setAddressInput] = useState({ street_address: "", city: "", state: "", zip: "" });
  const [addressResult, setAddressResult] = useState<any>(null);

  // States: Background Check (Manual)
```

```

const [bgCheckInput, setBgCheckInput] = useState({ first_name: "", last_name: "", state: "", date_of_birth: "" });
const [bgCheckResult, setBgCheckResult] = useState<any>(null);

// States: Background Check (Lead)
const [leadIdInput, setLeadIdInput] = useState("");
const [leadBgCheckResult, setLeadBgCheckResult] = useState<any>(null);

// Handlers
const handleCopyEmbed = (configId: string) => {
  const code = `<script src="${window.location.origin}/api/forms/embed/${configId}"></script>\n<div id="mtos-form"></div>`;
  navigator.clipboard.writeText(code);
  toast({
    title: "Copied to clipboard",
    description: "Embed code has been copied.",
  });
};

const handleValidateEmail = () => {
  if (!emailInput) return;
  validateEmail.mutate({ data: { email: emailInput } }, {
    onSuccess: (data) => setEmailResult(data),
    onError: () => toast({ title: "Error", description: "Failed to validate email", variant: "destructive" })
  });
};

const handleValidateAddress = () => {
  if (!addressInput.street_address || !addressInput.city || !addressInput.state || !addressInput.zip) {
    toast({ title: "Validation Error", description: "Please fill out all address fields.", variant: "destructive" });
    return;
  }
  validateAddress.mutate({ data: addressInput }, {
    onSuccess: (data) => setAddressResult(data),
    onError: () => toast({ title: "Error", description: "Failed to validate address", variant: "destructive" })
  });
};

const handleRunBgCheck = () => {
  if (!bgCheckInput.first_name || !bgCheckInput.last_name) {
    toast({ title: "Validation Error", description: "First and last name are required.", variant: "destructive" });
    return;
  }
  runBackgroundCheck.mutate({ data: bgCheckInput }, {
    onSuccess: (data) => setBgCheckResult(data),
    onError: () => toast({ title: "Error", description: "Failed to run background check", variant: "destructive" })
  });
};

const handleRunLeadBgCheck = () => {
  if (!leadIdInput || isNaN(Number(leadIdInput))) {
    toast({ title: "Validation Error", description: "Please enter a valid Lead ID.", variant: "destructive" });
    return;
  }
  runLeadBackgroundCheck.mutate({ id: Number(leadIdInput) }, {
    onSuccess: (data) => setLeadBgCheckResult(data),
    onError: () => toast({ title: "Error", description: "Failed to run background check for lead", variant: "destructive" })
  });
};

return (
  <div className="flex-1 space-y-8 p-8 pt-6 max-w-7xl mx-auto w-full">
    <div className="flex items-center justify-between">
      <div className="space-y-1">
        <h2 className="text-3xl font-bold tracking-tight">Form Engine</h2>
        <p className="text-muted-foreground">Generate and manage embeddable intake forms with TCPA + TrustedForm compliance.</p>
      </div>
    </div>

    <Tabs defaultValue="builder" className="w-full">
      <TabsList className="grid w-full grid-cols-3 mb-8 max-w-2xl">
        <TabsTrigger value="builder">Form Builder</TabsTrigger>
        <TabsTrigger value="validation">Validation Tools</TabsTrigger>
        <TabsTrigger value="background">Background Check</TabsTrigger>
      </TabsList>

      <TabsContent value="builder" className="space-y-6">
        <div className="grid gap-6 md:grid-cols-2 lg:grid-cols-3">
          {isLoadingConfigs ? (
            Array.from({ length: 6 }).map((_, i) => (
              <Card key={i} className="border-border/50">
                <CardHeader><Skeleton className="h-6 w-3/4" /></CardHeader>
                <CardContent className="space-y-4">
                  <Skeleton className="h-4 w-full" />
                  <Skeleton className="h-4 w-5/6" />
                </CardContent>
              </Card>
            ))
          ) : configs.length === 0 ? (
            <div className="col-span-full py-12 text-center text-muted-foreground bg-muted/20 rounded-lg border border-dashed">
              <p>No campaign configurations found.</p>
            </div>
          ) : (
            configs.map((config: FormConfig) => (
              <Card key={config.id} className="flex flex-col border-border/50 shadow-sm transition-all hover:shadow-md">
                <CardHeader className="pb-4">

```

```

<div className="flex items-start justify-between">
  <CardTitle className="text-xl">{config.label}</CardTitle>
</div>
<CardDescription className="flex items-center gap-2 mt-2">
  <Badge variant="secondary" className="font-mono text-xs">{config.id}</Badge>
</CardDescription>
</CardHeader>
<CardContent className="flex-1 space-y-4">
  {config.fields && config.fields.length > 0 && (
    <div className="space-y-2">
      <Label className="text-xs font-semibold text-muted-foreground uppercase tracking-wider">Extra Fields</
Label>
      <div className="flex flex-wrap gap-1">
        {config.fields.map(f => (
          <Badge key={f} variant="outline" className="bg-muted/50 text-xs font-normal">{f}</Badge>
        ))}
      </div>
    </div>
  )}
  {config.rules && config.rules.length > 0 && (
    <div className="space-y-2">
      <Label className="text-xs font-semibold text-muted-foreground uppercase tracking-wider">Validation
Rules</Label>
      <ul className="text-sm space-y-1 text-muted-foreground">
        {config.rules.map((rule, idx) => (
          <li key={idx} className="flex items-start gap-2">
            <CheckCircle2 className="h-3.5 w-3.5 mt-0.5 text-primary/60 shrink-0" />
            <span className="leading-tight">{rule}</span>
          </li>
        ))}
      </ul>
    </div>
  )}
</CardContent>
  {config.custom_fields && config.custom_fields.length > 0 && (
    <div className="px-6 pb-2">
      <Label className="text-xs font-semibold text-muted-foreground uppercase tracking-wider">Custom Fields</
Label>
      <div className="flex flex-wrap gap-1 mt-2">
        {config.custom_fields.map(cf => (
          <Badge key={cf.key} variant="outline" className="bg-blue-500/5 text-blue-600 border-blue-500/30 text-
xs font-normal">
            {cf.label}{cf.required ? "*" : ""}
          </Badge>
        ))}
      </div>
    </div>
  )}
<CardFooter className="bg-muted/20 border-t p-4 flex flex-col gap-2">
  <div className="flex gap-2 w-full">
    <FormPreviewDialog config={config} />
    <Button variant="default" className="flex-1" size="sm" onClick={() => handleCopyEmbed(config.id)}>
      <Copy className="h-4 w-4 mr-2" /> Embed
    </Button>
  </div>
  <FormEditDialog config={config} />
</CardFooter>
</Card>
  )}
</div>
</TabsContent>
<TabsContent value="validation" className="space-y-6">
  <div className="grid gap-6 md:grid-cols-2">
    <Card className="border-border/50 shadow-sm flex flex-col">
      <CardHeader>
        <div className="flex items-center gap-2">
          <div className="p-2 bg-primary/10 rounded-md text-primary">
            <Mail className="h-5 w-5" />
          </div>
          <div>
            <CardTitle className="text-lg">Email Validator</CardTitle>
            <CardDescription>Real-time inbox verification</CardDescription>
          </div>
        </div>
      </CardHeader>
      <CardContent className="space-y-4 flex-1">
        <div className="flex gap-2">
          <Input
            placeholder="Enter email address..."
            type="email"
            value={emailInput}
            onChange={(e) => setEmailInput(e.target.value)}
            onKeyDown={(e) => e.key === 'Enter' && handleValidateEmail()}
          />
          <Button onClick={handleValidateEmail} disabled={validateEmail.isPending}>
            {validateEmail.isPending ? "Validating..." : "Validate"}
          </Button>
        </div>
        <div>
          {emailResult && (
            <div className="mt-6 rounded-md border p-4 bg-muted/20 space-y-3 animate-in fade-in slide-in-from-bottom-2">
              <div className="flex items-center gap-2">
                <span className="text-sm font-medium text-muted-foreground w-16">Status:</span>
                {emailResult.valid ? (
                  <Badge className="bg-green-500/10 text-green-500 hover:bg-green-500/20 border-

```

```

green-500/20"><CheckCircle2 className="h-3 w-3 mr-1" /> Valid</Badge>
    ) : (
      <Badge className="bg-red-500/10 text-red-500 hover:bg-red-500/20 border-red-500/20"><XCircle
className="h-3 w-3 mr-1" /> Invalid</Badge>
    )}
  </div>

  {!emailResult.valid && emailResult.errors && emailResult.errors.length > 0 && (
    <div className="space-y-1">
      <span className="text-sm font-medium text-muted-foreground">Errors:</span>
      <ul className="text-sm text-red-500/80 bg-red-500/10 p-3 rounded-md list-disc list-inside pl-6">
        {emailResult.errors.map((e: string, i: number) => <li key={i}>{e}</li>)}
      </ul>
    </div>
  )}

  {emailResult.suggestion && (
    <div className="flex items-start gap-2 bg-blue-500/10 text-blue-600 p-3 rounded-md text-sm">
      <Info className="h-4 w-4 mt-0.5 shrink-0" />
      <div>
        <span className="font-medium">Did you mean?</span> {emailResult.suggestion}
      </div>
    </div>
  )}
</div>
)}
</CardContent>
</Card>

<Card className="border-border/50 shadow-sm flex flex-col">
  <CardHeader>
    <div className="flex items-center gap-2">
      <div className="p-2 bg-primary/10 rounded-md text-primary">
        <MapPin className="h-5 w-5" />
      </div>
      <div>
        <CardTitle className="text-lg">Address Validator</CardTitle>
        <CardDescription>USPS CASS certification check</CardDescription>
      </div>
    </div>
  </CardHeader>
  <CardContent className="space-y-4 flex-1">
    <div className="space-y-3">
      <div className="space-y-1.5">
        <Label className="text-xs">Street Address</Label>
        <Input
          placeholder="123 Main St"
          value={addressInput.street_address}
          onChange={(e) => setAddressInput({...addressInput, street_address: e.target.value})}
        />
      </div>
      <div className="grid grid-cols-6 gap-3">
        <div className="col-span-3 space-y-1.5">
          <Label className="text-xs">City</Label>
          <Input
            placeholder="City"
            value={addressInput.city}
            onChange={(e) => setAddressInput({...addressInput, city: e.target.value})}
          />
        </div>
        <div className="col-span-1 space-y-1.5">
          <Label className="text-xs">State</Label>
          <Select value={addressInput.state} onChange={(v) => setAddressInput({...addressInput, state: v})}>
            <SelectTrigger className="px-2"><SelectValue placeholder="ST" /></SelectTrigger>
            <SelectContent>
              {US_STATES.map(s => <SelectItem key={s} value={s}>{s}</SelectItem>)}
            </SelectContent>
          </Select>
        </div>
        <div className="col-span-2 space-y-1.5">
          <Label className="text-xs">ZIP</Label>
          <Input
            placeholder="12345"
            value={addressInput.zip}
            onChange={(e) => setAddressInput({...addressInput, zip: e.target.value})}
          />
        </div>
      </div>
      <Button className="w-full mt-2" onClick={handleValidateAddress} disabled={validateAddress.isPending}>
        {validateAddress.isPending ? "Validating..." : "Validate Address"}
      </Button>
    </div>

    {addressResult && (
      <div className="mt-4 rounded-md border p-4 bg-muted/20 space-y-3 animate-in fade-in slide-in-from-bottom-2">
        <div className="flex items-center gap-2">
          <span className="text-sm font-medium text-muted-foreground w-16">Status:</span>
          {addressResult.valid ? (
            <Badge className="bg-green-500/10 text-green-500 hover:bg-green-500/20 border-
green-500/20"><CheckCircle2 className="h-3 w-3 mr-1" /> Valid</Badge>
          ) : (
            <Badge className="bg-red-500/10 text-red-500 hover:bg-red-500/20 border-red-500/20"><XCircle
className="h-3 w-3 mr-1" /> Invalid</Badge>
          )}
        </div>

        {!addressResult.valid && addressResult.errors && addressResult.errors.length > 0 && (

```



```

        <div className="space-y-1">
          <span className="text-sm font-medium text-muted-foreground">Errors:</span>
          <ul className="text-sm text-red-500/80 bg-red-500/10 p-3 rounded-md list-disc list-inside pl-6">
            {addressResult.errors.map((e: string, i: number) => <li key={i}>{e}</li>)}
          </ul>
        </div>
      </div>
    </div>
  </div>
</CardContent>
</Card>
</div>
</TabsContent>

<TabsContent value="background" className="space-y-6">
  <div className="grid gap-6 md:grid-cols-2">
    <Card className="border-border/50 shadow-sm flex flex-col">
      <CardHeader>
        <div className="flex items-center gap-2">
          <div className="p-2 bg-primary/10 rounded-md text-primary">
            <Shield className="h-5 w-5" />
          </div>
          <div>
            <CardTitle className="text-lg">Manual Background Check</CardTitle>
            <CardDescription>Search public records and legal histories</CardDescription>
          </div>
        </div>
      </CardHeader>
      <CardContent className="space-y-4">
        <div className="grid grid-cols-2 gap-4">
          <div className="space-y-2">
            <Label>First Name</Label>
            <Input
              placeholder="First Name"
              value={bgCheckInput.first_name}
              onChange={(e) => setBgCheckInput({...bgCheckInput, first_name: e.target.value})}
            />
          </div>
          <div className="space-y-2">
            <Label>Last Name</Label>
            <Input
              placeholder="Last Name"
              value={bgCheckInput.last_name}
              onChange={(e) => setBgCheckInput({...bgCheckInput, last_name: e.target.value})}
            />
          </div>
          <div className="space-y-2">
            <Label>State (Optional)</Label>
            <Select value={bgCheckInput.state} onChange={(v) => setBgCheckInput({...bgCheckInput, state: v ===
"none" ? "" : v})}>
              <SelectTrigger><SelectValue placeholder="Select State" /></SelectTrigger>
              <SelectContent>
                <SelectItem value="none">Any State</SelectItem>
                {US_STATES.map(s => <SelectItem key={s} value={s}>{s}</SelectItem>)}
              </SelectContent>
            </Select>
          </div>
          <div className="space-y-2">
            <Label>Date of Birth (Optional)</Label>
            <Input
              type="date"
              value={bgCheckInput.date_of_birth}
              onChange={(e) => setBgCheckInput({...bgCheckInput, date_of_birth: e.target.value})}
            />
          </div>
        </div>
        <Button className="w-full" onClick={handleRunBgCheck} disabled={runBackgroundCheck.isPending}>
          {runBackgroundCheck.isPending ? "Running Check..." : "Run Background Check"}
        </Button>

        {bgCheckResult && (
          <div className="mt-6 border rounded-lg overflow-hidden animate-in fade-in duration-300">
            <div className="bg-muted/40 p-4 border-b space-y-3">
              <div className="flex items-center justify-between">
                <div className="flex items-center gap-2">
                  <span className="font-medium text-sm">Status:</span>
                  {getStatusBadge(bgCheckResult.status)}
                </div>
                <span className="text-xs text-muted-foreground font-mono">
                  Source: {bgCheckResult.source} • {new Date(bgCheckResult.checked_at).toLocaleDateString()}
                </span>
              </div>
              <p className="text-sm bg-background p-3 rounded border text-muted-foreground">
                {bgCheckResult.summary}
              </p>
            </div>

            {bgCheckResult.records && bgCheckResult.records.length > 0 && (
              <Table>
                <TableHeader>
                  <TableRow className="bg-muted/20 hover:bg-muted/20">
                    <TableHead>Type</TableHead>
                    <TableHead>Description</TableHead>
                    <TableHead>Date</TableHead>
                    <TableHead>Jurisdiction</TableHead>
                    <TableHead>Severity</TableHead>
                  </TableRow>

```

```

        </TableHeader>
        <TableBody>
          {bgCheckResult.records.map((rec: any, i: number) => (
            <TableRow key={i}>
              <TableCell className="font-medium text-xs">{rec.type}</TableCell>
              <TableCell className="text-xs">{rec.description}</TableCell>
              <TableCell className="text-xs whitespace-nowrap">{rec.date || "N/A"}</TableCell>
              <TableCell className="text-xs">{rec.jurisdiction || "N/A"}</TableCell>
              <TableCell>{getSeverityBadge(rec.severity)}</TableCell>
            </TableRow>
          ))}
        </TableBody>
      </Table>
    </div>
  </div>
)
</CardContent>
</Card>

<Card className="border-border/50 shadow-sm flex flex-col h-fit">
  <CardHeader>
    <div className="flex items-center gap-2">
      <div className="p-2 bg-primary/10 rounded-md text-primary">
        <Search className="h-5 w-5" />
      </div>
      <div>
        <CardTitle className="text-lg">Check Existing Lead</CardTitle>
        <CardDescription>Run screening on a lead already in the system</CardDescription>
      </div>
    </div>
  </CardHeader>
  <CardContent className="space-y-4">
    <div className="flex gap-2">
      <Input
        placeholder="Lead ID (e.g. 1234)"
        value={leadIdInput}
        onChange={(e) => setLeadIdInput(e.target.value)}
        onKeyDown={(e) => e.key === 'Enter' && handleRunLeadBgCheck()}
      />
      <Button onClick={handleRunLeadBgCheck} disabled={runLeadBackgroundCheck.isPending}>
        {runLeadBackgroundCheck.isPending ? "Checking..." : "Run Lead Check"}
      </Button>
    </div>

    {leadBgCheckResult && (
      <div className="mt-4 border rounded-lg overflow-hidden animate-in fade-in duration-300">
        <div className="bg-muted/40 p-4 border-b space-y-3">
          <div className="flex items-center justify-between">
            <div className="flex items-center gap-2">
              <span className="font-medium text-sm">Status:</span>
              {getStatusBadge(leadBgCheckResult.status)}
            </div>
            <span className="text-xs text-muted-foreground font-mono">
              {new Date(leadBgCheckResult.checked_at).toLocaleDateString()}
            </span>
          </div>
          <p className="text-sm bg-background p-3 rounded border text-muted-foreground">
            {leadBgCheckResult.summary}
          </p>
        </div>
      </div>

      {leadBgCheckResult.records && leadBgCheckResult.records.length > 0 && (
        <ScrollArea className="h-[200px]">
          <Table>
            <TableHeader className="sticky top-0 bg-background z-10">
              <TableRow className="bg-muted/20 hover:bg-muted/20">
                <TableHead>Type</TableHead>
                <TableHead>Severity</TableHead>
              </TableRow>
            </TableHeader>
            <TableBody>
              {leadBgCheckResult.records.map((rec: any, i: number) => (
                <TableRow key={i}>
                  <TableCell>
                    <div className="font-medium text-xs">{rec.type}</div>
                    <div className="text-[10px] text-muted-foreground mt-0.5 line-clamp-1">{rec.description}</div>
                  </TableCell>
                  <TableCell>{getSeverityBadge(rec.severity)}</TableCell>
                </TableRow>
              ))}
            </TableBody>
          </Table>
        </ScrollArea>
      </div>
    )}
  </div>
)
</CardContent>
</Card>
</div>
</div>
</div>
</div>
);
}

function FormPreviewDialog({ config }: { config: FormConfig }) {
  const [open, setOpen] = useState(false);

```

```

const [reloadKey, setReloadKey] = useState(0);
const apiBase = (import.meta.env.VITE_API_URL as string | undefined) || `${window.location.origin}/api`;
const previewSrc = `${apiBase.replace(/\/$/, "")}/forms/preview/${config.id}?k=${reloadKey}`;

return (
  <Dialog open={open} onOpenChange={setOpen}>
    <DialogTrigger asChild>
      <Button variant="outline" className="flex-1" size="sm">
        <Play className="h-4 w-4 mr-2" /> Preview
      </Button>
    </DialogTrigger>
    <DialogContent className="sm:max-w-[680px] h-[85vh] flex flex-col p-0">
      <DialogHeader className="p-6 pb-2 shrink-0 flex-row items-center justify-between space-y-0">
        <DialogTitle>Live Preview: {config.label}</DialogTitle>
        <Button size="sm" variant="ghost" onClick={() => setReloadKey(k => k + 1)}>
          <RefreshCw className="h-4 w-4 mr-1" /> Reload
        </Button>
      </DialogHeader>
      <div className="flex-1 p-6 pt-0 min-h-0">
        {open && (
          <iframe
            key={reloadKey}
            src={previewSrc}
            title={`Preview of ${config.label} intake form`}
            className="w-full h-full border rounded-md bg-white"
            sandbox="allow-forms allow-scripts"
            referrerPolicy="no-referrer"
            data-testid={`iframe-preview-${config.id}`}
          />
        )}
      </div>
    </DialogContent>
  </Dialog>
);
}

const FIELD_TYPES: CustomField["type"][] = ["text", "email", "tel", "date", "number", "select", "textarea", "checkbox"];

function FormEditDialog({ config }: { config: FormConfig }) {
  const { toast } = useToast();
  const queryClient = useQueryClient();
  const [open, setOpen] = useState(false);
  const [diagnoses, setDiagnoses] = useState<string[]>(config.valid_diagnoses || []);
  const [diagInput, setDiagInput] = useState("");
  const [introText, setIntroText] = useState(config.intro_text || "");
  const [active, setActive] = useState(config.active !== false);
  const [avgSettlementLow, setAvgSettlementLow] = useState<string>(config.avg_settlement_low?.toString() ?? "");
  const [avgSettlementHigh, setAvgSettlementHigh] = useState<string>(config.avg_settlement_high?.toString() ?? "");
  const [mdlStatus, setMdlStatus] = useState<string>(config.mdl_status ?? "");
  const [solMonths, setSolMonths] = useState<string>(config.sol_months?.toString() ?? "");
  const [customFields, setCustomFields] = useState<CustomField[]>(config.custom_fields || []);
  const [newField, setNewField] = useState<CustomField>({ key: "", label: "", type: "text", required: false });

  const updateConfig = useUpdateFormConfig();
  const addField = useAddCustomField();
  const removeField = useRemoveCustomField();

  const invalidate = () => queryClient.invalidateQueries({ queryKey: getGetFormConfigsQueryKey() });

  const addDiagnosis = () => {
    const v = diagInput.trim().toLowerCase();
    if (!v || diagnoses.includes(v)) return;
    setDiagnoses([...diagnoses, v]);
    setDiagInput("");
  };

  const handleSaveBase = () => {
    updateConfig.mutate(
      { tortId: config.id, data: {
        valid_diagnoses: diagnoses,
        intro_text: introText || null,
        active,
        avg_settlement_low: avgSettlementLow === "" ? null : Number(avgSettlementLow),
        avg_settlement_high: avgSettlementHigh === "" ? null : Number(avgSettlementHigh),
        mdl_status: mdlStatus || null,
        sol_months: solMonths === "" ? null : Number(solMonths),
      } },
      {
        onSuccess: () => {
          invalidate();
          toast({ title: "Saved", description: `${config.label} updated.` });
        },
        onError: (err: any) => toast({ title: "Save failed", description: err?.message || "Update failed", variant: "destructive" }),
      }
    );
  };

  const handleAddField = () => {
    const key = newField.key.trim().toLowerCase().replace(/^[a-z0-9_]/g, "_");
    if (!key || !newField.label.trim()) {
      toast({ title: "Field invalid", description: "Key and label are required", variant: "destructive" });
      return;
    }
    if (customFields.some(f => f.key === key)) {
      toast({ title: "Duplicate key", description: `Key "${key}" already exists`, variant: "destructive" });
      return;
    }
  };
}

```

```

    }
    const field: CustomField = {
      ...newField,
      key,
      options: newField.type === "select"
        ? (newField.options && newField.options.length > 0 ? newField.options : ["Option 1"])
        : undefined,
    };
    addField.mutate(
      { tortId: config.id, data: field },
      {
        onSuccess: (data) => {
          invalidate();
          // Prefer server response when present; otherwise append locally.
          // Functional updater avoids stale closure issues.
          const fromServer = (data as { custom_fields?: CustomField[] } | undefined)?.custom_fields;
          if (fromServer) setCustomFields(fromServer);
          else setCustomFields(prev => [...prev, field]);
          setNewField({ key: "", label: "", type: "text", required: false });
          toast({ title: "Field added", description: field.label });
        },
        onError: (err: any) => toast({ title: "Add failed", description: err?.message || "Failed to add field", variant:
"destructive" })),
      );
    };
  };

  const handleRemoveField = (key: string) => {
    removeField.mutate(
      { tortId: config.id, key },
      {
        onSuccess: (data) => {
          invalidate();
          const fromServer = (data as { custom_fields?: CustomField[] } | undefined)?.custom_fields;
          if (fromServer) setCustomFields(fromServer);
          else setCustomFields(prev => prev.filter(f => f.key !== key));
          toast({ title: "Field removed" });
        },
        onError: (err: any) => toast({ title: "Remove failed", description: err?.message, variant: "destructive" })),
      );
  };

  return (
    <Dialog open={open} onOpenChange={setOpen}>
      <DialogTrigger asChild>
        <Button variant="ghost" size="sm" className="w-full" data-testid={`button-edit-${config.id}`}>
          <Pencil className="h-4 w-4 mr-2" /> Edit Form
        </Button>
      </DialogTrigger>
      <DialogContent className="sm:max-w-[720px] max-h-[88vh] overflow-y-auto">
        <DialogHeader>
          <DialogTitle>Edit: {config.label}</DialogTitle>
          </DialogHeader>
        <div className="space-y-6">
          <div className="flex items-center justify-between rounded-md border p-3">
            <div>
              <Label>Active</Label>
              <p className="text-xs text-muted-foreground">Inactive campaigns return 404 on embed.</p>
            </div>
            <Switch checked={active} onCheckedChange={setActive} data-testid={`switch-active-${config.id}`} />
          </div>

          <div className="space-y-2">
            <Label>Form Intro Text</Label>
            <Textarea
              value={introText}
              onChange={(e) => setIntroText(e.target.value)}
              placeholder="Optional intro shown at the top of the embedded form."
              rows={2}
              data-testid={`textarea-intro-${config.id}`}
            />
          </div>

          <div className="rounded-md border p-3 space-y-3 bg-muted/10">
            <div className="text-xs font-semibold uppercase tracking-wider text-muted-foreground">Decision Engine inputs</div>
            <div className="grid grid-cols-2 gap-2">
              <div>
                <Label>Avg settlement LOW ($)</Label>
                <Input type="number" value={avgSettlementLow} onChange={e => setAvgSettlementLow(e.target.value)}
placeholder="e.g. 50000" data-testid={`input-settlement-low-${config.id}`} />
              </div>
              <div>
                <Label>Avg settlement HIGH ($)</Label>
                <Input type="number" value={avgSettlementHigh} onChange={e => setAvgSettlementHigh(e.target.value)}
placeholder="e.g. 250000" data-testid={`input-settlement-high-${config.id}`} />
              </div>
              <div>
                <Label>MDL status</Label>
                <Select value={mdlStatus || "_none"} onValueChange={v => setMdlStatus(v === "_none" ? "" : v)}>
                  <SelectTrigger data-testid={`select-mdl-${config.id}`}><SelectValue placeholder="-" /></SelectTrigger>
                  <SelectContent>
                    <SelectItem value="_none">— Unset —</SelectItem>
                    <SelectItem value="pre_mdl">Pre-MDL</SelectItem>
                    <SelectItem value="active_bellwether">Active bellwether</SelectItem>
                    <SelectItem value="settling">Settling</SelectItem>
                    <SelectItem value="closed">Closed</SelectItem>
                  </SelectContent>
                </div>
            </div>
          </div>
        </DialogContent>
      </Dialog>
    );
  );
}

```

```

        </SelectContent>
      </Select>
    </div>
  </div>
  <div>
    <Label>SQL window (months)</Label>
    <Input type="number" value={solMonths} onChange={e => setSolMonths(e.target.value)} placeholder="e.g. 24" data-
testid={`input-sol-${config.id}`} />
  </div>
</div>
<p className="text-xs text-muted-foreground">Used to score lead convexity and detect ruin flags.</p>
</div>

<div className="space-y-2">
  <Label>Valid Diagnoses (matched case-insensitively)</Label>
  <div className="flex gap-2">
    <Input
      value={diagInput}
      onChange={(e) => setDiagInput(e.target.value)}
      onKeyDown={(e) => { if (e.key === "Enter") { e.preventDefault(); addDiagnosis(); }}}
      placeholder="e.g. non-hodgkin lymphoma"
      data-testid={`input-diagnosis-${config.id}`}
    />
    <Button type="button" onClick={addDiagnosis} variant="outline">Add</Button>
  </div>
  <div className="flex flex-wrap gap-1.5 pt-2">
    {diagnoses.map((d) => (
      <Badge key={d} variant="secondary" className="gap-1.5">
        {d}
        <button type="button" onClick={() => setDiagnoses(diagnoses.filter(x => x !== d))} className="hover:text-
red-500" aria-label={`Remove ${d}`}>
          <XCircle className="h-3 w-3" />
        </button>
      </Badge>
    ))}
    {diagnoses.length === 0 && <span className="text-xs text-muted-foreground">No diagnoses configured.</span>}
  </div>
</div>

<div className="space-y-3">
  <div className="flex items-center justify-between">
    <Label>Custom Fields</Label>
    <Badge variant="outline">{customFields.length} field(s)</Badge>
  </div>
  <div className="space-y-2">
    {customFields.map((f) => (
      <div key={f.key} className="flex items-center justify-between rounded-md border p-3 text-sm">
        <div className="space-y-0.5">
          <div className="font-medium">{f.label} {f.required && <span className="text-red-500">*</span>}</div>
          ($ {f.options.length} options) : ""</div>
        </div>
        <Button type="button" size="icon" variant="ghost" onClick={() => handleRemoveField(f.key)} data-
testid={`button-remove-field-${f.key}`}>
          <Trash2 className="h-4 w-4 text-red-500" />
        </Button>
      </div>
    ))}
    {customFields.length === 0 && <p className="text-xs text-muted-foreground">No custom fields yet.</p>}
  </div>

  <div className="rounded-md border bg-muted/20 p-3 space-y-3">
    <div className="text-xs font-semibold text-muted-foreground uppercase tracking-wider">Add Custom Field</div>
    <div className="grid grid-cols-2 gap-2">
      <Input placeholder="Key (snake_case)" value={newField.key} onChange={(e) => setNewField({ ...newField, key:
e.target.value })} data-testid="input-field-key" />
      <Input placeholder="Label" value={newField.label} onChange={(e) => setNewField({ ...newField, label:
e.target.value })} data-testid="input-field-label" />
    </div>
    <div className="grid grid-cols-2 gap-2">
      <Select value={newField.type} onValueChange={(v) => setNewField({ ...newField, type: v as
CustomField["type"] })}>
        <SelectTrigger data-testid="select-field-type"><SelectValue /></SelectTrigger>
        <SelectContent>
          {FIELD_TYPES.map(t => <SelectItem key={t} value={t}>{t}</SelectItem>)}
        </SelectContent>
      </Select>
      <div className="flex items-center justify-between rounded-md border bg-background px-3">
        <Label className="text-sm">Required</Label>
        <Switch checked={newField.required} onCheckedChange={(c) => setNewField({ ...newField, required: c })} data-
testid="switch-field-required" />
      </div>
    </div>
    <Input placeholder="Placeholder (optional)" value={newField.placeholder || ""} onChange={(e) =>
setNewField({ ...newField, placeholder: e.target.value })} />
    {newField.type === "select" && (
      <Input
        placeholder="Options comma-separated (e.g. Yes,No,Maybe)"
        value={(newField.options || []).join(",")}
        onChange={(e) => setNewField({ ...newField, options: e.target.value.split(",").map(s =>
s.trim()).filter(Boolean) })}
      />
    )}
    <Button type="button" onClick={handleAddField} disabled={addField.isPending} data-testid="button-add-field">
      <Plus className="h-4 w-4 mr-2" /> Add Field
    </Button>
  </div>
</div>
</div>

```

```
    <div className="flex justify-end gap-2 pt-2 border-t">
      <Button variant="ghost" onClick={() => setOpen(false)}>Close</Button>
      <Button onClick={handleSaveBase} disabled={updateConfig.isPending} data-testid={`button-save-${config.id}`}>
        {updateConfig.isPending ? "Saving..." : "Save Changes"}
      </Button>
    </div>
  </div>
</DialogContent>
</Dialog>
);
}
```

[artifacts/mtos-crm/src/pages/integrations.tsx](#)

```
import { useState, useEffect, useMemo } from "react";
import { Card, CardContent } from "@components/ui/card";
import { Button } from "@components/ui/button";
import { Badge } from "@components/ui/badge";
import { Input } from "@components/ui/input";
import { Label } from "@components/ui/label";
import { Tabs, TabsContent, TabsList, TabsTrigger } from "@components/ui/tabs";
import { Dialog, DialogContent, DialogHeader, DialogTitle, DialogFooter, DialogDescription } from "@components/ui/dialog";
import { Select, SelectContent, SelectItem, SelectTrigger, SelectValue } from "@components/ui/select";
import { Table, TableBody, TableCell, TableHead, TableHeader, TableRow } from "@components/ui/table";
import { useToast } from "@hooks/use-toast";
import {
  Plug, Plus, TestTube, RefreshCw, Trash2, Settings, CheckCircle, XCircle,
  Star, Search, Sparkles, Brain, FileSignature, Phone, MessageSquare, Mail,
  Printer, ScanText, MapPin, ShieldCheck, UserCheck, CreditCard, Database,
  Calendar as CalendarIcon, Briefcase, TrendingUp, Bot, Globe, Zap, Shield,
} from "lucide-react";

interface Preset {
  provider: string;
  name: string;
  type: string;
  category: string;
  description: string;
  docs_url: string;
  fields: string[];
  score: number;
  recommended?: boolean;
  pricing: "free" | "freemium" | "usage" | "subscription" | "enterprise";
  notes?: string;
}

interface Integration {
  id: number;
  name: string;
  type: string;
  provider: string;
  status: string;
  api_url: string | null;
  api_key_hash: string | null;
  webhook_url: string | null;
  config: any;
  last_sync_at: string | null;
  sync_direction: string;
  created_at: string;
}

const categoryIcons: Record<string, typeof Plug> = {
  "AI / LLM": Brain,
  "E-Signature": FileSignature,
  "Voice AI Agent": Bot,
  "SMS & Telephony": Phone,
  "Email": Mail,
  "Fax": Printer,
  "OCR & Document AI": ScanText,
  "Address & Geo": MapPin,
  "Identity & KYC": UserCheck,
  "Background Check": ShieldCheck,
  "Phone Validation & Fraud": Phone,
  "TCPA Compliance": Shield,
  "File Storage": Database,
  "Payments": CreditCard,
  "Web Search": Search,
  "Calendar & Scheduling": CalendarIcon,
  "Legal CRM": Briefcase,
  "CRM": Briefcase,
  "Data Enrichment": TrendingUp,
  "Ad Platforms": TrendingUp,
  "Automation": Zap,
  "Lead Vendor & Routing": MessageSquare,
};

const pricingColors: Record<string, string> = {
  free: "bg-emerald-100 text-emerald-700 border-emerald-200",
  freemium: "bg-teal-100 text-teal-700 border-teal-200",
  usage: "bg-blue-100 text-blue-700 border-blue-200",
  subscription: "bg-purple-100 text-purple-700 border-purple-200",
  enterprise: "bg-amber-100 text-amber-700 border-amber-200",
};

function ScoreBadge({ score }: { score: number }) {
  const tone =
    score >= 9 ? "bg-emerald-100 text-emerald-700 border-emerald-200"
      : score >= 7 ? "bg-blue-100 text-blue-700 border-blue-200"
        : score >= 5 ? "bg-amber-100 text-amber-700 border-amber-200"
          : "bg-gray-100 text-gray-700 border-gray-200";
  return (
    <Badge variant="outline" className={tone} data-testid={`score-${score}`}>
      {score}/10
    </Badge>
  );
}

export default function Integrations() {
  const { toast } = useToast();
```

```

const [presets, setPresets] = useState<Preset[]>([]);
const [integrations, setIntegrations] = useState<Integration[]>([]);
const [loading, setLoading] = useState(true);
const [addOpen, setAddOpen] = useState(false);
const [customOpen, setCustomOpen] = useState(false);
const [selectedPreset, setSelectedPreset] = useState<Preset | null>(null);
const [formData, setFormData] = useState({
  name: "", api_key: "", api_url: "", webhook_url: "",
  client_id: "", client_secret: "", account_sid: "",
  sync_direction: "bidirectional", type: "custom", provider: "custom",
});
const [testing, setTesting] = useState<number | null>(null);
const [syncing, setSyncing] = useState<number | null>(null);
const [search, setSearch] = useState("");
const [categoryFilter, setCategoryFilter] = useState<string>("all");
const [recommendedOnly, setRecommendedOnly] = useState(false);

const fetchData = async () => {
  setLoading(true);
  try {
    const [presetsRes, integrationsRes] = await Promise.all([
      fetch("/api/integrations/presets"),
      fetch("/api/integrations"),
    ]);
    setPresets(await presetsRes.json());
    setIntegrations(await integrationsRes.json());
  } catch {
    toast({ title: "Failed to load integrations", variant: "destructive" });
  } finally {
    setLoading(false);
  }
};

useEffect(() => { fetchData(); }, []);

const categories = useMemo(() => {
  const cats = Array.from(new Set(presets.map(p => p.category))).sort();
  return cats;
}, [presets]);

const filteredPresets = useMemo(() => {
  let list = [...presets];
  if (categoryFilter !== "all") list = list.filter(p => p.category === categoryFilter);
  if (recommendedOnly) list = list.filter(p => p.recommended);
  if (search.trim()) {
    const q = search.toLowerCase();
    list = list.filter(p =>
      p.name.toLowerCase().includes(q) ||
      p.description.toLowerCase().includes(q) ||
      p.category.toLowerCase().includes(q),
    );
  }
  list.sort((a, b) => {
    if (a.recommended !== b.recommended) return a.recommended ? -1 : 1;
    return b.score - a.score;
  });
  return list;
}, [presets, search, categoryFilter, recommendedOnly]);

const groupedByCategory = useMemo(() => {
  const map = new Map<string, Preset[]>();
  filteredPresets.forEach(p => {
    const arr = map.get(p.category) || [];
    arr.push(p);
    map.set(p.category, arr);
  });
  return Array.from(map.entries());
}, [filteredPresets]);

const connectPreset = (p: Preset) => {
  setSelectedPreset(p);
  setFormData({
    name: p.name, api_key: "", api_url: "", webhook_url: "",
    client_id: "", client_secret: "", account_sid: "",
    sync_direction: "bidirectional", type: p.type, provider: p.provider,
  });
  setAddOpen(true);
};

const openCustom = (kind: "custom" | "vendor" | "webpage") => {
  setFormData({
    name: "", api_key: "", api_url: "", webhook_url: "",
    client_id: "", client_secret: "", account_sid: "",
    sync_direction: "bidirectional", type: kind, provider: kind,
  });
  setCustomOpen(true);
};

const saveIntegration = async (isCustom: boolean) => {
  const body = isCustom
    ? formData
    : { ...formData, type: selectedPreset?.type, provider: selectedPreset?.provider };
  const res = await fetch("/api/integrations", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify(body),
  });
};

```



```
if (res.ok) {
    toast({ title: "Connected", description: `${body.name} is now active` });
    setAddOpen(false);
    setCustomOpen(false);
    fetchData();
} else {
    toast({ title: "Failed to save", variant: "destructive" });
};

const testIntegration = async (id: number) => {
    setTesting(id);
    const res = await fetch(`/api/integrations/${id}/test`, { method: "POST" });
    const data = await res.json();
    toast({
        title: data.success ? "Connection verified" : "Connection failed",
        description: data.message || `Latency: ${data.latency_ms}ms`,
        variant: data.success ? "default" : "destructive",
    });
    setTesting(null);
    fetchData();
};

const syncIntegration = async (id: number) => {
    setSyncing(id);
    const res = await fetch(`/api/integrations/${id}/sync`, { method: "POST" });
    const data = await res.json();
    toast({ title: "Sync complete", description: `${data.records_synced} records (${data.direction})` });
    setSyncing(null);
    fetchData();
};

const deleteIntegration = async (id: number) => {
    if (!confirm("Disconnect this integration?")) return;
    await fetch(`/api/integrations/${id}`, { method: "DELETE" });
    toast({ title: "Disconnected" });
    fetchData();
};

return (


# <Plug className="h-7 w-7" /> Integrations Hub



Bring your own API keys for AI engines, e-signature, voice agents, telephony, and more.  
Pick a provider, paste your key, and MTOS handles the rest.



<Button variant="outline" onClick={() => openCustom("vendor")} data-testid="button-add-vendor"><Plus className="h-4 w-4 mr-1" /> Vendor API</Button>
<Button variant="outline" onClick={() => openCustom("webpage")} data-testid="button-add-webpage"><Plus className="h-4 w-4 mr-1" /> Web Page / URL</Button>
<Button variant="outline" onClick={() => openCustom("custom")} data-testid="button-add-custom"><Plus className="h-4 w-4 mr-1" /> Custom REST</Button>



/* Active connections */>
{integrations.length > 0 && (>
<Card>
<CardContent className="pt-6">
<div className="flex items-center gap-2 mb-4">
<CheckCircle className="h-5 w-5 text-green-600" />
<h2 className="text-xl font-semibold">Active Connections ({integrations.length}</h2>
</div>
<Table>
<TableHeader>
<TableRow>
<TableHead>Name</TableHead>
<TableHead>Provider</TableHead>
<TableHead>Type</TableHead>
<TableHead>Status</TableHead>
<TableHead>Last Sync</TableHead>
<TableHead className="text-right">Actions</TableHead>
</TableRow>
</TableHeader>
<TableBody>
{integrations.map((i) => (>
<TableRow key={i.id} data-testid={`row-integration-${i.id}`}>
<TableCell className="font-medium">{i.name}</TableCell>
<TableCell><Badge variant="outline">{i.provider}</Badge></TableCell>
<TableCell><Badge variant="outline">{i.type}</Badge></TableCell>
<TableCell>
{i.status === "active" ? (>
<Badge className="bg-green-100 text-green-700"><CheckCircle className="h-3 w-3 mr-1" />Active</Badge>
)<Badge className="bg-gray-100 text-gray-700"><XCircle className="h-3 w-3 mr-1" />Inactive</Badge>
)}
</TableCell>
</TableRow>
</TableBody>
</Table>


```

```

        <TableCell className="text-sm text-muted-foreground">
          {i.last_sync_at ? new Date(i.last_sync_at).toLocaleString() : "Never"}
        </TableCell>
        <TableCell className="text-right">
          <div className="flex justify-end gap-1">
            <Button variant="ghost" size="sm" onClick={() => testIntegration(i.id)} disabled={testing === i.id}
data-testid={`button-test-${i.id}`} />
            <TestTube className={`h-4 w-4 ${testing === i.id ? "animate-pulse" : ""}`} />
          </div>
          <Button variant="ghost" size="sm" onClick={() => syncIntegration(i.id)} disabled={syncing === i.id}
data-testid={`button-sync-${i.id}`} />
          <RefreshCw className={`h-4 w-4 ${syncing === i.id ? "animate-spin" : ""}`} />
          </Button>
          <Button variant="ghost" size="sm" onClick={() => deleteIntegration(i.id)} data-testid={`button-delete-
${i.id}`} />
          <Trash2 className="h-4 w-4 text-red-500" />
        </Button>
      </div>
    </TableCell>
  </TableRow>
</tbody>
</table>
</TableBody>
</Table>
</CardContent>
</Card>
)}

{/* Filters */}
<Card>
  <CardContent className="pt-6">
    <div className="flex flex-wrap gap-3 items-end">
      <div className="flex-1 min-w-[240px]">
        <Label className="text-xs">Search providers</Label>
        <div className="relative mt-1">
          <Search className="h-4 w-4 absolute left-3 top-1/2 -translate-y-1/2 text-muted-foreground" />
          <Input
            className="pl-9"
            placeholder="DocuSign, Claude, Twilio..."
            value={search}
            onChange={(e) => setSearch(e.target.value)}
            data-testid="input-search"
          />
        </div>
      </div>
      <div className="min-w-[220px]">
        <Label className="text-xs">Category</Label>
        <Select value={categoryFilter} onValueChange={setCategoryFilter}>
          <SelectTrigger className="mt-1" data-testid="select-category"><SelectValue /></SelectTrigger>
          <SelectContent>
            <SelectItem value="all">All Categories ({presets.length})</SelectItem>
            {categories.map(c => {
              const count = presets.filter(p => p.category === c).length;
              return <SelectItem key={c} value={c}>{c} ({count})</SelectItem>;
            })}
          </SelectContent>
        </Select>
      </div>
      <div>
        <Button
          variant={recommendedOnly ? "default" : "outline"}
          onClick={() => setRecommendedOnly(v => !v)}
          data-testid="button-recommended-toggle"
        >
          <Star className={`h-4 w-4 mr-1 ${recommendedOnly ? "fill-current" : ""}`} />
          Recommended only
        </Button>
      </div>
      <div className="text-xs text-muted-foreground mt-3">
        Showing {filteredPresets.length} of {presets.length} providers. Sorted by recommendation, then score.
      </div>
    </CardContent>
  </Card>

  {/* Provider grid grouped by category */}
  {loading ? (
    <div className="text-center py-12 text-muted-foreground">Loading providers...</div>
  ) : (
    <div className="space-y-8">
      {groupedByCategory.length === 0 && (
        <div className="text-center py-12 text-muted-foreground">No providers match your filters.</div>
      )}
      {groupedByCategory.map(([category, items]) => {
        const Icon = categoryIcons[category] || Plug;
        return (
          <div key={category}>
            <div className="flex items-center gap-2 mb-3">
              <Icon className="h-5 w-5 text-primary" />
              <h2 className="text-lg font-semibold">{category}</h2>
              <Badge variant="outline" className="ml-1">{items.length}</Badge>
            </div>
            <div className="grid gap-4 md:grid-cols-2 lg:grid-cols-3">
              {items.map((p) => {
                const isConnected = integrations.some(i => i.provider === p.provider);
                return (
                  <Card
                    key={p.provider}
                    className={`relative ${isConnected ? "border-green-300 bg-green-50/40" : ""} ${p.recommended ? "ring-1
ring-amber-200" : ""}`}

```

```

    data-testid={`card-preset-${p.provider}`}
  ) {
    {p.recommended && (
      <div className="absolute -top-2 -right-2">
        <Badge className="bg-amber-500 text-white border-amber-600 shadow-sm">
          <Sparkles className="h-3 w-3 mr-1" /> Recommended
        </Badge>
      </div>
    )}
    <CardContent className="pt-6">
      <div className="flex items-start justify-between gap-2">
        <div className="flex-1 min-w-0">
          <div className="font-semibold truncate">{p.name}</div>
          <div className="flex flex-wrap gap-1 mt-1">
            <ScoreBadge score={p.score} />
            <Badge variant="outline" className={`text-xs capitalize ${pricingColors[p.pricing]}`>
              {p.pricing}
            </Badge>
            {isConnected && (
              <Badge className="bg-green-100 text-green-700 text-xs">Connected</Badge>
            )}
          </div>
        </div>
        <div>
          <p className="text-sm text-muted-foreground mt-3 line-clamp-3">{p.description}</p>
          {p.notes && (
            <p className="text-xs italic text-muted-foreground/80 mt-2 border-l-2 border-amber-200 pl-2">
              {p.notes}
            </p>
          )}
        </div>
      </div>
      <div className="flex gap-2 mt-4">
        <Button
          size="sm"
          variant={isConnected ? "outline" : "default"}
          className="flex-1"
          onClick={() => connectPreset(p)}
          data-testid={`button-connect-${p.provider}`}
        >
          {isConnected ? "Reconfigure" : "Connect"}
        </Button>
        <Button size="sm" variant="ghost" onClick={() => window.open(p.docs_url, "_blank")}>
          Docs
        </Button>
      </div>
    </CardContent>
  </Card>
);
}}
</div>
</div>
);
}}
</div>
)
}
}

/* Preset connect dialog */
<Dialog open={addOpen} onOpenChange={setAddOpen}>
  <DialogContent className="sm:max-w-lg">
    <DialogHeader>
      <DialogTitle className="flex items-center gap-2">
        Connect {selectedPreset?.name}
        {selectedPreset && <ScoreBadge score={selectedPreset.score} />}
      </DialogTitle>
      <DialogDescription>{selectedPreset?.description}</DialogDescription>
    </DialogHeader>
    <div className="space-y-4 py-4">
      {selectedPreset?.notes && (
        <div className="rounded-md bg-amber-50 border border-amber-200 px-3 py-2 text-sm text-amber-900">
          <strong>Tip:</strong> {selectedPreset.notes}
        </div>
      )}
      <div className="grid gap-2">
        <Label>Display Name</Label>
        <Input
          value={formData.name}
          onChange={(e) => setFormData(d => ({ ...d, name: e.target.value })))}
          data-testid="input-name"
        />
      </div>
      {selectedPreset?.fields.includes("api_key") && (
        <div className="grid gap-2">
          <Label>API Key</Label>
          <Input
            type="password"
            placeholder="Paste your API key..."
            value={formData.api_key}
            onChange={(e) => setFormData(d => ({ ...d, api_key: e.target.value })))}
            data-testid="input-api-key"
          />
        </div>
      )}
      {selectedPreset?.fields.includes("account_sid") && (
        <div className="grid gap-2">
          <Label>Account SID / Account ID</Label>
          <Input
            placeholder="ACxxxxxxxxx..."
            value={formData.account_sid}

```

```

        onChange={(e) => setFormData(d => ({ ...d, account_sid: e.target.value })))}
        data-testid="input-account-sid"
      />
    </div>
  )}
  {selectedPreset?.fields.includes("client_id") && (
    <div className="grid gap-2">
      <Label>Client ID</Label>
      <Input
        placeholder="OAuth client ID"
        value={formData.client_id}
        onChange={(e) => setFormData(d => ({ ...d, client_id: e.target.value })))}
        data-testid="input-client-id"
      />
    </div>
  )}
  {selectedPreset?.fields.includes("client_secret") && (
    <div className="grid gap-2">
      <Label>Client Secret</Label>
      <Input
        type="password"
        placeholder="OAuth client secret"
        value={formData.client_secret}
        onChange={(e) => setFormData(d => ({ ...d, client_secret: e.target.value })))}
        data-testid="input-client-secret"
      />
    </div>
  )}
  {selectedPreset?.fields.includes("api_url") && (
    <div className="grid gap-2">
      <Label>API Base URL</Label>
      <Input
        placeholder="https://api.example.com/v1"
        value={formData.api_url}
        onChange={(e) => setFormData(d => ({ ...d, api_url: e.target.value })))}
        data-testid="input-api-url"
      />
    </div>
  )}
  {selectedPreset?.fields.includes("webhook_url") && (
    <div className="grid gap-2">
      <Label>Webhook URL</Label>
      <Input
        placeholder="https://hooks.zapier.com/..."
        value={formData.webhook_url}
        onChange={(e) => setFormData(d => ({ ...d, webhook_url: e.target.value })))}
        data-testid="input-webhook-url"
      />
    </div>
  )}
</div>
<DialogFooter>
  <Button variant="outline" onClick={() => setAddOpen(false)}>Cancel</Button>
  <Button onClick={() => saveIntegration(false)} data-testid="button-save-preset">Connect</Button>
</DialogFooter>
</DialogContent>
</Dialog>

{/* Custom dialog */}
<Dialog open={customOpen} onOpenChange={setCustomOpen}>
  <DialogContent className="sm:max-w-lg">
    <DialogHeader>
      <DialogTitle>
        {formData.type === "vendor" ? "Connect Vendor API"
          : formData.type === "webpage" ? "Connect Web Page / Service"
          : "Custom API Integration"}
      </DialogTitle>
      <DialogDescription>
        {formData.type === "vendor" ? "Connect to a vendor's API for lead exchange and reporting."
          : formData.type === "webpage" ? "Add a web page or external service connection."
          : "Configure a custom REST API connection."}
      </DialogDescription>
    </DialogHeader>
    <div className="space-y-4 py-4">
      <div className="grid gap-2">
        <Label>Name</Label>
        <Input
          placeholder={formData.type === "vendor" ? "Vendor Name" : "Integration Name"}
          value={formData.name}
          onChange={(e) => setFormData(d => ({ ...d, name: e.target.value })))}
        />
      </div>
      <div className="grid gap-2">
        <Label>Provider / Identifier</Label>
        <Input
          placeholder="e.g. my-custom-crm"
          value={formData.provider}
          onChange={(e) => setFormData(d => ({ ...d, provider: e.target.value })))}
        />
      </div>
      <div className="grid gap-2">
        <Label>{formData.type === "webpage" ? "URL" : "API Base URL"}</Label>
        <Input
          placeholder="https://api.example.com/v1"
          value={formData.api_url}
          onChange={(e) => setFormData(d => ({ ...d, api_url: e.target.value })))}
        />
      </div>
    </div>
  </DialogContent>

```

```

    </div>
    <div className="grid gap-2">
      <Label>API Key (optional)</Label>
      <Input
        type="password"
        placeholder="Enter API key..."
        value={formData.api_key}
        onChange={(e) => setFormData(d => ({ ...d, api_key: e.target.value })))}
      />
    </div>
    <div className="grid gap-2">
      <Label>Webhook URL (optional)</Label>
      <Input
        placeholder="https://hooks.example.com/..."
        value={formData.webhook_url}
        onChange={(e) => setFormData(d => ({ ...d, webhook_url: e.target.value })))}
      />
    </div>
    <div className="grid gap-2">
      <Label>Sync Direction</Label>
      <Select value={formData.sync_direction} onChange={(v) => setFormData(d => ({ ...d, sync_direction: v })))>
        <SelectTrigger><SelectValue /></SelectTrigger>
        <SelectContent>
          <SelectItem value="bidirectional">Bidirectional</SelectItem>
          <SelectItem value="push">Push (MTOS to External)</SelectItem>
          <SelectItem value="pull">Pull (External to MTOS)</SelectItem>
        </SelectContent>
      </Select>
    </div>
    </div>
    <DialogFooter>
      <Button variant="outline" onClick={() => setCustomOpen(false)}>Cancel</Button>
      <Button onClick={() => saveIntegration(true)}>Save</Button>
    </DialogFooter>
  </DialogContent>
</Dialog>
</div>
);
}

```

[artifacts/mtos-crm/src/pages/lead-detail.tsx](#)

```
import { useState, useCallback, useEffect, useRef } from "react";
import { useParams, Link } from "wouter";
import { useGetLead, getGetLeadQueryKey, useUpdateLead, useDeleteLead, useListDocuments, getListDocumentsQueryKey,
useQualifyLead, useGetFormConfigs, UpdateLeadBodyStatus } from "@workspace/api-client-react";
import { useToast } from "@hooks/use-toast";
import { useQueryClient } from "@tanstack/react-query";
import { Button } from "@components/ui/button";
import { Badge } from "@components/ui/badge";
import { Card, CardContent, CardHeader, CardTitle, CardDescription } from "@components/ui/card";
import { ConvexityCard } from "@components/convexity-card";
import { Skeleton } from "@components/ui/skeleton";
import { Textarea } from "@components/ui/textarea";
import { Tabs, TabsContent, TabsList, TabsTrigger } from "@components/ui/tabs";
import { Separator } from "@components/ui/separator";
import { format } from "date-fns";
import {
  ArrowLeft, Trash2, FileText, CheckCircle, XCircle, FileSignature,
  AlertTriangle, Download, Brain, Save, RefreshCw, Shield, User,
  MapPin, Phone, Mail, Building2, Stethoscope, Scale, Activity, Zap
} from "lucide-react";
import { EnvelopeTimeline } from "@components/envelope-timeline";
import { Dialog, DialogContent, DialogDescription, DialogFooter, DialogHeader, DialogTitle, DialogTrigger } from "@components/ui/dialog";

interface IntelligenceScore {
  completion_score: number;
  completion_grade: string;
  completion_details: string[];
  reliability_score: number;
  reliability_grade: string;
  reliability_details: string[];
  truthfulness_score: number;
  truthfulness_grade: string;
  truthfulness_details: string[];
  overall_assessment: string;
  recommended_action: string;
  scored_at: string;
}

interface LensScore {
  conversion_probability: number;
  risk_score: number;
  quality_tier: string;
  factors: Record<string, number>;
}

function FieldRow({ label, value, sensitive }: { label: string; value: any; sensitive?: boolean }) {
  const displayVal = value === null || value === undefined || value === "" ? null : String(value);
  return (
    <div className="flex flex-col py-2 border-b border-gray-100 last:border-0">
      <span className="text-xs font-medium text-muted-foreground uppercase tracking-wider">{label}</span>
      <span className={`text-sm mt-0.5 ${displayVal ? "text-foreground" : "text-muted-foreground italic"} ${sensitive ? "font-
mono" : ""}`}>
        {displayVal || "Not provided"}
      </span>
    </div>
  );
}

function ScoreBadge({ score, label, grade }: { score: number; label: string; grade: string }) {
  const color = score >= 75 ? "text-emerald-600" : score >= 50 ? "text-amber-600" : "text-red-600";
  const bg = score >= 75 ? "bg-emerald-50 border-emerald-200" : score >= 50 ? "bg-amber-50 border-amber-200" : "bg-red-50
border-red-200";
  const barColor = score >= 75 ? "bg-emerald-500" : score >= 50 ? "bg-amber-500" : "bg-red-500";
  return (
    <div className={`rounded-lg border p-3 ${bg}`}>
      <div className="flex items-center justify-between mb-1">
        <span className="text-xs font-medium text-gray-600">{label}</span>
        <span className={`text-lg font-bold ${color}`}>{score}</span>
      </div>
      <div className="w-full bg-gray-200 rounded-full h-1.5">
        <div className={`h-1.5 rounded-full transition-all duration-500 ${barColor}`} style={{ width: `${score}%` }} />
      </div>
      <span className={`text-[10px] font-medium ${color}`}>{grade}</span>
    </div>
  );
}

function ScoreGauge({ score, label, grade }: { score: number; label: string; grade: string }) {
  const color = score >= 75 ? "text-emerald-600" : score >= 50 ? "text-amber-600" : "text-red-600";
  const bg = score >= 75 ? "bg-emerald-50 border-emerald-200" : score >= 50 ? "bg-amber-50 border-amber-200" : "bg-red-50
border-red-200";
  const barColor = score >= 75 ? "bg-emerald-500" : score >= 50 ? "bg-amber-500" : "bg-red-500";
  return (
    <div className={`rounded-lg border p-4 ${bg}`}>
      <div className="flex items-center justify-between mb-2">
        <span className="text-sm font-semibold text-gray-700">{label}</span>
        <span className={`text-2xl font-bold ${color}`}>{score}</span>
      </div>
      <div className="w-full bg-gray-200 rounded-full h-2 mb-2">
        <div className={`h-2 rounded-full transition-all duration-500 ${barColor}`} style={{ width: `${score}%` }} />
      </div>
      <span className={`text-xs font-medium ${color}`}>{grade}</span>
    </div>
  );
}
```

```

    });
}

export default function LeadDetail() {
  const { id } = useParams<{ id: string }>();
  const leadId = parseInt(id, 10);
  const { toast } = useToast();
  const queryClient = useQueryClient();

  const [notes, setNotes] = useState<string | null>(null);
  const [notesLoaded, setNotesLoaded] = useState(false);
  const [savingNotes, setSavingNotes] = useState(false);
  const [intelligence, setIntelligence] = useState<IntelligenceScore | null>(null);
  const [scoringInProgress, setScoringInProgress] = useState(false);
  const [scoringError, setScoringError] = useState<string | null>(null);
  const [lensScore, setLensScore] = useState<LensScore | null>(null);
  const [lensLoading, setLensLoading] = useState(false);
  const autoTriggered = useRef(false);

  const { data: lead, isLoading } = useGetLead(leadId, {
    query: { enabled: !!leadId, queryKey: getGetLeadQueryKey(leadId) }
  });

  const { data: documents, isLoading: docsLoading } = useListDocuments({ lead_id: leadId }, {
    query: { enabled: !!leadId, queryKey: getListDocumentsQueryKey({ lead_id: leadId }) }
  });

  const updateLead = useUpdateLead();
  const deleteLead = useDeleteLead();
  const qualifyLead = useQualifyLead();

  if (lead && !notesLoaded) {
    setNotes(lead.notes || "");
    setNotesLoaded(true);
  }

  const handleStatusChange = async (newStatus: UpdateLeadBodyStatus) => {
    try {
      await updateLead.mutateAsync({ id: leadId, data: { status: newStatus } });
      queryClient.invalidateQueries({ queryKey: getGetLeadQueryKey(leadId) });
      toast({ title: "Status updated successfully" });
    } catch {
      toast({ title: "Unable to update status – please retry", variant: "destructive" });
    }
  };

  const handleRunGatekeeper = async () => {
    try {
      const res = await qualifyLead.mutateAsync({ id: leadId });
      queryClient.invalidateQueries({ queryKey: getGetLeadQueryKey(leadId) });
      toast({
        title: "Boolean Gatekeeper Complete",
        description: `Determination: ${res.status.toUpperCase()}`,
        variant: res.status === "rejected" ? "destructive" : "default"
      });
    } catch {
      toast({ title: "Gatekeeper evaluation failed – please retry", variant: "destructive" });
    }
  };

  const handleSaveNotes = async () => {
    setSavingNotes(true);
    try {
      const res = await fetch(`/api/leads/${leadId}/notes`, {
        method: "PATCH",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify({ notes: notes || "" })
      });
      if (!res.ok) throw new Error("Save failed");
      toast({ title: "Case notes saved" });
    } catch {
      toast({ title: "Unable to save notes – please retry", variant: "destructive" });
    } finally {
      setSavingNotes(false);
    }
  };

  const handleRunIntelligence = useCallback(async () => {
    setScoringInProgress(true);
    setScoringError(null);

    const maxRetries = 3;
    for (let attempt = 1; attempt <= maxRetries; attempt++) {
      try {
        const res = await fetch(`/api/leads/${leadId}/intelligence`, {
          method: "POST",
          headers: { "Content-Type": "application/json" },
        });
      } catch {
        if (!res.ok) {
          const errData = await res.json().catch(() => ({}));
          throw new Error(errData.error || `Scoring returned status ${res.status}`);
        }
        const data = await res.json();
        setIntelligence(data);
        setScoringInProgress(false);
        return;
      } catch (err: any) {

```

```

    if (attempt === maxRetries) {
      setScoringError(err.message || "Intelligence scoring unavailable after multiple attempts");
      setScoringInProgress(false);
    } else {
      await new Promise(r => setTimeout(r, 1000 * attempt));
    }
  }
}, [leadId]);

const fetchLensScore = useCallback(async () => {
  setLensLoading(true);
  try {
    const res = await fetch(`/api/analytics/predictive/lead/${leadId}`);
    if (res.ok) {
      setLensScore(await res.json());
    }
  } catch {
  } finally {
    setLensLoading(false);
  }
}, [leadId]);

useEffect(() => {
  if (lead && !autoTriggered.current) {
    autoTriggered.current = true;
    handleRunIntelligence();
    fetchLensScore();
  }
}, [lead, handleRunIntelligence, fetchLensScore]);

if (isLoading || !lead) {
  return (
    <div className="space-y-6">
      <Skeleton className="h-8 w-64" />
      <Skeleton className="h-[600px] w-full" />
    </div>
  );
}

const leadExt = lead as typeof lead & {
  medications?: string | null;
  npi_number?: string | null;
  physician_taxonomy?: string | null;
  npi_verified?: boolean | null;
  fraud_status?: string | null;
  fraud_score?: number | null;
  fraud_indicators?: string[] | string | null;
};

return (
  <div className="space-y-6">
    <div className="flex items-center justify-between">
      <div className="flex items-center gap-4">
        <Button variant="outline" size="icon" asChild>
          <Link href="/leads"><ArrowLeft className="h-4 w-4" /></Link>
        </Button>
      </div>
      <div>
        <h1 className="text-3xl font-bold tracking-tight">{lead.name}</h1>
        <div className="text-sm text-muted-foreground flex items-center gap-2 mt-1">
          <span className="font-mono">ID: {lead.id}</span>
          <span>.</span>
          <span>{lead.tort_type}</span>
          <span>.</span>
          <span>Intake {format(new Date(lead.created_at), "MMM d, yyyy")}</span>
        </div>
      </div>
    </div>
    <div className="flex items-center gap-2">
      {lead.status === "new" && (
        <Button onClick={handleRunGatekeeper} disabled={qualifyLead.isPending}>
          <Scale className="mr-2 h-4 w-4" />
          Run Gatekeeper
        </Button>
      )}
      {lead.status === "qualified" && (
        <Button onClick={() => handleStatusChange("signed")} className="bg-emerald-600 hover:bg-emerald-700">
          <FileSignature className="mr-2 h-4 w-4" />
          Execute Retainer
        </Button>
      )}
      <Button variant="outline" size="sm" onClick={() => window.open(`/api/leads/export?lead_id=${lead.id}`, "_blank")}>
        <Download className="mr-2 h-4 w-4" />
        Export
      </Button>
      <Badge variant={
        lead.status === "signed" ? "default" :
        lead.status === "qualified" ? "secondary" :
        lead.status === "rejected" ? "destructive" :
        lead.status === "review_required" ? "secondary" : "outline"
      } className={
        `text-sm px-3 py-1 ${lead.status === "review_required" ? "bg-amber-500/10 text-amber-600 border-amber-500/20" : lead.status === "signed" ? "bg-emerald-600" : ""}`
      }>
        {lead.status === "review_required" ? "REVIEW REQUIRED" : lead.status.toUpperCase()}
      </Badge>
    </div>
  </div>
);

```



```

{lead.rejection_reason && (
<div className="bg-destructive/10 text-destructive p-4 rounded-lg text-sm border border-destructive/20 flex items-start
gap-3">
  <AlertTriangle className="h-5 w-5 shrink-0 mt-0.5" />
  <div>
    <span className="font-semibold">Disposition Note:</span> {lead.rejection_reason}
  </div>
</div>
)}

<div className="grid gap-3 md:grid-cols-5">
  {scoringInProgress ? (
    <div>
      {[1,2,3].map(i => <Skeleton key={i} className="h-[72px] rounded-lg" />)}
      <Skeleton className="h-[72px] rounded-lg" />
      <Skeleton className="h-[72px] rounded-lg" />
    </div>
  ) : intelligence ? (
    <div>
      <ScoreBadge label="Completion" score={intelligence.completion_score} grade={intelligence.completion_grade} />
      <ScoreBadge label="Reliability" score={intelligence.reliability_score} grade={intelligence.reliability_grade} />
      <ScoreBadge label="Truthfulness" score={intelligence.truthfulness_score} grade={intelligence.truthfulness_grade} />
    </div>
  ) : scoringError ? (
    <div className="md:col-span-3 flex items-center gap-2 text-sm text-muted-foreground bg-gray-50 rounded-lg border p-3">
      <AlertTriangle className="h-4 w-4 text-amber-500" />
      <span>Intelligence scoring unavailable</span>
      <Button variant="ghost" size="sm" onClick={handleRunIntelligence} className="ml-auto text-xs">Retry</Button>
    </div>
  ) : (
    <div className="md:col-span-3" />
  )}
  {lensLoading ? (
    <div>
      <Skeleton className="h-[72px] rounded-lg" />
      <Skeleton className="h-[72px] rounded-lg" />
    </div>
  ) : lensScore ? (
    <div>
      <div className="rounded-lg border bg-slate-50 p-3 flex flex-col justify-center">
        <span className="text-xs text-muted-foreground font-medium">Conversion</span>
        <span className="text-xl font-bold text-slate-800">{Math.round(lensScore.conversion_probability)}%</span>
      </div>
      <div className={`rounded-lg border p-3 flex flex-col justify-center ${
        lensScore.quality_tier === "platinum" ? "bg-violet-50 border-violet-200" :
        lensScore.quality_tier === "gold" ? "bg-amber-50 border-amber-200" :
        lensScore.quality_tier === "silver" ? "bg-gray-50 border-gray-300" :
        "bg-orange-50 border-orange-200"
      }`>
        <span className="text-xs text-muted-foreground font-medium">Quality Tier</span>
        <span className={`text-xl font-bold capitalize ${
          lensScore.quality_tier === "platinum" ? "text-violet-700" :
          lensScore.quality_tier === "gold" ? "text-amber-700" :
          lensScore.quality_tier === "silver" ? "text-gray-600" :
          "text-orange-700"
        }`}>{lensScore.quality_tier}</span>
      </div>
    </div>
  ) : (
    <div className="md:col-span-2" />
  )}
</div>

{intelligence?.overall_assessment && (
  <div className="bg-slate-50 border rounded-lg p-4 flex items-start gap-3">
    <Brain className="h-5 w-5 text-primary shrink-0 mt-0.5" />
    <div className="space-y-1">
      <p className="text-sm leading-relaxed">{intelligence.overall_assessment}</p>
      <p className="text-xs text-muted-foreground"><span className="font-medium">Recommended:</span>
{intelligence.recommended_action}</p>
    </div>
    <Button variant="ghost" size="sm" onClick={handleRunIntelligence} className="shrink-0 ml-auto text-xs"
disabled={scoringInProgress}>
      Refresh
    </Button>
  </div>
)}

<Tabs defaultValue="profile" className="w-full">
  <TabsList className="grid w-full grid-cols-6">
    <TabsTrigger value="profile" className="flex items-center gap-2"><User className="h-4 w-4" />Claimant Profile</
TabsTrigger>
    <TabsTrigger value="medical" className="flex items-center gap-2"><Stethoscope className="h-4 w-4" />Medical &
Exposure</TabsTrigger>
    <TabsTrigger value="compliance" className="flex items-center gap-2"><Shield className="h-4 w-4" />Compliance</
TabsTrigger>
    <TabsTrigger value="documents" className="flex items-center gap-2"><FileText className="h-4 w-4" />Documents</
TabsTrigger>
    <TabsTrigger value="automation" className="flex items-center gap-2"><Zap className="h-4 w-4" />Automation</
TabsTrigger>
    <TabsTrigger value="intelligence" className="flex items-center gap-2"><Brain className="h-4 w-4" />AI Intelligence</
TabsTrigger>
  </TabsList>

  <TabsContent value="profile" className="space-y-6 mt-6">
    <div className="grid gap-6 md:grid-cols-2">

```

```

<Card>
  <CardHeader>
    <CardTitle className="flex items-center gap-2"><User className="h-4 w-4" />Personal Information</CardTitle>
  </CardHeader>
  <CardContent className="space-y-0">
    <FieldRow label="First Name" value={lead.first_name} />
    <FieldRow label="Last Name" value={lead.last_name} />
    <FieldRow label="Date of Birth" value={lead.date_of_birth} sensitive />
    <FieldRow label="Last 4 SSN" value={lead.last_4_ssn} sensitive />
  </CardContent>
</Card>

<Card>
  <CardHeader>
    <CardTitle className="flex items-center gap-2"><Phone className="h-4 w-4" />Contact Details</CardTitle>
  </CardHeader>
  <CardContent className="space-y-0">
    <FieldRow label="Primary Phone" value={lead.phone_primary || lead.phone} />
    <FieldRow label="Email Address" value={lead.email} />
    <FieldRow label="Lead Source" value={lead.source} />
    <FieldRow label="Acquisition Cost" value={lead.ad_spend ? `${Number(lead.ad_spend).toFixed(2)}` : null} />
  </CardContent>
</Card>

<Card>
  <CardHeader>
    <CardTitle className="flex items-center gap-2"><MapPin className="h-4 w-4" />Residential Address</CardTitle>
  </CardHeader>
  <CardContent className="space-y-0">
    <FieldRow label="Street Address" value={lead.street_address} />
    <FieldRow label="City" value={lead.city} />
    <FieldRow label="State" value={lead.state} />
    <FieldRow label="ZIP Code" value={lead.zip} />
  </CardContent>
</Card>

<Card>
  <CardHeader>
    <CardTitle className="flex items-center gap-2"><Building2 className="h-4 w-4" />Firm & Vendor Details</CardTitle>
  </CardHeader>
  <CardContent className="space-y-0">
    <FieldRow label="Referring Law Firm" value={lead.law_firm} />
    <FieldRow label="Client Identifier" value={lead.client_id} />
    <FieldRow label="Vendor ID" value={lead.vendor_id} />
    <FieldRow label="Pipeline Routing" value={lead.routing} />
  </CardContent>
</Card>
</div>

<Card>
  <CardHeader className="flex flex-row items-center justify-between space-y-0">
    <div>
      <CardTitle>Case Notes</CardTitle>
      <CardDescription>Internal memoranda and observations regarding this claimant</CardDescription>
    </div>
    <Button onClick={handleSaveNotes} disabled={savingNotes} size="sm" className="flex items-center gap-2">
      <Save className="h-4 w-4" />
      {savingNotes ? "Saving..." : "Save Notes"}
    </Button>
  </CardHeader>
  <CardContent>
    <Textarea
      placeholder="Enter case notes, observations, follow-up items, or internal memoranda..."
      className="min-h-[160px] resize-y"
      value={notes || ""}
      onChange={(e) => setNotes(e.target.value)}
    />
    <div className="text-xs text-muted-foreground mt-2">
      Last modified: {lead.updated_at ? format(new Date(lead.updated_at), "MMM d, yyyy 'at' h:mm a") : "-"}
    </div>
  </CardContent>
</Card>

<ConvexityCard
  leadId={lead.id}
  score={(lead as any).convexity_score ?? null}
  action={(lead as any).convexity_action ?? null}
  rationale={(lead as any).convexity_rationale ?? null}
  ruinFlags={((lead as any).convexity_ruin_flags ?? []) as string[]}
  missingFields={((lead as any).convexity_missing_fields ?? []) as string[]}
  contradictions={((lead as any).convexity_contradictions ?? []) as string[]}
  downsideUsd={(lead as any).convexity_downside_usd ?? null}
  upsideUsd={(lead as any).convexity_upside_usd ?? null}
  ratio={(lead as any).convexity_ratio ?? null}
  confidence={(lead as any).convexity_confidence ?? null}
  computedAt={(lead as any).convexity_computed_at ?? null}
/>
</TabsContent>

<TabsContent value="medical" className="space-y-6 mt-6">
  <div className="grid gap-6 md:grid-cols-2">
    <Card>
      <CardHeader>
        <CardTitle className="flex items-center gap-2"><Activity className="h-4 w-4" />Tort & Diagnosis</CardTitle>
      </CardHeader>
      <CardContent className="space-y-0">

```

```

        <FieldRow label="Tort Classification" value={lead.tort_type} />
        <FieldRow label="Primary Diagnosis" value={lead.diagnosis} />
        <FieldRow label="Diagnosis Date" value={lead.diagnosis_date} />
        <FieldRow label="Diagnosis Type" value={lead.diagnosis_type} />
        <FieldRow label="Current Medications" value={leadExt.medications} />
        <div className="flex items-center gap-3 py-3 border-b border-gray-100">
            {lead.diagnosis_confirmed ? <CheckCircle className="text-emerald-500 h-4 w-4" /> : <XCircle className="text-
red-500 h-4 w-4" />}
            <span className="text-sm">{lead.diagnosis_confirmed ? "Diagnosis confirmed by claimant" : "Diagnosis not
confirmed"}</span>
        </div>
    </CardContent>
</Card>

<Card>
    <CardHeader>
        <CardTitle className="flex items-center gap-2"><MapPin className="h-4 w-4" />Exposure Details</CardTitle>
    </CardHeader>
    <CardContent className="space-y-0">
        <FieldRow label="Exposure Location" value={lead.location_name} />
        <FieldRow label="Exposure Start" value={lead.exposure_start} />
        <FieldRow label="Exposure End" value={lead.exposure_end} />
        <div className="flex items-center gap-3 py-3 border-b border-gray-100">
            {lead.was_at_location ? <CheckCircle className="text-emerald-500 h-4 w-4" /> : <XCircle className="text-
red-500 h-4 w-4" />}
            <span className="text-sm">{lead.was_at_location ? "Location exposure verified" : "Location exposure not
verified"}</span>
        </div>
    </CardContent>
</Card>

<Card>
    <CardHeader>
        <CardTitle className="flex items-center gap-2"><Stethoscope className="h-4 w-4" />Treating Physician</CardTitle>
    </CardHeader>
    <CardContent className="space-y-0">
        <FieldRow label="First Name" value={lead.physician_first_name} />
        <FieldRow label="Last Name" value={lead.physician_last_name} />
        <FieldRow label="Full Address" value={lead.physician_full_address} />
        <FieldRow label="Contact Information" value={lead.physician_contact_info} />
        <FieldRow label="NPI Number" value={leadExt.npi_number} />
        <FieldRow label="Taxonomy Code" value={leadExt.physician_taxonomy} />
        <div className="flex items-center gap-3 py-3">
            {leadExt.npi_verified ? <CheckCircle className="text-emerald-500 h-4 w-4" /> : <XCircle className="text-
red-500 h-4 w-4" />}
            <span className="text-sm">{leadExt.npi_verified ? "NPI verified via NPPES registry" : "NPI not yet verified"}
</span>
        </div>
    </CardContent>
</Card>

<Card>
    <CardHeader>
        <CardTitle className="flex items-center gap-2"><Building2 className="h-4 w-4" />Hospital / Treatment Facility</
CardTitle>
    </CardHeader>
    <CardContent className="space-y-0">
        <FieldRow label="Facility Name" value={lead.hospital_name} />
        <FieldRow label="Fax Number" value={lead.hospital_fax} />
        <FieldRow label="Contact Information" value={lead.hospital_contact_info} />
    </CardContent>
</Card>

    <CustomFieldsCard tortType={lead.tort_type} customFields={{(lead as any).custom_fields}} />
</div>
</TabsContent>

<TabsContent value="compliance" className="space-y-6 mt-6">
    <div className="grid gap-6 md:grid-cols-2">
        <Card>
            <CardHeader>
                <CardTitle>Consent & Certification</CardTitle>
            </CardHeader>
            <CardContent className="space-y-3">
                <div className="flex items-center gap-3 py-2 border-b border-gray-100">
                    {lead.tcpa_consent ? <CheckCircle className="text-emerald-500 h-5 w-5" /> : <XCircle className="text-red-500
h-5 w-5" />}
                    <div>
                        <div className="font-medium text-sm">TCPA Consent</div>
                        <div className="text-xs text-muted-foreground">{lead.tcpa_consent ? "Express written consent obtained" :
"Consent not on file"}</div>
                    </div>
                    <FieldRow label="TrustedForm Certificate URL" value={lead.trustedform_cert_url} />
                    <FieldRow label="TrustedForm IP" value={lead.trustedform_ip} />
                    <FieldRow label="TrustedForm User Agent" value={lead.trustedform_user_agent} />
                    <FieldRow label="TrustedForm Timestamp" value={lead.trustedform_timestamp ? format(new
Date(lead.trustedform_timestamp), "MMM d, yyyy h:mm a") : null} />
                </CardContent>
            </Card>

            <Card>
                <CardHeader>
                    <CardTitle>Verification Status</CardTitle>
                </CardHeader>
                <CardContent className="space-y-3">
                    <div className="flex items-center gap-3 py-2 border-b border-gray-100">

```

```

        {lead.email_validation_status === "valid" ? <CheckCircle className="text-emerald-500 h-5 w-5" /> :
        lead.email_validation_status === "invalid" ? <XCircle className="text-red-500 h-5 w-5" /> :
        <AlertTriangle className="text-amber-500 h-5 w-5" />}
      </div>
      <div className="font-medium text-sm">Email Validation</div>
      <div className="text-xs text-muted-foreground">{lead.email_validation_status ?
lead.email_validation_status.charAt(0).toUpperCase() + lead.email_validation_status.slice(1) : "Not performed"}</div>
    </div>
    <div className="flex items-center gap-3 py-2 border-b border-gray-100">
      {lead.address_validation_status === "valid" ? <CheckCircle className="text-emerald-500 h-5 w-5" /> :
      lead.address_validation_status === "invalid" ? <XCircle className="text-red-500 h-5 w-5" /> :
      <AlertTriangle className="text-amber-500 h-5 w-5" />}
    </div>
    <div className="font-medium text-sm">Address Validation</div>
    <div className="text-xs text-muted-foreground">{lead.address_validation_status ?
lead.address_validation_status.charAt(0).toUpperCase() + lead.address_validation_status.slice(1) : "Not performed"}</div>
  </div>
  <div className="flex items-center gap-3 py-2 border-b border-gray-100">
    {lead.background_check_status === "clean" ? <CheckCircle className="text-emerald-500 h-5 w-5" /> :
    lead.background_check_status === "flagged" ? <AlertTriangle className="text-amber-500 h-5 w-5" /> :
    <XCircle className="text-muted-foreground h-5 w-5" />}
  </div>
  <div className="font-medium text-sm">Background Check</div>
  <div className="text-xs text-muted-foreground">{lead.background_check_status ?
lead.background_check_status.charAt(0).toUpperCase() + lead.background_check_status.slice(1) : "Not initiated"}</div>
</div>
</CardContent>
</Card>

<Card>
  <CardHeader>
    <CardTitle>Fraud Assessment</CardTitle>
  </CardHeader>
  <CardContent className="space-y-3">
    {leadExt.fraud_status && (
      <div className="flex items-center justify-between py-2 border-b border-gray-100">
        <span className="text-sm font-medium">Final Arbiter Decision</span>
        <Badge variant={
          leadExt.fraud_status === "ACCEPTED" ? "default" :
          leadExt.fraud_status === "REJECTED" ? "destructive" : "secondary"
        } className={leadExt.fraud_status === "ACCEPTED" ? "bg-emerald-500/10 text-emerald-600 border-emerald-500/20" : leadExt.fraud_status === "TO_BE_REVIEWED" ? "bg-amber-500/10 text-amber-600 border-amber-500/20" : ""}>
          {leadExt.fraud_status}
        </Badge>
      </div>
    )}
    {typeof leadExt.fraud_score === "number" && (
      <div className="flex items-center justify-between py-2 border-b border-gray-100">
        <span className="text-sm font-medium">Fraud Risk Score</span>
        <span className={`text-lg font-bold font-mono ${leadExt.fraud_score >= 60 ? "text-red-600" :
leadExt.fraud_score >= 20 ? "text-amber-600" : "text-emerald-600"}`}>
          {leadExt.fraud_score}/100
        </span>
      </div>
    )}
    {leadExt.fraud_indicators && (
      <div className="py-2">
        <span className="text-xs font-medium text-muted-foreground uppercase tracking-wider">Indicators</span>
        <p className="text-sm mt-1">{leadExt.fraud_indicators}</p>
      </div>
    )}
  </CardContent>
</Card>

{lead.background_check_data && (() => {
  try {
    const bgData = JSON.parse(lead.background_check_data as string) as {
      summary?: string;
      checked_at?: string;
      records?: Array<{
        type: string;
        description: string;
        date?: string;
        jurisdiction?: string;
        severity: string;
        role?: string;
      }>;
    };
    if (!bgData.records || bgData.records.length === 0) return null;
    const partyRecords = bgData.records.filter(r => r.role === "party");
    const mentionRecords = bgData.records.filter(r => r.role === "mentioned");
    return (
      <Card>
        <CardHeader>
          <CardTitle>Court Records</CardTitle>
          <CardDescription>
            {bgData.summary}
            {bgData.checked_at && ` - checked ${format(new Date(bgData.checked_at), "MMM d, yyyy")}`}
          </CardDescription>
        </CardHeader>
        <CardContent className="space-y-3">
          {partyRecords.length > 0 && (
            <div className="space-y-2">
              <div className="text-sm font-semibold text-red-600">Named Party ({partyRecords.length})</div>

```

```

        {partyRecords.map((rec, i) => (
          <div key={`party-${i}`} className="border border-red-200 bg-red-50 rounded-md p-3 space-y-1">
            <div className="flex items-center justify-between">
              <span className="font-medium text-sm">{rec.description}</span>
              <Badge variant="destructive" className="text-xs">{rec.severity === "high" ? "High" : "Medium"}</
Badge>
            </div>
            <div className="flex gap-4 text-xs text-muted-foreground">
              {rec.date && <span>Filed: {format(new Date(rec.date), "MMM d, yyyy")}</span>}
              {rec.jurisdiction && <span>{rec.jurisdiction}</span>}
            </div>
          </div>
        ))}
      </div>
    )}
    {mentionRecords.length > 0 && (
      <div className="space-y-2">
        <div className="text-sm font-semibold text-muted-foreground">Referenced Mentions
        ({mentionRecords.length})</div>
        {mentionRecords.map((rec, i) => (
          <div key={`mention-${i}`} className="border rounded-md p-3 space-y-1">
            <div className="flex items-center justify-between">
              <span className="text-sm">{rec.description}</span>
              <Badge variant="outline" className="text-xs">Low</Badge>
            </div>
            <div className="flex gap-4 text-xs text-muted-foreground">
              {rec.date && <span>Filed: {format(new Date(rec.date), "MMM d, yyyy")}</span>}
              {rec.jurisdiction && <span>{rec.jurisdiction}</span>}
            </div>
          </div>
        ))}
      </div>
    )}
  </CardContent>
</Card>
);
} catch { return null; }
})();
</div>
</TabsContent>

<TabsContent value="documents" className="space-y-6 mt-6">
  <Card>
    <CardHeader className="flex flex-row items-center justify-between space-y-0">
      <div>
        <CardTitle>Associated Documents</CardTitle>
        <CardDescription>All documents, agreements, and records linked to this claimant</CardDescription>
      </div>
    </CardHeader>
    <CardContent>
      {docsLoading ? (
        <Skeleton className="h-24 w-full" />
      ) : !documents || documents.length === 0 ? (
        <div className="text-center py-8 text-muted-foreground">
          <FileText className="h-8 w-8 mx-auto mb-2 opacity-50" />
          <p className="text-sm">No documents have been associated with this claimant record.</p>
        </div>
      ) : (
        <div className="space-y-2">
          {documents.map(doc => (
            <div key={doc.id} className="flex items-center justify-between py-3 border-b last:border-0">
              <div className="flex items-center gap-3">
                <FileText className="h-5 w-5 text-muted-foreground" />
                <div>
                  <div className="font-medium text-sm">{doc.file_name}</div>
                  <div className="text-xs text-muted-foreground">{doc.document_type} · ID: {doc.id}</div>
                </div>
              </div>
              <div className="flex items-center gap-2">
                {doc.signed ? (
                  <Badge className="bg-emerald-500/10 text-emerald-600 border-emerald-500/20">Executed</Badge>
                ) : (
                  <Badge variant="outline">Pending Execution</Badge>
                )}
              </div>
            </div>
          ))}
        </div>
      )}
    </CardContent>
  </Card>
</TabsContent>

<TabsContent value="automation" className="space-y-6 mt-6">
  <EnvelopeTimeline leadId={leadId} />
</TabsContent>

<TabsContent value="intelligence" className="space-y-6 mt-6">
  <Card>
    <CardHeader className="flex flex-row items-center justify-between space-y-0">
      <div>
        <CardTitle className="flex items-center gap-2"><Brain className="h-5 w-5" />Lead Intelligence Assessment</
CardTitle>
        <CardDescription>AI-powered analysis evaluating file completeness, data reliability, and claim veracity</
CardDescription>
      </div>
      <Button onClick={handleRunIntelligence} disabled={scoringInProgress} variant="outline" size="sm" className="flex

```

```

items-center gap-2">
    {scoringInProgress ? <RefreshCw className="h-4 w-4 animate-spin" /> : <RefreshCw className="h-4 w-4" />}
    {scoringInProgress ? "Re-analyzing..." : "Refresh Scores"}
  </Button>
</CardHeader>
<CardContent>
  {!intelligence && !scoringInProgress && !scoringError && (
    <div className="text-center py-12 text-muted-foreground">
      <Brain className="h-10 w-10 mx-auto mb-3 opacity-40" />
      <p className="text-sm">Intelligence assessment is initializing...</p>
    </div>
  )}

  {scoringError && (
    <div className="bg-destructive/10 text-destructive p-4 rounded-lg text-sm border border-destructive/20 flex
items-start gap-3">
      <AlertTriangle className="h-5 w-5 shrink-0 mt-0.5" />
      <div>
        <span className="font-semibold">Assessment Unavailable:</span> {scoringError}
        <div className="mt-2">
          <Button variant="outline" size="sm" onClick={handleRunIntelligence}>Retry Assessment</Button>
        </div>
      </div>
    </div>
  )}

  {scoringInProgress && (
    <div className="text-center py-12">
      <RefreshCw className="h-8 w-8 mx-auto mb-3 animate-spin text-primary" />
      <p className="text-sm text-muted-foreground">Evaluating claimant record across all available parameters...</p>
    </div>
  )}

  {intelligence && !scoringInProgress && (
    <div className="space-y-6">
      <div className="grid gap-4 md:grid-cols-3">
        <ScoreGauge score={intelligence.completion_score} label="Completion Score"
grade={intelligence.completion_grade} />
        <ScoreGauge score={intelligence.reliability_score} label="Reliability Score"
grade={intelligence.reliability_grade} />
        <ScoreGauge score={intelligence.truthfulness_score} label="Truthfulness Score"
grade={intelligence.truthfulness_grade} />
      </div>

      <Separator />

      <div className="bg-slate-50 border rounded-lg p-4 space-y-3">
        <h4 className="font-semibold text-sm">Overall Assessment</h4>
        <p className="text-sm leading-relaxed">{intelligence.overall_assessment}</p>
        <div className="bg-white border rounded-md p-3 mt-2">
          <span className="text-xs font-medium text-muted-foreground uppercase tracking-wider">Recommended Action</
span>
          <p className="text-sm mt-1 font-medium">{intelligence.recommended_action}</p>
        </div>
        <div className="text-xs text-muted-foreground">
          Scored: {format(new Date(intelligence.scored_at), "MMM d, yyyy 'at' h:mm a")}
        </div>
      </div>

      <div className="grid gap-4 md:grid-cols-3">
        <Card>
          <CardHeader className="pb-2">
            <CardTitle className="text-sm">Completion Findings</CardTitle>
          </CardHeader>
          <CardContent>
            <ul className="space-y-1.5">
              {intelligence.completion_details.map((d, i) => (
                <li key={i} className="text-xs text-muted-foreground flex items-start gap-1.5">
                  <span className={d.startsWith("Missing") ? "text-red-400" : "text-emerald-400"}>.</span>
                  {d}
                </li>
              )
            )}
            </ul>
          </CardContent>
        </Card>
        <Card>
          <CardHeader className="pb-2">
            <CardTitle className="text-sm">Reliability Findings</CardTitle>
          </CardHeader>
          <CardContent>
            <ul className="space-y-1.5">
              {intelligence.reliability_details.map((d, i) => (
                <li key={i} className="text-xs text-muted-foreground flex items-start gap-1.5">
                  <span className={d.includes("not") || d.includes("failed") || d.includes("No ") ? "text-
red-400" : "text-emerald-400"}>.</span>
                  {d}
                </li>
              )
            )}
            </ul>
          </CardContent>
        </Card>
        <Card>
          <CardHeader className="pb-2">
            <CardTitle className="text-sm">Truthfulness Findings</CardTitle>
          </CardHeader>
          <CardContent>
            <ul className="space-y-1.5">

```

```

        {intelligence.truthfulness_details.map((d, i) => (
          <li key={i} className="text-xs text-muted-foreground flex items-start gap-1.5">
            <span className={d.includes("REJECTED") || d.includes("critically") || d.includes("elevated")} ?
"text-red-400" : d.includes("not") || d.includes("gaps") ? "text-amber-400" : "text-emerald-400">.</span>
            {d}
          </li>
        ))}
      </ul>
    </CardContent>
  </Card>
</div>
</div>
)}
</CardContent>
</Card>
</TabsContent>
</Tabs>

<Card className="border-destructive/30">
  <CardHeader>
    <CardTitle className="text-destructive">Administrative Actions</CardTitle>
  </CardHeader>
  <CardContent>
    <Dialog>
      <DialogTrigger asChild>
        <Button variant="destructive" size="sm">
          <Trash2 className="mr-2 h-4 w-4" />
          Permanently Delete Record
        </Button>
      </DialogTrigger>
      <DialogContent>
        <DialogHeader>
          <DialogTitle>Confirm Permanent Deletion</DialogTitle>
          <DialogDescription>
            This action is irreversible. All data associated with this claimant – including documents,
            notes, and compliance records – will be permanently removed from the system.
          </DialogDescription>
        </DialogHeader>
        <DialogFooter>
          <Button variant="outline">Cancel</Button>
          <Button variant="destructive" onClick={async () => {
            try {
              await deleteLead.mutateAsync({ id: leadId });
              toast({ title: "Record permanently deleted" });
              window.location.href = "/leads";
            } catch {
              toast({ title: "Deletion failed – please retry", variant: "destructive" });
            }
          }} disabled={deleteLead.isPending}>
            {deleteLead.isPending ? "Deleting..." : "Confirm Deletion"}
          </Button>
        </DialogFooter>
      </DialogContent>
    </Dialog>
  </CardContent>
</Card>
</div>
);
}

```

```

function CustomFieldsCard({ tortType, customFields }: { tortType?: string | null; customFields?: Record<string, unknown> |
null }) {
  const { data: configsData } = useGetFormConfigs();
  const all = ((configsData as { tort_campaigns?: Array<{ id: string; label: string; custom_fields?: Array<{ key: string;
label: string; type: string }> }> }>).tort_campaigns) || []];
  const needle = (tortType || "").toLowerCase().trim();
  const cfg = all.find(c => c.id.toLowerCase() === needle || c.label.toLowerCase() === needle) || null;
  const cfgFields = (cfg?.custom_fields || []) as Array<{ key: string; label: string; type: string }>;
  const values = (customFields || {}) as Record<string, unknown>;
  const valueKeys = Object.keys(values);

  if (cfgFields.length === 0 && valueKeys.length === 0) return null;

  const seen = new Set<string>();
  const rows: Array<{ key: string; label: string; value: unknown }> = [];
  for (const f of cfgFields) {
    seen.add(f.key);
    rows.push({ key: f.key, label: f.label, value: values[f.key] });
  }
  for (const k of valueKeys) {
    if (!seen.has(k)) rows.push({ key: k, label: k.replace(/_/g, " "), value: values[k] });
  }

  const renderValue = (v: unknown): string => {
    if (v === null || v === undefined || v === "") return "-";
    if (typeof v === "boolean") return v ? "Yes" : "No";
    if (Array.isArray(v)) return v.join(", ");
    if (typeof v === "object") return JSON.stringify(v);
    return String(v);
  };

  return (
    <Card data-testid="card-custom-fields">
      <CardHeader>
        <CardTitle className="flex items-center gap-2">Custom Fields</CardTitle>
        <CardDescription>Form fields defined for the {tortType || "tort"} campaign.</CardDescription>
      </CardHeader>
    </Card>
  );
}

```

```

    <CardContent className="space-y-0">
      {rows.map(r => (
        <div key={r.key} className="flex flex-col py-2 border-b border-gray-100 last:border-0" data-testid={`row-cf-${r.key}`}`>
          <span className="text-xs font-medium text-muted-foreground uppercase tracking-wider">{r.label}</span>
          <span className={`text-sm mt-0.5 ${values[r.key] !== undefined && values[r.key] !== null && values[r.key] !== "" ?
"text-foreground" : "text-muted-foreground italic"}`>
            {renderValue(r.value)}
          </span>
        </div>
      ))}
    </CardContent>
  </Card>
);
}

```


[artifacts/mtos-crm/src/pages/lead-import.tsx](#)

```
import { useState, useEffect, useCallback } from "react";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import { Button } from "@components/ui/button";
import { Badge } from "@components/ui/badge";
import { Skeleton } from "@components/ui/skeleton";
import { useToast } from "@hooks/use-toast";
import { Upload, FileSpreadsheet, CheckCircle2, XCircle, AlertTriangle, Copy, RefreshCw, ArrowRight, ChevronDown, ChevronUp }
from "lucide-react";

interface ImportBatch {
  id: number;
  filename: string;
  status: string;
  total_rows: number;
  processed_rows: number;
  success_count: number;
  duplicate_count: number;
  error_count: number;
  conflict_count: number;
  created_at: string;
  completed_at: string | null;
}

interface PreviewResult {
  total_rows: number;
  headers: string[];
  column_mapping: Record<string, string>;
  unmapped_columns: string[];
  available_fields: string[];
  sample_leads: Record<string, any>[];
}

interface ImportRow {
  id: number;
  row_number: number;
  status: string;
  raw_data: Record<string, string>;
  lead_id: number | null;
  dedup_match_id: number | null;
  dedup_reason: string | null;
  error_message: string | null;
  conflict_details: any;
}

type Step = "upload" | "preview" | "importing" | "results";

export default function LeadImport() {
  const { toast } = useToast();
  const [step, setStep] = useState<Step>("upload");
  const [csvData, setCsvData] = useState<string>("");
  const [filename, setFilename] = useState<string>("");
  const [preview, setPreview] = useState<PreviewResult | null>(null);
  const [columnMapping, setColumnMapping] = useState<Record<string, string>>({});
  const [activeBatch, setActiveBatch] = useState<ImportBatch | null>(null);
  const [batches, setBatches] = useState<ImportBatch[]>([]);
  const [batchRows, setBatchRows] = useState<ImportRow[]>([]);
  const [loading, setLoading] = useState(false);
  const [expandedErrors, setExpandedErrors] = useState(false);
  const [expandedDupes, setExpandedDupes] = useState(false);

  const fetchBatches = useCallback(async () => {
    try {
      const res = await fetch("/api/lead-import/batches");
      if (res.ok) setBatches(await res.json());
    } catch {}
  }, []);

  useEffect(() => {
    fetchBatches();
  }, [fetchBatches]);

  const handleFileUpload = async (e: React.ChangeEvent<HTMLInputElement>) => {
    const file = e.target.files?.[0];
    if (!file) return;

    if (!file.name.endsWith(".csv")) {
      toast({ title: "Invalid file", description: "Please upload a CSV file.", variant: "destructive" });
      return;
    }

    if (file.size > 10 * 1024 * 1024) {
      toast({ title: "File too large", description: "Maximum file size is 10MB.", variant: "destructive" });
      return;
    }

    const text = await file.text();
    setCsvData(text);
    setFilename(file.name);
    setLoading(true);

    try {
      const res = await fetch("/api/lead-import/preview", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify({ csv_data: text }),
      });
    }
  }
}
```

```

});

if (!res.ok) {
  const err = await res.json();
  throw new Error(err.error || "Preview failed");
}

const data: PreviewResult = await res.json();
setPreview(data);
setColumnMapping(data.column_mapping);
setStep("preview");
} catch (err: any) {
  toast({ title: "Preview failed", description: err.message, variant: "destructive" });
} finally {
  setLoading(false);
}
};

const handleExecuteImport = async () => {
  setLoading(true);
  try {
    const res = await fetch("/api/lead-import/execute", {
      method: "POST",
      headers: { "Content-Type": "application/json" },
      body: JSON.stringify({
        csv_data: csvData,
        column_mapping: columnMapping,
        filename,
      })),
    });
  } catch (err: any) {
    const err = await res.json();
    throw new Error(err.error || "Import failed");
  }

  const data = await res.json();
  setActiveBatch({ ...data, filename, success_count: 0, duplicate_count: 0, error_count: 0, conflict_count: 0, created_at:
new Date().toISOString(), completed_at: null } as ImportBatch);
  setStep("importing");

  pollBatchStatus(data.batch_id);
} catch (err: any) {
  toast({ title: "Import failed", description: err.message, variant: "destructive" });
} finally {
  setLoading(false);
}
};

const pollBatchStatus = async (batchId: number) => {
  const poll = async () => {
    try {
      const res = await fetch(`/api/lead-import/batches/${batchId}`);
      if (!res.ok) return;
      const data = await res.json();
      const { rows, ...batch } = data;
      setActiveBatch(batch);
      setBatchRows(rows || []);

      if (batch.status === "completed") {
        setStep("results");
        fetchBatches();
        return;
      }
    } catch {}
    setTimeout(poll, 2000);
  };
  poll();
};

const handleViewBatch = async (batchId: number) => {
  try {
    const res = await fetch(`/api/lead-import/batches/${batchId}`);
    if (!res.ok) return;
    const data = await res.json();
    const { rows, ...batch } = data;
    setActiveBatch(batch);
    setBatchRows(rows || []);
    setStep("results");
  } catch {}
};

const resetImport = () => {
  setStep("upload");
  setCsvData("");
  setFilename("");
  setPreview(null);
  setColumnMapping({});
  setActiveBatch(null);
  setBatchRows([]);
  setExpandedErrors(false);
  setExpandedDups(false);
};

const statusBadge = (status: string) => {
  const map: Record<string, { variant: "default" | "secondary" | "destructive" | "outline"; label: string }> = {
    success: { variant: "default", label: "Imported" },
  }

```

```

duplicate: { variant: "secondary", label: "Duplicate" },
error: { variant: "destructive", label: "Error" },
rejected: { variant: "destructive", label: "Rejected" },
conflict: { variant: "outline", label: "Conflict" },
pending: { variant: "secondary", label: "Pending" },
};
const cfg = map[status] || { variant: "secondary" as const, label: status };
return <Badge variant={cfg.variant}>{cfg.label}</Badge>;
};

return (
  <div className="space-y-6">
    <div className="flex items-center justify-between">
      <div>
        <h1 className="text-3xl font-bold tracking-tight">Lead Import</h1>
        <p className="text-muted-foreground mt-1">
          Bulk import leads from CSV files with full validation, deduplication, and conflict checking.
        </p>
      </div>
      {step !== "upload" && (
        <Button variant="outline" onClick={resetImport}>
          <Upload className="h-4 w-4 mr-2" />
          New Import
        </Button>
      )}
    </div>

    {step === "upload" && (
      <div className="grid gap-6 md:grid-cols-2">
        <Card>
          <CardHeader>
            <CardTitle className="flex items-center gap-2">
              <FileSpreadsheet className="h-5 w-5" />
              Upload CSV
            </CardTitle>
          </CardHeader>
          <CardContent>
            <div className="border-2 border-dashed rounded-lg p-8 text-center">
              <Upload className="h-10 w-10 mx-auto text-muted-foreground mb-4" />
              <p className="text-sm text-muted-foreground mb-4">
                Drop a CSV file here or click to browse
              </p>
              <input
                type="file"
                accept=".csv"
                onChange={handleFileUpload}
                className="hidden"
                id="csv-upload"
              />
              <label htmlFor="csv-upload">
                <Button asChild variant="outline" disabled={loading}>
                  <span>{loading ? "Analyzing..." : "Choose File"}</span>
                </Button>
              </label>
            </div>
            <div className="mt-4 space-y-2 text-xs text-muted-foreground">
              <p>Each lead is individually validated through the full pipeline:</p>
              <ul className="list-disc pl-4 space-y-1">
                <li>Data integrity checks (required fields, format validation)</li>
                <li>Deduplication against existing CRM leads (email, phone)</li>
                <li>Conflict detection (geography, diagnosis-tort matching)</li>
                <li>Encryption of sensitive fields (SSN, DOB, medical data)</li>
                <li>Audit trail logging for every imported lead</li>
              </ul>
              <p className="font-medium mt-2">Maximum: 5,000 rows per import</p>
            </div>
          </CardContent>
        </Card>

        <Card>
          <CardHeader>
            <CardTitle>Import History</CardTitle>
          </CardHeader>
          <CardContent>
            {batches.length === 0 ? (
              <p className="text-sm text-muted-foreground">No previous imports</p>
            ) : (
              <div className="space-y-3">
                {batches.slice(0, 10).map(batch => (
                  <div
                    key={batch.id}
                    className="flex items-center justify-between p-3 rounded-lg border cursor-pointer hover:bg-muted/50"
                    onClick={() => handleViewBatch(batch.id)}
                  >
                    <div>
                      <p className="text-sm font-medium">{batch.filename}</p>
                      <p className="text-xs text-muted-foreground">
                        {new Date(batch.created_at).toLocaleDateString()} – {batch.total_rows} rows
                      </p>
                    </div>
                    <div className="flex items-center gap-2">
                      <span className="text-xs text-green-600">{batch.success_count} ok</span>
                      {batch.duplicate_count > 0 && <span className="text-xs text-amber-600">{batch.duplicate_count} dup</span>
                      {batch.error_count > 0 && <span className="text-xs text-red-600">{batch.error_count} err</span>}
                      <Badge variant={batch.status === "completed" ? "default" : "secondary"}>
                        {batch.status}
                      </Badge>
                    </div>
                  </div>
                )}
              </div>
            )}
          </CardContent>
        </Card>
      </div>
    )}
  </div>
);

```

```

        </Badge>
      </div>
    </div>
  )}}
</div>
</CardContent>
</Card>
</div>
))
{step === "preview" && preview && (
  <div>
    <Card>
      <CardHeader>
        <CardTitle>Column Mapping</CardTitle>
      </CardHeader>
      <CardContent>
        <p className="text-sm text-muted-foreground mb-4">
          {preview.total_rows} rows detected in <span className="font-medium">{filename}</span>.
          Review the column mapping below.
        </p>
        <div className="grid gap-2 md:grid-cols-2 lg:grid-cols-3">
          {preview.headers.map(header => (
            <div key={header} className="flex items-center gap-2 p-2 rounded border text-sm">
              <span className="font-mono text-xs bg-muted px-1.5 py-0.5 rounded flex-shrink-0">{header}</span>
              <ArrowRight className="h-3 w-3 text-muted-foreground flex-shrink-0" />
              {columnMapping[header] ? (
                <Badge variant="default" className="text-xs">{columnMapping[header]}</Badge>
              ) : (
                <Badge variant="outline" className="text-xs text-amber-600">unmapped</Badge>
              )}
            </div>
          )}}
        </div>
        {preview.unmapped_columns.length > 0 && (
          <p className="text-xs text-amber-600 mt-3">
            {preview.unmapped_columns.length} column(s) unmapped and will be skipped: {preview.unmapped_columns.join(",
          </p>
        )}
      </CardContent>
    </Card>

    <Card>
      <CardHeader>
        <CardTitle>Sample Preview (First 5 Rows)</CardTitle>
      </CardHeader>
      <CardContent>
        <div className="overflow-x-auto">
          <table className="w-full text-xs">
            <thead>
              <tr className="border-b">
                <th className="p-2 text-left font-medium">#</th>
                {Object.values(columnMapping).slice(0, 8).map(field => (
                  <th key={field} className="p-2 text-left font-medium">{field}</th>
                )}}
              </tr>
            </thead>
            <tbody>
              {preview.sample_leads.map((lead, i) => (
                <tr key={i} className="border-b">
                  <td className="p-2 text-muted-foreground">{i + 1}</td>
                  {Object.values(columnMapping).slice(0, 8).map(field => (
                    <td key={field} className="p-2 max-w-[150px] truncate">
                      {lead[field] || <span className="text-muted-foreground italic">—</span>}
                    </td>
                  )}}
                </tr>
              )}}
            </tbody>
          </table>
        </div>
      </CardContent>
    </Card>

    <div className="flex items-center gap-4">
      <Button onClick={handleExecuteImport} disabled={loading} size="lg">
        {loading ? "Starting Import..." : `Import ${preview.total_rows} Leads`}
      </Button>
      <Button variant="outline" onClick={resetImport}>Cancel</Button>
      <p className="text-xs text-muted-foreground">
        Each lead will be individually validated, deduped, and conflict-checked before insertion.
      </p>
    </div>
  </div>
))
{step === "importing" && activeBatch && (
  <Card>
    <CardHeader>
      <CardTitle className="flex items-center gap-2">
        <RefreshCw className="h-5 w-5 animate-spin" />
        Importing {filename}...
      </CardTitle>
    </CardHeader>
    <CardContent>

```

```

0}%` }}
<div className="space-y-4">
  <div className="w-full bg-muted rounded-full h-3">
    <div
      className="bg-primary h-3 rounded-full transition-all duration-500"
      style={{ width: ${activeBatch.total_rows > 0 ? (activeBatch.processed_rows / activeBatch.total_rows) * 100 :
0}%` }}
    />
  </div>
  <div className="flex items-center justify-between text-sm">
    <span>{activeBatch.processed_rows} / {activeBatch.total_rows} rows processed</span>
    <span className="text-muted-foreground">
      {activeBatch.total_rows > 0 ? Math.round((activeBatch.processed_rows / activeBatch.total_rows) * 100) : 0}%
    </span>
  </div>
  <div className="grid grid-cols-4 gap-4">
    <div className="text-center p-3 rounded-lg bg-green-50 border border-green-200">
      <p className="text-2xl font-bold text-green-700">{activeBatch.success_count}</p>
      <p className="text-xs text-green-600">Imported</p>
    </div>
    <div className="text-center p-3 rounded-lg bg-amber-50 border border-amber-200">
      <p className="text-2xl font-bold text-amber-700">{activeBatch.duplicate_count}</p>
      <p className="text-xs text-amber-600">Duplicates</p>
    </div>
    <div className="text-center p-3 rounded-lg bg-orange-50 border border-orange-200">
      <p className="text-2xl font-bold text-orange-700">{activeBatch.conflict_count}</p>
      <p className="text-xs text-orange-600">Conflicts</p>
    </div>
    <div className="text-center p-3 rounded-lg bg-red-50 border border-red-200">
      <p className="text-2xl font-bold text-red-700">{activeBatch.error_count}</p>
      <p className="text-xs text-red-600">Errors</p>
    </div>
  </div>
</div>
</CardContent>
</Card>
})
{step === "results" && activeBatch && (
  <div className="grid gap-4 md:grid-cols-5">
    <Card className="border-green-200 bg-green-50">
      <CardContent className="pt-6 text-center">
        <CheckCircle2 className="h-8 w-8 mx-auto text-green-600 mb-2" />
        <p className="text-3xl font-bold text-green-700">{activeBatch.success_count}</p>
        <p className="text-sm text-green-600">Successfully Imported</p>
      </CardContent>
    </Card>
    <Card className="border-amber-200 bg-amber-50">
      <CardContent className="pt-6 text-center">
        <Copy className="h-8 w-8 mx-auto text-amber-600 mb-2" />
        <p className="text-3xl font-bold text-amber-700">{activeBatch.duplicate_count}</p>
        <p className="text-sm text-amber-600">Duplicates Skipped</p>
      </CardContent>
    </Card>
    <Card className="border-orange-200 bg-orange-50">
      <CardContent className="pt-6 text-center">
        <AlertTriangle className="h-8 w-8 mx-auto text-orange-600 mb-2" />
        <p className="text-3xl font-bold text-orange-700">{activeBatch.conflict_count}</p>
        <p className="text-sm text-orange-600">Sent to Review</p>
      </CardContent>
    </Card>
    <Card className="border-red-200 bg-red-50">
      <CardContent className="pt-6 text-center">
        <XCircle className="h-8 w-8 mx-auto text-red-600 mb-2" />
        <p className="text-3xl font-bold text-red-700">{activeBatch.error_count}</p>
        <p className="text-sm text-red-600">Errors / Rejected</p>
      </CardContent>
    </Card>
    <Card>
      <CardContent className="pt-6 text-center">
        <FileSpreadsheet className="h-8 w-8 mx-auto text-muted-foreground mb-2" />
        <p className="text-3xl font-bold">{activeBatch.total_rows}</p>
        <p className="text-sm text-muted-foreground">Total Rows</p>
      </CardContent>
    </Card>
  </div>
  {batchRows.filter(r => r.status === "duplicate").length > 0 && (
    <Card>
      <CardHeader
        className="cursor-pointer"
        onClick={() => setExpandedDupes(!expandedDupes)}
      >
        <CardTitle className="flex items-center justify-between text-amber-700">
          <span className="flex items-center gap-2">
            <Copy className="h-5 w-5" />
            Duplicates ({batchRows.filter(r => r.status === "duplicate").length})
          </span>
          {expandedDupes ? <ChevronUp className="h-4 w-4" /> : <ChevronDown className="h-4 w-4" />}
        </CardTitle>
      </CardHeader>
      {expandedDupes && (
        <CardContent>
          <div className="space-y-2 max-h-[300px] overflow-y-auto">
            {batchRows.filter(r => r.status === "duplicate").map(row => (
              <div key={row.id} className="flex items-center justify-between p-2 rounded border text-sm">
                <span className="text-muted-foreground">Row {row.row_number}</span>

```

```

Name"] || "-"</span>
    <span className="font-mono text-xs">{(row.raw_data as any)?.name || (row.raw_data as any)?.["First
    <span className="text-xs text-amber-600">{row.dedup_reason}</span>
    {row.dedup_match_id && (
      <a href={"/leads/${row.dedup_match_id}`} className="text-xs text-primary underline">
        View Match #{row.dedup_match_id}
      </a>
    )}
  </div>
)}
</div>
</CardContent>
)}
</Card>
)}

{batchRows.filter(r => r.status === "error" || r.status === "rejected").length > 0 && (
  <Card>
    <CardHeader
      className="cursor-pointer"
      onClick={() => setExpandedErrors(!expandedErrors)}
    >
      <CardTitle className="flex items-center justify-between text-red-700">
        <span className="flex items-center gap-2">
          <XCircle className="h-5 w-5" />
          Errors & Rejections ({batchRows.filter(r => r.status === "error" || r.status === "rejected").length})
        </span>
        {expandedErrors ? <ChevronUp className="h-4 w-4" /> : <ChevronDown className="h-4 w-4" />}
      </CardTitle>
    </CardHeader>
    {expandedErrors && (
      <CardContent>
        <div className="space-y-2 max-h-[300px] overflow-y-auto">
          {batchRows.filter(r => r.status === "error" || r.status === "rejected").map(row => (
            <div key={row.id} className="p-2 rounded border text-sm space-y-1">
              <div className="flex items-center justify-between">
                <span className="text-muted-foreground">Row {row.row_number}</span>
                {statusBadge(row.status)}
              </div>
              <p className="text-xs text-red-600">{row.error_message}</p>
            </div>
          ))}
        </div>
      </CardContent>
    )}
  </Card>
)}
</div>
)}
);
}

```

[artifacts/mtos-crm/src/pages/lead-intake.tsx](#)

```
import { useState } from "react";
import { useLocation } from "wouter";
import { useForm } from "react-hook-form";
import { zodResolver } from "@hookform/resolvers/zod";
import * as z from "zod";
import { useCreateLead, useQualifyLead } from "@workspace/api-client-react";
import { useToast } from "@hooks/use-toast";
import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { Textarea } from "@components/ui/textarea";
import { Select, SelectContent, SelectItem, SelectTrigger, SelectValue } from "@components/ui/select";
import { Checkbox } from "@components/ui/checkbox";
import { Card, CardContent, CardHeader, CardTitle, CardDescription } from "@components/ui/card";
import { Form, FormControl, FormDescription, FormField, FormItem, FormLabel, FormMessage } from "@components/ui/form";
import { AlertTriangle } from "lucide-react";
import { Alert, AlertDescription, AlertTitle } from "@components/ui/alert";

const formSchema = z.object({
  first_name: z.string().min(2, "First name is required"),
  last_name: z.string().min(2, "Last name is required"),
  date_of_birth: z.string().min(1, "Date of birth is required"),
  street_address: z.string().min(2, "Street address is required"),
  city: z.string().min(2, "City is required"),
  state: z.string().min(2, "State is required"),
  zip: z.string().min(5, "Zip code is required"),
  phone_primary: z.string().min(10, "Phone number is required"),
  last_4_ssn: z.string().length(4, "Must be exactly 4 digits").regex(/^\d{4}$/, "Must be 4 digits"),
  email: z.string().email().optional().or(z.literal("")),
  tort_type: z.string().min(1, "Tort type is required"),
  diagnosis: z.string().min(2, "Diagnosis is required"),
  diagnosis_date: z.string().min(1, "Diagnosis date is required"),
  diagnosis_confirmed: z.boolean(),
  was_at_location: z.boolean(),
  location_name: z.string().optional(),
  physician_first_name: z.string().min(2, "Physician first name is required"),
  physician_last_name: z.string().min(2, "Physician last name is required"),
  physician_full_address: z.string().min(5, "Physician address is required"),
  physician_contact_info: z.string().min(5, "Physician contact info is required"),
  hospital_name: z.string().min(2, "Hospital name is required"),
  hospital_fax: z.string().min(10, "Hospital fax is required"),
  hospital_contact_info: z.string().min(5, "Hospital contact info is required"),
  source: z.string().optional(),
  notes: z.string().optional(),
  law_firm: z.string().optional(),
  client_id: z.string().optional(),
});

const STATES = [
  "AL", "AK", "AZ", "AR", "CA", "CO", "CT", "DE", "DC", "FL", "GA", "HI", "ID", "IL", "IN", "IA",
  "KS", "KY", "LA", "ME", "MD", "MA", "MI", "MN", "MS", "MO", "MT", "NE", "NV", "NH", "NJ", "NM",
  "NY", "NC", "ND", "OH", "OK", "OR", "PA", "RI", "SC", "SD", "TN", "TX", "UT", "VT", "VA", "WA",
  "WV", "WI", "WY"
];

const TORT_CAMPAIGNS = [
  "Camp Lejeune",
  "AFFF Firefighting Foam",
  "Necrotizing Enterocolitis",
  "Roundup",
  "Talcum Powder",
  "Asbestos",
  "Paraquat",
  "Zantac",
  "Hair Relaxer",
  "Tylenol"
];

export default function LeadIntake() {
  const [, setLocation] = useLocation();
  const { toast } = useToast();

  const createLead = useCreateLead();
  const qualifyLead = useQualifyLead();

  const form = useForm<z.infer<typeof formSchema>>({
    resolver: zodResolver(formSchema),
    defaultValues: {
      first_name: "",
      last_name: "",
      date_of_birth: "",
      street_address: "",
      city: "",
      state: "",
      zip: "",
      phone_primary: "",
      last_4_ssn: "",
      email: "",
      tort_type: "",
      diagnosis: "",
      diagnosis_date: "",
      diagnosis_confirmed: false,
      was_at_location: false,
      location_name: "",
      physician_first_name: "",
      physician_last_name: "",
    },
  });
```

```

    physician_full_address: "",
    physician_contact_info: "",
    hospital_name: "",
    hospital_fax: "",
    hospital_contact_info: "",
    source: "Manual Intake",
    notes: "",
    law_firm: "",
    client_id: "",
  },
});

const diagnosisConfirmed = form.watch("diagnosis_confirmed");
const wasAtLocation = form.watch("was_at_location");

const isDisqualified = !diagnosisConfirmed || !wasAtLocation;

async function onSubmit(values: z.infer<typeof formSchema>) {
  try {
    const newLead = await createLead.mutateAsync({
      data: {
        ...values,
        name: `${values.first_name} ${values.last_name}`,
        email: values.email || undefined,
        location_name: values.location_name || undefined,
        source: values.source || undefined,
        notes: values.notes || undefined,
        law_firm: values.law_firm || undefined,
        client_id: values.client_id || undefined,
      }
    });

    if ((newLead as any)?._conflict?.output_state === "REVIEW_REQUIRED") {
      toast({
        title: "Sent to Review Queue",
        description: (newLead as any)?._conflict.details?.join(". ") || "Lead flagged for manual review due to detected conflicts.",
        variant: "destructive",
      });
      setLocation("/review-queue");
      return;
    }

    const qualResult = await qualifyLead.mutateAsync({ id: newLead.id });

    toast({
      title: "Intake Submitted",
      description: `Lead created and evaluated. Result: ${qualResult.status.toUpperCase()}`,
      variant: qualResult.status === "rejected" ? "destructive" : "default",
    });

    setLocation(`/leads/${newLead.id}`);
  } catch (error: any) {
    const responseData = error?.response?.data;
    if (responseData?.output_state === "REJECT") {
      toast({
        title: "Lead Rejected",
        description: responseData.details?.join(". ") || responseData.error || "Lead failed conflict validation.",
        variant: "destructive",
      });
    } else if (responseData?.output_state === "REVIEW_REQUIRED") {
      toast({
        title: "Sent to Review",
        description: responseData.details?.join(". ") || "Lead flagged for manual review due to conflicts.",
      });
    } else {
      toast({
        title: "Error",
        description: responseData?.error || "Failed to submit intake form.",
        variant: "destructive",
      });
    }
  }
}

return (
  <div className="max-w-4xl mx-auto space-y-6 pb-20">
    <div>
      <h1 className="text-3xl font-bold tracking-tight">New Intake</h1>
      <p className="text-muted-foreground mt-2">
        Complete the intake form to run the Boolean Gatekeeper protocol.
      </p>
    </div>

    {isDisqualified && (
      <Alert variant="destructive">
        <AlertTriangle className="h-4 w-4" />
        <AlertTitle>Probable Disqualification</AlertTitle>
        <AlertDescription>
          Based on current inputs, this lead will likely fail the Boolean Gatekeeper protocol. You may still submit to record the rejection.
        </AlertDescription>
      </Alert>
    )}

    <Form {...form}>
      <form onSubmit={form.handleSubmit(onSubmit)} className="space-y-8">

```



```

{/* SECTION 1: Personal Information */}
<Card>
  <CardHeader>
    <CardTitle>Personal Information</CardTitle>
  </CardHeader>
  <CardContent>
    <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
      <FormField>
        control={form.control}
        name="first_name"
        render={({ field }) => (
          <FormItem>
            <FormLabel>First Name *</FormLabel>
            <FormControl>
              <Input placeholder="First Name" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      </FormField>
      <FormField>
        control={form.control}
        name="last_name"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Last Name *</FormLabel>
            <FormControl>
              <Input placeholder="Last Name" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      </FormField>
      <FormField>
        control={form.control}
        name="date_of_birth"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Date of Birth *</FormLabel>
            <FormControl>
              <Input type="date" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      </FormField>
      <FormField>
        control={form.control}
        name="phone_primary"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Phone *</FormLabel>
            <FormControl>
              <Input type="tel" placeholder="555-555-0100" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      </FormField>
      <FormField>
        control={form.control}
        name="email"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Email</FormLabel>
            <FormControl>
              <Input type="email" placeholder="jane@example.com" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      </FormField>
      <FormField>
        control={form.control}
        name="last_4_ssn"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Last 4 of SSN *</FormLabel>
            <FormControl>
              <Input placeholder="0000" maxLength={4} pattern="\d{4}" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      </FormField>
      <FormField>
        control={form.control}
        name="street_address"
        render={({ field }) => (
          <FormItem className="md:col-span-2">
            <FormLabel>Street Address *</FormLabel>
            <FormControl>
              <Input placeholder="123 Main St" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      </FormField>
    </div>
  </CardContent>
</Card>

```

```

    })
  />
  <FormField
    control={form.control}
    name="city"
    render={({ field }) => (
      <FormItem>
        <FormLabel>City *</FormLabel>
        <FormControl>
          <Input placeholder="City" {...field} />
        </FormControl>
        <FormMessage />
      </FormItem>
    )}
  />
  <div className="grid grid-cols-2 gap-6">
    <FormField
      control={form.control}
      name="state"
      render={({ field }) => (
        <FormItem>
          <FormLabel>State *</FormLabel>
          <Select onChange={field.onChange} defaultValue={field.value}>
            <FormControl>
              <SelectTrigger>
                <SelectValue placeholder="State" />
              </SelectTrigger>
            </FormControl>
            <SelectContent>
              {STATES.map((state) => (
                <SelectItem key={state} value={state}>
                  {state}
                </SelectItem>
              ))}
            </SelectContent>
          </Select>
          <FormMessage />
        </FormItem>
      )}
    />
    <FormField
      control={form.control}
      name="zip"
      render={({ field }) => (
        <FormItem>
          <FormLabel>Zip *</FormLabel>
          <FormControl>
            <Input placeholder="Zip" maxLength={10} {...field} />
          </FormControl>
          <FormMessage />
        </FormItem>
      )}
    />
  </div>
</div>
</CardContent>
</Card>

{/* SECTION 2: Medical Information */}
<Card>
  <CardHeader>
    <CardTitle>Medical Information</CardTitle>
  </CardHeader>
  <CardContent>
    <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
      <FormField
        control={form.control}
        name="tort_type"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Tort Campaign *</FormLabel>
            <Select onChange={field.onChange} defaultValue={field.value}>
              <FormControl>
                <SelectTrigger>
                  <SelectValue placeholder="Select campaign..." />
                </SelectTrigger>
              </FormControl>
              <SelectContent>
                {TORT_CAMPAIGNS.map((campaign) => (
                  <SelectItem key={campaign} value={campaign}>
                    {campaign}
                  </SelectItem>
                ))}
              </SelectContent>
            </Select>
            <FormMessage />
          </FormItem>
        )}
      />
      <div className="hidden md:block" />
      <FormField
        control={form.control}
        name="diagnosis"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Diagnosis *</FormLabel>

```

```

        <FormControl>
          <Input placeholder="e.g. Non-Hodgkin Lymphoma" {...field} />
        </FormControl>
      </FormMessage />
    </FormItem>
  )}
/>
<FormField
  control={form.control}
  name="diagnosis_date"
  render={({ field }) => (
    <FormItem>
      <FormLabel>Diagnosis Date *</FormLabel>
      <FormControl>
        <Input type="date" {...field} />
      </FormControl>
      <FormMessage />
    </FormItem>
  )}
/>

<FormField
  control={form.control}
  name="diagnosis_confirmed"
  render={({ field }) => (
    <FormItem className="flex flex-row items-start space-x-3 space-y-0 md:col-span-2">
      <FormControl>
        <Checkbox
          checked={field.value}
          onChange={field.onChange}
        />
      </FormControl>
      <div className="space-y-1 leading-none">
        <FormLabel>Medical diagnosis has been confirmed by a physician</FormLabel>
      </div>
    </FormItem>
  )}
/>

<FormField
  control={form.control}
  name="was_at_location"
  render={({ field }) => (
    <FormItem className="flex flex-row items-start space-x-3 space-y-0 md:col-span-2">
      <FormControl>
        <Checkbox
          checked={field.value}
          onChange={field.onChange}
        />
      </FormControl>
      <div className="space-y-1 leading-none">
        <FormLabel>Client was at the qualifying location for the required duration</FormLabel>
      </div>
    </FormItem>
  )}
/>

{wasAtLocation && (
  <FormField
    control={form.control}
    name="location_name"
    render={({ field }) => (
      <FormItem className="md:col-span-2">
        <FormLabel>Location Name</FormLabel>
        <FormControl>
          <Input placeholder="e.g. MCB Camp Lejeune" {...field} />
        </FormControl>
        <FormMessage />
      </FormItem>
    )}
  />
)}
</div>
</CardContent>
</Card>

/* SECTION 3: Physician Information */
<Card>
  <CardHeader>
    <CardTitle>Physician Information</CardTitle>
  </CardHeader>
  <CardContent>
    <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
      <FormField
        control={form.control}
        name="physician_first_name"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Physician First Name *</FormLabel>
            <FormControl>
              <Input placeholder="First Name" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      />
    </div>
  </CardContent>

```

```

        control={form.control}
        name="physician_last_name"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Physician Last Name *</FormLabel>
            <FormControl>
              <Input placeholder="Last Name" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      />
    <FormField>
      control={form.control}
      name="physician_full_address"
      render={({ field }) => (
        <FormItem className="md:col-span-2">
          <FormLabel>Physician Full Address *</FormLabel>
          <FormControl>
            <Textarea placeholder="Full Address" className="resize-none" {...field} />
          </FormControl>
          <FormMessage />
        </FormItem>
      )}
    </FormField>
    <FormField>
      control={form.control}
      name="physician_contact_info"
      render={({ field }) => (
        <FormItem className="md:col-span-2">
          <FormLabel>Physician Contact Info *</FormLabel>
          <FormControl>
            <Input placeholder="Phone or email" {...field} />
          </FormControl>
          <FormMessage />
        </FormItem>
      )}
    </FormField>
  </div>
</CardContent>
</Card>

{/* SECTION 4: Hospital Information */}
<Card className="border-l-4 border-l-destructive">
  <CardHeader>
    <CardTitle className="text-destructive">Hospital Information</CardTitle>
    <CardDescription className="text-destructive font-medium">
      All hospital fields are mandatory. Leads without complete hospital information will be rejected.
    </CardDescription>
  </CardHeader>
  <CardContent>
    <div className="grid grid-cols-1 md:grid-cols-2 gap-6">
      <FormField>
        control={form.control}
        name="hospital_name"
        render={({ field }) => (
          <FormItem className="md:col-span-2">
            <FormLabel>Hospital Name *</FormLabel>
            <FormControl>
              <Input placeholder="Hospital Name" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      </FormField>
      <FormField>
        control={form.control}
        name="hospital_fax"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Hospital Fax *</FormLabel>
            <FormControl>
              <Input type="tel" placeholder="555-555-0100" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      </FormField>
      <FormField>
        control={form.control}
        name="hospital_contact_info"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Hospital Contact Info *</FormLabel>
            <FormControl>
              <Input placeholder="Phone, email, or contact person" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      </FormField>
    </div>
  </CardContent>
</Card>

{/* BOTTOM SECTION */}
<Card>

```

```

<CardContent className="pt-6">
  <div className="grid grid-cols-1 gap-6">
    <div className="grid grid-cols-1 md:grid-cols-3 gap-6">
      <FormField
        control={form.control}
        name="source"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Source</FormLabel>
            <FormControl>
              <Input placeholder="Manual Intake" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      />
      <FormField
        control={form.control}
        name="law_firm"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Law Firm</FormLabel>
            <FormControl>
              <Input placeholder="Referring law firm" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      />
      <FormField
        control={form.control}
        name="client_id"
        render={({ field }) => (
          <FormItem>
            <FormLabel>Client ID</FormLabel>
            <FormControl>
              <Input placeholder="External client ID" {...field} />
            </FormControl>
            <FormMessage />
          </FormItem>
        )}
      />
    </div>
    <FormField
      control={form.control}
      name="notes"
      render={({ field }) => (
        <FormItem>
          <FormLabel>Notes</FormLabel>
          <FormControl>
            <Textarea placeholder="Additional notes..." className="min-h-[100px]" {...field} />
          </FormControl>
          <FormMessage />
        </FormItem>
      )}
    />
  </div>
</CardContent>
</Card>

<div className="flex justify-end gap-4">
  <Button type="button" variant="outline" onClick={() => setLocation("/leads")}>Cancel</Button>
  <Button type="submit" disabled={createLead.isPending || qualifyLead.isPending}>
    {createLead.isPending || qualifyLead.isPending ? "Processing..." : "Run Gatekeeper & Submit"}
  </Button>
</div>
</form>
</Form>
</div>
  );
}

```

[artifacts/mtos-crm/src/pages/leads.tsx](#)

```
import { useState } from "react";
import { Link } from "wouter";
import { useListLeads, getListLeadsQueryKey, ListLeadsStatus } from "@workspace/api-client-react";
import { Table, TableBody, TableCell, TableHead, TableHeader, TableRow } from "@components/ui/table";
import { Input } from "@components/ui/input";
import { Label } from "@components/ui/label";
import { Select, SelectContent, SelectItem, SelectTrigger, SelectValue } from "@components/ui/select";
import { Badge } from "@components/ui/badge";
import { Button } from "@components/ui/button";
import { Dialog, DialogContent, DialogHeader, DialogTitle, DialogFooter, DialogDescription } from "@components/ui/dialog";
import { Search, Plus, Download } from "lucide-react";
import { Skeleton } from "@components/ui/skeleton";
import { format } from "date-fns";

function ExportDialog({ open, onOpenChange }: { open: boolean; onOpenChange: (v: boolean) => void }) {
  const [exportStatus, setExportStatus] = useState("all");
  const [tortType, setTortType] = useState("");
  const [dateFrom, setDateFrom] = useState("");
  const [dateTo, setDateTo] = useState("");
  const [lawFirm, setLawFirm] = useState("");
  const [clientId, setClientId] = useState("");
  const [selectedFields, setSelectedFields] = useState("all");

  const handleExport = () => {
    const params = new URLSearchParams();
    if (exportStatus !== "all") params.set("status", exportStatus);
    if (tortType) params.set("tort_type", tortType);
    if (dateFrom) params.set("date_from", dateFrom);
    if (dateTo) params.set("date_to", dateTo);
    if (lawFirm) params.set("law_firm", lawFirm);
    if (clientId) params.set("client_id", clientId);
    if (selectedFields !== "all") params.set("fields", selectedFields);

    const url = `/api/leads/export?${params.toString()}`;
    window.open(url, "_blank");
    onOpenChange(false);
  };

  return (
    <Dialog open={open} onOpenChange={onOpenChange}>
      <DialogContent className="sm:max-w-lg">
        <DialogHeader>
          <DialogTitle>Export Leads</DialogTitle>
          <DialogDescription>Configure filters and download leads as CSV.</DialogDescription>
        </DialogHeader>
        <div className="grid gap-4 py-4">
          <div className="grid grid-cols-2 gap-4">
            <div className="grid gap-2">
              <Label>Status</Label>
              <Select value={exportStatus} onValueChange={setExportStatus}>
                <SelectTrigger><SelectValue /></SelectTrigger>
                <SelectContent>
                  <SelectItem value="all">All Statuses</SelectItem>
                  <SelectItem value="new">New</SelectItem>
                  <SelectItem value="qualified">Qualified</SelectItem>
                  <SelectItem value="signed">Signed</SelectItem>
                  <SelectItem value="review_required">Review Required</SelectItem>
                  <SelectItem value="rejected">Rejected</SelectItem>
                </SelectContent>
              </Select>
            </div>
            <div className="grid gap-2">
              <Label>Tort Type</Label>
              <Input placeholder="e.g. Camp Lejeune" value={tortType} onChange={(e) => setTortType(e.target.value)} />
            </div>
          </div>
          <div className="grid grid-cols-2 gap-4">
            <div className="grid gap-2">
              <Label>Date From</Label>
              <Input type="date" value={dateFrom} onChange={(e) => setDateFrom(e.target.value)} />
            </div>
            <div className="grid gap-2">
              <Label>Date To</Label>
              <Input type="date" value={dateTo} onChange={(e) => setDateTo(e.target.value)} />
            </div>
          </div>
          <div className="grid grid-cols-2 gap-4">
            <div className="grid gap-2">
              <Label>Law Firm</Label>
              <Input placeholder="Filter by law firm" value={lawFirm} onChange={(e) => setLawFirm(e.target.value)} />
            </div>
            <div className="grid gap-2">
              <Label>Client ID</Label>
              <Input placeholder="Filter by client ID" value={clientId} onChange={(e) => setClientId(e.target.value)} />
            </div>
          </div>
          <div className="grid gap-2">
            <Label>Fields</Label>
            <Select value={selectedFields} onValueChange={setSelectedFields}>
              <SelectTrigger><SelectValue /></SelectTrigger>
              <SelectContent>
                <SelectItem value="all">All Fields</SelectItem>
                <SelectItem value="id,name,email,phone,tort_type,status,created_at">Basic Info</SelectItem>
                <SelectItem value="id,name,email,phone,tort_type,status,vendor_id,law_firm,client_id,created_at">With Vendor Info</SelectItem>
              </SelectContent>
            </Select>
          </div>
        </div>
      </DialogContent>
    </Dialog>
  );
}
```

```

        <SelectItem
value="id,first_name,last_name,email,phone_primary,tort_type,status,diagnosis,diagnosis_date,created_at">Clinical</SelectItem>
        </SelectContent>
      </Select>
    </div>
  </div>
  <DialogFooter>
    <Button variant="outline" onClick={() => onOpenChange(false)}>Cancel</Button>
    <Button onClick={handleExport} className="flex items-center gap-2">
      <Download className="h-4 w-4" />
      Download CSV
    </Button>
  </DialogFooter>
</DialogContent>
</Dialog>
  );
}

export default function Leads() {
  const [search, setSearch] = useState("");
  const [status, setStatus] = useState<ListLeadsStatus | "all">("all");
  const [exportOpen, setExportOpen] = useState(false);

  const params = {
    ...(search ? { search } : {}),
    ...(status !== "all" ? { status: status as ListLeadsStatus } : {}),
  };

  const { data: leads, isLoading } = useListLeads(params, {
    query: { queryKey: getListLeadsQueryKey(params) }
  });

  return (
    <div className="space-y-6">
      <div className="flex items-center justify-between">
        <h1 className="text-3xl font-bold tracking-tight">Leads</h1>
        <div className="flex items-center gap-2">
          <Button variant="outline" onClick={() => setExportOpen(true)} className="flex items-center gap-2">
            <Download className="h-4 w-4" />
            Export
          </Button>
          <Button asChild>
            <Link href="/leads/new" className="flex items-center gap-2">
              <Plus className="h-4 w-4" />
              New Intake
            </Link>
          </Button>
        </div>
      </div>

      <ExportDialog open={exportOpen} onOpenChange={setExportOpen} />

      <div className="flex items-center gap-4">
        <div className="relative flex-1 max-w-sm">
          <Search className="absolute left-2.5 top-2.5 h-4 w-4 text-muted-foreground" />
          <Input
            type="search"
            placeholder="Search leads..."
            className="pl-8"
            value={search}
            onChange={(e) => setSearch(e.target.value)}
          />
        </div>
        <Select value={status} onValueChange={(val: any) => setStatus(val)}>
          <SelectTrigger className="w-[180px]">
            <SelectValue placeholder="Filter by status" />
          </SelectTrigger>
          <SelectContent>
            <SelectItem value="all">All Statuses</SelectItem>
            <SelectItem value="new">New</SelectItem>
            <SelectItem value="qualified">Qualified</SelectItem>
            <SelectItem value="signed">Signed</SelectItem>
            <SelectItem value="review_required">Review Required</SelectItem>
            <SelectItem value="rejected">Rejected</SelectItem>
          </SelectContent>
        </Select>
      </div>

      <div className="rounded-md border">
        <Table>
          <TableHeader>
            <TableRow>
              <TableHead>Name</TableHead>
              <TableHead>Tort</TableHead>
              <TableHead>Status</TableHead>
              <TableHead>Convexity</TableHead>
              <TableHead>Created</TableHead>
              <TableHead className="text-right">Actions</TableHead>
            </TableRow>
          </TableHeader>
          <TableBody>
            {isLoading ? (
              Array.from({ length: 5 }).map((_, i) => (
                <TableRow key={i}>
                  <TableCell><Skeleton className="h-4 w-32" /></TableCell>
                  <TableCell><Skeleton className="h-4 w-24" /></TableCell>
                  <TableCell><Skeleton className="h-6 w-20" /></TableCell>
                </TableRow>
              ))
            ) : (
              <Table>
                <TableHeader>
                  <TableRow>
                    <TableHead>Name</TableHead>
                    <TableHead>Tort</TableHead>
                    <TableHead>Status</TableHead>
                    <TableHead>Convexity</TableHead>
                    <TableHead>Created</TableHead>
                    <TableHead className="text-right">Actions</TableHead>
                  </TableRow>
                </TableHeader>
                <TableBody>
                  {isLoading ? (
                    Array.from({ length: 5 }).map((_, i) => (
                      <TableRow key={i}>
                        <TableCell><Skeleton className="h-4 w-32" /></TableCell>
                        <TableCell><Skeleton className="h-4 w-24" /></TableCell>
                        <TableCell><Skeleton className="h-6 w-20" /></TableCell>
                      </TableRow>
                    ))
                  ) : (
                    <Table>
                      <TableHeader>
                        <TableRow>
                          <TableHead>Name</TableHead>
                          <TableHead>Tort</TableHead>
                          <TableHead>Status</TableHead>
                          <TableHead>Convexity</TableHead>
                          <TableHead>Created</TableHead>
                          <TableHead className="text-right">Actions</TableHead>
                        </TableRow>
                      </TableHeader>
                      <TableBody>
                        {isLoading ? (
                          Array.from({ length: 5 }).map((_, i) => (
                            <TableRow key={i}>
                              <TableCell><Skeleton className="h-4 w-32" /></TableCell>
                              <TableCell><Skeleton className="h-4 w-24" /></TableCell>
                              <TableCell><Skeleton className="h-6 w-20" /></TableCell>
                            </TableRow>
                          ))
                        ) : (
                          <Table>
                            <TableHeader>
                              <TableRow>
                                <TableHead>Name</TableHead>
                                <TableHead>Tort</TableHead>
                                <TableHead>Status</TableHead>
                                <TableHead>Convexity</TableHead>
                                <TableHead>Created</TableHead>
                                <TableHead className="text-right">Actions</TableHead>
                              </TableRow>
                            </TableHeader>
                            <TableBody>
                              {isLoading ? (
                                Array.from({ length: 5 }).map((_, i) => (
                                  <TableRow key={i}>
                                    <TableCell><Skeleton className="h-4 w-32" /></TableCell>
                                    <TableCell><Skeleton className="h-4 w-24" /></TableCell>
                                    <TableCell><Skeleton className="h-6 w-20" /></TableCell>
                                  </TableRow>
                                ))
                              ) : (
                                <Table>
                                  <TableHeader>
                                    <TableRow>
                                      <TableHead>Name</TableHead>
                                      <TableHead>Tort</TableHead>
                                      <TableHead>Status</TableHead>
                                      <TableHead>Convexity</TableHead>
                                      <TableHead>Created</TableHead>
                                      <TableHead className="text-right">Actions</TableHead>
                                    </TableRow>
                                  </TableHeader>
                                  <TableBody>
                                    {isLoading ? (
                                      Array.from({ length: 5 }).map((_, i) => (
                                        <TableRow key={i}>
                                          <TableCell><Skeleton className="h-4 w-32" /></TableCell>
                                          <TableCell><Skeleton className="h-4 w-24" /></TableCell>
                                          <TableCell><Skeleton className="h-6 w-20" /></TableCell>
                                        </TableRow>
                                      ))
                                    ) : (
                                      <Table>
                                        <TableHeader>
                                          <TableRow>
                                            <TableHead>Name</TableHead>
                                            <TableHead>Tort</TableHead>
                                            <TableHead>Status</TableHead>
                                            <TableHead>Convexity</TableHead>
                                            <TableHead>Created</TableHead>
                                            <TableHead className="text-right">Actions</TableHead>
                                          </TableRow>
                                        </TableHeader>
                                        <TableBody>
                                          {isLoading ? (
                                            Array.from({ length: 5 }).map((_, i) => (
                                              <TableRow key={i}>
                                                <TableCell><Skeleton className="h-4 w-32" /></TableCell>
                                                <TableCell><Skeleton className="h-4 w-24" /></TableCell>
                                                <TableCell><Skeleton className="h-6 w-20" /></TableCell>
                                              </TableRow>
                                            ))
                                            </Table>
                                          </Table>
                                        </Table>
                                      </Table>
                                    </Table>
                                  </Table>
                                </Table>
                              </Table>
                            </Table>
                          </Table>
                        </Table>
                      </Table>
                    </Table>
                  </Table>
                </Table>
              </Table>
            )
          </TableBody>
        </Table>
      </div>
    </div>
  );
}

```

```

        <TableCell><Skeleton className="h-4 w-24" /></TableCell>
        <TableCell><Skeleton className="h-8 w-16 ml-auto" /></TableCell>
    </TableRow>
  ))
) : leads?.length === 0 ? (
  <TableRow>
    <TableCell colspan={6} className="h-24 text-center">
      No leads found.
    </TableCell>
  </TableRow>
) : (
  leads?.map((lead) => (
    <TableRow key={lead.id}>
      <TableCell className="font-medium">
        <div>{lead.name}</div>
        <div className="text-xs text-muted-foreground">{lead.email || lead.phone}</div>
      </TableCell>
      <TableCell>{lead.tort_type}</TableCell>
      <TableCell>
        <Badge variant={
          lead.status === "signed" ? "default" :
          lead.status === "qualified" ? "secondary" :
          lead.status === "rejected" ? "destructive" :
          lead.status === "review_required" ? "secondary" : "outline"
        } className={lead.status === "review_required" ? "bg-amber-500/10 text-amber-600 border-amber-500/20" : ""}>
          {lead.status === "review_required" ? "REVIEW" : lead.status.toUpperCase()}
        </Badge>
      </TableCell>
      <TableCell>
        {(() => {
          const c = (lead as any).convexity_score as string | null | undefined;
          const flags = (((lead as any).convexity_ruin_flags ?? []) as string[]).length;
          if (flags > 0) return <Badge className="bg-rose-100 text-rose-800 border-rose-300" data-testid={`badge-convexity-${lead.id}`}>RUIN</Badge>;
          if (c === "convex") return <Badge className="bg-emerald-100 text-emerald-800 border-emerald-300" data-testid={`badge-convexity-${lead.id}`}>Convex</Badge>;
          if (c === "concave") return <Badge className="bg-amber-100 text-amber-800 border-amber-300" data-testid={`badge-convexity-${lead.id}`}>Concave</Badge>;
          if (c === "neutral") return <Badge variant="secondary" data-testid={`badge-convexity-${lead.id}`}>Neutral</Badge>;
          return <Badge variant="outline" data-testid={`badge-convexity-${lead.id}`}></Badge>;
        })()}
      </TableCell>
      <TableCell className="text-muted-foreground text-sm font-mono">
        {format(new Date(lead.created_at), "yyyy-MM-dd")}
      </TableCell>
      <TableCell className="text-right">
        <Button variant="ghost" size="sm" asChild>
          <Link href={`/leads/${lead.id}`}>View</Link>
        </Button>
      </TableCell>
    </TableRow>
  ))
)
</TableBody>
</Table>
</div>
</div>
);
}

```


[artifacts/mtos-crm/src/pages/news.tsx](#)

```
import { useState, useEffect } from "react";
import { Card, CardContent, CardHeader, CardTitle, CardDescription } from "@components/ui/card";
import { Badge } from "@components/ui/badge";
import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { Skeleton } from "@components/ui/skeleton";
import { useToast } from "@hooks/use-toast";
import { Newspaper, ExternalLink, RefreshCw, Search, Clock } from "lucide-react";

interface Article {
  title: string;
  link: string;
  source: string;
  published: string;
  description: string;
  category: string;
}

const categoryLabels: Record<string, { label: string; color: string }> = {
  mass_tort: { label: "Mass Tort", color: "bg-red-100 text-red-700" },
  pharma: { label: "Pharmaceutical", color: "bg-purple-100 text-purple-700" },
  mdl: { label: "MDL", color: "bg-blue-100 text-blue-700" },
  product_liability: { label: "Product Liability", color: "bg-orange-100 text-orange-700" },
  toxic_tort: { label: "Toxic Tort", color: "bg-green-100 text-green-700" },
};

function timeAgo(dateStr: string): string {
  const now = Date.now();
  const date = new Date(dateStr).getTime();
  const diff = now - date;
  const mins = Math.floor(diff / 60000);
  if (mins < 60) return `${mins}m ago`;
  const hours = Math.floor(mins / 60);
  if (hours < 24) return `${hours}h ago`;
  const days = Math.floor(hours / 24);
  return `${days}d ago`;
}

export default function News() {
  const { toast } = useToast();
  const [articles, setArticles] = useState<Article[]>([]);
  const [loading, setLoading] = useState(true);
  const [search, setSearch] = useState("");
  const [filterCategory, setFilterCategory] = useState("all");

  const fetchNews = async () => {
    setLoading(true);
    try {
      const res = await fetch("/api/news/mass-tort");
      if (!res.ok) throw new Error("Failed");
      setArticles(await res.json());
    } catch {
      toast({ title: "Failed to load news", variant: "destructive" });
    } finally {
      setLoading(false);
    }
  };

  useEffect(() => { fetchNews(); }, []);

  const filtered = articles.filter(a => {
    if (filterCategory !== "all" && a.category !== filterCategory) return false;
    if (search && !a.title.toLowerCase().includes(search.toLowerCase()) && !
a.description.toLowerCase().includes(search.toLowerCase())) return false;
    return true;
  });

  return (
    <div className="space-y-6">
      <div className="flex items-center justify-between">
        <div>
          <h1 className="text-3xl font-bold tracking-tight">Mass Tort News</h1>
          <p className="text-muted-foreground">Real-time news from tier-one legal and mass tort sources.</p>
        </div>
        <Button variant="outline" onClick={fetchNews} disabled={loading}>
          <RefreshCw className={`h-4 w-4 mr-2 ${loading ? "animate-spin" : ""}` } />
          Refresh
        </Button>
      </div>
      <Card>
        <CardContent className="pt-6">
          <div className="flex gap-3 flex-wrap items-center">
            <div className="relative flex-1 min-w-[200px]">
              <Search className="absolute left-3 top-1/2 -translate-y-1/2 h-4 w-4 text-muted-foreground" />
              <Input placeholder="Search articles..." className="pl-10" value={search} onChange={(e) =>
setSearch(e.target.value)} />
            </div>
            <div className="flex gap-2 flex-wrap">
              <Badge variant={filterCategory === "all" ? "default" : "outline"} className="cursor-pointer" onClick={() =>
setFilterCategory("all")}>All</Badge>
              {Object.entries(categoryLabels).map(([key, cfg]) => (
                <Badge key={key} variant="outline" className={`cursor-pointer ${filterCategory === key ? cfg.color : ""}`}
onClick={() => setFilterCategory(key)}>{cfg.label}</Badge>
              ))}
            </div>
          </div>
        </CardContent>
      </Card>
    </div>
  );
}
```

```

    </div>
  </div>
</CardContent>
</Card>

{loading ? (
  <div className="space-y-4">
    {Array.from({ length: 5 }).map((_, i) => (
      <Card key={i}><CardContent className="pt-6"><Skeleton className="h-24" /></CardContent></Card>
    ))}
  </div>
) : filtered.length === 0 ? (
  <Card>
    <CardContent className="py-12 text-center text-muted-foreground">
      <Newspaper className="h-12 w-12 mx-auto mb-3" />
      <div>No articles found. Try adjusting your filters.</div>
    </CardContent>
  </Card>
) : (
  <div className="space-y-3">
    {filtered.map((article, i) => {
      const cat = categoryLabels[article.category];
      return (
        <Card key={i} className="hover:shadow-md transition-shadow cursor-pointer" onClick={() =>
          window.open(article.link, "_blank")}>
          <CardContent className="pt-5 pb-4">
            <div className="flex items-start justify-between gap-4">
              <div className="flex-1 min-w-0">
                <div className="flex items-center gap-2 mb-1 flex-wrap">
                  {cat && <Badge variant="outline" className={`text-xs ${cat.color}`}>{cat.label}</Badge>}
                  <span className="text-xs text-muted-foreground font-medium">{article.source}</span>
                </div>
                <span className="text-xs text-muted-foreground flex items-center gap-1">
                  <Clock className="h-3 w-3" />
                  {timeAgo(article.published)}
                </span>
              </div>
              <div>
                <h3 className="font-semibold text-base leading-snug line-clamp-2">{article.title}</h3>
                {article.description && (
                  <p className="text-sm text-muted-foreground mt-1 line-clamp-2">{article.description}</p>
                )}
              </div>
            </div>
            <ExternalLink className="h-4 w-4 text-muted-foreground flex-shrink-0 mt-1" />
          </CardContent>
        </Card>
      );
    })}
  </div>
);
</div>
})

{!loading && filtered.length > 0 && (
  <div className="text-center text-sm text-muted-foreground">
    Showing {filtered.length} articles from {new Set(filtered.map(a => a.source)).size} sources
  </div>
)}
</div>
);
}

```

[artifacts/mtos-crm/src/pages/not-found.tsx](#)

```
import { Card, CardContent } from "@components/ui/card";
import { AlertCircle } from "lucide-react";

export default function NotFound() {
  return (
    <div className="min-h-screen w-full flex items-center justify-center bg-gray-50">
      <Card className="w-full max-w-md mx-4">
        <CardContent className="pt-6">
          <div className="flex mb-4 gap-2">
            <AlertCircle className="h-8 w-8 text-red-500" />
            <h1 className="text-2xl font-bold text-gray-900">404 Page Not Found</h1>
          </div>

          <p className="mt-4 text-sm text-gray-600">
            Did you forget to add the page to the router?
          </p>
        </CardContent>
      </Card>
    </div>
  );
}
```

[artifacts/mtos-crm/src/pages/npi-lookup.tsx](#)

```
import { useState } from "react";
import { Link } from "wouter";
import { useSearchNpi, getSearchNpiQueryKey, SearchNpiParams, NpiProvider } from "@workspace/api-client-react";
import { Search, Stethoscope, ChevronDown, ChevronUp, MapPin, Phone, Building, User, Calendar } from "lucide-react";
import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { Label } from "@components/ui/label";
import { Select, SelectContent, SelectItem, SelectTrigger, SelectValue } from "@components/ui/select";
import { Card, CardContent, CardDescription, CardHeader, CardTitle } from "@components/ui/card";
import { Table, TableBody, TableCell, TableHead, TableHeader, TableRow } from "@components/ui/table";
import { Badge } from "@components/ui/badge";
import { Skeleton } from "@components/ui/skeleton";

const US_STATES = [
  "AL", "AK", "AZ", "AR", "CA", "CO", "CT", "DE", "DC", "FL", "GA",
  "HI", "ID", "IL", "IN", "IA", "KS", "KY", "LA", "ME", "MD", "MA",
  "MI", "MN", "MS", "MO", "MT", "NE", "NV", "NH", "NJ", "NM", "NY",
  "NC", "ND", "OH", "OK", "OR", "PA", "RI", "SC", "SD", "TN", "TX",
  "UT", "VT", "VA", "WA", "WV", "WI", "WY"
];

function ProviderRow({ provider }: { provider: NpiProvider }) {
  const [expanded, setExpanded] = useState(false);

  return (
    <TableRow
      className="cursor-pointer hover:bg-muted/50 transition-colors"
      onClick={() => setExpanded(!expanded)}
    >
      <TableCell className="font-mono text-xs">{provider.npi}</TableCell>
      <TableCell className="font-medium">{provider.name}</TableCell>
      <TableCell>
        {provider.credential ? (
          <Badge variant="secondary" className="text-xs">
            {provider.credential}
          </Badge>
        ) : (
          <span className="text-muted-foreground text-xs">N/A</span>
        )}
      </TableCell>
      <TableCell className="max-w-[200px] truncate" title={provider.specialty}>
        {provider.specialty || <span className="text-muted-foreground">Unknown</span>}
      </TableCell>
      <TableCell>
        {provider.city}, {provider.state}
      </TableCell>
      <TableCell>{provider.phone || "N/A"}</TableCell>
      <TableCell className="text-right">
        <div className="flex justify-end items-center gap-2">
          <a
            href={provider.npi_registry_url}
            target="_blank"
            rel="noreferrer"
            className="text-xs text-primary hover:underline font-medium"
            onClick={(e) => e.stopPropagation()}
          >
            View NPI Profile
          </a>
          {expanded ? <ChevronUp className="h-4 w-4 text-muted-foreground" /> : <ChevronDown className="h-4 w-4 text-muted-foreground" />}
        </div>
      </TableCell>
    </TableRow>
    {expanded && (
      <TableRow className="bg-muted/30">
        <TableCell colspan={7} className="p-0 border-b">
          <div className="p-4 grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6 animate-in fade-in slide-in-from-top-2 duration-200">
            <div className="space-y-3">
              <div className="flex items-start gap-2">
                <MapPin className="h-4 w-4 mt-0.5 text-muted-foreground" />
                <div>
                  <p className="text-sm font-medium">Primary Practice Address</p>
                  <p className="text-sm text-muted-foreground mt-1">
                    {provider.address_line_1}
                    {provider.address_line_2 && <<br />>{provider.address_line_2}</>}
                  <br />
                    {provider.city}, {provider.state} {provider.postal_code}
                  </p>
                </div>
              </div>
            </div>
            <div className="space-y-3">
              <div className="flex items-start gap-2">
                <Phone className="h-4 w-4 mt-0.5 text-muted-foreground" />
                <div>
                  <p className="text-sm font-medium">Contact Information</p>
                  <div className="text-sm text-muted-foreground mt-1 space-y-1">
                    <p>Phone: {provider.phone || "N/A"}</p>
                    <p>Fax: {provider.fax || "N/A"}</p>
                  </div>
                </div>
              </div>
            </div>
          </div>
        </TableCell>
      </TableRow>
    )}
  );
}
```

```

    </div>
    <div className="space-y-3">
      <div className="flex items-start gap-2">
        {provider.provider_type === "Individual" ? (
          <User className="h-4 w-4 mt-0.5 text-muted-foreground" />
        ) : (
          <Building className="h-4 w-4 mt-0.5 text-muted-foreground" />
        )}
      </div>
      <p className="text-sm font-medium">Provider Details</p>
      <div className="text-sm text-muted-foreground mt-1 space-y-1">
        <p>Type: {provider.provider_type}</p>
        {provider.gender && <p>Gender: {provider.gender}</p>}
      </div>
    </div>
    <div className="flex items-start gap-2 pt-2 border-t border-border/50">
      <Calendar className="h-4 w-4 mt-0.5 text-muted-foreground" />
      <div className="text-xs text-muted-foreground">
        <p>Enumerated: {provider.enumeration_date ? new Date(provider.enumeration_date).toLocaleDateString() :
"Unknown"}</p>
        <p>Last Updated: {provider.last_updated ? new Date(provider.last_updated).toLocaleDateString() : "Unknown"}
</p>
      </div>
    </div>
  </div>
</TableCell>
</TableRow>
</>
);
}

export default function NpiLookup() {
  const [formValues, setFormValues] = useState<SearchNpiParams>({
    npi_number: "",
    first_name: "",
    last_name: "",
    city: "",
    state: "",
    specialty: "",
    limit: "50"
  });

  const [searchParams, setSearchParams] = useState<SearchNpiParams | null>(null);

  const { data, isLoading, isError } = useSearchNpi(searchParams || {}, {
    query: {
      enabled: !!searchParams,
      queryKey: getSearchNpiQueryKey(searchParams || {})
    }
  });

  const handleSearch = (e: React.FormEvent) => {
    e.preventDefault();
    // Clean up empty strings
    const cleanedParams: SearchNpiParams = {};
    Object.entries(formValues).forEach(([key, value]) => {
      if (value && value.trim() !== "") {
        (cleanedParams as any)[key] = value.trim();
      }
    });
    setSearchParams(cleanedParams);
  };

  const handleClear = () => {
    setFormValues({
      npi_number: "",
      first_name: "",
      last_name: "",
      city: "",
      state: "",
      specialty: "",
      limit: "50"
    });
    setSearchParams(null);
  };

  return (
    <div className="flex-1 space-y-6 p-8 pt-6 max-w-7xl mx-auto w-full">
      <div className="flex items-center justify-between">
        <div className="space-y-1">
          <h2 className="text-3xl font-bold tracking-tight">NPI Provider Lookup</h2>
          <p className="text-muted-foreground">
            Search the national registry for doctors and medical organizations.
          </p>
        </div>
        <div className="flex items-center gap-2">
          <Stethoscope className="h-8 w-8 text-primary opacity-20" />
        </div>
      </div>

      <Card className="border-border/50 shadow-sm">
        <form onSubmit={handleSearch}>
          <CardContent className="pt-6">

```

```

<div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
  <div className="space-y-2">
    <Label htmlFor="npi_number">NPI Number</Label>
    <Input
      id="npi_number"
      placeholder="10-digit NPI"
      value={formValues.npi_number || ""}
      onChange={(e) => setFormValues({ ...formValues, npi_number: e.target.value })}
      className="font-mono"
    />
  </div>
  <div className="space-y-2">
    <Label htmlFor="first_name">First Name</Label>
    <Input
      id="first_name"
      placeholder="e.g. John"
      value={formValues.first_name || ""}
      onChange={(e) => setFormValues({ ...formValues, first_name: e.target.value })}
    />
  </div>
  <div className="space-y-2">
    <Label htmlFor="last_name">Last Name</Label>
    <Input
      id="last_name"
      placeholder="e.g. Smith"
      value={formValues.last_name || ""}
      onChange={(e) => setFormValues({ ...formValues, last_name: e.target.value })}
    />
  </div>
  <div className="space-y-2">
    <Label htmlFor="city">City</Label>
    <Input
      id="city"
      placeholder="e.g. Houston"
      value={formValues.city || ""}
      onChange={(e) => setFormValues({ ...formValues, city: e.target.value })}
    />
  </div>
  <div className="space-y-2">
    <Label htmlFor="state">State</Label>
    <Select
      value={formValues.state || ""}
      onValueChange={(val) => setFormValues({ ...formValues, state: val === "ALL" ? "" : val })}
    >
      <SelectTrigger id="state">
        <SelectValue placeholder="Any State" />
      </SelectTrigger>
      <SelectContent>
        <SelectItem value="ALL">Any State</SelectItem>
        {US_STATES.map((state) => (
          <SelectItem key={state} value={state}>{state}</SelectItem>
        ))}
      </SelectContent>
    </Select>
  </div>
  <div className="space-y-2">
    <Label htmlFor="specialty">Specialty (Taxonomy)</Label>
    <Input
      id="specialty"
      placeholder="e.g. Oncology, Internal Medicine"
      value={formValues.specialty || ""}
      onChange={(e) => setFormValues({ ...formValues, specialty: e.target.value })}
    />
  </div>
</div>

<div className="mt-8 flex flex-col items-center gap-3">
  <div className="flex w-full gap-3">
    <Button type="button" variant="outline" onClick={handleClear} className="w-1/4">
      Clear
    </Button>
    <Button type="submit" className="w-3/4" disabled={isLoading}>
      {isLoading ? (
        <span className="flex items-center gap-2">
          <div className="h-4 w-4 rounded-full border-background border-t-transparent animate-spin" />
          Searching...
        </span>
      ) : (
        <span className="flex items-center gap-2">
          <Search className="h-4 w-4" />
          Search Providers
        </span>
      )}
    </Button>
  </div>
  <p className="text-xs text-muted-foreground">
    Powered by CMS NPI Registry (npiregistry.cms.hhs.gov)
  </p>
</div>
</CardContent>
</form>
</Card>

<div className="space-y-4">
  {searchParams && data && (
    <div className="flex items-center justify-between">
      <h3 className="text-sm font-medium text-muted-foreground">

```


[artifacts/mtos-crm/src/pages/ocr-inbox.tsx](#)

```
import React, { useState, useRef } from "react";
import { Link, useLocation } from "wouter";
import { useListFaxResults, getListFaxResultsQueryKey, useUploadFax, useGetOcrQueueStats, getGetOcrQueueStatsQueryKey } from
"@workspace/api-client-react";
import { format } from "date-fns";
import { Search, Plus, AlertCircle, CheckCircle, Clock, UploadCloud, ChevronDown, ChevronRight } from "lucide-react";
import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import { Badge } from "@components/ui/badge";
import {
  Table,
  TableBody,
  TableCell,
  TableHead,
  TableHeader,
  TableRow,
} from "@components/ui/table";
import { Skeleton } from "@components/ui/skeleton";
import { Dialog, DialogContent, DialogHeader, DialogTitle, DialogTrigger } from "@components/ui/dialog";
import { Label } from "@components/ui/label";
import { useQueryClient } from "@tanstack/react-query";
import { toast } from "@hooks/use-toast";

function getStatusBadge(status: string) {
  switch (status) {
    case "pending":
      return <Badge variant="secondary" className="bg-slate-500/10 text-slate-500 hover:bg-slate-500/20">Pending</Badge>;
    case "processing":
      return <Badge variant="secondary" className="bg-amber-500/10 text-amber-500 hover:bg-amber-500/20"><Clock className="w-3
h-3 mr-1" />Processing</Badge>;
    case "done":
      return <Badge variant="secondary" className="bg-green-500/10 text-green-500 hover:bg-green-500/20"><CheckCircle
className="w-3 h-3 mr-1" />Done</Badge>;
    case "failed":
      return <Badge variant="destructive" className="bg-red-500/10 text-red-500 hover:bg-red-500/20"><AlertCircle
className="w-3 h-3 mr-1" />Failed</Badge>;
    default:
      return <Badge variant="outline">{status}</Badge>;
  }
}

export default function OcrInbox() {
  const [searchTerm, setSearchTerm] = useState("");
  const [uploadOpen, setUploadOpen] = useState(false);
  const [expandedRow, setExpandedRow] = useState<number | null>(null);

  const queryClient = useQueryClient();
  const fileInputRef = useRef<HTMLInputElement>(null);
  const [fileName, setFileName] = useState("");

  const { data: faxes = [], isLoading } = useListFaxResults();
  const { data: queueStats } = useGetOcrQueueStats({
    query: { refetchInterval: 5000 } as any
  });

  const uploadMutation = useUploadFax({
    mutation: {
      onSuccess: () => {
        toast({ title: "Fax uploaded successfully" });
        queryClient.invalidateQueries({ queryKey: getListFaxResultsQueryKey() });
        setUploadOpen(false);
        if (fileInputRef.current) fileInputRef.current.value = "";
        setFileName("");
      },
      onError: (err) => {
        toast({ title: "Failed to upload fax", variant: "destructive" });
        console.error(err);
      }
    }
  });

  const filteredFaxes = faxes.filter(f =>
    f.id.toString().includes(searchTerm) ||
    (f.rx_number && f.rx_number.toLowerCase().includes(searchTerm.toLowerCase())) ||
    (f.drug_name && f.drug_name.toLowerCase().includes(searchTerm.toLowerCase())) ||
    (f.source_file && f.source_file.toLowerCase().includes(searchTerm.toLowerCase()))
  );

  const handleUploadSubmit = (e: React.FormEvent) => {
    e.preventDefault();
    const file = fileInputRef.current?.files?.[0];
    if (!file) {
      toast({ title: "Please select a file", variant: "destructive" });
      return;
    }

    const finalFileName = fileName.trim() || file.name;
    const reader = new FileReader();
    reader.onload = (event) => {
      const base64String = event.target?.result as string;
      uploadMutation.mutate({
        data: {
          file_name: finalFileName,
          image_base64: base64String,
        }
      });
    }
  };
}
```



```

        mime_type: file.type
      }
    });
  });
  reader.readAsDataURL(file);
};

const toggleRow = (id: number) => {
  setExpandedRow(expandedRow === id ? null : id);
};

return (
  <div className="flex-1 space-y-6 p-8 pt-6">
    <div className="flex items-center justify-between">
      <div>
        <h2 className="text-3xl font-bold tracking-tight text-foreground">OCR Inbox</h2>
        <p className="text-muted-foreground mt-1">Process and manage incoming faxes.</p>
      </div>
      <div className="flex items-center space-x-2">
        <Dialog open={uploadOpen} onOpenChange={setUploadOpen}>
          <DialogTrigger asChild>
            <Button>
              <UploadCloud className="mr-2 h-4 w-4" />
              Upload Fax
            </Button>
          </DialogTrigger>
          <DialogContent>
            <DialogHeader>
              <DialogTitle>Upload Fax for OCR Processing</DialogTitle>
            </DialogHeader>
            <form onSubmit={handleUploadSubmit} className="space-y-4 pt-4">
              <div className="space-y-2">
                <Label htmlFor="file">Fax Image File</Label>
                <Input id="file" type="file" accept="image/*" ref={fileInputRef} required />
              </div>
              <div className="space-y-2">
                <Label htmlFor="filename">File Name Override (Optional)</Label>
                <Input
                  id="filename"
                  placeholder="E.g. patient-records.png"
                  value={fileName}
                  onChange={(e) => setFileName(e.target.value)}
                />
              </div>
              <Button type="submit" className="w-full" disabled={uploadMutation.isPending}>
                {uploadMutation.isPending ? "Uploading..." : "Submit for OCR"}
              </Button>
            </form>
          </DialogContent>
        </Dialog>
      </div>
    </div>

    <div className="grid gap-4 md:grid-cols-2 lg:grid-cols-4">
      {queueStats ? (
        ["pending", "processing", "done", "failed"].map((key) => (
          <Card key={key} className="bg-card">
            <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
              <CardTitle className="text-sm font-medium uppercase tracking-wider text-muted-foreground">
                {key}
              </CardTitle>
              {key === "pending" && <Clock className="h-4 w-4 text-slate-500" />}
              {key === "processing" && <Clock className="h-4 w-4 text-amber-500" />}
              {key === "done" && <CheckCircle className="h-4 w-4 text-green-500" />}
              {key === "failed" && <AlertCircle className="h-4 w-4 text-red-500" />}
            </CardHeader>
            <CardContent>
              <div className="text-2xl font-bold">{queueStats[key as keyof typeof queueStats] ?? 0}</div>
            </CardContent>
          </Card>
        ))
      ) : (
        Array.from({ length: 4 }).map((_, i) => (
          <Card key={i}>
            <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
              <Skeleton className="h-4 w-1/2" />
              <Skeleton className="h-4 w-4" />
            </CardHeader>
            <CardContent>
              <Skeleton className="h-8 w-1/4" />
            </CardContent>
          </Card>
        ))
      )
    </div>

    <Card className="border-border">
      <CardHeader className="flex flex-row items-center justify-between space-y-0 py-4">
        <CardTitle>Processed Faxes</CardTitle>
      <div className="relative w-64">
        <Search className="absolute left-2 top-2.5 h-4 w-4 text-muted-foreground" />
        <Input
          placeholder="Search records..."
          value={searchTerm}
          onChange={(e) => setSearchTerm(e.target.value)}
          className="pl-8"
        />
      </div>
    </Card>
  </div>

```

```

</div>
</CardHeader>
<CardContent className="p-0">
  <Table>
    <TableHeader className="bg-muted/50">
      <TableRow>
        <TableHead className="w-[40px]"></TableHead>
        <TableHead className="w-[100px]">ID</TableHead>
        <TableHead>Source File</TableHead>
        <TableHead>RX Number</TableHead>
        <TableHead>Drug Name</TableHead>
        <TableHead>Fill Date</TableHead>
        <TableHead>Qty</TableHead>
        <TableHead>Confidence</TableHead>
        <TableHead>Status</TableHead>
        <TableHead className="text-right">Created</TableHead>
      </TableRow>
    </TableHeader>
    <TableBody>
      {isLoading ? (
        Array.from({ length: 5 }).map((_, i) => (
          <TableRow key={i}>
            <TableCell><Skeleton className="h-4 w-4" /></TableCell>
            <TableCell><Skeleton className="h-4 w-10" /></TableCell>
            <TableCell><Skeleton className="h-4 w-32" /></TableCell>
            <TableCell><Skeleton className="h-4 w-20" /></TableCell>
            <TableCell><Skeleton className="h-4 w-24" /></TableCell>
            <TableCell><Skeleton className="h-4 w-20" /></TableCell>
            <TableCell><Skeleton className="h-4 w-10" /></TableCell>
            <TableCell><Skeleton className="h-4 w-12" /></TableCell>
            <TableCell><Skeleton className="h-6 w-20" /></TableCell>
            <TableCell className="text-right"><Skeleton className="h-4 w-24 ml-auto" /></TableCell>
          </TableRow>
        ))
      ) : filteredFaxes.length > 0 ? (
        filteredFaxes.map((fax) => (
          <React.Fragment key={fax.id}>
            <TableRow
              className="cursor-pointer hover:bg-muted/50 transition-colors"
              onClick={() => toggleRow(fax.id)}
            >
              <TableCell>
                {expandedRow === fax.id ? <ChevronDown className="h-4 w-4" /> : <ChevronRight className="h-4 w-4" />}
              </TableCell>
              <TableCell className="font-mono text-xs text-muted-foreground">
                #{fax.id}
              </TableCell>
              <TableCell className="font-medium max-w-[200px] truncate" title={fax.source_file || ""}>
                {fax.source_file || "Unknown"}
              </TableCell>
              <TableCell>
                {fax.rx_number || "-"}
              </TableCell>
              <TableCell>
                {fax.drug_name || "-"}
              </TableCell>
              <TableCell>
                {fax.fill_date || "-"}
              </TableCell>
              <TableCell>
                {fax.quantity || "-"}
              </TableCell>
              <TableCell>
                {fax.confidence ? `${Math.round(fax.confidence * 100)}%` : "-"}
              </TableCell>
              <TableCell>
                {getStatusBadge(fax.status)}
              </TableCell>
              <TableCell className="text-right text-muted-foreground text-sm">
                {format(new Date(fax.created_at), "MMM d, yyyy")}
              </TableCell>
            </TableRow>
            {expandedRow === fax.id && (
              <TableRow className="bg-muted/20">
                <TableCell colspan={10} className="p-4">
                  <div className="space-y-4">
                    <div>
                      <h4 className="text-sm font-medium mb-2">Raw Extracted Text</h4>
                      <pre className="bg-muted p-4 rounded-md text-xs whitespace-pre-wrap font-mono overflow-auto max-h-[300px] border border-border">
                        {fax.raw_text || "No text extracted"}
                      </pre>
                    </div>
                    <div>
                      {fax.error && (
                        <div>
                          <h4 className="text-sm font-medium text-red-500 mb-2">Error Details</h4>
                          <p className="text-sm text-red-500">{fax.error}</p>
                        </div>
                      )}
                    </div>
                  </TableCell>
                </TableRow>
              )}
            </React.Fragment>
          ) : (
            <TableRow>

```

```
        <TableCell colSpan={10} className="h-24 text-center text-muted-foreground">
            No faxes found.
        </TableCell>
    </TableRow>
  </TableBody>
</Table>
</CardContent>
</Card>
</div>
  );
}
```

[artifacts/mtos-crm/src/pages/paralegals.tsx](#)

```
import { useState } from "react";
import { useListParalegals, useCreateParalegal, useGetParalegalLeaderboard, getListParalegalsQueryKey,
  getGetParalegalLeaderboardQueryKey } from "@workspace/api-client-react";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import { Skeleton } from "@components/ui/skeleton";
import { Button } from "@components/ui/button";
import { Progress } from "@components/ui/progress";
import { Dialog, DialogContent, DialogHeader, DialogTitle, DialogTrigger } from "@components/ui/dialog";
import { Input } from "@components/ui/input";
import { Label } from "@components/ui/label";
import { Plus, User } from "lucide-react";
import { useToast } from "@hooks/use-toast";
import { useQueryClient } from "@tanstack/react-query";

export default function Paralegals() {
  const { data: paralegals, isLoading: paralegalsLoading } = useListParalegals({
    query: { queryKey: getListParalegalsQueryKey() }
  });

  const { data: leaderboard, isLoading: leaderboardLoading } = useGetParalegalLeaderboard({
    query: { queryKey: getGetParalegalLeaderboardQueryKey() }
  });

  const [isAddOpen, setIsAddOpen] = useState(false);
  const [newParalegal, setNewParalegal] = useState({ name: "", email: "", role: "" });

  const createParalegal = useCreateParalegal();
  const { toast } = useToast();
  const queryClient = useQueryClient();

  const handleAddParalegal = async (e: React.FormEvent) => {
    e.preventDefault();
    try {
      await createParalegal.mutateAsync({ data: newParalegal });
      toast({ title: "Paralegal added successfully" });
      setIsAddOpen(false);
      setNewParalegal({ name: "", email: "", role: "" });
      queryClient.invalidateQueries({ queryKey: getListParalegalsQueryKey() });
      queryClient.invalidateQueries({ queryKey: getGetParalegalLeaderboardQueryKey() });
    } catch (err) {
      toast({ title: "Failed to add paralegal", variant: "destructive" });
    }
  };

  return (
    <div className="space-y-6">
      <div className="flex items-center justify-between">
        <h1 className="text-3xl font-bold tracking-tight">Paralegal Management</h1>
        <Dialog open={isAddOpen} onOpenChange={setIsAddOpen}>
          <DialogTrigger asChild>
            <Button>
              <Plus className="mr-2 h-4 w-4" />
              Add Paralegal
            </Button>
          </DialogTrigger>
        </Dialog>
      </div>
      <DialogContent>
        <DialogHeader>
          <DialogTitle>Add New Paralegal</DialogTitle>
        </DialogHeader>
        <form onSubmit={handleAddParalegal} className="space-y-4 pt-4">
          <div className="space-y-2">
            <Label htmlFor="name">Name</Label>
            <Input id="name" required value={newParalegal.name} onChange={e => setNewParalegal({...newParalegal, name:
e.target.value}}) />
          </div>
          <div className="space-y-2">
            <Label htmlFor="email">Email</Label>
            <Input id="email" type="email" value={newParalegal.email} onChange={e => setNewParalegal({...newParalegal,
email: e.target.value}}) />
          </div>
          <div className="space-y-2">
            <Label htmlFor="role">Role</Label>
            <Input id="role" value={newParalegal.role} onChange={e => setNewParalegal({...newParalegal, role:
e.target.value}} placeholder="e.g. Senior Paralegal" />
          </div>
          <div className="flex justify-end gap-2 pt-4">
            <Button type="button" variant="outline" onClick={() => setIsAddOpen(false)}>Cancel</Button>
            <Button type="submit" disabled={createParalegal.isPending}>
              {createParalegal.isPending ? "Adding..." : "Add Paralegal"}
            </Button>
          </div>
        </form>
      </DialogContent>
    </Dialog>
  </div>

  <div className="grid gap-4 md:grid-cols-2 lg:grid-cols-3 xl:grid-cols-4">
    {paralegalsLoading ? (
      <>
        <Skeleton className="h-40 w-full" />
        <Skeleton className="h-40 w-full" />
        <Skeleton className="h-40 w-full" />
        <Skeleton className="h-40 w-full" />
      </>
    ) : (
```

```

(paralegals || []).map((paralegal) => {
  const row = leaderboard?.find(r => r.id === paralegal.id);
  const convRate = row?.conversion_rate || 0;
  return (
    <Card key={paralegal.id}>
      <CardHeader className="pb-2">
        <div className="flex items-center gap-2">
          <div className="h-8 w-8 rounded-full bg-primary/10 flex items-center justify-center text-primary">
            <User className="h-4 w-4" />
          </div>
          <div>
            <CardTitle className="text-base">{paralegal.name}</CardTitle>
            <div className="text-xs text-muted-foreground">{paralegal.role}</div>
          </div>
        </div>
      </CardHeader>
      <CardContent className="space-y-4">
        <div className="grid grid-cols-2 gap-2 text-sm">
          <div>
            <div className="text-muted-foreground">Assigned</div>
            <div className="font-medium text-lg">{paralegal.total_assigned}</div>
          </div>
          <div>
            <div className="text-muted-foreground">Signed</div>
            <div className="font-medium text-lg text-primary">{paralegal.signed_cases}</div>
          </div>
        </div>
        <div className="space-y-1">
          <div className="flex items-center justify-between text-xs">
            <span className="text-muted-foreground">Conversion</span>
            <span className="font-medium">{convRate.toFixed(1)}%</span>
          </div>
          <Progress value={convRate} className="h-2" />
        </div>
      </CardContent>
    </Card>
  );
});
</div>

<Card>
  <CardHeader>
    <CardTitle>Leaderboard</CardTitle>
  </CardHeader>
  <CardContent>
    {leaderboardLoading ? (
      <div className="space-y-4">
        <Skeleton className="h-10 w-full" />
        <Skeleton className="h-10 w-full" />
        <Skeleton className="h-10 w-full" />
      </div>
    ) : (
      <div className="overflow-x-auto">
        <table className="w-full text-sm text-left">
          <thead className="text-xs text-muted-foreground uppercase bg-muted/50 border-b">
            <tr>
              <th className="px-4 py-3 w-16">Rank</th>
              <th className="px-4 py-3">Name</th>
              <th className="px-4 py-3">Role</th>
              <th className="px-4 py-3 text-right">Assigned</th>
              <th className="px-4 py-3 text-right">Qualified</th>
              <th className="px-4 py-3 text-right">Signed</th>
              <th className="px-4 py-3 text-right">Rejected</th>
              <th className="px-4 py-3 text-right">Conv Rate</th>
            </tr>
          </thead>
          <tbody>
            {((leaderboard || []).map((row, idx) => (
              <tr key={row.id} className={`border-b last:border-0 hover:bg-muted/50 transition-colors ${idx === 0 ? 'bg-primary/5' : ''}`}>
                <td className="px-4 py-3 font-medium">#{idx + 1}</td>
                <td className="px-4 py-3 font-medium">{row.name}</td>
                <td className="px-4 py-3 text-muted-foreground">{row.role}</td>
                <td className="px-4 py-3 text-right">{row.total_assigned}</td>
                <td className="px-4 py-3 text-right">{row.qualified}</td>
                <td className="px-4 py-3 text-right font-bold text-primary">{row.signed}</td>
                <td className="px-4 py-3 text-right text-destructive">{row.rejected}</td>
                <td className="px-4 py-3 text-right font-medium">{row.conversion_rate.toFixed(1)}%</td>
              </tr>
            ))}
          </tbody>
        </table>
        {(!leaderboard || leaderboard.length === 0) && (
          <div className="text-center py-6 text-muted-foreground">
            No data available.
          </div>
        )}
      </div>
    )}
  </CardContent>
</Card>
</div>
);
}

```

[artifacts/mtos-crm/src/pages/pipeline.tsx](#)

```
import { useGetConversionFunnel, useGetPipelineTrend, useGetTortBreakdown, getGetConversionFunnelQueryKey,
getGetPipelineTrendQueryKey, getGetTortBreakdownQueryKey } from "@workspace/api-client-react";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import { Skeleton } from "@components/ui/skeleton";
import { BarChart, Bar, XAxis, YAxis, CartesianGrid, Tooltip, ResponsiveContainer, LineChart, Line, Legend } from "recharts";
import { format } from "date-fns";

export default function Pipeline() {
  const { data: funnel, isLoading: funnelLoading } = useGetConversionFunnel({
    query: { queryKey: getGetConversionFunnelQueryKey() }
  });

  const { data: trend, isLoading: trendLoading } = useGetPipelineTrend({
    query: { queryKey: getGetPipelineTrendQueryKey() }
  });

  const { data: torts, isLoading: tortsLoading } = useGetTortBreakdown({
    query: { queryKey: getGetTortBreakdownQueryKey() }
  });

  return (
    <div className="space-y-6">
      <div className="flex items-center justify-between">
        <h1 className="text-3xl font-bold tracking-tight">Pipeline View</h1>
      </div>

      <Card>
        <CardHeader>
          <CardTitle>Conversion Funnel</CardTitle>
        </CardHeader>
        <CardContent>
          {funnelLoading ? (
            <Skeleton className="h-[300px] w-full" />
          ) : (
            <div className="h-[300px] w-full">
              <ResponsiveContainer width="100%" height="100%">
                <BarChart data={funnel || []} layout="vertical" margin={{ top: 5, right: 30, left: 40, bottom: 5 }}>
                  <CartesianGrid strokeDasharray="3 3" horizontal={false} stroke="hsl(var(--border))" />
                  <XAxis type="number" stroke="hsl(var(--muted-foreground))" fontSize={12} tickLine={false} axisLine={false} />
                  <YAxis dataKey="stage" type="category" stroke="hsl(var(--foreground))" fontSize={12} tickLine={false}
axisLine={false} width={120} />
                  <Tooltip
                    cursor={{fill: 'hsl(var(--muted))'}}
                    contentStyle={{ backgroundColor: 'hsl(var(--popover))', borderColor: 'hsl(var(--border))' }}
                  />
                  <Bar dataKey="count" fill="hsl(var(--primary))" radius={[0, 4, 4, 0]}}>
                    {(funnel || []).map((entry, index) => (
                      <Cell key={ `cell-${index}` } fill={entry.color || `hsl(var(--chart-${(index % 5) + 1})` } />
                    ))}
                  </Bar>
                </BarChart>
              </ResponsiveContainer>
            </div>
          )}
        </CardContent>
      </Card>

      <Card>
        <CardHeader>
          <CardTitle>Pipeline Trend</CardTitle>
        </CardHeader>
        <CardContent>
          {trendLoading ? (
            <Skeleton className="h-[300px] w-full" />
          ) : (
            <div className="h-[300px] w-full">
              <ResponsiveContainer width="100%" height="100%">
                <LineChart data={trend || []}>
                  <CartesianGrid strokeDasharray="3 3" vertical={false} stroke="hsl(var(--border))" />
                  <XAxis
                    dataKey="date"
                    stroke="hsl(var(--muted-foreground))"
                    fontSize={12}
                    tickLine={false}
                    axisLine={false}
                    tickFormatter={(value) => format(new Date(value), "MMM d")}
                  />
                  <YAxis stroke="hsl(var(--muted-foreground))" fontSize={12} tickLine={false} axisLine={false} />
                  <Tooltip
                    contentStyle={{ backgroundColor: 'hsl(var(--popover))', borderColor: 'hsl(var(--border))' }}
                    labelFormatter={(value) => format(new Date(value), "MMM d, yyyy")}
                  />
                  <Legend />
                  <Line type="monotone" dataKey="total" stroke="hsl(var(--chart-3))" strokeWidth={2} dot={false} name="Total
Leads" />
                  <Line type="monotone" dataKey="qualified" stroke="hsl(var(--chart-2))" strokeWidth={2} dot={false}
name="Qualified" />
                  <Line type="monotone" dataKey="signed" stroke="hsl(var(--chart-1))" strokeWidth={2} dot={false}
name="Signed" />
                </LineChart>
              </ResponsiveContainer>
            </div>
          )}
        </CardContent>
      </Card>
    </div>
  );
}
```

```

<Card>
  <CardHeader>
    <CardTitle>Tort Type Breakdown</CardTitle>
  </CardHeader>
  <CardContent>
    {tortsLoading ? (
      <div className="space-y-4">
        <Skeleton className="h-10 w-full" />
        <Skeleton className="h-10 w-full" />
        <Skeleton className="h-10 w-full" />
      </div>
    ) : (
      <div className="overflow-x-auto">
        <table className="w-full text-sm text-left">
          <thead className="text-xs text-muted-foreground uppercase bg-muted/50 border-b">
            <tr>
              <th className="px-4 py-3">Tort Type</th>
              <th className="px-4 py-3">Total</th>
              <th className="px-4 py-3">Qualified</th>
              <th className="px-4 py-3">Signed</th>
              <th className="px-4 py-3">Rejected</th>
              <th className="px-4 py-3">Avg Ad Spend</th>
            </tr>
          </thead>
          <tbody>
            {(torts || []).map((row, idx) => (
              <tr key={idx} className="border-b last:border-0 hover:bg-muted/50 transition-colors">
                <td className="px-4 py-3 font-medium">{row.tort_type}</td>
                <td className="px-4 py-3">{row.total}</td>
                <td className="px-4 py-3">{row.qualified}</td>
                <td className="px-4 py-3 font-bold text-primary">{row.signed}</td>
                <td className="px-4 py-3 text-destructive">{row.rejected}</td>
                <td className="px-4 py-3">${row.avg_ad_spend?.toFixed(2) || '0.00'}</td>
              </tr>
            ))}
          </tbody>
        </table>
        {(torts || torts.length === 0) && (
          <div className="text-center py-6 text-muted-foreground">
            No data available.
          </div>
        )}
      </div>
    )}
  </CardContent>
</Card>
</div>
);
}

function Cell(props: any) {
  return <path {...props} />;
}

```

[artifacts/mtos-crm/src/pages/predictive.tsx](#)

```
import { useState, useEffect } from "react";
import { Card, CardContent, CardHeader, CardTitle, CardDescription } from "@components/ui/card";
import { Button } from "@components/ui/button";
import { Badge } from "@components/ui/badge";
import { Input } from "@components/ui/input";
import { Table, TableBody, TableCell, TableHead, TableHeader, TableRow } from "@components/ui/table";
import { Skeleton } from "@components/ui/skeleton";
import { useToast } from "@hooks/use-toast";
import { TrendingUp, TrendingDown, Search, RefreshCw, ArrowUpRight, ArrowDownRight } from "lucide-react";

interface PredictiveScore {
  lead_id: number;
  conversion_probability: number;
  risk_score: number;
  quality_tier: string;
  factors: { name: string; impact: number; description: string }[];
}

interface TortPrediction {
  tort_type: string;
  avg_conversion: number;
  avg_risk: number;
  count: number;
}

interface ModelStats {
  total_training_samples: number;
  feature_weights: Record<string, number>;
  model_accuracy: number;
  last_trained: string;
}

const tierColors: Record<string, string> = {
  platinum: "bg-purple-100 text-purple-800 border-purple-300",
  gold: "bg-yellow-100 text-yellow-800 border-yellow-300",
  silver: "bg-gray-100 text-gray-700 border-gray-300",
  bronze: "bg-orange-100 text-orange-800 border-orange-300",
  unqualified: "bg-red-100 text-red-800 border-red-300",
};

export default function Predictive() {
  const { toast } = useToast();
  const [batchScores, setBatchScores] = useState<PredictiveScore[]>([]);
  const [tortPredictions, setTortPredictions] = useState<TortPrediction[]>([]);
  const [modelStats, setModelStats] = useState<ModelStats | null>(null);
  const [singleScore, setSingleScore] = useState<PredictiveScore | null>(null);
  const [searchId, setSearchId] = useState("");
  const [loading, setLoading] = useState(true);
  const [searching, setSearching] = useState(false);

  const fetchData = async () => {
    setLoading(true);
    try {
      const [batchRes, tortRes, modelRes] = await Promise.all([
        fetch("/api/analytics/predictive/batch?limit=25"),
        fetch("/api/analytics/predictive/by-tort"),
        fetch("/api/analytics/predictive/model"),
      ]);
      setBatchScores(await batchRes.json());
      setTortPredictions(await tortRes.json());
      setModelStats(await modelRes.json());
    } catch {
      toast({ title: "Failed to load predictions", variant: "destructive" });
    } finally {
      setLoading(false);
    }
  };

  useEffect(() => { fetchData(); }, []);

  const searchLead = async () => {
    if (!searchId) return;
    setSearching(true);
    try {
      const res = await fetch(`/api/analytics/predictive/lead/${searchId}`);
      if (!res.ok) throw new Error("Not found");
      setSingleScore(await res.json());
    } catch {
      toast({ title: "Lead not found", variant: "destructive" });
    } finally {
      setSearching(false);
    }
  };

  if (loading) {
    return (
      <div className="space-y-6">
        <h1 className="text-3xl font-bold tracking-tight">Praxis AI Analytics</h1>
        <div className="grid gap-4 md:grid-cols-3">
          {Array.from({ length: 3 }).map((_, i) => <Card key={i}><CardContent className="pt-6"><Skeleton className="h-20" /></CardContent></Card>)}
        </div>
      </div>
    );
  }
}
```



```

const avgConversion = batchScores.length > 0 ? Math.round(batchScores.reduce((s, b) => s + b.conversion_probability, 0) / batchScores.length) : 0;
const avgRisk = batchScores.length > 0 ? Math.round(batchScores.reduce((s, b) => s + b.risk_score, 0) / batchScores.length) : 0;
const tierCounts: Record<string, number> = {};
batchScores.forEach(b => { tierCounts[b.quality_tier] = (tierCounts[b.quality_tier] || 0) + 1; });

return (
  <div className="space-y-6">
    <div className="flex items-center justify-between">
      <div>
        <h1 className="text-3xl font-bold tracking-tight">Praxis AI Analytics</h1>
        <p className="text-muted-foreground">Predictive scoring and quality analysis across your lead pipeline.</p>
      </div>
      <Button variant="outline" onClick={fetchData}><RefreshCw className="h-4 w-4 mr-2" />Refresh</Button>
    </div>

    <div className="grid gap-4 md:grid-cols-4">
      <Card>
        <CardHeader className="pb-2"><CardTitle className="text-sm text-muted-foreground">Avg Conversion Score</CardTitle></CardHeader>
        <CardContent>
          <div className="flex items-center gap-2">
            <TrendingUp className="h-5 w-5 text-green-600" />
            <span className="text-3xl font-bold">{avgConversion}%</span>
          </div>
        </CardContent>
      </Card>
      <Card>
        <CardHeader className="pb-2"><CardTitle className="text-sm text-muted-foreground">Avg Risk Score</CardTitle></CardHeader>
        <CardContent>
          <div className="flex items-center gap-2">
            <TrendingDown className="h-5 w-5 text-red-600" />
            <span className="text-3xl font-bold">{avgRisk}%</span>
          </div>
        </CardContent>
      </Card>
      <Card>
        <CardHeader className="pb-2"><CardTitle className="text-sm text-muted-foreground">Model Accuracy</CardTitle></CardHeader>
        <CardContent><div className="text-3xl font-bold">{modelStats ? Math.round(modelStats.model_accuracy * 100) : 0}%</div></CardContent>
      </Card>
      <Card>
        <CardHeader className="pb-2"><CardTitle className="text-sm text-muted-foreground">Training Samples</CardTitle></CardHeader>
        <CardContent><div className="text-3xl font-bold">{modelStats?.total_training_samples || 0}</div></CardContent>
      </Card>
      <Card>
        <CardHeader>
          <CardTitle>Score a Lead</CardTitle>
        </CardHeader>
        <CardContent>
          <div className="flex gap-2 max-w-md">
            <Input placeholder="Lead ID" value={searchId} onChange={(e) => setSearchId(e.target.value)} onKeyDown={(e) => e.key === "Enter" && searchLead()} />
            <Button onClick={searchLead} disabled={searching}><Search className="h-4 w-4 mr-2" />{searching ? "..." : "Score"}</Button>
          </div>
          <div>
            {singleScore && (
              <div className="mt-4 p-4 border rounded-lg">
                <div className="flex items-center gap-4 mb-3">
                  <Badge className={tierColors[singleScore.quality_tier] || ""}>{singleScore.quality_tier.toUpperCase()}</Badge>
                  <span className="text-sm">Lead #{singleScore.lead_id}</span>
                </div>
                <div className="grid gap-4 md:grid-cols-2 mb-3">
                  <div className="flex items-center gap-2">
                    <ArrowUpRight className="h-4 w-4 text-green-600" />
                    <span className="text-sm">Conversion: <strong>{singleScore.conversion_probability}%</strong></span>
                  </div>
                  <div className="flex items-center gap-2">
                    <ArrowDownRight className="h-4 w-4 text-red-600" />
                    <span className="text-sm">Risk: <strong>{singleScore.risk_score}%</strong></span>
                  </div>
                </div>
                <div className="space-y-1">
                  {singleScore.factors.map((f, i) => (
                    <div key={i} className="flex items-center gap-2 text-sm">
                      <span className={f.impact > 0 ? "text-green-600" : f.impact < 0 ? "text-red-600" : "text-gray-500"}>
                        {f.impact > 0 ? "+" : f.impact < 0 ? "-" : "~"}
                      </span>
                      <span>{f.name}: {f.description}</span>
                    </div>
                  ))}
                </div>
              </div>
            )}
          </div>
        </CardContent>
      </Card>
    </div>
  </div>
)

```

```

    <CardTitle>Quality Distribution</CardTitle>
    <CardDescription>Lead quality tiers across recent leads</CardDescription>
  </CardHeader>
  <CardContent className="space-y-3">
    {Object.entries(tierCounts).sort((a, b) => b[1] - a[1]).map(([tier, count]) => (
      <div key={tier} className="flex items-center justify-between">
        <Badge className={tierColors[tier]} || "">{tier.toUpperCase()}</Badge>
        <div className="flex items-center gap-2">
          <div className="w-32 bg-gray-100 rounded-full h-2">
            <div className="h-2 rounded-full bg-primary" style={{ width: `${(count / batchScores.length) * 100}%` }} />
          </div>
          <span className="text-sm font-bold w-8 text-right">{count}</span>
        </div>
      </div>
    ))}
    {Object.keys(tierCounts).length === 0 && <div className="text-sm text-muted-foreground">No data yet</div>}
  </CardContent>
</Card>

<Card>
  <CardHeader>
    <CardTitle>By Tort Type</CardTitle>
    <CardDescription>Average conversion and risk by tort</CardDescription>
  </CardHeader>
  <CardContent>
    <Table>
      <TableHeader>
        <TableRow>
          <TableHead>Tort</TableHead>
          <TableHead>Leads</TableHead>
          <TableHead>Conv %</TableHead>
          <TableHead>Risk %</TableHead>
        </TableRow>
      </TableHeader>
      <TableBody>
        {tortPredictions.sort((a, b) => b.avg_conversion - a.avg_conversion).map((t) => (
          <TableRow key={t.tort_type}>
            <TableCell className="font-medium text-sm">{t.tort_type}</TableCell>
            <TableCell>{t.count}</TableCell>
            <TableCell className="text-green-600 font-bold">{t.avg_conversion}%</TableCell>
            <TableCell className="text-red-600 font-bold">{t.avg_risk}%</TableCell>
          </TableRow>
        ))}
      </TableBody>
    </Table>
  </CardContent>
</Card>
</div>

<Card>
  <CardHeader>
    <CardTitle>Recent Lead Predictions</CardTitle>
    <CardDescription>Predictive scores for the most recent {batchScores.length} leads</CardDescription>
  </CardHeader>
  <CardContent>
    <Table>
      <TableHeader>
        <TableRow>
          <TableHead>Lead ID</TableHead>
          <TableHead>Quality Tier</TableHead>
          <TableHead>Conversion %</TableHead>
          <TableHead>Risk %</TableHead>
          <TableHead>Top Factor</TableHead>
        </TableRow>
      </TableHeader>
      <TableBody>
        {batchScores.map((s) => (
          <TableRow key={s.lead_id}>
            <TableCell className="font-mono">#{s.lead_id}</TableCell>
            <TableCell><Badge className={tierColors[s.quality_tier]} || "">{s.quality_tier.toUpperCase()}</Badge></
TableCell>
            <TableCell className={s.conversion_probability >= 50 ? "text-green-600 font-bold" : ""}>
              {s.conversion_probability}%</TableCell>
            <TableCell className={s.risk_score >= 50 ? "text-red-600 font-bold" : ""}>{s.risk_score}%</TableCell>
            <TableCell className="text-sm">{s.factors[0]?.name || "-"}</TableCell>
          </TableRow>
        ))}
      </TableBody>
    </Table>
  </CardContent>
</Card>
</div>
);
}

```

[artifacts/mtos-crm/src/pages/review-queue.tsx](#)

```
import { useState } from "react";
import { useQueryClient } from "@tanstack/react-query";
import { formatDistanceToNow } from "date-fns";
import { ChevronDown, ChevronRight, Check, X, Search, AlertTriangle } from "lucide-react";

import {
  useListReviewQueue,
  useGetReviewQueueStats,
  useResolveReviewItem,
  getListReviewQueueQueryKey,
  getGetReviewQueueStatsQueryKey,
} from "@workspace/api-client-react";
import type { ReviewQueueItem } from "@workspace/api-client-react";

import { Button } from "@components/ui/button";
import { Badge } from "@components/ui/badge";
import {
  Select,
  SelectContent,
  SelectItem,
  SelectTrigger,
  SelectValue,
} from "@components/ui/select";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import {
  Table,
  TableBody,
  TableCell,
  TableHead,
  TableHeader,
  TableRow,
} from "@components/ui/table";
import { Skeleton } from "@components/ui/skeleton";
import { useToast } from "@hooks/use-toast";

function getConflictBadge(conflictType: string) {
  switch (conflictType) {
    case "LOGICAL_CONFLICT":
      return <Badge variant="secondary" className="bg-blue-500/10 text-blue-500 hover:bg-blue-500/20">{conflictType}</Badge>;
    case "RULE_OVERRIDE_CONFLICT":
      return <Badge variant="secondary" className="bg-purple-500/10 text-purple-500 hover:bg-purple-500/20">{conflictType}</
Badge>;
    case "DATA_INTEGRITY_CONFLICT":
      return <Badge variant="secondary" className="bg-red-500/10 text-red-500 hover:bg-red-500/20">{conflictType}</Badge>;
    case "AI_CLASSIFICATION_CONFLICT":
      return <Badge variant="secondary" className="bg-amber-500/10 text-amber-500 hover:bg-amber-500/20">{conflictType}</Badge>;
    case "EXECUTION_FAILURE":
      return <Badge variant="secondary" className="bg-orange-500/10 text-orange-500 hover:bg-orange-500/20">{conflictType}</
Badge>;
    default:
      return <Badge variant="outline">{conflictType}</Badge>;
  }
}

function getSeverityBadge(severity: string) {
  switch (severity) {
    case "critical":
      return <Badge variant="destructive">{severity}</Badge>;
    case "high":
      return <Badge variant="secondary" className="bg-orange-500/10 text-orange-500 hover:bg-orange-500/20">{severity}</Badge>;
    case "medium":
      return <Badge variant="secondary" className="bg-yellow-500/10 text-yellow-500 hover:bg-yellow-500/20">{severity}</Badge>;
    case "low":
      return <Badge variant="secondary" className="bg-gray-500/10 text-gray-500 hover:bg-gray-500/20">{severity}</Badge>;
    default:
      return <Badge variant="outline">{severity}</Badge>;
  }
}

function getFailsafeText(mode: string) {
  switch (mode) {
    case "SAFE_FAIL":
      return "Auto-Reject";
    case "REVIEW_FAIL":
      return "Manual Review";
    case "HARD_BLOCK":
      return "Blocked";
    default:
      return mode;
  }
}

function ReviewQueueRow({ item }: { item: ReviewQueueItem }) {
  const [expanded, setExpanded] = useState(false);
  const queryClient = useQueryClient();
  const { toast } = useToast();
  const resolveMutation = useResolveReviewItem();

  const handleResolve = (resolution: "accepted" | "rejected") => {
    resolveMutation.mutate(
      {
        id: item.id,
        data: { resolution, resolved_by: "admin" },
      },
      {

```

```

onSuccess: () => {
  toast({
    title: `Item ${resolution}`,
    description: `Review item ${item.id} has been ${resolution}.`,
  });
  queryClient.invalidateQueries({
    queryKey: getListReviewQueueQueryKey(),
  });
  queryClient.invalidateQueries({
    queryKey: getGetReviewQueueStatsQueryKey(),
  });
},
onError: () => {
  toast({
    title: "Error",
    description: "Failed to resolve item.",
    variant: "destructive",
  });
},
});
};
return (
  <
    <TableRow className="group cursor-pointer hover:bg-muted/50" onClick={() => setExpanded(!expanded)}>
      <TableCell>
        <Button variant="ghost" size="icon" className="h-6 w-6">
          {expanded ? <ChevronDown className="h-4 w-4" /> : <ChevronRight className="h-4 w-4" />}
        </Button>
      </TableCell>
      <TableCell className="font-mono text-xs text-muted-foreground">
        {item.id}
      </TableCell>
      <TableCell>
        <span className="font-medium">{item.entity_type}</span> / {item.entity_id}
      </TableCell>
      <TableCell>{getConflictBadge(item.conflict_type)}</TableCell>
      <TableCell>{getSeverityBadge(item.severity)}</TableCell>
      <TableCell className="text-sm">{getFailsafeText(item.failsafe_mode)}</TableCell>
      <TableCell className="max-w-[200px] truncate" title={item.summary}>
        {item.summary}
      </TableCell>
      <TableCell>
        <Badge variant={item.resolution === "pending" ? "outline" : "secondary"}>
          {item.resolution}
        </Badge>
      </TableCell>
      <TableCell className="text-right text-sm">{item.retry_count}</TableCell>
      <TableCell className="text-right text-xs text-muted-foreground whitespace-nowrap">
        {formatDistanceToNow(new Date(item.created_at), { addSuffix: true })}
      </TableCell>
      <TableCell className="text-right">
        {item.resolution === "pending" ? (
          <div className="flex justify-end gap-2" onClick={(e) => e.stopPropagation()}>
            <Button
              variant="outline"
              size="sm"
              className="h-8 text-green-600 hover:text-green-700 hover:bg-green-50 dark:hover:bg-green-500/10"
              onClick={() => handleResolve("accepted")}
              disabled={resolveMutation.isPending}
            >
              <Check className="mr-1 h-3 w-3" /> Accept
            </Button>
            <Button
              variant="outline"
              size="sm"
              className="h-8 text-red-600 hover:text-red-700 hover:bg-red-50 dark:hover:bg-red-500/10"
              onClick={() => handleResolve("rejected")}
              disabled={resolveMutation.isPending}
            >
              <X className="mr-1 h-3 w-3" /> Reject
            </Button>
          </div>
        ) : null}
      </TableCell>
    </TableRow>
    {expanded && (
      <TableRow>
        <TableCell colSpan={11} className="bg-muted/30 p-0">
          <div className="p-6 grid grid-cols-2 gap-6">
            <div className="space-y-4">
              <div>
                <h4 className="font-medium text-sm mb-1 text-muted-foreground">Full Summary</h4>
                <p className="text-sm leading-relaxed">{item.summary}</p>
              </div>
              <div>
                <h4 className="font-medium text-sm mb-1 text-muted-foreground">Source Module</h4>
                <p className="text-sm font-mono bg-muted px-2 py-1 rounded inline-block">
                  {item.source_module}
                </p>
              </div>
            </div>
            {item.resolution !== "pending" && (
              <div className="bg-background rounded-md border p-4 space-y-2 mt-4">
                <h4 className="font-medium text-sm text-foreground">Resolution Details</h4>
                <div className="text-sm text-muted-foreground">
                  <span className="font-medium">Resolved By:</span> {item.resolved_by || "System"}
                </div>
              </div>
            )}
          </div>
        </TableCell>
      </TableRow>
    )}
  </

```

```

    </div>
    {item.resolved_at && (
      <div className="text-sm text-muted-foreground">
        <span className="font-medium">Resolved At:</span> {new Date(item.resolved_at).toLocaleString()}
      </div>
    )}
    {item.resolution_notes && (
      <div className="text-sm mt-2">
        <span className="font-medium text-muted-foreground block mb-1">Notes:</span>
        {item.resolution_notes}
      </div>
    )}
  </div>
)}
</div>
<div className="space-y-4">
  <div>
    <h4 className="font-medium text-sm mb-1 text-muted-foreground">Details</h4>
    <div className="bg-muted rounded-md p-4 overflow-auto max-h-[300px]">
      <pre className="text-xs font-mono text-foreground">
        {JSON.stringify(item.details || {}, null, 2)}
      </pre>
    </div>
  </div>
  <div>
    {item.resolution === "pending" && (item.severity === "critical" || item.severity === "high") && (
      <div className="border border-red-200 bg-red-50 dark:bg-red-900/10 rounded-md p-4">
        <h4 className="font-medium text-sm mb-2 text-red-700 dark:text-red-400 flex items-center gap-2">
          <AlertTriangle className="h-4 w-4" /> Escalation
        </h4>
        <p className="text-xs text-red-600 dark:text-red-500 mb-3">
          For suspected fraud or criminal activity, escalate to FBI tip line.
        </p>
        <Button
          variant="destructive"
          size="sm"
          onClick={(e) => {
            e.stopPropagation();
            const apiBase = import.meta.env.BASE_URL.replace(/\/$/, "");
            fetch(`${apiBase}/api/forms/escalate/fbi`, {
              method: "POST",
              headers: { "Content-Type": "application/json" },
              body: JSON.stringify({
                lead_id: item.entity_id,
                reason: `Review queue escalation: ${item.conflict_type} - ${item.summary}`,
                fraud_indicators: item.details,
              }),
            })
              .then(r => r.json())
              .then(d => {
                toast({
                  title: "Escalated to FBI",
                  description: "Case logged. Submit tip at tips.fbi.gov",
                });
                window.open("https://tips.fbi.gov/", "_blank");
              })
              .catch(() => {
                toast({
                  title: "Escalation failed",
                  description: "Could not log escalation. Please try again.",
                  variant: "destructive",
                });
              });
          })
        >
          <AlertTriangle className="mr-1 h-3 w-3" /> Send to FBI
        </Button>
      </div>
    )}
  </div>
</div>
</TableCell>
</TableRow>
)}
</>
);
}

```

```

export default function ReviewQueue() {
  const [resolution, setResolution] = useState("pending");
  const [conflictType, setConflictType] = useState("all");
  const [severity, setSeverity] = useState("all");

  const { data: stats, isLoading: statsLoading } = useGetReviewQueueStats();

  const queryParams = {
    ...(resolution !== "all" && { resolution }),
    ...(conflictType !== "all" && { conflict_type: conflictType }),
    ...(severity !== "all" && { severity }),
  };

  const { data: queueItems = [], isLoading: queueLoading } = useListReviewQueue(queryParams);

  const pendingCount = stats?.total_pending || 0;

  const criticalCount = stats?.by_severity?.find((s: any) => s.severity === "critical")?.count || 0;
  const highCount = stats?.by_severity?.find((s: any) => s.severity === "high")?.count || 0;
}

```

```

const resolvedCount = stats?.by_resolution?.reduce((acc: number, r: any) => {
  if (r.resolution !== "pending") return acc + (r.count || 0);
  return acc;
}, 0) || 0;

return (
  <div className="flex flex-col gap-6 p-8 max-w-7xl mx-auto w-full">
    <div className="flex items-center justify-between">
      <div>
        <h1 className="text-3xl font-bold tracking-tight">Review Queue</h1>
        <p className="text-muted-foreground mt-2">
          Manage conflicts, execution failures, and flagged items requiring human intervention.
        </p>
      </div>
    </div>

    <div className="grid gap-4 md:grid-cols-4">
      <Card>
        <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
          <CardTitle className="text-sm font-medium">Pending Review</CardTitle>
        </CardHeader>
        <CardContent>
          {statsLoading ? (
            <Skeleton className="h-8 w-16" />
          ) : (
            <div className={`text-2xl font-bold ${pendingCount > 0 ? "text-amber-500" : ""}`}>
              {pendingCount}
            </div>
          )}
        </CardContent>
      </Card>
      <Card>
        <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
          <CardTitle className="text-sm font-medium text-red-500">Critical Conflicts</CardTitle>
        </CardHeader>
        <CardContent>
          {statsLoading ? (
            <Skeleton className="h-8 w-16" />
          ) : (
            <div className="text-2xl font-bold text-red-500">{criticalCount}</div>
          )}
        </CardContent>
      </Card>
      <Card>
        <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
          <CardTitle className="text-sm font-medium text-orange-500">High Severity</CardTitle>
        </CardHeader>
        <CardContent>
          {statsLoading ? (
            <Skeleton className="h-8 w-16" />
          ) : (
            <div className="text-2xl font-bold text-orange-500">{highCount}</div>
          )}
        </CardContent>
      </Card>
      <Card>
        <CardHeader className="flex flex-row items-center justify-between space-y-0 pb-2">
          <CardTitle className="text-sm font-medium">Resolved Today</CardTitle>
        </CardHeader>
        <CardContent>
          {statsLoading ? (
            <Skeleton className="h-8 w-16" />
          ) : (
            <div className="text-2xl font-bold text-muted-foreground">{resolvedCount}</div>
          )}
        </CardContent>
      </Card>
    </div>

    <div className="flex flex-col gap-4 bg-card rounded-lg border shadow-sm">
      <div className="p-4 border-b flex flex-wrap gap-4 items-center bg-muted/20">
        <div className="flex items-center gap-2 text-sm font-medium text-muted-foreground">
          <Search className="h-4 w-4" />
          Filters
        </div>
      </div>

      <Select value={resolution} onChange={setResolution}>
        <SelectTrigger className="w-[180px] h-9 bg-background">
          <SelectValue placeholder="Resolution" />
        </SelectTrigger>
        <SelectContent>
          <SelectItem value="all">All Resolutions</SelectItem>
          <SelectItem value="pending">Pending</SelectItem>
          <SelectItem value="accepted">Accepted</SelectItem>
          <SelectItem value="rejected">Rejected</SelectItem>
          <SelectItem value="escalated">Escalated</SelectItem>
        </SelectContent>
      </Select>

      <Select value={conflictType} onChange={setConflictType}>
        <SelectTrigger className="w-[240px] h-9 bg-background">
          <SelectValue placeholder="Conflict Type" />
        </SelectTrigger>
        <SelectContent>
          <SelectItem value="all">All Types</SelectItem>
          <SelectItem value="LOGICAL_CONFLICT">Logical Conflict</SelectItem>
          <SelectItem value="RULE_OVERRIDE_CONFLICT">Rule Override</SelectItem>
        </SelectContent>
      </Select>
    </div>
  </div>
)

```

```

        <SelectItem value="DATA_INTEGRITY_CONFLICT">Data Integrity</SelectItem>
        <SelectItem value="AI_CLASSIFICATION_CONFLICT">AI Classification</SelectItem>
        <SelectItem value="EXECUTION_FAILURE">Execution Failure</SelectItem>
      </SelectContent>
    </Select>

    <Select value={severity} onChange={setSeverity}>
      <SelectTrigger className="w-[160px] h-9 bg-background">
        <SelectValue placeholder="Severity" />
      </SelectTrigger>
      <SelectContent>
        <SelectItem value="all">All Severities</SelectItem>
        <SelectItem value="critical">Critical</SelectItem>
        <SelectItem value="high">High</SelectItem>
        <SelectItem value="medium">Medium</SelectItem>
        <SelectItem value="low">Low</SelectItem>
      </SelectContent>
    </Select>
  </div>

  <div className="overflow-x-auto">
    <Table>
      <TableHeader>
        <TableRow className="bg-muted/50 hover:bg-muted/50">
          <TableHead className="w-[40px]"></TableHead>
          <TableHead className="w-[80px]">ID</TableHead>
          <TableHead className="w-[150px]">Entity</TableHead>
          <TableHead className="w-[180px]">Conflict Type</TableHead>
          <TableHead className="w-[100px]">Severity</TableHead>
          <TableHead className="w-[140px]">Mode</TableHead>
          <TableHead className="min-w-[200px]">Summary</TableHead>
          <TableHead className="w-[100px]">Status</TableHead>
          <TableHead className="w-[80px] text-right">Retries</TableHead>
          <TableHead className="w-[120px] text-right">Created</TableHead>
          <TableHead className="w-[180px] text-right">Actions</TableHead>
        </TableRow>
      </TableHeader>
      <TableBody>
        {queueLoading ? (
          Array.from({ length: 5 }).map((_, i) => (
            <TableRow key={i}>
              <TableCell><Skeleton className="h-4 w-4" /></TableCell>
              <TableCell><Skeleton className="h-4 w-8" /></TableCell>
              <TableCell><Skeleton className="h-4 w-24" /></TableCell>
              <TableCell><Skeleton className="h-6 w-24 rounded-full" /></TableCell>
              <TableCell><Skeleton className="h-6 w-16 rounded-full" /></TableCell>
              <TableCell><Skeleton className="h-4 w-20" /></TableCell>
              <TableCell><Skeleton className="h-4 w-full max-w-[200px]" /></TableCell>
              <TableCell><Skeleton className="h-6 w-20 rounded-full" /></TableCell>
              <TableCell className="text-right"><Skeleton className="h-4 w-8 ml-auto" /></TableCell>
              <TableCell className="text-right"><Skeleton className="h-4 w-16 ml-auto" /></TableCell>
              <TableCell><Skeleton className="h-8 w-24 ml-auto" /></TableCell>
            </TableRow>
          ))
        ) : queueItems.length === 0 ? (
          <TableRow>
            <TableCell colSpan={11} className="h-32 text-center text-muted-foreground">
              No review items match the current filters.
            </TableCell>
          </TableRow>
        ) : (
          queueItems.map((item) => (
            <ReviewQueueRow key={item.id} item={item as ReviewQueueItem} />
          ))
        )
      </TableBody>
    </Table>
  </div>
</div>
);
}

```

[artifacts/mtos-crm/src/pages/security.tsx](#)

```
import { useState, useEffect, useCallback } from "react";
import { Card, CardContent, CardHeader, CardTitle, CardDescription } from "@components/ui/card";
import { Table, TableBody, TableCell, TableHead, TableHeader, TableRow } from "@components/ui/table";
import { Button } from "@components/ui/button";
import { Badge } from "@components/ui/badge";
import { Input } from "@components/ui/input";
import { Label } from "@components/ui/label";
import { Dialog, DialogContent, DialogHeader, DialogTitle, DialogFooter, DialogDescription } from "@components/ui/dialog";
import { Skeleton } from "@components/ui/skeleton";
import { useToast } from "@hooks/use-toast";
import { Shield, ShieldAlert, ShieldOff, ShieldCheck, Ban, Brain, RefreshCw, Trash2 } from "lucide-react";
import { format } from "date-fns";

interface SecurityAlert {
  id: number;
  type: string;
  severity: string;
  source_ip: string | null;
  user_agent: string | null;
  request_path: string | null;
  request_method: string | null;
  details: string;
  payload_sample: string | null;
  ai_analysis: string | null;
  status: string;
  blocked: boolean;
  created_at: string;
}

interface BlockedIp {
  id: number;
  ip: string;
  reason: string;
  blocked_until: string | null;
  auto_blocked: boolean;
  alert_count: number;
  created_at: string;
}

interface SecurityStats {
  threat_level: string;
  total_alerts_24h: number;
  critical_alerts_24h: number;
  alerts_last_hour: number;
  blocked_ips: number;
  by_severity: { severity: string; count: number }[];
  by_type: { type: string; count: number }[];
}

const severityColor: Record<string, string> = {
  critical: "bg-red-600 text-white",
  high: "bg-orange-500 text-white",
  medium: "bg-yellow-500 text-black",
  low: "bg-blue-400 text-white",
};

const threatLevelConfig: Record<string, { icon: typeof ShieldAlert; color: string; bg: string; label: string }> = {
  critical: { icon: ShieldOff, color: "text-red-600", bg: "bg-red-50 border-red-200", label: "CRITICAL THREAT" },
  high: { icon: ShieldAlert, color: "text-orange-600", bg: "bg-orange-50 border-orange-200", label: "HIGH THREAT" },
  elevated: { icon: Shield, color: "text-yellow-600", bg: "bg-yellow-50 border-yellow-200", label: "ELEVATED" },
  normal: { icon: ShieldCheck, color: "text-green-600", bg: "bg-green-50 border-green-200", label: "NORMAL" },
};

export default function Security() {
  const { toast } = useToast();
  const [stats, setStats] = useState<SecurityStats | null>(null);
  const [alerts, setAlerts] = useState<SecurityAlert[]>([]);
  const [blockedIps, setBlockedIps] = useState<BlockedIp[]>([]);
  const [loading, setLoading] = useState(true);
  const [blockDialogOpen, setBlockDialogOpen] = useState(false);
  const [blockIp, setBlockIp] = useState("");
  const [blockReason, setBlockReason] = useState("");
  const [blockDuration, setBlockDuration] = useState("24");
  const [analyzing, setAnalyzing] = useState(false);
  const [analysisResult, setAnalysisResult] = useState<any>(null);
  const [analysisDialogOpen, setAnalysisDialogOpen] = useState(false);

  const fetchData = useCallback(async () => {
    try {
      const [statsRes, alertsRes, blockedRes] = await Promise.all([
        fetch("/api/security/stats"),
        fetch("/api/security/alerts?limit=50"),
        fetch("/api/security/blocked-ips"),
      ]);
      setStats(await statsRes.json());
      setAlerts(await alertsRes.json());
      setBlockedIps(await blockedRes.json());
    } catch {
      toast({ title: "Failed to load security data", variant: "destructive" });
    } finally {
      setLoading(false);
    }
  }, [toast]);

  useEffect(() => { fetchData(); }, [fetchData]);
```



```

useEffect(() => {
  const interval = setInterval(fetchData, 30000);
  return () => clearInterval(interval);
}, [fetchData]);

const handleBlockIp = async () => {
  if (!blockIp || !blockReason) return;
  try {
    await fetch("/api/security/block-ip", {
      method: "POST",
      headers: { "Content-Type": "application/json" },
      body: JSON.stringify({ ip: blockIp, reason: blockReason, duration_hours: parseInt(blockDuration) }),
    });
    toast({ title: "IP blocked" });
    setBlockDialogOpen(false);
    setBlockIp("");
    setBlockReason("");
    fetchData();
  } catch {
    toast({ title: "Failed to block IP", variant: "destructive" });
  }
};

const handleUnblockIp = async (ip: string) => {
  try {
    await fetch(`/api/security/blocked-ips/${encodeURIComponent(ip)}`, { method: "DELETE" });
    toast({ title: "IP unblocked" });
    fetchData();
  } catch {
    toast({ title: "Failed to unblock IP", variant: "destructive" });
  }
};

const handleDismiss = async (id: number) => {
  try {
    await fetch(`/api/security/alerts/${id}/dismiss`, { method: "PATCH" });
    fetchData();
  } catch {
    toast({ title: "Failed to dismiss alert", variant: "destructive" });
  }
};

const handleAiAnalysis = async () => {
  setAnalyzing(true);
  try {
    const res = await fetch("/api/security/analyze", {
      method: "POST",
      headers: { "Content-Type": "application/json" },
      body: JSON.stringify({}),
    });
    const data = await res.json();
    setAnalysisResult(data);
    setAnalysisDialogOpen(true);
    fetchData();
  } catch {
    toast({ title: "AI analysis failed", variant: "destructive" });
  } finally {
    setAnalyzing(false);
  }
};

const tlConfig = threatLevelConfig[stats?.threat_level || "normal"];
const ThreatIcon = tlConfig.icon;

if (loading) {
  return (
    <div className="space-y-6">
      <h1 className="text-3xl font-bold tracking-tight">Security Center</h1>
      <div className="grid gap-4 md:grid-cols-4">
        {Array.from({ length: 4 }).map((_, i) => (
          <Card key={i}><CardContent className="pt-6"><Skeleton className="h-16" /></CardContent></Card>
        ))}
      </div>
    </div>
  );
}

return (
  <div className="space-y-6">
    <div className="flex items-center justify-between">
      <h1 className="text-3xl font-bold tracking-tight">Security Center</h1>
      <div className="flex items-center gap-2">
        <Button variant="outline" onClick={fetchData}>
          <RefreshCw className="mr-2 h-4 w-4" />
          Refresh
        </Button>
        <Button onClick={handleAiAnalysis} disabled={analyzing}>
          <Brain className="mr-2 h-4 w-4" />
          {analyzing ? "Analyzing..." : "AI Threat Analysis"}
        </Button>
        <Button variant="destructive" onClick={() => setBlockDialogOpen(true)}>
          <Ban className="mr-2 h-4 w-4" />
          Block IP
        </Button>
      </div>
    </div>
  </div>
);

```

```

<Card className={`border-2 ${tlConfig.bg}`}>
  <CardContent className="pt-6">
    <div className="flex items-center gap-4">
      <ThreatIcon className={`h-12 w-12 ${tlConfig.color}`} />
      <div>
        <div className={`text-2xl font-bold ${tlConfig.color}`}>{tlConfig.label}</div>
        <div className="text-sm text-muted-foreground">
          {stats?.total_alerts_24h || 0} alerts in the last 24 hours
          {stats?.critical_alerts_24h ? ` (${stats.critical_alerts_24h} critical)` : ""}
        </div>
      </div>
    </div>
  </CardContent>
</Card>

<div className="grid gap-4 md:grid-cols-4">
  <Card>
    <CardHeader className="pb-2"><CardTitle className="text-sm font-medium text-muted-foreground">Alerts (24h)</CardTitle></CardHeader>
    <CardContent><div className="text-3xl font-bold">{stats?.total_alerts_24h || 0}</div></CardContent>
  </Card>
  <Card>
    <CardHeader className="pb-2"><CardTitle className="text-sm font-medium text-muted-foreground">Critical (24h)</CardTitle></CardHeader>
    <CardContent><div className="text-3xl font-bold text-red-600">{stats?.critical_alerts_24h || 0}</div></CardContent>
  </Card>
  <Card>
    <CardHeader className="pb-2"><CardTitle className="text-sm font-medium text-muted-foreground">Last Hour</CardTitle></CardHeader>
    <CardContent><div className="text-3xl font-bold">{stats?.alerts_last_hour || 0}</div></CardContent>
  </Card>
  <Card>
    <CardHeader className="pb-2"><CardTitle className="text-sm font-medium text-muted-foreground">Blocked IPs</CardTitle></CardHeader>
    <CardContent><div className="text-3xl font-bold text-orange-600">{stats?.blocked_ips || 0}</div></CardContent>
  </Card>
</div>

{stats && (stats.by_type.length > 0 || stats.by_severity.length > 0) && (
  <div className="grid gap-4 md:grid-cols-2">
    <Card>
      <CardHeader><CardTitle>By Attack Type</CardTitle></CardHeader>
      <CardContent>
        <div className="space-y-2">
          {stats.by_type.map((t) => (
            <div key={t.type} className="flex items-center justify-between">
              <span className="text-sm font-mono">{t.type.replace(/_/g, " ")}</span>
              <Badge variant="outline">{t.count}</Badge>
            </div>
          ))}
          {stats.by_type.length === 0 && <div className="text-sm text-muted-foreground">No attacks detected</div>}
        </div>
      </CardContent>
    </Card>
    <Card>
      <CardHeader><CardTitle>By Severity</CardTitle></CardHeader>
      <CardContent>
        <div className="space-y-2">
          {stats.by_severity.map((s) => (
            <div key={s.severity} className="flex items-center justify-between">
              <Badge className={severityColor[s.severity] || ""}>{s.severity.toUpperCase()}</Badge>
              <span className="font-bold">{s.count}</span>
            </div>
          ))}
          {stats.by_severity.length === 0 && <div className="text-sm text-muted-foreground">No alerts</div>}
        </div>
      </CardContent>
    </Card>
  </div>
)}

{blockedIps.length > 0 && (
  <Card>
    <CardHeader>
      <CardTitle>Blocked IPs</CardTitle>
      <CardDescription>Currently blocked IP addresses</CardDescription>
    </CardHeader>
    <CardContent>
      <Table>
        <TableHeader>
          <TableRow>
            <TableHead>IP Address</TableHead>
            <TableHead>Reason</TableHead>
            <TableHead>Blocked Until</TableHead>
            <TableHead>Source</TableHead>
            <TableHead>Alerts</TableHead>
            <TableHead className="text-right">Actions</TableHead>
          </TableRow>
        </TableHeader>
        <TableBody>
          {blockedIps.map((ip) => (
            <TableRow key={ip.id}>
              <TableCell className="font-mono font-bold">{ip.ip}</TableCell>
              <TableCell className="max-w-[300px] truncate">{ip.reason}</TableCell>
              <TableCell className="font-mono text-sm">
                {ip.blocked_until ? format(new Date(ip.blocked_until), "yyyy-MM-dd HH:mm") : "Permanent"}
              </TableCell>
            </TableRow>
          ))}
        </TableBody>
      </Table>
    </CardContent>
  </Card>
)

```

```

        </TableCell>
        <TableCell>
          <Badge variant={ip.auto_blocked ? "destructive" : "secondary"}>
            {ip.auto_blocked ? "AUTO" : "MANUAL"}
          </Badge>
        </TableCell>
        <TableCell>{ip.alert_count}</TableCell>
        <TableCell className="text-right">
          <Button variant="ghost" size="sm" onClick={() => handleUnblockIp(ip.ip)}>
            <Trash2 className="h-4 w-4" />
          </Button>
        </TableCell>
      </TableRow>
    </Table>
  </TableBody>
</Table>
</CardContent>
</Card>
)}

<Card>
  <CardHeader>
    <CardTitle>Recent Security Alerts</CardTitle>
    <CardDescription>Last 50 security events detected by the IDS</CardDescription>
  </CardHeader>
  <CardContent>
    {alerts.length === 0 ? (
      <div className="text-center py-8 text-muted-foreground">
        <ShieldCheck className="h-12 w-12 mx-auto mb-3 text-green-500" />
        <div className="text-lg font-medium">All Clear</div>
        <div>No security alerts detected. The system is being monitored.</div>
      </div>
    ) : (
      <Table>
        <TableHeader>
          <TableRow>
            <TableHead>Severity</TableHead>
            <TableHead>Type</TableHead>
            <TableHead>Source IP</TableHead>
            <TableHead>Target</TableHead>
            <TableHead>Details</TableHead>
            <TableHead>Status</TableHead>
            <TableHead>Time</TableHead>
            <TableHead className="text-right">Actions</TableHead>
          </TableRow>
        </TableHeader>
        <TableBody>
          {alerts.map((alert) => (
            <TableRow key={alert.id} className={alert.severity === "critical" ? "bg-red-50" : ""}>
              <TableCell>
                <Badge className={severityColor[alert.severity] || ""}>
                  {alert.severity.toUpperCase()}
                </Badge>
              </TableCell>
              <TableCell className="font-mono text-sm">{alert.type.replace(/_/g, " ")}</TableCell>
              <TableCell className="font-mono text-sm">{alert.source_ip || "-"}</TableCell>
              <TableCell className="font-mono text-xs max-w-[150px] truncate">
                {alert.request_method} {alert.request_path}
              </TableCell>
              <TableCell className="max-w-[200px] truncate text-sm">{alert.details}</TableCell>
              <TableCell>
                <Badge variant={
                  alert.status === "new" ? "outline" :
                  alert.status === "analyzed" ? "secondary" :
                  "default"
                }>
                  {alert.status.toUpperCase()}
                </Badge>
              </TableCell>
              <TableCell className="font-mono text-xs">
                {format(new Date(alert.created_at), "HH:mm:ss")}
              </TableCell>
              <TableCell className="text-right">
                {alert.status === "new" && (
                  <Button variant="ghost" size="sm" onClick={() => handleDismiss(alert.id)}>
                    Dismiss
                  </Button>
                )}
              </TableCell>
            </TableRow>
          )
        </TableBody>
      </Table>
    )}
  </CardContent>
</Card>

<Dialog open={blockDialogOpen} onOpenChange={setBlockDialogOpen}>
  <DialogContent>
    <DialogHeader>
      <DialogTitle>Block IP Address</DialogTitle>
      <DialogDescription>Manually block an IP address from accessing the system.</DialogDescription>
    </DialogHeader>
    <div className="grid gap-4 py-4">
      <div className="grid gap-2">
        <Label>IP Address</Label>
        <Input placeholder="192.168.1.1" value={blockIp} onChange={(e) => setBlockIp(e.target.value)} />
      </div>
    </div>
  </DialogContent>
</Dialog>

```

```

    </div>
    <div className="grid gap-2">
      <Label>Reason</Label>
      <Input placeholder="Suspicious activity" value={blockReason} onChange={(e) => setBlockReason(e.target.value)} />
    </div>
    <div className="grid gap-2">
      <Label>Duration (hours)</Label>
      <Input type="number" value={blockDuration} onChange={(e) => setBlockDuration(e.target.value)} />
    </div>
  </Dialog>
  <DialogFooter>
    <Button variant="outline" onClick={() => setBlockDialogOpen(false)}>Cancel</Button>
    <Button variant="destructive" onClick={handleBlockIp} disabled={!blockIp || !blockReason}>
      Block IP
    </Button>
  </DialogFooter>
</DialogContent>
</Dialog>

<Dialog open={analysisDialogOpen} onOpenChange={setAnalysisDialogOpen}>
  <DialogContent className="sm:max-w-lg">
    <DialogHeader>
      <DialogTitle>AI Threat Analysis</DialogTitle>
      <DialogDescription>AI-powered analysis of recent security events.</DialogDescription>
    </DialogHeader>
    {analysisResult?.analysis ? (
      <div className="space-y-4 py-4">
        <div className="flex items-center gap-2">
          <Badge className={severityColor[analysisResult.analysis.threat_level] || ""}>
            {analysisResult.analysis.threat_level?.toUpperCase()}
          </Badge>
          <span className="font-medium">{analysisResult.analysis.attack_type}</span>
        </div>
        <div>
          <div className="text-sm font-medium text-muted-foreground mb-1">Summary</div>
          <div className="text-sm">{analysisResult.analysis.summary}</div>
        </div>
        <div>
          <div className="text-sm font-medium text-muted-foreground mb-1">Recommended Action</div>
          <div className="text-sm font-medium">{analysisResult.analysis.recommended_action}</div>
        </div>
        {analysisResult.analysis.indicators?.length > 0 && (
          <div>
            <div className="text-sm font-medium text-muted-foreground mb-1">Indicators</div>
            <ul className="list-disc pl-4 text-sm space-y-1">
              {analysisResult.analysis.indicators.map((ind: string, i: number) => (
                <li key={i}>{ind}</li>
              ))}
            </ul>
          </div>
        )}
        <div className="text-xs text-muted-foreground">
          Analyzed {analysisResult.alert_count} alert(s)
          {analysisResult.analysis.is_automated ? " - Automated attack detected" : ""}
        </div>
      </div>
    ) : (
      <div className="py-4 text-center text-muted-foreground">
        {analysisResult?.message || "No threats to analyze."}
      </div>
    )}
    <DialogFooter>
      <Button onClick={() => setAnalysisDialogOpen(false)}>Close</Button>
    </DialogFooter>
  </DialogContent>
</Dialog>
</div>
  );
}

```

[artifacts/mtos-crm/src/pages/template-assignments.tsx](#)

```
import { useState, useEffect, useMemo } from "react";
import { Card, CardContent, CardHeader, CardTitle } from "@components/ui/card";
import { Switch } from "@components/ui/switch";
import { Badge } from "@components/ui/badge";
import { useToast } from "@hooks/use-toast";
import { Grid3x3 } from "lucide-react";

interface Template { id: number; name: string; template_type: string; active: boolean }
interface Buyer { id: number; name: string }
interface Assignment {
  id: number;
  template_id: number;
  buyer_id: number | null;
  tort_type: string | null;
  enabled: boolean;
}

interface FormConfig { id: string; label: string; active: boolean }

export default function TemplateAssignmentsPage() {
  const { toast } = useToast();
  const [templates, setTemplates] = useState<Template[]>([]);
  const [buyers, setBuyers] = useState<Buyer[]>([]);
  const [torts, setTorts] = useState<FormConfig[]>([]);
  const [assignments, setAssignments] = useState<Assignment[]>([]);
  const [loading, setLoading] = useState(true);
  const [savingKey, setSavingKey] = useState<string | null>(null);
  const [selectedBuyer, setSelectedBuyer] = useState<number | null>(null);

  const load = async () => {
    setLoading(true);
    try {
      const [tRes, bRes, aRes, fRes] = await Promise.all([
        fetch("/api/document-templates"),
        fetch("/api/buyers"),
        fetch("/api/document-templates/assignments/all"),
        fetch("/api/forms"),
      ]);
      setTemplates(await tRes.json());
      setBuyers(await bRes.json());
      setAssignments(await aRes.json());
      const formsBody = await fRes.json().catch(() => []);
      setTorts(Array.isArray(formsBody) ? formsBody.filter((f: FormConfig) => f.active) : []);
    } finally {
      setLoading(false);
    }
  };

  useEffect(() => { load(); }, []);

  const lookup = useMemo(() => {
    const m = new Map<string, Assignment>();
    for (const a of assignments) {
      const k = `${a.template_id}|${a.buyer_id ?? "null"}|${a.tort_type ?? "null"}`;
      m.set(k, a);
    }
    return m;
  }, [assignments]);

  const resolved = (templateId: number, templateActive: boolean, tortId: string): boolean => {
    if (selectedBuyer) {
      const buyerKey = `${templateId}|${selectedBuyer}|${tortId}`;
      const buyerHit = lookup.get(buyerKey);
      if (buyerHit) return buyerHit.enabled;
    }
    const tortKey = `${templateId}|null|${tortId}`;
    const tortHit = lookup.get(tortKey);
    if (tortHit) return tortHit.enabled;
    return templateActive;
  };

  const toggle = async (templateId: number, tortId: string, current: boolean) => {
    const key = `${templateId}|${tortId}|${selectedBuyer ?? "global"}`;
    setSavingKey(key);
    try {
      const res = await fetch("/api/document-templates/assignments", {
        method: "POST",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify({
          template_id: templateId,
          buyer_id: selectedBuyer,
          tort_type: tortId,
          enabled: !current,
          send_order: 0,
        }),
      });
    } catch {
      const body = await res.json().catch(() => ({}));
      toast({ title: "Save failed", description: body.error || `HTTP ${res.status}`, variant: "destructive" });
      return;
    }
    await load();
  } finally {
    setSavingKey(null);
  }
};
```

```

return (
  <div className="p-6 space-y-6">
    <div>
      <h1 className="text-2xl font-bold flex items-center gap-2">
        <Grid3x3 className="h-6 w-6" /> Template Assignments
      </h1>
      <p className="text-sm text-muted-foreground">
        Toggle which templates are sent for each (buyer, tort) combination. Empty cell = uses template default.
      </p>
    </div>

    <Card>
      <CardHeader>
        <CardTitle>Scope</CardTitle>
      </CardHeader>
      <CardContent>
        <div className="flex items-center gap-3 flex-wrap">
          <button
            onClick={() => setSelectedBuyer(null)}
            className={`px-3 py-1.5 rounded-md text-sm border ${selectedBuyer === null ? "bg-primary text-primary-foreground" : "bg-background"}`}
          >
            Global (all buyers)
          </button>
          {buyers.map((b) => (
            <button
              key={b.id}
              onClick={() => setSelectedBuyer(b.id)}
              className={`px-3 py-1.5 rounded-md text-sm border ${selectedBuyer === b.id ? "bg-primary text-primary-foreground" : "bg-background"}`}
            >
              {b.name}
            </button>
          ))}
        </div>
        <p className="text-xs text-muted-foreground mt-2">
          {selectedBuyer
            ? "Buyer-specific overrides take precedence over global. Cells without a buyer override fall back to global."
            : "Global rules apply to all buyers unless that buyer overrides them."}
        </p>
      </CardContent>
    </Card>

    <Card>
      <CardHeader>
        <CardTitle>{selectedBuyer ? `Matrix – ${buyers.find((b) => b.id === selectedBuyer)?.name}` : "Matrix – Global"}</CardTitle>
      </CardHeader>
      <CardContent className="overflow-x-auto">
        {loading ? (
          <p className="text-muted-foreground">Loading...</p>
        ) : templates.length === 0 || torts.length === 0 ? (
          <p className="text-sm text-muted-foreground">
            {templates.length === 0 ? "No templates configured. " : ""}
            {torts.length === 0 ? "No active torts in Form Engine. " : ""}
          </p>
        ) : (
          <table className="w-full text-sm border-collapse">
            <thead>
              <tr>
                <th className="text-left p-2 font-medium border-b sticky left-0 bg-background">Template</th>
                {torts.map((t) => (
                  <th key={t.id} className="p-2 font-medium border-b text-center min-w-[120px]">{t.label}</th>
                ))}
              </tr>
            </thead>
            <tbody>
              {templates.map((tpl) => (
                <tr key={tpl.id} className="border-b">
                  <td className="p-2 sticky left-0 bg-background">
                    <div className="font-medium">{tpl.name}</div>
                    <div className="text-xs text-muted-foreground">
                      <Badge variant="outline" className="text-xs">{tpl.template_type}</Badge>
                    </div>
                  </td>
                  {torts.map((tort) => {
                    const enabled = resolved(tpl.id, tpl.active, tort.id);
                    const key = `${tpl.id}|${tort.id}|${selectedBuyer ?? "global"}`;
                    const cellKey = selectedBuyer
                      ? `${tpl.id}|${selectedBuyer}|${tort.id}`
                      : `${tpl.id}|null|${tort.id}`;
                    const hasOverride = lookup.has(cellKey);
                    return (
                      <td key={tort.id} className="p-2 text-center">
                        <div className="flex flex-col items-center gap-1">
                          <Switch
                            checked={enabled}
                            disabled={savingKey === key}
                            onChange={() => toggle(tpl.id, tort.id, enabled)}
                          />
                          <span className="text-[10px] text-muted-foreground">
                            {hasOverride ? "explicit" : "inherited"}
                          </span>
                        </div>
                      </td>
                    );
                  })}
                </tr>
              ))}
            </tbody>
          </table>
        )}
      </CardContent>
    </Card>
  </div>
);

```

```
        </tr>
      )})
    </tbody>
  </table>
  )}
</CardContent>
</Card>
</div>
  );
}
```

[artifacts/mtos-crm/src/pages/timeline.tsx](#)

```
import { useState } from "react";
import { Card, CardContent, CardHeader, CardTitle, CardDescription } from "@components/ui/card";
import { Button } from "@components/ui/button";
import { Badge } from "@components/ui/badge";
import { Input } from "@components/ui/input";
import { useToast } from "@hooks/use-toast";
import { Clock, Search, Calendar, Stethoscope, Shield, FileText, AlertTriangle, User } from "lucide-react";

interface TimelineEvent {
  date: string;
  event: string;
  category: string;
  source: string;
  details?: string;
}

interface TimelineData {
  lead_id: number;
  lead_name: string;
  tort_type: string;
  events: TimelineEvent[];
  summary: { total_events: number; categories: string[]; date_range: { start: string; end: string } | null };
}

const categoryConfig: Record<string, { icon: typeof Clock; color: string; bg: string }> = {
  personal: { icon: User, color: "text-blue-600", bg: "bg-blue-100 border-blue-300" },
  exposure: { icon: AlertTriangle, color: "text-red-600", bg: "bg-red-100 border-red-300" },
  diagnosis: { icon: Stethoscope, color: "text-purple-600", bg: "bg-purple-100 border-purple-300" },
  legal: { icon: Shield, color: "text-green-600", bg: "bg-green-100 border-green-300" },
  verification: { icon: Shield, color: "text-teal-600", bg: "bg-teal-100 border-teal-300" },
  compliance: { icon: Shield, color: "text-indigo-600", bg: "bg-indigo-100 border-indigo-300" },
  document: { icon: FileText, color: "text-orange-600", bg: "bg-orange-100 border-orange-300" },
};

export default function Timeline() {
  const { toast } = useToast();
  const [leadId, setLeadId] = useState("");
  const [timeline, setTimeline] = useState<TimelineData | null>(null);
  const [loading, setLoading] = useState(false);

  const loadTimeline = async () => {
    if (!leadId) return;
    setLoading(true);
    try {
      const res = await fetch(`/api/timeline/lead/${leadId}`);
      if (!res.ok) throw new Error("Failed");
      setTimeline(await res.json());
    } catch {
      toast({ title: "Failed to load timeline", variant: "destructive" });
    } finally {
      setLoading(false);
    }
  };

  return (
    <div className="space-y-6">
      <h1 className="text-3xl font-bold tracking-tight">Medical Timeline Builder</h1>
      <p className="text-muted-foreground">Generate exposure and medical timelines from lead data, OCR results, and NPI records.</p>

      <Card>
        <CardContent className="pt-6">
          <div className="flex gap-2 max-w-md">
            <Input placeholder="Enter Lead ID" value={leadId} onChange={(e) => setLeadId(e.target.value)} onKeyDown={(e) => e.key === "Enter" && loadTimeline()} />
            <Button onClick={loadTimeline} disabled={loading}>
              <Search className="h-4 w-4 mr-2" />
              {loading ? "Loading..." : "Build Timeline"}
            </Button>
          </div>
        </CardContent>
      </Card>

      {timeline && (
        <>
          <Card>
            <CardHeader>
              <CardTitle>{timeline.lead_name}</CardTitle>
              <CardDescription>
                {timeline.tort_type} | {timeline.summary.total_events} events
                {timeline.summary.date_range && ` | ${timeline.summary.date_range.start} to ${timeline.summary.date_range.end}`}
              </CardDescription>
            </CardHeader>
            <CardContent>
              <div className="flex gap-2 flex-wrap">
                {timeline.summary.categories.map((cat) => {
                  const cfg = categoryConfig[cat] || { color: "text-gray-600", bg: "bg-gray-100" };
                  return (
                    <Badge key={cat} variant="outline" className={cfg.bg}>
                      {cat.charAt(0).toUpperCase() + cat.slice(1)}
                    </Badge>
                  );
                })}
              </div>
            </CardContent>
          </Card>
        </>
      )}
    </div>
  );
}
```



```

</Card>
<Card>
  <CardHeader>
    <CardTitle>Timeline</CardTitle>
  </CardHeader>
  <CardContent>
    <div className="relative">
      <div className="absolute left-6 top-0 bottom-0 w-0.5 bg-gray-200" />
      <div className="space-y-6">
        {timeline.events.map((event, i) => {
border-gray-300" });
          const cfg = categoryConfig[event.category] || { icon: Clock, color: "text-gray-600", bg: "bg-gray-100
          const Icon = cfg.icon;
          return (
            <div key={i} className="relative flex items-start gap-4 pl-2">
              <div className={`relative z-10 flex h-9 w-9 items-center justify-center rounded-full border-2 bg-white
${cfg.bg}`}>
                <Icon className={`h-4 w-4 ${cfg.color}`} />
              </div>
              <div className="flex-1 min-w-0">
                <div className="flex items-center gap-2 flex-wrap">
                  <span className="font-mono text-sm text-muted-foreground">{event.date}</span>
                  <Badge variant="outline" className="text-xs">{event.category}</Badge>
                  <Badge variant="secondary" className="text-xs">{event.source}</Badge>
                </div>
                <div className="font-medium mt-1">{event.event}</div>
                {event.details && <div className="text-sm text-muted-foreground mt-0.5">{event.details}</div>}
              </div>
            </div>
          );
        })}
      </div>
    </div>
    {timeline.events.length === 0 && (
      <div className="text-center py-8 text-muted-foreground">
        <Calendar className="h-12 w-12 mx-auto mb-3" />
        <div>No timeline events found for this lead.</div>
      </div>
    )}
  </CardContent>
</Card>
</>
</div>
);
}

```

[artifacts/mtos-crm/src/pages/vendors.tsx](#)

```
import { useState } from "react";
import {
  useListVendors,
  useCreateVendor,
  useUpdateVendor,
  useDeleteVendor,
  getListVendorsQueryKey,
} from "@workspace/api-client-react";
import { useQueryClient } from "@tanstack/react-query";
import { Table, TableBody, TableCell, TableHead, TableHeader, TableRow } from "@components/ui/table";
import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { Label } from "@components/ui/label";
import { Badge } from "@components/ui/badge";
import { Select, SelectContent, SelectItem, SelectTrigger, SelectValue } from "@components/ui/select";
import { Dialog, DialogContent, DialogHeader, DialogTitle, DialogFooter, DialogDescription } from "@components/ui/dialog";
import { Skeleton } from "@components/ui/skeleton";
import { useToast } from "@hooks/use-toast";
import { Plus, Pencil, Trash2 } from "lucide-react";
import { format } from "date-fns";

interface VendorForm {
  name: string;
  contact_name: string;
  contact_email: string;
  contact_phone: string;
  type: string;
  status: string;
  notes: string;
}

const emptyForm: VendorForm = {
  name: "",
  contact_name: "",
  contact_email: "",
  contact_phone: "",
  type: "lead_gen",
  status: "active",
  notes: "",
};

export default function Vendors() {
  const { data: vendors, isLoading } = useListVendors({
    query: { queryKey: getListVendorsQueryKey() },
  });
  const queryClient = useQueryClient();
  const { toast } = useToast();
  const [dialogOpen, setDialogOpen] = useState(false);
  const [deleteDialogOpen, setDeleteDialogOpen] = useState(false);
  const [editingId, setEditingId] = useState<number | null>(null);
  const [deletingId, setDeletingId] = useState<number | null>(null);
  const [form, setForm] = useState<VendorForm>(emptyForm);

  const createMutation = useCreateVendor({
    mutation: {
      onSuccess: () => {
        queryClient.invalidateQueries({ queryKey: getListVendorsQueryKey() });
        setDialogOpen(false);
        setForm(emptyForm);
        toast({ title: "Vendor created" });
      },
    },
  });

  const updateMutation = useUpdateVendor({
    mutation: {
      onSuccess: () => {
        queryClient.invalidateQueries({ queryKey: getListVendorsQueryKey() });
        setDialogOpen(false);
        setEditingId(null);
        setForm(emptyForm);
        toast({ title: "Vendor updated" });
      },
    },
  });

  const deleteMutation = useDeleteVendor({
    mutation: {
      onSuccess: () => {
        queryClient.invalidateQueries({ queryKey: getListVendorsQueryKey() });
        setDeleteDialogOpen(false);
        setDeletingId(null);
        toast({ title: "Vendor deleted" });
      },
    },
  });

  const handleSubmit = () => {
    if (!form.name.trim()) return;
    const body = {
      name: form.name,
      contact_name: form.contact_name || undefined,
      contact_email: form.contact_email || undefined,
      contact_phone: form.contact_phone || undefined,
      type: form.type,
    };
  };
}
```



```

        onClick={() => { setDeletingId(vendor.id); setDeleteDialogOpen(true); }}
      >
        <Trash2 className="h-4 w-4 text-destructive" />
      </Button>
    </div>
  </TableCell>
</TableRow>
))
</TableBody>
</Table>
</div>

<Dialog open={dialogOpen} onOpenChange={setDialogOpen}>
  <DialogContent className="sm:max-w-lg">
    <DialogHeader>
      <DialogTitle>{editingId ? "Edit Vendor" : "Add Vendor"}</DialogTitle>
      <DialogDescription>
        {editingId ? "Update vendor details." : "Create a new vendor."}
      </DialogDescription>
    </DialogHeader>
    <div className="grid gap-4 py-4">
      <div className="grid gap-2">
        <Label htmlFor="name">Name </Label>
        <Input id="name" value={form.name} onChange={(e) => setForm({ ...form, name: e.target.value })} />
      </div>
      <div className="grid grid-cols-2 gap-4">
        <div className="grid gap-2">
          <Label htmlFor="contact_name">Contact Name</Label>
          <Input id="contact_name" value={form.contact_name} onChange={(e) => setForm({ ...form, contact_name:
e.target.value })} />
        </div>
        <div className="grid gap-2">
          <Label htmlFor="contact_email">Contact Email</Label>
          <Input id="contact_email" type="email" value={form.contact_email} onChange={(e) => setForm({ ...form,
contact_email: e.target.value })} />
        </div>
      </div>
      <div className="grid grid-cols-2 gap-4">
        <div className="grid gap-2">
          <Label htmlFor="contact_phone">Contact Phone</Label>
          <Input id="contact_phone" value={form.contact_phone} onChange={(e) => setForm({ ...form, contact_phone:
e.target.value })} />
        </div>
        <div className="grid gap-2">
          <Label htmlFor="type">Type</Label>
          <Select value={form.type} onValueChange={(val) => setForm({ ...form, type: val })}>
            <SelectTrigger>
              <SelectValue />
            </SelectTrigger>
            <SelectContent>
              <SelectItem value="lead_gen">Lead Gen</SelectItem>
              <SelectItem value="law_firm">Law Firm</SelectItem>
              <SelectItem value="marketing">Marketing</SelectItem>
              <SelectItem value="referral">Referral</SelectItem>
              <SelectItem value="other">Other</SelectItem>
            </SelectContent>
          </Select>
        </div>
      </div>
      <div className="grid gap-2">
        <Label htmlFor="status">Status</Label>
        <Select value={form.status} onValueChange={(val) => setForm({ ...form, status: val })}>
          <SelectTrigger>
            <SelectValue />
          </SelectTrigger>
          <SelectContent>
            <SelectItem value="active">Active</SelectItem>
            <SelectItem value="inactive">Inactive</SelectItem>
            <SelectItem value="suspended">Suspended</SelectItem>
          </SelectContent>
        </Select>
      </div>
      <div className="grid gap-2">
        <Label htmlFor="notes">Notes</Label>
        <Input id="notes" value={form.notes} onChange={(e) => setForm({ ...form, notes: e.target.value })} />
      </div>
    </DialogContent>
    <DialogFooter>
      <Button variant="outline" onClick={() => setDialogOpen(false)}>Cancel</Button>
      <Button onClick={handleSubmit} disabled={!form.name.trim()}>
        {editingId ? "Save Changes" : "Create Vendor"}
      </Button>
    </DialogFooter>
  </Dialog>

  <Dialog open={deleteDialogOpen} onOpenChange={setDeleteDialogOpen}>
    <DialogContent>
      <DialogHeader>
        <DialogTitle>Delete Vendor</DialogTitle>
        <DialogDescription>
          Are you sure you want to delete this vendor? This action cannot be undone.
        </DialogDescription>
      </DialogHeader>
      <DialogFooter>
        <Button variant="outline" onClick={() => setDeleteDialogOpen(false)}>Cancel</Button>
      </DialogFooter>
    </DialogContent>
  </Dialog>

```

```
        <Button
          variant="destructive"
          onClick={() => deletingId && deleteMutation.mutate({ id: deletingId })}
        >
          Delete
        </Button>
      </DialogFooter>
    </DialogContent>
  </Dialog>
</div>
);
}
```

[artifacts/mtos-crm/src/pages/workflow-settings.tsx](#)

```
import { useState, useEffect } from "react";
import { Card, CardContent, CardHeader, CardTitle, CardDescription } from "@components/ui/card";
import { Button } from "@components/ui/button";
import { Input } from "@components/ui/input";
import { Label } from "@components/ui/label";
import { Switch } from "@components/ui/switch";
import { Badge } from "@components/ui/badge";
import { Textarea } from "@components/ui/textarea";
import { useToast } from "@hooks/use-toast";
import { Settings, Save, AlertTriangle } from "lucide-react";

interface ProviderOption { id: number; name: string; provider: string; adapter_implemented: boolean }
interface ProviderOptions { esign: ProviderOption[]; fax: ProviderOption[]; email: ProviderOption[] }

interface Settings {
  scope: string;
  esign_provider_integration_id: number | null;
  fax_provider_integration_id: number | null;
  email_provider_integration_id: number | null;
  default_email_from_name: string | null;
  default_email_from_address: string | null;
  default_fax_from_number: string | null;
  auto_send_on_approval: boolean;
  auto_fax_doctor_on_signed_hipaa: boolean;
  esign_resend_after_days: number;
  esign_expire_after_days: number;
  max_send_retries: number;
  notify_on_failure_email: string | null;
  notes: string | null;
}

const DEFAULTS: Settings = {
  scope: "global",
  esign_provider_integration_id: null,
  fax_provider_integration_id: null,
  email_provider_integration_id: null,
  default_email_from_name: null,
  default_email_from_address: null,
  default_fax_from_number: null,
  auto_send_on_approval: true,
  auto_fax_doctor_on_signed_hipaa: true,
  esign_resend_after_days: 3,
  esign_expire_after_days: 30,
  max_send_retries: 3,
  notify_on_failure_email: null,
  notes: null,
};

export default function WorkflowSettingsPage() {
  const { toast } = useToast();
  const [s, setS] = useState<Settings>(DEFAULTS);
  const [options, setOptions] = useState<ProviderOptions>({ esign: [], fax: [], email: [] });
  const [loading, setLoading] = useState(true);
  const [saving, setSaving] = useState(false);

  useEffect(() => {
    (async () => {
      try {
        const [sRes, oRes] = await Promise.all([
          fetch("/api/workflow-settings/global"),
          fetch("/api/workflow-settings/_options/providers"),
        ]);
        if (sRes.ok) {
          const row = await sRes.json();
          setS({ ...DEFAULTS, ...row });
        }
        if (oRes.ok) setOptions(await oRes.json());
      } finally {
        setLoading(false);
      }
    })();
  }, []);

  const save = async () => {
    setSaving(true);
    try {
      const res = await fetch("/api/workflow-settings", {
        method: "PUT",
        headers: { "Content-Type": "application/json" },
        body: JSON.stringify(s),
      });
      if (!res.ok) {
        const body = await res.json().catch(() => ({}));
        toast({ title: "Save failed", description: body.error || `HTTP ${res.status}`, variant: "destructive" });
        return;
      }
      toast({ title: "Workflow settings saved" });
    } finally {
      setSaving(false);
    }
  };

  const ProviderPicker = ({
    label, value, options: opts, onChange, hint,
  }: { label: string; value: number | null; options: ProviderOption[]; onChange: (v: number | null) => void; hint?: string })
```

```

=> (
  <div>
    <Label>{label}</Label>
    <select
      className="w-full h-10 rounded-md border border-input bg-background px-3 text-sm"
      value={value ?? ""}
      onChange={(e) => onChange(e.target.value ? Number(e.target.value) : null)}
    >
      <option value="">- none configured </option>
      {opts.map((o) => (
        <option key={o.id} value={o.id}>
          {o.name} ({o.provider}){!o.adapter_implemented ? " - NO ADAPTER" : ""}
        </option>
      ))}
    </select>
    {hint && <p className="text-xs text-muted-foreground mt-1">{hint}</p>}
    {opts.length === 0 && (
      <div className="flex items-center gap-1 text-xs text-amber-600 mt-1">
        <AlertTriangle className="h-3 w-3" /> No active integrations of this type. Add one on the Integrations page.
      </div>
    )}
  </div>
);

if (loading) return <div className="p-6">Loading...</div>;

return (
  <div className="p-6 space-y-6 max-w-4xl">
    <div>
      <h1 className="text-2xl font-bold flex items-center gap-2">
        <Settings className="h-6 w-6" /> Workflow Settings
      </h1>
      <p className="text-sm text-muted-foreground">
        Global defaults for the document automation workflow. Per-buyer overrides live on the Buyers page.
      </p>
    </div>

    <Card>
      <CardHeader>
        <CardTitle>Provider Selection</CardTitle>
        <CardDescription>Pick which active integration to use for each step. These are pulled from your Integrations page.</CardDescription>
      </CardHeader>
      <CardContent className="space-y-4">
        <ProviderPicker
          label="E-Signature Provider"
          value={s.esign_provider_integration_id}
          options={options.esign}
          onChange={(v) => setS({ ...s, esign_provider_integration_id: v })}
          hint="Used to send retainer / HIPAA / affidavit packets to leads on approval."
        />
        <ProviderPicker
          label="Fax Provider"
          value={s.fax_provider_integration_id}
          options={options.fax}
          onChange={(v) => setS({ ...s, fax_provider_integration_id: v })}
          hint="Used to send medical records requests to the lead's doctor after the HIPAA is signed."
        />
        <ProviderPicker
          label="Email Provider"
          value={s.email_provider_integration_id}
          options={options.email}
          onChange={(v) => setS({ ...s, email_provider_integration_id: v })}
          hint="Used for paralegal failure notifications and any non-e-signature email."
        />
      </CardContent>
    </Card>

    <Card>
      <CardHeader>
        <CardTitle>Defaults</CardTitle>
        <CardDescription>Used when a buyer does not override.</CardDescription>
      </CardHeader>
      <CardContent className="grid grid-cols-2 gap-3">
        <div>
          <Label>Email From Name</Label>
          <Input value={s.default_email_from_name ?? ""} onChange={(e) => setS({ ...s, default_email_from_name:
e.target.value || null })} />
        </div>
        <div>
          <Label>Email From Address</Label>
          <Input value={s.default_email_from_address ?? ""} onChange={(e) => setS({ ...s, default_email_from_address:
e.target.value || null })} />
        </div>
        <div>
          <Label>Fax From Number</Label>
          <Input value={s.default_fax_from_number ?? ""} onChange={(e) => setS({ ...s, default_fax_from_number:
e.target.value || null })} />
        </div>
        <div>
          <Label>Notify on Failure (email)</Label>
          <Input value={s.notify_on_failure_email ?? ""} onChange={(e) => setS({ ...s, notify_on_failure_email:
e.target.value || null })} />
        </div>
      </CardContent>
    </Card>
  </div>
);

```

```

<Card>
  <CardHeader>
    <CardTitle>Automation</CardTitle>
  </CardHeader>
  <CardContent className="space-y-4">
    <div className="flex items-center justify-between rounded-md border p-3">
      <div>
        <Label>Auto-send packets on lead approval</Label>
        <p className="text-xs text-muted-foreground">When a lead is qualified, automatically send all active templates.</p>
      </div>
      <Switch checked={s.auto_send_on_approval} onCheckedChange={(v) => setS({ ...s, auto_send_on_approval: v })} />
    </div>
    <div className="flex items-center justify-between rounded-md border p-3">
      <div>
        <Label>Auto-fax doctor when HIPAA is signed</Label>
        <p className="text-xs text-muted-foreground">Once a HIPAA template returns signed, fax the medical records request to the lead's doctor on file.</p>
      </div>
      <Switch checked={s.auto_fax_doctor_on_signed_hipaa} onCheckedChange={(v) => setS({ ...s, auto_fax_doctor_on_signed_hipaa: v })} />
    </div>

    <div className="grid grid-cols-3 gap-3">
      <div>
        <Label>Resend after (days)</Label>
        <Input type="number" min={0} value={s.esign_resend_after_days} onChange={(e) => setS({ ...s, esign_resend_after_days: Number(e.target.value) })} />
      </div>
      <div>
        <Label>Expire after (days)</Label>
        <Input type="number" min={1} value={s.esign_expire_after_days} onChange={(e) => setS({ ...s, esign_expire_after_days: Number(e.target.value) })} />
      </div>
      <div>
        <Label>Max retries</Label>
        <Input type="number" min={0} max={10} value={s.max_send_retries} onChange={(e) => setS({ ...s, max_send_retries: Number(e.target.value) })} />
      </div>
    </div>

    <div>
      <Label>Notes</Label>
      <Textarea rows={2} value={s.notes ?? ""} onChange={(e) => setS({ ...s, notes: e.target.value || null })} />
    </div>
  </CardContent>
</Card>

  <div className="flex justify-between items-center">
    <Badge variant="outline">Scope: global</Badge>
    <Button onClick={save} disabled={saving}>
      <Save className="mr-2 h-4 w-4" />{saving ? "Saving..." : "Save Settings"}
    </Button>
  </div>
);
}

```


artifacts/mtos-crm/tsconfig.json

```
{
  "extends": "../../tsconfig.base.json",
  "include": ["src/**/*.ts"],
  "exclude": ["node_modules", "build", "dist", "**/*.test.ts"],
  "compilerOptions": {
    "noEmit": true,
    "jsx": "preserve",
    "lib": ["esnext", "dom", "dom.iterable"],
    "resolveJsonModule": true,
    "allowImportingTsExtensions": true,
    "moduleResolution": "bundler",
    "types": ["node", "vite/client"],
    "paths": {
      "@/*": [".src/*"]
    }
  },
  "references": [
    {
      "path": "../../lib/api-client-react"
    }
  ]
}
```

[artifacts/mtos-crm/vite.config.ts](#)

```
import { defineConfig } from "vite";
import react from "@vitejs/plugin-react";
import tailwindcss from "@tailwindcss/vite";
import path from "path";
import runtimeErrorOverlay from "@replit/vite-plugin-runtime-error-modal";

const rawPort = process.env.PORT;

if (!rawPort) {
  throw new Error(
    "PORT environment variable is required but was not provided.",
  );
}

const port = Number(rawPort);

if (Number.isNaN(port) || port <= 0) {
  throw new Error(`Invalid PORT value: "${rawPort}"`);
}

const basePath = process.env.BASE_PATH;

if (!basePath) {
  throw new Error(
    "BASE_PATH environment variable is required but was not provided.",
  );
}

export default defineConfig({
  base: basePath,
  plugins: [
    react(),
    tailwindcss(),
    runtimeErrorOverlay(),
    ...(process.env.NODE_ENV !== "production" &&
      process.env.REPL_ID !== undefined
        ? [
            await import("@replit/vite-plugin-cartographer").then((m) =>
              m.cartographer({
                root: path.resolve(import.meta.dirname, ".."),
              })
            ),
            await import("@replit/vite-plugin-dev-banner").then((m) =>
              m.devBanner(),
            ),
          ]
        : []),
  ],
  resolve: {
    alias: {
      "@": path.resolve(import.meta.dirname, "src"),
      "@assets": path.resolve(import.meta.dirname, "..", "..", "attached_assets"),
    },
    dedupe: ["react", "react-dom"],
  },
  root: path.resolve(import.meta.dirname),
  build: {
    outDir: path.resolve(import.meta.dirname, "dist/public"),
    emptyOutDir: true,
  },
  server: {
    port,
    host: "0.0.0.0",
    allowedHosts: true,
    fs: {
      strict: true,
      deny: ["**/*.css"],
    },
  },
  preview: {
    port,
    host: "0.0.0.0",
    allowedHosts: true,
  },
});
```

lib/api-client-react/package.json

```
{
  "name": "@workspace/api-client-react",
  "version": "0.0.0",
  "private": true,
  "type": "module",
  "exports": {
    ".": "./src/index.ts"
  },
  "dependencies": {
    "@tanstack/react-query": "catalog:"
  },
  "peerDependencies": {
    "react": ">=18"
  }
}
```

[lib/api-client-react/src/custom-fetch.ts](#)

```
export type CustomFetchOptions = RequestInit & {
  responseType?: "json" | "text" | "blob" | "auto";
};

export type ErrorType<T = unknown> = ApiError<T>;

export type BodyType<T> = T;

export type AuthTokenGetter = () => Promise<string | null> | string | null;

const NO_BODY_STATUS = new Set([204, 205, 304]);
const DEFAULT_JSON_ACCEPT = "application/json, application/problem+json";

// -----
// Module-level configuration
// -----

let _baseUrl: string | null = null;
let _authTokenGetter: AuthTokenGetter | null = null;

/**
 * Set a base URL that is prepended to every relative request URL
 * (i.e. paths that start with `/`).
 *
 * Useful for Expo bundles that need to call a remote API server.
 * Pass `null` to clear the base URL.
 */
export function setBaseUrl(url: string | null): void {
  _baseUrl = url ? url.replace(/\/+$/, "") : null;
}

/**
 * Register a getter that supplies a bearer auth token. Before every fetch
 * the getter is invoked; when it returns a non-null string, an
 * `Authorization: Bearer <token>` header is attached to the request.
 *
 * Useful for Expo bundles making token-gated API calls.
 * Pass `null` to clear the getter.
 *
 * NOTE: This function should never be used in web applications where session
 * token cookies are automatically associated with API calls by the browser.
 */
export function setAuthTokenGetter(getter: AuthTokenGetter | null): void {
  _authTokenGetter = getter;
}

function isRequest(input: RequestInfo | URL): input is Request {
  return typeof Request !== "undefined" && input instanceof Request;
}

function resolveMethod(input: RequestInfo | URL, explicitMethod?: string): string {
  if (explicitMethod) return explicitMethod.toUpperCase();
  if (isRequest(input)) return input.method.toUpperCase();
  return "GET";
}

// Use loose check for URL – some runtimes (e.g. React Native) polyfill URL
// differently, so `instanceof URL` can fail.
function isUrl(input: RequestInfo | URL): input is URL {
  return typeof URL !== "undefined" && input instanceof URL;
}

function applyBaseUrl(input: RequestInfo | URL): RequestInfo | URL {
  if (!_baseUrl) return input;
  const url = resolveUrl(input);
  // Only prepend to relative paths (starting with /)
  if (!url.startsWith("/")) return input;

  const absolute = `${_baseUrl}${url}`;
  if (typeof input === "string") return absolute;
  if (isUrl(input)) return new URL(absolute);
  return new Request(absolute, input as Request);
}

function resolveUrl(input: RequestInfo | URL): string {
  if (typeof input === "string") return input;
  if (isUrl(input)) return input.toString();
  return input.url;
}

function mergeHeaders(...sources: Array<HeadersInit | undefined>): Headers {
  const headers = new Headers();

  for (const source of sources) {
    if (!source) continue;
    new Headers(source).forEach((value, key) => {
      headers.set(key, value);
    });
  }

  return headers;
}

function getMediaType(headers: Headers): string | null {
  const value = headers.get("content-type");
```

```

    return value ? value.split("; ", 1)[0].trim().toLowerCase() : null;
}

function isJsonMediaType(mediaType: string | null): boolean {
    return mediaType === "application/json" || Boolean(mediaType?.endsWith("+json"));
}

function isTextMediaType(mediaType: string | null): boolean {
    return Boolean(
        mediaType &&
        (mediaType.startsWith("text/") ||
            mediaType === "application/xml" ||
            mediaType === "text/xml" ||
            mediaType.endsWith("+xml") ||
            mediaType === "application/x-www-form-urlencoded"),
    );
}

// Use strict equality: in browsers, `response.body` is `null` when the
// response genuinely has no content. In React Native, `response.body` is
// always `undefined` because the ReadableStream API is not implemented –
// even when the response carries a full payload readable via `.text()` or
// `.json()`. Loose equality (`== null`) matches both `null` and `undefined`,
// which causes every React Native response to be treated as empty.
function hasNoBody(response: Response, method: string): boolean {
    if (method === "HEAD") return true;
    if (NO_BODY_STATUS.has(response.status)) return true;
    if (response.headers.get("content-length") === "0") return true;
    if (response.body === null) return true;
    return false;
}

function stripBom(text: string): string {
    return text.charCodeAt(0) === 0xfeff ? text.slice(1) : text;
}

function looksLikeJson(text: string): boolean {
    const trimmed = text.trimStart();
    return trimmed.startsWith("{") || trimmed.startsWith("[");
}

function getStringField(value: unknown, key: string): string | undefined {
    if (!value || typeof value !== "object") return undefined;

    const candidate = (value as Record<string, unknown>)[key];
    if (typeof candidate !== "string") return undefined;

    const trimmed = candidate.trim();
    return trimmed === "" ? undefined : trimmed;
}

function truncate(text: string, maxLength = 300): string {
    return text.length > maxLength ? `${text.slice(0, maxLength - 1)}...` : text;
}

function buildErrorMessage(response: Response, data: unknown): string {
    const prefix = `HTTP ${response.status} ${response.statusText}`;

    if (typeof data === "string") {
        const text = data.trim();
        return text ? `${prefix}: ${truncate(text)}` : prefix;
    }

    const title = getStringField(data, "title");
    const detail = getStringField(data, "detail");
    const message =
        getStringField(data, "message") ??
        getStringField(data, "error_description") ??
        getStringField(data, "error");

    if (title && detail) return `${prefix}: ${title} – ${detail}`;
    if (detail) return `${prefix}: ${detail}`;
    if (message) return `${prefix}: ${message}`;
    if (title) return `${prefix}: ${title}`;

    return prefix;
}

export class ApiError<T = unknown> extends Error {
    readonly name = "ApiError";
    readonly status: number;
    readonly statusText: string;
    readonly data: T | null;
    readonly headers: Headers;
    readonly response: Response;
    readonly method: string;
    readonly url: string;

    constructor(
        response: Response,
        data: T | null,
        requestInfo: { method: string; url: string },
    ) {
        super(buildErrorMessage(response, data));
        Object.setPrototypeOf(this, new.target.prototype);

        this.status = response.status;
    }
}

```

```

    this.statusText = response.statusText;
    this.data = data;
    this.headers = response.headers;
    this.response = response;
    this.method = requestInfo.method;
    this.url = response.url || requestInfo.url;
  }
}

export class ResponseParseError extends Error {
  readonly name = "ResponseParseError";
  readonly status: number;
  readonly statusText: string;
  readonly headers: Headers;
  readonly response: Response;
  readonly method: string;
  readonly url: string;
  readonly rawBody: string;
  readonly cause: unknown;

  constructor(
    response: Response,
    rawBody: string,
    cause: unknown,
    requestInfo: { method: string; url: string },
  ) {
    super(
      `Failed to parse response from ${requestInfo.method} ${response.url || requestInfo.url} ` +
        `(${response.status} ${response.statusText}) as JSON`,
    );
    Object.setPrototypeOf(this, new.target.prototype);

    this.status = response.status;
    this.statusText = response.statusText;
    this.headers = response.headers;
    this.response = response;
    this.method = requestInfo.method;
    this.url = response.url || requestInfo.url;
    this.rawBody = rawBody;
    this.cause = cause;
  }
}

async function parseJsonBody(
  response: Response,
  requestInfo: { method: string; url: string },
): Promise<unknown> {
  const raw = await response.text();
  const normalized = stripBom(raw);

  if (normalized.trim() === "") {
    return null;
  }

  try {
    return JSON.parse(normalized);
  } catch (cause) {
    throw new ResponseParseError(response, raw, cause, requestInfo);
  }
}

async function parseErrorBody(response: Response, method: string): Promise<unknown> {
  if (hasNoBody(response, method)) {
    return null;
  }

  const mediaType = getMediaType(response.headers);

  // Fall back to text when blob() is unavailable (e.g. some React Native builds).
  if (mediaType && !isJsonMediaType(mediaType) && !isTextMediaType(mediaType)) {
    return typeof response.blob === "function" ? response.blob() : response.text();
  }

  const raw = await response.text();
  const normalized = stripBom(raw);
  const trimmed = normalized.trim();

  if (trimmed === "") {
    return null;
  }

  if (isJsonMediaType(mediaType) || looksLikeJson(normalized)) {
    try {
      return JSON.parse(normalized);
    } catch {
      return raw;
    }
  }

  return raw;
}

function inferResponseType(response: Response): "json" | "text" | "blob" {
  const mediaType = getMediaType(response.headers);

  if (isJsonMediaType(mediaType)) return "json";
  if (isTextMediaType(mediaType) || mediaType == null) return "text";
}

```

```

    return "blob";
}

async function parseSuccessBody(
  response: Response,
  responseType: "json" | "text" | "blob" | "auto",
  requestInfo: { method: string; url: string },
): Promise<unknown> {
  if (hasNoBody(response, requestInfo.method)) {
    return null;
  }

  const effectiveType =
    responseType === "auto" ? inferResponseType(response) : responseType;

  switch (effectiveType) {
    case "json":
      return parseJsonBody(response, requestInfo);

    case "text": {
      const text = await response.text();
      return text === "" ? null : text;
    }

    case "blob":
      if (typeof response.blob !== "function") {
        throw new TypeError(
          "Blob responses are not supported in this runtime. " +
            "Use responseType \"json\" or \"text\" instead.",
        );
      }
      return response.blob();
  }
}

export async function customFetch<T = unknown>(
  input: RequestInfo | URL,
  options: CustomFetchOptions = {},
): Promise<T> {
  input = applyBaseUrl(input);
  const { responseType = "auto", headers: headersInit, ...init } = options;

  const method = resolveMethod(input, init.method);

  if (init.body !== null && (method === "GET" || method === "HEAD")) {
    throw new TypeError(`customFetch: ${method} requests cannot have a body.`);
  }

  const headers = mergeHeaders(isRequest(input) ? input.headers : undefined, headersInit);

  if (
    typeof init.body === "string" &&
    !headers.has("content-type") &&
    looksLikeJson(init.body)
  ) {
    headers.set("content-type", "application/json");
  }

  if (responseType === "json" && !headers.has("accept")) {
    headers.set("accept", DEFAULT_JSON_ACCEPT);
  }

  // Attach bearer token when an auth getter is configured and no
  // Authorization header has been explicitly provided.
  if (_authTokenGetter && !headers.has("authorization")) {
    const token = await _authTokenGetter();
    if (token) {
      headers.set("authorization", `Bearer ${token}`);
    }
  }

  const requestInfo = { method, url: resolveUrl(input) };
  const response = await fetch(input, { ...init, method, headers });

  if (!response.ok) {
    const errorData = await parseErrorBody(response, method);
    throw new ApiError(response, errorData, requestInfo);
  }

  return (await parseSuccessBody(response, responseType, requestInfo)) as T;
}

```

[lib/api-client-react/src/generated/api.schemas.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
export interface HealthStatus {
  status: string;
}

export interface ErrorResponse {
  error: string;
}

export type LeadStatus = (typeof LeadStatus)[keyof typeof LeadStatus];

export const LeadStatus = {
  new: "new",
  qualified: "qualified",
  signed: "signed",
  rejected: "rejected",
  review_required: "review_required",
} as const;

export type LeadRouting = (typeof LeadRouting)[keyof typeof LeadRouting] | null;

export const LeadRouting = {
  hot: "hot",
  warm: "warm",
  cold: "cold",
} as const;

export interface Lead {
  id: number;
  name: string;
  email?: string;
  phone?: string;
  tort_type: string;
  first_name?: string | null;
  last_name?: string | null;
  date_of_birth?: string | null;
  street_address?: string | null;
  city?: string | null;
  state?: string | null;
  zip?: string | null;
  phone_primary?: string | null;
  last_4_ssn?: string | null;
  diagnosis?: string | null;
  diagnosis_date?: string | null;
  physician_first_name?: string | null;
  physician_last_name?: string | null;
  physician_full_address?: string | null;
  physician_contact_info?: string | null;
  hospital_name?: string | null;
  hospital_fax?: string | null;
  hospital_contact_info?: string | null;
  tcpa_consent?: boolean | null;
  trustedform_cert_url?: string | null;
  trustedform_ip?: string | null;
  trustedform_user_agent?: string | null;
  trustedform_timestamp?: string | null;
  email_validation_status?: string | null;
  address_validation_status?: string | null;
  background_check_status?: string | null;
  background_check_data?: string | null;
  exposure_start?: string | null;
  exposure_end?: string | null;
  diagnosis_confirmed: boolean;
  diagnosis_type?: string | null;
  was_at_location: boolean;
  location_name?: string | null;
  status: LeadStatus;
  rejection_reason?: string | null;
  notes?: string | null;
  ad_spend?: number | null;
  source?: string | null;
  assigned_to?: number | null;
  routing?: LeadRouting;
  vendor_id?: number | null;
  law_firm?: string | null;
  client_id?: string | null;
  created_at: string;
  updated_at: string;
}

export interface CreateLeadBody {
  name: string;
  email?: string;
  phone?: string;
  tort_type: string;
  first_name: string;
  last_name: string;
  date_of_birth: string;
  street_address: string;
```



```

city: string;
state: string;
zip: string;
phone_primary: string;
last_4_ssn: string;
diagnosis: string;
diagnosis_date: string;
physician_first_name: string;
physician_last_name: string;
physician_full_address: string;
physician_contact_info: string;
hospital_name: string;
hospital_fax: string;
hospital_contact_info: string;
exposure_start?: string | null;
exposure_end?: string | null;
diagnosis_confirmed: boolean;
diagnosis_type?: string | null;
was_at_location: boolean;
location_name?: string | null;
notes?: string | null;
ad_spend?: number | null;
source?: string | null;
vendor_id?: number | null;
law_firm?: string | null;
client_id?: string | null;
}

export type UpdateLeadBodyStatus =
  (typeof UpdateLeadBodyStatus)[keyof typeof UpdateLeadBodyStatus];

export const UpdateLeadBodyStatus = {
  new: "new",
  qualified: "qualified",
  signed: "signed",
  rejected: "rejected",
  review_required: "review_required",
} as const;

export interface UpdateLeadBody {
  name?: string;
  email?: string;
  phone?: string;
  tort_type?: string;
  first_name?: string;
  last_name?: string;
  date_of_birth?: string;
  street_address?: string;
  city?: string;
  state?: string;
  zip?: string;
  phone_primary?: string;
  last_4_ssn?: string;
  diagnosis?: string;
  diagnosis_date?: string;
  physician_first_name?: string;
  physician_last_name?: string;
  physician_full_address?: string;
  physician_contact_info?: string;
  hospital_name?: string;
  hospital_fax?: string;
  hospital_contact_info?: string;
  tcpa_consent?: boolean;
  trustedform_cert_url?: string;
  exposure_start?: string | null;
  exposure_end?: string | null;
  diagnosis_confirmed?: boolean;
  diagnosis_type?: string | null;
  was_at_location?: boolean;
  location_name?: string | null;
  status?: UpdateLeadBodyStatus;
  rejection_reason?: string | null;
  notes?: string | null;
  ad_spend?: number | null;
  source?: string | null;
  vendor_id?: number | null;
  law_firm?: string | null;
  client_id?: string | null;
}

export interface Vendor {
  id: number;
  name: string;
  contact_name?: string | null;
  contact_email?: string | null;
  contact_phone?: string | null;
  type: string;
  status: string;
  notes?: string | null;
  created_at: string;
  updated_at: string;
}

export interface CreateVendorBody {
  name: string;
  contact_name?: string | null;
  contact_email?: string | null;

```

```

    contact_phone?: string | null;
    type?: string;
    status?: string;
    notes?: string | null;
}

export interface UpdateVendorBody {
    name?: string;
    contact_name?: string | null;
    contact_email?: string | null;
    contact_phone?: string | null;
    type?: string;
    status?: string;
    notes?: string | null;
}

export type QualificationResultStatus =
    (typeof QualificationResultStatus)[keyof typeof QualificationResultStatus];

export const QualificationResultStatus = {
    qualified: "qualified",
    rejected: "rejected",
} as const;

export interface QualificationResult {
    lead_id: number;
    qualified: boolean;
    status: QualificationResultStatus;
    reason: string;
    gates_passed: string[];
    gates_failed: string[];
}

export type DocumentDocumentType =
    (typeof DocumentDocumentType)[keyof typeof DocumentDocumentType];

export const DocumentDocumentType = {
    retainer: "retainer",
    medical_record: "medical_record",
    intake_form: "intake_form",
    other: "other",
} as const;

export interface Document {
    id: number;
    lead_id: number;
    document_type: DocumentDocumentType;
    file_name: string;
    file_url?: string | null;
    signed: boolean;
    signed_at?: string | null;
    notes?: string | null;
    created_at: string;
}

export type CreateDocumentBodyDocumentType =
    (typeof CreateDocumentBodyDocumentType)[keyof typeof CreateDocumentBodyDocumentType];

export const CreateDocumentBodyDocumentType = {
    retainer: "retainer",
    medical_record: "medical_record",
    intake_form: "intake_form",
    other: "other",
} as const;

export interface CreateDocumentBody {
    lead_id: number;
    document_type: CreateDocumentBodyDocumentType;
    file_name: string;
    file_url?: string | null;
    signed: boolean;
    signed_at?: string | null;
    notes?: string | null;
}

export type UpdateDocumentBodyDocumentType =
    (typeof UpdateDocumentBodyDocumentType)[keyof typeof UpdateDocumentBodyDocumentType];

export const UpdateDocumentBodyDocumentType = {
    retainer: "retainer",
    medical_record: "medical_record",
    intake_form: "intake_form",
    other: "other",
} as const;

export interface UpdateDocumentBody {
    document_type?: UpdateDocumentBodyDocumentType;
    file_name?: string;
    file_url?: string | null;
    signed?: boolean;
    signed_at?: string | null;
    notes?: string | null;
}

export interface DashboardStats {
    total_leads: number;
    qualified_leads: number;
}

```

```

    signed_retainers: number;
    rejected_leads: number;
    new_leads: number;
    total_ad_spend: number;
    cpsr?: number | null;
    qualification_rate: number;
    conversion_rate: number;
    leads_today: number;
    leads_this_week: number;
}

export type PipelineBreakdownByStatusItem = {
    status: string;
    count: number;
};

export type PipelineBreakdownByTortTypeItem = {
    tort_type: string;
    count: number;
    qualified: number;
    signed: number;
};

export interface PipelineBreakdown {
    by_status: PipelineBreakdownByStatusItem[];
    by_tort_type: PipelineBreakdownByTortTypeItem[];
}

export interface ActivityItem {
    id: number;
    lead_id: number;
    lead_name: string;
    event_type: string;
    description: string;
    occurred_at: string;
    tort_type: string;
}

export type CaseData = { [key: string]: unknown };

export type CaseStatus = (typeof CaseStatus)[keyof typeof CaseStatus];

export const CaseStatus = {
    open: "open",
    processing: "processing",
    analyzed: "analyzed",
    failed: "failed",
} as const;

export interface Case {
    id: string;
    data?: CaseData;
    status: CaseStatus;
    created_at: string;
    updated_at: string;
}

export interface CaseDocument {
    id: number;
    case_id: string;
    path: string;
    file_hash?: string | null;
    file_name: string;
    content_type?: string | null;
    created_at: string;
}

export type CaseAnalysisFeatures = { [key: string]: unknown } | null;

export interface CaseAnalysis {
    id: number;
    case_id: string;
    features?: CaseAnalysisFeatures;
    score?: number | null;
    ai_model?: string | null;
    raw_text_length?: number | null;
    created_at: string;
}

export type AuditLogEntryDetails = { [key: string]: unknown } | null;

export interface AuditLogEntry {
    id: number;
    entity_type: string;
    entity_id: string;
    action: string;
    details?: AuditLogEntryDetails;
    ip_address?: string | null;
    occurred_at: string;
}

export interface CaseDetail {
    case: Case;
    documents: CaseDocument[];
    analyses: CaseAnalysis[];
    audit_trail: AuditLogEntry[];
}

```

```

export interface CreateCaseResponse {
  case_id: string;
  status: string;
  job_id: number;
}

export interface UploadCaseFileBody {
  file_name: string;
  content: string;
  content_type?: string | null;
}

export interface QueueStats {
  [key: string]: number;
}

export type FaxResultStatus =
  (typeof FaxResultStatus)[keyof typeof FaxResultStatus];

export const FaxResultStatus = {
  pending: "pending",
  processing: "processing",
  done: "done",
  failed: "failed",
} as const;

export interface FaxResult {
  id: number;
  job_id?: number | null;
  source_file: string;
  vault_path: string;
  rx_number: string;
  drug_name: string;
  fill_date: string;
  quantity: string;
  confidence: number;
  raw_text: string;
  status: FaxResultStatus;
  error?: string | null;
  created_at: string;
  processed_at?: string | null;
}

export interface UploadFaxBody {
  file_name: string;
  image_base64: string;
  mime_type?: string | null;
}

export interface UploadFaxResponse {
  fax_result_id: number;
  status: string;
  job_id: number;
  vault_path: string;
}

export type ReviewQueueItemDetails = { [key: string]: unknown };

export interface ReviewQueueItem {
  id: number;
  entity_type: string;
  entity_id: string;
  conflict_type: string;
  severity: string;
  failsafe_mode: string;
  source_module: string;
  summary: string;
  details?: ReviewQueueItemDetails;
  resolution: string;
  resolution_notes?: string | null;
  resolved_by?: string | null;
  resolved_at?: string | null;
  retry_count: number;
  created_at: string;
}

export type ReviewQueueStatsByResolutionItem = {
  resolution?: string;
  count?: number;
};

export type ReviewQueueStatsByConflictTypeItem = {
  conflict_type?: string;
  count?: number;
};

export type ReviewQueueStatsBySeverityItem = {
  severity?: string;
  count?: number;
};

export type ReviewQueueStatsByFailsafeModeItem = {
  failsafe_mode?: string;
  count?: number;
};

```

```

export interface ReviewQueueStats {
  total_pending: number;
  by_resolution: ReviewQueueStatsByResolutionItem[];
  by_conflict_type: ReviewQueueStatsByConflictTypeItem[];
  by_severity: ReviewQueueStatsBySeverityItem[];
  by_failsafe_mode: ReviewQueueStatsByFailsafeModeItem[];
}

export type ResolveReviewBodyResolution =
  (typeof ResolveReviewBodyResolution)[keyof typeof ResolveReviewBodyResolution];

export const ResolveReviewBodyResolution = {
  accepted: "accepted",
  rejected: "rejected",
  escalated: "escalated",
} as const;

export interface ResolveReviewBody {
  resolution: ResolveReviewBodyResolution;
  resolution_notes?: string;
  resolved_by?: string;
}

export interface NpiProvider {
  npi: string;
  provider_type: string;
  name: string;
  first_name?: string;
  last_name?: string;
  credential?: string;
  gender?: string;
  specialty: string;
  address_line_1?: string;
  address_line_2?: string;
  city: string;
  state: string;
  postal_code?: string;
  phone: string;
  fax?: string;
  enumeration_date?: string;
  last_updated?: string;
  npi_registry_url: string;
}

export interface NpiSearchResponse {
  results: NpiProvider[];
  result_count: number;
}

export interface Paralegal {
  id: number;
  name: string;
  email?: string | null;
  role: string;
  active_cases: number;
  signed_cases: number;
  total_assigned: number;
  created_at: string;
  updated_at: string;
}

export interface CreateParalegalBody {
  name: string;
  email?: string | null;
  role?: string | null;
}

export interface ParalegalDetail {
  id: number;
  name: string;
  email?: string | null;
  role: string;
  leads: Lead[];
  signed_cases: number;
  active_cases: number;
  total_assigned: number;
  conversion_rate: number;
}

export type ParalegalPerformanceBreakdownItem = {
  status: string;
  count: number;
};

export interface ParalegalPerformance {
  paralegal: Paralegal;
  breakdown: ParalegalPerformanceBreakdownItem[];
  total_assigned: number;
  signed: number;
  qualified: number;
  conversion_rate: number;
}

export type AnalyticsOverviewLeads = { [key: string]: unknown };
export type AnalyticsOverviewCases = { [key: string]: unknown };

```

```

export type AnalyticsOverviewAnalysis = { [key: string]: unknown };
export type AnalyticsOverviewFaxes = { [key: string]: unknown };

export interface AnalyticsOverview {
  leads: AnalyticsOverviewLeads;
  cases: AnalyticsOverviewCases;
  analysis: AnalyticsOverviewAnalysis;
  faxes: AnalyticsOverviewFaxes;
}

export interface PipelineTrendPoint {
  date: string;
  total: number;
  qualified: number;
  signed: number;
}

export interface FunnelStage {
  stage: string;
  count: number;
  color: string;
}

export interface TortBreakdownRow {
  tort_type: string;
  total: number;
  qualified: number;
  signed: number;
  rejected: number;
  avg_ad_spend: number;
}

export interface ParalegalLeaderboardRow {
  id: number;
  name: string;
  role: string;
  total_assigned: number;
  signed: number;
  qualified: number;
  rejected: number;
  conversion_rate: number;
}

export type AuditSummaryByEntityItem = {
  entity_type: string;
  count: number;
};

export type AuditSummaryByActionItem = {
  action: string;
  count: number;
};

export interface AuditSummary {
  by_entity: AuditSummaryByEntityItem[];
  by_action: AuditSummaryByActionItem[];
  total_events: number;
  last_24h: number;
  last_7d: number;
}

export type CustomFieldType =
  (typeof CustomFieldType)[keyof typeof CustomFieldType];

export const CustomFieldType = {
  text: "text",
  email: "email",
  tel: "tel",
  date: "date",
  number: "number",
  select: "select",
  textarea: "textarea",
  checkbox: "checkbox",
} as const;

export interface CustomField {
  key: string;
  label: string;
  type: CustomFieldType;
  required: boolean;
  placeholder?: string | null;
  helper_text?: string | null;
  options?: string[] | null;
  max_length?: number | null;
}

export type FormConfigRejectionConditionsItem = { [key: string]: unknown };

export interface FormConfig {
  id: string;
  label: string;
  category?: string;
  active?: boolean;
  intro_text?: string | null;
  fields: string[];
  extra_fields?: string[];
}

```

```

    exposure_fields?: string[];
    rules: string[];
    valid_diagnoses?: string[];
    custom_fields?: CustomField[];
    rejection_conditions?: FormConfigRejectionConditionsItem[];
    avg_settlement_low?: number | null;
    avg_settlement_high?: number | null;
    mdl_status?: string | null;
    sol_months?: number | null;
    updated_at?: string | null;
    updated_by?: string | null;
}

export interface FormConfigUpdate {
    label?: string;
    category?: string;
    active?: boolean;
    intro_text?: string | null;
    valid_diagnoses?: string[];
    extra_fields?: string[];
    exposure_fields?: string[];
    custom_fields?: CustomField[];
    avg_settlement_low?: number | null;
    avg_settlement_high?: number | null;
    mdl_status?: string | null;
    sol_months?: number | null;
}

export type NpiVerifyResultTaxonomyMatch = {
    matched?: boolean;
    physician_specialty?: string;
    expected_specialties?: string[];
    diagnosis_category?: string | null;
    fraud_indicators?: string[];
    confidence?: string;
} | null;

export interface NpiVerifyResult {
    npi_found: boolean;
    npi_number?: string | null;
    specialty?: string | null;
    taxonomy_match?: NpiVerifyResultTaxonomyMatch;
    error?: string | null;
}

export type FraudCheckResultTortValidation = {
    valid?: boolean;
    tort_id?: string | null;
    errors?: string[];
    diagnosis_match?: boolean;
    category?: string | null;
};

export type FraudCheckResultFraudResultIndicatorsItem = {
    type?: string;
    description?: string;
    severity?: string;
};

export type FraudCheckResultFraudResult = {
    status?: string;
    fraud_score?: number;
    indicators?: FraudCheckResultFraudResultIndicatorsItem[];
    summary?: string;
};

export interface FraudCheckResult {
    tort_validation: FraudCheckResultTortValidation;
    fraud_result: FraudCheckResultFraudResult;
}

export interface FormConfigList {
    tort_campaigns: FormConfig[];
}

export interface EmailValidationResult {
    valid: boolean;
    errors: string[];
    suggestion?: string | null;
}

export interface AddressValidationResult {
    valid: boolean;
    errors: string[];
}

export interface FormSubmission {
    first_name: string;
    last_name: string;
    date_of_birth: string;
    street_address: string;
    city: string;
    state: string;
    zip: string;
    phone_primary: string;
    email: string;
    last_4_ssn: string;
}

```

```

diagnosis: string;
diagnosis_date: string;
diagnosis_confirmed?: boolean;
was_at_location?: boolean;
location_name?: string | null;
physician_first_name: string;
physician_last_name: string;
physician_full_address: string;
physician_contact_info: string;
hospital_name: string;
hospital_fax: string;
hospital_contact_info: string;
tort_type: string;
tcpa_consent: boolean;
trustedform_cert_url: string;
trustedform_ip?: string | null;
trustedform_user_agent?: string | null;
trustedform_timestamp?: string | null;
exposure_start?: string | null;
exposure_end?: string | null;
source?: string | null;
notes?: string | null;
}

export type FormSubmissionResultBackgroundCheck = {
  status?: string;
  summary?: string;
} | null;

export type FormSubmissionResultPipelineItem = {
  name?: string;
  status?: string;
  errors?: string[];
};

export interface FormSubmissionResult {
  status: string;
  lead_id: number;
  lead_status: string;
  tort_id?: string | null;
  fraud_score?: number;
  fraud_status?: string;
  npi_verified?: boolean;
  background_check?: FormSubmissionResultBackgroundCheck;
  output_state: string;
  pipeline?: FormSubmissionResultPipelineItem[];
}

export type FormRejectionPipelineItem = { [key: string]: unknown };

export interface FormRejection {
  status: string;
  errors: string[];
  action: string;
  details?: string[] | null;
  pipeline?: FormRejectionPipelineItem[] | null;
  failed_step?: string | null;
  fraud_score?: number | null;
  fraud_summary?: string | null;
}

export type BackgroundCheckResultStatus =
  (typeof BackgroundCheckResultStatus)[keyof typeof BackgroundCheckResultStatus];

export const BackgroundCheckResultStatus = {
  clean: "clean",
  flagged: "flagged",
  not_found: "not_found",
  error: "error",
} as const;

export type BackgroundCheckResultRecordsItemSeverity =
  (typeof BackgroundCheckResultRecordsItemSeverity)[keyof typeof BackgroundCheckResultRecordsItemSeverity];

export const BackgroundCheckResultRecordsItemSeverity = {
  low: "low",
  medium: "medium",
  high: "high",
} as const;

export type BackgroundCheckResultRecordsItem = {
  type: string;
  description: string;
  date?: string | null;
  jurisdiction?: string | null;
  severity: BackgroundCheckResultRecordsItemSeverity;
};

export interface BackgroundCheckResult {
  status: BackgroundCheckResultStatus;
  source: string;
  checked_at: string;
  records: BackgroundCheckResultRecordsItem[];
  summary: string;
}

export type ListLeadsParams = {

```



```

status?: ListLeadsStatus;
tort_type?: string;
search?: string;
vendor_id?: number;
law_firm?: string;
client_id?: string;
date_from?: string;
date_to?: string;
};

export type ListLeadsStatus =
  (typeof ListLeadsStatus)[keyof typeof ListLeadsStatus];

export const ListLeadsStatus = {
  new: "new",
  qualified: "qualified",
  signed: "signed",
  rejected: "rejected",
  review_required: "review_required",
} as const;

export type ExportLeadsParams = {
  lead_id?: number;
  status?: string;
  tort_type?: string;
  vendor_id?: number;
  law_firm?: string;
  client_id?: string;
  date_from?: string;
  date_to?: string;
  source?: string;
  fields?: string;
};

export type ListDocumentsParams = {
  lead_id?: number;
};

export type CreateCaseBody = { [key: string]: unknown };

export type ListReviewQueueParams = {
  resolution?: string;
  conflict_type?: string;
  severity?: string;
  entity_type?: string;
  limit?: string;
};

export type SearchNpiParams = {
  npi_number?: string;
  first_name?: string;
  last_name?: string;
  city?: string;
  state?: string;
  specialty?: string;
  limit?: string;
};

export type ValidateEmailBody = {
  email: string;
};

export type ValidateAddressBody = {
  street_address: string;
  city: string;
  state: string;
  zip: string;
};

export type RunBackgroundCheckBody = {
  first_name: string;
  last_name: string;
  state?: string;
  date_of_birth?: string;
};

export type GetTortCategories200ItemTortsItem = {
  id: string;
  label: string;
};

export type GetTortCategories200Item = {
  category: string;
  torts: GetTortCategories200ItemTortsItem[];
};

export type VerifyNpiBody = {
  physician_first_name: string;
  physician_last_name: string;
  diagnosis: string;
};

export type RunFraudCheckBodyLeadData = { [key: string]: unknown };

export type RunFraudCheckBody = {
  tort_type: string;
  diagnosis: string;
};

```

```
    physician_first_name?: string;
    physician_last_name?: string;
    lead_data?: RunFraudCheckBodyLeadData;
};

export type EscalateToFbiBodyFraudIndicatorsItem = { [key: string]: unknown };

export type EscalateToFbiBody = {
    lead_id: string;
    reason: string;
    fraud_indicators?: EscalateToFbiBodyFraudIndicatorsItem[];
};

export type EscalateToFbi200 = {
    status: string;
    fbi_tip_url: string;
    message: string;
};

export type GetAuditTrailParams = {
    limit?: number;
    entity_type?: string;
    action?: string;
};
```

lib/api-client-react/src/generated/api.ts

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import { useMutation, useQuery } from "@tanstack/react-query";
import type {
  MutationFunction,
  QueryFunction,
  QueryKey,
  UseMutationOptions,
  UseMutationResult,
  UseQueryOptions,
  UseQueryResult,
} from "@tanstack/react-query";

import type {
  ActivityItem,
  AddressValidationResult,
  AnalyticsOverview,
  AuditLogEntry,
  AuditSummary,
  BackgroundCheckResult,
  Case,
  CaseDetail,
  CreateCaseBody,
  CreateCaseResponse,
  CreateDocumentBody,
  CreateLeadBody,
  CreateParalegalBody,
  CreateVendorBody,
  CustomField,
  DashboardStats,
  Document,
  EmailValidationResult,
  ErrorResponse,
  EscalateToFbi200,
  EscalateToFbiBody,
  ExportLeadsParams,
  FaxResult,
  FormConfig,
  FormConfigList,
  FormConfigUpdate,
  FormRejection,
  FormSubmission,
  FormSubmissionResult,
  FraudCheckResult,
  FunnelStage,
  GetAuditTrailParams,
  GetTortCategories200Item,
  HealthStatus,
  Lead,
  ListDocumentsParams,
  ListLeadsParams,
  ListReviewQueueParams,
  NpiProvider,
  NpiSearchResponse,
  NpiVerifyResult,
  Paralegal,
  ParalegalDetail,
  ParalegalLeaderboardRow,
  ParalegalPerformance,
  PipelineBreakdown,
  PipelineTrendPoint,
  QualificationResult,
  QueueStats,
  ResolveReviewBody,
  ReviewQueueItem,
  ReviewQueueStats,
  RunBackgroundCheckBody,
  RunFraudCheckBody,
  SearchNpiParams,
  TortBreakdownRow,
  UpdateDocumentBody,
  UpdateLeadBody,
  UpdateVendorBody,
  UploadCaseFileBody,
  UploadFaxBody,
  UploadFaxResponse,
  ValidateAddressBody,
  ValidateEmailBody,
  Vendor,
  VerifyNpiBody,
} from "../api.schemas";

import { customFetch } from "../custom-fetch";
import type { ErrorType, BodyType } from "../custom-fetch";

type AwaitedInput<T> = PromiseLike<T> | T;

type Awaited<O> = O extends AwaitedInput<infer T> ? T : never;

type SecondParameter<T extends (...args: never) => unknown> = Parameters<T>[1];
```

```

/**
 * @summary Health check
 */
export const getHealthCheckUrl = () => {
  return `/api/healthz`;
};

export const healthCheck = async (
  options?: RequestInit,
): Promise<HealthStatus> => {
  return customFetch<HealthStatus>(getHealthCheckUrl(), {
    ...options,
    method: "GET",
  });
};

export const getHealthCheckQueryKey = () => {
  return [`/api/healthz`] as const;
};

export const getHealthCheckQueryOptions = <
  TData = Awaited<ReturnType<typeof healthCheck>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof healthCheck>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getHealthCheckQueryKey();

  const queryFn: QueryFunction<Awaited<ReturnType<typeof healthCheck>>> = ({
    signal,
  }) => healthCheck({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof healthCheck>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type HealthCheckQueryResult = NonNullable<
  Awaited<ReturnType<typeof healthCheck>>
>;
export type HealthCheckQueryError = ErrorType<unknown>;

/**
 * @summary Health check
 */
export function useHealthCheck<
  TData = Awaited<ReturnType<typeof healthCheck>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof healthCheck>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getHealthCheckQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary List all leads
 */
export const getListLeadsUrl = (params?: ListLeadsParams) => {
  const normalizedParams = new URLSearchParams();

  Object.entries(params || {}).forEach(([key, value]) => {
    if (value !== undefined) {
      normalizedParams.append(key, value === null ? "null" : value.toString());
    }
  });

  const stringifiedParams = normalizedParams.toString();

  return stringifiedParams.length > 0
    ? `/api/leads?${stringifiedParams}`
    : `/api/leads`;
};

export const listLeads = async (
  params?: ListLeadsParams,

```

```

    options?: RequestInit,
  ): Promise<Lead[]> => {
    return customFetch<Lead[]>(getListLeadsUrl(params), {
      ...options,
      method: "GET",
    });
  });

export const getListLeadsQueryKey = (params?: ListLeadsParams) => {
  return ['/api/leads', ...(params ? [params] : [])] as const;
};

export const getListLeadsQueryOptions = <
  TData = Awaited<ReturnType<typeof listLeads>>,
  TError = ErrorType<unknown>,
>(
  params?: ListLeadsParams,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof listLeads>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getListLeadsQueryKey(params);

  const queryFn: QueryFunction<Awaited<ReturnType<typeof listLeads>>> = ({
    signal,
  }) => listLeads(params, { signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof listLeads>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type ListLeadsQueryResult = NonNullable<
  Awaited<ReturnType<typeof listLeads>>
>;
export type ListLeadsQueryError = ErrorType<unknown>;

/**
 * @summary List all leads
 */

export function useListLeads<
  TData = Awaited<ReturnType<typeof listLeads>>,
  TError = ErrorType<unknown>,
>(
  params?: ListLeadsParams,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof listLeads>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getListLeadsQueryOptions(params, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Create a new lead (intake)
 */
export const getCreateLeadUrl = () => {
  return `/api/leads`;
};

export const createLead = async (
  createLeadBody: CreateLeadBody,
  options?: RequestInit,
): Promise<Lead> => {
  return customFetch<Lead>(getCreateLeadUrl(), {
    ...options,
    method: "POST",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(createLeadBody),
  });
};

export const getCreateLeadMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<

```

```

    Awaited<ReturnType<typeof createLead>>,
    Error,
    { data: BodyType<CreateLeadBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof createLead>>,
  Error,
  { data: BodyType<CreateLeadBody> },
  TContext
> => {
  const mutationKey = ["createLead"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
    : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof createLead>>,
    { data: BodyType<CreateLeadBody> }
  > = (props) => {
    const { data } = props ?? {};

    return createLead(data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type CreateLeadMutationResult = NonNullable<
  Awaited<ReturnType<typeof createLead>>
>;
export type CreateLeadMutationBody = BodyType<CreateLeadBody>;
export type CreateLeadMutationError = ErrorType<unknown>;

/**
 * @summary Create a new lead (intake)
 */
export const useCreateLead = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof createLead>>,
    Error,
    { data: BodyType<CreateLeadBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof createLead>>,
  Error,
  { data: BodyType<CreateLeadBody> },
  TContext
> => {
  return useMutation(getCreateLeadMutationOptions(options));
};

/**
 * @summary Get a single lead
 */
export const getGetLeadUrl = (id: number) => {
  return `/api/leads/${id}`;
};

export const getLead = async (
  id: number,
  options?: RequestInit,
): Promise<Lead> => {
  return customFetch<Lead>(getGetLeadUrl(id), {
    ...options,
    method: "GET",
  });
};

export const getGetLeadQueryKey = (id: number) => {
  return [`/api/leads/${id}`] as const;
};

export const getGetLeadQueryOptions = <
  TData = Awaited<ReturnType<typeof getLead>>,
  TError = ErrorType<ErrorResponse>,
>(
  id: number,
  options?: {
    query?: UseQueryOptions<Awaited<ReturnType<typeof getLead>>, TError, TData>;
    request?: SecondParameter<typeof customFetch>;
  },
) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetLeadQueryKey(id);

```

```

const queryFn: QueryFunction<Awaited<ReturnType<typeof getLead>>> = ({
  signal,
}) => getLead(id, { signal, ...requestOptions });

return {
  queryKey,
  queryFn,
  enabled: !!id,
  ...queryOptions,
} as UseQueryOptions<Awaited<ReturnType<typeof getLead>>, TError, TData> & {
  queryKey: QueryKey;
};
};

export type GetLeadQueryResult = NonNullable<
  Awaited<ReturnType<typeof getLead>>
>;
export type GetLeadQueryError = ErrorType<ErrorResponse>;

/**
 * @summary Get a single lead
 */

export function useGetLead<
  TData = Awaited<ReturnType<typeof getLead>>,
  TError = ErrorType<ErrorResponse>,
>({
  id: number,
  options?: {
    query?: UseQueryOptions<Awaited<ReturnType<typeof getLead>>, TError, TData>;
    request?: SecondParameter<typeof customFetch>;
  },
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetLeadQueryOptions(id, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Update lead status or fields
 */
export const getUpdateLeadUrl = (id: number) => {
  return `/api/leads/${id}`;
};

export const updateLead = async (
  id: number,
  updateLeadBody: UpdateLeadBody,
  options?: RequestInit,
): Promise<Lead> => {
  return customFetch<Lead>(getUpdateLeadUrl(id), {
    ...options,
    method: "PATCH",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(updateLeadBody),
  });
};

export const getUpdateLeadMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>({options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof updateLead>>,
    TError,
    { id: number; data: BodyType<UpdateLeadBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof updateLead>>,
  TError,
  { id: number; data: BodyType<UpdateLeadBody> },
  TContext
> => {
  const mutationKey = ["updateLead"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof updateLead>>,
    { id: number; data: BodyType<UpdateLeadBody> }
  > = (props) => {
    const { id, data } = props ?? {};
    return updateLead(id, data, requestOptions);
  };
};

```

```

    return { mutationFn, ...mutationOptions };
};

export type UpdateLeadMutationResult = NonNullable<
  Awaited<ReturnType<typeof updateLead>>
>;
export type UpdateLeadMutationBody = BodyType<UpdateLeadBody>;
export type UpdateLeadMutationError = ErrorType<unknown>;

/**
 * @summary Update lead status or fields
 */
export const useUpdateLead = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof updateLead>>,
    TError,
    { id: number; data: BodyType<UpdateLeadBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof updateLead>>,
  TError,
  { id: number; data: BodyType<UpdateLeadBody> },
  TContext
> => {
  return useMutation(getUpdateLeadMutationOptions(options));
};

/**
 * @summary Delete a lead
 */
export const getDeleteLeadUrl = (id: number) => {
  return `/api/leads/${id}`;
};

export const deleteLead = async (
  id: number,
  options?: RequestInit,
): Promise<void> => {
  return customFetch<void>(getDeleteLeadUrl(id), {
    ...options,
    method: "DELETE",
  });
};

export const getDeleteLeadMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof deleteLead>>,
    TError,
    { id: number },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof deleteLead>>,
  TError,
  { id: number },
  TContext
> => {
  const mutationKey = ["deleteLead"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
      ? options
      : { ...options, mutation: { ...options.mutation, mutationKey } }
    : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof deleteLead>>,
    { id: number }
  > = (props) => {
    const { id } = props ?? {};

    return deleteLead(id, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type DeleteLeadMutationResult = NonNullable<
  Awaited<ReturnType<typeof deleteLead>>
>;

export type DeleteLeadMutationError = ErrorType<unknown>;

/**
 * @summary Delete a lead
 */
export const useDeleteLead = <

```



```

    TError = ErrorType<unknown>,
    TContext = unknown,
  >(options?: {
    mutation?: UseMutationOptions<
      Awaited<ReturnType<typeof deleteLead>>,
      TError,
      { id: number },
      TContext
    >;
    request?: SecondParameter<typeof customFetch>;
  }): UseMutationResult<
    Awaited<ReturnType<typeof deleteLead>>,
    TError,
    { id: number },
    TContext
  > => {
    return useMutation(getDeleteLeadMutationOptions(options));
  };
}

/**
 * @summary Run Boolean Gatekeeper qualification on a lead
 */
export const getQualifyLeadUrl = (id: number) => {
  return `/api/leads/${id}/qualify`;
};

export const qualifyLead = async (
  id: number,
  options?: RequestInit,
): Promise<QualificationResult> => {
  return customFetch<QualificationResult>(getQualifyLeadUrl(id), {
    ...options,
    method: "POST",
  });
});

export const getQualifyLeadMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof qualifyLead>>,
    TError,
    { id: number },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof qualifyLead>>,
  TError,
  { id: number },
  TContext
> => {
  const mutationKey = ["qualifyLead"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof qualifyLead>>,
    { id: number }
  > = (props) => {
    const { id } = props ?? {};

    return qualifyLead(id, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type QualifyLeadMutationResult = NonNullable<
  Awaited<ReturnType<typeof qualifyLead>>
>;

export type QualifyLeadMutationError = ErrorType<unknown>;

/**
 * @summary Run Boolean Gatekeeper qualification on a lead
 */
export const useQualifyLead = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof qualifyLead>>,
    TError,
    { id: number },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof qualifyLead>>,
  TError,
  { id: number },
  TContext
> => {
  const mutationKey = ["qualifyLead"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof qualifyLead>>,
    { id: number }
  > = (props) => {
    const { id } = props ?? {};

    return qualifyLead(id, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

```

```

    { id: number },
    TContext
> => {
    return useMutation(getQualifyLeadMutationOptions(options));
};

/**
 * @summary Export leads as CSV
 */
export const getExportLeadsUrl = (params?: ExportLeadsParams) => {
    const normalizedParams = new URLSearchParams();

    Object.entries(params || {}).forEach(([key, value]) => {
        if (value !== undefined) {
            normalizedParams.append(key, value === null ? "null" : value.toString());
        }
    });

    const stringifiedParams = normalizedParams.toString();

    return stringifiedParams.length > 0
        ? `/api/leads/export?${stringifiedParams}`
        : `/api/leads/export`;
};

export const exportLeads = async (
    params?: ExportLeadsParams,
    options?: RequestInit,
): Promise<string> => {
    return customFetch<string>(getExportLeadsUrl(params), {
        ...options,
        method: "GET",
    });
};

export const getExportLeadsQueryKey = (params?: ExportLeadsParams) => {
    return [`/api/leads/export`, ...(params ? [params] : [])] as const;
};

export const getExportLeadsQueryOptions = <
    TData = Awaited<ReturnType<typeof exportLeads>>,
    TError = ErrorType<unknown>,
>(
    params?: ExportLeadsParams,
    options?: {
        query?: UseQueryOptions<
            Awaited<ReturnType<typeof exportLeads>>,
            TError,
            TData
        >;
        request?: SecondParameter<typeof customFetch>;
    },
) => {
    const { query: queryOptions, request: requestOptions } = options ?? {};

    const queryKey = queryOptions?.queryKey ?? getExportLeadsQueryKey(params);

    const queryFn: QueryFunction<Awaited<ReturnType<typeof exportLeads>>> = ({
        signal,
    }) => exportLeads(params, { signal, ...requestOptions });

    return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
        Awaited<ReturnType<typeof exportLeads>>,
        TError,
        TData
    > & { queryKey: QueryKey };
};

export type ExportLeadsQueryResult = NonNullable<
    Awaited<ReturnType<typeof exportLeads>>
>;
export type ExportLeadsQueryError = ErrorType<unknown>;

/**
 * @summary Export leads as CSV
 */

export function useExportLeads<
    TData = Awaited<ReturnType<typeof exportLeads>>,
    TError = ErrorType<unknown>,
>(
    params?: ExportLeadsParams,
    options?: {
        query?: UseQueryOptions<
            Awaited<ReturnType<typeof exportLeads>>,
            TError,
            TData
        >;
        request?: SecondParameter<typeof customFetch>;
    },
): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
    const queryOptions = getExportLeadsQueryOptions(params, options);

    const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
        queryKey: QueryKey;
    };
};

```

```

    return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary List all vendors
 */
export const getListVendorsUrl = () => {
    return `/api/vendors`;
};

export const listVendors = async (options?: RequestInit): Promise<Vendor[]> => {
    return customFetch<Vendor[]>(getListVendorsUrl(), {
        ...options,
        method: "GET",
    });
};

export const getListVendorsQueryKey = () => {
    return [`/api/vendors`] as const;
};

export const getListVendorsQueryOptions = <
    TData = Awaited<ReturnType<typeof listVendors>>,
    TError = ErrorType<unknown>,
>(options?: {
    query?: UseQueryOptions<
        Awaited<ReturnType<typeof listVendors>>,
        TError,
        TData
    >;
    request?: SecondParameter<typeof customFetch>;
}) => {
    const { query: queryOptions, request: requestOptions } = options ?? {};

    const queryKey = queryOptions?.queryKey ?? getListVendorsQueryKey();

    const queryFn: QueryFunction<Awaited<ReturnType<typeof listVendors>>> = ({
        signal,
    }) => listVendors({ signal, ...requestOptions });

    return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
        Awaited<ReturnType<typeof listVendors>>,
        TError,
        TData
    > & { queryKey: QueryKey };
};

export type ListVendorsQueryResult = NonNullable<
    Awaited<ReturnType<typeof listVendors>>
>;
export type ListVendorsQueryError = ErrorType<unknown>;

/**
 * @summary List all vendors
 */

export function useListVendors<
    TData = Awaited<ReturnType<typeof listVendors>>,
    TError = ErrorType<unknown>,
>(options?: {
    query?: UseQueryOptions<
        Awaited<ReturnType<typeof listVendors>>,
        TError,
        TData
    >;
    request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
    const queryOptions = getListVendorsQueryOptions(options);

    const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
        queryKey: QueryKey;
    };

    return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Create a new vendor
 */
export const getCreateVendorUrl = () => {
    return `/api/vendors`;
};

export const createVendor = async (
    createVendorBody: CreateVendorBody,
    options?: RequestInit,
): Promise<Vendor> => {
    return customFetch<Vendor>(getCreateVendorUrl(), {
        ...options,
        method: "POST",
        headers: { "Content-Type": "application/json", ...options?.headers },
        body: JSON.stringify(createVendorBody),
    });
};

export const getCreateVendorMutationOptions = <
    TError = ErrorType<unknown>,

```

```

TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof createVendor>>,
    Error,
    { data: BodyType<CreateVendorBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof createVendor>>,
  Error,
  { data: BodyType<CreateVendorBody> },
  TContext
> => {
  const mutationKey = ["createVendor"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof createVendor>>,
    { data: BodyType<CreateVendorBody> }
  > = (props) => {
    const { data } = props ?? {};

    return createVendor(data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type CreateVendorMutationResult = NonNullable<
  Awaited<ReturnType<typeof createVendor>>
>;
export type CreateVendorMutationBody = BodyType<CreateVendorBody>;
export type CreateVendorMutationError = ErrorType<unknown>;

/**
 * @summary Create a new vendor
 */
export const useCreateVendor = <
  Error = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof createVendor>>,
    Error,
    { data: BodyType<CreateVendorBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof createVendor>>,
  Error,
  { data: BodyType<CreateVendorBody> },
  TContext
> => {
  return useMutation(getCreateVendorMutationOptions(options));
};

/**
 * @summary Get a single vendor
 */
export const getGetVendorUrl = (id: number) => {
  return `/api/vendors/${id}`;
};

export const getVendor = async (
  id: number,
  options?: RequestInit,
): Promise<Vendor> => {
  return customFetch<Vendor>(getGetVendorUrl(id), {
    ...options,
    method: "GET",
  });
};

export const getGetVendorQueryKey = (id: number) => {
  return [`/api/vendors/${id}`] as const;
};

export const getGetVendorQueryOptions = <
  TData = Awaited<ReturnType<typeof getVendor>>,
  TError = ErrorType<unknown>,
>(
  id: number,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getVendor>>,
      TError,
      TData
    >;
  }
);

```

```

    request?: SecondParameter<typeof customFetch>;
  },
) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetVendorQueryKey(id);

  const queryFn: QueryFunction<Awaited<ReturnType<typeof getVendor>>> = ({
    signal,
  }) => getVendor(id, { signal, ...requestOptions });

  return {
    queryKey,
    queryFn,
    enabled: !!id,
    ...queryOptions,
  } as UseQueryOptions<Awaited<ReturnType<typeof getVendor>>, TError, TData> & {
    queryKey: QueryKey;
  }
};

export type GetVendorQueryResult = NonNullable<
  Awaited<ReturnType<typeof getVendor>>
>;
export type GetVendorQueryError = ErrorType<unknown>;

/**
 * @summary Get a single vendor
 */

export function useGetVendor<
  TData = Awaited<ReturnType<typeof getVendor>>,
  TError = ErrorType<unknown>,
>({
  id: number,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getVendor>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetVendorQueryOptions(id, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Update a vendor
 */
export const getUpdateVendorUrl = (id: number) => {
  return `/api/vendors/${id}`;
};

export const updateVendor = async (
  id: number,
  updateVendorBody: UpdateVendorBody,
  options?: RequestInit,
): Promise<Vendor> => {
  return customFetch<Vendor>(getUpdateVendorUrl(id), {
    ...options,
    method: "PATCH",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(updateVendorBody),
  });
};

export const getUpdateVendorMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>({options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof updateVendor>>,
    TError,
    { id: number; data: BodyType<UpdateVendorBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof updateVendor>>,
  TError,
  { id: number; data: BodyType<UpdateVendorBody> },
  TContext
> => {
  const mutationKey = ["updateVendor"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }

```

```

      : { mutation: { mutationKey }, request: undefined };

const mutationFn: MutationFunction<
  Awaited<ReturnType<typeof updateVendor>>,
  { id: number; data: BodyType<UpdateVendorBody> }
> = (props) => {
  const { id, data } = props ?? {};

  return updateVendor(id, data, requestOptions);
};

return { mutationFn, ...mutationOptions };
};

export type UpdateVendorMutationResult = NonNullable<
  Awaited<ReturnType<typeof updateVendor>>
>;
export type UpdateVendorMutationBody = BodyType<UpdateVendorBody>;
export type UpdateVendorMutationError = ErrorType<unknown>;

/**
 * @summary Update a vendor
 */
export const useUpdateVendor = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof updateVendor>>,
    TError,
    { id: number; data: BodyType<UpdateVendorBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof updateVendor>>,
  TError,
  { id: number; data: BodyType<UpdateVendorBody> },
  TContext
> => {
  return useMutation(getUpdateVendorMutationOptions(options));
};

/**
 * @summary Delete a vendor
 */
export const getDeleteVendorUrl = (id: number) => {
  return `/api/vendors/${id}`;
};

export const deleteVendor = async (
  id: number,
  options?: RequestInit,
): Promise<void> => {
  return customFetch<void>(getDeleteVendorUrl(id), {
    ...options,
    method: "DELETE",
  });
};

export const getDeleteVendorMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof deleteVendor>>,
    TError,
    { id: number },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof deleteVendor>>,
  TError,
  { id: number },
  TContext
> => {
  const mutationKey = ["deleteVendor"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof deleteVendor>>,
    { id: number }
  > = (props) => {
    const { id } = props ?? {};

    return deleteVendor(id, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

```

```

export type DeleteVendorMutationResult = NonNullable<
  Awaited<ReturnType<typeof deleteVendor>>
>;

export type DeleteVendorMutationError = ErrorType<unknown>;

/**
 * @summary Delete a vendor
 */
export const useDeleteVendor = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof deleteVendor>>,
    TError,
    { id: number },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof deleteVendor>>,
  TError,
  { id: number },
  TContext
> => {
  return useMutation(getDeleteVendorMutationOptions(options));
};

/**
 * @summary List all documents
 */
export const getListDocumentsUrl = (params?: ListDocumentsParams) => {
  const normalizedParams = new URLSearchParams();

  Object.entries(params || {}).forEach(([key, value]) => {
    if (value !== undefined) {
      normalizedParams.append(key, value === null ? "null" : value.toString());
    }
  });

  const stringifiedParams = normalizedParams.toString();

  return stringifiedParams.length > 0
    ? `/api/documents?${stringifiedParams}`
    : `/api/documents`;
};

export const listDocuments = async (
  params?: ListDocumentsParams,
  options?: RequestInit,
): Promise<Document[]> => {
  return customFetch<Document[]>(getListDocumentsUrl(params), {
    ...options,
    method: "GET",
  });
};

export const getListDocumentsQueryKey = (params?: ListDocumentsParams) => {
  return [ `/api/documents`, ...(params ? [params] : []) ] as const;
};

export const getListDocumentsQueryOptions = <
  TData = Awaited<ReturnType<typeof listDocuments>>,
  TError = ErrorType<unknown>,
>(
  params?: ListDocumentsParams,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof listDocuments>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getListDocumentsQueryKey(params);

  const queryFn: QueryFunction<Awaited<ReturnType<typeof listDocuments>>> = ({
    signal,
  }) => listDocuments(params, { signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof listDocuments>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type ListDocumentsQueryResult = NonNullable<
  Awaited<ReturnType<typeof listDocuments>>
>;

export type ListDocumentsQueryError = ErrorType<unknown>;

```

```

/**
 * @summary List all documents
 */

export function useListDocuments<
  TData = Awaited<ReturnType<typeof listDocuments>>,
  TError = ErrorType<unknown>,
>({
  params?: ListDocumentsParams,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof listDocuments>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getListDocumentsQueryOptions(params, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Upload or register a document
 */
export const getCreateDocumentUrl = () => {
  return `/api/documents`;
};

export const createDocument = async (
  createDocumentBody: CreateDocumentBody,
  options?: RequestInit,
): Promise<Document> => {
  return customFetch<Document>(getCreateDocumentUrl(), {
    ...options,
    method: "POST",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(createDocumentBody),
  });
};

export const getCreateDocumentMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>({options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof createDocument>>,
    TError,
    { data: BodyType<CreateDocumentBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof createDocument>>,
  TError,
  { data: BodyType<CreateDocumentBody> },
  TContext
> => {
  const mutationKey = ["createDocument"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof createDocument>>,
    { data: BodyType<CreateDocumentBody> }
  > = (props) => {
    const { data } = props ?? {};

    return createDocument(data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type CreateDocumentMutationResult = NonNullable<
  Awaited<ReturnType<typeof createDocument>>
>;
export type CreateDocumentMutationBody = BodyType<CreateDocumentBody>;
export type CreateDocumentMutationError = ErrorType<unknown>;

/**
 * @summary Upload or register a document
 */
export const useCreateDocument = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>({options?: {

```



```

mutation?: UseMutationOptions<
  Awaited<ReturnType<typeof createDocument>>,
  Error,
  { data: BodyType<CreateDocumentBody> },
  TContext
>;
request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof createDocument>>,
  Error,
  { data: BodyType<CreateDocumentBody> },
  TContext
> => {
  return useMutation(getCreateDocumentMutationOptions(options));
};

/**
 * @summary Update document metadata or status
 */
export const getUpdateDocumentUrl = (id: number) => {
  return `/api/documents/${id}`;
};

export const updateDocument = async (
  id: number,
  updateDocumentBody: UpdateDocumentBody,
  options?: RequestInit,
): Promise<Document> => {
  return customFetch<Document>(getUpdateDocumentUrl(id), {
    ...options,
    method: "PATCH",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(updateDocumentBody),
  });
};

export const getUpdateDocumentMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof updateDocument>>,
    Error,
    { id: number; data: BodyType<UpdateDocumentBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof updateDocument>>,
  TError,
  { id: number; data: BodyType<UpdateDocumentBody> },
  TContext
> => {
  const mutationKey = ["updateDocument"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof updateDocument>>,
    { id: number; data: BodyType<UpdateDocumentBody> }
  > = (props) => {
    const { id, data } = props ?? {};

    return updateDocument(id, data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type UpdateDocumentMutationResult = NonNullable<
  Awaited<ReturnType<typeof updateDocument>>
>;
export type UpdateDocumentMutationBody = BodyType<UpdateDocumentBody>;
export type UpdateDocumentMutationError = ErrorType<unknown>;

/**
 * @summary Update document metadata or status
 */
export const useUpdateDocument = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof updateDocument>>,
    Error,
    { id: number; data: BodyType<UpdateDocumentBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof updateDocument>>,
  TError,

```

```

    { id: number; data: BodyType<UpdateDocumentBody> },
    TContext
  > => {
    return useMutation(getUpdateDocumentMutationOptions(options));
  };
}

/**
 * @summary Delete a document
 */
export const getDeleteDocumentUrl = (id: number) => {
  return `/api/documents/${id}`;
};

export const deleteDocument = async (
  id: number,
  options?: RequestInit,
): Promise<void> => {
  return customFetch<void>(getDeleteDocumentUrl(id), {
    ...options,
    method: "DELETE",
  });
};

export const getDeleteDocumentMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof deleteDocument>>,
    TError,
    { id: number },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof deleteDocument>>,
  TError,
  { id: number },
  TContext
> => {
  const mutationKey = ["deleteDocument"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof deleteDocument>>,
    { id: number }
  > = (props) => {
    const { id } = props ?? {};

    return deleteDocument(id, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type DeleteDocumentMutationResult = NonNullable<
  Awaited<ReturnType<typeof deleteDocument>>
>;

export type DeleteDocumentMutationError = ErrorType<unknown>;

/**
 * @summary Delete a document
 */
export const useDeleteDocument = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof deleteDocument>>,
    TError,
    { id: number },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof deleteDocument>>,
  TError,
  { id: number },
  TContext
> => {
  return useMutation(getDeleteDocumentMutationOptions(options));
};

/**
 * @summary Get pipeline summary stats
 */
export const getGetDashboardStatsUrl = () => {
  return `/api/dashboard/stats`;
};

```

```

export const getDashboardStats = async (
  options?: RequestInit,
): Promise<DashboardStats> => {
  return customFetch<DashboardStats>(getGetDashboardStatsUrl(), {
    ...options,
    method: "GET",
  });
};

export const getGetDashboardStatsQueryKey = () => {
  return [`/api/dashboard/stats`] as const;
};

export const getGetDashboardStatsQueryOptions = <
  TData = Awaited<ReturnType<typeof getDashboardStats>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getDashboardStats>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetDashboardStatsQueryKey();

  const queryFn: QueryFunction<
    Awaited<ReturnType<typeof getDashboardStats>>
  > = ({ signal }) => getDashboardStats({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof getDashboardStats>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetDashboardStatsQueryResult = NonNullable<
  Awaited<ReturnType<typeof getDashboardStats>>
>;
export type GetDashboardStatsQueryError = ErrorType<unknown>;

/**
 * @summary Get pipeline summary stats
 */

export function useGetDashboardStats<
  TData = Awaited<ReturnType<typeof getDashboardStats>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getDashboardStats>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetDashboardStatsQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Get lead counts grouped by status and tort type
 */
export const getGetPipelineBreakdownUrl = () => {
  return [`/api/dashboard/pipeline`] as const;
};

export const getPipelineBreakdown = async (
  options?: RequestInit,
): Promise<PipelineBreakdown> => {
  return customFetch<PipelineBreakdown>(getGetPipelineBreakdownUrl(), {
    ...options,
    method: "GET",
  });
};

export const getGetPipelineBreakdownQueryKey = () => {
  return [`/api/dashboard/pipeline`] as const;
};

export const getGetPipelineBreakdownQueryOptions = <
  TData = Awaited<ReturnType<typeof getPipelineBreakdown>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getPipelineBreakdown>>,
    TError,
    TData
  >;
});

```

```

    request?: SecondParameter<typeof customFetch>;
  }) => {
    const { query: queryOptions, request: requestOptions } = options ?? {};

    const queryKey = queryOptions?.queryKey ?? getGetPipelineBreakdownQueryKey();

    const queryFn: QueryFunction<
      Awaited<ReturnType<typeof getPipelineBreakdown>>
    > = ({ signal }) => getPipelineBreakdown({ signal, ...requestOptions });

    return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
      Awaited<ReturnType<typeof getPipelineBreakdown>>,
      TError,
      TData
    > & { queryKey: QueryKey };
  };

export type GetPipelineBreakdownQueryResult = NonNullable<
  Awaited<ReturnType<typeof getPipelineBreakdown>>
>;
export type GetPipelineBreakdownQueryError = ErrorType<unknown>;

/**
 * @summary Get lead counts grouped by status and tort type
 */

export function useGetPipelineBreakdown<
  TData = Awaited<ReturnType<typeof getPipelineBreakdown>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getPipelineBreakdown>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetPipelineBreakdownQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Get recent lead activity feed
 */
export const getGetRecentActivityUrl = () => {
  return `/api/dashboard/recent-activity`;
};

export const getRecentActivity = async (
  options?: RequestInit,
): Promise<ActivityItem[]> => {
  return customFetch<ActivityItem[]>(getGetRecentActivityUrl(), {
    ...options,
    method: "GET",
  });
};

export const getGetRecentActivityQueryKey = () => {
  return [`/api/dashboard/recent-activity`] as const;
};

export const getGetRecentActivityQueryOptions = <
  TData = Awaited<ReturnType<typeof getRecentActivity>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getRecentActivity>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetRecentActivityQueryKey();

  const queryFn: QueryFunction<
    Awaited<ReturnType<typeof getRecentActivity>>
  > = ({ signal }) => getRecentActivity({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof getRecentActivity>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetRecentActivityQueryResult = NonNullable<
  Awaited<ReturnType<typeof getRecentActivity>>
>;
export type GetRecentActivityQueryError = ErrorType<unknown>;

```

```

/**
 * @summary Get recent lead activity feed
 */

export function useGetRecentActivity<
  TData = Awaited<ReturnType<typeof getRecentActivity>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getRecentActivity>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetRecentActivityQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary List all cases
 */
export const getListCasesUrl = () => {
  return `/api/cases`;
};

export const listCases = async (options?: RequestInit): Promise<Case[]> => {
  return customFetch<Case[]>(getListCasesUrl(), {
    ...options,
    method: "GET",
  });
};

export const getListCasesQueryKey = () => {
  return ['/api/cases'] as const;
};

export const getListCasesQueryOptions = <
  TData = Awaited<ReturnType<typeof listCases>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<Awaited<ReturnType<typeof listCases>>, TError, TData>;
  request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getListCasesQueryKey();

  const queryFn: QueryFunction<Awaited<ReturnType<typeof listCases>>> = ({
    signal,
  }) => listCases({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof listCases>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type ListCasesQueryResult = NonNullable<
  Awaited<ReturnType<typeof listCases>>
>;
export type ListCasesQueryError = ErrorType<unknown>;

/**
 * @summary List all cases
 */

export function useListCases<
  TData = Awaited<ReturnType<typeof listCases>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<Awaited<ReturnType<typeof listCases>>, TError, TData>;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getListCasesQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Submit a new case to the processing queue
 */
export const getCreateCaseUrl = () => {
  return `/api/cases`;
};

export const createCase = async (

```

```

    createCaseBody: CreateCaseBody,
    options?: RequestInit,
  ): Promise<CreateCaseResponse> => {
    return customFetch<CreateCaseResponse>(getCreateCaseUrl(), {
      ...options,
      method: "POST",
      headers: { "Content-Type": "application/json", ...options?.headers },
      body: JSON.stringify(createCaseBody),
    });
  });

export const getCreateCaseMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof createCase>>,
    TError,
    { data: BodyType<CreateCaseBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof createCase>>,
  TError,
  { data: BodyType<CreateCaseBody> },
  TContext
> => {
  const mutationKey = ["createCase"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof createCase>>,
    { data: BodyType<CreateCaseBody> }
  > = (props) => {
    const { data } = props ?? {};

    return createCase(data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type CreateCaseMutationResult = NonNullable<
  Awaited<ReturnType<typeof createCase>>
>;
export type CreateCaseMutationBody = BodyType<CreateCaseBody>;
export type CreateCaseMutationError = ErrorType<unknown>;

/**
 * @summary Submit a new case to the processing queue
 */
export const useCreateCase = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof createCase>>,
    TError,
    { data: BodyType<CreateCaseBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof createCase>>,
  TError,
  { data: BodyType<CreateCaseBody> },
  TContext
> => {
  return useMutation(getCreateCaseMutationOptions(options));
};

/**
 * @summary Get job queue statistics by status
 */
export const getGetQueueStatsUrl = () => {
  return `/api/cases/worker/queue-stats`;
};

export const getQueueStats = async (
  options?: RequestInit,
): Promise<QueueStats> => {
  return customFetch<QueueStats>(getGetQueueStatsUrl(), {
    ...options,
    method: "GET",
  });
};

export const getGetQueueStatsQueryKey = () => {
  return [ `/api/cases/worker/queue-stats` ] as const;
};

```

```

export const getGetQueueStatsQueryOptions = <
  TData = Awaited<ReturnType<typeof getQueueStats>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getQueueStats>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetQueueStatsQueryKey();

  const queryFn: QueryFunction<Awaited<ReturnType<typeof getQueueStats>>> = ({
    signal,
  }) => getQueueStats({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof getQueueStats>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetQueueStatsQueryResult = NonNullable<
  Awaited<ReturnType<typeof getQueueStats>>
>;
export type GetQueueStatsQueryError = ErrorType<unknown>;

/**
 * @summary Get job queue statistics by status
 */
export function useGetQueueStats<
  TData = Awaited<ReturnType<typeof getQueueStats>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getQueueStats>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetQueueStatsQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Get full case detail with documents, analyses, and audit trail
 */
export const getGetCaseUrl = (id: string) => {
  return `/api/cases/${id}`;
};

export const getCase = async (
  id: string,
  options?: RequestInit,
): Promise<CaseDetail> => {
  return customFetch<CaseDetail>(getGetCaseUrl(id), {
    ...options,
    method: "GET",
  });
};

export const getGetCaseQueryKey = (id: string) => {
  return [`/api/cases/${id}`] as const;
};

export const getGetCaseQueryOptions = <
  TData = Awaited<ReturnType<typeof getCase>>,
  TError = ErrorType<ErrorResponse>,
>(
  id: string,
  options?: {
    query?: UseQueryOptions<Awaited<ReturnType<typeof getCase>>, TError, TData>;
    request?: SecondParameter<typeof customFetch>;
  },
  request?: SecondParameter<typeof customFetch>;
) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetCaseQueryKey(id);

  const queryFn: QueryFunction<Awaited<ReturnType<typeof getCase>>> = ({
    signal,
  }) => getCase(id, { signal, ...requestOptions });

  return {
    queryKey,
  };
};

```

```

    queryFn,
    enabled: !!id,
    ...queryOptions,
  } as UseQueryOptions<Awaited<ReturnType<typeof getCase>>, TError, TData> & {
    queryKey: QueryKey;
  }
};

export type GetCaseQueryResult = NonNullable<
  Awaited<ReturnType<typeof getCase>>
>;
export type GetCaseQueryError = ErrorType<ErrorResponse>;

/**
 * @summary Get full case detail with documents, analyses, and audit trail
 */

export function useGetCase<
  TData = Awaited<ReturnType<typeof getCase>>,
  TError = ErrorType<ErrorResponse>,
>({
  id: string,
  options?: {
    query?: UseQueryOptions<Awaited<ReturnType<typeof getCase>>, TError, TData>;
    request?: SecondParameter<typeof customFetch>;
  },
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetCaseQueryOptions(id, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Upload a document to the case vault
 */
export const getUploadCaseFileUrl = (id: string) => {
  return `/api/cases/${id}/upload`;
};

export const uploadCaseFile = async (
  id: string,
  uploadCaseFileBody: UploadCaseFileBody,
  options?: RequestInit,
): Promise<CreateCaseResponse> => {
  return customFetch<CreateCaseResponse>(getUploadCaseFileUrl(id), {
    ...options,
    method: "POST",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(uploadCaseFileBody),
  });
};

export const getUploadCaseFileMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>({options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof uploadCaseFile>>,
    TError,
    { id: string; data: BodyType<UploadCaseFileBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof uploadCaseFile>>,
  TError,
  { id: string; data: BodyType<UploadCaseFileBody> },
  TContext
> => {
  const mutationKey = ["uploadCaseFile"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
        options.mutation.mutationKey
      ? options
      : { ...options, mutation: { ...options.mutation, mutationKey } }
    : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof uploadCaseFile>>,
    { id: string; data: BodyType<UploadCaseFileBody> }
  > = (props) => {
    const { id, data } = props ?? {};

    return uploadCaseFile(id, data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type UploadCaseFileMutationResult = NonNullable<
  Awaited<ReturnType<typeof uploadCaseFile>>
>;

```



```

export type UploadCaseFileMutationBody = BodyType<UploadCaseFileBody>;
export type UploadCaseFileMutationError = ErrorType<unknown>;

/**
 * @summary Upload a document to the case vault
 */
export const useUploadCaseFile = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof uploadCaseFile>>,
    TError,
    { id: string; data: BodyType<UploadCaseFileBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof uploadCaseFile>>,
  TError,
  { id: string; data: BodyType<UploadCaseFileBody> },
  TContext
> => {
  return useMutation(getUploadCaseFileMutationOptions(options));
};

/**
 * @summary Queue AI extraction and scoring for all case documents
 */
export const getAnalyzeCaseFilesUrl = (id: string) => {
  return `/api/cases/${id}/analyze`;
};

export const analyzeCaseFiles = async (
  id: string,
  options?: RequestInit,
): Promise<CreateCaseResponse> => {
  return customFetch<CreateCaseResponse>(getAnalyzeCaseFilesUrl(id), {
    ...options,
    method: "POST",
  });
};

export const getAnalyzeCaseFilesMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof analyzeCaseFiles>>,
    TError,
    { id: string },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof analyzeCaseFiles>>,
  TError,
  { id: string },
  TContext
> => {
  const mutationKey = ["analyzeCaseFiles"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
      ? options
      : { ...options, mutation: { ...options.mutation, mutationKey } }
    : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof analyzeCaseFiles>>,
    { id: string }
  > = (props) => {
    const { id } = props ?? {};

    return analyzeCaseFiles(id, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type AnalyzeCaseFilesMutationResult = NonNullable<
  Awaited<ReturnType<typeof analyzeCaseFiles>>
>;

export type AnalyzeCaseFilesMutationError = ErrorType<unknown>;

/**
 * @summary Queue AI extraction and scoring for all case documents
 */
export const useAnalyzeCaseFiles = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof analyzeCaseFiles>>,
    TError,

```

```

    { id: string },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof analyzeCaseFiles>>,
  TError,
  { id: string },
  TContext
> => {
  return useMutation(getAnalyzeCaseFilesMutationOptions(options));
};

/**
 * @summary List review queue items with optional filters
 */
export const getListReviewQueueUrl = (params?: ListReviewQueueParams) => {
  const normalizedParams = new URLSearchParams();

  Object.entries(params || {}).forEach(([key, value]) => {
    if (value !== undefined) {
      normalizedParams.append(key, value === null ? "null" : value.toString());
    }
  });

  const stringifiedParams = normalizedParams.toString();

  return stringifiedParams.length > 0
    ? `/api/review-queue?${stringifiedParams}`
    : `/api/review-queue`;
};

export const listReviewQueue = async (
  params?: ListReviewQueueParams,
  options?: RequestInit,
): Promise<ReviewQueueItem[]> => {
  return customFetch<ReviewQueueItem[]>(getListReviewQueueUrl(params), {
    ...options,
    method: "GET",
  });
};

export const getListReviewQueueQueryKey = (params?: ListReviewQueueParams) => {
  return [ `/api/review-queue`, ...(params ? [params] : []) ] as const;
};

export const getListReviewQueueQueryOptions = <
  TData = Awaited<ReturnType<typeof listReviewQueue>>,
  TError = ErrorType<unknown>,
>(
  params?: ListReviewQueueParams,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof listReviewQueue>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getListReviewQueueQueryKey(params);

  const queryFn: QueryFunction<Awaited<ReturnType<typeof listReviewQueue>>> = ({
    signal,
  }) => listReviewQueue(params, { signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof listReviewQueue>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type ListReviewQueueQueryResult = NonNullable<
  Awaited<ReturnType<typeof listReviewQueue>>
>;
export type ListReviewQueueQueryError = ErrorType<unknown>;

/**
 * @summary List review queue items with optional filters
 */
export function useListReviewQueue<
  TData = Awaited<ReturnType<typeof listReviewQueue>>,
  TError = ErrorType<unknown>,
>(
  params?: ListReviewQueueParams,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof listReviewQueue>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
)

```

```

): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getListReviewQueueQueryOptions(params, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Get review queue statistics
 */
export const getGetReviewQueueStatsUrl = () => {
  return `/api/review-queue/stats`;
};

export const getReviewQueueStats = async (
  options?: RequestInit,
): Promise<ReviewQueueStats> => {
  return customFetch<ReviewQueueStats>(getGetReviewQueueStatsUrl(), {
    ...options,
    method: "GET",
  });
};

export const getGetReviewQueueStatsQueryKey = () => {
  return [`/api/review-queue/stats`] as const;
};

export const getGetReviewQueueStatsQueryOptions = <
  TData = Awaited<ReturnType<typeof getReviewQueueStats>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getReviewQueueStats>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetReviewQueueStatsQueryKey();

  const queryFn: QueryFunction<
    Awaited<ReturnType<typeof getReviewQueueStats>>
  > = ({ signal }) => getReviewQueueStats({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof getReviewQueueStats>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetReviewQueueStatsQueryResult = NonNullable<
  Awaited<ReturnType<typeof getReviewQueueStats>>
>;
export type GetReviewQueueStatsQueryError = ErrorType<unknown>;

/**
 * @summary Get review queue statistics
 */
export function useGetReviewQueueStats<
  TData = Awaited<ReturnType<typeof getReviewQueueStats>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getReviewQueueStats>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetReviewQueueStatsQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Resolve a review queue item
 */
export const getResolveReviewItemUrl = (id: number) => {
  return `/api/review-queue/${id}`;
};

export const resolveReviewItem = async (
  id: number,
  resolveReviewBody: ResolveReviewBody,
  options?: RequestInit,
): Promise<ReviewQueueItem> => {

```

```

    return customFetch<ReviewQueueItem>(getResolveReviewItemUrl(id), {
      ...options,
      method: "PATCH",
      headers: { "Content-Type": "application/json", ...options?.headers },
      body: JSON.stringify(resolveReviewBody),
    });
  });
};

export const getResolveReviewItemMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof resolveReviewItem>>,
    TError,
    { id: number; data: BodyType<ResolveReviewBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof resolveReviewItem>>,
  TError,
  { id: number; data: BodyType<ResolveReviewBody> },
  TContext
> => {
  const mutationKey = ["resolveReviewItem"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof resolveReviewItem>>,
    { id: number; data: BodyType<ResolveReviewBody> }
  > = (props) => {
    const { id, data } = props ?? {};

    return resolveReviewItem(id, data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type ResolveReviewItemMutationResult = NonNullable<
  Awaited<ReturnType<typeof resolveReviewItem>>
>;
export type ResolveReviewItemMutationBody = BodyType<ResolveReviewBody>;
export type ResolveReviewItemMutationError = ErrorType<unknown>;

/**
 * @summary Resolve a review queue item
 */
export const useResolveReviewItem = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof resolveReviewItem>>,
    TError,
    { id: number; data: BodyType<ResolveReviewBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof resolveReviewItem>>,
  TError,
  { id: number; data: BodyType<ResolveReviewBody> },
  TContext
> => {
  return useMutation(getResolveReviewItemMutationOptions(options));
};

/**
 * @summary Search the NPI Registry for doctors and providers
 */
export const getSearchNpiUrl = (params?: SearchNpiParams) => {
  const normalizedParams = new URLSearchParams();

  Object.entries(params || {}).forEach(([key, value]) => {
    if (value !== undefined) {
      normalizedParams.append(key, value === null ? "null" : value.toString());
    }
  });
};

const stringifiedParams = normalizedParams.toString();

return stringifiedParams.length > 0
  ? `/api/npi/search?${stringifiedParams}`
  : `/api/npi/search`;
};

export const searchNpi = async (
  params?: SearchNpiParams,
  options?: RequestInit,

```

```

): Promise<NpiSearchResponse> => {
  return customFetch<NpiSearchResponse>(getSearchNpiUrl(params), {
    ...options,
    method: "GET",
  });
});

export const getSearchNpiQueryKey = (params?: SearchNpiParams) => {
  return [ `/api/npi/search`, ...(params ? [params] : []) ] as const;
};

export const getSearchNpiQueryOptions = <
  TData = Awaited<ReturnType<typeof searchNpi>>,
  TError = ErrorType<ErrorResponse>,
>(
  params?: SearchNpiParams,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof searchNpi>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getSearchNpiQueryKey(params);

  const queryFn: QueryFunction<Awaited<ReturnType<typeof searchNpi>>> = ({
    signal,
  }) => searchNpi(params, { signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof searchNpi>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type SearchNpiQueryResult = NonNullable<
  Awaited<ReturnType<typeof searchNpi>>
>;
export type SearchNpiQueryError = ErrorType<ErrorResponse>;

/**
 * @summary Search the NPI Registry for doctors and providers
 */

export function useSearchNpi<
  TData = Awaited<ReturnType<typeof searchNpi>>,
  TError = ErrorType<ErrorResponse>,
>(
  params?: SearchNpiParams,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof searchNpi>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getSearchNpiQueryOptions(params, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Look up a single provider by NPI number
 */
export const getLookupNpiUrl = (npi: string) => {
  return `/api/npi/lookup/${npi}`;
};

export const lookupNpi = async (
  npi: string,
  options?: RequestInit,
): Promise<NpiProvider> => {
  return customFetch<NpiProvider>(getLookupNpiUrl(npi), {
    ...options,
    method: "GET",
  });
};

export const getLookupNpiQueryKey = (npi: string) => {
  return [ `/api/npi/lookup/${npi}` ] as const;
};

export const getLookupNpiQueryOptions = <
  TData = Awaited<ReturnType<typeof lookupNpi>>,
  TError = ErrorType<ErrorResponse>,
>(

```

```

    napi: string,
    options?: {
      query?: UseQueryOptions<
        Awaited<ReturnType<typeof lookupNpi>>,
        TError,
        TData
      >;
      request?: SecondParameter<typeof customFetch>;
    },
  ) => {
    const { query: queryOptions, request: requestOptions } = options ?? {};

    const queryKey = queryOptions?.queryKey ?? getLookupNpiQueryKey(npi);

    const queryFn: QueryFunction<Awaited<ReturnType<typeof lookupNpi>>> = ({
      signal,
    }) => lookupNpi(npi, { signal, ...requestOptions });

    return {
      queryKey,
      queryFn,
      enabled: !!npi,
      ...queryOptions,
    } as UseQueryOptions<Awaited<ReturnType<typeof lookupNpi>>, TError, TData> & {
      queryKey: QueryKey;
    };
  };

export type LookupNpiQueryResult = NonNullable<
  Awaited<ReturnType<typeof lookupNpi>>
>;
export type LookupNpiQueryError = ErrorType<ErrorResponse>;

/**
 * @summary Look up a single provider by NPI number
 */

export function useLookupNpi<
  TData = Awaited<ReturnType<typeof lookupNpi>>,
  TError = ErrorType<ErrorResponse>,
>(
  napi: string,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof lookupNpi>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getLookupNpiQueryOptions(npi, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Get all tort campaign form configurations
 */
export const getGetFormConfigsUrl = () => {
  return `/api/forms/config`;
};

export const getFormConfigs = async (
  options?: RequestInit,
): Promise<FormConfigList> => {
  return customFetch<FormConfigList>(getGetFormConfigsUrl(), {
    ...options,
    method: "GET",
  });
};

export const getGetFormConfigsQueryKey = () => {
  return [`/api/forms/config`] as const;
};

export const getGetFormConfigsQueryOptions = <
  TData = Awaited<ReturnType<typeof getFormConfigs>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getFormConfigs>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetFormConfigsQueryKey();

  const queryFn: QueryFunction<Awaited<ReturnType<typeof getFormConfigs>>> = ({
    signal,
  }) => {

```

```

    }) => getFormConfigs({ signal, ...requestOptions });

    return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
      Awaited<ReturnType<typeof getFormConfigs>>,
      Error,
      TData
    > & { queryKey: QueryKey };
  };

export type GetFormConfigsQueryResult = NonNullable<
  Awaited<ReturnType<typeof getFormConfigs>>
>;
export type GetFormConfigsQueryError = ErrorType<unknown>;

/**
 * @summary Get all tort campaign form configurations
 */
export function useGetFormConfigs<
  TData = Awaited<ReturnType<typeof getFormConfigs>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getFormConfigs>>,
    Error,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetFormConfigsQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Get a single tort form configuration
 */
export const getGetFormConfigUrl = (tortId: string) => {
  return `/api/forms/config/${tortId}`;
};

export const getFormConfig = async (
  tortId: string,
  options?: RequestInit,
): Promise<FormConfig> => {
  return customFetch<FormConfig>(getGetFormConfigUrl(tortId), {
    ...options,
    method: "GET",
  });
};

export const getGetFormConfigQueryKey = (tortId: string) => {
  return [`/api/forms/config/${tortId}`] as const;
};

export const getGetFormConfigQueryOptions = <
  TData = Awaited<ReturnType<typeof getFormConfig>>,
  TError = ErrorType<unknown>,
>(
  tortId: string,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getFormConfig>>,
      Error,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetFormConfigQueryKey(tortId);

  const queryFn: QueryFunction<Awaited<ReturnType<typeof getFormConfig>>> = ({
    signal,
  }) => getFormConfig(tortId, { signal, ...requestOptions });

  return {
    queryKey,
    queryFn,
    enabled: !!tortId,
    ...queryOptions,
  } as UseQueryOptions<
    Awaited<ReturnType<typeof getFormConfig>>,
    Error,
    TData
  > & { queryKey: QueryKey };
};

export type GetFormConfigQueryResult = NonNullable<
  Awaited<ReturnType<typeof getFormConfig>>
>;
export type GetFormConfigQueryError = ErrorType<unknown>;

```

```

/**
 * @summary Get a single tort form configuration
 */
export function useGetFormConfig<
  TData = Awaited<ReturnType<typeof getFormConfig>>,
  TError = ErrorType<unknown>,
>({
  tortId: string,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getFormConfig>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetFormConfigQueryOptions(tortId, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Update a tort form configuration (admin only)
 */
export const getUpdateFormConfigUrl = (tortId: string) => {
  return `/api/forms/config/${tortId}`;
};

export const updateFormConfig = async (
  tortId: string,
  formConfigUpdate: FormConfigUpdate,
  options?: RequestInit,
): Promise<FormConfig> => {
  return customFetch<FormConfig>(getUpdateFormConfigUrl(tortId), {
    ...options,
    method: "PUT",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(formConfigUpdate),
  });
};

export const getUpdateFormConfigMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>({options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof updateFormConfig>>,
    TError,
    { tortId: string; data: BodyType<FormConfigUpdate> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}}): UseMutationOptions<
  Awaited<ReturnType<typeof updateFormConfig>>,
  TError,
  { tortId: string; data: BodyType<FormConfigUpdate> },
  TContext
> => {
  const mutationKey = ["updateFormConfig"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof updateFormConfig>>,
    { tortId: string; data: BodyType<FormConfigUpdate> }
  > = (props) => {
    const { tortId, data } = props ?? {};

    return updateFormConfig(tortId, data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type UpdateFormConfigMutationResult = NonNullable<
  Awaited<ReturnType<typeof updateFormConfig>>
>;
export type UpdateFormConfigMutationBody = BodyType<FormConfigUpdate>;
export type UpdateFormConfigMutationError = ErrorType<unknown>;

/**
 * @summary Update a tort form configuration (admin only)
 */
export const useUpdateFormConfig = <
  TError = ErrorType<unknown>,

```



```

TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof updateFormConfig>>,
    Error,
    { tortId: string; data: BodyType<FormConfigUpdate> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof updateFormConfig>>,
  Error,
  { tortId: string; data: BodyType<FormConfigUpdate> },
  TContext
> => {
  return useMutation(getUpdateFormConfigMutationOptions(options));
};

/**
 * @summary Add a custom field to a tort form (admin only)
 */
export const getAddCustomFieldUrl = (tortId: string) => {
  return `/api/forms/config/${tortId}/fields`;
};

export const addCustomField = async (
  tortId: string,
  customField: CustomField,
  options?: RequestInit,
): Promise<FormConfig> => {
  return customFetch<FormConfig>(getAddCustomFieldUrl(tortId), {
    ...options,
    method: "POST",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(customField),
  });
};

export const getAddCustomFieldMutationOptions = <
  Error = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof addCustomField>>,
    Error,
    { tortId: string; data: BodyType<CustomField> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof addCustomField>>,
  Error,
  { tortId: string; data: BodyType<CustomField> },
  TContext
> => {
  const mutationKey = ["addCustomField"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
      ? options
      : { ...options, mutation: { ...options.mutation, mutationKey } }
    : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof addCustomField>>,
    { tortId: string; data: BodyType<CustomField> }
  > = (props) => {
    const { tortId, data } = props ?? {};

    return addCustomField(tortId, data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type AddCustomFieldMutationResult = NonNullable<
  Awaited<ReturnType<typeof addCustomField>>
>;
export type AddCustomFieldMutationBody = BodyType<CustomField>;
export type AddCustomFieldMutationError = ErrorType<unknown>;

/**
 * @summary Add a custom field to a tort form (admin only)
 */
export const useAddCustomField = <
  Error = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof addCustomField>>,
    Error,
    { tortId: string; data: BodyType<CustomField> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<

```

```

    Awaited<ReturnType<typeof addCustomField>>,
    TError,
    { tortId: string; data: BodyType<CustomField> },
    TContext
  > => {
    return useMutation(getAddCustomFieldMutationOptions(options));
  };

/**
 * @summary Remove a custom field from a tort form (admin only)
 */
export const getRemoveCustomFieldUrl = (tortId: string, key: string) => {
  return `/api/forms/config/${tortId}/fields/${key}`;
};

export const removeCustomField = async (
  tortId: string,
  key: string,
  options?: RequestInit,
): Promise<FormConfig> => {
  return customFetch<FormConfig>(getRemoveCustomFieldUrl(tortId, key), {
    ...options,
    method: "DELETE",
  });
};

export const getRemoveCustomFieldMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof removeCustomField>>,
    TError,
    { tortId: string; key: string },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof removeCustomField>>,
  TError,
  { tortId: string; key: string },
  TContext
> => {
  const mutationKey = ["removeCustomField"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof removeCustomField>>,
    { tortId: string; key: string }
  > = (props) => {
    const { tortId, key } = props ?? {};

    return removeCustomField(tortId, key, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type RemoveCustomFieldMutationResult = NonNullable<
  Awaited<ReturnType<typeof removeCustomField>>
>;

export type RemoveCustomFieldMutationError = ErrorType<unknown>;

/**
 * @summary Remove a custom field from a tort form (admin only)
 */
export const useRemoveCustomField = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof removeCustomField>>,
    TError,
    { tortId: string; key: string },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof removeCustomField>>,
  TError,
  { tortId: string; key: string },
  TContext
> => {
  return useMutation(getRemoveCustomFieldMutationOptions(options));
};

/**
 * @summary Validate email (RFC + typo detection)
 */
export const getValidateEmailUrl = () => {

```

```

    return `/api/forms/validate/email`;
  };

export const validateEmail = async (
  validateEmailBody: ValidateEmailBody,
  options?: RequestInit,
): Promise<EmailValidationResult> => {
  return customFetch<EmailValidationResult>(getValidateEmailUrl(), {
    ...options,
    method: "POST",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(validateEmailBody),
  });
};

export const getValidateEmailMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof validateEmail>>,
    TError,
    { data: BodyType<ValidateEmailBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof validateEmail>>,
  TError,
  { data: BodyType<ValidateEmailBody> },
  TContext
> => {
  const mutationKey = ["validateEmail"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof validateEmail>>,
    { data: BodyType<ValidateEmailBody> }
  > = (props) => {
    const { data } = props ?? {};

    return validateEmail(data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type ValidateEmailMutationResult = NonNullable<
  Awaited<ReturnType<typeof validateEmail>>
>;
export type ValidateEmailMutationBody = BodyType<ValidateEmailBody>;
export type ValidateEmailMutationError = ErrorType<unknown>;

/**
 * @summary Validate email (RFC + typo detection)
 */
export const useValidateEmail = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof validateEmail>>,
    TError,
    { data: BodyType<ValidateEmailBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof validateEmail>>,
  TError,
  { data: BodyType<ValidateEmailBody> },
  TContext
> => {
  return useMutation(getValidateEmailMutationOptions(options));
};

/**
 * @summary Validate address (US format)
 */
export const getValidateAddressUrl = () => {
  return `/api/forms/validate/address`;
};

export const validateAddress = async (
  validateAddressBody: ValidateAddressBody,
  options?: RequestInit,
): Promise<AddressValidationResult> => {
  return customFetch<AddressValidationResult>(getValidateAddressUrl(), {
    ...options,
    method: "POST",
    headers: { "Content-Type": "application/json", ...options?.headers },
  });
};

```

```

    body: JSON.stringify(validateAddressBody),
  });
};

export const getValidateAddressMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof validateAddress>>,
    TError,
    { data: BodyType<ValidateAddressBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof validateAddress>>,
  TError,
  { data: BodyType<ValidateAddressBody> },
  TContext
> => {
  const mutationKey = ["validateAddress"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
    : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof validateAddress>>,
    { data: BodyType<ValidateAddressBody> }
  > = (props) => {
    const { data } = props ?? {};

    return validateAddress(data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type ValidateAddressMutationResult = NonNullable<
  Awaited<ReturnType<typeof validateAddress>>
>;
export type ValidateAddressMutationBody = BodyType<ValidateAddressBody>;
export type ValidateAddressMutationError = ErrorType<unknown>;

/**
 * @summary Validate address (US format)
 */
export const useValidateAddress = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof validateAddress>>,
    TError,
    { data: BodyType<ValidateAddressBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof validateAddress>>,
  TError,
  { data: BodyType<ValidateAddressBody> },
  TContext
> => {
  return useMutation(getValidateAddressMutationOptions(options));
};

/**
 * @summary Submit intake form with full compliance validation
 */
export const getSubmitFormUrl = () => {
  return `/api/forms/submit`;
};

export const submitForm = async (
  formSubmission: FormSubmission,
  options?: RequestInit,
): Promise<FormSubmissionResult> => {
  return customFetch<FormSubmissionResult>(getSubmitFormUrl(), {
    ...options,
    method: "POST",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(formSubmission),
  });
};

export const getSubmitFormMutationOptions = <
  TError = ErrorType<FormRejection>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof submitForm>>,
    TError,

```

```

    { data: BodyType<FormSubmission> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof submitForm>>,
  TError,
  { data: BodyType<FormSubmission> },
  TContext
> => {
  const mutationKey = ["submitForm"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
    : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof submitForm>>,
    { data: BodyType<FormSubmission> }
  > = (props) => {
    const { data } = props ?? {};

    return submitForm(data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type SubmitFormMutationResult = NonNullable<
  Awaited<ReturnType<typeof submitForm>>
>;
export type SubmitFormMutationBody = BodyType<FormSubmission>;
export type SubmitFormMutationError = ErrorType<FormRejection>;

/**
 * @summary Submit intake form with full compliance validation
 */
export const useSubmitForm = <
  TError = ErrorType<FormRejection>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof submitForm>>,
    TError,
    { data: BodyType<FormSubmission> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof submitForm>>,
  TError,
  { data: BodyType<FormSubmission> },
  TContext
> => {
  return useMutation(getSubmitFormMutationOptions(options));
};

/**
 * @summary Get embeddable JavaScript form for a tort campaign
 */
export const getGetFormEmbedUrl = (tortId: string) => {
  return `/api/forms/embed/${tortId}`;
};

export const getFormEmbed = async (
  tortId: string,
  options?: RequestInit,
): Promise<string> => {
  return customFetch<string>(getGetFormEmbedUrl(tortId), {
    ...options,
    method: "GET",
  });
};

export const getGetFormEmbedQueryKey = (tortId: string) => {
  return [`/api/forms/embed/${tortId}`] as const;
};

export const getGetFormEmbedQueryOptions = <
  TData = Awaited<ReturnType<typeof getFormEmbed>>,
  TError = ErrorType<unknown>,
>(
  tortId: string,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getFormEmbed>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

```

```

const queryKey = queryOptions?.queryKey ?? getGetFormEmbedQueryKey(tortId);

const queryFn: QueryFunction<Awaited<ReturnType<typeof getFormEmbed>>> = ({
  signal,
}) => getFormEmbed(tortId, { signal, ...requestOptions });

return {
  queryKey,
  queryFn,
  enabled: !!tortId,
  ...queryOptions,
} as UseQueryOptions<
  Awaited<ReturnType<typeof getFormEmbed>>,
  TError,
  TData
> & { queryKey: QueryKey };
};

export type GetFormEmbedQueryResult = NonNullable<
  Awaited<ReturnType<typeof getFormEmbed>>
>;
export type GetFormEmbedQueryError = ErrorType<unknown>;

/**
 * @summary Get embeddable JavaScript form for a tort campaign
 */

export function useGetFormEmbed<
  TData = Awaited<ReturnType<typeof getFormEmbed>>,
  TError = ErrorType<unknown>,
>({
  tortId: string,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getFormEmbed>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetFormEmbedQueryOptions(tortId, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Run background check on a person
 */
export const getRunBackgroundCheckUrl = () => {
  return `/api/forms/background-check`;
};

export const runBackgroundCheck = async (
  runBackgroundCheckBody: RunBackgroundCheckBody,
  options?: RequestInit,
): Promise<BackgroundCheckResult> => {
  return customFetch<BackgroundCheckResult>(getRunBackgroundCheckUrl(), {
    ...options,
    method: "POST",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(runBackgroundCheckBody),
  });
};

export const getRunBackgroundCheckMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>({options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof runBackgroundCheck>>,
    TError,
    { data: BodyType<RunBackgroundCheckBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof runBackgroundCheck>>,
  TError,
  { data: BodyType<RunBackgroundCheckBody> },
  TContext
> => {
  const mutationKey = ["runBackgroundCheck"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof runBackgroundCheck>>,

```

```

    { data: BodyType<RunBackgroundCheckBody> }
  > = (props) => {
    const { data } = props ?? {};

    return runBackgroundCheck(data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type RunBackgroundCheckMutationResult = NonNullable<
  Awaited<ReturnType<typeof runBackgroundCheck>>
>;
export type RunBackgroundCheckMutationBody = BodyType<RunBackgroundCheckBody>;
export type RunBackgroundCheckMutationError = ErrorType<unknown>;

/**
 * @summary Run background check on a person
 */
export const useRunBackgroundCheck = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof runBackgroundCheck>>,
    TError,
    { data: BodyType<RunBackgroundCheckBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof runBackgroundCheck>>,
  TError,
  { data: BodyType<RunBackgroundCheckBody> },
  TContext
> => {
  return useMutation(getRunBackgroundCheckMutationOptions(options));
};

/**
 * @summary Run background check on an existing lead
 */
export const getRunLeadBackgroundCheckUrl = (id: number) => {
  return `/api/forms/background-check/lead/${id}`;
};

export const runLeadBackgroundCheck = async (
  id: number,
  options?: RequestInit,
): Promise<BackgroundCheckResult> => {
  return customFetch<BackgroundCheckResult>(getRunLeadBackgroundCheckUrl(id), {
    ...options,
    method: "POST",
  });
};

export const getRunLeadBackgroundCheckMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof runLeadBackgroundCheck>>,
    TError,
    { id: number },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof runLeadBackgroundCheck>>,
  TError,
  { id: number },
  TContext
> => {
  const mutationKey = ["runLeadBackgroundCheck"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
      ? options
      : { ...options, mutation: { ...options.mutation, mutationKey } }
    : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof runLeadBackgroundCheck>>,
    { id: number }
  > = (props) => {
    const { id } = props ?? {};

    return runLeadBackgroundCheck(id, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type RunLeadBackgroundCheckMutationResult = NonNullable<
  Awaited<ReturnType<typeof runLeadBackgroundCheck>>
>;

```

```

export type RunLeadBackgroundCheckMutationError = ErrorType<unknown>;

/**
 * @summary Run background check on an existing lead
 */
export const useRunLeadBackgroundCheck = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof runLeadBackgroundCheck>>,
    TError,
    { id: number },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof runLeadBackgroundCheck>>,
  TError,
  { id: number },
  TContext
> => {
  return useMutation(getRunLeadBackgroundCheckMutationOptions(options));
};

/**
 * @summary Get all tort categories with grouped torts
 */
export const getGetTortCategoriesUrl = () => {
  return `/api/forms/categories`;
};

export const getTortCategories = async (
  options?: RequestInit,
): Promise<GetTortCategories200Item[]> => {
  return customFetch<GetTortCategories200Item[]>(getGetTortCategoriesUrl(), {
    ...options,
    method: "GET",
  });
};

export const getGetTortCategoriesQueryKey = () => {
  return [`/api/forms/categories`] as const;
};

export const getGetTortCategoriesQueryOptions = <
  TData = Awaited<ReturnType<typeof getTortCategories>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getTortCategories>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetTortCategoriesQueryKey();

  const queryFn: QueryFunction<
    Awaited<ReturnType<typeof getTortCategories>>
  > = ({ signal }) => getTortCategories({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof getTortCategories>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetTortCategoriesQueryResult = NonNullable<
  Awaited<ReturnType<typeof getTortCategories>>
>;
export type GetTortCategoriesQueryError = ErrorType<unknown>;

/**
 * @summary Get all tort categories with grouped torts
 */
export function useGetTortCategories<
  TData = Awaited<ReturnType<typeof getTortCategories>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getTortCategories>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetTortCategoriesQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };
};

```



```

    return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Verify physician NPI and match taxonomy to diagnosis
 */
export const getVerifyNpiUrl = () => {
    return `/api/forms/npi-verify`;
};

export const verifyNpi = async (
    verifyNpiBody: VerifyNpiBody,
    options?: RequestInit,
): Promise<NpiVerifyResult> => {
    return customFetch<NpiVerifyResult>(getVerifyNpiUrl(), {
        ...options,
        method: "POST",
        headers: { "Content-Type": "application/json", ...options?.headers },
        body: JSON.stringify(verifyNpiBody),
    });
};

export const getVerifyNpiMutationOptions = <
    TError = ErrorType<unknown>,
    TContext = unknown,
>(options?: {
    mutation?: UseMutationOptions<
        Awaited<ReturnType<typeof verifyNpi>>,
        TError,
        { data: BodyType<VerifyNpiBody> },
        TContext
    >;
    request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
    Awaited<ReturnType<typeof verifyNpi>>,
    TError,
    { data: BodyType<VerifyNpiBody> },
    TContext
> => {
    const mutationKey = ["verifyNpi"];
    const { mutation: mutationOptions, request: requestOptions } = options
        ? options.mutation &&
          "mutationKey" in options.mutation &&
          options.mutation.mutationKey
        ? options
        : { ...options, mutation: { ...options.mutation, mutationKey } }
    : { mutation: { mutationKey }, request: undefined };

    const mutationFn: MutationFunction<
        Awaited<ReturnType<typeof verifyNpi>>,
        { data: BodyType<VerifyNpiBody> }
    > = (props) => {
        const { data } = props ?? {};

        return verifyNpi(data, requestOptions);
    };

    return { mutationFn, ...mutationOptions };
};

export type VerifyNpiMutationResult = NonNullable<
    Awaited<ReturnType<typeof verifyNpi>>
>;
export type VerifyNpiMutationBody = BodyType<VerifyNpiBody>;
export type VerifyNpiMutationError = ErrorType<unknown>;

/**
 * @summary Verify physician NPI and match taxonomy to diagnosis
 */
export const useVerifyNpi = <
    TError = ErrorType<unknown>,
    TContext = unknown,
>(options?: {
    mutation?: UseMutationOptions<
        Awaited<ReturnType<typeof verifyNpi>>,
        TError,
        { data: BodyType<VerifyNpiBody> },
        TContext
    >;
    request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
    Awaited<ReturnType<typeof verifyNpi>>,
    TError,
    { data: BodyType<VerifyNpiBody> },
    TContext
> => {
    return useMutation(getVerifyNpiMutationOptions(options));
};

/**
 * @summary Run fraud detection on claim data
 */
export const getRunFraudCheckUrl = () => {
    return `/api/forms/fraud-check`;
};

```

```

export const runFraudCheck = async (
  runFraudCheckBody: RunFraudCheckBody,
  options?: RequestInit,
): Promise<FraudCheckResult> => {
  return customFetch<FraudCheckResult>(getRunFraudCheckUrl(), {
    ...options,
    method: "POST",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(runFraudCheckBody),
  });
};

export const getRunFraudCheckMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof runFraudCheck>>,
    TError,
    { data: BodyType<RunFraudCheckBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof runFraudCheck>>,
  TError,
  { data: BodyType<RunFraudCheckBody> },
  TContext
> => {
  const mutationKey = ["runFraudCheck"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
      ? options
      : { ...options, mutation: { ...options.mutation, mutationKey } }
    : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof runFraudCheck>>,
    { data: BodyType<RunFraudCheckBody> }
  > = (props) => {
    const { data } = props ?? {};

    return runFraudCheck(data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type RunFraudCheckMutationResult = NonNullable<
  Awaited<ReturnType<typeof runFraudCheck>>
>;
export type RunFraudCheckMutationBody = BodyType<RunFraudCheckBody>;
export type RunFraudCheckMutationError = ErrorType<unknown>;

/**
 * @summary Run fraud detection on claim data
 */
export const useRunFraudCheck = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof runFraudCheck>>,
    TError,
    { data: BodyType<RunFraudCheckBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof runFraudCheck>>,
  TError,
  { data: BodyType<RunFraudCheckBody> },
  TContext
> => {
  return useMutation(getRunFraudCheckMutationOptions(options));
};

/**
 * @summary Escalate case to FBI tip line and log event
 */
export const getEscalateToFbiUrl = () => {
  return `/api/forms/escalate/fbi`;
};

export const escalateToFbi = async (
  escalateToFbiBody: EscalateToFbiBody,
  options?: RequestInit,
): Promise<EscalateToFbi200> => {
  return customFetch<EscalateToFbi200>(getEscalateToFbiUrl(), {
    ...options,
    method: "POST",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(escalateToFbiBody),
  });
};

```

```

export const getEscalateToFbiMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof escalateToFbi>>,
    TError,
    { data: BodyType<EscalateToFbiBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof escalateToFbi>>,
  TError,
  { data: BodyType<EscalateToFbiBody> },
  TContext
> => {
  const mutationKey = ["escalateToFbi"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof escalateToFbi>>,
    { data: BodyType<EscalateToFbiBody> }
  > = (props) => {
    const { data } = props ?? {};

    return escalateToFbi(data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type EscalateToFbiMutationResult = NonNullable<
  Awaited<ReturnType<typeof escalateToFbi>>
>;
export type EscalateToFbiMutationBody = BodyType<EscalateToFbiBody>;
export type EscalateToFbiMutationError = ErrorType<unknown>;

/**
 * @summary Escalate case to FBI tip line and log event
 */
export const useEscalateToFbi = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof escalateToFbi>>,
    TError,
    { data: BodyType<EscalateToFbiBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof escalateToFbi>>,
  TError,
  { data: BodyType<EscalateToFbiBody> },
  TContext
> => {
  return useMutation(getEscalateToFbiMutationOptions(options));
};

/**
 * @summary List all paralegals
 */
export const getListParalegalsUrl = () => {
  return `/api/paralegals`;
};

export const listParalegals = async (
  options?: RequestInit,
): Promise<Paralegal[]> => {
  return customFetch<Paralegal[]>(getListParalegalsUrl(), {
    ...options,
    method: "GET",
  });
};

export const getListParalegalsQueryKey = () => {
  return [`/api/paralegals`] as const;
};

export const getListParalegalsQueryOptions = <
  TData = Awaited<ReturnType<typeof listParalegals>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof listParalegals>>,
    TError,
    TData
  >;
});

```

```

    request?: SecondParameter<typeof customFetch>;
  }) => {
    const { query: queryOptions, request: requestOptions } = options ?? {};

    const queryKey = queryOptions?.queryKey ?? getListParalegalsQueryKey();

    const queryFn: QueryFunction<Awaited<ReturnType<typeof listParalegals>>> = ({
      signal,
    }) => listParalegals({ signal, ...requestOptions });

    return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
      Awaited<ReturnType<typeof listParalegals>>,
      Error,
      TData
    > & { queryKey: QueryKey };
  };

export type ListParalegalsQueryResult = NonNullable<
  Awaited<ReturnType<typeof listParalegals>>
>;
export type ListParalegalsQueryError = ErrorType<unknown>;

/**
 * @summary List all paralegals
 */
export function useListParalegals<
  TData = Awaited<ReturnType<typeof listParalegals>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof listParalegals>>,
    Error,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getListParalegalsQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Create a new paralegal
 */
export const getCreateParalegalUrl = () => {
  return `/api/paralegals`;
};

export const createParalegal = async (
  createParalegalBody: CreateParalegalBody,
  options?: RequestInit,
): Promise<Paralegal> => {
  return customFetch<Paralegal>(getCreateParalegalUrl(), {
    ...options,
    method: "POST",
    headers: { "Content-Type": "application/json", ...options?.headers },
    body: JSON.stringify(createParalegalBody),
  });
};

export const getCreateParalegalMutationOptions = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof createParalegal>>,
    Error,
    { data: BodyType<CreateParalegalBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof createParalegal>>,
  TError,
  { data: BodyType<CreateParalegalBody> },
  TContext
> => {
  const mutationKey = ["createParalegal"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof createParalegal>>,
    { data: BodyType<CreateParalegalBody> }
  > = (props) => {
    const { data } = props ?? {};

```

```

    return createParalegal(data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type CreateParalegalMutationResult = NonNullable<
  Awaited<ReturnType<typeof createParalegal>>
>;
export type CreateParalegalMutationBody = BodyType<CreateParalegalBody>;
export type CreateParalegalMutationError = ErrorType<unknown>;

/**
 * @summary Create a new paralegal
 */
export const useCreateParalegal = <
  TError = ErrorType<unknown>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof createParalegal>>,
    TError,
    { data: BodyType<CreateParalegalBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof createParalegal>>,
  TError,
  { data: BodyType<CreateParalegalBody> },
  TContext
> => {
  return useMutation(getCreateParalegalMutationOptions(options));
};

/**
 * @summary Get paralegal detail with assigned leads
 */
export const getGetParalegalUrl = (id: number) => {
  return `/api/paralegals/${id}`;
};

export const getParalegal = async (
  id: number,
  options?: RequestInit,
): Promise<ParalegalDetail> => {
  return customFetch<ParalegalDetail>(getGetParalegalUrl(id), {
    ...options,
    method: "GET",
  });
};

export const getGetParalegalQueryKey = (id: number) => {
  return [`/api/paralegals/${id}`] as const;
};

export const getGetParalegalQueryOptions = <
  TData = Awaited<ReturnType<typeof getParalegal>>,
  TError = ErrorType<ErrorResponse>,
>(
  id: number,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getParalegal>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetParalegalQueryKey(id);

  const queryFn: QueryFunction<Awaited<ReturnType<typeof getParalegal>>> = ({
    signal,
  }) => getParalegal(id, { signal, ...requestOptions });

  return {
    queryKey,
    queryFn,
    enabled: !!id,
    ...queryOptions,
  } as UseQueryOptions<
    Awaited<ReturnType<typeof getParalegal>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetParalegalQueryResult = NonNullable<
  Awaited<ReturnType<typeof getParalegal>>
>;
export type GetParalegalQueryError = ErrorType<ErrorResponse>;

/**
 * @summary Get paralegal detail with assigned leads

```

```

*/
export function useGetParalegal<
  TData = Awaited<ReturnType<typeof getParalegal>>,
  TError = ErrorType<ErrorResponse>,
>(
  id: number,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getParalegal>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetParalegalQueryOptions(id, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Get paralegal performance breakdown
 */
export const getGetParalegalPerformanceUrl = (id: number) => {
  return `/api/paralegals/${id}/performance`;
};

export const getParalegalPerformance = async (
  id: number,
  options?: RequestInit,
): Promise<ParalegalPerformance> => {
  return customFetch<ParalegalPerformance>(getGetParalegalPerformanceUrl(id), {
    ...options,
    method: "GET",
  });
};

export const getGetParalegalPerformanceQueryKey = (id: number) => {
  return [ `/api/paralegals/${id}/performance` ] as const;
};

export const getGetParalegalPerformanceQueryOptions = <
  TData = Awaited<ReturnType<typeof getParalegalPerformance>>,
  TError = ErrorType<unknown>,
>(
  id: number,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getParalegalPerformance>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey =
    queryOptions?.queryKey ?? getGetParalegalPerformanceQueryKey(id);

  const queryFn: QueryFunction<
    Awaited<ReturnType<typeof getParalegalPerformance>>
  > = ({ signal }) =>
    getParalegalPerformance(id, { signal, ...requestOptions });

  return {
    queryKey,
    queryFn,
    enabled: !!id,
    ...queryOptions,
  } as UseQueryOptions<
    Awaited<ReturnType<typeof getParalegalPerformance>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetParalegalPerformanceQueryResult = NonNullable<
  Awaited<ReturnType<typeof getParalegalPerformance>>
>;
export type GetParalegalPerformanceQueryError = ErrorType<unknown>;

/**
 * @summary Get paralegal performance breakdown
 */
export function useGetParalegalPerformance<
  TData = Awaited<ReturnType<typeof getParalegalPerformance>>,
  TError = ErrorType<unknown>,
>(
  id: number,
  options?: {

```

```

    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getParalegalPerformance>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetParalegalPerformanceQueryOptions(id, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Full analytics overview across all modules
 */
export const getGetAnalyticsOverviewUrl = () => {
  return `/api/analytics/overview`;
};

export const getAnalyticsOverview = async (
  options?: RequestInit,
): Promise<AnalyticsOverview> => {
  return customFetch<AnalyticsOverview>(getGetAnalyticsOverviewUrl(), {
    ...options,
    method: "GET",
  });
};

export const getGetAnalyticsOverviewQueryKey = () => {
  return [ `/api/analytics/overview` ] as const;
};

export const getGetAnalyticsOverviewQueryOptions = <
  TData = Awaited<ReturnType<typeof getAnalyticsOverview>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getAnalyticsOverview>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetAnalyticsOverviewQueryKey();

  const queryFn: QueryFunction<
    Awaited<ReturnType<typeof getAnalyticsOverview>>
  > = ({ signal }) => getAnalyticsOverview({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof getAnalyticsOverview>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetAnalyticsOverviewQueryResult = NonNullable<
  Awaited<ReturnType<typeof getAnalyticsOverview>>
>;
export type GetAnalyticsOverviewQueryError = ErrorType<unknown>;

/**
 * @summary Full analytics overview across all modules
 */
export function useGetAnalyticsOverview<
  TData = Awaited<ReturnType<typeof getAnalyticsOverview>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getAnalyticsOverview>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetAnalyticsOverviewQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Daily pipeline trend data
 */
export const getGetPipelineTrendUrl = () => {
  return `/api/analytics/pipeline-trend`;
};

```

```

};

export const getPipelineTrend = async (
  options?: RequestInit,
): Promise<PipelineTrendPoint[]> => {
  return customFetch<PipelineTrendPoint[]>(getGetPipelineTrendUrl(), {
    ...options,
    method: "GET",
  });
};

export const getGetPipelineTrendQueryKey = () => {
  return [`/api/analytics/pipeline-trend`] as const;
};

export const getGetPipelineTrendQueryOptions = <
  TData = Awaited<ReturnType<typeof getPipelineTrend>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getPipelineTrend>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetPipelineTrendQueryKey();

  const queryFn: QueryFunction<
    Awaited<ReturnType<typeof getPipelineTrend>>
  > = ({ signal }) => getPipelineTrend({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof getPipelineTrend>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetPipelineTrendQueryResult = NonNullable<
  Awaited<ReturnType<typeof getPipelineTrend>>
>;
export type GetPipelineTrendQueryError = ErrorType<unknown>;

/**
 * @summary Daily pipeline trend data
 */

export function useGetPipelineTrend<
  TData = Awaited<ReturnType<typeof getPipelineTrend>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getPipelineTrend>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetPipelineTrendQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Conversion funnel stages
 */
export const getGetConversionFunnelUrl = () => {
  return [`/api/analytics/conversion-funnel`];
};

export const getConversionFunnel = async (
  options?: RequestInit,
): Promise<FunnelStage[]> => {
  return customFetch<FunnelStage[]>(getGetConversionFunnelUrl(), {
    ...options,
    method: "GET",
  });
};

export const getGetConversionFunnelQueryKey = () => {
  return [`/api/analytics/conversion-funnel`] as const;
};

export const getGetConversionFunnelQueryOptions = <
  TData = Awaited<ReturnType<typeof getConversionFunnel>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getConversionFunnel>>,
    TError,

```



```

    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetConversionFunnelQueryKey();

  const queryFn: QueryFunction<
    Awaited<ReturnType<typeof getConversionFunnel>>
  > = ({ signal }) => getConversionFunnel({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof getConversionFunnel>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetConversionFunnelQueryResult = NonNullable<
  Awaited<ReturnType<typeof getConversionFunnel>>
>;
export type GetConversionFunnelQueryError = ErrorType<unknown>;

/**
 * @summary Conversion funnel stages
 */
export function useGetConversionFunnel<
  TData = Awaited<ReturnType<typeof getConversionFunnel>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getConversionFunnel>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetConversionFunnelQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Breakdown by tort type
 */
export const getGetTortBreakdownUrl = () => {
  return `/api/analytics/tort-breakdown`;
};

export const getTortBreakdown = async (
  options?: RequestInit,
): Promise<TortBreakdownRow[]> => {
  return customFetch<TortBreakdownRow[]>(getGetTortBreakdownUrl(), {
    ...options,
    method: "GET",
  });
};

export const getGetTortBreakdownQueryKey = () => {
  return [`/api/analytics/tort-breakdown`] as const;
};

export const getGetTortBreakdownQueryOptions = <
  TData = Awaited<ReturnType<typeof getTortBreakdown>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getTortBreakdown>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetTortBreakdownQueryKey();

  const queryFn: QueryFunction<
    Awaited<ReturnType<typeof getTortBreakdown>>
  > = ({ signal }) => getTortBreakdown({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof getTortBreakdown>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetTortBreakdownQueryResult = NonNullable<
  Awaited<ReturnType<typeof getTortBreakdown>>
>;

```

```

export type GetTortBreakdownQueryError = ErrorType<unknown>;

/**
 * @summary Breakdown by tort type
 */

export function useGetTortBreakdown<
  TData = Awaited<ReturnType<typeof getTortBreakdown>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getTortBreakdown>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetTortBreakdownQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Paralegal performance leaderboard
 */
export const getGetParalegalLeaderboardUrl = () => {
  return `/api/analytics/paralegal-leaderboard`;
};

export const getParalegalLeaderboard = async (
  options?: RequestInit,
): Promise<ParalegalLeaderboardRow[]> => {
  return customFetch<ParalegalLeaderboardRow[]>(<
    getGetParalegalLeaderboardUrl(),
    {
      ...options,
      method: "GET",
    }
  >);
};

export const getGetParalegalLeaderboardQueryKey = () => {
  return [`/api/analytics/paralegal-leaderboard`] as const;
};

export const getGetParalegalLeaderboardQueryOptions = <
  TData = Awaited<ReturnType<typeof getParalegalLeaderboard>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getParalegalLeaderboard>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey =
    queryOptions?.queryKey ?? getGetParalegalLeaderboardQueryKey();

  const queryFn: QueryFunction<
    Awaited<ReturnType<typeof getParalegalLeaderboard>>
  > = ({ signal }) => getParalegalLeaderboard({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof getParalegalLeaderboard>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetParalegalLeaderboardQueryResult = NonNullable<
  Awaited<ReturnType<typeof getParalegalLeaderboard>>
>;
export type GetParalegalLeaderboardQueryError = ErrorType<unknown>;

/**
 * @summary Paralegal performance leaderboard
 */

export function useGetParalegalLeaderboard<
  TData = Awaited<ReturnType<typeof getParalegalLeaderboard>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getParalegalLeaderboard>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetParalegalLeaderboardQueryOptions(options);

```

```

const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
  queryKey: QueryKey;
};

return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Full audit trail with optional filters
 */
export const getGetAuditTrailUrl = (params?: GetAuditTrailParams) => {
  const normalizedParams = new URLSearchParams();

  Object.entries(params || {}).forEach(([key, value]) => {
    if (value !== undefined) {
      normalizedParams.append(key, value === null ? "null" : value.toString());
    }
  });

  const stringifiedParams = normalizedParams.toString();

  return stringifiedParams.length > 0
    ? `/api/compliance/audit-trail?${stringifiedParams}`
    : `/api/compliance/audit-trail`;
};

export const getAuditTrail = async (
  params?: GetAuditTrailParams,
  options?: RequestInit,
): Promise<AuditLogEntry[]> => {
  return customFetch<AuditLogEntry[]>(getGetAuditTrailUrl(params), {
    ...options,
    method: "GET",
  });
};

export const getGetAuditTrailQueryKey = (params?: GetAuditTrailParams) => {
  return [`/api/compliance/audit-trail`, ...(params ? [params] : [])] as const;
};

export const getGetAuditTrailQueryOptions = <
  TData = Awaited<ReturnType<typeof getAuditTrail>>,
  TError = ErrorType<unknown>,
>(
  params?: GetAuditTrailParams,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getAuditTrail>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetAuditTrailQueryKey(params);

  const queryFn: QueryFunction<Awaited<ReturnType<typeof getAuditTrail>>> = ({
    signal,
  }) => getAuditTrail(params, { signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof getAuditTrail>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetAuditTrailQueryResult = NonNullable<
  Awaited<ReturnType<typeof getAuditTrail>>
>;
export type GetAuditTrailQueryError = ErrorType<unknown>;

/**
 * @summary Full audit trail with optional filters
 */
export function useGetAuditTrail<
  TData = Awaited<ReturnType<typeof getAuditTrail>>,
  TError = ErrorType<unknown>,
>(
  params?: GetAuditTrailParams,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getAuditTrail>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetAuditTrailQueryOptions(params, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };
}

```

```

    };

    return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Audit trail summary with counts
 */
export const getGetAuditSummaryUrl = () => {
    return `/api/compliance/audit-summary`;
};

export const getAuditSummary = async (
    options?: RequestInit,
): Promise<AuditSummary> => {
    return customFetch<AuditSummary>(getGetAuditSummaryUrl(), {
        ...options,
        method: "GET",
    });
};

export const getGetAuditSummaryQueryKey = () => {
    return [`/api/compliance/audit-summary`] as const;
};

export const getGetAuditSummaryQueryOptions = <
    TData = Awaited<ReturnType<typeof getAuditSummary>>,
    TError = ErrorType<unknown>,
>(options?: {
    query?: UseQueryOptions<
        Awaited<ReturnType<typeof getAuditSummary>>,
        TError,
        TData
    >;
    request?: SecondParameter<typeof customFetch>;
}) => {
    const { query: queryOptions, request: requestOptions } = options ?? {};

    const queryKey = queryOptions?.queryKey ?? getGetAuditSummaryQueryKey();

    const queryFn: QueryFunction<Awaited<ReturnType<typeof getAuditSummary>>> = ({
        signal,
    }) => getAuditSummary({ signal, ...requestOptions });

    return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
        Awaited<ReturnType<typeof getAuditSummary>>,
        TError,
        TData
    > & { queryKey: QueryKey };
};

export type GetAuditSummaryQueryResult = NonNullable<
    Awaited<ReturnType<typeof getAuditSummary>>
>;
export type GetAuditSummaryQueryError = ErrorType<unknown>;

/**
 * @summary Audit trail summary with counts
 */

export function useGetAuditSummary<
    TData = Awaited<ReturnType<typeof getAuditSummary>>,
    TError = ErrorType<unknown>,
>(options?: {
    query?: UseQueryOptions<
        Awaited<ReturnType<typeof getAuditSummary>>,
        TError,
        TData
    >;
    request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
    const queryOptions = getGetAuditSummaryQueryOptions(options);

    const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
        queryKey: QueryKey;
    };

    return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Upload a fax image for OCR processing
 */
export const getUploadFaxUrl = () => {
    return `/api/ocr/upload`;
};

export const uploadFax = async (
    uploadFaxBody: UploadFaxBody,
    options?: RequestInit,
): Promise<UploadFaxResponse> => {
    return customFetch<UploadFaxResponse>(getUploadFaxUrl(), {
        ...options,
        method: "POST",
        headers: { "Content-Type": "application/json", ...options?.headers },
        body: JSON.stringify(uploadFaxBody),
    });
};

```

```

};

export const getUploadFaxMutationOptions = <
  TError = ErrorType<ErrorResponse>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof uploadFax>>,
    TError,
    { data: BodyType<UploadFaxBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationOptions<
  Awaited<ReturnType<typeof uploadFax>>,
  TError,
  { data: BodyType<UploadFaxBody> },
  TContext
> => {
  const mutationKey = ["uploadFax"];
  const { mutation: mutationOptions, request: requestOptions } = options
    ? options.mutation &&
      "mutationKey" in options.mutation &&
      options.mutation.mutationKey
    ? options
    : { ...options, mutation: { ...options.mutation, mutationKey } }
  : { mutation: { mutationKey }, request: undefined };

  const mutationFn: MutationFunction<
    Awaited<ReturnType<typeof uploadFax>>,
    { data: BodyType<UploadFaxBody> }
  > = (props) => {
    const { data } = props ?? {};

    return uploadFax(data, requestOptions);
  };

  return { mutationFn, ...mutationOptions };
};

export type UploadFaxMutationResult = NonNullable<
  Awaited<ReturnType<typeof uploadFax>>
>;
export type UploadFaxMutationBody = BodyType<UploadFaxBody>;
export type UploadFaxMutationError = ErrorType<ErrorResponse>;

/**
 * @summary Upload a fax image for OCR processing
 */
export const useUploadFax = <
  TError = ErrorType<ErrorResponse>,
  TContext = unknown,
>(options?: {
  mutation?: UseMutationOptions<
    Awaited<ReturnType<typeof uploadFax>>,
    TError,
    { data: BodyType<UploadFaxBody> },
    TContext
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseMutationResult<
  Awaited<ReturnType<typeof uploadFax>>,
  TError,
  { data: BodyType<UploadFaxBody> },
  TContext
> => {
  return useMutation(getUploadFaxMutationOptions(options));
};

/**
 * @summary List all Legora Grid results
 */
export const getListFaxResultsUrl = () => {
  return `/api/ocr/results`;
};

export const listFaxResults = async (
  options?: RequestInit,
): Promise<FaxResult[]> => {
  return customFetch<FaxResult[]>(getListFaxResultsUrl(), {
    ...options,
    method: "GET",
  });
};

export const getListFaxResultsQueryKey = () => {
  return [`/api/ocr/results`] as const;
};

export const getListFaxResultsQueryOptions = <
  TData = Awaited<ReturnType<typeof listFaxResults>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof listFaxResults>>,
    TError,
    TData
  >

```

```

>;
request?: SecondParameter<typeof customFetch>;
}) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getListFaxResultsQueryKey();

  const queryFn: QueryFunction<Awaited<ReturnType<typeof listFaxResults>>> = ({
    signal,
  }) => listFaxResults({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof listFaxResults>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type ListFaxResultsQueryResult = NonNullable<
  Awaited<ReturnType<typeof listFaxResults>>
>;
export type ListFaxResultsQueryError = ErrorType<unknown>;

/**
 * @summary List all Legora Grid results
 */

export function useListFaxResults<
  TData = Awaited<ReturnType<typeof listFaxResults>>,
  TError = ErrorType<unknown>,
>(options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof listFaxResults>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getListFaxResultsQueryOptions(options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary Get a single Legora Grid result
 */
export const getGetFaxResultUrl = (id: number) => {
  return `/api/ocr/results/${id}`;
};

export const getFaxResult = async (
  id: number,
  options?: RequestInit,
): Promise<FaxResult> => {
  return customFetch<FaxResult>(getGetFaxResultUrl(id), {
    ...options,
    method: "GET",
  });
};

export const getGetFaxResultQueryKey = (id: number) => {
  return [`/api/ocr/results/${id}`] as const;
};

export const getGetFaxResultQueryOptions = <
  TData = Awaited<ReturnType<typeof getFaxResult>>,
  TError = ErrorType<ErrorResponse>,
>(
  id: number,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getFaxResult>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
) => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetFaxResultQueryKey(id);

  const queryFn: QueryFunction<Awaited<ReturnType<typeof getFaxResult>>> = ({
    signal,
  }) => getFaxResult(id, { signal, ...requestOptions });

  return {
    queryKey,
    queryFn,
    enabled: !!id,
    ...queryOptions,
  } as UseQueryOptions<
    Awaited<ReturnType<typeof getFaxResult>>,

```

```

    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetFaxResultQueryResult = NonNullable<
  Awaited<ReturnType<typeof getFaxResult>>
>;
export type GetFaxResultQueryError = ErrorType<ErrorResponse>;

/**
 * @summary Get a single Legora Grid result
 */

export function useGetFaxResult<
  TData = Awaited<ReturnType<typeof getFaxResult>>,
  TError = ErrorType<ErrorResponse>,
>({
  id: number,
  options?: {
    query?: UseQueryOptions<
      Awaited<ReturnType<typeof getFaxResult>>,
      TError,
      TData
    >;
    request?: SecondParameter<typeof customFetch>;
  },
}): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
  const queryOptions = getGetFaxResultQueryOptions(id, options);

  const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
    queryKey: QueryKey;
  };

  return { ...query, queryKey: queryOptions.queryKey };
}

/**
 * @summary OCR fax processing queue stats
 */
export const getGetOcrQueueStatsUrl = () => {
  return `/api/ocr/queue-stats`;
};

export const getOcrQueueStats = async (
  options?: RequestInit,
): Promise<QueueStats> => {
  return customFetch<QueueStats>(getGetOcrQueueStatsUrl(), {
    ...options,
    method: "GET",
  });
};

export const getGetOcrQueueStatsQueryKey = () => {
  return [`/api/ocr/queue-stats`] as const;
};

export const getGetOcrQueueStatsQueryOptions = <
  TData = Awaited<ReturnType<typeof getOcrQueueStats>>,
  TError = ErrorType<unknown>,
>({options?: {
  query?: UseQueryOptions<
    Awaited<ReturnType<typeof getOcrQueueStats>>,
    TError,
    TData
  >;
  request?: SecondParameter<typeof customFetch>;
}} => {
  const { query: queryOptions, request: requestOptions } = options ?? {};

  const queryKey = queryOptions?.queryKey ?? getGetOcrQueueStatsQueryKey();

  const queryFn: QueryFunction<
    Awaited<ReturnType<typeof getOcrQueueStats>>
  > = ({ signal }) => getOcrQueueStats({ signal, ...requestOptions });

  return { queryKey, queryFn, ...queryOptions } as UseQueryOptions<
    Awaited<ReturnType<typeof getOcrQueueStats>>,
    TError,
    TData
  > & { queryKey: QueryKey };
};

export type GetOcrQueueStatsQueryResult = NonNullable<
  Awaited<ReturnType<typeof getOcrQueueStats>>
>;
export type GetOcrQueueStatsQueryError = ErrorType<unknown>;

/**
 * @summary OCR fax processing queue stats
 */

export function useGetOcrQueueStats<
  TData = Awaited<ReturnType<typeof getOcrQueueStats>>,
  TError = ErrorType<unknown>,
>({options?: {
  query?: UseQueryOptions<

```

```

        Awaited<ReturnType<typeof get0crQueueStats>>,
        TError,
        TData
    >;
    request?: SecondParameter<typeof customFetch>;
  }): UseQueryResult<TData, TError> & { queryKey: QueryKey } {
    const queryOptions = getGet0crQueueStatsQueryOptions(options);

    const query = useQuery(queryOptions) as UseQueryResult<TData, TError> & {
      queryKey: QueryKey;
    };

    return { ...query, queryKey: queryOptions.queryKey };
  }

```


[lib/api-client-react/src/index.ts](#)

```
export * from "./generated/api";
export * from "./generated/api.schemas";
export { setBaseUrl, setAuthTokenGetter } from "./custom-fetch";
export type { AuthTokenGetter } from "./custom-fetch";
```

lib/api-client-react/tsconfig.json

```
{
  "extends": "../../tsconfig.base.json",
  "compilerOptions": {
    "composite": true,
    "declarationMap": true,
    "emitDeclarationOnly": true,
    "outDir": "dist",
    "rootDir": "src",
    "lib": ["dom", "es2022"]
  },
  "include": ["src"]
}
```

lib/api-spec/openapi.yaml

```
openapi: 3.1.0
info:
  # Do not change the title, if the title changes, the import paths will be broken
  title: Api
  version: 0.1.0
  description: Mass Tort Operating System API
servers:
  - url: /api
    description: Base API path
tags:
  - name: health
    description: Health operations
  - name: leads
    description: Lead management
  - name: documents
    description: Document management
  - name: cases
    description: Distributed case processing pipeline
  - name: worker
    description: Worker queue status
  - name: dashboard
    description: Dashboard analytics
  - name: ocr
    description: Fax OCR inbox – Legora Grid extraction
  - name: review-queue
    description: Conflict resolution review queue
  - name: npi
    description: NPI Registry doctor lookup
  - name: paralegals
    description: Paralegal management and assignment
  - name: analytics
    description: ROI tracking, conversion funnels, and performance analytics
  - name: compliance
    description: Audit trail and compliance tracking
  - name: forms
    description: Form engine – embeddable intake forms with compliance validation
  - name: vendors
    description: Vendor management
paths:
  /healthz:
    get:
      operationId: healthCheck
      tags: [health]
      summary: Health check
      responses:
        "200":
          description: Healthy
          content:
            application/json:
              schema:
                $ref: "#/components/schemas/HealthStatus"
  /leads:
    get:
      operationId: listLeads
      tags: [leads]
      summary: List all leads
      parameters:
        - name: status
          in: query
          required: false
          schema:
            type: string
            enum: [new, qualified, signed, rejected, review_required]
        - name: tort_type
          in: query
          required: false
          schema:
            type: string
        - name: search
          in: query
          required: false
          schema:
            type: string
        - name: vendor_id
          in: query
          required: false
          schema:
            type: integer
        - name: law_firm
          in: query
          required: false
          schema:
            type: string
        - name: client_id
          in: query
          required: false
          schema:
            type: string
        - name: date_from
          in: query
          required: false
          schema:
            type: string
        - name: date_to
```

```

      in: query
      required: false
      schema:
        type: string
    responses:
      "200":
        description: List of leads
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: "#/components/schemas/Lead"
  post:
    operationId: createLead
    tags: [leads]
    summary: Create a new lead (intake)
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/CreateLeadBody"
    responses:
      "201":
        description: Lead created
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/Lead"

/leads/{id}:
  get:
    operationId: getLead
    tags: [leads]
    summary: Get a single lead
    parameters:
      - name: id
        in: path
        required: true
        schema:
          type: integer
    responses:
      "200":
        description: Lead details
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/Lead"
      "404":
        description: Not found
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/ErrorResponse"
  patch:
    operationId: updateLead
    tags: [leads]
    summary: Update lead status or fields
    parameters:
      - name: id
        in: path
        required: true
        schema:
          type: integer
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/UpdateLeadBody"
    responses:
      "200":
        description: Updated lead
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/Lead"
  delete:
    operationId: deleteLead
    tags: [leads]
    summary: Delete a lead
    parameters:
      - name: id
        in: path
        required: true
        schema:
          type: integer
    responses:
      "204":
        description: Deleted

/leads/{id}/qualify:
  post:
    operationId: qualifyLead
    tags: [leads]
    summary: Run Boolean Gatekeeper qualification on a lead

```

```

parameters:
  - name: id
    in: path
    required: true
    schema:
      type: integer
responses:
  "200":
    description: Qualification result
    content:
      application/json:
        schema:
          $ref: "#/components/schemas/QualificationResult"

/leads/export:
  get:
    operationId: exportLeads
    tags: [leads]
    summary: Export leads as CSV
    parameters:
      - name: lead_id
        in: query
        required: false
        schema:
          type: integer
      - name: status
        in: query
        required: false
        schema:
          type: string
      - name: tort_type
        in: query
        required: false
        schema:
          type: string
      - name: vendor_id
        in: query
        required: false
        schema:
          type: integer
      - name: law_firm
        in: query
        required: false
        schema:
          type: string
      - name: client_id
        in: query
        required: false
        schema:
          type: string
      - name: date_from
        in: query
        required: false
        schema:
          type: string
      - name: date_to
        in: query
        required: false
        schema:
          type: string
      - name: source
        in: query
        required: false
        schema:
          type: string
      - name: fields
        in: query
        required: false
        schema:
          type: string
    responses:
      "200":
        description: CSV file
        content:
          text/csv:
            schema:
              type: string

/vendors:
  get:
    operationId: listVendors
    tags: [vendors]
    summary: List all vendors
    responses:
      "200":
        description: List of vendors
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: "#/components/schemas/Vendor"
  post:
    operationId: createVendor
    tags: [vendors]
    summary: Create a new vendor
    requestBody:

```

```

    required: true
    content:
      application/json:
        schema:
          $ref: "#/components/schemas/CreateVendorBody"
  responses:
    "201":
      description: Vendor created
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/Vendor"

/vendors/{id}:
  get:
    operationId: getVendor
    tags: [vendors]
    summary: Get a single vendor
    parameters:
      - name: id
        in: path
        required: true
        schema:
          type: integer
    responses:
      "200":
        description: Vendor details
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/Vendor"
  patch:
    operationId: updateVendor
    tags: [vendors]
    summary: Update a vendor
    parameters:
      - name: id
        in: path
        required: true
        schema:
          type: integer
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/UpdateVendorBody"
    responses:
      "200":
        description: Updated vendor
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/Vendor"
  delete:
    operationId: deleteVendor
    tags: [vendors]
    summary: Delete a vendor
    parameters:
      - name: id
        in: path
        required: true
        schema:
          type: integer
    responses:
      "204":
        description: Vendor deleted

/documents:
  get:
    operationId: listDocuments
    tags: [documents]
    summary: List all documents
    parameters:
      - name: lead_id
        in: query
        required: false
        schema:
          type: integer
    responses:
      "200":
        description: List of documents
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: "#/components/schemas/Document"
  post:
    operationId: createDocument
    tags: [documents]
    summary: Upload or register a document
    requestBody:
      required: true
      content:
        application/json:
          schema:

```

```

        $ref: "#/components/schemas/CreateDocumentBody"
responses:
  "201":
    description: Document created
    content:
      application/json:
        schema:
          $ref: "#/components/schemas/Document"
/documents/{id}:
  patch:
    operationId: updateDocument
    tags: [documents]
    summary: Update document metadata or status
    parameters:
      - name: id
        in: path
        required: true
        schema:
          type: integer
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/UpdateDocumentBody"
    responses:
      "200":
        description: Updated document
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/Document"
  delete:
    operationId: deleteDocument
    tags: [documents]
    summary: Delete a document
    parameters:
      - name: id
        in: path
        required: true
        schema:
          type: integer
    responses:
      "204":
        description: Deleted
/dashboard/stats:
  get:
    operationId: getDashboardStats
    tags: [dashboard]
    summary: Get pipeline summary stats
    responses:
      "200":
        description: Dashboard statistics
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/DashboardStats"
/dashboard/pipeline:
  get:
    operationId: getPipelineBreakdown
    tags: [dashboard]
    summary: Get lead counts grouped by status and tort type
    responses:
      "200":
        description: Pipeline breakdown
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/PipelineBreakdown"
/dashboard/recent-activity:
  get:
    operationId: getRecentActivity
    tags: [dashboard]
    summary: Get recent lead activity feed
    responses:
      "200":
        description: Recent activity items
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: "#/components/schemas/ActivityItem"
/cases:
  get:
    operationId: listCases
    tags: [cases]
    summary: List all cases
    responses:
      "200":
        description: List of cases
        content:

```

```

        application/json:
          schema:
            type: array
            items:
              $ref: "#/components/schemas/Case"
  post:
    operationId: createCase
    tags: [cases]
    summary: Submit a new case to the processing queue
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            additionalProperties: true
    responses:
      "201":
        description: Case queued
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/CreateCaseResponse"

/cases/worker/queue-stats:
  get:
    operationId: getQueueStats
    tags: [worker]
    summary: Get job queue statistics by status
    responses:
      "200":
        description: Queue stats
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/QueueStats"

/cases/{id}:
  get:
    operationId: getCase
    tags: [cases]
    summary: Get full case detail with documents, analyses, and audit trail
    parameters:
      - name: id
        in: path
        required: true
        schema:
          type: string
    responses:
      "200":
        description: Case detail
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/CaseDetail"
      "404":
        description: Not found
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/ErrorResponse"

/cases/{id}/upload:
  post:
    operationId: uploadCaseFile
    tags: [cases]
    summary: Upload a document to the case vault
    parameters:
      - name: id
        in: path
        required: true
        schema:
          type: string
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/UploadCaseFileBody"
    responses:
      "200":
        description: File queued for ingestion
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/CreateCaseResponse"

/cases/{id}/analyze:
  post:
    operationId: analyzeCaseFiles
    tags: [cases]
    summary: Queue AI extraction and scoring for all case documents
    parameters:
      - name: id
        in: path
        required: true
        schema:

```



```

        type: string
responses:
  "200":
    description: Analysis queued
    content:
      application/json:
        schema:
          $ref: "#/components/schemas/CreateCaseResponse"
/review-queue:
  get:
    operationId: listReviewQueue
    tags: [review-queue]
    summary: List review queue items with optional filters
    parameters:
      - name: resolution
        in: query
        required: false
        schema:
          type: string
      - name: conflict_type
        in: query
        required: false
        schema:
          type: string
      - name: severity
        in: query
        required: false
        schema:
          type: string
      - name: entity_type
        in: query
        required: false
        schema:
          type: string
      - name: limit
        in: query
        required: false
        schema:
          type: string
    responses:
      "200":
        description: Review queue items
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: "#/components/schemas/ReviewQueueItem"
/review-queue/stats:
  get:
    operationId: getReviewQueueStats
    tags: [review-queue]
    summary: Get review queue statistics
    responses:
      "200":
        description: Review queue stats
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/ReviewQueueStats"
/review-queue/{id}:
  patch:
    operationId: resolveReviewItem
    tags: [review-queue]
    summary: Resolve a review queue item
    parameters:
      - name: id
        in: path
        required: true
        schema:
          type: integer
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/ResolveReviewBody"
    responses:
      "200":
        description: Resolved review item
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/ReviewQueueItem"
/npi/search:
  get:
    operationId: searchNpi
    tags: [npi]
    summary: Search the NPI Registry for doctors and providers
    parameters:
      - name: npi_number
        in: query
        required: false

```

```

    schema:
      type: string
  - name: first_name
    in: query
    required: false
    schema:
      type: string
  - name: last_name
    in: query
    required: false
    schema:
      type: string
  - name: city
    in: query
    required: false
    schema:
      type: string
  - name: state
    in: query
    required: false
    schema:
      type: string
  - name: specialty
    in: query
    required: false
    schema:
      type: string
  - name: limit
    in: query
    required: false
    schema:
      type: string
  responses:
    "200":
      description: NPI search results
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/NpiSearchResponse"
    "400":
      description: Bad request
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/ErrorResponse"

/npi/lookup/{npi}:
  get:
    operationId: lookupNpi
    tags: [npi]
    summary: Look up a single provider by NPI number
    parameters:
      - name: npi
        in: path
        required: true
        schema:
          type: string
    responses:
      "200":
        description: Provider detail
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/NpiProvider"
      "404":
        description: NPI not found
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/ErrorResponse"

/forms/config:
  get:
    operationId: getFormConfigs
    tags: [forms]
    summary: Get all tort campaign form configurations
    responses:
      "200":
        description: Form configurations
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/FormConfigList"

/forms/config/{tortId}:
  get:
    operationId: getFormConfig
    tags: [forms]
    summary: Get a single tort form configuration
    parameters:
      - name: tortId
        in: path
        required: true
        schema:
          type: string
    responses:
      "200":

```

```

        description: Form configuration
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/FormConfig"
    put:
      operationId: updateFormConfig
      tags: [forms]
      summary: Update a tort form configuration (admin only)
      parameters:
        - name: tortId
          in: path
          required: true
          schema:
            type: string
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/FormConfigUpdate"
      responses:
        "200":
          description: Updated configuration
          content:
            application/json:
              schema:
                $ref: "#/components/schemas/FormConfig"

/forms/config/{tortId}/fields:
  post:
    operationId: addCustomField
    tags: [forms]
    summary: Add a custom field to a tort form (admin only)
    parameters:
      - name: tortId
        in: path
        required: true
        schema:
          type: string
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/CustomField"
    responses:
      "200":
        description: Updated configuration
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/FormConfig"

/forms/config/{tortId}/fields/{key}:
  delete:
    operationId: removeCustomField
    tags: [forms]
    summary: Remove a custom field from a tort form (admin only)
    parameters:
      - name: tortId
        in: path
        required: true
        schema:
          type: string
      - name: key
        in: path
        required: true
        schema:
          type: string
    responses:
      "200":
        description: Updated configuration
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/FormConfig"

/forms/validate/email:
  post:
    operationId: validateEmail
    tags: [forms]
    summary: Validate email (RFC + typo detection)
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            properties:
              email:
                type: string
            required: [email]
    responses:
      "200":
        description: Email validation result
        content:

```

```

        application/json:
          schema:
            $ref: "#/components/schemas/EmailValidationResult"
/forms/validate/address:
  post:
    operationId: validateAddress
    tags: [forms]
    summary: Validate address (US format)
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            properties:
              street_address:
                type: string
              city:
                type: string
              state:
                type: string
              zip:
                type: string
            required: [street_address, city, state, zip]
    responses:
      "200":
        description: Address validation result
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/AddressValidationResult"
/forms/submit:
  post:
    operationId: submitForm
    tags: [forms]
    summary: Submit intake form with full compliance validation
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/FormSubmission"
    responses:
      "201":
        description: Lead accepted
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/FormSubmissionResult"
      "422":
        description: Validation failed
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/FormRejection"
/forms/embed/{tortId}:
  get:
    operationId: getFormEmbed
    tags: [forms]
    summary: Get embeddable JavaScript form for a tort campaign
    parameters:
      - name: tortId
        in: path
        required: true
        schema:
          type: string
    responses:
      "200":
        description: JavaScript embed code
        content:
          application/javascript:
            schema:
              type: string
/forms/background-check:
  post:
    operationId: runBackgroundCheck
    tags: [forms]
    summary: Run background check on a person
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            properties:
              first_name:
                type: string
              last_name:
                type: string
              state:
                type: string
              date_of_birth:
                type: string

```

```

        required: [first_name, last_name]
responses:
  "200":
    description: Background check result
    content:
      application/json:
        schema:
          $ref: "#/components/schemas/BackgroundCheckResult"

/forms/background-check/lead/{id}:
  post:
    operationId: runLeadBackgroundCheck
    tags: [forms]
    summary: Run background check on an existing lead
    parameters:
      - name: id
        in: path
        required: true
        schema:
          type: integer
    responses:
      "200":
        description: Background check result
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/BackgroundCheckResult"

/forms/categories:
  get:
    operationId: getTortCategories
    tags: [forms]
    summary: Get all tort categories with grouped torts
    responses:
      "200":
        description: Tort categories
        content:
          application/json:
            schema:
              type: array
              items:
                type: object
                properties:
                  category:
                    type: string
                  torts:
                    type: array
                    items:
                      type: object
                      properties:
                        id:
                          type: string
                        label:
                          type: string
                        required: [id, label]
                        required: [category, torts]

/forms/npi-verify:
  post:
    operationId: verifyNpi
    tags: [forms]
    summary: Verify physician NPI and match taxonomy to diagnosis
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            properties:
              physician_first_name:
                type: string
              physician_last_name:
                type: string
              diagnosis:
                type: string
            required: [physician_first_name, physician_last_name, diagnosis]
    responses:
      "200":
        description: NPI verification result
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/NpiVerifyResult"

/forms/fraud-check:
  post:
    operationId: runFraudCheck
    tags: [forms]
    summary: Run fraud detection on claim data
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            properties:
              tort_type:

```

```

        type: string
      diagnosis:
        type: string
      physician_first_name:
        type: string
      physician_last_name:
        type: string
      lead_data:
        type: object
      required: [tort_type, diagnosis]
    responses:
      "200":
        description: Fraud check result
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/FraudCheckResult"

/forms/escalate/fbi:
  post:
    operationId: escalateToFbi
    tags: [forms]
    summary: Escalate case to FBI tip line and log event
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            properties:
              lead_id:
                type: string
              reason:
                type: string
              fraud_indicators:
                type: array
              items:
                type: object
            required: [lead_id, reason]
    responses:
      "200":
        description: Escalation result
        content:
          application/json:
            schema:
              type: object
              properties:
                status:
                  type: string
                fbi_tip_url:
                  type: string
                message:
                  type: string
              required: [status, fbi_tip_url, message]

/paralegals:
  get:
    operationId: listParalegals
    tags: [paralegals]
    summary: List all paralegals
    responses:
      "200":
        description: Paralegals list
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: "#/components/schemas/Paralegal"
  post:
    operationId: createParalegal
    tags: [paralegals]
    summary: Create a new paralegal
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/CreateParalegalBody"
    responses:
      "201":
        description: Paralegal created
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/Paralegal"

/paralegals/{id}:
  get:
    operationId: getParalegal
    tags: [paralegals]
    summary: Get paralegal detail with assigned leads
    parameters:
      - name: id
        in: path
        required: true
        schema:

```

```

      type: integer
    responses:
      "200":
        description: Paralegal detail
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/ParalegalDetail"
      "404":
        description: Not found
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/ErrorResponse"

/paralegals/{id}/performance:
  get:
    operationId: getParalegalPerformance
    tags: [paralegals]
    summary: Get paralegal performance breakdown
    parameters:
      - name: id
        in: path
        required: true
        schema:
          type: integer
    responses:
      "200":
        description: Performance stats
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/ParalegalPerformance"

/analytics/overview:
  get:
    operationId: getAnalyticsOverview
    tags: [analytics]
    summary: Full analytics overview across all modules
    responses:
      "200":
        description: Analytics overview
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/AnalyticsOverview"

/analytics/pipeline-trend:
  get:
    operationId: getPipelineTrend
    tags: [analytics]
    summary: Daily pipeline trend data
    responses:
      "200":
        description: Pipeline trend
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: "#/components/schemas/PipelineTrendPoint"

/analytics/conversion-funnel:
  get:
    operationId: getConversionFunnel
    tags: [analytics]
    summary: Conversion funnel stages
    responses:
      "200":
        description: Funnel stages
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: "#/components/schemas/FunnelStage"

/analytics/tort-breakdown:
  get:
    operationId: getTortBreakdown
    tags: [analytics]
    summary: Breakdown by tort type
    responses:
      "200":
        description: Tort breakdown
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: "#/components/schemas/TortBreakdownRow"

/analytics/paralegal-leaderboard:
  get:
    operationId: getParalegalLeaderboard
    tags: [analytics]
    summary: Paralegal performance leaderboard

```

```

responses:
  "200":
    description: Leaderboard
    content:
      application/json:
        schema:
          type: array
          items:
            $ref: "#/components/schemas/ParalegalLeaderboardRow"

/compliance/audit-trail:
  get:
    operationId: getAuditTrail
    tags: [compliance]
    summary: Full audit trail with optional filters
    parameters:
      - name: limit
        in: query
        required: false
        schema:
          type: integer
      - name: entity_type
        in: query
        required: false
        schema:
          type: string
      - name: action
        in: query
        required: false
        schema:
          type: string
    responses:
      "200":
        description: Audit trail
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: "#/components/schemas/AuditLogEntry"

/compliance/audit-summary:
  get:
    operationId: getAuditSummary
    tags: [compliance]
    summary: Audit trail summary with counts
    responses:
      "200":
        description: Audit summary
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/AuditSummary"

/ocr/upload:
  post:
    operationId: uploadFax
    tags: [ocr]
    summary: Upload a fax image for OCR processing
    requestBody:
      required: true
      content:
        application/json:
          schema:
            $ref: "#/components/schemas/UploadFaxBody"
    responses:
      "200":
        description: Fax queued for processing
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/UploadFaxResponse"
      "400":
        description: Bad request
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/ErrorResponse"

/ocr/results:
  get:
    operationId: listFaxResults
    tags: [ocr]
    summary: List all Legora Grid results
    responses:
      "200":
        description: Legora Grid results
        content:
          application/json:
            schema:
              type: array
              items:
                $ref: "#/components/schemas/FaxResult"

/ocr/results/{id}:
  get:
    operationId: getFaxResult

```



```

tags: [ocr]
summary: Get a single Legora Grid result
parameters:
  - name: id
    in: path
    required: true
    schema:
      type: integer
responses:
  "200":
    description: Legora Grid result
    content:
      application/json:
        schema:
          $ref: "#/components/schemas/FaxResult"
  "404":
    description: Not found
    content:
      application/json:
        schema:
          $ref: "#/components/schemas/ErrorResponse"

/ocr/queue-stats:
  get:
    operationId: getOcrQueueStats
    tags: [ocr]
    summary: OCR fax processing queue stats
    responses:
      "200":
        description: OCR queue stats
        content:
          application/json:
            schema:
              $ref: "#/components/schemas/QueueStats"

components:
  schemas:
    HealthStatus:
      type: object
      properties:
        status:
          type: string
          required: [status]

    ErrorResponse:
      type: object
      properties:
        error:
          type: string
          required: [error]

    Lead:
      type: object
      properties:
        id:
          type: integer
        name:
          type: string
        email:
          type: string
        phone:
          type: string
        tort_type:
          type: string
        first_name:
          type: string
          nullable: true
        last_name:
          type: string
          nullable: true
        date_of_birth:
          type: string
          nullable: true
        street_address:
          type: string
          nullable: true
        city:
          type: string
          nullable: true
        state:
          type: string
          nullable: true
        zip:
          type: string
          nullable: true
        phone_primary:
          type: string
          nullable: true
        last_4_ssn:
          type: string
          nullable: true
        diagnosis:
          type: string
          nullable: true
        diagnosis_date:
          type: string
          nullable: true

```

```
physician_first_name:
  type: string
  nullable: true
physician_last_name:
  type: string
  nullable: true
physician_full_address:
  type: string
  nullable: true
physician_contact_info:
  type: string
  nullable: true
hospital_name:
  type: string
  nullable: true
hospital_fax:
  type: string
  nullable: true
hospital_contact_info:
  type: string
  nullable: true
tcpa_consent:
  type: boolean
  nullable: true
trustedform_cert_url:
  type: string
  nullable: true
trustedform_ip:
  type: string
  nullable: true
trustedform_user_agent:
  type: string
  nullable: true
trustedform_timestamp:
  type: string
  nullable: true
email_validation_status:
  type: string
  nullable: true
address_validation_status:
  type: string
  nullable: true
background_check_status:
  type: string
  nullable: true
background_check_data:
  type: string
  nullable: true
exposure_start:
  type: string
  nullable: true
exposure_end:
  type: string
  nullable: true
diagnosis_confirmed:
  type: boolean
diagnosis_type:
  type: string
  nullable: true
was_at_location:
  type: boolean
location_name:
  type: string
  nullable: true
status:
  type: string
  enum: [new, qualified, signed, rejected, review_required]
rejection_reason:
  type: string
  nullable: true
notes:
  type: string
  nullable: true
ad_spend:
  type: number
  nullable: true
source:
  type: string
  nullable: true
assigned_to:
  type: integer
  nullable: true
routing:
  type: string
  nullable: true
  enum: [hot, warm, cold]
vendor_id:
  type: integer
  nullable: true
law_firm:
  type: string
  nullable: true
client_id:
  type: string
  nullable: true
created_at:
  type: string
```

```

    updated_at:
      type: string
required:
  - id
  - name
  - tort_type
  - diagnosis_confirmed
  - was_at_location
  - status
  - created_at
  - updated_at

CreateLeadBody:
  type: object
  properties:
    name:
      type: string
    email:
      type: string
    phone:
      type: string
    tort_type:
      type: string
    first_name:
      type: string
    last_name:
      type: string
    date_of_birth:
      type: string
    street_address:
      type: string
    city:
      type: string
    state:
      type: string
    zip:
      type: string
    phone_primary:
      type: string
    last_4_ssn:
      type: string
    diagnosis:
      type: string
    diagnosis_date:
      type: string
    physician_first_name:
      type: string
    physician_last_name:
      type: string
    physician_full_address:
      type: string
    physician_contact_info:
      type: string
    hospital_name:
      type: string
    hospital_fax:
      type: string
    hospital_contact_info:
      type: string
    exposure_start:
      type: string
      nullable: true
    exposure_end:
      type: string
      nullable: true
    diagnosis_confirmed:
      type: boolean
    diagnosis_type:
      type: string
      nullable: true
    was_at_location:
      type: boolean
    location_name:
      type: string
      nullable: true
    notes:
      type: string
      nullable: true
    ad_spend:
      type: number
      nullable: true
    source:
      type: string
      nullable: true
    vendor_id:
      type: integer
      nullable: true
    law_firm:
      type: string
      nullable: true
    client_id:
      type: string
      nullable: true
required:
  - name
  - tort_type
  - diagnosis_confirmed

```

- was_at_location
- first_name
- last_name
- date_of_birth
- street_address
- city
- state
- zip
- phone_primary
- last_4_ssn
- diagnosis
- diagnosis_date
- physician_first_name
- physician_last_name
- physician_full_address
- physician_contact_info
- hospital_name
- hospital_fax
- hospital_contact_info

UpdateLeadBody:

type: object
properties:
 name:
 type: string
 email:
 type: string
 phone:
 type: string
 tort_type:
 type: string
 first_name:
 type: string
 last_name:
 type: string
 date_of_birth:
 type: string
 street_address:
 type: string
 city:
 type: string
 state:
 type: string
 zip:
 type: string
 phone_primary:
 type: string
 last_4_ssn:
 type: string
 diagnosis:
 type: string
 diagnosis_date:
 type: string
 physician_first_name:
 type: string
 physician_last_name:
 type: string
 physician_full_address:
 type: string
 physician_contact_info:
 type: string
 hospital_name:
 type: string
 hospital_fax:
 type: string
 hospital_contact_info:
 type: string
 tcpa_consent:
 type: boolean
 trustedform_cert_url:
 type: string
 exposure_start:
 type: string
 nullable: true
 exposure_end:
 type: string
 nullable: true
 diagnosis_confirmed:
 type: boolean
 diagnosis_type:
 type: string
 nullable: true
 was_at_location:
 type: boolean
 location_name:
 type: string
 nullable: true
 status:
 type: string
 enum: [new, qualified, signed, rejected, review_required]
 rejection_reason:
 type: string
 nullable: true
 notes:
 type: string
 nullable: true
 ad_spend:

```
    type: number
    nullable: true
  source:
    type: string
    nullable: true
  vendor_id:
    type: integer
    nullable: true
  law_firm:
    type: string
    nullable: true
  client_id:
    type: string
    nullable: true
```

```
Vendor:
  type: object
  properties:
    id:
      type: integer
    name:
      type: string
    contact_name:
      type: string
      nullable: true
    contact_email:
      type: string
      nullable: true
    contact_phone:
      type: string
      nullable: true
    type:
      type: string
    status:
      type: string
    notes:
      type: string
      nullable: true
    created_at:
      type: string
    updated_at:
      type: string
  required:
    - id
    - name
    - type
    - status
    - created_at
    - updated_at
```

```
CreateVendorBody:
  type: object
  properties:
    name:
      type: string
    contact_name:
      type: string
      nullable: true
    contact_email:
      type: string
      nullable: true
    contact_phone:
      type: string
      nullable: true
    type:
      type: string
    status:
      type: string
    notes:
      type: string
      nullable: true
  required:
    - name
```

```
UpdateVendorBody:
  type: object
  properties:
    name:
      type: string
    contact_name:
      type: string
      nullable: true
    contact_email:
      type: string
      nullable: true
    contact_phone:
      type: string
      nullable: true
    type:
      type: string
    status:
      type: string
    notes:
      type: string
      nullable: true
```

```
QualificationResult:
```

```

type: object
properties:
  lead_id:
    type: integer
  qualified:
    type: boolean
  status:
    type: string
    enum: [qualified, rejected]
  reason:
    type: string
  gates_passed:
    type: array
    items:
      type: string
  gates_failed:
    type: array
    items:
      type: string
  required:
    - lead_id
    - qualified
    - status
    - reason
    - gates_passed
    - gates_failed

Document:
type: object
properties:
  id:
    type: integer
  lead_id:
    type: integer
  document_type:
    type: string
    enum: [retainer, medical_record, intake_form, other]
  file_name:
    type: string
  file_url:
    type: string
    nullable: true
  signed:
    type: boolean
  signed_at:
    type: string
    nullable: true
  notes:
    type: string
    nullable: true
  created_at:
    type: string
  required:
    - id
    - lead_id
    - document_type
    - file_name
    - signed
    - created_at

CreateDocumentBody:
type: object
properties:
  lead_id:
    type: integer
  document_type:
    type: string
    enum: [retainer, medical_record, intake_form, other]
  file_name:
    type: string
  file_url:
    type: string
    nullable: true
  signed:
    type: boolean
  signed_at:
    type: string
    nullable: true
  notes:
    type: string
    nullable: true
  required:
    - lead_id
    - document_type
    - file_name
    - signed

UpdateDocumentBody:
type: object
properties:
  document_type:
    type: string
    enum: [retainer, medical_record, intake_form, other]
  file_name:
    type: string
  file_url:
    type: string

```

```

    nullable: true
signed:
  type: boolean
signed_at:
  type: string
  nullable: true
notes:
  type: string
  nullable: true
DashboardStats:
  type: object
  properties:
    total_leads:
      type: integer
    qualified_leads:
      type: integer
    signed_retainers:
      type: integer
    rejected_leads:
      type: integer
    new_leads:
      type: integer
    total_ad_spend:
      type: number
    cpsr:
      type: number
      nullable: true
    qualification_rate:
      type: number
    conversion_rate:
      type: number
    leads_today:
      type: integer
    leads_this_week:
      type: integer
  required:
    - total_leads
    - qualified_leads
    - signed_retainers
    - rejected_leads
    - new_leads
    - total_ad_spend
    - qualification_rate
    - conversion_rate
    - leads_today
    - leads_this_week
PipelineBreakdown:
  type: object
  properties:
    by_status:
      type: array
      items:
        type: object
        properties:
          status:
            type: string
          count:
            type: integer
          required: [status, count]
    by_tort_type:
      type: array
      items:
        type: object
        properties:
          tort_type:
            type: string
          count:
            type: integer
          qualified:
            type: integer
          signed:
            type: integer
          required: [tort_type, count, qualified, signed]
  required: [by_status, by_tort_type]
ActivityItem:
  type: object
  properties:
    id:
      type: integer
    lead_id:
      type: integer
    lead_name:
      type: string
    event_type:
      type: string
    description:
      type: string
    occurred_at:
      type: string
    tort_type:
      type: string
  required:
    - id
    - lead_id

```

- lead_name
- event_type
- description
- occurred_at
- tort_type

Case:

```

type: object
properties:
  id:
    type: string
  data:
    type: object
    additionalProperties: true
  status:
    type: string
    enum: [open, processing, analyzed, failed]
  created_at:
    type: string
  updated_at:
    type: string
required: [id, status, created_at, updated_at]

```

CaseDocument:

```

type: object
properties:
  id:
    type: integer
  case_id:
    type: string
  path:
    type: string
  file_hash:
    type: string
    nullable: true
  file_name:
    type: string
  content_type:
    type: string
    nullable: true
  created_at:
    type: string
required: [id, case_id, path, file_name, created_at]

```

CaseAnalysis:

```

type: object
properties:
  id:
    type: integer
  case_id:
    type: string
  features:
    type: object
    additionalProperties: true
    nullable: true
  score:
    type: number
    nullable: true
  ai_model:
    type: string
    nullable: true
  raw_text_length:
    type: number
    nullable: true
  created_at:
    type: string
required: [id, case_id, created_at]

```

AuditLogEntry:

```

type: object
properties:
  id:
    type: integer
  entity_type:
    type: string
  entity_id:
    type: string
  action:
    type: string
  details:
    type: object
    additionalProperties: true
    nullable: true
  ip_address:
    type: string
    nullable: true
  occurred_at:
    type: string
required: [id, entity_type, entity_id, action, occurred_at]

```

CaseDetail:

```

type: object
properties:
  case:
    $ref: "#/components/schemas/Case"
  documents:
    type: array

```



```

    items:
      $ref: "#/components/schemas/CaseDocument"
  analyses:
    type: array
    items:
      $ref: "#/components/schemas/CaseAnalysis"
  audit_trail:
    type: array
    items:
      $ref: "#/components/schemas/AuditLogEntry"
  required: [case, documents, analyses, audit_trail]

CreateCaseResponse:
  type: object
  properties:
    case_id:
      type: string
    status:
      type: string
    job_id:
      type: integer
  required: [case_id, status, job_id]

UploadCaseFileBody:
  type: object
  properties:
    file_name:
      type: string
    content:
      type: string
    content_type:
      type: string
      nullable: true
  required: [file_name, content]

QueueStats:
  type: object
  additionalProperties:
    type: integer

FaxResult:
  type: object
  properties:
    id:
      type: integer
    job_id:
      type: integer
      nullable: true
    source_file:
      type: string
    vault_path:
      type: string
    rx_number:
      type: string
    drug_name:
      type: string
    fill_date:
      type: string
    quantity:
      type: string
    confidence:
      type: number
    raw_text:
      type: string
    status:
      type: string
      enum: [pending, processing, done, failed]
    error:
      type: string
      nullable: true
    created_at:
      type: string
    processed_at:
      type: string
      nullable: true
  required: [id, source_file, vault_path, rx_number, drug_name, fill_date, quantity, confidence, raw_text, status,
created_at]

UploadFaxBody:
  type: object
  properties:
    file_name:
      type: string
    image_base64:
      type: string
    mime_type:
      type: string
      nullable: true
  required: [file_name, image_base64]

UploadFaxResponse:
  type: object
  properties:
    fax_result_id:
      type: integer
    status:
      type: string

```

```

    job_id:
      type: integer
    vault_path:
      type: string
    required: [fax_result_id, status, job_id, vault_path]

ReviewQueueItem:
  type: object
  properties:
    id:
      type: integer
    entity_type:
      type: string
    entity_id:
      type: string
    conflict_type:
      type: string
    severity:
      type: string
    failsafe_mode:
      type: string
    source_module:
      type: string
    summary:
      type: string
    details:
      type: object
    resolution:
      type: string
    resolution_notes:
      type: string
      nullable: true
    resolved_by:
      type: string
      nullable: true
    resolved_at:
      type: string
      nullable: true
    retry_count:
      type: integer
    created_at:
      type: string
    required: [id, entity_type, entity_id, conflict_type, severity, failsafe_mode, source_module, summary, resolution,
retry_count, created_at]

ReviewQueueStats:
  type: object
  properties:
    total_pending:
      type: integer
    by_resolution:
      type: array
      items:
        type: object
        properties:
          resolution:
            type: string
          count:
            type: integer
    by_conflict_type:
      type: array
      items:
        type: object
        properties:
          conflict_type:
            type: string
          count:
            type: integer
    by_severity:
      type: array
      items:
        type: object
        properties:
          severity:
            type: string
          count:
            type: integer
    by_failsafe_mode:
      type: array
      items:
        type: object
        properties:
          failsafe_mode:
            type: string
          count:
            type: integer
    required: [total_pending, by_resolution, by_conflict_type, by_severity, by_failsafe_mode]

ResolveReviewBody:
  type: object
  properties:
    resolution:
      type: string
      enum: [accepted, rejected, escalated]
    resolution_notes:
      type: string
    resolved_by:

```

```

    type: string
    required: [resolution]

NpiProvider:
  type: object
  properties:
    npi:
      type: string
    provider_type:
      type: string
    name:
      type: string
    first_name:
      type: string
    last_name:
      type: string
    credential:
      type: string
    gender:
      type: string
    specialty:
      type: string
    address_line_1:
      type: string
    address_line_2:
      type: string
    city:
      type: string
    state:
      type: string
    postal_code:
      type: string
    phone:
      type: string
    fax:
      type: string
    enumeration_date:
      type: string
    last_updated:
      type: string
    npi_registry_url:
      type: string
    required: [npi, provider_type, name, specialty, city, state, phone, npi_registry_url]

NpiSearchResponse:
  type: object
  properties:
    results:
      type: array
    items:
      $ref: "#/components/schemas/NpiProvider"
    result_count:
      type: integer
    required: [results, result_count]

Paralegal:
  type: object
  properties:
    id:
      type: integer
    name:
      type: string
    email:
      type: string
      nullable: true
    role:
      type: string
    active_cases:
      type: integer
    signed_cases:
      type: integer
    total_assigned:
      type: integer
    created_at:
      type: string
    updated_at:
      type: string
    required: [id, name, role, active_cases, signed_cases, total_assigned, created_at, updated_at]

CreateParalegalBody:
  type: object
  properties:
    name:
      type: string
    email:
      type: string
      nullable: true
    role:
      type: string
      nullable: true
    required: [name]

ParalegalDetail:
  type: object
  properties:
    id:
      type: integer

```

```

    name:
      type: string
    email:
      type: string
      nullable: true
    role:
      type: string
    leads:
      type: array
      items:
        $ref: "#/components/schemas/Lead"
    signed_cases:
      type: integer
    active_cases:
      type: integer
    total_assigned:
      type: integer
    conversion_rate:
      type: integer
    required: [id, name, role, leads, signed_cases, active_cases, total_assigned, conversion_rate]

ParalegalPerformance:
  type: object
  properties:
    paralegal:
      $ref: "#/components/schemas/Paralegal"
    breakdown:
      type: array
      items:
        type: object
        properties:
          status:
            type: string
          count:
            type: integer
          required: [status, count]
    total_assigned:
      type: integer
    signed:
      type: integer
    qualified:
      type: integer
    conversion_rate:
      type: integer
    required: [paralegal, breakdown, total_assigned, signed, qualified, conversion_rate]

AnalyticsOverview:
  type: object
  properties:
    leads:
      type: object
      additionalProperties: true
    cases:
      type: object
      additionalProperties: true
    analysis:
      type: object
      additionalProperties: true
    faxes:
      type: object
      additionalProperties: true
    required: [leads, cases, analysis, faxes]

PipelineTrendPoint:
  type: object
  properties:
    date:
      type: string
    total:
      type: integer
    qualified:
      type: integer
    signed:
      type: integer
    required: [date, total, qualified, signed]

FunnelStage:
  type: object
  properties:
    stage:
      type: string
    count:
      type: integer
    color:
      type: string
    required: [stage, count, color]

TortBreakdownRow:
  type: object
  properties:
    tort_type:
      type: string
    total:
      type: integer
    qualified:
      type: integer
    signed:

```

```

    type: integer
  rejected:
    type: integer
  avg_ad_spend:
    type: number
  required: [tort_type, total, qualified, signed, rejected, avg_ad_spend]

ParalegalLeaderboardRow:
  type: object
  properties:
    id:
      type: integer
    name:
      type: string
    role:
      type: string
    total_assigned:
      type: integer
    signed:
      type: integer
    qualified:
      type: integer
    rejected:
      type: integer
    conversion_rate:
      type: integer
  required: [id, name, role, total_assigned, signed, qualified, rejected, conversion_rate]

AuditSummary:
  type: object
  properties:
    by_entity:
      type: array
      items:
        type: object
        properties:
          entity_type:
            type: string
          count:
            type: integer
        required: [entity_type, count]
    by_action:
      type: array
      items:
        type: object
        properties:
          action:
            type: string
          count:
            type: integer
        required: [action, count]
    total_events:
      type: integer
    last_24h:
      type: integer
    last_7d:
      type: integer
  required: [by_entity, by_action, total_events, last_24h, last_7d]

CustomField:
  type: object
  properties:
    key:
      type: string
    label:
      type: string
    type:
      type: string
      enum: [text, email, tel, date, number, select, textarea, checkbox]
    required:
      type: boolean
    placeholder:
      type: string
      nullable: true
    helper_text:
      type: string
      nullable: true
    options:
      type: array
      nullable: true
      items:
        type: string
    max_length:
      type: integer
      nullable: true
  required: [key, label, type, required]

FormConfig:
  type: object
  properties:
    id:
      type: string
    label:
      type: string
    category:
      type: string
    active:

```

```

    type: boolean
  intro_text:
    type: string
    nullable: true
  fields:
    type: array
    items:
      type: string
  extra_fields:
    type: array
    items:
      type: string
  exposure_fields:
    type: array
    items:
      type: string
  rules:
    type: array
    items:
      type: string
  valid_diagnoses:
    type: array
    items:
      type: string
  custom_fields:
    type: array
    items:
      $ref: "#/components/schemas/CustomField"
  rejection_conditions:
    type: array
    items:
      type: object
  avg_settlement_low:
    type: number
    nullable: true
  avg_settlement_high:
    type: number
    nullable: true
  mdl_status:
    type: string
    nullable: true
  sol_months:
    type: integer
    nullable: true
  updated_at:
    type: string
    nullable: true
  updated_by:
    type: string
    nullable: true
  required: [id, label, fields, rules]

FormConfigUpdate:
  type: object
  properties:
    label:
      type: string
    category:
      type: string
    active:
      type: boolean
    intro_text:
      type: string
      nullable: true
    valid_diagnoses:
      type: array
      items:
        type: string
    extra_fields:
      type: array
      items:
        type: string
    exposure_fields:
      type: array
      items:
        type: string
    custom_fields:
      type: array
      items:
        $ref: "#/components/schemas/CustomField"
    avg_settlement_low:
      type: number
      nullable: true
    avg_settlement_high:
      type: number
      nullable: true
    mdl_status:
      type: string
      nullable: true
    sol_months:
      type: integer
      nullable: true

NpiVerifyResult:
  type: object
  properties:
    npi_found:

```

```

    type: boolean
  npi_number:
    type: string
    nullable: true
  specialty:
    type: string
    nullable: true
  taxonomy_match:
    type: object
    nullable: true
  properties:
    matched:
      type: boolean
    physician_specialty:
      type: string
    expected_specialties:
      type: array
      items:
        type: string
    diagnosis_category:
      type: string
      nullable: true
    fraud_indicators:
      type: array
      items:
        type: string
    confidence:
      type: string
  error:
    type: string
    nullable: true
  required: [npi_found]

FraudCheckResult:
  type: object
  properties:
    tort_validation:
      type: object
      properties:
        valid:
          type: boolean
        tort_id:
          type: string
          nullable: true
        errors:
          type: array
          items:
            type: string
        diagnosis_match:
          type: boolean
        category:
          type: string
          nullable: true
    fraud_result:
      type: object
      properties:
        status:
          type: string
        fraud_score:
          type: integer
        indicators:
          type: array
          items:
            type: object
            properties:
              type:
                type: string
              description:
                type: string
              severity:
                type: string
        summary:
          type: string
    required: [tort_validation, fraud_result]

FormConfigList:
  type: object
  properties:
    tort_campaigns:
      type: array
      items:
        $ref: "#/components/schemas/FormConfig"
    required: [tort_campaigns]

EmailValidationResult:
  type: object
  properties:
    valid:
      type: boolean
    errors:
      type: array
      items:
        type: string
    suggestion:
      type: string
      nullable: true
    required: [valid, errors]

```

```
AddressValidationResult:
  type: object
  properties:
    valid:
      type: boolean
    errors:
      type: array
      items:
        type: string
  required: [valid, errors]
```

```
FormSubmission:
  type: object
  properties:
    first_name:
      type: string
    last_name:
      type: string
    date_of_birth:
      type: string
    street_address:
      type: string
    city:
      type: string
    state:
      type: string
    zip:
      type: string
    phone_primary:
      type: string
    email:
      type: string
    last_4_ssn:
      type: string
    diagnosis:
      type: string
    diagnosis_date:
      type: string
    diagnosis_confirmed:
      type: boolean
    was_at_location:
      type: boolean
    location_name:
      type: string
      nullable: true
    physician_first_name:
      type: string
    physician_last_name:
      type: string
    physician_full_address:
      type: string
    physician_contact_info:
      type: string
    hospital_name:
      type: string
    hospital_fax:
      type: string
    hospital_contact_info:
      type: string
    tort_type:
      type: string
    tcpa_consent:
      type: boolean
    trustedform_cert_url:
      type: string
    trustedform_ip:
      type: string
      nullable: true
    trustedform_user_agent:
      type: string
      nullable: true
    trustedform_timestamp:
      type: string
      nullable: true
    exposure_start:
      type: string
      nullable: true
    exposure_end:
      type: string
      nullable: true
    source:
      type: string
      nullable: true
    notes:
      type: string
      nullable: true
  required:
    - first_name
    - last_name
    - date_of_birth
    - street_address
    - city
    - state
    - zip
    - phone_primary
    - email
```


- last_4_ssn
- diagnosis
- diagnosis_date
- physician_first_name
- physician_last_name
- physician_full_address
- physician_contact_info
- hospital_name
- hospital_fax
- hospital_contact_info
- tort_type
- tcpa_consent
- trustedform_cert_url

FormSubmissionResult:

type: object
properties:
 status:
 type: string
 lead_id:
 type: integer
 lead_status:
 type: string
 tort_id:
 type: string
 nullable: true
 fraud_score:
 type: integer
 fraud_status:
 type: string
 npi_verified:
 type: boolean
 background_check:
 type: object
 nullable: true
 properties:
 status:
 type: string
 summary:
 type: string
 output_state:
 type: string
 pipeline:
 type: array
 items:
 type: object
 properties:
 name:
 type: string
 status:
 type: string
 errors:
 type: array
 items:
 type: string
 required: [status, lead_id, lead_status, output_state]

FormRejection:

type: object
properties:
 status:
 type: string
 errors:
 type: array
 items:
 type: string
 action:
 type: string
 details:
 type: array
 items:
 type: string
 nullable: true
 pipeline:
 type: array
 items:
 type: object
 nullable: true
 failed_step:
 type: string
 nullable: true
 fraud_score:
 type: integer
 nullable: true
 fraud_summary:
 type: string
 nullable: true
 required: [status, errors, action]

BackgroundCheckResult:

type: object
properties:
 status:
 type: string
 enum: [clean, flagged, not_found, error]
 source:
 type: string

```
checked_at:
  type: string
records:
  type: array
  items:
    type: object
    properties:
      type:
        type: string
      description:
        type: string
      date:
        type: string
        nullable: true
      jurisdiction:
        type: string
        nullable: true
      severity:
        type: string
        enum: [low, medium, high]
    required: [type, description, severity]
summary:
  type: string
required: [status, source, checked_at, records, summary]
```

lib/api-spec/orval.config.ts

```
import { defineConfig, InputTransformerFn } from "orval";
import path from "path";

const root = path.resolve(__dirname, "..", "..");
const apiClientReactSrc = path.resolve(root, "lib", "api-client-react", "src");
const apiZodSrc = path.resolve(root, "lib", "api-zod", "src");

// Our exports make assumptions about the title of the API being "Api" (i.e. generated output is `api.ts`).
const titleTransformer: InputTransformerFn = (config) => {
  config.info ??= {};
  config.info.title = "Api";

  return config;
};

export default defineConfig({
  "api-client-react": {
    input: {
      target: "./openapi.yaml",
      override: {
        transformer: titleTransformer,
      },
    },
    output: {
      workspace: apiClientReactSrc,
      target: "generated",
      client: "react-query",
      mode: "split",
      baseUrl: "/api",
      clean: true,
      prettier: true,
      override: {
        fetch: {
          includeHttpResponseReturnType: false,
        },
        mutator: {
          path: path.resolve(apiClientReactSrc, "custom-fetch.ts"),
          name: "customFetch",
        },
      },
    },
  },
  zod: {
    input: {
      target: "./openapi.yaml",
      override: {
        transformer: titleTransformer,
      },
    },
    output: {
      workspace: apiZodSrc,
      client: "zod",
      target: "generated",
      schemas: { path: "generated/types", type: "typescript" },
      mode: "split",
      clean: true,
      prettier: true,
      override: {
        zod: {
          coerce: {
            query: ['boolean', 'number', 'string'],
            param: ['boolean', 'number', 'string'],
            body: ['bigint', 'date'],
            response: ['bigint', 'date'],
          },
        },
      },
      useDates: true,
      useBigInt: true,
    },
  },
});
```

lib/api-spec/package.json

```
{
  "name": "@workspace/api-spec",
  "version": "0.0.0",
  "private": true,
  "scripts": {
    "codegen": "orval --config ./orval.config.ts"
  },
  "devDependencies": {
    "orval": "^8.5.2"
  }
}
```

lib/api-zod/package.json

```
{
  "name": "@workspace/api-zod",
  "version": "0.0.0",
  "private": true,
  "type": "module",
  "exports": {
    ".": "./src/index.ts"
  },
  "dependencies": {
    "zod": "catalog:"
  }
}
```

lib/api-zod/src/generated/api.ts

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import * as zod from "zod";

/**
 * @summary Health check
 */
export const HealthCheckResponse = zod.object({
  status: zod.string(),
});

/**
 * @summary List all leads
 */
export const ListLeadsQueryParams = zod.object({
  status: zod
    .enum(["new", "qualified", "signed", "rejected", "review_required"])
    .optional(),
  tort_type: zod.coerce.string().optional(),
  search: zod.coerce.string().optional(),
  vendor_id: zod.coerce.number().optional(),
  law_firm: zod.coerce.string().optional(),
  client_id: zod.coerce.string().optional(),
  date_from: zod.coerce.string().optional(),
  date_to: zod.coerce.string().optional(),
});

export const ListLeadsResponseItem = zod.object({
  id: zod.number(),
  name: zod.string(),
  email: zod.string().optional(),
  phone: zod.string().optional(),
  tort_type: zod.string(),
  first_name: zod.string().nullish(),
  last_name: zod.string().nullish(),
  date_of_birth: zod.string().nullish(),
  street_address: zod.string().nullish(),
  city: zod.string().nullish(),
  state: zod.string().nullish(),
  zip: zod.string().nullish(),
  phone_primary: zod.string().nullish(),
  last_4_ssn: zod.string().nullish(),
  diagnosis: zod.string().nullish(),
  diagnosis_date: zod.string().nullish(),
  physician_first_name: zod.string().nullish(),
  physician_last_name: zod.string().nullish(),
  physician_full_address: zod.string().nullish(),
  physician_contact_info: zod.string().nullish(),
  hospital_name: zod.string().nullish(),
  hospital_fax: zod.string().nullish(),
  hospital_contact_info: zod.string().nullish(),
  tcpa_consent: zod.boolean().nullish(),
  trustedform_cert_url: zod.string().nullish(),
  trustedform_ip: zod.string().nullish(),
  trustedform_user_agent: zod.string().nullish(),
  trustedform_timestamp: zod.string().nullish(),
  email_validation_status: zod.string().nullish(),
  address_validation_status: zod.string().nullish(),
  background_check_status: zod.string().nullish(),
  background_check_data: zod.string().nullish(),
  exposure_start: zod.string().nullish(),
  exposure_end: zod.string().nullish(),
  diagnosis_confirmed: zod.boolean(),
  diagnosis_type: zod.string().nullish(),
  was_at_location: zod.boolean(),
  location_name: zod.string().nullish(),
  status: zod.enum([
    "new",
    "qualified",
    "signed",
    "rejected",
    "review_required",
  ]),
  rejection_reason: zod.string().nullish(),
  notes: zod.string().nullish(),
  ad_spend: zod.number().nullish(),
  source: zod.string().nullish(),
  assigned_to: zod.number().nullish(),
  routing: zod.enum(["hot", "warm", "cold"]).nullish(),
  vendor_id: zod.number().nullish(),
  law_firm: zod.string().nullish(),
  client_id: zod.string().nullish(),
  created_at: zod.string(),
  updated_at: zod.string(),
});

export const ListLeadsResponse = zod.array(ListLeadsResponseItem);

/**
 * @summary Create a new lead (intake)
 */
```

```

export const CreateLeadBody = zod.object({
  name: zod.string(),
  email: zod.string().optional(),
  phone: zod.string().optional(),
  tort_type: zod.string(),
  first_name: zod.string(),
  last_name: zod.string(),
  date_of_birth: zod.string(),
  street_address: zod.string(),
  city: zod.string(),
  state: zod.string(),
  zip: zod.string(),
  phone_primary: zod.string(),
  last_4_ssn: zod.string(),
  diagnosis: zod.string(),
  diagnosis_date: zod.string(),
  physician_first_name: zod.string(),
  physician_last_name: zod.string(),
  physician_full_address: zod.string(),
  physician_contact_info: zod.string(),
  hospital_name: zod.string(),
  hospital_fax: zod.string(),
  hospital_contact_info: zod.string(),
  exposure_start: zod.string().nullish(),
  exposure_end: zod.string().nullish(),
  diagnosis_confirmed: zod.boolean(),
  diagnosis_type: zod.string().nullish(),
  was_at_location: zod.boolean(),
  location_name: zod.string().nullish(),
  notes: zod.string().nullish(),
  ad_spend: zod.number().nullish(),
  source: zod.string().nullish(),
  vendor_id: zod.number().nullish(),
  law_firm: zod.string().nullish(),
  client_id: zod.string().nullish(),
});

/**
 * @summary Get a single lead
 */
export const GetLeadParams = zod.object({
  id: zod.coerce.number(),
});

export const GetLeadResponse = zod.object({
  id: zod.number(),
  name: zod.string(),
  email: zod.string().optional(),
  phone: zod.string().optional(),
  tort_type: zod.string(),
  first_name: zod.string().nullish(),
  last_name: zod.string().nullish(),
  date_of_birth: zod.string().nullish(),
  street_address: zod.string().nullish(),
  city: zod.string().nullish(),
  state: zod.string().nullish(),
  zip: zod.string().nullish(),
  phone_primary: zod.string().nullish(),
  last_4_ssn: zod.string().nullish(),
  diagnosis: zod.string().nullish(),
  diagnosis_date: zod.string().nullish(),
  physician_first_name: zod.string().nullish(),
  physician_last_name: zod.string().nullish(),
  physician_full_address: zod.string().nullish(),
  physician_contact_info: zod.string().nullish(),
  hospital_name: zod.string().nullish(),
  hospital_fax: zod.string().nullish(),
  hospital_contact_info: zod.string().nullish(),
  tcpa_consent: zod.boolean().nullish(),
  trustedform_cert_url: zod.string().nullish(),
  trustedform_ip: zod.string().nullish(),
  trustedform_user_agent: zod.string().nullish(),
  trustedform_timestamp: zod.string().nullish(),
  email_validation_status: zod.string().nullish(),
  address_validation_status: zod.string().nullish(),
  background_check_status: zod.string().nullish(),
  background_check_data: zod.string().nullish(),
  exposure_start: zod.string().nullish(),
  exposure_end: zod.string().nullish(),
  diagnosis_confirmed: zod.boolean(),
  diagnosis_type: zod.string().nullish(),
  was_at_location: zod.boolean(),
  location_name: zod.string().nullish(),
  status: zod.enum([
    "new",
    "qualified",
    "signed",
    "rejected",
    "review_required",
  ]),
  rejection_reason: zod.string().nullish(),
  notes: zod.string().nullish(),
  ad_spend: zod.number().nullish(),
  source: zod.string().nullish(),
  assigned_to: zod.number().nullish(),
  routing: zod.enum(["hot", "warm", "cold"]).nullish(),
  vendor_id: zod.number().nullish(),
});

```

```

    law_firm: zod.string().nullish(),
    client_id: zod.string().nullish(),
    created_at: zod.string(),
    updated_at: zod.string(),
  });

/**
 * @summary Update lead status or fields
 */
export const UpdateLeadParams = zod.object({
  id: zod.coerce.number(),
});

export const UpdateLeadBody = zod.object({
  name: zod.string().optional(),
  email: zod.string().optional(),
  phone: zod.string().optional(),
  tort_type: zod.string().optional(),
  first_name: zod.string().optional(),
  last_name: zod.string().optional(),
  date_of_birth: zod.string().optional(),
  street_address: zod.string().optional(),
  city: zod.string().optional(),
  state: zod.string().optional(),
  zip: zod.string().optional(),
  phone_primary: zod.string().optional(),
  last_4_ssn: zod.string().optional(),
  diagnosis: zod.string().optional(),
  diagnosis_date: zod.string().optional(),
  physician_first_name: zod.string().optional(),
  physician_last_name: zod.string().optional(),
  physician_full_address: zod.string().optional(),
  physician_contact_info: zod.string().optional(),
  hospital_name: zod.string().optional(),
  hospital_fax: zod.string().optional(),
  hospital_contact_info: zod.string().optional(),
  tcpa_consent: zod.boolean().optional(),
  trustedform_cert_url: zod.string().optional(),
  exposure_start: zod.string().nullish(),
  exposure_end: zod.string().nullish(),
  diagnosis_confirmed: zod.boolean().optional(),
  diagnosis_type: zod.string().nullish(),
  was_at_location: zod.boolean().optional(),
  location_name: zod.string().nullish(),
  status: zod
    .enum(["new", "qualified", "signed", "rejected", "review_required"])
    .optional(),
  rejection_reason: zod.string().nullish(),
  notes: zod.string().nullish(),
  ad_spend: zod.number().nullish(),
  source: zod.string().nullish(),
  vendor_id: zod.number().nullish(),
  law_firm: zod.string().nullish(),
  client_id: zod.string().nullish(),
});

export const UpdateLeadResponse = zod.object({
  id: zod.number(),
  name: zod.string(),
  email: zod.string().optional(),
  phone: zod.string().optional(),
  tort_type: zod.string(),
  first_name: zod.string().nullish(),
  last_name: zod.string().nullish(),
  date_of_birth: zod.string().nullish(),
  street_address: zod.string().nullish(),
  city: zod.string().nullish(),
  state: zod.string().nullish(),
  zip: zod.string().nullish(),
  phone_primary: zod.string().nullish(),
  last_4_ssn: zod.string().nullish(),
  diagnosis: zod.string().nullish(),
  diagnosis_date: zod.string().nullish(),
  physician_first_name: zod.string().nullish(),
  physician_last_name: zod.string().nullish(),
  physician_full_address: zod.string().nullish(),
  physician_contact_info: zod.string().nullish(),
  hospital_name: zod.string().nullish(),
  hospital_fax: zod.string().nullish(),
  hospital_contact_info: zod.string().nullish(),
  tcpa_consent: zod.boolean().nullish(),
  trustedform_cert_url: zod.string().nullish(),
  trustedform_ip: zod.string().nullish(),
  trustedform_user_agent: zod.string().nullish(),
  trustedform_timestamp: zod.string().nullish(),
  email_validation_status: zod.string().nullish(),
  address_validation_status: zod.string().nullish(),
  background_check_status: zod.string().nullish(),
  background_check_data: zod.string().nullish(),
  exposure_start: zod.string().nullish(),
  exposure_end: zod.string().nullish(),
  diagnosis_confirmed: zod.boolean(),
  diagnosis_type: zod.string().nullish(),
  was_at_location: zod.boolean(),
  location_name: zod.string().nullish(),
  status: zod.enum([
    "new",

```



```

    "qualified",
    "signed",
    "rejected",
    "review_required",
  ]),
  rejection_reason: zod.string().nullish(),
  notes: zod.string().nullish(),
  ad_spend: zod.number().nullish(),
  source: zod.string().nullish(),
  assigned_to: zod.number().nullish(),
  routing: zod.enum(["hot", "warm", "cold"]).nullish(),
  vendor_id: zod.number().nullish(),
  law_firm: zod.string().nullish(),
  client_id: zod.string().nullish(),
  created_at: zod.string(),
  updated_at: zod.string(),
});

/**
 * @summary Delete a lead
 */
export const DeleteLeadParams = zod.object({
  id: zod.coerce.number(),
});

/**
 * @summary Run Boolean Gatekeeper qualification on a lead
 */
export const QualifyLeadParams = zod.object({
  id: zod.coerce.number(),
});

export const QualifyLeadResponse = zod.object({
  lead_id: zod.number(),
  qualified: zod.boolean(),
  status: zod.enum(["qualified", "rejected"]),
  reason: zod.string(),
  gates_passed: zod.array(zod.string()),
  gates_failed: zod.array(zod.string()),
});

/**
 * @summary Export leads as CSV
 */
export const ExportLeadsQueryParams = zod.object({
  lead_id: zod.coerce.number().optional(),
  status: zod.coerce.string().optional(),
  tort_type: zod.coerce.string().optional(),
  vendor_id: zod.coerce.number().optional(),
  law_firm: zod.coerce.string().optional(),
  client_id: zod.coerce.string().optional(),
  date_from: zod.coerce.string().optional(),
  date_to: zod.coerce.string().optional(),
  source: zod.coerce.string().optional(),
  fields: zod.coerce.string().optional(),
});

/**
 * @summary List all vendors
 */
export const ListVendorsResponseItem = zod.object({
  id: zod.number(),
  name: zod.string(),
  contact_name: zod.string().nullish(),
  contact_email: zod.string().nullish(),
  contact_phone: zod.string().nullish(),
  type: zod.string(),
  status: zod.string(),
  notes: zod.string().nullish(),
  created_at: zod.string(),
  updated_at: zod.string(),
});
export const ListVendorsResponse = zod.array(ListVendorsResponseItem);

/**
 * @summary Create a new vendor
 */
export const CreateVendorBody = zod.object({
  name: zod.string(),
  contact_name: zod.string().nullish(),
  contact_email: zod.string().nullish(),
  contact_phone: zod.string().nullish(),
  type: zod.string().optional(),
  status: zod.string().optional(),
  notes: zod.string().nullish(),
});

/**
 * @summary Get a single vendor
 */
export const GetVendorParams = zod.object({
  id: zod.coerce.number(),
});

export const GetVendorResponse = zod.object({
  id: zod.number(),
  name: zod.string(),

```

```

    contact_name: zod.string().nullish(),
    contact_email: zod.string().nullish(),
    contact_phone: zod.string().nullish(),
    type: zod.string(),
    status: zod.string(),
    notes: zod.string().nullish(),
    created_at: zod.string(),
    updated_at: zod.string(),
  });

  /**
   * @summary Update a vendor
   */
  export const UpdateVendorParams = zod.object({
    id: zod.coerce.number(),
  });

  export const UpdateVendorBody = zod.object({
    name: zod.string().optional(),
    contact_name: zod.string().nullish(),
    contact_email: zod.string().nullish(),
    contact_phone: zod.string().nullish(),
    type: zod.string().optional(),
    status: zod.string().optional(),
    notes: zod.string().nullish(),
  });

  export const UpdateVendorResponse = zod.object({
    id: zod.number(),
    name: zod.string(),
    contact_name: zod.string().nullish(),
    contact_email: zod.string().nullish(),
    contact_phone: zod.string().nullish(),
    type: zod.string(),
    status: zod.string(),
    notes: zod.string().nullish(),
    created_at: zod.string(),
    updated_at: zod.string(),
  });

  /**
   * @summary Delete a vendor
   */
  export const DeleteVendorParams = zod.object({
    id: zod.coerce.number(),
  });

  /**
   * @summary List all documents
   */
  export const ListDocumentsQueryParams = zod.object({
    lead_id: zod.coerce.number().optional(),
  });

  export const ListDocumentsResponseItem = zod.object({
    id: zod.number(),
    lead_id: zod.number(),
    document_type: zod.enum([
      "retainer",
      "medical_record",
      "intake_form",
      "other",
    ]),
    file_name: zod.string(),
    file_url: zod.string().nullish(),
    signed: zod.boolean(),
    signed_at: zod.string().nullish(),
    notes: zod.string().nullish(),
    created_at: zod.string(),
  });
  export const ListDocumentsResponse = zod.array(ListDocumentsResponseItem);

  /**
   * @summary Upload or register a document
   */
  export const CreateDocumentBody = zod.object({
    lead_id: zod.number(),
    document_type: zod.enum([
      "retainer",
      "medical_record",
      "intake_form",
      "other",
    ]),
    file_name: zod.string(),
    file_url: zod.string().nullish(),
    signed: zod.boolean(),
    signed_at: zod.string().nullish(),
    notes: zod.string().nullish(),
  });

  /**
   * @summary Update document metadata or status
   */
  export const UpdateDocumentParams = zod.object({
    id: zod.coerce.number(),
  });

```

```

export const UpdateDocumentBody = zod.object({
  document_type: zod
    .enum(["retainer", "medical_record", "intake_form", "other"])
    .optional(),
  file_name: zod.string().optional(),
  file_url: zod.string().nullish(),
  signed: zod.boolean().optional(),
  signed_at: zod.string().nullish(),
  notes: zod.string().nullish(),
});

export const UpdateDocumentResponse = zod.object({
  id: zod.number(),
  lead_id: zod.number(),
  document_type: zod.enum([
    "retainer",
    "medical_record",
    "intake_form",
    "other",
  ]),
  file_name: zod.string(),
  file_url: zod.string().nullish(),
  signed: zod.boolean(),
  signed_at: zod.string().nullish(),
  notes: zod.string().nullish(),
  created_at: zod.string(),
});

/**
 * @summary Delete a document
 */
export const DeleteDocumentParams = zod.object({
  id: zod.coerce.number(),
});

/**
 * @summary Get pipeline summary stats
 */
export const GetDashboardStatsResponse = zod.object({
  total_leads: zod.number(),
  qualified_leads: zod.number(),
  signed_retainers: zod.number(),
  rejected_leads: zod.number(),
  new_leads: zod.number(),
  total_ad_spend: zod.number(),
  cpsr: zod.number().nullish(),
  qualification_rate: zod.number(),
  conversion_rate: zod.number(),
  leads_today: zod.number(),
  leads_this_week: zod.number(),
});

/**
 * @summary Get lead counts grouped by status and tort type
 */
export const GetPipelineBreakdownResponse = zod.object({
  by_status: zod.array(
    zod.object({
      status: zod.string(),
      count: zod.number(),
    })
  ),
  by_tort_type: zod.array(
    zod.object({
      tort_type: zod.string(),
      count: zod.number(),
      qualified: zod.number(),
      signed: zod.number(),
    })
  ),
});

/**
 * @summary Get recent lead activity feed
 */
export const GetRecentActivityResponseItem = zod.object({
  id: zod.number(),
  lead_id: zod.number(),
  lead_name: zod.string(),
  event_type: zod.string(),
  description: zod.string(),
  occurred_at: zod.string(),
  tort_type: zod.string(),
});
export const GetRecentActivityResponse = zod.array(
  GetRecentActivityResponseItem,
);

/**
 * @summary List all cases
 */
export const ListCasesResponseItem = zod.object({
  id: zod.string(),
  data: zod.record(zod.string(), zod.unknown()).optional(),
  status: zod.enum(["open", "processing", "analyzed", "failed"]),
  created_at: zod.string(),
  updated_at: zod.string(),
});

```

```

});
export const ListCasesResponse = zod.array(ListCasesResponseItem);

/**
 * @summary Submit a new case to the processing queue
 */
export const CreateCaseBody = zod.record(zod.string(), zod.unknown());

/**
 * @summary Get job queue statistics by status
 */
export const GetQueueStatsResponse = zod.record(zod.string(), zod.number());

/**
 * @summary Get full case detail with documents, analyses, and audit trail
 */
export const GetCaseParams = zod.object({
  id: zod.coerce.string(),
});

export const GetCaseResponse = zod.object({
  case: zod.object({
    id: zod.string(),
    data: zod.record(zod.string(), zod.unknown()).optional(),
    status: zod.enum(["open", "processing", "analyzed", "failed"]),
    created_at: zod.string(),
    updated_at: zod.string(),
  }),
  documents: zod.array(
    zod.object({
      id: zod.number(),
      case_id: zod.string(),
      path: zod.string(),
      file_hash: zod.string().nullish(),
      file_name: zod.string(),
      content_type: zod.string().nullish(),
      created_at: zod.string(),
    })
  ),
  analyses: zod.array(
    zod.object({
      id: zod.number(),
      case_id: zod.string(),
      features: zod.record(zod.string(), zod.unknown()).nullish(),
      score: zod.number().nullish(),
      ai_model: zod.string().nullish(),
      raw_text_length: zod.number().nullish(),
      created_at: zod.string(),
    })
  ),
  audit_trail: zod.array(
    zod.object({
      id: zod.number(),
      entity_type: zod.string(),
      entity_id: zod.string(),
      action: zod.string(),
      details: zod.record(zod.string(), zod.unknown()).nullish(),
      ip_address: zod.string().nullish(),
      occurred_at: zod.string(),
    })
  ),
});

/**
 * @summary Upload a document to the case vault
 */
export const UploadCaseFileParams = zod.object({
  id: zod.coerce.string(),
});

export const UploadCaseFileBody = zod.object({
  file_name: zod.string(),
  content: zod.string(),
  content_type: zod.string().nullish(),
});

export const UploadCaseFileResponse = zod.object({
  case_id: zod.string(),
  status: zod.string(),
  job_id: zod.number(),
});

/**
 * @summary Queue AI extraction and scoring for all case documents
 */
export const AnalyzeCaseFilesParams = zod.object({
  id: zod.coerce.string(),
});

export const AnalyzeCaseFilesResponse = zod.object({
  case_id: zod.string(),
  status: zod.string(),
  job_id: zod.number(),
});

/**
 * @summary List review queue items with optional filters

```

```

*/
export const ListReviewQueueQueryParams = zod.object({
  resolution: zod.coerce.string().optional(),
  conflict_type: zod.coerce.string().optional(),
  severity: zod.coerce.string().optional(),
  entity_type: zod.coerce.string().optional(),
  limit: zod.coerce.string().optional(),
});

export const ListReviewQueueResponseItem = zod.object({
  id: zod.number(),
  entity_type: zod.string(),
  entity_id: zod.string(),
  conflict_type: zod.string(),
  severity: zod.string(),
  failsafe_mode: zod.string(),
  source_module: zod.string(),
  summary: zod.string(),
  details: zod.object({}).passthrough().optional(),
  resolution: zod.string(),
  resolution_notes: zod.string().nullish(),
  resolved_by: zod.string().nullish(),
  resolved_at: zod.string().nullish(),
  retry_count: zod.number(),
  created_at: zod.string(),
});

export const ListReviewQueueResponse = zod.array(ListReviewQueueResponseItem);

/**
 * @summary Get review queue statistics
 */
export const GetReviewQueueStatsResponse = zod.object({
  total_pending: zod.number(),
  by_resolution: zod.array(
    zod.object({
      resolution: zod.string().optional(),
      count: zod.number().optional(),
    })
  ),
  by_conflict_type: zod.array(
    zod.object({
      conflict_type: zod.string().optional(),
      count: zod.number().optional(),
    })
  ),
  by_severity: zod.array(
    zod.object({
      severity: zod.string().optional(),
      count: zod.number().optional(),
    })
  ),
  by_failsafe_mode: zod.array(
    zod.object({
      failsafe_mode: zod.string().optional(),
      count: zod.number().optional(),
    })
  ),
});

/**
 * @summary Resolve a review queue item
 */
export const ResolveReviewItemParams = zod.object({
  id: zod.coerce.number(),
});

export const ResolveReviewItemBody = zod.object({
  resolution: zod.enum(["accepted", "rejected", "escalated"]),
  resolution_notes: zod.string().optional(),
  resolved_by: zod.string().optional(),
});

export const ResolveReviewItemResponse = zod.object({
  id: zod.number(),
  entity_type: zod.string(),
  entity_id: zod.string(),
  conflict_type: zod.string(),
  severity: zod.string(),
  failsafe_mode: zod.string(),
  source_module: zod.string(),
  summary: zod.string(),
  details: zod.object({}).passthrough().optional(),
  resolution: zod.string(),
  resolution_notes: zod.string().nullish(),
  resolved_by: zod.string().nullish(),
  resolved_at: zod.string().nullish(),
  retry_count: zod.number(),
  created_at: zod.string(),
});

/**
 * @summary Search the NPI Registry for doctors and providers
 */
export const SearchNpiQueryParams = zod.object({
  npi_number: zod.coerce.string().optional(),
  first_name: zod.coerce.string().optional(),
  last_name: zod.coerce.string().optional(),
});

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    city: zod.coerce.string().optional(),
    state: zod.coerce.string().optional(),
    specialty: zod.coerce.string().optional(),
    limit: zod.coerce.string().optional(),
  });

export const SearchNpiResponse = zod.object({
  results: zod.array(
    zod.object({
      npi: zod.string(),
      provider_type: zod.string(),
      name: zod.string(),
      first_name: zod.string().optional(),
      last_name: zod.string().optional(),
      credential: zod.string().optional(),
      gender: zod.string().optional(),
      specialty: zod.string(),
      address_line_1: zod.string().optional(),
      address_line_2: zod.string().optional(),
      city: zod.string(),
      state: zod.string(),
      postal_code: zod.string().optional(),
      phone: zod.string(),
      fax: zod.string().optional(),
      enumeration_date: zod.string().optional(),
      last_updated: zod.string().optional(),
      npi_registry_url: zod.string(),
    })
  ),
  result_count: zod.number(),
});

/**
 * @summary Look up a single provider by NPI number
 */
export const LookupNpiParams = zod.object({
  npi: zod.coerce.string(),
});

export const LookupNpiResponse = zod.object({
  npi: zod.string(),
  provider_type: zod.string(),
  name: zod.string(),
  first_name: zod.string().optional(),
  last_name: zod.string().optional(),
  credential: zod.string().optional(),
  gender: zod.string().optional(),
  specialty: zod.string(),
  address_line_1: zod.string().optional(),
  address_line_2: zod.string().optional(),
  city: zod.string(),
  state: zod.string(),
  postal_code: zod.string().optional(),
  phone: zod.string(),
  fax: zod.string().optional(),
  enumeration_date: zod.string().optional(),
  last_updated: zod.string().optional(),
  npi_registry_url: zod.string(),
});

/**
 * @summary Get all tort campaign form configurations
 */
export const GetFormConfigsResponse = zod.object({
  tort_campaigns: zod.array(
    zod.object({
      id: zod.string(),
      label: zod.string(),
      category: zod.string().optional(),
      active: zod.boolean().optional(),
      intro_text: zod.string().nullish(),
      fields: zod.array(zod.string()),
      extra_fields: zod.array(zod.string()).optional(),
      exposure_fields: zod.array(zod.string()).optional(),
      rules: zod.array(zod.string()),
      valid_diagnoses: zod.array(zod.string()).optional(),
      custom_fields: zod
        .array(
          zod.object({
            key: zod.string(),
            label: zod.string(),
            type: zod.enum([
              "text",
              "email",
              "tel",
              "date",
              "number",
              "select",
              "textarea",
              "checkbox",
            ]),
            required: zod.boolean(),
            placeholder: zod.string().nullish(),
            helper_text: zod.string().nullish(),
            options: zod.array(zod.string()).nullish(),
            max_length: zod.number().nullish(),
          })
        )
        .array(),
    })
  ),
});

```

```

    )
    .optional(),
    rejection_conditions: zod.array(zod.object({}).passthrough()).optional(),
    avg_settlement_low: zod.number().nullish(),
    avg_settlement_high: zod.number().nullish(),
    mdl_status: zod.string().nullish(),
    sol_months: zod.number().nullish(),
    updated_at: zod.string().nullish(),
    updated_by: zod.string().nullish(),
  }},
),
});

/**
 * @summary Get a single tort form configuration
 */
export const GetFormConfigParams = zod.object({
  tortId: zod.coerce.string(),
});

export const GetFormConfigResponse = zod.object({
  id: zod.string(),
  label: zod.string(),
  category: zod.string().optional(),
  active: zod.boolean().optional(),
  intro_text: zod.string().nullish(),
  fields: zod.array(zod.string()),
  extra_fields: zod.array(zod.string()).optional(),
  exposure_fields: zod.array(zod.string()).optional(),
  rules: zod.array(zod.string()),
  valid_diagnoses: zod.array(zod.string()).optional(),
  custom_fields: zod
    .array(
      zod.object({
        key: zod.string(),
        label: zod.string(),
        type: zod.enum([
          "text",
          "email",
          "tel",
          "date",
          "number",
          "select",
          "textarea",
          "checkbox",
        ]),
        required: zod.boolean(),
        placeholder: zod.string().nullish(),
        helper_text: zod.string().nullish(),
        options: zod.array(zod.string()).nullish(),
        max_length: zod.number().nullish(),
      })
    )
    .optional(),
  rejection_conditions: zod.array(zod.object({}).passthrough()).optional(),
  avg_settlement_low: zod.number().nullish(),
  avg_settlement_high: zod.number().nullish(),
  mdl_status: zod.string().nullish(),
  sol_months: zod.number().nullish(),
  updated_at: zod.string().nullish(),
  updated_by: zod.string().nullish(),
});

/**
 * @summary Update a tort form configuration (admin only)
 */
export const UpdateFormConfigParams = zod.object({
  tortId: zod.coerce.string(),
});

export const UpdateFormConfigBody = zod.object({
  label: zod.string().optional(),
  category: zod.string().optional(),
  active: zod.boolean().optional(),
  intro_text: zod.string().nullish(),
  valid_diagnoses: zod.array(zod.string()).optional(),
  extra_fields: zod.array(zod.string()).optional(),
  exposure_fields: zod.array(zod.string()).optional(),
  custom_fields: zod
    .array(
      zod.object({
        key: zod.string(),
        label: zod.string(),
        type: zod.enum([
          "text",
          "email",
          "tel",
          "date",
          "number",
          "select",
          "textarea",
          "checkbox",
        ]),
        required: zod.boolean(),
        placeholder: zod.string().nullish(),
        helper_text: zod.string().nullish(),
        options: zod.array(zod.string()).nullish(),

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        max_length: zod.number().nullish(),
    }},
)
.optional(),
avg_settlement_low: zod.number().nullish(),
avg_settlement_high: zod.number().nullish(),
mdl_status: zod.string().nullish(),
sol_months: zod.number().nullish(),
});

export const UpdateFormConfigResponse = zod.object({
  id: zod.string(),
  label: zod.string(),
  category: zod.string().optional(),
  active: zod.boolean().optional(),
  intro_text: zod.string().nullish(),
  fields: zod.array(zod.string()),
  extra_fields: zod.array(zod.string()).optional(),
  exposure_fields: zod.array(zod.string()).optional(),
  rules: zod.array(zod.string()),
  valid_diagnoses: zod.array(zod.string()).optional(),
  custom_fields: zod
    .array(
      zod.object({
        key: zod.string(),
        label: zod.string(),
        type: zod.enum([
          "text",
          "email",
          "tel",
          "date",
          "number",
          "select",
          "textarea",
          "checkbox",
        ]),
        required: zod.boolean(),
        placeholder: zod.string().nullish(),
        helper_text: zod.string().nullish(),
        options: zod.array(zod.string()).nullish(),
        max_length: zod.number().nullish(),
      })
    )
    .optional(),
  rejection_conditions: zod.array(zod.object({}).passthrough()).optional(),
  avg_settlement_low: zod.number().nullish(),
  avg_settlement_high: zod.number().nullish(),
  mdl_status: zod.string().nullish(),
  sol_months: zod.number().nullish(),
  updated_at: zod.string().nullish(),
  updated_by: zod.string().nullish(),
});

/**
 * @summary Add a custom field to a tort form (admin only)
 */
export const AddCustomFieldParams = zod.object({
  tortId: zod.coerce.string(),
});

export const AddCustomFieldBody = zod.object({
  key: zod.string(),
  label: zod.string(),
  type: zod.enum([
    "text",
    "email",
    "tel",
    "date",
    "number",
    "select",
    "textarea",
    "checkbox",
  ]),
  required: zod.boolean(),
  placeholder: zod.string().nullish(),
  helper_text: zod.string().nullish(),
  options: zod.array(zod.string()).nullish(),
  max_length: zod.number().nullish(),
});

export const AddCustomFieldResponse = zod.object({
  id: zod.string(),
  label: zod.string(),
  category: zod.string().optional(),
  active: zod.boolean().optional(),
  intro_text: zod.string().nullish(),
  fields: zod.array(zod.string()),
  extra_fields: zod.array(zod.string()).optional(),
  exposure_fields: zod.array(zod.string()).optional(),
  rules: zod.array(zod.string()),
  valid_diagnoses: zod.array(zod.string()).optional(),
  custom_fields: zod
    .array(
      zod.object({
        key: zod.string(),
        label: zod.string(),
        type: zod.enum([

```



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        "text",
        "email",
        "tel",
        "date",
        "number",
        "select",
        "textarea",
        "checkbox",
    ]),
    required: zod.boolean(),
    placeholder: zod.string().nullish(),
    helper_text: zod.string().nullish(),
    options: zod.array(zod.string()).nullish(),
    max_length: zod.number().nullish(),
  })),
  ).optional(),
  rejection_conditions: zod.array(zod.object({}).passthrough()).optional(),
  avg_settlement_low: zod.number().nullish(),
  avg_settlement_high: zod.number().nullish(),
  mdl_status: zod.string().nullish(),
  sol_months: zod.number().nullish(),
  updated_at: zod.string().nullish(),
  updated_by: zod.string().nullish(),
});

/**
 * @summary Remove a custom field from a tort form (admin only)
 */
export const RemoveCustomFieldParams = zod.object({
  tortId: zod.coerce.string(),
  key: zod.coerce.string(),
});

export const RemoveCustomFieldResponse = zod.object({
  id: zod.string(),
  label: zod.string(),
  category: zod.string().optional(),
  active: zod.boolean().optional(),
  intro_text: zod.string().nullish(),
  fields: zod.array(zod.string()),
  extra_fields: zod.array(zod.string()).optional(),
  exposure_fields: zod.array(zod.string()).optional(),
  rules: zod.array(zod.string()),
  valid_diagnoses: zod.array(zod.string()).optional(),
  custom_fields: zod
    .array(
      zod.object({
        key: zod.string(),
        label: zod.string(),
        type: zod.enum([
          "text",
          "email",
          "tel",
          "date",
          "number",
          "select",
          "textarea",
          "checkbox",
        ]),
        required: zod.boolean(),
        placeholder: zod.string().nullish(),
        helper_text: zod.string().nullish(),
        options: zod.array(zod.string()).nullish(),
        max_length: zod.number().nullish(),
      })
    )
    .optional(),
  rejection_conditions: zod.array(zod.object({}).passthrough()).optional(),
  avg_settlement_low: zod.number().nullish(),
  avg_settlement_high: zod.number().nullish(),
  mdl_status: zod.string().nullish(),
  sol_months: zod.number().nullish(),
  updated_at: zod.string().nullish(),
  updated_by: zod.string().nullish(),
});

/**
 * @summary Validate email (RFC + typo detection)
 */
export const ValidateEmailBody = zod.object({
  email: zod.string(),
});

export const ValidateEmailResponse = zod.object({
  valid: zod.boolean(),
  errors: zod.array(zod.string()),
  suggestion: zod.string().nullish(),
});

/**
 * @summary Validate address (US format)
 */
export const ValidateAddressBody = zod.object({
  street_address: zod.string(),
  city: zod.string(),
  state: zod.string(),
});

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    zip: zod.string(),
  });

export const ValidateAddressResponse = zod.object({
  valid: zod.boolean(),
  errors: zod.array(zod.string()),
});

/**
 * @summary Submit intake form with full compliance validation
 */
export const SubmitFormBody = zod.object({
  first_name: zod.string(),
  last_name: zod.string(),
  date_of_birth: zod.string(),
  street_address: zod.string(),
  city: zod.string(),
  state: zod.string(),
  zip: zod.string(),
  phone_primary: zod.string(),
  email: zod.string(),
  last_4_ssn: zod.string(),
  diagnosis: zod.string(),
  diagnosis_date: zod.string(),
  diagnosis_confirmed: zod.boolean().optional(),
  was_at_location: zod.boolean().optional(),
  location_name: zod.string().nullish(),
  physician_first_name: zod.string(),
  physician_last_name: zod.string(),
  physician_full_address: zod.string(),
  physician_contact_info: zod.string(),
  hospital_name: zod.string(),
  hospital_fax: zod.string(),
  hospital_contact_info: zod.string(),
  tort_type: zod.string(),
  tcpa_consent: zod.boolean(),
  trustedform_cert_url: zod.string(),
  trustedform_ip: zod.string().nullish(),
  trustedform_user_agent: zod.string().nullish(),
  trustedform_timestamp: zod.string().nullish(),
  exposure_start: zod.string().nullish(),
  exposure_end: zod.string().nullish(),
  source: zod.string().nullish(),
  notes: zod.string().nullish(),
});

/**
 * @summary Get embeddable JavaScript form for a tort campaign
 */
export const GetFormEmbedParams = zod.object({
  tortId: zod.coerce.string(),
});

/**
 * @summary Run background check on a person
 */
export const RunBackgroundCheckBody = zod.object({
  first_name: zod.string(),
  last_name: zod.string(),
  state: zod.string().optional(),
  date_of_birth: zod.string().optional(),
});

export const RunBackgroundCheckResponse = zod.object({
  status: zod.enum(["clean", "flagged", "not_found", "error"]),
  source: zod.string(),
  checked_at: zod.string(),
  records: zod.array(
    zod.object({
      type: zod.string(),
      description: zod.string(),
      date: zod.string().nullish(),
      jurisdiction: zod.string().nullish(),
      severity: zod.enum(["low", "medium", "high"]),
    })
  ),
  summary: zod.string(),
});

/**
 * @summary Run background check on an existing lead
 */
export const RunLeadBackgroundCheckParams = zod.object({
  id: zod.coerce.number(),
});

export const RunLeadBackgroundCheckResponse = zod.object({
  status: zod.enum(["clean", "flagged", "not_found", "error"]),
  source: zod.string(),
  checked_at: zod.string(),
  records: zod.array(
    zod.object({
      type: zod.string(),
      description: zod.string(),
      date: zod.string().nullish(),
      jurisdiction: zod.string().nullish(),
      severity: zod.enum(["low", "medium", "high"]),
    })
  ),
});

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    }},
  ),
  summary: zod.string(),
});

/**
 * @summary Get all tort categories with grouped torts
 */
export const GetTortCategoriesResponseItem = zod.object({
  category: zod.string(),
  torts: zod.array(
    zod.object({
      id: zod.string(),
      label: zod.string(),
    })
  ),
});

export const GetTortCategoriesResponse = zod.array(
  GetTortCategoriesResponseItem,
);

/**
 * @summary Verify physician NPI and match taxonomy to diagnosis
 */
export const VerifyNpiBody = zod.object({
  physician_first_name: zod.string(),
  physician_last_name: zod.string(),
  diagnosis: zod.string(),
});

export const VerifyNpiResponse = zod.object({
  npi_found: zod.boolean(),
  npi_number: zod.string().nullish(),
  specialty: zod.string().nullish(),
  taxonomy_match: zod
    .object({
      matched: zod.boolean().optional(),
      physician_specialty: zod.string().optional(),
      expected_specialties: zod.array(zod.string()).optional(),
      diagnosis_category: zod.string().nullish(),
      fraud_indicators: zod.array(zod.string()).optional(),
      confidence: zod.string().optional(),
    })
    .nullish(),
  error: zod.string().nullish(),
});

/**
 * @summary Run fraud detection on claim data
 */
export const RunFraudCheckBody = zod.object({
  tort_type: zod.string(),
  diagnosis: zod.string(),
  physician_first_name: zod.string().optional(),
  physician_last_name: zod.string().optional(),
  lead_data: zod.object({}).passthrough().optional(),
});

export const RunFraudCheckResponse = zod.object({
  tort_validation: zod.object({
    valid: zod.boolean().optional(),
    tort_id: zod.string().nullish(),
    errors: zod.array(zod.string()).optional(),
    diagnosis_match: zod.boolean().optional(),
    category: zod.string().nullish(),
  }),
  fraud_result: zod.object({
    status: zod.string().optional(),
    fraud_score: zod.number().optional(),
    indicators: zod
      .array(
        zod.object({
          type: zod.string().optional(),
          description: zod.string().optional(),
          severity: zod.string().optional(),
        })
      )
      .optional(),
    summary: zod.string().optional(),
  }),
});

/**
 * @summary Escalate case to FBI tip line and log event
 */
export const EscalateToFbiBody = zod.object({
  lead_id: zod.string(),
  reason: zod.string(),
  fraud_indicators: zod.array(zod.object({}).passthrough()).optional(),
});

export const EscalateToFbiResponse = zod.object({
  status: zod.string(),
  fbi_tip_url: zod.string(),
  message: zod.string(),
});

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/**
 * @summary List all paralegals
 */
export const ListParalegalsResponseItem = zod.object({
  id: zod.number(),
  name: zod.string(),
  email: zod.string().nullish(),
  role: zod.string(),
  active_cases: zod.number(),
  signed_cases: zod.number(),
  total_assigned: zod.number(),
  created_at: zod.string(),
  updated_at: zod.string(),
});
export const ListParalegalsResponse = zod.array(ListParalegalsResponseItem);

/**
 * @summary Create a new paralegal
 */
export const CreateParalegalBody = zod.object({
  name: zod.string(),
  email: zod.string().nullish(),
  role: zod.string().nullish(),
});

/**
 * @summary Get paralegal detail with assigned leads
 */
export const GetParalegalParams = zod.object({
  id: zod.coerce.number(),
});

export const GetParalegalResponse = zod.object({
  id: zod.number(),
  name: zod.string(),
  email: zod.string().nullish(),
  role: zod.string(),
  leads: zod.array(
    zod.object({
      id: zod.number(),
      name: zod.string(),
      email: zod.string().optional(),
      phone: zod.string().optional(),
      tort_type: zod.string(),
      first_name: zod.string().nullish(),
      last_name: zod.string().nullish(),
      date_of_birth: zod.string().nullish(),
      street_address: zod.string().nullish(),
      city: zod.string().nullish(),
      state: zod.string().nullish(),
      zip: zod.string().nullish(),
      phone_primary: zod.string().nullish(),
      last_4_ssn: zod.string().nullish(),
      diagnosis: zod.string().nullish(),
      diagnosis_date: zod.string().nullish(),
      physician_first_name: zod.string().nullish(),
      physician_last_name: zod.string().nullish(),
      physician_full_address: zod.string().nullish(),
      physician_contact_info: zod.string().nullish(),
      hospital_name: zod.string().nullish(),
      hospital_fax: zod.string().nullish(),
      hospital_contact_info: zod.string().nullish(),
      tcpa_consent: zod.boolean().nullish(),
      trustedform_cert_url: zod.string().nullish(),
      trustedform_ip: zod.string().nullish(),
      trustedform_user_agent: zod.string().nullish(),
      trustedform_timestamp: zod.string().nullish(),
      email_validation_status: zod.string().nullish(),
      address_validation_status: zod.string().nullish(),
      background_check_status: zod.string().nullish(),
      background_check_data: zod.string().nullish(),
      exposure_start: zod.string().nullish(),
      exposure_end: zod.string().nullish(),
      diagnosis_confirmed: zod.boolean(),
      diagnosis_type: zod.string().nullish(),
      was_at_location: zod.boolean(),
      location_name: zod.string().nullish(),
      status: zod.enum([
        "new",
        "qualified",
        "signed",
        "rejected",
        "review_required",
      ]),
      rejection_reason: zod.string().nullish(),
      notes: zod.string().nullish(),
      ad_spend: zod.number().nullish(),
      source: zod.string().nullish(),
      assigned_to: zod.number().nullish(),
      routing: zod.enum(["hot", "warm", "cold"]).nullish(),
      vendor_id: zod.number().nullish(),
      law_firm: zod.string().nullish(),
      client_id: zod.string().nullish(),
      created_at: zod.string(),
      updated_at: zod.string(),
    })
  ),
});

```

```

    signed_cases: zod.number(),
    active_cases: zod.number(),
    total_assigned: zod.number(),
    conversion_rate: zod.number(),
  });

/**
 * @summary Get paralegal performance breakdown
 */
export const GetParalegalPerformanceParams = zod.object({
  id: zod.coerce.number(),
});

export const GetParalegalPerformanceResponse = zod.object({
  paralegal: zod.object({
    id: zod.number(),
    name: zod.string(),
    email: zod.string().nullish(),
    role: zod.string(),
    active_cases: zod.number(),
    signed_cases: zod.number(),
    total_assigned: zod.number(),
    created_at: zod.string(),
    updated_at: zod.string(),
  }),
  breakdown: zod.array(
    zod.object({
      status: zod.string(),
      count: zod.number(),
    })
  ),
  total_assigned: zod.number(),
  signed: zod.number(),
  qualified: zod.number(),
  conversion_rate: zod.number(),
});

/**
 * @summary Full analytics overview across all modules
 */
export const GetAnalyticsOverviewResponse = zod.object({
  leads: zod.record(zod.string(), zod.unknown()),
  cases: zod.record(zod.string(), zod.unknown()),
  analysis: zod.record(zod.string(), zod.unknown()),
  faxes: zod.record(zod.string(), zod.unknown()),
});

/**
 * @summary Daily pipeline trend data
 */
export const GetPipelineTrendResponseItem = zod.object({
  date: zod.string(),
  total: zod.number(),
  qualified: zod.number(),
  signed: zod.number(),
});
export const GetPipelineTrendResponse = zod.array(GetPipelineTrendResponseItem);

/**
 * @summary Conversion funnel stages
 */
export const GetConversionFunnelResponseItem = zod.object({
  stage: zod.string(),
  count: zod.number(),
  color: zod.string(),
});
export const GetConversionFunnelResponse = zod.array(
  GetConversionFunnelResponseItem
);

/**
 * @summary Breakdown by tort type
 */
export const GetTortBreakdownResponseItem = zod.object({
  tort_type: zod.string(),
  total: zod.number(),
  qualified: zod.number(),
  signed: zod.number(),
  rejected: zod.number(),
  avg_ad_spend: zod.number(),
});
export const GetTortBreakdownResponse = zod.array(GetTortBreakdownResponseItem);

/**
 * @summary Paralegal performance leaderboard
 */
export const GetParalegalLeaderboardResponseItem = zod.object({
  id: zod.number(),
  name: zod.string(),
  role: zod.string(),
  total_assigned: zod.number(),
  signed: zod.number(),
  qualified: zod.number(),
  rejected: zod.number(),
  conversion_rate: zod.number(),
});
export const GetParalegalLeaderboardResponse = zod.array(

```

```

    GetParalegalLeaderboardResponseItem,
  );

  /**
   * @summary Full audit trail with optional filters
   */
  export const GetAuditTrailQueryParams = zod.object({
    limit: zod.coerce.number().optional(),
    entity_type: zod.coerce.string().optional(),
    action: zod.coerce.string().optional(),
  });

  export const GetAuditTrailResponseItem = zod.object({
    id: zod.number(),
    entity_type: zod.string(),
    entity_id: zod.string(),
    action: zod.string(),
    details: zod.record(zod.string(), zod.unknown()).nullish(),
    ip_address: zod.string().nullish(),
    occurred_at: zod.string(),
  });
  export const GetAuditTrailResponse = zod.array(GetAuditTrailResponseItem);

  /**
   * @summary Audit trail summary with counts
   */
  export const GetAuditSummaryResponse = zod.object({
    by_entity: zod.array(
      zod.object({
        entity_type: zod.string(),
        count: zod.number(),
      })
    ),
    by_action: zod.array(
      zod.object({
        action: zod.string(),
        count: zod.number(),
      })
    ),
    total_events: zod.number(),
    last_24h: zod.number(),
    last_7d: zod.number(),
  });

  /**
   * @summary Upload a fax image for OCR processing
   */
  export const UploadFaxBody = zod.object({
    file_name: zod.string(),
    image_base64: zod.string(),
    mime_type: zod.string().nullish(),
  });

  export const UploadFaxResponse = zod.object({
    fax_result_id: zod.number(),
    status: zod.string(),
    job_id: zod.number(),
    vault_path: zod.string(),
  });

  /**
   * @summary List all Legora Grid results
   */
  export const ListFaxResultsResponseItem = zod.object({
    id: zod.number(),
    job_id: zod.number().nullish(),
    source_file: zod.string(),
    vault_path: zod.string(),
    rx_number: zod.string(),
    drug_name: zod.string(),
    fill_date: zod.string(),
    quantity: zod.string(),
    confidence: zod.number(),
    raw_text: zod.string(),
    status: zod.enum(["pending", "processing", "done", "failed"]),
    error: zod.string().nullish(),
    created_at: zod.string(),
    processed_at: zod.string().nullish(),
  });
  export const ListFaxResultsResponse = zod.array(ListFaxResultsResponseItem);

  /**
   * @summary Get a single Legora Grid result
   */
  export const GetFaxResultParams = zod.object({
    id: zod.coerce.number(),
  });

  export const GetFaxResultResponse = zod.object({
    id: zod.number(),
    job_id: zod.number().nullish(),
    source_file: zod.string(),
    vault_path: zod.string(),
    rx_number: zod.string(),
    drug_name: zod.string(),
    fill_date: zod.string(),
    quantity: zod.string(),
  });

```

```
confidence: zod.number(),
raw_text: zod.string(),
status: zod.enum(["pending", "processing", "done", "failed"]),
error: zod.string().nullish(),
created_at: zod.string(),
processed_at: zod.string().nullish(),
});

/**
 * @summary OCR fax processing queue stats
 */
export const GetOcrQueueStatsResponse = zod.record(zod.string(), zod.number());
```

[lib/api-zod/src/generated/types/activityItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface ActivityItem {
  id: number;
  lead_id: number;
  lead_name: string;
  event_type: string;
  description: string;
  occurred_at: string;
  tort_type: string;
}
```


[lib/api-zod/src/generated/types/addressValidationResult.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface AddressValidationResult {
  valid: boolean;
  errors: string[];
}
```

[lib/api-zod/src/generated/types/analyticsOverview.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { AnalyticsOverviewAnalysis } from "../analyticsOverviewAnalysis";
import type { AnalyticsOverviewCases } from "../analyticsOverviewCases";
import type { AnalyticsOverviewFaxes } from "../analyticsOverviewFaxes";
import type { AnalyticsOverviewLeads } from "../analyticsOverviewLeads";

export interface AnalyticsOverview {
  leads: AnalyticsOverviewLeads;
  cases: AnalyticsOverviewCases;
  analysis: AnalyticsOverviewAnalysis;
  faxes: AnalyticsOverviewFaxes;
}
```

[lib/api-zod/src/generated/types/analyticsOverviewAnalysis.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type AnalyticsOverviewAnalysis = { [key: string]: unknown };
```

[lib/api-zod/src/generated/types/analyticsOverviewCases.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
export type AnalyticsOverviewCases = { [key: string]: unknown };
```

[lib/api-zod/src/generated/types/analyticsOverviewFaxes.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
export type AnalyticsOverviewFaxes = { [key: string]: unknown };
```

[lib/api-zod/src/generated/types/analyticsOverviewLeads.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type AnalyticsOverviewLeads = { [key: string]: unknown };
```

[lib/api-zod/src/generated/types/auditLogEntry.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { AuditLogEntryDetails } from "../auditLogEntryDetails";

export interface AuditLogEntry {
  id: number;
  entity_type: string;
  entity_id: string;
  action: string;
  details?: AuditLogEntryDetails;
  ip_address?: string | null;
  occurred_at: string;
}
```

[lib/api-zod/src/generated/types/auditLogEntryDetails.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
export type AuditLogEntryDetails = { [key: string]: unknown } | null;
```


[lib/api-zod/src/generated/types/auditSummary.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { AuditSummaryByActionItem } from "../auditSummaryByActionItem";
import type { AuditSummaryByEntityItem } from "../auditSummaryByEntityItem";

export interface AuditSummary {
  by_entity: AuditSummaryByEntityItem[];
  by_action: AuditSummaryByActionItem[];
  total_events: number;
  last_24h: number;
  last_7d: number;
}
```

[lib/api-zod/src/generated/types/auditSummaryByActionItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type AuditSummaryByActionItem = {
  action: string;
  count: number;
};
```

[lib/api-zod/src/generated/types/auditSummaryByEntityItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type AuditSummaryByEntityItem = {
  entity_type: string;
  count: number;
};
```

[lib/api-zod/src/generated/types/backgroundCheckResult.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { BackgroundCheckResultRecordsItem } from "../backgroundCheckResultRecordsItem";
import type { BackgroundCheckResultStatus } from "../backgroundCheckResultStatus";

export interface BackgroundCheckResult {
  status: BackgroundCheckResultStatus;
  source: string;
  checked_at: string;
  records: BackgroundCheckResultRecordsItem[];
  summary: string;
}
```

[lib/api-zod/src/generated/types/backgroundCheckResultRecordsItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { BackgroundCheckResultRecordsItemSeverity } from "../backgroundCheckResultRecordsItemSeverity";

export type BackgroundCheckResultRecordsItem = {
  type: string;
  description: string;
  date?: string | null;
  jurisdiction?: string | null;
  severity: BackgroundCheckResultRecordsItemSeverity;
};
```

[lib/api-zod/src/generated/types/backgroundCheckResultRecordsItemSeverity.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type BackgroundCheckResultRecordsItemSeverity =
  (typeof BackgroundCheckResultRecordsItemSeverity)[keyof typeof BackgroundCheckResultRecordsItemSeverity];

export const BackgroundCheckResultRecordsItemSeverity = {
  low: "low",
  medium: "medium",
  high: "high",
} as const;
```

[lib/api-zod/src/generated/types/backgroundCheckResultStatus.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type BackgroundCheckResultStatus =
  (typeof BackgroundCheckResultStatus)[keyof typeof BackgroundCheckResultStatus];

export const BackgroundCheckResultStatus = {
  clean: "clean",
  flagged: "flagged",
  not_found: "not_found",
  error: "error",
} as const;
```

lib/api-zod/src/generated/types/case.ts

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { CaseData } from "../caseData";
import type { CaseStatus } from "../caseStatus";

export interface Case {
  id: string;
  data?: CaseData;
  status: CaseStatus;
  created_at: string;
  updated_at: string;
}
```


lib/api-zod/src/generated/types/caseAnalysis.ts

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { CaseAnalysisFeatures } from "../caseAnalysisFeatures";

export interface CaseAnalysis {
  id: number;
  case_id: string;
  features?: CaseAnalysisFeatures;
  score?: number | null;
  ai_model?: string | null;
  raw_text_length?: number | null;
  created_at: string;
}
```

[lib/api-zod/src/generated/types/caseAnalysisFeatures.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
export type CaseAnalysisFeatures = { [key: string]: unknown } | null;
```

[lib/api-zod/src/generated/types/caseData.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type CaseData = { [key: string]: unknown };
```

[lib/api-zod/src/generated/types/caseDetail.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { AuditLogEntry } from "../auditLogEntry";
import type { Case } from "../case";
import type { CaseAnalysis } from "../caseAnalysis";
import type { CaseDocument } from "../caseDocument";

export interface CaseDetail {
  case: Case;
  documents: CaseDocument[];
  analyses: CaseAnalysis[];
  audit_trail: AuditLogEntry[];
}
```

[lib/api-zod/src/generated/types/caseDocument.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface CaseDocument {
  id: number;
  case_id: string;
  path: string;
  file_hash?: string | null;
  file_name: string;
  content_type?: string | null;
  created_at: string;
}
```

[lib/api-zod/src/generated/types/caseStatus.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type CaseStatus = (typeof CaseStatus)[keyof typeof CaseStatus];

export const CaseStatus = {
  open: "open",
  processing: "processing",
  analyzed: "analyzed",
  failed: "failed",
} as const;
```

[lib/api-zod/src/generated/types/createCaseBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type CreateCaseBody = { [key: string]: unknown };
```

[lib/api-zod/src/generated/types/createCaseResponse.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface CreateCaseResponse {
  case_id: string;
  status: string;
  job_id: number;
}
```


[lib/api-zod/src/generated/types/createDocumentBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { CreateDocumentBodyDocumentType } from "../createDocumentBodyDocumentType";

export interface CreateDocumentBody {
  lead_id: number;
  document_type: CreateDocumentBodyDocumentType;
  file_name: string;
  file_url?: string | null;
  signed: boolean;
  signed_at?: string | null;
  notes?: string | null;
}
```

[lib/api-zod/src/generated/types/createDocumentBodyDocumentType.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type CreateDocumentBodyDocumentType =
  (typeof CreateDocumentBodyDocumentType)[keyof typeof CreateDocumentBodyDocumentType];

export const CreateDocumentBodyDocumentType = {
  retainer: "retainer",
  medical_record: "medical_record",
  intake_form: "intake_form",
  other: "other",
} as const;
```

lib/api-zod/src/generated/types/createLeadBody.ts

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface CreateLeadBody {
  name: string;
  email?: string;
  phone?: string;
  tort_type: string;
  first_name: string;
  last_name: string;
  date_of_birth: string;
  street_address: string;
  city: string;
  state: string;
  zip: string;
  phone_primary: string;
  last_4_ssn: string;
  diagnosis: string;
  diagnosis_date: string;
  physician_first_name: string;
  physician_last_name: string;
  physician_full_address: string;
  physician_contact_info: string;
  hospital_name: string;
  hospital_fax: string;
  hospital_contact_info: string;
  exposure_start?: string | null;
  exposure_end?: string | null;
  diagnosis_confirmed: boolean;
  diagnosis_type?: string | null;
  was_at_location: boolean;
  location_name?: string | null;
  notes?: string | null;
  ad_spend?: number | null;
  source?: string | null;
  vendor_id?: number | null;
  law_firm?: string | null;
  client_id?: string | null;
}
```

[lib/api-zod/src/generated/types/createParalegalBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface CreateParalegalBody {
  name: string;
  email?: string | null;
  role?: string | null;
}
```

[lib/api-zod/src/generated/types/createVendorBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface CreateVendorBody {
  name: string;
  contact_name?: string | null;
  contact_email?: string | null;
  contact_phone?: string | null;
  type?: string;
  status?: string;
  notes?: string | null;
}
```

[lib/api-zod/src/generated/types/customField.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { CustomFieldType } from "../customFieldType";

export interface CustomField {
  key: string;
  label: string;
  type: CustomFieldType;
  required: boolean;
  placeholder?: string | null;
  helper_text?: string | null;
  options?: string[] | null;
  max_length?: number | null;
}
```

[lib/api-zod/src/generated/types/customFieldType.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type CustomFieldType =
  (typeof CustomFieldType)[keyof typeof CustomFieldType];

export const CustomFieldType = {
  text: "text",
  email: "email",
  tel: "tel",
  date: "date",
  number: "number",
  select: "select",
  textarea: "textarea",
  checkbox: "checkbox",
} as const;
```

[lib/api-zod/src/generated/types/dashboardStats.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
```

```
export interface DashboardStats {
  total_leads: number;
  qualified_leads: number;
  signed_retainers: number;
  rejected_leads: number;
  new_leads: number;
  total_ad_spend: number;
  cpsr?: number | null;
  qualification_rate: number;
  conversion_rate: number;
  leads_today: number;
  leads_this_week: number;
}
```


[lib/api-zod/src/generated/types/document.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { DocumentDocumentType } from "../documentDocumentType";

export interface Document {
  id: number;
  lead_id: number;
  document_type: DocumentDocumentType;
  file_name: string;
  file_url?: string | null;
  signed: boolean;
  signed_at?: string | null;
  notes?: string | null;
  created_at: string;
}
```

[lib/api-zod/src/generated/types/documentDocumentType.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type DocumentDocumentType =
  (typeof DocumentDocumentType)[keyof typeof DocumentDocumentType];

export const DocumentDocumentType = {
  retainer: "retainer",
  medical_record: "medical_record",
  intake_form: "intake_form",
  other: "other",
} as const;
```

[lib/api-zod/src/generated/types/emailValidationResult.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface EmailValidationResult {
  valid: boolean;
  errors: string[];
  suggestion?: string | null;
}
```

[lib/api-zod/src/generated/types/errorResponse.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface ErrorResponse {
  error: string;
}
```

[lib/api-zod/src/generated/types/escalateToFbi200.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type EscalateToFbi200 = {
  status: string;
  fbi_tip_url: string;
  message: string;
};
```

[lib/api-zod/src/generated/types/escalateToFbiBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { EscalateToFbiBodyFraudIndicatorsItem } from "./escalateToFbiBodyFraudIndicatorsItem";

export type EscalateToFbiBody = {
  lead_id: string;
  reason: string;
  fraud_indicators?: EscalateToFbiBodyFraudIndicatorsItem[];
};
```

[lib/api-zod/src/generated/types/escalateToFbiBodyFraudIndicatorsItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type EscalateToFbiBodyFraudIndicatorsItem = { [key: string]: unknown };
```

[lib/api-zod/src/generated/types/exportLeadsParams.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type ExportLeadsParams = {
  lead_id?: number;
  status?: string;
  tort_type?: string;
  vendor_id?: number;
  law_firm?: string;
  client_id?: string;
  date_from?: string;
  date_to?: string;
  source?: string;
  fields?: string;
};
```


[lib/api-zod/src/generated/types/faxResult.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { FaxResultStatus } from "../faxResultStatus";

export interface FaxResult {
  id: number;
  job_id?: number | null;
  source_file: string;
  vault_path: string;
  rx_number: string;
  drug_name: string;
  fill_date: string;
  quantity: string;
  confidence: number;
  raw_text: string;
  status: FaxResultStatus;
  error?: string | null;
  created_at: string;
  processed_at?: string | null;
}
```

[lib/api-zod/src/generated/types/faxResultStatus.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type FaxResultStatus =
  (typeof FaxResultStatus)[keyof typeof FaxResultStatus];

export const FaxResultStatus = {
  pending: "pending",
  processing: "processing",
  done: "done",
  failed: "failed",
} as const;
```

[lib/api-zod/src/generated/types/formConfig.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { CustomField } from "../customField";
import type { FormConfigRejectionConditionsItem } from "../formConfigRejectionConditionsItem";

export interface FormConfig {
  id: string;
  label: string;
  category?: string;
  active?: boolean;
  intro_text?: string | null;
  fields: string[];
  extra_fields?: string[];
  exposure_fields?: string[];
  rules: string[];
  valid_diagnoses?: string[];
  custom_fields?: CustomField[];
  rejection_conditions?: FormConfigRejectionConditionsItem[];
  avg_settlement_low?: number | null;
  avg_settlement_high?: number | null;
  mdl_status?: string | null;
  sol_months?: number | null;
  updated_at?: string | null;
  updated_by?: string | null;
}
```

[lib/api-zod/src/generated/types/formConfigList.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { FormConfig } from "../formConfig";

export interface FormConfigList {
  tort_campaigns: FormConfig[];
}
```

[lib/api-zod/src/generated/types/formConfigRejectionConditionsItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type FormConfigRejectionConditionsItem = { [key: string]: unknown };
```

[lib/api-zod/src/generated/types/formConfigUpdate.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { CustomField } from "../customField";

export interface FormConfigUpdate {
  label?: string;
  category?: string;
  active?: boolean;
  intro_text?: string | null;
  valid_diagnoses?: string[];
  extra_fields?: string[];
  exposure_fields?: string[];
  custom_fields?: CustomField[];
  avg_settlement_low?: number | null;
  avg_settlement_high?: number | null;
  mdl_status?: string | null;
  sol_months?: number | null;
}
```

[lib/api-zod/src/generated/types/formRejection.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { FormRejectionPipelineItem } from "../formRejectionPipelineItem";

export interface FormRejection {
  status: string;
  errors: string[];
  action: string;
  details?: string[] | null;
  pipeline?: FormRejectionPipelineItem[] | null;
  failed_step?: string | null;
  fraud_score?: number | null;
  fraud_summary?: string | null;
}
```

[lib/api-zod/src/generated/types/formRejectionPipelineItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
export type FormRejectionPipelineItem = { [key: string]: unknown };
```


[lib/api-zod/src/generated/types/formSubmission.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface FormSubmission {
  first_name: string;
  last_name: string;
  date_of_birth: string;
  street_address: string;
  city: string;
  state: string;
  zip: string;
  phone_primary: string;
  email: string;
  last_4_ssn: string;
  diagnosis: string;
  diagnosis_date: string;
  diagnosis_confirmed?: boolean;
  was_at_location?: boolean;
  location_name?: string | null;
  physician_first_name: string;
  physician_last_name: string;
  physician_full_address: string;
  physician_contact_info: string;
  hospital_name: string;
  hospital_fax: string;
  hospital_contact_info: string;
  tort_type: string;
  tcpa_consent: boolean;
  trustedform_cert_url: string;
  trustedform_ip?: string | null;
  trustedform_user_agent?: string | null;
  trustedform_timestamp?: string | null;
  exposure_start?: string | null;
  exposure_end?: string | null;
  source?: string | null;
  notes?: string | null;
}
```

[lib/api-zod/src/generated/types/formSubmissionResult.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { FormSubmissionResultBackgroundCheck } from "../formSubmissionResultBackgroundCheck";
import type { FormSubmissionResultPipelineItem } from "../formSubmissionResultPipelineItem";

export interface FormSubmissionResult {
  status: string;
  lead_id: number;
  lead_status: string;
  tort_id?: string | null;
  fraud_score?: number;
  fraud_status?: string;
  napi_verified?: boolean;
  background_check?: FormSubmissionResultBackgroundCheck;
  output_state: string;
  pipeline?: FormSubmissionResultPipelineItem[];
}
```

[lib/api-zod/src/generated/types/formSubmissionResultBackgroundCheck.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type FormSubmissionResultBackgroundCheck = {
  status?: string;
  summary?: string;
} | null;
```

[lib/api-zod/src/generated/types/formSubmissionResultPipelineItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type FormSubmissionResultPipelineItem = {
  name?: string;
  status?: string;
  errors?: string[];
};
```

[lib/api-zod/src/generated/types/fraudCheckResult.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { FraudCheckResultFraudResult } from "../fraudCheckResultFraudResult";
import type { FraudCheckResultTortValidation } from "../fraudCheckResultTortValidation";

export interface FraudCheckResult {
  tort_validation: FraudCheckResultTortValidation;
  fraud_result: FraudCheckResultFraudResult;
}
```

[lib/api-zod/src/generated/types/fraudCheckResultFraudResult.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { FraudCheckResultFraudResultIndicatorsItem } from "../fraudCheckResultFraudResultIndicatorsItem";

export type FraudCheckResultFraudResult = {
  status?: string;
  fraud_score?: number;
  indicators?: FraudCheckResultFraudResultIndicatorsItem[];
  summary?: string;
};
```

[lib/api-zod/src/generated/types/fraudCheckResultFraudResultIndicatorsItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type FraudCheckResultFraudResultIndicatorsItem = {
  type?: string;
  description?: string;
  severity?: string;
};
```

[lib/api-zod/src/generated/types/fraudCheckResultTortValidation.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type FraudCheckResultTortValidation = {
  valid?: boolean;
  tort_id?: string | null;
  errors?: string[];
  diagnosis_match?: boolean;
  category?: string | null;
};
```


[lib/api-zod/src/generated/types/funnelStage.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface FunnelStage {
  stage: string;
  count: number;
  color: string;
}
```

[lib/api-zod/src/generated/types/getAuditTrailParams.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type GetAuditTrailParams = {
  limit?: number;
  entity_type?: string;
  action?: string;
};
```

[lib/api-zod/src/generated/types/getTortCategories200Item.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { GetTortCategories200ItemTortsItem } from "./getTortCategories200ItemTortsItem";

export type GetTortCategories200Item = {
  category: string;
  torts: GetTortCategories200ItemTortsItem[];
};
```

[lib/api-zod/src/generated/types/getTortCategories200ItemTortsItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type GetTortCategories200ItemTortsItem = {
  id: string;
  label: string;
};
```

[lib/api-zod/src/generated/types/healthStatus.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface HealthStatus {
  status: string;
}
```

[lib/api-zod/src/generated/types/index.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export * from "./activityItem";
export * from "./addressValidationResult";
export * from "./analyticsOverview";
export * from "./analyticsOverviewAnalysis";
export * from "./analyticsOverviewCases";
export * from "./analyticsOverviewFaxes";
export * from "./analyticsOverviewLeads";
export * from "./auditLogEntry";
export * from "./auditLogEntryDetails";
export * from "./auditSummary";
export * from "./auditSummaryByActionItem";
export * from "./auditSummaryByEntityItem";
export * from "./backgroundCheckResult";
export * from "./backgroundCheckResultRecordsItem";
export * from "./backgroundCheckResultRecordsItemSeverity";
export * from "./backgroundCheckResultStatus";
export * from "./case";
export * from "./caseAnalysis";
export * from "./caseAnalysisFeatures";
export * from "./caseData";
export * from "./caseDetail";
export * from "./caseDocument";
export * from "./caseStatus";
export * from "./createCaseBody";
export * from "./createCaseResponse";
export * from "./createDocumentBody";
export * from "./createDocumentBodyDocumentType";
export * from "./createLeadBody";
export * from "./createParalegalBody";
export * from "./createVendorBody";
export * from "./customField";
export * from "./customFieldType";
export * from "./dashboardStats";
export * from "./document";
export * from "./documentDocumentType";
export * from "./emailValidationResult";
export * from "./errorResponse";
export * from "./escalateToFbi200";
export * from "./escalateToFbiBody";
export * from "./escalateToFbiBodyFraudIndicatorsItem";
export * from "./exportLeadsParams";
export * from "./faxResult";
export * from "./faxResultStatus";
export * from "./formConfig";
export * from "./formConfigList";
export * from "./formConfigRejectionConditionsItem";
export * from "./formConfigUpdate";
export * from "./formRejection";
export * from "./formRejectionPipelineItem";
export * from "./formSubmission";
export * from "./formSubmissionResult";
export * from "./formSubmissionResultBackgroundCheck";
export * from "./formSubmissionResultPipelineItem";
export * from "./fraudCheckResult";
export * from "./fraudCheckResultFraudResult";
export * from "./fraudCheckResultFraudResultIndicatorsItem";
export * from "./fraudCheckResultTortValidation";
export * from "./funnelStage";
export * from "./getAuditTrailParams";
export * from "./getTortCategories200Item";
export * from "./getTortCategories200ItemTortsItem";
export * from "./healthStatus";
export * from "./lead";
export * from "./leadRouting";
export * from "./leadStatus";
export * from "./listDocumentsParams";
export * from "./listLeadsParams";
export * from "./listLeadsStatus";
export * from "./listReviewQueueParams";
export * from "./npiProvider";
export * from "./npiSearchResponse";
export * from "./npiVerifyResult";
export * from "./npiVerifyResultTaxonomyMatch";
export * from "./paralegal";
export * from "./paralegalDetail";
export * from "./paralegalLeaderboardRow";
export * from "./paralegalPerformance";
export * from "./paralegalPerformanceBreakdownItem";
export * from "./pipelineBreakdown";
export * from "./pipelineBreakdownByStatusItem";
export * from "./pipelineBreakdownByTortTypeItem";
export * from "./pipelineTrendPoint";
export * from "./qualificationResult";
export * from "./qualificationResultStatus";
export * from "./queueStats";
export * from "./resolveReviewBody";
export * from "./resolveReviewBodyResolution";
```

```
export * from "./reviewQueueItem";
export * from "./reviewQueueItemDetails";
export * from "./reviewQueueStats";
export * from "./reviewQueueStatsByConflictTypeItem";
export * from "./reviewQueueStatsByFailsafeModeItem";
export * from "./reviewQueueStatsByResolutionItem";
export * from "./reviewQueueStatsBySeverityItem";
export * from "./runBackgroundCheckBody";
export * from "./runFraudCheckBody";
export * from "./runFraudCheckBodyLeadData";
export * from "./searchNpiParams";
export * from "./tortBreakdownRow";
export * from "./updateDocumentBody";
export * from "./updateDocumentBodyDocumentType";
export * from "./updateLeadBody";
export * from "./updateLeadBodyStatus";
export * from "./updateVendorBody";
export * from "./uploadCaseFileBody";
export * from "./uploadFaxBody";
export * from "./uploadFaxResponse";
export * from "./validateAddressBody";
export * from "./validateEmailBody";
export * from "./vendor";
export * from "./verifyNpiBody";
```

lib/api-zod/src/generated/types/lead.ts

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { LeadRouting } from "../leadRouting";
import type { LeadStatus } from "../leadStatus";

export interface Lead {
  id: number;
  name: string;
  email?: string;
  phone?: string;
  tort_type: string;
  first_name?: string | null;
  last_name?: string | null;
  date_of_birth?: string | null;
  street_address?: string | null;
  city?: string | null;
  state?: string | null;
  zip?: string | null;
  phone_primary?: string | null;
  last_4_ssn?: string | null;
  diagnosis?: string | null;
  diagnosis_date?: string | null;
  physician_first_name?: string | null;
  physician_last_name?: string | null;
  physician_full_address?: string | null;
  physician_contact_info?: string | null;
  hospital_name?: string | null;
  hospital_fax?: string | null;
  hospital_contact_info?: string | null;
  tcpa_consent?: boolean | null;
  trustedform_cert_url?: string | null;
  trustedform_ip?: string | null;
  trustedform_user_agent?: string | null;
  trustedform_timestamp?: string | null;
  email_validation_status?: string | null;
  address_validation_status?: string | null;
  background_check_status?: string | null;
  background_check_data?: string | null;
  exposure_start?: string | null;
  exposure_end?: string | null;
  diagnosis_confirmed: boolean;
  diagnosis_type?: string | null;
  was_at_location: boolean;
  location_name?: string | null;
  status: LeadStatus;
  rejection_reason?: string | null;
  notes?: string | null;
  ad_spend?: number | null;
  source?: string | null;
  assigned_to?: number | null;
  routing?: LeadRouting;
  vendor_id?: number | null;
  law_firm?: string | null;
  client_id?: string | null;
  created_at: string;
  updated_at: string;
}
```


[lib/api-zod/src/generated/types/leadRouting.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type LeadRouting = (typeof LeadRouting)[keyof typeof LeadRouting] | null;

export const LeadRouting = {
  hot: "hot",
  warm: "warm",
  cold: "cold",
} as const;
```

[lib/api-zod/src/generated/types/leadStatus.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type LeadStatus = (typeof LeadStatus)[keyof typeof LeadStatus];

export const LeadStatus = {
  new: "new",
  qualified: "qualified",
  signed: "signed",
  rejected: "rejected",
  review_required: "review_required",
} as const;
```

[lib/api-zod/src/generated/types/listDocumentsParams.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type ListDocumentsParams = {
  lead_id?: number;
};
```

[lib/api-zod/src/generated/types/listLeadsParams.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { ListLeadsStatus } from "../listLeadsStatus";

export type ListLeadsParams = {
  status?: ListLeadsStatus;
  tort_type?: string;
  search?: string;
  vendor_id?: number;
  law_firm?: string;
  client_id?: string;
  date_from?: string;
  date_to?: string;
};
```

[lib/api-zod/src/generated/types/listLeadsStatus.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type ListLeadsStatus =
  (typeof ListLeadsStatus)[keyof typeof ListLeadsStatus];

export const ListLeadsStatus = {
  new: "new",
  qualified: "qualified",
  signed: "signed",
  rejected: "rejected",
  review_required: "review_required",
} as const;
```

[lib/api-zod/src/generated/types/listReviewQueueParams.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type ListReviewQueueParams = {
  resolution?: string;
  conflict_type?: string;
  severity?: string;
  entity_type?: string;
  limit?: string;
};
```

[lib/api-zod/src/generated/types/npiProvider.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
```

```
export interface NpiProvider {
  npi: string;
  provider_type: string;
  name: string;
  first_name?: string;
  last_name?: string;
  credential?: string;
  gender?: string;
  specialty: string;
  address_line_1?: string;
  address_line_2?: string;
  city: string;
  state: string;
  postal_code?: string;
  phone: string;
  fax?: string;
  enumeration_date?: string;
  last_updated?: string;
  npi_registry_url: string;
}
```

[lib/api-zod/src/generated/types/npiSearchResponse.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { NpiProvider } from "../npiProvider";

export interface NpiSearchResponse {
  results: NpiProvider[];
  result_count: number;
}
```


[lib/api-zod/src/generated/types/npiVerifyResult.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { NpiVerifyResultTaxonomyMatch } from "../npiVerifyResultTaxonomyMatch";

export interface NpiVerifyResult {
  npi_found: boolean;
  npi_number?: string | null;
  specialty?: string | null;
  taxonomy_match?: NpiVerifyResultTaxonomyMatch;
  error?: string | null;
}
```

[lib/api-zod/src/generated/types/npiVerifyResultTaxonomyMatch.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type NpiVerifyResultTaxonomyMatch = {
  matched?: boolean;
  physician_specialty?: string;
  expected_specialties?: string[];
  diagnosis_category?: string | null;
  fraud_indicators?: string[];
  confidence?: string;
} | null;
```

[lib/api-zod/src/generated/types/paralegal.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
```

```
export interface Paralegal {
  id: number;
  name: string;
  email?: string | null;
  role: string;
  active_cases: number;
  signed_cases: number;
  total_assigned: number;
  created_at: string;
  updated_at: string;
}
```

[lib/api-zod/src/generated/types/paralegalDetail.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { Lead } from "../lead";

export interface ParalegalDetail {
  id: number;
  name: string;
  email?: string | null;
  role: string;
  leads: Lead[];
  signed_cases: number;
  active_cases: number;
  total_assigned: number;
  conversion_rate: number;
}
```

[lib/api-zod/src/generated/types/paralegalLeaderboardRow.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface ParalegalLeaderboardRow {
  id: number;
  name: string;
  role: string;
  total_assigned: number;
  signed: number;
  qualified: number;
  rejected: number;
  conversion_rate: number;
}
```

[lib/api-zod/src/generated/types/paralegalPerformance.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { Paralegal } from "../paralegal";
import type { ParalegalPerformanceBreakdownItem } from "../paralegalPerformanceBreakdownItem";

export interface ParalegalPerformance {
  paralegal: Paralegal;
  breakdown: ParalegalPerformanceBreakdownItem[];
  total_assigned: number;
  signed: number;
  qualified: number;
  conversion_rate: number;
}
```

[lib/api-zod/src/generated/types/paralegalPerformanceBreakdownItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type ParalegalPerformanceBreakdownItem = {
  status: string;
  count: number;
};
```

[lib/api-zod/src/generated/types/pipelineBreakdown.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { PipelineBreakdownByStatusItem } from "./pipelineBreakdownByStatusItem";
import type { PipelineBreakdownByTortTypeItem } from "./pipelineBreakdownByTortTypeItem";

export interface PipelineBreakdown {
  by_status: PipelineBreakdownByStatusItem[];
  by_tort_type: PipelineBreakdownByTortTypeItem[];
}
```


[lib/api-zod/src/generated/types/pipelineBreakdownByStatusItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type PipelineBreakdownByStatusItem = {
  status: string;
  count: number;
};
```

[lib/api-zod/src/generated/types/pipelineBreakdownByTortTypeItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type PipelineBreakdownByTortTypeItem = {
  tort_type: string;
  count: number;
  qualified: number;
  signed: number;
};
```

[lib/api-zod/src/generated/types/pipelineTrendPoint.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface PipelineTrendPoint {
  date: string;
  total: number;
  qualified: number;
  signed: number;
}
```

[lib/api-zod/src/generated/types/qualificationResult.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { QualificationResultStatus } from "../qualificationResultStatus";

export interface QualificationResult {
  lead_id: number;
  qualified: boolean;
  status: QualificationResultStatus;
  reason: string;
  gates_passed: string[];
  gates_failed: string[];
}
```

[lib/api-zod/src/generated/types/qualificationResultStatus.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type QualificationResultStatus =
  (typeof QualificationResultStatus)[keyof typeof QualificationResultStatus];

export const QualificationResultStatus = {
  qualified: "qualified",
  rejected: "rejected",
} as const;
```

[lib/api-zod/src/generated/types/queueStats.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface QueueStats {
  [key: string]: number;
}
```

[lib/api-zod/src/generated/types/resolveReviewBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { ResolveReviewBodyResolution } from "../resolveReviewBodyResolution";

export interface ResolveReviewBody {
  resolution: ResolveReviewBodyResolution;
  resolution_notes?: string;
  resolved_by?: string;
}
```

[lib/api-zod/src/generated/types/resolveReviewBodyResolution.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type ResolveReviewBodyResolution =
  (typeof ResolveReviewBodyResolution)[keyof typeof ResolveReviewBodyResolution];

export const ResolveReviewBodyResolution = {
  accepted: "accepted",
  rejected: "rejected",
  escalated: "escalated",
} as const;
```


[lib/api-zod/src/generated/types/reviewQueueItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { ReviewQueueItemDetails } from "../reviewQueueItemDetails";

export interface ReviewQueueItem {
  id: number;
  entity_type: string;
  entity_id: string;
  conflict_type: string;
  severity: string;
  failsafe_mode: string;
  source_module: string;
  summary: string;
  details?: ReviewQueueItemDetails;
  resolution: string;
  resolution_notes?: string | null;
  resolved_by?: string | null;
  resolved_at?: string | null;
  retry_count: number;
  created_at: string;
}
```

[lib/api-zod/src/generated/types/reviewQueueItemDetails.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
export type ReviewQueueItemDetails = { [key: string]: unknown };
```

[lib/api-zod/src/generated/types/reviewQueueStats.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { ReviewQueueStatsByConflictTypeItem } from "../reviewQueueStatsByConflictTypeItem";
import type { ReviewQueueStatsByFailsafeModeItem } from "../reviewQueueStatsByFailsafeModeItem";
import type { ReviewQueueStatsByResolutionItem } from "../reviewQueueStatsByResolutionItem";
import type { ReviewQueueStatsBySeverityItem } from "../reviewQueueStatsBySeverityItem";

export interface ReviewQueueStats {
  total_pending: number;
  by_resolution: ReviewQueueStatsByResolutionItem[];
  by_conflict_type: ReviewQueueStatsByConflictTypeItem[];
  by_severity: ReviewQueueStatsBySeverityItem[];
  by_failsafe_mode: ReviewQueueStatsByFailsafeModeItem[];
}
```

[lib/api-zod/src/generated/types/reviewQueueStatsByConflictTypeItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type ReviewQueueStatsByConflictTypeItem = {
  conflict_type?: string;
  count?: number;
};
```

[lib/api-zod/src/generated/types/reviewQueueStatsByFailsafeModeItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type ReviewQueueStatsByFailsafeModeItem = {
  failsafe_mode?: string;
  count?: number;
};
```

[lib/api-zod/src/generated/types/reviewQueueStatsByResolutionItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type ReviewQueueStatsByResolutionItem = {
  resolution?: string;
  count?: number;
};
```

[lib/api-zod/src/generated/types/reviewQueueStatsBySeverityItem.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type ReviewQueueStatsBySeverityItem = {
  severity?: string;
  count?: number;
};
```

[lib/api-zod/src/generated/types/runBackgroundCheckBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type RunBackgroundCheckBody = {
  first_name: string;
  last_name: string;
  state?: string;
  date_of_birth?: string;
};
```


[lib/api-zod/src/generated/types/runFraudCheckBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { RunFraudCheckBodyLeadData } from "../runFraudCheckBodyLeadData";

export type RunFraudCheckBody = {
  tort_type: string;
  diagnosis: string;
  physician_first_name?: string;
  physician_last_name?: string;
  lead_data?: RunFraudCheckBodyLeadData;
};
```

[lib/api-zod/src/generated/types/runFraudCheckBodyLeadData.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
export type RunFraudCheckBodyLeadData = { [key: string]: unknown };
```

[lib/api-zod/src/generated/types/searchNpiParams.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type SearchNpiParams = {
  npi_number?: string;
  first_name?: string;
  last_name?: string;
  city?: string;
  state?: string;
  specialty?: string;
  limit?: string;
};
```

[lib/api-zod/src/generated/types/tortBreakdownRow.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface TortBreakdownRow {
  tort_type: string;
  total: number;
  qualified: number;
  signed: number;
  rejected: number;
  avg_ad_spend: number;
}
```

lib/api-zod/src/generated/types/updateDocumentBody.ts

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { UpdateDocumentBodyDocumentType } from "../updateDocumentBodyDocumentType";

export interface UpdateDocumentBody {
  document_type?: UpdateDocumentBodyDocumentType;
  file_name?: string;
  file_url?: string | null;
  signed?: boolean;
  signed_at?: string | null;
  notes?: string | null;
}
```

[lib/api-zod/src/generated/types/updateDocumentBodyDocumentType.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type UpdateDocumentBodyDocumentType =
  (typeof UpdateDocumentBodyDocumentType)[keyof typeof UpdateDocumentBodyDocumentType];

export const UpdateDocumentBodyDocumentType = {
  retainer: "retainer",
  medical_record: "medical_record",
  intake_form: "intake_form",
  other: "other",
} as const;
```

[lib/api-zod/src/generated/types/updateLeadBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */
import type { UpdateLeadBodyStatus } from "../updateLeadBodyStatus";

export interface UpdateLeadBody {
  name?: string;
  email?: string;
  phone?: string;
  tort_type?: string;
  first_name?: string;
  last_name?: string;
  date_of_birth?: string;
  street_address?: string;
  city?: string;
  state?: string;
  zip?: string;
  phone_primary?: string;
  last_4_ssn?: string;
  diagnosis?: string;
  diagnosis_date?: string;
  physician_first_name?: string;
  physician_last_name?: string;
  physician_full_address?: string;
  physician_contact_info?: string;
  hospital_name?: string;
  hospital_fax?: string;
  hospital_contact_info?: string;
  tcpa_consent?: boolean;
  trustedform_cert_url?: string;
  exposure_start?: string | null;
  exposure_end?: string | null;
  diagnosis_confirmed?: boolean;
  diagnosis_type?: string | null;
  was_at_location?: boolean;
  location_name?: string | null;
  status?: UpdateLeadBodyStatus;
  rejection_reason?: string | null;
  notes?: string | null;
  ad_spend?: number | null;
  source?: string | null;
  vendor_id?: number | null;
  law_firm?: string | null;
  client_id?: string | null;
}
```

[lib/api-zod/src/generated/types/updateLeadBodyStatus.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type UpdateLeadBodyStatus =
  (typeof UpdateLeadBodyStatus)[keyof typeof UpdateLeadBodyStatus];

export const UpdateLeadBodyStatus = {
  new: "new",
  qualified: "qualified",
  signed: "signed",
  rejected: "rejected",
  review_required: "review_required",
} as const;
```


[lib/api-zod/src/generated/types/updateVendorBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface UpdateVendorBody {
  name?: string;
  contact_name?: string | null;
  contact_email?: string | null;
  contact_phone?: string | null;
  type?: string;
  status?: string;
  notes?: string | null;
}
```

[lib/api-zod/src/generated/types/uploadCaseFileBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface UploadCaseFileBody {
  file_name: string;
  content: string;
  content_type?: string | null;
}
```

[lib/api-zod/src/generated/types/uploadFaxBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface UploadFaxBody {
  file_name: string;
  image_base64: string;
  mime_type?: string | null;
}
```

[lib/api-zod/src/generated/types/uploadFaxResponse.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface UploadFaxResponse {
  fax_result_id: number;
  status: string;
  job_id: number;
  vault_path: string;
}
```

[lib/api-zod/src/generated/types/validateAddressBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type ValidateAddressBody = {
  street_address: string;
  city: string;
  state: string;
  zip: string;
};
```

[lib/api-zod/src/generated/types/validateEmailBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type ValidateEmailBody = {
  email: string;
};
```

[lib/api-zod/src/generated/types/vendor.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export interface Vendor {
  id: number;
  name: string;
  contact_name?: string | null;
  contact_email?: string | null;
  contact_phone?: string | null;
  type: string;
  status: string;
  notes?: string | null;
  created_at: string;
  updated_at: string;
}
```

[lib/api-zod/src/generated/types/verifyNpiBody.ts](#)

```
/**
 * Generated by orval v8.5.3 0<Bz
 * Do not edit manually.
 * Api
 * Mass Tort Operating System API
 * OpenAPI spec version: 0.1.0
 */

export type VerifyNpiBody = {
  physician_first_name: string;
  physician_last_name: string;
  diagnosis: string;
};
```


[lib/api-zod/src/index.ts](#)

```
export * from "./generated/api";  
export * from "./generated/types";
```

lib/api-zod/tsconfig.json

```
{
  "extends": "../../tsconfig.base.json",
  "compilerOptions": {
    "composite": true,
    "declarationMap": true,
    "emitDeclarationOnly": true,
    "outDir": "dist",
    "rootDir": "src"
  },
  "include": ["src"]
}
```

lib/db/drizzle.config.ts

```
import { defineConfig } from "drizzle-kit";
import path from "path";

if (!process.env.DATABASE_URL) {
  throw new Error("DATABASE_URL, ensure the database is provisioned");
}

export default defineConfig({
  schema: path.join(__dirname, "../src/schema/index.ts"),
  dialect: "postgresql",
  dbCredentials: {
    url: process.env.DATABASE_URL,
  },
});
```

lib/db/package.json

```
{
  "name": "@workspace/db",
  "version": "0.0.0",
  "private": true,
  "type": "module",
  "exports": {
    ".": "./src/index.ts",
    "./schema": "./src/schema/index.ts"
  },
  "scripts": {
    "push": "drizzle-kit push --config ./drizzle.config.ts",
    "push-force": "drizzle-kit push --force --config ./drizzle.config.ts"
  },
  "dependencies": {
    "drizzle-orm": "catalog:",
    "drizzle-zod": "^0.8.3",
    "pg": "^8.20.0",
    "zod": "catalog:"
  },
  "devDependencies": {
    "@types/node": "catalog:",
    "@types/pg": "^8.18.0",
    "drizzle-kit": "^0.31.9"
  }
}
```

[lib/db/src/index.ts](#)

```
import { drizzle } from "drizzle-orm/node-postgres";
import pg from "pg";
import * as schema from "../schema";

const { Pool } = pg;

if (!process.env.DATABASE_URL) {
  throw new Error(
    "DATABASE_URL must be set. Did you forget to provision a database?",
  );
}

export const pool = new Pool({ connectionString: process.env.DATABASE_URL });
export const db = drizzle(pool, { schema });

export * from "../schema";
```

[lib/db/src/schema/analysis.ts](#)

```
import {
  pgTable,
  serial,
  varchar,
  jsonb,
  real,
  timestamp,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";
import { casesTable } from "../cases";

export const analysisTable = pgTable("analysis", {
  id: serial("id").primaryKey(),
  case_id: varchar("case_id", { length: 36 })
    .notNull()
    .references(() => casesTable.id, { onDelete: "cascade" }),
  features: jsonb("features"),
  score: real("score"),
  ai_model: varchar("ai_model", { length: 100 }),
  raw_text_length: real("raw_text_length"),
  created_at: timestamp("created_at").defaultNow().notNull(),
});

export const insertAnalysisSchema = createInsertSchema(analysisTable).omit({
  id: true,
  created_at: true,
});

export type InsertAnalysis = z.infer<typeof insertAnalysisSchema>;
export type Analysis = typeof analysisTable.$inferSelect;
```

lib/db/src/schema/audit_log.ts

```
import {
  pgTable,
  serial,
  varchar,
  jsonb,
  text,
  timestamp,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const auditLogTable = pgTable("audit_log", {
  id: serial("id").primaryKey(),
  entity_type: varchar("entity_type", { length: 50 }).notNull(),
  entity_id: varchar("entity_id", { length: 100 }).notNull(),
  action: varchar("action", { length: 50 }).notNull(),
  details: jsonb("details"),
  ip_address: varchar("ip_address", { length: 45 }),
  user_agent: text("user_agent"),
  occurred_at: timestamp("occurred_at").defaultNow().notNull(),
});

export const insertAuditLogSchema = createInsertSchema(auditLogTable).omit({
  id: true,
  occurred_at: true,
});

export type InsertAuditLog = z.infer<typeof insertAuditLogSchema>;
export type AuditLog = typeof auditLogTable.$inferSelect;
```

[lib/db/src/schema/buyers.ts](#)

```
import { pgTable, serial, varchar, text, timestamp, boolean, integer, jsonb } from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const buyersTable = pgTable("buyers", {
  id: serial("id").primaryKey(),
  name: varchar("name", { length: 255 }).notNull().unique(),
  legal_name: varchar("legal_name", { length: 255 }),
  contact_name: varchar("contact_name", { length: 255 }),
  contact_email: varchar("contact_email", { length: 255 }),
  contact_phone: varchar("contact_phone", { length: 50 }),
  notes: text("notes"),
  esign_provider_integration_id: integer("esign_provider_integration_id"),
  fax_provider_integration_id: integer("fax_provider_integration_id"),
  email_provider_integration_id: integer("email_provider_integration_id"),
  default_email_from_name: varchar("default_email_from_name", { length: 255 }),
  default_email_from_address: varchar("default_email_from_address", { length: 255 }),
  default_fax_from_number: varchar("default_fax_from_number", { length: 50 }),
  active: boolean("active").notNull().default(true),
  metadata: jsonb("metadata").$type<Record<string, unknown>>().notNull().default({}),
  created_at: timestamp("created_at").defaultNow().notNull(),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export const insertBuyerSchema = createInsertSchema(buyersTable).omit({
  id: true,
  created_at: true,
  updated_at: true,
});

export type Buyer = typeof buyersTable.$inferSelect;
export type InsertBuyer = z.infer<typeof insertBuyerSchema>;
```


lib/db/src/schema/case_documents.ts

```
import {
  pgTable,
  serial,
  integer,
  varchar,
  text,
  timestamp,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";
import { casesTable } from "../cases";

export const caseDocumentsTable = pgTable("case_documents", {
  id: serial("id").primaryKey(),
  case_id: varchar("case_id", { length: 36 })
    .notNull()
    .references(() => casesTable.id, { onDelete: "cascade" }),
  path: text("path").notNull(),
  file_hash: varchar("file_hash", { length: 64 }),
  file_name: varchar("file_name", { length: 255 }).notNull(),
  content_type: varchar("content_type", { length: 100 }),
  size_bytes: integer("size_bytes"),
  created_at: timestamp("created_at").defaultNow().notNull(),
});

export const insertCaseDocumentSchema = createInsertSchema(caseDocumentsTable).omit({
  id: true,
  size_bytes: true,
  created_at: true,
});

export type InsertCaseDocument = z.infer<typeof insertCaseDocumentSchema>;
export type CaseDocument = typeof caseDocumentsTable.$inferSelect;
```

[lib/db/src/schema/cases.ts](#)

```
import {
  pgTable,
  varchar,
  jsonb,
  timestamp,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const casesTable = pgTable("cases", {
  id: varchar("id", { length: 36 }).primaryKey(),
  data: jsonb("data"),
  status: varchar("status", { length: 50 }).notNull().default("open"),
  created_at: timestamp("created_at").defaultNow().notNull(),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export const insertCaseSchema = createInsertSchema(casesTable).omit({
  created_at: true,
  updated_at: true,
});

export type InsertCase = z.infer<typeof insertCaseSchema>;
export type Case = typeof casesTable.$inferSelect;
```

lib/db/src/schema/conversations.ts

```
import { pgTable, serial, text, timestamp } from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const conversations = pgTable("conversations", {
  id: serial("id").primaryKey(),
  title: text("title").notNull(),
  createdAt: timestamp("created_at", { withTimezone: true }).defaultNow().notNull(),
});

export const insertConversationSchema = createInsertSchema(conversations).omit({
  id: true,
  createdAt: true,
});

export type Conversation = typeof conversations.$inferSelect;
export type InsertConversation = z.infer<typeof insertConversationSchema>;
```

[lib/db/src/schema/decision_engine_settings.ts](#)

```
import { pgTable, integer, decimal, boolean, timestamp } from "drizzle-orm/pg-core";

// Singleton row (id = 1) holding global Decision Engine thresholds.
export const decisionEngineSettingsTable = pgTable("decision_engine_settings", {
  id: integer("id").primaryKey().default(1),
  concentration_warning_pct: integer("concentration_warning_pct").notNull().default(40),
  ruin_auto_flag: boolean("ruin_auto_flag").notNull().default(false),
  default_attorney_hourly_cost: decimal("default_attorney_hourly_cost", { precision: 8, scale: 2 }).notNull().default("250.00"),
  default_hours_per_lead: decimal("default_hours_per_lead", { precision: 6, scale: 2 }).notNull().default("8.00"),
  convex_ratio_threshold: decimal("convex_ratio_threshold", { precision: 6, scale: 2 }).notNull().default("5.00"),
  concave_ratio_threshold: decimal("concave_ratio_threshold", { precision: 6, scale: 2 }).notNull().default("1.50"),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export type DecisionEngineSettings = typeof decisionEngineSettingsTable.$inferSelect;
```

lib/db/src/schema/document_envelopes.ts

```
import { pgTable, serial, varchar, text, timestamp, integer, jsonb } from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";
import { leadsTable } from "../leads";
import { documentTemplatesTable } from "../document_templates";
import { documentsTable } from "../documents";

/**
 * One row per envelope sent through an e-sign provider.
 * Lifecycle: created !' sent !' delivered !' viewed !' signed (or declined / expired / voided / error).
 */
export const documentEnvelopesTable = pgTable("document_envelopes", {
  id: serial("id").primaryKey(),
  lead_id: integer("lead_id")
    .notNull()
    .references(() => leadsTable.id, { onDelete: "cascade" }),
  template_id: integer("template_id").references(() => documentTemplatesTable.id, {
    onDelete: "set null",
  }),
  document_id: integer("document_id").references(() => documentsTable.id, {
    onDelete: "set null",
  }),
  provider: varchar("provider", { length: 50 }).notNull(),
  provider_integration_id: integer("provider_integration_id"),
  external_envelope_id: varchar("external_envelope_id", { length: 255 }),
  signer_name: varchar("signer_name", { length: 255 }),
  signer_email: varchar("signer_email", { length: 255 }),
  status: varchar("status", { length: 30 }).notNull().default("created"),
  status_detail: text("status_detail"),
  sent_at: timestamp("sent_at"),
  delivered_at: timestamp("delivered_at"),
  viewed_at: timestamp("viewed_at"),
  signed_at: timestamp("signed_at"),
  declined_at: timestamp("declined_at"),
  signed_pdf_path: text("signed_pdf_path"),
  events: jsonb("events").$type<EnvelopeEvent[]>().notNull().default([]),
  metadata: jsonb("metadata").$type<Record<string, unknown>>().notNull().default({}),
  last_error: text("last_error"),
  created_at: timestamp("created_at").defaultNow().notNull(),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export interface EnvelopeEvent {
  type: string;
  at: string;
  raw?: Record<string, unknown>;
}

export const insertDocumentEnvelopeSchema = createInsertSchema(documentEnvelopesTable).omit({
  id: true,
  created_at: true,
  updated_at: true,
});

export type DocumentEnvelope = typeof documentEnvelopesTable.$inferSelect;
export type InsertDocumentEnvelope = z.infer<typeof insertDocumentEnvelopeSchema>;
```

lib/db/src/schema/document_templates.ts

```
import { pgTable, serial, varchar, text, timestamp, boolean, integer, jsonb } from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

/**
 * Reusable document template uploaded once and used many times.
 * source = "pdf"    !' static uploaded PDF, sent as-is (with merge fields if specified)
 * source = "ai"     !' drafted on-demand via drafting-ai.ts using the lead context
 */
export const documentTemplatesTable = pgTable("document_templates", {
  id: serial("id").primaryKey(),
  name: varchar("name", { length: 255 }).notNull(),
  template_type: varchar("template_type", { length: 50 }).notNull(),
  description: text("description"),
  source: varchar("source", { length: 20 }).notNull().default("pdf"),
  storage_path: text("storage_path"),
  ai_prompt: text("ai_prompt"),
  requires_signature: boolean("requires_signature").notNull().default(true),
  signer_role: varchar("signer_role", { length: 50 }).notNull().default("lead"),
  merge_fields: jsonb("merge_fields").$type<string[]>().notNull().default([]),
  delivery_subject: varchar("delivery_subject", { length: 255 }),
  delivery_message: text("delivery_message"),
  triggers_med_records_request: boolean("triggers_med_records_request").notNull().default(false),
  active: boolean("active").notNull().default(true),
  uploaded_by_user_id: integer("uploaded_by_user_id"),
  created_at: timestamp("created_at").defaultNow().notNull(),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export const insertDocumentTemplateSchema = createInsertSchema(documentTemplatesTable).omit({
  id: true,
  created_at: true,
  updated_at: true,
});

export type DocumentTemplate = typeof documentTemplatesTable.$inferSelect;
export type InsertDocumentTemplate = z.infer<typeof insertDocumentTemplateSchema>;
```

[lib/db/src/schema/documents.ts](#)

```
import {
  pgTable,
  serial,
  integer,
  varchar,
  text,
  boolean,
  timestamp,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";
import { leadsTable } from "../leads";

export const documentsTable = pgTable("documents", {
  id: serial("id").primaryKey(),
  lead_id: integer("lead_id")
    .notNull()
    .references(() => leadsTable.id, { onDelete: "cascade" }),
  document_type: varchar("document_type", { length: 50 }).notNull(),
  file_name: varchar("file_name", { length: 255 }).notNull(),
  file_url: text("file_url"),
  signed: boolean("signed").notNull().default(false),
  signed_at: timestamp("signed_at"),
  notes: text("notes"),
  created_at: timestamp("created_at").defaultNow().notNull(),
});

export const insertDocumentSchema = createInsertSchema(documentsTable).omit({
  id: true,
  created_at: true,
});

export type InsertDocument = z.infer<typeof insertDocumentSchema>;
export type Document = typeof documentsTable.$inferSelect;
```

[lib/db/src/schema/fax_results.ts](#)

```
import {
  pgTable,
  serial,
  integer,
  text,
  real,
  timestamp,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const faxResultsTable = pgTable("fax_results", {
  id: serial("id").primaryKey(),
  job_id: integer("job_id"),
  source_file: text("source_file").notNull(),
  vault_path: text("vault_path").notNull(),
  rx_number: text("rx_number").default("").notNull(),
  drug_name: text("drug_name").default("").notNull(),
  fill_date: text("fill_date").default("").notNull(),
  quantity: text("quantity").default("").notNull(),
  confidence: real("confidence").default(0).notNull(),
  raw_text: text("raw_text").default("").notNull(),
  status: text("status").default("pending").notNull(),
  error: text("error"),
  created_at: timestamp("created_at").defaultNow().notNull(),
  processed_at: timestamp("processed_at"),
});

export const insertFaxResultSchema = createInsertSchema(faxResultsTable).omit({
  id: true,
  created_at: true,
});

export type InsertFaxResult = z.infer<typeof insertFaxResultSchema>;
export type FaxResult = typeof faxResultsTable.$inferSelect;
```


lib/db/src/schema/form_configurations.ts

```
import {
  pgTable,
  varchar,
  jsonb,
  timestamp,
  integer,
  boolean,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const formConfigurationsTable = pgTable("form_configurations", {
  id: varchar("id", { length: 100 }).primaryKey(),
  label: varchar("label", { length: 255 }).notNull(),
  category: varchar("category", { length: 50 }).notNull(),
  valid_diagnoses: jsonb("valid_diagnoses").$type<string[]>().notNull().default([]),
  exposure_fields: jsonb("exposure_fields").$type<string[]>().notNull().default([]),
  extra_fields: jsonb("extra_fields").$type<string[]>().notNull().default([]),
  custom_fields: jsonb("custom_fields").$type<CustomField[]>().notNull().default([]),
  rules: jsonb("rules").$type<string[]>().notNull().default([]),
  rejection_conditions: jsonb("rejection_conditions").$type<string[]>().notNull().default([]),
  required_exposure: boolean("required_exposure").notNull().default(false),
  intro_text: varchar("intro_text", { length: 1000 }),
  active: boolean("active").notNull().default(true),
  // Decision Engine inputs
  avg_settlement_low: integer("avg_settlement_low"),
  avg_settlement_high: integer("avg_settlement_high"),
  expected_duration_months: integer("expected_duration_months"),
  mdl_status: varchar("mdl_status", { length: 30 }),
  sol_months: integer("sol_months"),
  updated_by: integer("updated_by"),
  created_at: timestamp("created_at").defaultNow().notNull(),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export type CustomFieldType =
  | "text"
  | "email"
  | "tel"
  | "date"
  | "number"
  | "select"
  | "textarea"
  | "checkbox";

export interface CustomField {
  key: string;
  label: string;
  type: CustomFieldType;
  required: boolean;
  placeholder?: string;
  helper_text?: string;
  options?: string[];
  max_length?: number;
}

export const customFieldSchema = z.object({
  key: z.string().min(1).max(100).regex(/^[a-z][a-z0-9_]*$/, "key must be snake_case"),
  label: z.string().min(1).max(255),
  type: z.enum(["text", "email", "tel", "date", "number", "select", "textarea", "checkbox"]),
  required: z.boolean(),
  placeholder: z.string().max(255).optional(),
  helper_text: z.string().max(500).optional(),
  options: z.array(z.string().max(100)).optional(),
  max_length: z.number().int().positive().optional(),
});

export const insertFormConfigurationSchema = createInsertSchema(formConfigurationsTable).omit({
  created_at: true,
  updated_at: true,
});

export type FormConfiguration = typeof formConfigurationsTable.$inferSelect;
export type InsertFormConfiguration = z.infer<typeof insertFormConfigurationSchema>;
```

lib/db/src/schema/image_objects.ts

```
import {
  pgTable,
  serial,
  integer,
  varchar,
  text,
  boolean,
  timestamp,
  jsonb,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const imageObjectsTable = pgTable("image_objects", {
  id: serial("id").primaryKey(),
  original_filename: text("original_filename").notNull(),
  mime_type: varchar("mime_type", { length: 100 }).notNull(),
  file_size: integer("file_size").notNull(),
  width: integer("width"),
  height: integer("height"),
  source_type: varchar("source_type", { length: 50 }).notNull().default("upload"),
  vault_path: text("vault_path").notNull(),
  checksum_sha256: varchar("checksum_sha256", { length: 64 }).notNull(),
  ocr_status: varchar("ocr_status", { length: 20 }).notNull().default("pending"),
  ocr_text: text("ocr_text"),
  ocr_confidence: integer("ocr_confidence"),
  ocr_extracted_fields: jsonb("ocr_extracted_fields"),
  lead_id: integer("lead_id"),
  document_id: integer("document_id"),
  fax_result_id: integer("fax_result_id"),
  case_id: varchar("case_id", { length: 100 }),
  is_sensitive: boolean("is_sensitive").notNull().default(true),
  access_classification: varchar("access_classification", { length: 20 }).notNull().default("confidential"),
  retention_days: integer("retention_days").default(2555),
  metadata: jsonb("metadata"),
  created_by: varchar("created_by", { length: 255 }).notNull().default("system"),
  created_at: timestamp("created_at").defaultNow().notNull(),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export const insertImageObjectSchema = createInsertSchema(imageObjectsTable).omit({
  id: true,
  created_at: true,
  updated_at: true,
});

export type InsertImageObject = z.infer<typeof insertImageObjectSchema>;
export type ImageObject = typeof imageObjectsTable.$inferSelect;
```

[lib/db/src/schema/import_batches.ts](#)

```
import {
  pgTable,
  serial,
  integer,
  varchar,
  text,
  timestamp,
  jsonb,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const importBatchesTable = pgTable("import_batches", {
  id: serial("id").primaryKey(),
  filename: text("filename").notNull(),
  status: varchar("status", { length: 20 }).notNull().default("pending"),
  total_rows: integer("total_rows").notNull().default(0),
  processed_rows: integer("processed_rows").notNull().default(0),
  success_count: integer("success_count").notNull().default(0),
  duplicate_count: integer("duplicate_count").notNull().default(0),
  error_count: integer("error_count").notNull().default(0),
  conflict_count: integer("conflict_count").notNull().default(0),
  column_mapping: jsonb("column_mapping"),
  created_by: varchar("created_by", { length: 255 }).notNull().default("system"),
  created_at: timestamp("created_at").defaultNow().notNull(),
  completed_at: timestamp("completed_at"),
});

export const importRowsTable = pgTable("import_rows", {
  id: serial("id").primaryKey(),
  batch_id: integer("batch_id").notNull(),
  row_number: integer("row_number").notNull(),
  status: varchar("status", { length: 20 }).notNull().default("pending"),
  raw_data: jsonb("raw_data").notNull(),
  lead_id: integer("lead_id"),
  dedup_match_id: integer("dedup_match_id"),
  dedup_reason: text("dedup_reason"),
  error_message: text("error_message"),
  conflict_details: jsonb("conflict_details"),
  created_at: timestamp("created_at").defaultNow().notNull(),
  processed_at: timestamp("processed_at"),
});

export const insertImportBatchSchema = createInsertSchema(importBatchesTable).omit({
  id: true,
  created_at: true,
  completed_at: true,
});

export const insertImportRowSchema = createInsertSchema(importRowsTable).omit({
  id: true,
  created_at: true,
  processed_at: true,
});

export type InsertImportBatch = z.infer<typeof insertImportBatchSchema>;
export type ImportBatch = typeof importBatchesTable.$inferSelect;
export type InsertImportRow = z.infer<typeof insertImportRowSchema>;
export type ImportRow = typeof importRowsTable.$inferSelect;
```

[lib/db/src/schema/index.ts](#)

```
export * from "./leads";
export * from "./documents";
export * from "./cases";
export * from "./analysis";
export * from "./job_queue";
export * from "./audit_log";
export * from "./case_documents";
export * from "./fax_results";
export * from "./paralegals";
export * from "./review_queue";
export * from "./vendors";
export * from "./security_alerts";
export * from "./conversations";
export * from "./messages";
export * from "./users";
export * from "./integrations";
export * from "./image_objects";
export * from "./import_batches";
export * from "./form_configurations";
export * from "./lead_sources";
export * from "./decision_engine_settings";
export * from "./buyers";
export * from "./document_templates";
export * from "./template_assignments";
export * from "./document_envelopes";
export * from "./workflow_settings";
export { refreshTokensTable } from "./users";
export { securityNotificationsTable } from "./security_alerts";
```

[lib/db/src/schema/integrations.ts](#)

```
import { pgTable, serial, text, timestamp, boolean, jsonb } from "drizzle-orm/pg-core";

export const integrationsTable = pgTable("integrations", {
  id: serial("id").primaryKey(),
  name: text("name").notNull(),
  type: text("type").notNull(),
  provider: text("provider").notNull(),
  status: text("status").notNull().default("inactive"),
  api_url: text("api_url"),
  api_key_hash: text("api_key_hash"),
  webhook_url: text("webhook_url"),
  config: jsonb("config"),
  last_sync_at: timestamp("last_sync_at"),
  sync_direction: text("sync_direction").default("bidirectional"),
  field_mapping: jsonb("field_mapping"),
  created_at: timestamp("created_at").defaultNow(),
  updated_at: timestamp("updated_at").defaultNow(),
});
```

[lib/db/src/schema/job_queue.ts](#)

```
import {
  pgTable,
  serial,
  varchar,
  jsonb,
  text,
  timestamp,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const jobQueueTable = pgTable("job_queue", {
  id: serial("id").primaryKey(),
  job_type: varchar("job_type", { length: 50 }).notNull(),
  payload: jsonb("payload").notNull(),
  status: varchar("status", { length: 20 }).notNull().default("pending"),
  error: text("error"),
  attempts: serial("attempts"),
  created_at: timestamp("created_at").defaultNow().notNull(),
  started_at: timestamp("started_at"),
  completed_at: timestamp("completed_at"),
});

export const insertJobSchema = createInsertSchema(jobQueueTable).omit({
  id: true,
  attempts: true,
  created_at: true,
  started_at: true,
  completed_at: true,
});

export type InsertJob = z.infer<typeof insertJobSchema>;
export type Job = typeof jobQueueTable.$inferSelect;
```

[lib/db/src/schema/lead_sources.ts](#)

```
import { pgTable, serial, varchar, decimal, timestamp, boolean } from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const leadSourcesTable = pgTable("lead_sources", {
  id: serial("id").primaryKey(),
  name: varchar("name", { length: 255 }).notNull().unique(),
  channel: varchar("channel", { length: 50 }).notNull().default("other"),
  cost_per_lead: decimal("cost_per_lead", { precision: 10, scale: 2 }),
  historical_qualified_rate: decimal("historical_qualified_rate", { precision: 5, scale: 4 }),
  historical_retained_rate: decimal("historical_retained_rate", { precision: 5, scale: 4 }),
  active: boolean("active").notNull().default(true),
  created_at: timestamp("created_at").defaultNow().notNull(),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export const insertLeadSourceSchema = createInsertSchema(leadSourcesTable).omit({
  id: true,
  created_at: true,
  updated_at: true,
});

export type LeadSource = typeof leadSourcesTable.$inferSelect;
export type InsertLeadSource = z.infer<typeof insertLeadSourceSchema>;
```

[lib/db/src/schema/leads.ts](#)

```
import {
  pgTable,
  serial,
  integer,
  varchar,
  text,
  boolean,
  decimal,
  timestamp,
  date,
  jsonb,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const leadsTable = pgTable("leads", {
  id: serial("id").primaryKey(),
  name: varchar("name", { length: 255 }).notNull(),
  email: varchar("email", { length: 255 }),
  phone: text("phone"),
  tort_type: varchar("tort_type", { length: 100 }).notNull(),
  exposure_start: date("exposure_start"),
  exposure_end: date("exposure_end"),
  diagnosis_confirmed: boolean("diagnosis_confirmed").notNull().default(false),
  diagnosis_type: varchar("diagnosis_type", { length: 255 }),
  was_at_location: boolean("was_at_location").notNull().default(false),
  location_name: varchar("location_name", { length: 255 }),
  status: varchar("status", { length: 20 }).notNull().default("new"),
  rejection_reason: text("rejection_reason"),
  notes: text("notes"),
  ad_spend: decimal("ad_spend", { precision: 10, scale: 2 }),
  source: varchar("source", { length: 100 }),
  assigned_to: integer("assigned_to"),
  routing: varchar("routing", { length: 20 }).default("cold"),

  first_name: varchar("first_name", { length: 255 }),
  last_name: varchar("last_name", { length: 255 }),
  date_of_birth: text("date_of_birth"),
  street_address: text("street_address"),
  city: varchar("city", { length: 100 }),
  state: varchar("state", { length: 2 }),
  zip: varchar("zip", { length: 10 }),
  phone_primary: text("phone_primary"),
  last_4_ssn: text("last_4_ssn"),

  diagnosis: text("diagnosis"),
  diagnosis_date: text("diagnosis_date"),

  physician_first_name: varchar("physician_first_name", { length: 255 }),
  physician_last_name: varchar("physician_last_name", { length: 255 }),
  physician_full_address: text("physician_full_address"),
  physician_contact_info: text("physician_contact_info"),

  hospital_name: varchar("hospital_name", { length: 500 }),
  hospital_fax: varchar("hospital_fax", { length: 50 }),
  hospital_contact_info: text("hospital_contact_info"),

  tcpa_consent: boolean("tcpa_consent").default(false),
  trustedform_cert_url: text("trustedform_cert_url"),
  trustedform_ping_url: text("trustedform_ping_url"),
  trustedform_cert_token: text("trustedform_cert_token"),
  trustedform_ip: varchar("trustedform_ip", { length: 45 }),
  trustedform_user_agent: text("trustedform_user_agent"),
  trustedform_timestamp: timestamp("trustedform_timestamp"),

  medications: text("medications"),
  npi_verified: boolean("npi_verified"),
  npi_number: varchar("npi_number", { length: 10 }),
  physician_taxonomy: varchar("physician_taxonomy", { length: 255 }),
  fraud_score: integer("fraud_score"),
  fraud_status: varchar("fraud_status", { length: 20 }),
  fraud_indicators: text("fraud_indicators"),

  email_validation_status: varchar("email_validation_status", { length: 20 }),
  address_validation_status: varchar("address_validation_status", { length: 20 }),
  background_check_status: varchar("background_check_status", { length: 20 }),
  background_check_data: text("background_check_data"),

  vendor_id: integer("vendor_id"),
  law_firm: varchar("law_firm", { length: 255 }),
  client_id: varchar("client_id", { length: 100 }),
  buyer_id: integer("buyer_id"),
  created_by_user_id: integer("created_by_user_id"),

  custom_fields: jsonb("custom_fields").$type<Record<string, unknown>>().notNull().default({}),

  // Decision Engine – convexity scoring (computed on create/update)
  convexity_score: varchar("convexity_score", { length: 20 }),
  convexity_action: varchar("convexity_action", { length: 20 }),
  convexity_rationale: text("convexity_rationale"),
  convexity_ruin_flags: jsonb("convexity_ruin_flags").$type<string[]>().notNull().default([]),
  convexity_downside_usd: decimal("convexity_downside_usd", { precision: 12, scale: 2 }),
  convexity_upside_usd: decimal("convexity_upside_usd", { precision: 12, scale: 2 }),
  convexity_ratio: decimal("convexity_ratio", { precision: 10, scale: 4 }),
  convexity_confidence: varchar("convexity_confidence", { length: 10 }),
```



```
convexity_missing_fields: jsonb("convexity_missing_fields").$type<string[]>().notNull().default([]),
convexity_contradictions: jsonb("convexity_contradictions").$type<string[]>().notNull().default([]),
convexity_computed_at: timestamp("convexity_computed_at"),

created_at: timestamp("created_at").defaultNow().notNull(),
updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export const insertLeadSchema = createInsertSchema(leadsTable).omit({
  id: true,
  created_at: true,
  updated_at: true,
});

export type InsertLead = z.infer<typeof insertLeadSchema>;
export type Lead = typeof leadsTable.$inferSelect;
```

[lib/db/src/schema/messages.ts](#)

```
import { integer, pgTable, serial, text, timestamp } from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

import { conversations } from "../conversations";

export const messages = pgTable("messages", {
  id: serial("id").primaryKey(),
  conversationId: integer("conversation_id")
    .notNull()
    .references(() => conversations.id, { onDelete: "cascade" }),
  role: text("role").notNull(),
  content: text("content").notNull(),
  createdAt: timestamp("created_at", { withTimezone: true }).defaultNow().notNull(),
});

export const insertMessageSchema = createInsertSchema(messages).omit({
  id: true,
  createdAt: true,
});

export type Message = typeof messages.$inferSelect;
export type InsertMessage = z.infer<typeof insertMessageSchema>;
```

[lib/db/src/schema/paralegals.ts](#)

```
import {
  pgTable,
  serial,
  varchar,
  integer,
  timestamp,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const paralegalsTable = pgTable("paralegals", {
  id: serial("id").primaryKey(),
  name: varchar("name", { length: 255 }).notNull(),
  email: varchar("email", { length: 255 }),
  role: varchar("role", { length: 100 }).notNull().default("Paralegal"),
  active_cases: integer("active_cases").notNull().default(0),
  signed_cases: integer("signed_cases").notNull().default(0),
  total_assigned: integer("total_assigned").notNull().default(0),
  created_at: timestamp("created_at").defaultNow().notNull(),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export const insertParalegalSchema = createInsertSchema(paralegalsTable).omit({
  id: true,
  created_at: true,
  updated_at: true,
});

export type InsertParalegal = z.infer<typeof insertParalegalSchema>;
export type Paralegal = typeof paralegalsTable.$inferSelect;
```

lib/db/src/schema/review_queue.ts

```
import {
  pgTable,
  serial,
  varchar,
  text,
  jsonb,
  timestamp,
  integer,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const reviewQueueTable = pgTable("review_queue", {
  id: serial("id").primaryKey(),
  entity_type: varchar("entity_type", { length: 50 }).notNull(),
  entity_id: varchar("entity_id", { length: 100 }).notNull(),
  conflict_type: varchar("conflict_type", { length: 50 }).notNull(),
  severity: varchar("severity", { length: 20 }).notNull().default("medium"),
  failsafe_mode: varchar("failsafe_mode", { length: 20 }).notNull(),
  source_module: varchar("source_module", { length: 100 }).notNull(),
  summary: text("summary").notNull(),
  details: jsonb("details"),
  resolution: varchar("resolution", { length: 20 }).default("pending"),
  resolution_notes: text("resolution_notes"),
  resolved_by: varchar("resolved_by", { length: 100 }),
  resolved_at: timestamp("resolved_at"),
  retry_count: integer("retry_count").notNull().default(0),
  created_at: timestamp("created_at").defaultNow().notNull(),
});

export const insertReviewQueueSchema = createInsertSchema(reviewQueueTable).omit({
  id: true,
  created_at: true,
});

export type InsertReviewQueue = z.infer<typeof insertReviewQueueSchema>;
export type ReviewQueueItem = typeof reviewQueueTable.$inferSelect;
```

lib/db/src/schema/security_alerts.ts

```
import {
  pgTable,
  serial,
  varchar,
  text,
  timestamp,
  boolean,
  integer,
  jsonb,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const securityAlertsTable = pgTable("security_alerts", {
  id: serial("id").primaryKey(),
  type: varchar("type", { length: 50 }).notNull(),
  severity: varchar("severity", { length: 20 }).notNull(),
  source_ip: varchar("source_ip", { length: 45 }),
  user_agent: text("user_agent"),
  request_path: text("request_path"),
  request_method: varchar("request_method", { length: 10 }),
  details: text("details").notNull(),
  payload_sample: text("payload_sample"),
  ai_analysis: text("ai_analysis"),
  status: varchar("status", { length: 20 }).notNull().default("new"),
  blocked: boolean("blocked").default(false),
  country: varchar("country", { length: 50 }),
  metadata: jsonb("metadata"),
  created_at: timestamp("created_at").defaultNow().notNull(),
});

export const blockedIpsTable = pgTable("blocked_ips", {
  id: serial("id").primaryKey(),
  ip: varchar("ip", { length: 45 }).notNull().unique(),
  reason: text("reason").notNull(),
  blocked_until: timestamp("blocked_until"),
  auto_blocked: boolean("auto_blocked").default(true),
  alert_count: integer("alert_count").default(1),
  created_at: timestamp("created_at").defaultNow().notNull(),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export const insertSecurityAlertSchema = createInsertSchema(securityAlertsTable).omit({
  id: true,
  created_at: true,
});

export const insertBlockedIpSchema = createInsertSchema(blockedIpsTable).omit({
  id: true,
  created_at: true,
  updated_at: true,
});

export type SecurityAlert = typeof securityAlertsTable.$inferSelect;
export type InsertSecurityAlert = z.infer<typeof insertSecurityAlertSchema>;
export type BlockedIp = typeof blockedIpsTable.$inferSelect;
export type InsertBlockedIp = z.infer<typeof insertBlockedIpSchema>;

export const securityNotificationsTable = pgTable("security_notifications", {
  id: serial("id").primaryKey(),
  alert_id: integer("alert_id"),
  severity: varchar("severity", { length: 20 }).notNull(),
  title: varchar("title", { length: 255 }).notNull(),
  message: text("message").notNull(),
  channel: varchar("channel", { length: 20 }).notNull().default("in_app"),
  webhook_url: text("webhook_url"),
  delivered: boolean("delivered").notNull().default(false),
  acknowledged: boolean("acknowledged").notNull().default(false),
  acknowledged_by: varchar("acknowledged_by", { length: 255 }),
  created_at: timestamp("created_at").defaultNow().notNull(),
});
```

lib/db/src/schema/template_assignments.ts

```
import { pgTable, serial, varchar, text, timestamp, boolean, integer, primaryKey } from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";
import { documentTemplatesTable } from "../document_templates";
import { buyersTable } from "../buyers";

/**
 * Per-(buyer, tort) toggle for whether a given template should be sent.
 * Three-tier resolution at send time:
 * 1. Row exists for (buyer_id, tort_type, template_id) !' use enabled flag
 * 2. Row exists for (buyer_id=NULL, tort_type, template_id) !' tort-wide default
 * 3. Falls back to template default (active=true means send by default)
 */
export const templateAssignmentsTable = pgTable("template_assignments", {
  id: serial("id").primaryKey(),
  template_id: integer("template_id")
    .notNull()
    .references(() => documentTemplatesTable.id, { onDelete: "cascade" }),
  buyer_id: integer("buyer_id").references(() => buyersTable.id, { onDelete: "cascade" }),
  tort_type: varchar("tort_type", { length: 100 }),
  enabled: boolean("enabled").notNull().default(true),
  send_order: integer("send_order").notNull().default(0),
  notes: text("notes"),
  updated_by_user_id: integer("updated_by_user_id"),
  created_at: timestamp("created_at").defaultNow().notNull(),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export const insertTemplateAssignmentSchema = createInsertSchema(templateAssignmentsTable).omit({
  id: true,
  created_at: true,
  updated_at: true,
});

export type TemplateAssignment = typeof templateAssignmentsTable.$inferSelect;
export type InsertTemplateAssignment = z.infer<typeof insertTemplateAssignmentSchema>;
```

[lib/db/src/schema/users.ts](#)

```
import { pgTable, serial, varchar, text, timestamp, integer, boolean } from "drizzle-orm/pg-core";

export const usersTable = pgTable("mtos_users", {
  id: serial("id").primaryKey(),
  email: varchar("email", { length: 255 }).notNull().unique(),
  name: varchar("name", { length: 255 }).notNull(),
  role: varchar("role", { length: 20 }).notNull().default("viewer"),
  password_hash: text("password_hash").notNull(),
  token_version: integer("token_version").notNull().default(0),
  totp_secret: text("totp_secret"),
  mfa_enabled: boolean("mfa_enabled").notNull().default(false),
  failed_login_attempts: integer("failed_login_attempts").notNull().default(0),
  locked_until: timestamp("locked_until"),
  created_at: timestamp("created_at").defaultNow().notNull(),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export const refreshTokensTable = pgTable("refresh_tokens", {
  id: serial("id").primaryKey(),
  user_id: integer("user_id").notNull(),
  token_hash: text("token_hash").notNull(),
  expires_at: timestamp("expires_at").notNull(),
  revoked: boolean("revoked").notNull().default(false),
  replaced_by: text("replaced_by"),
  created_at: timestamp("created_at").defaultNow().notNull(),
});
```

[lib/db/src/schema/vendors.ts](#)

```
import {
  pgTable,
  serial,
  varchar,
  text,
  timestamp,
} from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

export const vendorsTable = pgTable("vendors", {
  id: serial("id").primaryKey(),
  name: varchar("name", { length: 255 }).notNull(),
  contact_name: varchar("contact_name", { length: 255 }),
  contact_email: varchar("contact_email", { length: 255 }),
  contact_phone: varchar("contact_phone", { length: 50 }),
  type: varchar("type", { length: 50 }).notNull().default("lead_gen"),
  status: varchar("status", { length: 20 }).notNull().default("active"),
  notes: text("notes"),
  created_at: timestamp("created_at").defaultNow().notNull(),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export const insertVendorSchema = createInsertSchema(vendorsTable).omit({
  id: true,
  created_at: true,
  updated_at: true,
});

export type InsertVendor = z.infer<typeof insertVendorSchema>;
export type Vendor = typeof vendorsTable.$inferSelect;
```


lib/db/src/schema/workflow_settings.ts

```
import { pgTable, serial, varchar, text, timestamp, integer, boolean, jsonb } from "drizzle-orm/pg-core";
import { createInsertSchema } from "drizzle-zod";
import { z } from "zod/v4";

/**
 * Singleton-style settings for the auto-document workflow.
 * One row per "scope" – scope="global" is the default; scope="buyer:<id>" overrides per buyer.
 * Buyer-level rows take precedence; missing fields fall through to global.
 */
export const workflowSettingsTable = pgTable("workflow_settings", {
  id: serial("id").primaryKey(),
  scope: varchar("scope", { length: 100 }).notNull().unique(),
  esign_provider_integration_id: integer("esign_provider_integration_id"),
  fax_provider_integration_id: integer("fax_provider_integration_id"),
  email_provider_integration_id: integer("email_provider_integration_id"),
  default_email_from_name: varchar("default_email_from_name", { length: 255 }),
  default_email_from_address: varchar("default_email_from_address", { length: 255 }),
  default_fax_from_number: varchar("default_fax_from_number", { length: 50 }),
  auto_send_on_approval: boolean("auto_send_on_approval").notNull().default(true),
  auto_fax_doctor_on_signed_hipaa: boolean("auto_fax_doctor_on_signed_hipaa").notNull().default(true),
  esign_resend_after_days: integer("esign_resend_after_days").notNull().default(3),
  esign_expire_after_days: integer("esign_expire_after_days").notNull().default(14),
  max_send_retries: integer("max_send_retries").notNull().default(3),
  notify_on_failure_email: varchar("notify_on_failure_email", { length: 255 }),
  med_records_cover_letter_template_id: integer("med_records_cover_letter_template_id"),
  metadata: jsonb("metadata").$type<Record<string, unknown>>().notNull().default({}),
  notes: text("notes"),
  updated_by_user_id: integer("updated_by_user_id"),
  created_at: timestamp("created_at").defaultNow().notNull(),
  updated_at: timestamp("updated_at").defaultNow().notNull(),
});

export const insertWorkflowSettingsSchema = createInsertSchema(workflowSettingsTable).omit({
  id: true,
  created_at: true,
  updated_at: true,
});

export type WorkflowSettings = typeof workflowSettingsTable.$inferSelect;
export type InsertWorkflowSettings = z.infer<typeof insertWorkflowSettingsSchema>;
```

lib/db/tsconfig.json

```
{
  "extends": "../../tsconfig.base.json",
  "compilerOptions": {
    "composite": true,
    "declarationMap": true,
    "emitDeclarationOnly": true,
    "outDir": "dist",
    "rootDir": "src",
    "types": ["node"]
  },
  "include": ["src"]
}
```

lib/integrations-anthropic-ai/package.json

```
{
  "name": "@workspace/integrations-anthropic-ai",
  "version": "0.0.0",
  "private": true,
  "type": "module",
  "exports": {
    ".": "./src/index.ts",
    "./batch": "./src/batch/index.ts"
  },
  "dependencies": {
    "@anthropic-ai/sdk": "^0.78.0",
    "p-limit": "^7.3.0",
    "p-retry": "^7.1.1"
  },
  "devDependencies": {
    "@types/node": "catalog:"
  }
}
```

[lib/integrations-anthropic-ai/src/batch/index.ts](#)

```
export {  
  batchProcess,  
  batchProcessWithSSE,  
  isRateLimitError,  
  type BatchOptions,  
} from "./utils";
```

lib/integrations-anthropic-ai/src/batch/utils.ts

```
import pLimit from "p-limit";
import pRetry, { AbortError } from "p-retry";

/**
 * Batch Processing Utilities
 *
 * Generic batch processing with built-in rate limiting and automatic retries.
 * Use for any task that requires processing multiple items through an LLM or external API.
 *
 * USAGE:
 * ```typescript
 * import { batchProcess } from "@workspace/integrations-anthropic-ai/batch";
 * import { anthropic } from "@workspace/integrations-anthropic-ai";
 *
 * const results = await batchProcess(
 *   artworks,
 *   async (artwork) => {
 *     const message = await anthropic.messages.create({
 *       model: "claude-sonnet-4-6",
 *       max_tokens: 8192,
 *       messages: [{ role: "user", content: `Categorize: ${artwork.name}` }],
 *     });
 *     const block = message.content[0];
 *     return block.type === "text" ? block.text : "";
 *   },
 *   { concurrency: 2, retries: 5 }
 * );
 */

export interface BatchOptions {
  concurrency?: number;
  retries?: number;
  minTimeout?: number;
  maxTimeout?: number;
  onProgress?: (completed: number, total: number, item: unknown) => void;
}

export function isRateLimitError(error: unknown): boolean {
  const errorMsg = error instanceof Error ? error.message : String(error);
  return (
    errorMsg.includes("429") ||
    errorMsg.includes("RATELIMIT_EXCEEDED") ||
    errorMsg.toLowerCase().includes("quota") ||
    errorMsg.toLowerCase().includes("rate limit")
  );
}

export async function batchProcess<T, R>(
  items: T[],
  processor: (item: T, index: number) => Promise<R>,
  options: BatchOptions = {}
): Promise<R[]> {
  const {
    concurrency = 2,
    retries = 7,
    minTimeout = 2000,
    maxTimeout = 128000,
    onProgress,
  } = options;

  const limit = pLimit(concurrency);
  let completed = 0;

  const promises = items.map((item, index) =>
    limit(() =>
      pRetry(
        async () => {
          try {
            const result = await processor(item, index);
            completed++;
            onProgress?.(completed, items.length, item);
            return result;
          } catch (error: unknown) {
            if (isRateLimitError(error)) {
              throw error;
            }
            throw new AbortError(
              error instanceof Error ? error : new Error(String(error))
            );
          }
        },
        { retries, minTimeout, maxTimeout, factor: 2 }
      )
    )
  );

  return Promise.all(promises);
}

export async function batchProcessWithSSE<T, R>(
  items: T[],
  processor: (item: T, index: number) => Promise<R>,
  sendEvent: (event: { type: string; [key: string]: unknown }) => void,
  options: Omit<BatchOptions, "concurrency" | "onProgress"> = {}
)
```

```

): Promise<R[]> {
  const { retries = 5, minTimeout = 1000, maxTimeout = 15000 } = options;

  sendEvent({ type: "started", total: items.length });

  const results: R[] = [];
  let errors = 0;

  for (let index = 0; index < items.length; index++) {
    const item = items[index];
    sendEvent({ type: "processing", index, item });

    try {
      const result = await pRetry(
        () => processor(item, index),
        {
          retries,
          minTimeout,
          maxTimeout,
          factor: 2,
          onFailedAttempt: (error) => {
            if (!isRateLimitError(error)) {
              throw new AbortError(
                error instanceof Error ? error : new Error(String(error))
              );
            }
          }
        }
      );
      results.push(result);
      sendEvent({ type: "progress", index, result });
    } catch (error) {
      errors++;
      results.push(undefined as R);
      sendEvent({
        type: "progress",
        index,
        error: error instanceof Error ? error.message : "Processing failed",
      });
    }
  }

  sendEvent({ type: "complete", processed: items.length, errors });
  return results;
}

```

[lib/integrations-anthropic-ai/src/client.ts](#)

```
import Anthropic from "@anthropic-ai/sdk";

if (!process.env.AI_INTEGRATIONS_ANTHROPIC_BASE_URL) {
  throw new Error(
    "AI_INTEGRATIONS_ANTHROPIC_BASE_URL must be set. Did you forget to provision the Anthropic AI integration?",
  );
}

if (!process.env.AI_INTEGRATIONS_ANTHROPIC_API_KEY) {
  throw new Error(
    "AI_INTEGRATIONS_ANTHROPIC_API_KEY must be set. Did you forget to provision the Anthropic AI integration?",
  );
}

export const anthropic = new Anthropic({
  apiKey: process.env.AI_INTEGRATIONS_ANTHROPIC_API_KEY,
  baseUrl: process.env.AI_INTEGRATIONS_ANTHROPIC_BASE_URL,
});
```

[lib/integrations-anthropic-ai/src/index.ts](#)

```
export { anthropic } from "./client";  
export { batchProcess, batchProcessWithSSE, isRateLimitError, type BatchOptions } from "./batch";
```


lib/integrations-anthropic-ai/tsconfig.json

```
{
  "extends": "../../tsconfig.base.json",
  "compilerOptions": {
    "composite": true,
    "declarationMap": true,
    "emitDeclarationOnly": true,
    "outDir": "dist",
    "rootDir": "src",
    "types": ["node"]
  },
  "include": ["src"]
}
```

[lib/integrations/anthropic ai integrations/src/server/batch/index.ts](#)

```
export {  
  batchProcess,  
  batchProcessWithSSE,  
  isRateLimitError,  
  type BatchOptions,  
} from "./utils";
```

lib/integrations/anthropic ai integrations/src/server/batch/utils.ts

```
import pLimit from "p-limit";
import pRetry from "p-retry";

/**
 * Batch Processing Utilities
 *
 * This module provides a generic batch processing function with built-in
 * rate limiting and automatic retries. Use it for any task that requires
 * processing multiple items through an LLM or external API.
 *
 * USAGE:
 * ```typescript
 * import { batchProcess, isRateLimitError } from "../replit_integrations/batch";
 *
 * const results = await batchProcess(
 *   artworks,
 *   async (artwork) => {
 *     // Your custom LLM logic here
 *     const response = await openai.chat.completions.create({
 *       model: "gpt-5.1",
 *       messages: [{ role: "user", content: `Categorize: ${artwork.name}` }],
 *       response_format: { type: "json_object" },
 *     });
 *     return JSON.parse(response.choices[0]?.message?.content || "{}");
 *   },
 *   { concurrency: 2, retries: 5 }
 * );
 * ```
 */

export interface BatchOptions {
  /** Max concurrent requests (default: 2) */
  concurrency?: number;
  /** Max retry attempts for rate limit errors (default: 7) */
  retries?: number;
  /** Initial retry delay in ms (default: 2000) */
  minTimeout?: number;
  /** Max retry delay in ms (default: 128000) */
  maxTimeout?: number;
  /** Callback for progress updates */
  onProgress?: (completed: number, total: number, item: unknown) => void;
}

/**
 * Check if an error is a rate limit or quota violation.
 * Use this in custom error handling if needed.
 */
export function isRateLimitError(error: unknown): boolean {
  const errorMsg = error instanceof Error ? error.message : String(error);
  return (
    errorMsg.includes("429") ||
    errorMsg.includes("RATELIMIT_EXCEEDED") ||
    errorMsg.toLowerCase().includes("quota") ||
    errorMsg.toLowerCase().includes("rate limit")
  );
}

/**
 * Process items in batches with rate limiting and automatic retries.
 *
 * @param items - Array of items to process
 * @param processor - Async function to process each item (write your LLM logic here)
 * @param options - Concurrency and retry settings
 * @returns Promise resolving to array of results in the same order as input
 */
export async function batchProcess<T, R>(
  items: T[],
  processor: (item: T, index: number) => Promise<R>,
  options: BatchOptions = {}
): Promise<R[]> {
  const {
    concurrency = 2,
    retries = 7,
    minTimeout = 2000,
    maxTimeout = 128000,
    onProgress,
  } = options;

  const limit = pLimit(concurrency);
  let completed = 0;

  const promises = items.map((item, index) =>
    limit(() =>
      pRetry(
        async () => {
          try {
            const result = await processor(item, index);
            completed++;
            onProgress?.(completed, items.length, item);
            return result;
          } catch (error: unknown) {
            if (isRateLimitError(error)) {
              throw error; // Rethrow to trigger p-retry
            }
            // For non-rate-limit errors, abort immediately
          }
        }
      )
    )
  );
}
```

```

        throw new pRetry.AbortError(
            error instanceof Error ? error : new Error(String(error))
        );
    }
    },
    { retries, minTimeout, maxTimeout, factor: 2 }
)
);
}

return Promise.all(promises);
}

/**
 * Process items sequentially with SSE progress streaming.
 * Use this when you need real-time progress updates to the client.
 *
 * @param items - Array of items to process
 * @param processor - Async function to process each item
 * @param sendEvent - Function to send SSE events to the client
 * @param options - Retry settings (concurrency is always 1 for sequential)
 */
export async function batchProcessWithSSE<T, R>({
    items: T[],
    processor: (item: T, index: number) => Promise<R>,
    sendEvent: (event: { type: string; [key: string]: unknown }) => void,
    options: Omit<BatchOptions, "concurrency" | "onProgress"> = {}
}): Promise<R[]> {
    const { retries = 5, minTimeout = 1000, maxTimeout = 15000 } = options;

    sendEvent({ type: "started", total: items.length });

    const results: R[] = [];
    let errors = 0;

    for (let index = 0; index < items.length; index++) {
        const item = items[index];
        sendEvent({ type: "processing", index, item });

        try {
            const result = await pRetry(
                () => processor(item, index),
                {
                    retries,
                    minTimeout,
                    maxTimeout,
                    factor: 2,
                    onFailedAttempt: (error) => {
                        if (!isRateLimitError(error)) {
                            throw new pRetry.AbortError(
                                error instanceof Error ? error : new Error(String(error))
                            );
                        }
                    },
                },
            );
            results.push(result);
            sendEvent({ type: "progress", index, result });
        } catch (error) {
            errors++;
            results.push(undefined as R); // Placeholder for failed items
            sendEvent({
                type: "progress",
                index,
                error: error instanceof Error ? error.message : "Processing failed",
            });
        }
    }

    sendEvent({ type: "complete", processed: items.length, errors });
    return results;
}

```

package.json

```
{
  "name": "workspace",
  "version": "0.0.0",
  "license": "MIT",
  "scripts": {
    "preinstall": "sh -c 'rm -f package-lock.json yarn.lock; case \"$npm_config_user_agent\" in pnpm/*) ;; *) echo \"Use pnpm instead\" >&2; exit 1 ;; esac'",
    "build": "pnpm run typecheck && pnpm -r --if-present run build",
    "typecheck:libs": "tsc --build",
    "typecheck": "pnpm run typecheck:libs && pnpm -r --filter \"./artifacts/**\" --filter \"./scripts\" --if-present run typecheck"
  },
  "private": true,
  "pnpm": {
    "onlyBuiltDependencies": [
      "sharp",
      "esbuild",
      "@img/sharp-libvips-linux-x64"
    ]
  },
  "devDependencies": {
    "prettier": "^3.8.1",
    "typescript": "~5.9.2"
  },
  "dependencies": {
    "pdfkit": "^0.18.0"
  }
}
```

pnpm-workspace.yaml

```
# =====
# SECURITY: Minimum release age for npm packages (supply-chain attack defense)
# =====
#
# This setting requires that any npm package version must have been published
# for at least 1 day (1440 minutes) before pnpm will allow installing it.
# This is a critical defense against supply-chain attacks. In most cases,
# malicious npm releases are discovered and pulled within hours, so a 1-day
# delay provides a strong safety buffer.
#
# DO NOT DISABLE THIS SETTING. Removing or setting it to 0 is considered
# extremely dangerous and leaves the entire workspace vulnerable to supply-
# chain attacks, which have been the #1 vector for npm ecosystem compromises.
#
# If you absolutely need to install a package before the 1-day window has
# passed (e.g. an urgent security bugfix), you can add it to the
# `minimumReleaseAgeExclude` allowlist below. Only consider doing this for
# packages released by trusted organizations with an impeccable security
# posture (e.g. Replit packages, react from Meta, typescript from Microsoft). Even then,
# remove the exclusion once the 1-day window has passed.
#
# Example:
#   minimumReleaseAgeExclude:
#     - react
#     - typescript
#
# =====
minimumReleaseAge: 1440

minimumReleaseAgeExclude:
  # Exclude @replit scoped packages from the minimum release age check.
  # These are published by Replit and trusted – the supply-chain attack vector
  # this setting guards against does not apply to our own packages.
  - '@replit/*'
  - stripe-replit-sync

packages:
  - artifacts/*
  - lib/*
  - lib/integrations/*
  - scripts

catalog:
  '@replit/vite-plugin-cartographer': ^0.5.1
  '@replit/vite-plugin-dev-banner': ^0.1.1
  '@replit/vite-plugin-runtime-error-modal': ^0.0.6
  '@tailwindcss/vite': ^4.1.14
  '@tanstack/react-query': ^5.90.21
  '@types/node': ^25.3.3
  '@types/react': ^19.2.0
  '@types/react-dom': ^19.2.0
  '@vitejs/plugin-react': ^5.0.4
  class-variance-authority: ^0.7.1
  clsx: ^2.1.1
  drizzle-orm: ^0.45.1
  framer-motion: ^12.23.24
  lucide-react: ^0.545.0
  # Must be this exact version because expo requires it
  react: 19.1.0
  # Must be this exact version because expo requires it
  react-dom: 19.1.0
  tailwind-merge: ^3.3.1
  tailwindcss: ^4.1.14
  tsx: ^4.21.0
  vite: ^7.3.0
  zod: ^3.25.76

autoInstallPeers: false

onlyBuiltDependencies:
  - '@swc/core'
  - esbuild
  - msw
  - unrs-resolver

overrides:
  # replit uses linux-x64 only, we can exclude all other platforms
  'esbuild>@esbuild/darwin-arm64': "-"
  'esbuild>@esbuild/darwin-x64': "-"
  'esbuild>@esbuild/freebsd-arm64': "-"
  'esbuild>@esbuild/freebsd-x64': "-"
  'esbuild>@esbuild/linux-arm': "-"
  'esbuild>@esbuild/linux-arm64': "-"
  'esbuild>@esbuild/linux-ia32': "-"
  'esbuild>@esbuild/linux-loong64': "-"
  'esbuild>@esbuild/linux-mips64el': "-"
  'esbuild>@esbuild/linux-ppc64': "-"
  'esbuild>@esbuild/linux-riscv64': "-"
  'esbuild>@esbuild/linux-s390x': "-"
  'esbuild>@esbuild/netbsd-arm64': "-"
  'esbuild>@esbuild/netbsd-x64': "-"
  'esbuild>@esbuild/openbsd-arm64': "-"
  'esbuild>@esbuild/openbsd-x64': "-"
  'esbuild>@esbuild/sunos-x64': "-"
  'esbuild>@esbuild/win32-arm64': "-"
```

```

"esbuild>@esbuild/win32-ia32": "-."
"esbuild>@esbuild/win32-x64": "-."
"esbuild>@esbuild/aix-ppc64": "-."
"esbuild>@esbuild/android-arm": "-."
"esbuild>@esbuild/android-arm64": "-."
"esbuild>@esbuild/android-x64": "-."
"esbuild>@esbuild/openharmony-arm64": "-."
"lightningcss>lightningcss-android-arm64": "-."
"lightningcss>lightningcss-darwin-arm64": "-."
"lightningcss>lightningcss-darwin-x64": "-."
"lightningcss>lightningcss-freebsd-x64": "-."
"lightningcss>lightningcss-linux-arm-gnueabi": "-."
"lightningcss>lightningcss-linux-arm64-gnu": "-."
"lightningcss>lightningcss-linux-arm64-musl": "-."
"lightningcss>lightningcss-linux-x64-musl": "-."
"lightningcss>lightningcss-win32-arm64-msvc": "-."
"lightningcss>lightningcss-win32-x64-msvc": "-."
"@tailwindcss/oxide>@tailwindcss/oxide-android-arm64": "-."
"@tailwindcss/oxide>@tailwindcss/oxide-darwin-arm64": "-."
"@tailwindcss/oxide>@tailwindcss/oxide-darwin-x64": "-."
"@tailwindcss/oxide>@tailwindcss/oxide-freebsd-x64": "-."
"@tailwindcss/oxide>@tailwindcss/oxide-linux-arm-gnueabi": "-."
"@tailwindcss/oxide>@tailwindcss/oxide-linux-arm64-gnu": "-."
"@tailwindcss/oxide>@tailwindcss/oxide-linux-arm64-musl": "-."
"@tailwindcss/oxide>@tailwindcss/oxide-win32-arm64-msvc": "-."
"@tailwindcss/oxide>@tailwindcss/oxide-win32-x64-msvc": "-."
"@tailwindcss/oxide>@tailwindcss/oxide-linux-x64-musl": "-."
"rollup>@rollup/rollup-android-arm-eabi": "-."
"rollup>@rollup/rollup-android-arm64": "-."
"rollup>@rollup/rollup-darwin-arm64": "-."
"rollup>@rollup/rollup-darwin-x64": "-."
"rollup>@rollup/rollup-freebsd-arm64": "-."
"rollup>@rollup/rollup-freebsd-x64": "-."
"rollup>@rollup/rollup-linux-arm-gnueabi": "-."
"rollup>@rollup/rollup-linux-arm-musleabi": "-."
"rollup>@rollup/rollup-linux-arm64-gnu": "-."
"rollup>@rollup/rollup-linux-arm64-musl": "-."
"rollup>@rollup/rollup-linux-loong64-gnu": "-."
"rollup>@rollup/rollup-linux-loong64-musl": "-."
"rollup>@rollup/rollup-linux-ppc64-gnu": "-."
"rollup>@rollup/rollup-linux-ppc64-musl": "-."
"rollup>@rollup/rollup-linux-riscv64-gnu": "-."
"rollup>@rollup/rollup-linux-riscv64-musl": "-."
"rollup>@rollup/rollup-linux-s390x-gnu": "-."
"rollup>@rollup/rollup-linux-x64-musl": "-."
"rollup>@rollup/rollup-openbsd-x64": "-."
"rollup>@rollup/rollup-openharmony-arm64": "-."
"rollup>@rollup/rollup-win32-arm64-msvc": "-."
"rollup>@rollup/rollup-win32-ia32-msvc": "-."
"rollup>@rollup/rollup-win32-x64-gnu": "-."
"rollup>@rollup/rollup-win32-x64-msvc": "-."
"@expo/ngrok-bin>@expo/ngrok-bin-darwin-arm64": "-."
"@expo/ngrok-bin>@expo/ngrok-bin-darwin-x64": "-."
"@expo/ngrok-bin>@expo/ngrok-bin-freebsd-ia32": "-."
"@expo/ngrok-bin>@expo/ngrok-bin-freebsd-x64": "-."
"@expo/ngrok-bin>@expo/ngrok-bin-linux-arm64": "-."
"@expo/ngrok-bin>@expo/ngrok-bin-linux-arm": "-."
"@expo/ngrok-bin>@expo/ngrok-bin-linux-ia32": "-."
"@expo/ngrok-bin>@expo/ngrok-bin-sunos-x64": "-."
"@expo/ngrok-bin>@expo/ngrok-bin-win32-ia32": "-."
"@expo/ngrok-bin>@expo/ngrok-bin-win32-x64": "-."
# drizzle-kit uses esbuild internally on an older version that's vulnerable, this overrides it
"@esbuild-kit/esm-loader": "npm:tsx@^4.21.0"
esbuild: "0.27.3"

```

[replit.md](#)

Overview

This project is a full-stack Mass Tort Operating System (MTOS), a distributed case processing CRM designed for mass tort law firms. It aims to streamline the entire legal process from lead intake and document management to case analysis, compliance, and distributed processing. The system incorporates AI for medical document extraction, advanced validation engines, and robust conflict resolution. It features a sophisticated form engine for compliant lead generation across 24 tort campaigns and a comprehensive analytics suite. The core business vision is to provide a scalable, efficient, and compliant solution to manage complex mass tort litigation, significantly improving operational efficiency and compliance for law firms.

User Preferences

I prefer clear and direct communication. I value a development process that emphasizes iterative development and early feedback. Please ask for my approval before implementing any major architectural changes or significant feature additions. I appreciate detailed explanations for complex technical decisions.

System Architecture

The project is structured as a pnpm monorepo using TypeScript, targeting Node.js 24. The backend is an Express 5 API, integrated with a PostgreSQL database using Drizzle ORM. API codegen is handled by Orval from an OpenAPI specification. The UI/UX is built with React and Vite.

Core Architectural Patterns & Design Decisions:

- * **Monorepo Structure**: Facilitates shared code, consistent tooling, and simplified dependency management across services.
- * **Distributed Architecture**: Employs a PostgreSQL job queue for asynchronous task processing by a dedicated worker process, enabling scalable and fault-tolerant case analysis and document ingestion.
- * **Deterministic Scoring Engine**: A rule-based scoring system provides transparent and auditable case qualification.
- * **File Vault**: Secure storage for case-related documents with SHA-256 hashing for integrity verification.
- * **AI Integration**: Utilizes Anthropic Claude for medical document extraction and OCR, enhancing automated data processing.
- * **Conflict Resolution & Error Fallback**: A robust system detects conflicts, provides fail-safe modes, and an error fallback mechanism with retry logic, ensuring system stability and data quality. All failures are logged to an audit trail and a review queue.
- * **Form Engine**: A comprehensive engine for TCPA and TrustedForm compliant lead generation, featuring a tort-based form builder, embeddable JS script, live validation, background checks, and a 10-step submission pipeline with a "Final Arbiter."
- * **Taxonomy Engine**: Matches NPI provider taxonomies with medical diagnoses to identify potential mismatches.
- * **Fraud Engine**: Flags potential fraud indicators, deferring to the Final Arbiter for resolution.
- * **Compliance & Auditability**: Extensive logging of all actions and conflicts to an 'audit_log' table, along with compliance-specific fields in lead records.

Key Features & Implementations:

- * **Dashboards**: Overview of pipeline stats, CPSR, pipeline charts, and activity feeds.
- * **Lead & Case Management**: Comprehensive features for lead intake, qualification, detailed views, document association, and distributed case processing with AI analysis and audit trails.
- * **Document Management**: Centralized document view, OCR inbox, AI document analysis for structured data extraction, PDF redaction/highlighting, side-by-side document review, medical timeline builder, and AI document drafting.
- * **Paralegal Management**: Tools for managing paralegal teams, performance tracking, and leaderboards.
- * **Analytics**: KPIs, conversion funnels, trend charts, and tort breakdown analysis, including Praxis AI Predictive Analytics for scoring.
- * **Vendor Management**: Full CRUD for vendors (lead gen, law firm, marketing, referral types) with status tracking.
- * **CSV Lead Import**: Bulk lead ingestion with auto-column mapping, preview, deduplication, conflict detection, encryption of sensitive fields, and detailed error tracking.
- * **Form Engine**: DB-backed 'form_configurations' table seeded from TORT_REGISTRY on boot (auto-refreshes registry-managed fields when 'updated_by' IS NULL, never overwrites admin edits). Admins can edit valid diagnoses and add/remove custom fields per tort via the Form Engine UI. Custom field values flow through embed forms !' 'leads.custom_fields' JSONB !' lead detail (CustomFieldsCard) !' CSV export (JSON-serialized). Real iframe live preview at '/api/forms/preview/:tortId'. 31 torts including 2026 MDL updates: Roundup (16 dx), AFFF (13 dx incl. UC/hypothyroid/breast cancer per MDL 2873), and new Tepezza (MDL 3079).
- * **Role-Based Access Control (RBAC)**: JWT (HS256) authentication with role hierarchy (admin > attorney > paralegal > viewer) and granular route-level access control.
- * **Security Infrastructure**: Comprehensive security layer protecting ePHI/PII data, including token revocation, refresh tokens, MFA/TOTP, password complexity, account lockout, row-level isolation, field-level encryption with key rotation and AAD, security headers, rate limiting, security event alerting, path traversal protection, information disclosure prevention, business logic hardening, Intrusion Detection System (IDS), and AI threat analysis.
- * **Auto-Document Workflow (MTOS)**: Lead qualification automatically dispatches an e-sign packet (HIPAA + Affidavit + Retainer) per per-buyer/per-tort template toggles; on signature webhook return, an automatic doctor fax for medical records is queued. Vendor-agnostic via 'provider-router' and the existing 'integrations' table (Dropbox Sign + DocuSign for e-sign, Telnyx for fax, SendGrid for email). Admin pages: Buyers, Document Templates (with PDF upload to App Storage), Template Assignments matrix, Workflow Settings (vendor pickers). Lead detail has an "Automation" tab showing envelope + fax timeline. Webhook routes are mounted PUBLIC (before auth middleware) and strictly verify HMAC signatures before mutating state – bad/missing signature !' 200 ack, no-op. Worker handlers are idempotent (skip if envelope already exists for lead+template) and wrap every adapter call in try/catch so the envelope/fax row always reflects the real outcome.

External Dependencies

- * **Database**: PostgreSQL
- * **ORM**: Drizzle ORM
- * **AI**: Anthropic Claude
- * **Image Processing**: Sharp
- * **Validation**: Zod
- * **API Codegen**: Orval
- * **Background Checks**: CourtListener, OFAC sanctions list
- * **NPI Lookup**: NPPES API (CMS NPI Registry)
- * **Security**: Helmet.js, express-rate-limit
- * **News Feeds**: Google News RSS, Yahoo Finance RSS, MarketWatch RSS, CNBC RSS

